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(54) **ORGANIC ELECTROLUMINESCENT
COMPOUND, A PLURALITY OF HOST
MATERIALS AND ORGANIC
ELECTROLUMINESCENT COMPRISING
THE SAME**

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(57) **ABSTRACT**

The present disclosure relates to a plurality of host materials comprising at least one first host compound represented by Formula 1 and at least one second host compound represented by Formula 2, wherein the first host compound and the second compound are different from each other; an organic electroluminescent compound represented by Formula 11; and an organic electroluminescent device comprising the same. It is possible to produce an organic electroluminescent device having improved driving voltage, efficiency, and/or lifetime properties, by comprising the organic electroluminescent compound according to the present disclosure or a specific combination of compounds according to the present disclosure as host materials.

**ORGANIC ELECTROLUMINESCENT
COMPOUND, A PLURALITY OF HOST
MATERIALS AND ORGANIC
ELECTROLUMINESCENT COMPRISING
THE SAME**

TECHNICAL FIELD

[0001] The present disclosure relates to an organic electroluminescent compound, a plurality of host materials, and an organic electroluminescent device comprising the same.

BACKGROUND ART

[0002] In 1987, Tang et al. of Eastman Kodak first developed a small-molecule green organic electroluminescent device (OLED) of TPD/Alq₃ bilayer consisting of a light-emitting layer and a charge transport layer. Since then, research on OLEDs has been rapidly carried out, and OLEDs have been commercialized. At present, phosphorescent materials, which provide excellent luminous efficiency in realizing panels, are mainly used in OLEDs. In various applications such as TVs and lighting, the lifetime of OLEDs is often insufficient, and higher OLED efficiency is still required. Generally, the lifetime of an OLED decreases as its luminance increases. Thus, OLEDs with high luminous efficiency and/or long lifetime are essential for long-term use and high-resolution displays.

[0003] In order to improve luminous efficiency, driving voltage, and/or lifetime, various materials or concepts for an organic layer of an organic electroluminescent device have been proposed, but these have not proved to be satisfactory in practical use. In addition, there is a continuous demand for the development of an organic electroluminescent material with enhanced performance, such as improved driving voltage, luminous efficiency, power efficiency, and/or lifetime properties, as compared to a previously disclosed combination of specific compounds.

[0004] Meanwhile, Korean Patent Application Laid-Open No. 2006-0108642 (published on Oct. 18, 2006) discloses asymmetric monoanthracene derivatives, Korean Patent Application Laid-Open No. 2015-0141271 (published on Dec. 18, 2015) discloses an organic electroluminescent device comprising a combination of anthracene derivatives, and Korean Patent Application Laid-Open No. 2020-0034649 (published on Mar. 31, 2020) discloses a compound containing an anthracene structure as a host material. However, the aforementioned references do not specifically disclose a specific compound or the host materials comprising a specific combination of compounds according to the present disclosure. In addition, there is a continuing need to develop light-emitting materials with improved performance, such as improved driving voltage, luminous efficiency, power efficiency, and/or lifetime properties, compared to previously disclosed compounds.

DISCLOSURE OF INVENTION

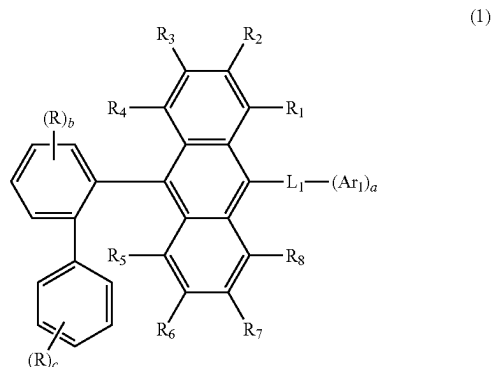
Technical Problem

[0005] The objective of the present disclosure is to provide an organic electroluminescent compound having a novel

structure suitable for applying to an organic electroluminescent device. Another objective of the present disclosure is to provide a plurality of host materials capable of producing an organic electroluminescent device having low driving voltage, high efficiency, and/or improved lifetime properties. Another objective of the present disclosure is to provide an organic electroluminescent device having low driving voltage, high efficiency, and/or improved lifetime properties by comprising a compound according to the present disclosure as a single host material or a specific combination of compounds according to the present disclosure as a plurality of host materials.

Solution to Problem

[0006] As a result of intensive studies conducted to solve the above technical problems, the present inventors have found that the above objective can be achieved through a plurality of host materials comprising at least one first host compound represented by the following Formula 1 and at least one second host compound represented by the following Formula 2, wherein the first host compound and the second compound are different from each other; or an organic electroluminescent compound represented by the following Formula 11; and a organic electroluminescent device comprising the same.



[0007] In Formula 1,

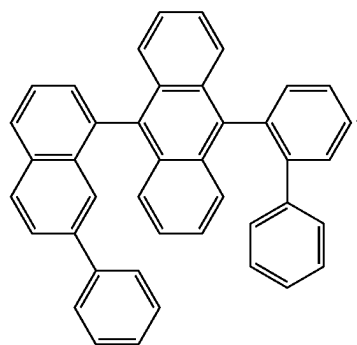
[0008] L₁ represents a single bond, a phenylene unsubstituted or substituted with deuterium, a naphthylene unsubstituted or substituted with deuterium, or a phenanthrenylene unsubstituted or substituted with deuterium;

[0009] Ar₁ each independently represents a substituted or unsubstituted (C₆-C₃₀)aryl, or a substituted or unsubstituted (3- to 30-membered)heteroaryl;

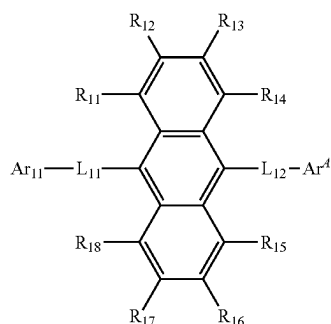
[0010] R and R₁ to R₈ each independently represent hydrogen or deuterium; and

[0011] a represents an integer of 1 or 2, b represents an integer of 4, and c represents an integer of 5; and each Ar₁ may be the same as or different from each other;

[0012] with a proviso that Formula 1 is not represented as

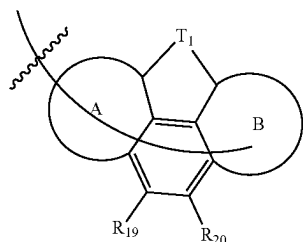


(2)



[0013] In Formula 2,

[0014] Ar^d represents a substituted or unsubstituted (C_6-C_{30})aryl, a substituted or unsubstituted (13- to 30-membered)heteroaryl, or the following Formula A-1:



(A-1)

[0015] wherein T_1 represents O, S, or CR_aR_b ;

[0016] Ring A and Ring B each independently represent a substituted or unsubstituted (C_6-C_{30})arene, or a substituted or unsubstituted (3- to 30-membered) heteroarene;

[0017] Ar_{11} represents a substituted or unsubstituted (C_6-C_{30})aryl, or a substituted or unsubstituted (13- to 30-membered)heteroaryl;

[0018] R_{11} to R_{18} each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1-C_{30})alkyl, a substituted or unsubstituted (C_2-C_{30})alkenyl, a substituted or unsubstituted (C_6-C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3-C_{30})cycloalkyl, a substituted or unsubstituted (C_3-C_{30})cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C_1-C_{30})alkoxy, a substituted or unsubstituted

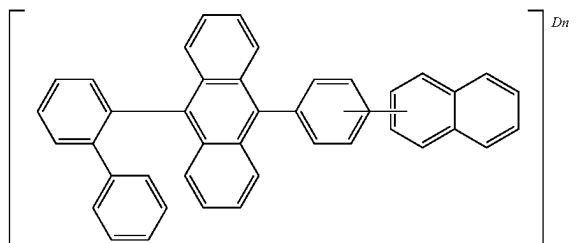
tri(C_1-C_{30})alkylsilyl, a substituted or unsubstituted di(C_1-C_{30})alkyl(C_6-C_{30})arylsilyl, a substituted or unsubstituted (C_1-C_{30})alkyldi(C_6-C_{30})arylsilyl, a substituted or unsubstituted tri(C_6-C_{30})arylsilyl, a substituted or unsubstituted fused ring group of a (C_3-C_{30}) aliphatic ring(s) and a (C_6-C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$;

[0019] R_{19} and R_{20} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1-C_{30})alkyl, a substituted or unsubstituted (C_2-C_{30})alkenyl, a substituted or unsubstituted (C_6-C_{30})aryl, a substituted or unsubstituted (3- to 30-membered) heteroaryl, a substituted or unsubstituted (C_3-C_{30})cycloalkyl, a substituted or unsubstituted (C_3-C_{30})cycloalkenyl, a substituted or unsubstituted (3- to 7-membered) heterocycloalkyl, a substituted or unsubstituted (C_1-C_{30})alkoxy, a substituted or unsubstituted tri(C_1-C_{30})alkylsilyl, a substituted or unsubstituted di(C_1-C_{30})alkyl(C_6-C_{30})arylsilyl, a substituted or unsubstituted tri(C_6-C_{30})arylsilyl, a substituted or unsubstituted fused ring group of a (C_3-C_{30}) aliphatic ring(s) and a (C_6-C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$;

[0020] R_a and R_b each independently represent a substituted or unsubstituted (C_1-C_{30})alkyl, or a substituted or unsubstituted (C_6-C_{30})aryl, or may be linked to each other to form a ring(s);

[0021] L_{11} to L_{13} each independently represent a single bond, a substituted or unsubstituted (C_6-C_{30})arylene, or a substituted or unsubstituted (3- to 30-membered) heteroarylene; and

[0022] Ar_{13} and Ar_{14} each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1-C_{30})alkyl, a substituted or unsubstituted (C_2-C_{30})alkenyl, a substituted or unsubstituted (C_6-C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3-C_{30})cycloalkyl, a substituted or unsubstituted (C_3-C_{30})cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C_1-C_{30})alkoxy, a substituted or unsubstituted tri(C_1-C_{30})alkylsilyl, a substituted or unsubstituted di(C_1-C_{30})alkyl(C_6-C_{30})arylsilyl, a substituted or unsubstituted (C_1-C_{30})alkyldi(C_6-C_{30})arylsilyl, a substituted or unsubstituted tri(C_6-C_{30})arylsilyl, or a substituted or unsubstituted fused ring group of a (C_3-C_{30}) aliphatic ring(s) and a (C_6-C_{30}) aromatic ring(s).



[0023] In Formula 11, D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to 28.

Advantageous Effects of Invention

[0024] The organic electroluminescent compound according to the present disclosure exhibits performance suitable for use in an organic electroluminescent device. In addition, by comprising the compound according to the present disclosure as a single host material, or a specific combination of compounds according to the present disclosure as a plurality of host materials, an organic electroluminescent device can be produced which has lower driving voltage, higher efficiency, and/or improved lifetime properties compared to a conventional organic electroluminescent device, and a display device or lighting device using the same can be produced.

MODE FOR THE INVENTION

[0025] Hereinafter, the present disclosure will be described in detail. However, the following description is intended to explain the present disclosure and is not meant in any way to restrict the scope of the present disclosure.

[0026] The present disclosure relates to a plurality of host materials comprising at least one first host compound represented by Formula 1 and at least one second host compound represented by Formula 2, wherein the first host compound and the second compound are different from each other; and an organic electroluminescent device comprising the host materials. In addition, the present disclosure relates to an organic electroluminescent compound represented by Formula 11, and an organic electroluminescent device comprising the same. Furthermore, the present disclosure relates to an organic electroluminescent device comprising a compound represented by Formula 1 as a host material(s).

[0027] The “organic electroluminescent compound” in the present disclosure is a compound that may be used in an organic electroluminescent device, and may be comprised in any material layer constituting an organic electroluminescent device, as necessary.

[0028] The “an organic electroluminescent material” in the present disclosure is a material that may be used in an organic electroluminescent device, and may comprise at least one compound. If necessary, the organic electroluminescent material may be comprised in any layer constituting an organic electroluminescent device. For example, the organic electroluminescent material may be a hole injection material, a hole transport material, a hole auxiliary material, a light-emitting auxiliary material, an electron-blocking material, a light-emitting material (including a host material and a dopant material), an electron buffer material, a hole-blocking material, an electron transport material, an electron injection material, etc.

[0029] The “plurality of organic electroluminescent materials” in the present disclosure is an organic electroluminescent material that is a combination of at least two compounds, which may be comprised in any layer constituting an organic electroluminescent device. It may be both a material before being comprised in an organic electroluminescent device (e.g., before vapor deposition) and a material after being comprised in an organic electroluminescent device (e.g., after vapor deposition). For example, the plurality of organic electroluminescent materials may be a combination of at least two compounds, which may be

comprised in at least one layer(s) of a hole injection layer, a hole transport layer, a hole auxiliary layer, a light-emitting auxiliary layer, an electron-blocking layer, a light-emitting layer, an electron buffer layer, a hole-blocking layer, an electron transport layer, and an electron injection layer. At least two of these compounds may be comprised in the same layer or different layers, and may be mixture-deposited, co-deposited, or separately deposited.

[0030] The “plurality of host materials” in the present disclosure is an organic electroluminescent material in which two or more host materials are combined. It may be both a material before being comprised in an organic electroluminescent device (e.g., before vapor deposition) and a material after being comprised in an organic electroluminescent device (e.g., after vapor deposition). The plurality of host materials of the present disclosure may be comprised in any light-emitting layer constituting an organic electroluminescent device. At least two compounds comprised in the plurality of host materials may be comprised together in one light-emitting layer or may each be comprised in different light-emitting layers. When two or more host materials are comprised in one layer, for example, they may be mixture-evaporated to form a layer, or they may be separately and simultaneously co-evaporated to form a layer.

[0031] Herein, the “(C_1 - C_{30})alkyl” is meant to be a linear or branched alkyl having 1 to 30 carbon atoms constituting a chain, in which the number of carbon atoms is preferably 1 to 20, and more preferably 1 to 10. The above alkyl may include methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, tert-butyl, sec-butyl, etc. Herein, the “(C_3 - C_{30})cycloalkyl” is meant to be a mono- or polycyclic hydrocarbon having 3 to 30 ring backbone carbon atoms, in which the number of ring backbone carbon atoms is preferably 3 to 20, and more preferably 3 to 7. The above cycloalkyl may include cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cyclopentylmethyl, cyclohexylmethyl, etc.

[0032] Herein, the “(C_6 - C_{30})aryl”, “(C_6 - C_{30})arylene”, or “(C_6 - C_{30})arene” is meant to be a monocyclic or fused ring-type radical derived from an aromatic hydrocarbon having 6 to 30 ring backbone carbon atoms that can be partially saturated, in which the number of ring backbone carbon atoms is preferably 6 to 20, and more preferably 6 to 15. The above aryl may comprise a spiro structure. The above aryl may include phenyl, biphenyl, terphenyl, quaterphenyl, naphthyl, binaphthyl, phenylnaphthyl, naphthylphenyl, fluorenyl, phenylfluorenyl, dimethylfluorenyl, diphenylfluorenyl, benzofluorenyl, diphenylbenzofluorenyl, dibenzofluorenyl, phenanthrenyl, benzophenanthrenyl, phenylphenanthrenyl, anthracenyl, benzoanthracenyl, indenyl, triphenylenyl, pyrenyl, tetracenyl, perylenyl, chrysenyl, benzochrysenyl, naphthacenyl, fluoranthrenyl, benzofluoranthrenyl, tolyl, xylyl, mesityl, cumenyl, spiro[fluorene-fluorenyl], spiro[fluorene-benzofluorenyl], azulenyl, tetramethyl-dihydrophenanthrenyl, etc. Specifically, the above aryl may include o-tolyl, m-tolyl, p-tolyl, 2,3-xylyl, 3,4-xylyl, 2,5-xylyl, mesityl, o-cumenyl, m-cumenyl, p-cumenyl, p-tert-butylphenyl, p-(2-phenylpropyl)phenyl, 4'-methylbiphenyl, 4"-tert-butyl-p-terphenyl-4-yl, o-biphenyl, m-biphenyl, p-biphenyl, o-terphenyl, m-terphenyl-4-yl, m-terphenyl-3-yl, m-terphenyl-2-yl, p-terphenyl-4-yl, p-terphenyl-3-yl, p-terphenyl-2-yl, m-quaterphenyl, 1-naphthyl, 2-naphthyl, 1-fluorenyl, 2-fluorenyl, 3-fluorenyl, 4-fluorenyl, 9-fluorenyl, 9,9-dimethyl-1-fluorenyl, 9,9-dimethyl-2-fluorenyl, 9,9-dimethyl-3-fluorenyl, 9,9-dimethyl-4-fluore-

nyl, 9,9-diphenyl-1-fluorenyl, 9,9-diphenyl-2-fluorenyl, 9,9-diphenyl-3-fluorenyl, 9,9-diphenyl-4-fluorenyl, 1-anthryl, 2-anthryl, 9-anthryl, 1-phenanthryl, 2-phenanthryl, 3-phenanthryl, 4-phenanthryl, 9-phenanthryl, 1-chrysenyl, 2-chrysenyl, 3-chrysenyl, 4-chrysenyl, 5-chrysenyl, 6-chrysenyl, benzo[c]phenanthryl, benzo[g]chrysenyl, 1-triphenylenyl, 2-triphenylenyl, 3-triphenylenyl, 4-triphenylenyl, 3-fluoranthenyl, 4-fluoranthenyl, 8-fluoranthenyl, 9-fluoranthenyl, benzo[fluoranthenyl, 11,11-dimethyl-1-benzo[a]fluorenyl, 11,11-dimethyl-2-benzo[a]fluorenyl, 11,11-dimethyl-3-benzo[a]fluorenyl, 11,11-dimethyl-4-benzo[a]fluorenyl, 11,11-dimethyl-5-benzo[a]fluorenyl, 11,11-dimethyl-6-benzo[a]fluorenyl, 11,11-dimethyl-7-benzo[a]fluorenyl, 11,11-dimethyl-8-benzo[a]fluorenyl, 11,11-dimethyl-9-benzo[a]fluorenyl, 11,11-dimethyl-10-benzo[a]fluorenyl, 11,11-dimethyl-1-benzo[b]fluorenyl, 11,11-dimethyl-2-benzo[b]fluorenyl, 11,11-dimethyl-3-benzo[b]fluorenyl, 11,11-dimethyl-4-benzo[b]fluorenyl, 11,11-dimethyl-5-benzo[b]fluorenyl, 11,11-dimethyl-6-benzo[b]fluorenyl, 11,11-dimethyl-7-benzo[b]fluorenyl, 11,11-dimethyl-8-benzo[b]fluorenyl, 11,11-dimethyl-9-benzo[b]fluorenyl, 11,11-dimethyl-10-benzo[b]fluorenyl, 11,11-dimethyl-1-benzo[c]fluorenyl, 11,11-dimethyl-2-benzo[c]fluorenyl, 11,11-dimethyl-3-benzo[c]fluorenyl, 11,11-dimethyl-4-benzo[c]fluorenyl, 11,11-dimethyl-5-benzo[c]fluorenyl, 11,11-dimethyl-6-benzo[c]fluorenyl, 11,11-dimethyl-7-benzo[c]fluorenyl, 11,11-dimethyl-8-benzo[c]fluorenyl, 11,11-dimethyl-9-benzo[c]fluorenyl, 11,11-dimethyl-10-benzo[c]fluorenyl, 11,11-diphenyl-1-benzo[a]fluorenyl, 11,11-diphenyl-2-benzo[a]fluorenyl, 11,11-diphenyl-3-benzo[a]fluorenyl, 11,11-diphenyl-4-benzo[a]fluorenyl, 11,11-diphenyl-5-benzo[a]fluorenyl, 11,11-diphenyl-6-benzo[a]fluorenyl, 11,11-diphenyl-7-benzo[a]fluorenyl, 11,11-diphenyl-8-benzo[a]fluorenyl, 11,11-diphenyl-9-benzo[a]fluorenyl, 11,11-diphenyl-10-benzo[a]fluorenyl, 11,11-diphenyl-1-benzo[b]fluorenyl, 11,11-diphenyl-2-benzo[b]fluorenyl, 11,11-diphenyl-3-benzo[b]fluorenyl, 11,11-diphenyl-4-benzo[b]fluorenyl, 11,11-diphenyl-5-benzo[b]fluorenyl, 11,11-diphenyl-6-benzo[b]fluorenyl, 11,11-diphenyl-7-benzo[b]fluorenyl, 11,11-diphenyl-8-benzo[b]fluorenyl, 11,11-diphenyl-9-benzo[b]fluorenyl, 11,11-diphenyl-10-benzo[b]fluorenyl, 11,11-diphenyl-1-benzo[c]fluorenyl, 11,11-diphenyl-2-benzo[c]fluorenyl, 11,11-diphenyl-3-benzo[c]fluorenyl, 11,11-diphenyl-4-benzo[c]fluorenyl, 11,11-diphenyl-5-benzo[c]fluorenyl, 11,11-diphenyl-6-benzo[c]fluorenyl, 11,11-diphenyl-7-benzo[c]fluorenyl, 11,11-diphenyl-8-benzo[c]fluorenyl, 11,11-diphenyl-9-benzo[c]fluorenyl, 11,11-diphenyl-10-benzo[c]fluorenyl, 9,9,10,10-tetramethyl-9,10-dihydro-1-phenanthrenyl, 9,9,10,10-tetramethyl-9,10-dihydro-2-phenanthrenyl, 9,9,10,10-tetramethyl-9,10-dihydro-3-phenanthrenyl, 9,9,10,10-tetramethyl-9,10-dihydro-4-phenanthrenyl, etc.

[0033] Herein, the “(3- to 30-membered)heteroaryl”, “(3- to 30-membered) heteroarylene”, or “(3- to 30-membered) heteroarene” is meant to be an aryl or arylene group having 3 to 30 ring backbone atoms, and comprising at least one heteroatom selected from the group consisting of B, N, O, S, Si, and P, in which the number of ring backbone atoms is preferably 3 to 30, and more preferably 5 to 20. The number of the heteroatoms is preferably 1 to 4. The above heteroaryl or heteroarylene may be a monocyclic ring, or a fused ring condensed with at least one benzene ring, and may be partially saturated. In addition, the above heteroaryl or

heteroarylene may be one formed by linking at least one heteroaryl or aryl group to a heteroaryl group via a single bond(s), and may comprise a spiro structure. The above heteroaryl may include a monocyclic ring-type heteroaryl such as furyl, thiophenyl, pyrrolyl, imidazolyl, pyrazolyl, thiazolyl, thiadiazolyl, isothiazolyl, isoxazolyl, oxazolyl, oxadiazolyl, triazinyl, tetrazinyl, triazolyl, tetrazolyl, furazanyl, pyridyl, pyrazinyl, pyrimidinyl, and pyridazinyl, and a fused ring-type heteroaryl such as benzofuranyl, benzothiophenyl, isobenzofuranyl, dibenzofuranyl, dibenzothiophenyl, benzofuroquinolinyl, benzofuroquinazolinyl, benzofuronaphthyridinyl, benzofuropyrimidinyl, naphthofuropyrimidinyl, benzothienonaphthyridinyl, benzothienopyrimidinyl, benzothienopyrazinyl, naphthothienopyrazinyl, pyrazinoindolyl, benzopyrimidinolyl, benzofuropyrazinyl, naphthofuropyrazinyl, benzothienopyrazinyl, naphthothienopyrazinyl, pyrazinoindolyl, benzopyrazinoindolyl, benzimidazolyl, benzothiazolyl, benzoisothiazolyl, benzoisoxazolyl, benzoxazolyl, imidazopyridinyl, isoindolyl, indolyl, benzoindolyl, indazolyl, benzothiadiazolyl, quinolyl, isoquinolyl, cinnolinyl, quinazolinyl, quinoxalinyl, carbazolyl, azacarbazolyl, benzocarbazolyl, dibenzocarbazolyl, phenoxazinyl, phenanthridinyl, benzodioxolyl, indolizidinyl, acridinyl, silafluorenyl, germafluorenyl, benzotriazolyl, phenazinyl, imidazopyridyl, chromenoquinazolinyl, thiochromenoquinazolinyl, dimethylbenzopyrimidinyl, indolocarbazolyl, indenocarbazolyl, etc. More specifically, the above heteroaryl may include 1-pyrrolyl, 2-pyrrolyl, 3-pyrrolyl, 2-pyridyl, 3-pyridyl, 4-pyridyl, 2-pyrimidinyl, 4-pyrimidinyl, 5-pyrimidinyl, 6-pyrimidinyl, 1,2,3-triazin-4-yl, 1,2,4-triazin-3-yl, 1,3,5-triazin-2-yl, 1-imidazolyl, 2-imidazolyl, 1-pyrazolyl, 1-indolidinyl, 2-indolidinyl, 3-indolidinyl, 5-indolidinyl, 6-indolidinyl, 7-indolidinyl, 8-indolidinyl, 2-imidazopyridyl, 3-imidazopyridyl, 5-imidazopyridyl, 6-imidazopyridyl, 7-imidazopyridyl, 8-imidazopyridyl, 1-indolyl, 2-indolyl, 3-indolyl, 4-indolyl, 5-indolyl, 6-indolyl, 7-indolyl, 1-isoindolyl, 2-isoindolyl, 3-isoindolyl, 4-isoindolyl, 5-isoindolyl, 6-isoindolyl, 7-isoindolyl, 2-furyl, 3-furyl, 2-benzofuranyl, 3-benzofuranyl, 4-benzofuranyl, 5-benzofuranyl, 6-benzofuranyl, 7-benzofuranyl, 1-isobenzofuranyl, 3-isobenzofuranyl, 4-isobenzofuranyl, 5-isobenzofuranyl, 6-isobenzofuranyl, 7-isobenzofuranyl, 2-quinolyl, 3-quinolyl, 4-quinolyl, 5-quinolyl, 6-quinolyl, 7-quinolyl, 8-quinolyl, 1-isoquinolyl, 3-isoquinolyl, 4-isoquinolyl, 5-isoquinolyl, 6-isoquinolyl, 7-isoquinolyl, 8-isoquinolyl, 2-quinoxalinyl, 5-quinoxalinyl, 6-quinoxalinyl, 1-carbazolyl, 2-carbazolyl, 3-carbazolyl, 4-carbazolyl, 9-carbazolyl, azacarbazol-1-yl, azacarbazol-2-yl, azacarbazol-3-yl, azacarbazol-4-yl, azacarbazol-5-yl, azacarbazol-6-yl, azacarbazol-7-yl, azacarbazol-8-yl, azacarbazol-9-yl, 1-phenanthridinyl, 2-phenanthridinyl, 3-phenanthridinyl, 4-phenanthridinyl, 6-phenanthridinyl, 7-phenanthridinyl, 8-phenanthridinyl, 9-phenanthridinyl, 10-phenanthridinyl, 1-acridinyl, 2-acridinyl, 3-acridinyl, 4-acridinyl, 9-acridinyl, 2-oxazolyl, 4-oxazolyl, 5-oxazolyl, 2-oxadiazolyl, 5-oxadiazolyl, 3-furazanyl, 2-thienyl, 3-thienyl, 2-methylpyrrol-1-yl, 2-methylpyrrol-3-yl, 2-methylpyrrol-4-yl, 2-methylpyrrol-5-yl, 3-methylpyrrol-1-yl, 3-methylpyrrol-2-yl, 3-methylpyrrol-4-yl, 3-methylpyrrol-5-yl, 2-tert-butylpyrrol-4-yl, 3-(2-phenylpropyl) pyrrol-1-yl, 2-methyl-1-indolyl, 4-methyl-1-indolyl, 2-methyl-3-indolyl, 4-methyl-3-indolyl, 2-tert-butyl-1-indolyl, 4-tert-butyl-1-indolyl, 2-tert-butyl-3-indolyl, 4-tert-butyl-3-indolyl, 1-dibenzofuranyl,

2-dibenzofuranyl, 3-dibenzofuranyl, 4-dibenzofuranyl, 1-dibenzothiophenyl, 2-dibenzothiophenyl, 3-dibenzothiophenyl, 4-dibenzothiophenyl, 1-naphtho-[1,2-b]-benzofuranyl, 2-naphtho-[1,2-b]-benzofuranyl, 3-naphtho-[1,2-b]-benzofuranyl, 4-naphtho-[1,2-b]-benzofuranyl, 5-naphtho-[1,2-b]-benzofuranyl, 6-naphtho-[1,2-b]-benzofuranyl, 7-naphtho-[1,2-b]-benzofuranyl, 8-naphtho-[1,2-b]-benzofuranyl, 9-naphtho-[1,2-b]-benzofuranyl, 10-naphtho-[1,2-b]-benzofuranyl, 1-naphtho-[2,3-b]-benzofuranyl, 2-naphtho-[2,3-b]-benzofuranyl, 3-naphtho-[2,3-b]-benzofuranyl, 4-naphtho-[2,3-b]-benzofuranyl, 5-naphtho-[2,3-b]-benzofuranyl, 6-naphtho-[2,3-b]-benzofuranyl, 7-naphtho-[2,3-b]-benzofuranyl, 8-naphtho-[2,3-b]-benzofuranyl, 9-naphtho-[2,3-b]-benzofuranyl, 10-naphtho-[2,3-b]-benzofuranyl, 1-naphtho-[2,1-b]-benzofuranyl, 2-naphtho-[2,1-b]-benzofuranyl, 3-naphtho-[2,1-b]-benzofuranyl, 4-naphtho-[2,1-b]-benzofuranyl, 5-naphtho-[2,1-b]-benzofuranyl, 6-naphtho-[2,1-b]-benzofuranyl, 7-naphtho-[2,1-b]-benzofuranyl, 8-naphtho-[2,1-b]-benzofuranyl, 9-naphtho-[2,1-b]-benzofuranyl, 10-naphtho-[2,1-b]-benzofuranyl, 1-naphtho-[1,2-b]-benzothiophenyl, 2-naphtho-[1,2-b]-benzothiophenyl, 3-naphtho-[1,2-b]-benzothiophenyl, 4-naphtho-[1,2-b]-benzothiophenyl, 5-naphtho-[1,2-b]-benzothiophenyl, 6-naphtho-[1,2-b]-benzothiophenyl, 7-naphtho-[1,2-b]-benzothiophenyl, 8-naphtho-[1,2-b]-benzothiophenyl, 9-naphtho-[1,2-b]-benzothiophenyl, 10-naphtho-[1,2-b]-benzothiophenyl, 1-naphtho-[2,3-b]-benzothiophenyl, 2-naphtho-[2,3-b]-benzothiophenyl, 3-naphtho-[2,3-b]-benzothiophenyl, 4-naphtho-[2,3-b]-benzothiophenyl, 5-naphtho-[2,3-b]-benzothiophenyl, 6-naphtho-[2,3-b]-benzothiophenyl, 7-naphtho-[2,3-b]-benzothiophenyl, 8-naphtho-[2,3-b]-benzothiophenyl, 9-naphtho-[2,3-b]-benzothiophenyl, 10-naphtho-[2,3-b]-benzothiophenyl, 1-naphtho-[2,1-b]-benzothiophenyl, 2-naphtho-[2,1-b]-benzothiophenyl, 3-naphtho-[2,1-b]-benzothiophenyl, 4-naphtho-[2,1-b]-benzothiophenyl, 5-naphtho-[2,1-b]-benzothiophenyl, 6-naphtho-[2,1-b]-benzothiophenyl, 7-naphtho-[2,1-b]-benzothiophenyl, 8-naphtho-[2,1-b]-benzothiophenyl, 9-naphtho-[2,1-b]-benzothiophenyl, 10-naphtho-[2,1-b]-benzothiophenyl, 2-benzofuro[3,2-d]pyrimidinyl, 6-benzofuro[3,2-d]pyrimidinyl, 7-benzofuro[3,2-d]pyrimidinyl, 8-benzofuro[3,2-d]pyrimidinyl, 9-benzofuro[3,2-d]pyrimidinyl, 2-benzothio[3,2-d]pyrimidinyl, 6-benzothio[3,2-d]pyrimidinyl, 7-benzothio[3,2-d]pyrimidinyl, 8-benzothio[3,2-d]pyrimidinyl, 9-benzothio[3,2-d]pyrimidinyl, 2-benzofuro[3,2-d]pyrazinyl, 6-benzofuro[3,2-d]pyrazinyl, 7-benzofuro[3,2-d]pyrazinyl, 8-benzofuro[3,2-d]pyrazinyl, 9-benzofuro[3,2-d]pyrazinyl, 2-benzothio[3,2-d]pyrazinyl, 6-benzothio[3,2-d]pyrazinyl, 7-benzothio[3,2-d]pyrazinyl, 8-benzothio[3,2-d]pyrazinyl, 9-benzothio[3,2-d]pyrazinyl, 1-silafluorenyl, 2-silafluorenyl, 3-silafluorenyl, 4-silafluorenyl, 1-germafluorenyl, 2-germafluorenyl, 3-germafluorenyl, 4-germafluorenyl, etc.

[0034] Herein, “a fused ring group of a (C_3 - C_{30}) aliphatic ring(s) and a (C_6 - C_{30}) aromatic ring(s)” is meant to be a functional group of a ring in which at least one aliphatic ring having 3 to 30 ring backbone carbon atoms, preferably 3 to 25 ring backbone carbon atoms, more preferably 3 to 18 ring backbone carbon atoms, is fused with at least one aromatic ring having 6 to 30 ring backbone carbon atoms, preferably 6 to 25 ring backbone carbon atoms, more preferably 6 to 18 ring backbone carbon atoms. Specific examples of the fused ring group include a fused ring group of one or more benzene and one or more cyclohexane, or a fused ring group of one or more naphthalene and one or more cyclopentane, etc. Herein, the carbon atom of the fused ring group of a (C_3 - C_{30}) aliphatic ring(s) and a (C_6 - C_{30}) aromatic ring(s) may be replaced with one or more heteroatoms selected

from B, N, O, S, Si, and P, preferably one or more heteroatoms selected from N, O, and S. Herein, “halogen” includes F, Cl, Br, and I.

[0035] In addition, “ortho-” (“o-”), “meta-” (“m-”), and “para-” (“p-”) are prefixes which represent the relative positions of substituents respectively. The prefix “ortho-” indicates that two substituents are adjacent to each other; for example, when two substituents in a benzene derivative occupy positions 1 and 2, this is called an “ortho-” configuration. The prefix “meta-” indicates that two substituents are at positions 1 and 3; for example, when two substituents in a benzene derivative occupy positions 1 and 3, this is called a “meta-” configuration. The prefix “para-” indicates that two substituents are at positions 1 and 4; for example, when two substituents in a benzene derivative occupy positions 1 and 4, this is called a “para-” configuration.

[0036] Herein, “a ring formed by being linked to an adjacent substituent” means that at least two adjacent substituents are linked to or fused with each other to form a substituted or unsubstituted, mono- or polycyclic, (3- to 30-membered) alicyclic or aromatic ring, or a combination thereof. Preferably, the ring may be a substituted or unsubstituted, mono- or polycyclic, (5- to 25-membered) alicyclic or aromatic ring, or a combination thereof. In addition, the ring may contain at least one heteroatom selected from B, N, O, S, Si, and P, preferably at least one heteroatom selected from N, O, and S. According to one embodiment of the present disclosure, the number of the ring backbone atoms may be 5- to 20-membered, and according to another embodiment of the present disclosure, the number of the ring backbone atoms may be 5- to 15-membered. For example, the ring may be a substituted or unsubstituted dibenzothiophene ring, a substituted or unsubstituted dibenzofuran ring, a substituted or unsubstituted naphthalene ring, a substituted or unsubstituted phenanthrene ring, a substituted or unsubstituted fluorene ring, a substituted or unsubstituted benzofluorene ring, a substituted or unsubstituted benzothiophene ring, a substituted or unsubstituted benzofuran ring, a substituted or unsubstituted indole ring, a substituted or unsubstituted indene ring, a substituted or unsubstituted benzene ring, or a substituted or unsubstituted carbazole ring, etc.

[0037] Herein, “substituted” in the expression “substituted or unsubstituted” means that a hydrogen atom in a certain functional group is replaced with another atom or another functional group, i.e., a substituent. The substituent also includes those in which two or more substituents are linked. For example, the substituent formed by linking two or more substituents may be pyridine-triazine. That is, pyridine-triazine may be interpreted as a heteroaryl, or a substituent(s) with two heteroaryls linked. In the formulas of the present disclosure, the substituent(s) of the substituted alkyl, the substituted alkenyl, the substituted aryl, the substituted arylene, the substituted heteroaryl, the substituted heteroarylene, the substituted cycloalkyl, the substituted cycloalkenyl, the substituted heterocycloalkyl, the substituted alkoxy, the substituted trialkylsilyl, the substituted dialkylarylsilyl, the substituted alkylidiarylsilyl, the substituted triarylsilyl, the substituted fused ring group of an aliphatic ring(s) and an aromatic ring(s), the substituted arene, and the substituted heteroarene each independently are substituted with at least one selected from the group consisting of deuterium, a halogen, a cyano, carboxyl, nitro, hydroxyl, phosphine oxide, (C_1 - C_{30})alkyl, halo(C_1 - C_{30})al-

kyl, (C₂-C₃₀)alkenyl, (C₂-C₃₀)alkynyl, (C₁-C₃₀)alkoxy, (C₁-C₃₀)alkylthio, (C₃-C₃₀)cycloalkyl, (C₃-C₃₀)cycloalkenyl, (3- to 7-membered)heterocycloalkyl, (C₆-C₃₀)aryloxy, (C₆-C₃₀)arylthio, (3- to 30-membered)heteroaryl, (C₆-C₃₀)aryl, tri(C₁-C₃₀)alkylsilyl, tri(C₆-C₃₀)arylsilyl, di(C₁-C₃₀)alkyl(C₆-C₃₀)arylsilyl, (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, amino, mono- or di(C₁-C₃₀)alkylamino, mono- or di(C₂-C₃₀)alkenylamino, mono- or di(C₆-C₃₀)arylamino, mono- or di(3- to 30-membered)heteroarylamino, (C₁-C₃₀)alkyl(C₂-C₃₀)alkenylamino, (C₁-C₃₀)alkyl(C₆-C₃₀)arylamino, (C₁-C₃₀)alkyl(3- to 30-membered)heteroarylamino, (C₂-C₃₀)alkenyl(C₆-C₃₀)arylamino, (C₂-C₃₀)alkenyl(3- to 30-membered)heteroarylamino, (C₆-C₃₀)aryl(3- to 30-membered)heteroarylamino, (C₁-C₃₀)alkylcarbonyl, (C₁-C₃₀)alkoxycarbonyl, (C₆-C₃₀)arylcarbonyl, di(C₆-C₃₀)arylboronyl, (C₆-C₃₀)arylphosphinyl, di(C₁-C₃₀)alkylboronyl, (C₁-C₃₀)alkyl(C₆-C₃₀)arylboronyl, (C₆-C₃₀)aryl(C₁-C₃₀)alkyl, (C₁-C₃₀)alkyl(C₆-C₃₀)aryl, and a combination thereof, which may be further substituted with deuterium. According to one embodiment of the present disclosure, the substituent(s) each independently may be at least one selected from the group consisting of deuterium, (C₁-C₂₀)alkyl unsubstituted or substituted with deuterium, and (C₆-C₂₅)aryl unsubstituted or substituted with deuterium. According to another embodiment of the present disclosure, the substituent(s) each independently may be at least one selected from the group consisting of deuterium, (C₁-C₁₀)alkyl unsubstituted or substituted with deuterium, and (C₆-C₁₈)aryl unsubstituted or substituted with deuterium. For example, the substituent(s) each independently may be at least one selected from the group consisting of deuterium, methyl, phenyl, naphthyl, biphenyl, and phenanthrenyl, which may be substituted with deuterium.

[0038] Herein, if a substituent is not indicated in the formula or compound structure, it may mean that all possible positions for the substituent are hydrogen or deuterium. That is, in the case of deuterium, it is an isotope of hydrogen, and some hydrogen atoms may be the isotope deuterium, and in this case, the content of deuterium may be 0% to 100%. In cases where a substituent is not indicated in the formula or compound structure in the present disclosure, if the substituent is not explicitly excluded, such as 0% deuterium, 100% hydrogen, and all substituents being hydrogen, hydrogen and deuterium may be used together in a compound. Deuterium is one of the isotopes of hydrogen and an element with a deuteron, consisting of one proton and one neutron, as its nucleus. It can be represented as hydrogen-2, whose element symbol can also be written as D or ²H. Isotopes are atoms with the same atomic number (Z) but different mass numbers (A), and can also be interpreted as elements with the same number of protons but different numbers of neutrons.

[0039] Herein, “a combination thereof” refers to a combination of one or more elements from the corresponding list to form a known or chemically stable arrangement that can be envisioned by one skilled in the art from the corresponding list. For example, alkyl and deuterium can be combined to form a partially or fully deuterated alkyl group; halogen and alkyl can be combined to form a halogenated alkyl substituent; and halogen, alkyl, and aryl can be combined to form a halogenated arylalkyl. For example, a preferred combination of substituents includes up to 50 atoms that are not hydrogen or deuterium, up to 40 atoms that are not hydrogen or deuterium, or up to 30 atoms that are not

hydrogen or deuterium, but in many cases, a preferred combination of substituents may comprise up to 20 atoms that are not hydrogen or deuterium.

[0040] In the formulas of the present disclosure, when there are multiple substituents represented by the same symbol, each substituent represented by the same symbol may be the same as or different from each other.

[0041] According to one embodiment of the present disclosure, at least one of Formula 1 and Formula 2 may contain deuterium.

[0042] In Formula 1, L₁ represents a single bond, a phenylene unsubstituted or substituted with deuterium, a naphthylene unsubstituted or substituted with deuterium, a biphenylene unsubstituted or substituted with deuterium, or a phenanthrenylene unsubstituted or substituted with deuterium.

[0043] In Formula 1, Ar₁ each independently represents a substituted or unsubstituted (C₆-C₃₀) aryl, or a substituted or unsubstituted (3- to 30-membered)heteroaryl. According to one embodiment of the present disclosure, Ar₁ each independently represents a substituted or unsubstituted (C₆-C₂₅) aryl. According to another embodiment of the present disclosure, Ar₁ each independently represents a (C₆-C₁₈)aryl unsubstituted or substituted with deuterium. For example, Ar₁ each independently may be a phenyl unsubstituted or substituted with deuterium, a biphenyl unsubstituted or substituted with deuterium, a terphenyl unsubstituted or substituted with deuterium, a naphthyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, or a combination thereof.

[0044] In Formula 1, R and R₁ to R₈ each independently represent hydrogen or deuterium.

[0045] In Formula 1, a represents an integer of 1 or 2, b represents an integer of 4, and c represents an integer of 5; each Ar₁ may be the same as or different from each other.

[0046] In Formula 2, Ar₁₁ represents a substituted or unsubstituted (C₆-C₃₀)aryl, or a substituted or unsubstituted (13- to 30-membered)heteroaryl. According to one embodiment of the present disclosure, Ar₁₁ represents a substituted or unsubstituted (C₆-C₂₅)aryl, or a substituted or unsubstituted (13- to 25-membered)heteroaryl. According to another embodiment of the present disclosure, Ar₁₁ represents a substituted or unsubstituted (C₆-C₂₅)aryl, or a substituted or unsubstituted (13- to 17-membered)heteroaryl. Their substituents may be at least one selected from the group consisting of deuterium, (C₁-C₁₀)alkyl, (C₆-C₃₀)aryl, and a combination thereof. For example, Ar₁₁ may be a phenyl, a phenyl substituted with a phenanthrenyl(s), a naphthyl, a naphthylphenyl, a phenylphenyl, a biphenyl, a phenanthrenyl, a terphenyl, a triphenylenyl, a dimethylfluorenyl, a diphenylfluorenyl, a dimethylbenzofluorenyl, a spirobifluorenyl, a carbazoyl unsubstituted or substituted with a phenyl(s), a benzocarbazoyl, a dibenzofuranyl unsubstituted or substituted with a phenyl(s), a dibenzothiophenyl unsubstituted or substituted with a phenyl(s), a benzonaphthofuranyl, or a benzonaphthothiophenyl, etc., which may be further substituted with deuterium.

[0047] In Formula 2, R₁₁ to R₁₈ each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C₁-C₃₀)alkyl, a substituted or unsubstituted (C₂-C₃₀)alkenyl, a substituted or unsubstituted (C₆-C₃₀)aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C₃-C₃₀)cycloalkyl, a substituted or unsubstituted (C₃-C₃₀)

cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C₁-C₃₀)alkoxy, a substituted or unsubstituted tri(C₁-C₃₀)alkylsilyl, a substituted or unsubstituted di(C₁-C₃₀)alkyl (C₆-C₃₀)arylsilyl, a substituted or unsubstituted (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, a substituted or unsubstituted tri (C₆-C₃₀)arylsilyl, a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s), or -L₁₃-N(Ar₁₃)(Ar₁₄). For example, R₁₁ to R₁₈ each independently may represent hydrogen or deuterium.

[0048] In Formula 2, Ar⁴ represents a substituted or unsubstituted (C₆-C₃₀)aryl, a substituted or unsubstituted (13- to 30-membered)heteroaryl, or the above Formula A-1. According to one embodiment of the present disclosure, Ar⁴ represents a substituted or unsubstituted (C₆-C₂₅)aryl, a substituted or unsubstituted (13- to 25-membered)heteroaryl, or the above Formula A-1. According to another embodiment of the present disclosure, Ar⁴ represents a (C₆-C₂₀)aryl unsubstituted or substituted with deuterium or a (C₆-C₃₀)aryl(s), or the above Formula A-1. For example, Ar⁴ may be Formula A-1, or a phenyl, a phenyl substituted with a phenanthrenyl(s), a naphthyl, a phenylnaphthyl, a naphthylphenyl, a biphenyl, a phenanthrenyl, or a terphenyl, etc., which may be further substituted with deuterium.

[0049] In Formula A-1, T₁ represents O, S, or CR_aR_b.

[0050] In Formula A-1, Ring A and Ring B each independently represent a substituted or unsubstituted (C₆-C₃₀) arene, or a substituted or unsubstituted (3- to 30-membered) heteroarene. According to one embodiment of the present disclosure, Ring A and Ring B each independently represent a substituted or unsubstituted (C₆-C₂₅)arene. According to another embodiment of the present disclosure, Ring A and Ring B each independently represent (C₆-C₁₈)arene unsubstituted or substituted with deuterium. For example, Ring A and Ring B each independently may be a substituted or unsubstituted benzene or naphthalene ring, and its substituents may be at least one selected from the group consisting of deuterium, phenyl, naphthyl, biphenyl, and a combination thereof.

[0051] In Formula A-1, R₁₉ and R₂₀ each independently represent the position linked to L₁₂, or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C₁-C₃₀)alkyl, a substituted or unsubstituted (C₂-C₃₀)alkenyl, a substituted or unsubstituted (C₆-C₃₀)aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C₃-C₃₀)cycloalkyl, a substituted or unsubstituted (C₃-C₃₀)cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C₁-C₃₀)alkoxy, a substituted or unsubstituted tri(C₁-C₃₀)alkylsilyl, a substituted or unsubstituted di(C₁-C₃₀)alkyl(C₆-C₃₀)arylsilyl, a substituted or unsubstituted (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, a substituted or unsubstituted tri(C₆-C₃₀)arylsilyl, a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s), or -L₁₃-N(Ar₁₃)(Ar₁₄). According to one embodiment of the present disclosure, R₁₉ and R₂₀ each independently represent the position linked to L₁₂, or represent hydrogen, deuterium, or a substituted or unsubstituted (C₆-C₂₅)aryl. According to another embodiment of the present disclosure, R₁₉ and R₂₀ each independently represent the position linked to L₁₂, or represent hydrogen, deuterium, or a (C₆-C₁₈)aryl unsubstituted or substituted with deuterium. For example, R₁₉ and R₂₀ each independently represent the position linked to L₁₂, or rep-

resent hydrogen, deuterium, a phenyl, a naphthyl, or a biphenyl, etc., which may be further substituted with deuterium.

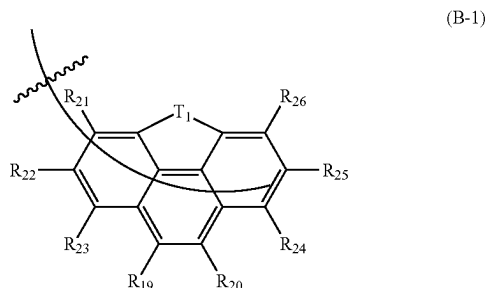
[0052] R_a and R_b each independently represent a substituted or unsubstituted (C₁-C₃₀)alkyl, or a substituted or unsubstituted (C₆-C₃₀)aryl, or may be linked to each other to form a ring(s). According to one embodiment of the present disclosure, R_a and R_b each independently represent a substituted or unsubstituted (C₁-C₂₀)alkyl. According to another embodiment of the present disclosure, R_a and R_b each independently represent a substituted or unsubstituted (C₁-C₁₀)alkyl. For example, R_a and R_b each independently may be a methyl unsubstituted or substituted with deuterium.

[0053] L₁₁ to L₁₃ each independently represent a single bond, a substituted or unsubstituted (C₆-C₃₀)arylene, or a substituted or unsubstituted (3- to 30-membered) heteroarylene. According to one embodiment of the present disclosure, L₁₁ to L₁₃ each independently represent a single bond, a substituted or unsubstituted (C₆-C₂₅)arylene, or a substituted or unsubstituted (5- to 25-membered)heteroarylene. According to another embodiment of the present disclosure, L₁₁ to L₁₃ each independently represent a single bond; a (C₆-C₁₈)arylene unsubstituted or substituted with deuterium, a (C₁-C₃₀)alkyl(s) and/or a (C₆-C₃₀)aryl(s); or a (5- to 20-membered)heteroarylene unsubstituted or substituted with deuterium. For example, L₁₁ to L₁₃ each independently represent a single bond, a phenylene unsubstituted or substituted with a methyl(s) or a phenyl(s), a naphthylene unsubstituted or substituted with a phenyl(s), a biphenylene, a phenanthrenylene, a carbazolyene, a dibenzofuranylene, or a dibenzothiophenylene, etc., which may be further substituted with deuterium.

[0054] Ar₁₃ and Ar₁₄ each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C₁-C₃₀)alkyl, a substituted or unsubstituted (C₂-C₃₀)alkenyl, a substituted or unsubstituted (C₆-C₃₀)aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C₃-C₃₀)cycloalkyl, a substituted or unsubstituted (C₃-C₃₀)cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C₁-C₃₀)alkoxy, a substituted or unsubstituted tri(C₁-C₃₀)alkylsilyl, a substituted or unsubstituted di(C₁-C₃₀)alkyl(C₆-C₃₀)arylsilyl, a substituted or unsubstituted (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, a substituted or unsubstituted tri(C₆-C₃₀)arylsilyl, or a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s).

[0055] According to one embodiment of the present disclosure, Ar₁₁ and Ar⁴ each independently may be selected from a phenyl unsubstituted or substituted with deuterium, a biphenyl unsubstituted or substituted with deuterium, a terphenyl unsubstituted or substituted with deuterium, a naphthyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, or a combination thereof. According to another embodiment of the present disclosure, Ar₁₁ may be selected from a phenyl unsubstituted or substituted with deuterium, a biphenyl unsubstituted or substituted with deuterium, a terphenyl unsubstituted or substituted with deuterium, a naphthyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, or a combination thereof; and Ar⁴ may represent a dibenzofuranyl unsubstituted or substituted with deuterium, or a dibenzothiophenyl unsubstituted or substituted with deuterium.

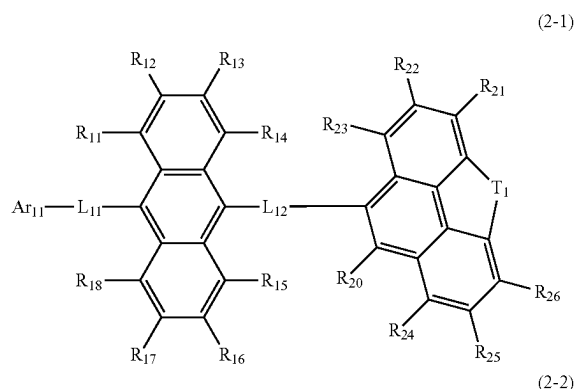
[0056] According to one embodiment of the present disclosure, Ar^4 is represented by the following Formula B-1:



[0057] In Formula B-1, R_{21} to R_{26} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_2 - C_{30})alkenyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3 - C_{30})cycloalkyl, a substituted or unsubstituted (C_3 - C_{30})cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C_1 - C_{30})alkoxy, a substituted or unsubstituted tri(C_1 - C_{30})alkylsilyl, a substituted or unsubstituted di(C_1 - C_{30})alkyl(C_6 - C_{30})arylsilyl, a substituted or unsubstituted (C_1 - C_{30})alkyldi(C_6 - C_{30})arylsilyl, a substituted or unsubstituted tri(C_6 - C_{30})arylsilyl, a substituted or unsubstituted fused ring group of a (C_3 - C_{30}) aliphatic ring(s) and a (C_6 - C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$, or may be linked to the adjacent substituent to form a ring(s). According to one embodiment of the present disclosure, R_{21} to R_{26} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, or a substituted or unsubstituted (C_6 - C_{25})aryl, or may be linked to the adjacent substituent to form a substituted or unsubstituted, mono- or polycyclic, (3- to 30-membered) alicyclic or aromatic ring, or a combination thereof. According to another embodiment of the present disclosure, R_{21} to R_{26} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, or a substituted or unsubstituted (C_6 - C_{18})aryl, or may be linked to the adjacent substituent to form a substituted or unsubstituted, mono- or polycyclic, (3- to 20-membered) aromatic ring. For example, R_{21} to R_{26} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, or a (C_6 - C_{18})aryl unsubstituted or substituted with deuterium, or may be linked to the adjacent substituent to form a benzene ring unsubstituted or substituted with deuterium.

[0058] In Formula B-1, T_1 , R_{19} , R_{20} , L_{13} , Ar_{13} , and Ar_{14} are as defined in Formula 2 above.

[0059] Formula 2 may be represented by the following Formula 2-1 or 2-2.



[0060] In Formulas 2-1 and 2-2,

[0061] R_{21} to R_{26} each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_2 - C_{30})alkenyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3 - C_{30})cycloalkyl, a substituted or unsubstituted (C_3 - C_{30})cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C_1 - C_{30})alkoxy, a substituted or unsubstituted tri(C_1 - C_{30})alkylsilyl, a substituted or unsubstituted di(C_1 - C_{30})alkyl(C_6 - C_{30})arylsilyl, a substituted or unsubstituted (C_1 - C_{30})alkyldi(C_6 - C_{30})arylsilyl, a substituted or unsubstituted tri(C_6 - C_{30})arylsilyl, a substituted or unsubstituted fused ring group of a (C_3 - C_{30}) aliphatic ring(s) and a (C_6 - C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$, or may be linked to the adjacent substituent to form a ring(s);

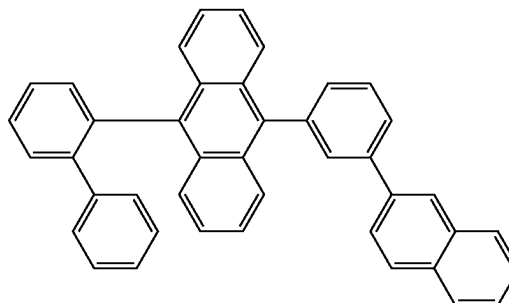
[0062] R_{27} to R_{29} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3 - C_{30})cycloalkyl, a substituted or unsubstituted (C_1 - C_{30})alkoxy, a substituted or unsubstituted tri(C_1 - C_{30})alkylsilyl, a substituted or unsubstituted di(C_1 - C_{30})alkyl(C_6 - C_{30})arylsilyl, a substituted or unsubstituted (C_1 - C_{30})alkyldi(C_6 - C_{30})arylsilyl, a substituted or unsubstituted tri(C_6 - C_{30})arylsilyl, a substituted or unsubstituted fused ring group of a (C_3 - C_{30}) aliphatic ring(s) and a (C_6 - C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$, or may be linked to the adjacent substituent to form a ring(s); and

[0063] T_1 , R_{11} to R_{20} , L_{11} to L_{13} , Ar_{11} , Ar_{13} , and Ar_{14} are as defined in Formula 2 above.

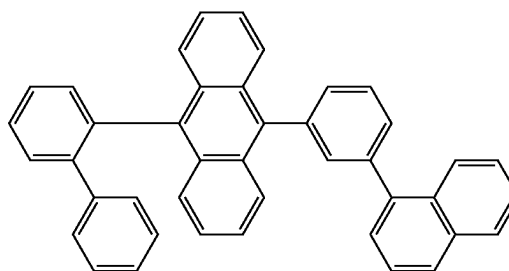
[0064] The compound represented by Formula 1 may be more specifically at least one selected from the group consisting of the following compounds, but is not limited thereto.

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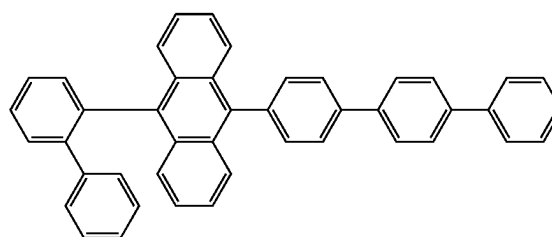
H1-6



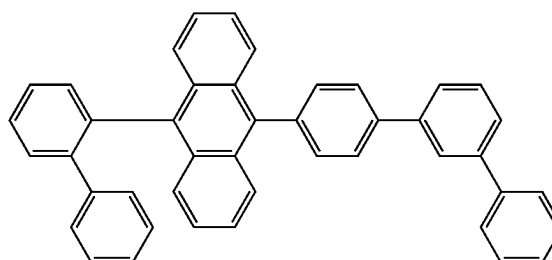
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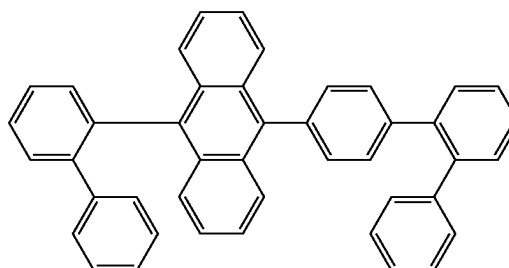
H1-8



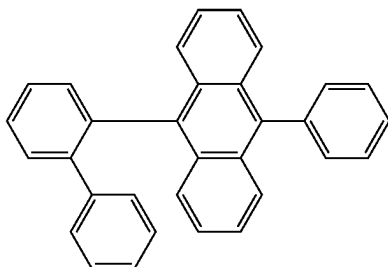
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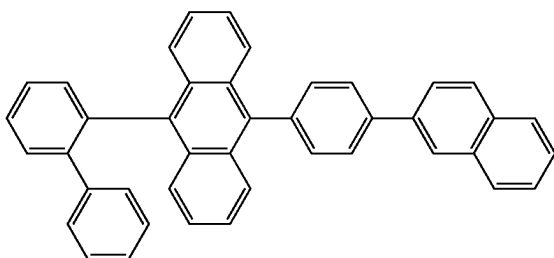
H1-10



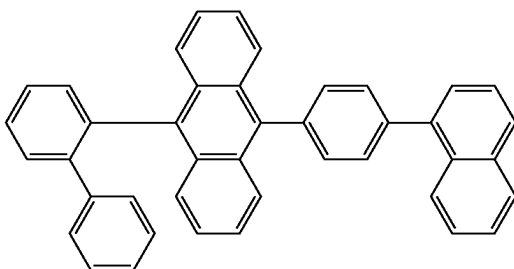
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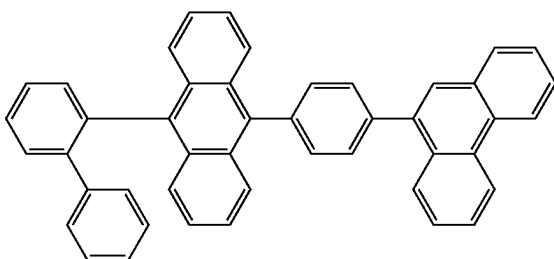
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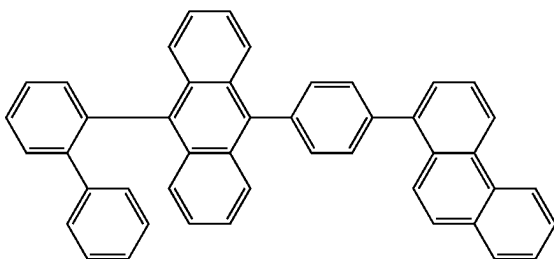
H1-3



H1-4

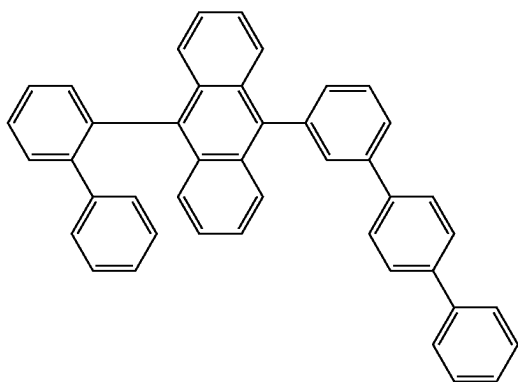


H1-5

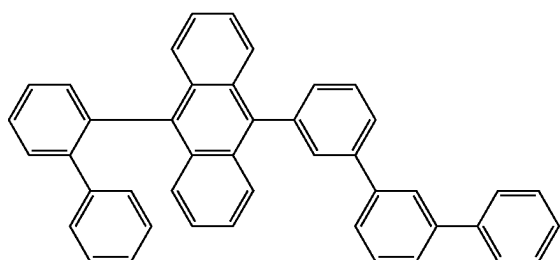


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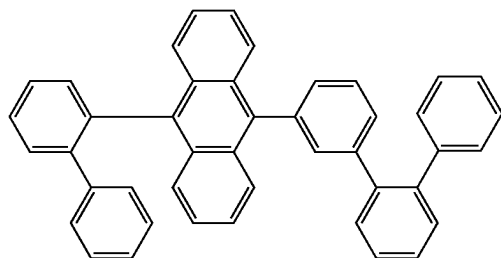
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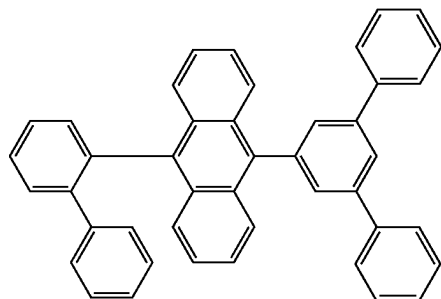
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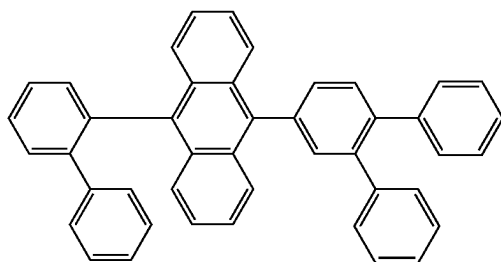
H1-13



H1-14

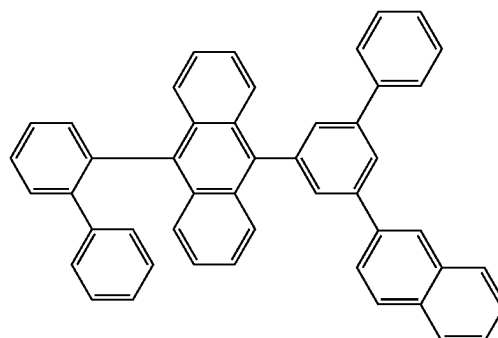


H1-15

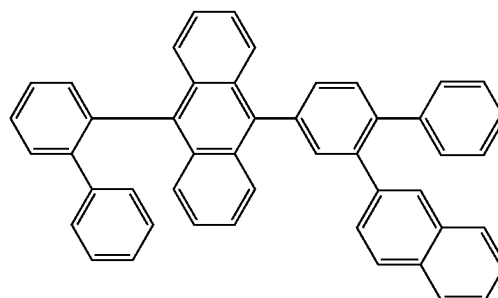


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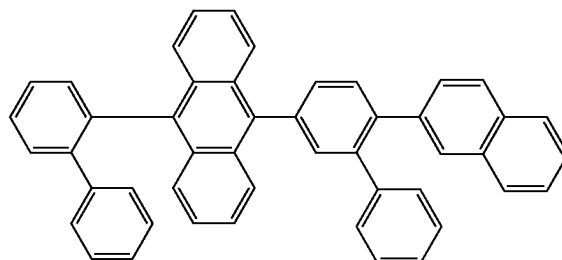
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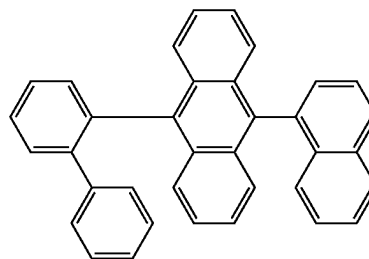
H1-17



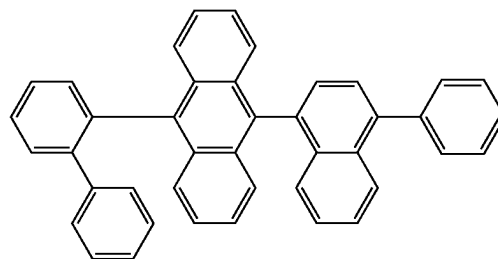
H1-18



H1-19

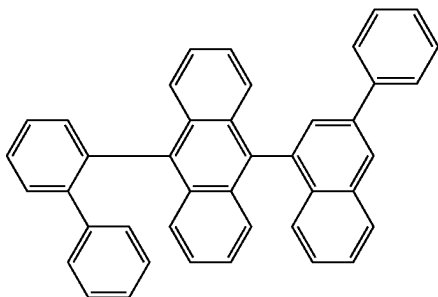


H1-20

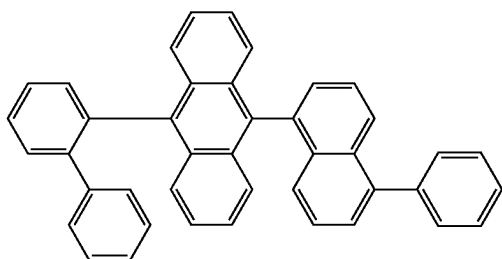


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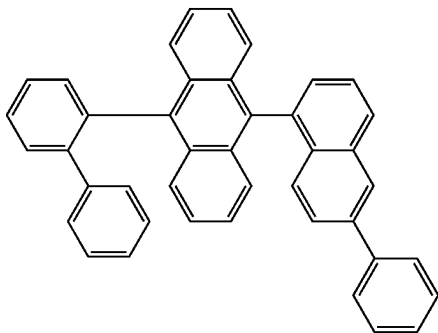
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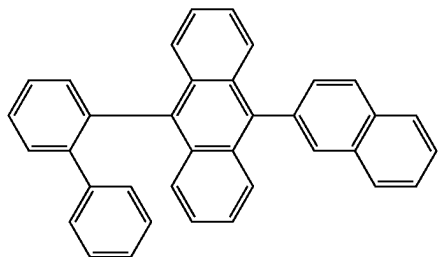
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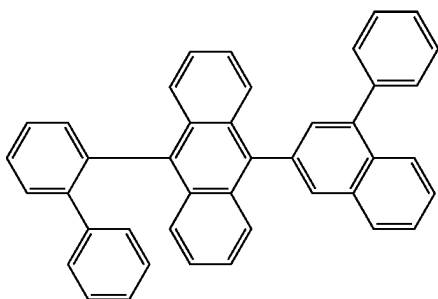
H1-23



H1-24

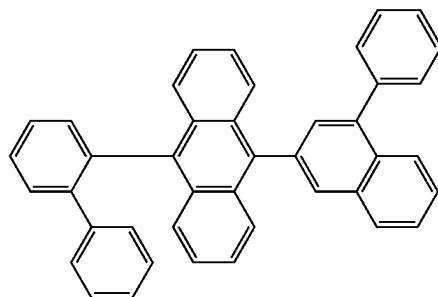


H1-25

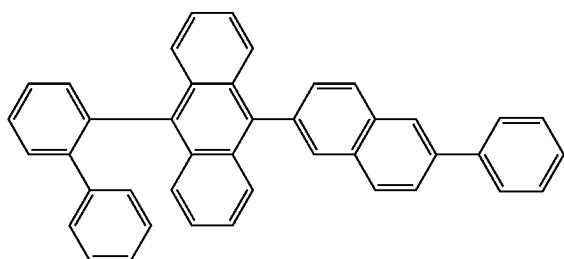


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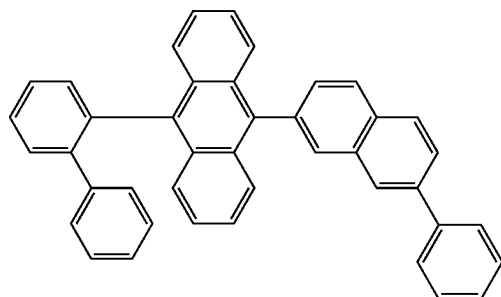
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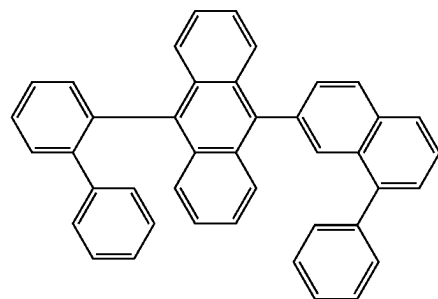
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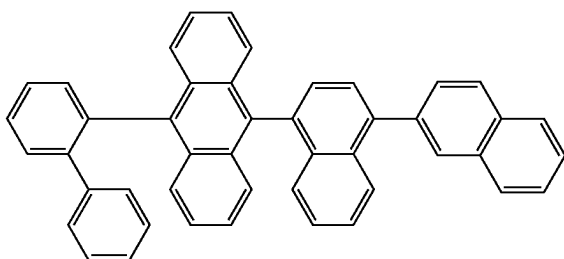
H1-28



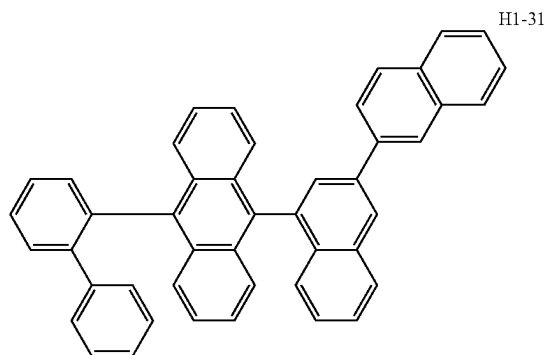
H1-29



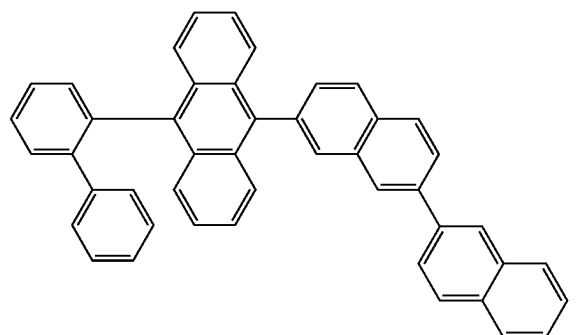
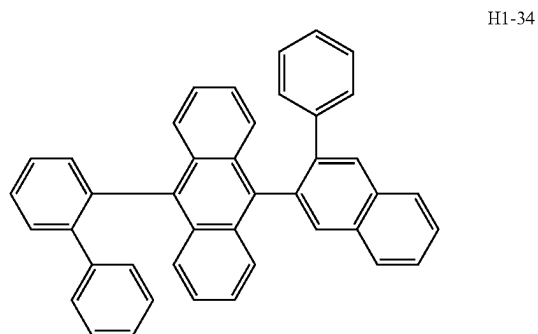
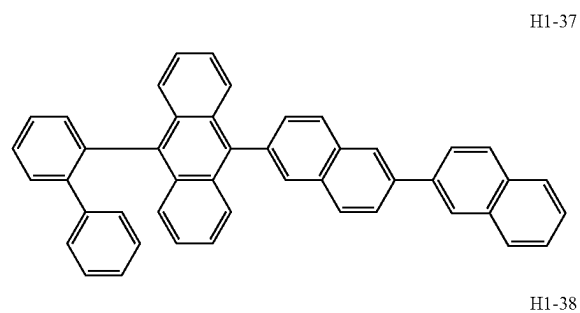
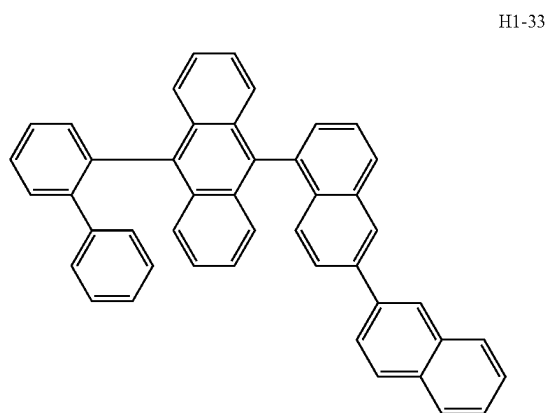
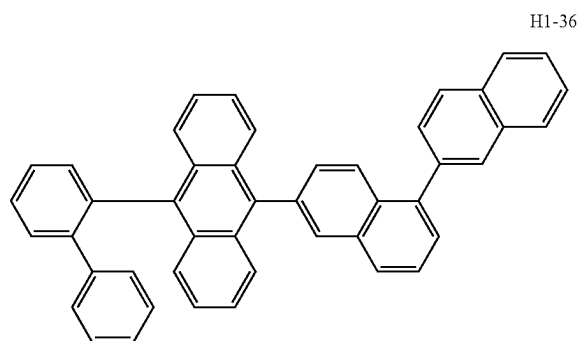
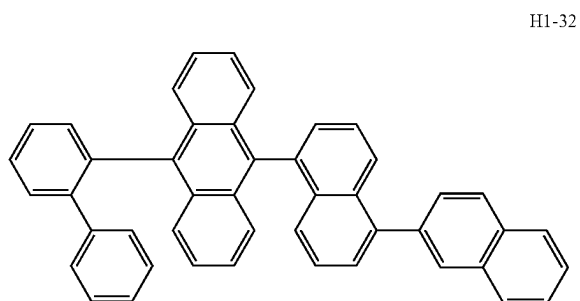
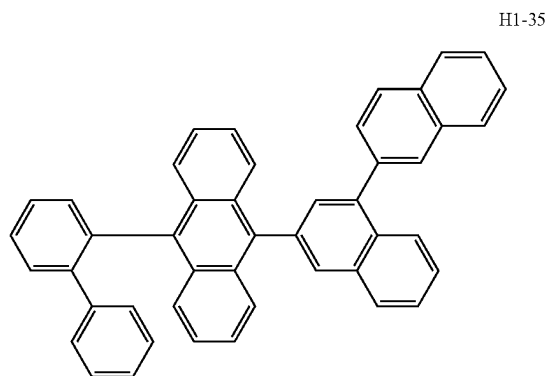
H1-30



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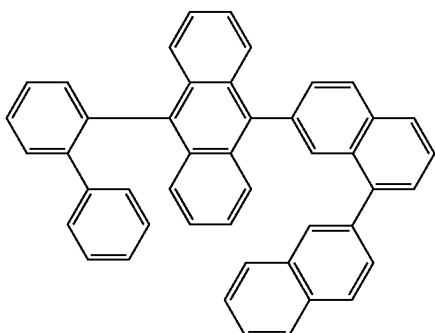


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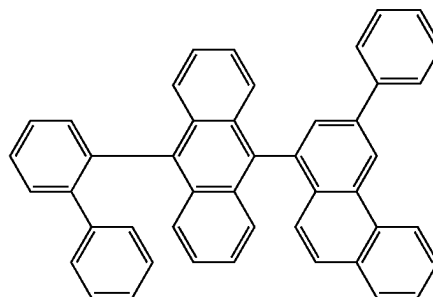
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H1-39



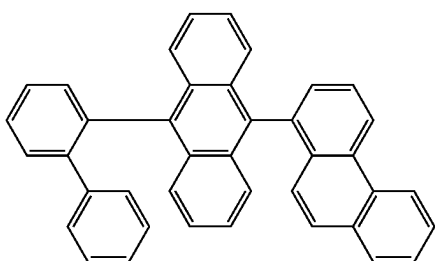
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H1-44

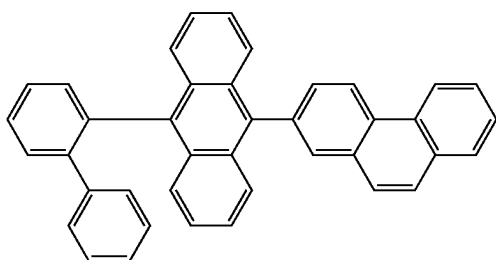


H1-45

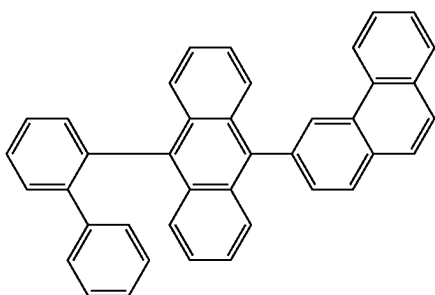
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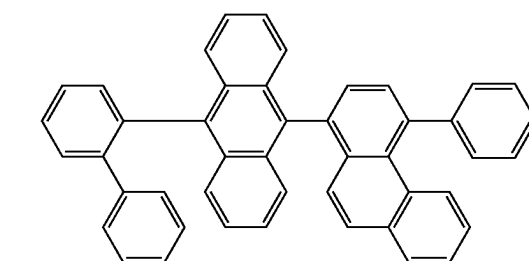
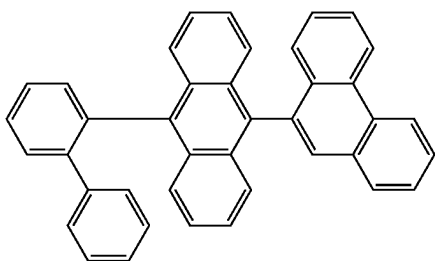
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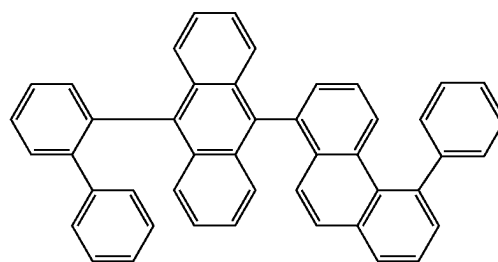
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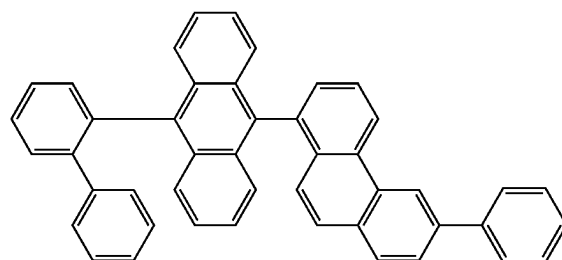
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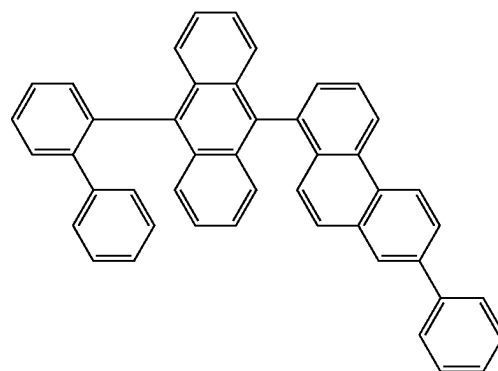
H1-46



H1-47

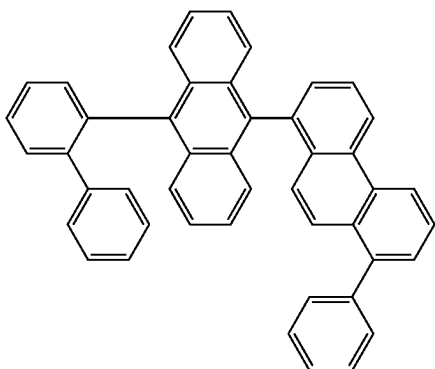


H1-48

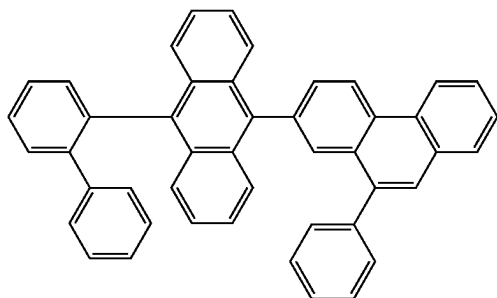


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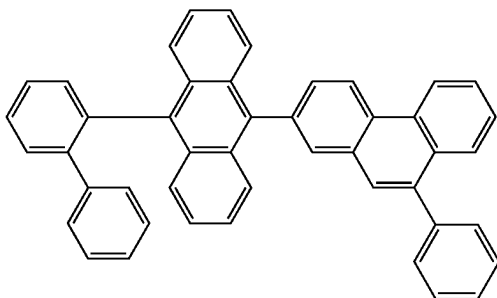
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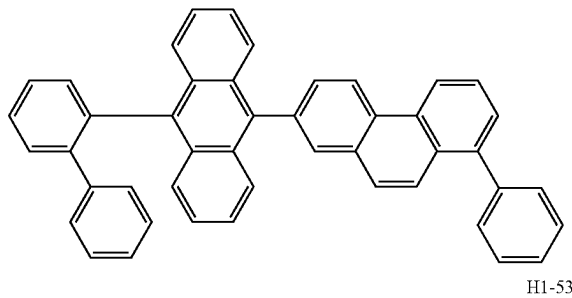
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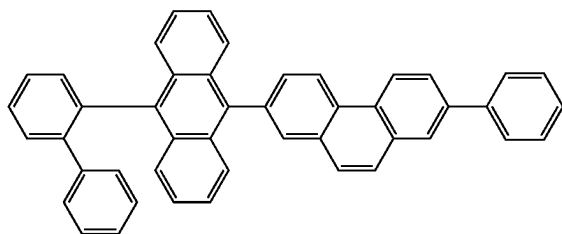
H1-51



H1-52

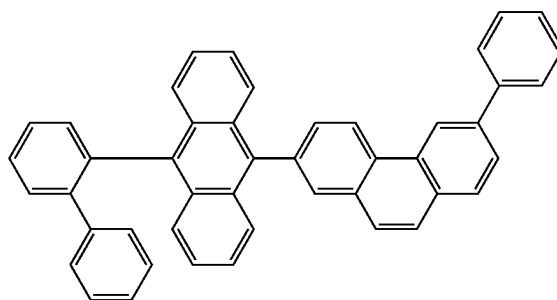


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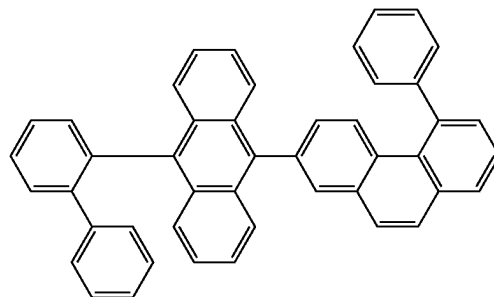


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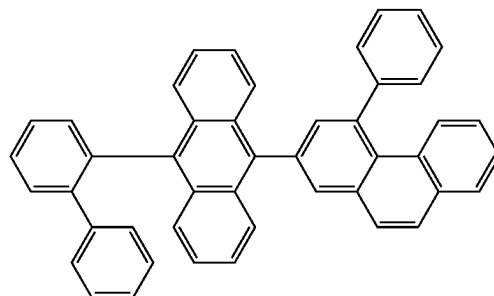
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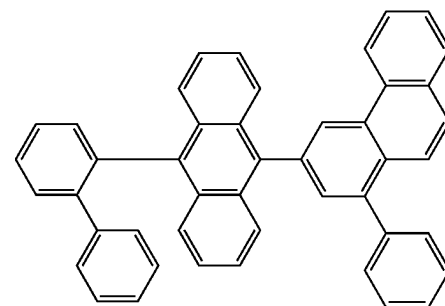
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H1-56

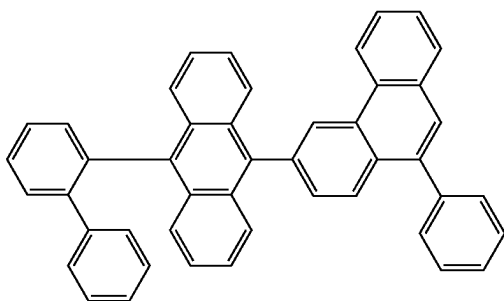


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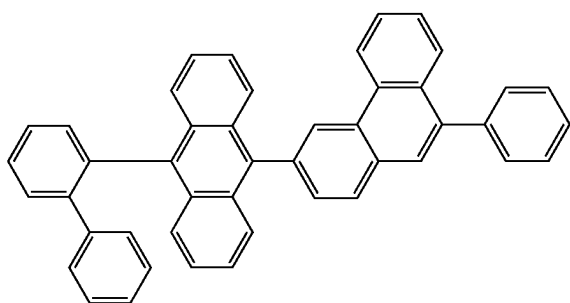


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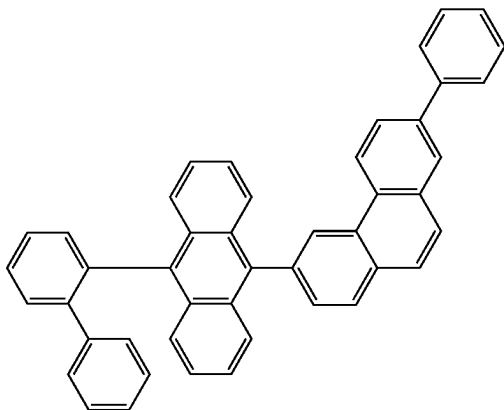
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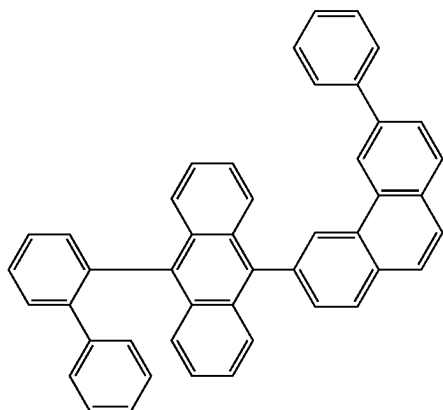
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H1-60

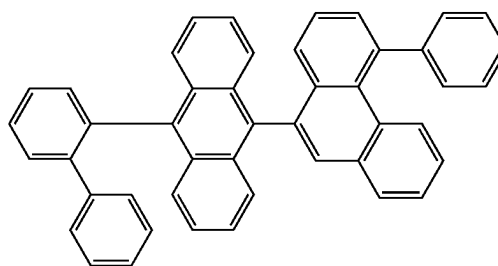


H1-61

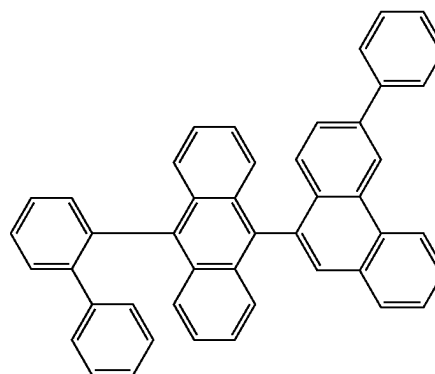


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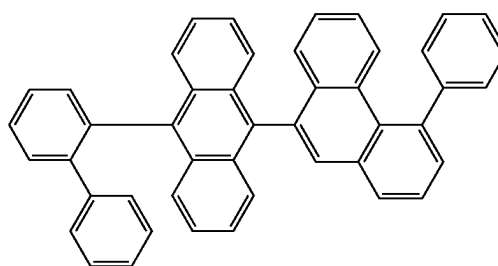
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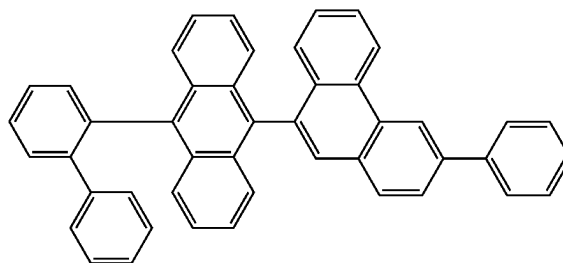
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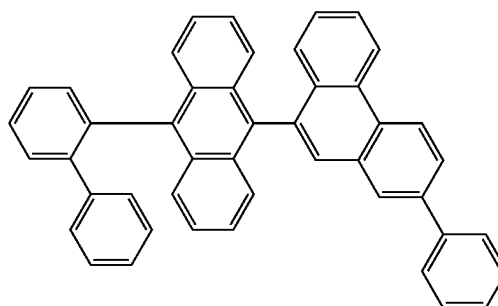
H1-64



H1-65

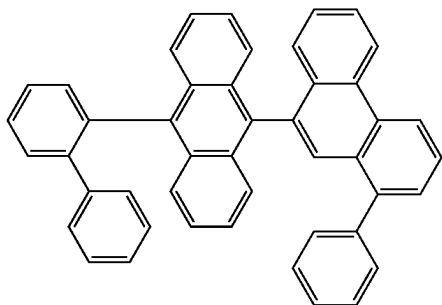


H1-66



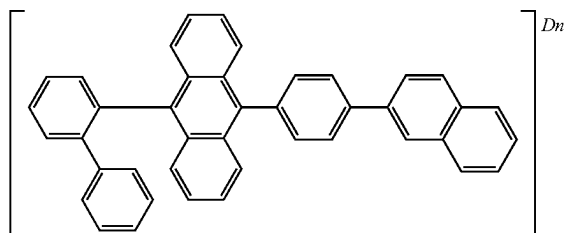
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H1-67

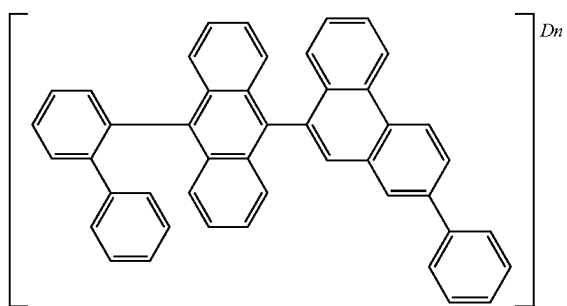


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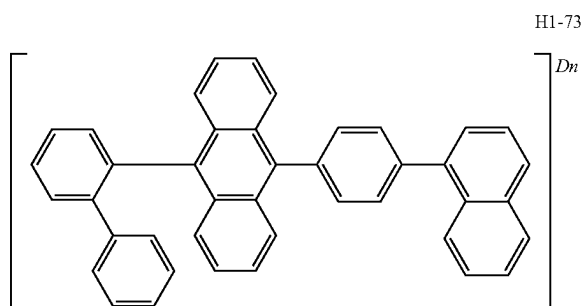
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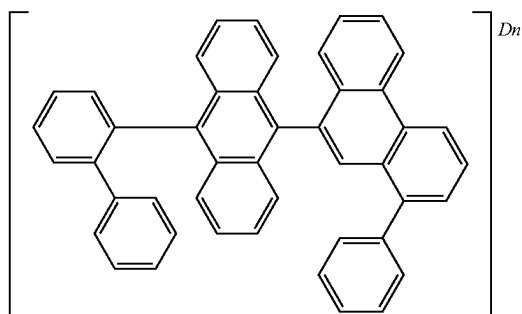
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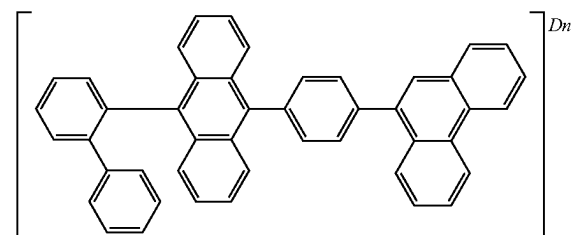
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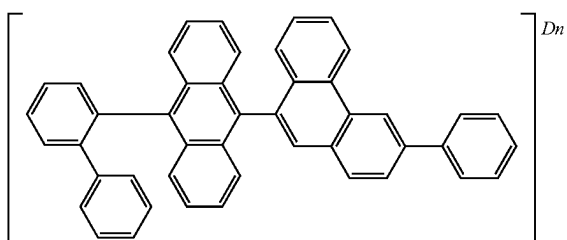
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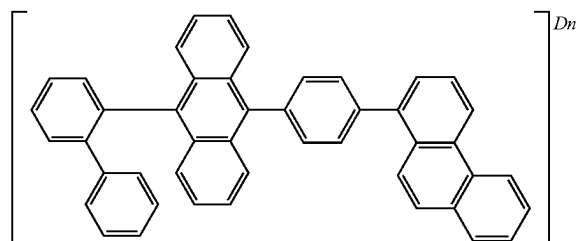
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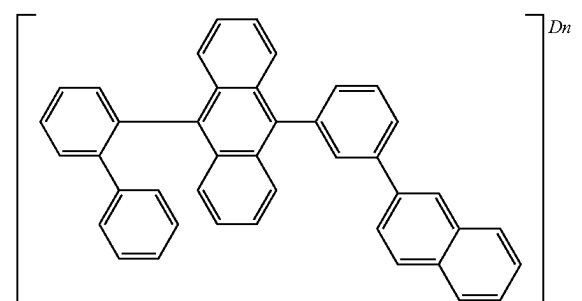
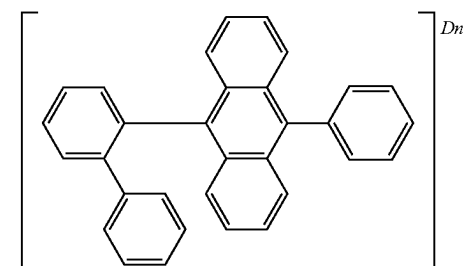
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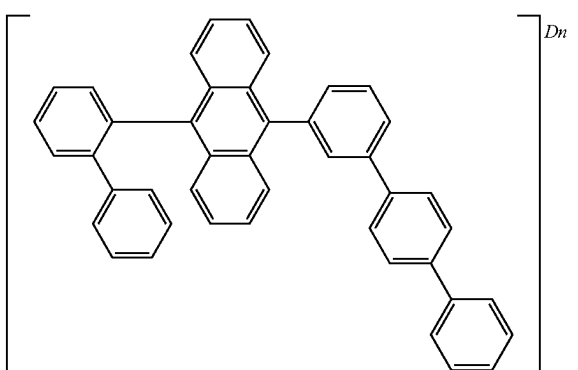
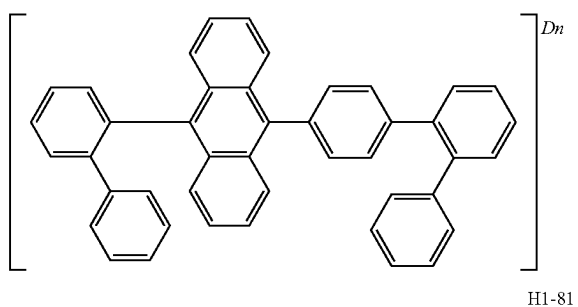
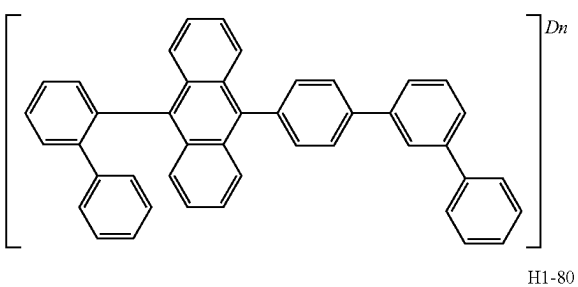
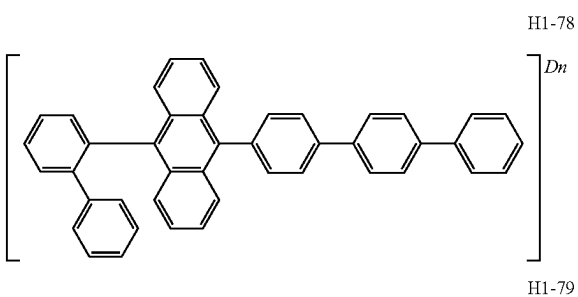
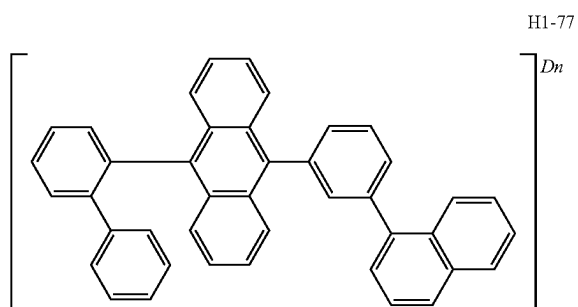
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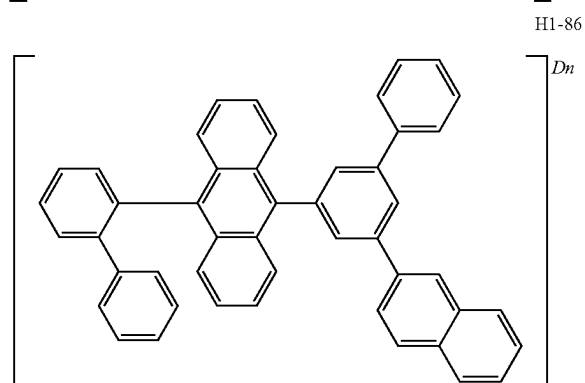
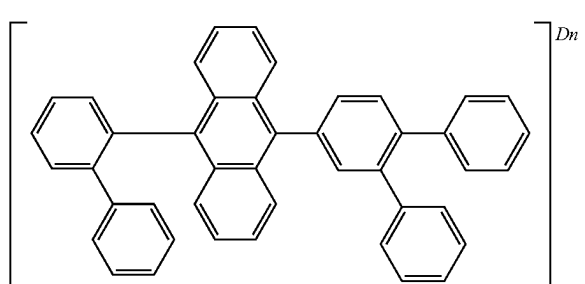
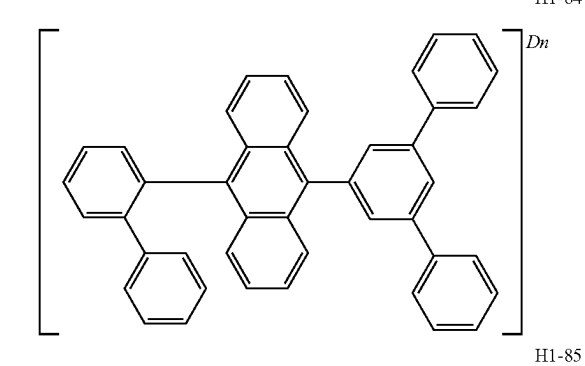
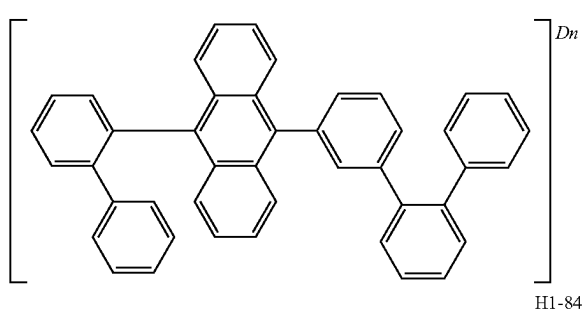
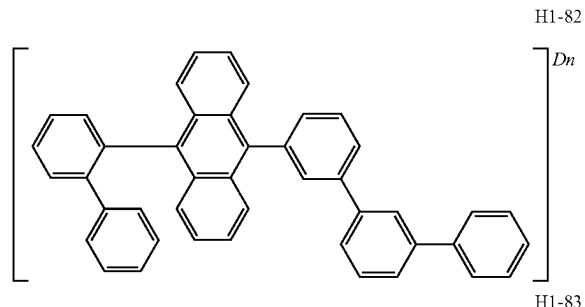
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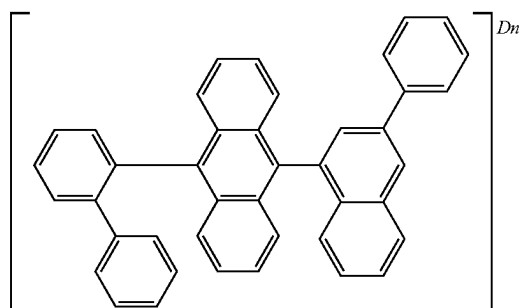
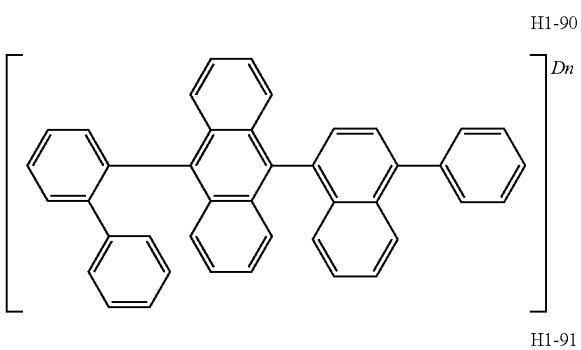
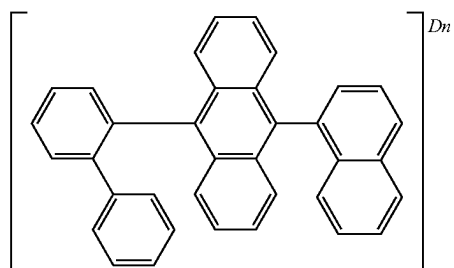
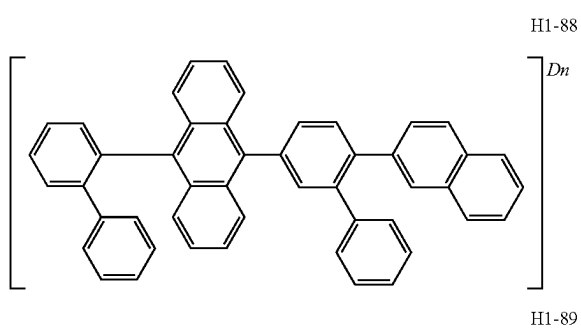
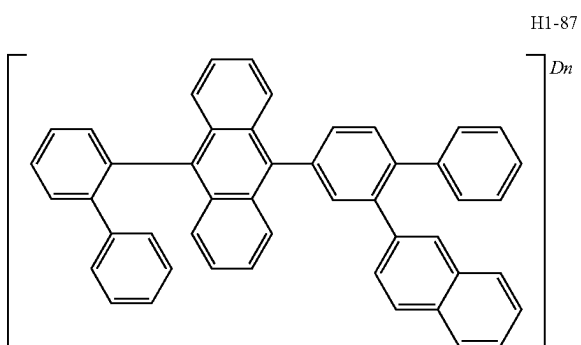
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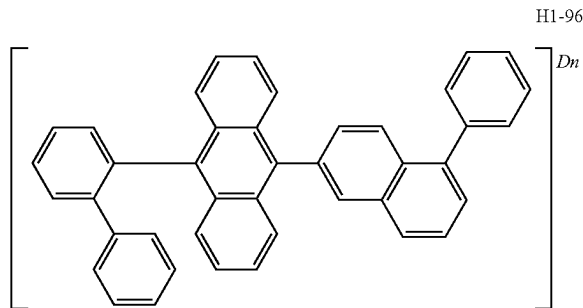
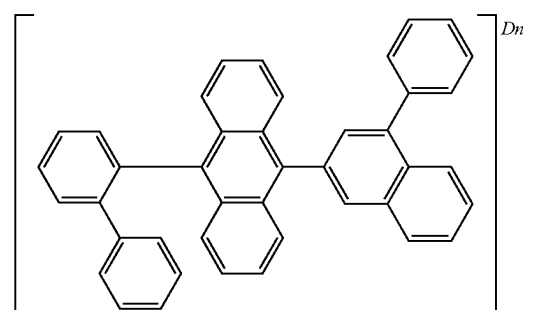
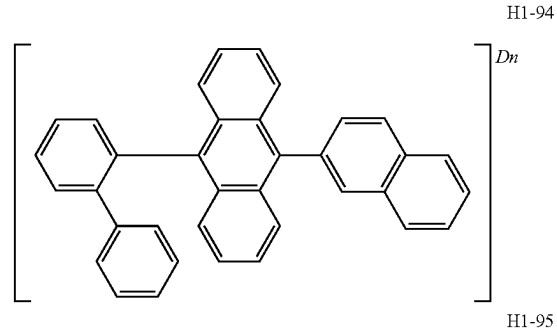
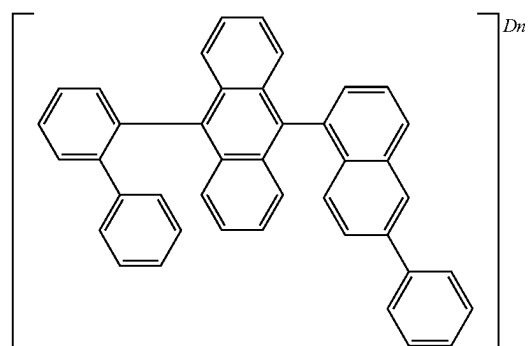
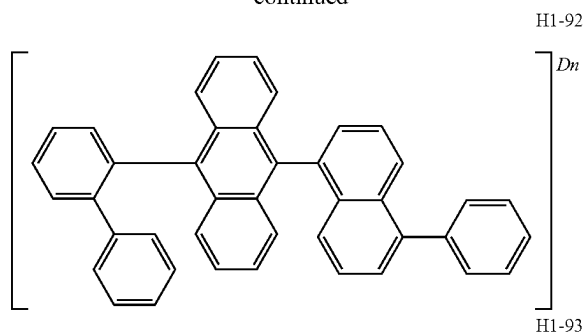
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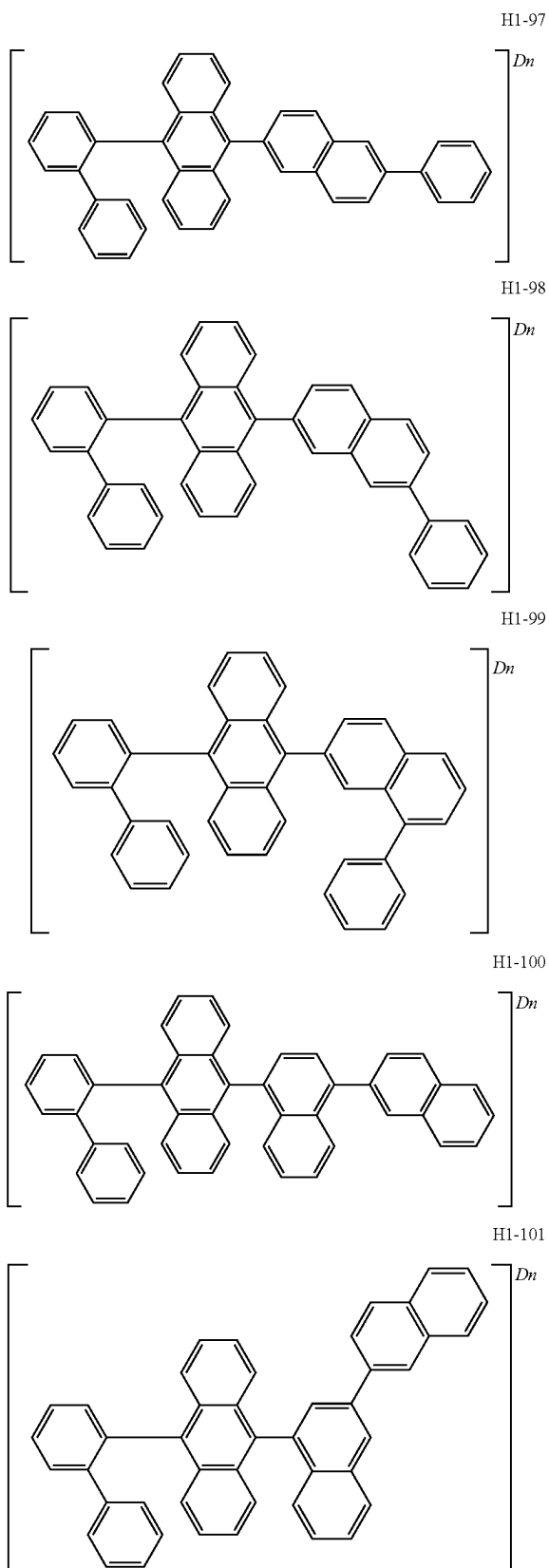
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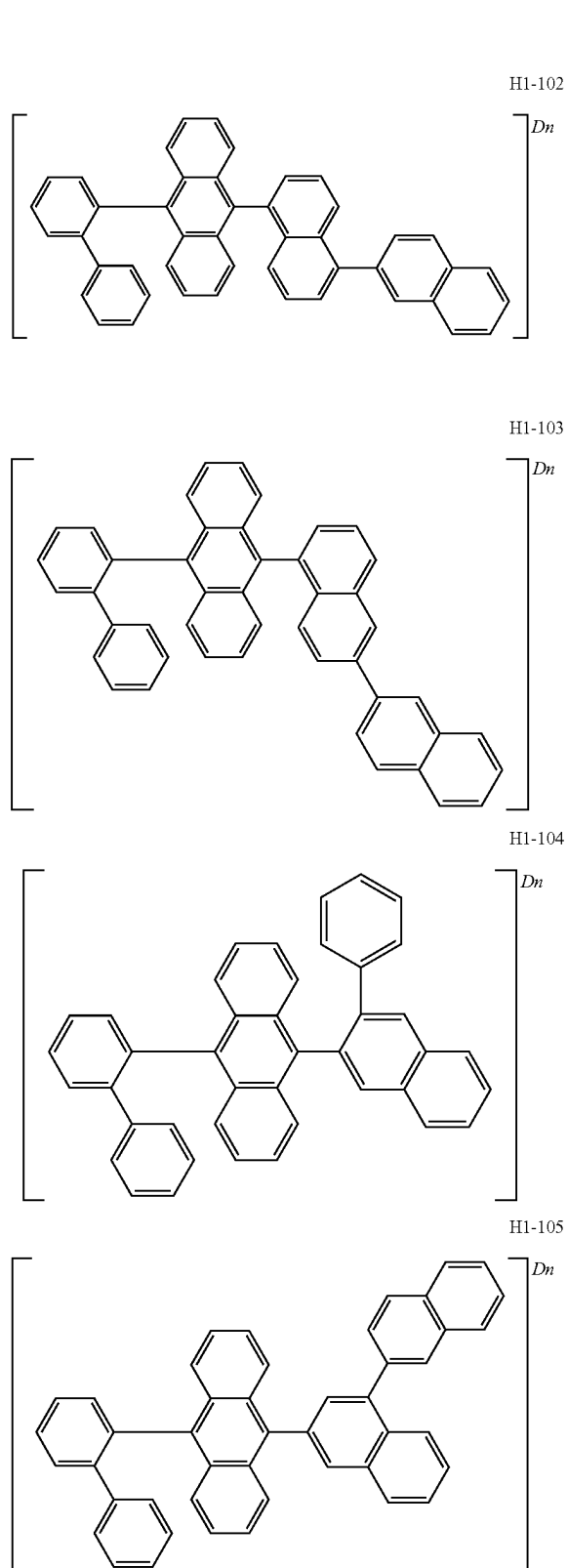
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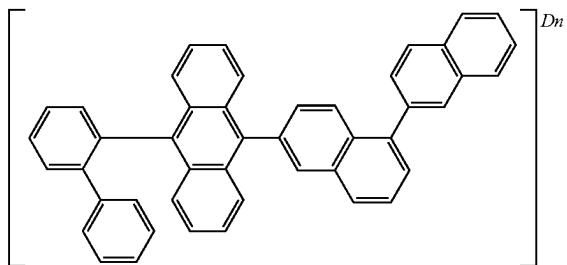


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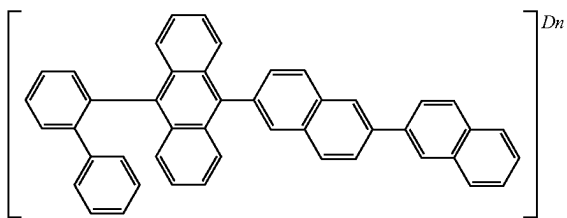


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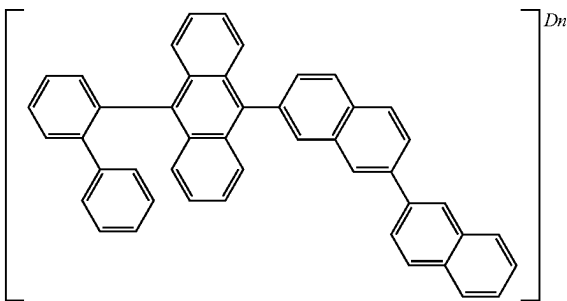
H1-106



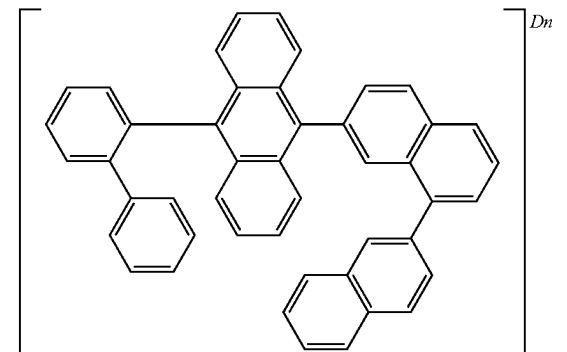
H1-107



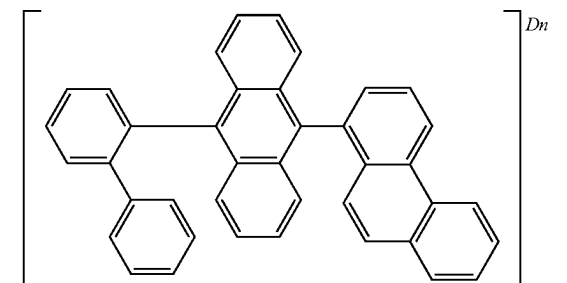
H1-108



H1-109

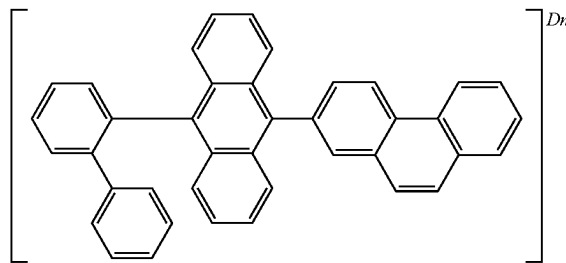


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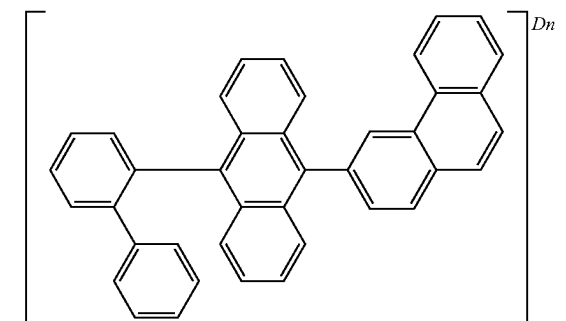


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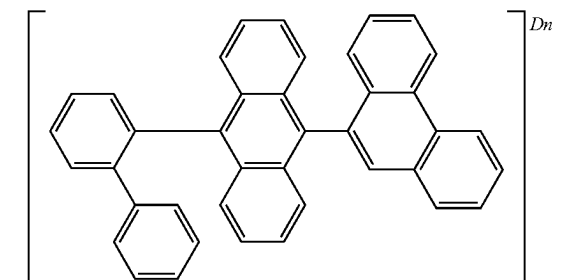
H1-111



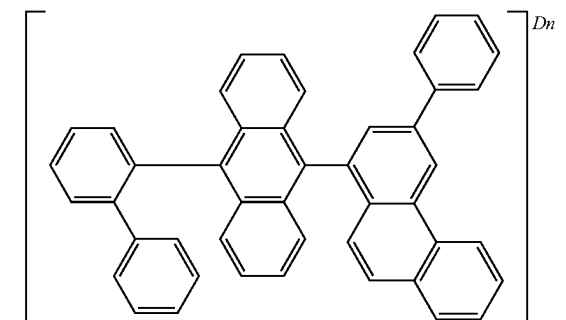
H1-112



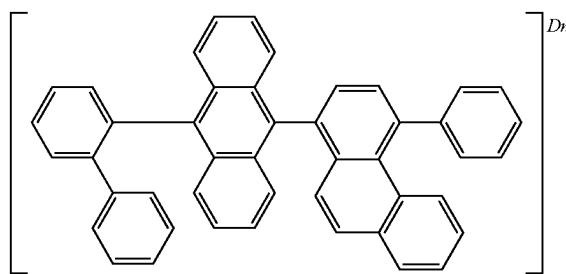
H1-113



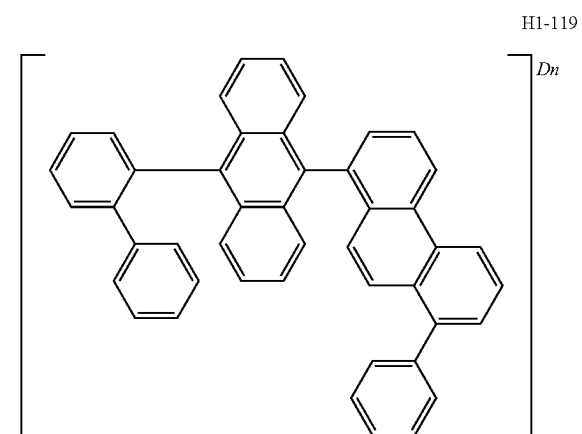
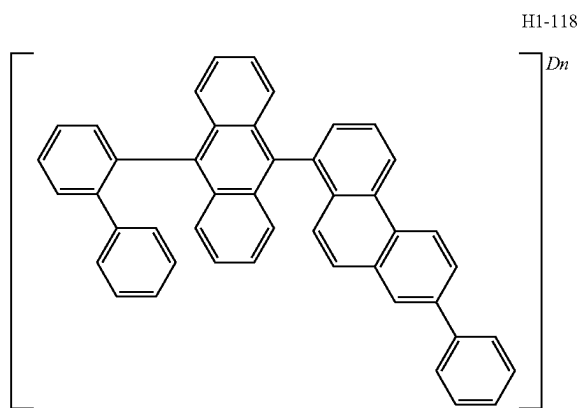
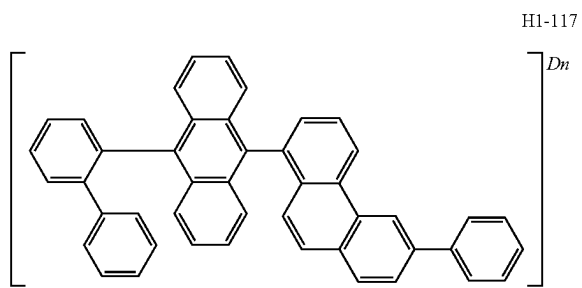
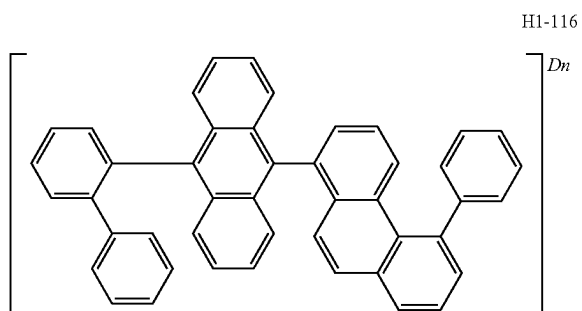
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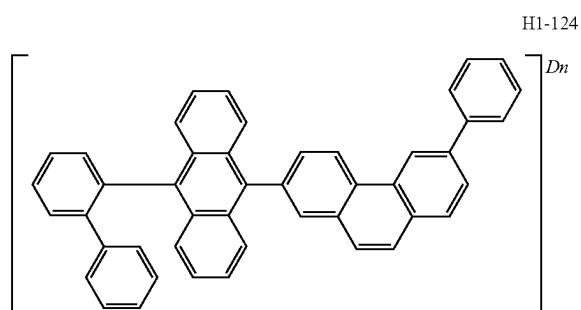
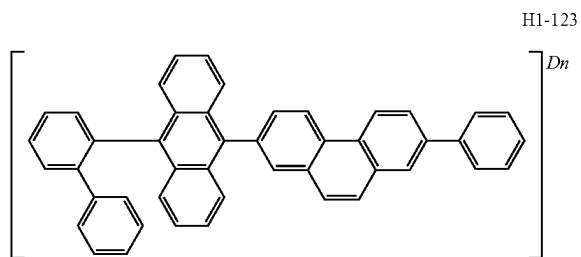
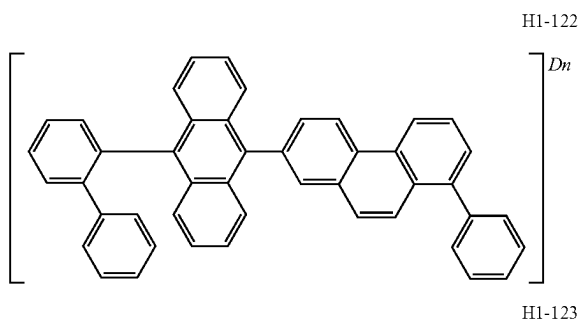
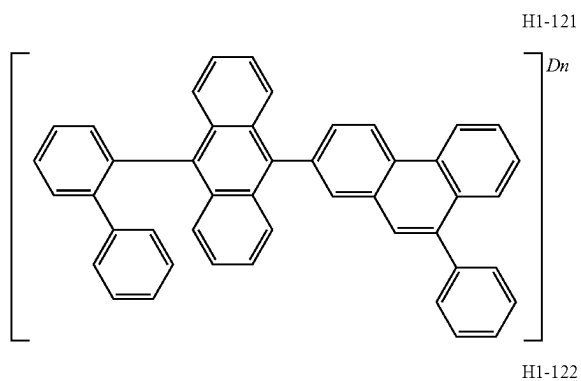
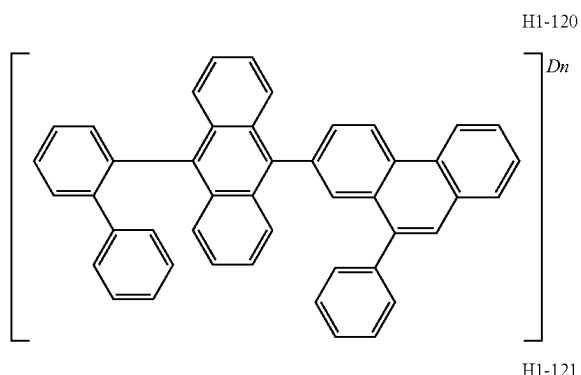
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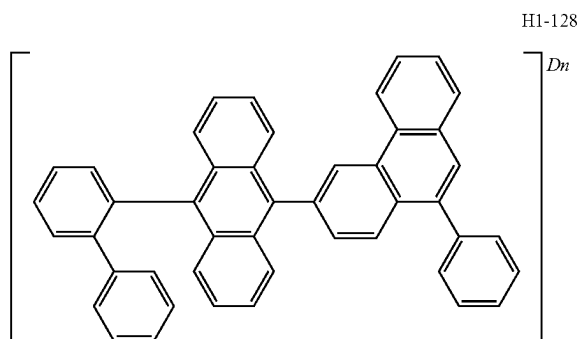
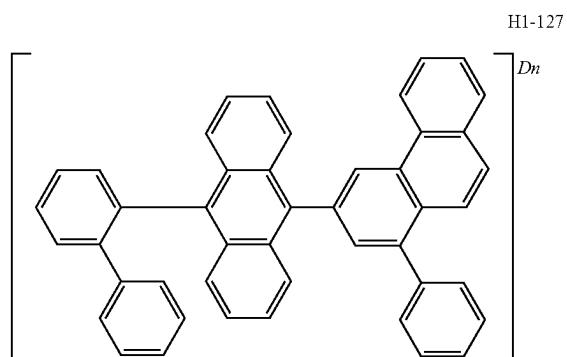
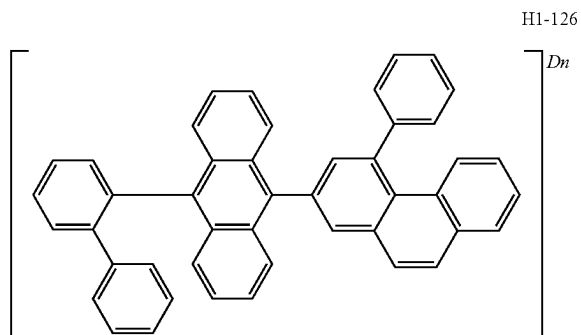
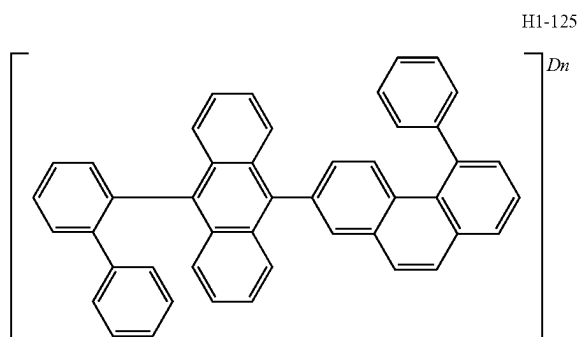
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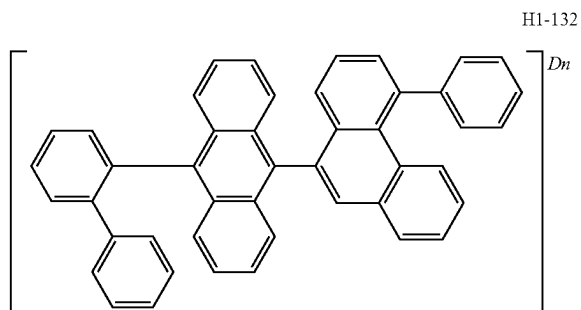
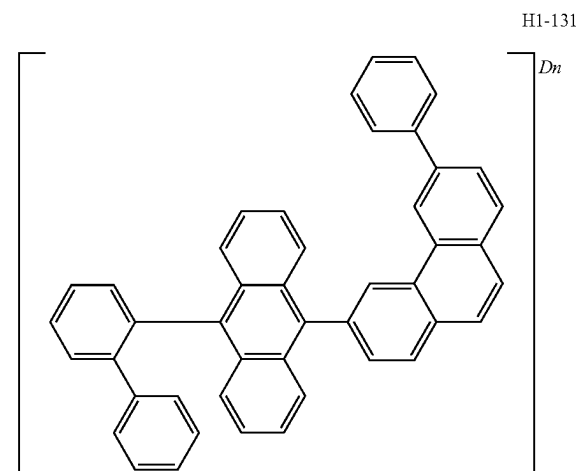
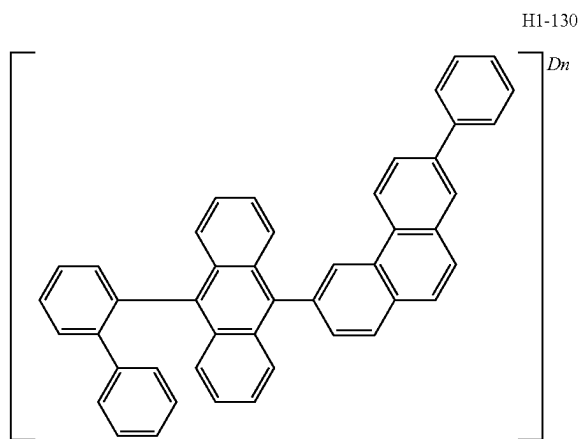
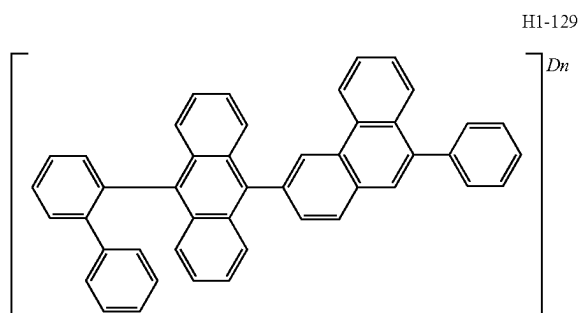
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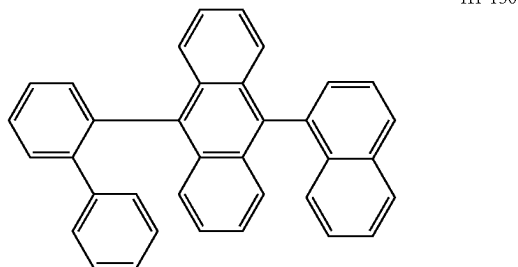
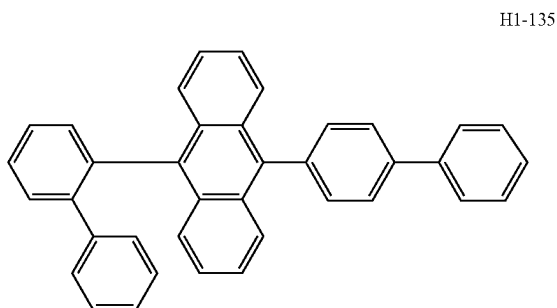
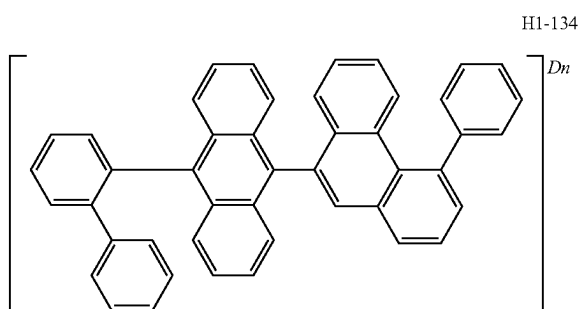
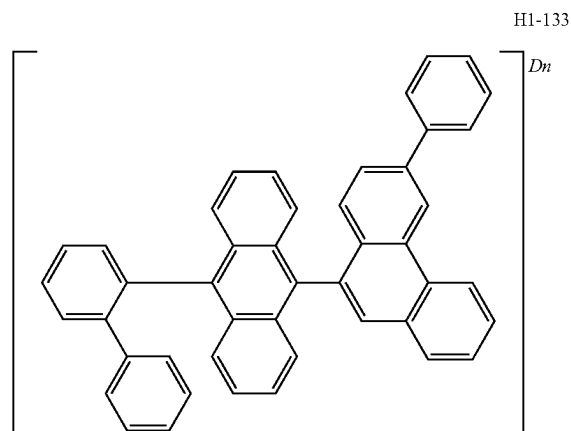
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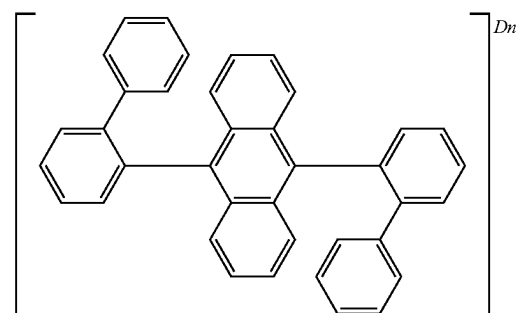
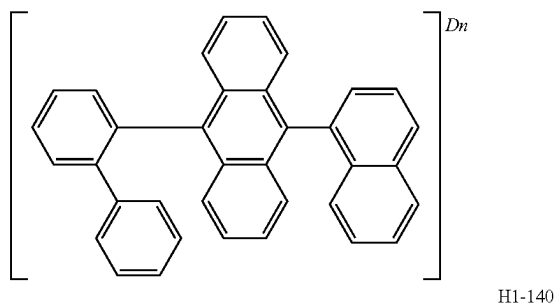
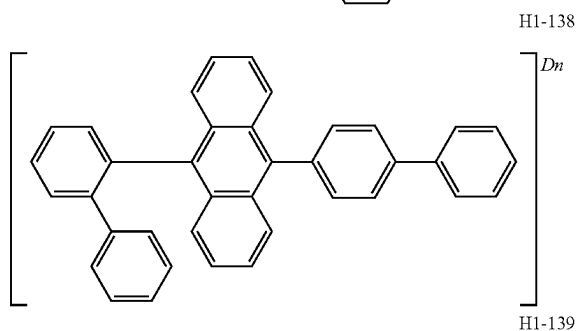
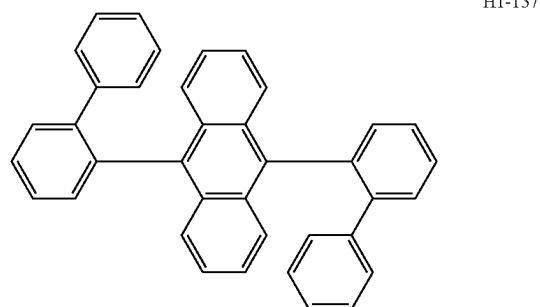
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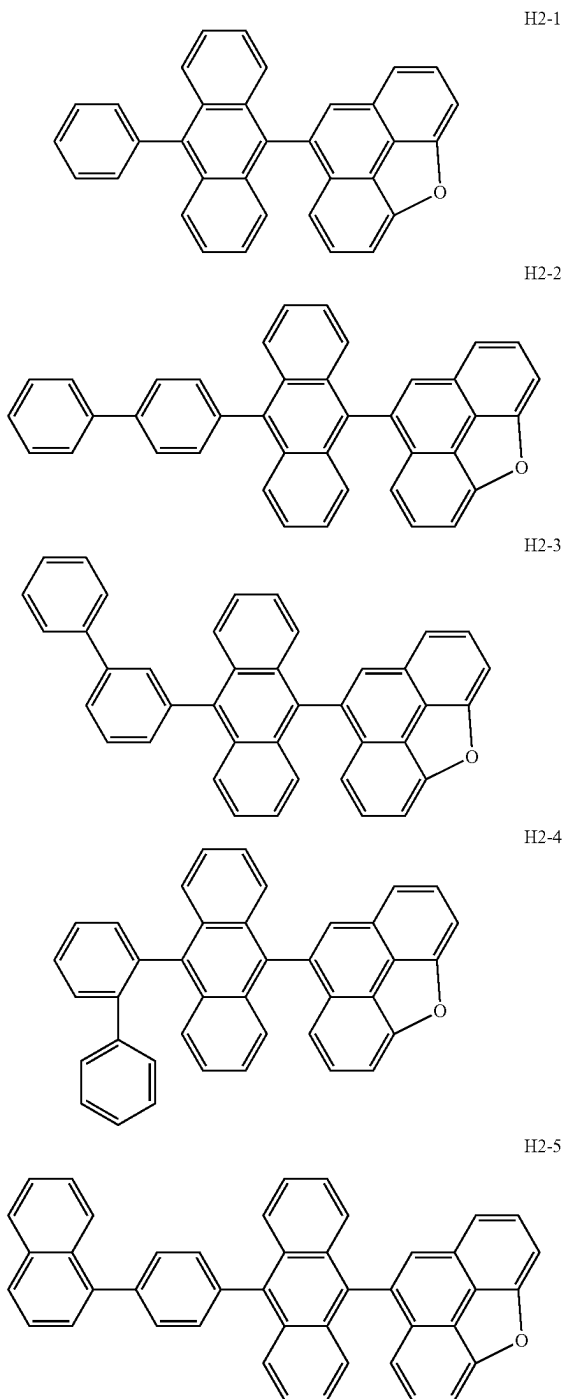
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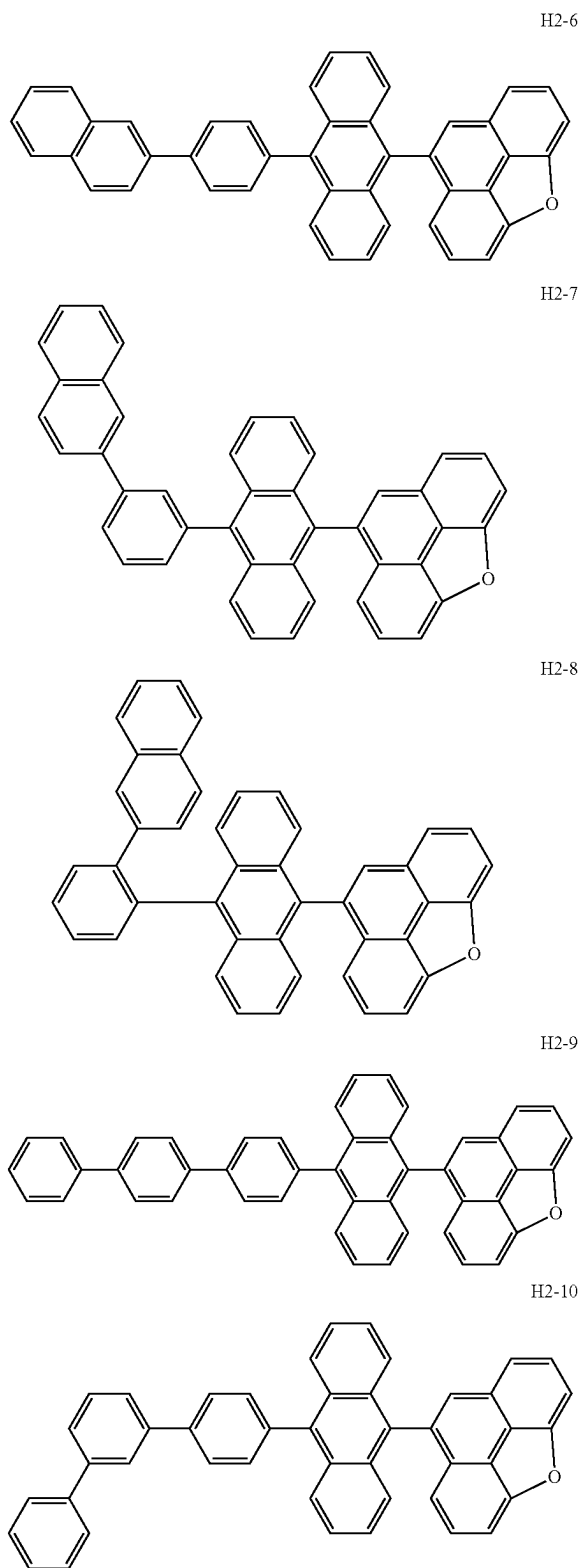
[0065] In the above compounds, D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to the maximum number of hydrogens in the compound. Specifically, n is a minimum of 1 and an integer up to the maximum number of hydrogens in the compound. According to one embodiment of the present disclosure, in one compound represented by Formula 1, when deuterium is included, the deuterium substitution rate is preferably about 100% or less, more preferably about 95% or less, still more preferably about 90% or less, and still more preferably about 85% or less of the total number of hydrogens. The compound of Formula 1 substituted with the above deuterium substitution rate can increase the bond dissociation energy

according to deuteration, thereby enhancing the stability of the compound, and an organic electroluminescent device comprising the compound can exhibit improved lifetime properties.

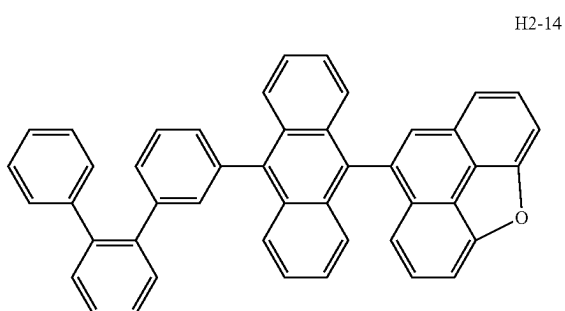
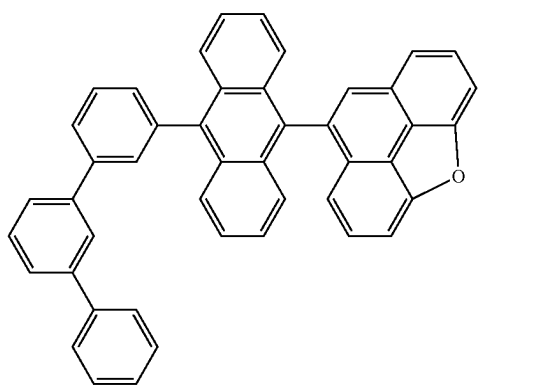
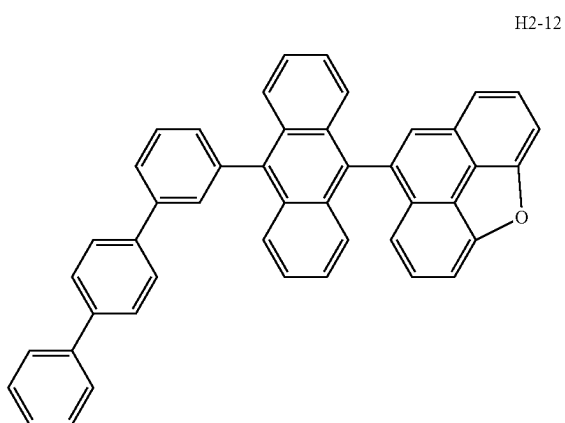
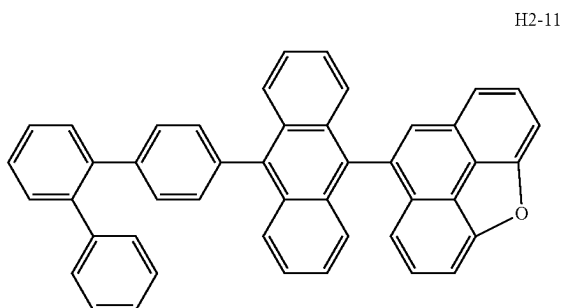
[0066] The compound represented by Formula 2 may be more specifically at least one selected from the group consisting of the following compounds, but is not limited thereto.



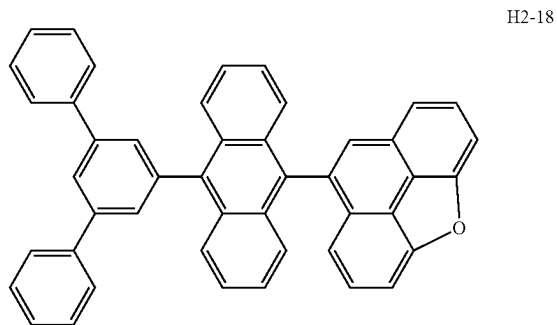
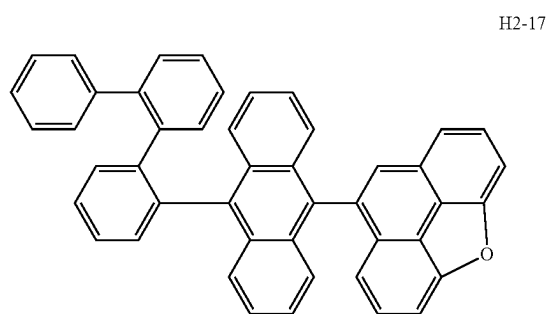
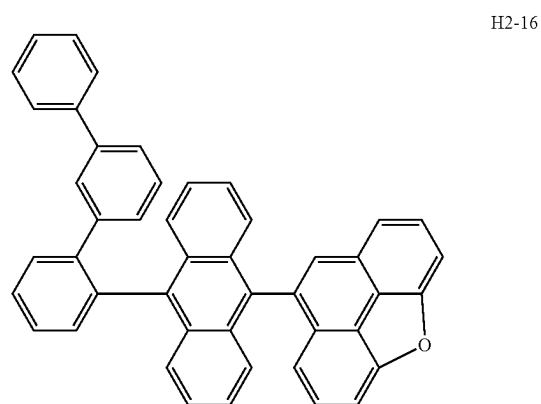
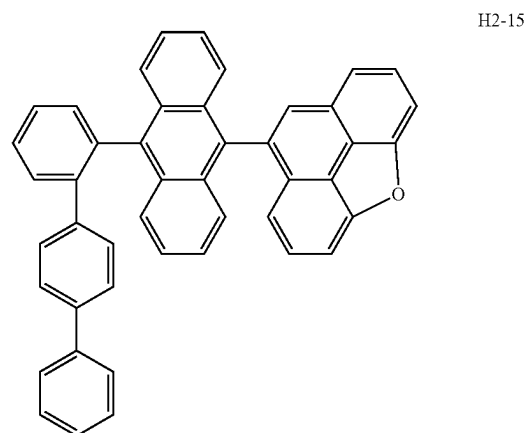
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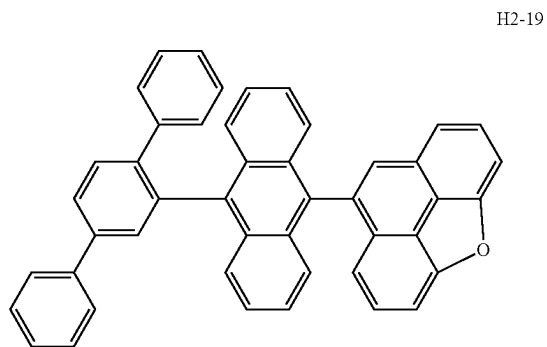
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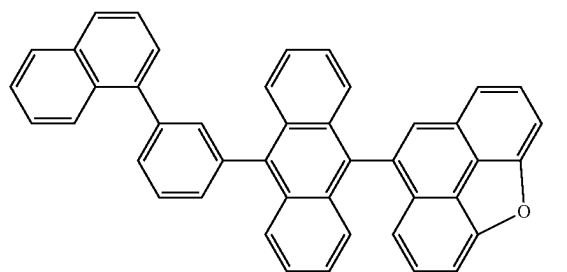
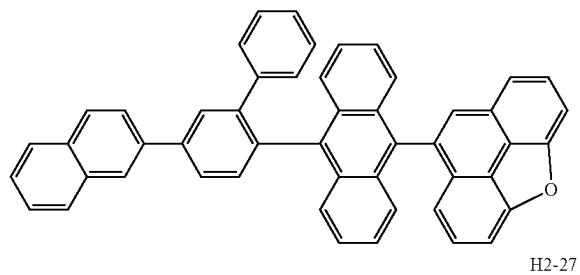
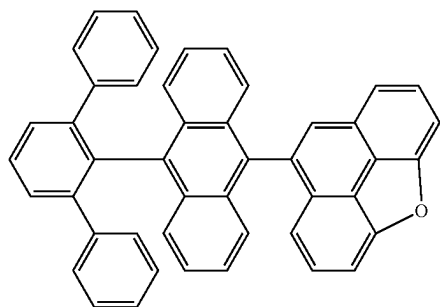
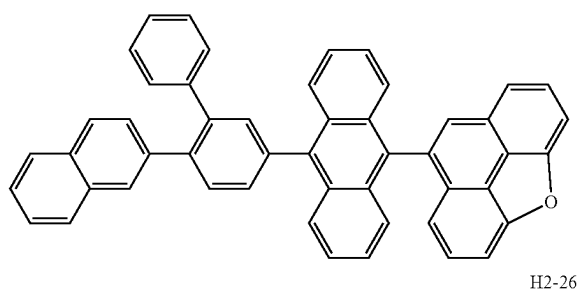
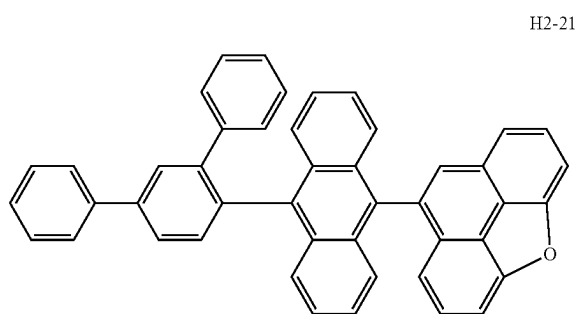
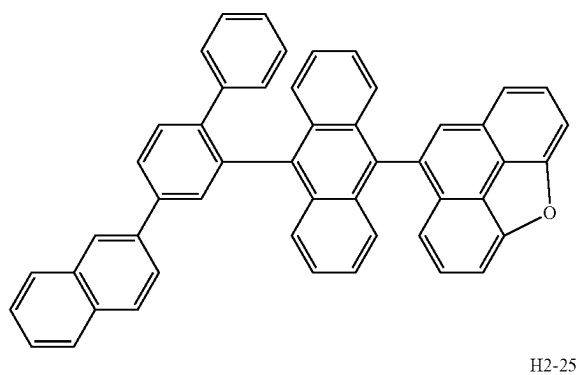
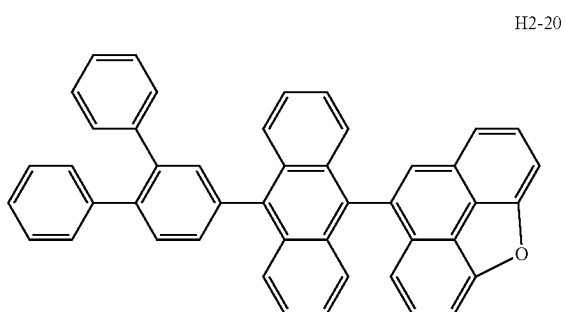
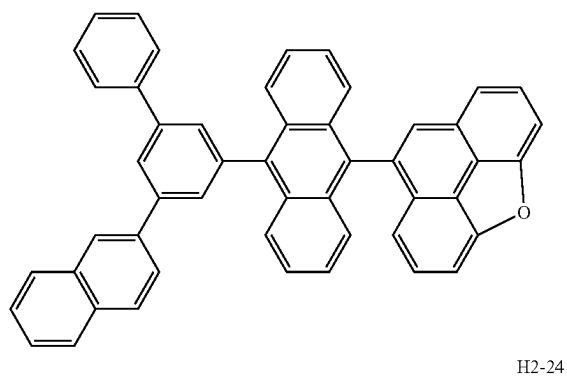
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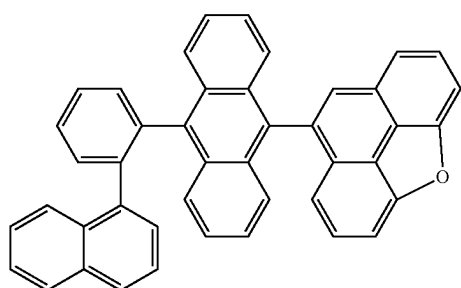
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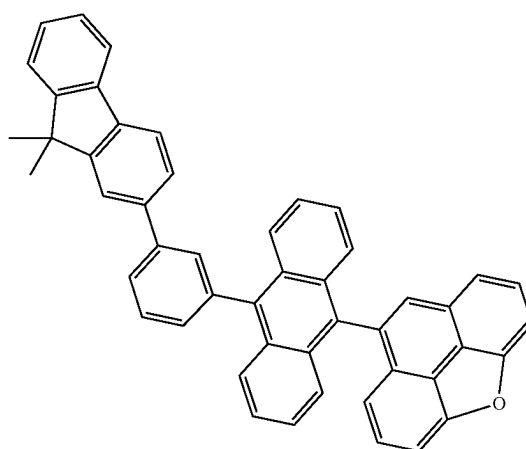


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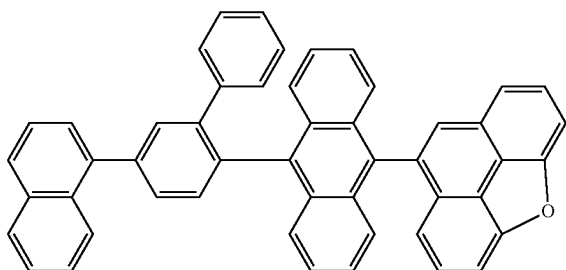


H2-28

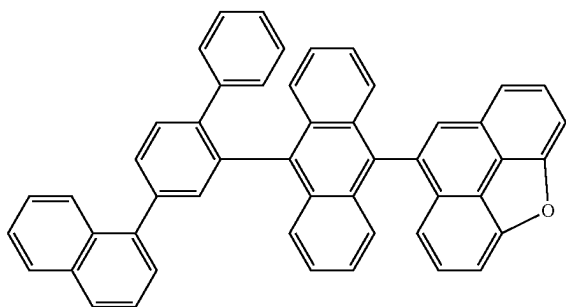
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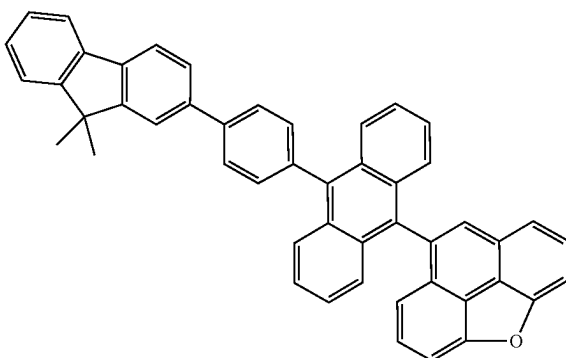
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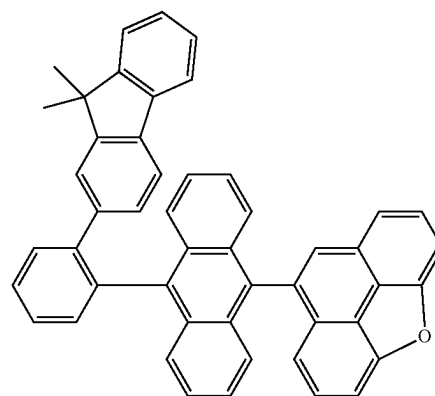
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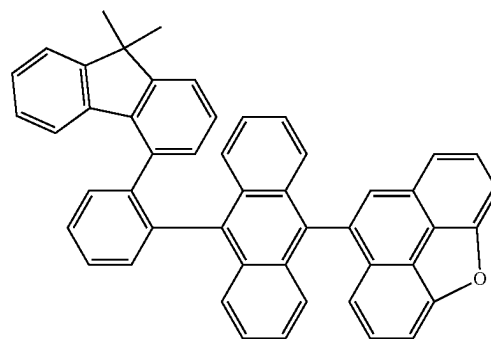
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H2-31



H2-33



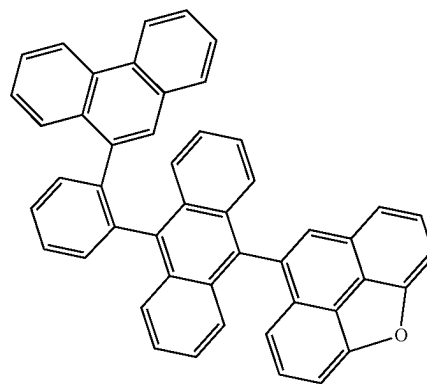
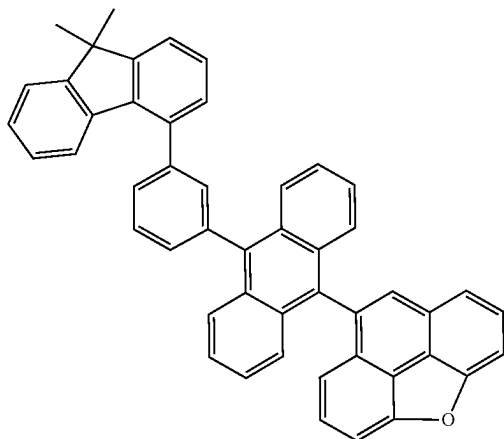
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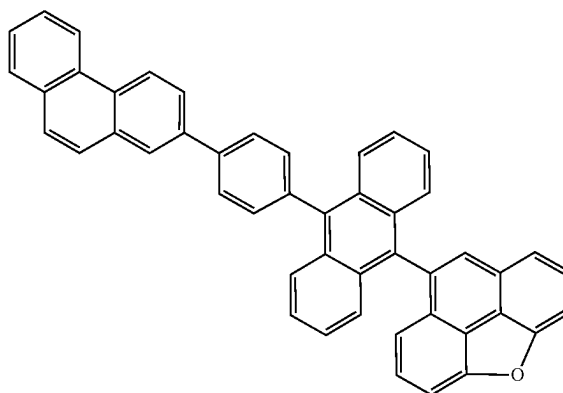
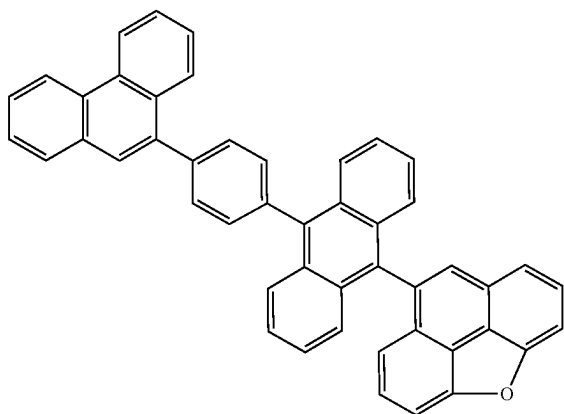
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H2-35



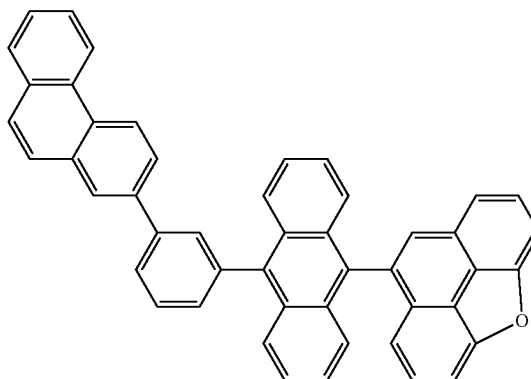
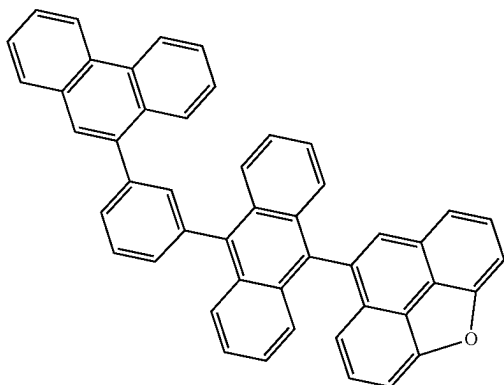
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H2-36



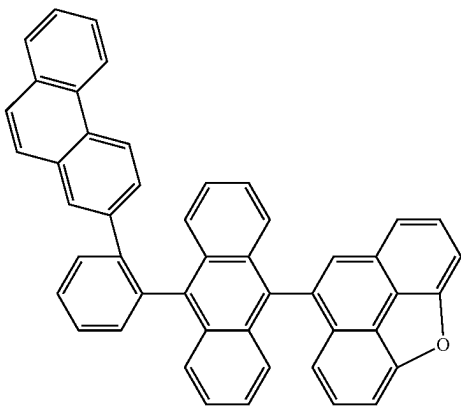
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H2-40



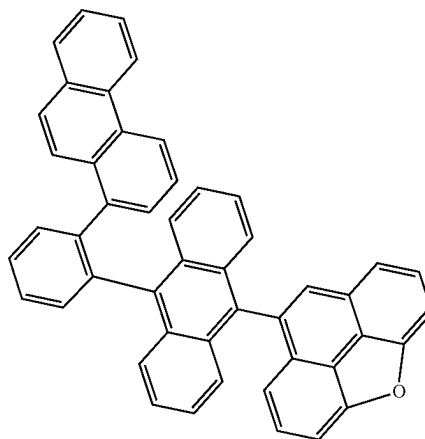
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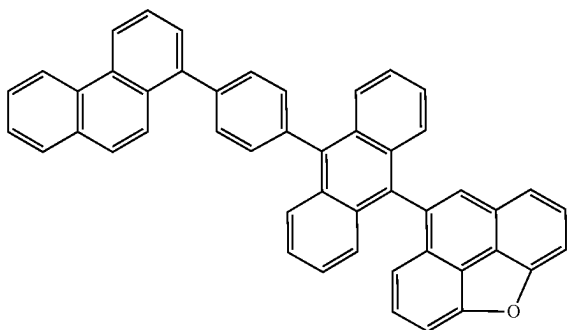


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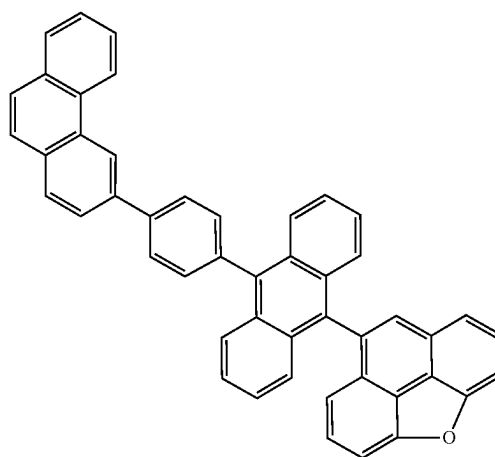
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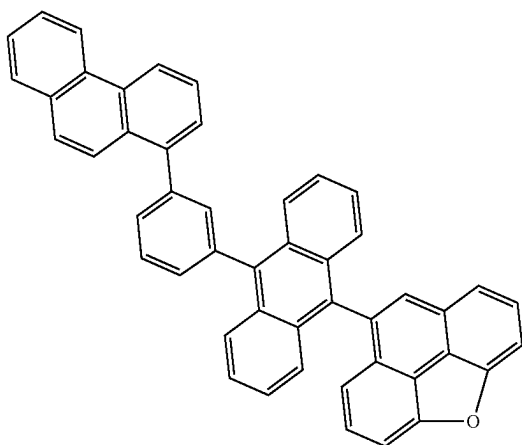
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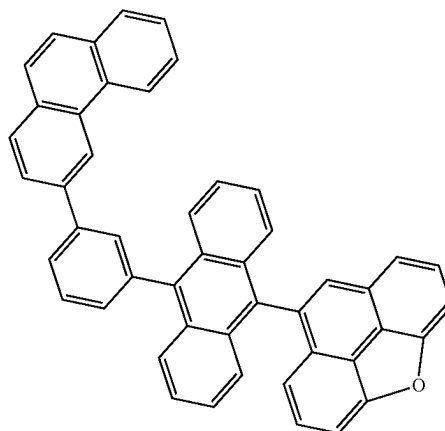
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H2-43

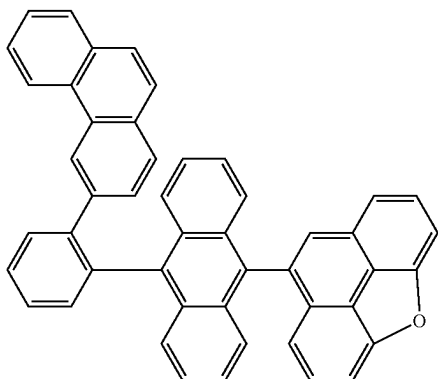


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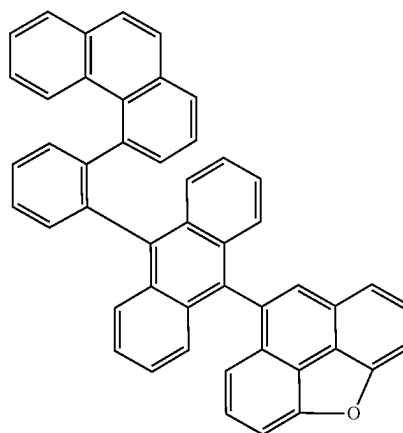
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H2-47



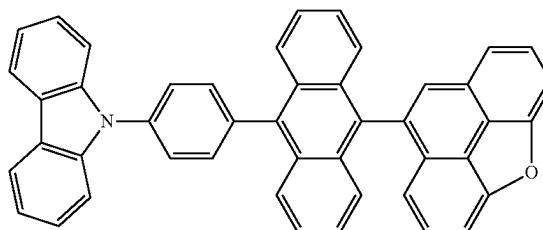
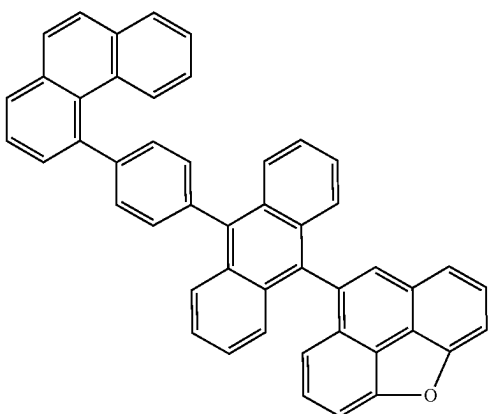
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H2-50



H2-48

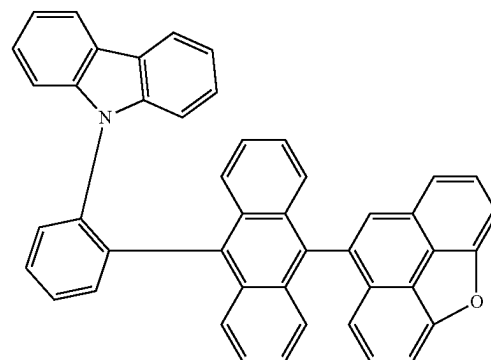
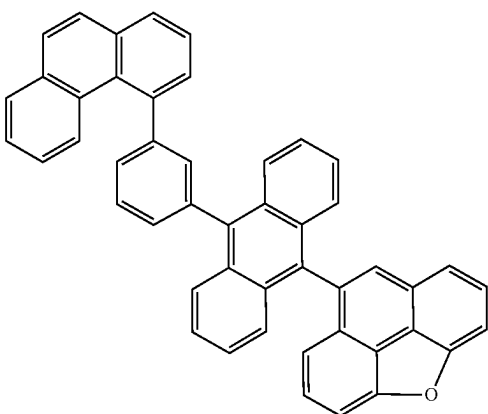
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H2-52

H2-49

H2-53

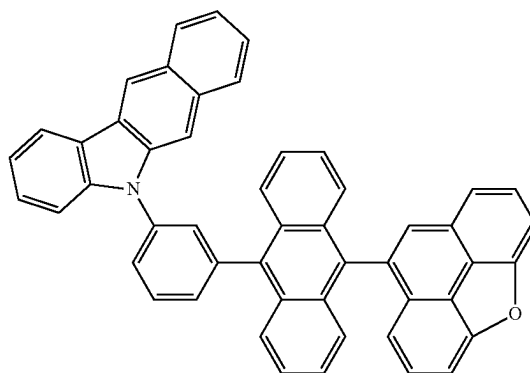
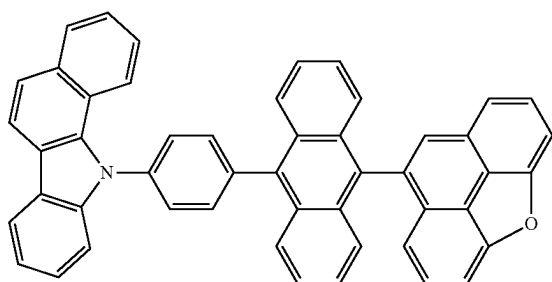


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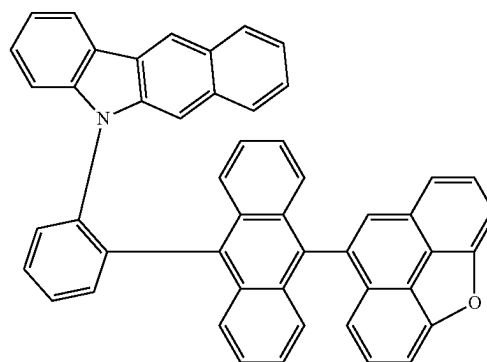
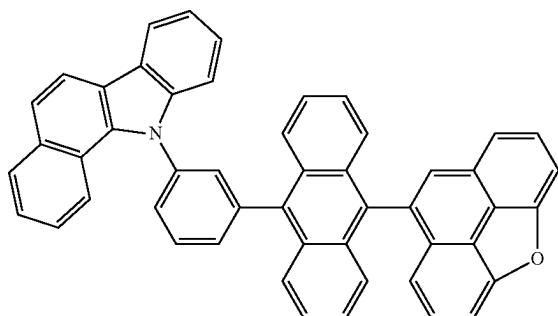
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H2-54



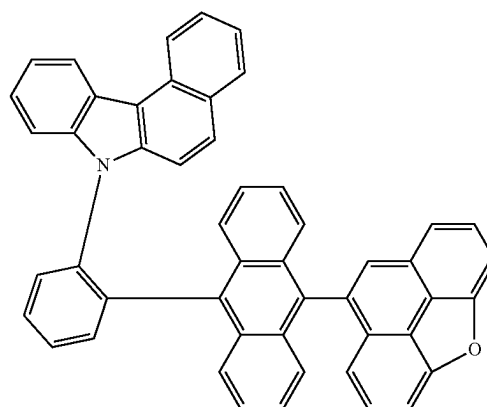
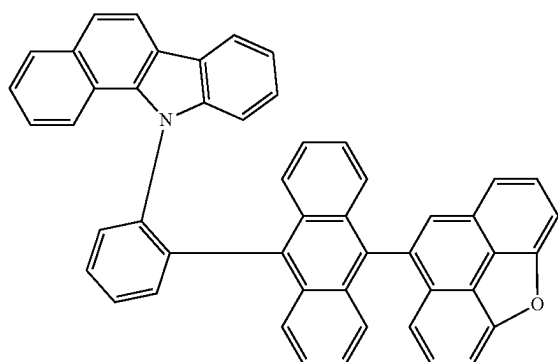
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H2-55



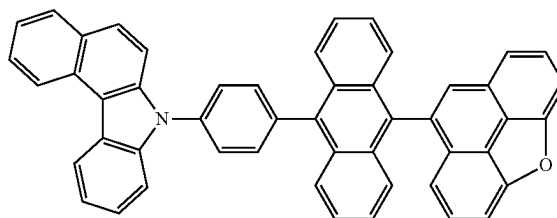
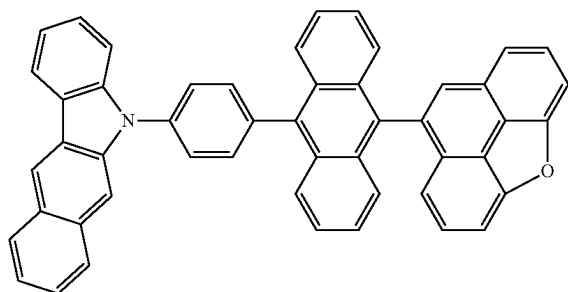
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H2-60



H2-57

H2-61

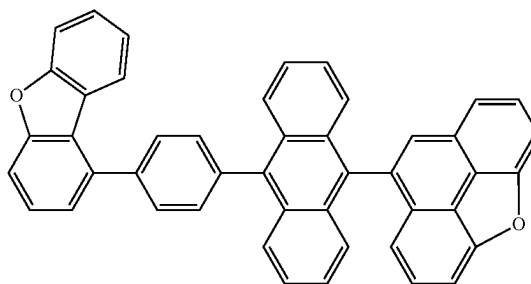
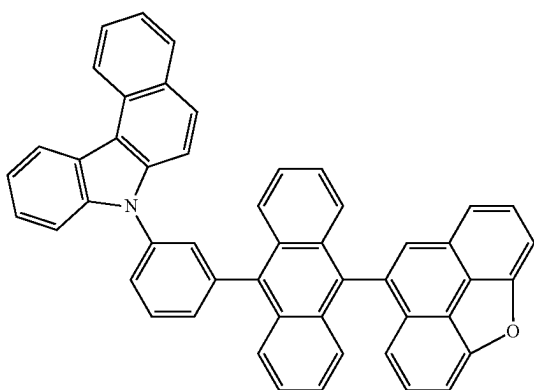


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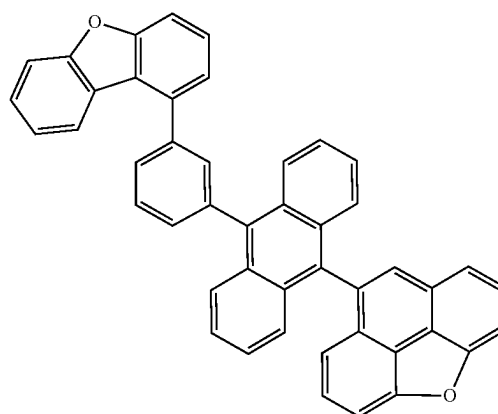
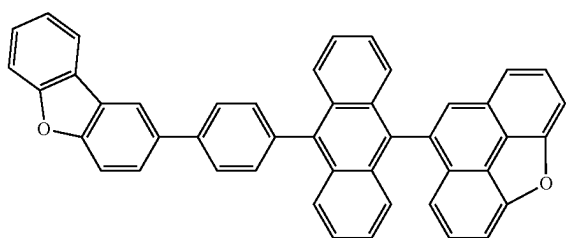
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H2-62



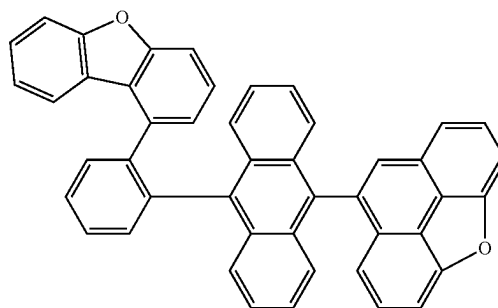
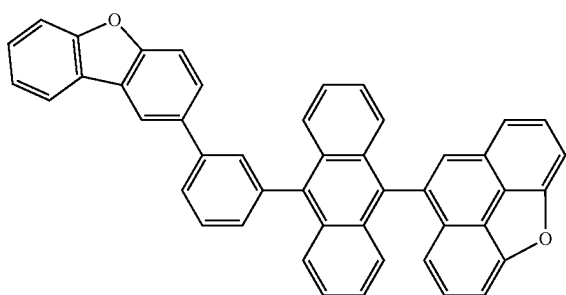
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H2-63



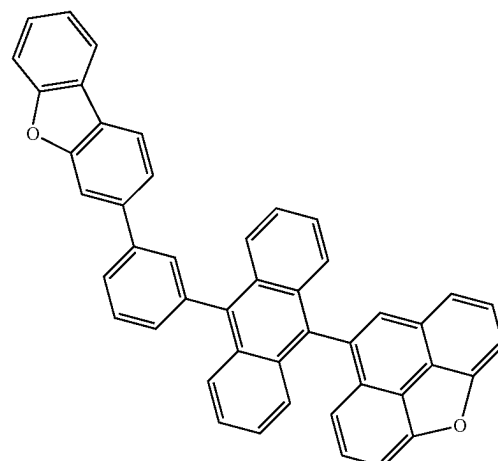
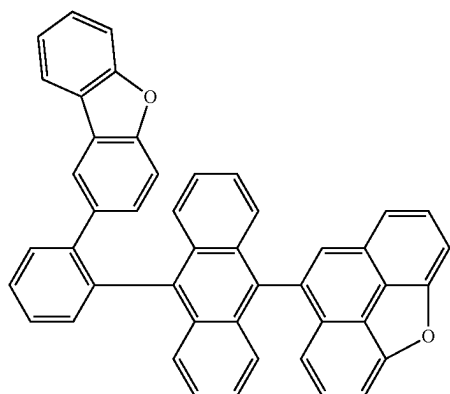
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H2-64



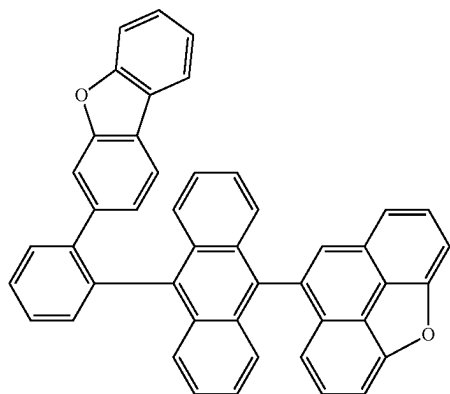
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H2-65



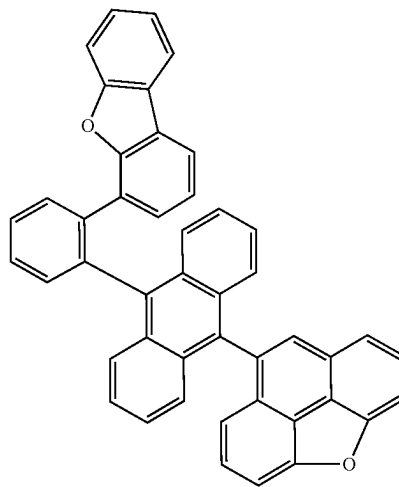
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H2-70

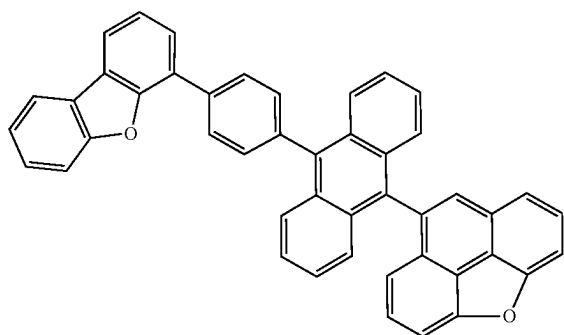


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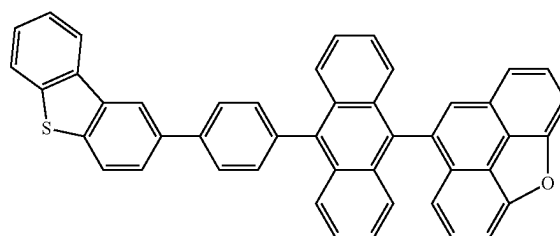
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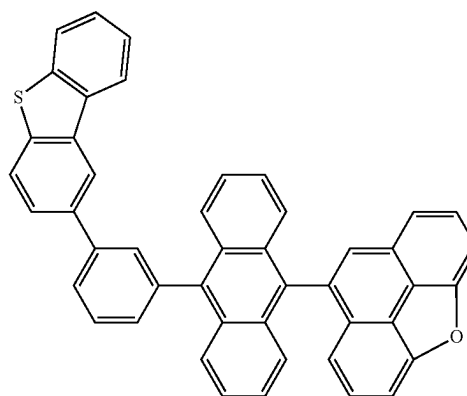
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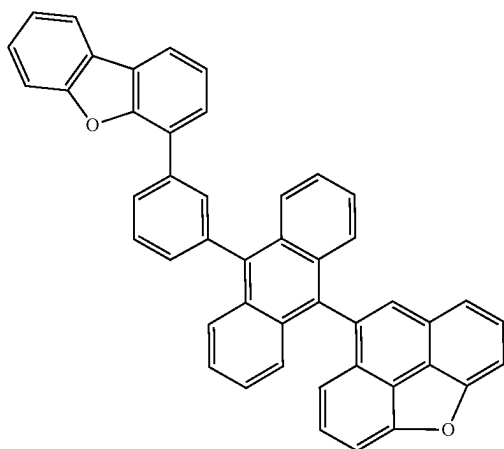
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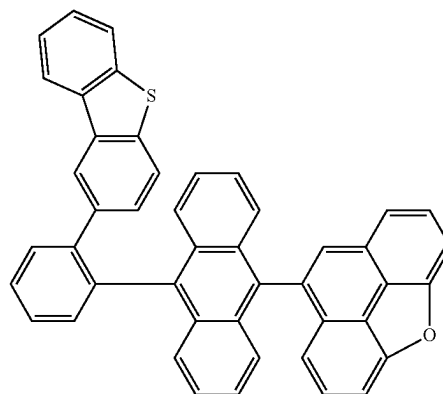
H2-75



H2-72

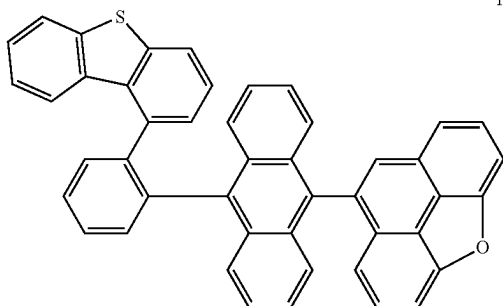


H2-76

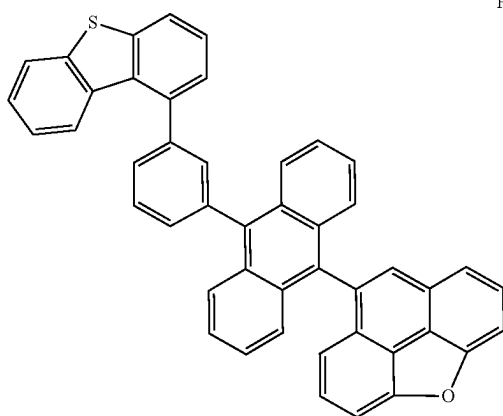


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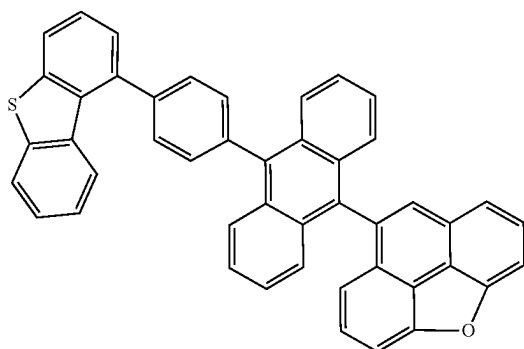
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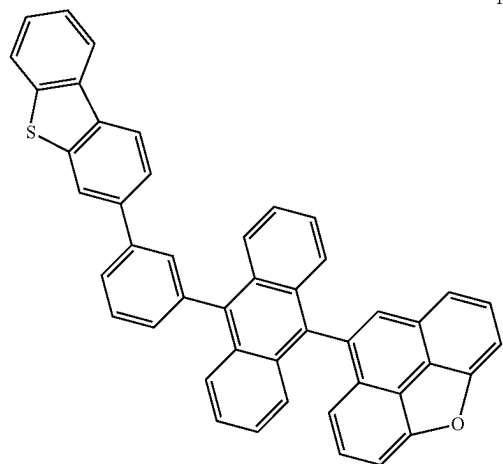
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H2-79

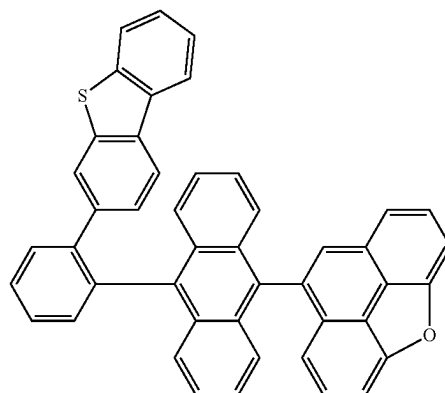


H2-80

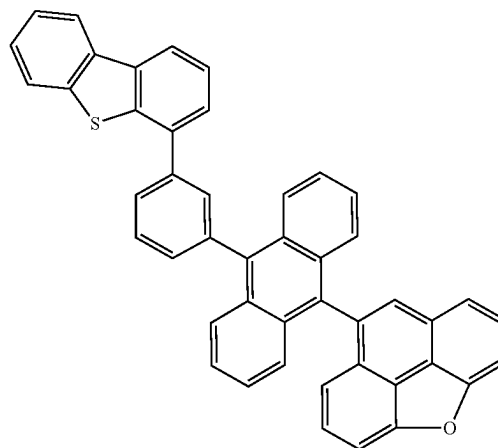


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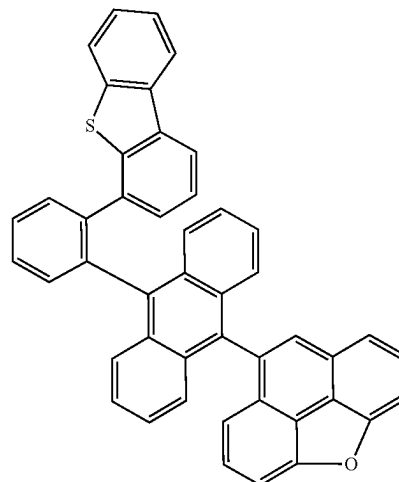
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H2-82

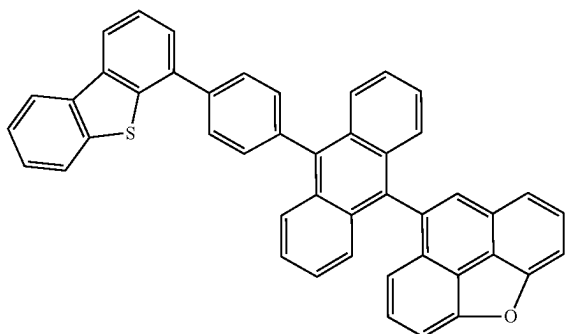


H2-83

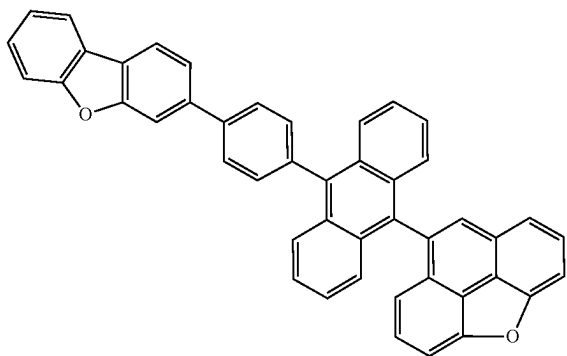


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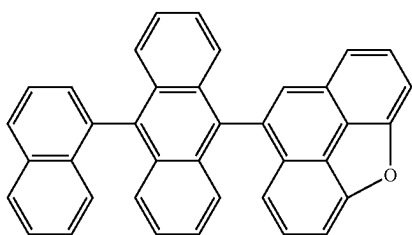
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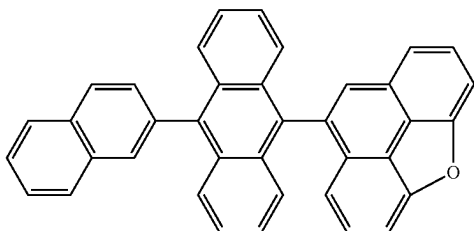
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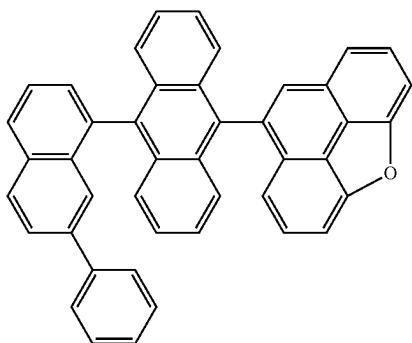
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H2-87

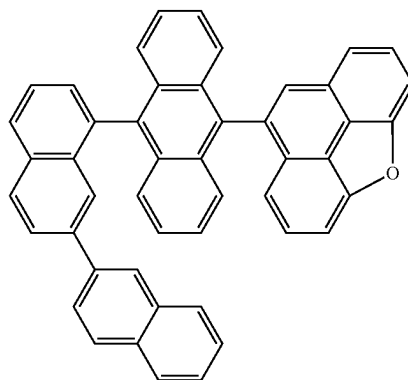


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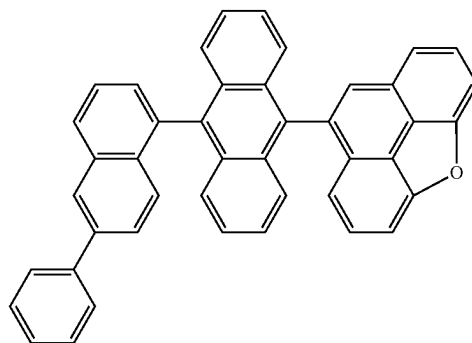


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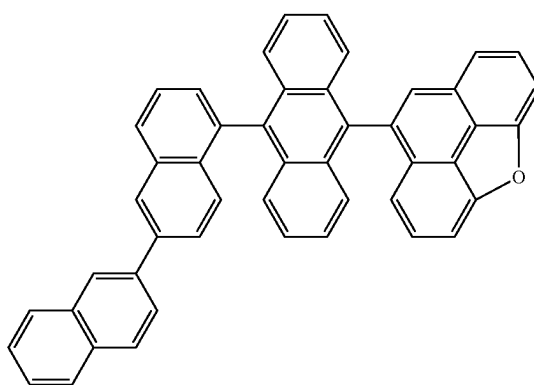
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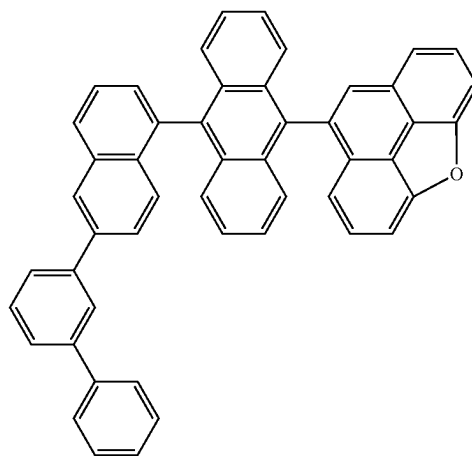
H2-90



H2-91

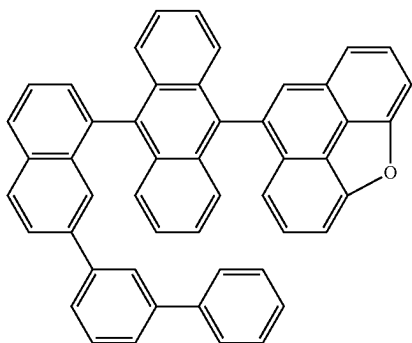


H2-92



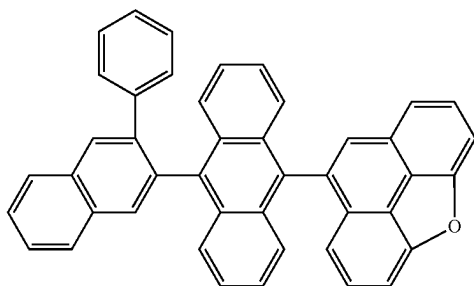
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H2-93

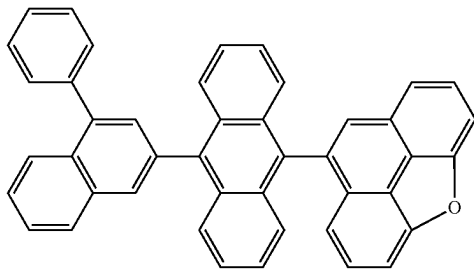


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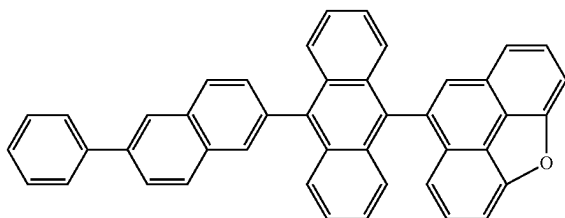
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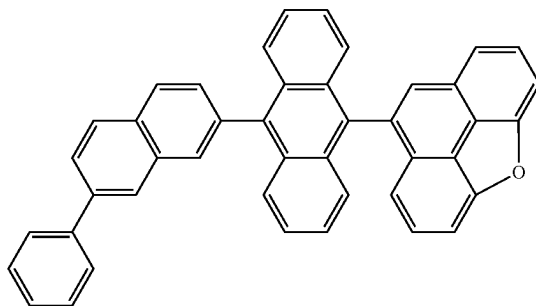
H2-99



H2-100

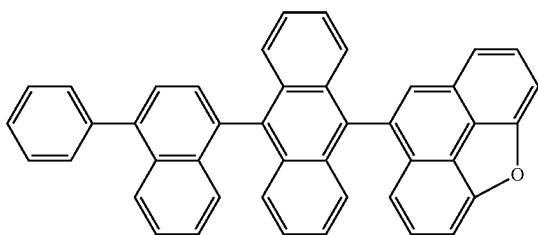


H2-101

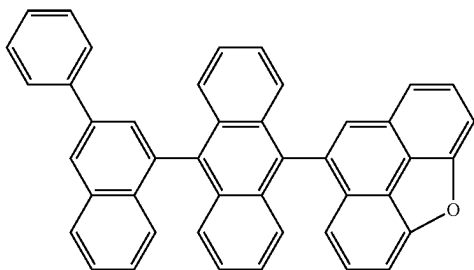


H2-102

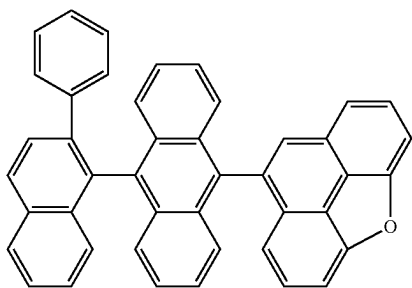
H2-94



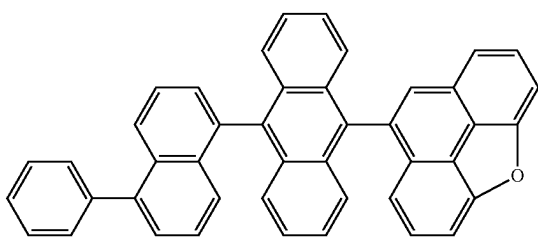
H2-95



H2-96

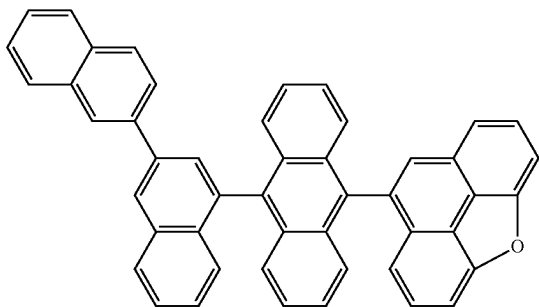


H2-97



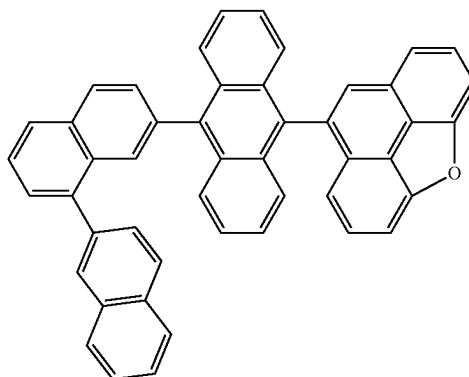
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H2-103

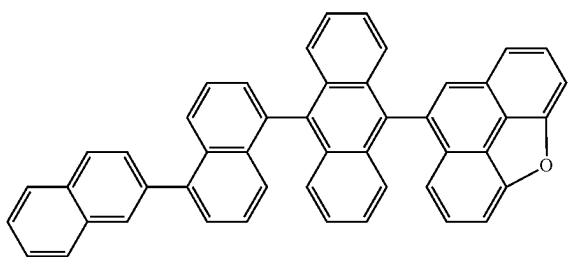


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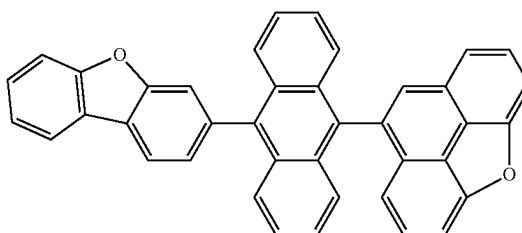
H2-107



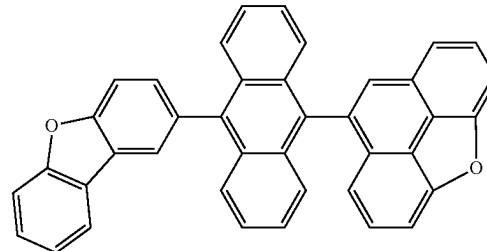
H2-104



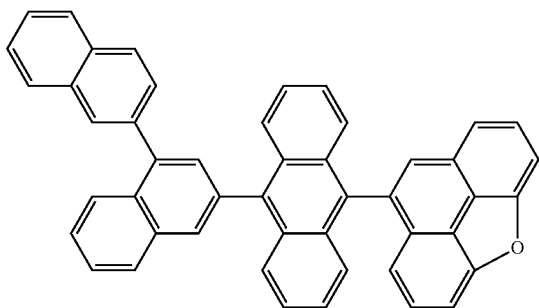
H2-108



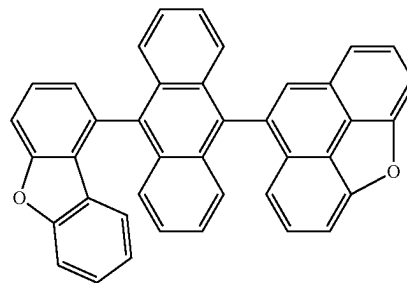
H2-109



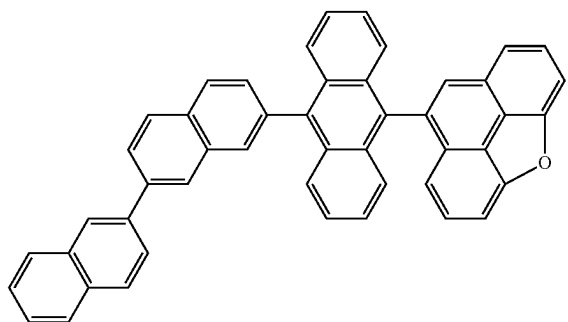
H2-105



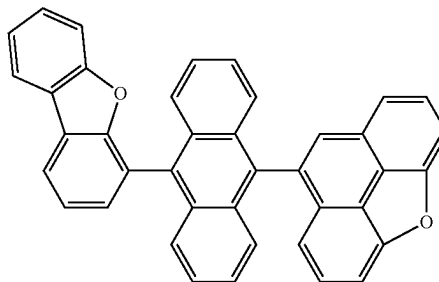
H2-110



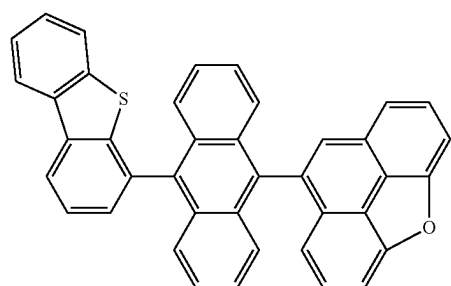
H2-106



H2-111

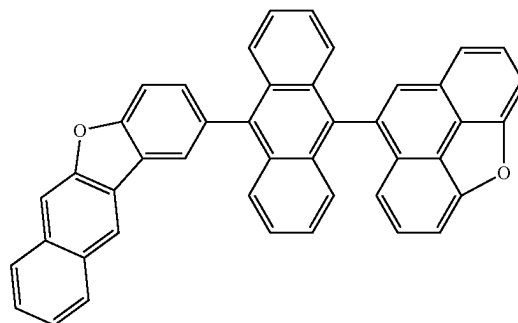


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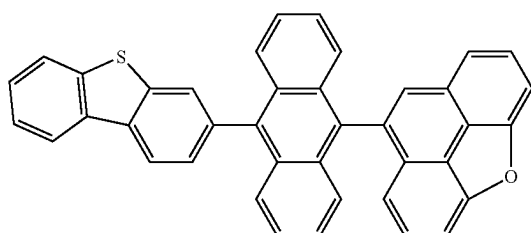


H2-112

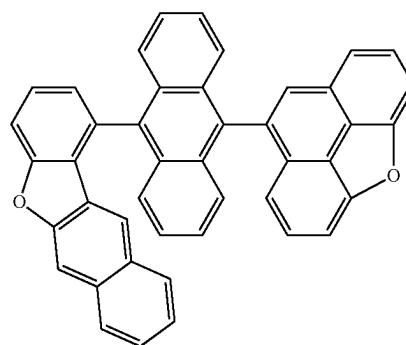
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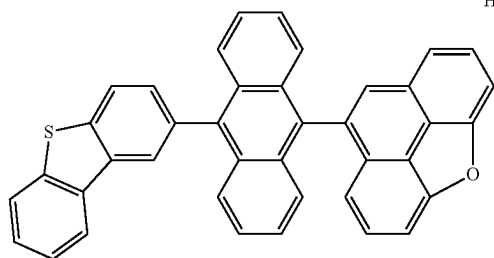
H2-117



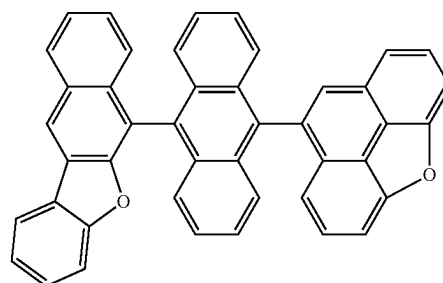
H2-113



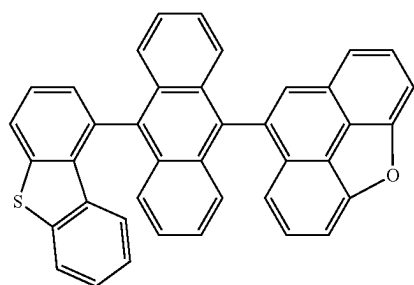
H2-118



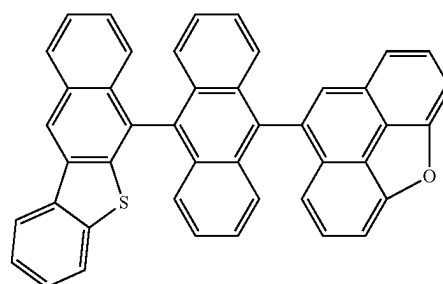
H2-114



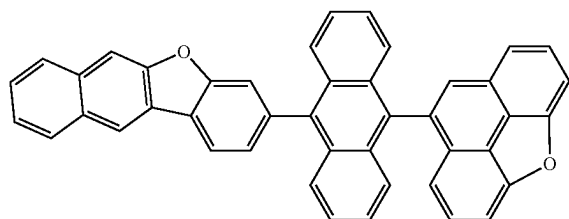
H2-119



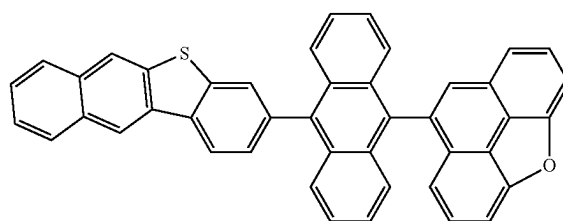
H2-115



H2-120

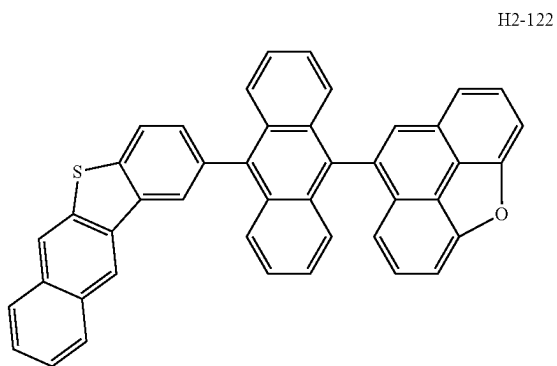


H2-116

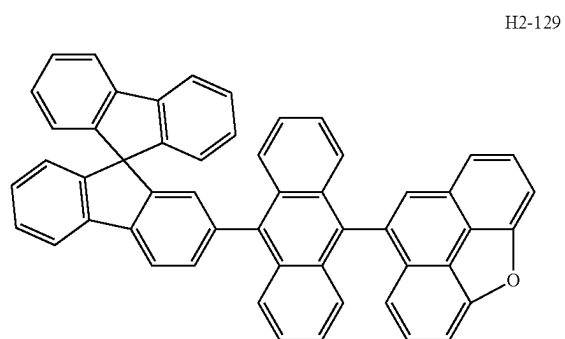
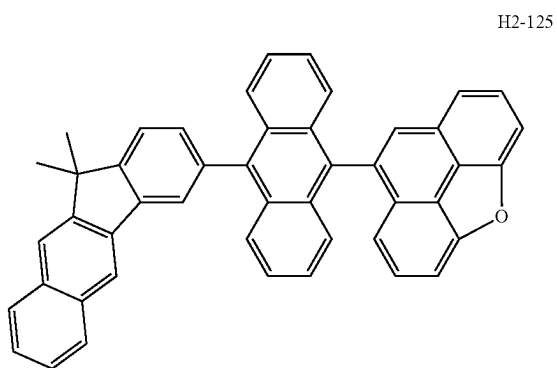
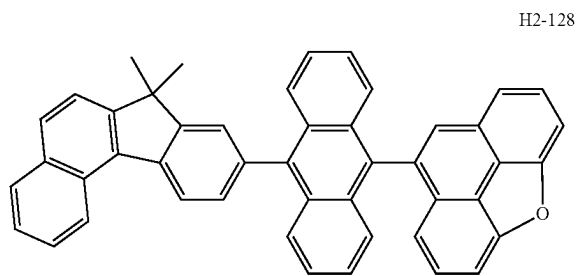
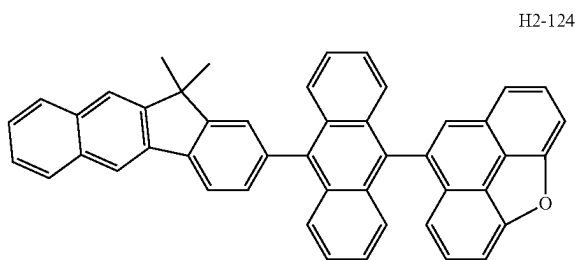
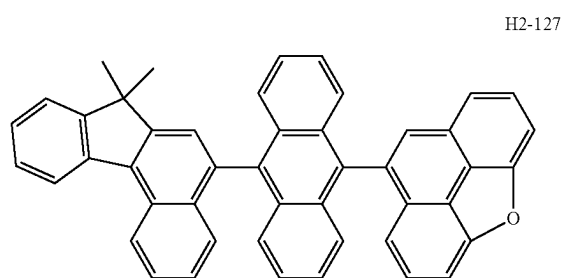
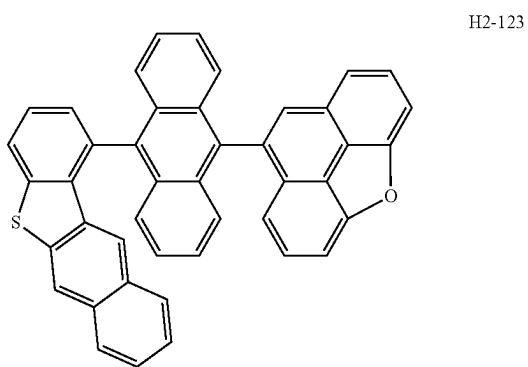
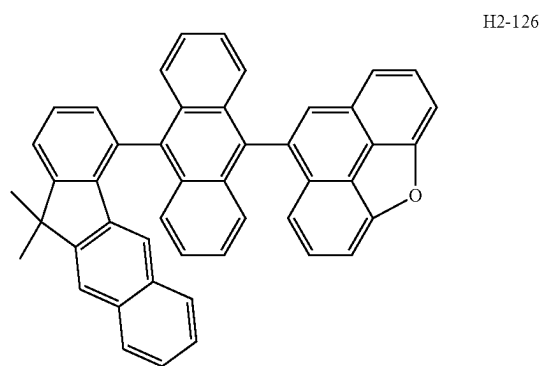


H2-121

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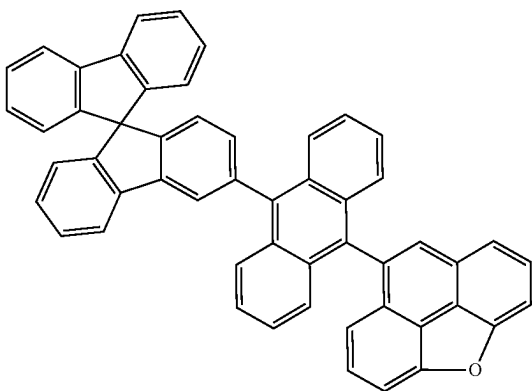


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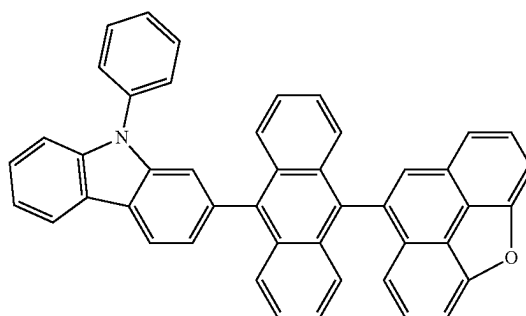
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H2-130

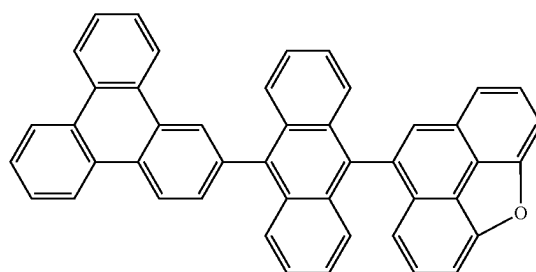


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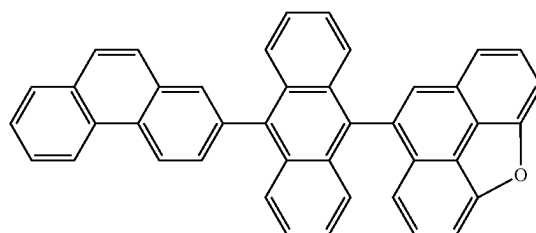
H2-134



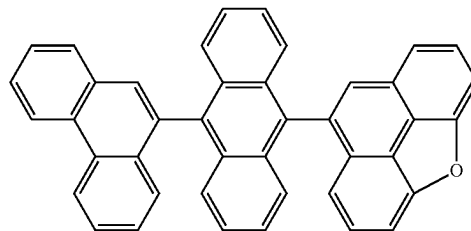
H2-135



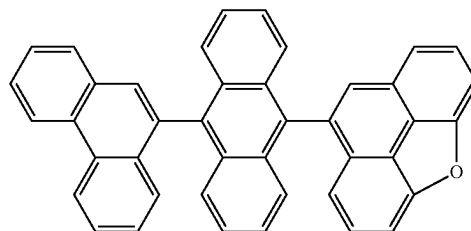
H2-136



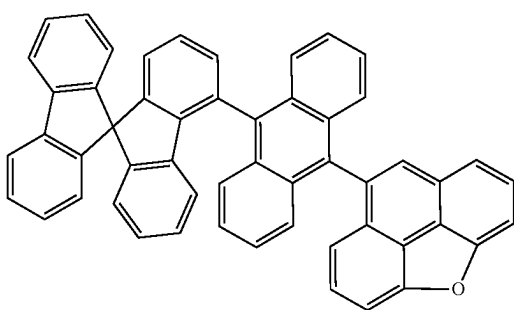
H2-137



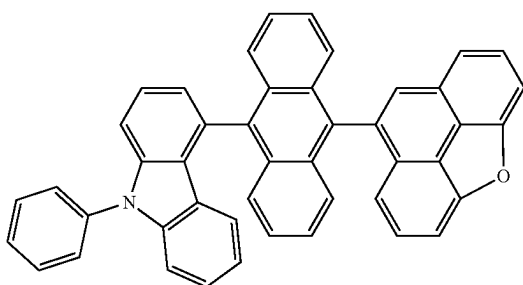
H2-138



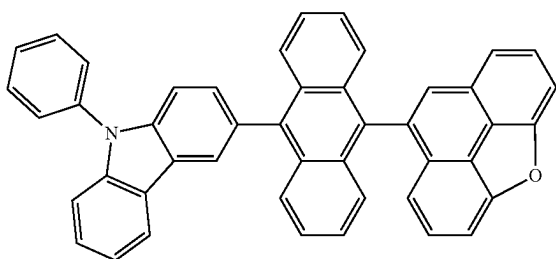
H2-131



H2-132

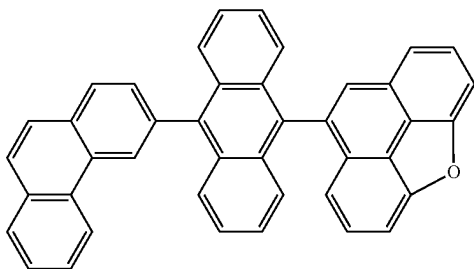


H2-133

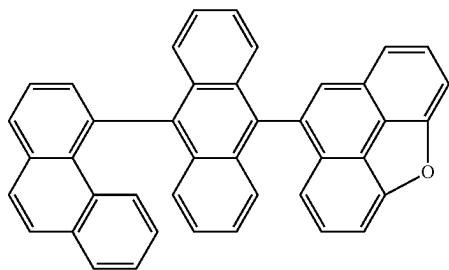


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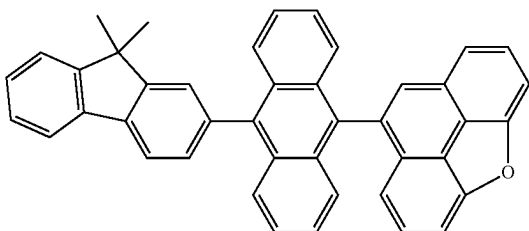
H2-139



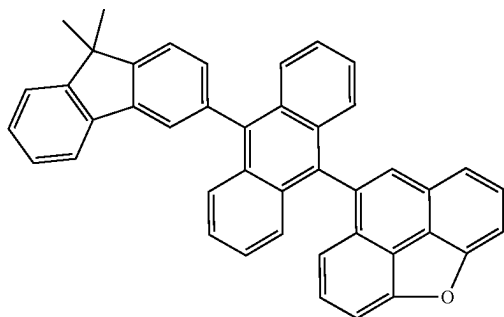
H2-140



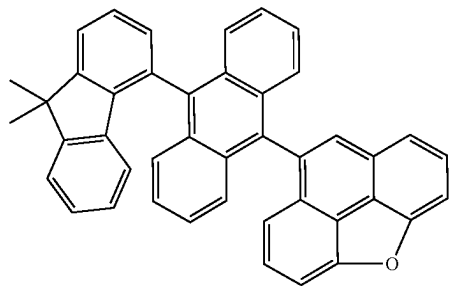
H2-141



H2-142

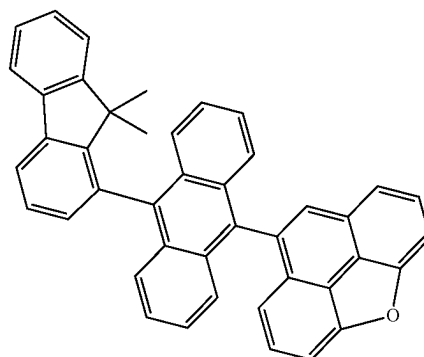


H2-143

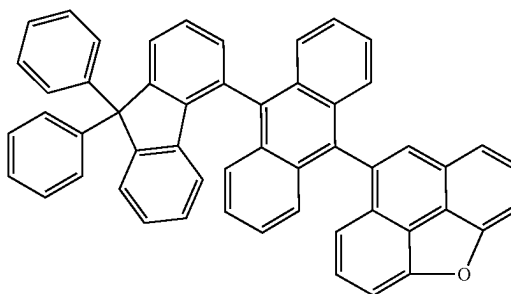


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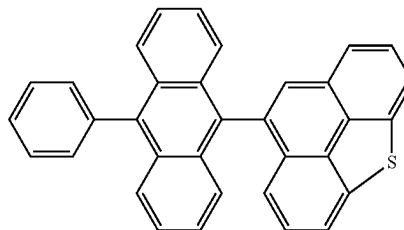
H2-144



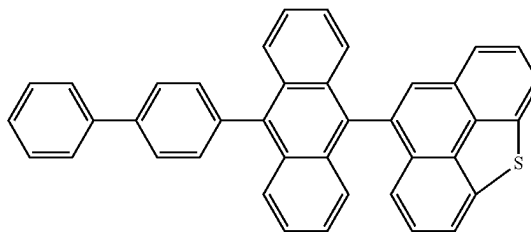
H2-145



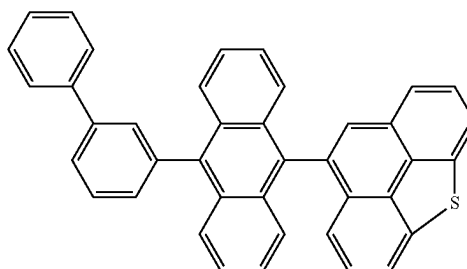
H2-146



H2-147

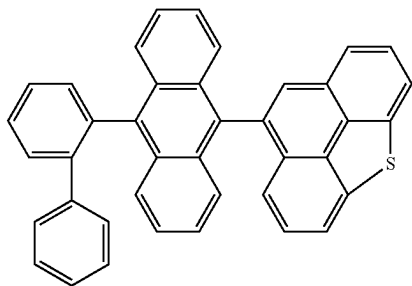


H2-148



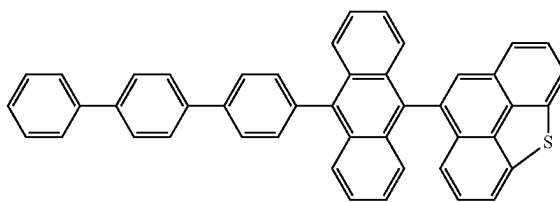
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H2-149



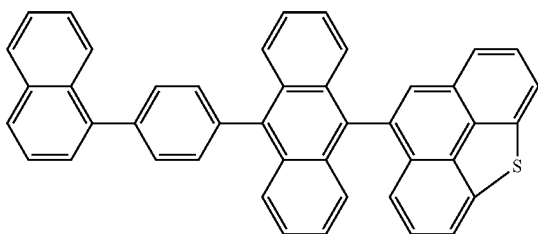
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H2-154

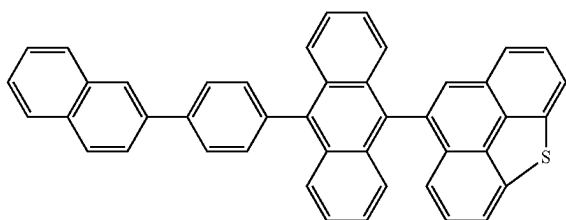


H2-155

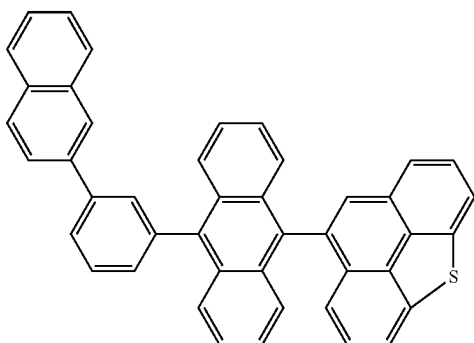
H2-150



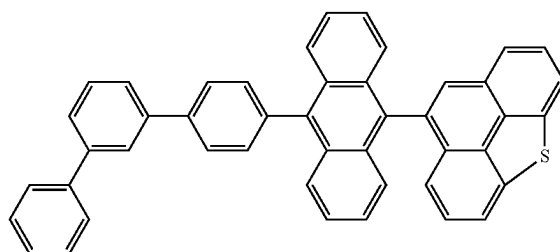
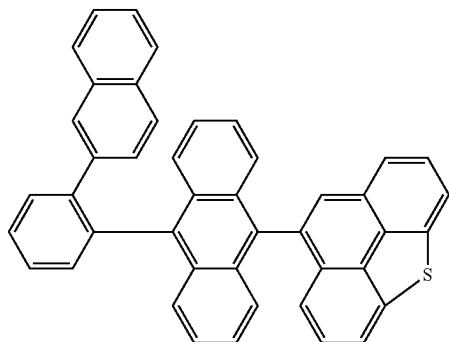
H2-151



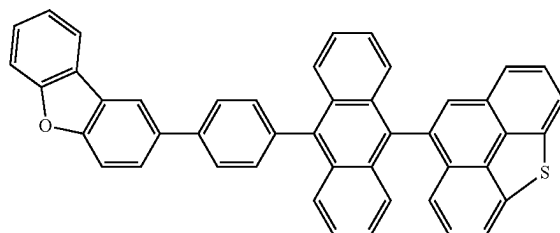
H2-152



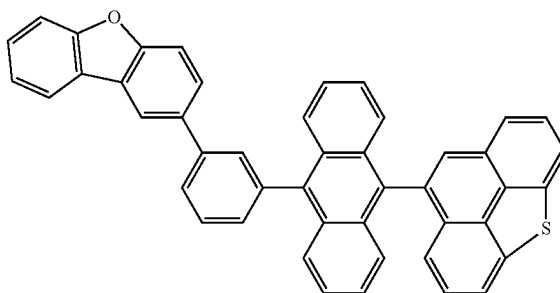
H2-153



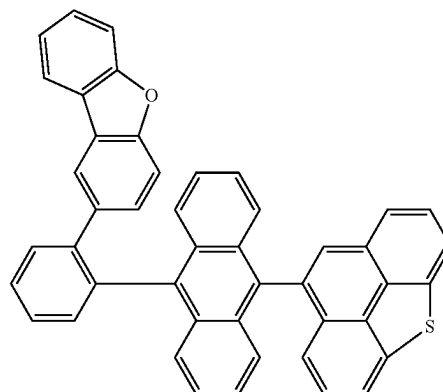
H2-157



H2-158

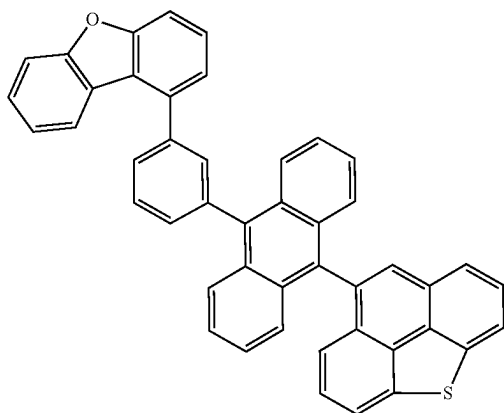


H2-159



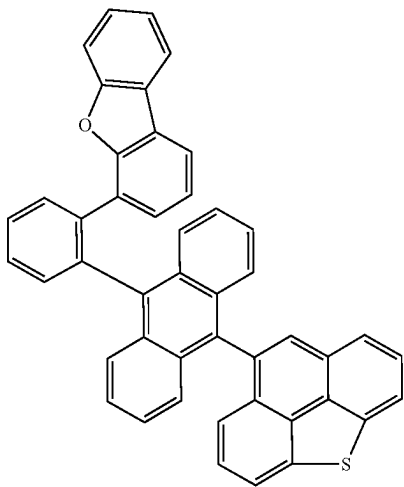
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H2-159

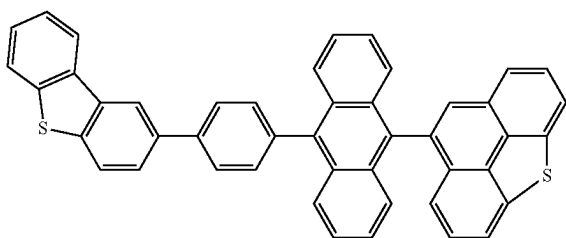


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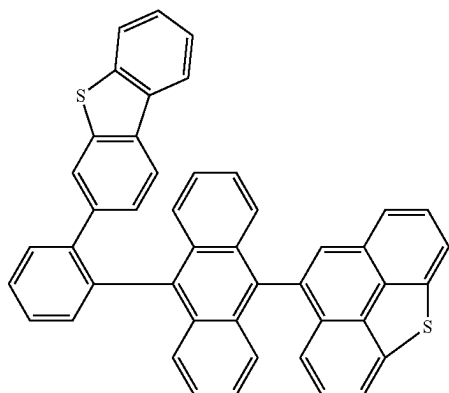
H2-166



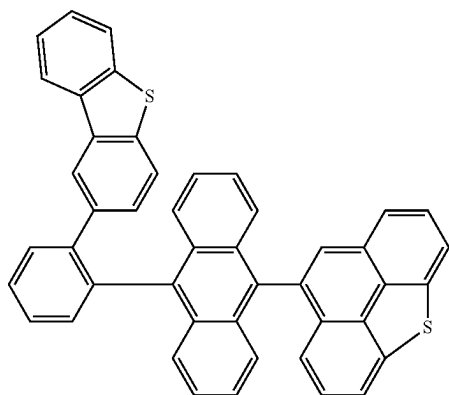
H2-167



H2-168

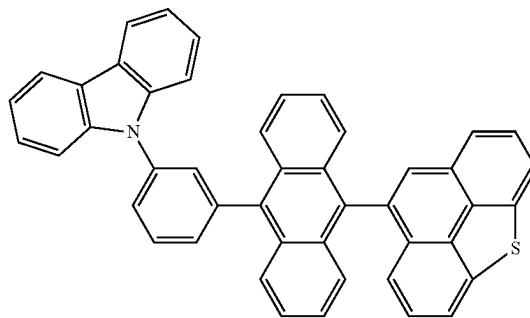


H2-169

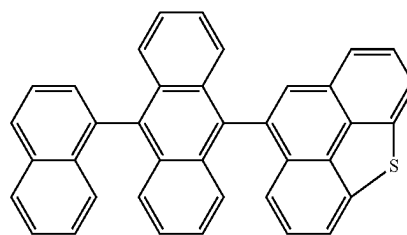


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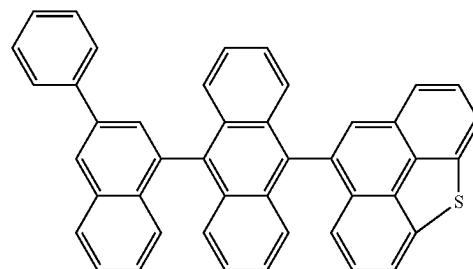
H2-170



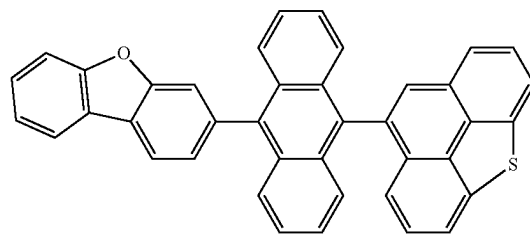
H2-171



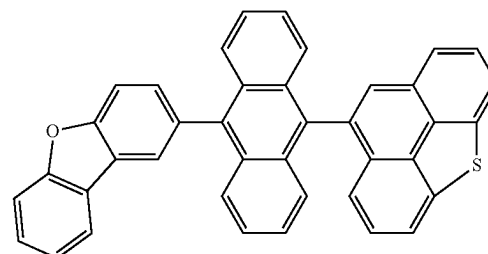
H2-172



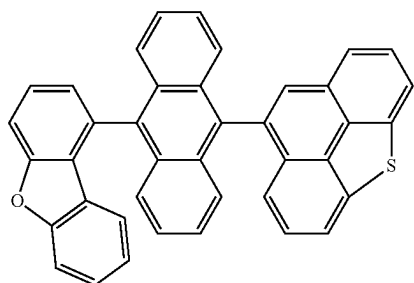
H2-173



H2-174

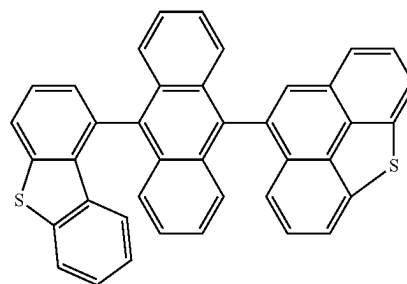


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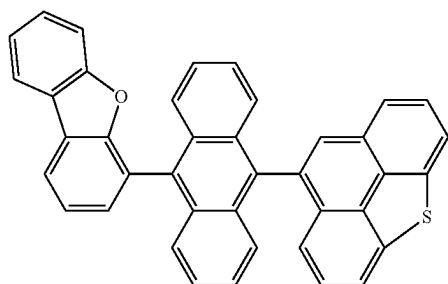


H2-175

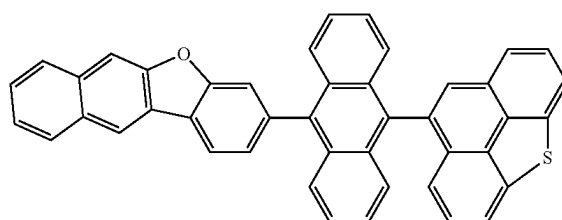
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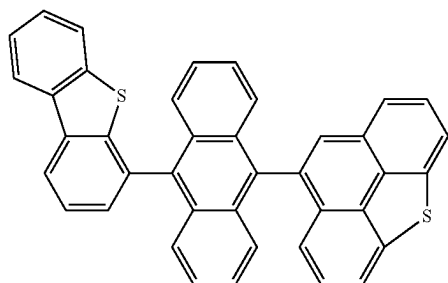
H2-180



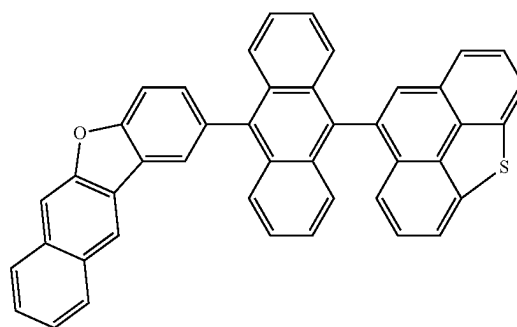
H2-176



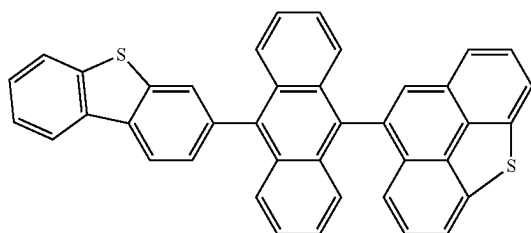
H2-181



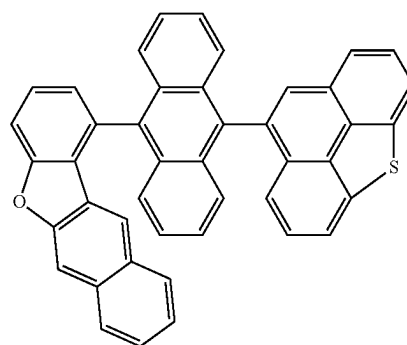
H2-177



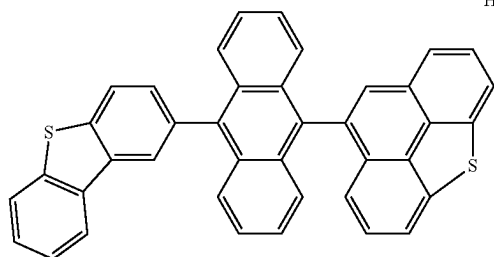
H2-182



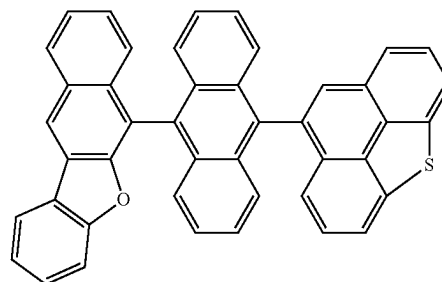
H2-178



H2-183

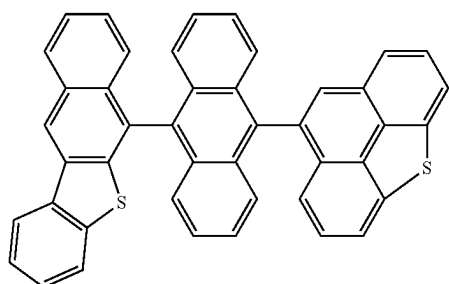


H2-179

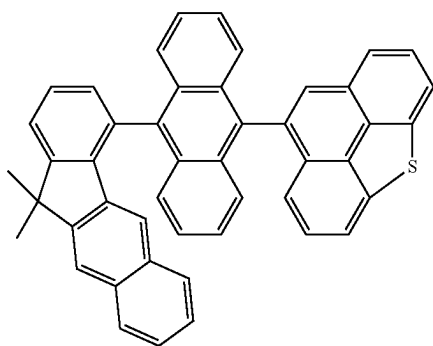


H2-184

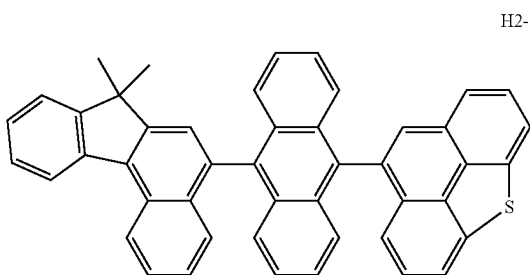
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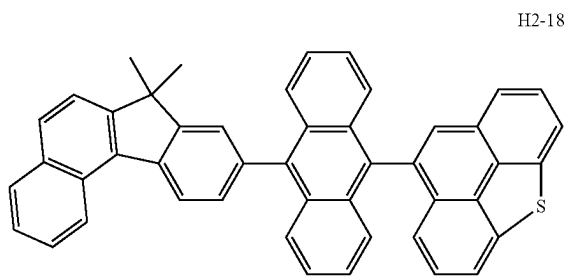
H2-185



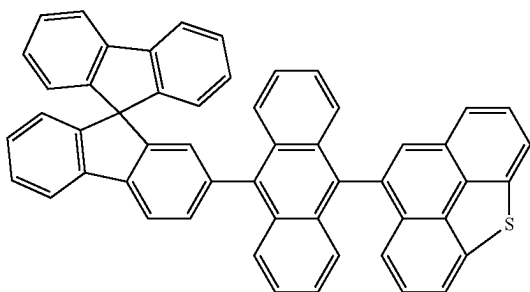
H2-186



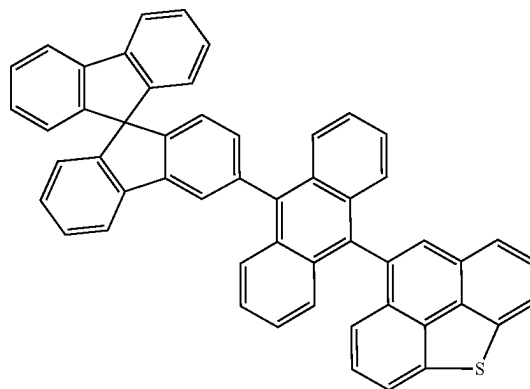
H2-187



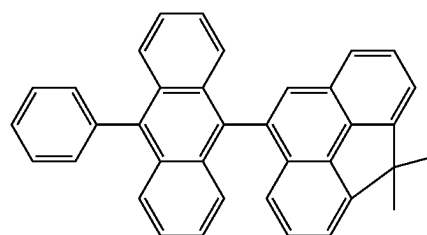
H2-188



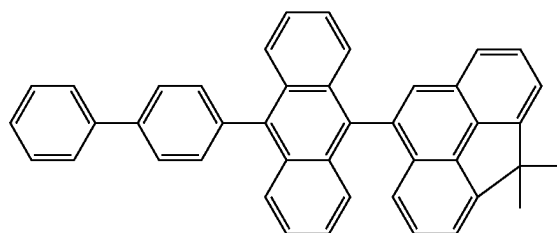
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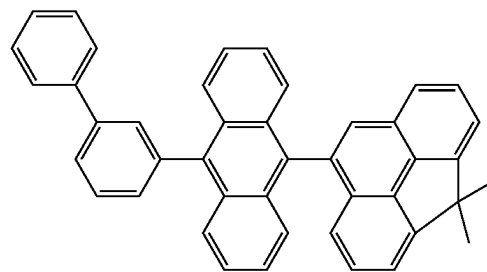
H2-190



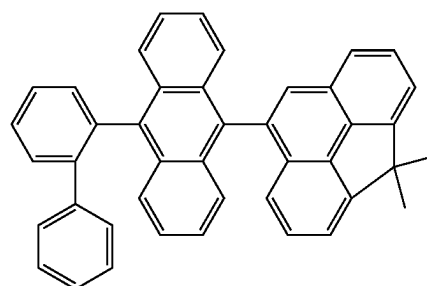
H2-191



H2-192



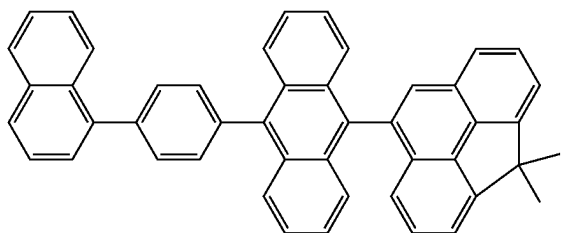
H2-193



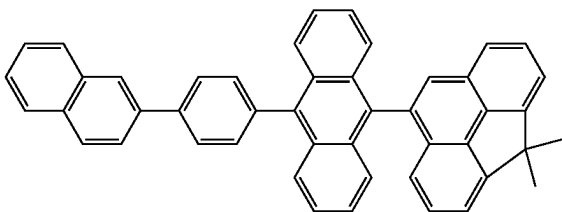
H2-194

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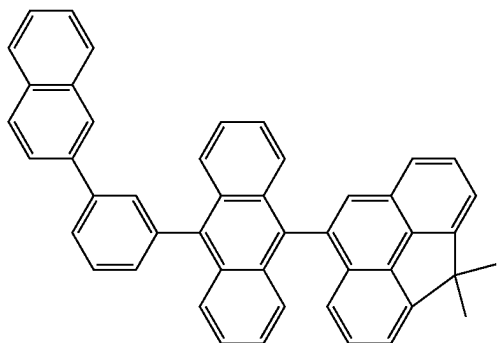
H2-195



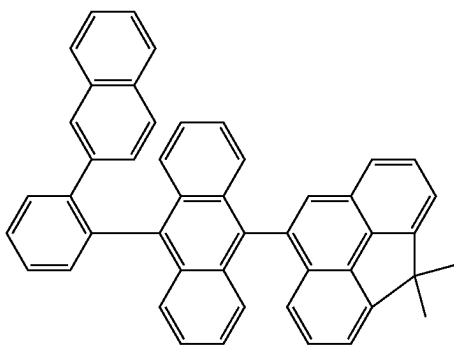
H2-196



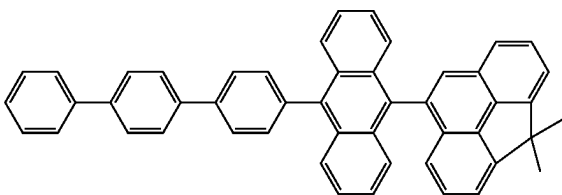
H2-197



H2-198

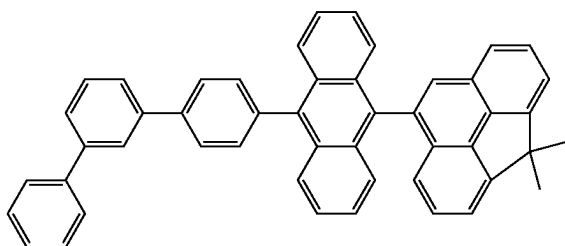


H2-199

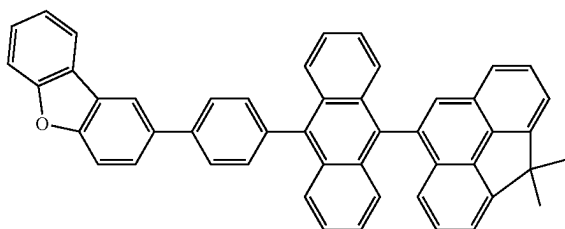


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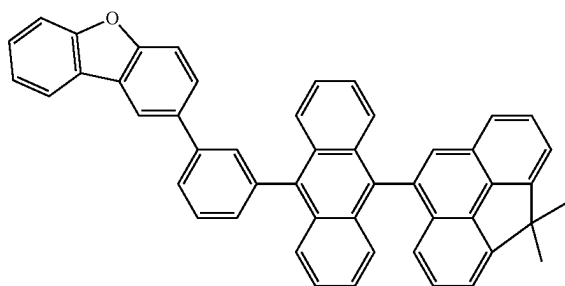
H2-200



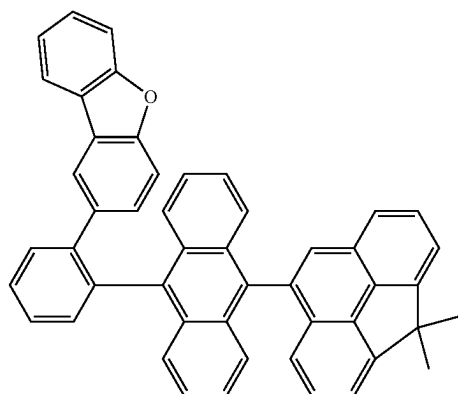
H2-201



H2-202

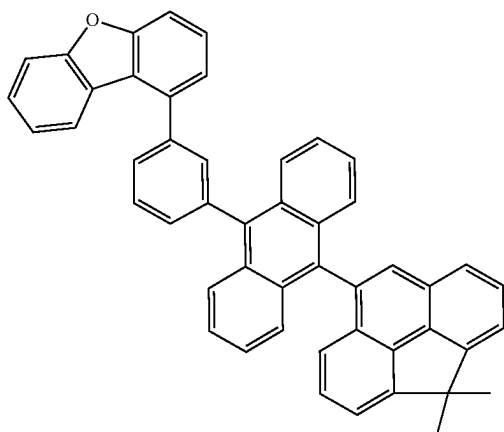


H2-203



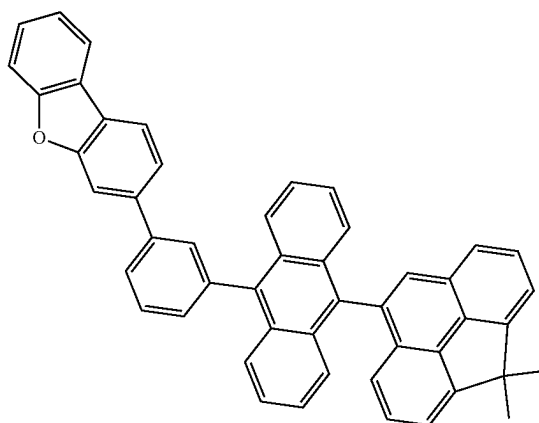
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H2-204



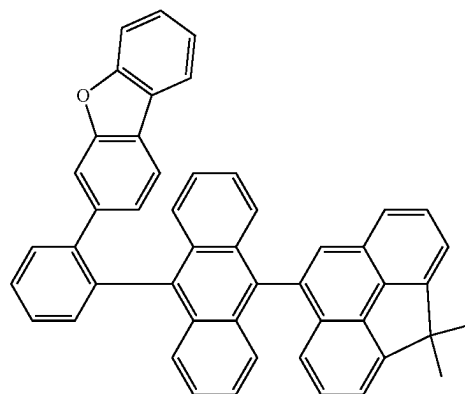
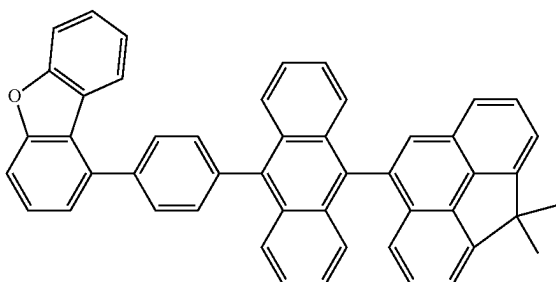
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H2-207



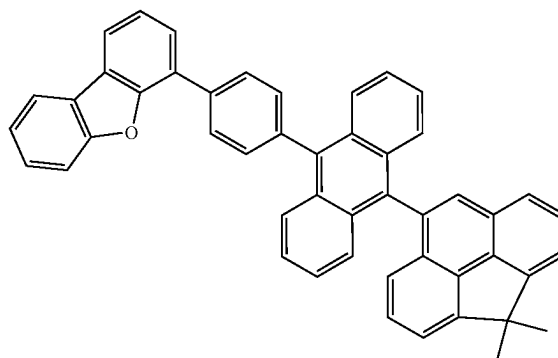
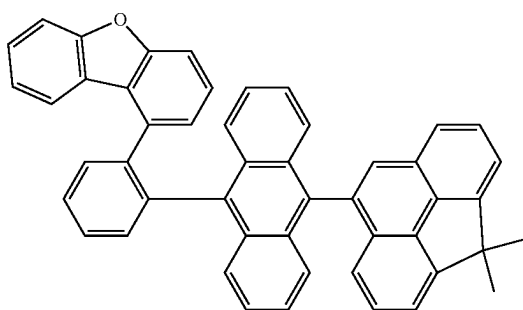
H2-208

H2-205



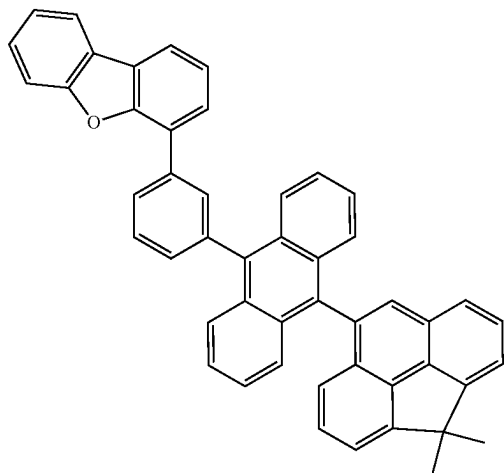
H2-209

H2-206



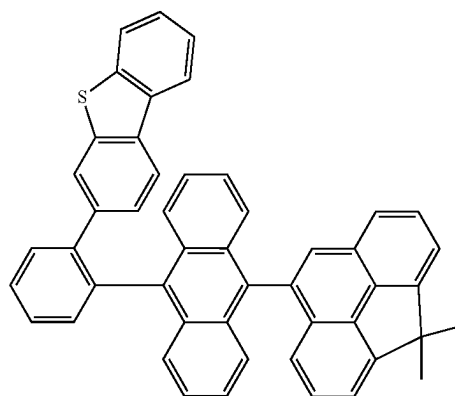
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H2-210

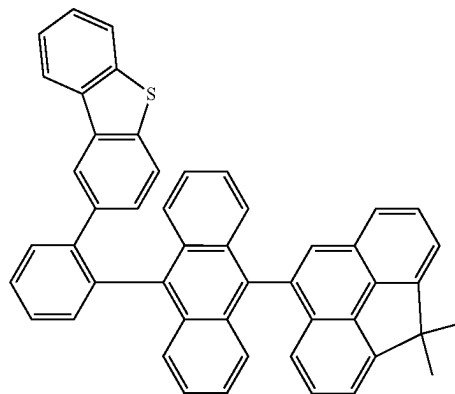


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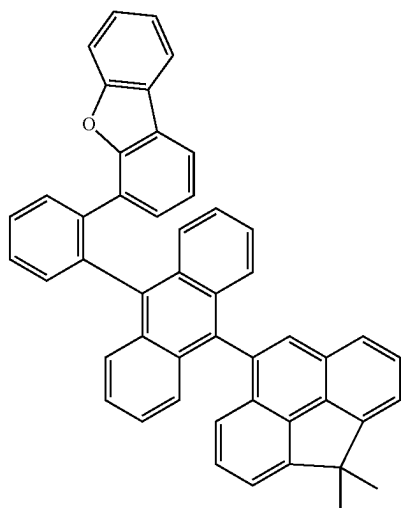
H2-213



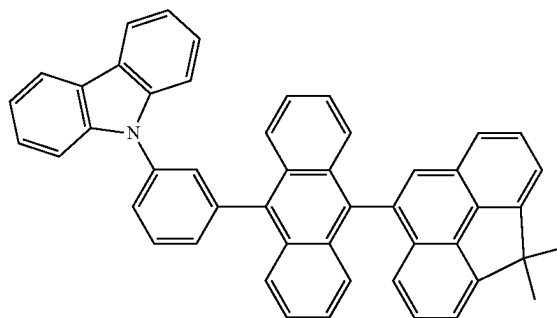
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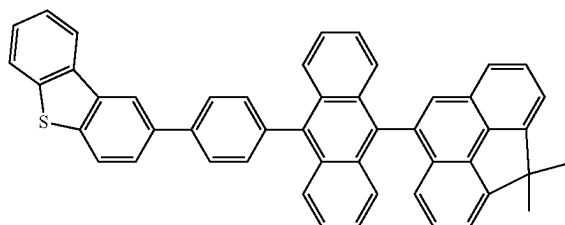
H2-211



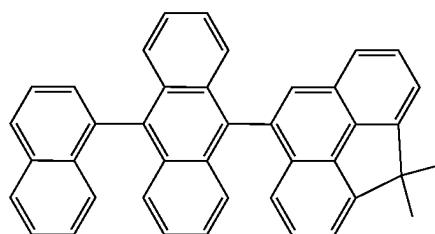
H2-215



H2-212

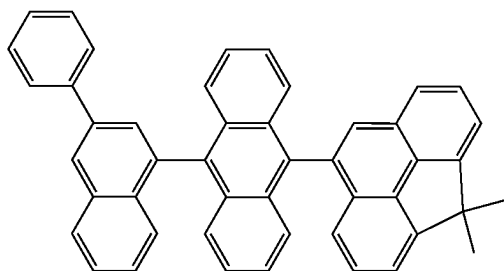


H2-216

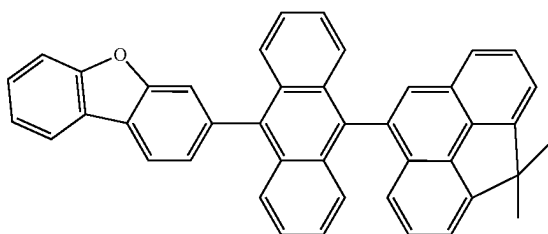


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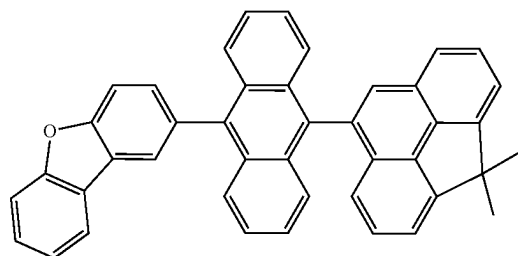
H2-217



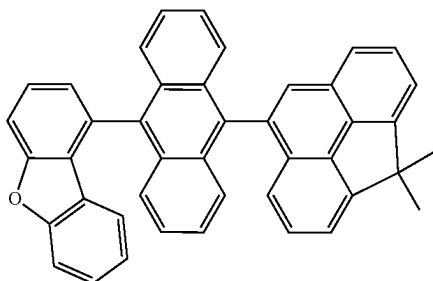
H2-218



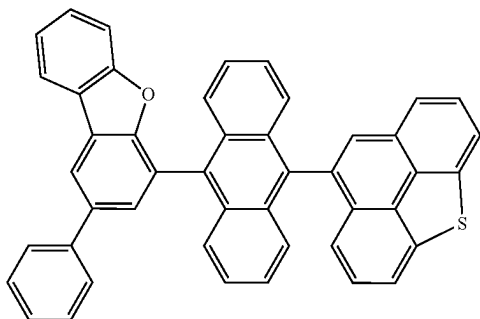
H2-219



H2-220

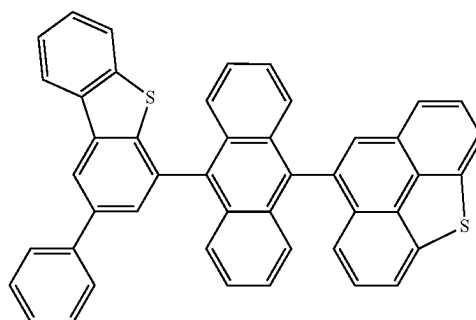


H2-221

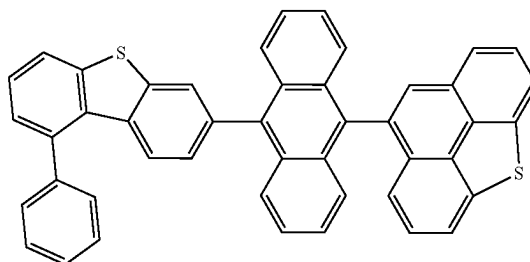


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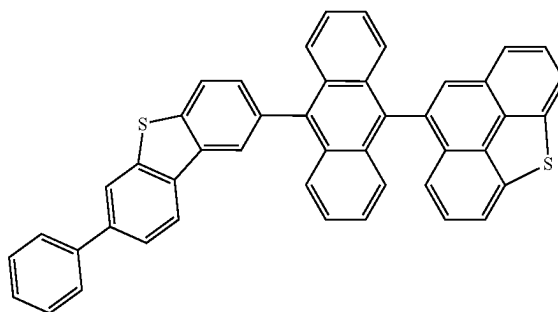
H2-222



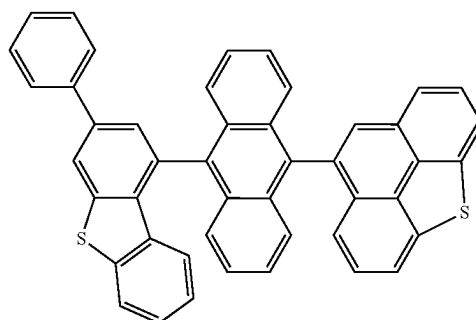
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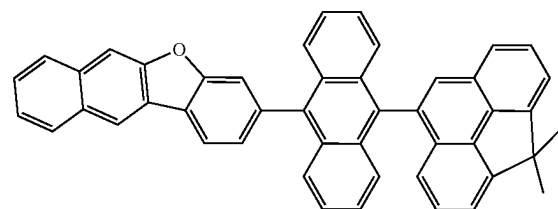
H2-224



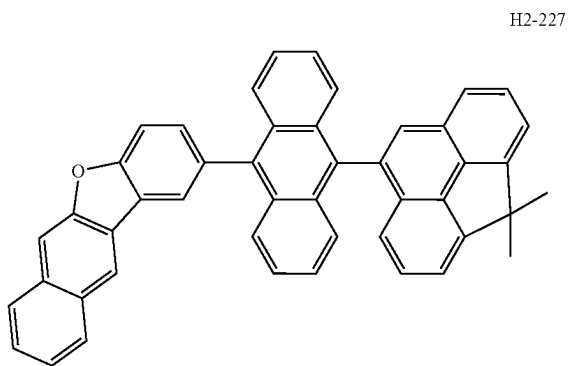
H2-225



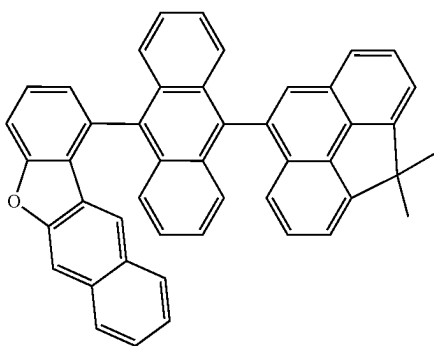
H2-226



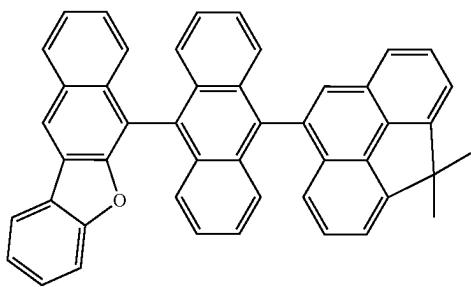
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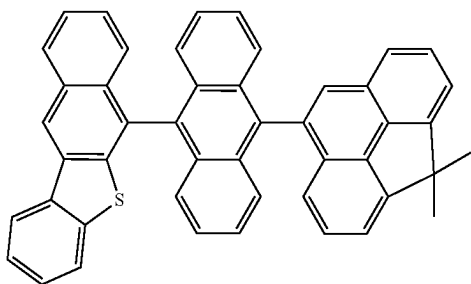
H2-228



H2-229

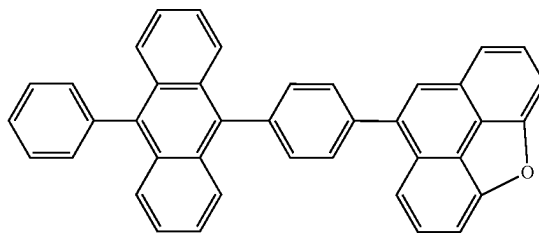


H2-230

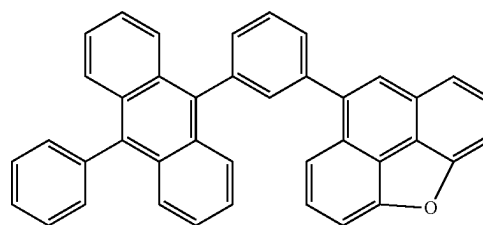


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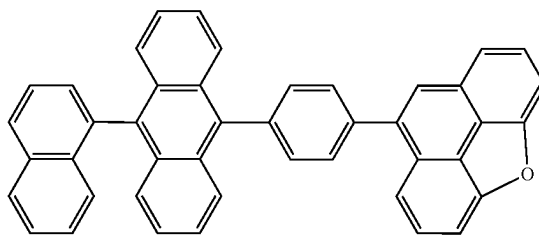
H2-231



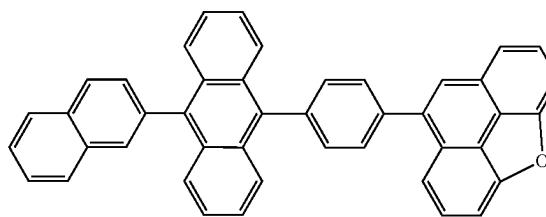
H2-232



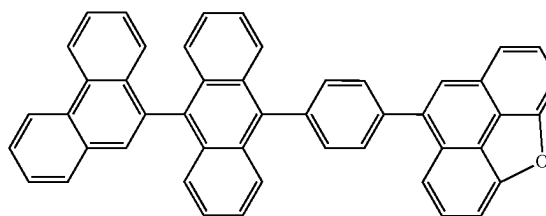
H2-233



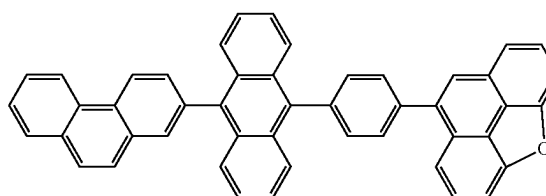
H2-234



H2-235

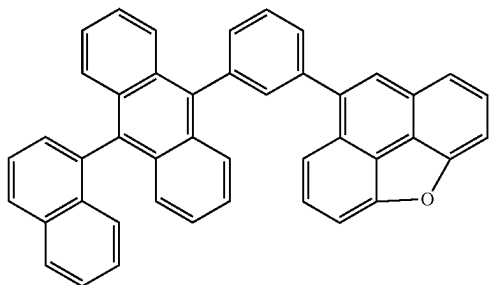


H2-236

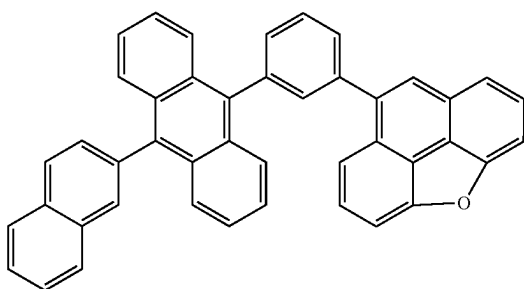


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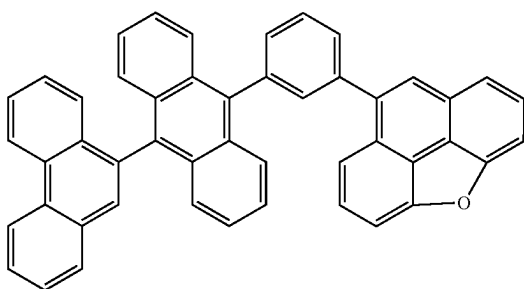
H2-237



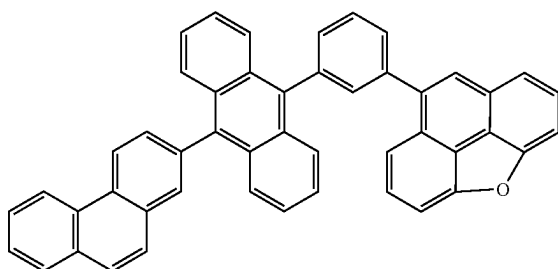
H2-238



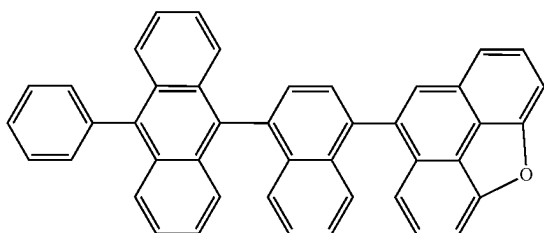
H2-239



H2-240

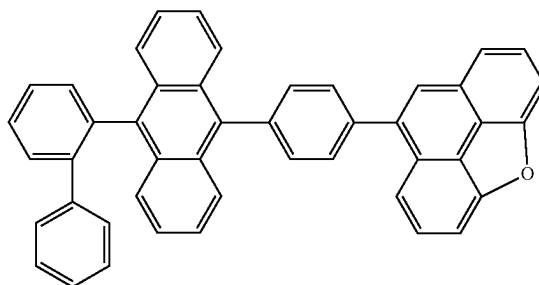


H2-241

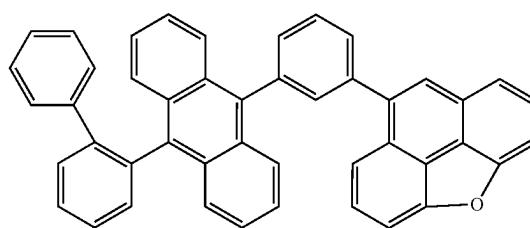


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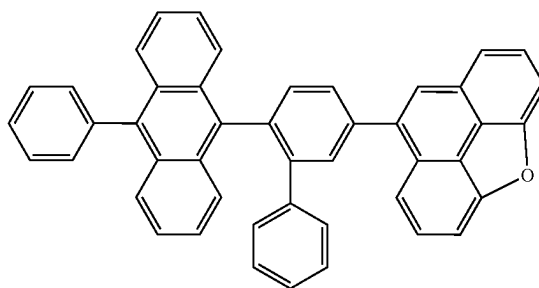
H2-242



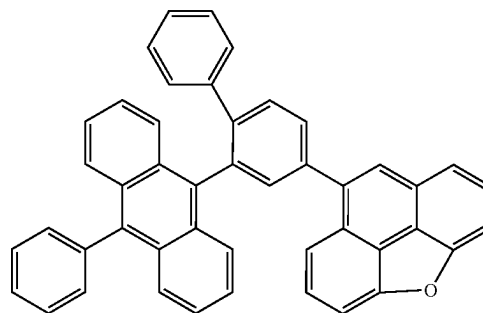
H2-243



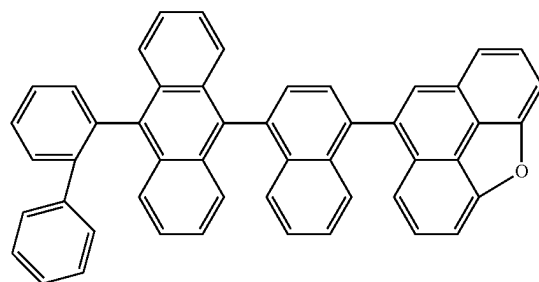
H2-244



H2-245

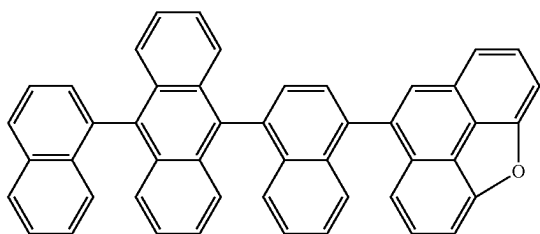


H2-246

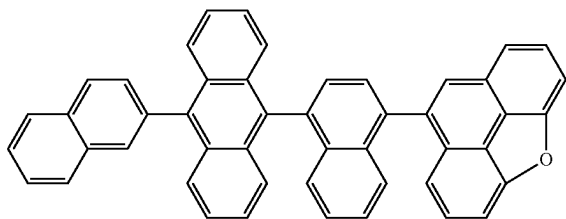


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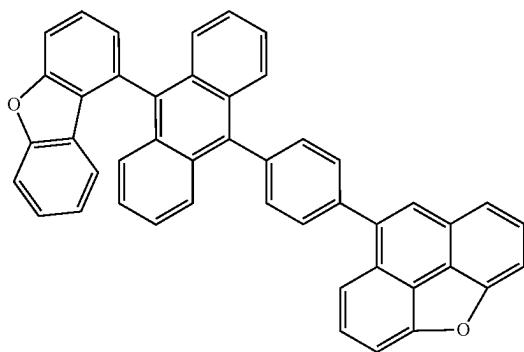
H2-247



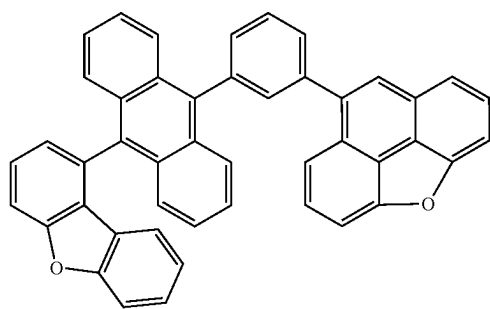
H2-248



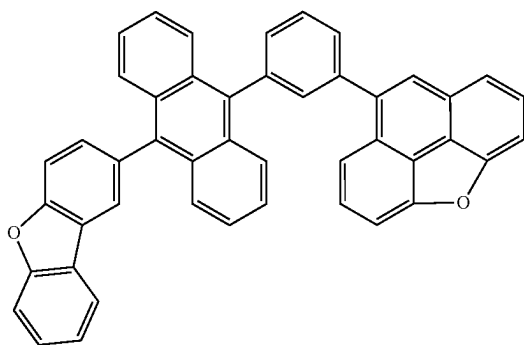
H2-249



H2-250

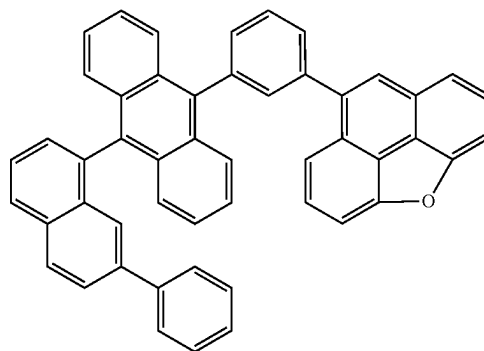


H2-251

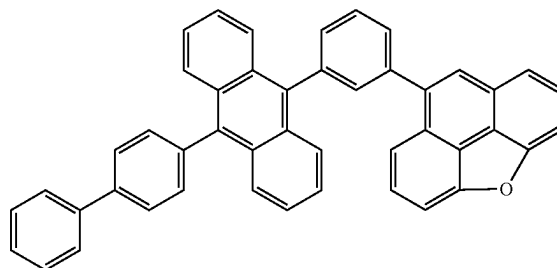


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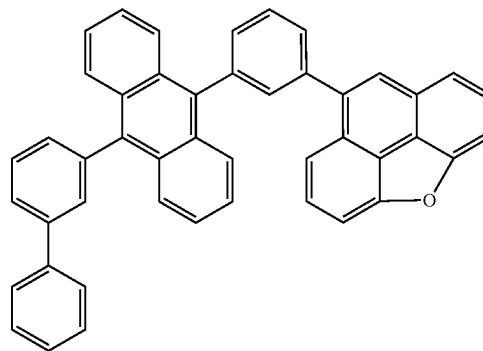
H2-252



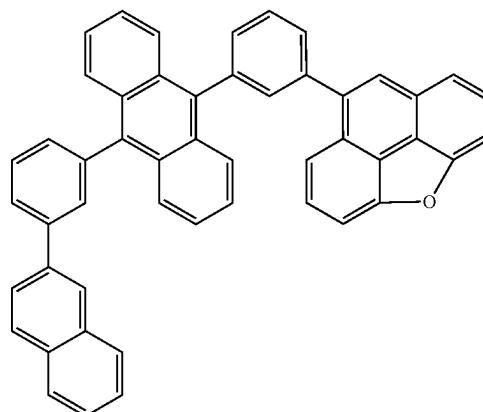
H2-253



H2-254

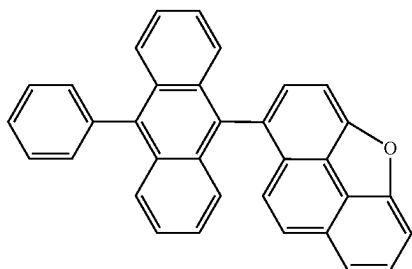


H2-255

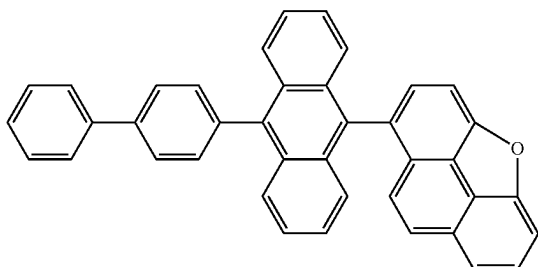


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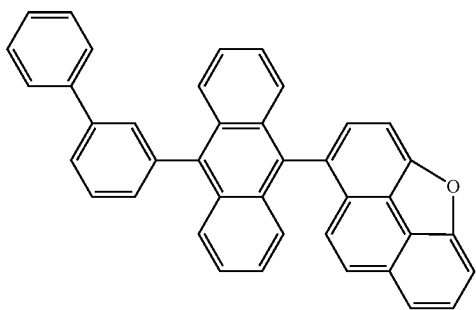
H2-256



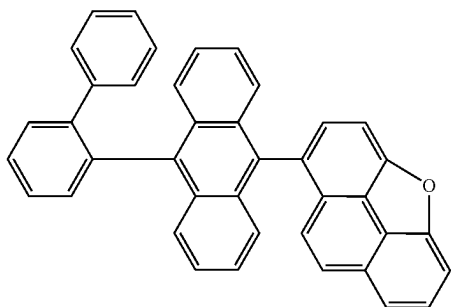
H2-257



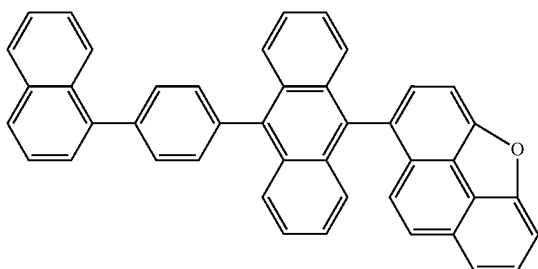
H2-258



H2-259

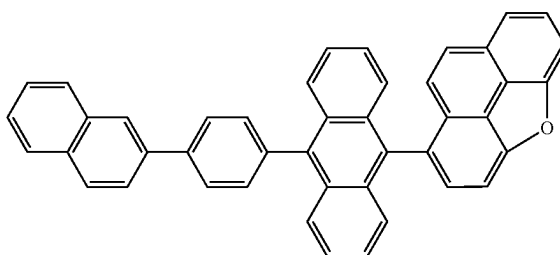


H2-260

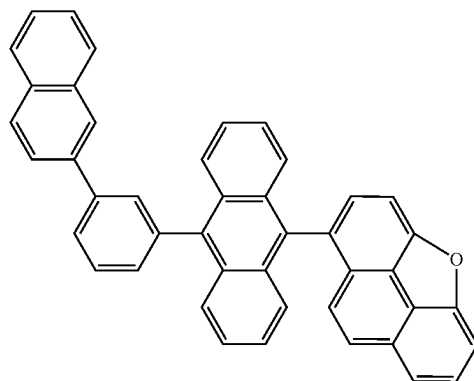


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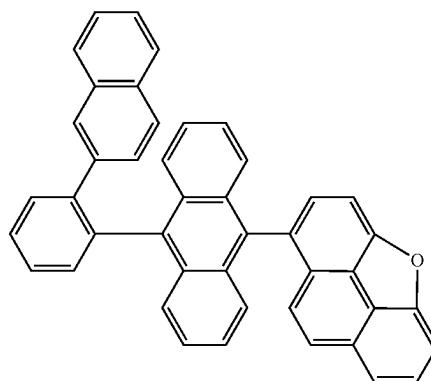
H2-261



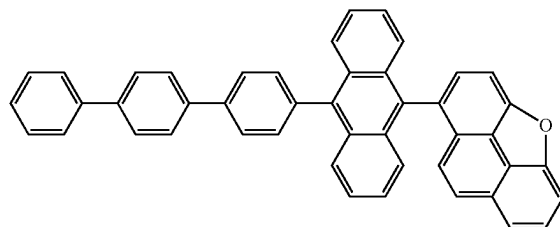
H2-262



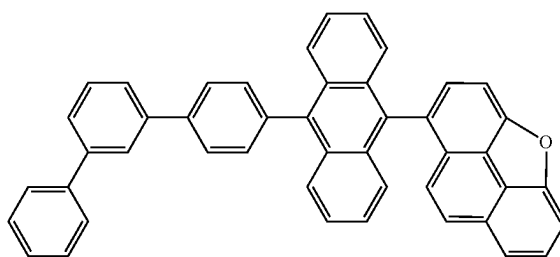
H2-263



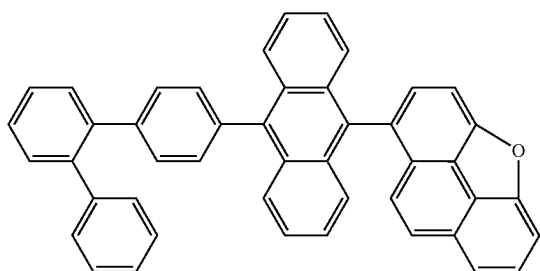
H2-264



H2-265

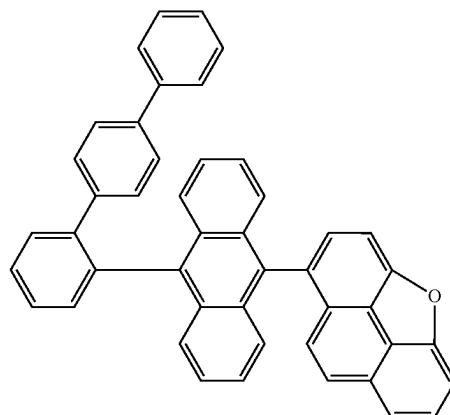


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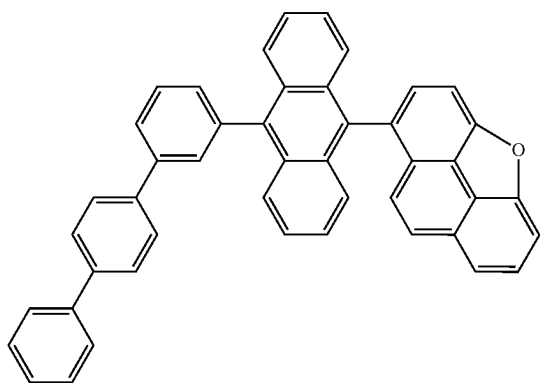
H2-266

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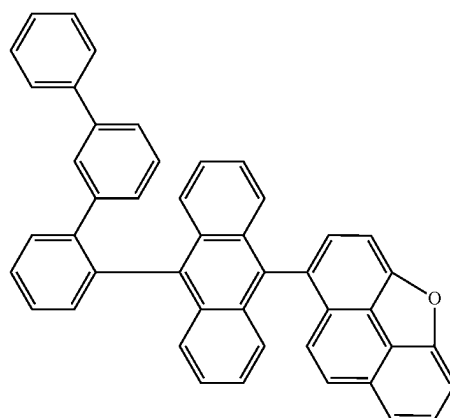


H2-270

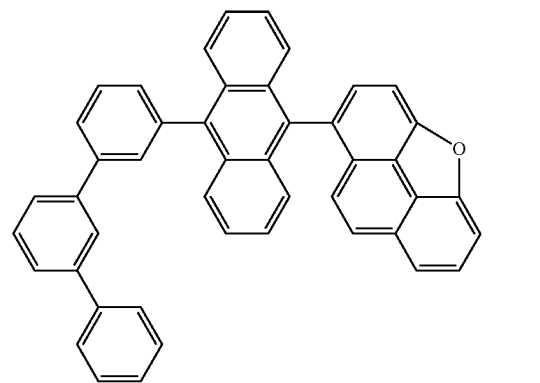
H2-267



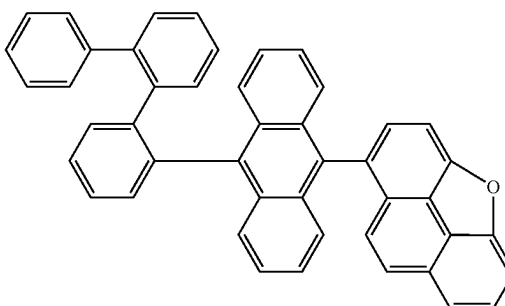
H2-271



H2-268

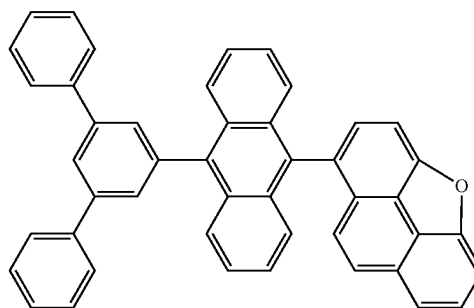
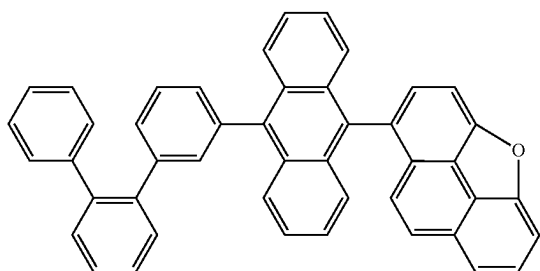


H2-272



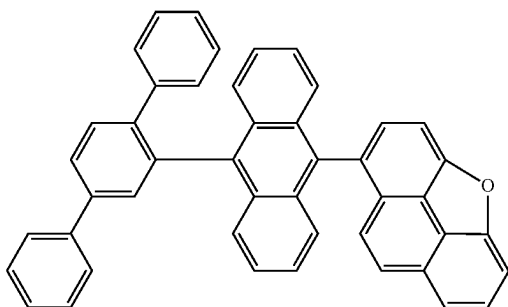
H2-273

H2-269



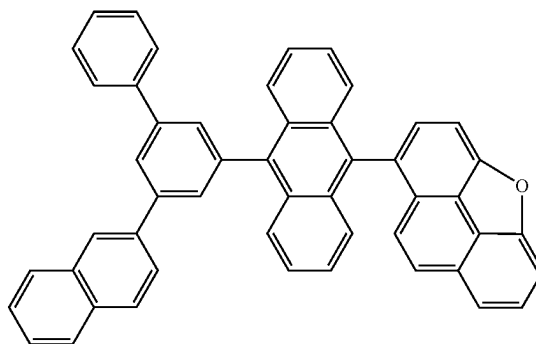
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H2-274

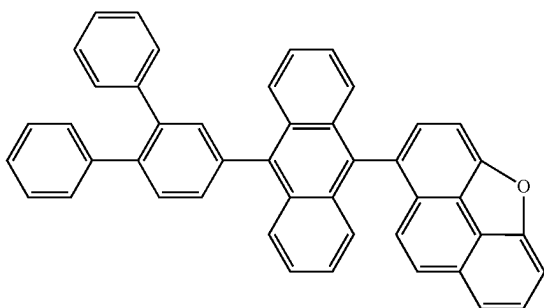


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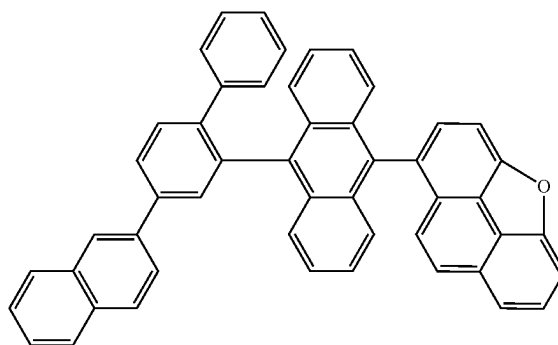
H2-278



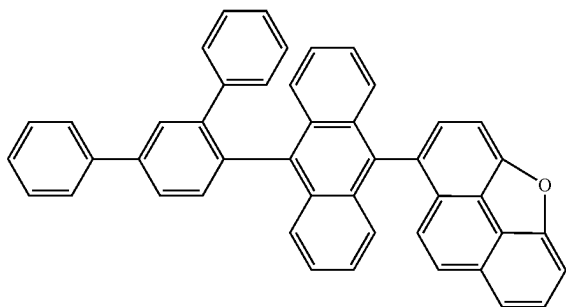
H2-275



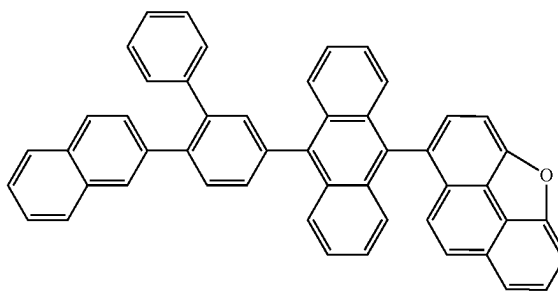
H2-279



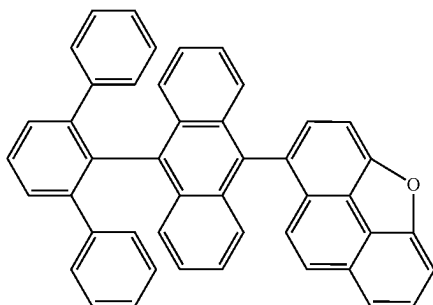
H2-276



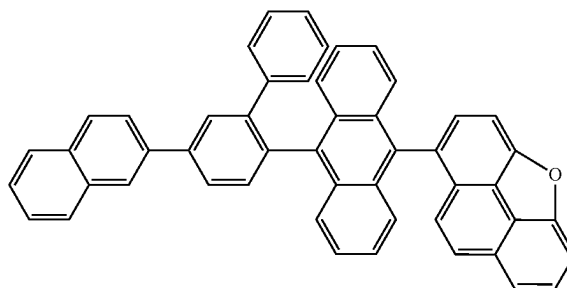
H2-280



H2-277

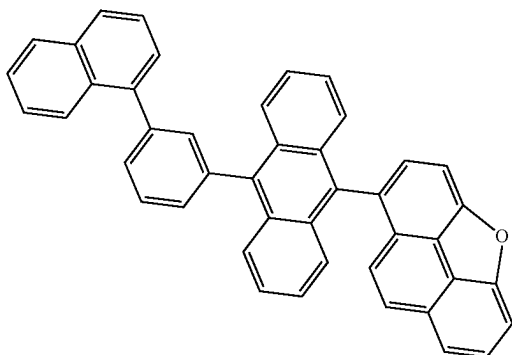


H2-281

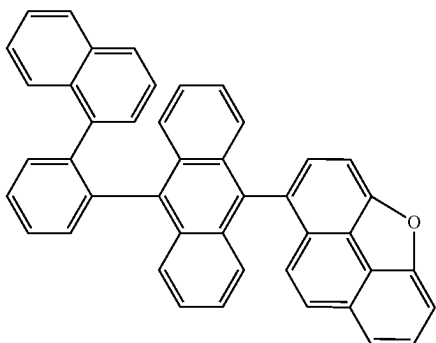


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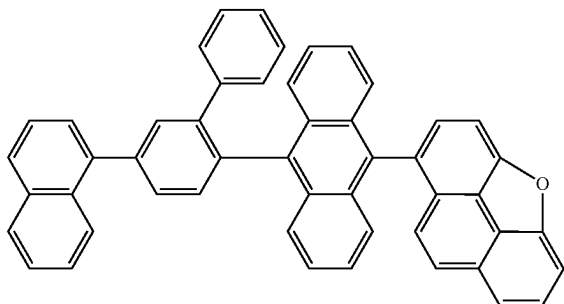
H2-282



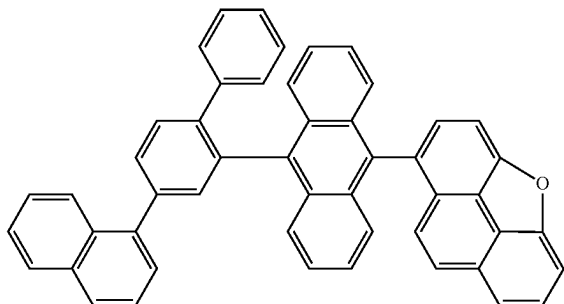
H2-283



H2-284

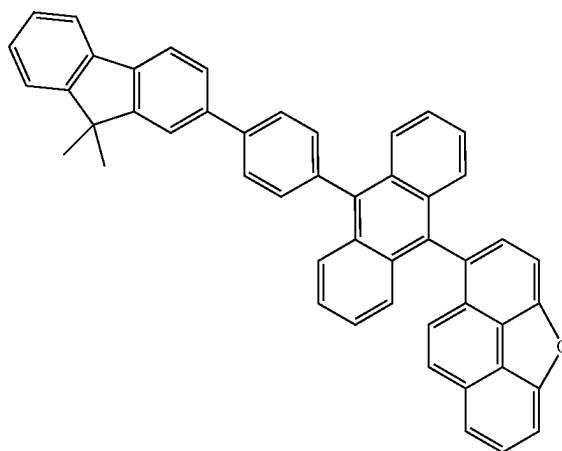


H2-285

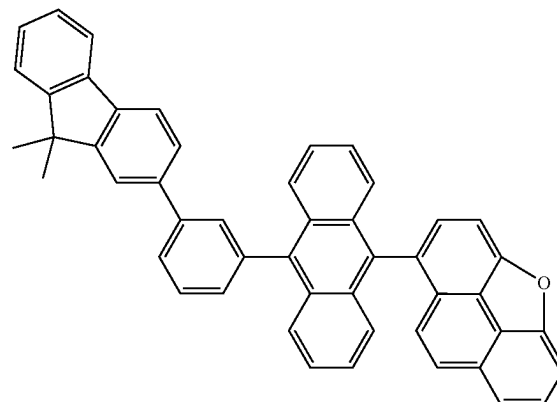


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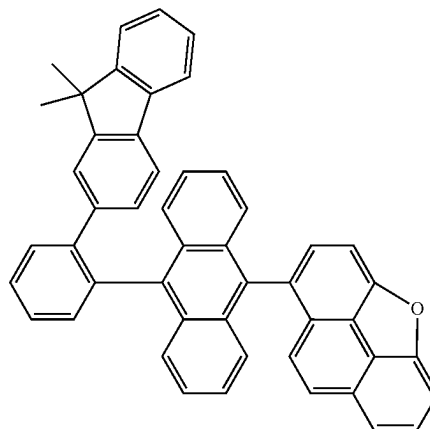
H2-286



H2-287

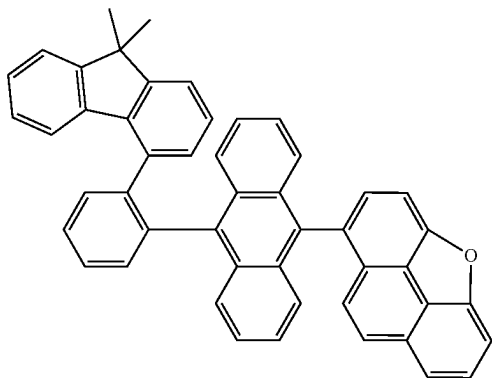


H2-288



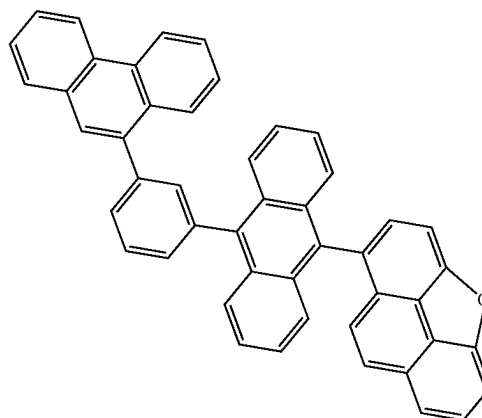
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H2-289

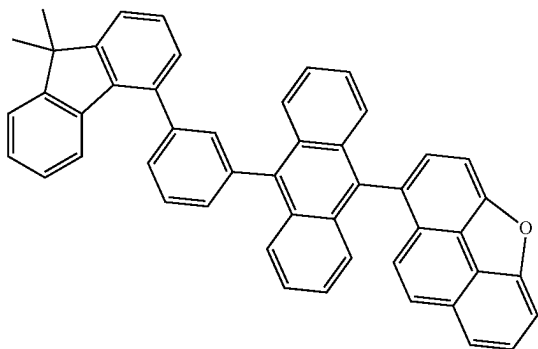


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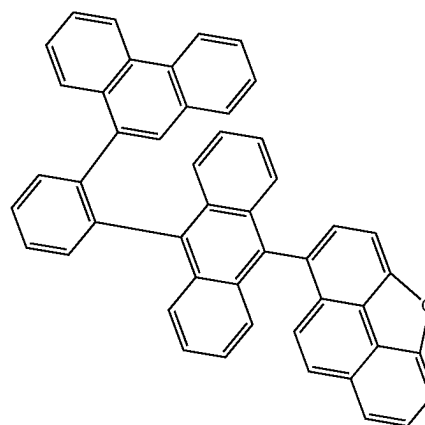
H2-292



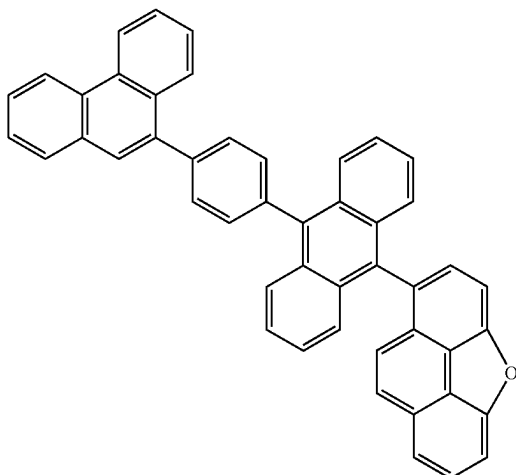
H2-290



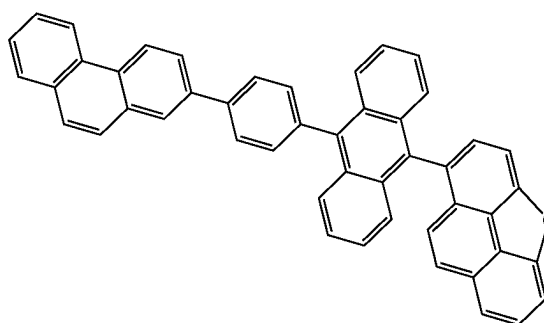
H2-293



H2-291

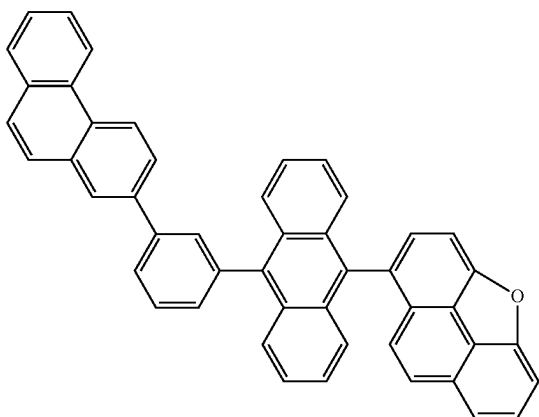


H2-294



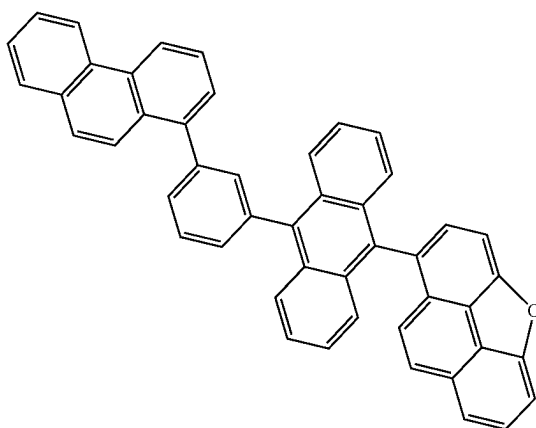
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H2-295

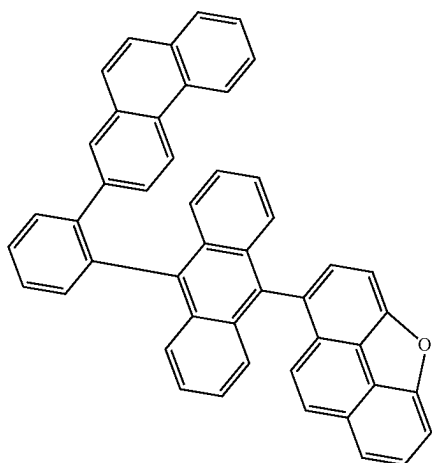


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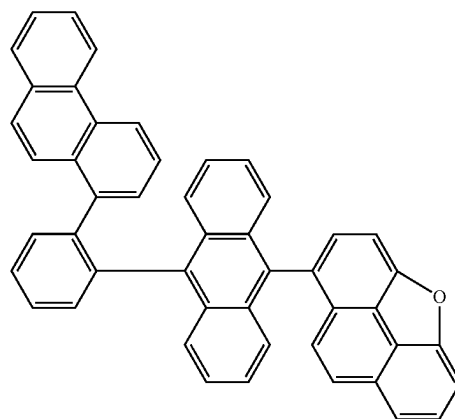
H2-298



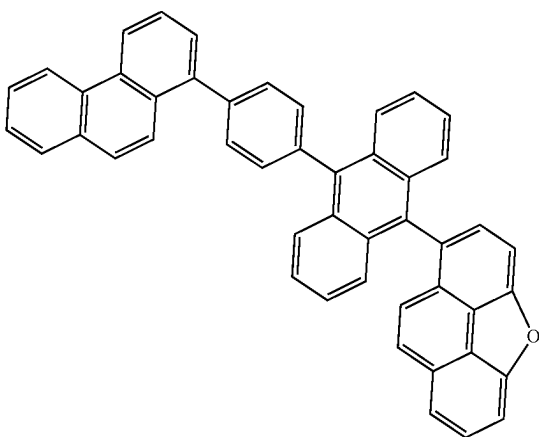
H2-296



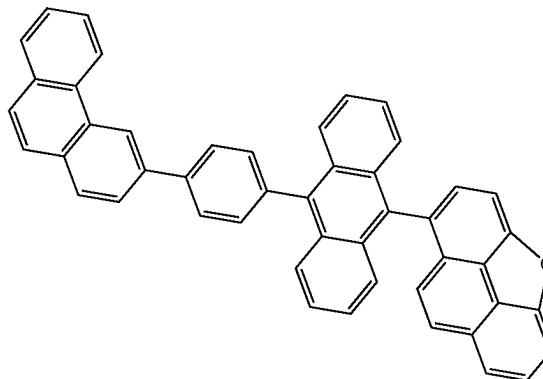
H2-299



H2-297

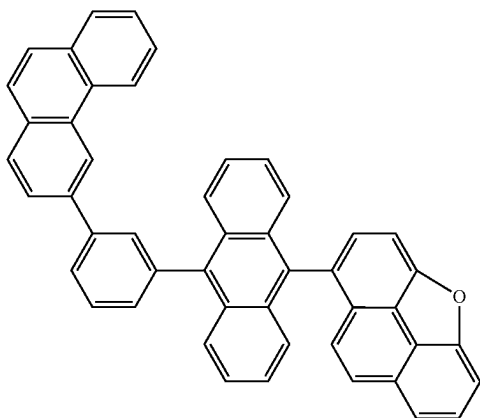


H2-300

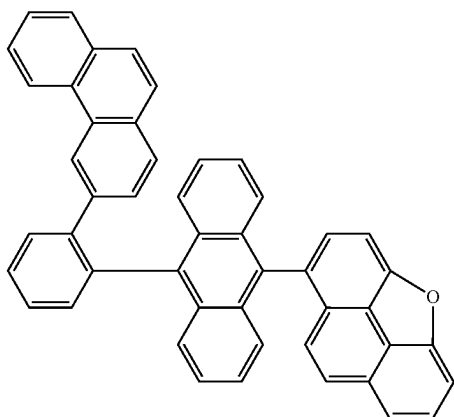


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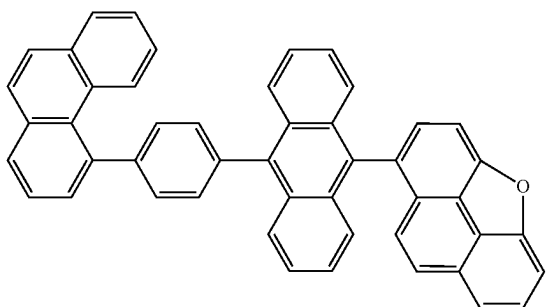
H2-301



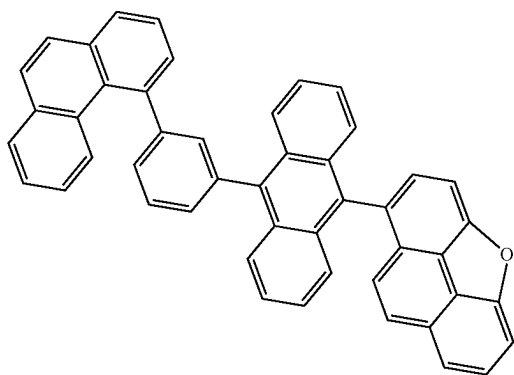
H2-302



H2-303

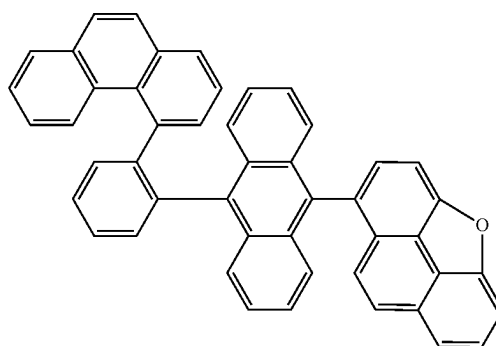


H2-304

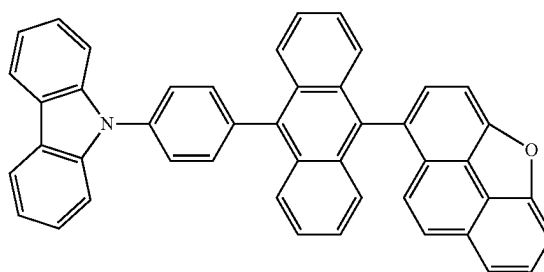


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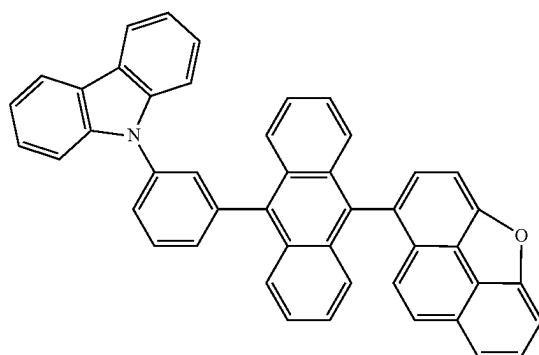
H2-305



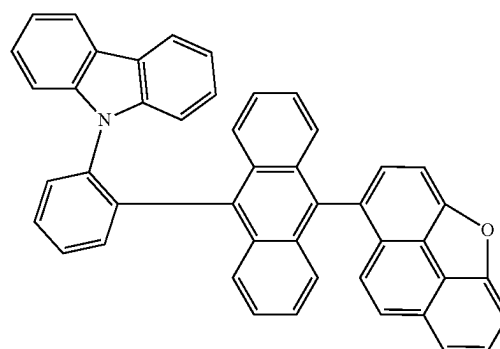
H2-306



H2-307

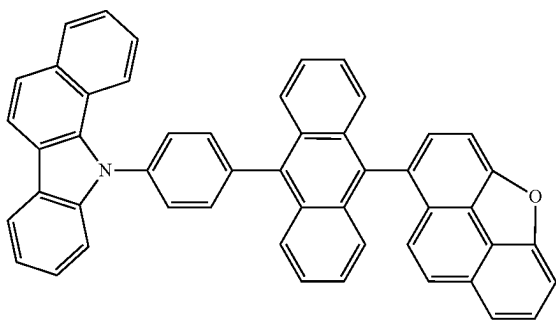


H2-308



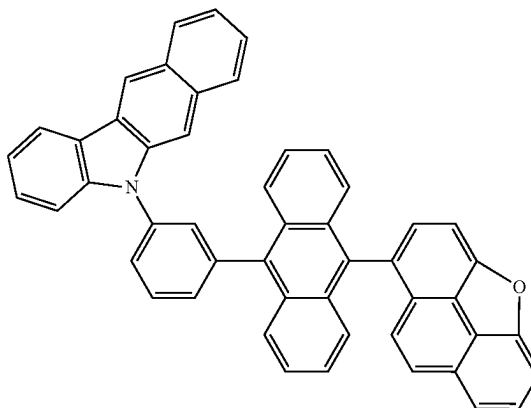
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H2-309



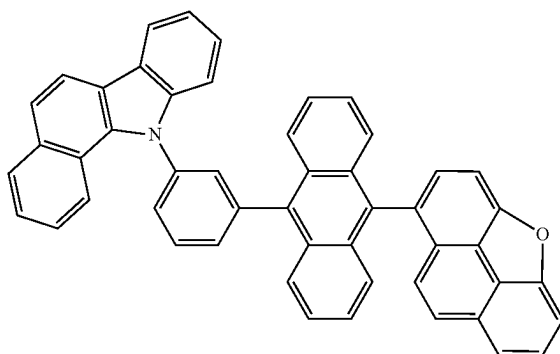
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H2-313

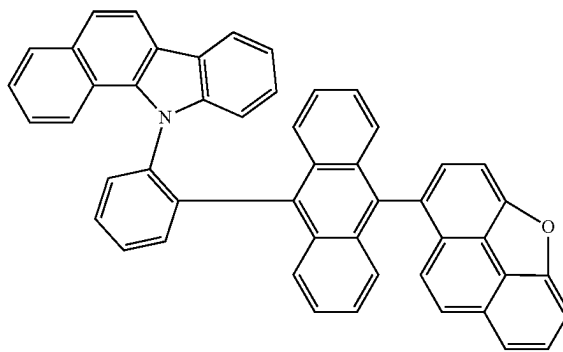


H2-314

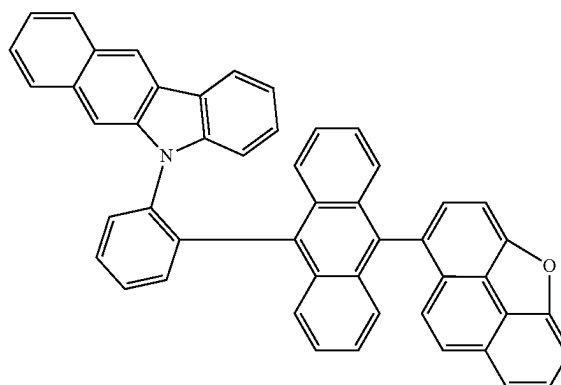
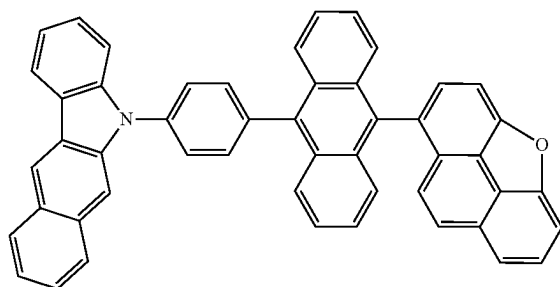
H2-310



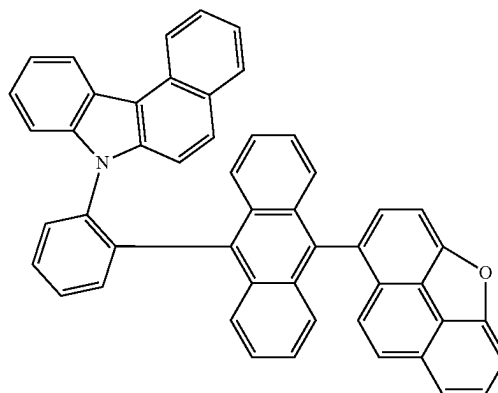
H2-311



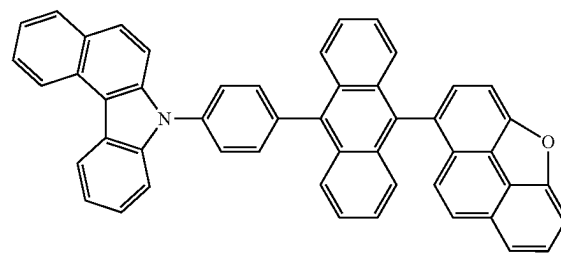
H2-312



H2-315

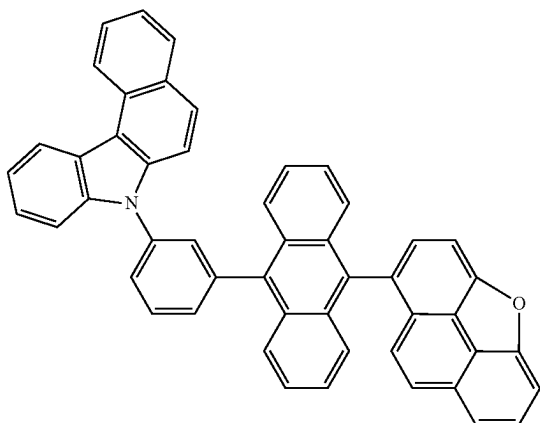


H2-316

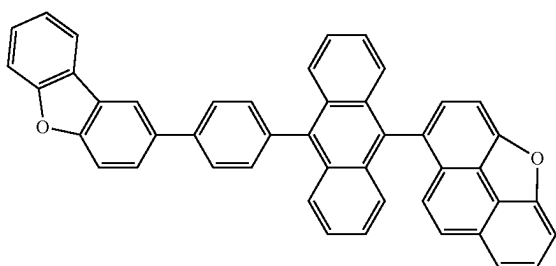


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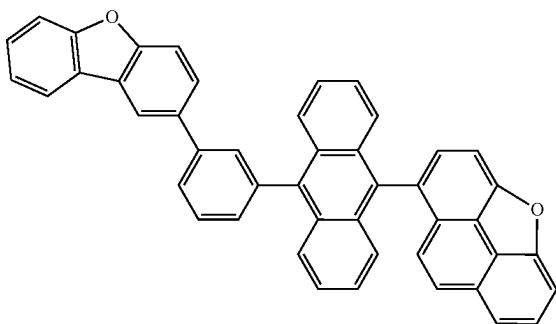
H2-317



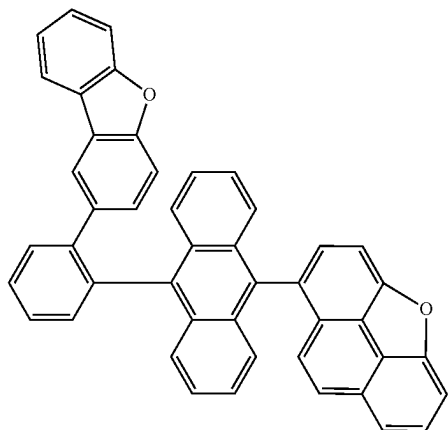
H2-318



H2-319

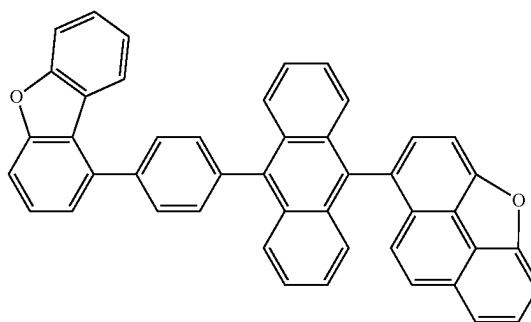


H2-320

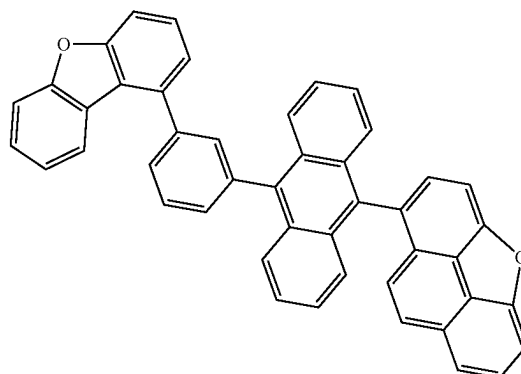


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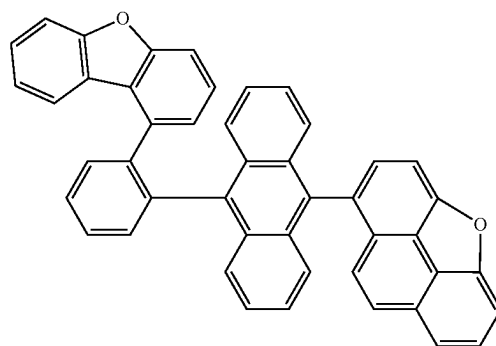
H2-321



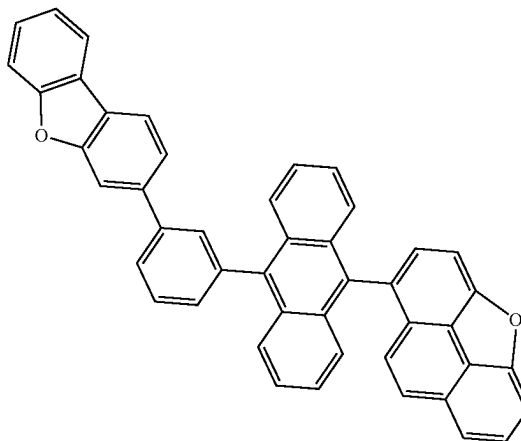
H2-322



H2-323

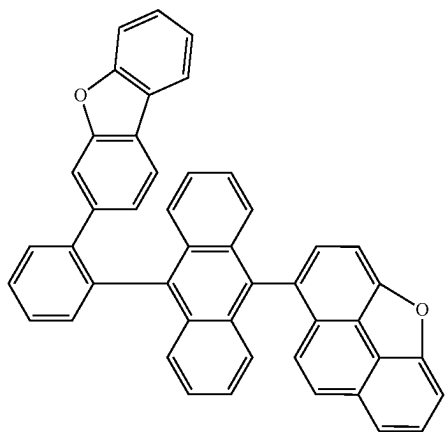


H2-324

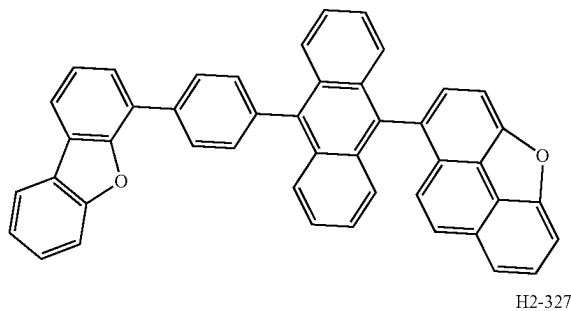


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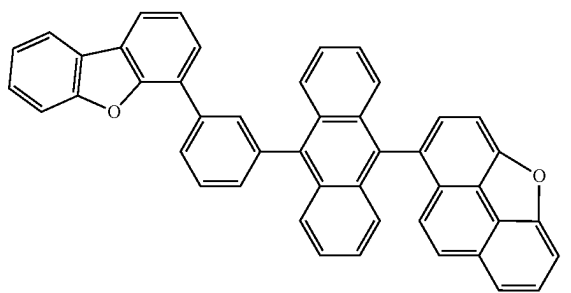
H2-325



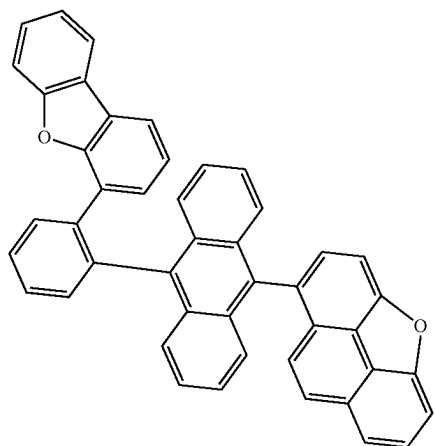
H2-326



H2-327

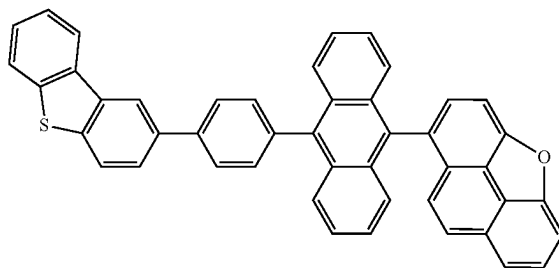


H2-328

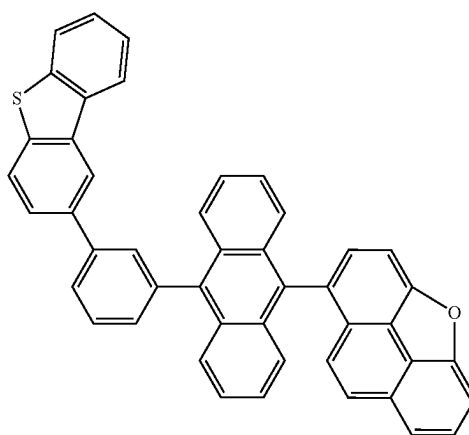


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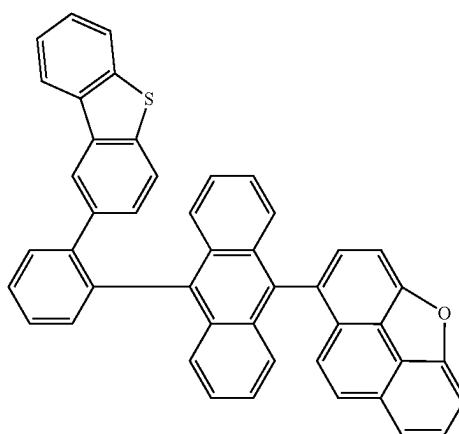
H2-329



H2-330

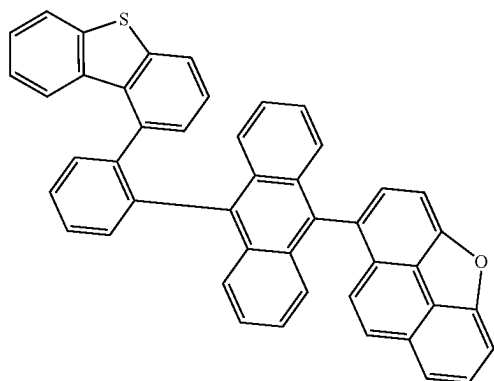


H2-331

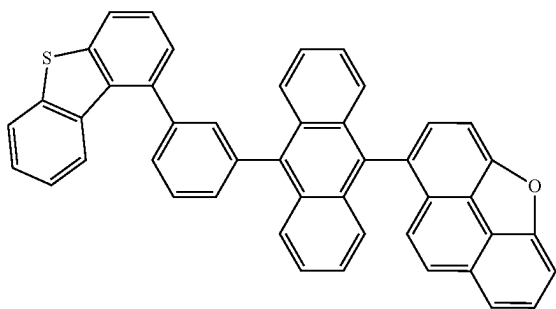


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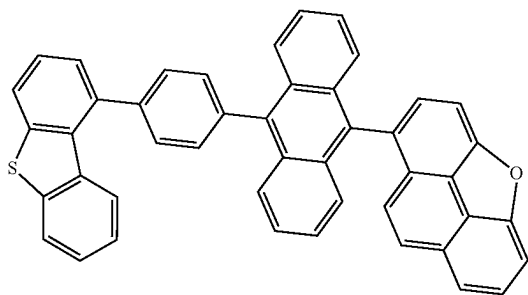
H2-332



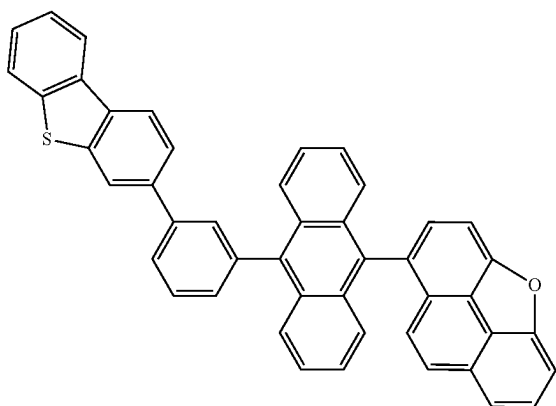
H2-333



H2-334

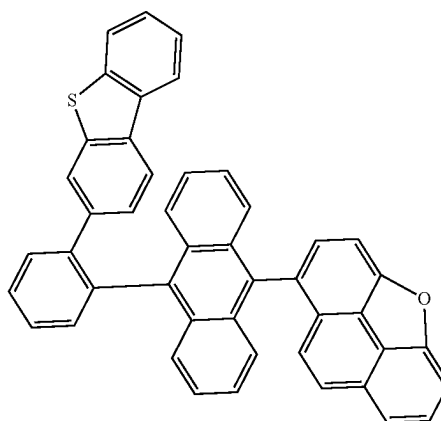


H2-335

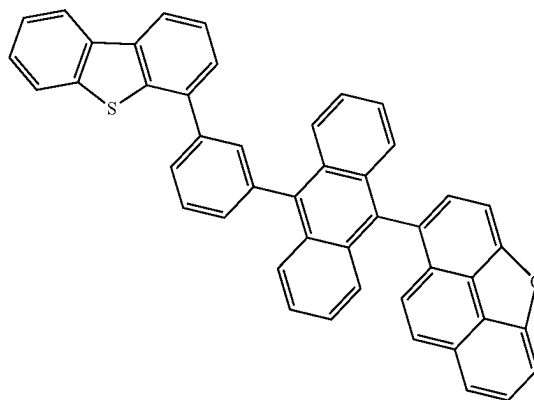


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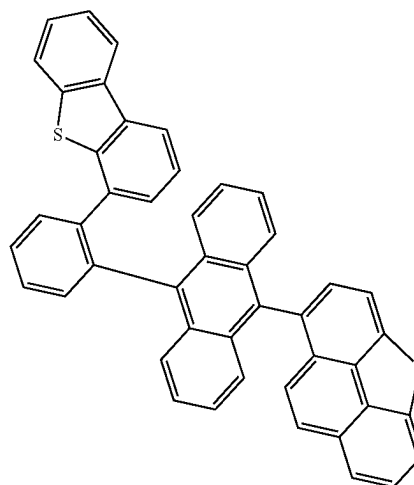
H2-336



H2-337

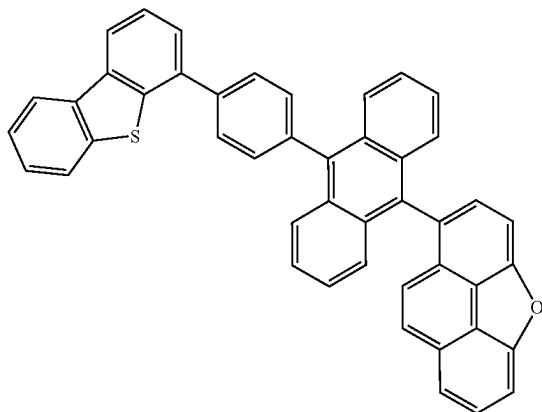


H2-338

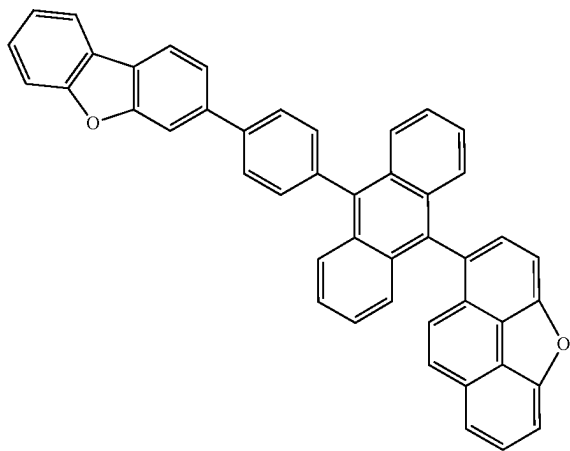


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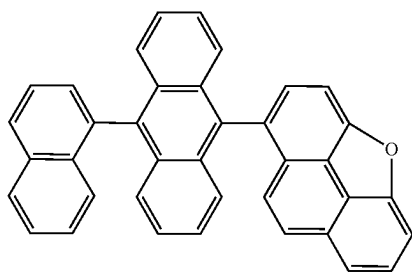
H2-339



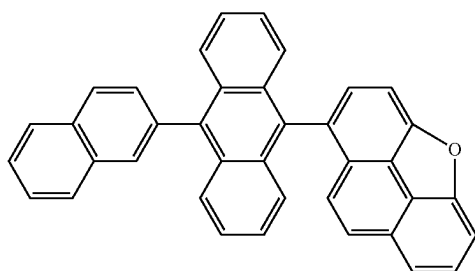
H2-340



H2-341

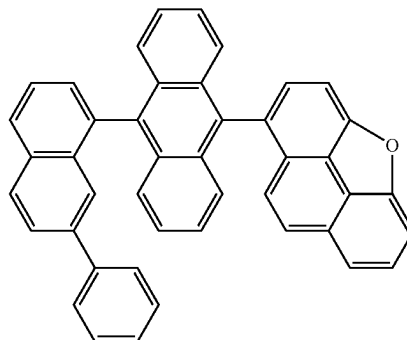


H2-342

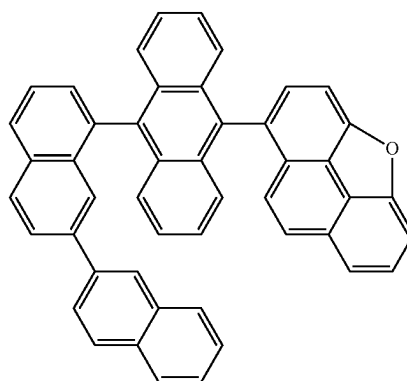


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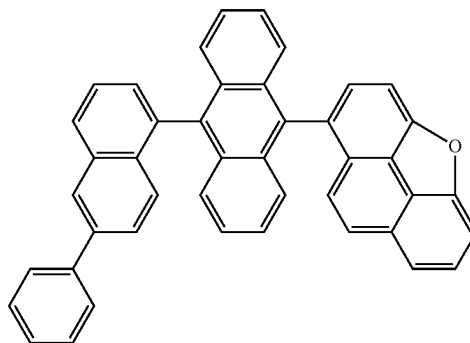
H2-343



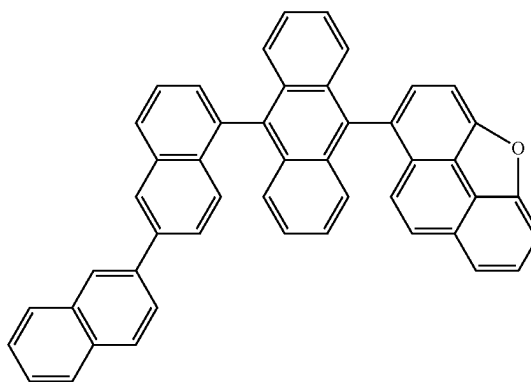
H2-344



H2-345

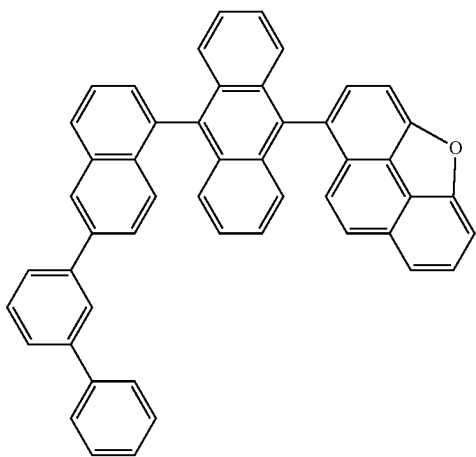


H2-346

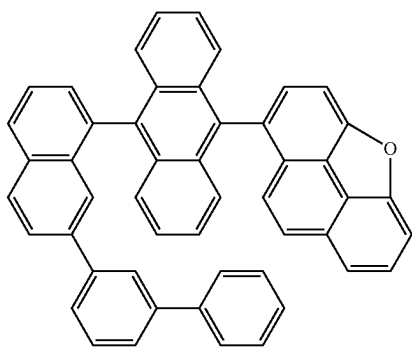


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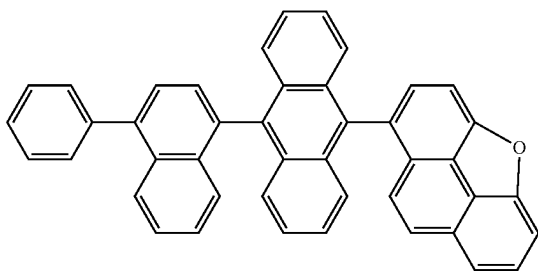
H2-347



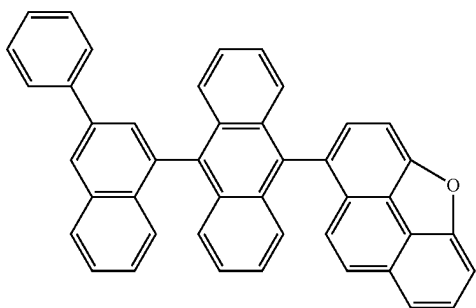
H2-348



H2-349

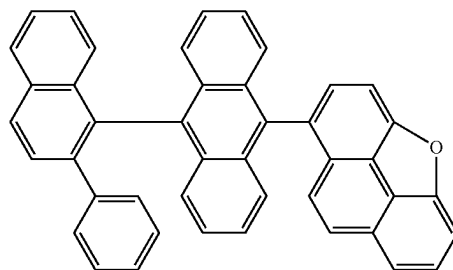


H2-350

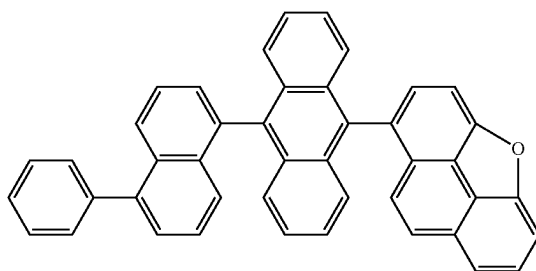


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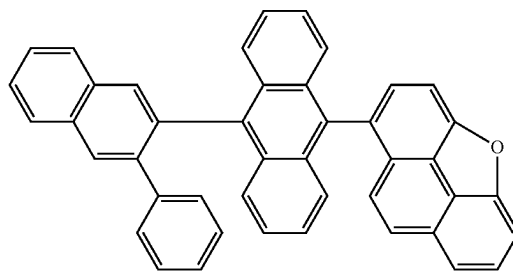
H2-351



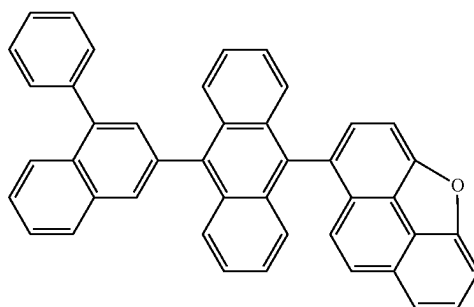
H2-352



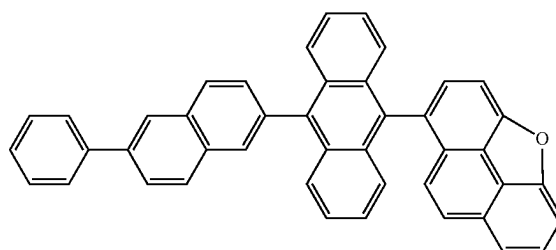
H2-353



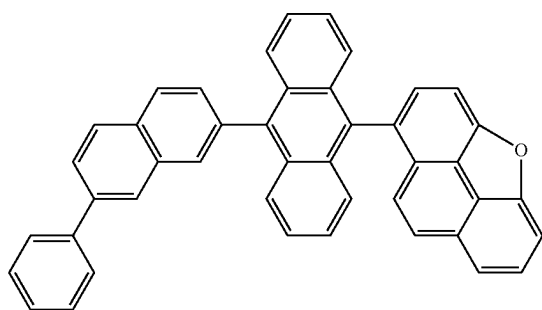
H2-354



H2-355

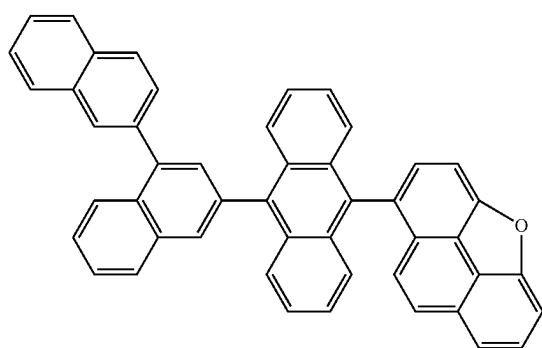


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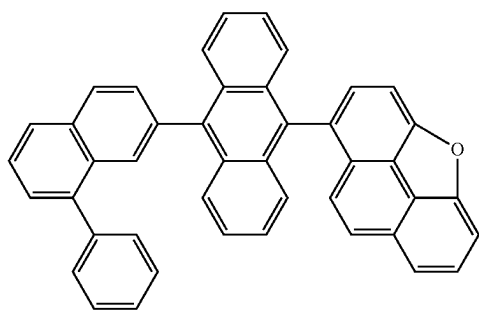


H2-356

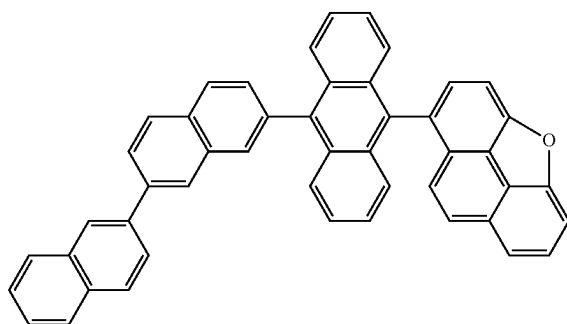
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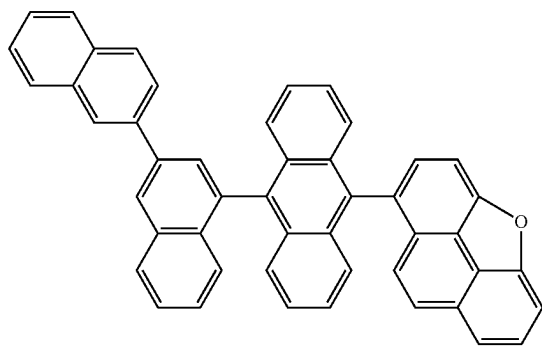
H2-360



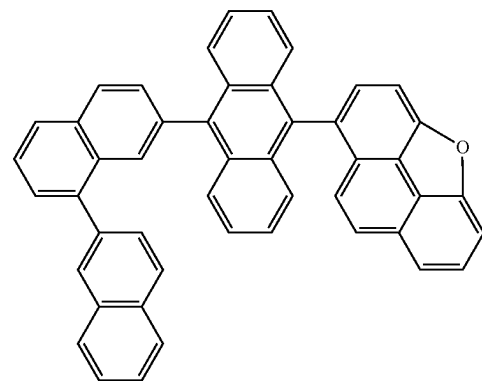
H2-357



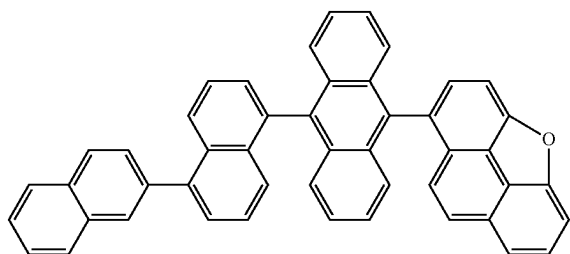
H2-361



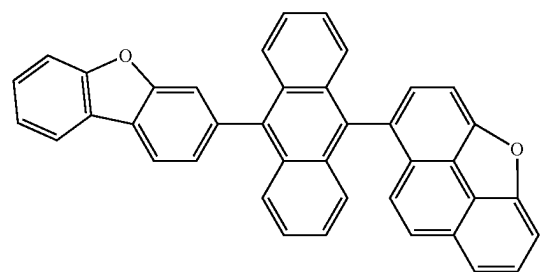
H2-358



H2-362



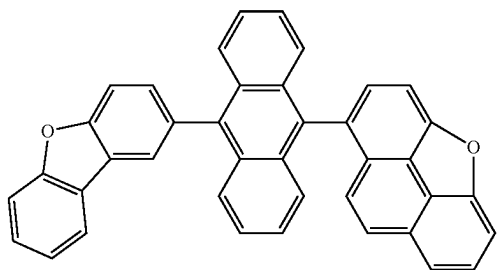
H2-359



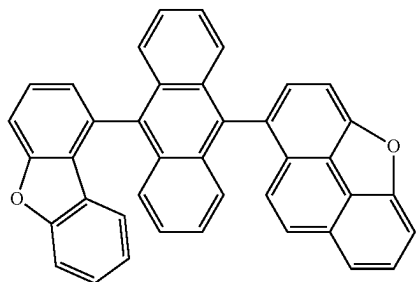
H2-363

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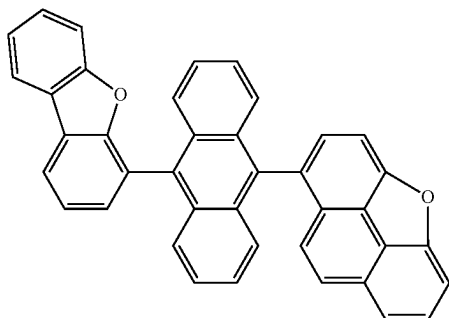
H2-364



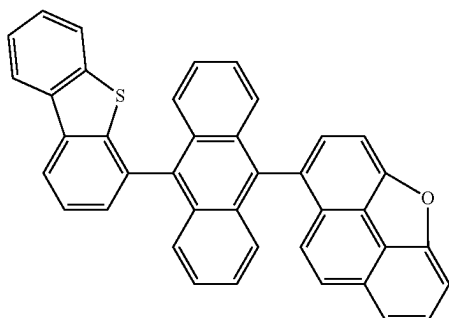
H2-365



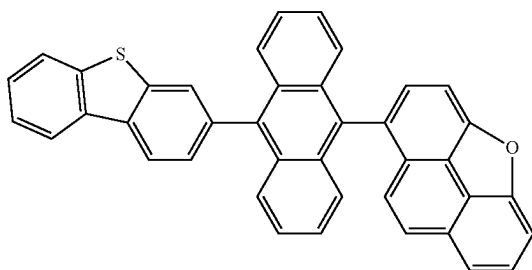
H2-366



H2-367

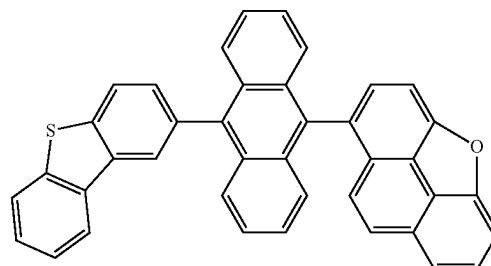


H2-368

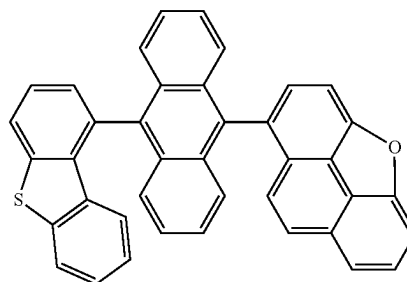


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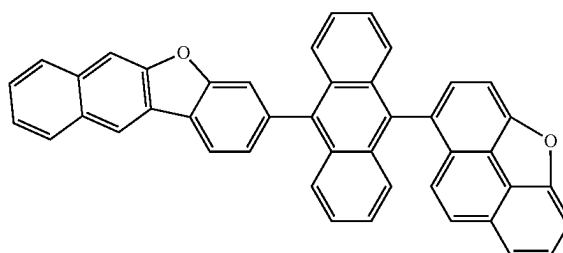
H2-369



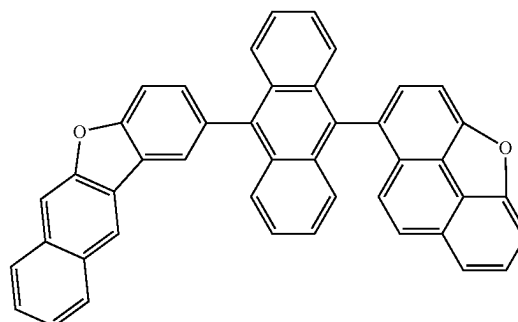
H2-370



H2-371

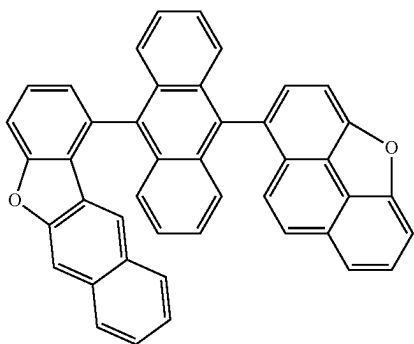


H2-372



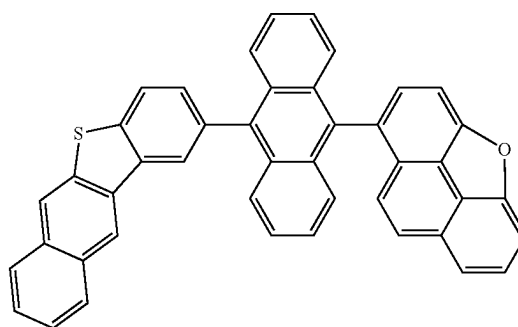
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H2-373

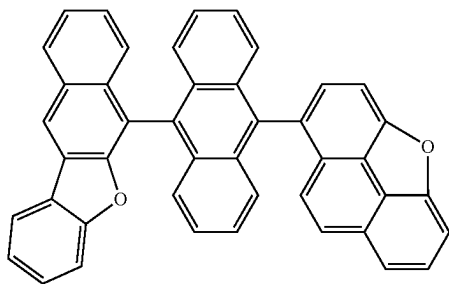


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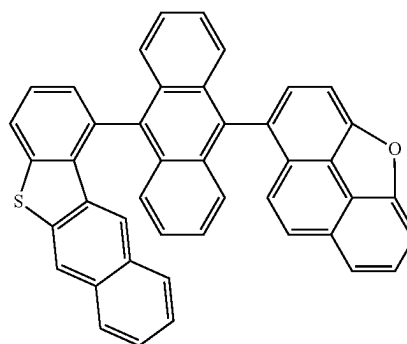
H2-377



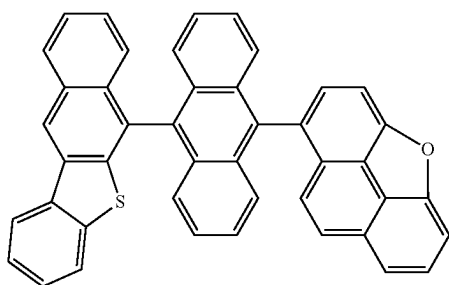
H2-374



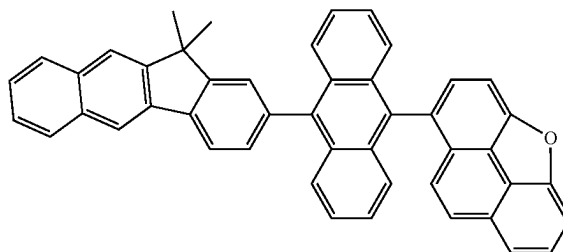
H2-378



H2-375

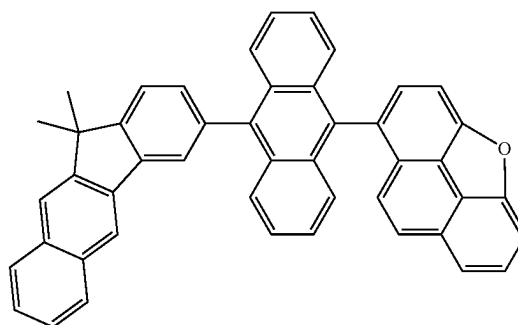
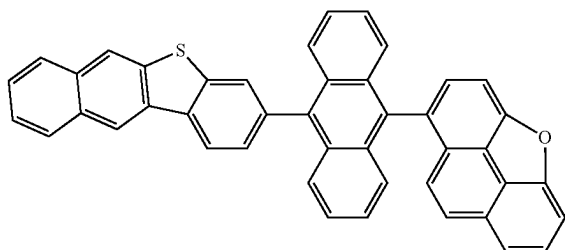


H2-379



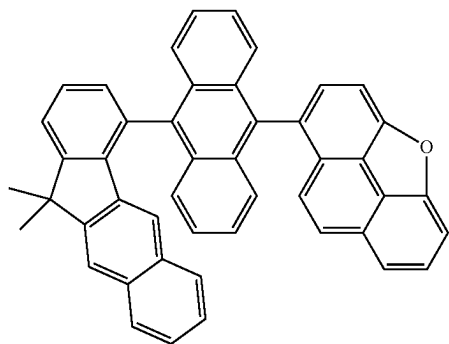
H2-380

H2-376



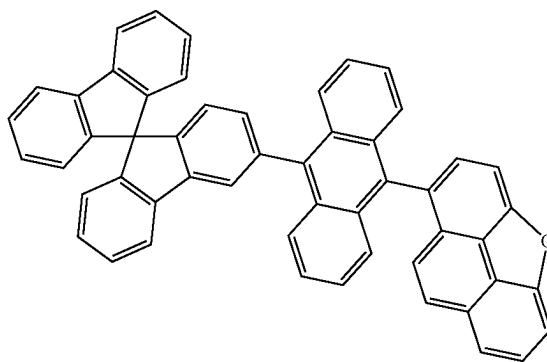
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H2-381

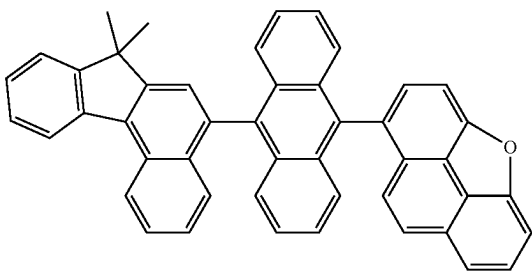


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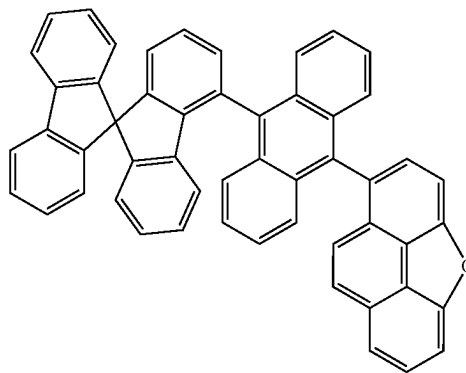
H2-385



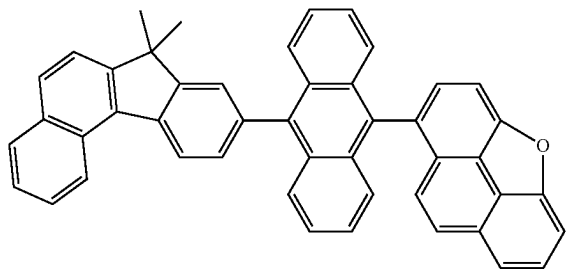
H2-382



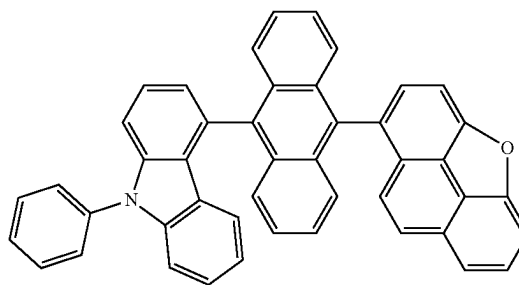
H2-386



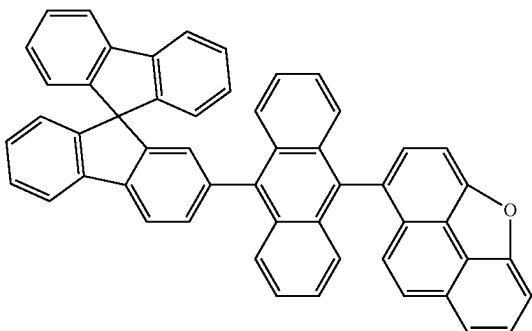
H2-383



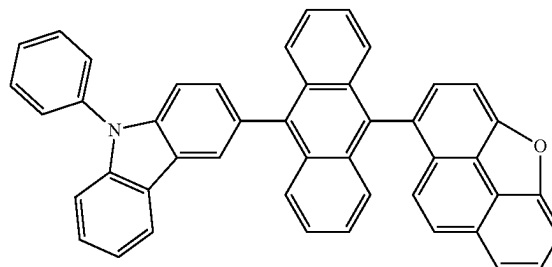
H2-387



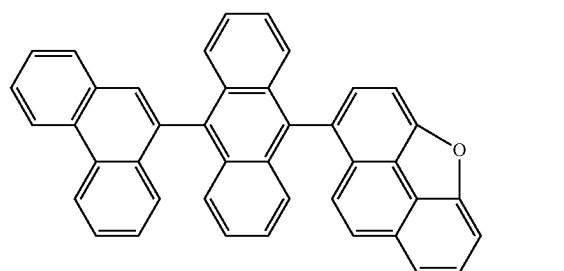
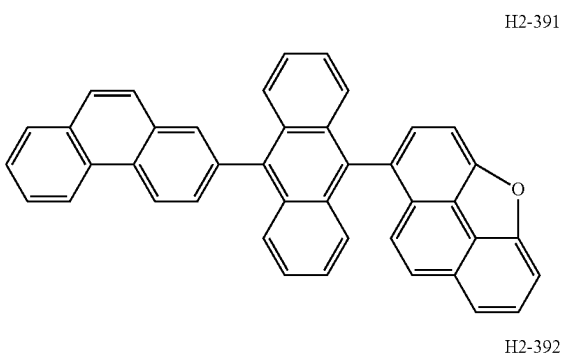
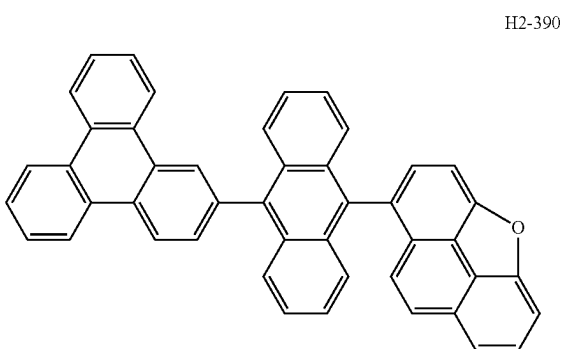
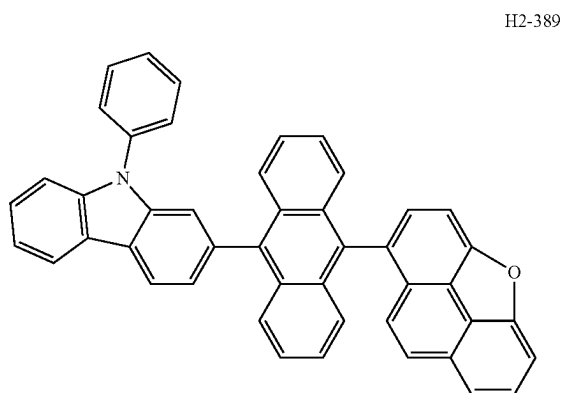
H2-384



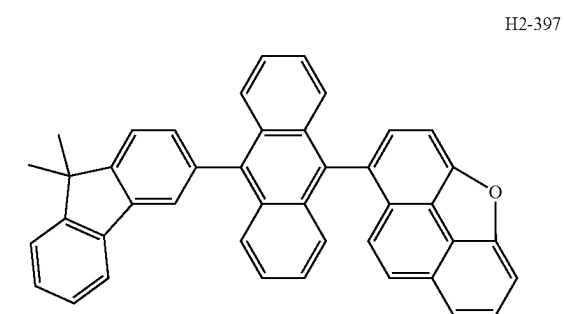
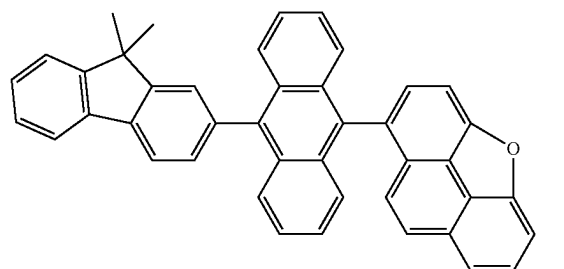
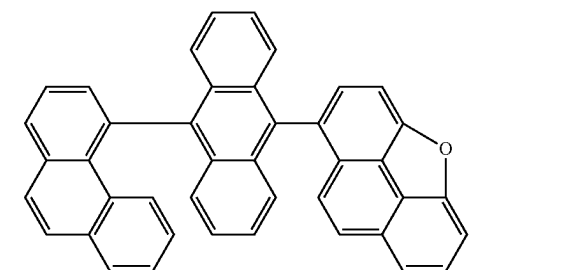
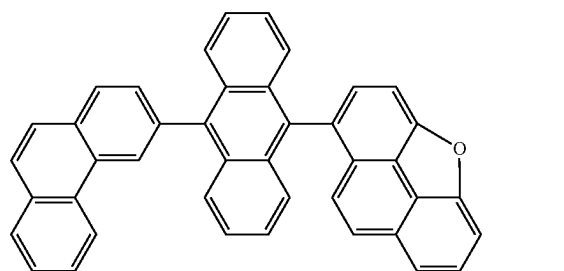
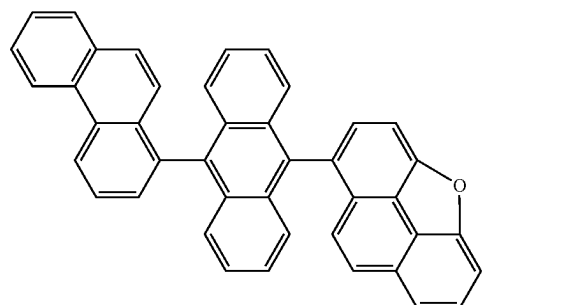
H2-388



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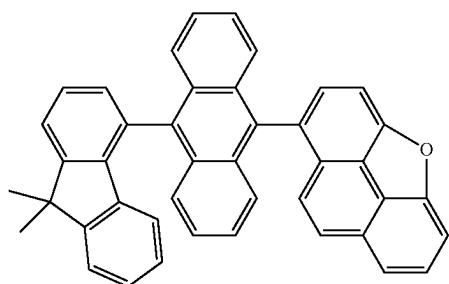


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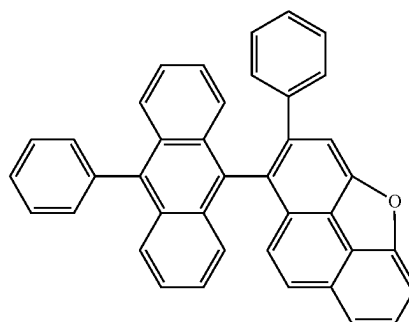
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H2-398



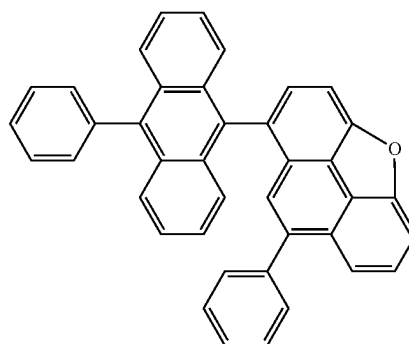
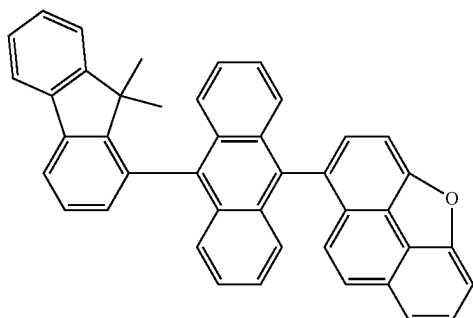
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H2-402



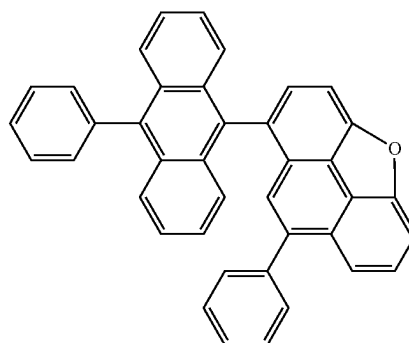
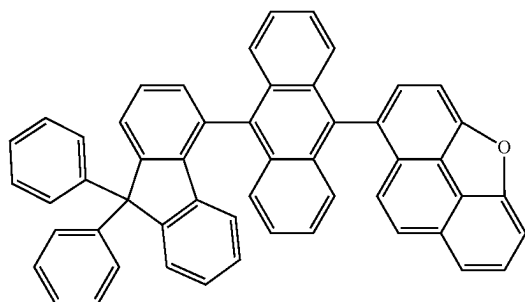
H2-403

H2-399



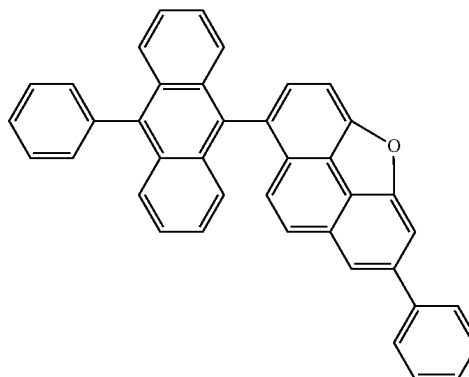
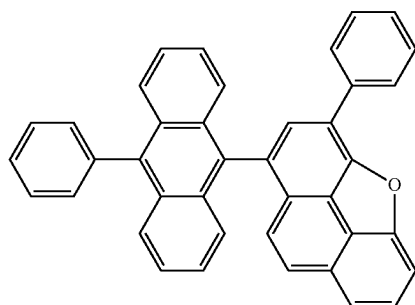
H2-404

H2-400



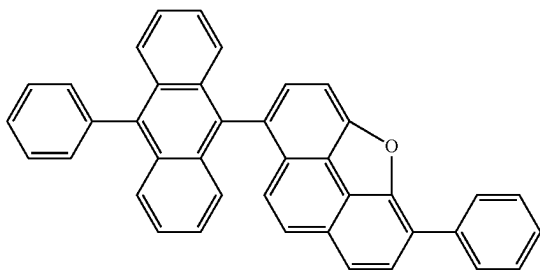
H2-405

H2-401

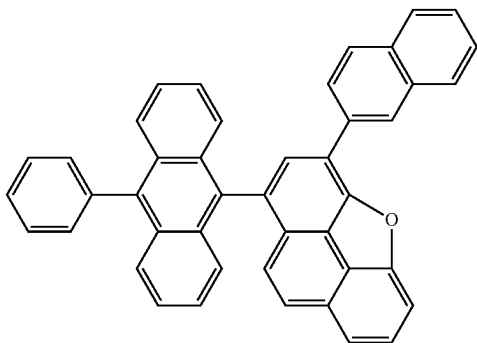


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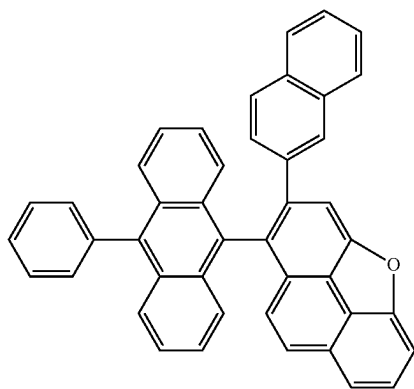
H2-406



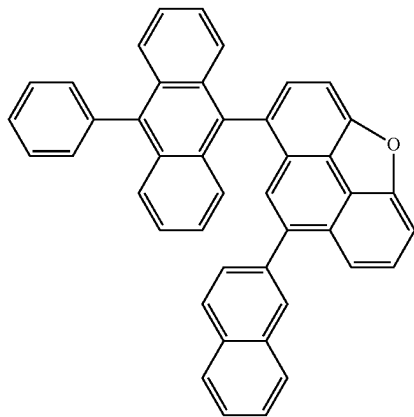
H2-407



H2-408

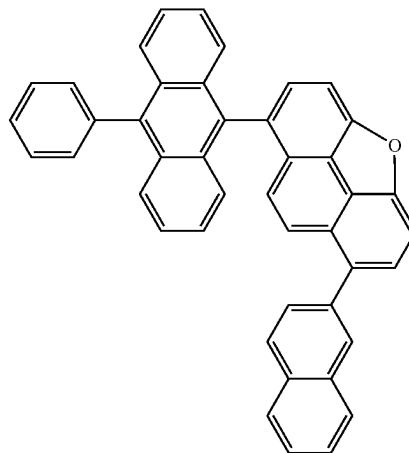


H2-409

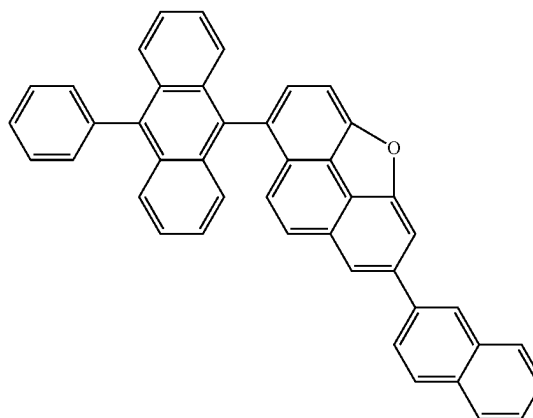


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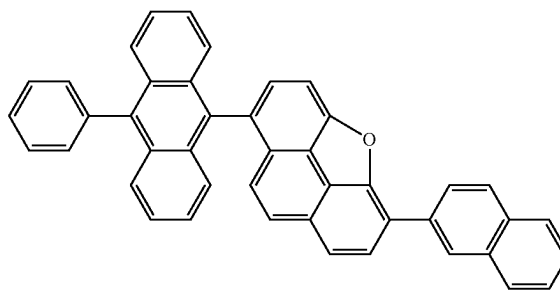
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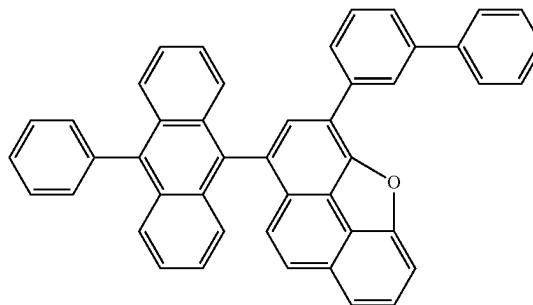
H2-411



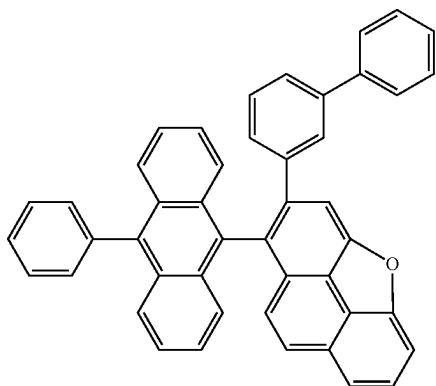
H2-412



H2-413

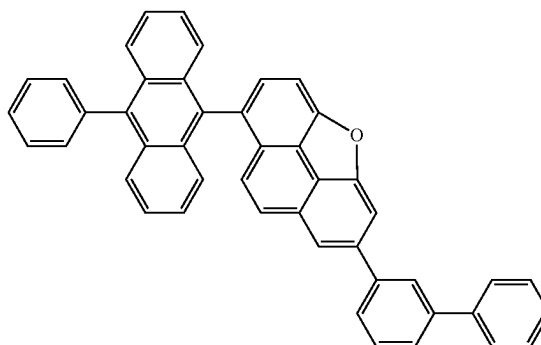


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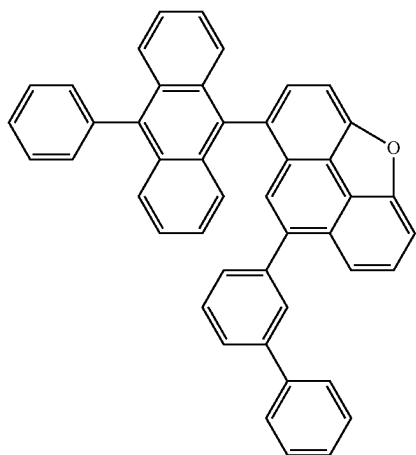


H2-414

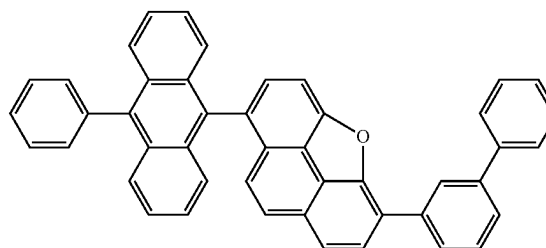
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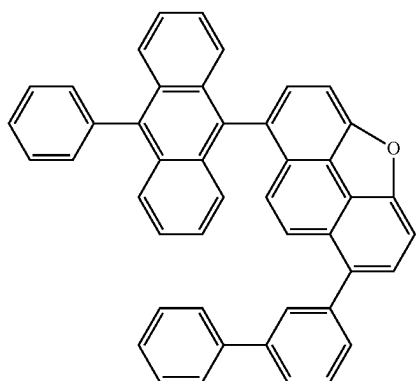
H2-417



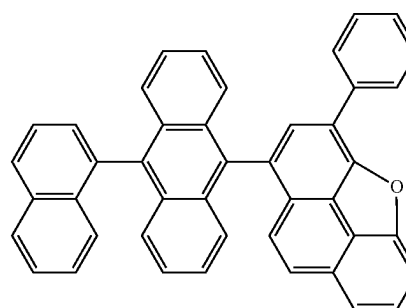
H2-415



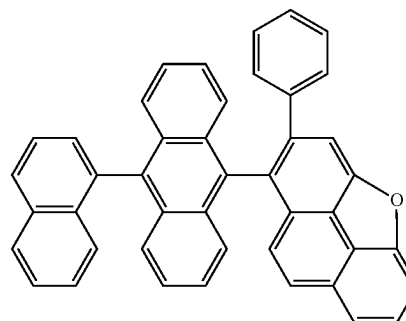
H2-418



H2-416

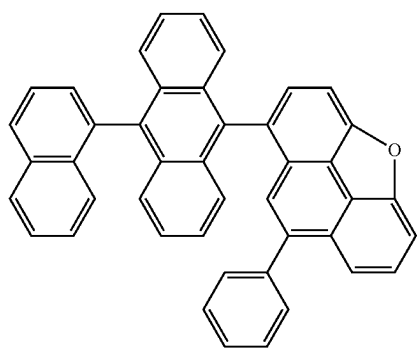


H2-419

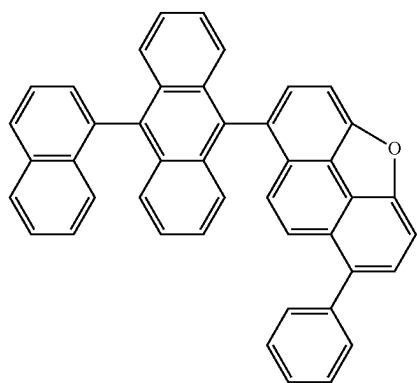


H2-420

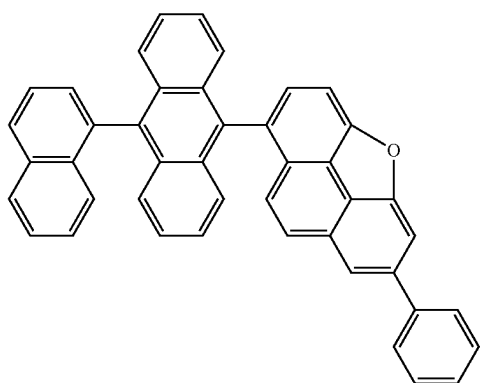
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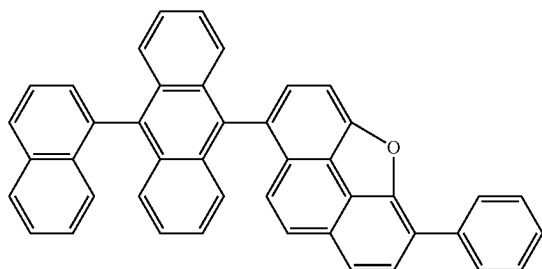
H2-421



H2-422

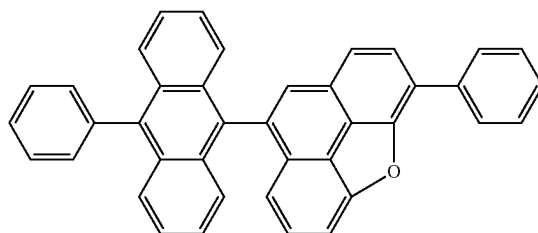


H2-423

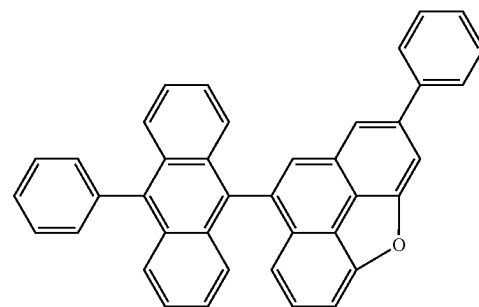


H2-424

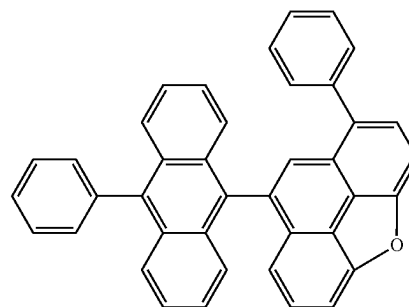
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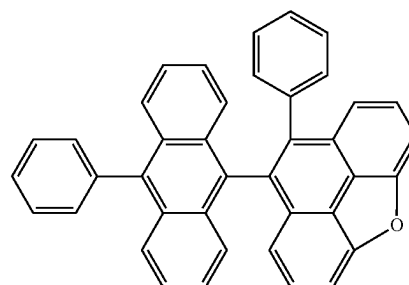
H2-425



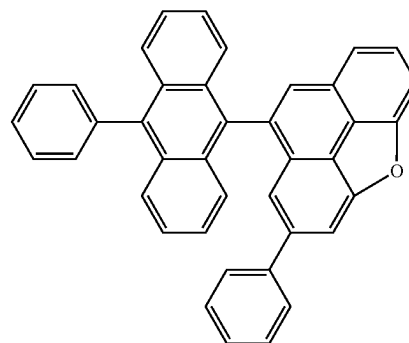
H2-426



H2-427

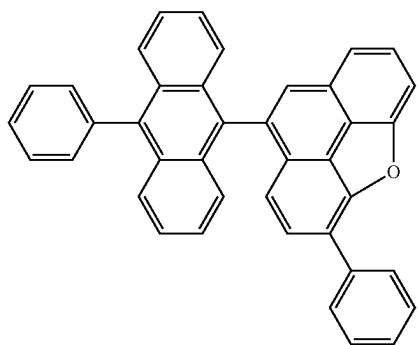


H2-428



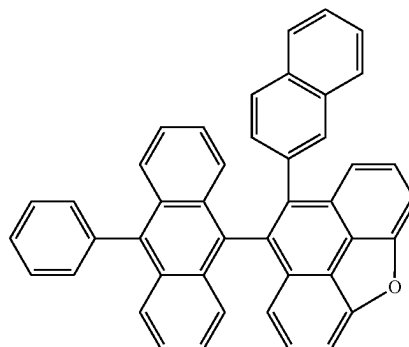
H2-429

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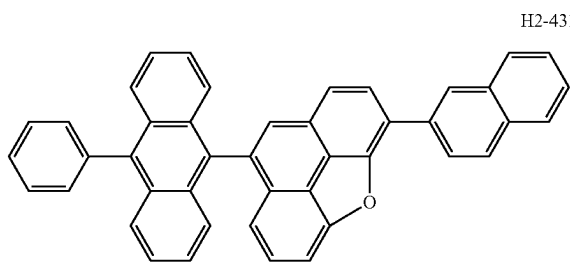


H2-430

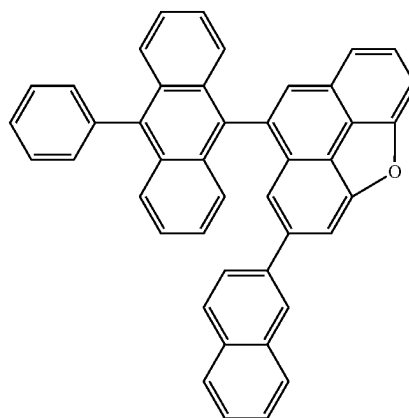
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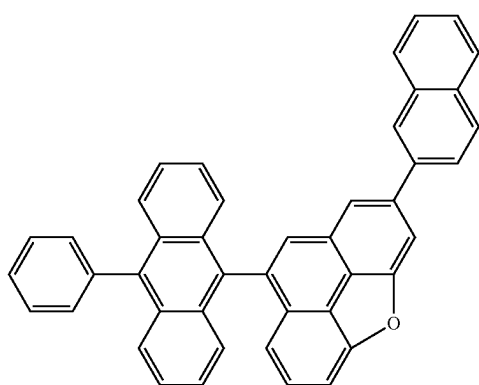
H2-434



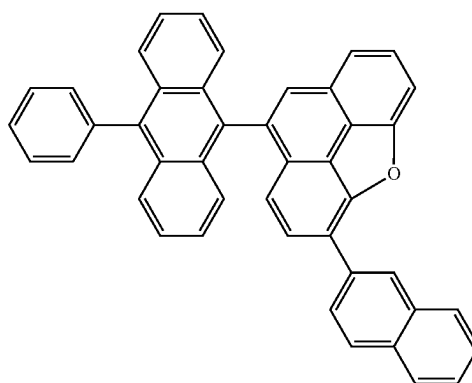
H2-431



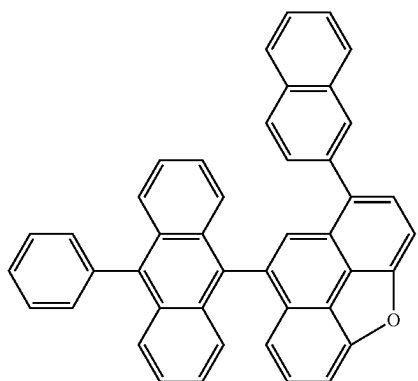
H2-435



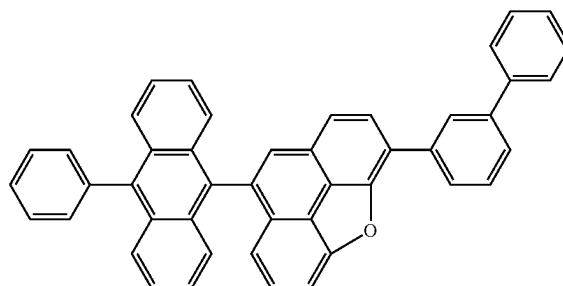
H2-432



H2-436

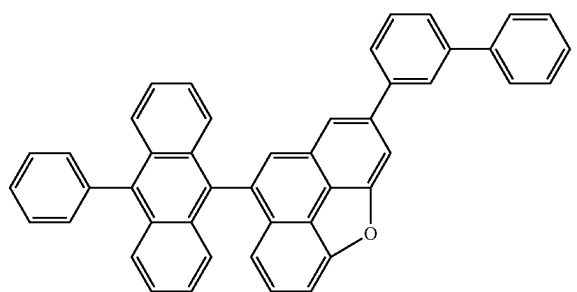


H2-433



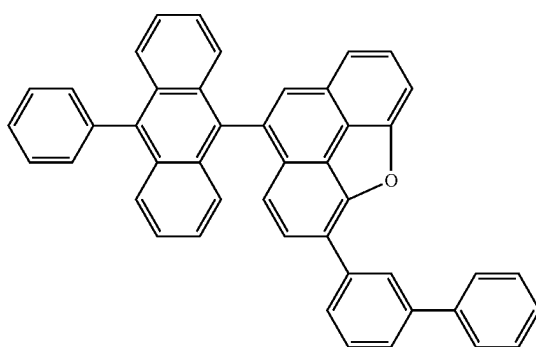
H2-437

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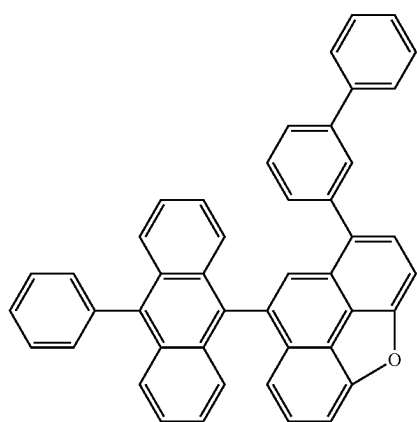
H2-438

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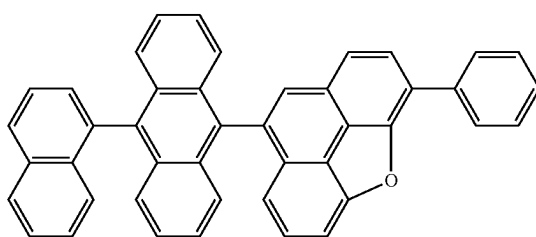


H2-442

H2-439

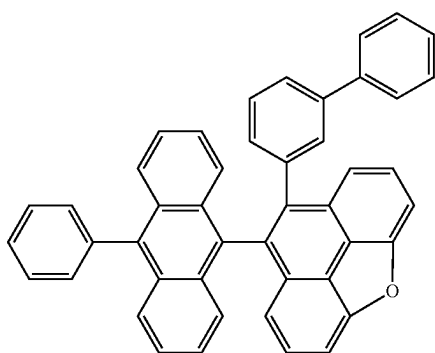


H2-443

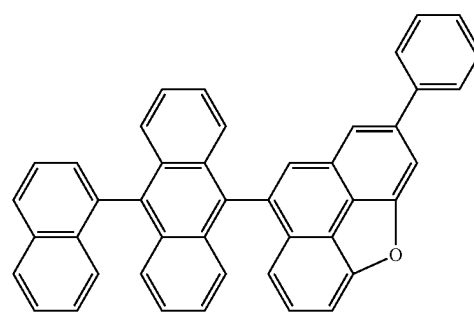


H2-444

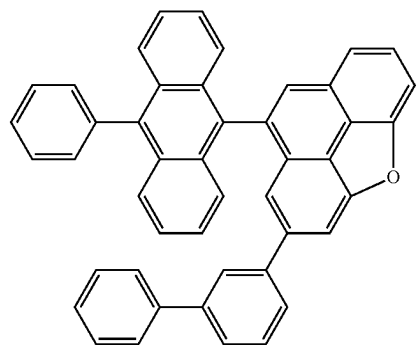
H2-440



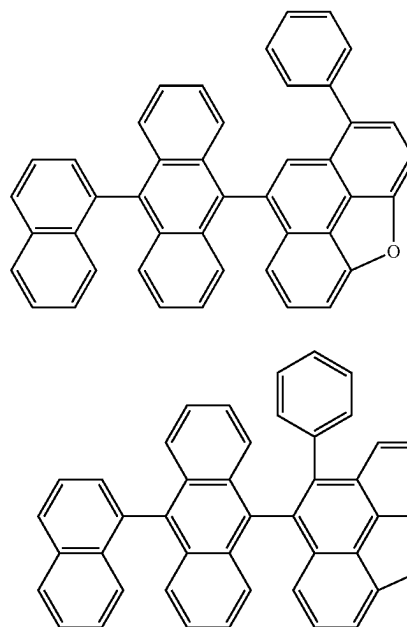
H2-445



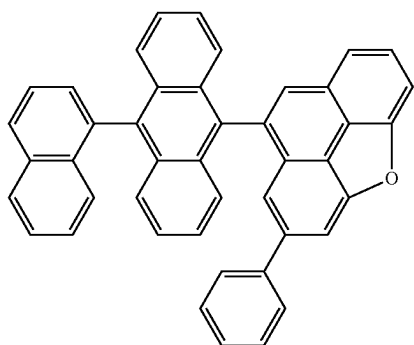
H2-441



H2-446

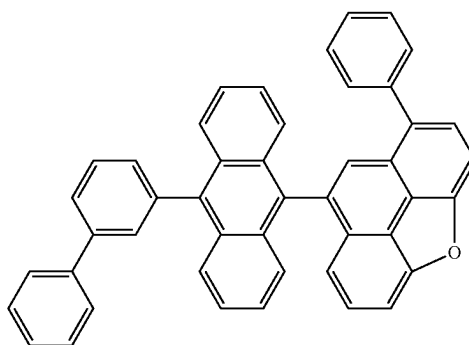


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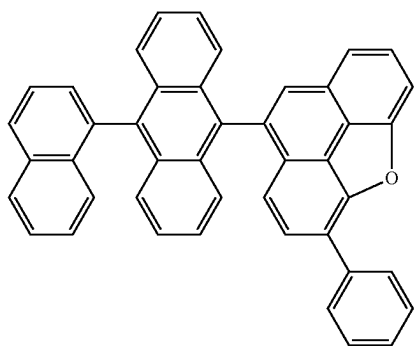


H2-447

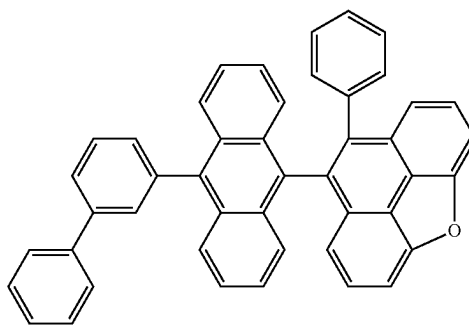
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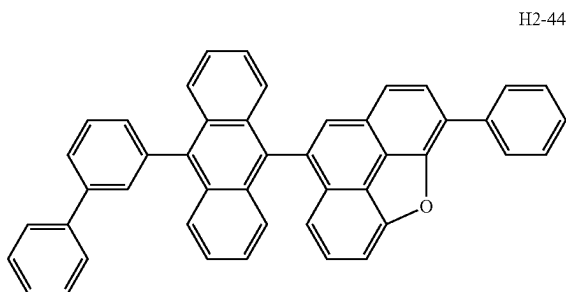
H2-451



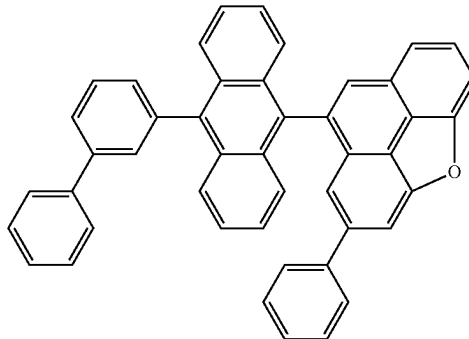
H2-448



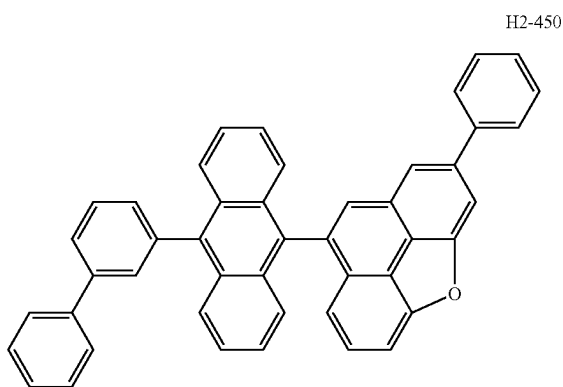
H2-452



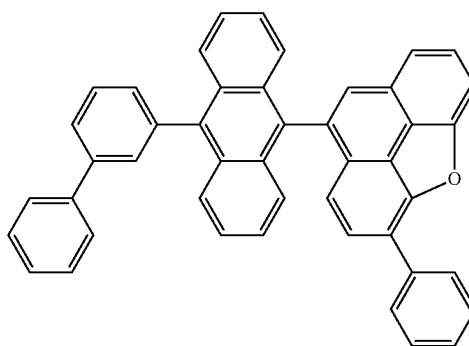
H2-449



H2-453

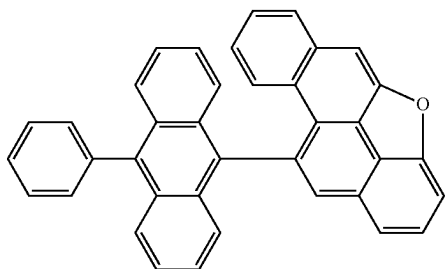


H2-450

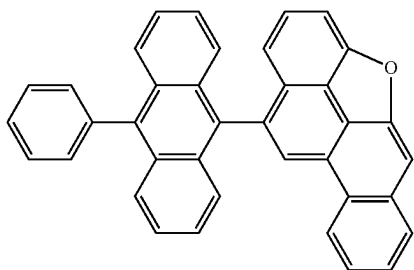


H2-454

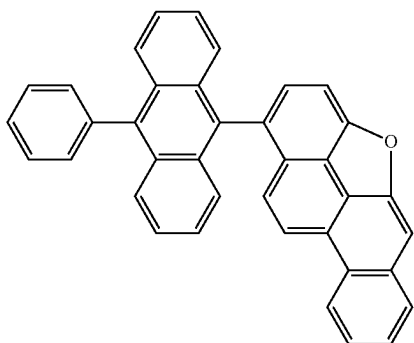
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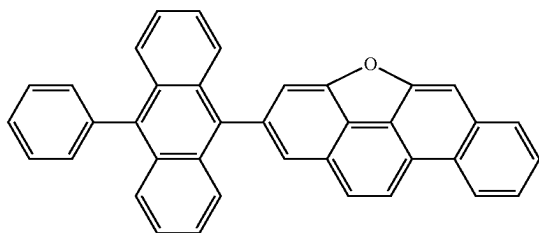
H2-455



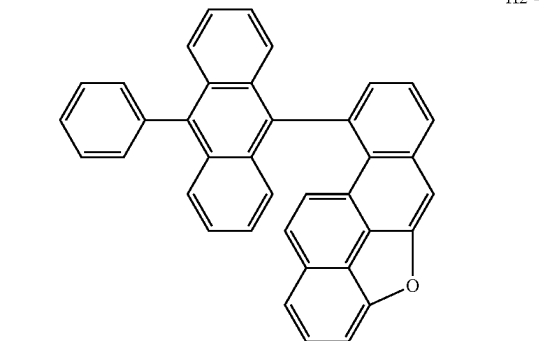
H2-456



H2-457

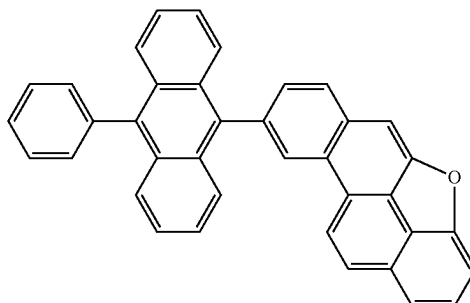


H2-458

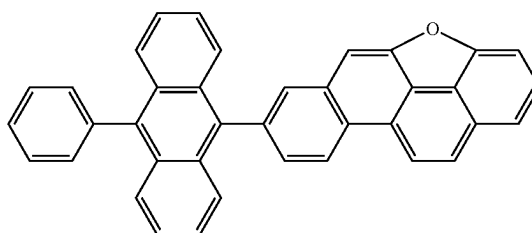


H2-459

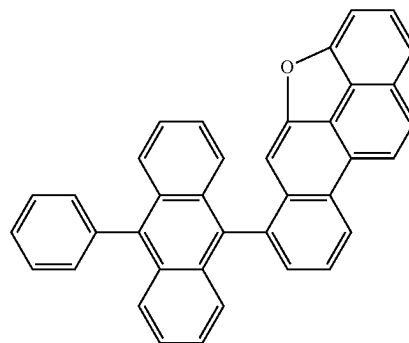
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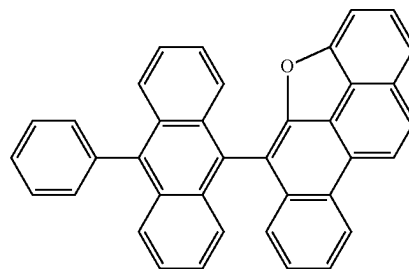
H2-460



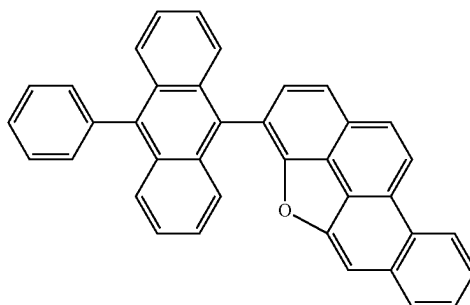
H2-461



H2-462



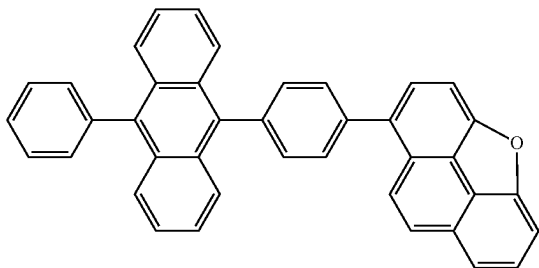
H2-463



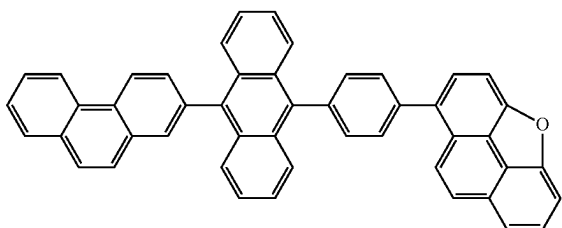
H2-464

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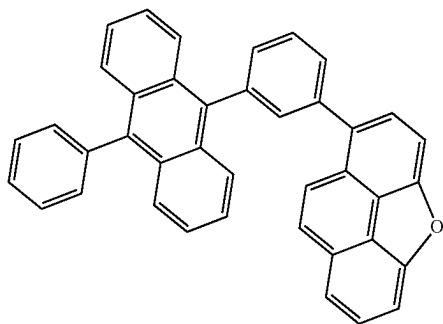
H2-465



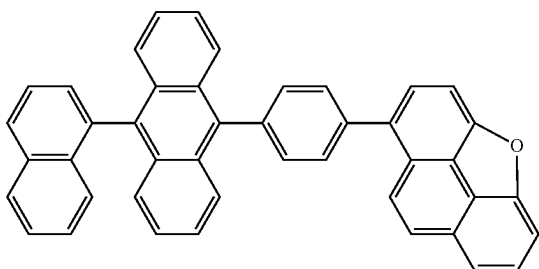
H2-466



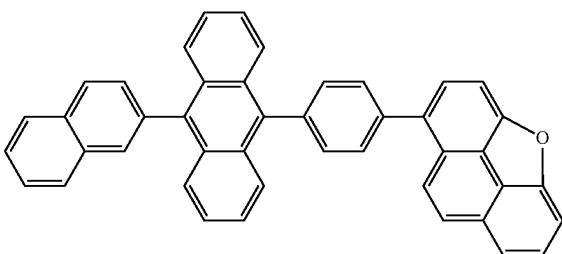
H2-467



H2-468

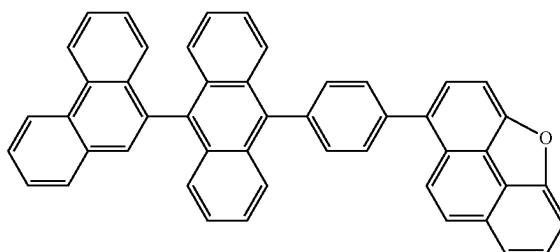


H2-469

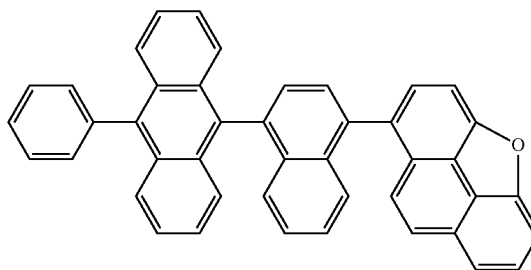


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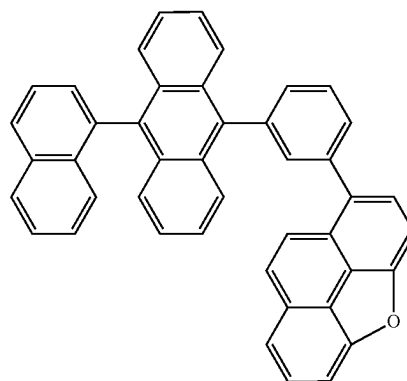
H2-470



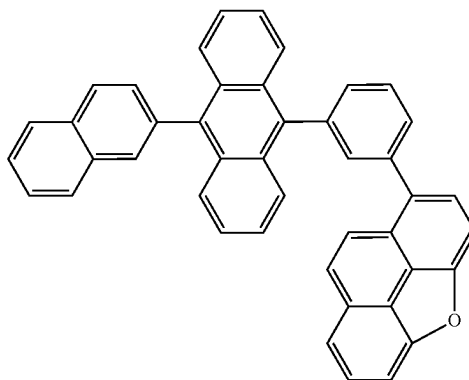
H2-471



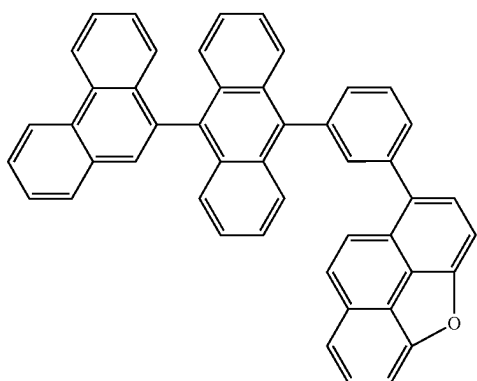
H2-472



H2-473

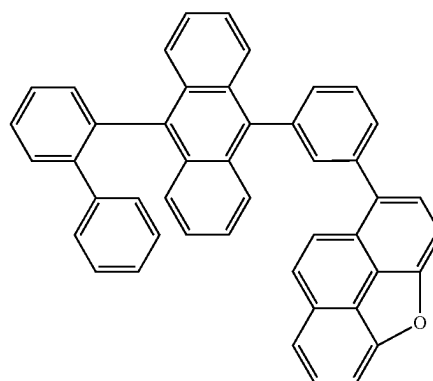


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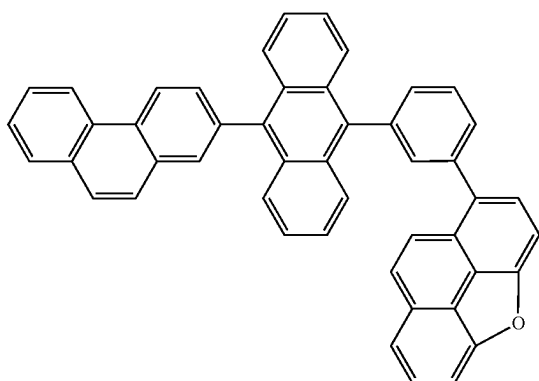


H2-474

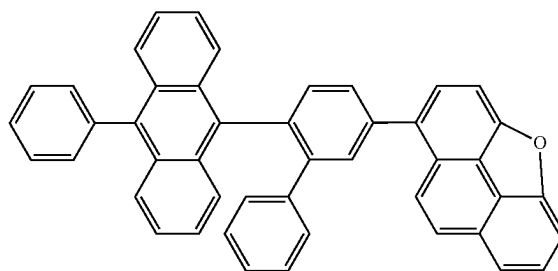
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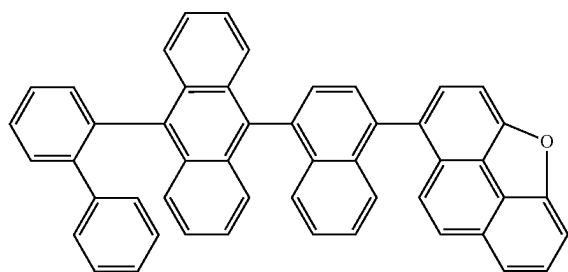
H2-478



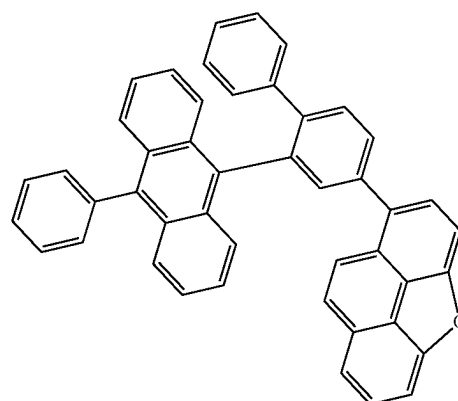
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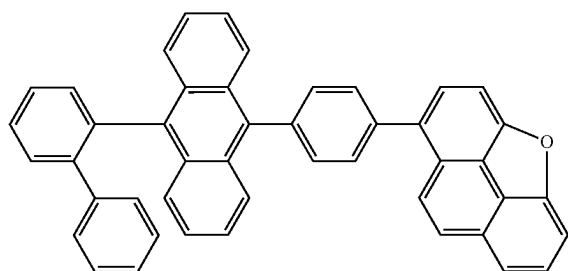
H2-479



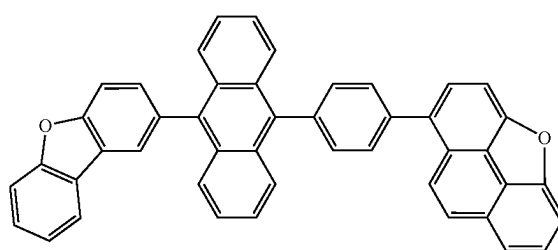
H2-476



H2-480



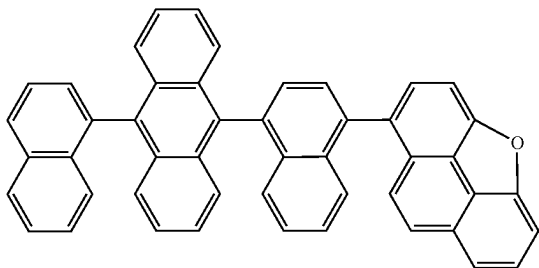
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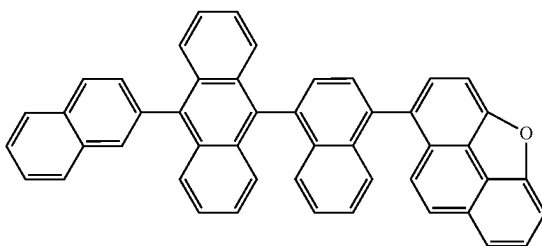
H2-481

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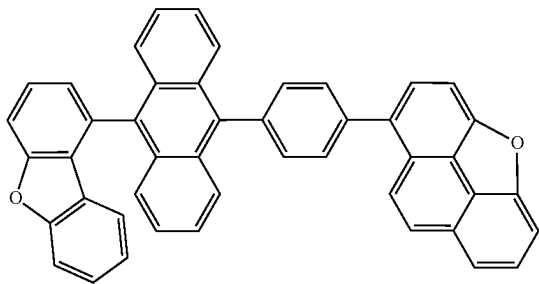
H2-482



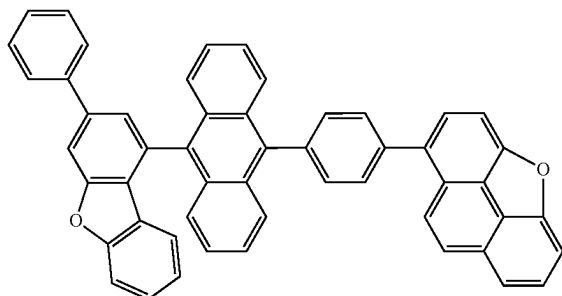
H2-483



H2-484

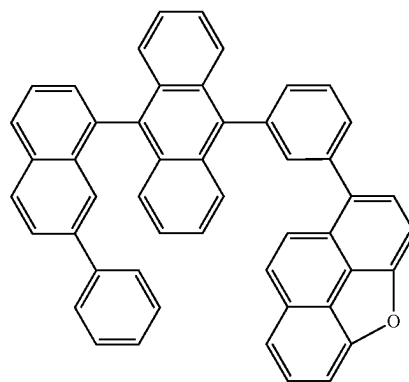


H2-485

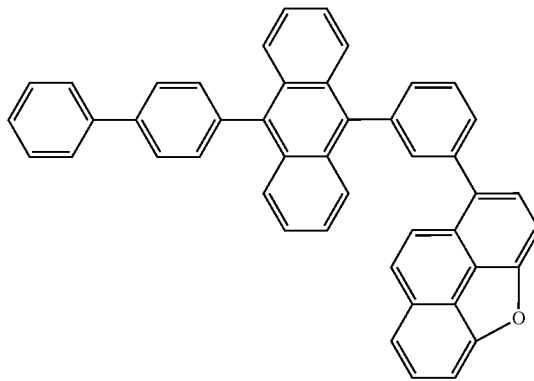


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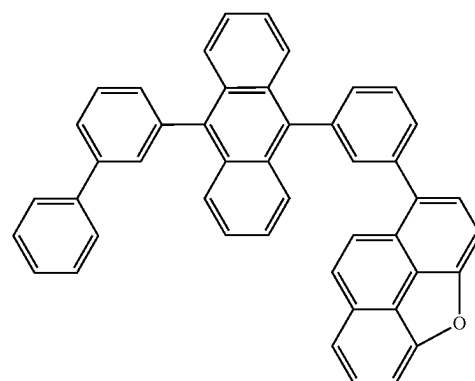
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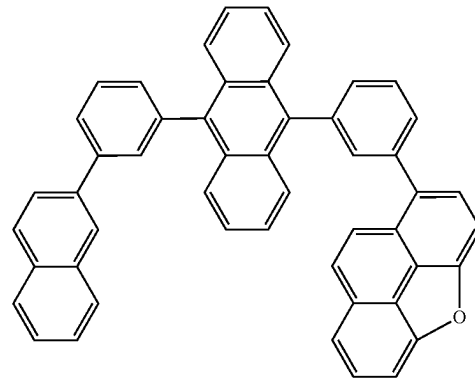
H2-487



H2-488

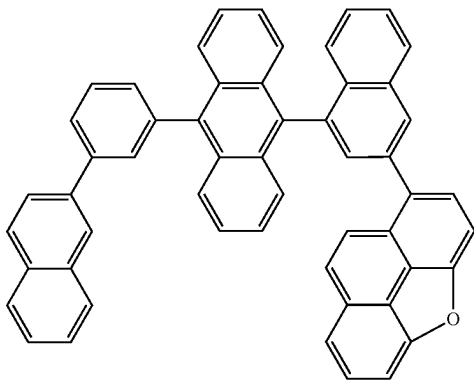


H2-489

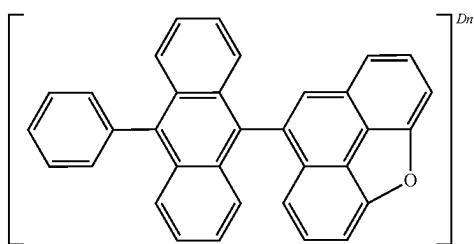


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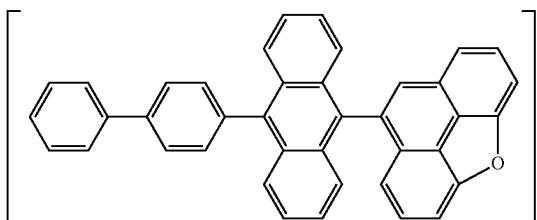
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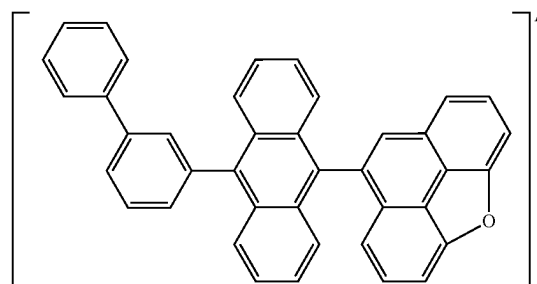
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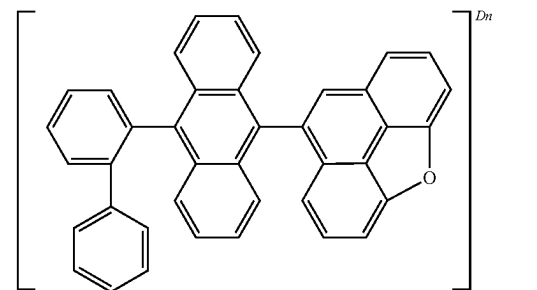
H2-492



H2-493

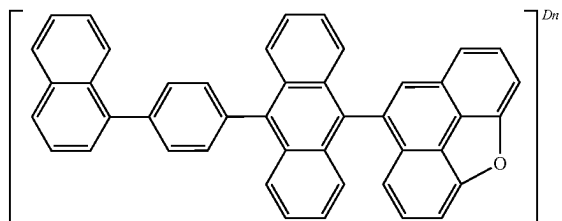


H2-494

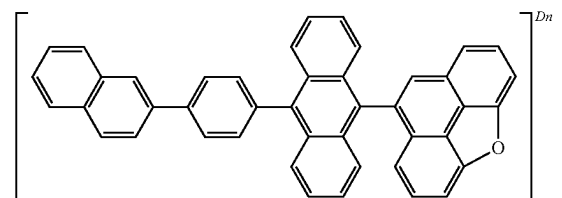


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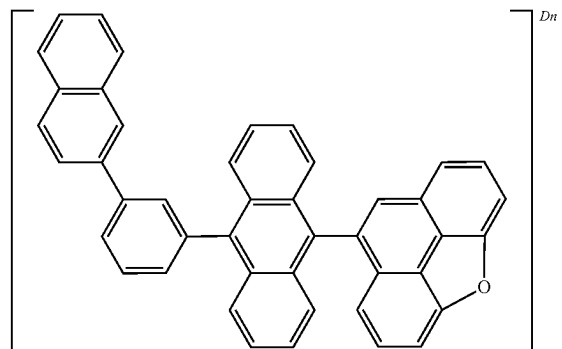
H2-495



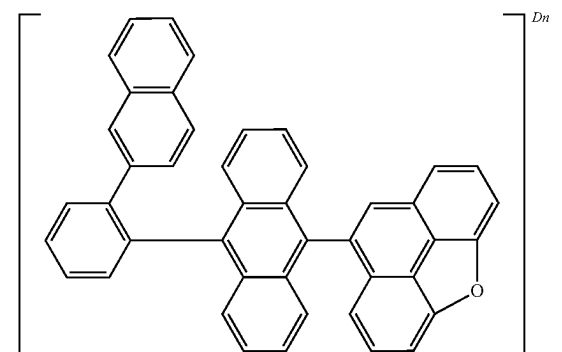
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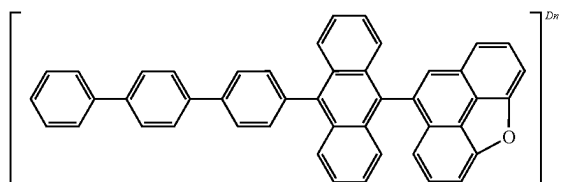
H2-497



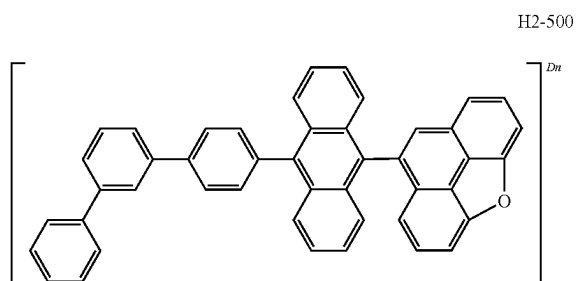
H2-498



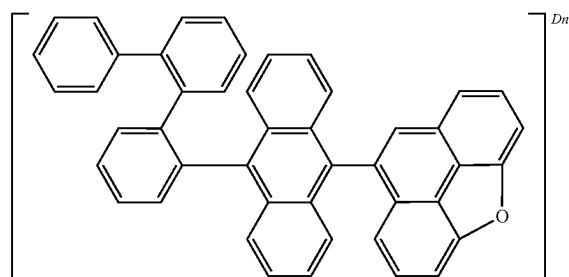
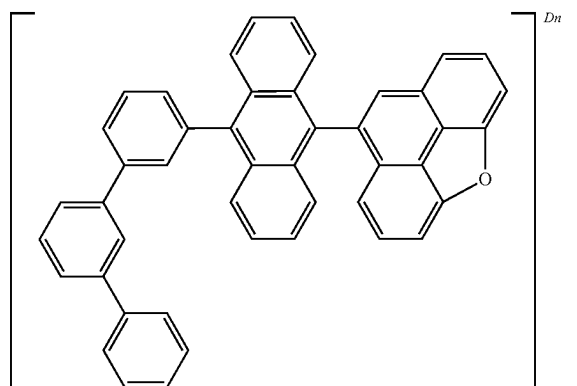
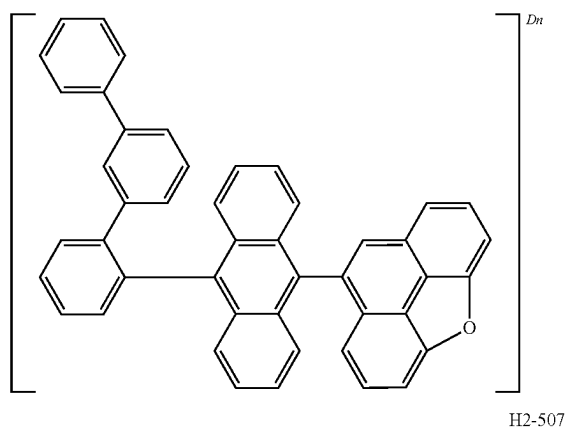
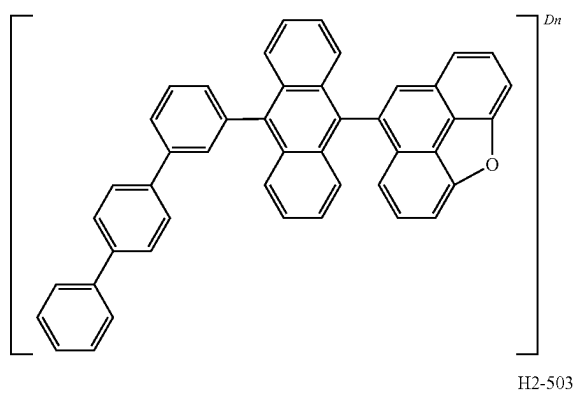
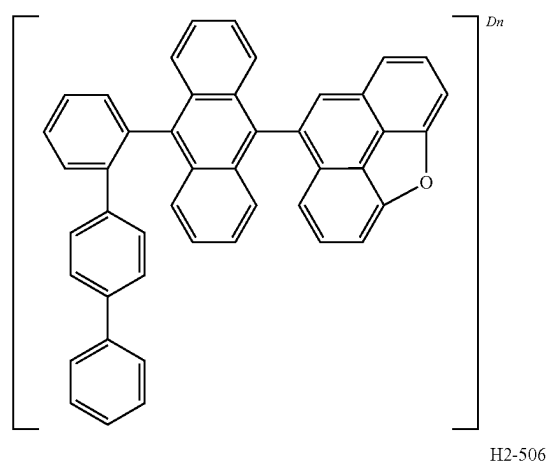
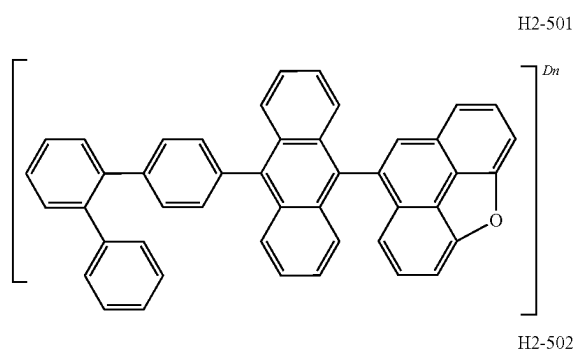
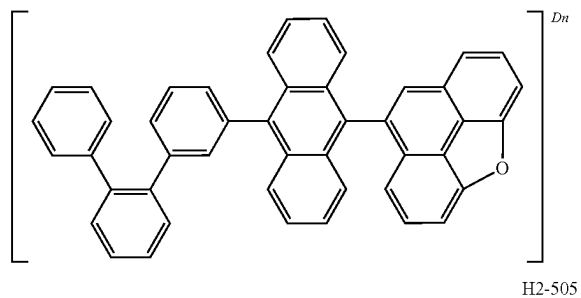
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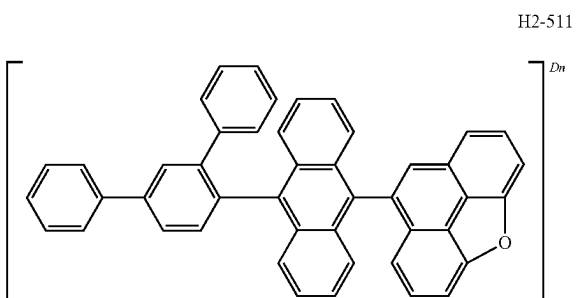
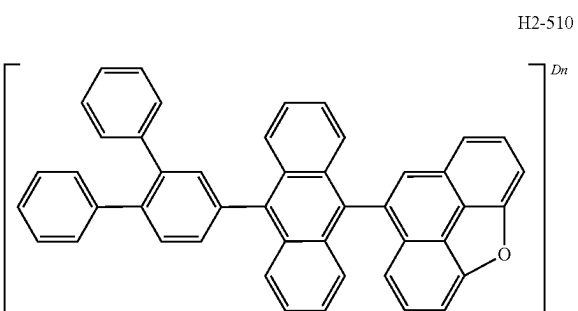
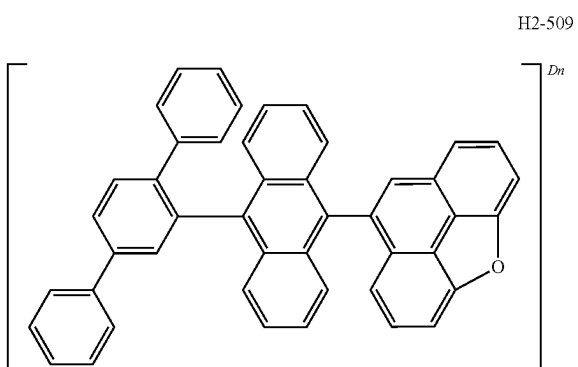
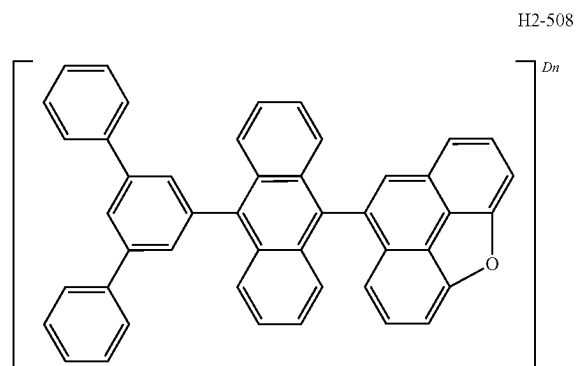
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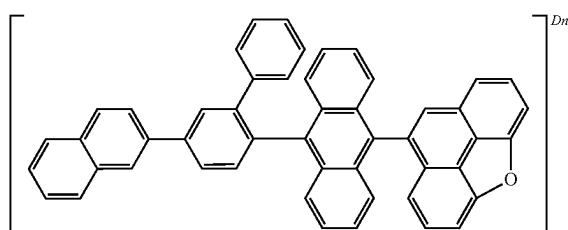
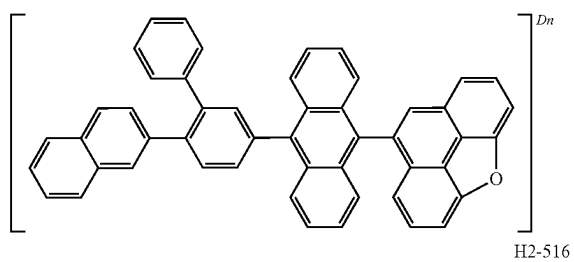
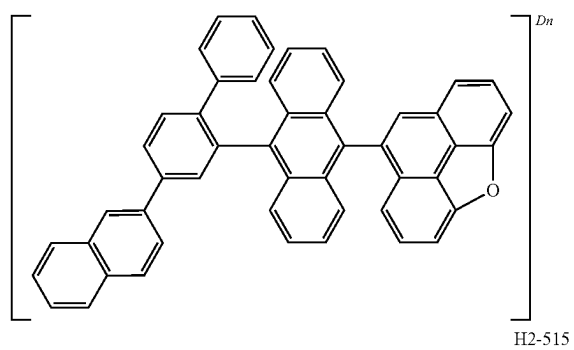
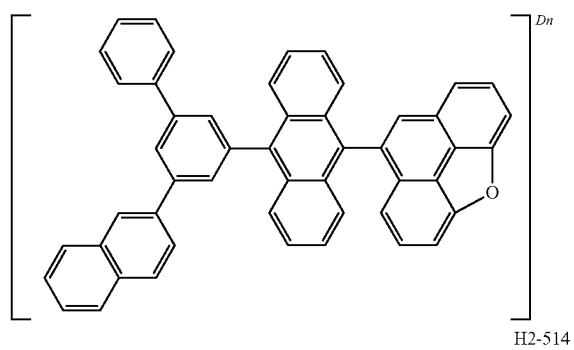
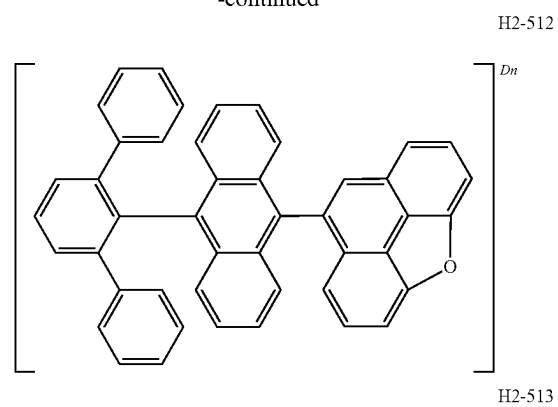
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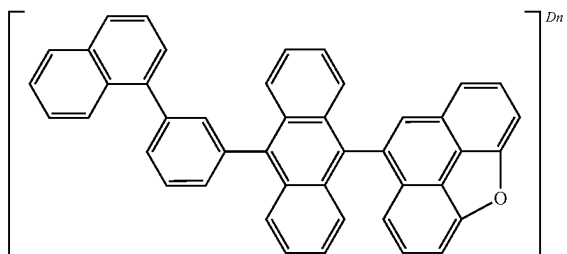


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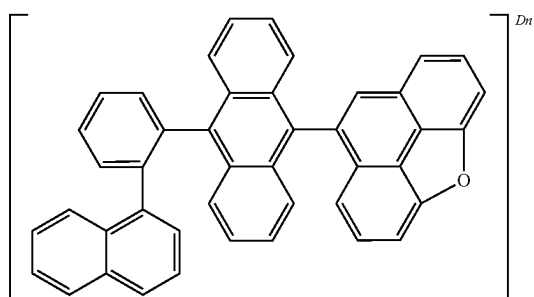


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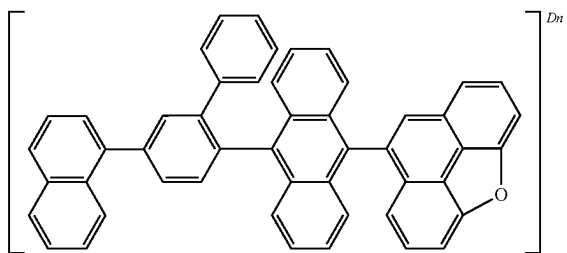
H2-517



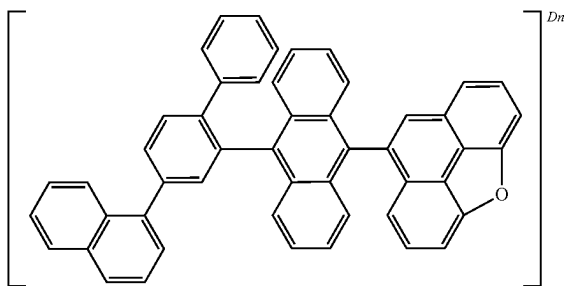
H2-518



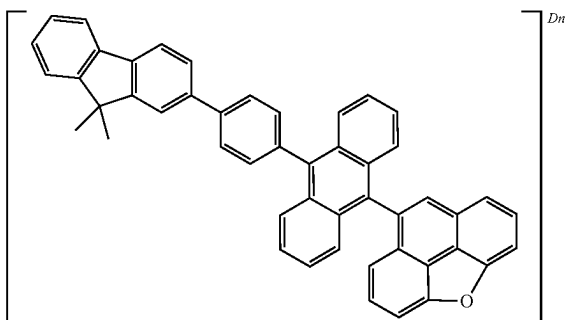
H2-519



H2-520

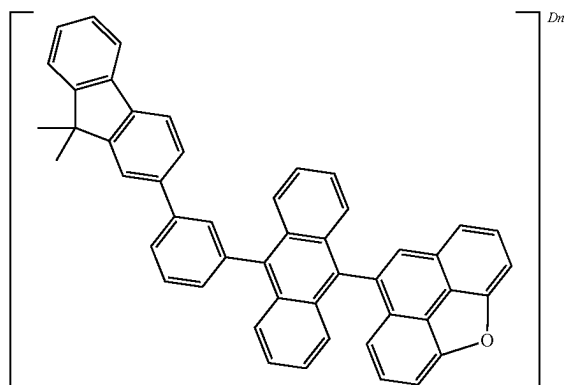


H2-521

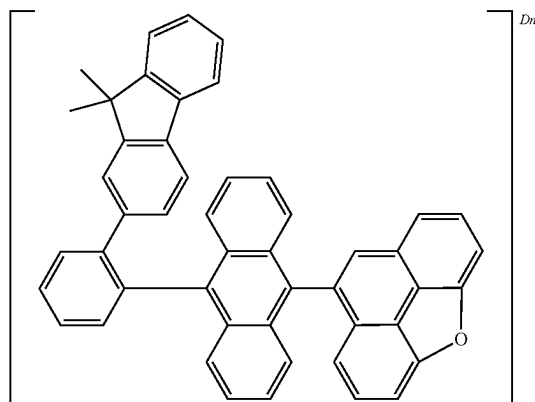


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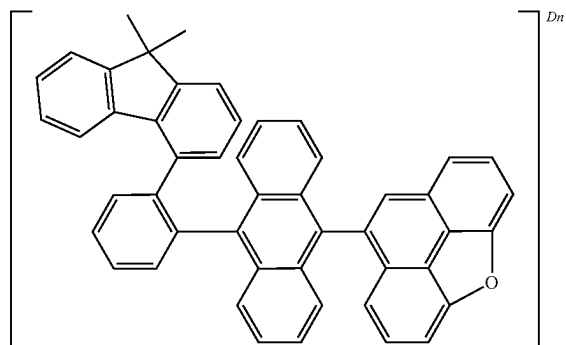
H2-522



H2-523

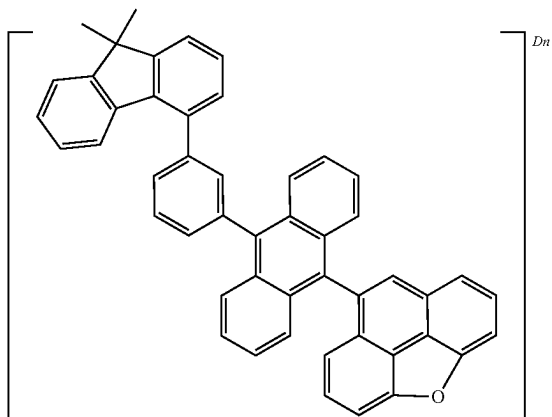


H2-524



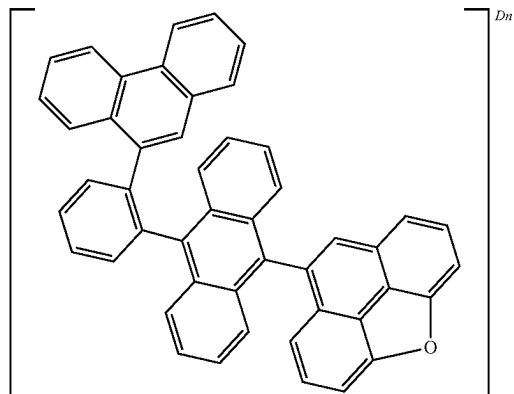
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H2-525

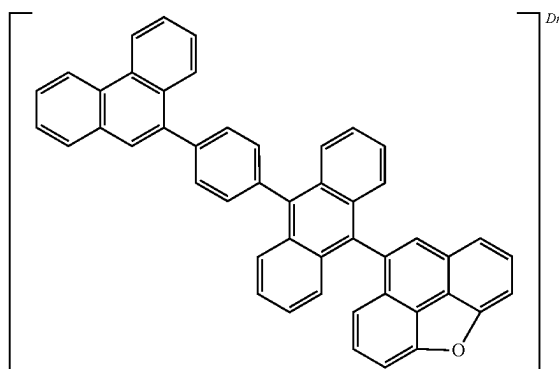


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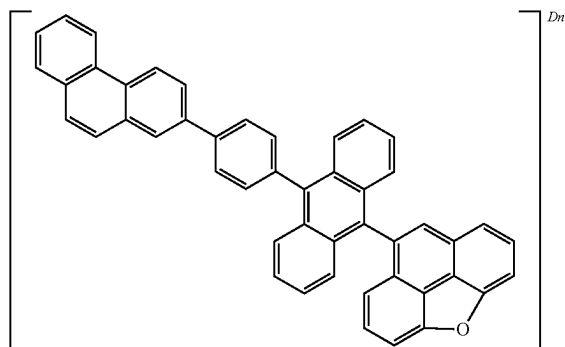
H2-528



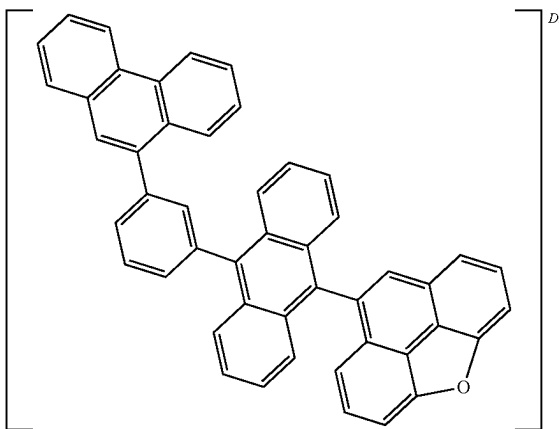
H2-526



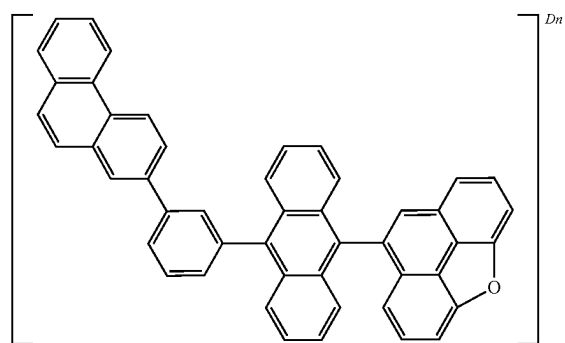
H2-529



H2-527

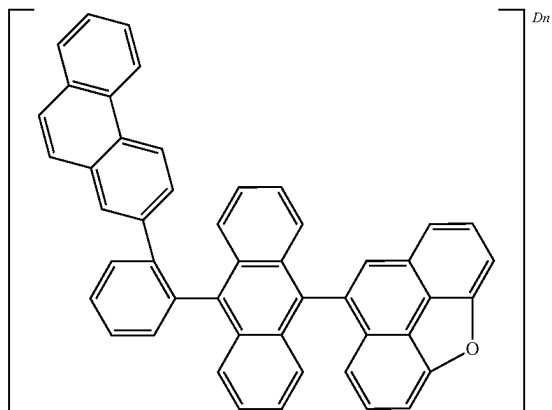


H2-530



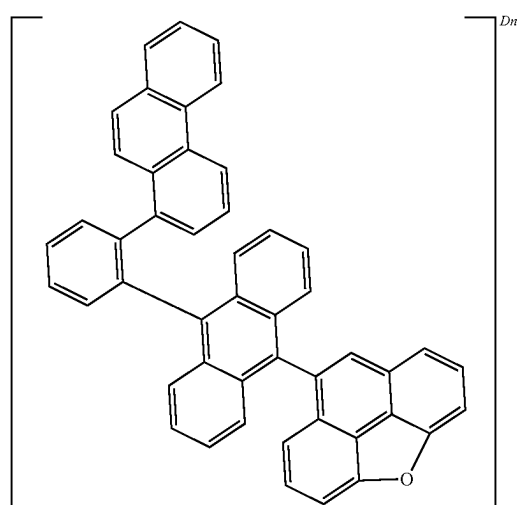
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H2-531

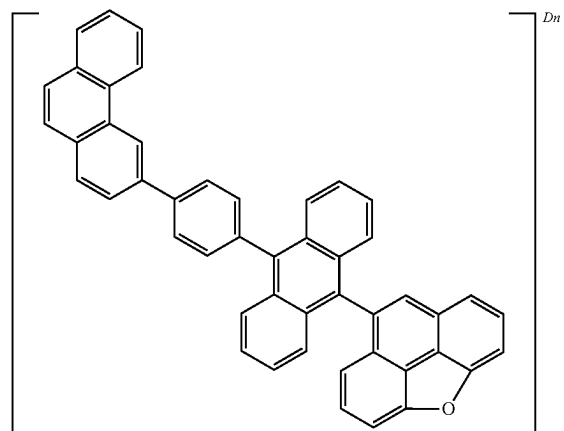
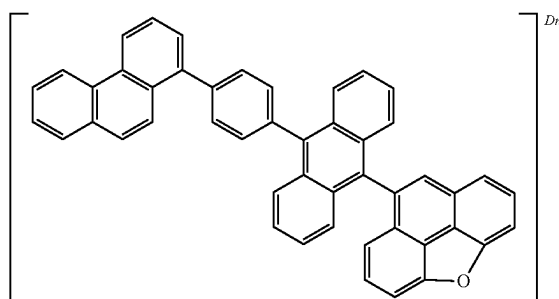


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H2-534

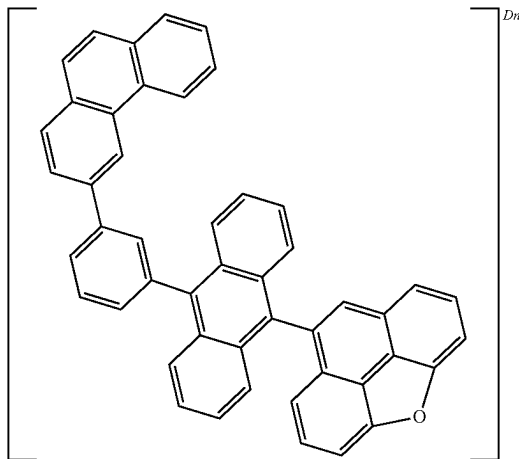
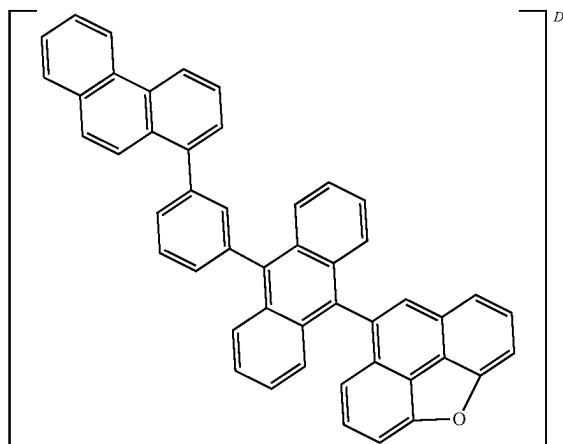


H2-532



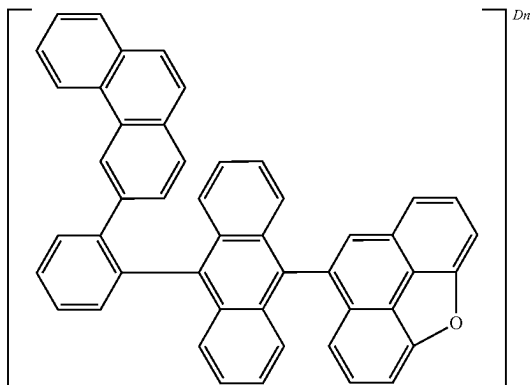
H2-536

H2-533



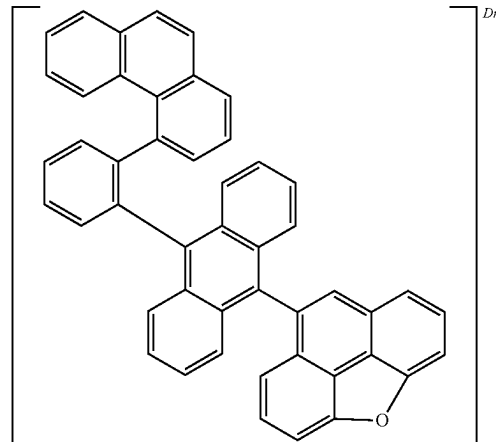
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H2-537



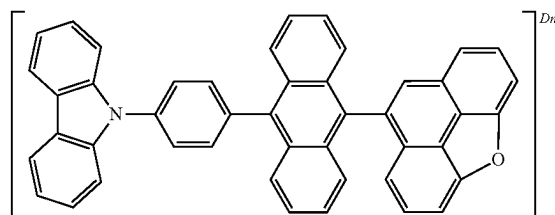
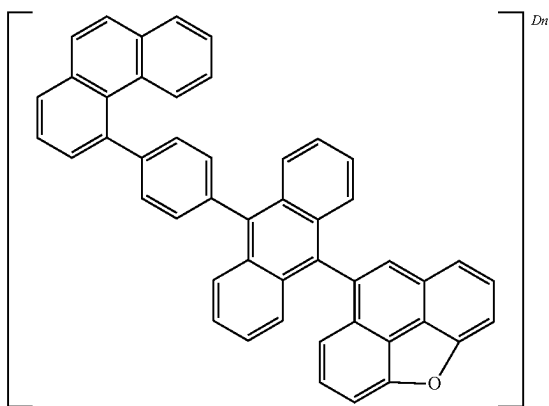
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H2-540

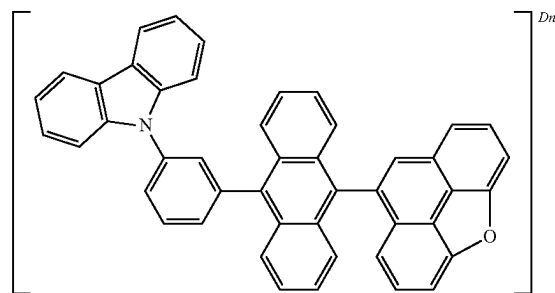


H2-541

H2-538

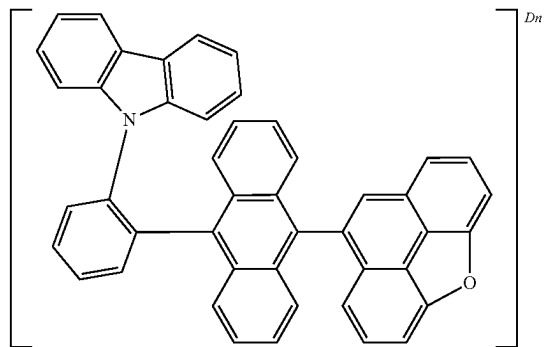
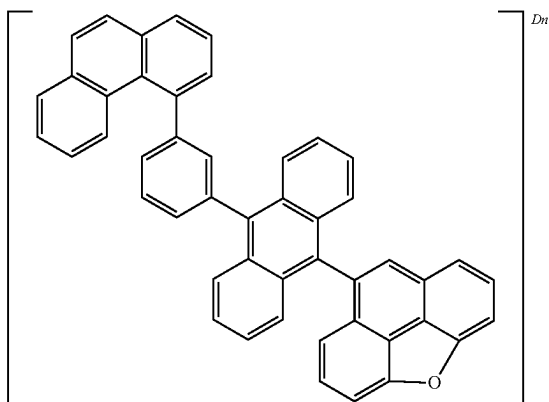


H2-542



H2-543

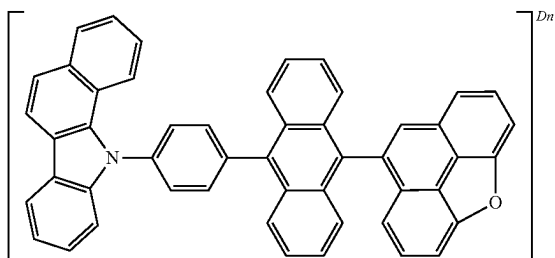
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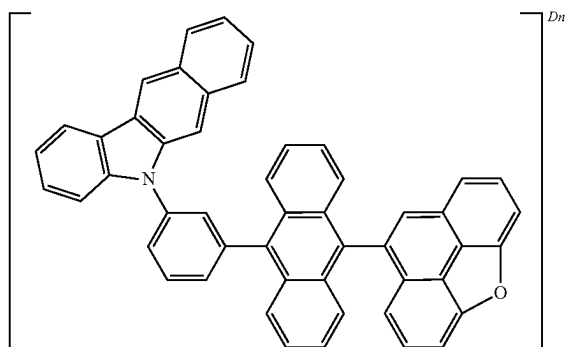
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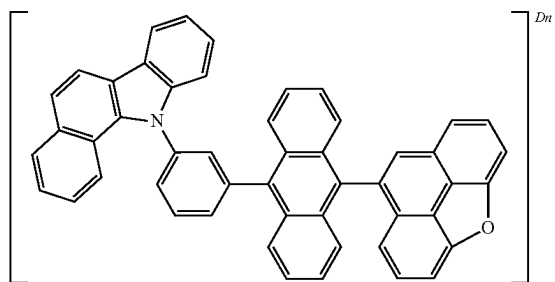
H2-544



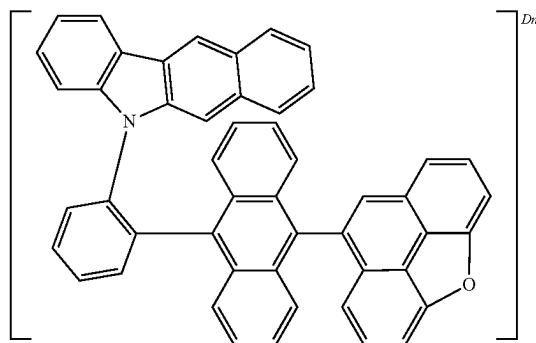
H2-548



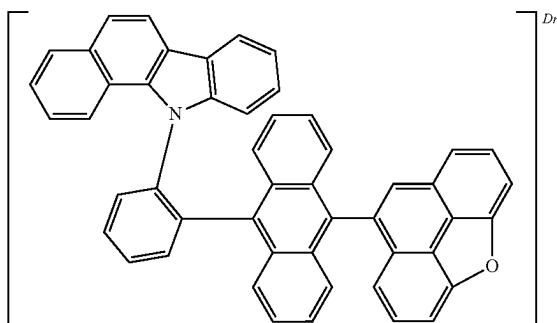
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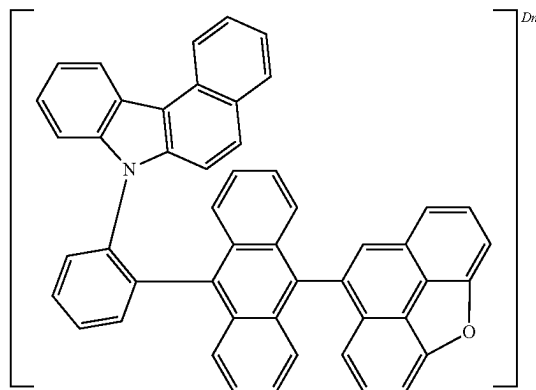
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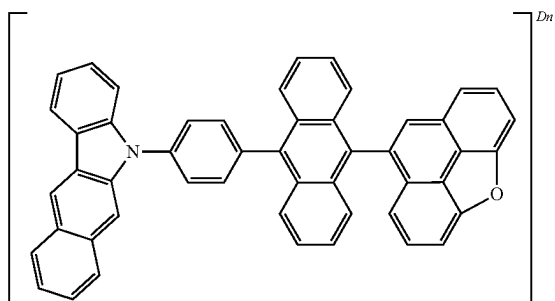
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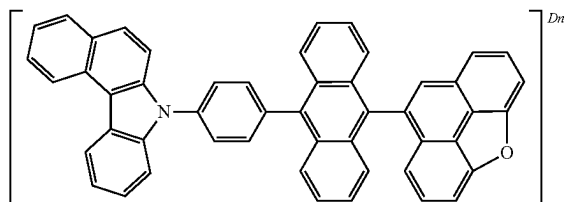
H2-550



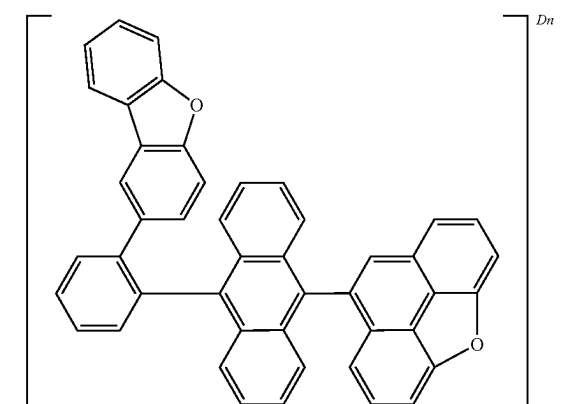
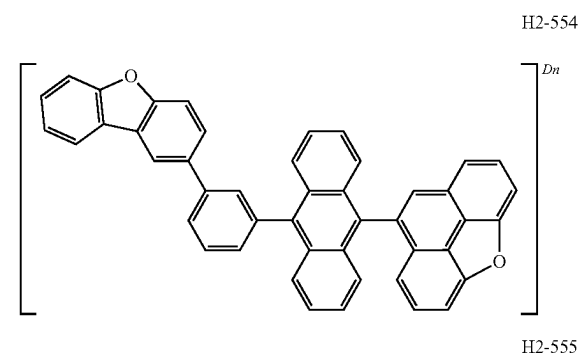
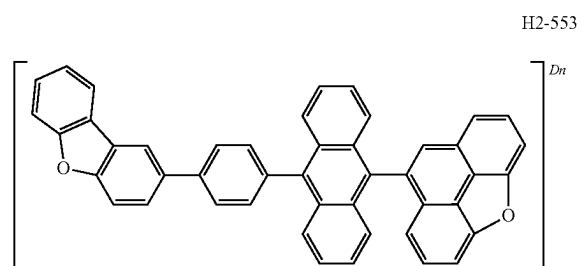
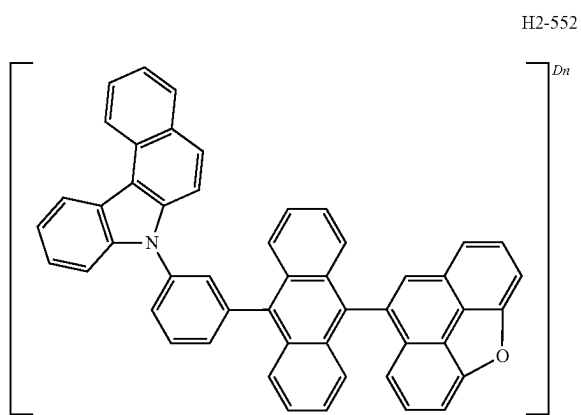
H2-547



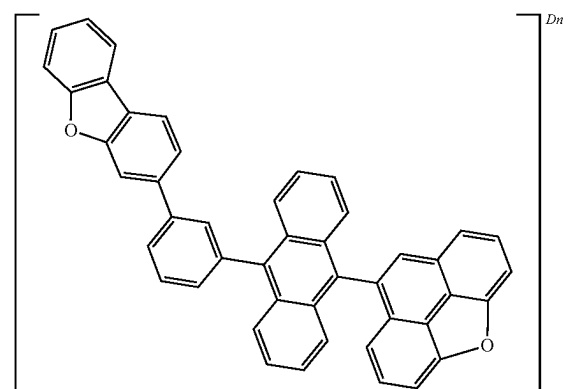
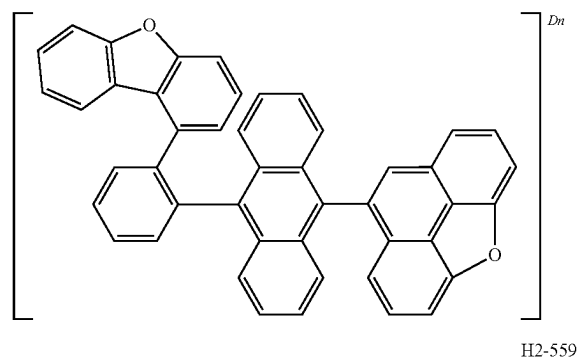
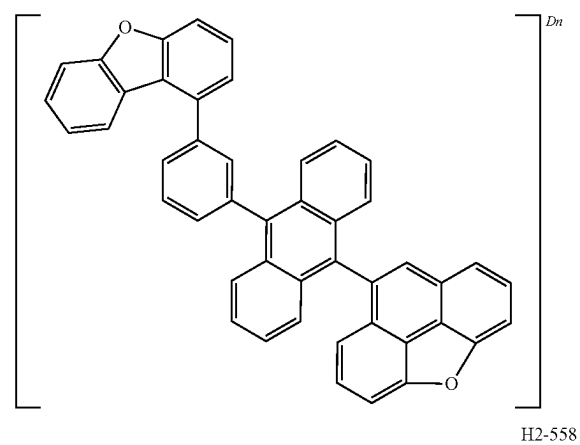
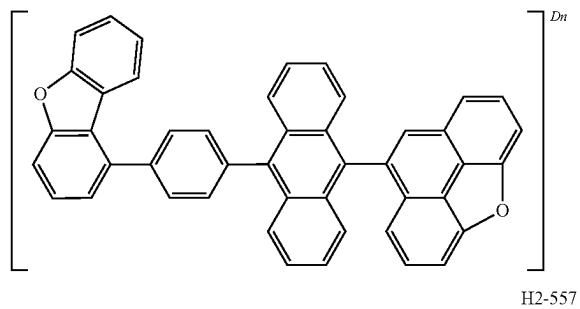
H2-551



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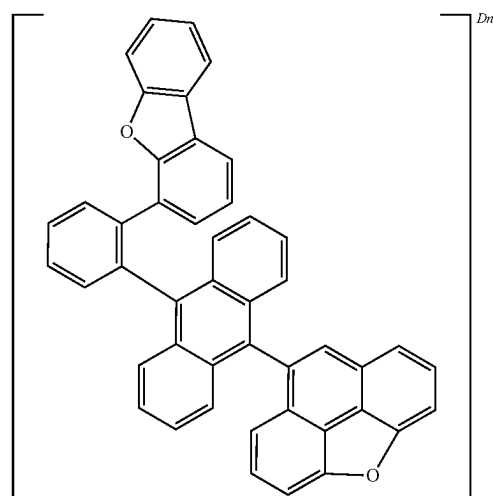


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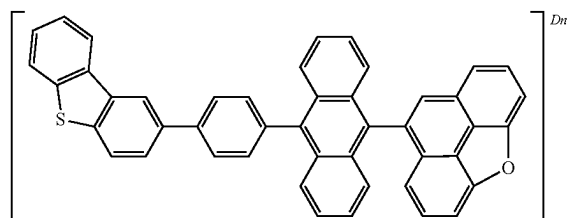


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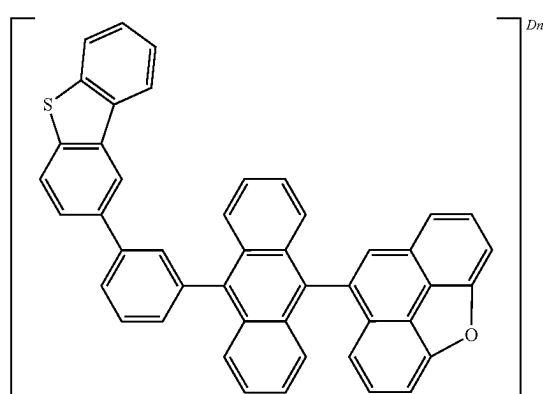
H2-563



H2-564

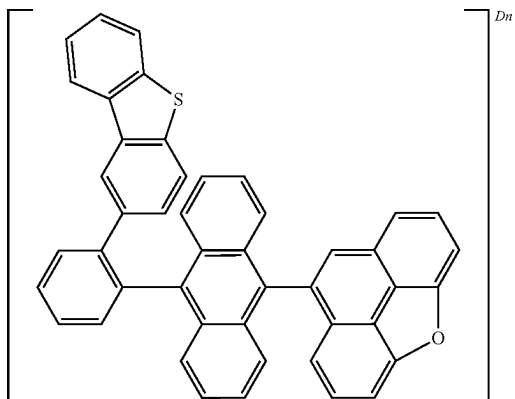


H2-565

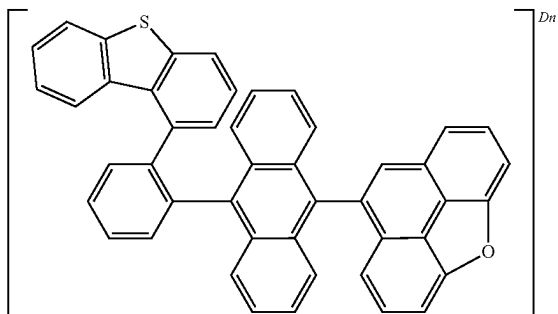


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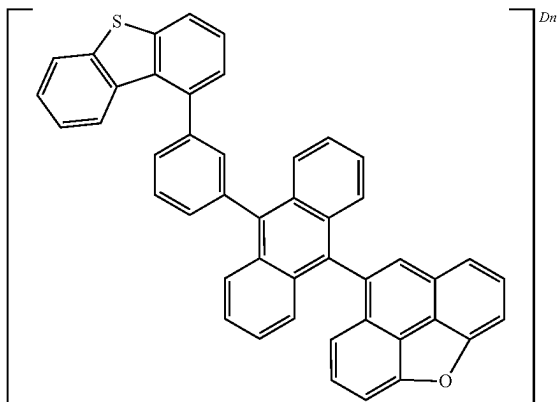
H2-566



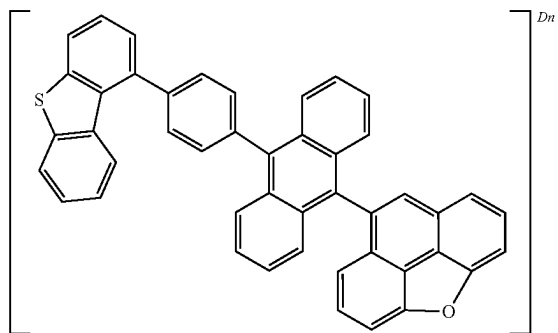
H2-567



H2-568

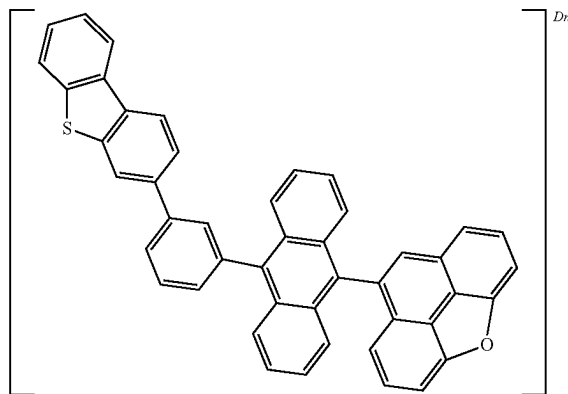


H2-569

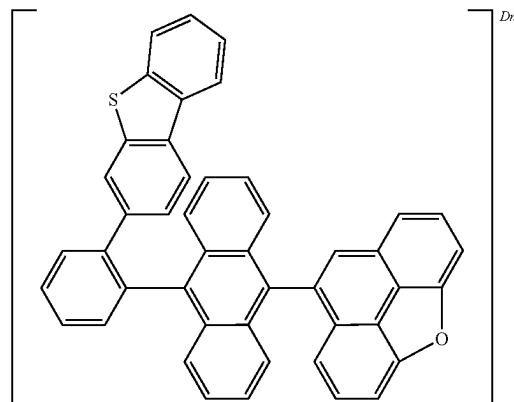


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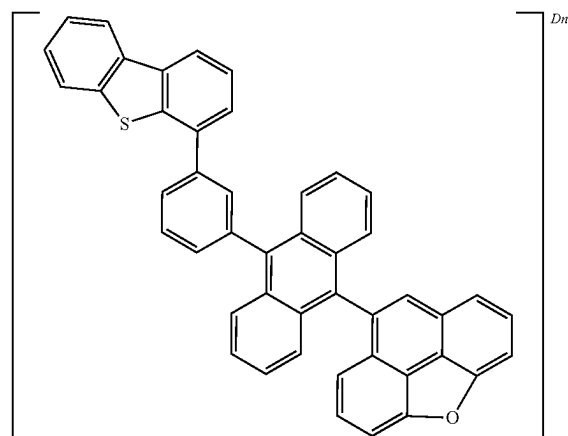
H2-570



H2-571

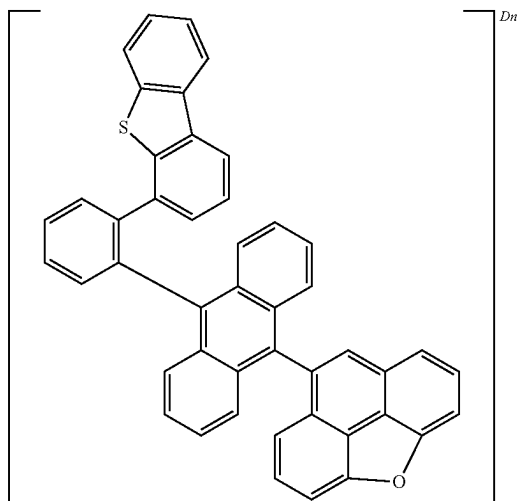


H2-572

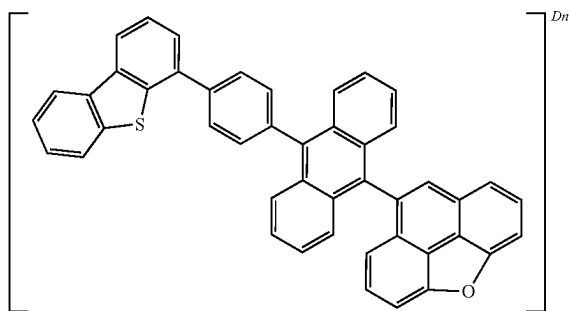


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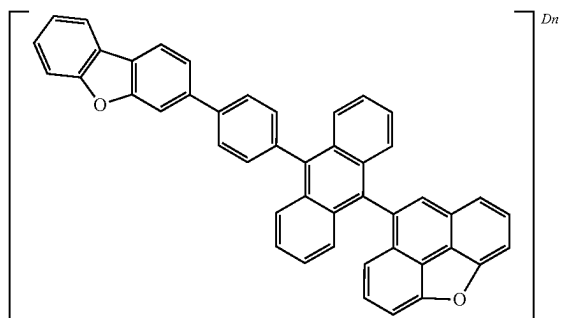
H2-573



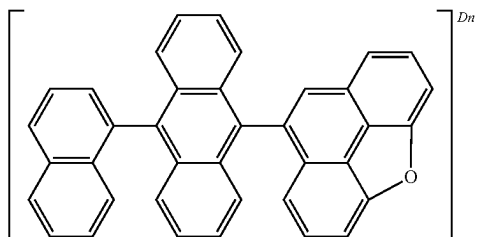
H2-574



H2-575

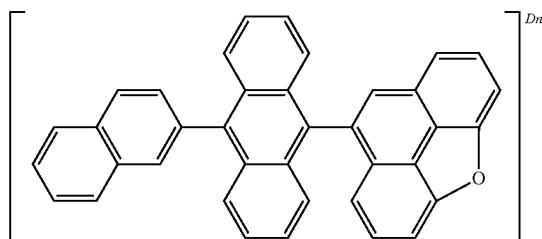


H2-576

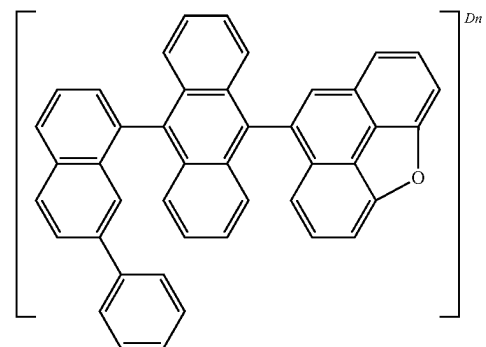


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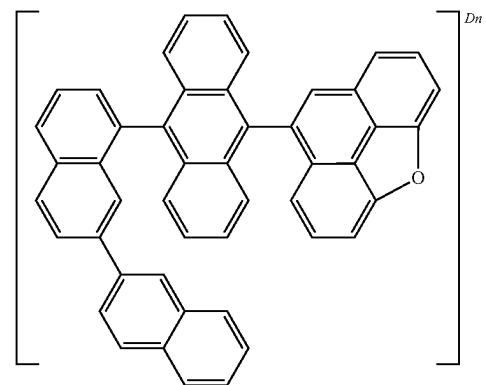
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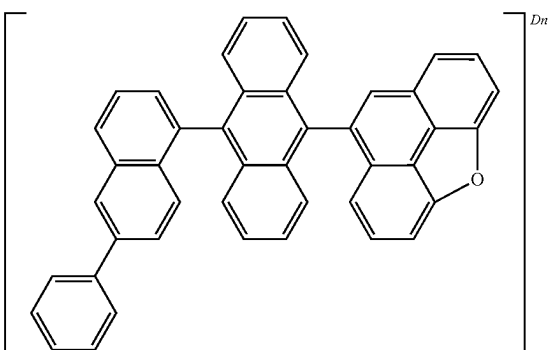
H2-578



H2-579

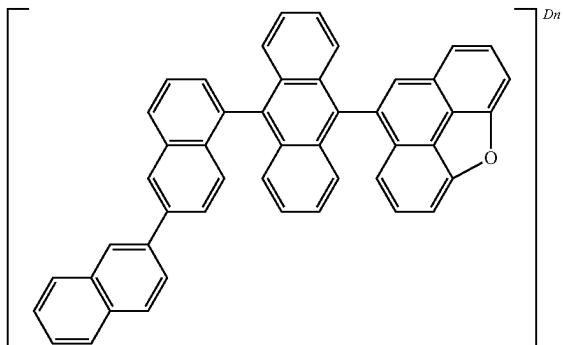


H2-580

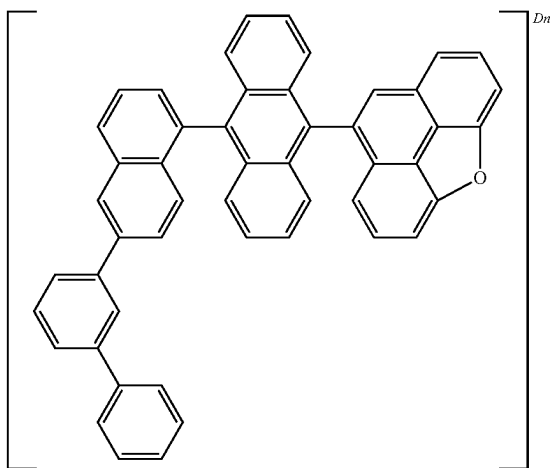


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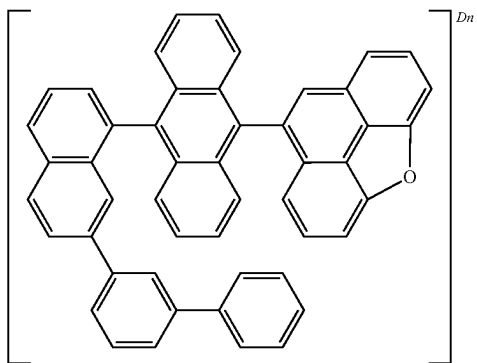
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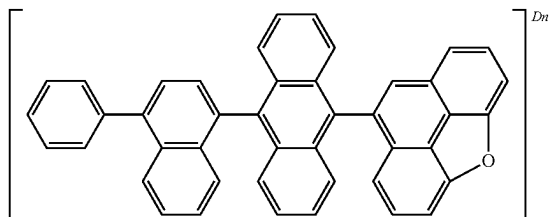
H2-582



H2-583

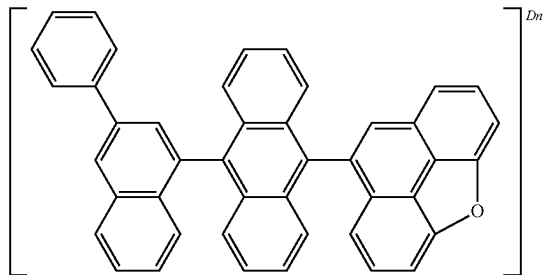


H2-584

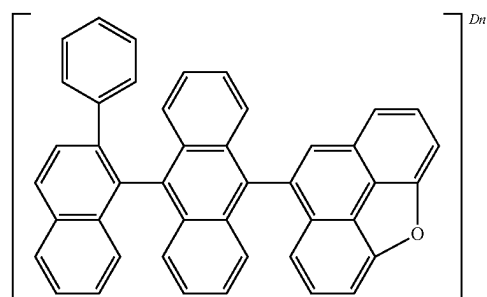


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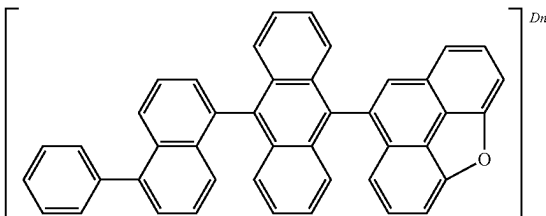
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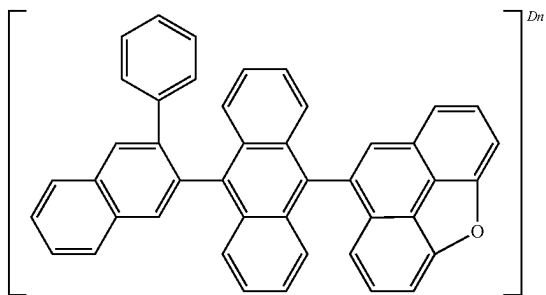
H2-586



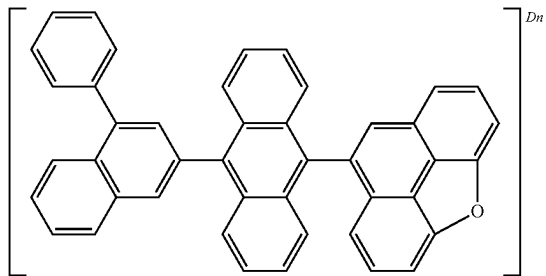
H2-587



H2-588

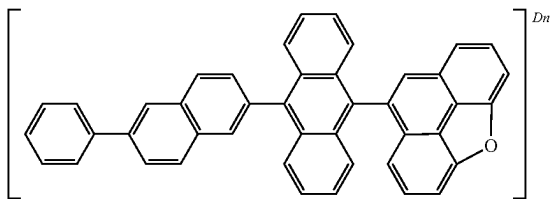


H2-589

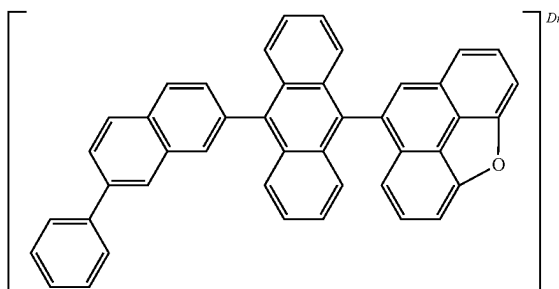


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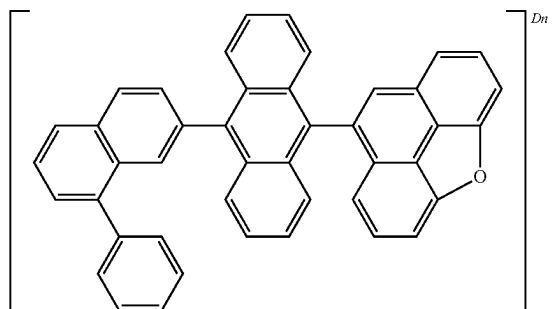
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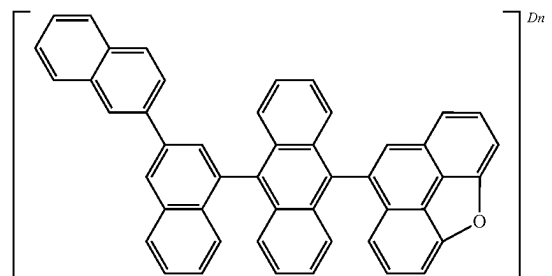
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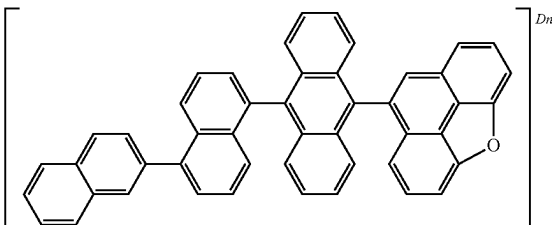
H2-592



H2-593

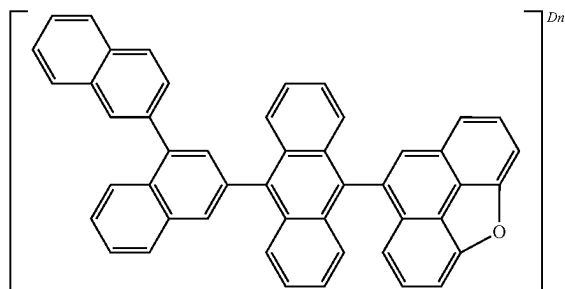


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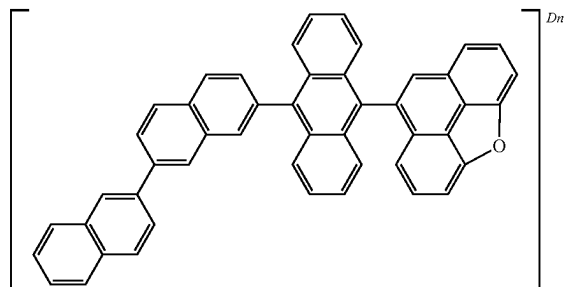


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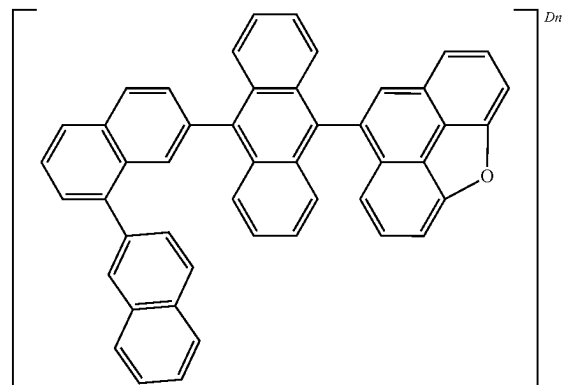
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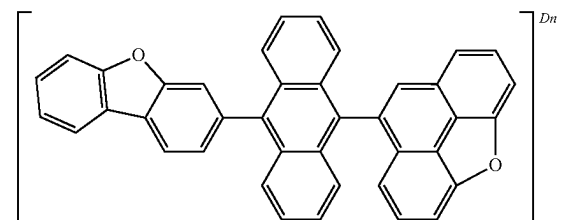
H2-596



H2-597

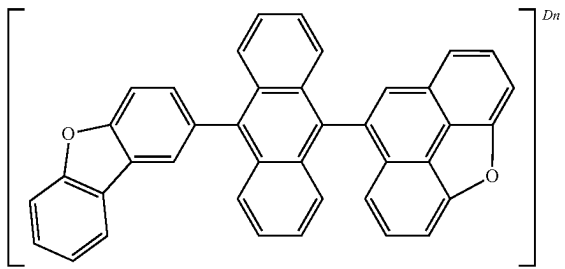


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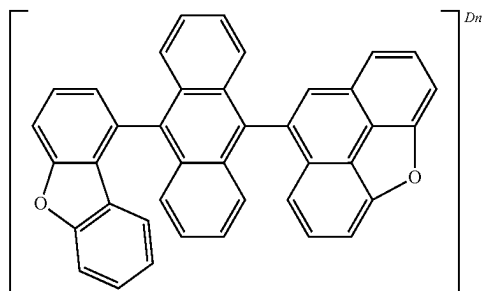


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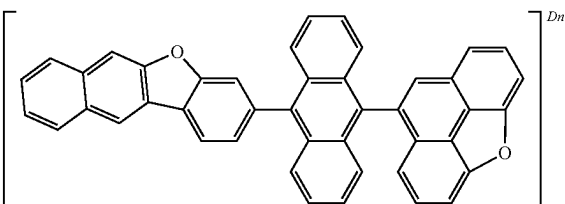
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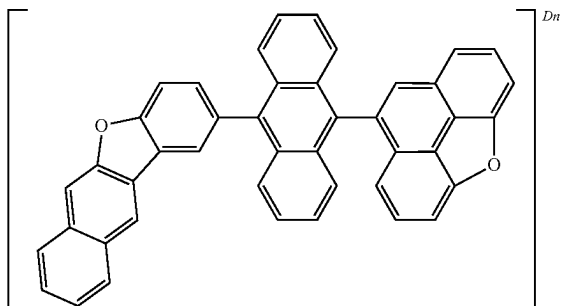
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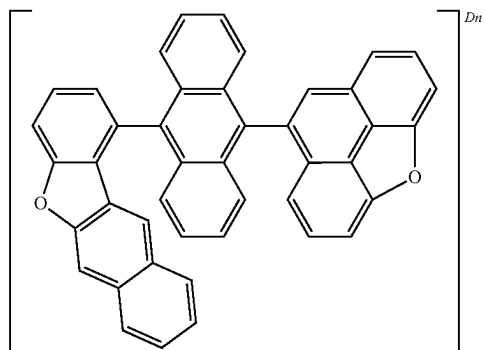
H2-601



H2-602

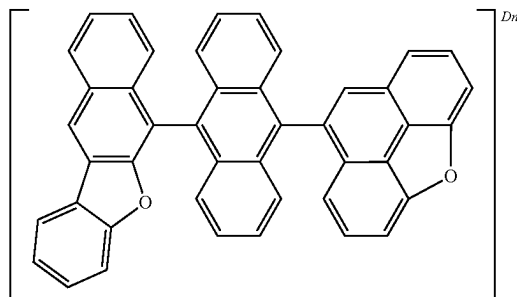


H2-603

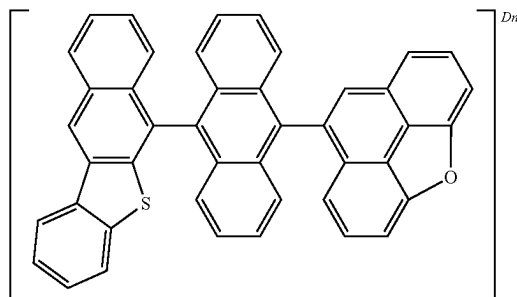


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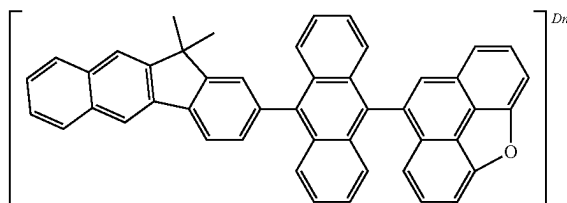
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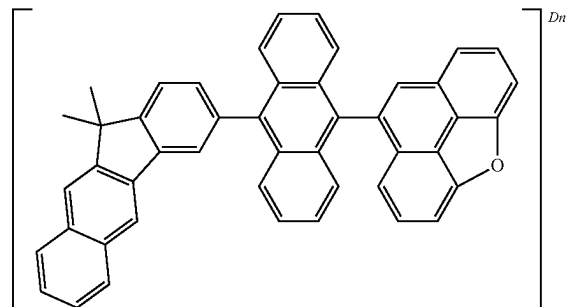
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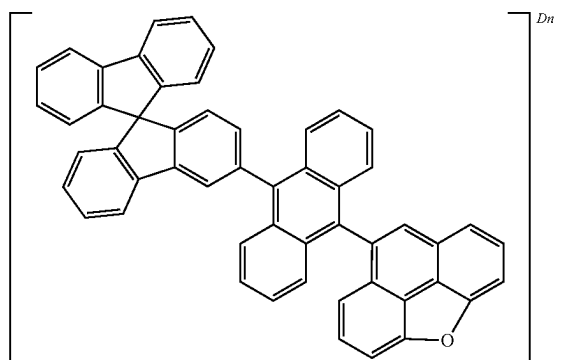
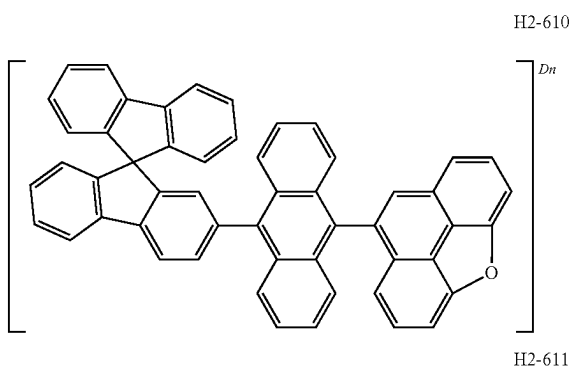
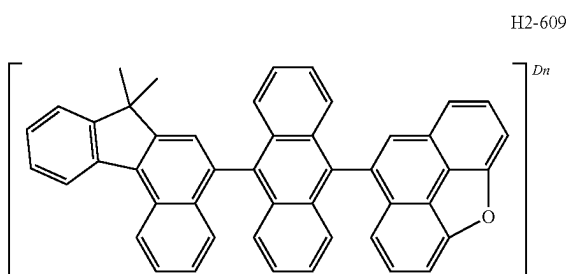
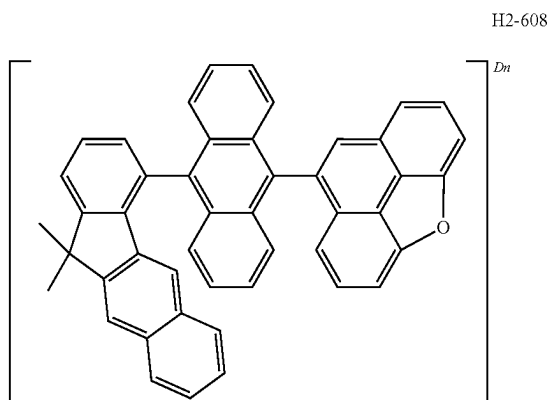
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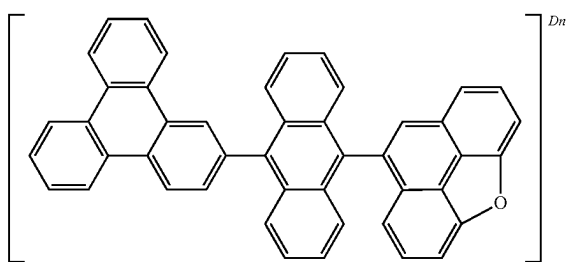
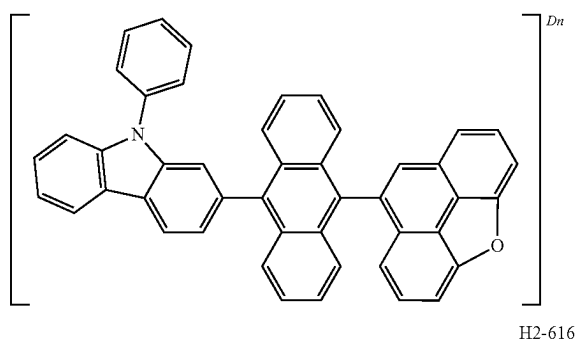
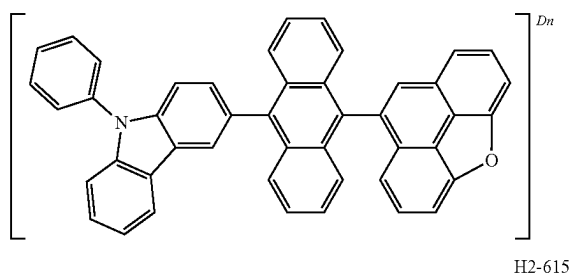
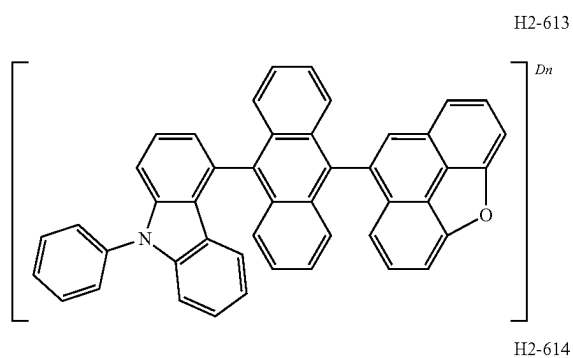
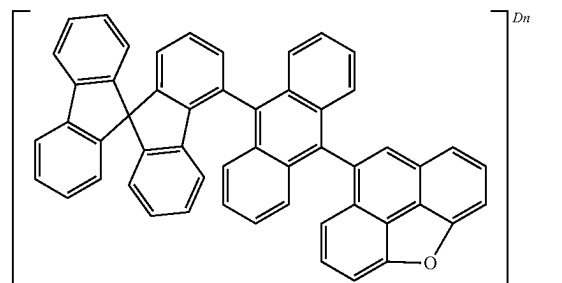
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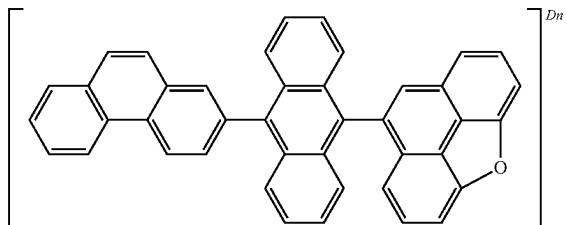


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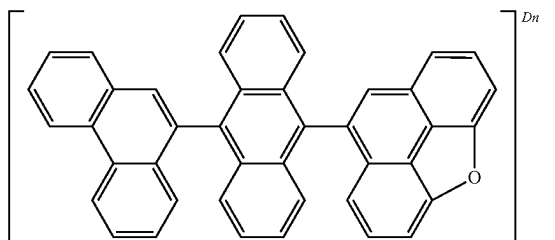


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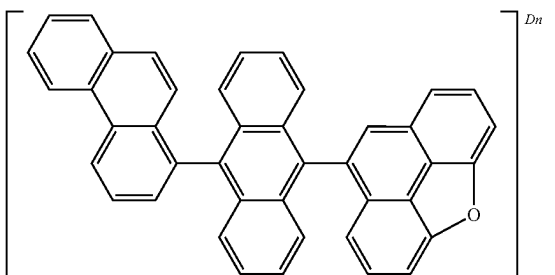
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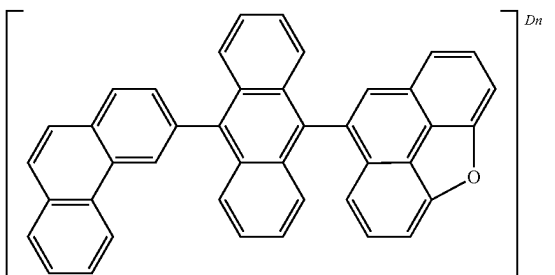
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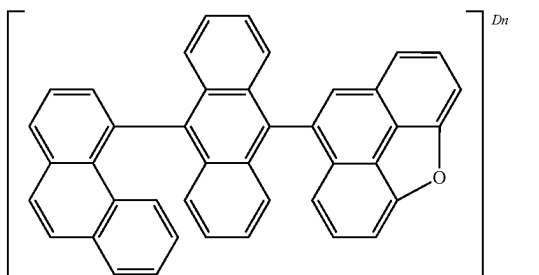
H2-619



H2-620

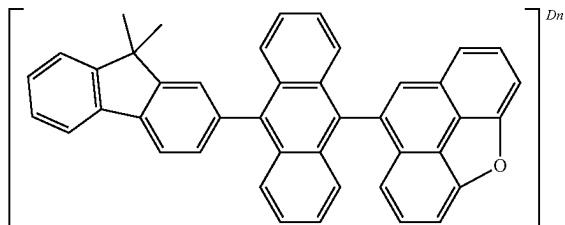


H2-621

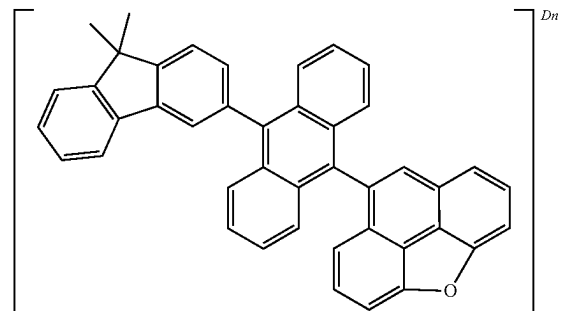


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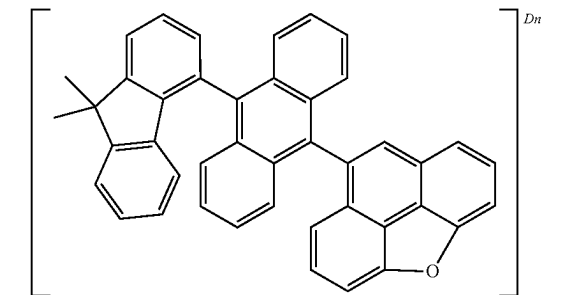
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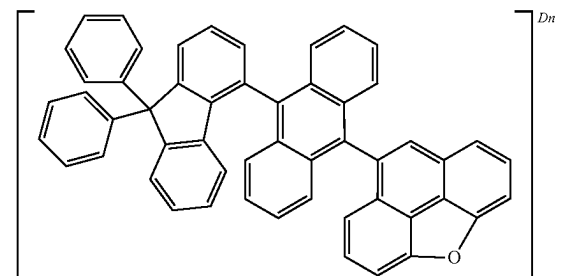
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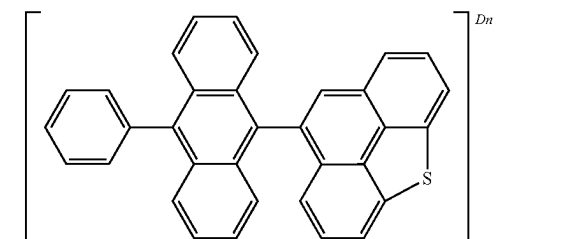
H2-624



H2-625

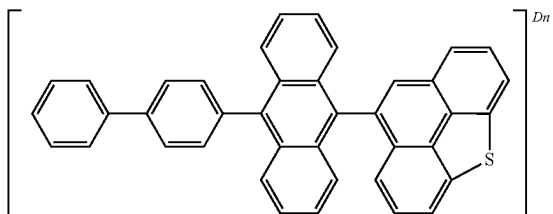


H2-626

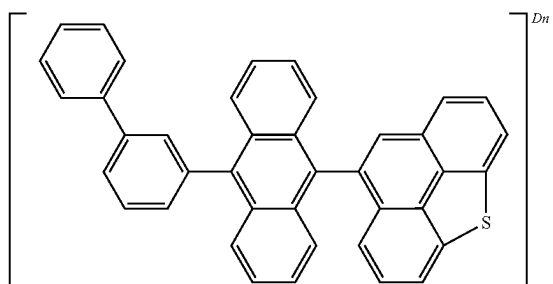


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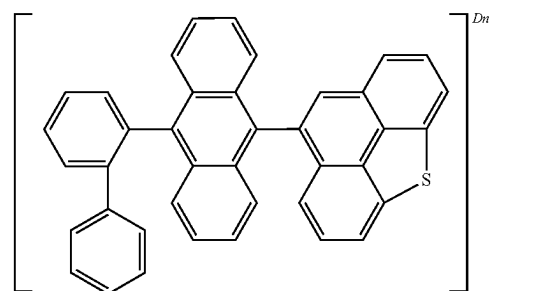
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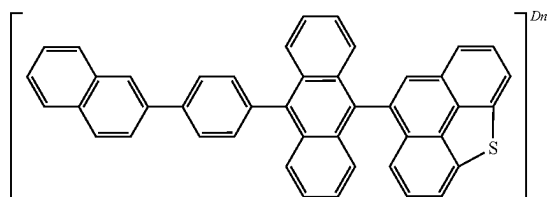
H2-628



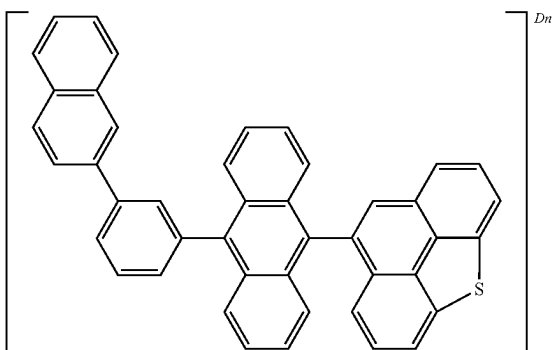
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H2-630

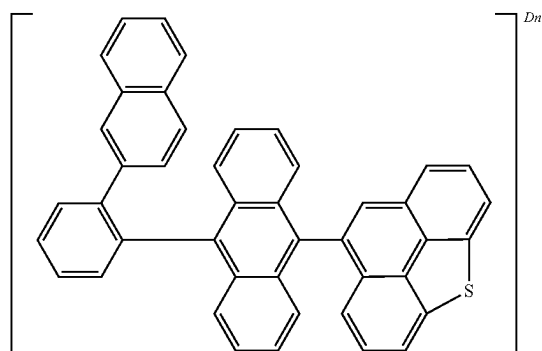


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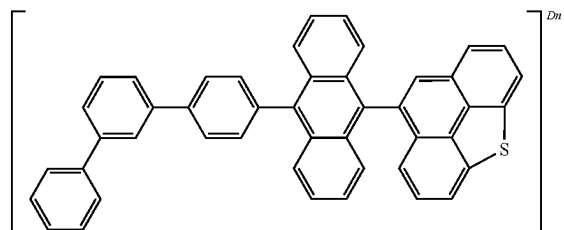


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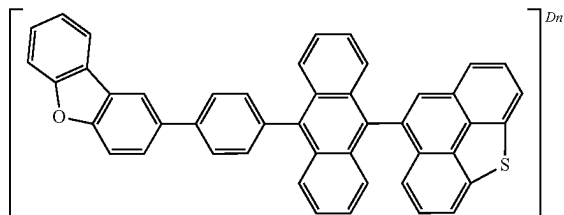
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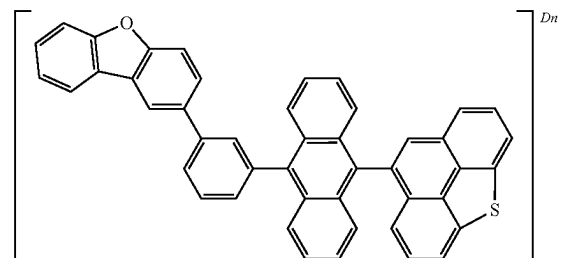
H2-633



H2-634

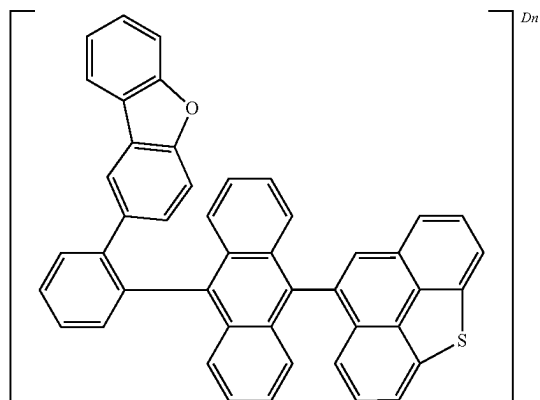


H2-635

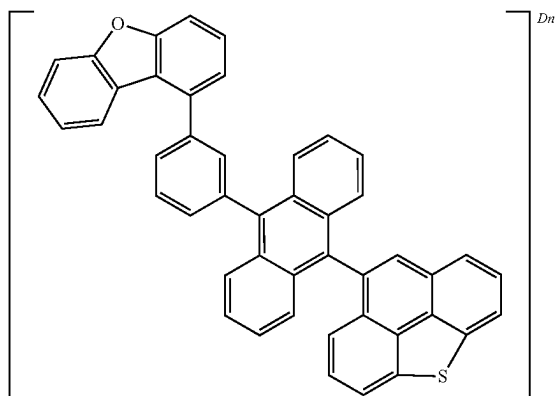


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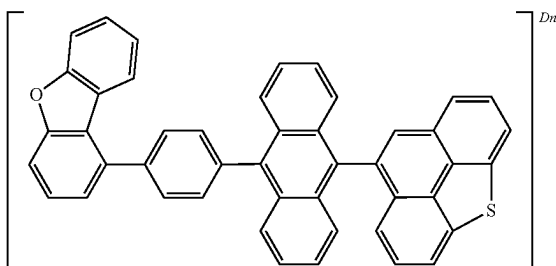
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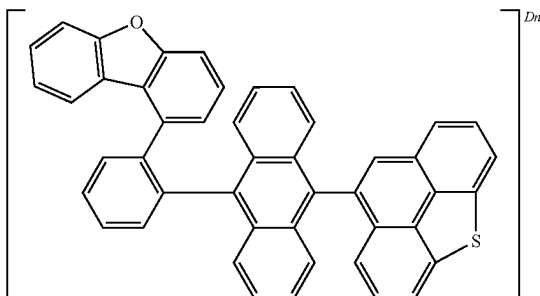
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H2-638

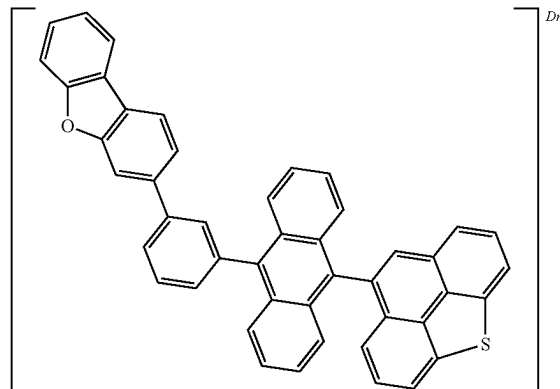


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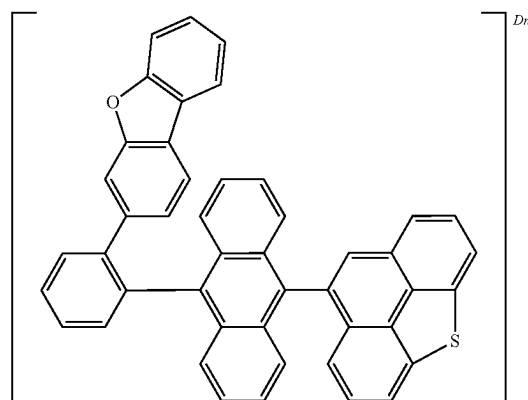


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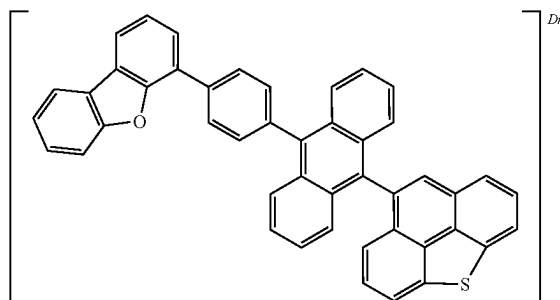
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H2-641

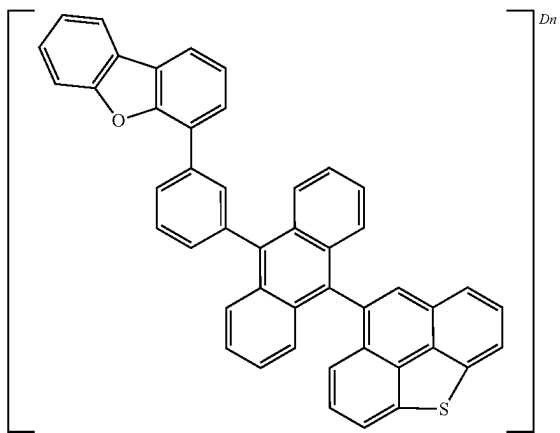


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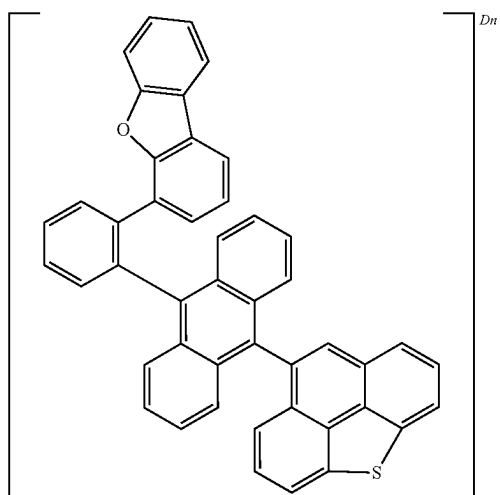


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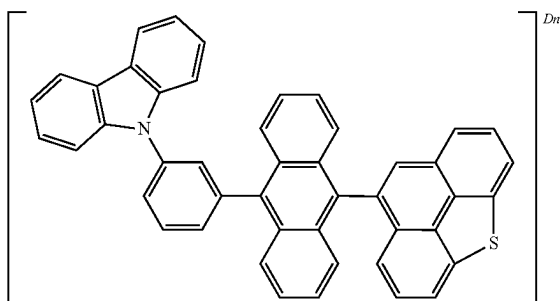
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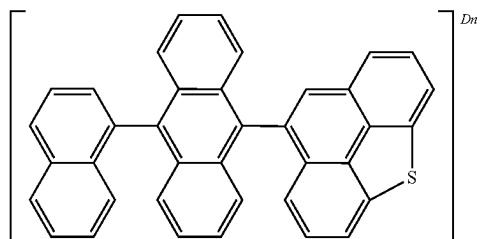
H2-644



H2-645

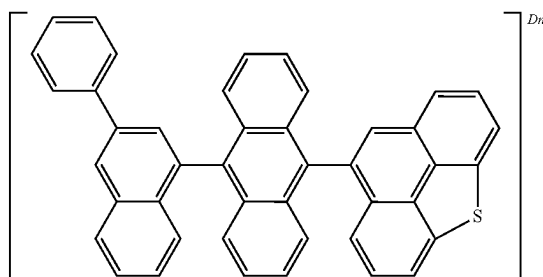


H2-646

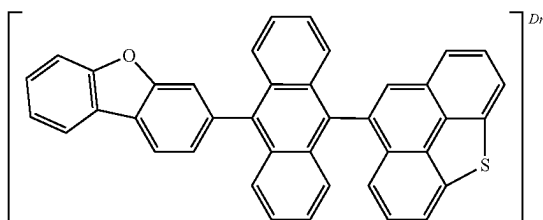


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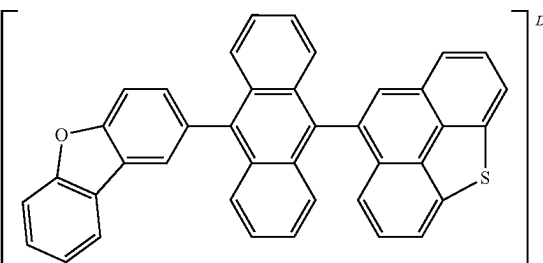
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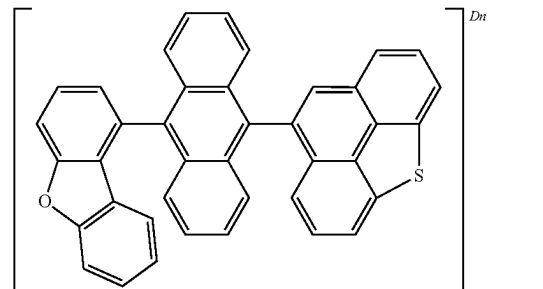
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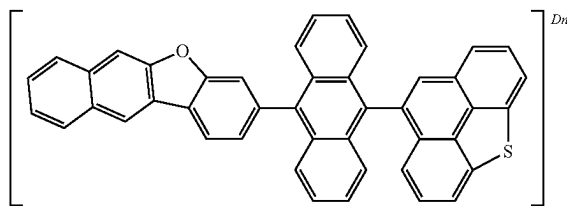
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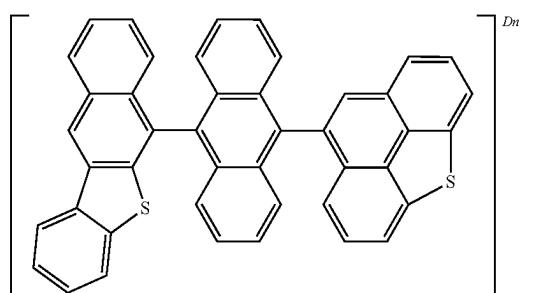
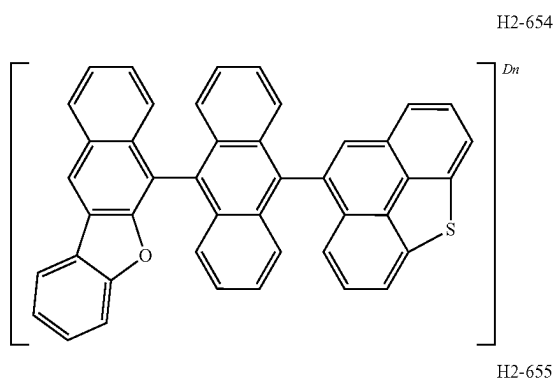
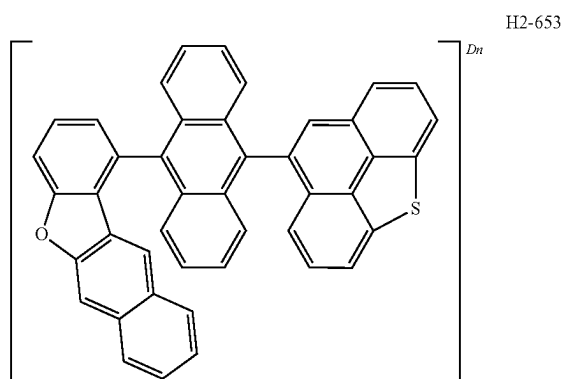
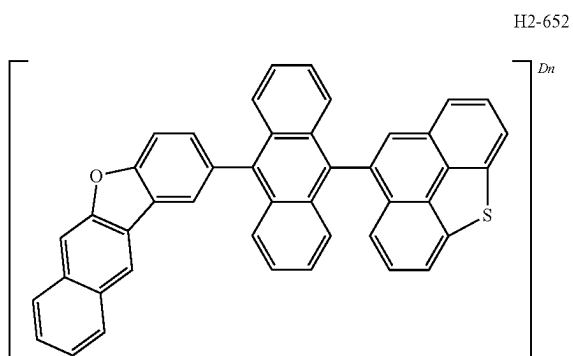
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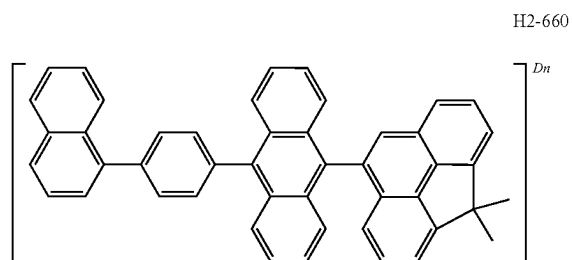
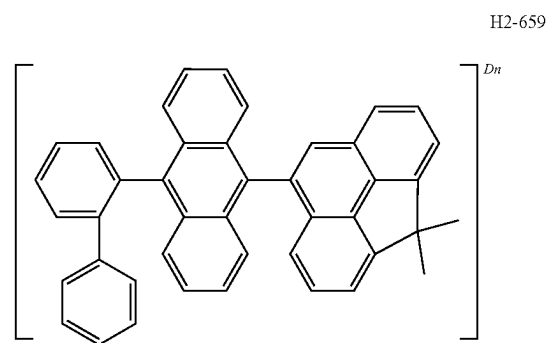
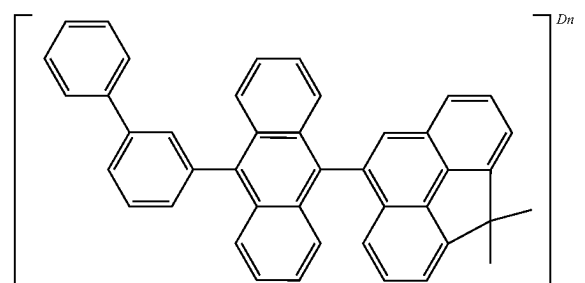
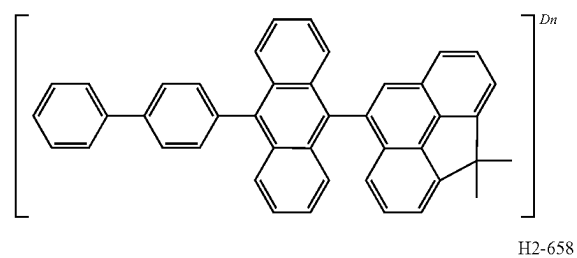
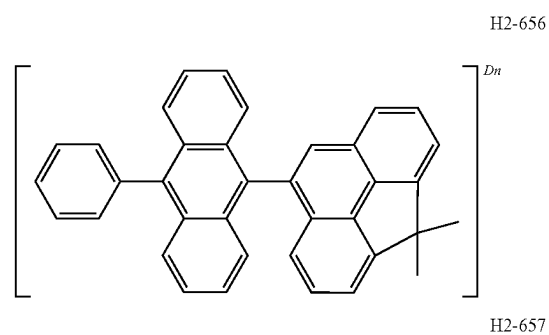
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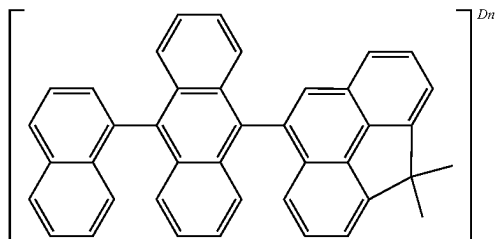


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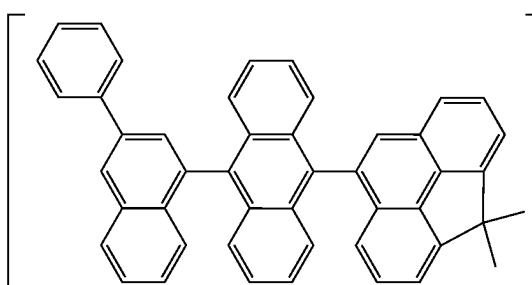


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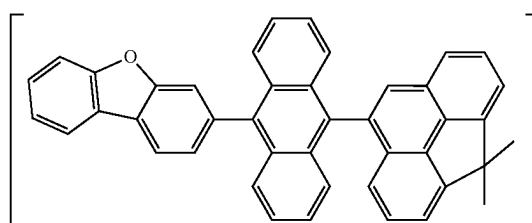
H2-661



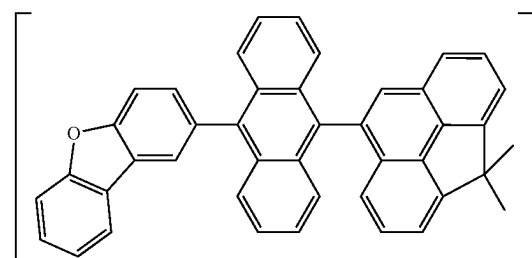
H2-662



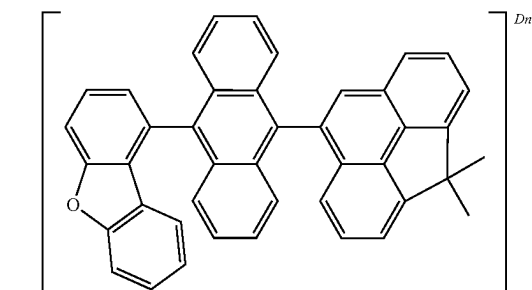
H2-663



H2-664

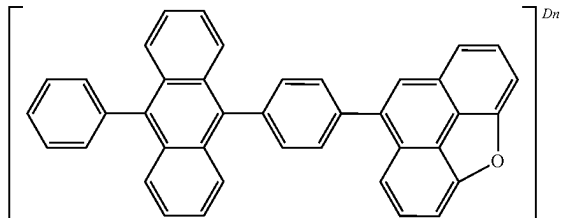


H2-665

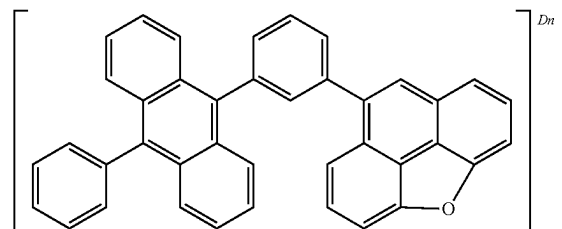


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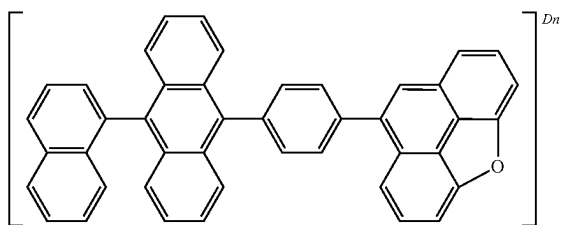
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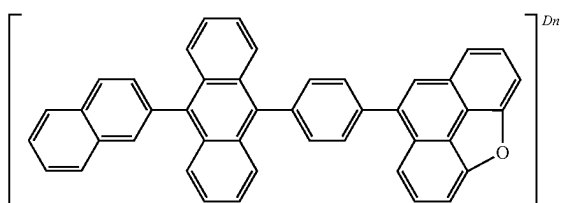
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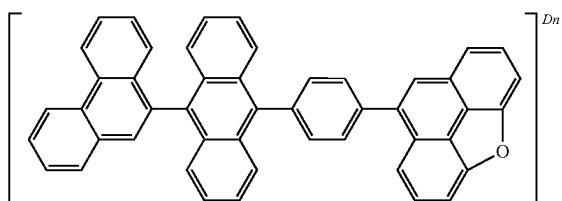
H2-668



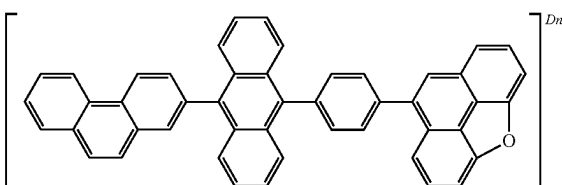
H2-669



H2-670

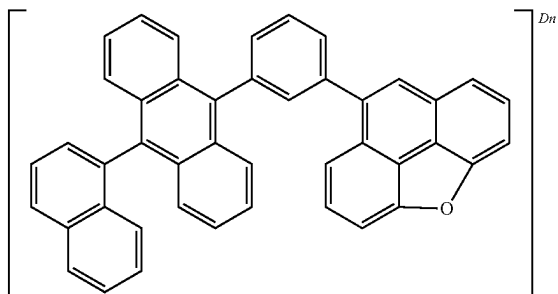


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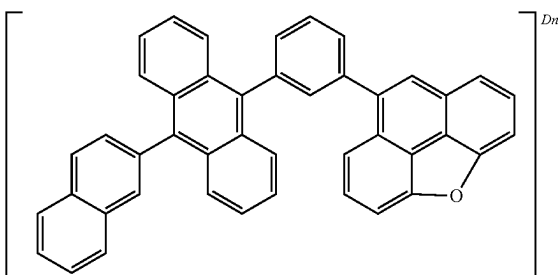


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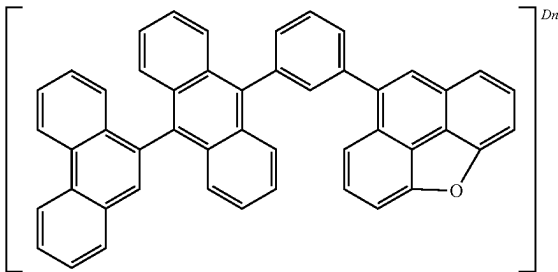
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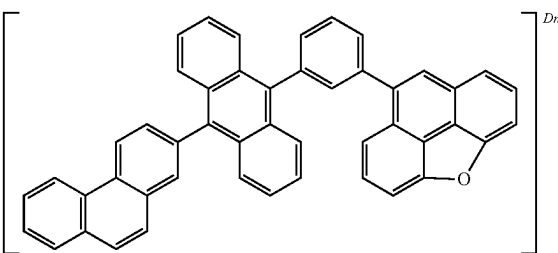
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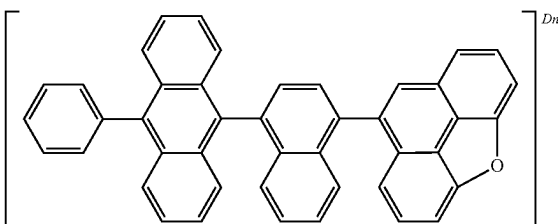
H2-674



H2-675

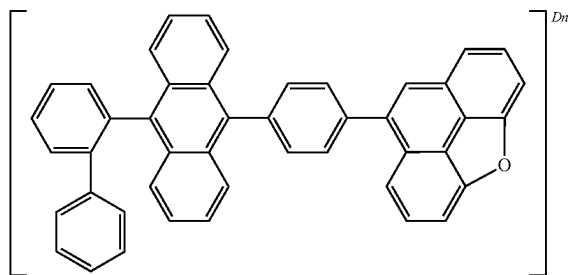


H2-676

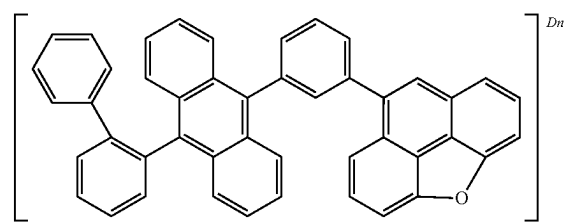


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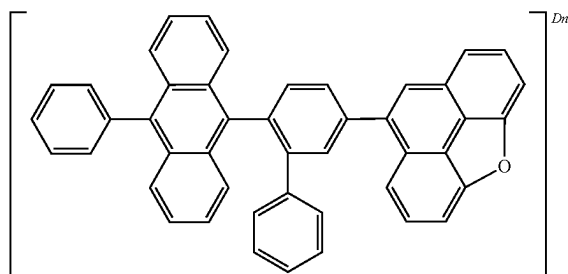
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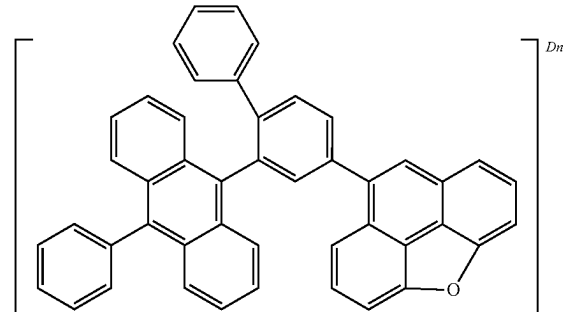
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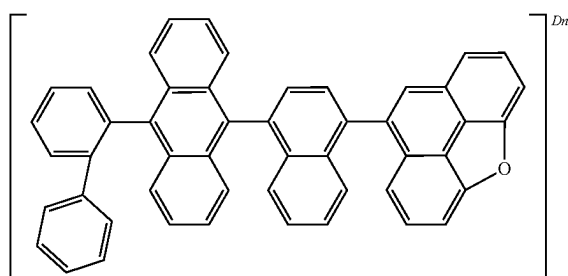
H2-679



H2-680

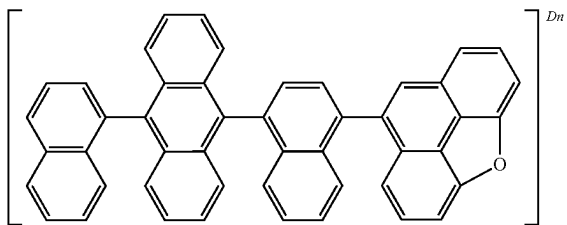


H2-681

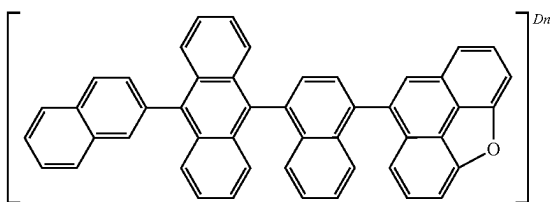


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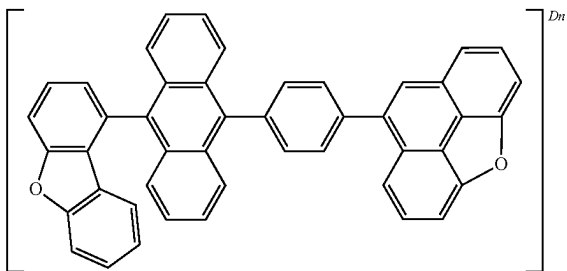
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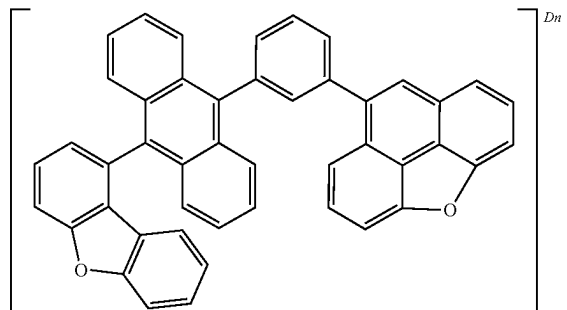
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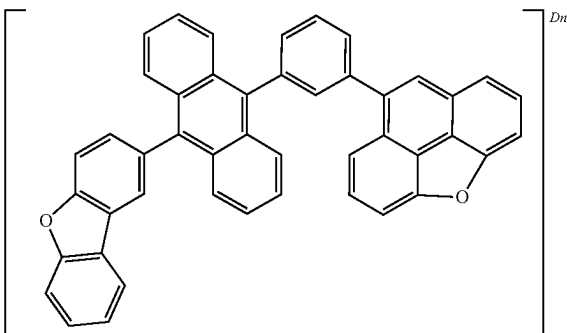
H2-684



H2-685

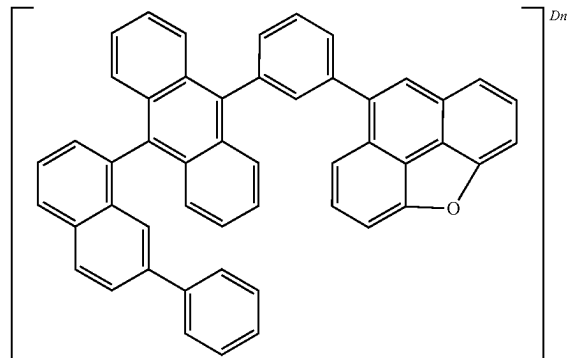


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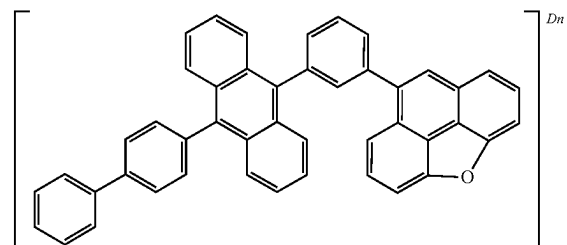


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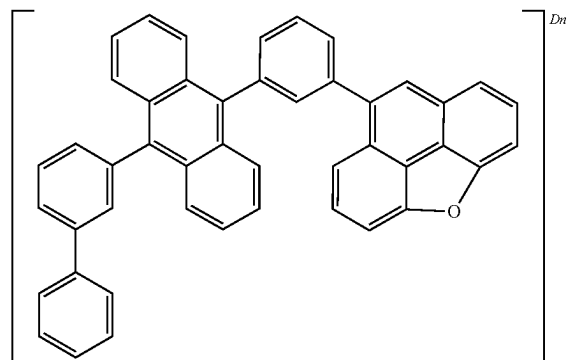
H2-687



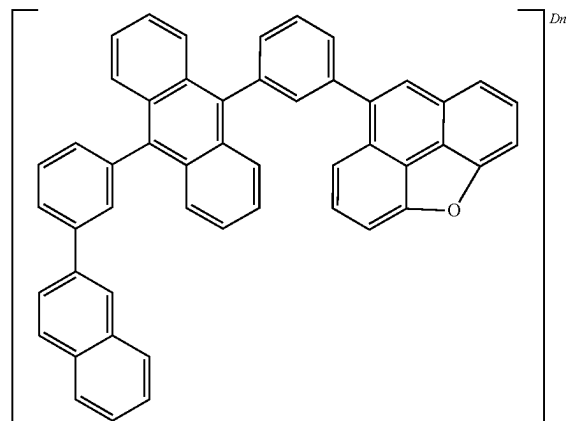
H2-688



H2-689

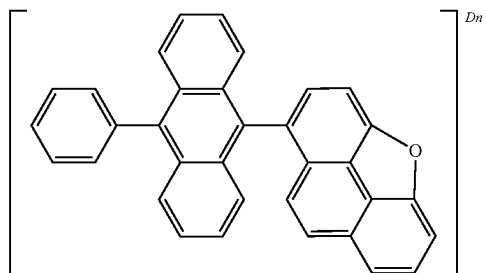


H2-690



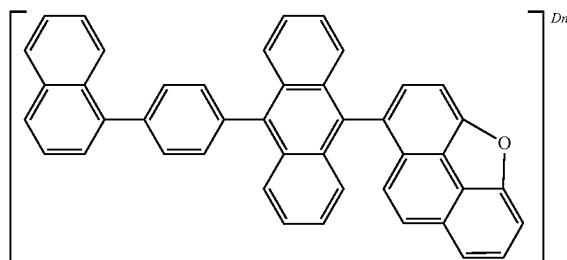
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H2-691



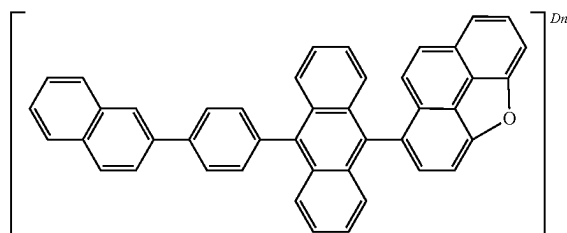
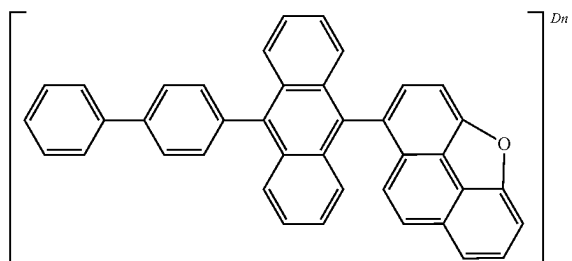
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H2-695



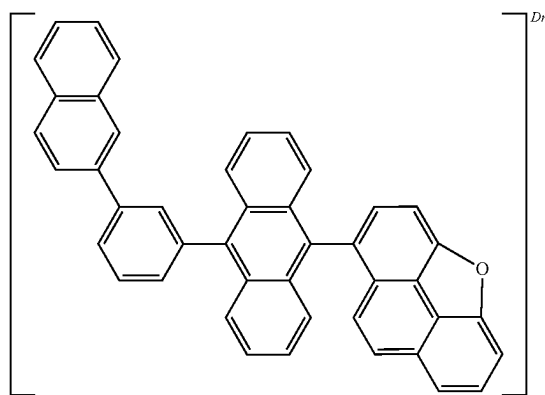
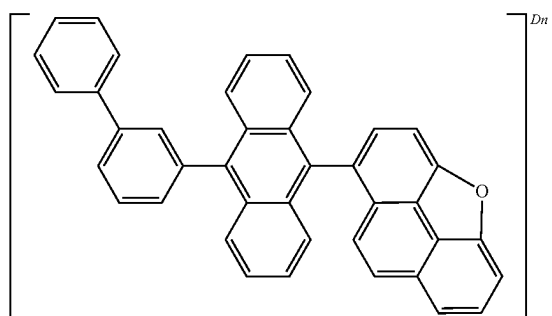
H2-696

H2-692



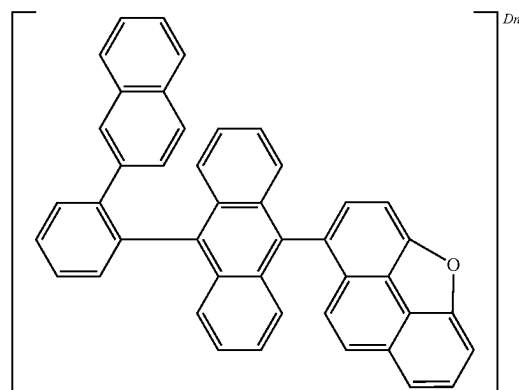
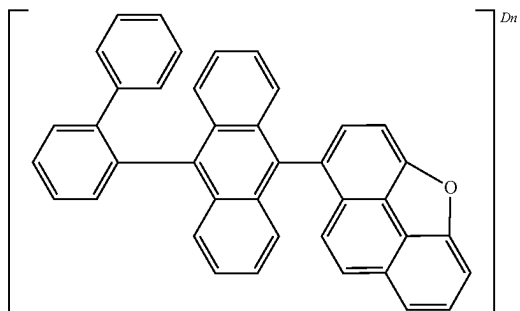
H2-697

H2-693



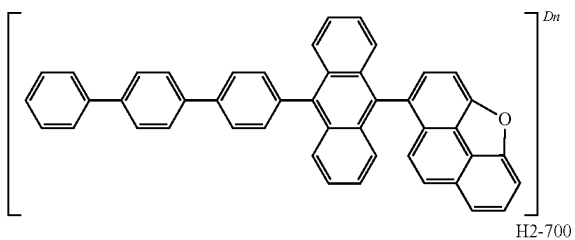
H2-698

H2-694

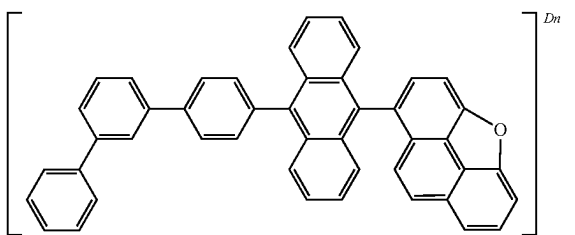


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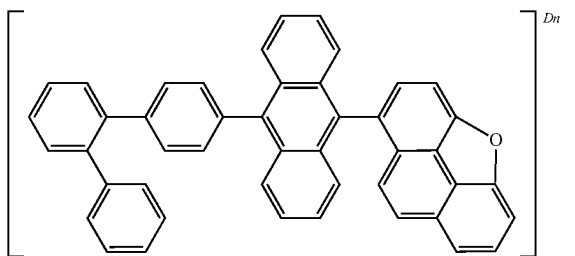
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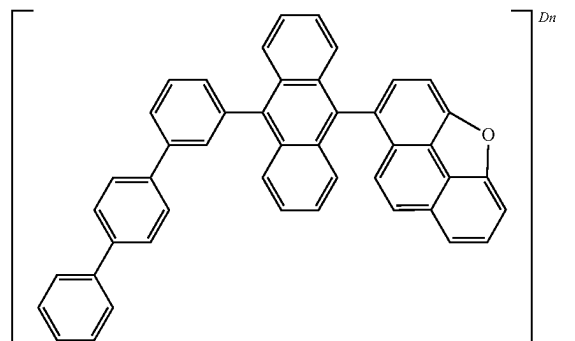
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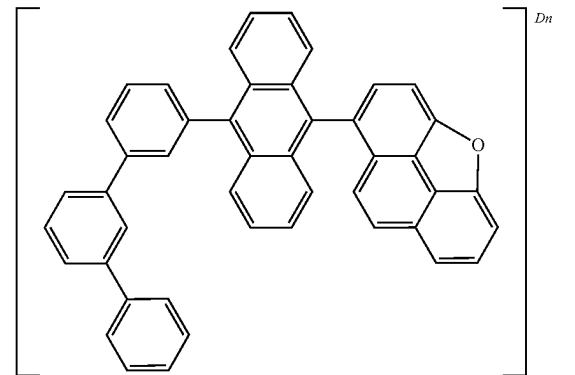
H2-701



H2-702

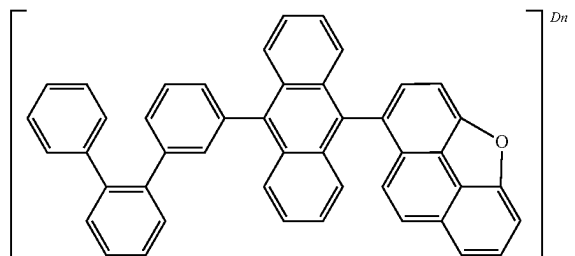


H2-703

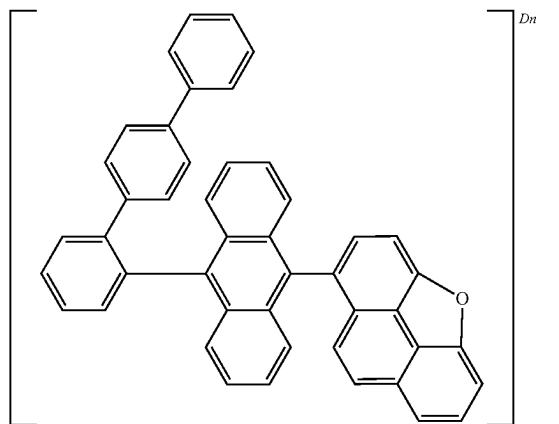


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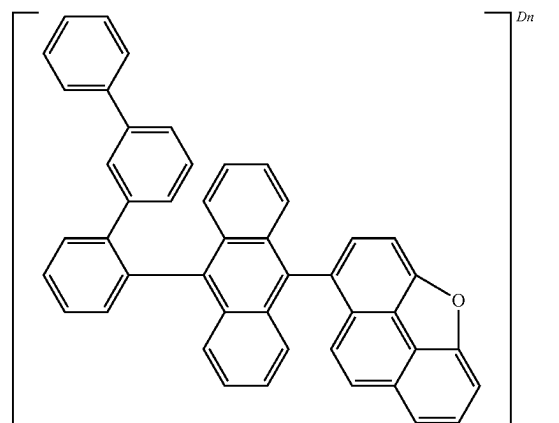
H2-704



H2-705



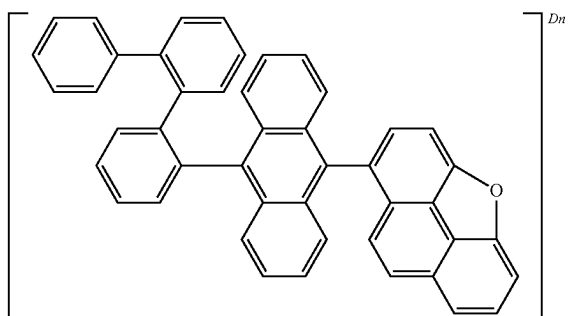
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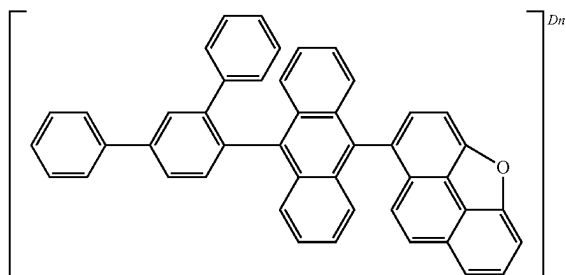
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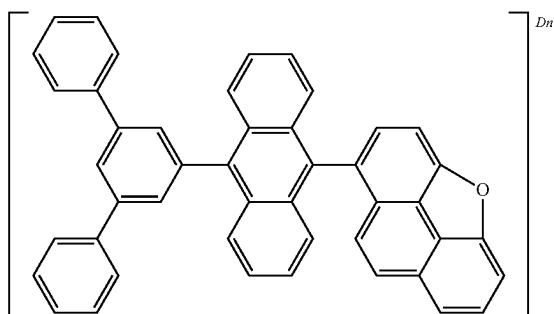
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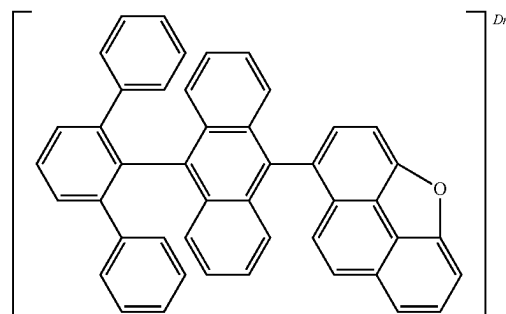
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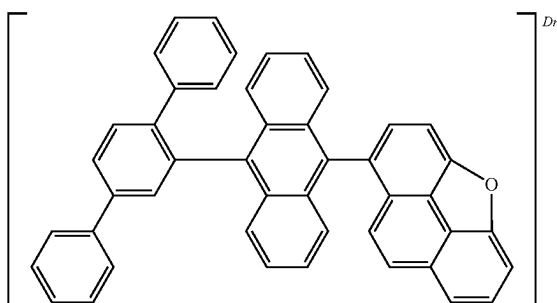
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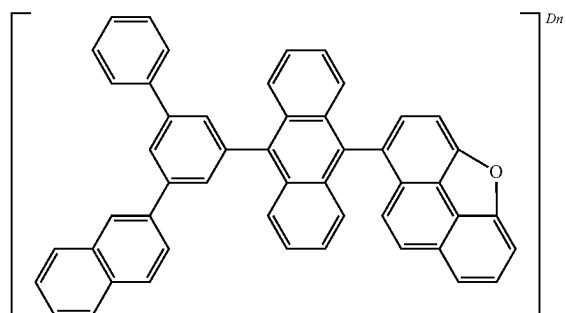
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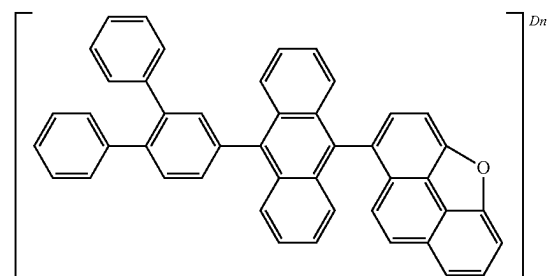
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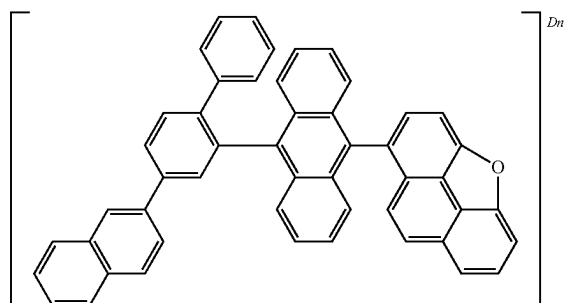
H2-713



H2-710



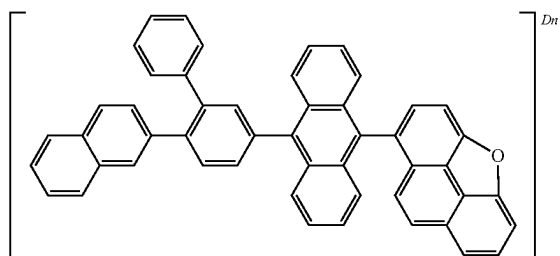
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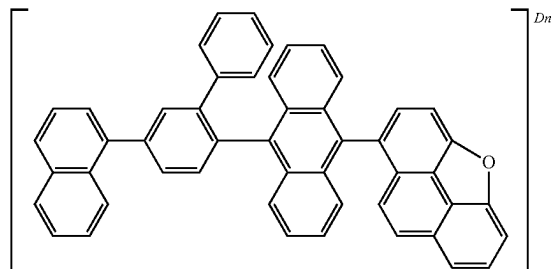
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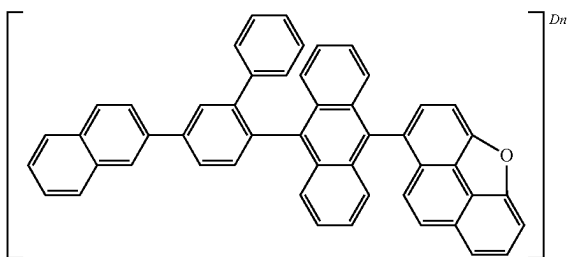
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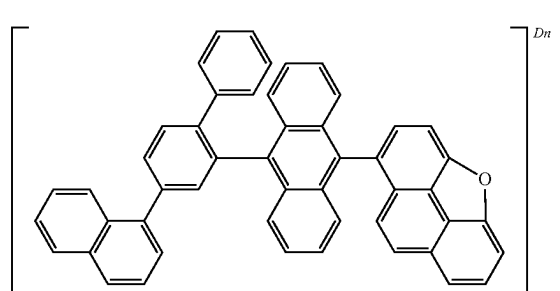
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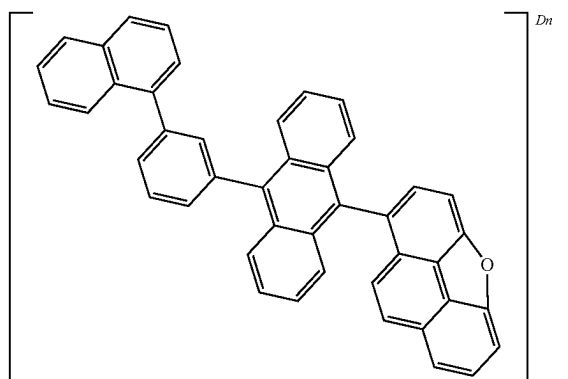
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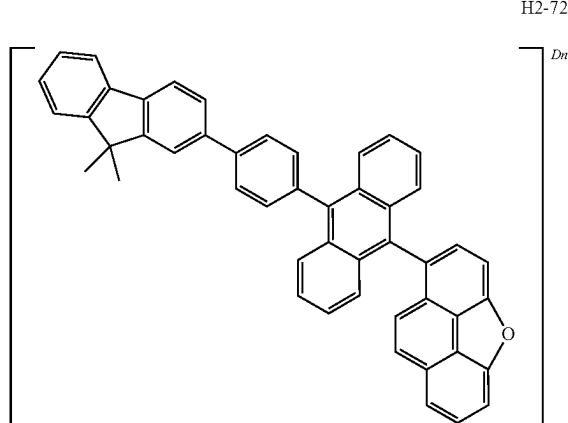
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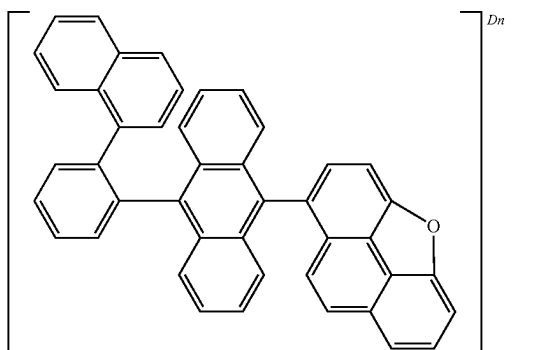
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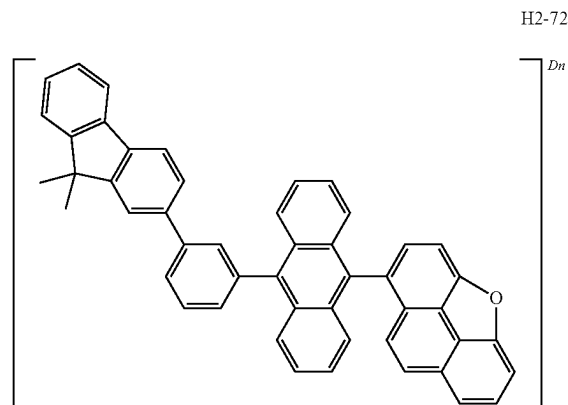
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H2-718

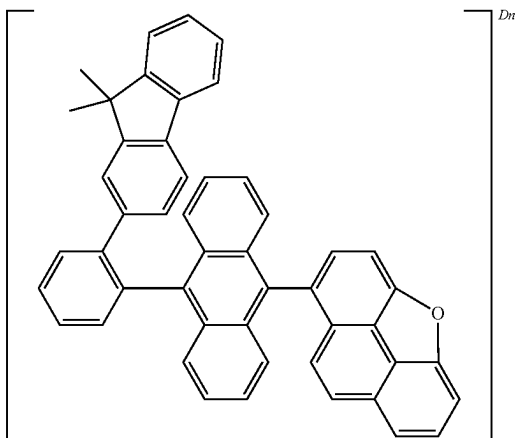


H2-722



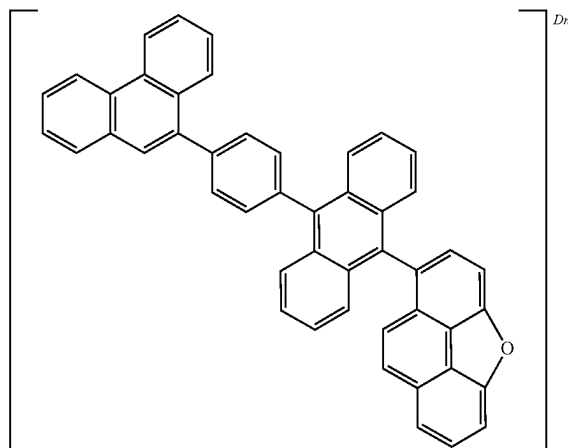
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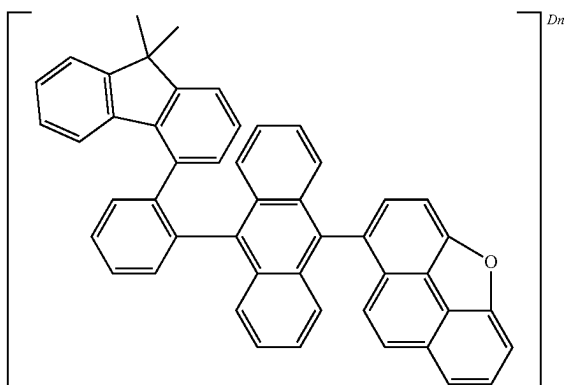


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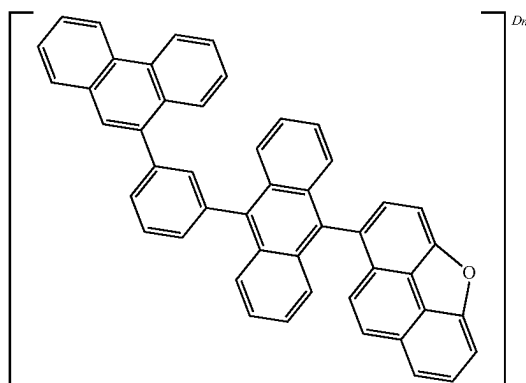
H2-726



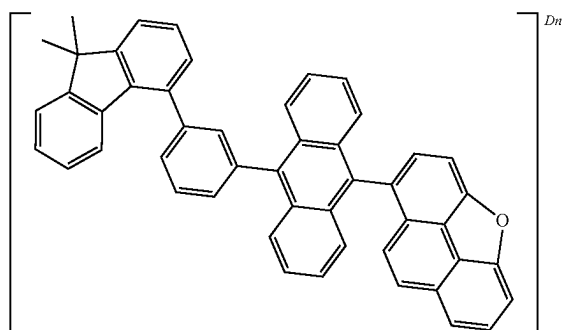
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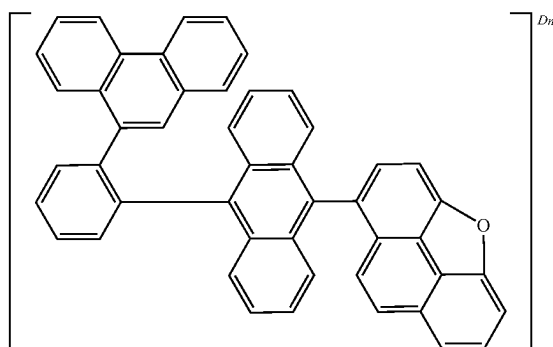
H2-727



H2-725

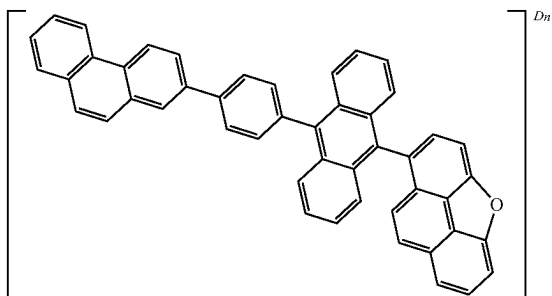


H2-728



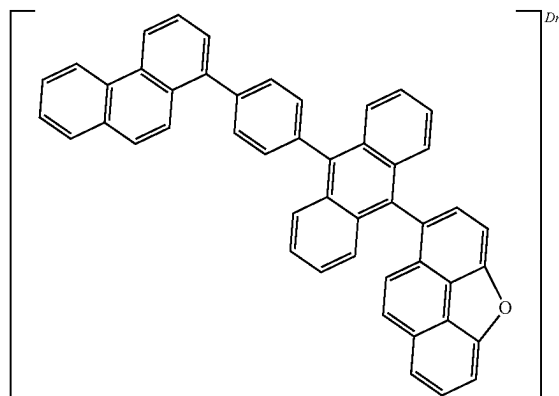
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H2-729

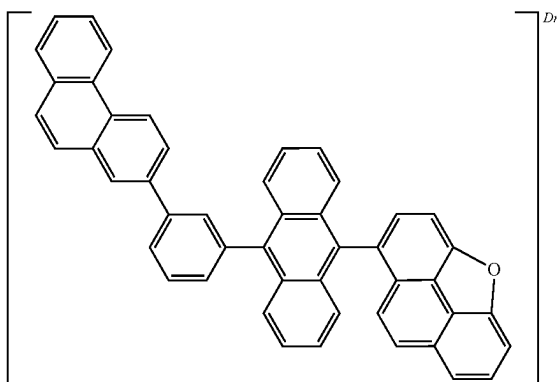


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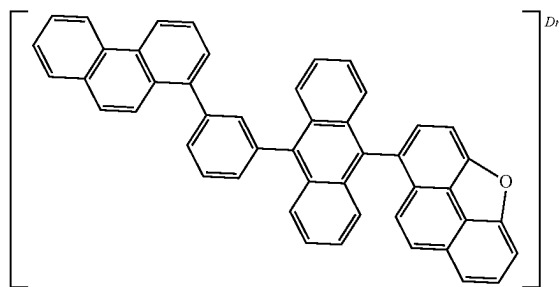
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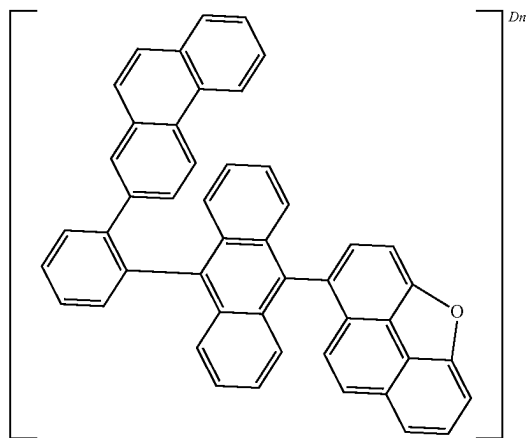
H2-730



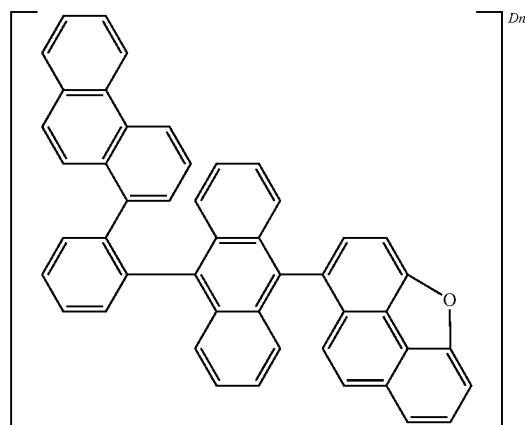
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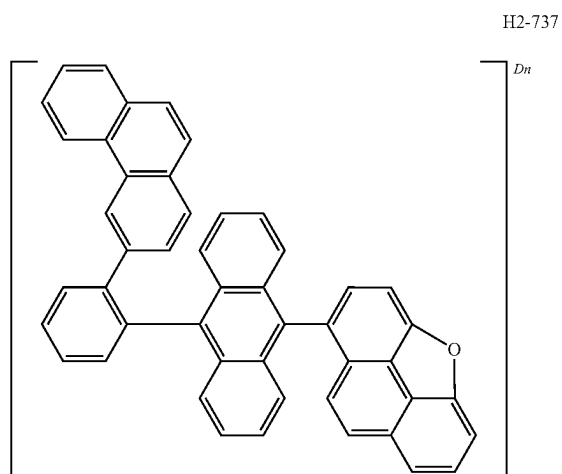
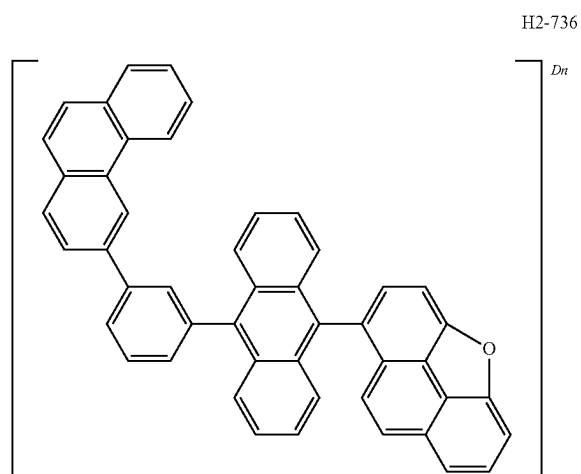
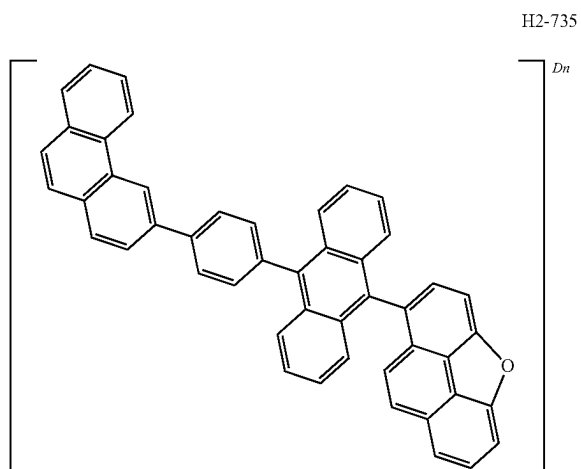
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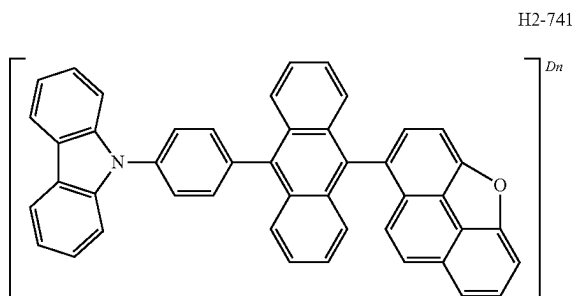
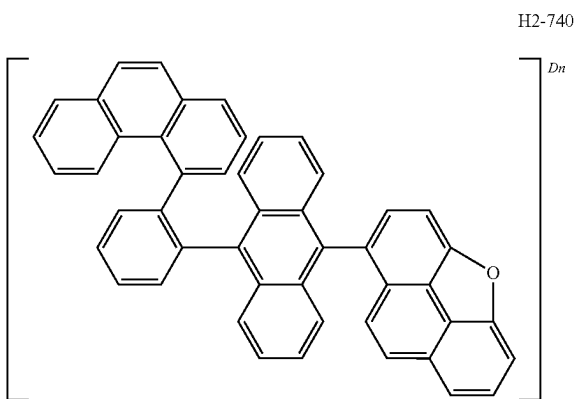
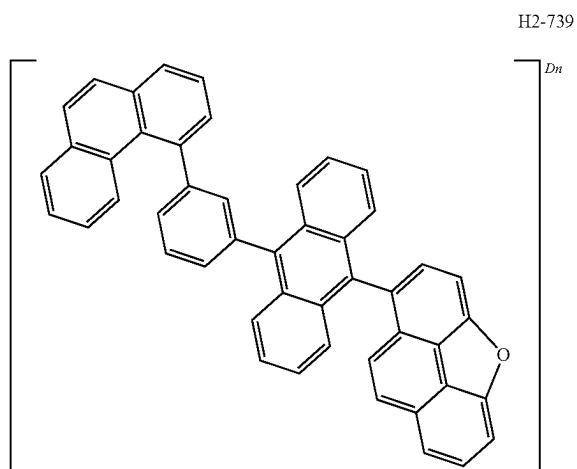
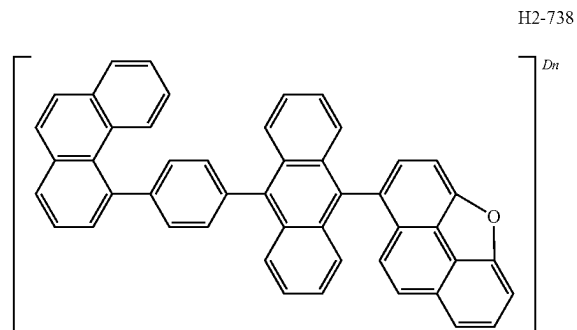
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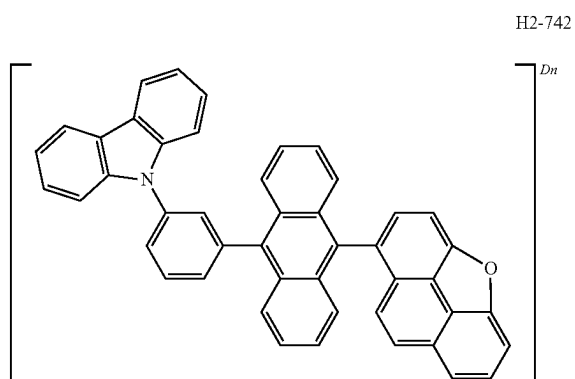
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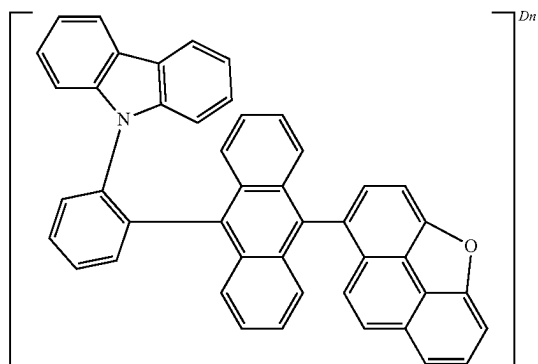
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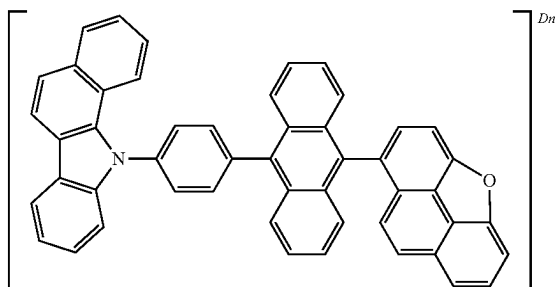
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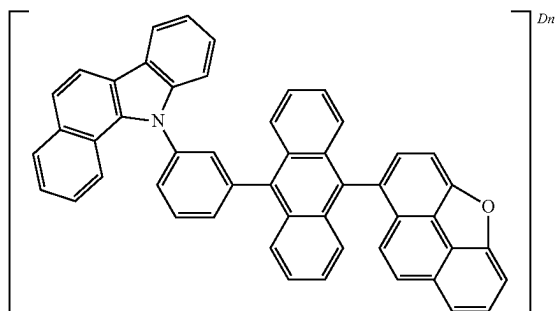
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H2-744

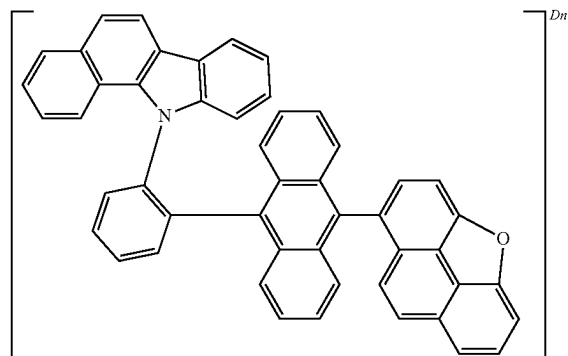


H2-745

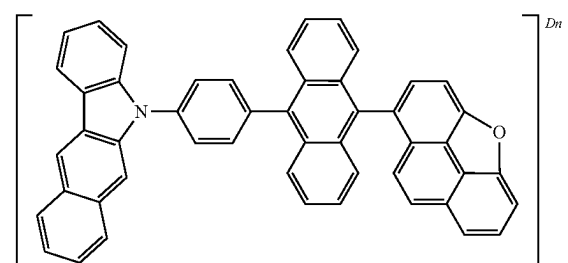


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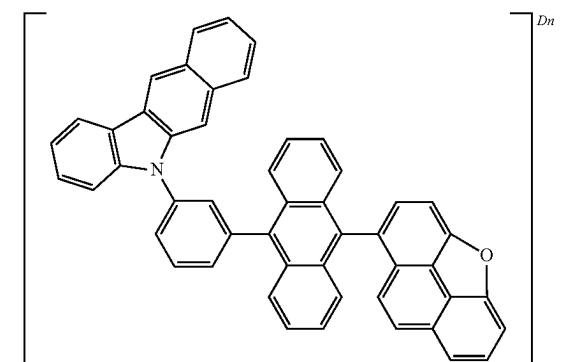
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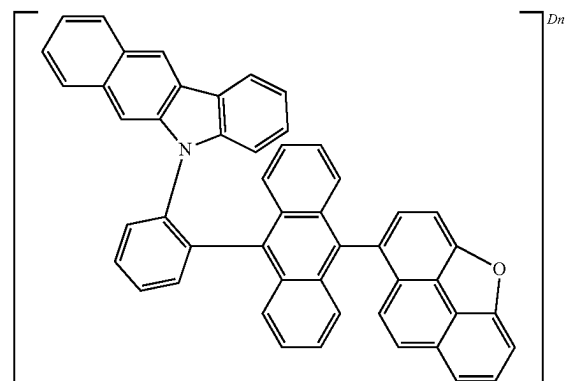
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H2-748



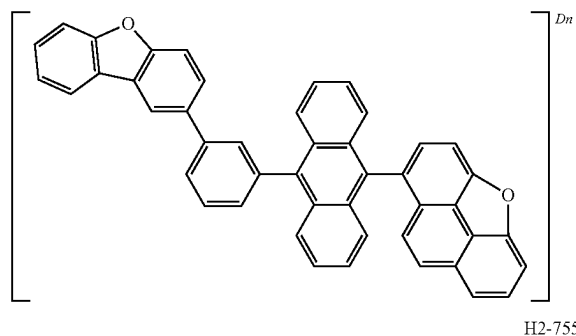
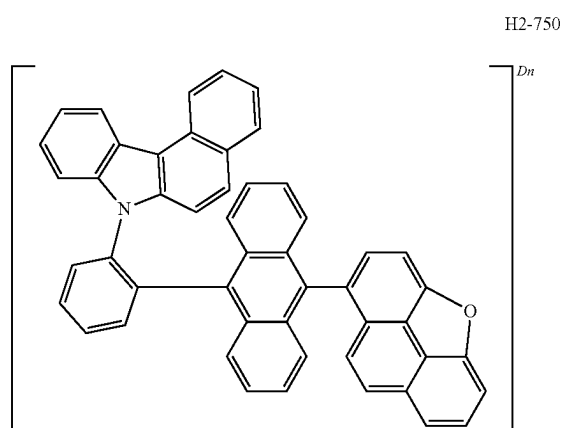
H2-749



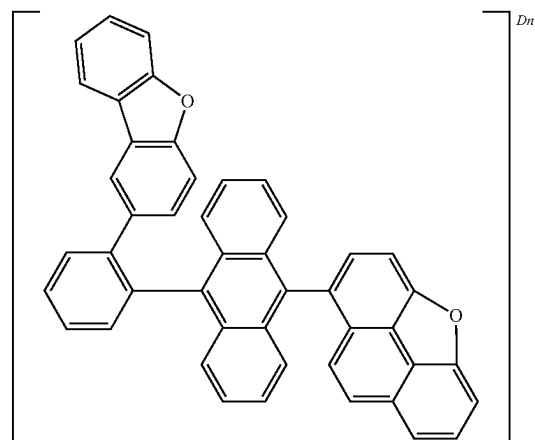
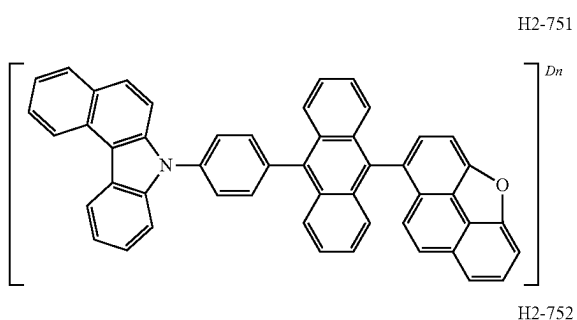
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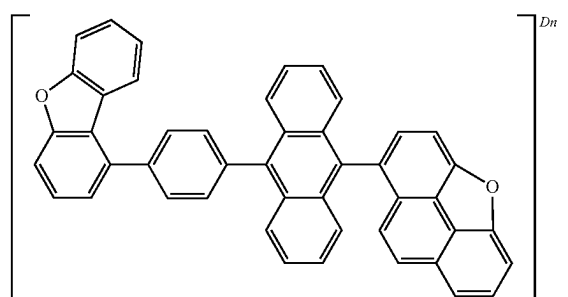
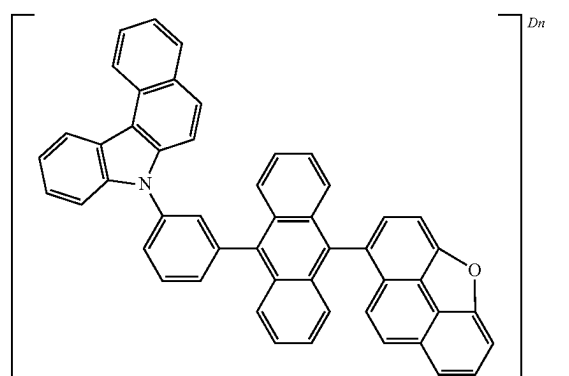
H2-754



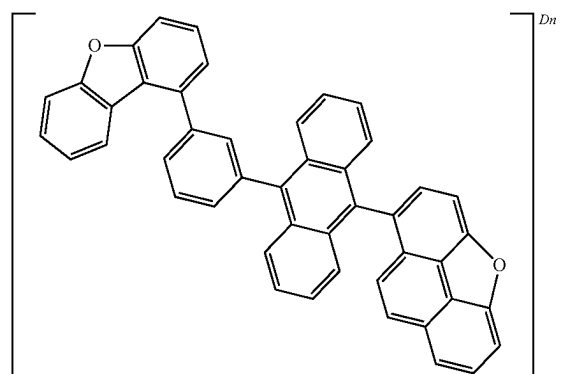
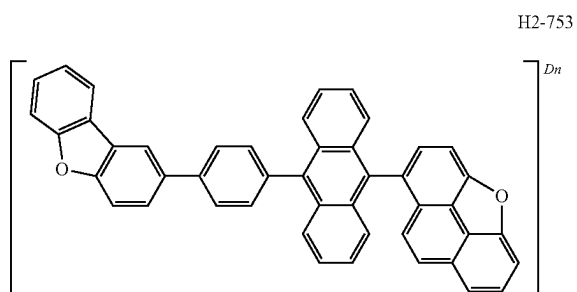
H2-755



H2-756

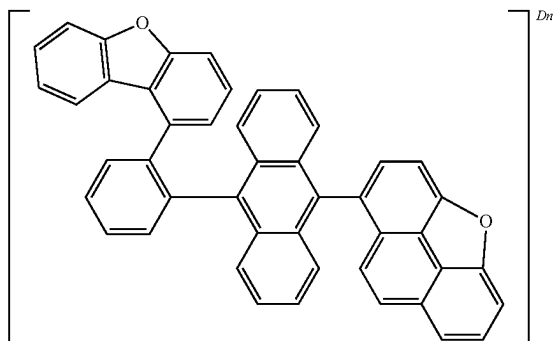


H2-757

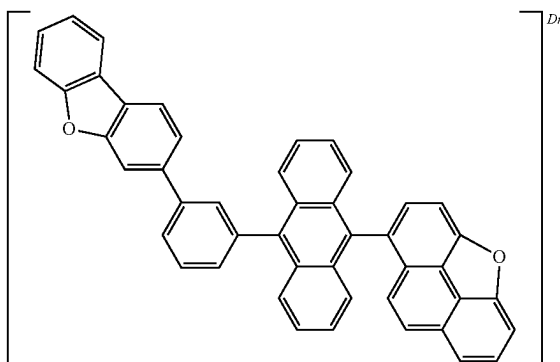


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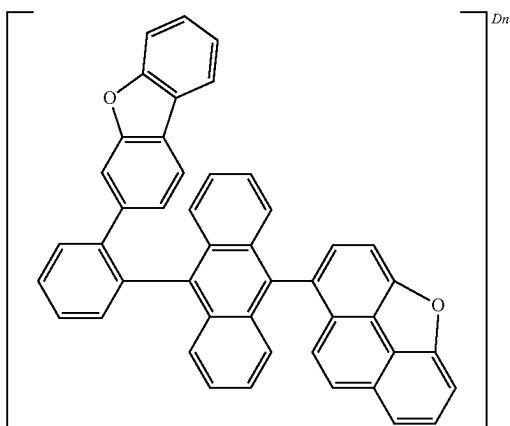
H2-758



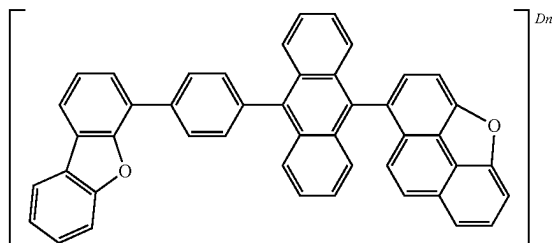
H2-759



H2-760

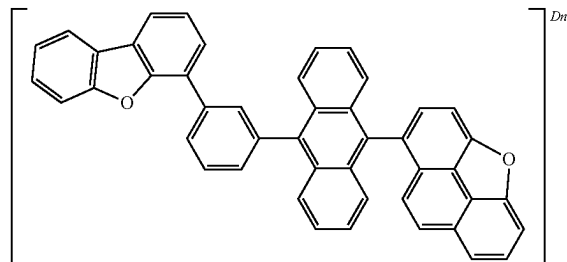


H2-761

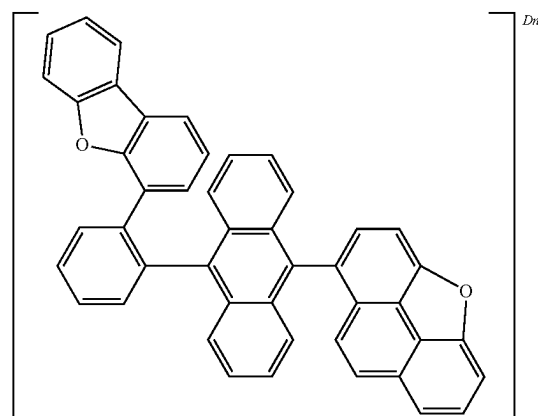


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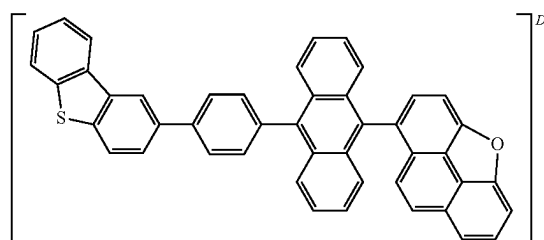
H2-762



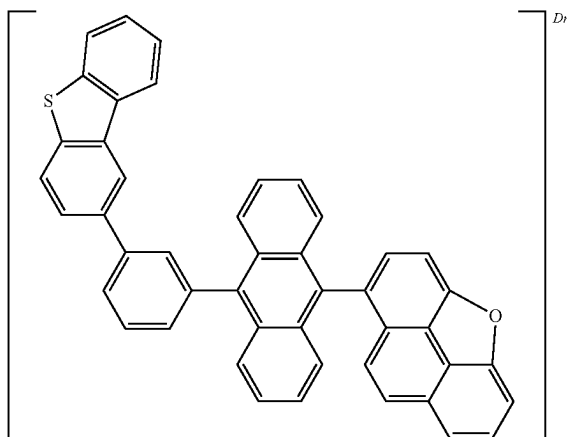
H2-763



H2-764

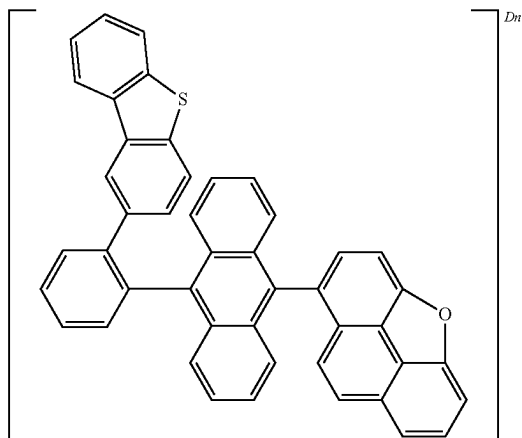


H2-765

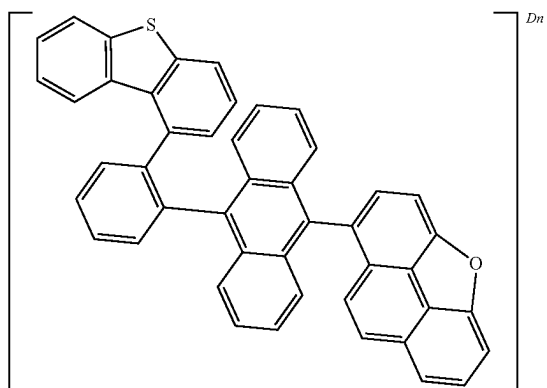


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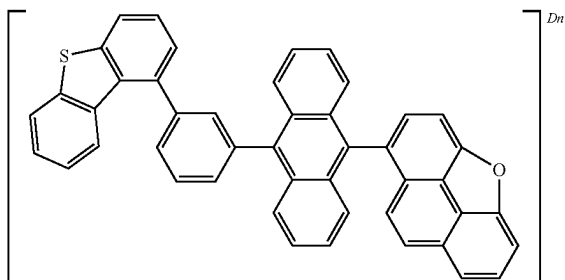
H2-766



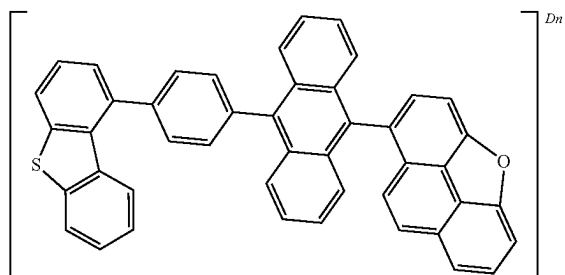
H2-767



H2-768

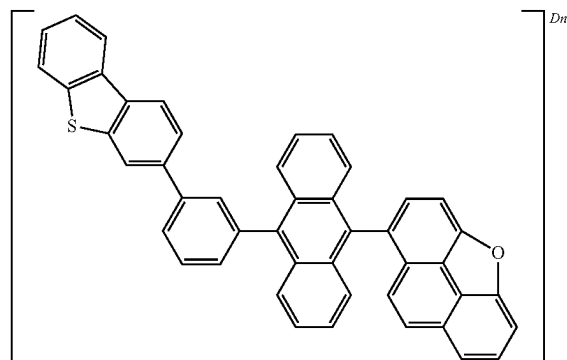


H2-769

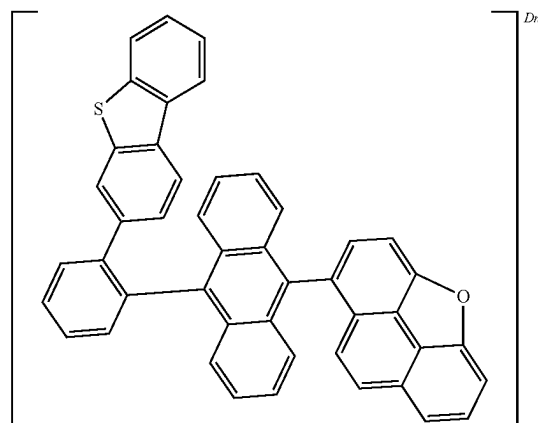


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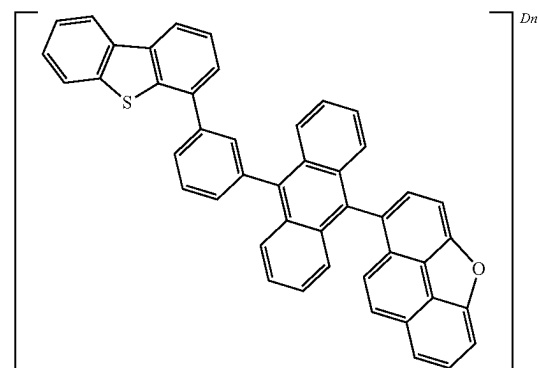
H2-770



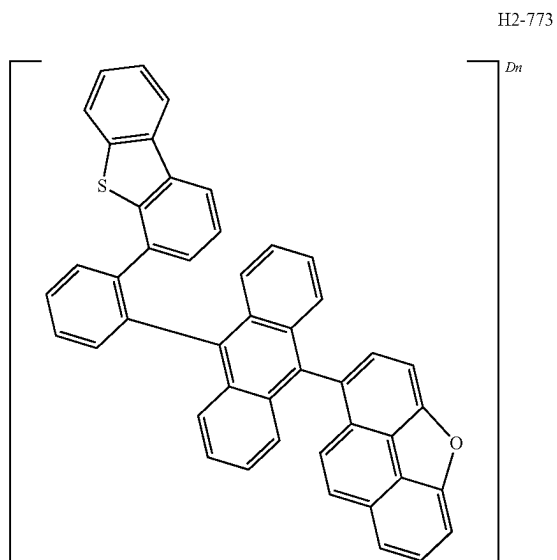
H2-771



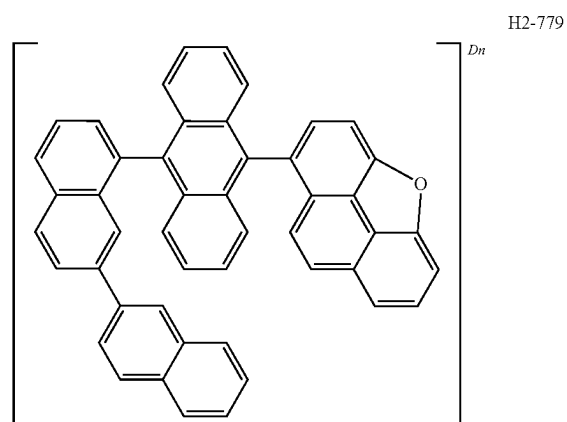
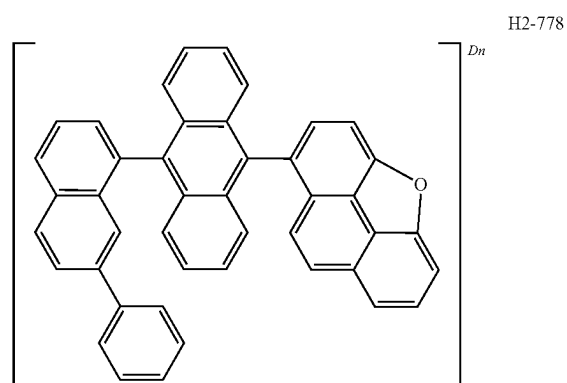
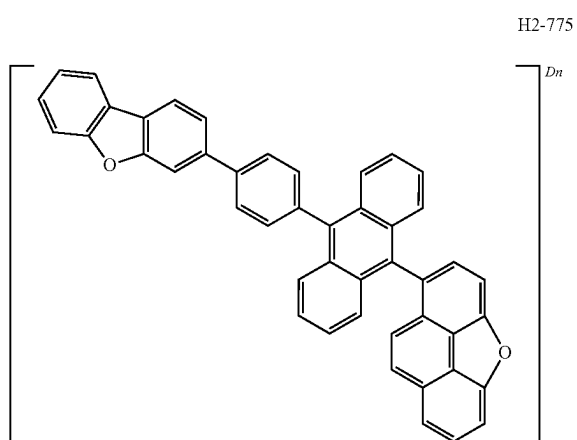
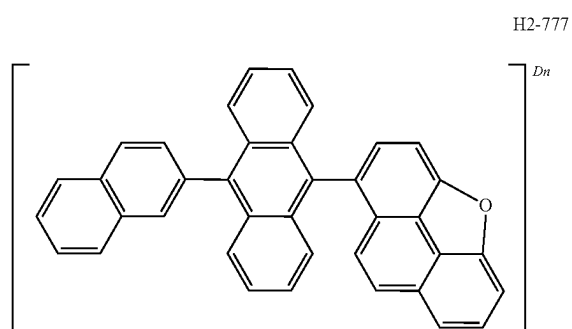
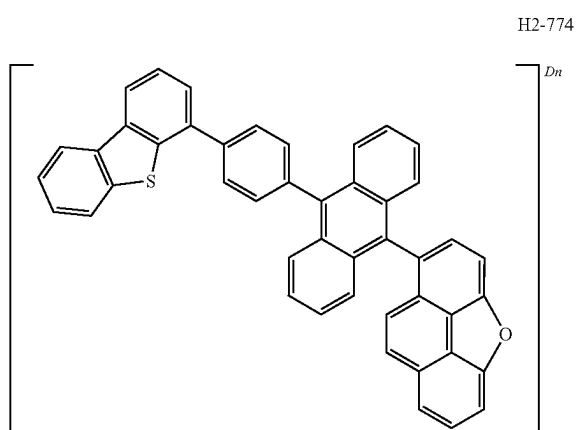
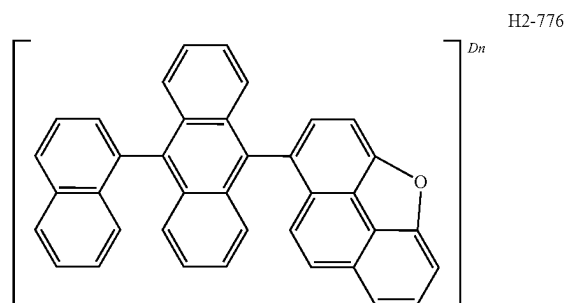
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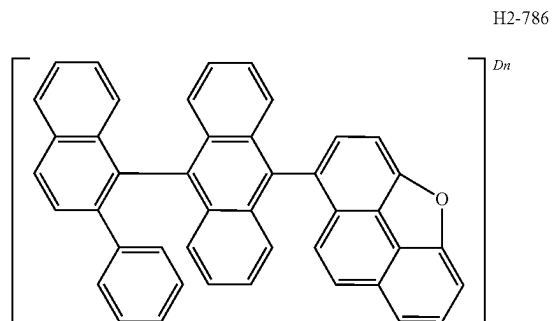
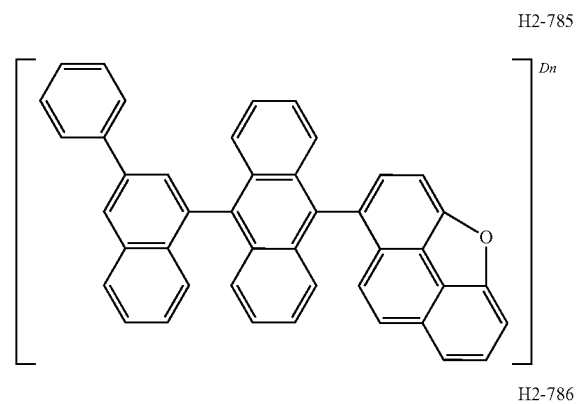
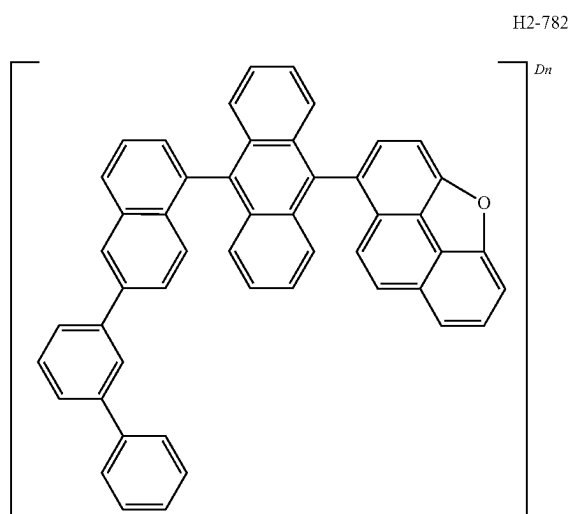
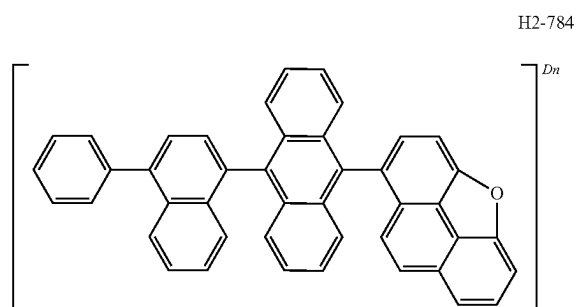
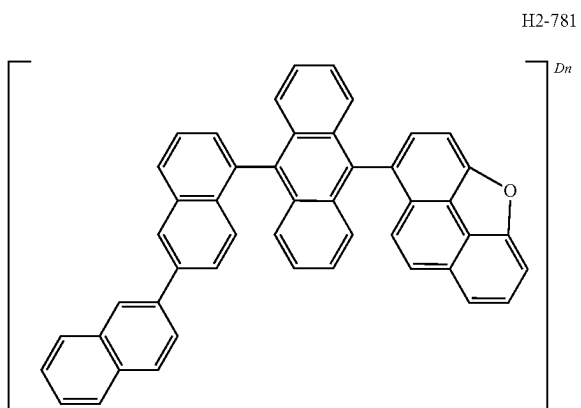
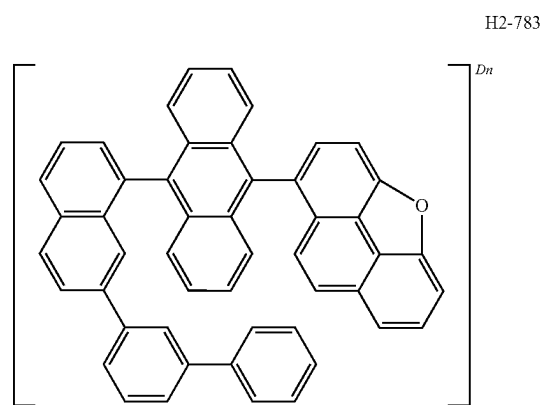
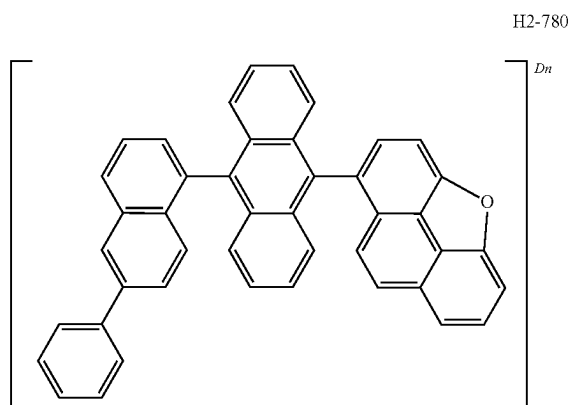


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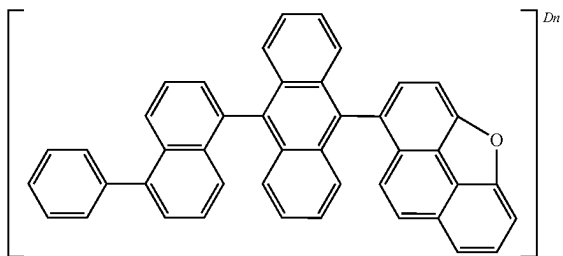
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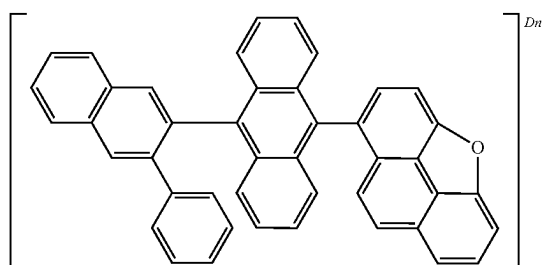


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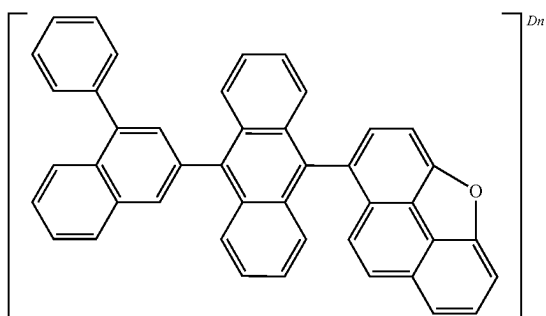
H2-787



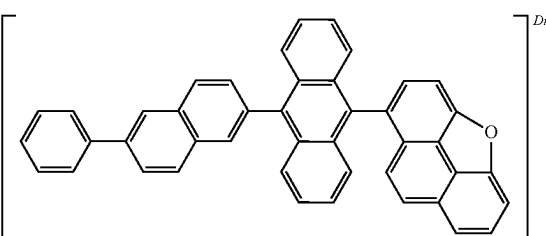
H2-788



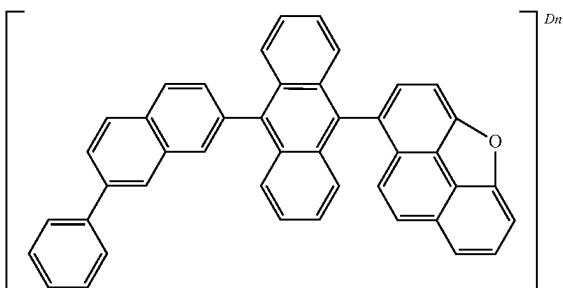
H2-789



H2-790

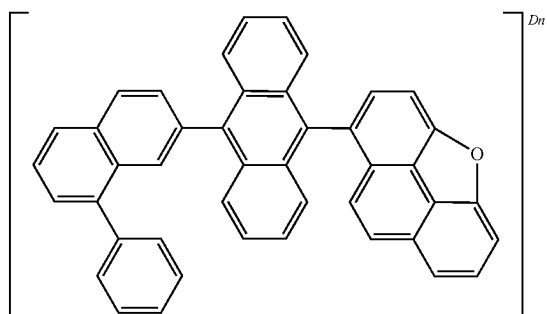


H2-791

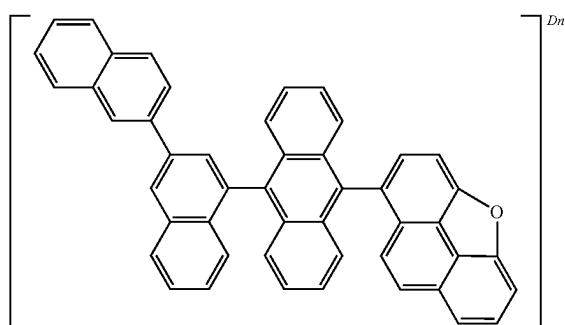


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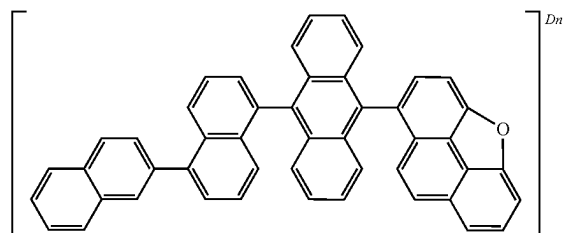
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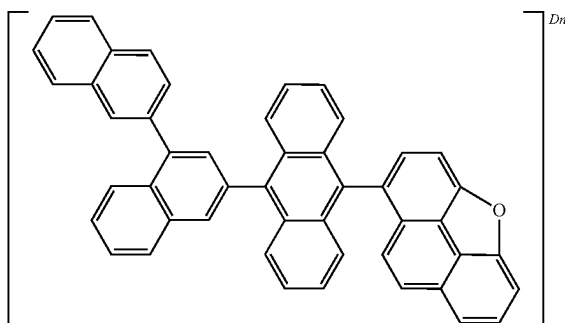
H2-793



H2-794

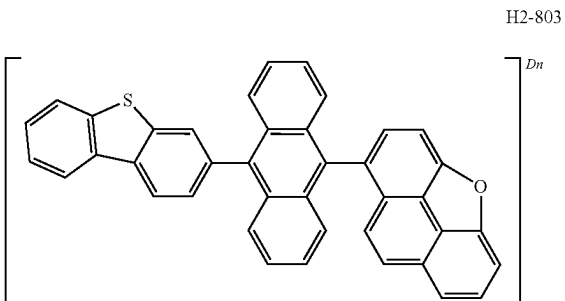
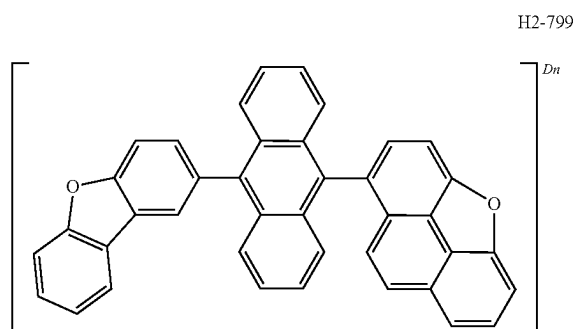
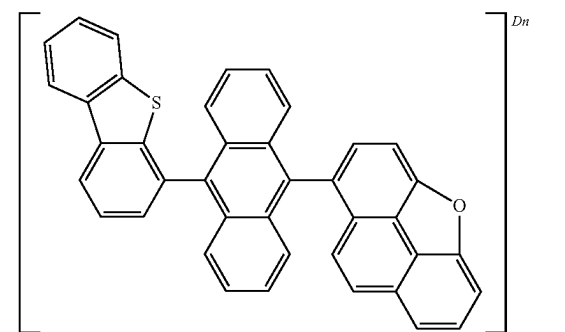
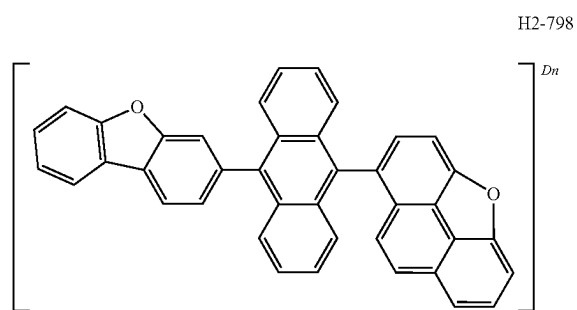
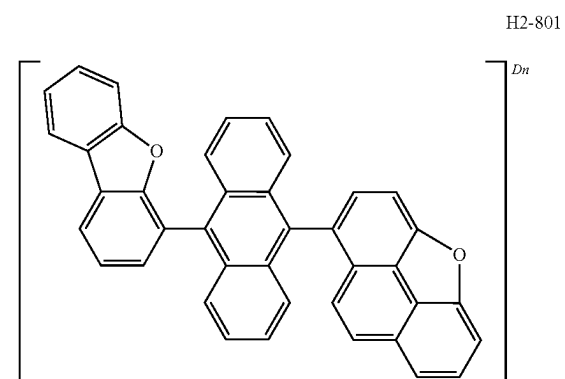
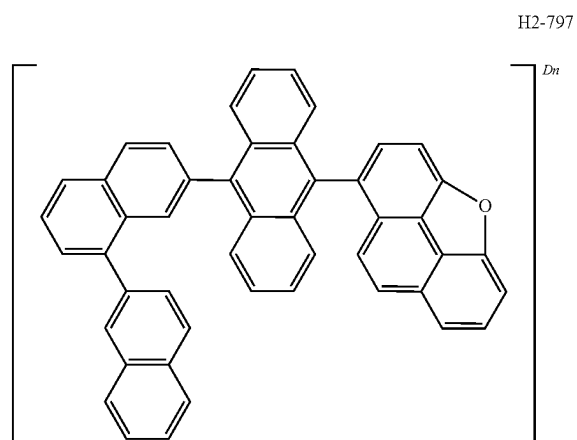
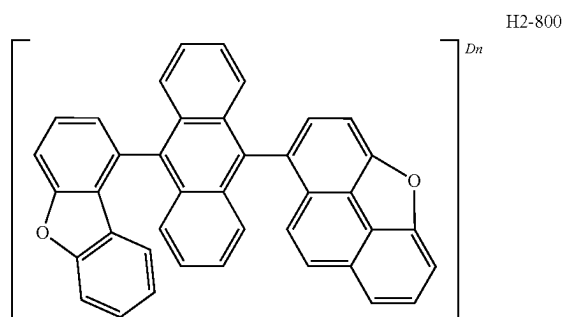
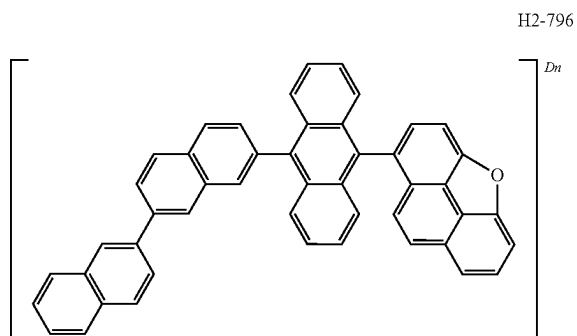


H2-795



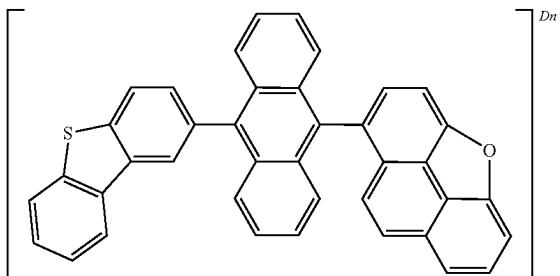
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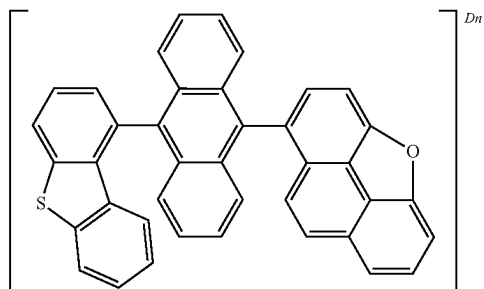


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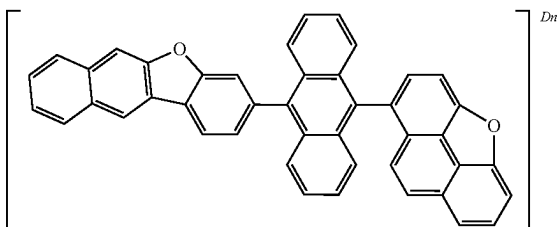
H2-804



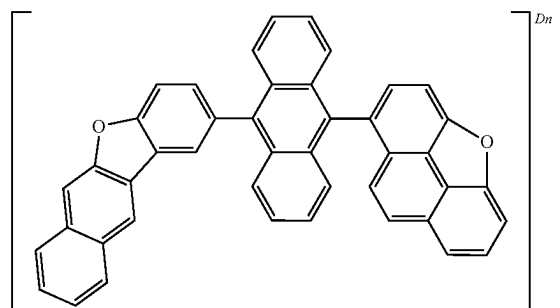
H2-805



H2-806

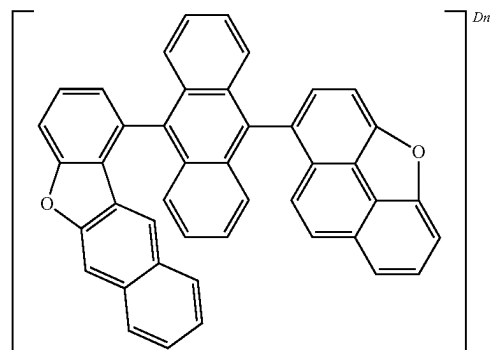


H2-807

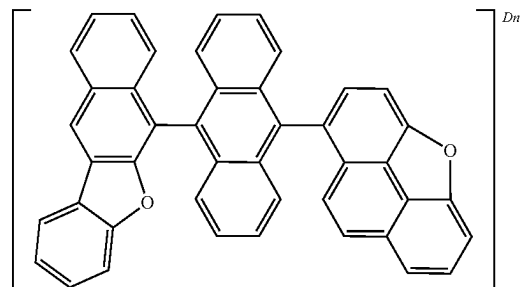


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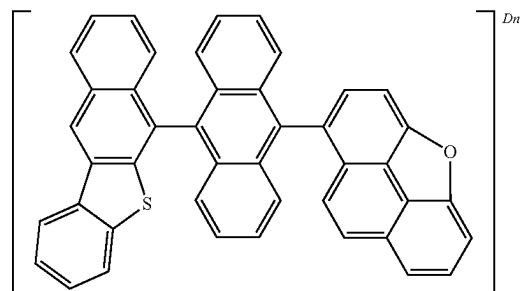
H2-808



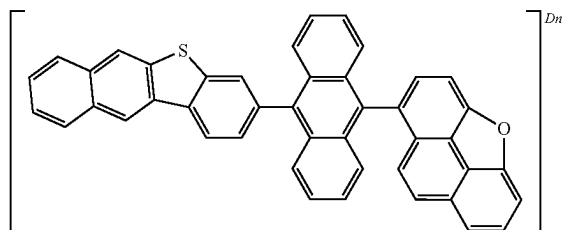
H2-809



H2-810



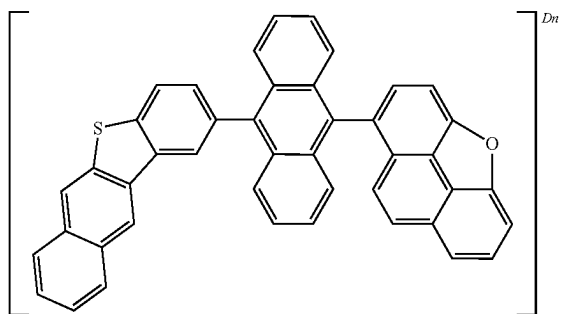
H2-811



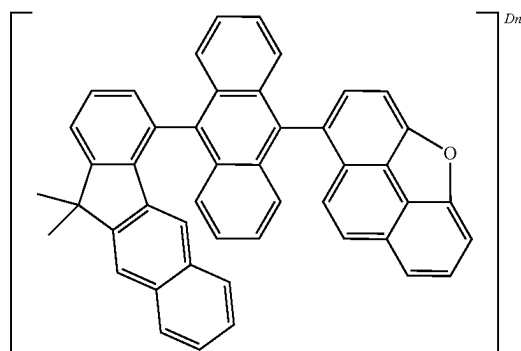
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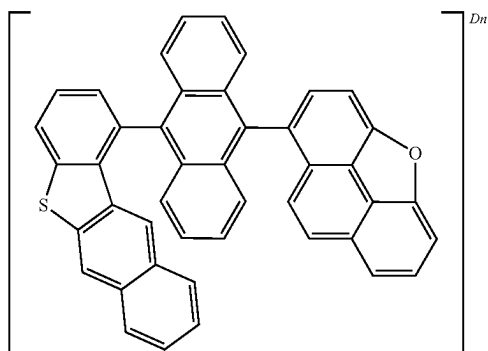
H2-812



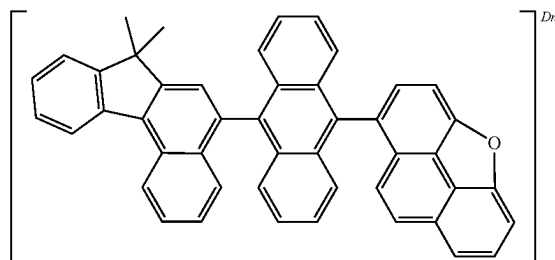
H2-816



H2-813

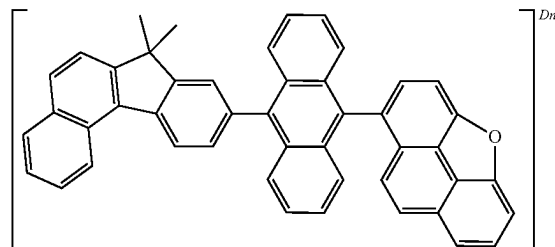
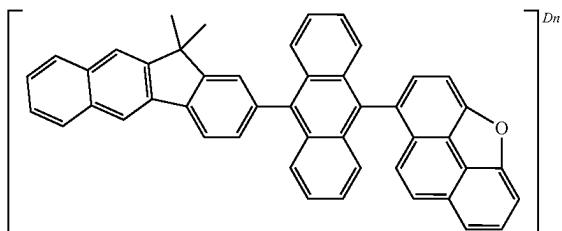


H2-817



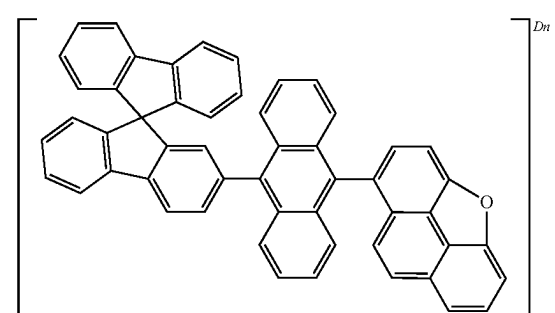
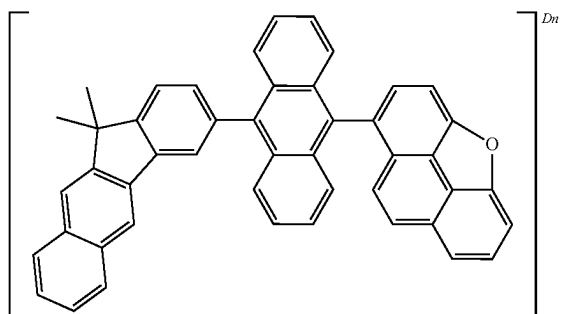
H2-818

H2-814



H2-819

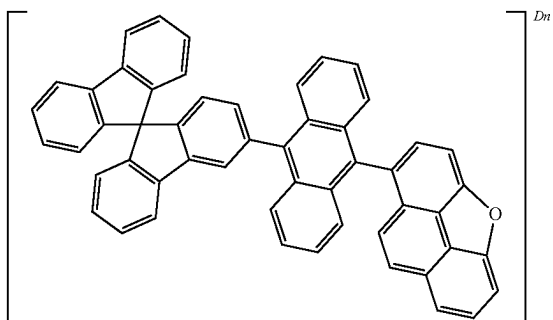
H2-815



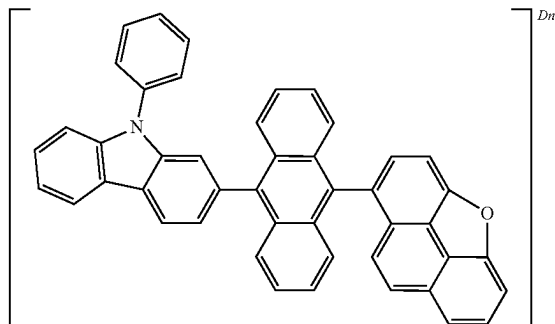
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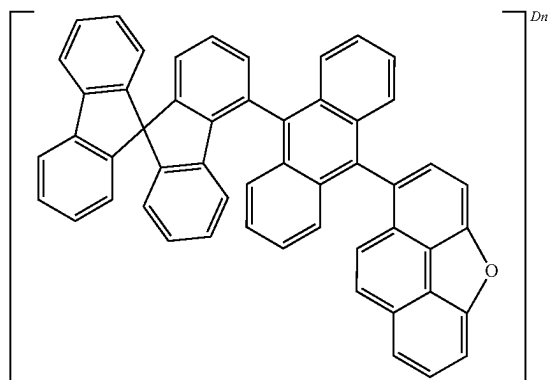
H2-820



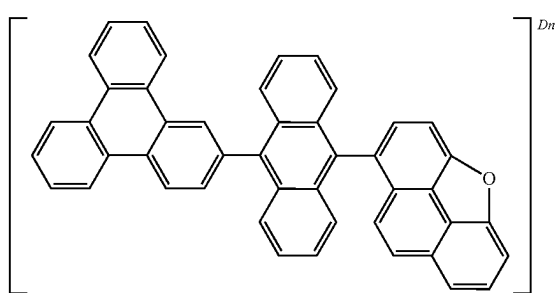
H2-824



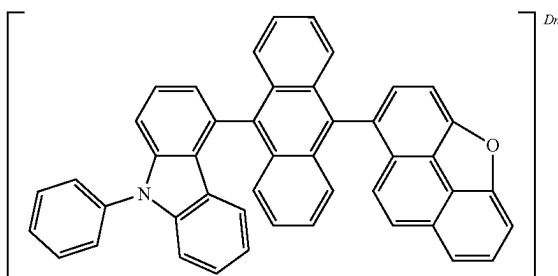
H2-821



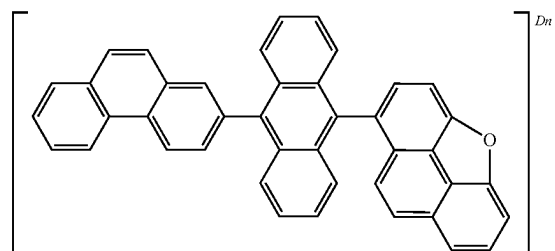
H2-825



H2-822

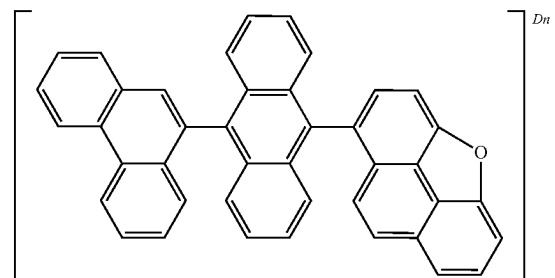
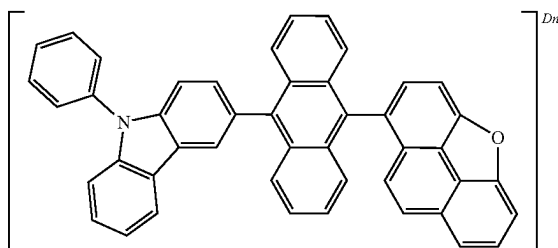


H2-826



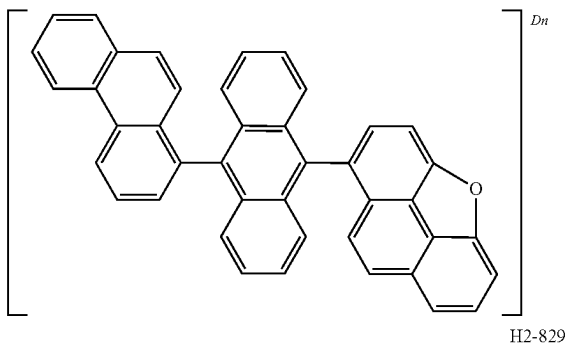
H2-827

H2-823

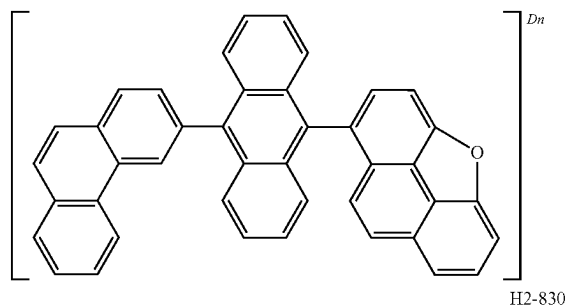


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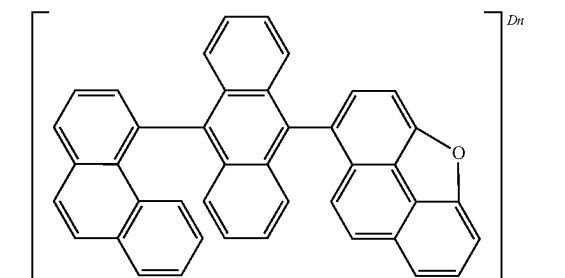
H2-828



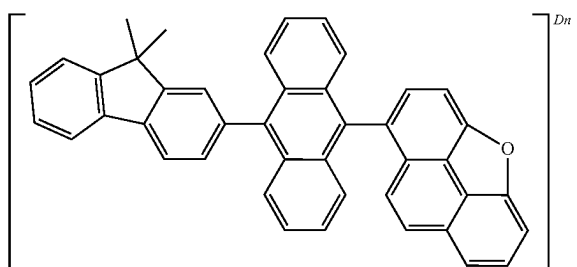
H2-829



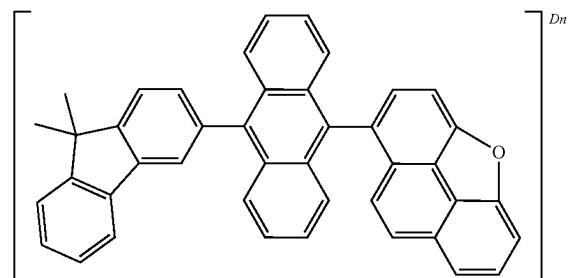
H2-830



H2-831

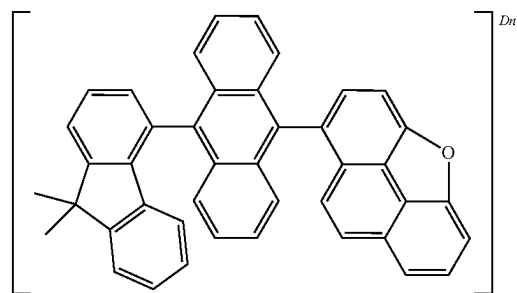


H2-832

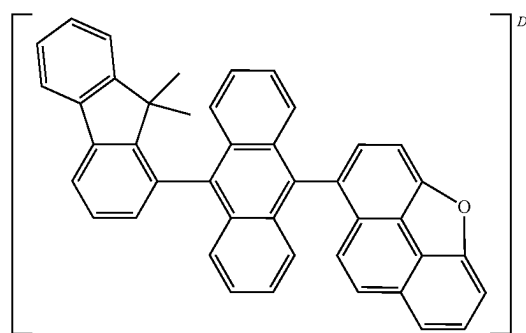


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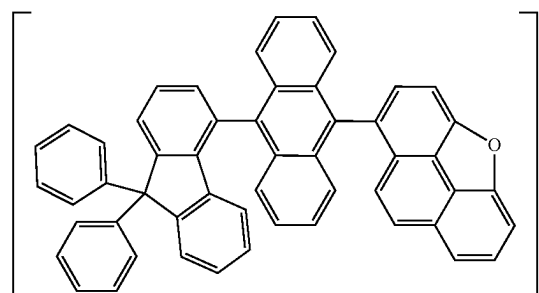
H2-833



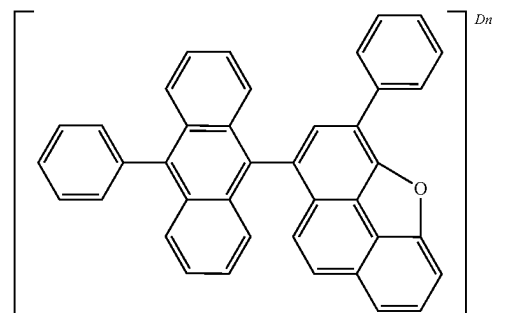
H2-834



H2-835

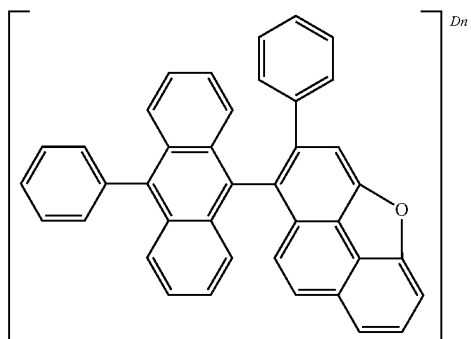


H2-836



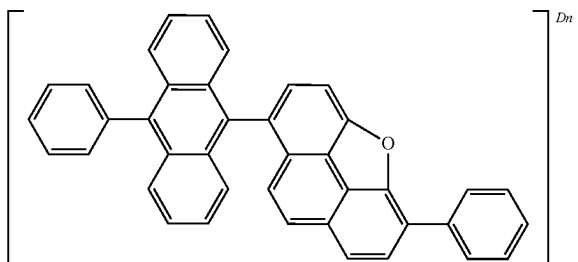
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H2-837



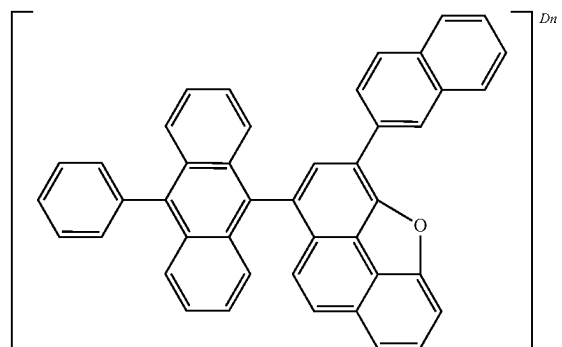
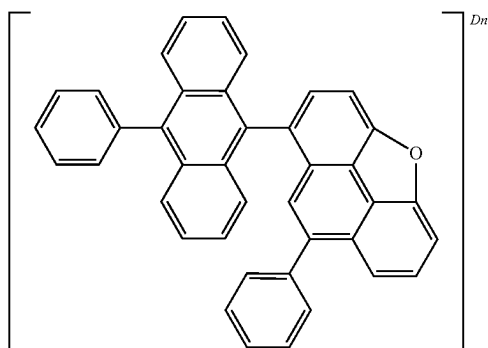
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H2-841



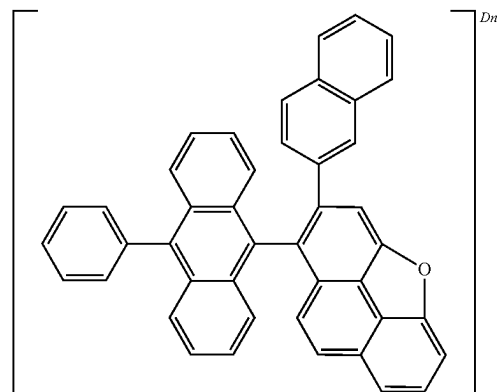
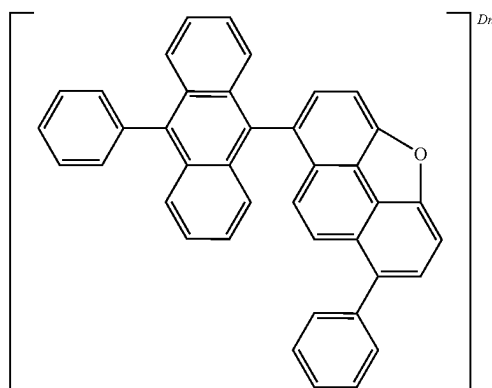
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H2-838



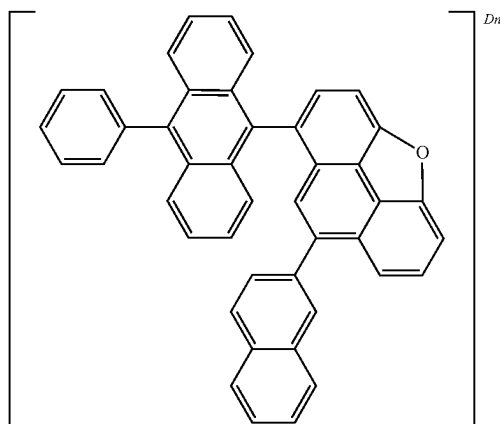
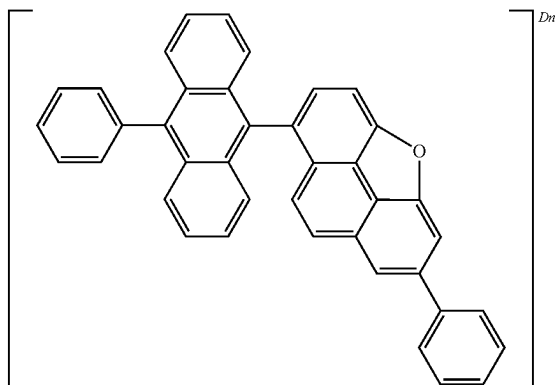
H2-843

H2-839



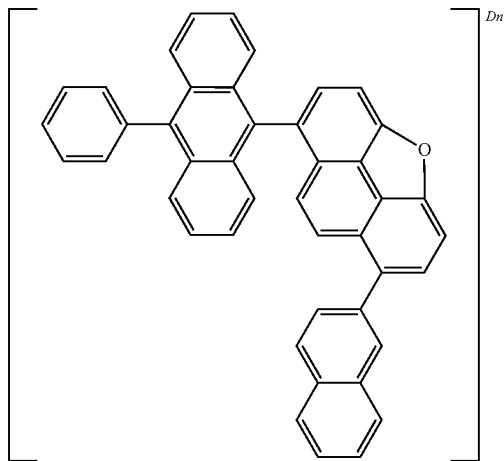
H2-844

H2-840



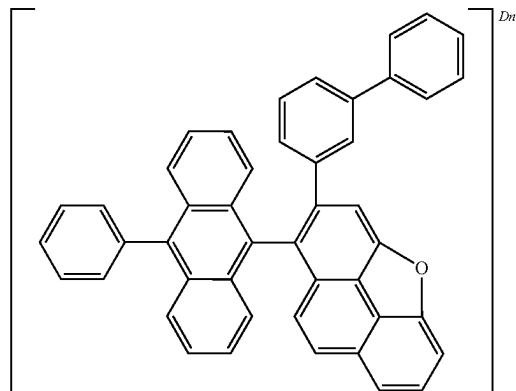
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H2-845

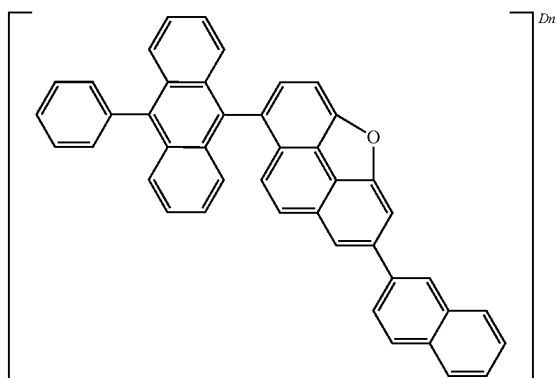


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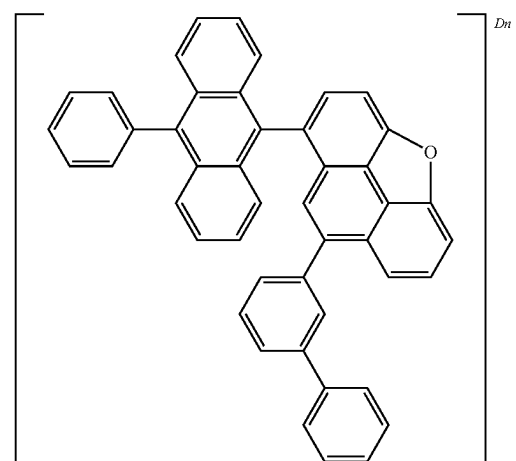
H2-849



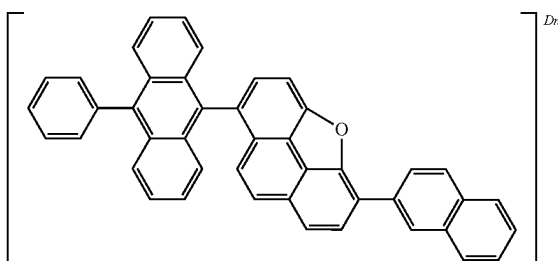
H2-846



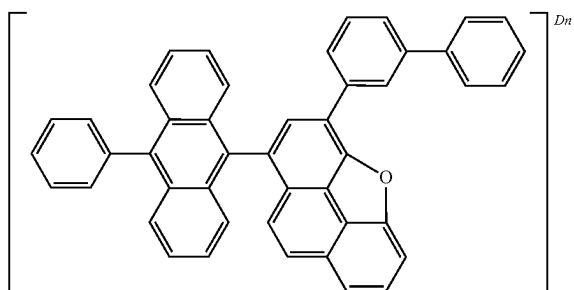
H2-847



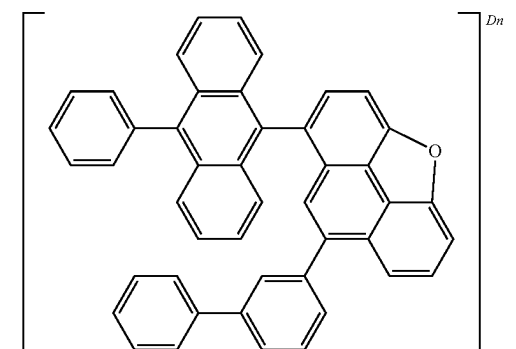
H2-850



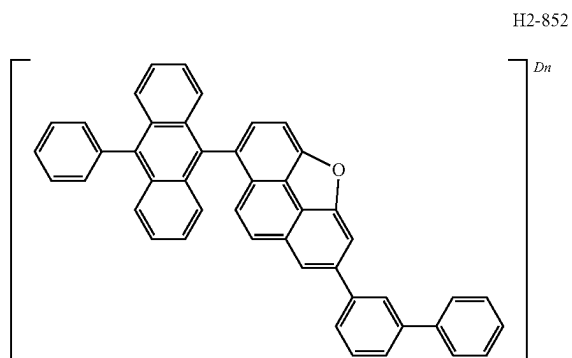
H2-848



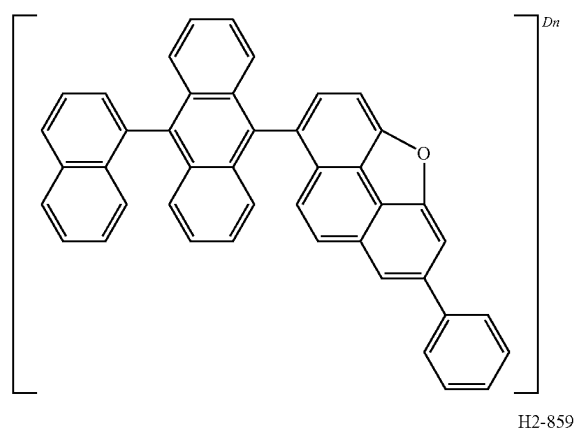
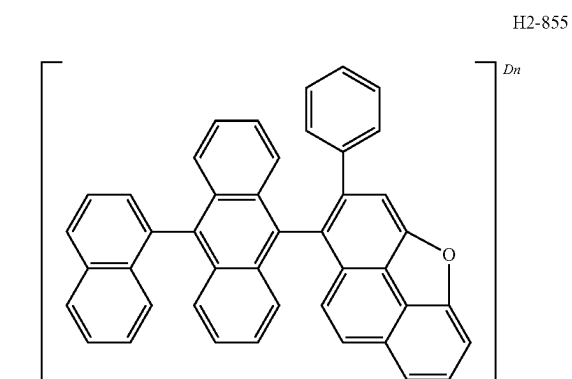
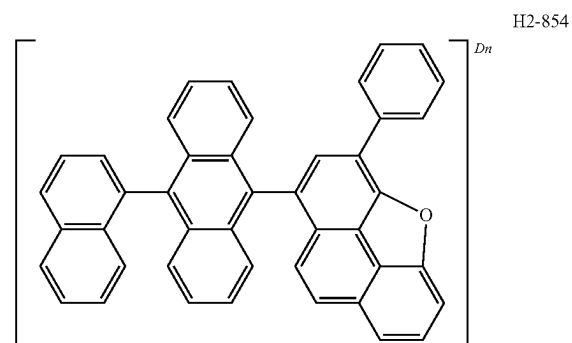
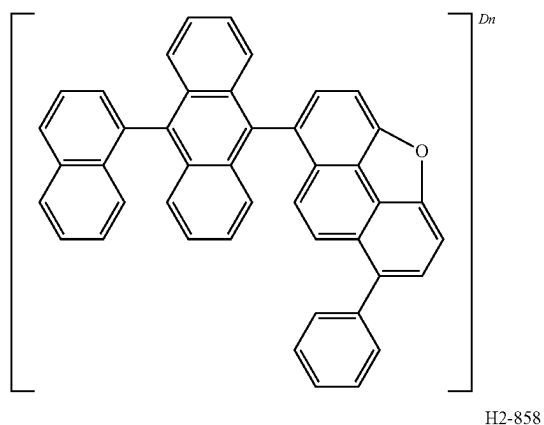
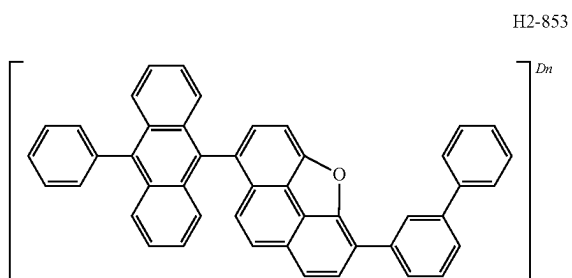
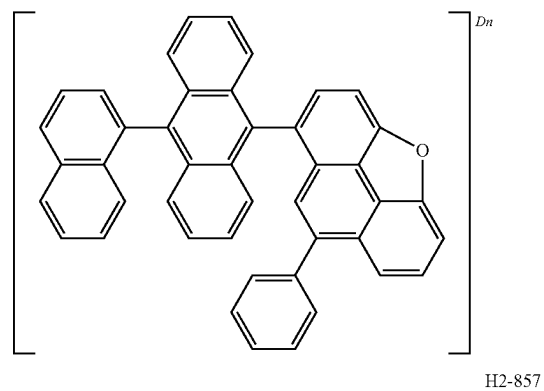
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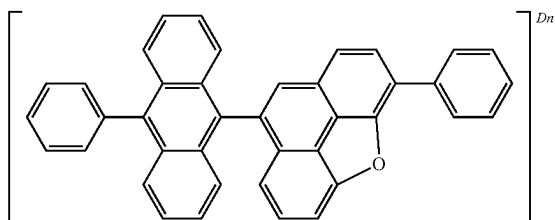


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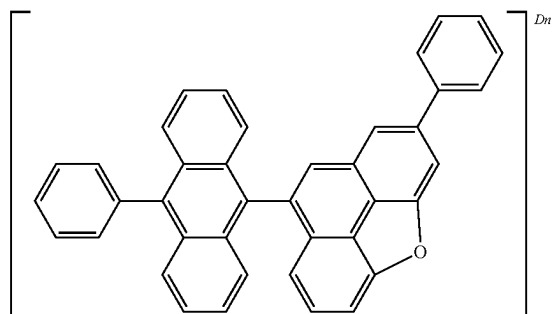


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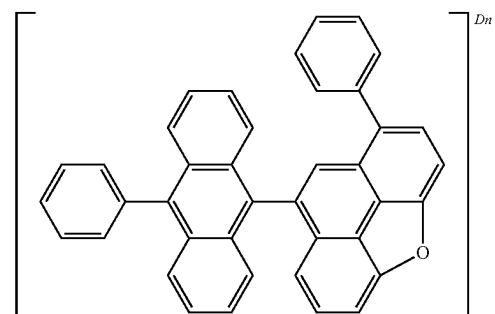
H2-860



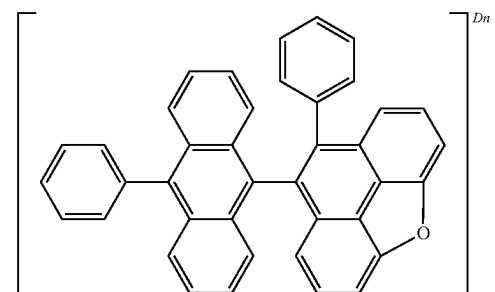
H2-861



H2-862

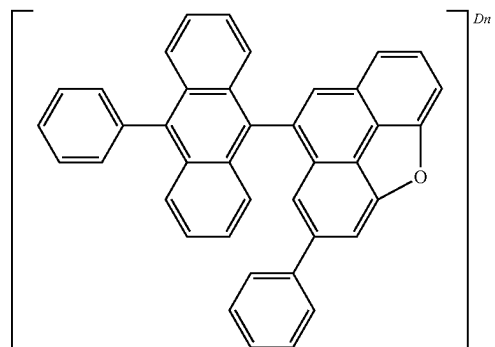


H2-863

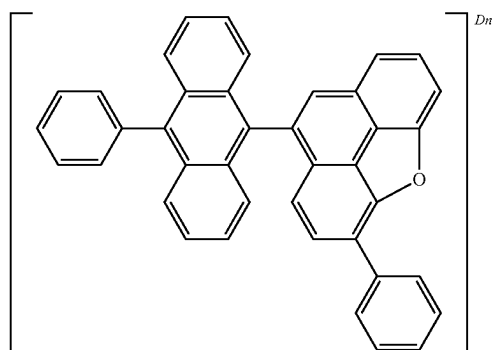


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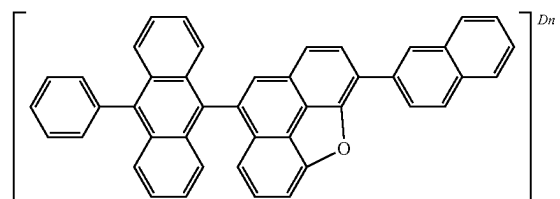
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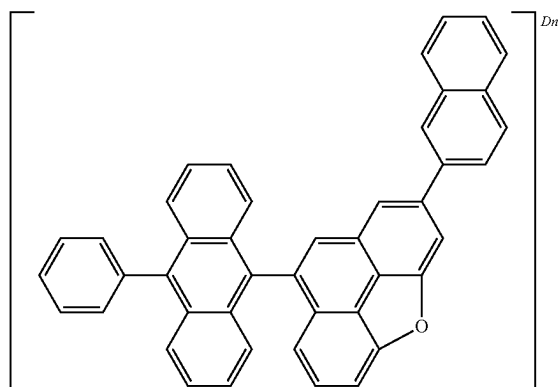
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H2-866

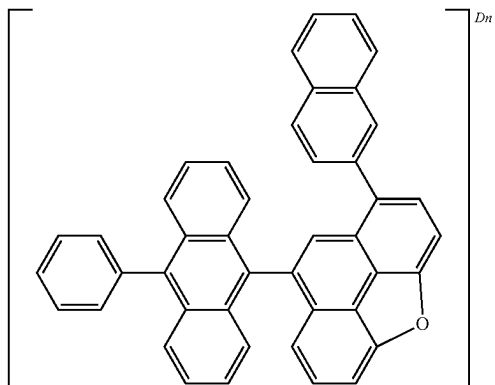


H2-867



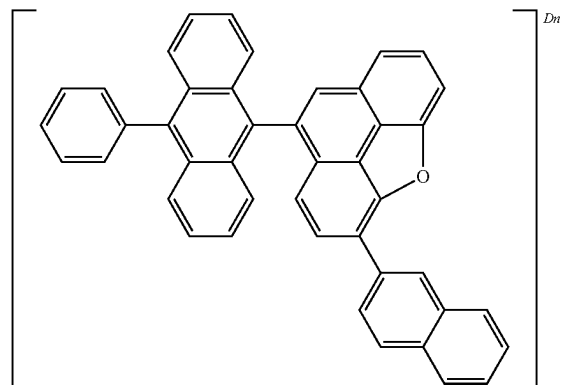
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H2-868



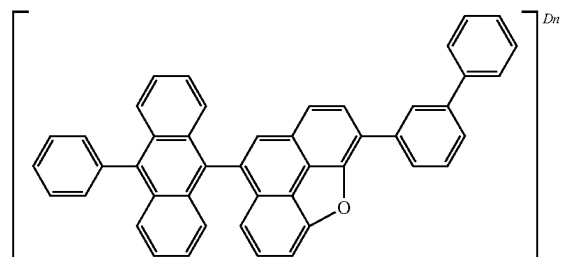
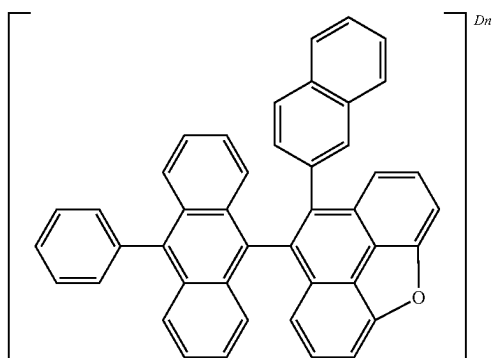
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H2-871

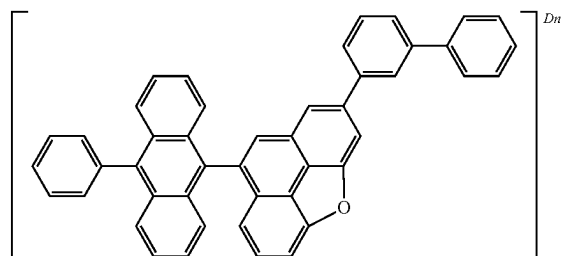


H2-872

H2-869

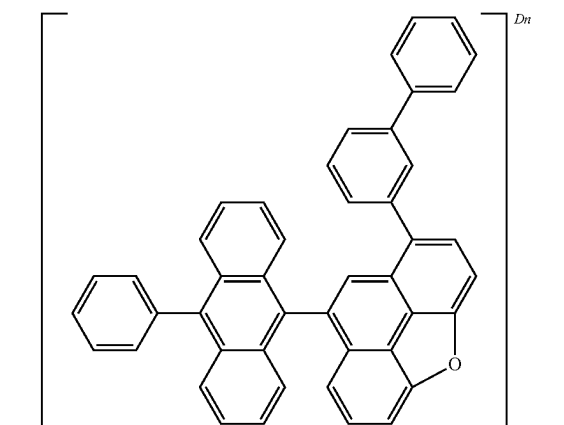
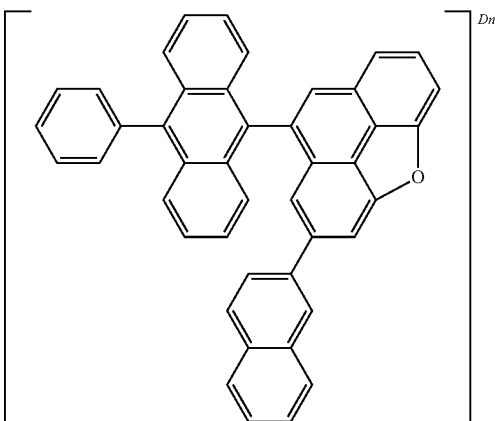


H2-873

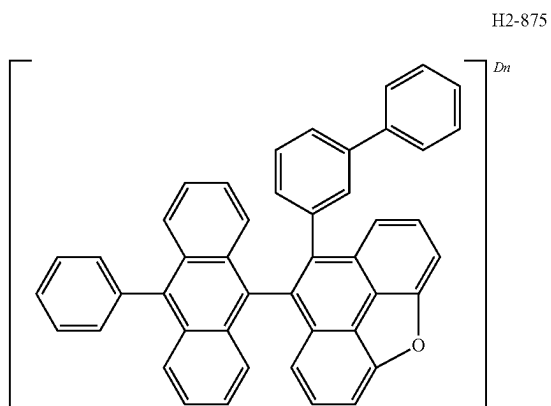


H2-874

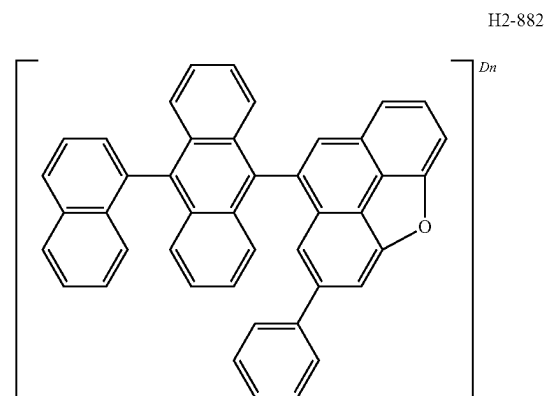
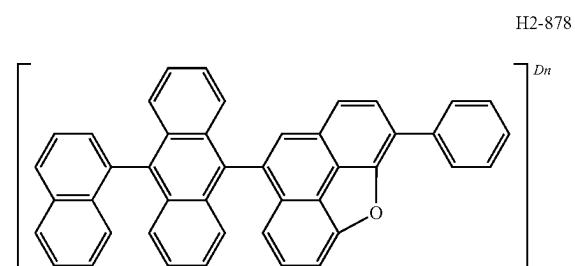
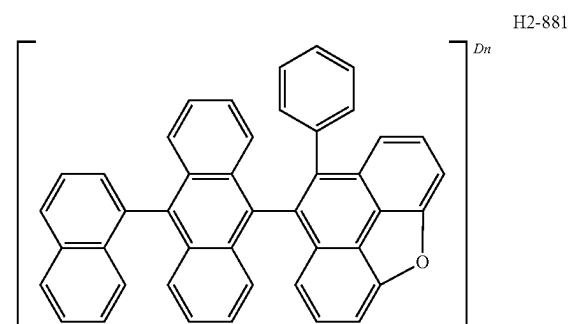
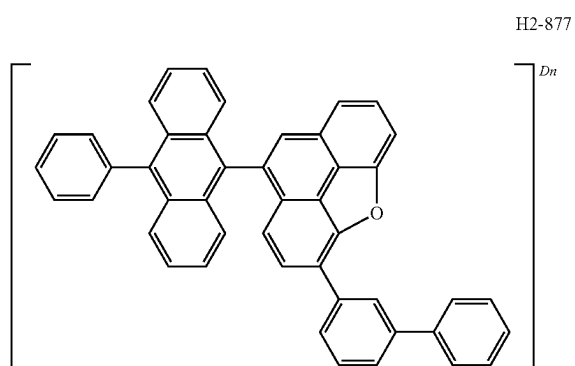
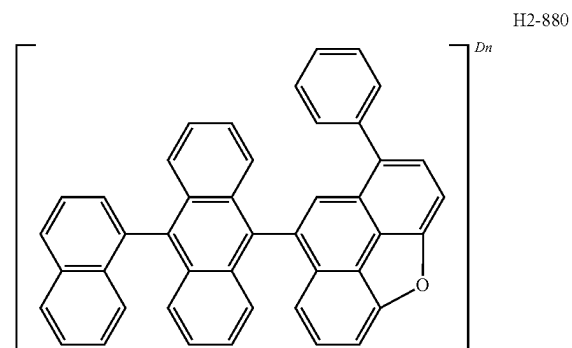
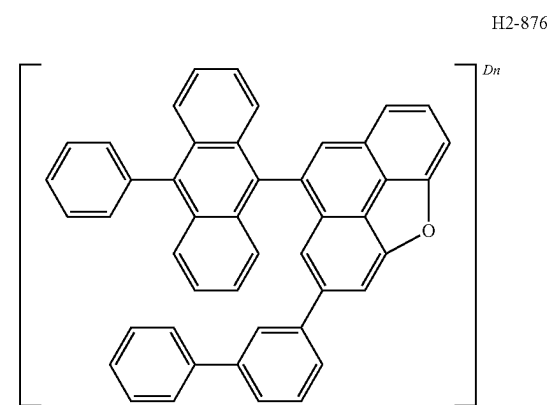
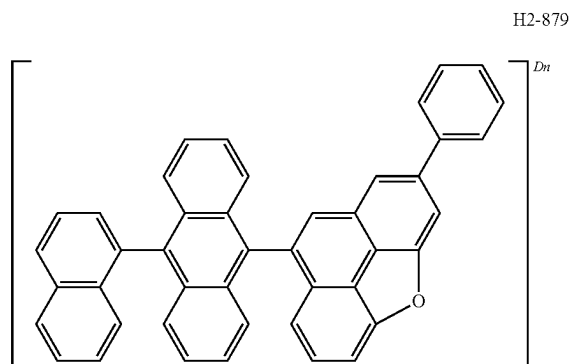
H2-870



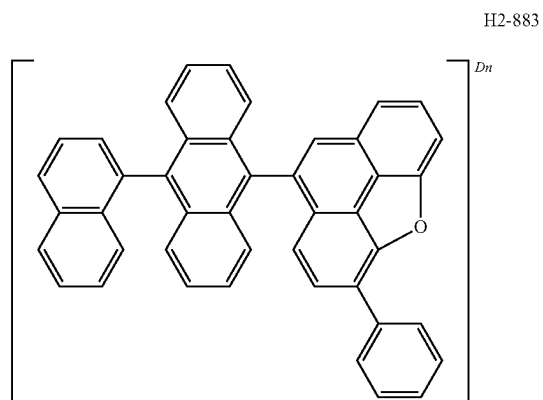
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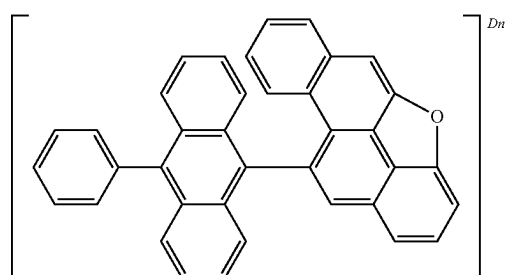
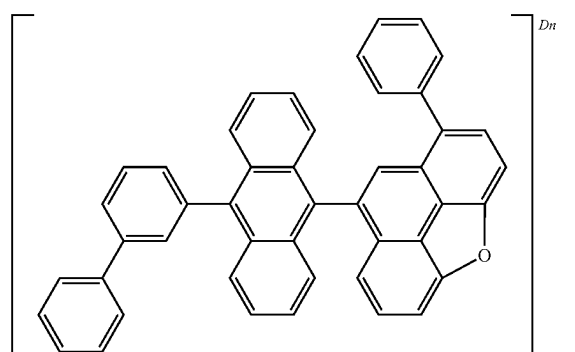
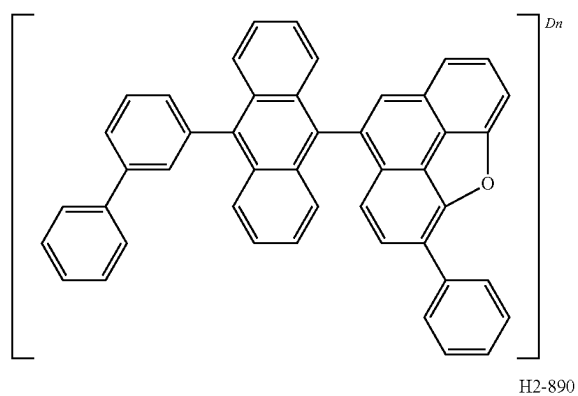
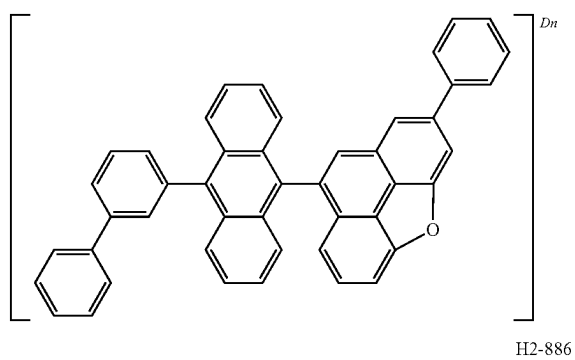
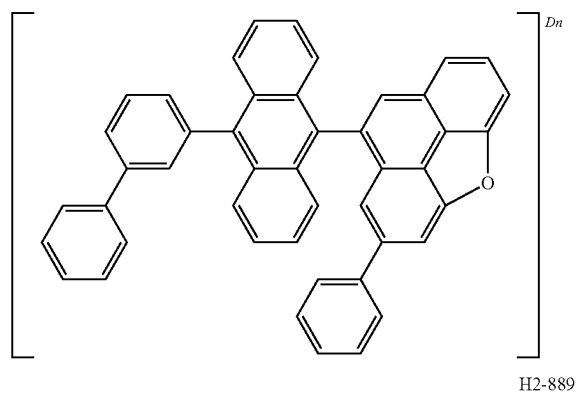
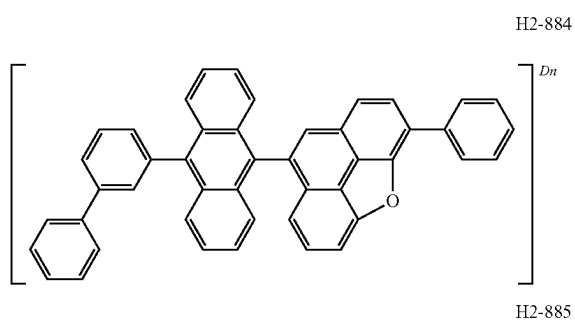
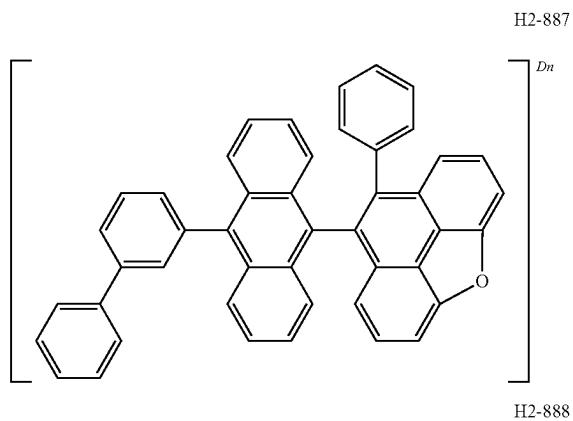
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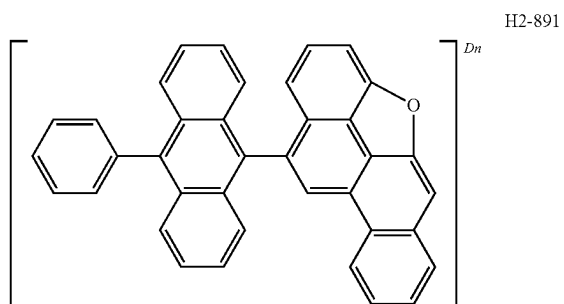
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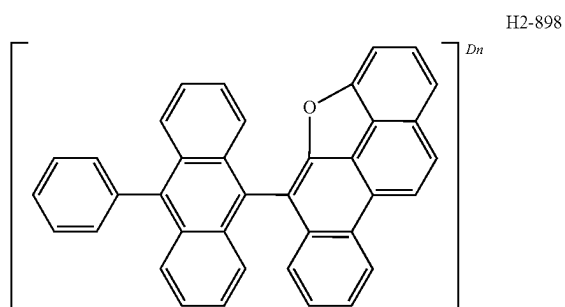
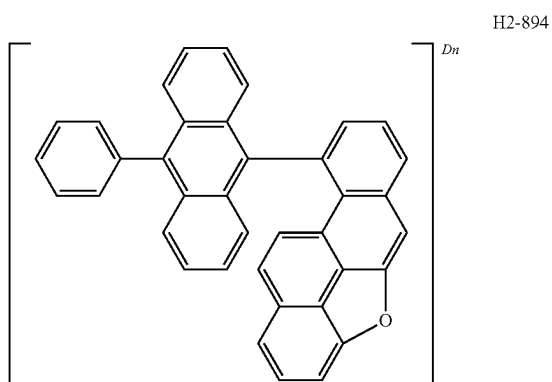
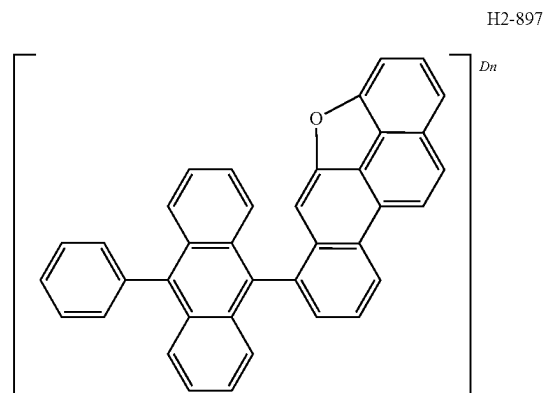
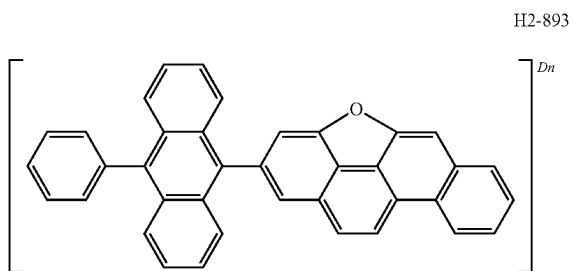
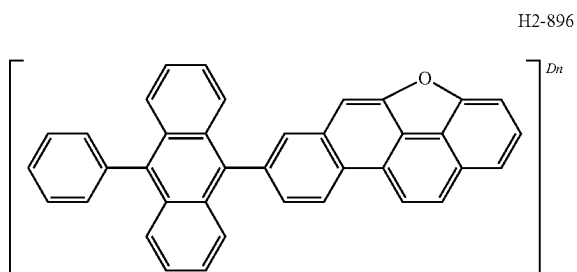
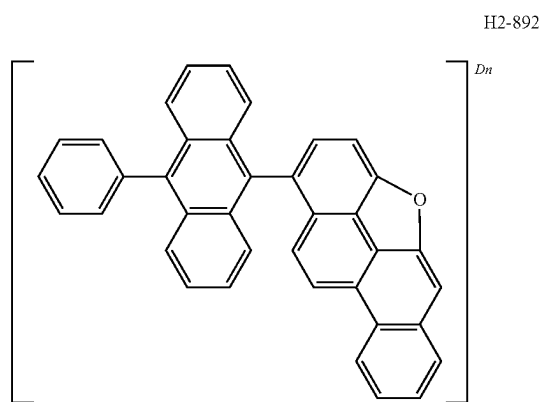
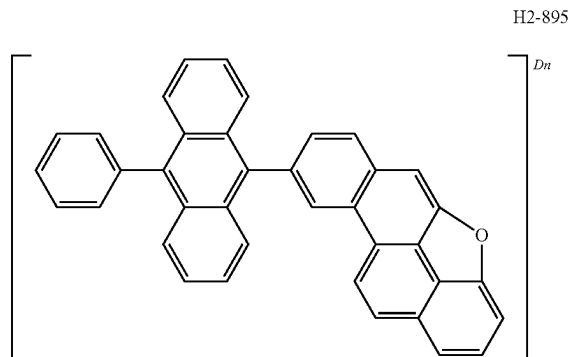
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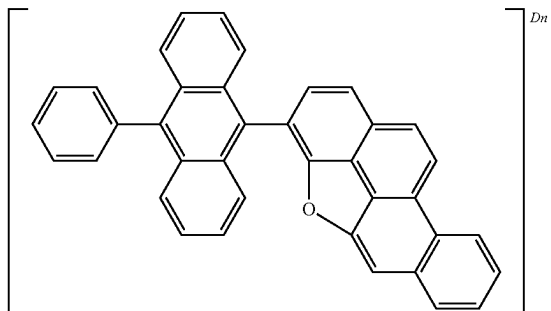


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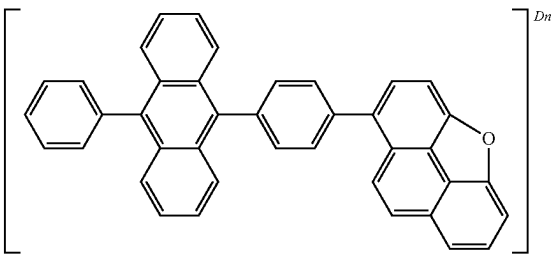


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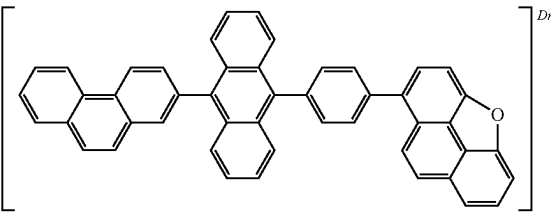
H2-899



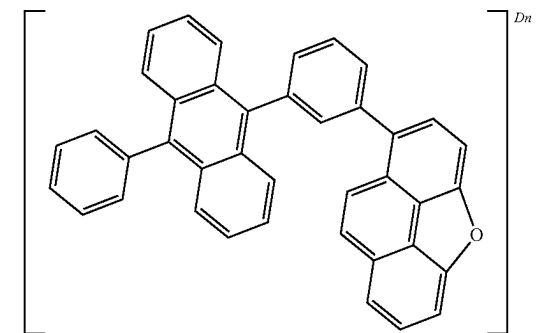
H2-900



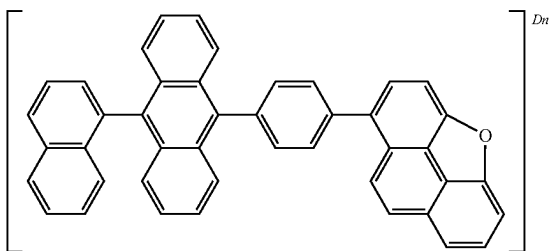
H2-901



H2-902

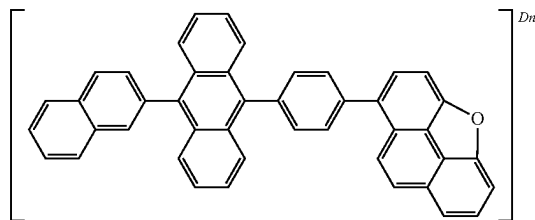


H2-903

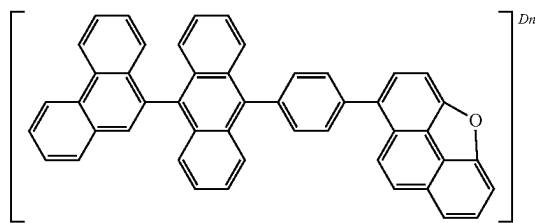


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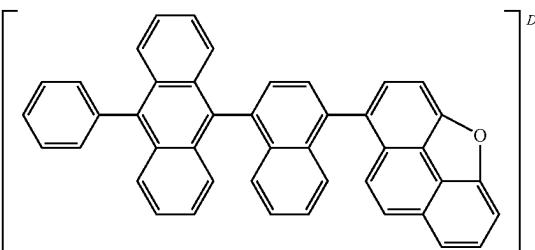
H2-904



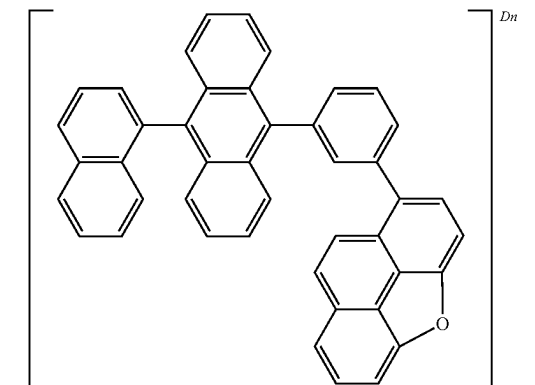
H2-905



H2-906

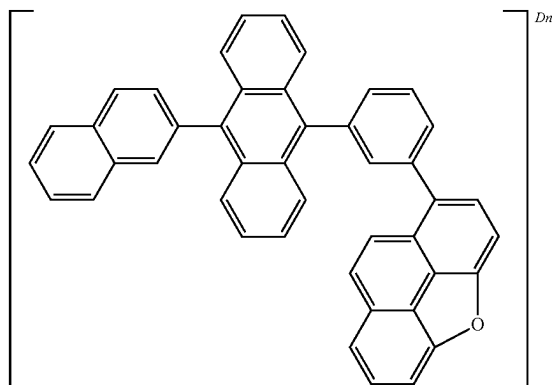


H2-907

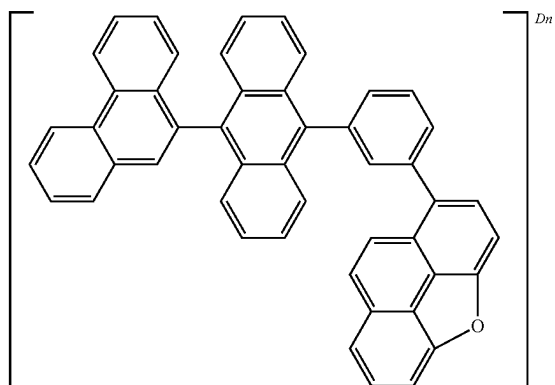


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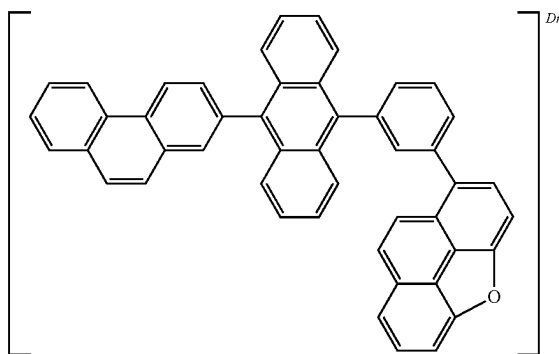
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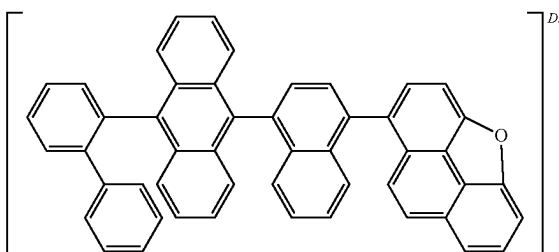
H2-909



H2-910

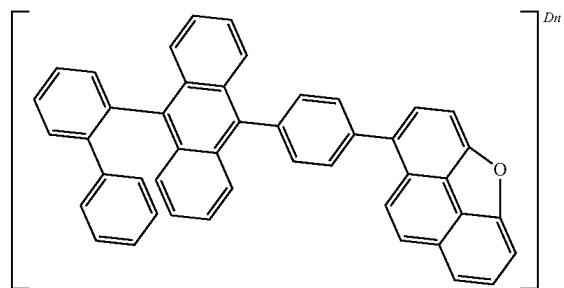


H2-911

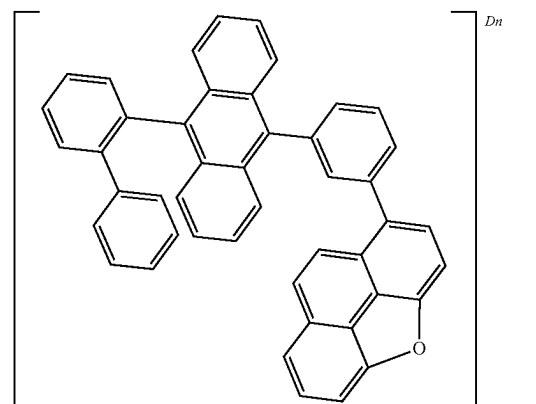


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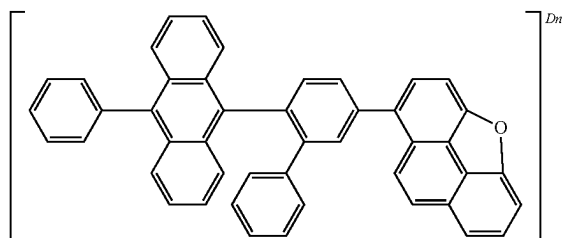
H2-912



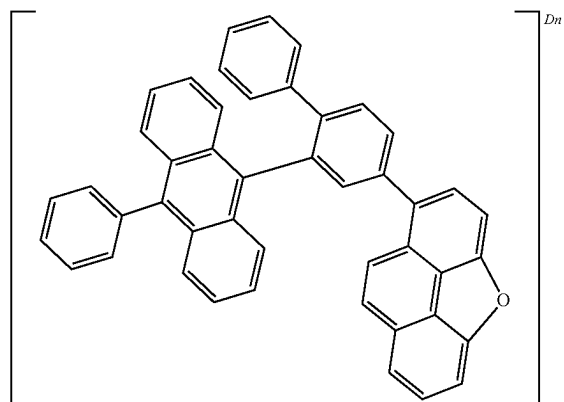
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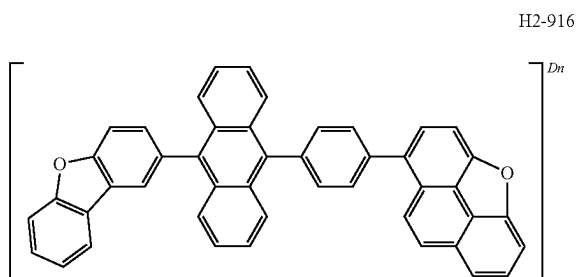
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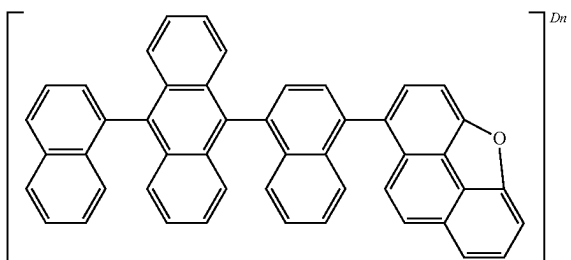
H2-915



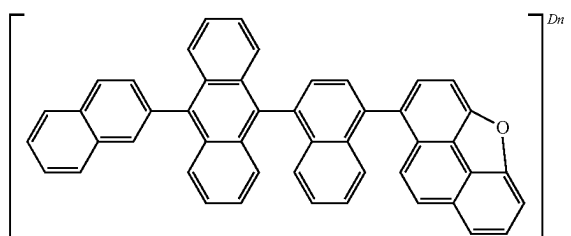
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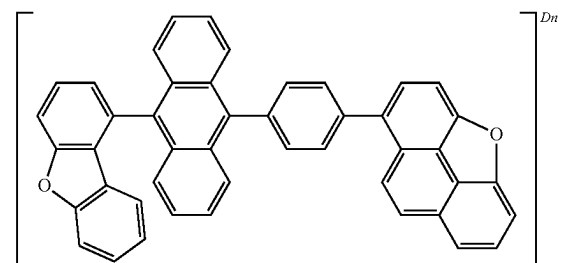
H2-917



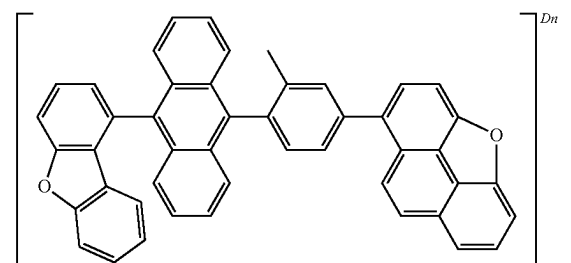
H2-918



H2-919

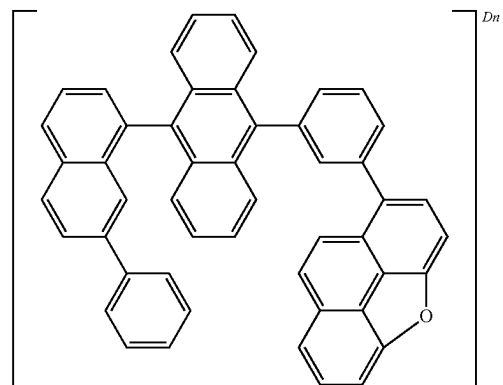


H2-920

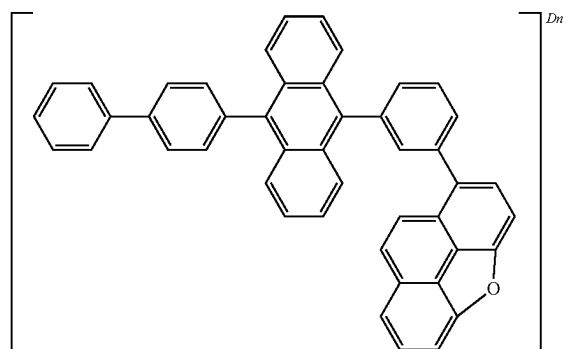


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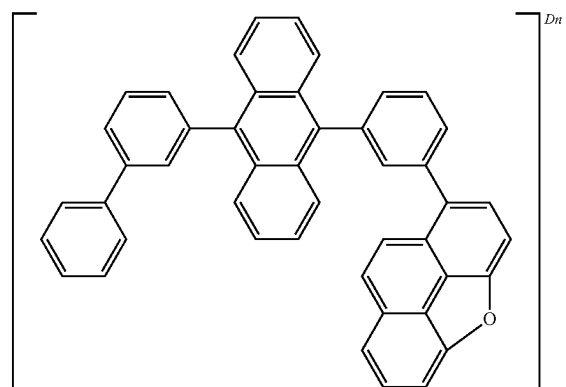
H2-921



H2-922

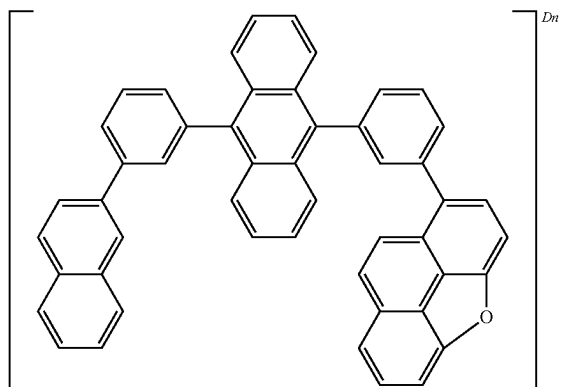


H2-923

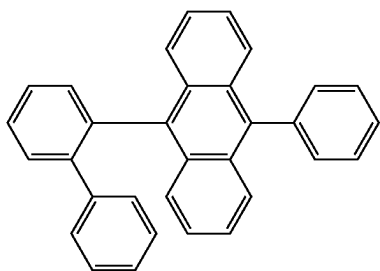


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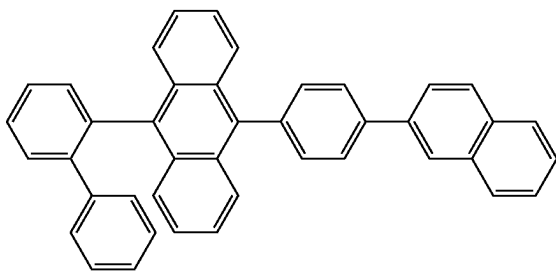
H2-924



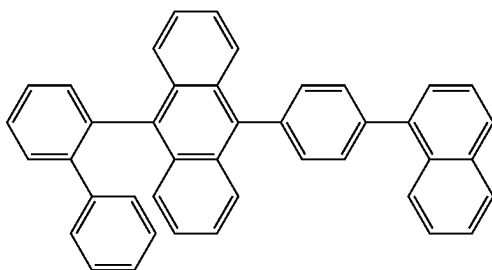
H1-1



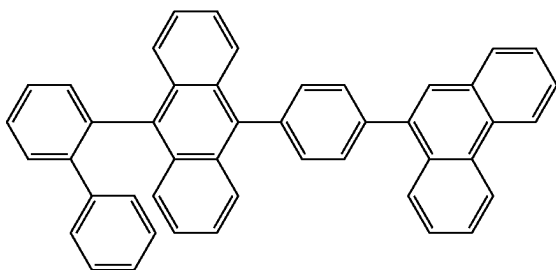
H1-2



H1-3

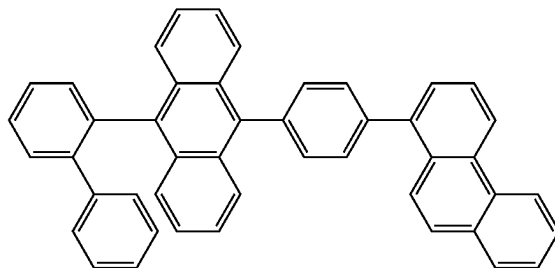


H1-4

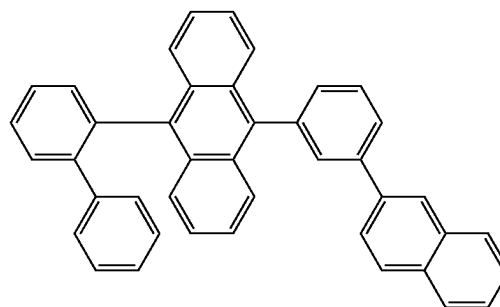


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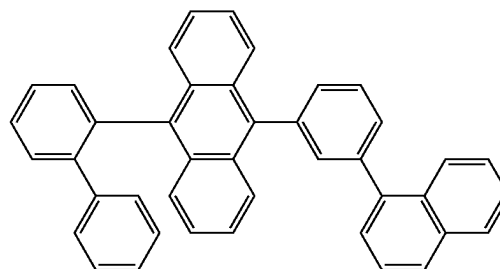
H1-5



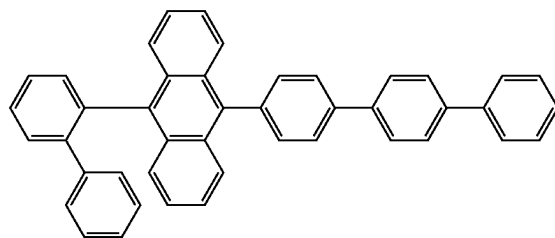
H1-6



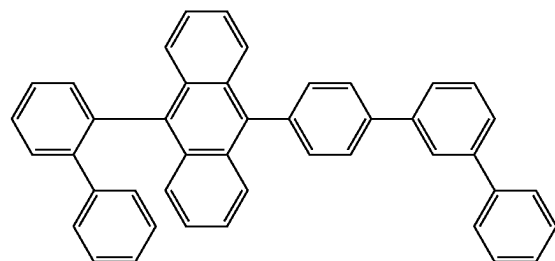
H1-7



H1-8

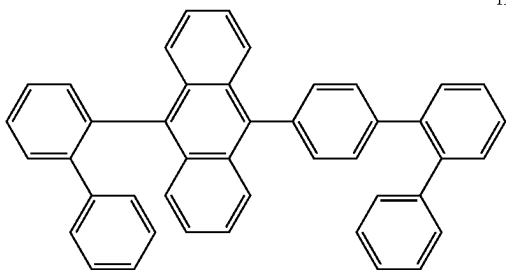


H1-9

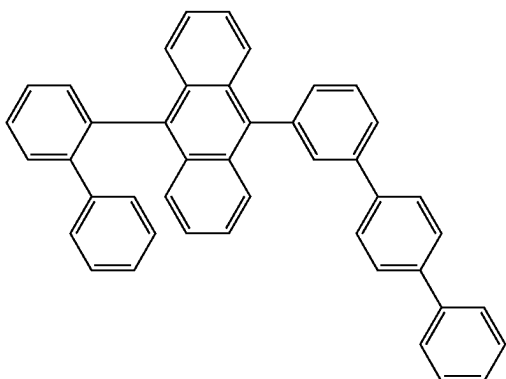


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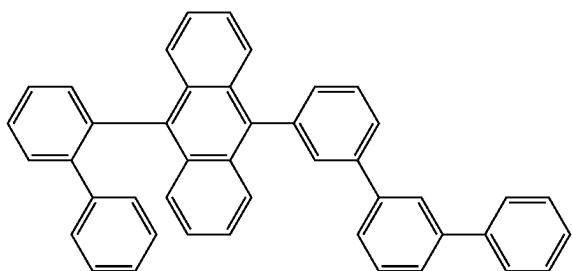
H1-10



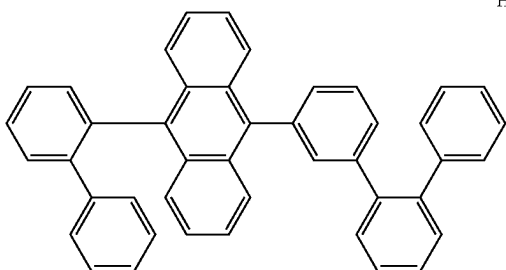
H1-11



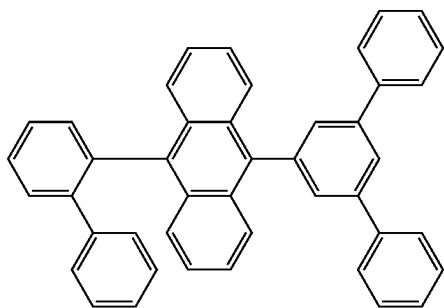
H1-12



H1-13

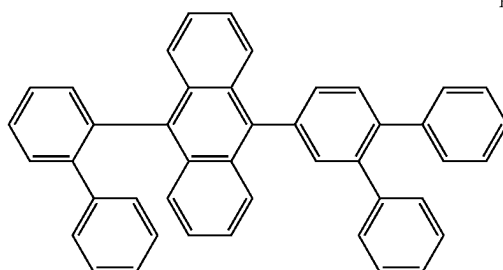


H1-14

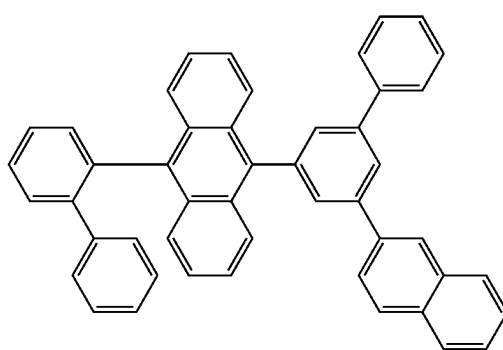


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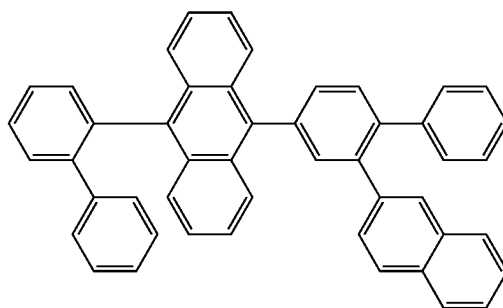
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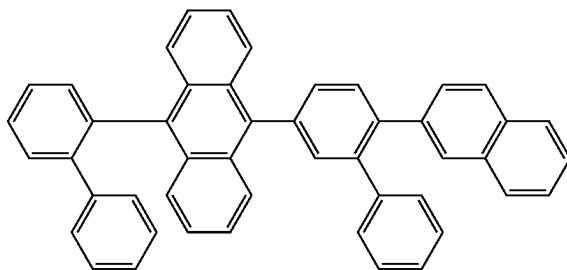
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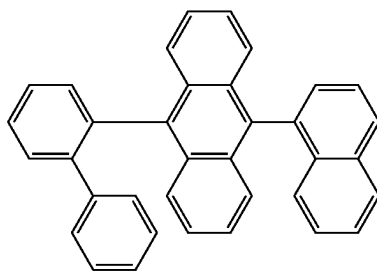
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H1-18

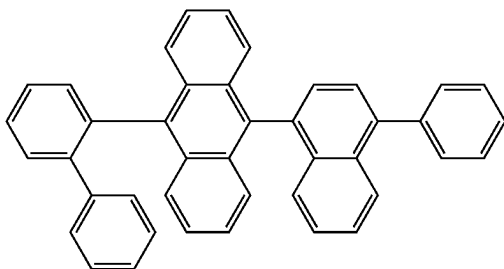


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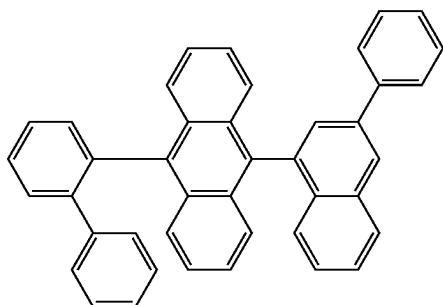


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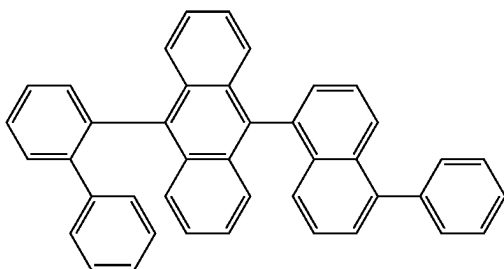
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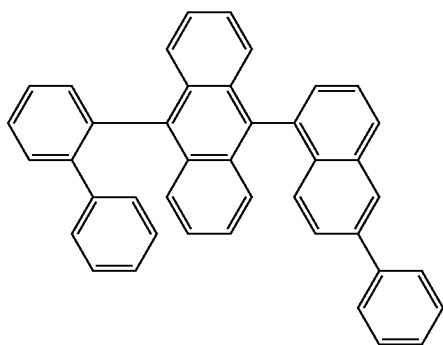
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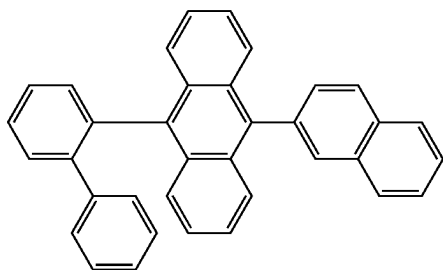
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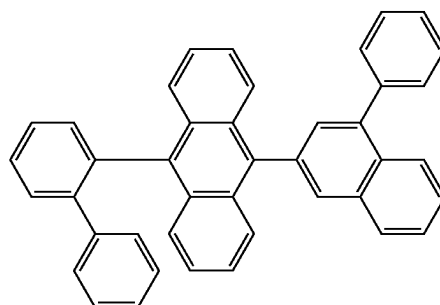


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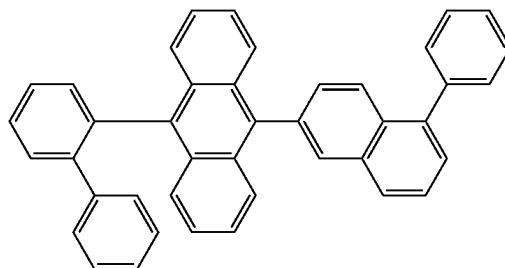


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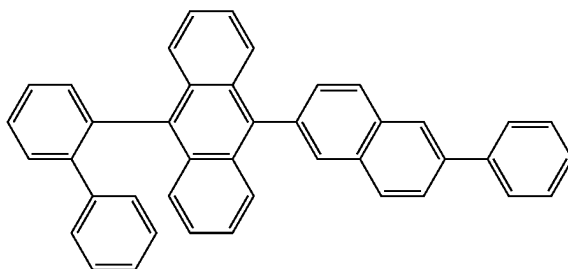
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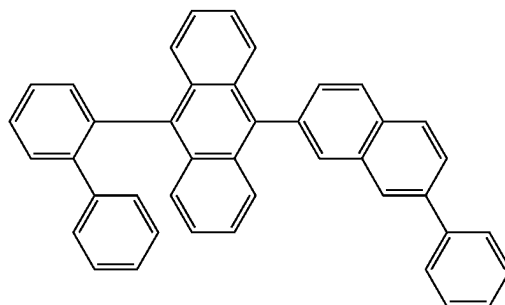
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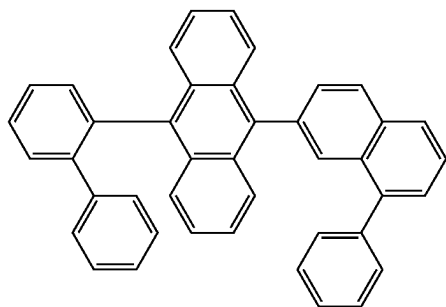
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H1-28

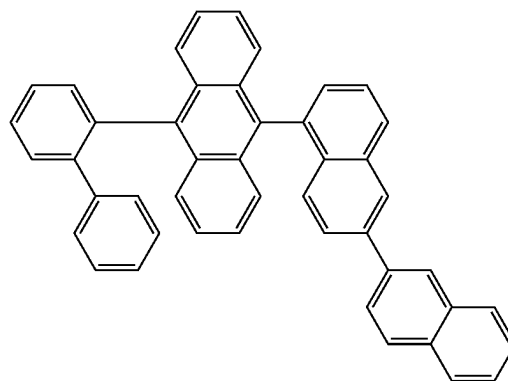


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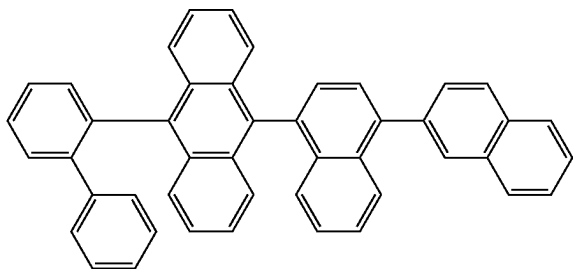


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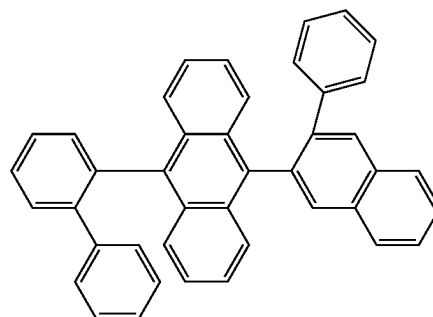
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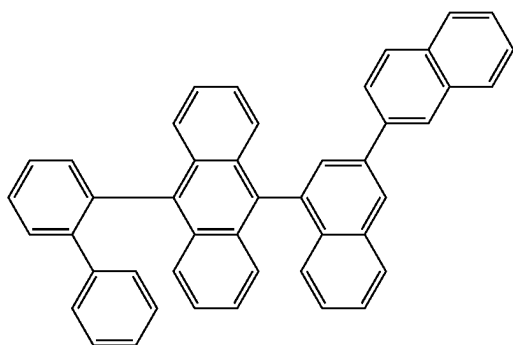
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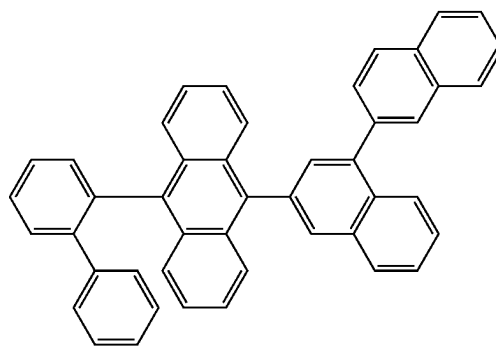
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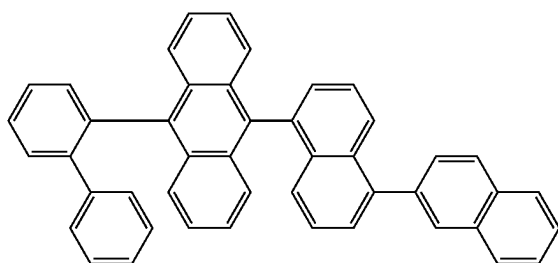
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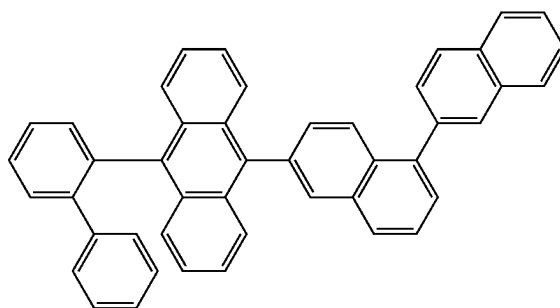
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H1-35



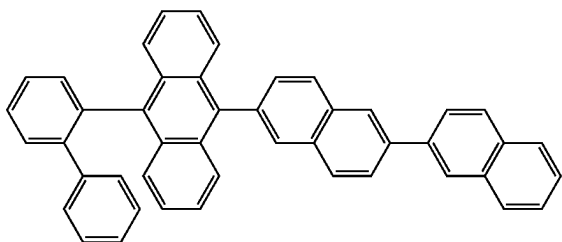
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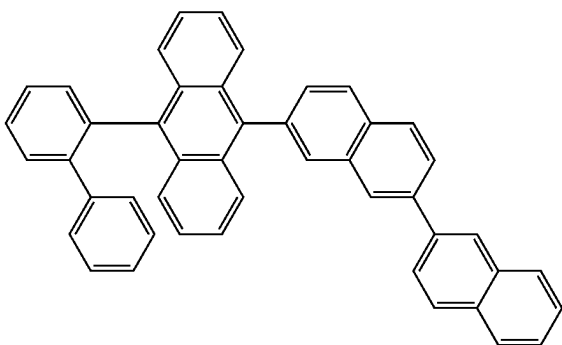
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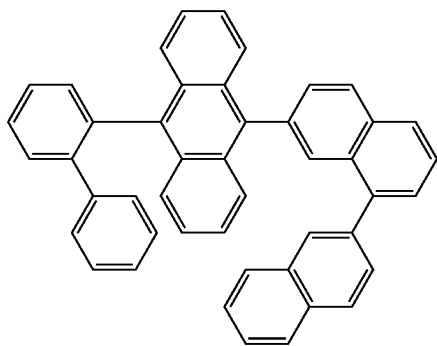
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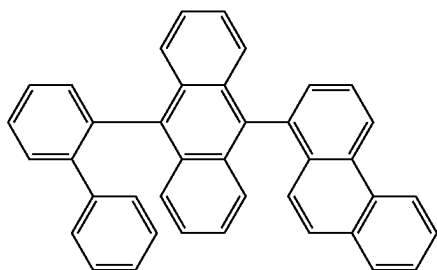
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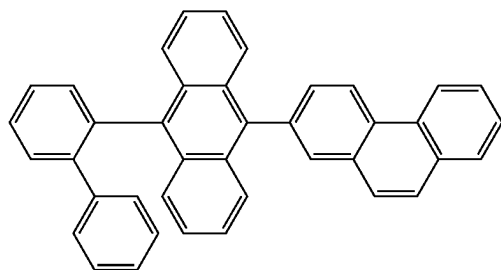
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H1-40

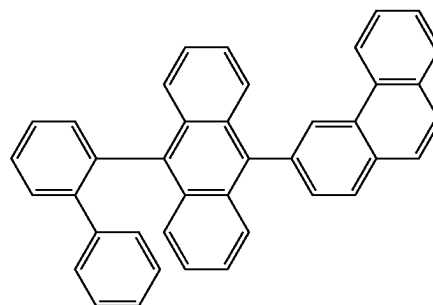


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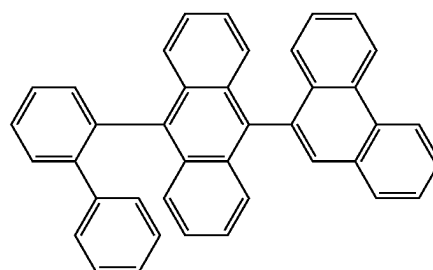


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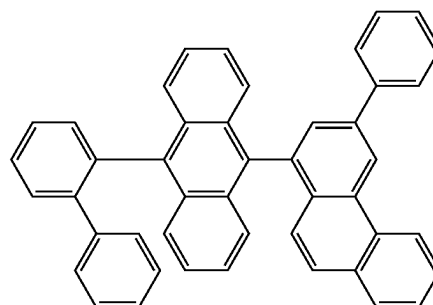
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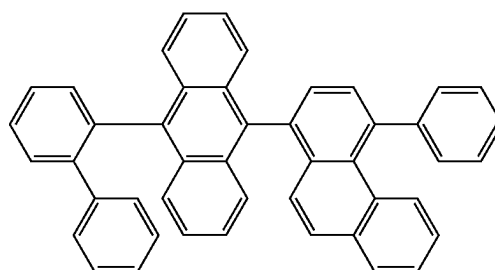
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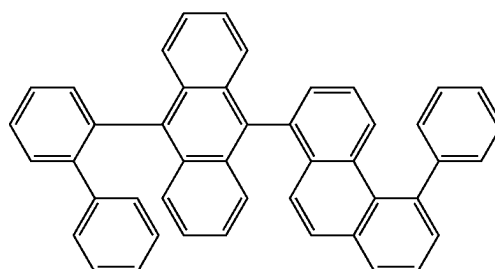
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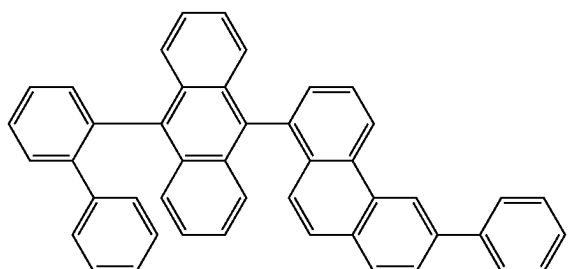
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H1-46

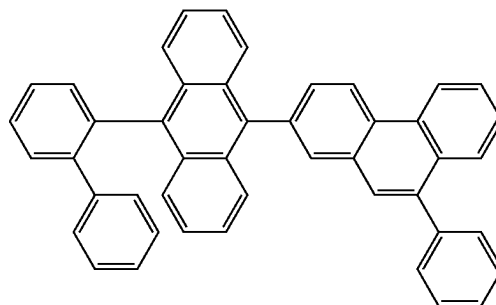


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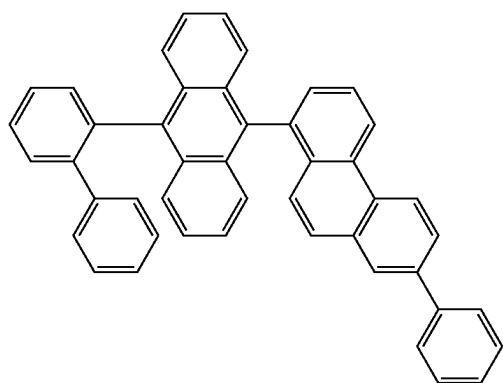
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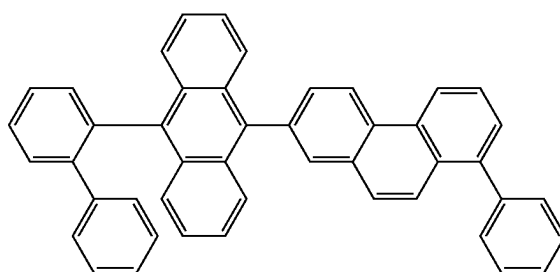


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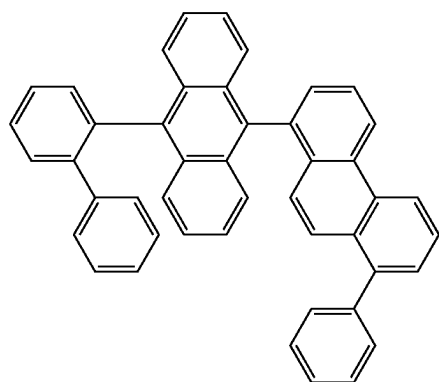
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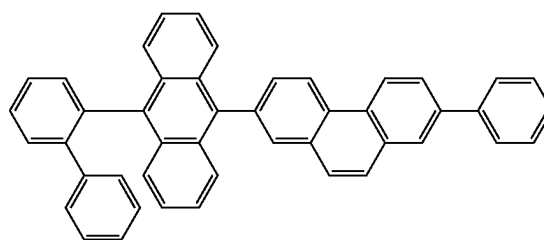
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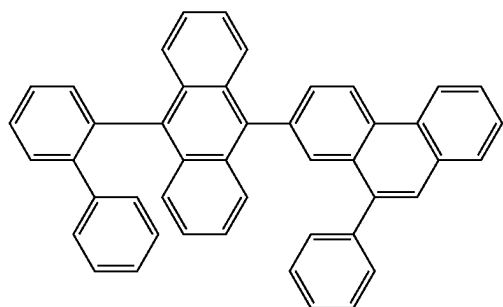
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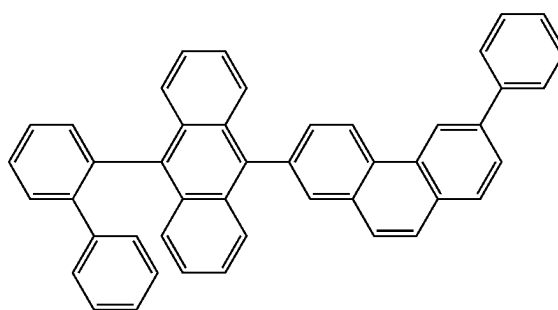
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H1-54



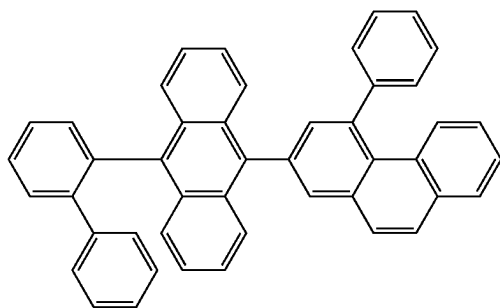
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H1-55

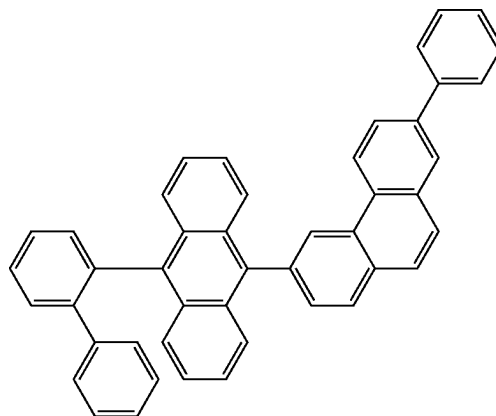
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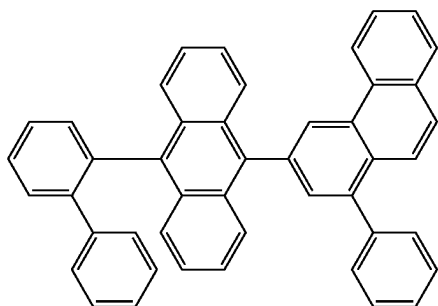


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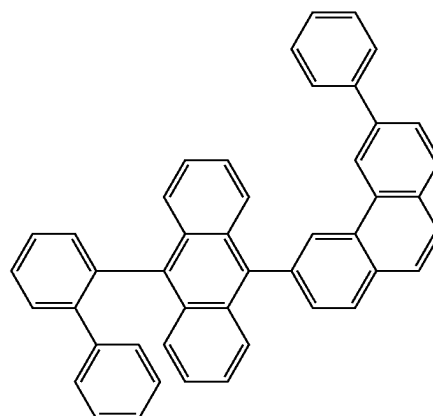
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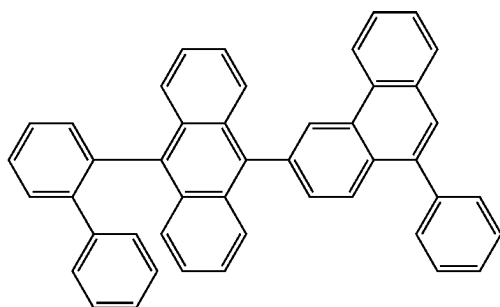
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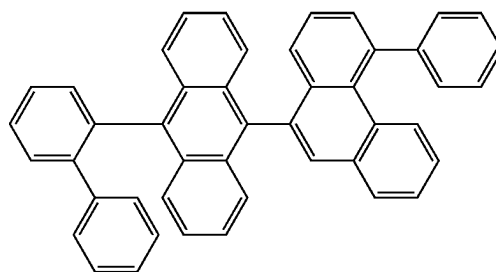
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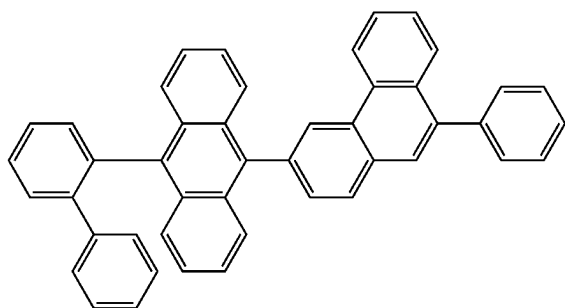
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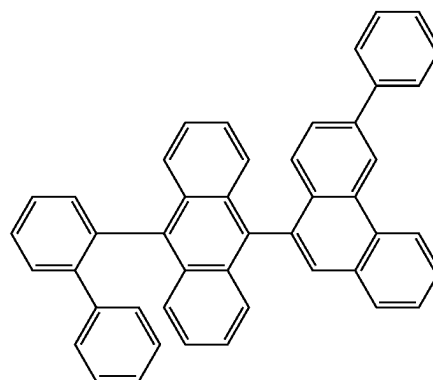
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H1-59

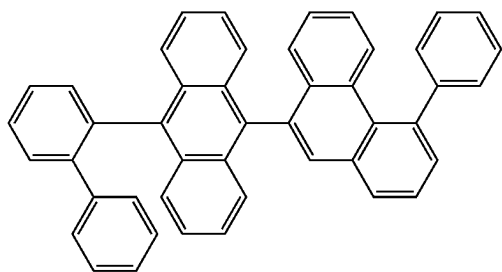


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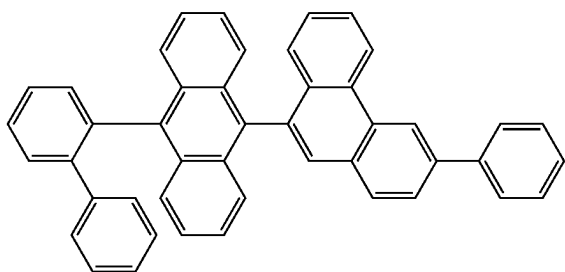


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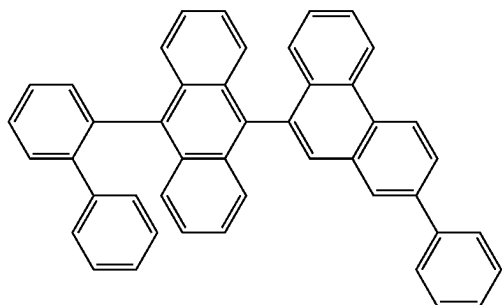
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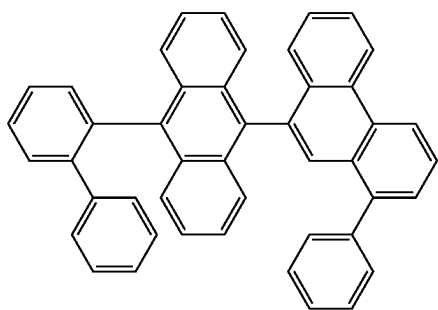
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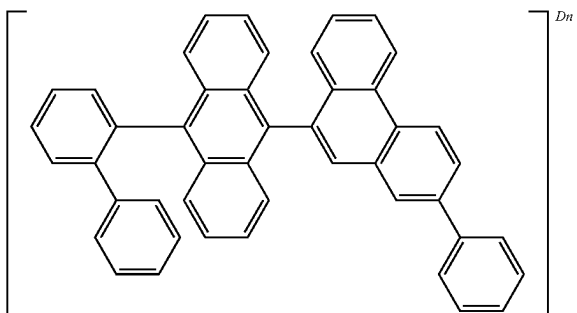
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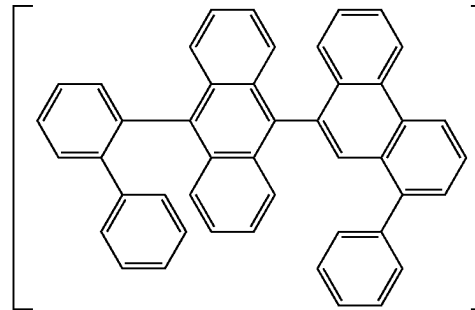
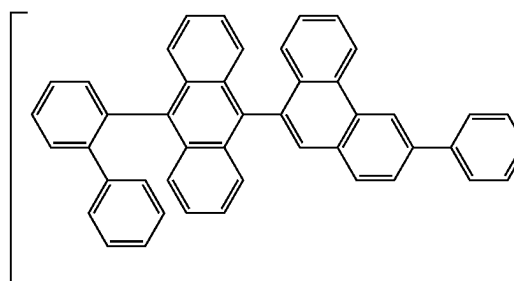
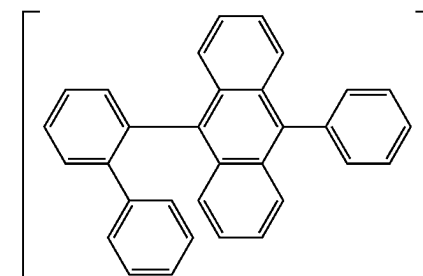
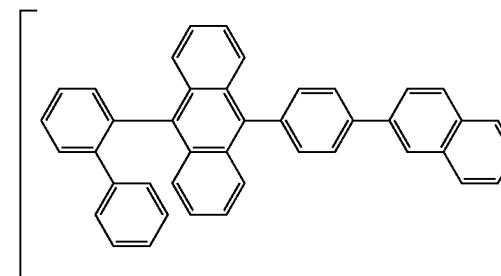
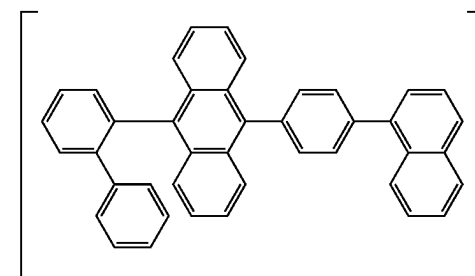
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H1-68

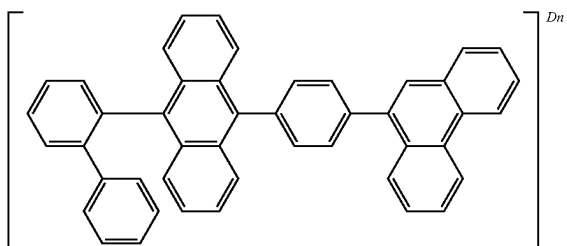


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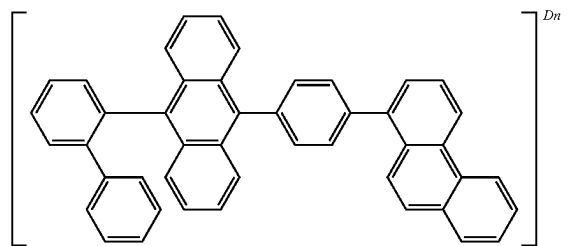
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DnH1-73
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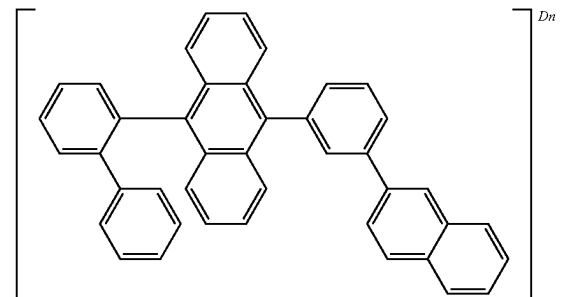
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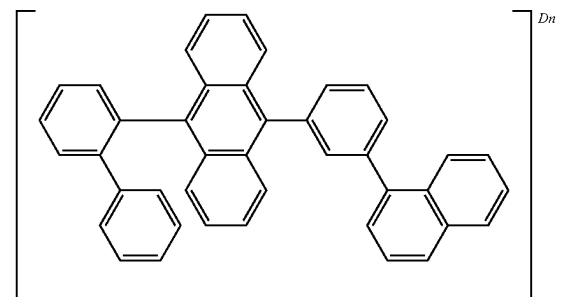
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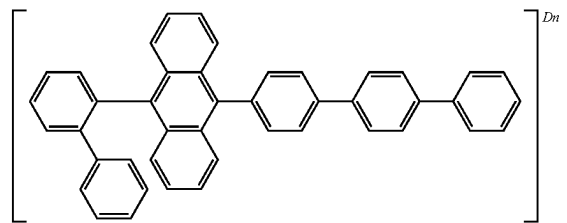
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H1-77

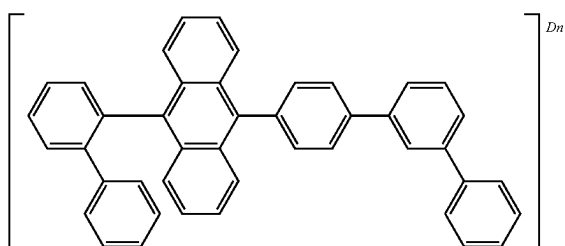


H1-78

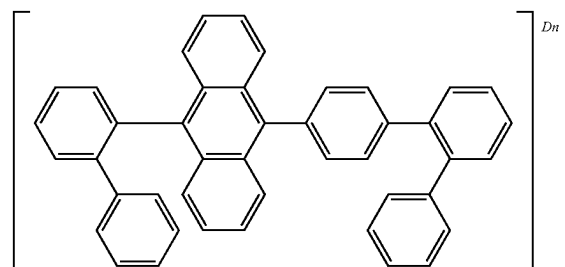


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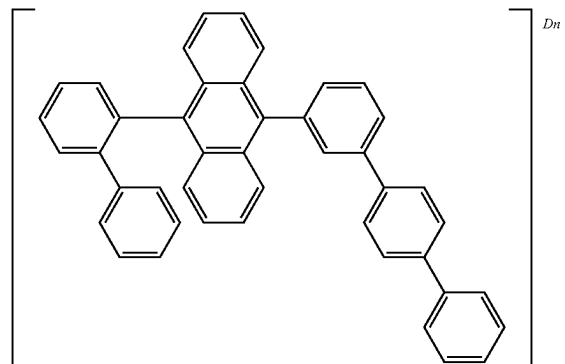
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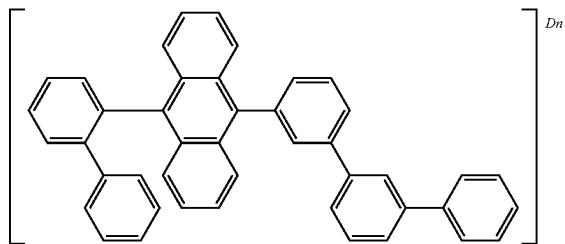
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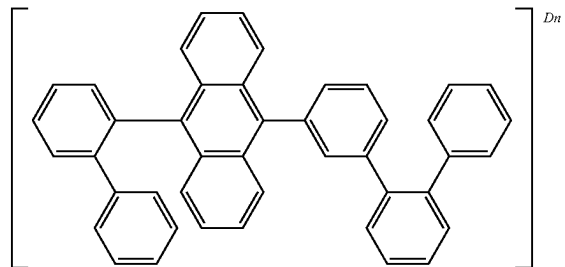
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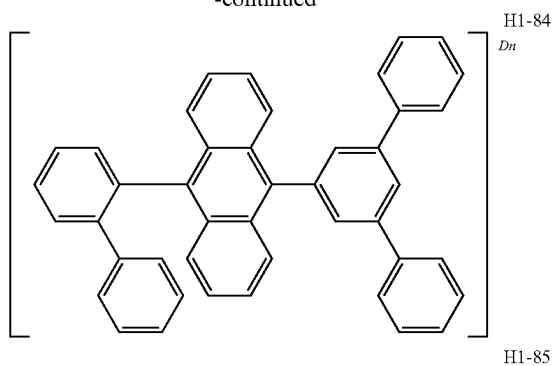
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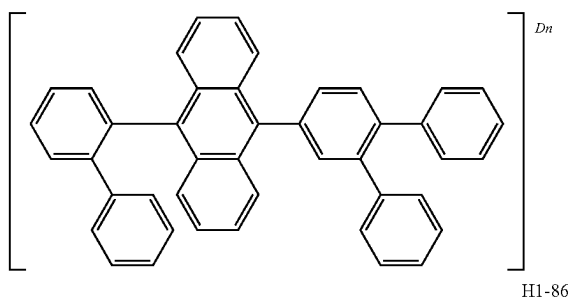
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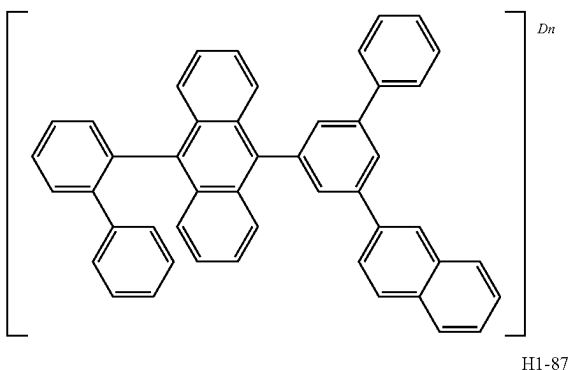
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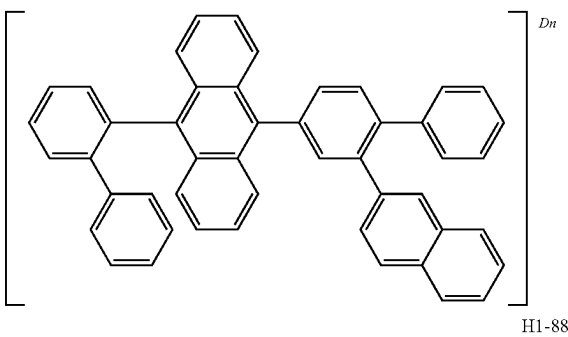
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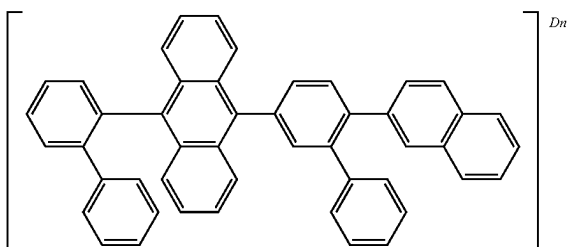
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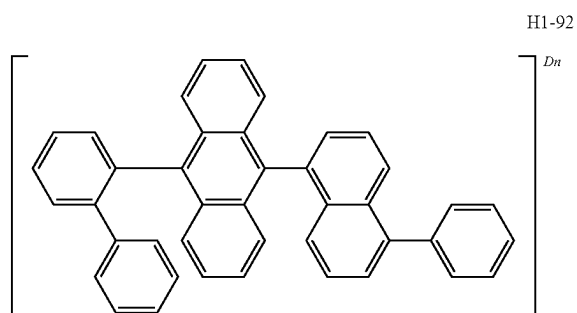
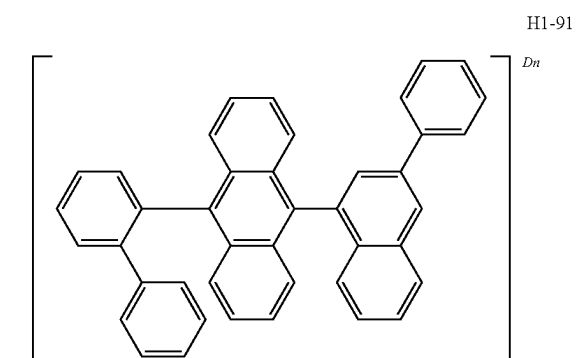
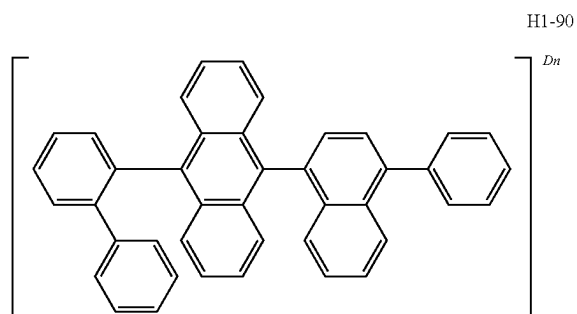
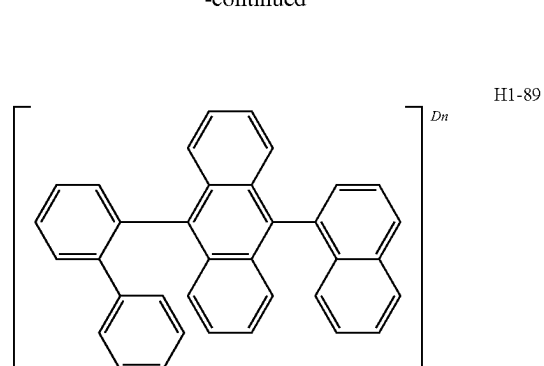
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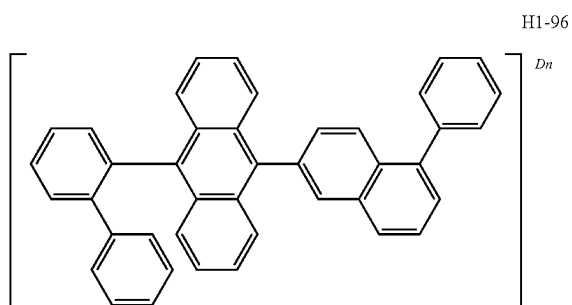
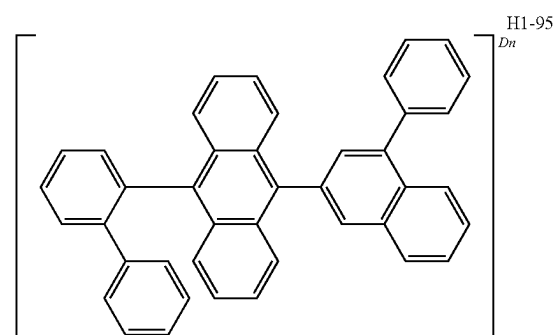
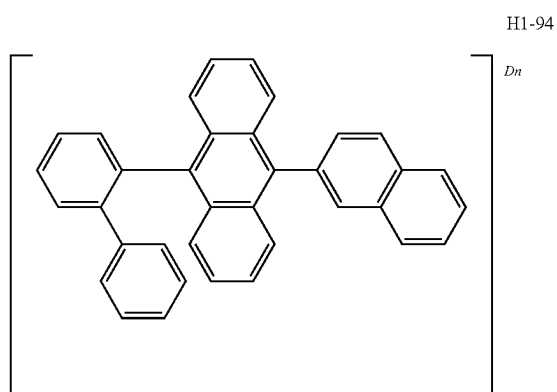
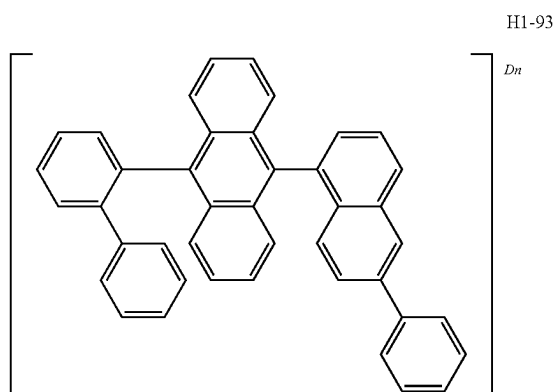
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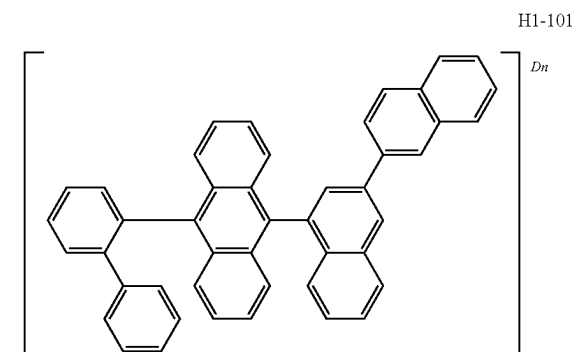
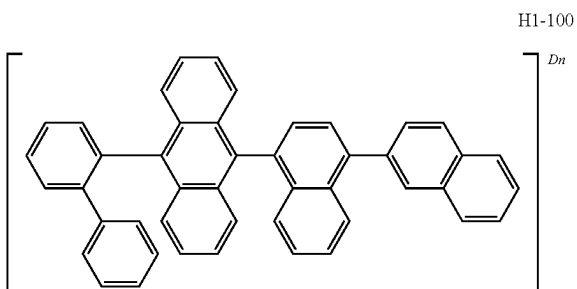
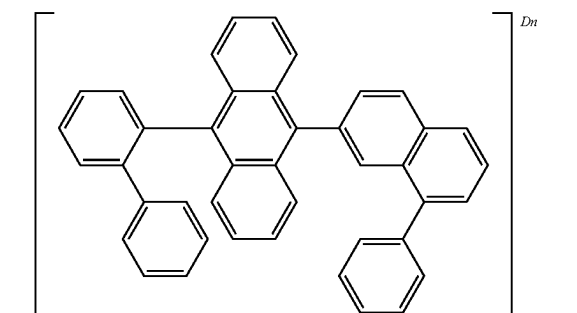
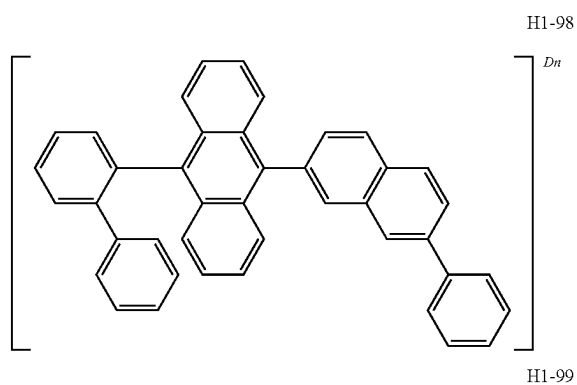
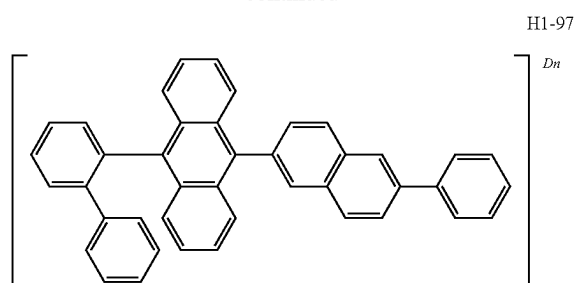
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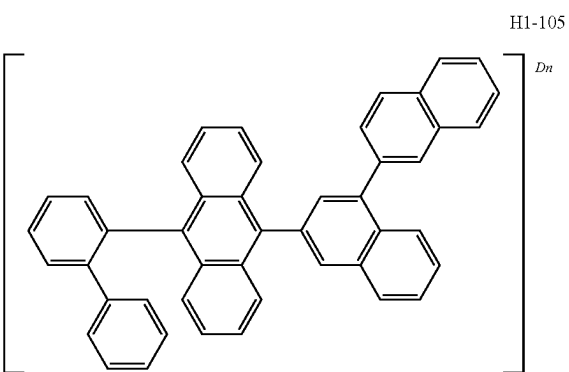
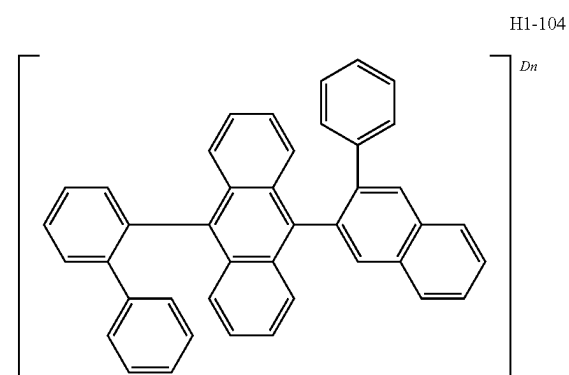
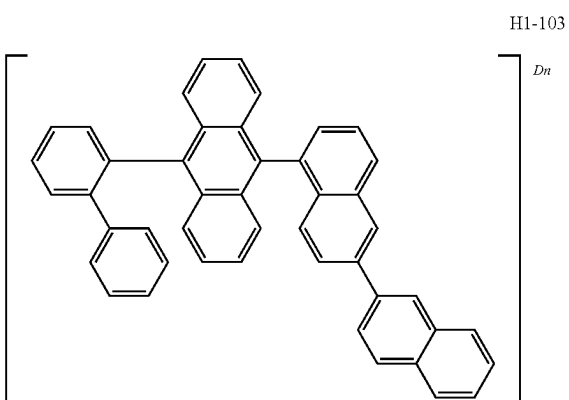
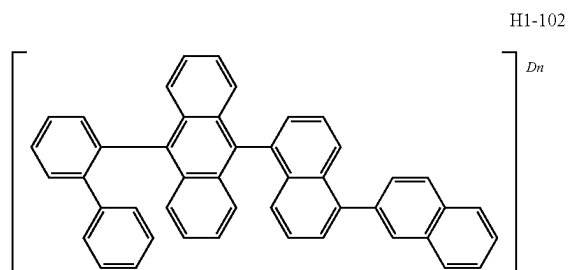
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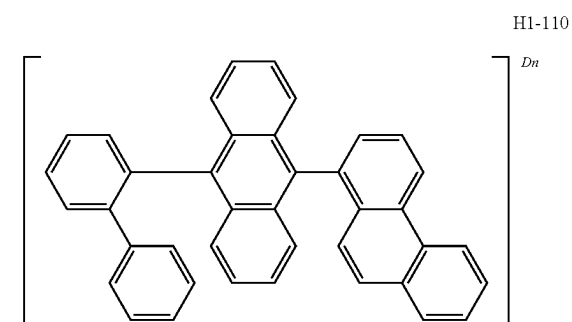
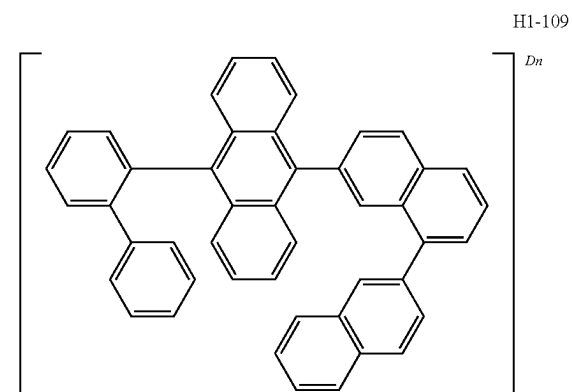
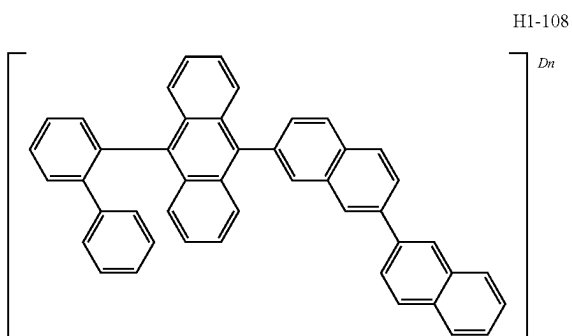
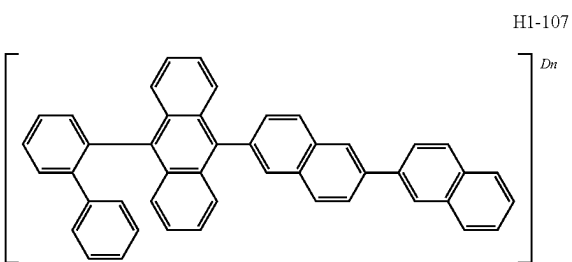
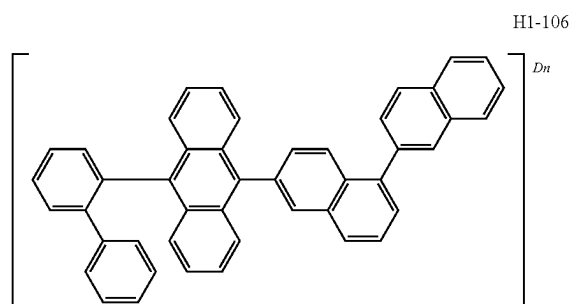
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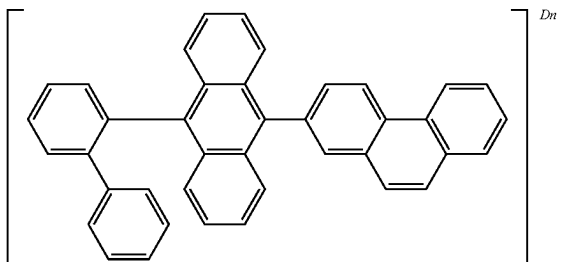


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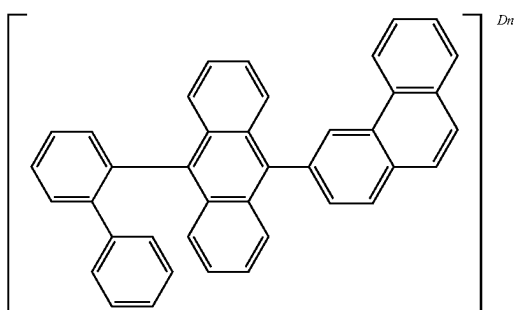


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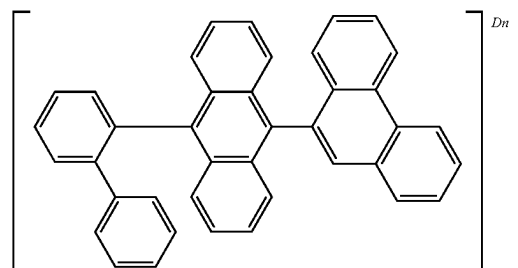
H1-111



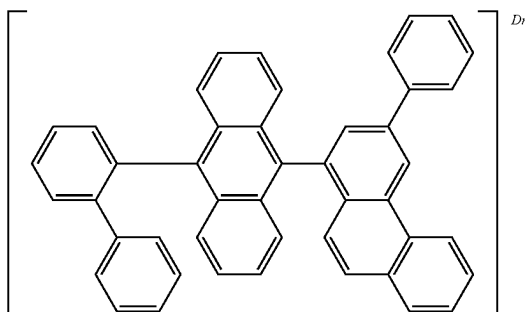
H1-112



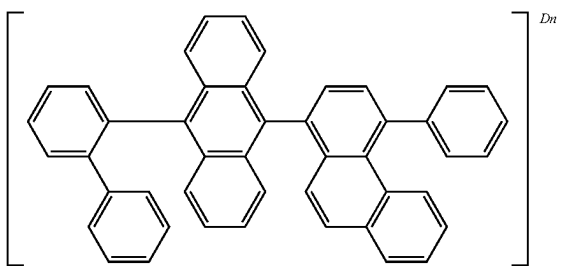
H1-113



H1-114

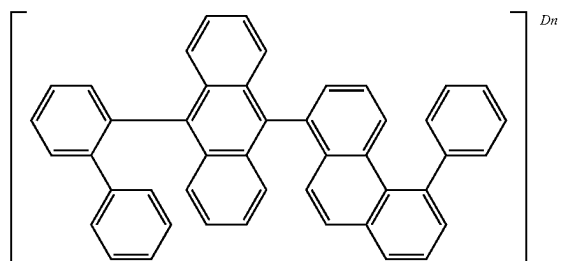


H1-115

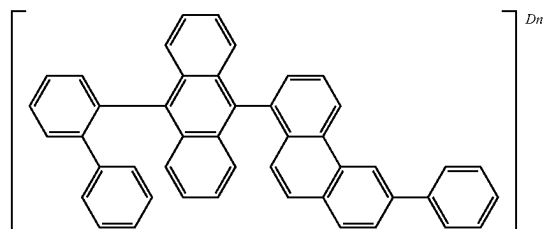


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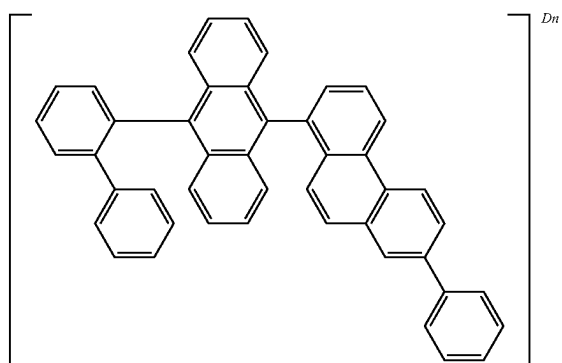
H1-116



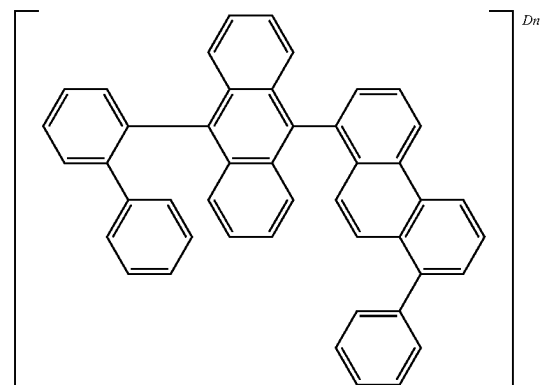
H1-117



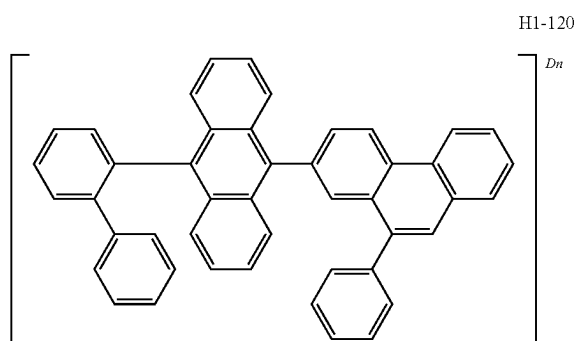
H1-118



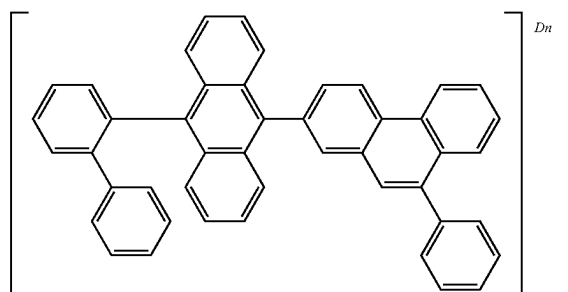
H1-119



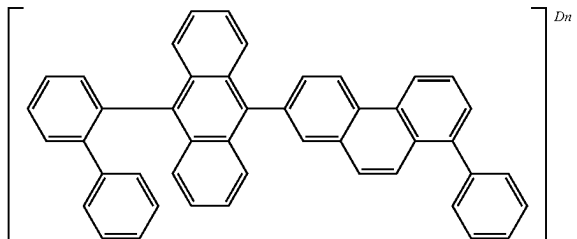
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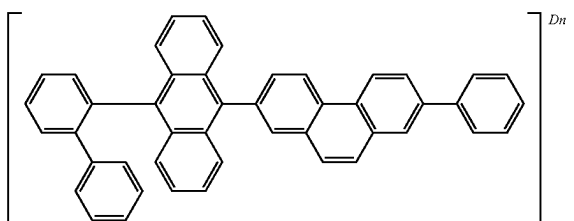
H1-121



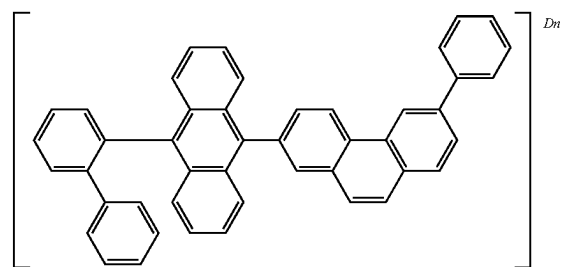
H1-122



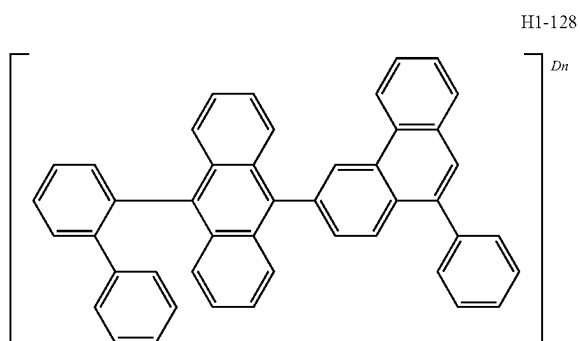
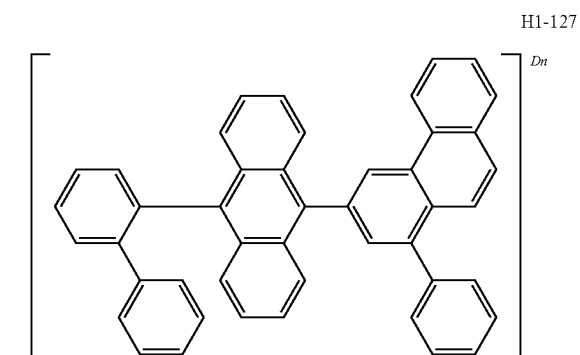
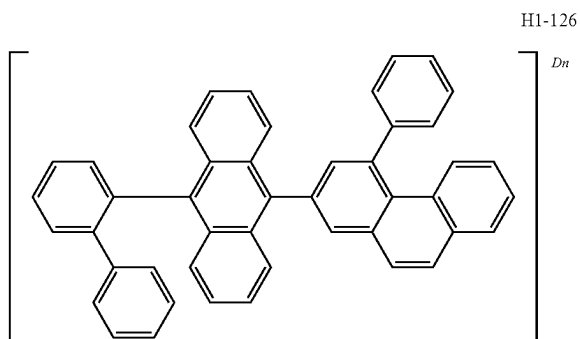
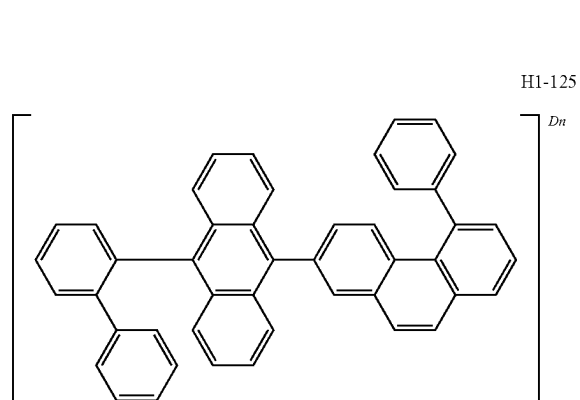
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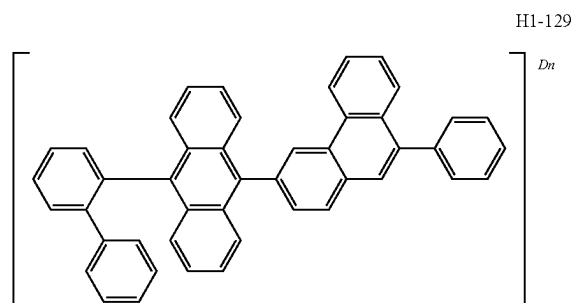
H1-124



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H1-130

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H1-131

D_n

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H1-132

D_n

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H1-133

D_n

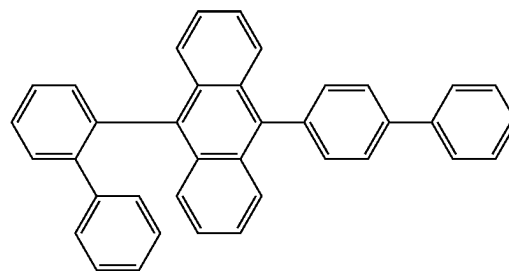
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H1-134

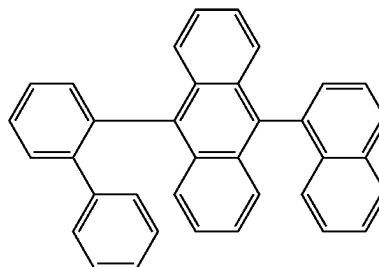
D_n

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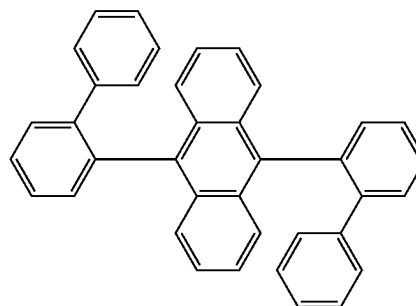
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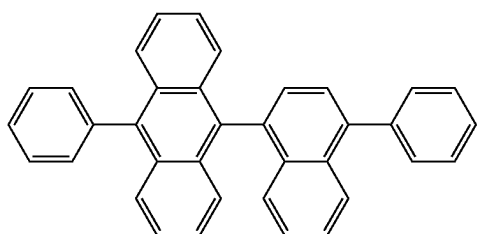
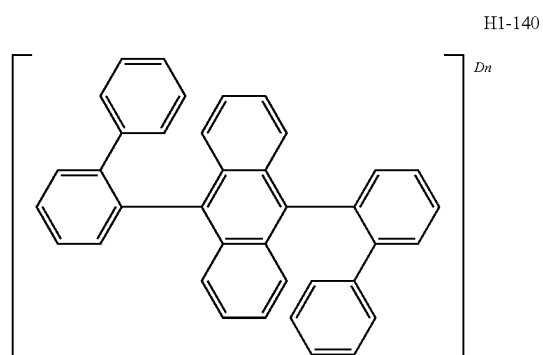
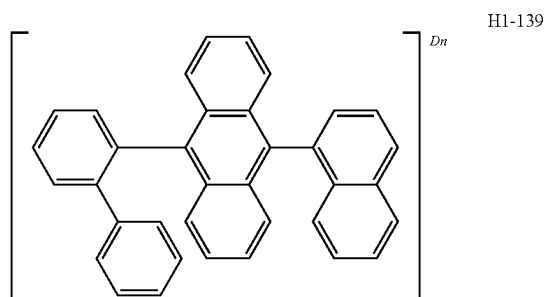
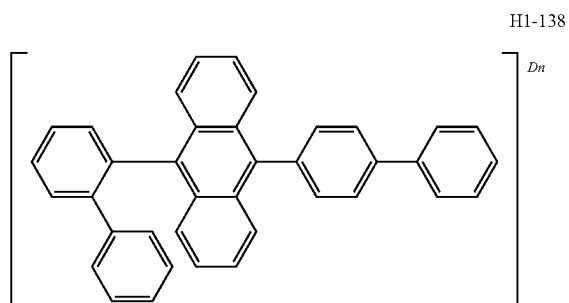
H1-136



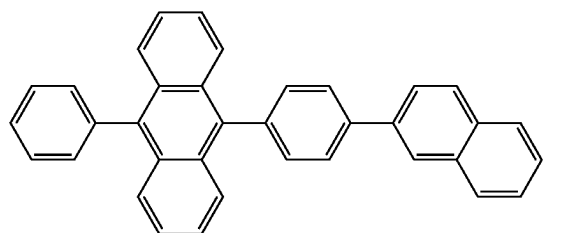
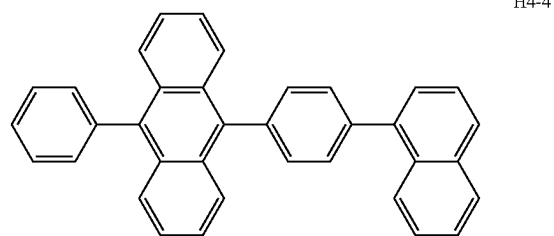
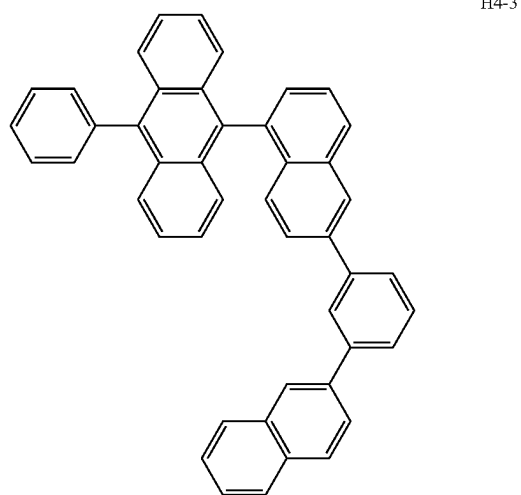
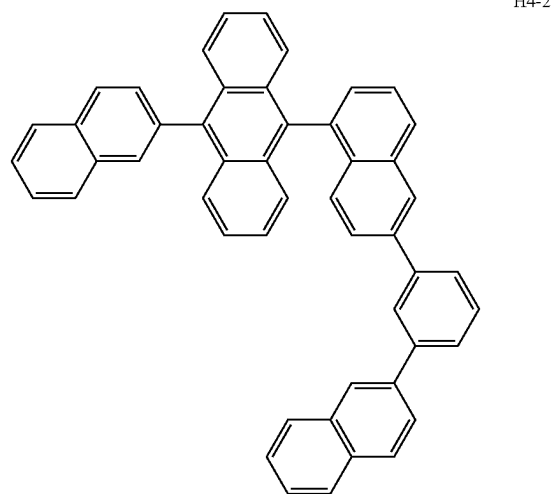
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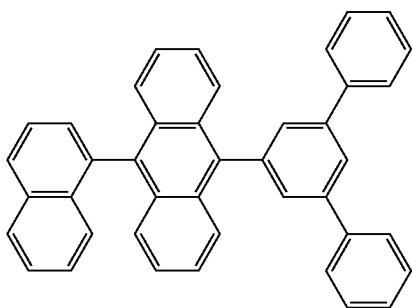
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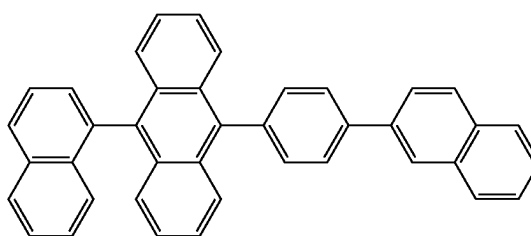


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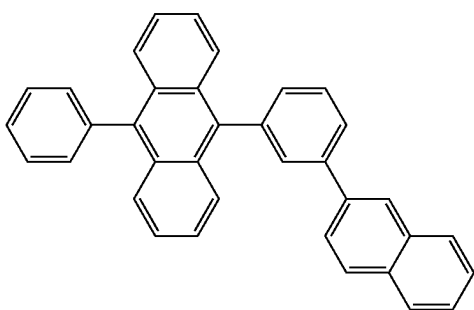


H4-6

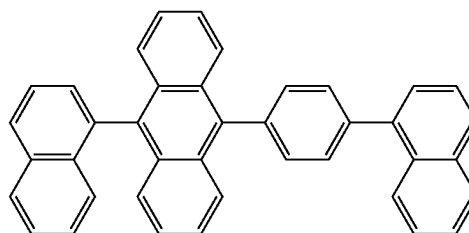
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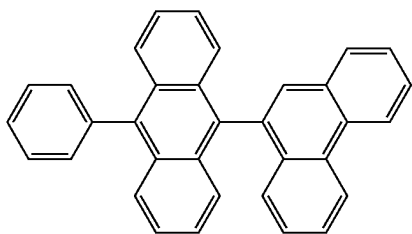
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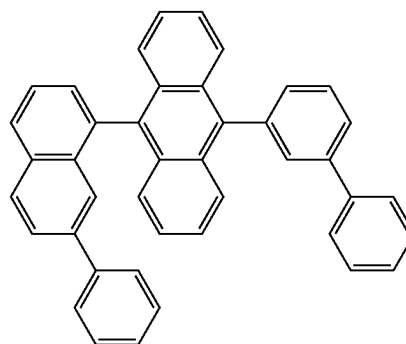
H4-7



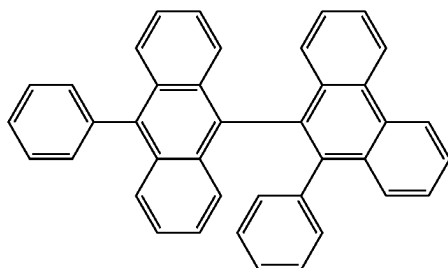
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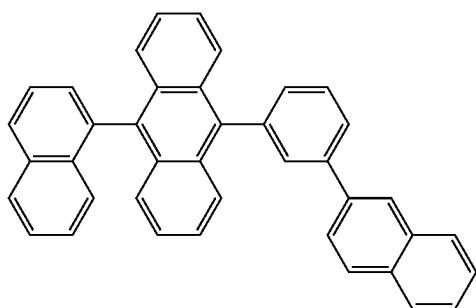
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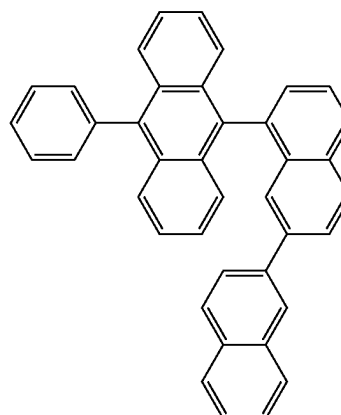
H4-13



H4-9



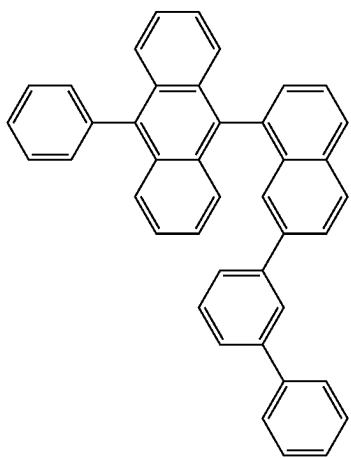
H4-10



H4-14

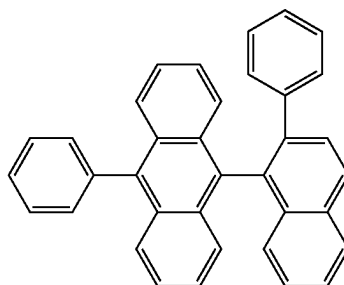
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H4-15

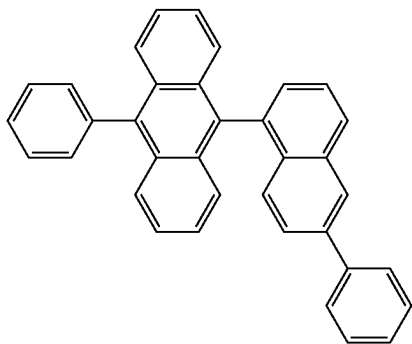


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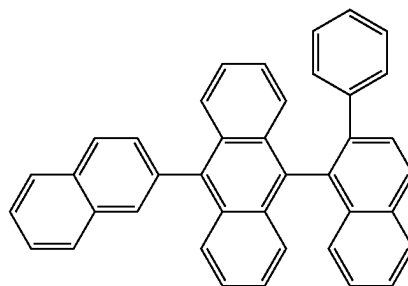
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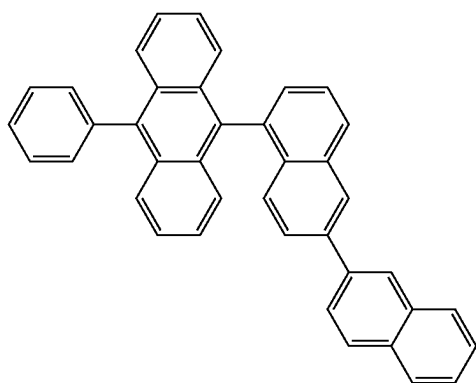
H4-16



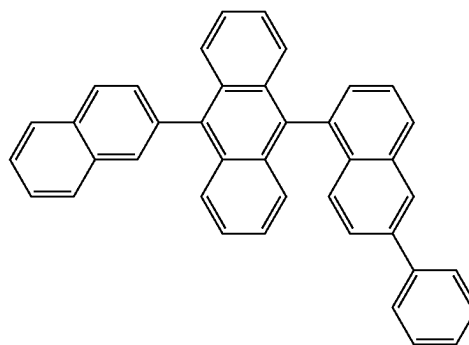
H4-20



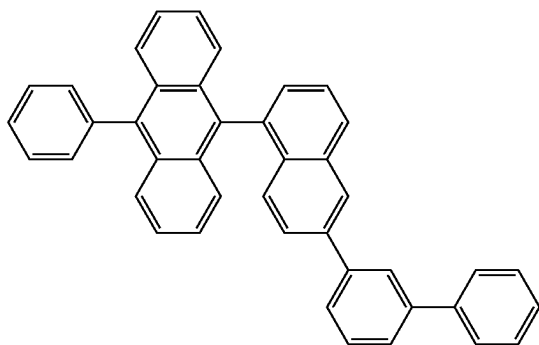
H4-17



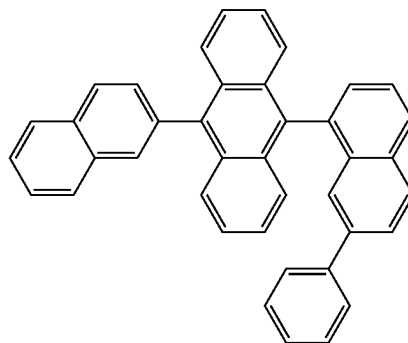
H4-21



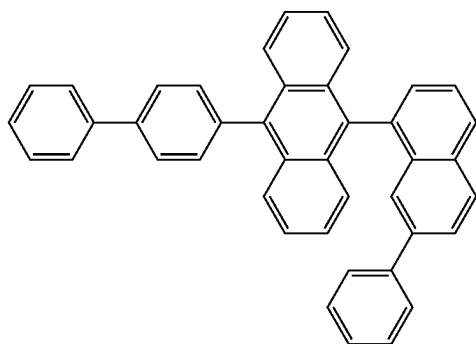
H4-18



H4-22

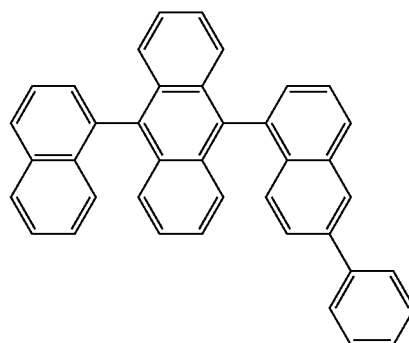


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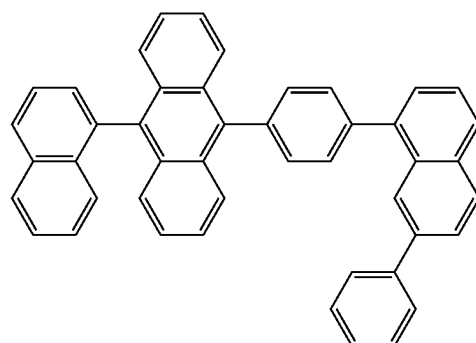
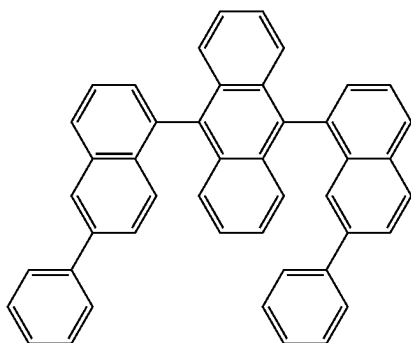
H4-23

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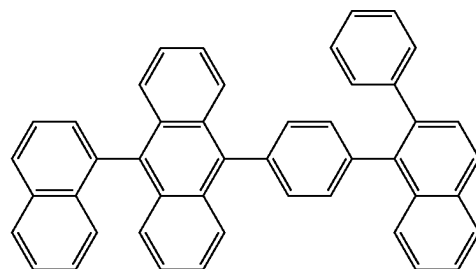
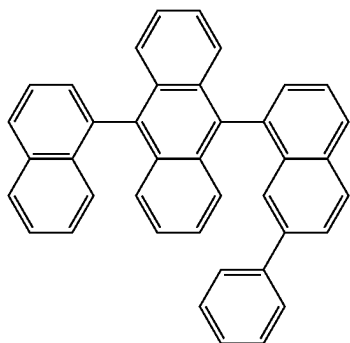
H4-27

H4-24



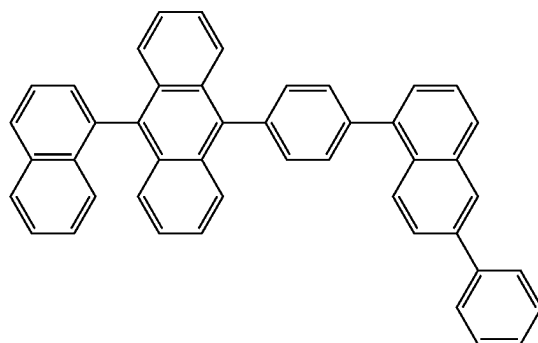
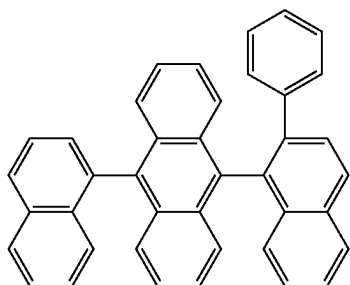
H4-28

H4-25

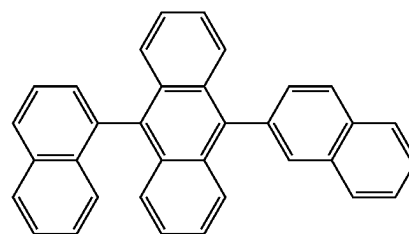


H4-29

H4-26

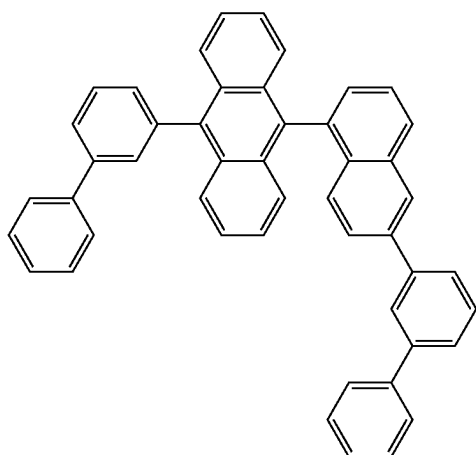


H4-30



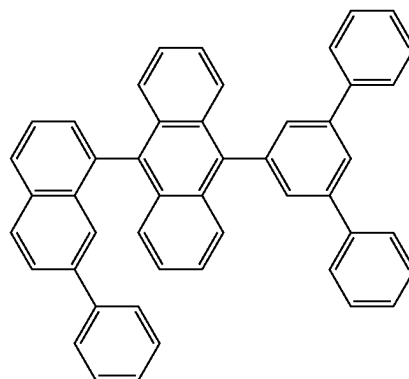
h4-31

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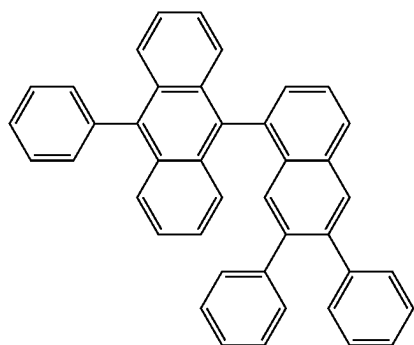


H4-32

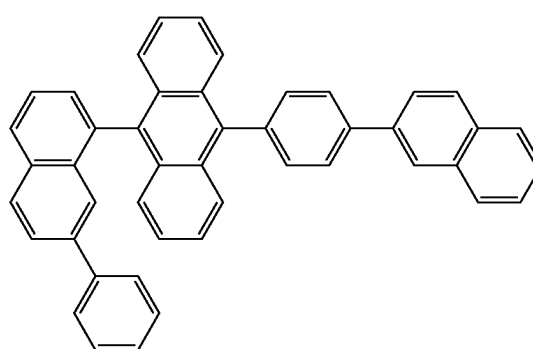
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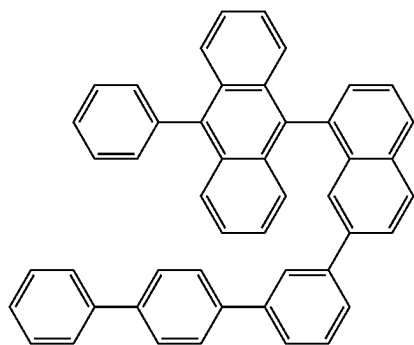
H4-36



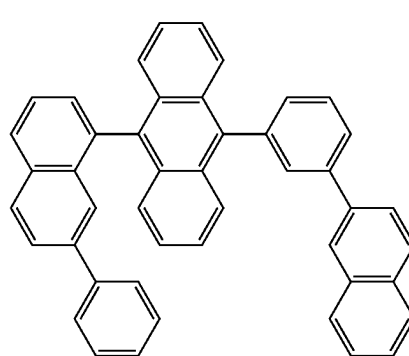
H4-33



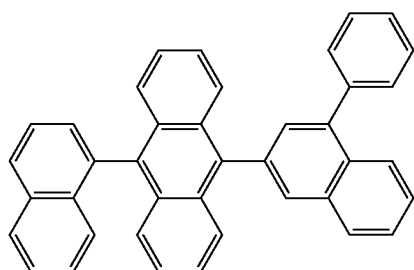
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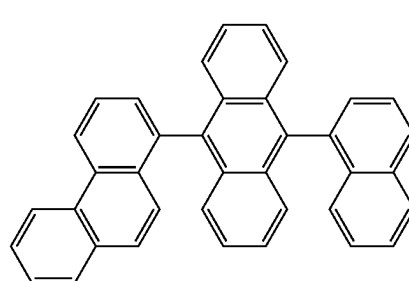
H4-34



H4-38

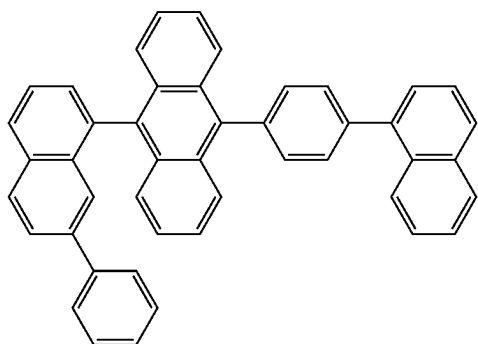


H4-35



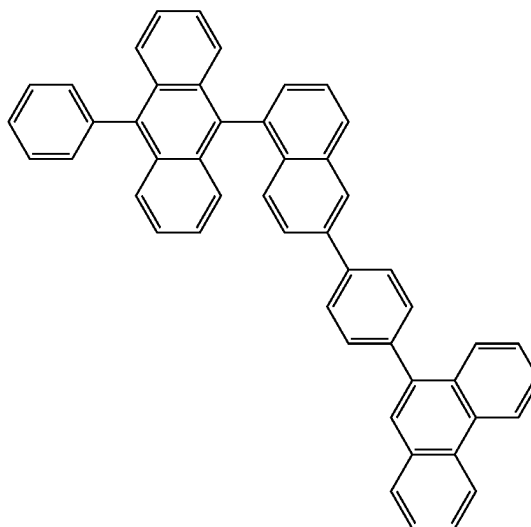
H4-39

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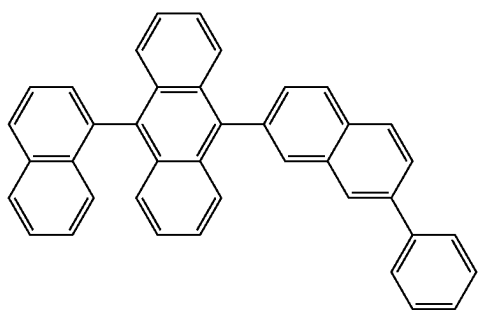


H4-40

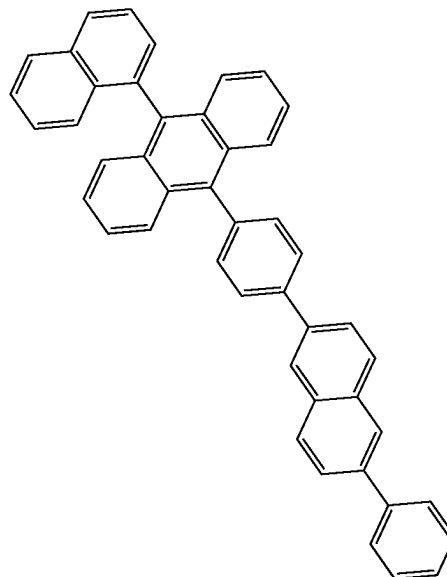
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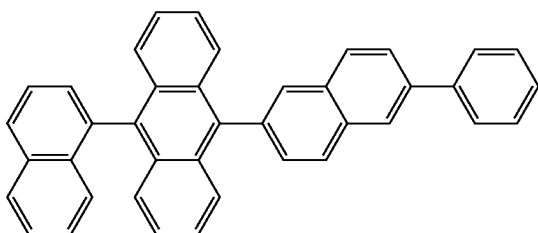
H4-44



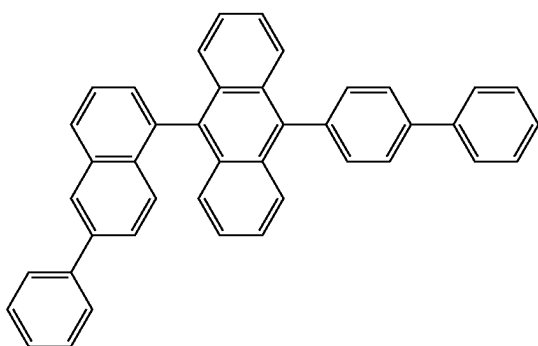
H4-41



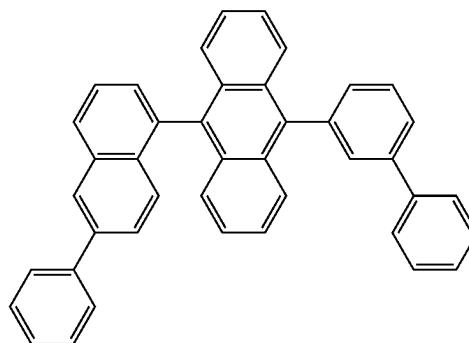
H4-45



H4-42



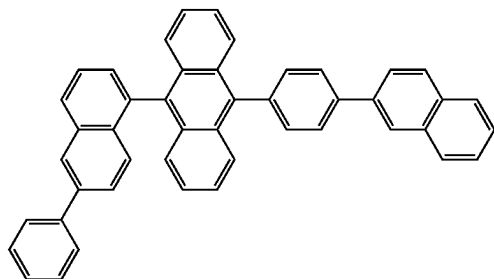
H4-43



H4-46

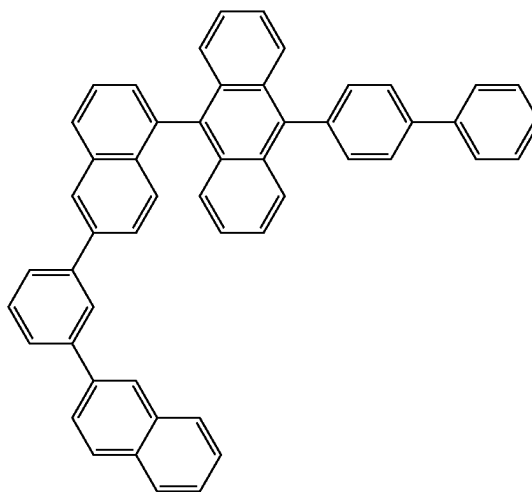
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H4-47

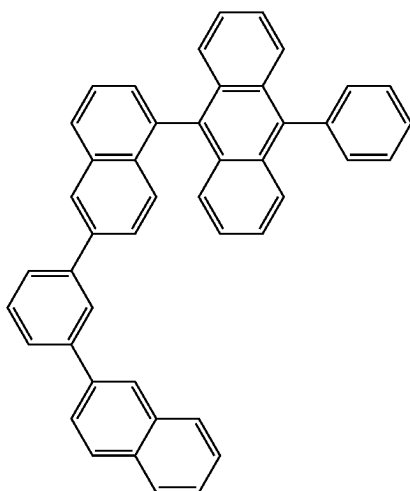


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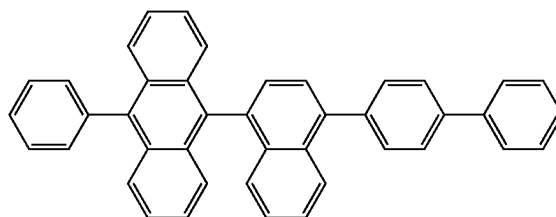
H4-50



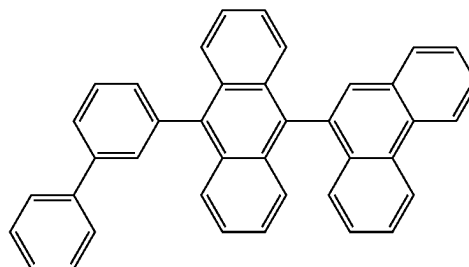
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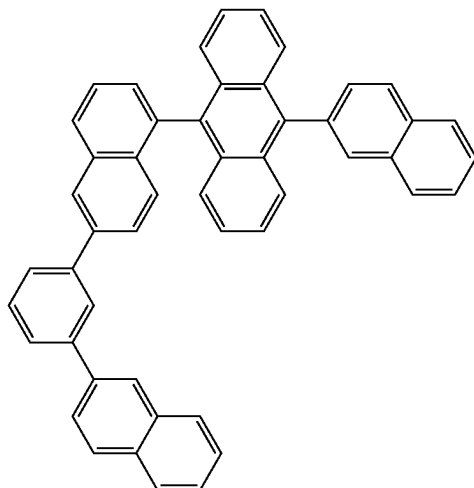
H4-51



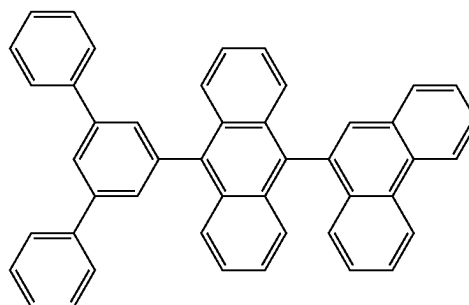
H4-52



H4-49

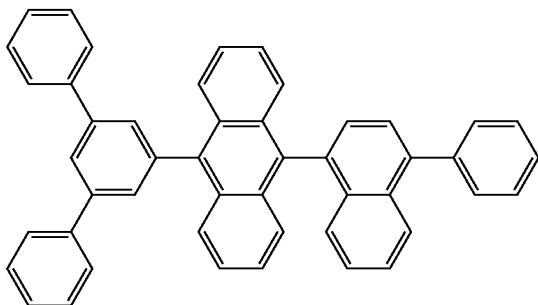


H4-53

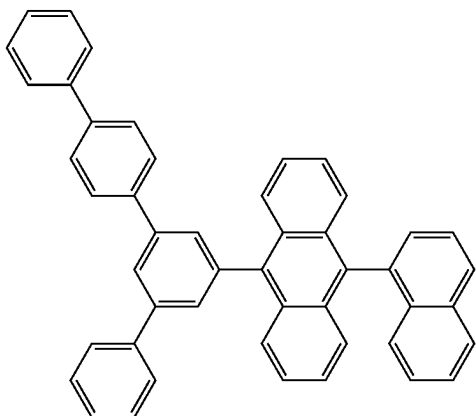


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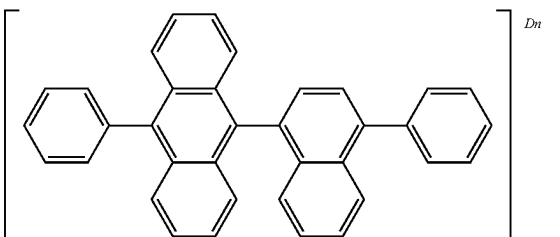
H4-54



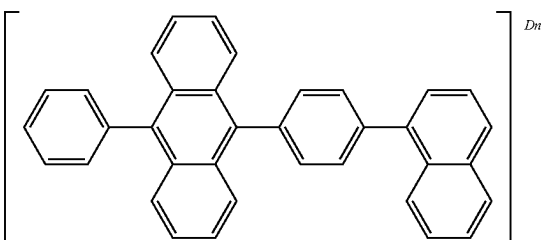
H4-55



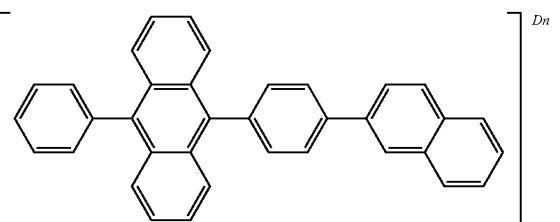
H4-56



H4-57

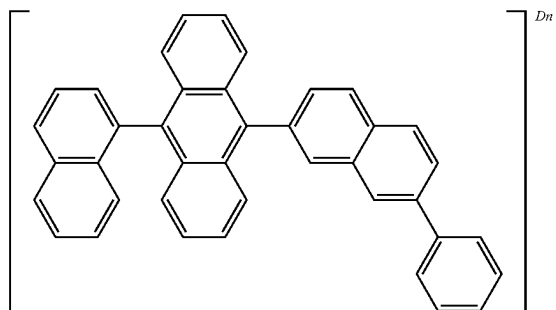


H4-58

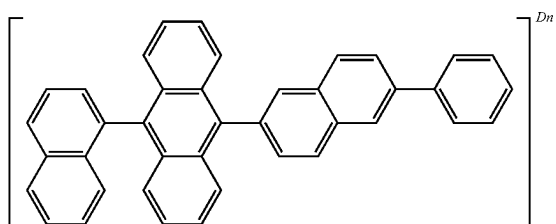


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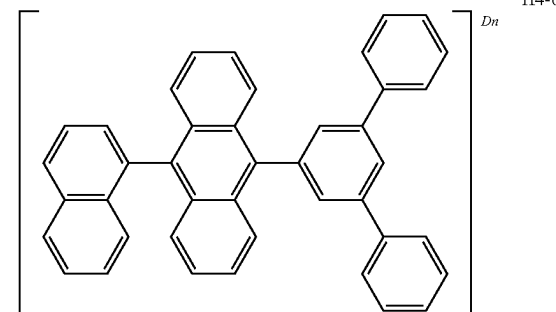
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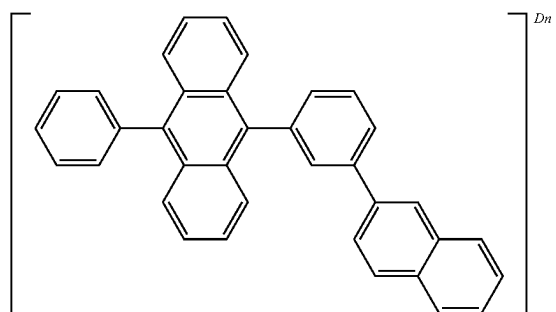
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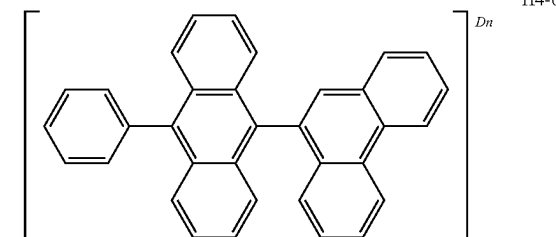
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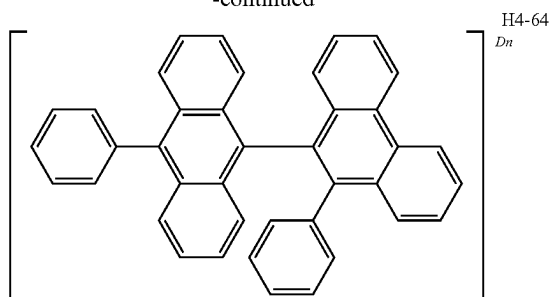
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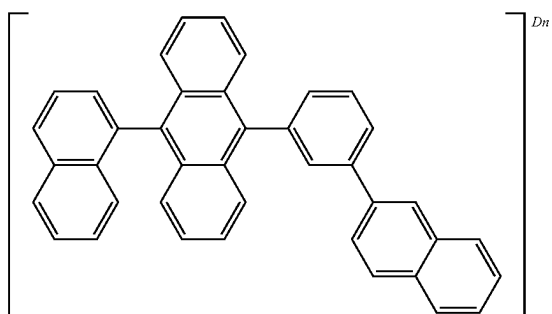
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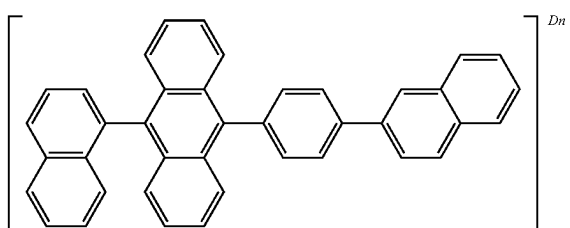
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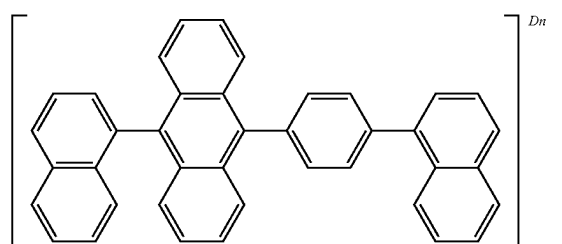
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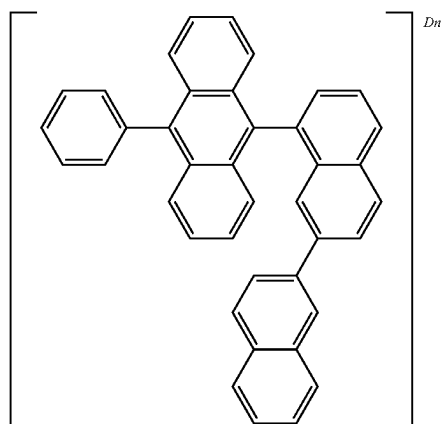
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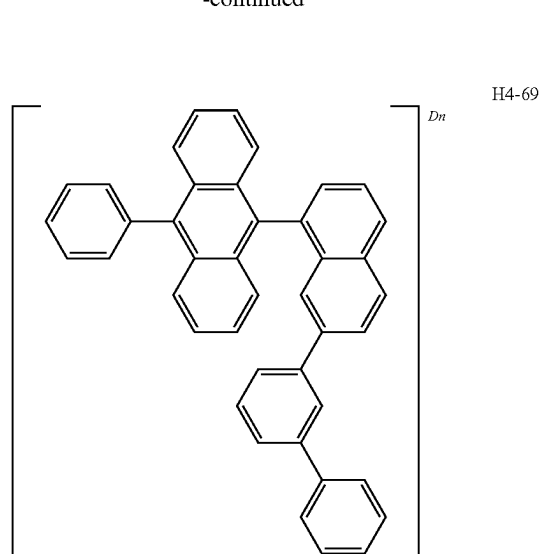
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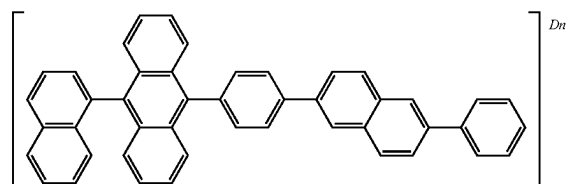
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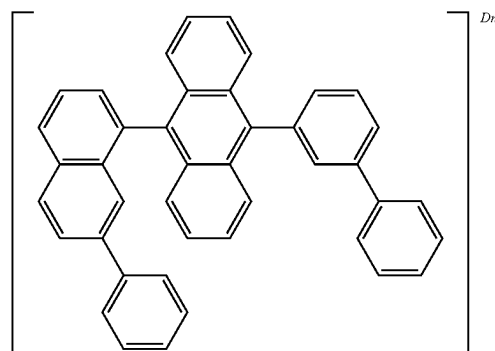
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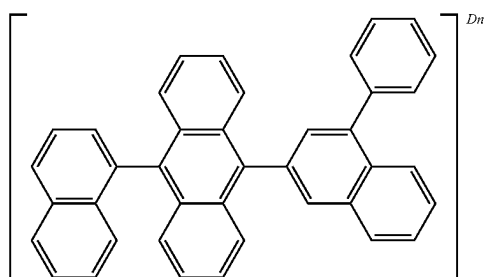
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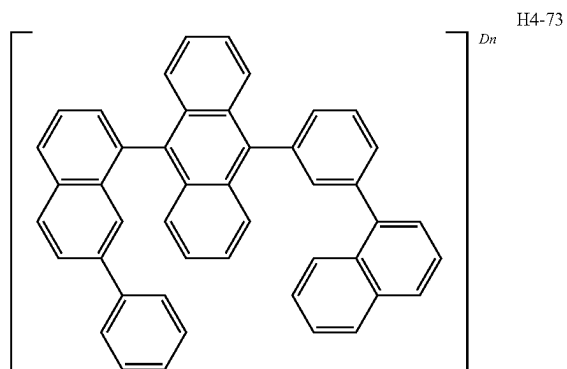
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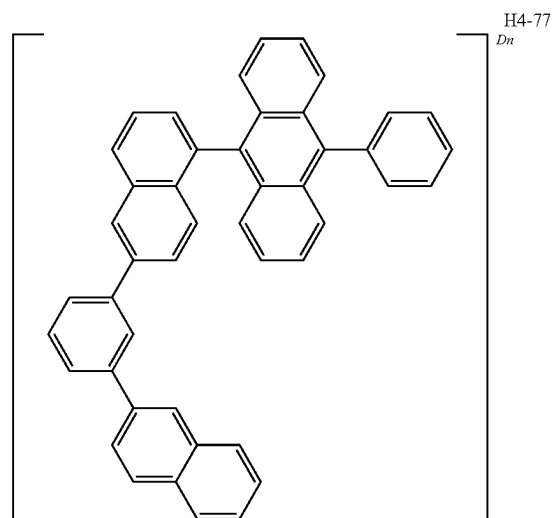
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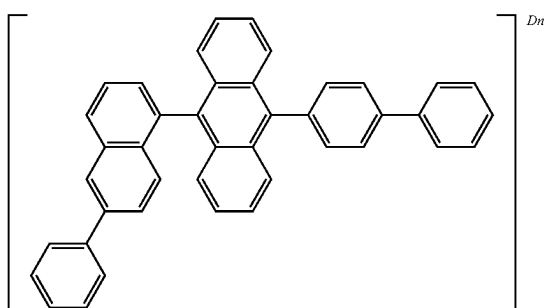
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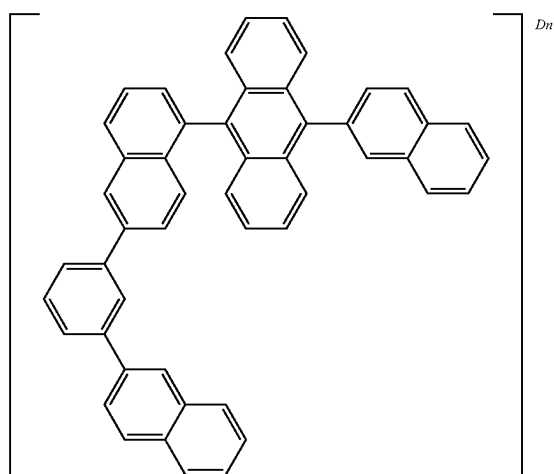
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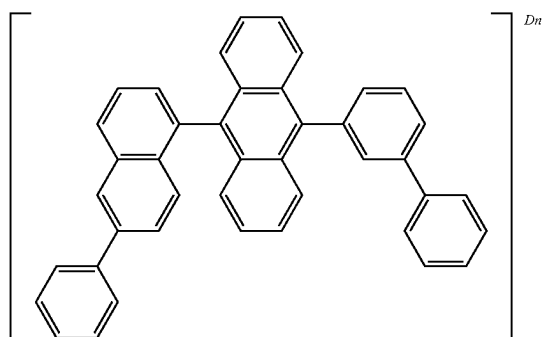
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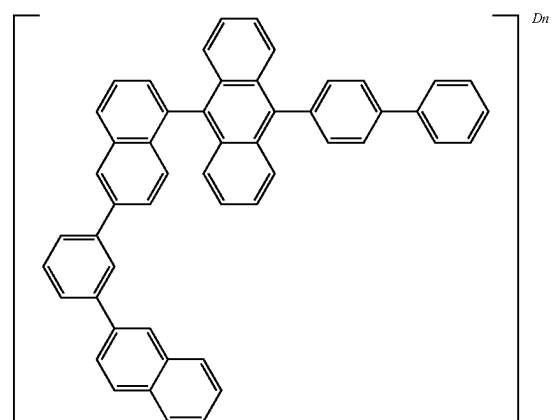
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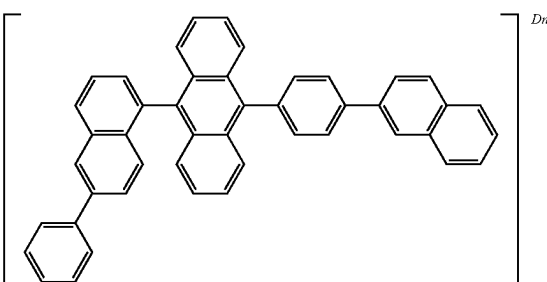
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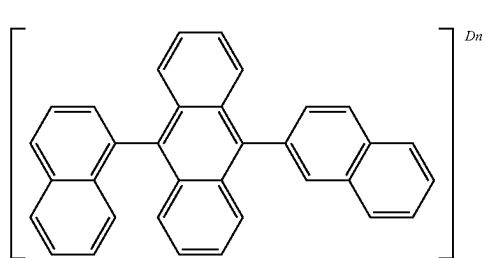
H4-79



H4-76

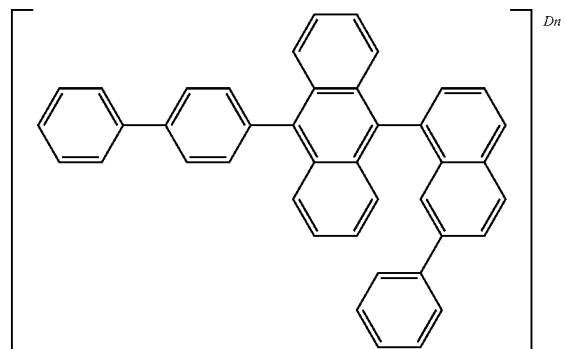


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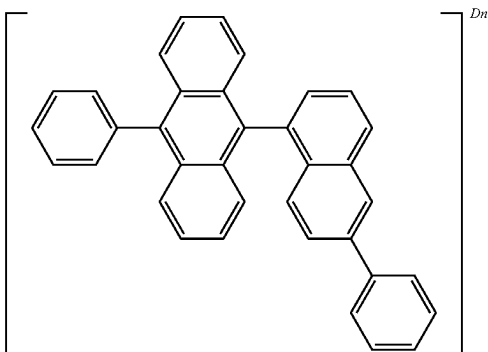


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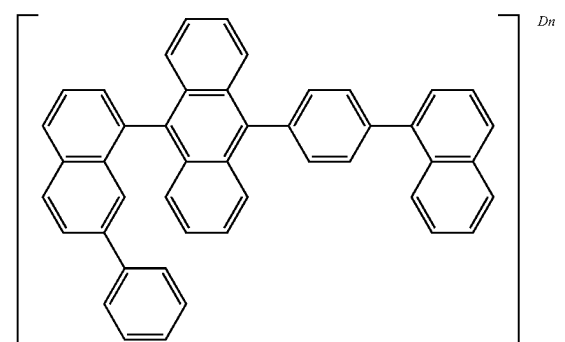
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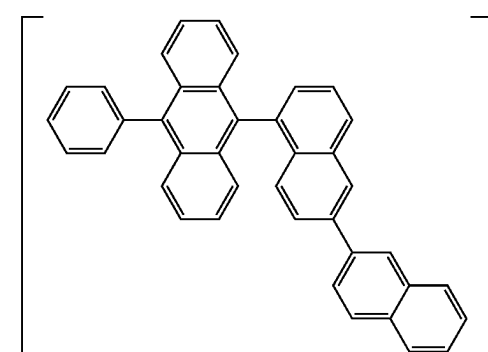
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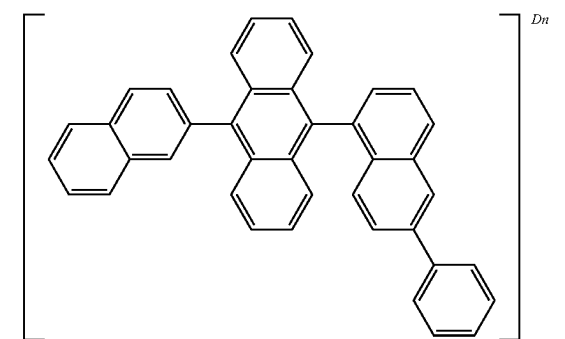
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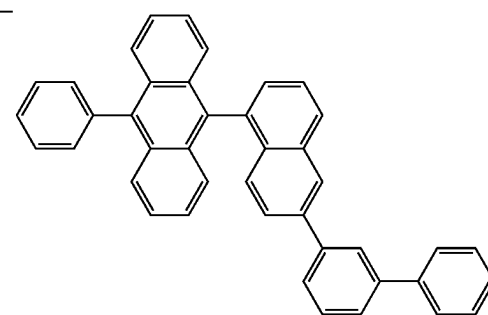
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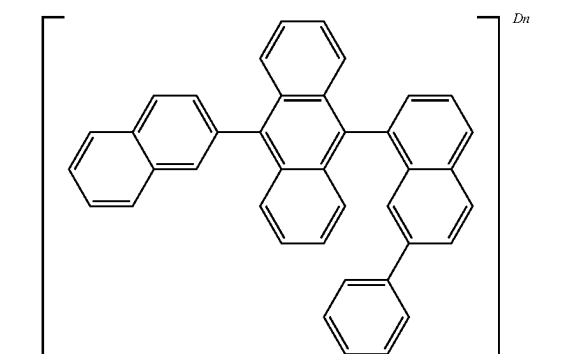
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H4-86



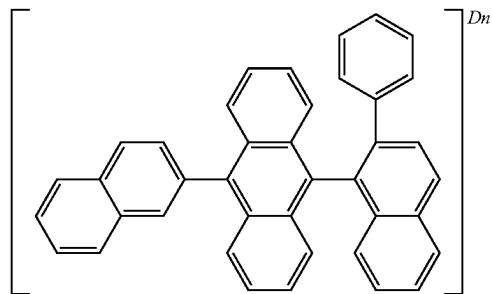
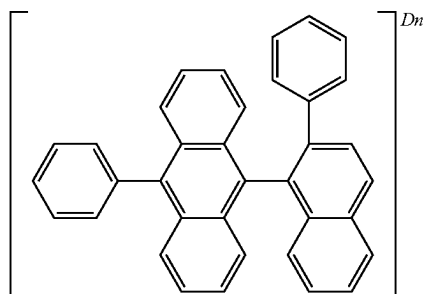
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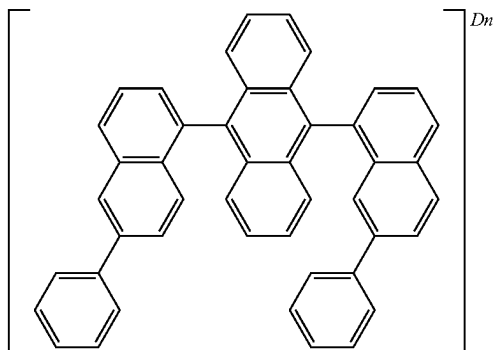
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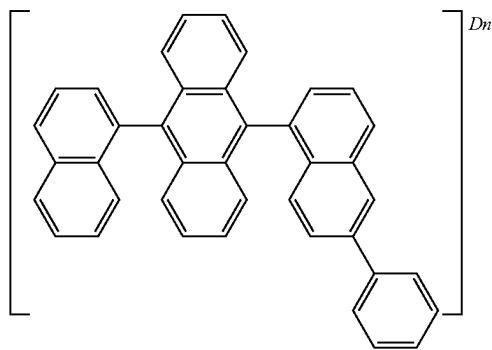
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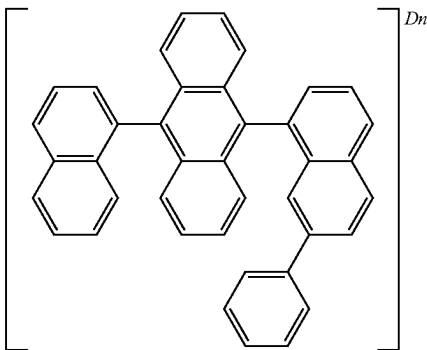
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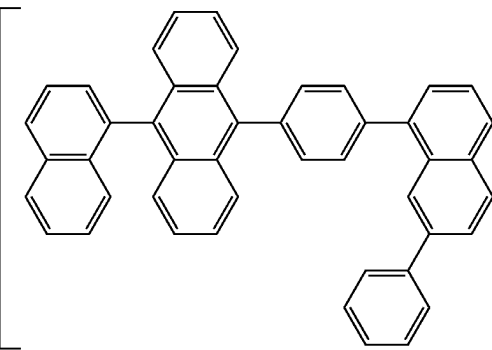
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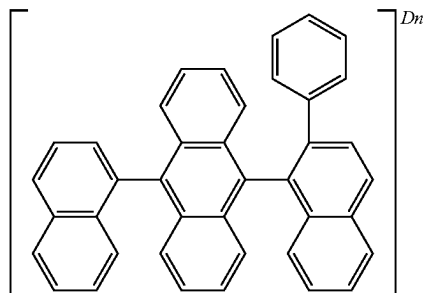
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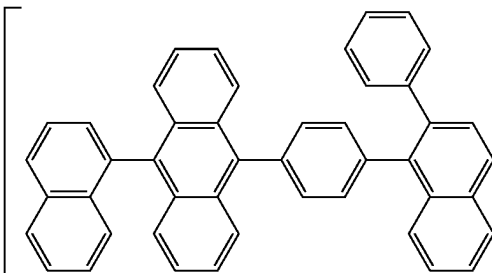
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H4-91

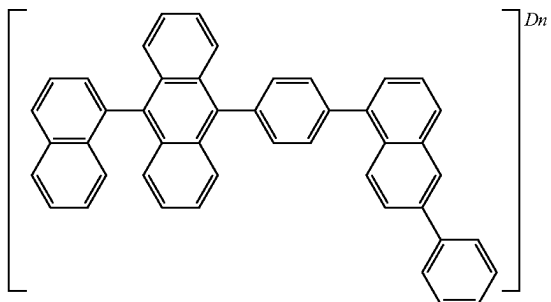


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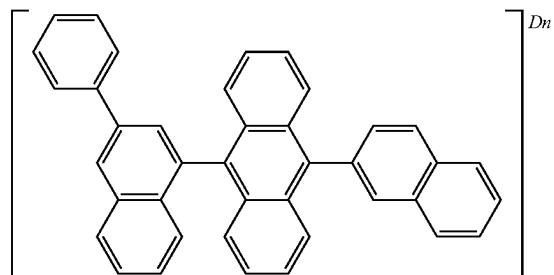
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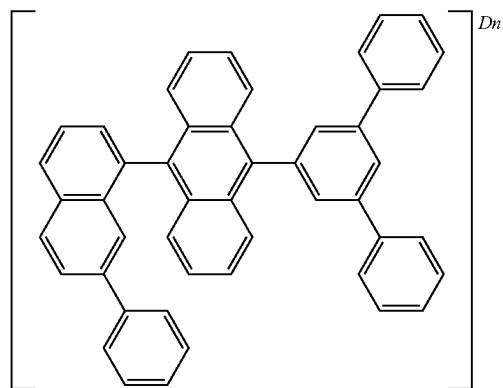
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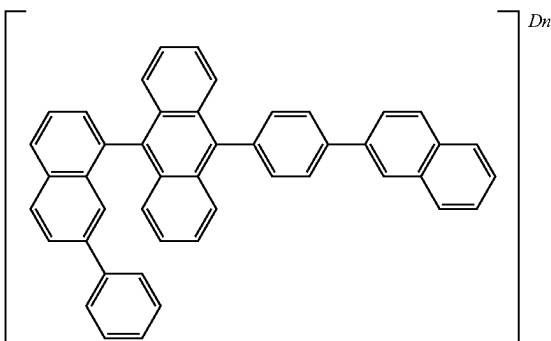


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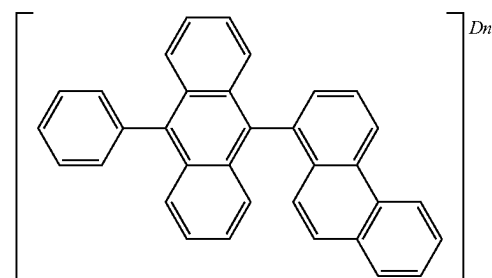
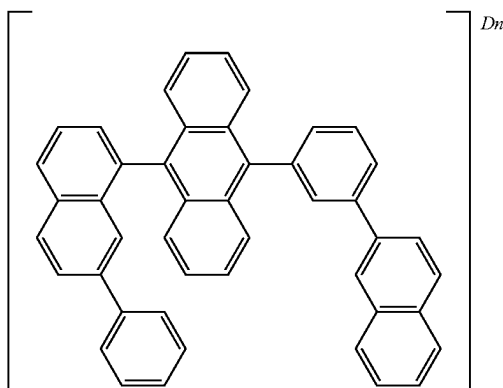
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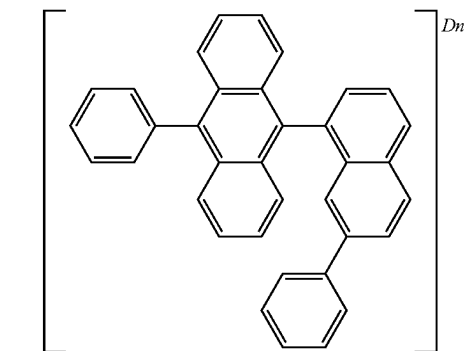
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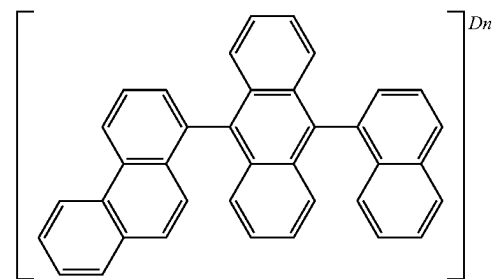
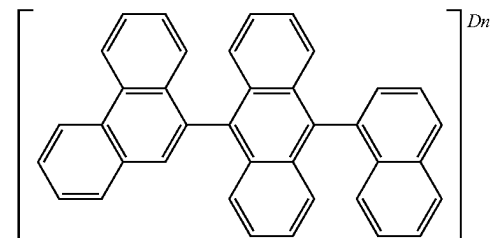
H4-99



H4-103

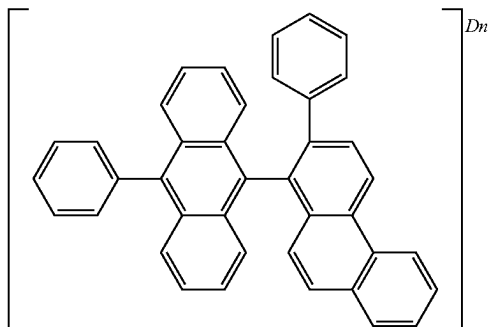


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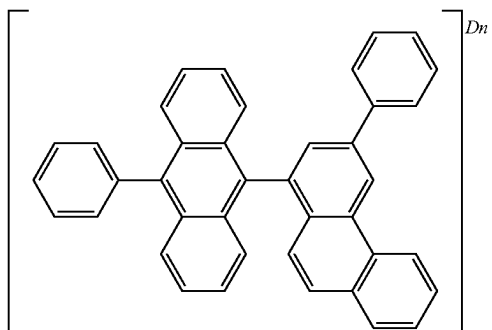


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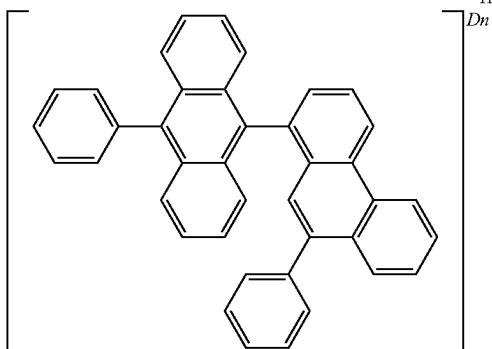
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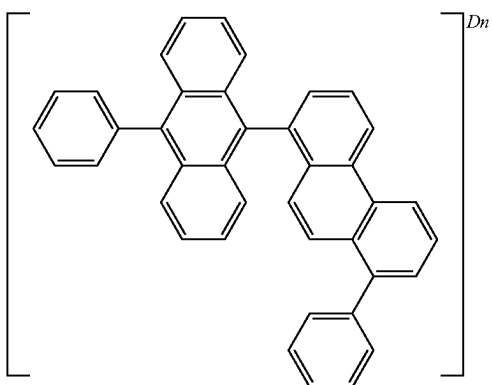
H4-106



H4-107

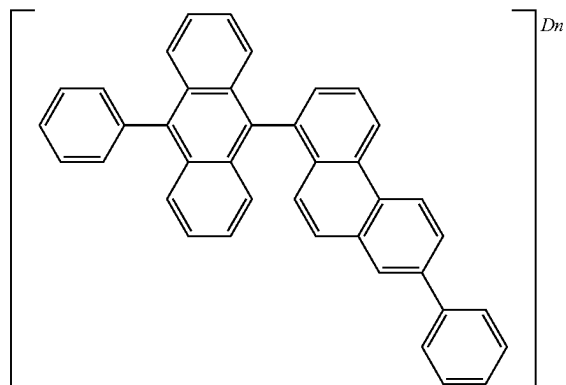


H4-108

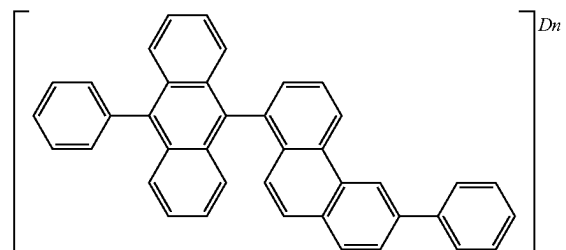


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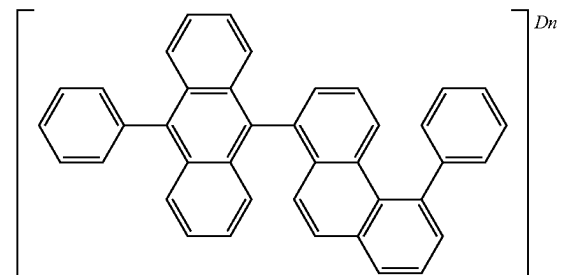
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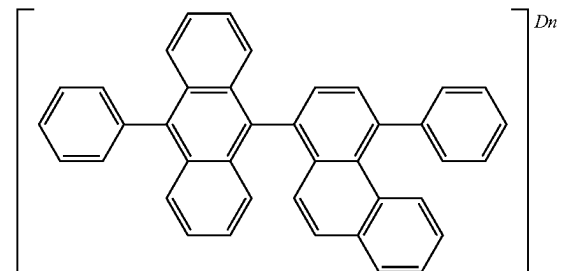
H4-112



H4-111

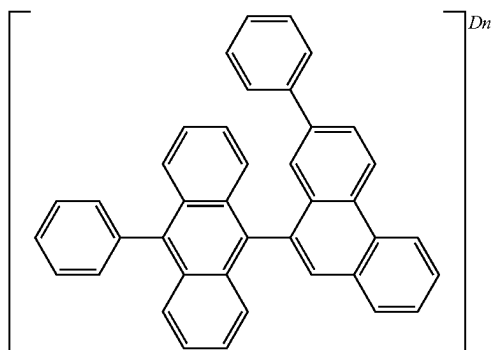


H4-112



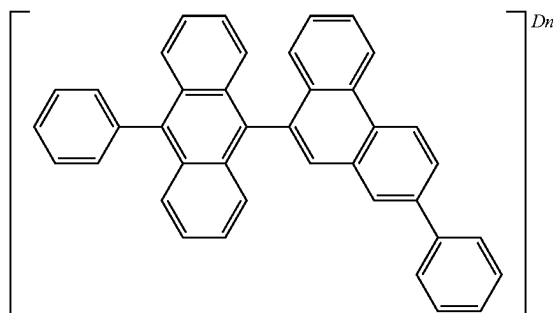
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H4-113

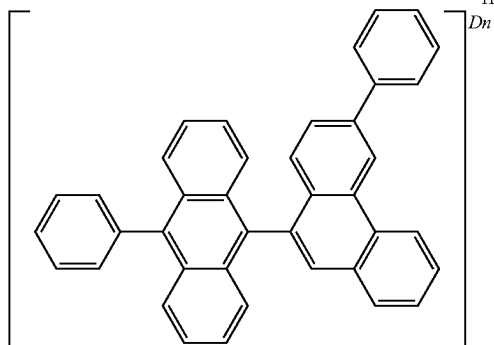


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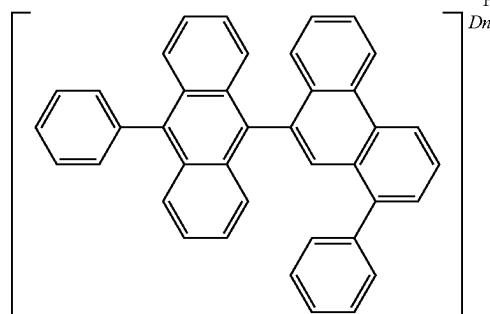
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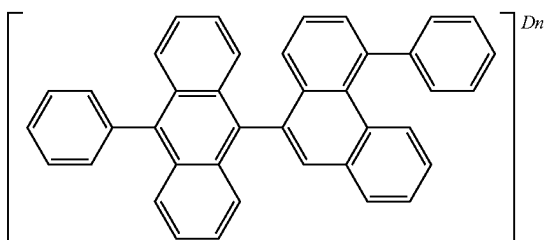
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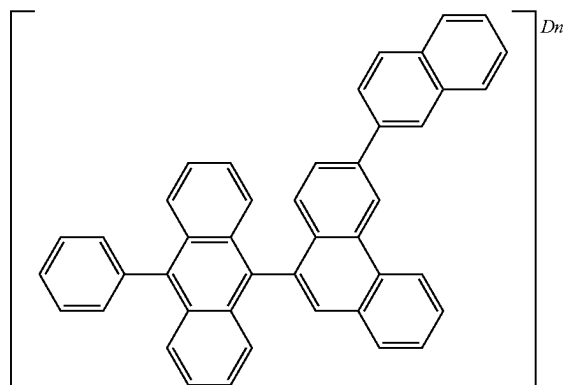
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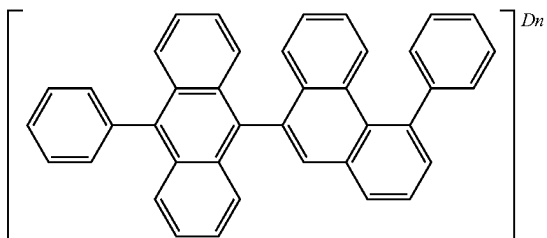
H4-115



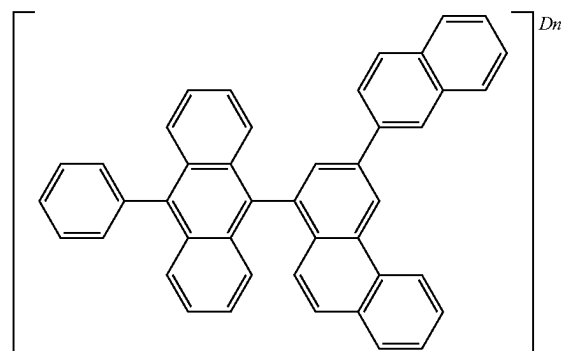
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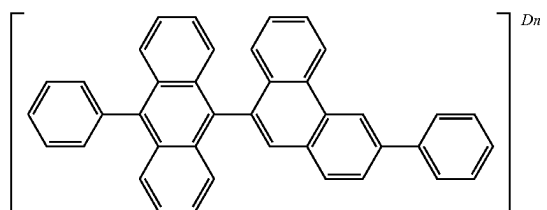
H4-116



H4-121

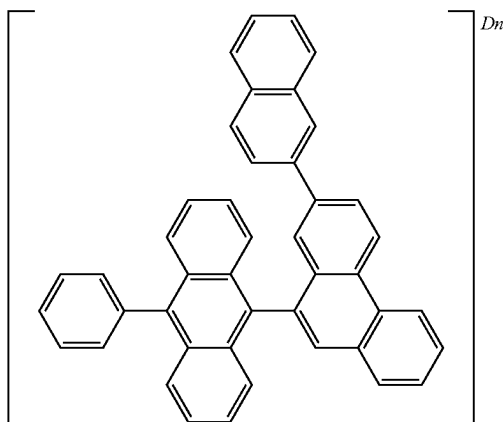


H4-117

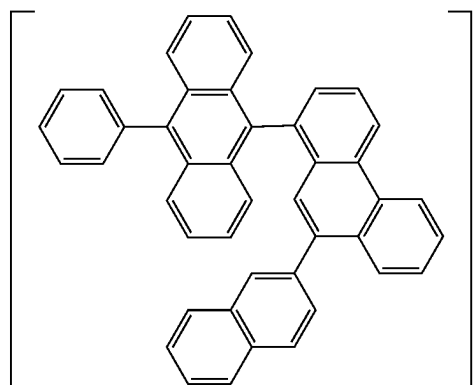


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H4-122



H4-123

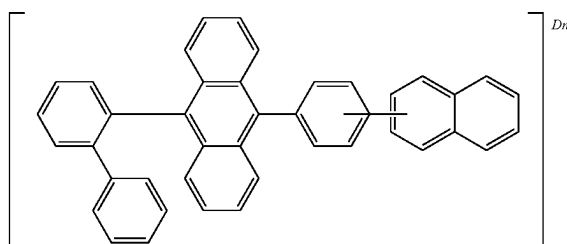


[0067] In the above compounds, D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to the maximum number of hydrogens in the compound. Specifically, n is a minimum of 1 and an integer up to the maximum number of hydrogens in the compound. According to one embodiment of the present disclosure, in one compound represented by Formula 2, when deuterium is included, the deuterium substitution rate is preferably about 100% or less, more preferably about 95% or less, still more preferably about 90% or less, and still more preferably about 85% or less of the total number of hydrogens. The compound of Formula 2 substituted with the above deuterium substitution rate can increase the bond dissociation energy according to deuteration, thereby enhancing the stability of the compound, and an organic electroluminescent device comprising the compound can exhibit improved lifetime properties.

[0068] Hereinafter, an organic electroluminescent compound according to one embodiment is described.

[0069] An organic electroluminescent compound according to one embodiment of the present disclosure is represented by the following Formula 11.

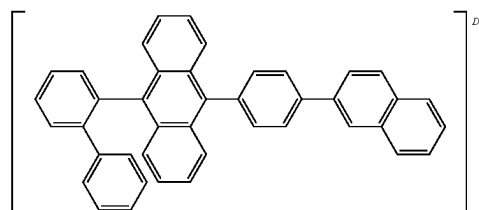
(11)



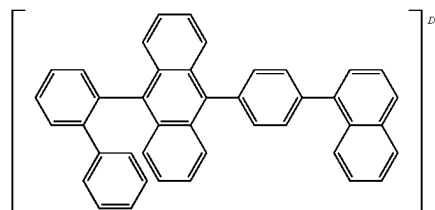
[0070] In Formula 11, D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to 28. According to one embodiment of the present disclosure, n is an integer from 8 to 27. According to another embodiment of the present disclosure, the deuterium substitution rate may be preferably about 100% or less, more preferably about 95% or less, still more preferably about 90% or less, and still more preferably about 85% or less of the total number of hydrogens. The compound of Formula 11 substituted with the above deuterium substitution rate can increase the bond dissociation energy according to deuteration, thereby enhancing the stability of the compound, and an organic electroluminescent device comprising the compound can exhibit improved lifetime properties.

[0071] The compound represented by Formula 11 may be at least one selected from the group consisting of the following compounds, but is not limited thereto.

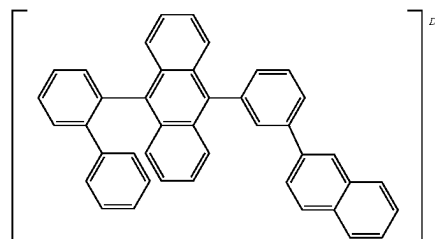
H1-72

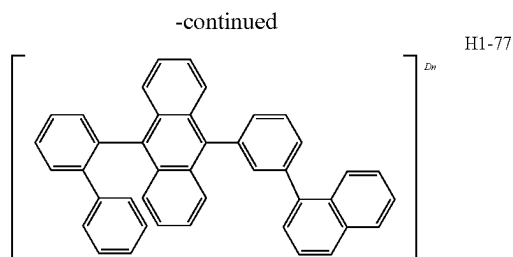


H1-73



H1-76





[0072] The compounds represented by Formula 1, Formula 2, and Formula 11 according to the present disclosure can be prepared by way of synthetic methods known to one skilled in the art. For example, the compounds of the present disclosure can be prepared by referring to Korean Patent Application Laid-Open No. 2023-0112551, etc., but is not limited thereto. The compounds substituted with deuterium among the compounds of the present disclosure can be prepared by referring to Korean Patent Nos. 10-2283849, 10-1427457, etc., but is not limited thereto.

[0073] Hereinafter, an organic electroluminescent device using the aforementioned plurality of host materials or organic electroluminescent compound is described.

[0074] According to one embodiment of the present disclosure, an organic electroluminescent device comprises an anode; a cathode; and at least one organic layer between the anode and the cathode, wherein the organic layer comprises at least one light-emitting layer or at least two light-emitting layers, wherein at least one layer of the light-emitting layers comprises a plurality of host materials comprising at least one first host compound represented by Formula 1 and at least one second host compound represented by Formula 2, and wherein the first host compound and the second compound are different from each other. The plurality of host materials may be included in the same organic layer, e.g., a light-emitting layer, or may be included in different light-emitting layers.

[0075] According to another embodiment of the present disclosure, an organic electroluminescent device comprises an anode; a cathode; and at least one organic layer between the anode and the cathode, wherein the organic layer comprises at least one light-emitting layer or at least two light-emitting layers, wherein at least one layer of the light-emitting layers comprises the organic electroluminescent compound represented by Formula 11. The compounds may be included in the same organic layer, e.g., a light-emitting layer, or may be included in different light-emitting layers.

[0076] According to one embodiment, the organic electroluminescent material of the present disclosure comprises an organic electroluminescent compound represented by Formula 11, and the organic electroluminescent material may be included in an organic layer, e.g., a light-emitting layer.

[0077] According to another embodiment of the present disclosure, an organic electroluminescent device comprises an anode; a cathode; and at least two light-emitting layers between the anode and the cathode, wherein at least one layer of the light-emitting layers comprises the compound represented by Formula 1 as a host material(s).

[0078] The organic layer comprises a light-emitting layer and may further comprise at least one layer selected from the group consisting of a hole injection layer, a hole transport

layer, a hole auxiliary layer, a light-emitting auxiliary layer, an electron transport layer, an electron injection layer, an interlayer, a hole-blocking layer, an electron-blocking layer, and an electron buffer layer. The organic layer may further include an amine-based compound and/or azine-based compound in addition to the light-emitting material of the present disclosure. Specifically, the hole injection layer, the hole transport layer, the hole auxiliary layer, the light-emitting layer, the light-emitting auxiliary layer, or the electron-blocking layer may comprise an amine-based compound, e.g., an arylamine-based compound, a styrylamine-based compound, etc., as a hole injection material, a hole transport material, a hole auxiliary material, a light-emitting material, a light-emitting auxiliary material, and an electron-blocking material. In addition, the electron transport layer, the electron injection layer, the electron buffer layer, and the hole-blocking layer may include an azine-based compound as an electron transport material, an electron injection material, an electron buffer material, and a hole-blocking material. Furthermore, the organic layer may further comprise at least one metal selected from the group consisting of metals of Group 1, metals of Group 2, transition metals of the 4th period, transition metals of the 5th period, lanthanides, and organic metals of the d-transition elements of the Periodic Table, or at least one complex compound comprising a metal.

[0079] The plurality of host materials or the organic electroluminescent compound according to one embodiment of the present disclosure may be used as a light-emitting material for a white organic light-emitting device. The white organic light-emitting device has been suggested to have various structures such as a side-by-side structure or a stacking structure depending on the arrangement of R (red), G (green) or YG (yellow-green), and B (blue) light-emitting parts, or color conversion material (CCM) method, etc. The plurality of host materials or the organic electroluminescent compound according to one embodiment of the present disclosure may also be used in an organic electroluminescent device comprising a quantum dot (QD).

[0080] A hole injection layer, a hole transport layer, an electron-blocking layer, or a combination thereof can be used between the anode and the light-emitting layer. The hole injection layer may be multi-layers in order to lower the hole injection barrier (or hole injection voltage) from the anode to the hole transport layer or the electron-blocking layer, wherein each of the multi-layers may use two compounds simultaneously. In addition, the hole injection layer may be doped with a p-dopant. The electron-blocking layer is located between the hole transport layer (or hole injection layer) and the light-emitting layer, and can prevent light leakage by blocking the overflow of electrons from the light-emitting layer and confining excitons within the light-emitting layer. The hole transport layer or the electron-blocking layer may be multi-layers, wherein a plurality of compounds may be used in each of the multi-layers.

[0081] An electron buffer layer, a hole-blocking layer, an electron transport layer, an electron injection layer, or a combination thereof can be used between the light-emitting layer and the cathode. The electron buffer layer may be multi-layers in order to control the injection of the electron and improve the interfacial properties between the light-emitting layer and the electron injection layer, wherein each of the multi-layers may use two compounds simultaneously. The hole-blocking layer is located between the electron

transport layer (or electron injection layer) and the light-emitting layer, and can improve the probability of recombination of electrons and holes in the light-emitting layer by preventing holes from reaching the cathode. The hole-blocking layer or the electron transport layer may be multi-layers, wherein a plurality of compounds may be used in each of the multi-layers. In addition, the electron injection layer may be doped with an n-dopant.

[0082] The light-emitting auxiliary layer may be placed between the anode and the light-emitting layer, or between the cathode and the light-emitting layer. When the light-emitting auxiliary layer is placed between the anode and the light-emitting layer, it can be used for promoting the injection and/or transport of hole, or for preventing the overflow of electrons. When the light-emitting auxiliary layer is placed between the cathode and the light-emitting layer, it can be used for promoting the injection and/or transport of electron, or for preventing the overflow of holes. In addition, the hole auxiliary layer may be placed between the hole transport layer (or hole injection layer) and the light-emitting layer, and may be effective to promote or block the hole transport rate (or hole injection rate), thereby enabling the charge balance to be controlled. When an organic electroluminescent device includes two or more hole transport layers, the hole transport layer, which is further included, may be used as a hole auxiliary layer or an electron-blocking layer. The light-emitting auxiliary layer, the hole auxiliary layer, or the electron-blocking layer may be provided to improve the efficiency and/or lifetime properties of the organic electroluminescent device.

[0083] In the organic electroluminescent device of the present disclosure, at least one layer selected from the group consisting of a chalcogenide layer, a metal halide layer, and a metal oxide layer (hereinafter, "a surface layer") may be placed on an inner surface(s) of one or both electrode(s). Specifically, a chalcogenide (including oxides) layer of silicon or aluminum may be placed on an anode surface of an electroluminescent medium layer, and a metal halide layer or a metal oxide layer may be placed on a cathode surface of an electroluminescent medium layer. Such a surface layer provides operation stability for the organic electroluminescent device. For example, the chalcogenide includes SiO_x ($1 \leq x \leq 2$), AlO_x ($1 \leq x \leq 1.5$), SiON , SiAlON , etc.; the metal halide includes LiF , MgF_2 , CaF_2 , a rare earth metal fluoride, etc.; and the metal oxide includes Cs_2O , Li_2O , MgO , SrO , BaO , CaO , etc.

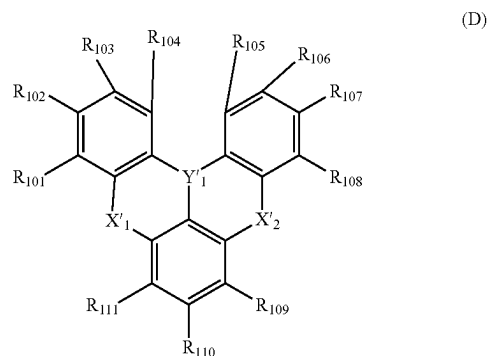
[0084] In addition, in the organic electroluminescent device of the present disclosure, a mixed region of an electron transport compound and a reductive dopant, or a mixed region of a hole transport compound and an oxidative dopant may be placed on at least one surface of a pair of electrodes. In this way, the electron transport compound is reduced to an anion, and thus it becomes easier to inject and transport electrons from the mixed region to the light-emitting medium. Furthermore, the hole transport compound is oxidized to a cation, and thus it becomes easier to inject and transport holes from the mixed region to the light-emitting medium. Specifically, the oxidative dopant includes various Lewis acids and acceptor compounds; and the reductive dopant includes alkali metals, alkali metal compounds, alkaline earth metals, rare earth metals, and mixtures thereof. The reductive dopant layer may be employed as a charge-generating layer to produce an organic

electroluminescent device having two or more light-emitting layers and emitting white light.

[0085] An organic electroluminescent device according to one embodiment may further include one or more dopants in the light-emitting layer.

[0086] The dopant comprised in the organic electroluminescent device of the present disclosure may be at least one phosphorescent or fluorescent dopant, and preferably, a fluorescent dopant. The phosphorescent dopant material is not particularly limited, but may be a complex compound of a metal selected from the group consisting of iridium (Ir), osmium (Os), copper (Cu), and platinum (Pt), preferably an ortho-metallated complex compound of a metal selected from the group consisting of iridium (Ir), osmium (Os), copper (Cu), and platinum (Pt), and more preferably an ortho-metallated iridium complex compound.

[0087] A compound represented by the following Formula D may be used as a dopant included in the organic electroluminescent device of the present disclosure, but is not limited thereto.



[0088] In Formula D,

[0089] R_{101} to R_{111} each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3 - C_{30})cycloalkyl, a substituted or unsubstituted (C_1 - C_{30})alkoxy, a substituted or unsubstituted tri(C_1 - C_{30})alkylsilyl, a substituted or unsubstituted di(C_1 - C_{30})alkyl(C_6 - C_{30})arylsilyl, a substituted or unsubstituted (C_1 - C_{30})alkyldi(C_6 - C_{30})arylsilyl, a substituted or unsubstituted tri(C_6 - C_{30})arylsilyl, or $-L'_4-N-(Ar'_4)$ (Ar'_5), or may be linked to the adjacent substituent to form a ring(s);

[0090] Y'_1 represents B;

[0091] X'_1 and X'_2 each independently represent NR' ;

[0092] R' each independently represents hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3 - C_{30})cycloalkyl, a substituted or unsubstituted (C_1 - C_{30})alkoxy, a substituted or unsubstituted tri(C_1 - C_{30})alkylsilyl, a substituted or unsubstituted di(C_1 - C_{30})alkyl(C_6 - C_{30})arylsilyl, a substituted or unsubstituted (C_1 - C_{30})alkyldi(C_6 - C_{30})arylsilyl, or $-L'_4-N-(Ar'_4)$

(Ar'₅); or may be linked to at least one of R₁₀₁, R₁₀₈, R₁₀₉, and R₁₁₁ to form a ring(s);

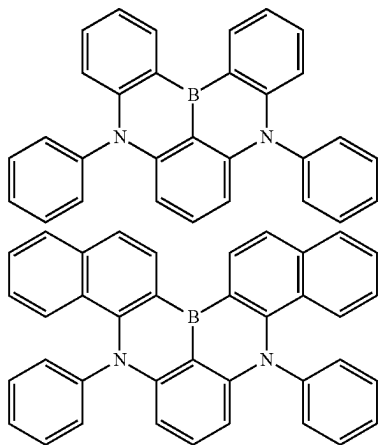
[0093] L'₄ each independently represents a single bond, a substituted or unsubstituted (C₆-C₃₀)arylene, or a substituted or unsubstituted (3- to 30-membered) heteroarylene; and

[0094] Ar'₄ and Ar'₅ each independently represent hydrogen, a substituted or unsubstituted (C₁-C₃₀)alkyl, a substituted or unsubstituted (C₂-C₃₀)alkenyl, a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s), a substituted or unsubstituted (C₆-C₃₀)aryl, or a substituted or unsubstituted (3- to 30-membered)heteroaryl.

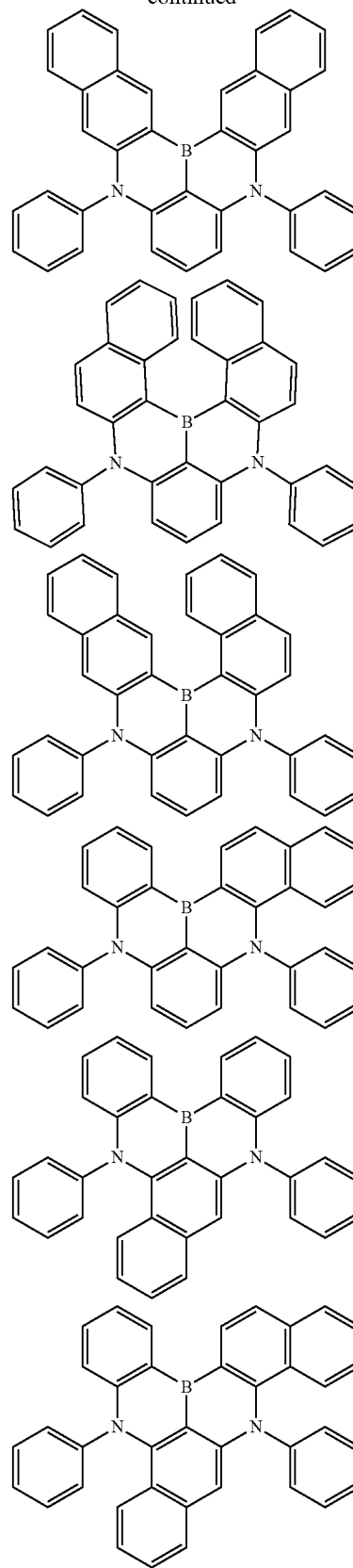
[0095] Preferably, R₁₀₁ to R₁₁₁ each independently represent hydrogen, deuterium, a substituted or unsubstituted (C₁-C₂₀)alkyl, a substituted or unsubstituted (C₆-C₂₅)aryl, a substituted or unsubstituted (5- to 20-membered)heteroaryl, or -L'₄-N—(Ar'₄)(Ar'₅), or may be linked to the adjacent substituent to form a ring(s).

[0096] More preferably, R₁₀₁ to R₁₁₁ each independently represent hydrogen; deuterium; an unsubstituted (C₁-C₁₀) alkyl; a (C₆-C₁₈)aryl unsubstituted or substituted with at least one of a (C₁-C₁₀)alkyl(s), a (13- to 18-membered) hetero-aryl(s), and a di(C₆-C₁₈)arylamino(s); a (5- to 18-membered)heteroaryl unsubstituted or substituted with at least one (C₁-C₁₀)alkyl(s); or -L'₄-N—(Ar'₄)(Ar'₅); or may be linked to the adjacent substituent to form a ring(s). For example, R₁₀₁ to R₁₁₁ each independently may be hydrogen, methyl, tert-butyl, a substituted or unsubstituted phenyl, a biphenyl, a terphenyl, a triphenylenyl, a carbazolyl, a phenoxazinyl, a phenothiazinyl, a dimethylacridinyl, a dimethylxanthenyl, a diphenylamino unsubstituted or substituted with at least one of a methyl(s) and a diphenylamino(s), a phenylnaphthylamino, a dibiphenylamino, a phenylamino substituted with a phenylcarbazolyl(s) or a dibenzofuranyl (s), or a (17- to 21-membered)heteroaryl substituted with at least one of a methyl(s) and a phenyl(s), or may be linked to adjacent substituent to form a benzene ring, an indole ring substituted with at least one of a phenyl(s) and a diphenylamino(s), a benzofuran ring, a benzothiophene ring, or a 19-membered hetero ring substituted with a methyl(s). The substituent of the above substituted phenyl may be at least one of a methyl(s), a carbazolyl(s), a dibenzofuranyl(s), a diphenylamino(s), a phenoxazinyl(s), a phenothiazinyl(s), and a dimethylacridinyl(s).

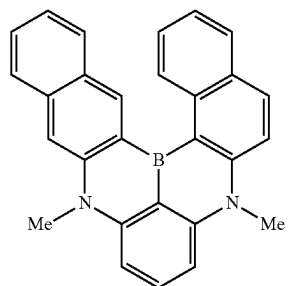
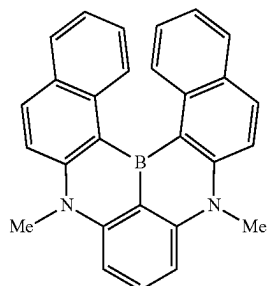
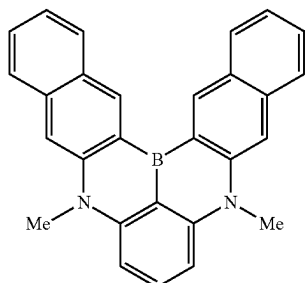
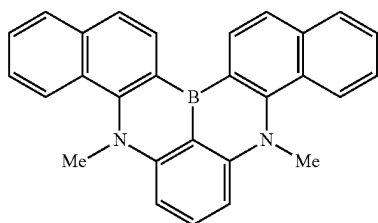
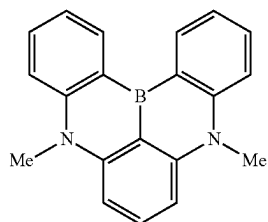
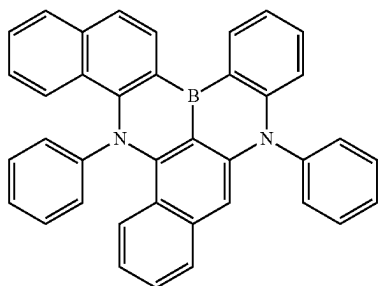
[0097] The specific examples of the dopant compound are as follows, but are not limited thereto.



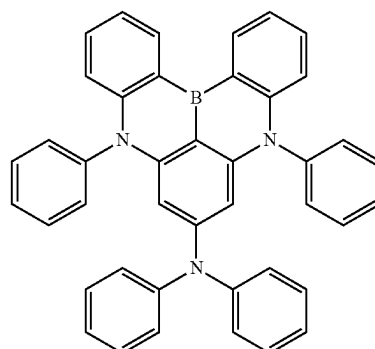
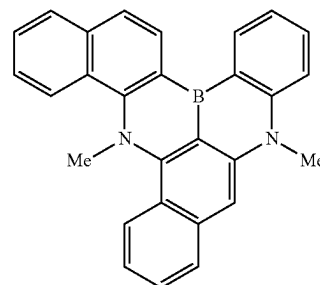
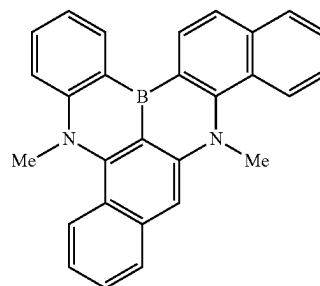
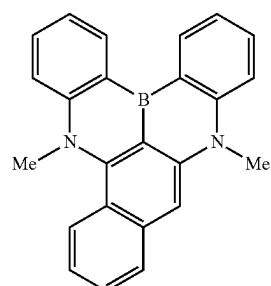
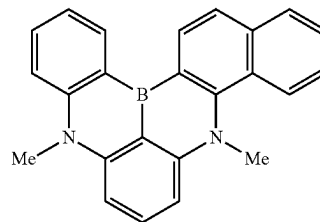
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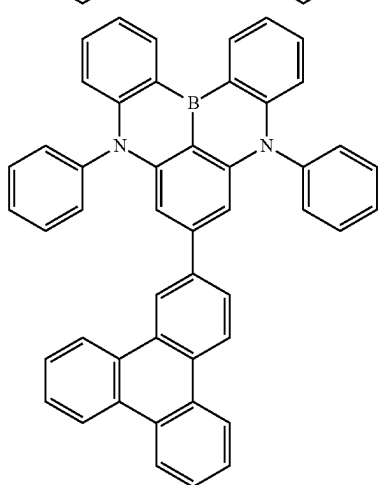
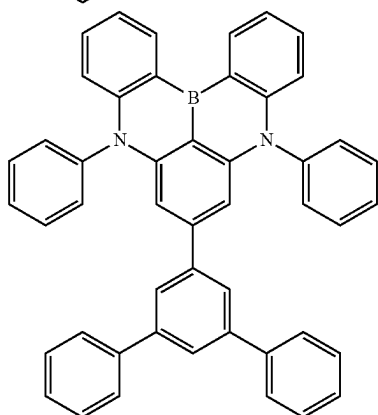
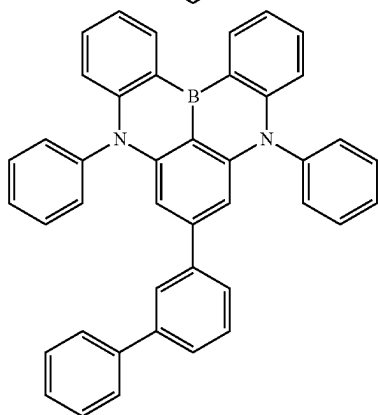
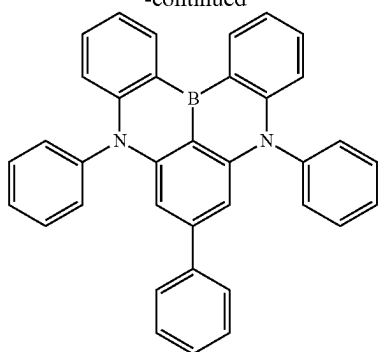
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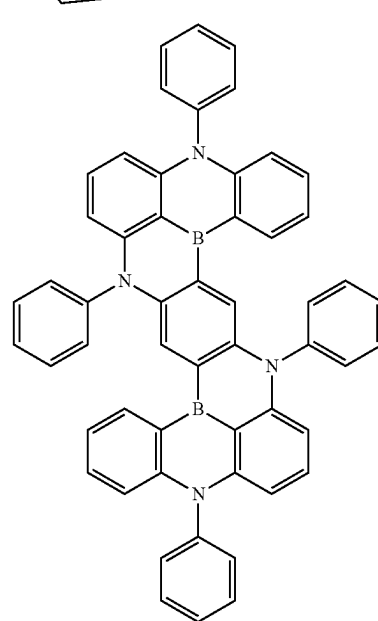
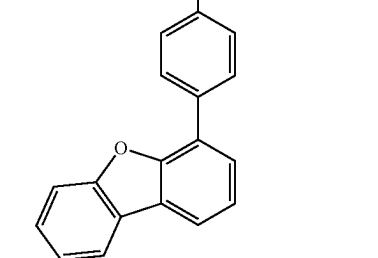
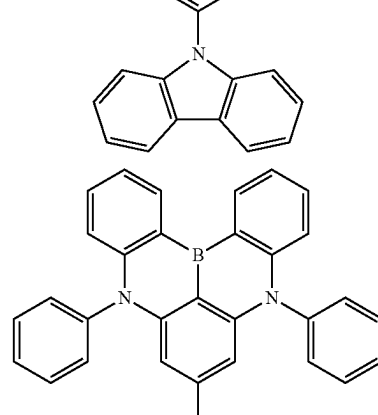
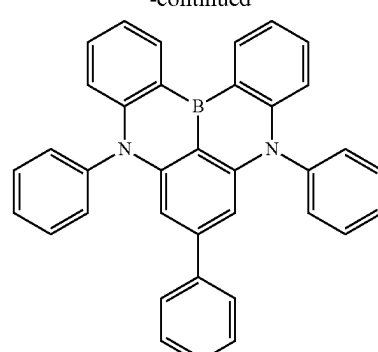
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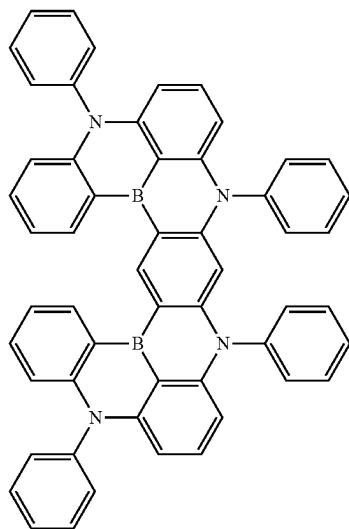
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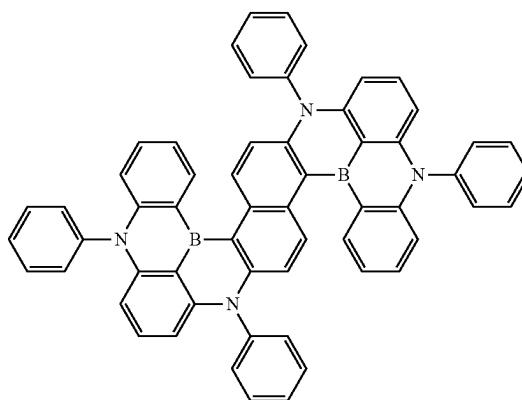
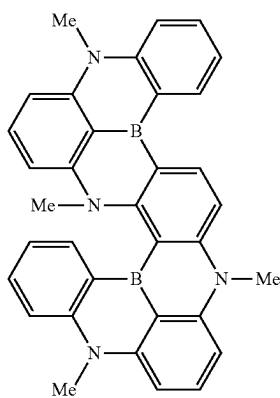
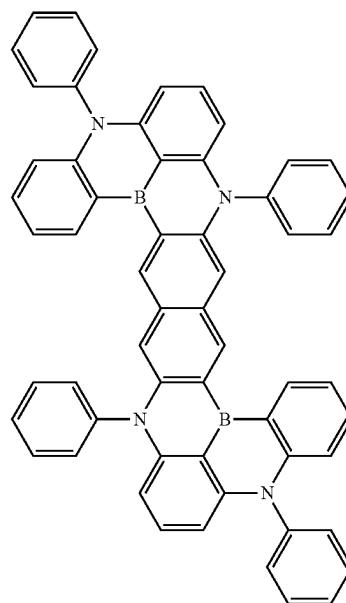
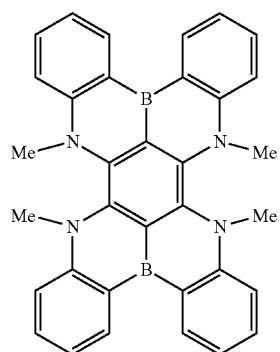
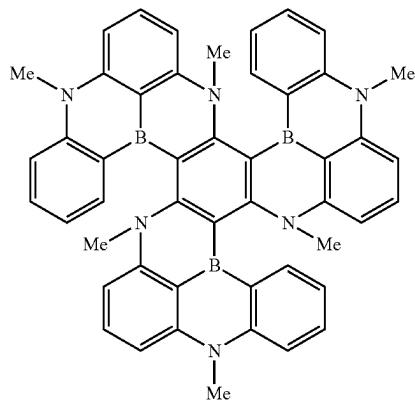
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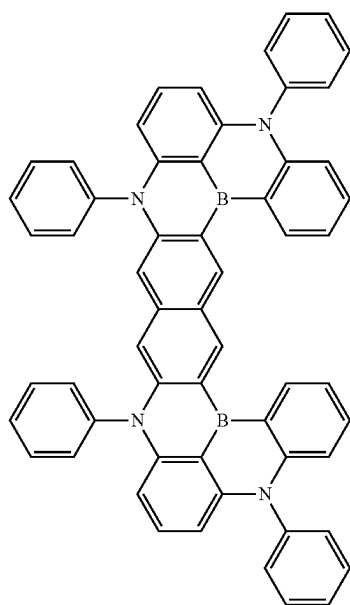
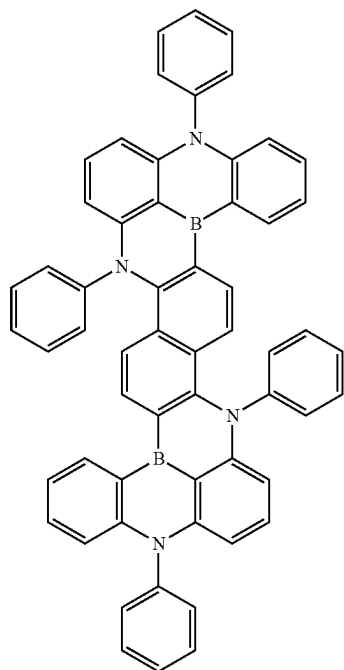
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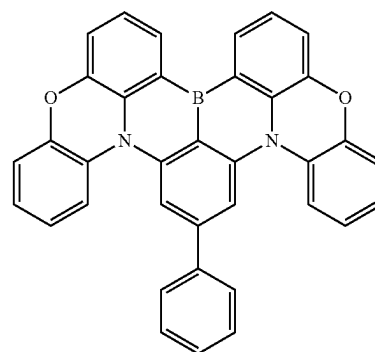
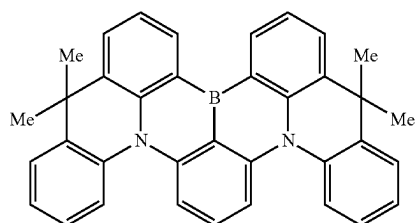
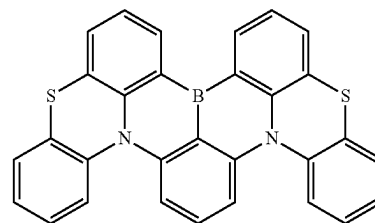
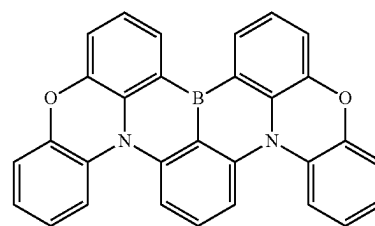
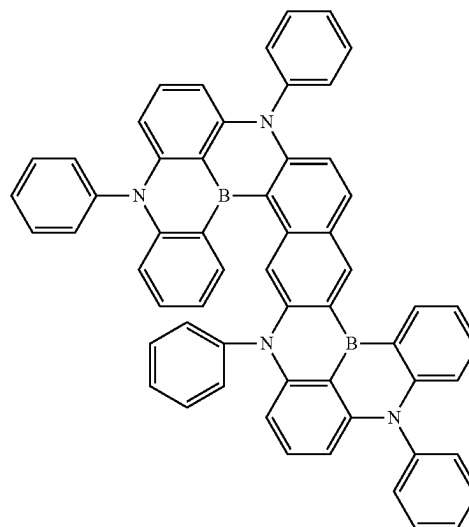
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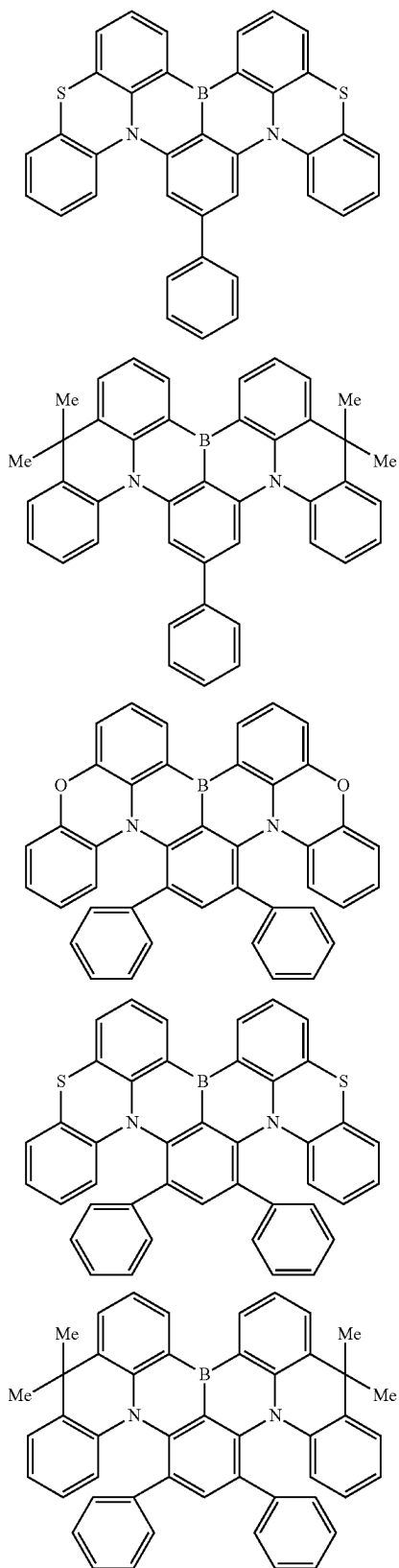
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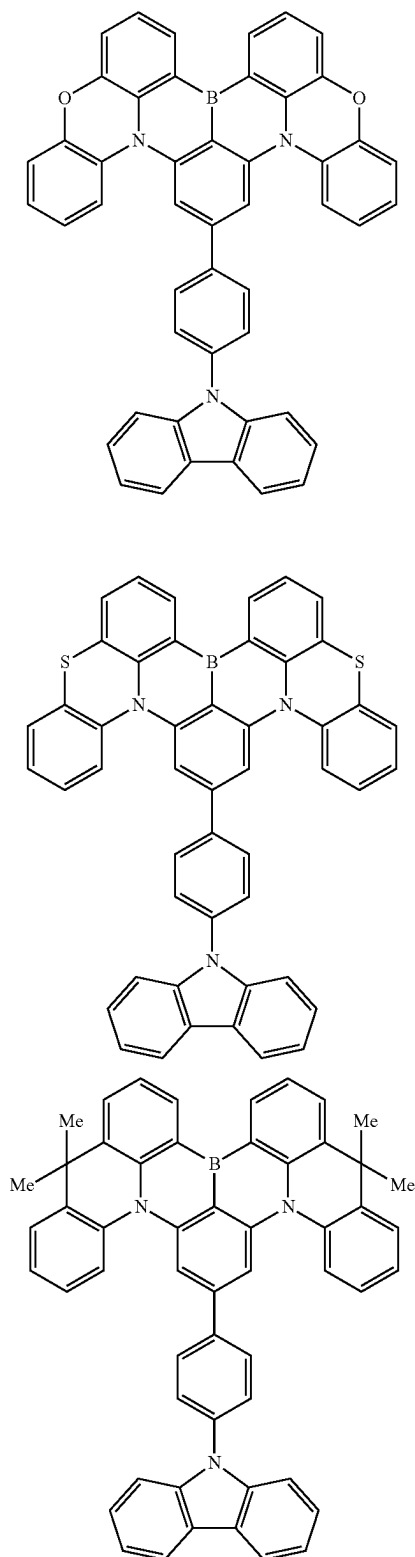
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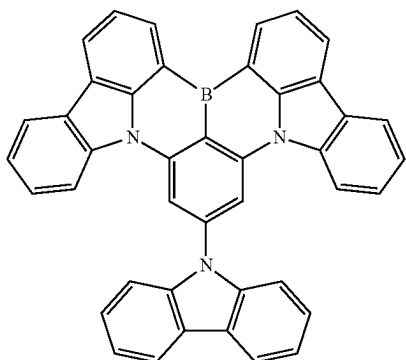
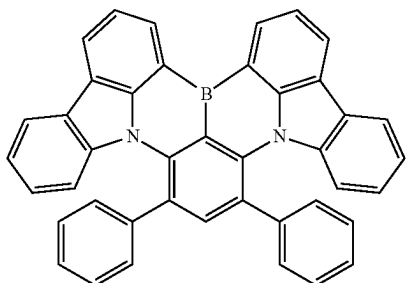
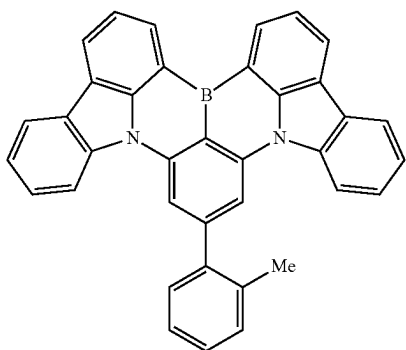
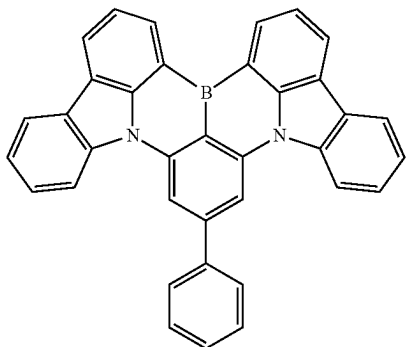
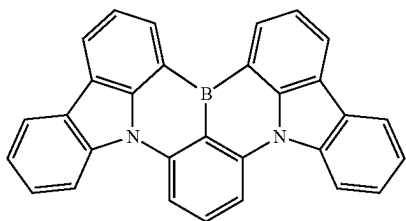
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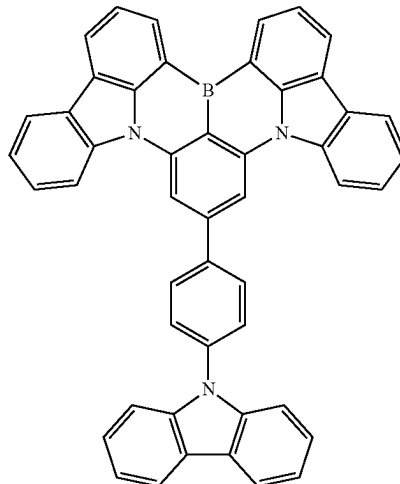
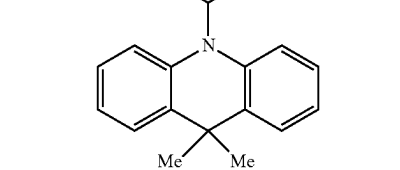
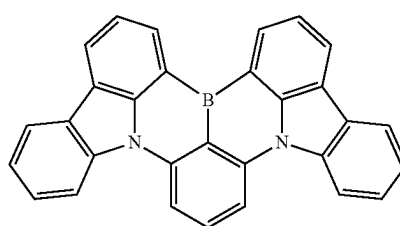
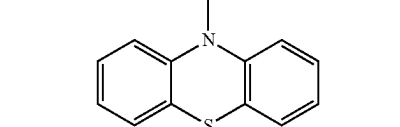
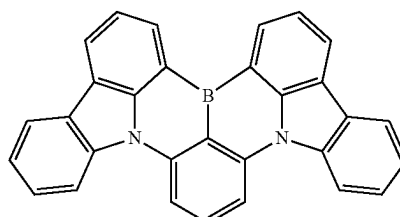
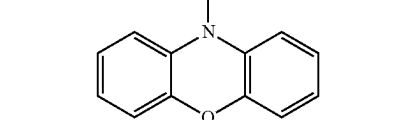
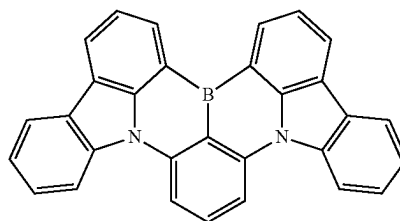
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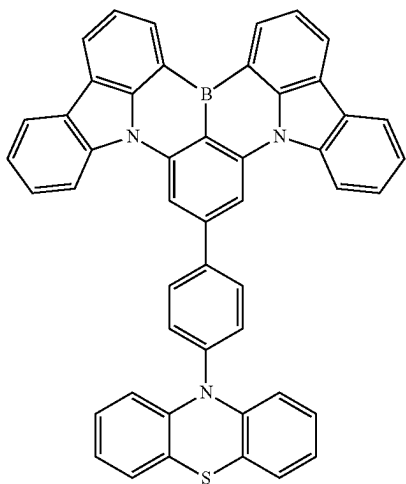
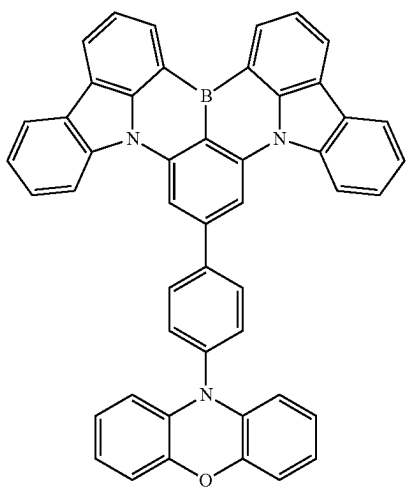
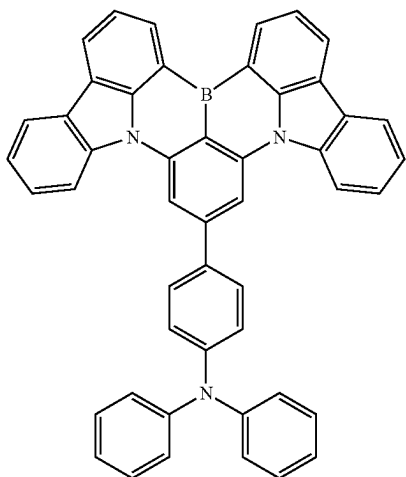
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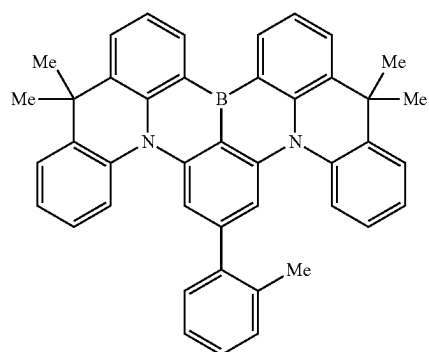
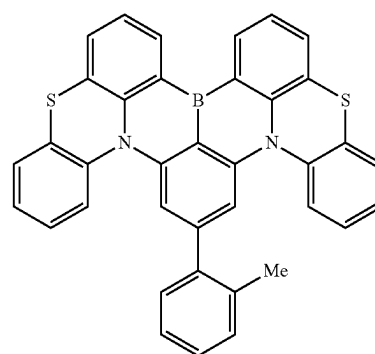
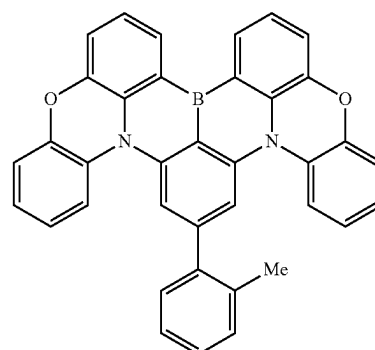
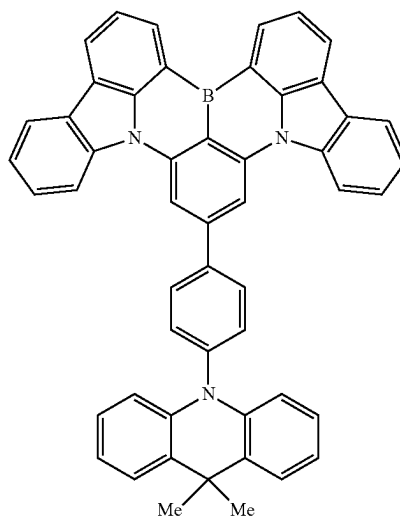
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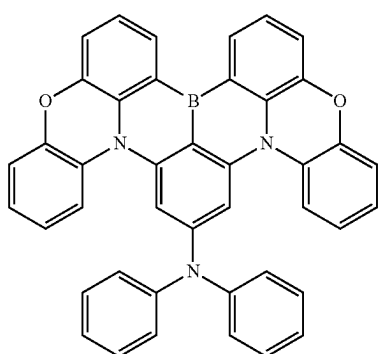
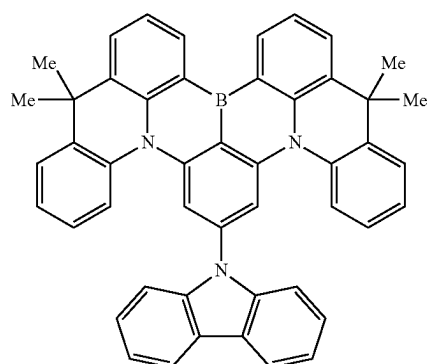
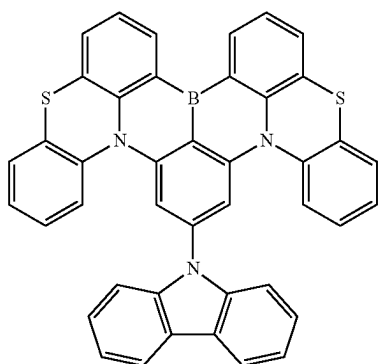
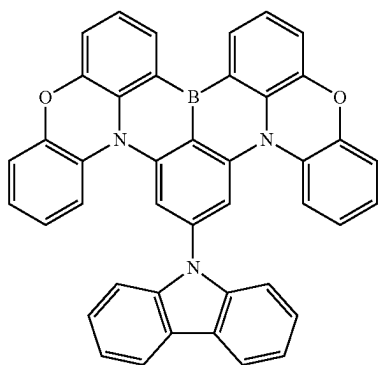
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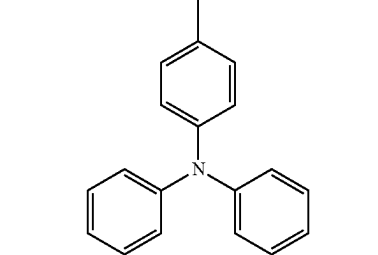
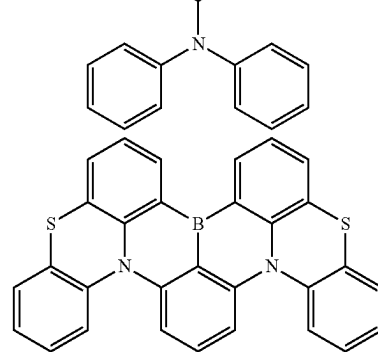
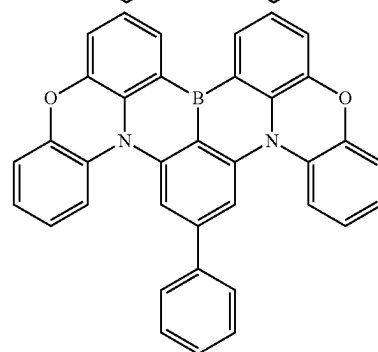
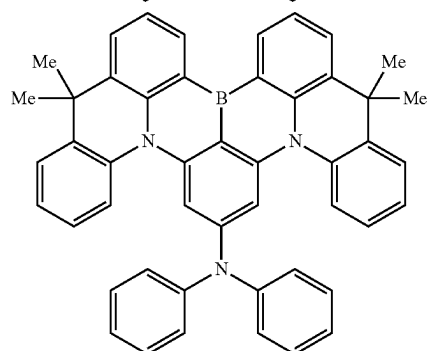
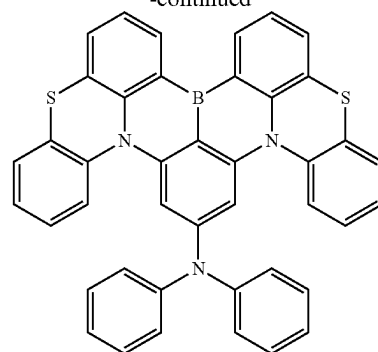
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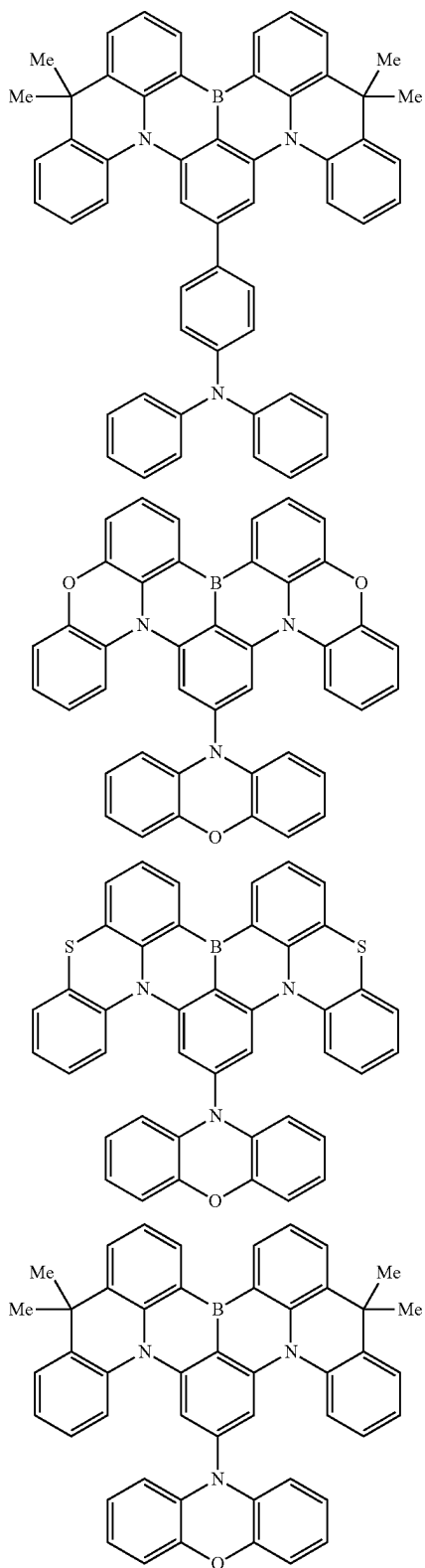
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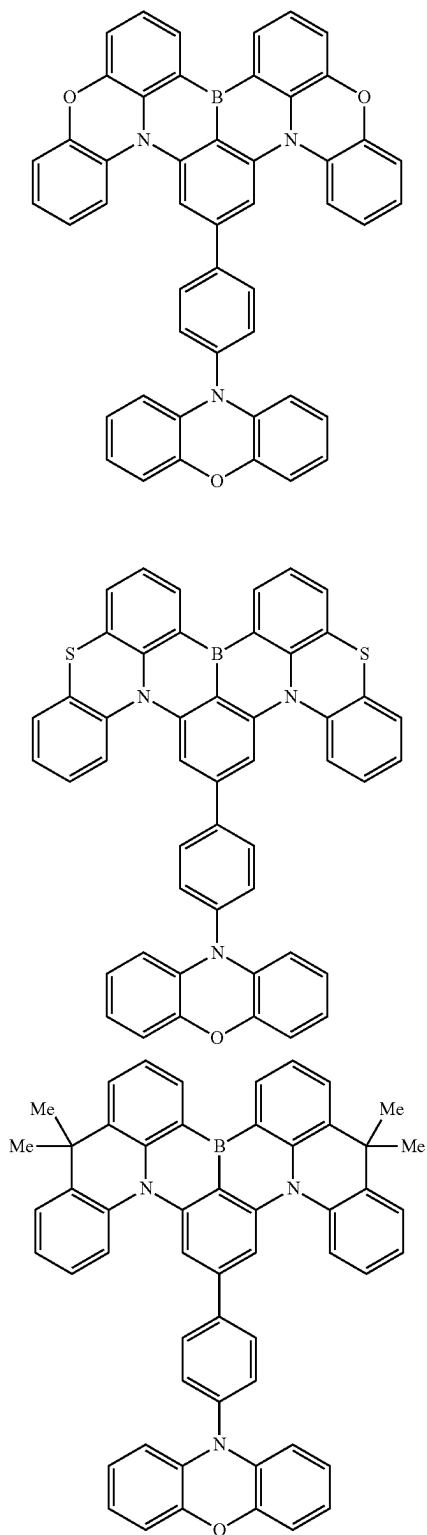
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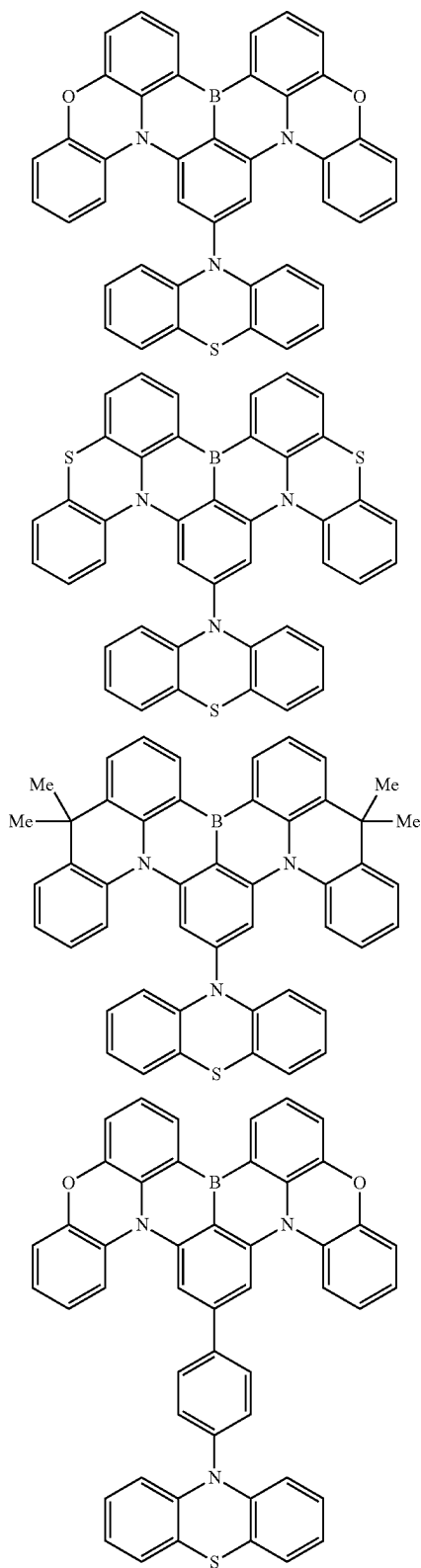
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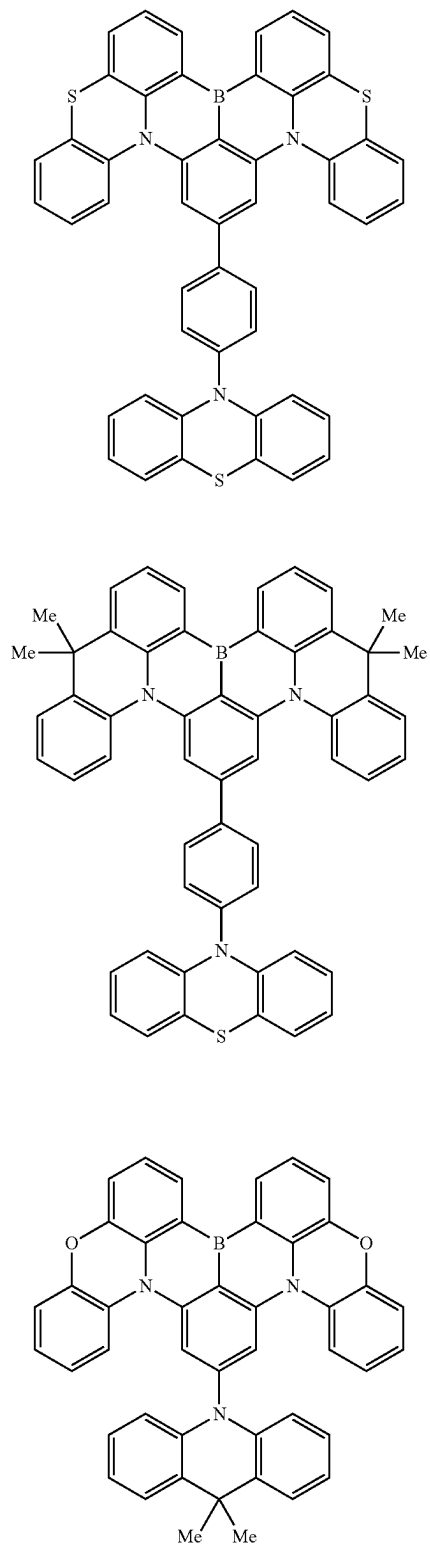
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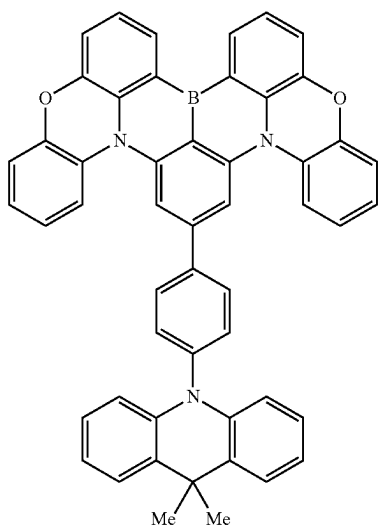
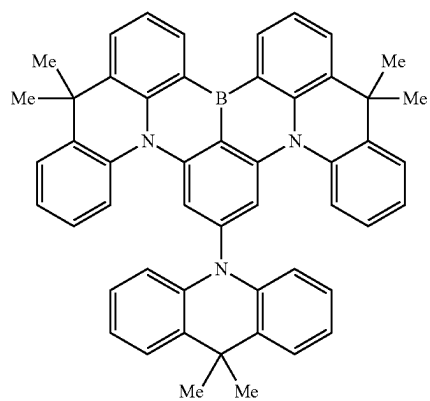
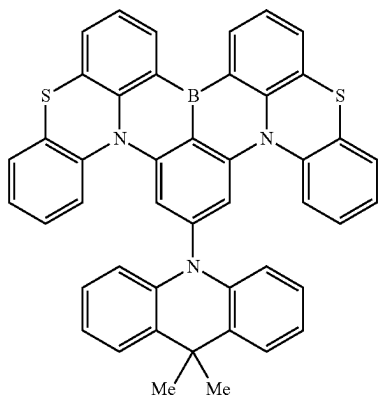
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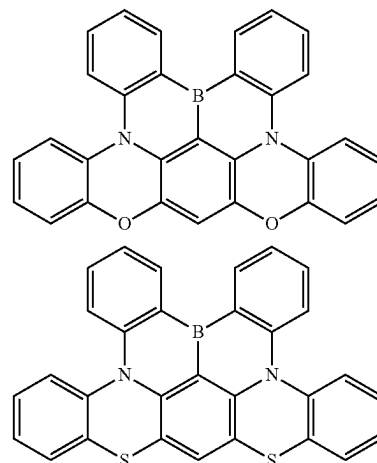
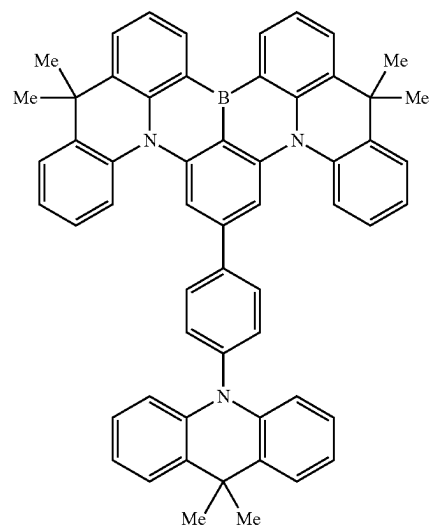
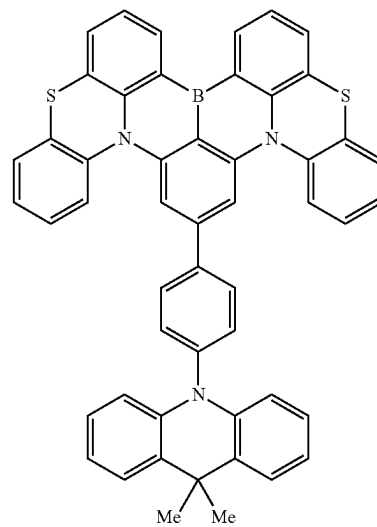
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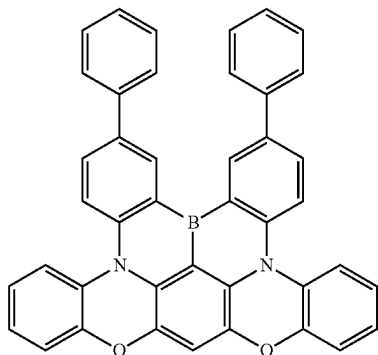
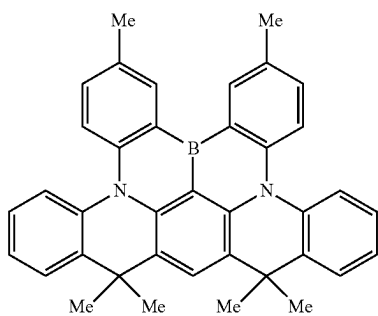
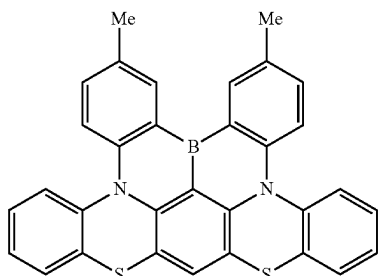
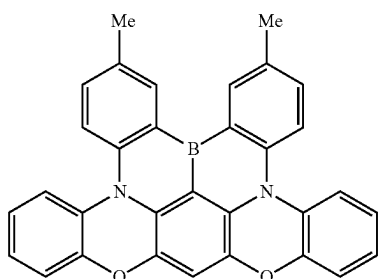
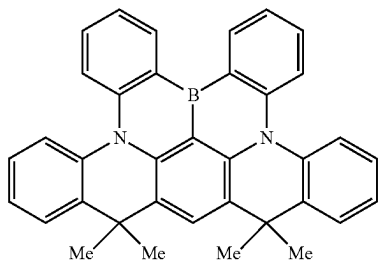
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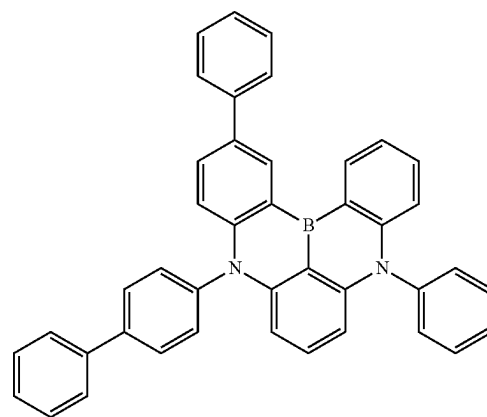
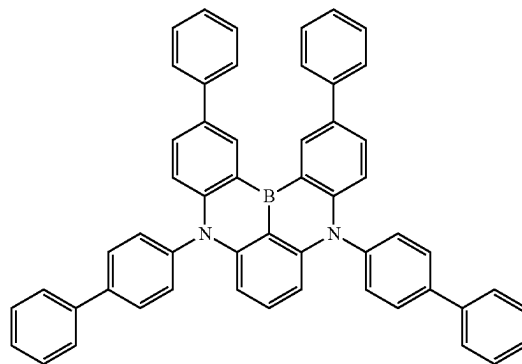
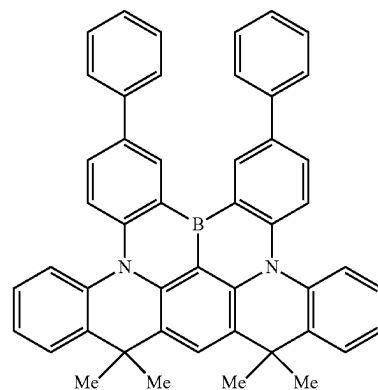
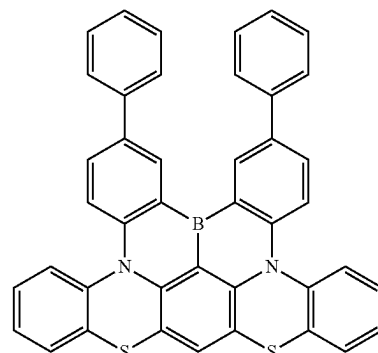
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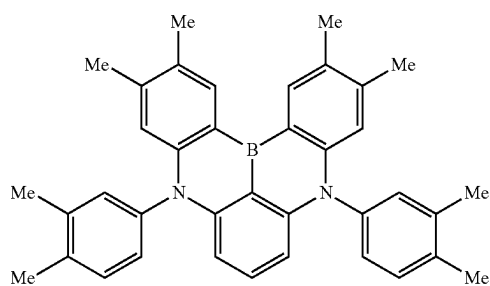
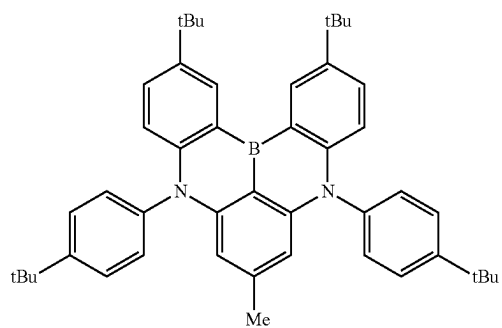
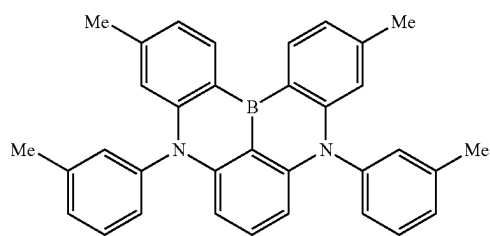
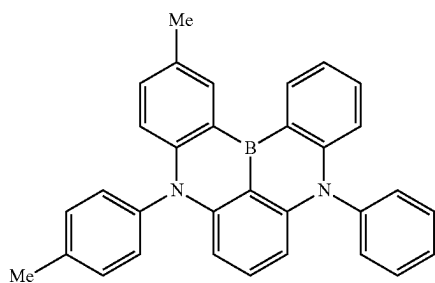
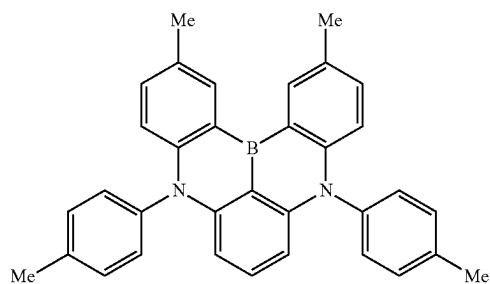
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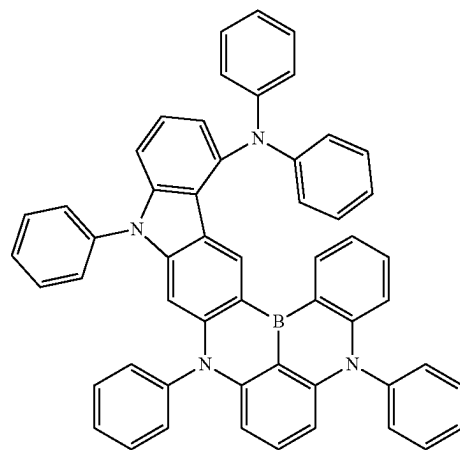
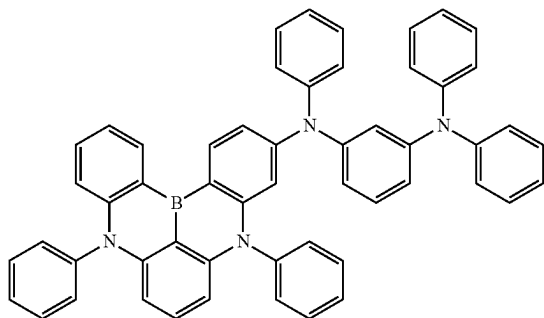
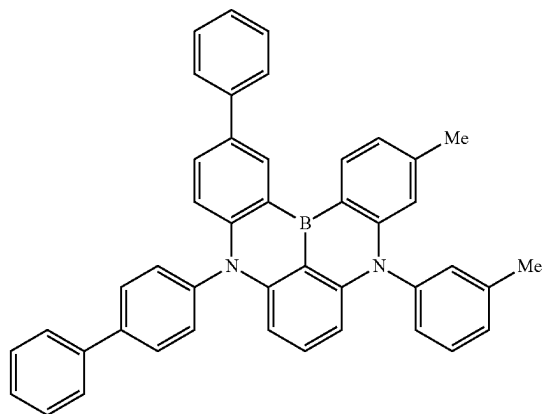
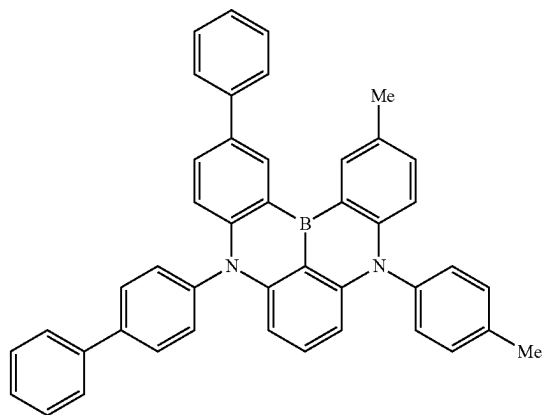
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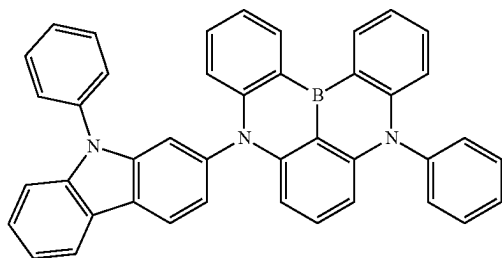
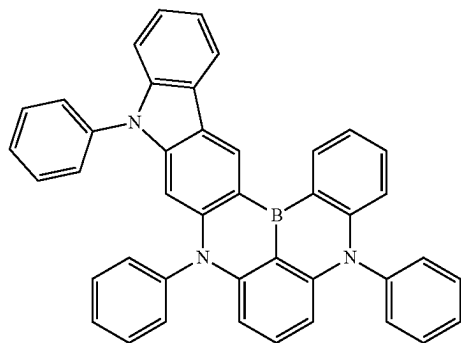
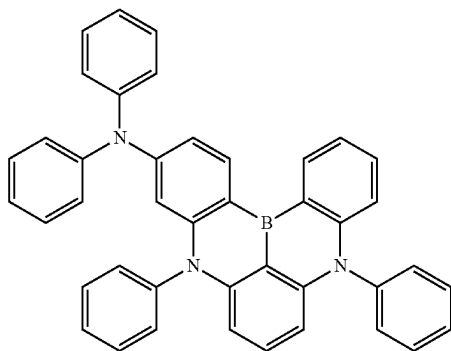
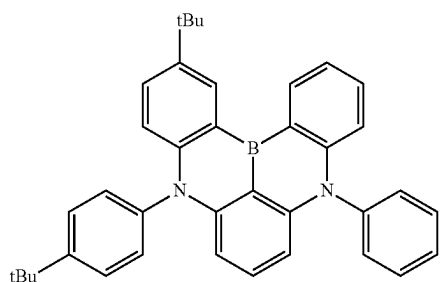
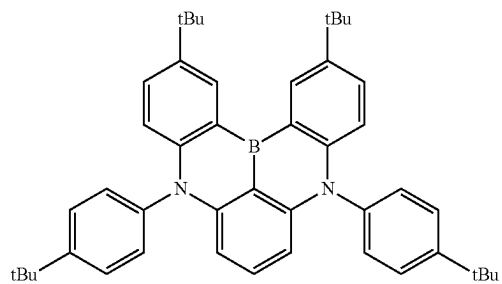
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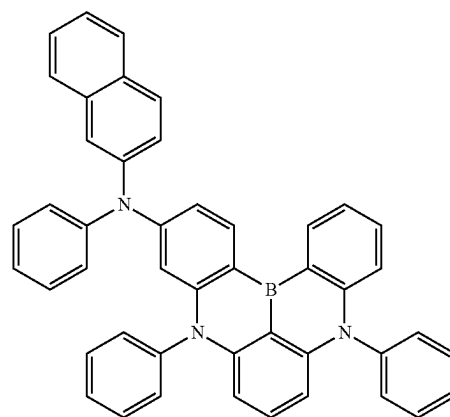
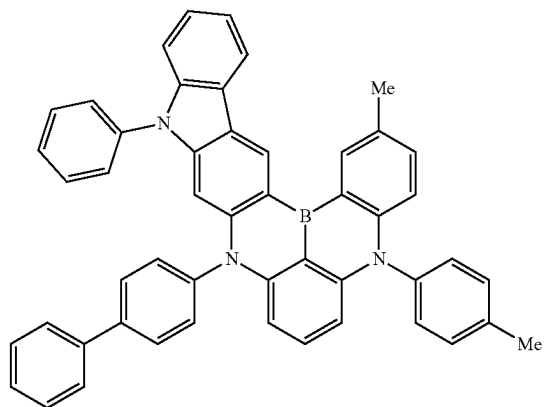
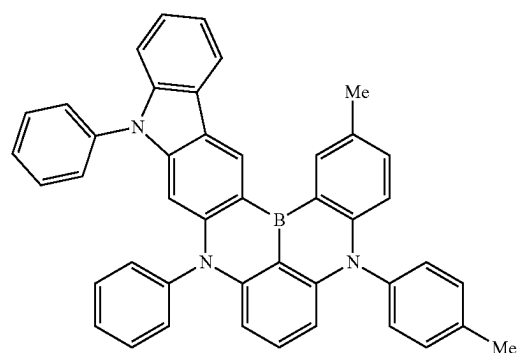
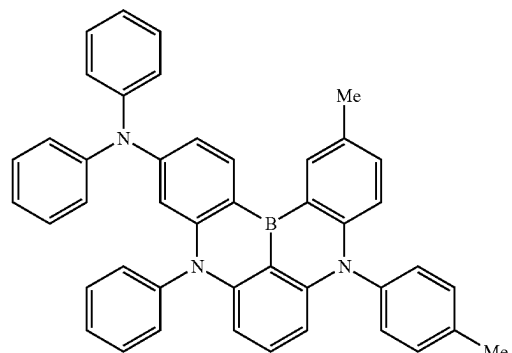
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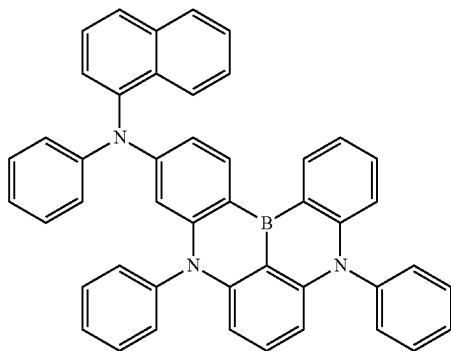
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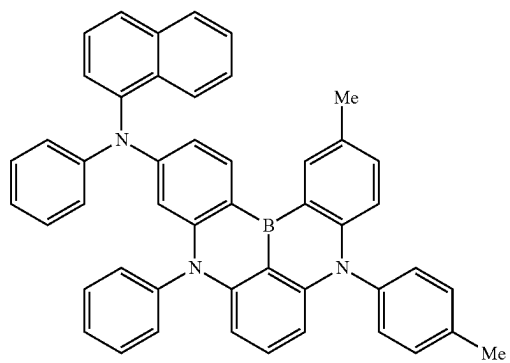
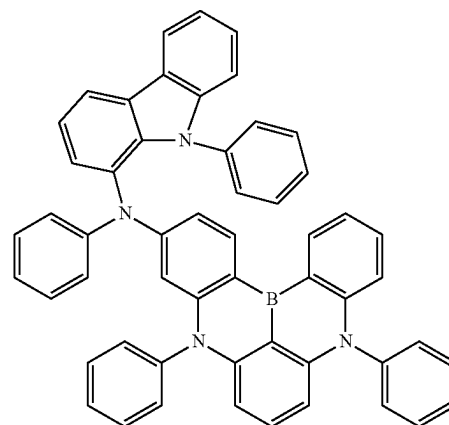
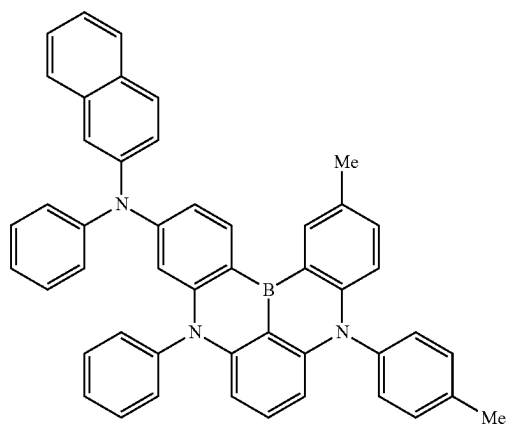
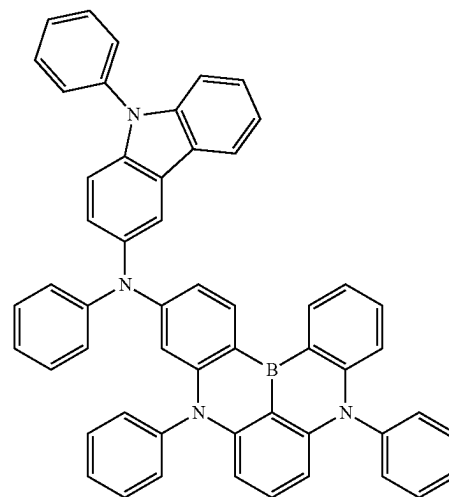
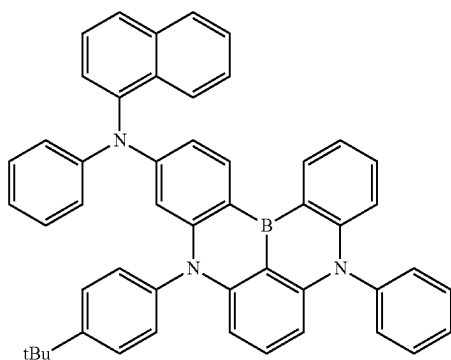
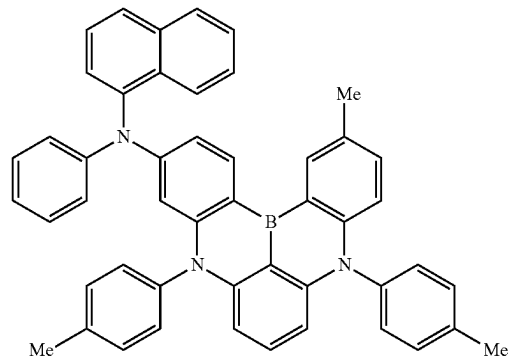
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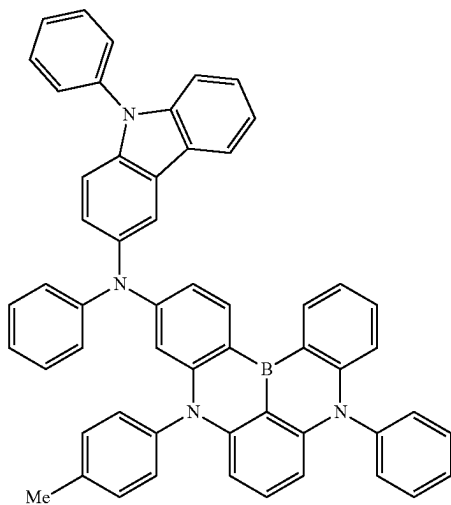
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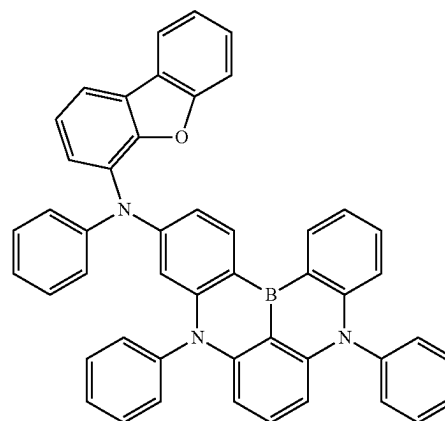
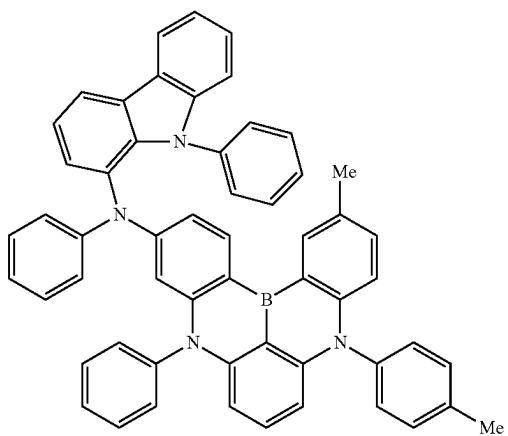
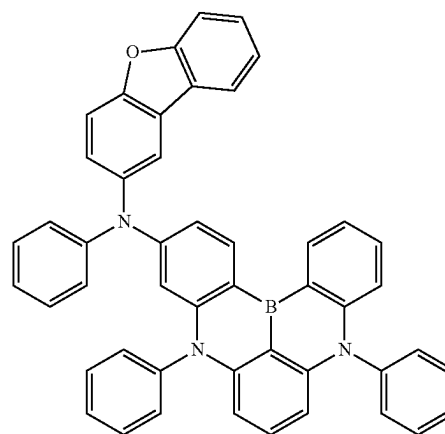
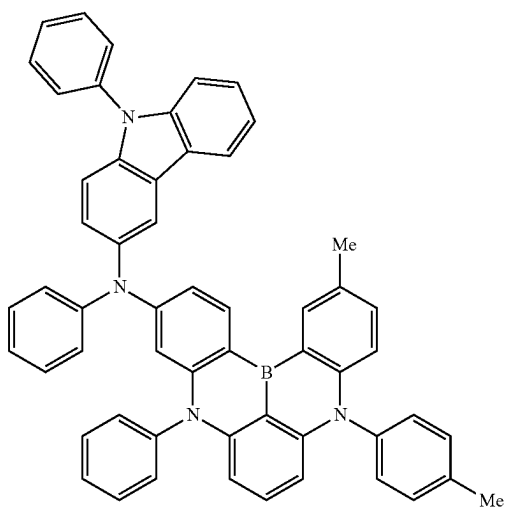
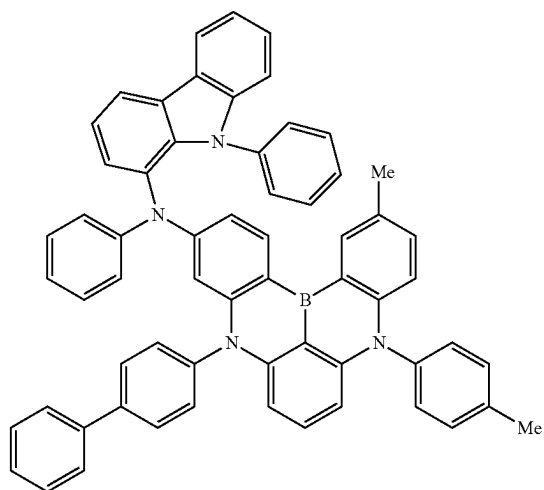
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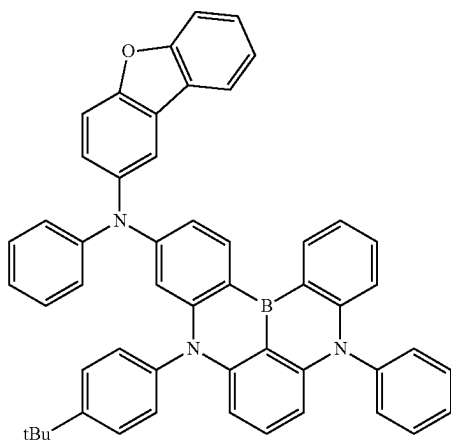
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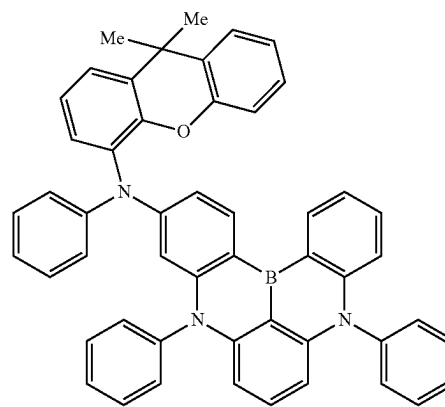
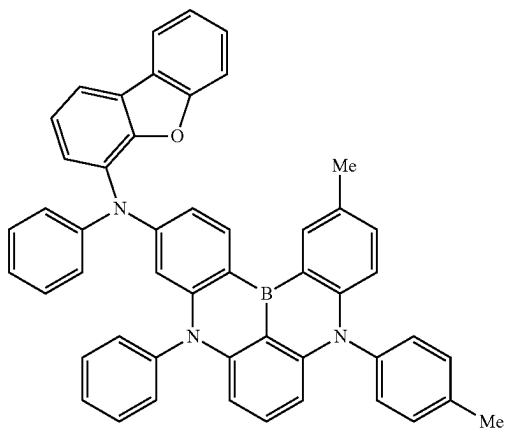
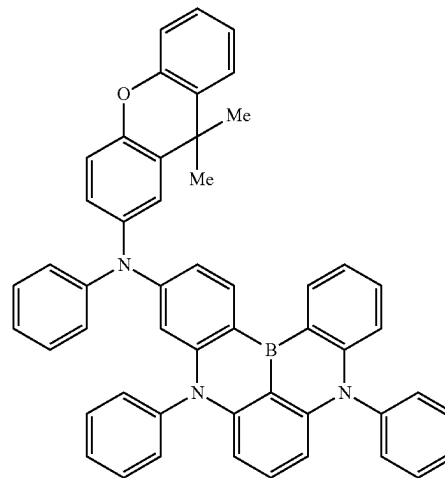
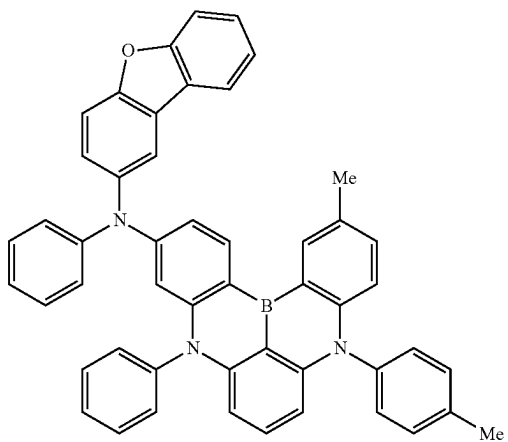
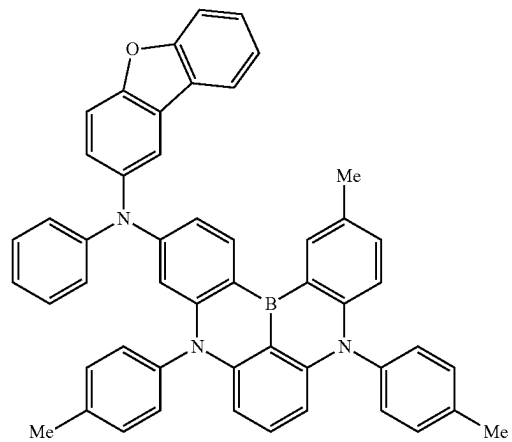
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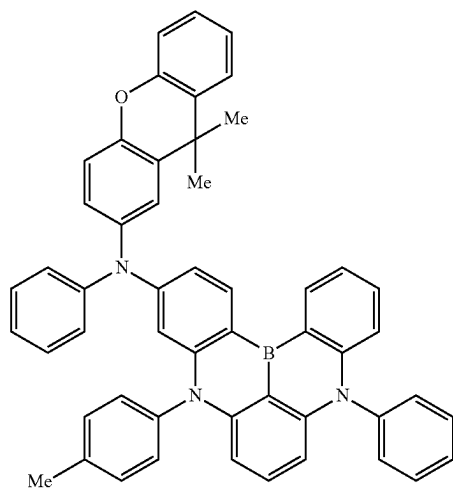
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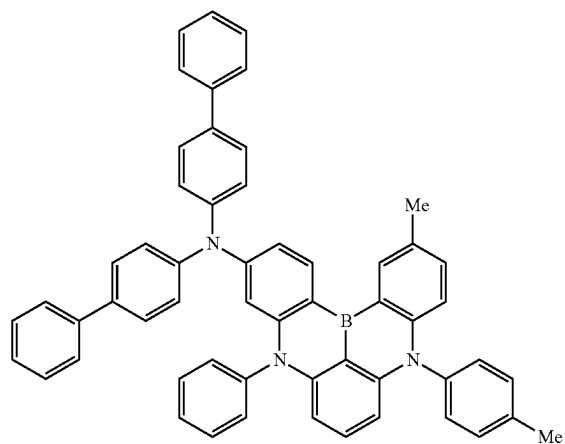
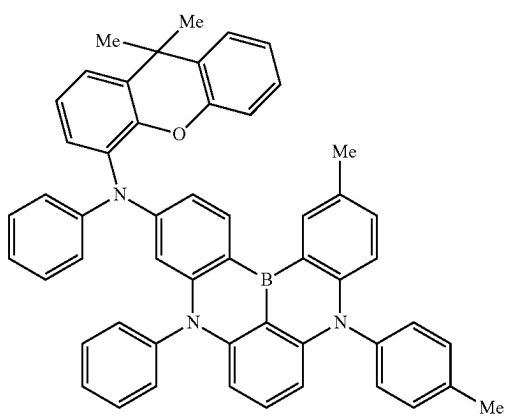
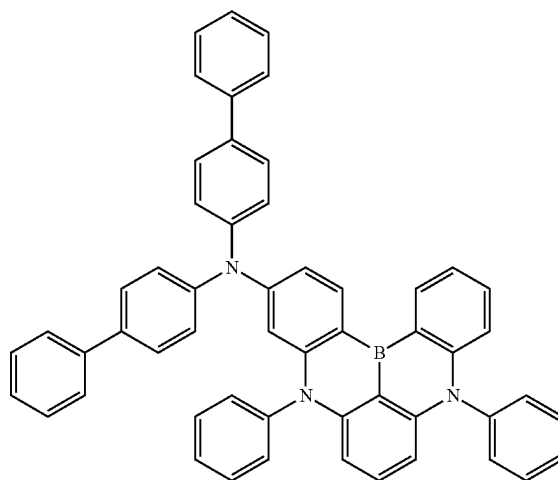
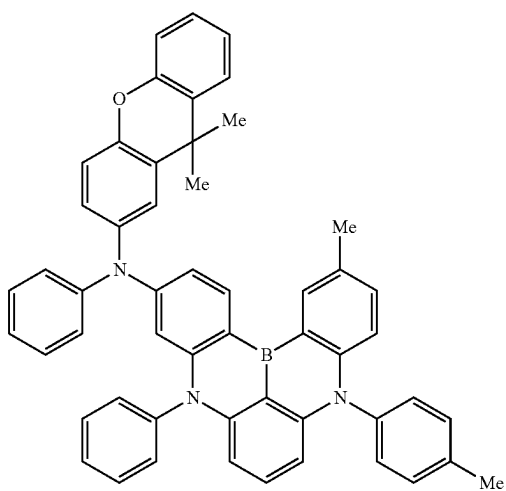
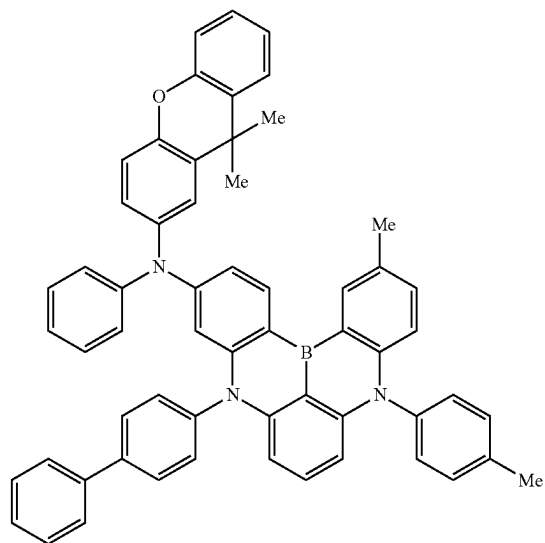
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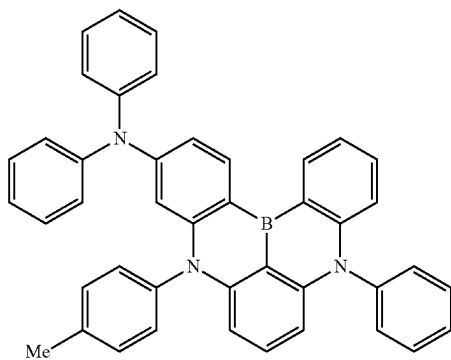
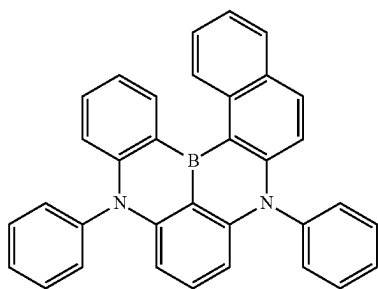
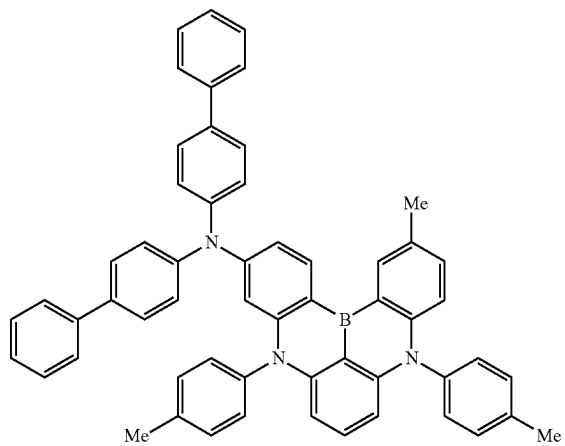
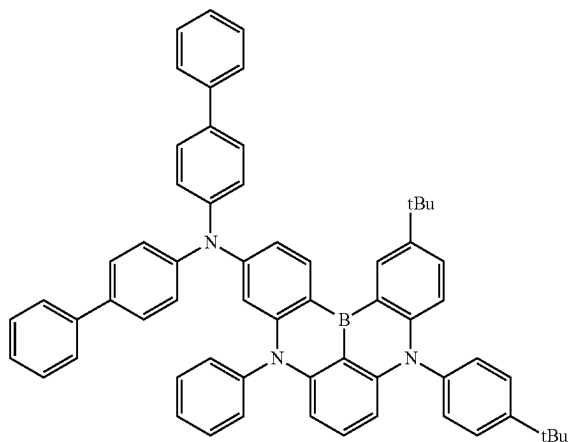
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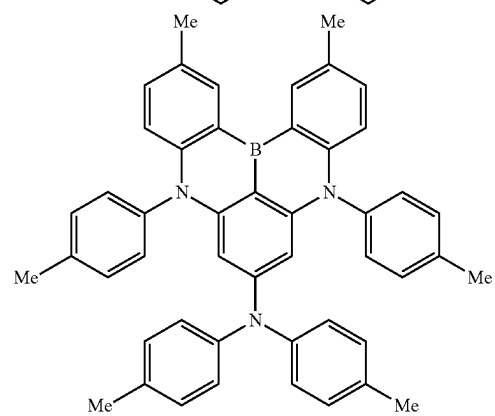
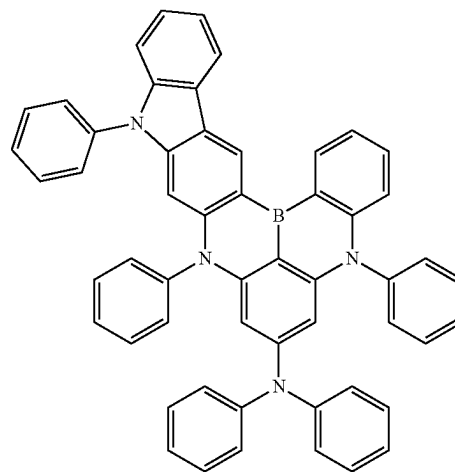
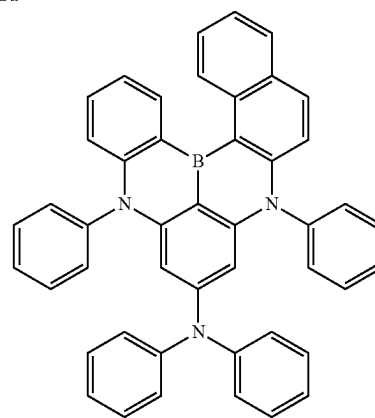
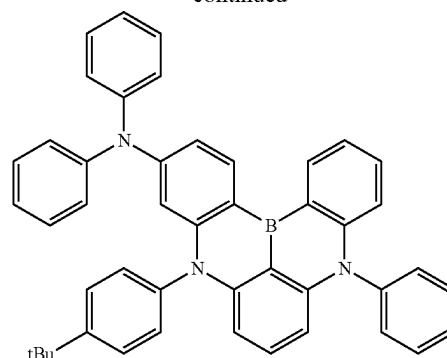
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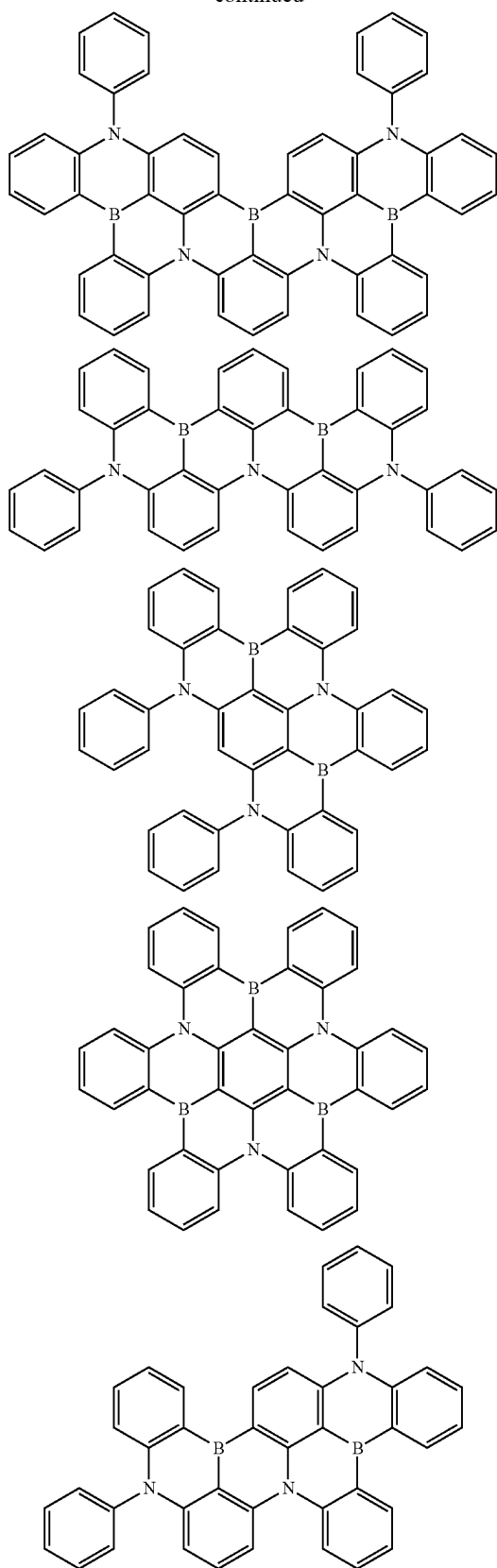
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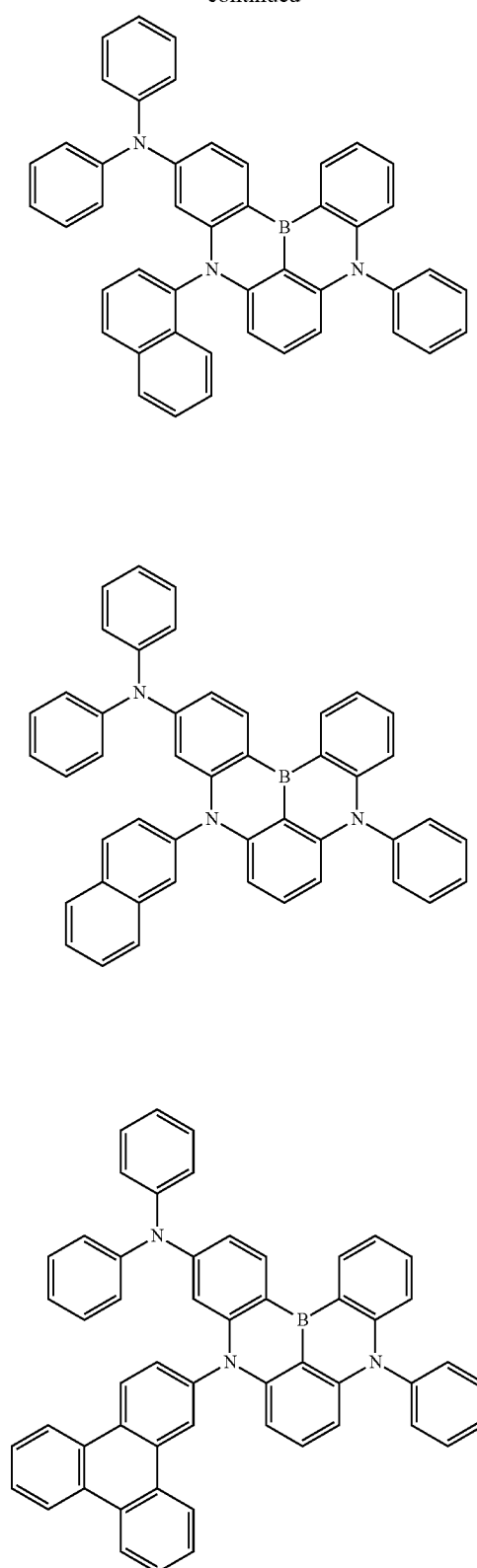
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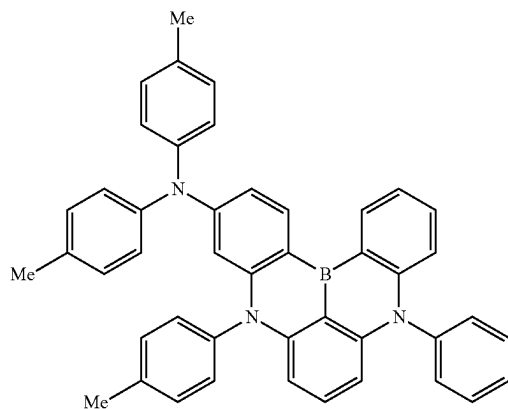
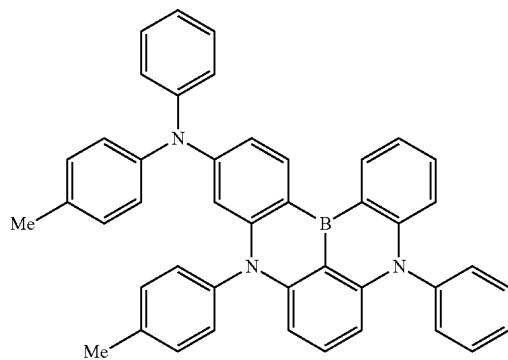
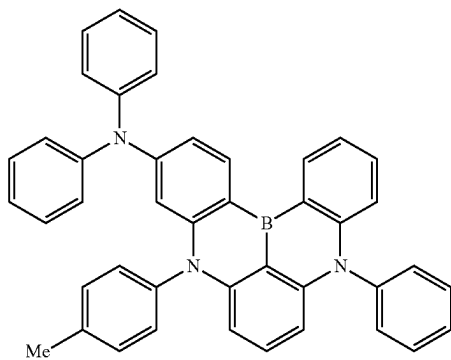
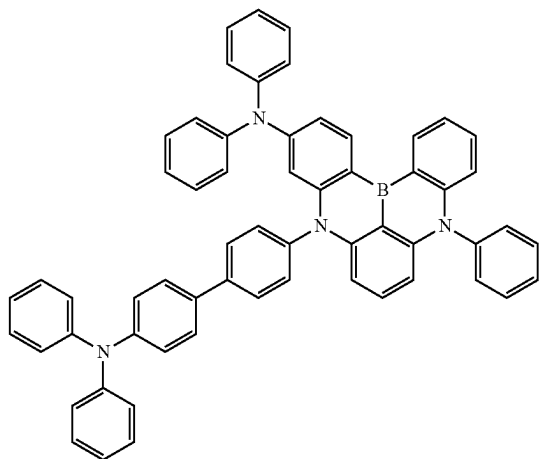
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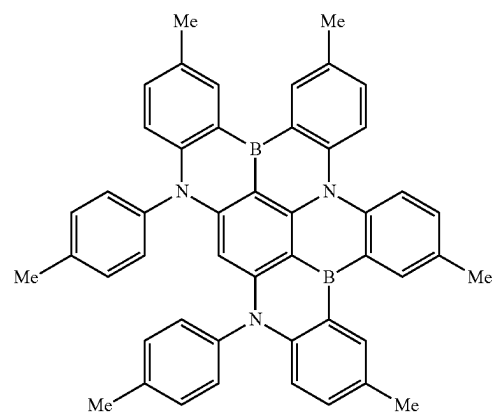
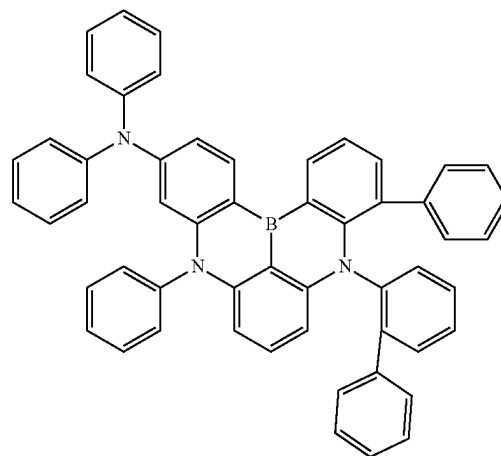
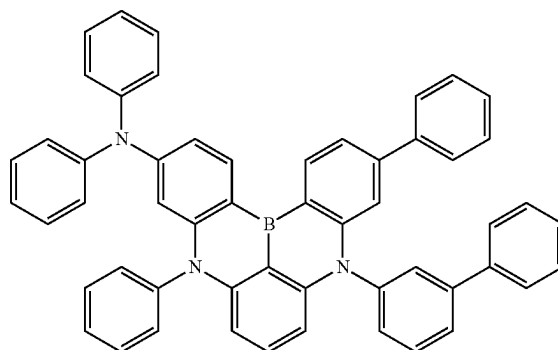
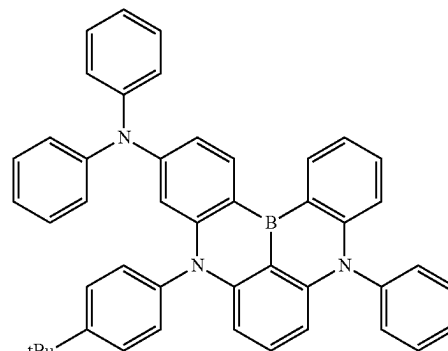
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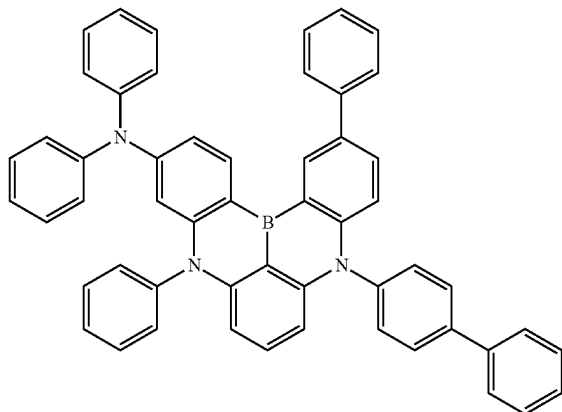
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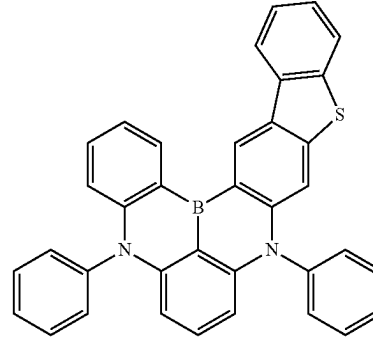
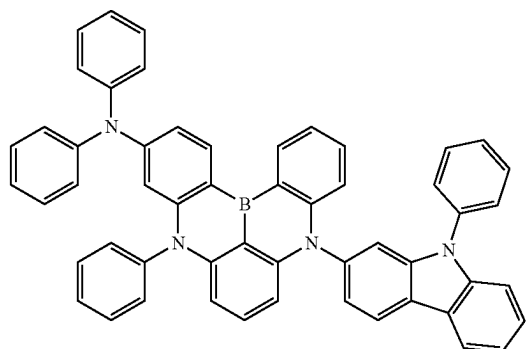
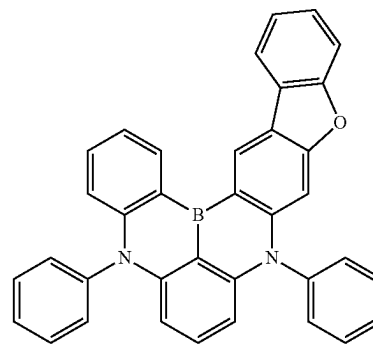
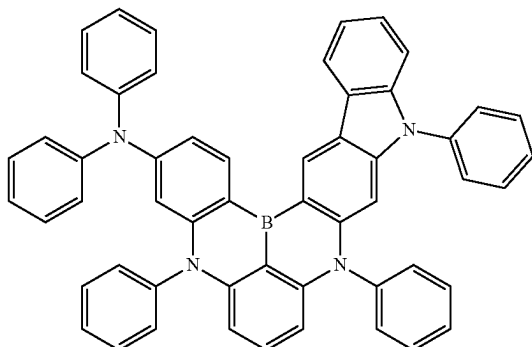
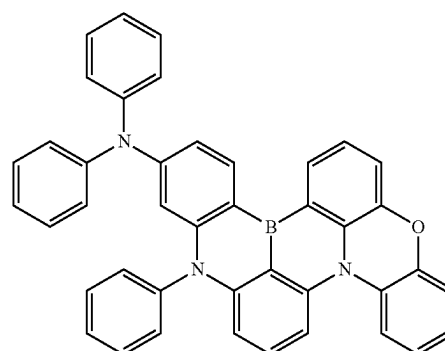
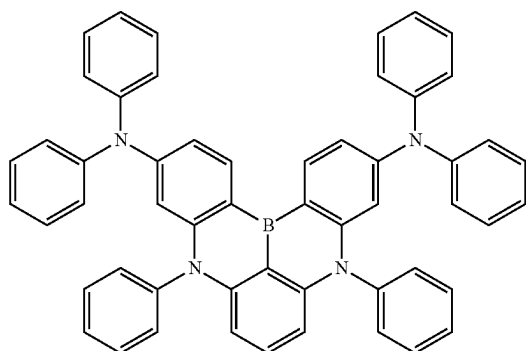
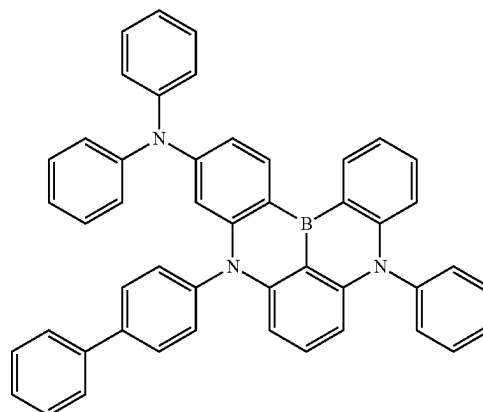
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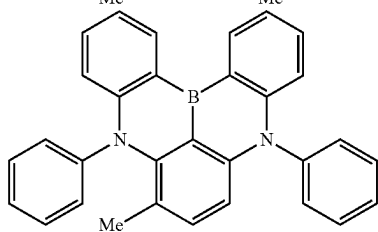
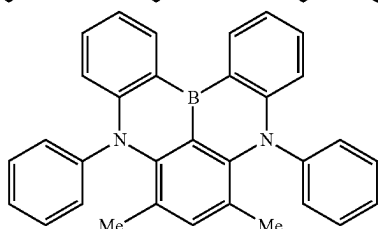
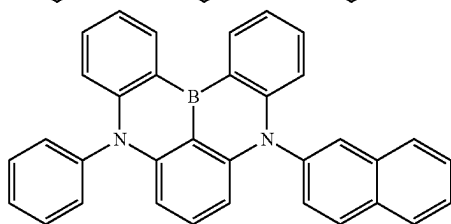
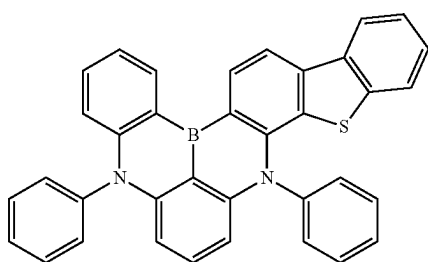
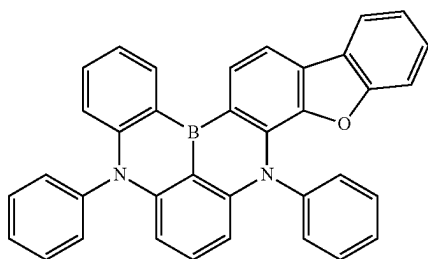
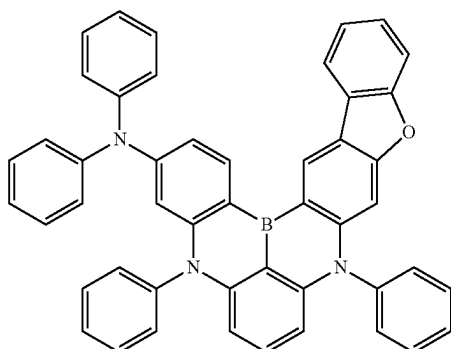
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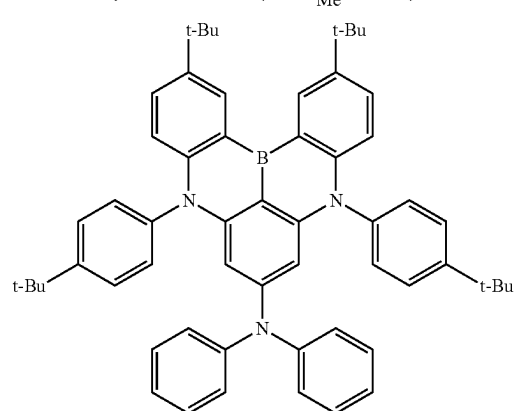
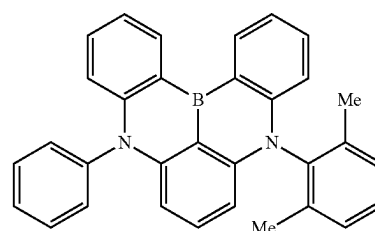
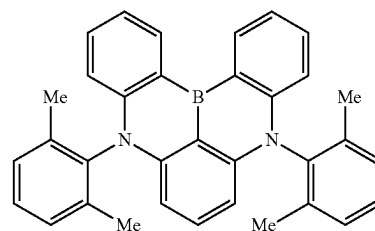
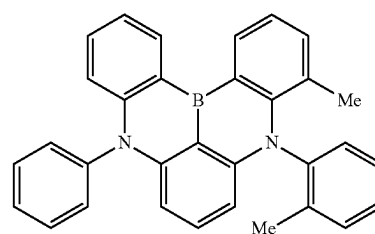
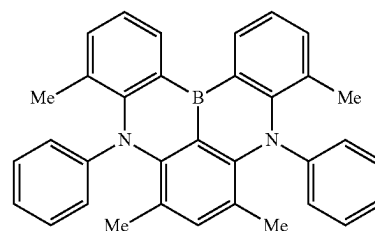
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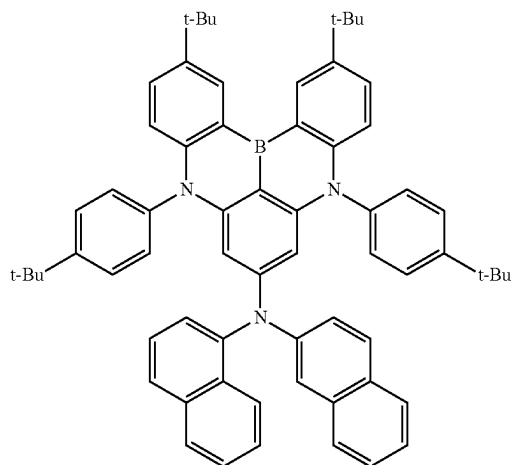
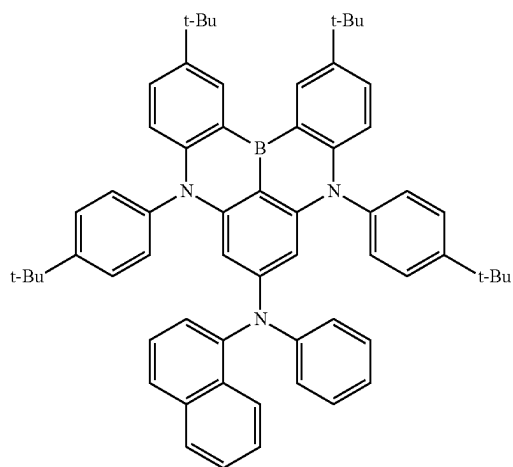
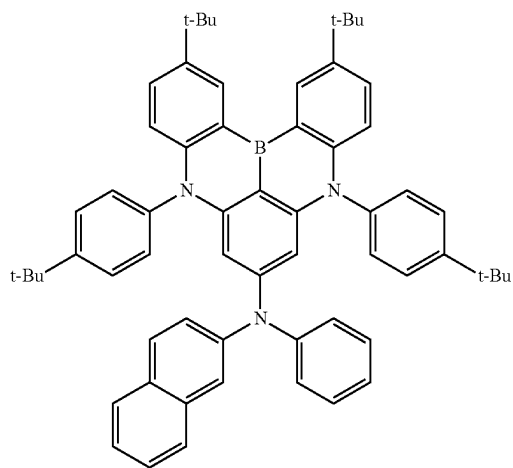
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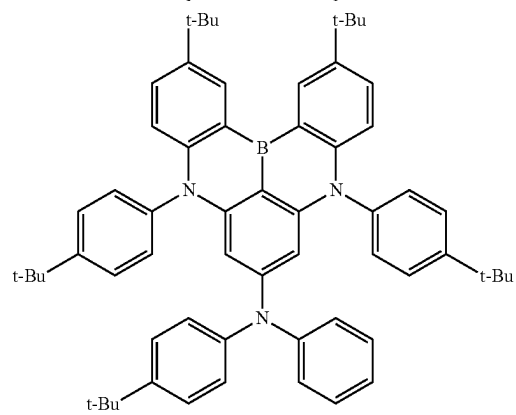
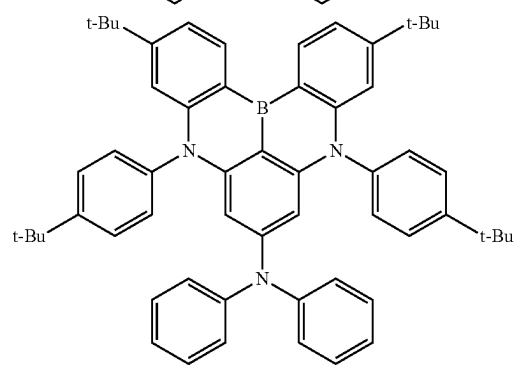
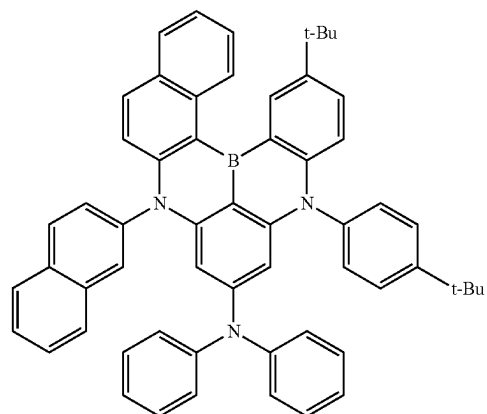
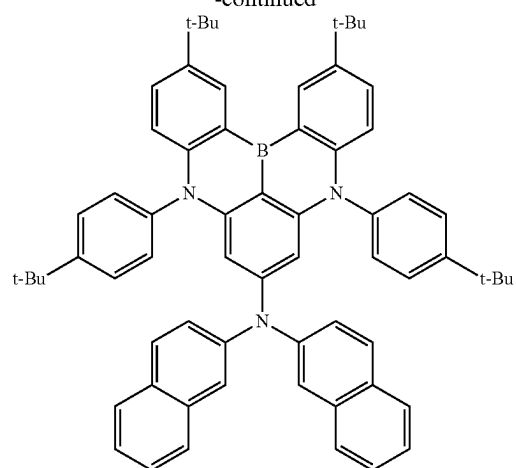
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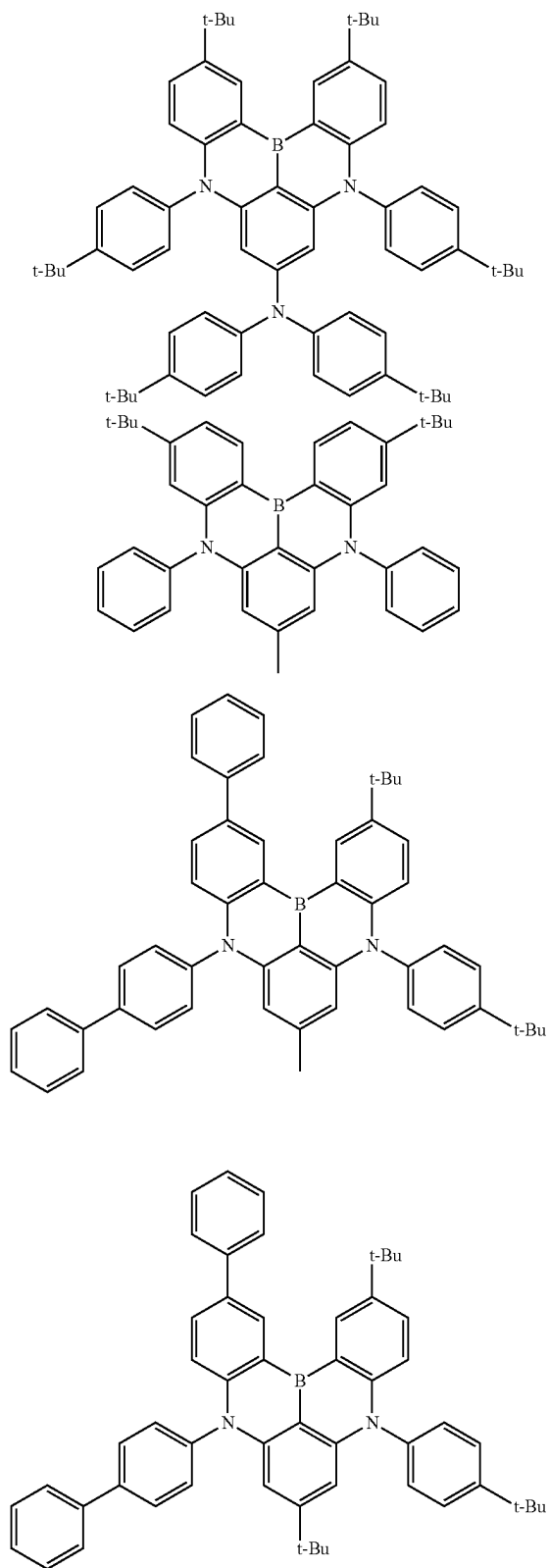
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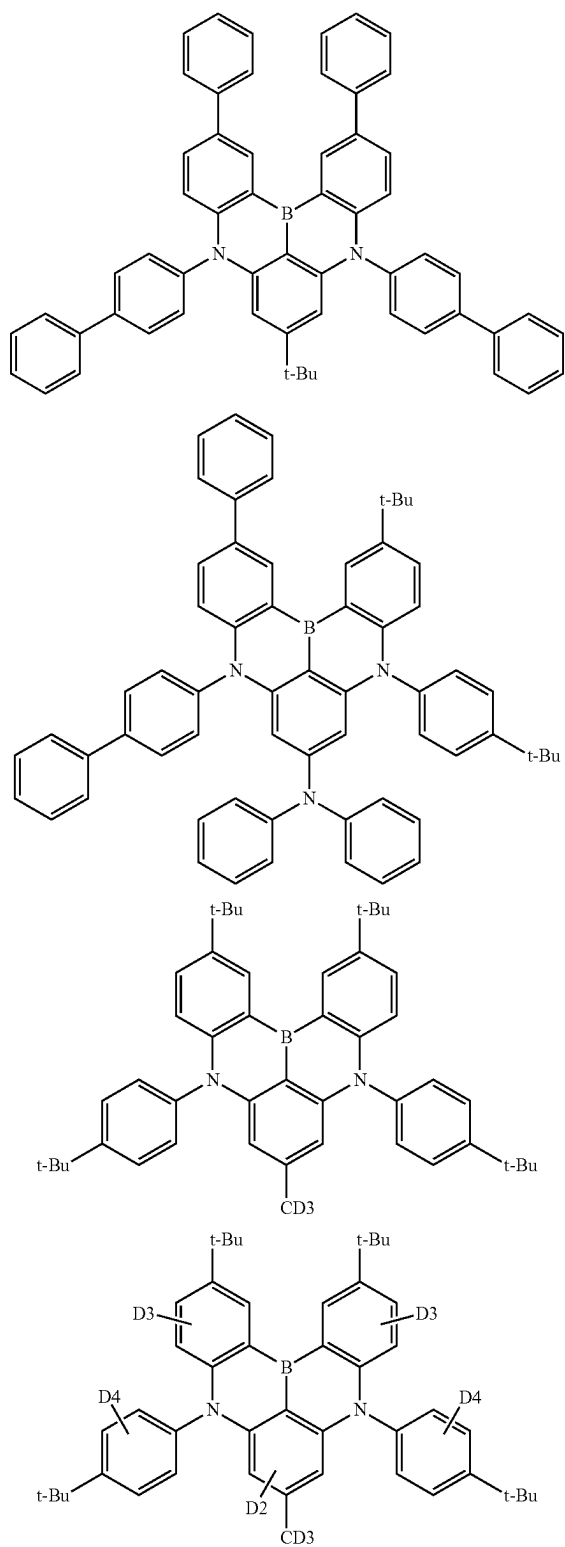
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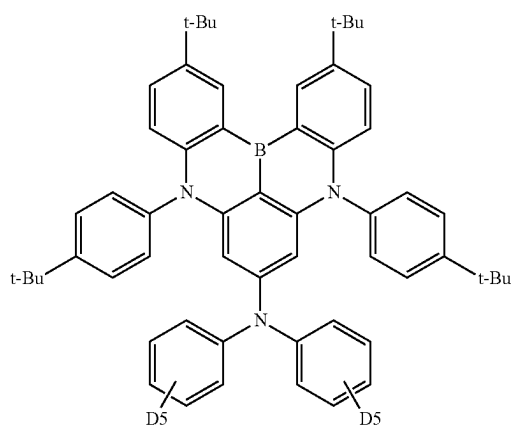
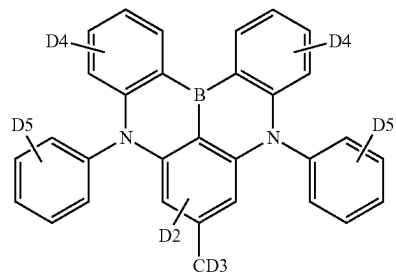
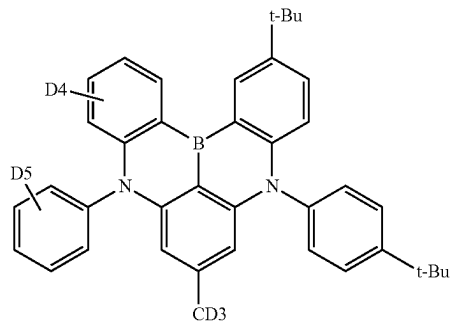
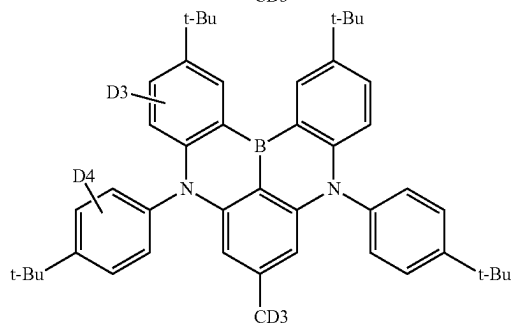
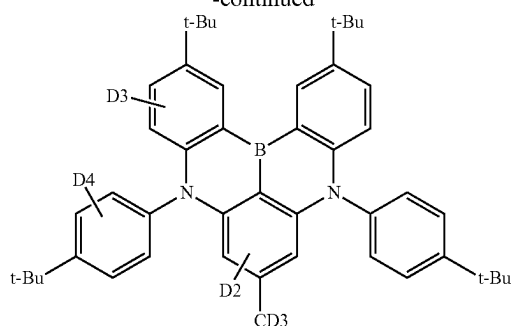
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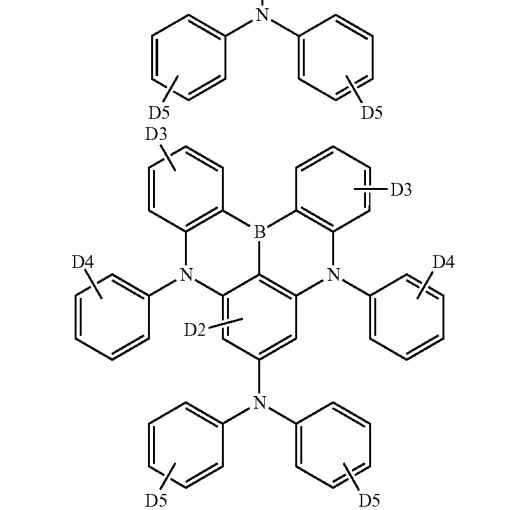
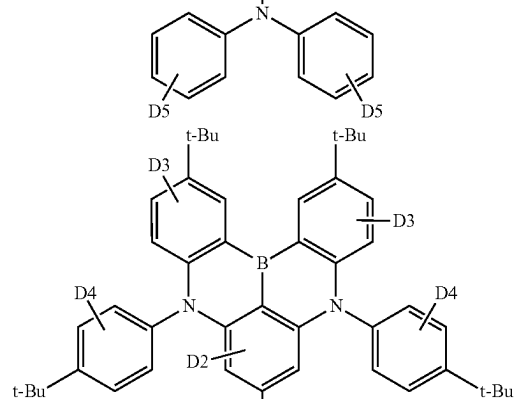
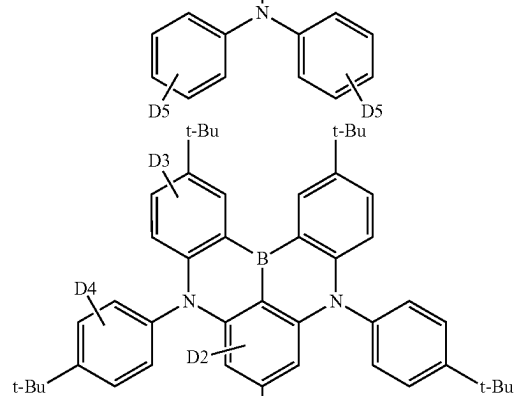
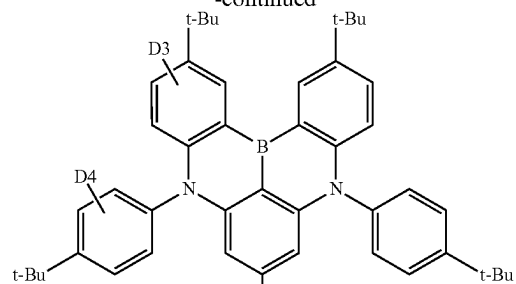
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[0098] In the above compounds, D2 to D5 each signify that 2 to 5 hydrogens are replaced with deuterium.

[0099] The formation of each layer of the organic electroluminescent device of the present disclosure can be accomplished by applying any one of dry film-forming methods such as vacuum deposition, sputtering, plasma, ion plating methods, etc., or wet film-forming methods such as spin coating, dip coating, flow coating methods, etc. When using a wet film-forming method, a thin film can be formed by dissolving or diffusing materials forming each layer into any suitable solvent such as ethanol, chloroform, tetrahydrofuran, dioxane, etc. The solvent can be any one where the materials forming each layer can be dissolved or diffused, and where there are no problems in film-formation capability.

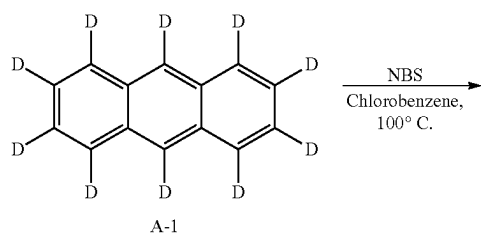
[0100] According to one embodiment of the present disclosure, when forming a layer of the first host material and the second host material, the layer can be formed by the methods listed above, and often can be formed by a co-deposition or mixed deposition process. Co-deposition is a method for mixing two or more materials into each individual crucible source and applying current to both cells simultaneously to evaporate the materials. Mixed deposition is a method for mixing two or more materials in one crucible source before deposition and then applying current to one cell to evaporate the materials.

[0101] According to one embodiment of the present disclosure, when the first host material and the second host material are present in the same layer or different layers in the organic electroluminescent device, the two host compounds may be individually formed into films. For example, the second host material may be deposited after depositing the first host material.

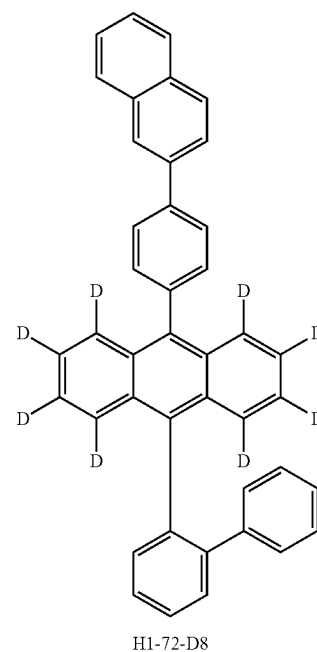
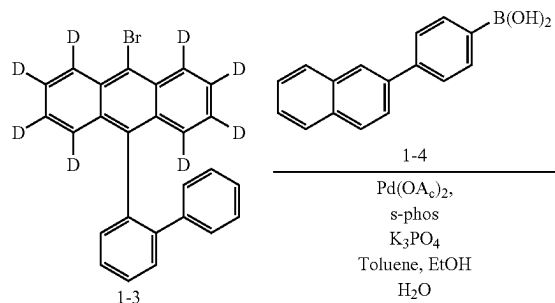
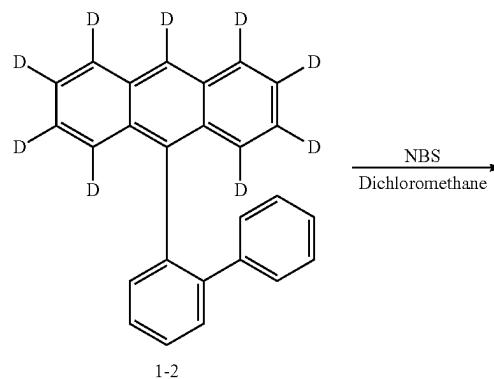
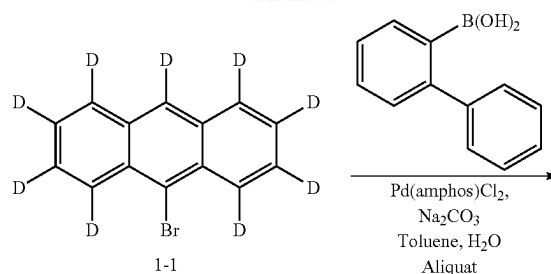
[0102] According to one embodiment of the present disclosure, a display system comprising the plurality of host materials comprising the first host compound represented by Formula 1 and the second host compound represented by Formula 2; or the organic electroluminescent compound represented by Formula 11 can be provided. In addition, it is possible to produce a display system, e.g., a display system for smart phones, tablets, notebooks, PCs, TVs, or cars; or a lighting system, e.g., an outdoor or indoor lighting system, by using the organic electroluminescent device of the present disclosure.

[0103] Hereinafter, the preparation method of the compound according to the present disclosure and the properties thereof, as well as the driving voltage, conversion efficiency, and lifetime properties of the OLED according to the present disclosure will be explained. However, the following examples are only provided to explain the characteristics of the compound and OLED according to the present disclosure for a detailed understanding of the present disclosure, and the present disclosure is not limited to the following examples.

Example 1: Preparation of Compound H1-72-D8



-continued



Synthesis of Compound 1-1

[0104] In a flask, Compound A-1 (47 g, 250 mmol) was dissolved in chlorobenzene (1.3 L), and N-bromosuccinimide (NBS) (49 g, 275 mmol) was added thereto, and then the mixture was reacted at 100° C. for 7 hours. After completion of the reaction, the mixture was neutralized with K_2CO_3 and sodium thiosulfate aqueous solution, the organic layer was extracted with dichloromethane, and treated with $MgSO_4$. The organic layer was filtered and concentrated, and the resulting mixture was purified by column chromatography to obtain Compound 1-1 (49.6 g, yield: 74%).

Synthesis of Compound 1-2

[0105] In a 3 L of round bottom flask (RBF), Compound 1-1 (49.6 g, 186 mmol), [1,1'-biphenyl]-2-ylboronic acid (44.3 g, 224 mmol), bis(di-tert-butyl(4-dimethylamino-phenyl)phosphine)dichloropalladium (II) (13.1 g, 18.6 mmol), Aliquat (7.5 g, 18.6 mmol), sodium carbonate (39.4 g, 372 mmol), 1.2 L of toluene, and 372 mL of distilled water were added, and the mixture was stirred under reflux at 140° C. for 4 hours. After completion of the reaction, the mixture was cooled to room temperature, and an organic layer was extracted with dichloromethane, and treated with $MgSO_4$. The organic layer was filtered and concentrated, and the resulting mixture was purified by column chromatography to obtain Compound 1-2 (54 g, yield: 85%).

Synthesis of Compound 1-3

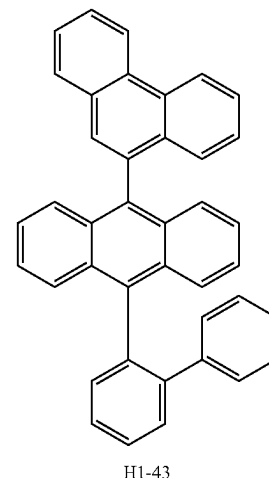
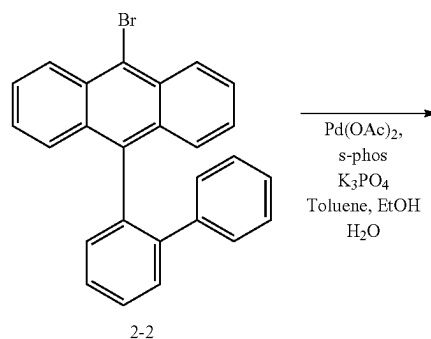
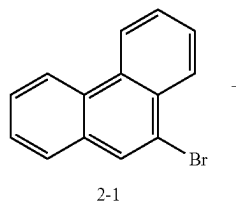
[0106] In a flask, Compound 1-2 (54 g, 159 mmol) was dissolved in dichloromethane (1.1 L), and N-bromosuccinimide (42.5 g, 239 mmol) was added thereto, and then the mixture was reacted at 35° C. for 20 hours. After completion of the reaction, the mixture was neutralized with K_2CO_3 and sodium thiosulfate aqueous solution, the organic layer was extracted with dichloromethane and treated with $MgSO_4$. The organic layer was filtered and concentrated, and the resulting mixture was purified by column chromatography to obtain Compound 1-3 (50 g, yield: 77%).

Synthesis of Compound H1-72-D8

[0107] In a flask, Compound 1-3 (12.4 g, 30 mmol), Compound 1-4 (10 g, 30 mmol), $Pd(OAc)_2$ (337 mg, 1.5 mmol), SPhos (1.2 g, 3 mmol), K_3PO_4 (16 g, 75 mmol), toluene (150 mL), ethanol (40 mL), and distilled water (40 mL) were added, and the mixture was stirred under reflux for 3 hours. The mixture was cooled to room temperature, and an organic layer was extracted with ethyl acetate, and dried with magnesium sulfate. The organic layer was distilled under reduced pressure and separated by column chromatography to obtain Compound H1-72-D8 (13 g, yield: 80%).

	MW	M.P.
H1-72-D8	540.69	248° C.

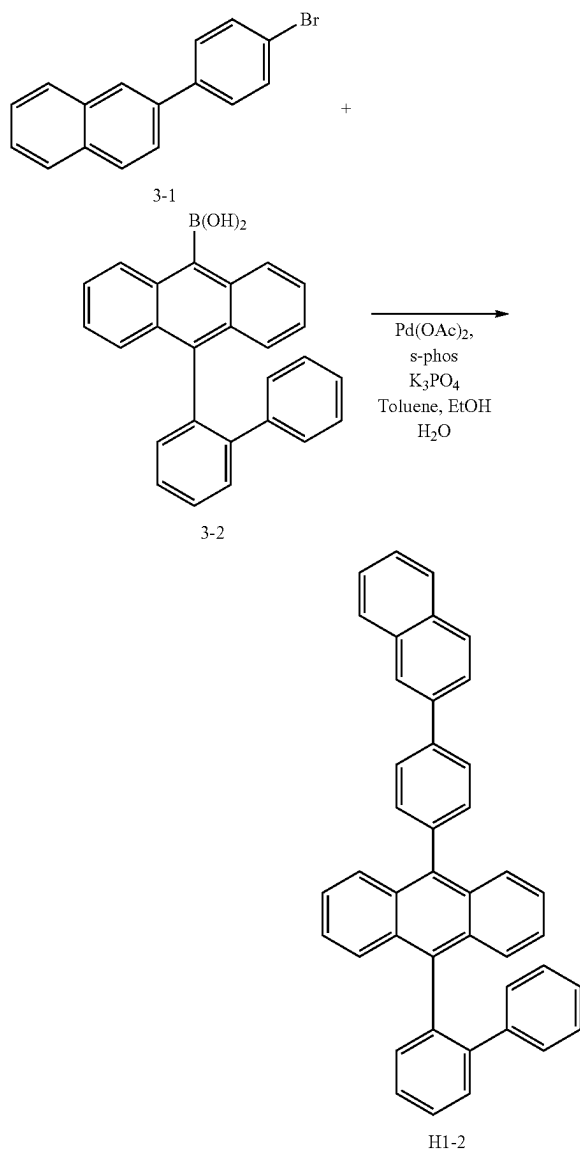
Example 2: Preparation of Compound H1-43



[0108] In a flask, Compound 2-1 (11 g, 42 mmol), Compound 2-2 (15 g, 40 mmol), $Pd(OAc)_2$ (449 mg, 2.0 mmol), SPhos (1.7 g, 4.0 mmol), K_3PO_4 (22 g, 105 mmol), toluene (200 mL), ethanol (40 mL), and distilled water (40 mL) were added, and the mixture was stirred under reflux for 3 hours. The mixture was cooled to room temperature, and an organic layer was extracted with ethyl acetate, and dried with magnesium sulfate. The organic layer was distilled under reduced pressure and separated by column chromatography to obtain Compound H1-43 (6.4 g, yield: 32%).

	MW	M.P.
H1-43	506.65	244° C.

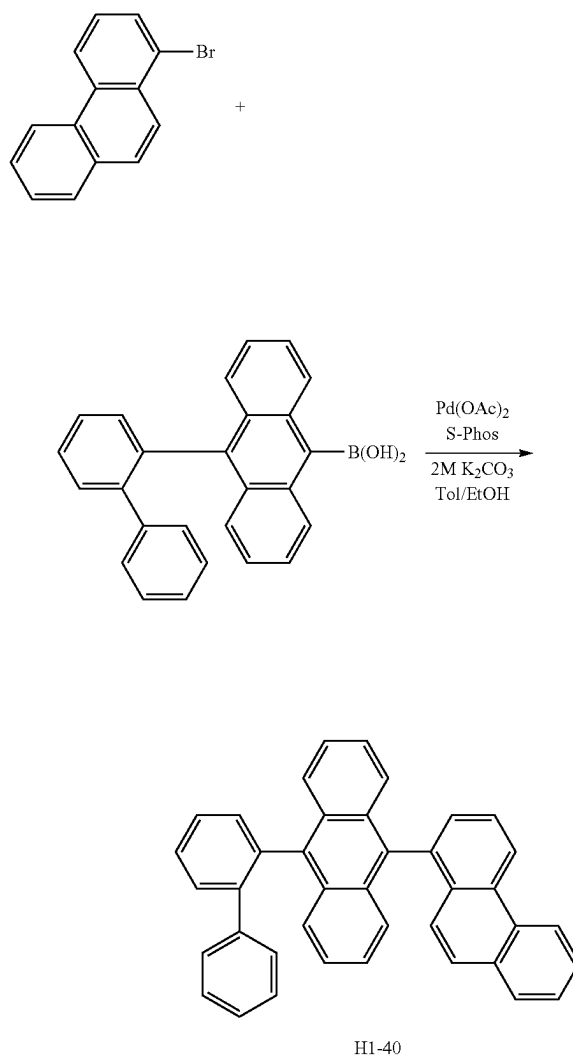
Example 3: Preparation of Compound H1-2



[0109] In a flask, Compound 3-1 (20 g, 71 mmol), Compound 3-2 (24 g, 64 mmol), Pd(OAc)₂ (797 mg, 3.55 mmol), SPhos (2.9 g, 7.1 mmol), K₃PO₄ (38 g, 178 mmol), toluene (360 mL), ethanol (90 mL), and distilled water (90 mL) were added, and the mixture was stirred under reflux for 5 hours. The mixture was cooled to room temperature, and an organic layer was extracted with ethyl acetate (EA), and dried with magnesium sulfate. The organic layer was distilled under reduced pressure and separated by column chromatography to obtain Compound H1-2 (24 g, yield: 70%).

	MW
H1-2	532.69

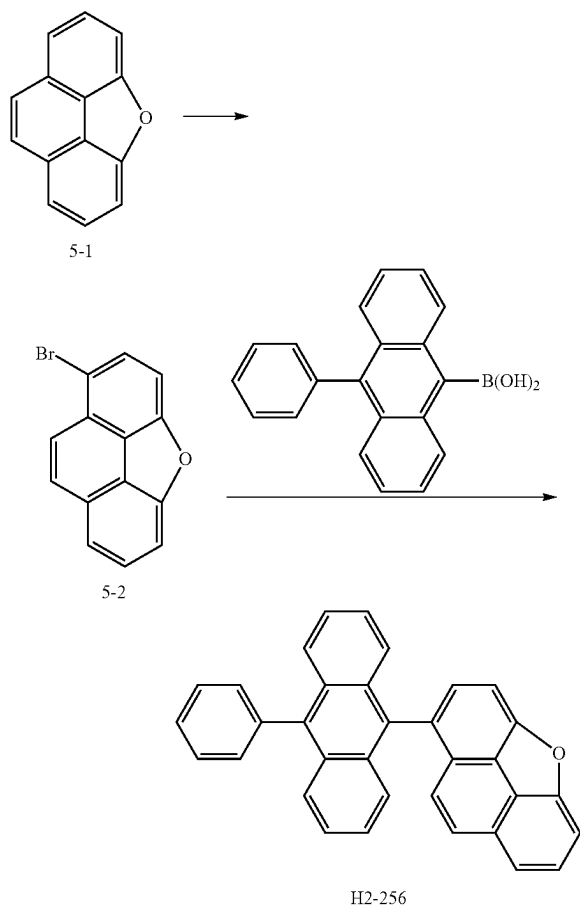
Example 4: Preparation of Compound H1-40



[0110] In a flask, 1-bromophenanthrene (15 g, 58.33 mmol), (10-([1,1'-biphenyl]-2-yl)anthracen-9-yl)boronic acid (24.01 g, 64.17 mmol), Pd(OAc)₂ (0.65 g, 2.91 mmol), SPhos (2.39 g, 5.83 mmol), K₂CO₃ (20.15 g, 145.8 mmol), 400 mL of toluene, 100 mL of distilled water, and 50 mL of ethanol were added, and the mixture was stirred under reflux. After 2 hours, the mixture was cooled to room temperature. Distilled water was added, and the organic layer was extracted with EA. Magnesium sulfate was added to the organic layer, dried, and filtered under reduced pressure. The organic layer was distilled under reduced pressure and separated by column chromatography to obtain Compound H1-40 (14.0 g, yield: 47.37%).

	MW	M.P.
H1-40	506.65	194.5° C.

Example 5: Preparation of Compound H2-256



Synthesis of Compound 5-2

[0111] Compound 5-1 (30 g, 156.07 mmol) was dissolved in 900 mL of methylene chloride (MC), and NBS (30.5 g, 171.67 mmol) was added. The mixture was stirred under reflux for 2 hours, cooled to room temperature, and stirred for 15 hours. Aqueous sodium thiosulfate solution was added to the mixture and stirred. The organic layer was separated and neutralized by adding Na_2CO_3 aqueous solution to the organic layer. The organic layer was separated, dried with magnesium sulfate, filtered under reduced pressure, and distilled under reduced pressure. The residue was separated by column chromatography to obtain Compound 5-2 (36 g, yield: 85.10%).

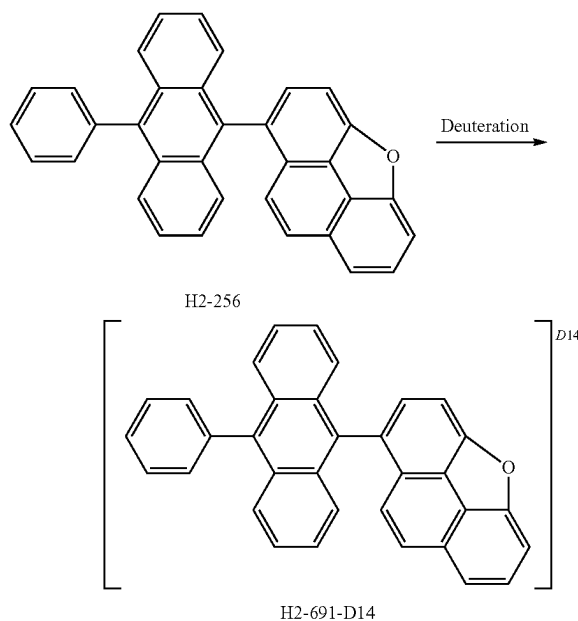
Synthesis of Compound H2-256

[0112] In a flask, Compound 5-2 (20 g, 73.77 mmol), (10-phenylanthracen-9-yl) boronic acid (24.19 g, 81.14 mmol), $\text{Pd}(\text{OAc})_2$ (0.66 g, 2.95 mmol), SPhos (3.63 g, 8.85 mmol), 400 mL of toluene, K_3PO_4 (39.1 g, 184.4 mmol), 90 mL of distilled water, and 90 mL of ethanol were added, and the mixture was stirred under reflux. After 4 hours, the mixture was cooled to room temperature. The organic layer was extracted with EA and washed with distilled water. The organic layer was dried with magnesium sulfate, and filtered

under reduced pressure. The organic layer was distilled under reduced pressure and separated by column chromatography to obtain Compound H2-256 (26 g, yield: 79.38%).

	MW	M.P.
H2-256	444.53	270.2° C.

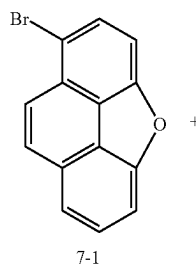
Example 6: Preparation of Compound H2-691-D14

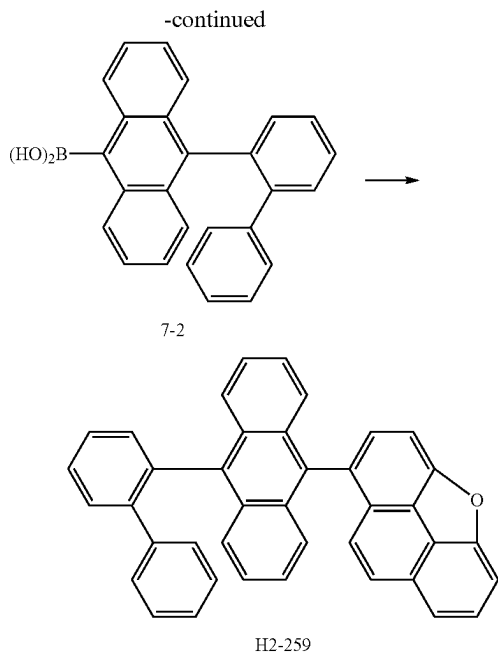


[0113] Compound H2-256 was synthesized by the deuteration method disclosed in Korean Patent No. 10-2283849 or 10-1427457, etc. to obtain Compound H2-691-D14 (16.4 g, yield: 72.34%, MS: $[\text{M}+\text{H}]^+=459.3$).

	MW	M.P.
H2-691-D14	458.3	268.9° C.

Example 7: Preparation of Compound H2-259

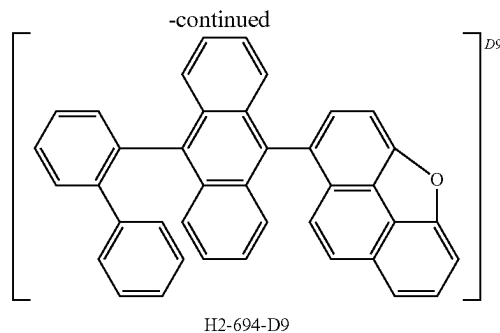
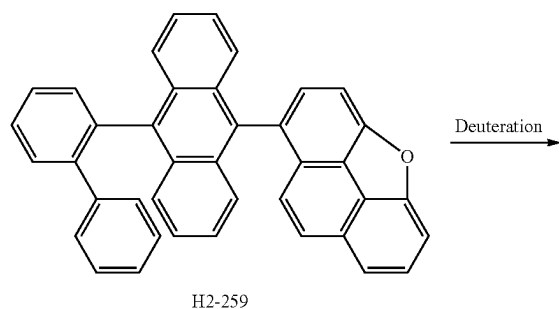




[0114] In a flask, Compound 7-1 (16 g, 59.01 mmol), Compound 7-2 (24.29 g, 64.91 mmol), Pd(OAc)₂ (0.53 g, 2.36 mmol), SPhos (2.42 g, 5.90 mmol), 400 mL of toluene, K₃PO₄ (25.05 g, 118.03 mmol), 80 mL of distilled water, and 40 mL of ethanol were added, and the mixture was stirred under reflux. After 2 hours, the mixture was cooled to room temperature. The organic layer was extracted with EA and washed with distilled water. The organic layer was dried with magnesium sulfate, and filtered under reduced pressure. The organic layer was distilled under reduced pressure and separated by column chromatography to obtain Compound H2-259 (14.4 g, yield: 46.87%).

	MW	M.P.
H2-259	520.63	250.4° C.

Example 8: Preparation of Compound H2-694-D9



[0115] Compound H2-259 was synthesized by the deuteration method disclosed in Korean Patent No. 10-2283849 or 10-1427457, etc. to obtain Compound H2-694-D9 (7.4 g, yield: 60.70%, MS: [M+H]⁺=530.1).

	MW	M.P.
H2-694-D9	529.1	240.1° C.

Device Examples 1-1 to 1-6: Producing an OLED Comprising the Compound According to the Present Disclosure as Hosts

[0116] An OLED according to the present disclosure was produced. A transparent electrode indium tin oxide (ITO) thin film (10Ω/sq) on a glass substrate for an OLED (GEO-MATEC CO., LTD.) was subjected to an ultrasonic washing with acetone and isopropyl alcohol, sequentially, and then was stored in isopropyl alcohol. The ITO substrate was then mounted on a substrate holder of a vacuum vapor deposition apparatus. Compound HI was introduced into a cell of the vacuum vapor deposition apparatus, and Compound HT-1 was introduced into another cell of the vacuum vapor deposition apparatus. The two materials were evaporated at different rates, and Compound HI was deposited in a doping amount of 5 wt % based on the total amount of Compound HI and Compound HT-1 to form a hole injection layer having a thickness of 10 nm. Compound HT-1 was then deposited on the hole injection layer to form a first hole transport layer having a thickness of 80 nm. Compound HT-2 was then introduced into another cell of the vacuum vapor deposition apparatus and was evaporated by applying an electric current to the cell, thereby forming a second hole transport layer having a thickness of 15 nm on the first hole transport layer. After forming the hole injection layer and the hole transport layers, a light-emitting layer was formed thereon as follows. The compounds shown in Table 1 below were introduced into two cells of the vacuum vapor deposition apparatus at a ratio of 1:1 as hosts, and Compound BD was introduced into another cell as a dopant. The host materials were evaporated, and the dopant material was simultaneously evaporated at a different rate, and the dopant was deposited in a doping amount of 2 wt % based on the total amount of the hosts and the dopant to form a light-emitting layer having a thickness of 22.5 nm on the second hole transport layer. Compound ET-1 was deposited on the light-emitting layer with a thickness of 5 nm as an electron

buffer layer. Compound EI-1 and Compound EI-2 were then added to two other cells and evaporated at a ratio of 2:1 to deposit an electron transport layer having a thickness of 25 nm on the electron buffer layer. In two other cells, Compound Yb (ytterbium) and Compound LiF (lithium fluoride) were added and evaporated at a rate of 2:1 (Yb:LiF) to deposit an electron injection layer with a thickness of 1 nm on the electron transport layer, and an Al cathode having a thickness of 80 nm was deposited on the electron injection layer by another vacuum vapor deposition apparatus. Thus, an OLED was produced. All the materials used for producing the OLED were purified by vacuum sublimation at 10⁻⁶ Torr.

Comparative Example 1-1: Producing an OLED
Comprising a Comparative Compound as a Host

[0117] An OLED was produced in the same manner as in Device Example 1-1, except that only the compound shown in Table 1 below was used as a host of the light-emitting layer.

[0118] The driving voltage, conversion efficiency, and the time taken for luminance to decrease from 100% to 95% (lifetime; T₉₅) at a luminance of 1,000 nit of the OLEDs produced in Device Examples 1-1 to 1-6 and Comparative Example 1-1 are provided in Table 1 below. Herein, the conversion efficiency [Eff/Y] means the current efficiency [cd/A] divided by the Y coordinate value of CIE.

TABLE 1

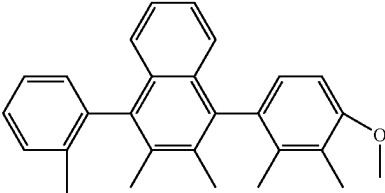
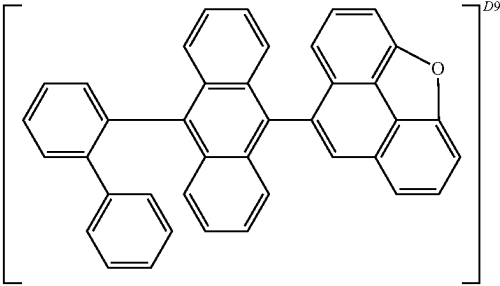
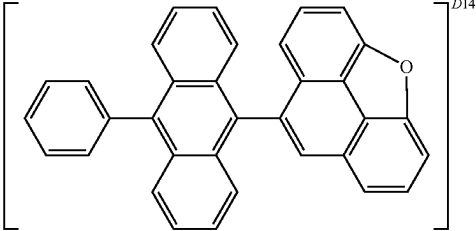
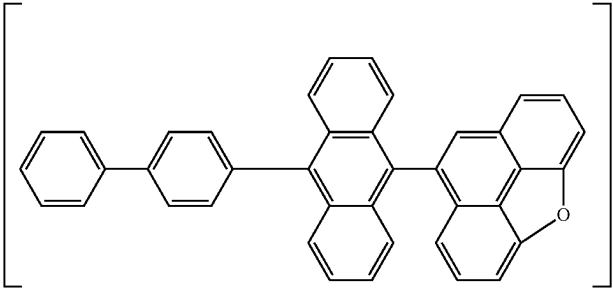
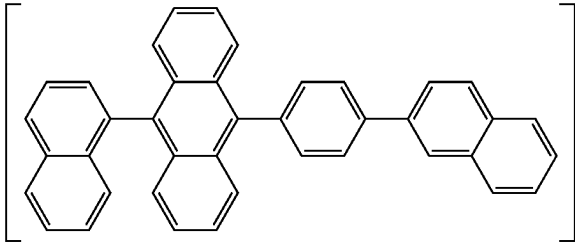
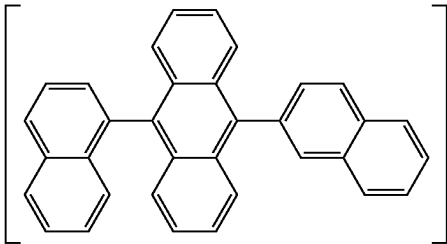
		Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T ₉₅) [hr]
First Host	Second Host			
Device Example 1-1	H1-2 — 	3.7	78	87
	H2-259			
Device Example 1-2	H1-2 — 	3.7	77	107
	H2-694-D9			
Device Example 1-3	H1-2 — 	3.7	82	102
	H2-691-D14			

TABLE 1-continued

	First Host	Second Host		Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T ₉₅) [hr]
Device Example 1-4	H1-2 —		D20	3.7	87	83
		H2-492-D20				
Device Example 1-5	H1-2 —		D24	3.9	90	114
		H2-66-D24				
Device Example 1-6	H1-2 —		D21	3.9	90	111
		H4-80-D21				
Comparative Example 1-1	H1-2 —	—		4.0	81	67

Device Example 2-1: Producing an OLED
Depositing the Compound According to the Present
Disclosure as a Single Host

[0119] An OLED was produced in the same manner as in Device Example 1-1, except that the compound shown in Table 2 below was used as a single host of the light-emitting layer.

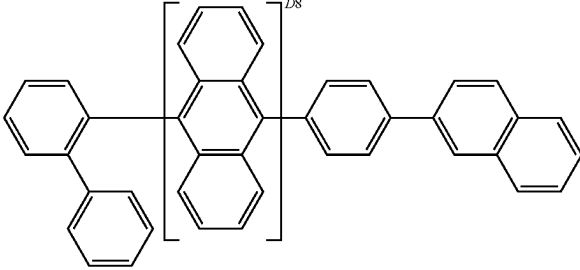
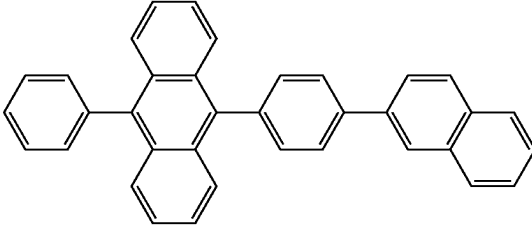
Comparative Examples 2-1 and 2-2: Producing an
OLED Depositing a Comparative Compound as a
Single Host

[0120] An OLED was produced in the same manner as in Device Example 2-1, except that the compound shown in Table 2 below was used as a single host of the light-emitting layer.

[0121] The driving voltage, conversion efficiency, and the time taken for luminance to decrease from 100% to 95% (lifetime; T_{95}) at a luminance of 1,000 nit of the OLEDs produced in Device Example 2-1 and Comparative Examples 2-1 and 2-2 are provided in Table 2 below. Herein, the conversion efficiency [Eff/Y] means the current efficiency [cd/A] divided by the Y coordinate value of CIE.

transport layer having a thickness of 15 nm on the first hole transport layer. After forming the hole injection layer and the hole transport layers, a light-emitting layer was formed thereon as follows. Compound H1-2 was introduced into one cell of the vacuum vapor deposition apparatus as a host, and Compound BD was introduced into another cell as a dopant. The two materials were evaporated at a different

TABLE 2

	Host	Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T_{95}) [hr]
Device Example 2-1	 H1-72-D8	4.0	81	124
Comparative Example 2-1	H1-2 —	4.0	81	67
Comparative Example 2-2	 H4-5	4.1	79	52

Device Example 3-1: Producing a Tandem OLED
Comprising the Compound According to the
Present Disclosure as a Host

[0122] An OLED according to the present disclosure was produced. A transparent electrode indium tin oxide (ITO) thin film (10Ω/sq) on a glass substrate for an OLED (GEO-MATEC CO., LTD.) was subjected to an ultrasonic washing with acetone and isopropyl alcohol, sequentially, and then was stored in isopropyl alcohol. The ITO substrate was then mounted on a substrate holder of a vacuum vapor deposition apparatus. Compound HI was introduced into a cell of the vacuum vapor deposition apparatus, and Compound HT-1 was introduced into another cell of the vacuum vapor deposition apparatus. The two materials were evaporated at different rates, and Compound HI was deposited in a doping amount of 5 wt % based on the total amount of Compound HI and Compound HT-1 to form a hole injection layer having a thickness of 10 nm. Compound HT-1 was then deposited on the hole injection layer to form a first hole transport layer having a thickness of 80 nm. Compound HT-2 was then introduced into another cell of the vacuum vapor deposition apparatus and was evaporated by applying an electric current to the cell, thereby forming a second hole

rate, and the dopant was deposited in a doping amount of 2 wt % based on the total amount of the host and the dopant to form a first light-emitting layer having a thickness of 22.5 nm on the second hole transport layer. Compound ET-1 as a first hole-blocking layer material was then deposited on the first light-emitting layer to form a first hole-blocking layer having a thickness of 5 nm. On the first hole-blocking layer, a first electron transport layer having a thickness of 12 nm was deposited using Compound EI-1 as an electron transport layer material. Thereafter, Li was deposited in an amount of 1 wt % on Compound n-CGL to form a n-type charge generation layer having a thickness of 12 nm on the first electron transport layer. Compound HI was then introduced into a cell of the vacuum vapor deposition apparatus, and Compound HT-1 was introduced into another cell of the vacuum vapor deposition apparatus. The two materials were evaporated at different rates, and Compound HI was deposited in a doping amount of 15 wt % based on the total amount of Compound HI and Compound HT-1 to form a p-type charge generation layer having a thickness of 10 nm. Compound HT-1 was then deposited to a thickness of 80 nm to form a third hole transport layer, and then compound HT-2 was deposited to form a fourth hole transport layer having a

thickness of 15 nm. A second light-emitting layer was then formed thereon as follows. Compound H1-2 was introduced into one cell of the vacuum vapor deposition apparatus as a host, and Compound BD was introduced into another cell as a dopant. The two materials were evaporated at a different rate, and the dopant was deposited in a doping amount of 2 wt % based on the total amount of the host and the dopant to form a second light-emitting layer having a thickness of 22.5 nm on the fourth hole transport layer. Compound ET-1 was deposited on the second light-emitting layer to form a second hole-blocking layer having a thickness of 5 nm. Compound EI-1 and Compound EI-2 were added to two other cells of the vacuum vapor deposition apparatus and evaporated at a ratio of 2:1 to deposit a second electron transport layer having a thickness of 25 nm. Compound Yb and Compound LiF were then added to two other cells and evaporated at a rate of 2:1 (Yb:LiF) to deposit an electron injection layer having a thickness of 1 nm on the second electron transport layer, and an Al cathode having a thickness of 80 nm was deposited on the electron injection layer by another vacuum vapor deposition apparatus. Thus, an OLED was produced. All the materials used for producing the OLED were purified by vacuum sublimation at 10⁻⁶ Torr.

[0123] The driving voltage, conversion efficiency, and the time taken for luminance to decrease from 100% to 95% (lifetime; T₉₅) at a luminance of 1,000 nit of the OLED produced in Device Example 3-1 are provided in Table 3 below.

TABLE 3				
	Host	Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T ₉₅) [hr]
Device Example 3-1	H1-2	7.7	116	185

Device Examples 5-1 to 5-6: Producing an OLED Comprising the Compound According to the Present Disclosure as Hosts

[0124] An OLED was produced in the same manner as in Device Example 1-1, except that the compounds shown in Table 4 below were used as hosts of the light-emitting layer.

Comparative Examples 5-1 to 5-3: Producing an OLED Comprising a Comparative Compound as a Host

[0125] An OLED was produced in the same manner as in Device Example 5-1, except that the compound shown in Table 4 below was used as a host of the light-emitting layer. [0126] The driving voltage, conversion efficiency, and the time taken for luminance to decrease from 100% to 95% (lifetime; T₉₅) at a luminance of 1,000 nit of the OLEDs produced in Device Examples 5-1 to 5-6 and Comparative Examples 5-1 to 5-3 are provided in Table 4 below. Herein, the conversion efficiency [Eff/Y] means the current efficiency [cd/A] divided by the Y coordinate value of CIE.

TABLE 4

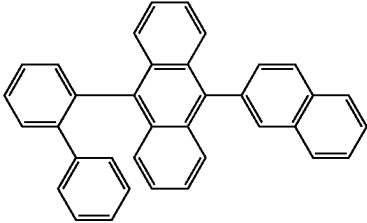
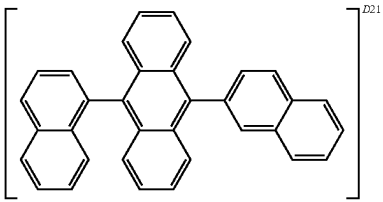
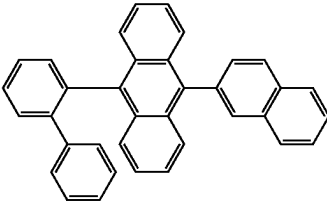
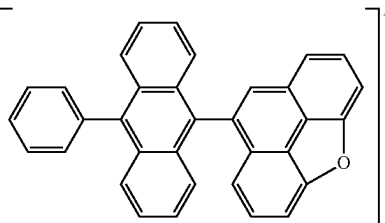
	First Host	Second Host	Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T ₉₅) [hr]
Device Example 5-1			3.9	89	123
	H1-24	H4-80-D21			
Device Example 5-2			3.6	87	91
	H1-24	H2-491-D14			

TABLE 4-continued

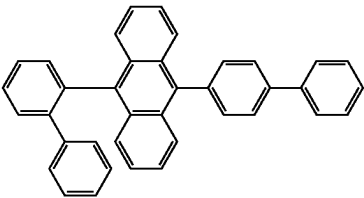
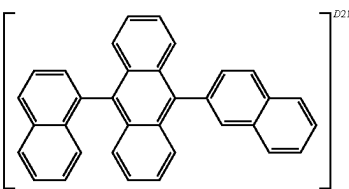
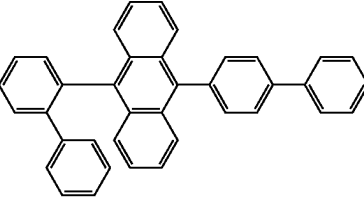
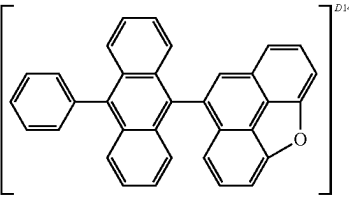
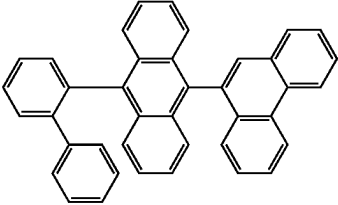
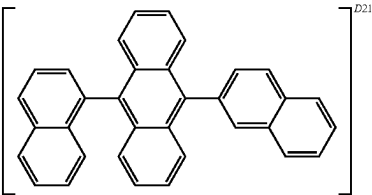
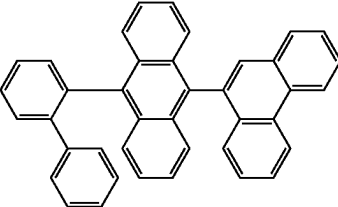
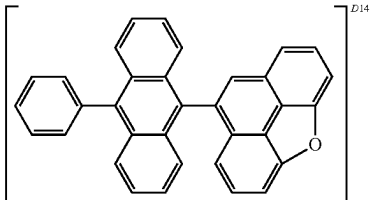
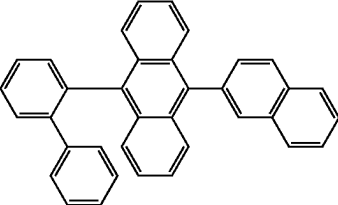
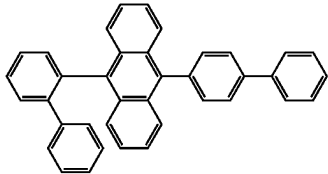
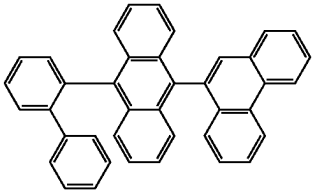
	First Host	Second Host	Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T ₉₅) [hr]
Device Example 5-3	 H1-135	 H4-80-D21	3.9	86	136
Device Example 5-4	 H1-135	 H2-491-D14	3.6	84	96
Device Example 5-5	 H1-43	 H4-80-D21	3.8	89	140
Device Example 5-6	 H1-43	 H2-491-D14	3.6	90	120
Comparative Example 5-1	 H1-24	—	4.0	83	91

TABLE 4-continued

	First Host	Second Host	Driving Voltage [V]	Conversion Efficiency [Eff/Y]	Lifetime (T ₉₅) [hr]
Compara- tive Example 5-2	 H1-135	—	4.0	78	71
Compara- tive Example 5-3	 H1-43	—	3.9	87	90

[0127] From Tables 1 to 4 above, it can be confirmed that the organic electroluminescent device comprising the compound or a specific combination of compounds according to the present disclosure as a host material(s) exhibits significantly improved lifetime properties while exhibiting an equivalent or lower driving voltage and/or equivalent or

higher current efficiency compared to the organic electroluminescent device comprising the comparative compound as a host material.

[0128] The compounds used in the above Device Examples and Comparative Examples are shown in Table 5 below.

TABLE 5

Hole Injection Layer/
Hole Transport Layer/
Charge Generation Layer

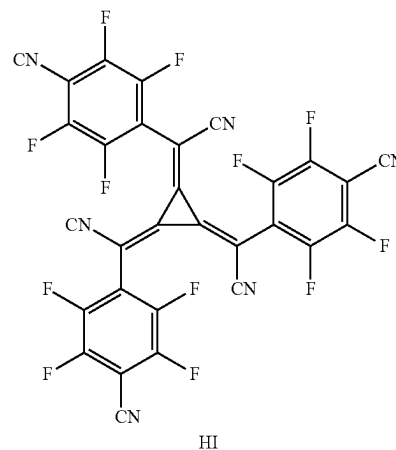
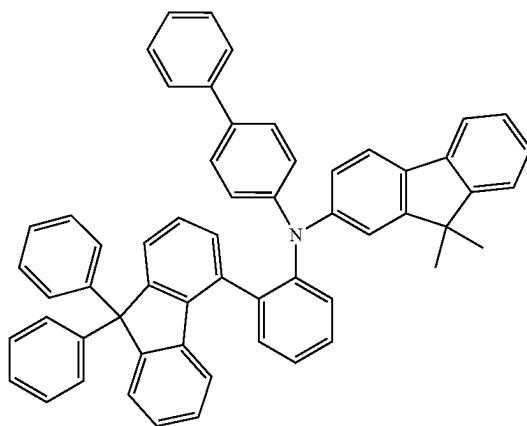
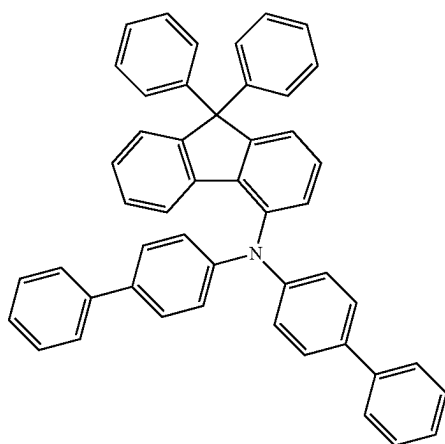


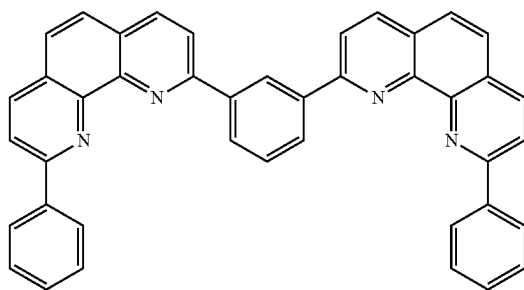
TABLE 5-continued



HT-1



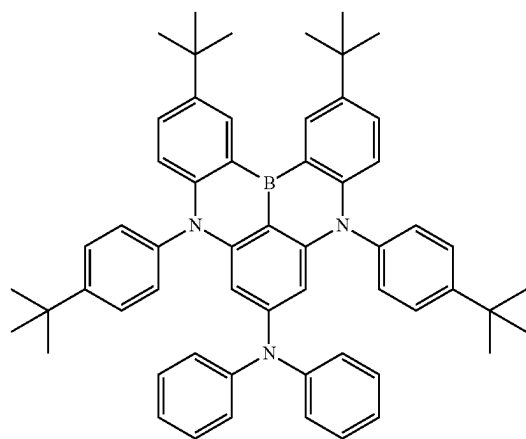
HT-2



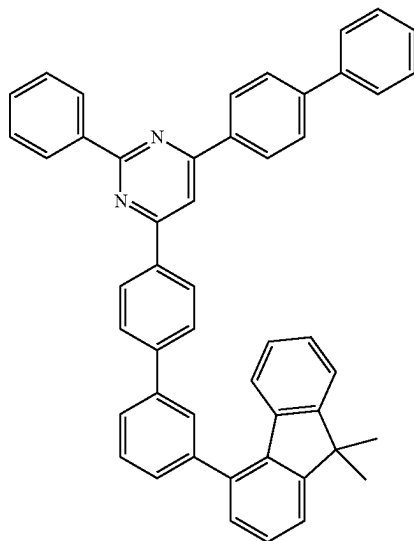
n-CGL

TABLE 5-continued

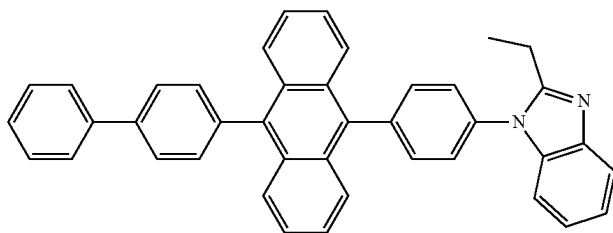
Light-Emitting Layer



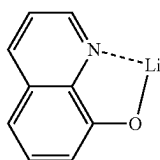
BD

Electron Buffer Layer/
Hole-Blocking Layer

ET-1

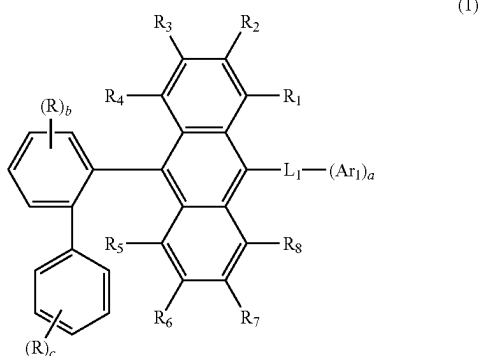
Electron Transport Layer/
Electron Injection Layer

EI-1



EI-2

1. A plurality of host materials comprising at least one first host compound and at least one second host compound, wherein the first host compound is represented by the following Formula 1, the second host compound is represented by the following Formula 2, and the first host compound and the second compound are different from each other:



in Formula 1,

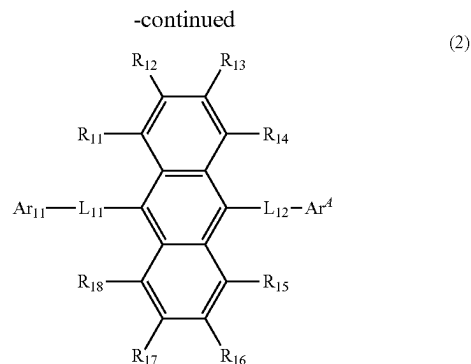
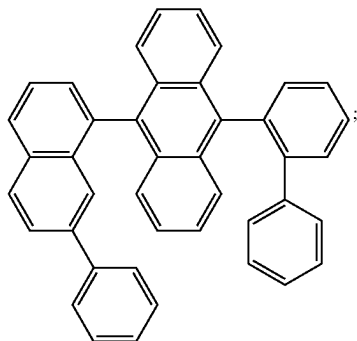
L_1 represents a single bond, a phenylene unsubstituted or substituted with deuterium, a naphthylene unsubstituted or substituted with deuterium, or a phenanthrylene unsubstituted or substituted with deuterium;

Ar_1 each independently represents a substituted or unsubstituted (C_6 - C_{30})aryl, or a substituted or unsubstituted (3- to 30-membered)heteroaryl;

R and R_1 to R_8 each independently represent hydrogen or deuterium; and

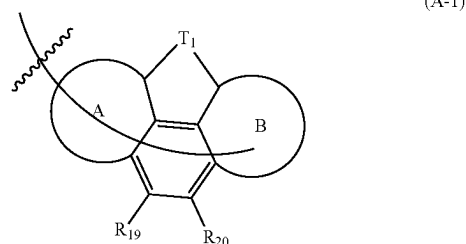
a represents an integer of 1 or 2, b represents an integer of 4, and c represents an integer of 5; and each Ar_1 may be the same as or different from each other;

with a proviso that Formula 1 is not represented as



in Formula 2,

Ar^4 represents a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (13- to 30-membered)heteroaryl, or the following Formula A-1:



wherein T_1 represents O, S, or CR_aR_b ;

Ring A and Ring B each independently represent a substituted or unsubstituted (C_6 - C_{30})arene, or a substituted or unsubstituted (3- to 30-membered) heteroarene;

Ar_{11} represents a substituted or unsubstituted (C_6 - C_{30})aryl, or a substituted or unsubstituted (13- to 30-membered)heteroaryl;

R_{11} to R_{18} each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_2 - C_{30})alkenyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C_3 - C_{30})cycloalkyl, a substituted or unsubstituted (C_3 - C_{30})cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C_1 - C_{30})alkoxy, a substituted or unsubstituted tri(C_1 - C_{30})alkylsilyl, a substituted or unsubstituted di(C_1 - C_{30})alkyl(C_6 - C_{30})arylsilyl, a substituted or unsubstituted (C_1 - C_{30})alkyldi(C_6 - C_{30})arylsilyl, a substituted or unsubstituted tri(C_6 - C_{30})arylsilyl, a substituted or unsubstituted fused ring group of a (C_3 - C_{30})aliphatic ring(s) and a (C_6 - C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$;

R_{19} and R_{20} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1 - C_{30})alkyl, a substituted or unsubstituted (C_2 - C_{30})alkenyl, a substituted or unsubstituted (C_6 - C_{30})aryl, a substituted or unsubstituted (3- to 30-membered) het-

eroaryl, a substituted or unsubstituted (C₃-C₃₀)cycloalkyl, a substituted or unsubstituted (C₃-C₃₀)cycloalkenyl, a substituted or unsubstituted (3- to 7-membered) heterocycloalkyl, a substituted or unsubstituted (C₁-C₃₀)alkoxy, a substituted or unsubstituted tri(C₁-C₃₀)alkylsilyl, a substituted or unsubstituted di(C₁-C₃₀)alkyl(C₆-C₃₀)arylsilyl, a substituted or unsubstituted (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, a substituted or unsubstituted tri(C₆-C₃₀)arylsilyl, a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s), or -L₁₃-N(Ar₁₃)(Ar₁₄);

R_a and R_b each independently represent a substituted or unsubstituted (C₁-C₃₀)alkyl, or a substituted or unsubstituted (C₆-C₃₀)aryl, or may be linked to each other to form a ring(s);

L₁₁ to L₁₃ each independently represent a single bond, a substituted or unsubstituted (C₆-C₃₀)arylene, or a substituted or unsubstituted (3- to 30-membered) heteroarylene; and

Ar₁₃ and Ar₁₄ each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C₁-C₃₀)alkyl, a substituted or unsubstituted (C₂-C₃₀)alkenyl, a substituted or unsubstituted (C₆-C₃₀)aryl, a substituted or unsubstituted (3- to 30-membered)heteroaryl, a substituted or unsubstituted (C₃-C₃₀)cycloalkyl, a substituted or unsubstituted (C₃-C₃₀)cycloalkenyl, a substituted or unsubstituted (3- to 7-membered)heterocycloalkyl, a substituted or unsubstituted (C₁-C₃₀)alkoxy, a substituted or unsubstituted tri(C₁-C₃₀)alkylsilyl, a substituted or unsubstituted di(C₁-C₃₀)alkyl(C₆-C₃₀)arylsilyl, a substituted or unsubstituted (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, a substituted or unsubstituted tri(C₆-C₃₀)arylsilyl, or a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s).

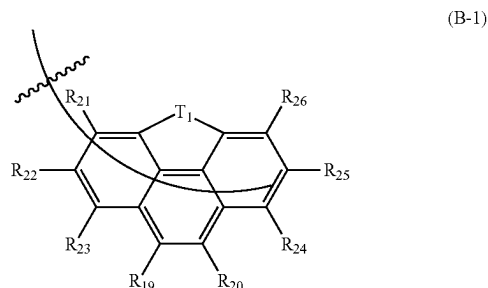
2. The plurality of host materials according to claim 1, wherein Ar₁ each independently represents a phenyl unsubstituted or substituted with deuterium, a biphenyl unsubstituted or substituted with deuterium, a terphenyl unsubstituted or substituted with deuterium, a naphthyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, or a combination thereof.

3. The plurality of host materials according to claim 1, wherein Ar₁₁ and Ar⁴ each independently are selected from a phenyl unsubstituted or substituted with deuterium, a biphenyl unsubstituted or substituted with deuterium, a terphenyl unsubstituted or substituted with deuterium, a naphthyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, or a combination thereof.

4. The plurality of host materials according to claim 1, wherein Ar₁₁ is selected from a phenyl unsubstituted or substituted with deuterium, a biphenyl unsubstituted or substituted with deuterium, a terphenyl unsubstituted or substituted with deuterium, a naphthyl unsubstituted or substituted with deuterium, a phenanthrenyl unsubstituted or substituted with deuterium, or a combination thereof; and

Ar⁴ represents a dibenzofuranyl unsubstituted or substituted with deuterium, or a dibenzothiophenyl unsubstituted or substituted with deuterium.

5. The plurality of host materials according to claim 1, wherein Ar⁴ is represented by the following Formula B-1:

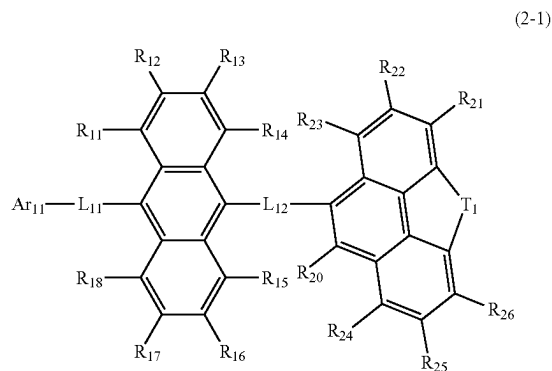


in Formula B-1,

R₂₁ to R₂₆ each independently represent the position linked to L₁₂, or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C₁-C₃₀)alkyl, a substituted or unsubstituted (C₂-C₃₀)alkenyl, a substituted or unsubstituted (C₆-C₃₀)aryl, a substituted or unsubstituted (3- to 30-membered) heteroaryl, a substituted or unsubstituted (C₃-C₃₀)cycloalkyl, a substituted or unsubstituted (C₃-C₃₀)cycloalkenyl, a substituted or unsubstituted (3- to 7-membered) heterocycloalkyl, a substituted or unsubstituted (C₁-C₃₀)alkoxy, a substituted or unsubstituted tri(C₁-C₃₀)alkylsilyl, a substituted or unsubstituted di(C₁-C₃₀)alkyl(C₆-C₃₀)arylsilyl, a substituted or unsubstituted (C₁-C₃₀)alkyldi(C₆-C₃₀)arylsilyl, a substituted or unsubstituted tri(C₆-C₃₀)arylsilyl, a substituted or unsubstituted fused ring group of a (C₃-C₃₀) aliphatic ring(s) and a (C₆-C₃₀) aromatic ring(s), or -L₁₃-N(Ar₁₃)(Ar₁₄), or may be linked to the adjacent substituent to form a ring(s); and

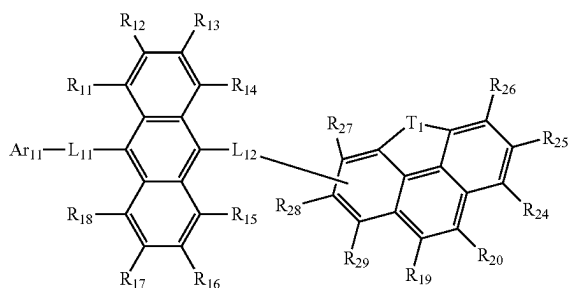
T₁, R₁₉, R₂₀, L₁₃, Ar₁₃, and Ar₁₄ are as defined in claim 1.

6. The plurality of host materials according to claim 1, wherein Formula 2 is represented by the following Formula 2-1 or 2-2:



-continued

(2-2)



in Formulas 2-1 and 2-2,

R_{21} to R_{26} each independently represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1-C_{30}) alkyl, a substituted or unsubstituted (C_2-C_{30}) alkenyl, a substituted or unsubstituted (C_6-C_{30}) aryl, a substituted or unsubstituted (3- to 30-membered) heteroaryl, a substituted or unsubstituted (C_3-C_{30}) cycloalkyl, a substituted or unsubstituted (C_3-C_{30}) cycloalkenyl, a substituted or unsubstituted (3- to 7-membered) heterocycloalkyl, a substituted or unsubstituted (C_1-C_{30}) alkoxy, a substituted or unsubstituted $tri(C_1-C_{30})$ alkylsilyl, a substituted or unsubstituted $di(C_1-C_{30})$ alkyl (C_6-C_{30}) arylsilyl, a substituted or unsubstituted (C_1-C_{30}) alkyldi (C_6-C_{30}) arylsilyl, a substituted or unsubstituted $tri(C_6-C_{30})$ arylsilyl, a substituted or unsubstituted fused ring group of a (C_3-C_{30}) aliphatic ring(s) and a (C_6-C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$, or may be linked to the adjacent substituent to form a ring(s);

R_{27} to R_{29} each independently represent the position linked to L_{12} , or represent hydrogen, deuterium, a halogen, a cyano, a substituted or unsubstituted (C_1-C_{30}) alkyl, a substituted or unsubstituted (C_6-C_{30}) aryl, a substituted or unsubstituted (3- to 30-membered) heteroaryl, a substituted or unsubstituted (C_3-C_{30}) cycloalkyl, a substituted or unsubstituted (C_1-C_{30}) alkoxy, a substituted or unsubstituted $tri(C_1-C_{30})$ alkylsilyl, a substituted or unsubstituted $di(C_1-C_{30})$ alkyl (C_6-C_{30}) arylsilyl, a substituted or unsubstituted (C_1-C_{30}) alkyldi (C_6-C_{30}) arylsilyl, a substituted or unsubstituted $tri(C_6-C_{30})$ arylsilyl, a substituted or unsubstituted fused ring

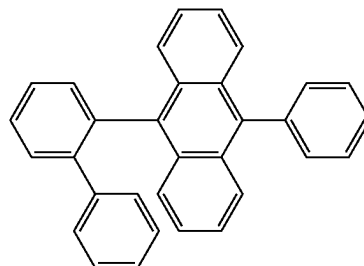
group of a (C_3-C_{30}) aliphatic ring(s) and a (C_6-C_{30}) aromatic ring(s), or $-L_{13}-N(Ar_{13})(Ar_{14})$, or may be linked to the adjacent substituent to form a ring(s); and T_1 , R_{11} to R_{20} , L_{11} to L_{13} , Ar_{11} , Ar_{13} , and Ar_{14} are as defined in claim 1.

7. The plurality of host materials according to claim 1, wherein at least one of Formula 1 and Formula 2 contains deuterium.

8. The plurality of host materials according to claim 1, wherein the substituted alkyl, the substituted alkenyl, the substituted aryl, the substituted arylene, the substituted heteroaryl, the substituted heteroarylene, the substituted cycloalkyl, the substituted cycloalkenyl, the substituted heterocycloalkyl, the substituted alkoxy, the substituted trialkylsilyl, the substituted dialkylarylsilyl, the substituted alkyldiarylsilyl, the substituted triarylsilyl, the substituted fused ring group of an aliphatic ring(s) and an aromatic ring(s), the substituted arene, and the substituted heteroarene each independently are substituted with at least one selected from the group consisting of deuterium, a halogen, a cyano, carboxyl, nitro, hydroxyl, phosphine oxide, (C_1-C_{30}) alkyl, halo (C_1-C_{30}) alkyl, (C_2-C_{30}) alkenyl, (C_2-C_{30}) alkynyl, (C_1-C_{30}) alkoxy, (C_1-C_{30}) alkylthio, (C_3-C_{30}) cycloalkyl, (C_3-C_{30}) cycloalkenyl, (3- to 7-membered) heterocycloalkyl, (C_6-C_{30}) aryloxy, (C_6-C_{30}) arylthio, (3- to 30-membered) heteroaryl, (C_6-C_{30}) aryl, $tri(C_1-C_{30})$ alkylsilyl, $tri(C_6-C_{30})$ arylsilyl, $di(C_1-C_{30})$ alkyl (C_6-C_{30}) arylsilyl, (C_1-C_{30}) alkyldi (C_6-C_{30}) arylsilyl, amino, mono- or $di(C_1-C_{30})$ alkylamino, mono- or $di(C_2-C_{30})$ alkenylamino, mono- or $di(C_6-C_{30})$ arylamino, mono- or $di(3- to 30-membered)$ heteroarylamino, (C_1-C_{30}) alkyl (C_2-C_{30}) alkenylamino, (C_1-C_{30}) alkyl (C_6-C_{30}) arylamino, (C_1-C_{30}) alkyl(3- to 30-membered) heteroarylamino, (C_2-C_{30}) alkenyl (C_6-C_{30}) arylamino, (C_2-C_{30}) alkenyl(3- to 30-membered) heteroarylamino, (C_6-C_{30}) aryl(3- to 30-membered) heteroarylamino, (C_1-C_{30}) alkylcarbonyl, (C_1-C_{30}) alkoxycarbonyl, (C_6-C_{30}) arylcarbonyl, $di(C_6-C_{30})$ arylboronyl, (C_6-C_{30}) arylphosphinyl, $di(C_1-C_{30})$ alkylboronyl, (C_1-C_{30}) alkyl (C_6-C_{30}) arylboronyl, (C_6-C_{30}) aryl (C_1-C_{30}) alkyl, (C_1-C_{30}) alkyl (C_6-C_{30}) aryl, and a combination thereof, which may be further substituted with deuterium.

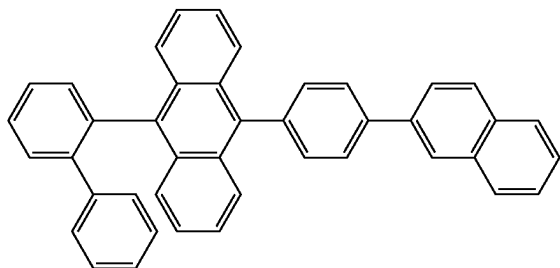
9. The plurality of host materials according to claim 1, wherein the compound represented by Formula 1 is at least one selected from the following compounds:

H1-1

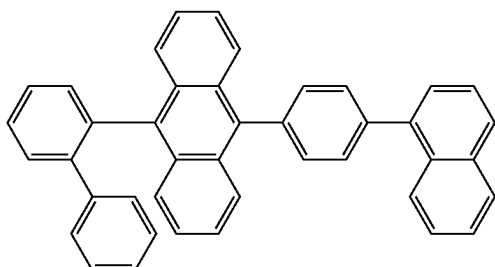


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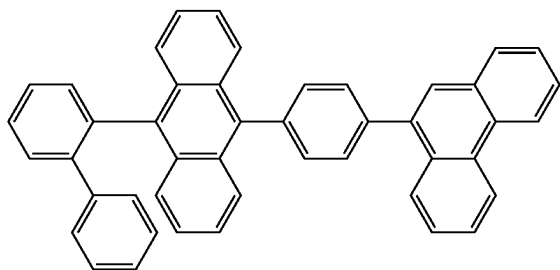
H1-2



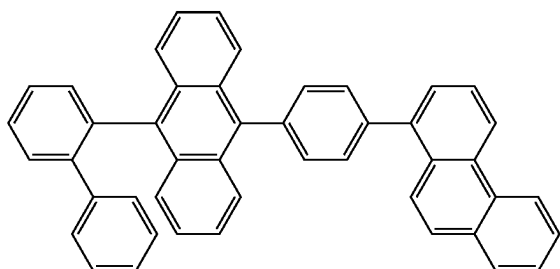
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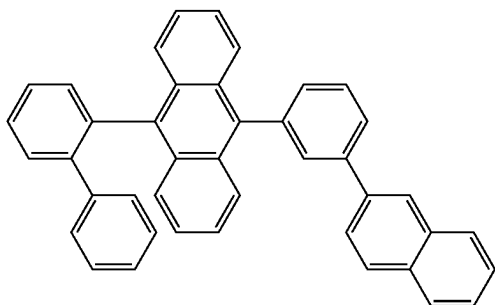
H1-4



H1-5

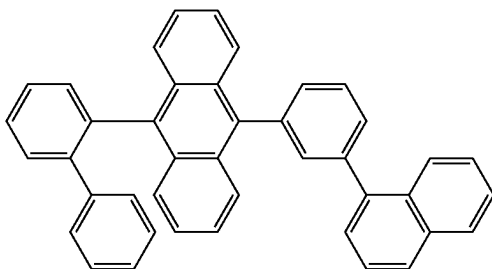


H1-6

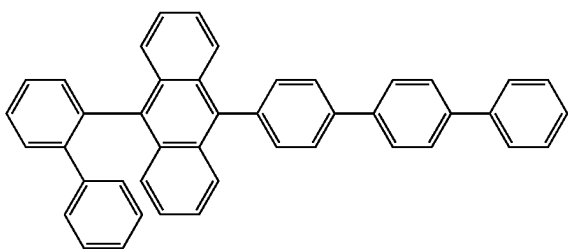


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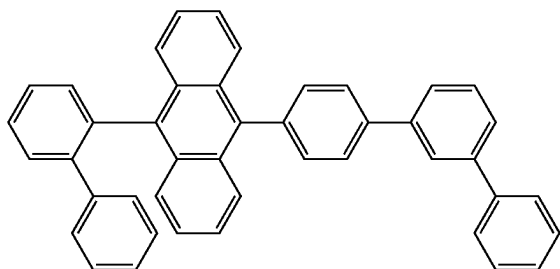
H1-7



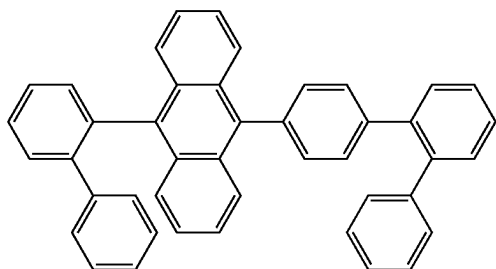
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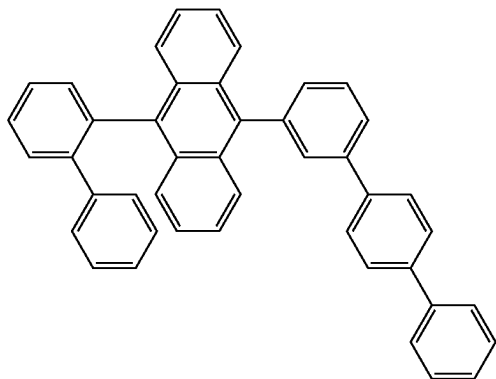
H1-9



H1-10

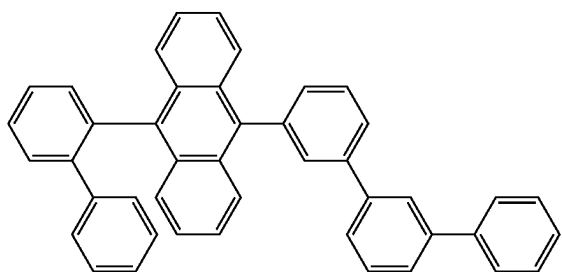


H1-11

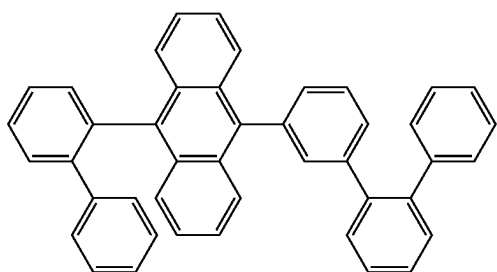


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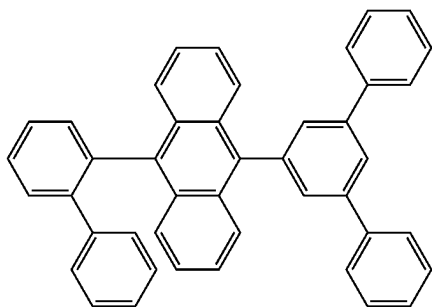
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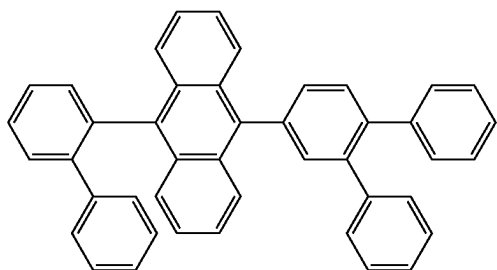
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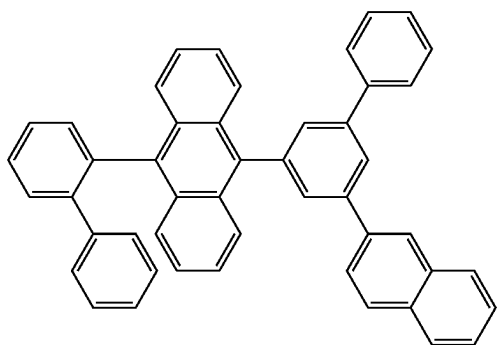
H1-14



H1-15

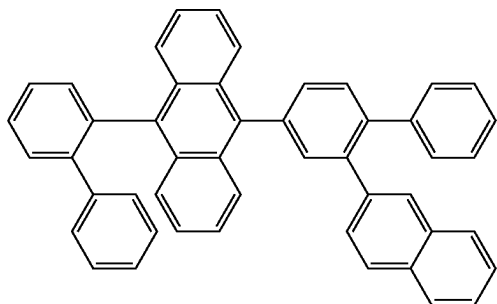


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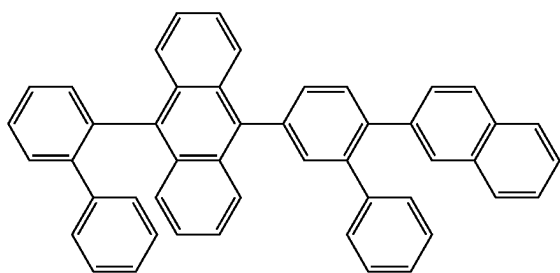


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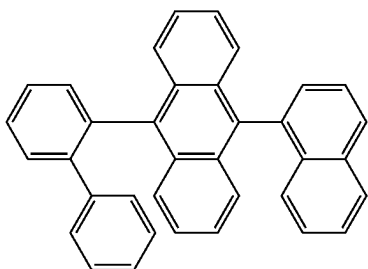
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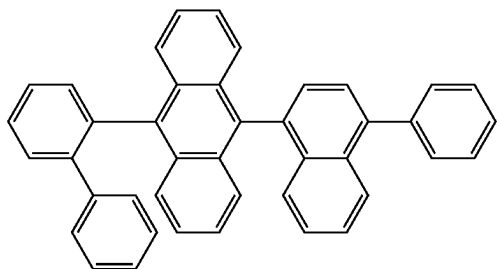
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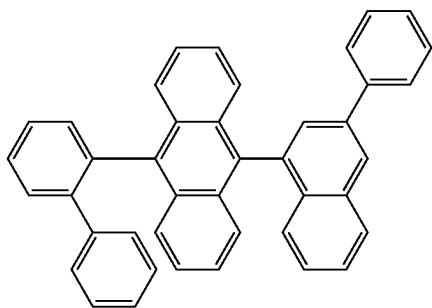
H1-19



H1-20

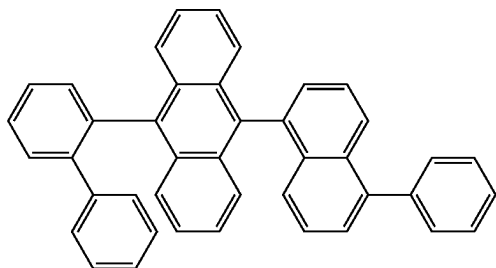


H1-21

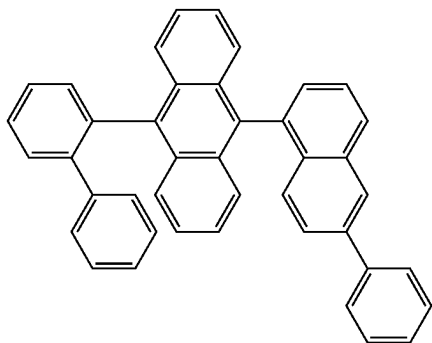


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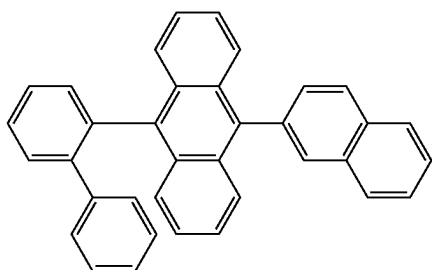
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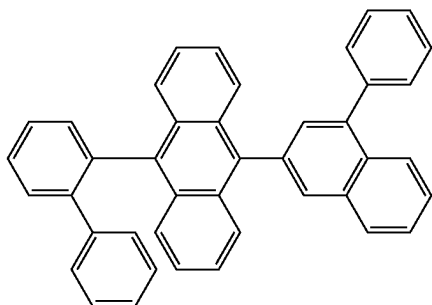
H1-23



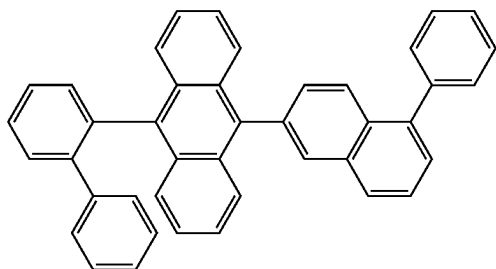
H1-24



H1-25

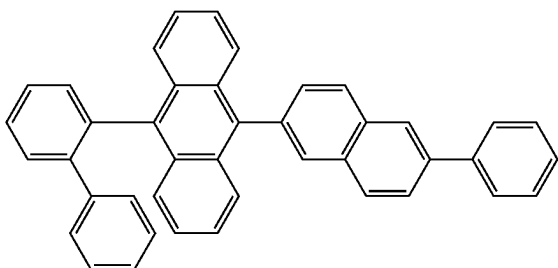


H1-26

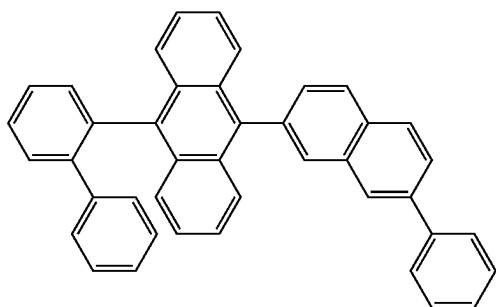


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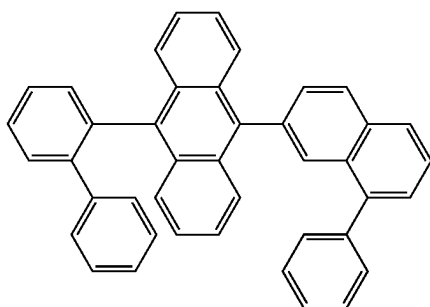
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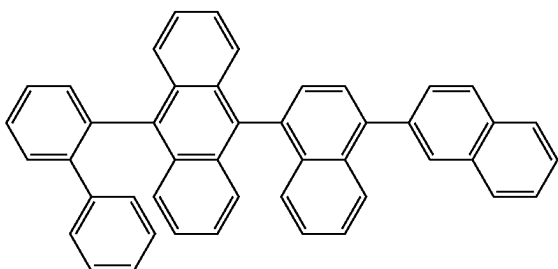
H1-28



H1-29

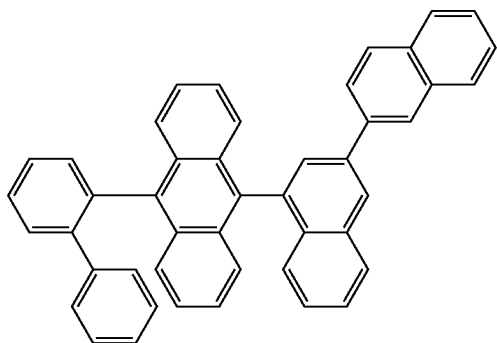


H1-30

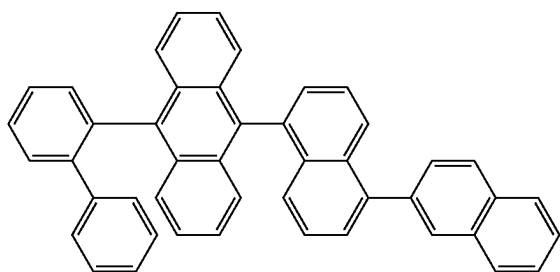


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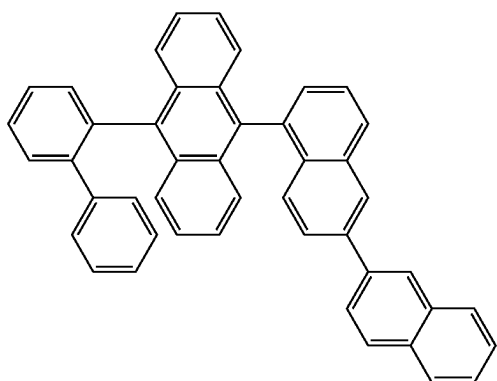
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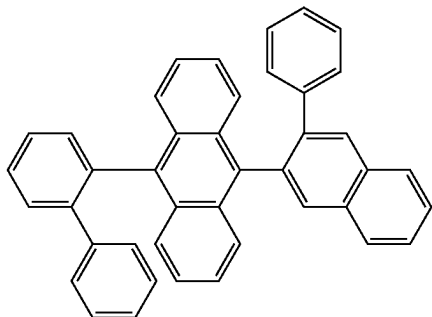
H1-32



H1-33

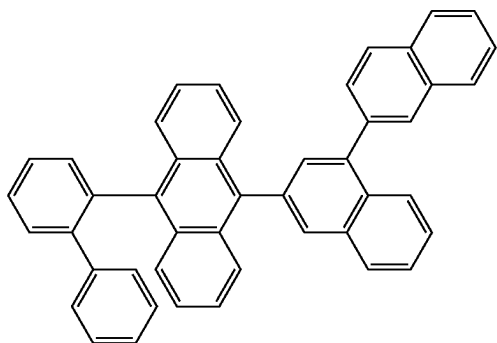


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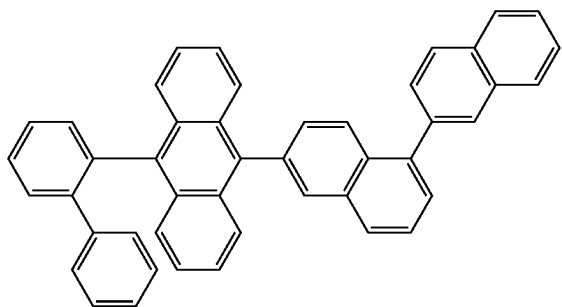


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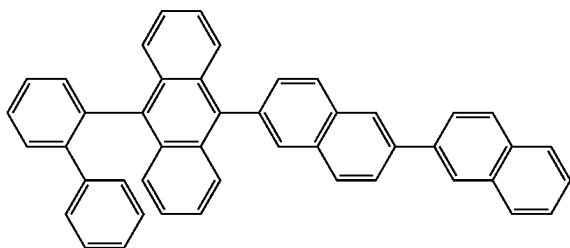
H1-35



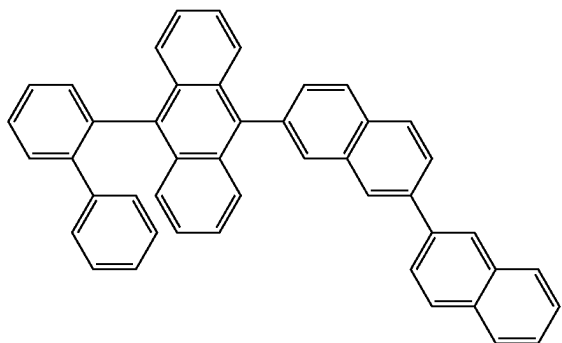
H1-36



H1-37

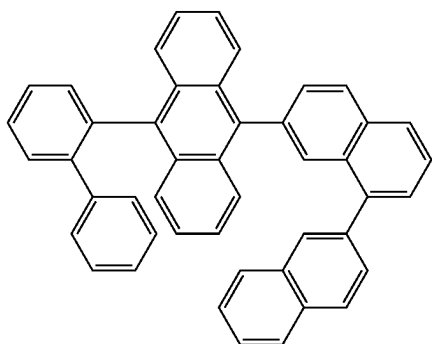


H1-38

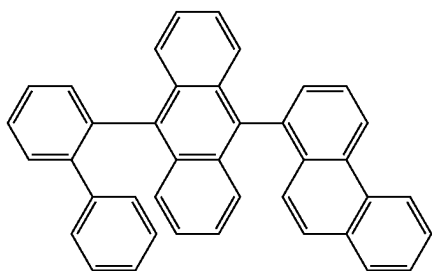


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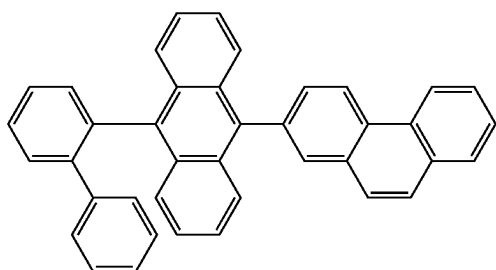
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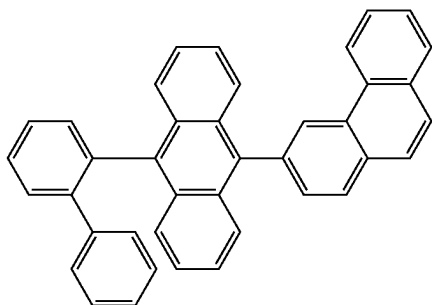
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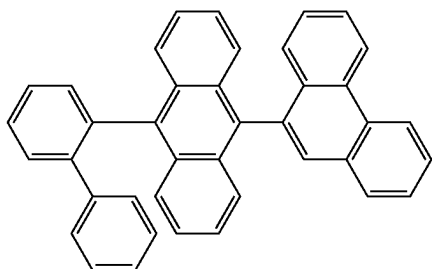
H1-41



H1-42

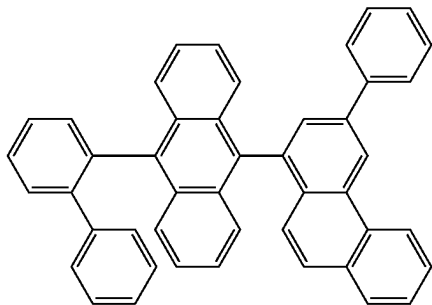


H1-43

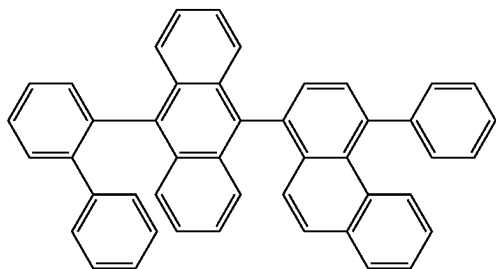


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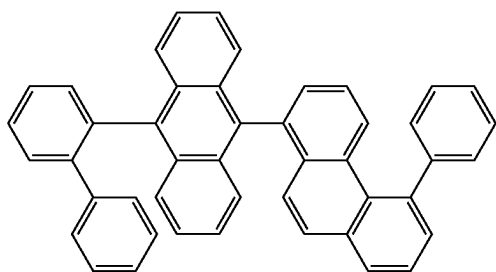
H1-44



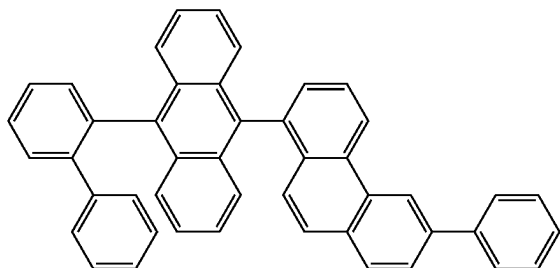
H1-45



H1-46

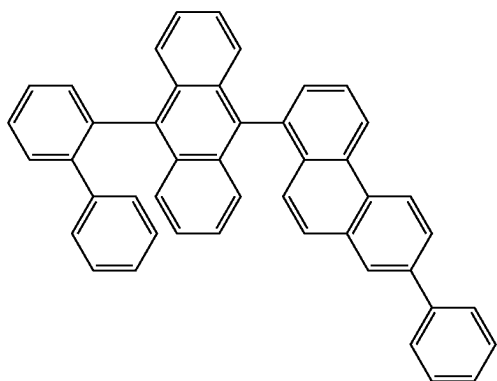


H1-47

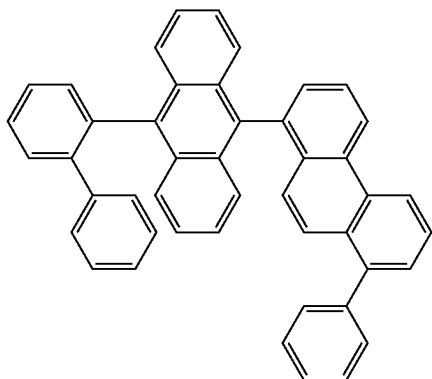


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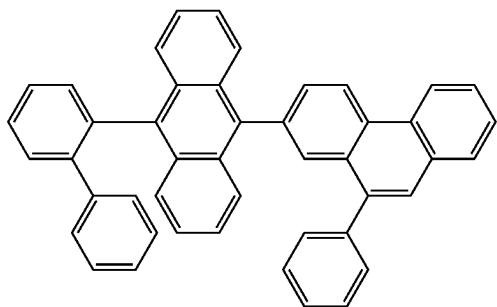
H1-48



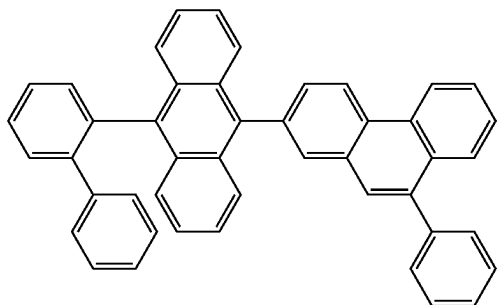
H1-49



H1-50

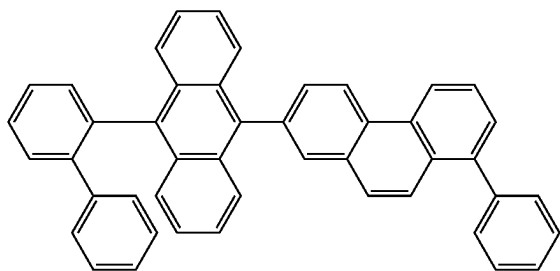


H1-51

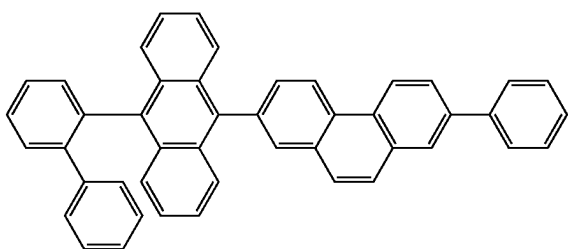


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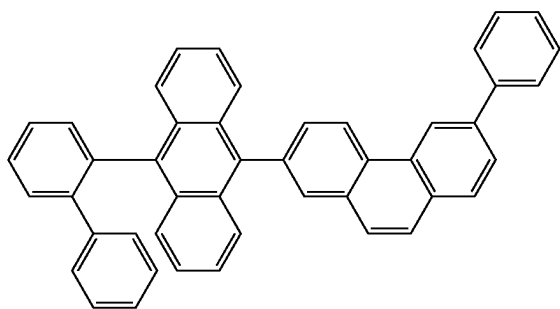
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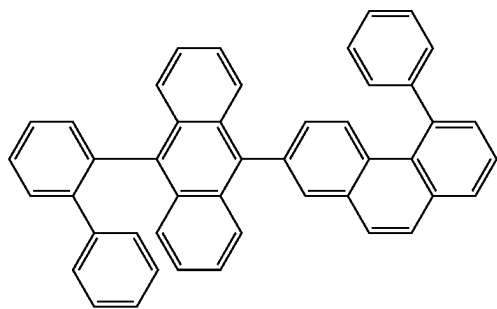
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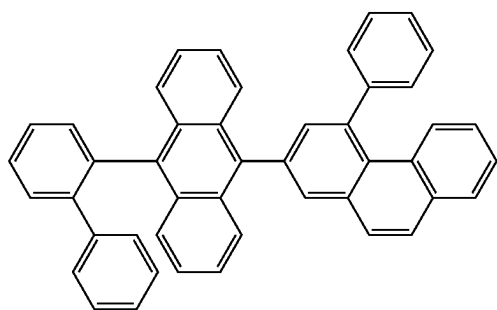
H1-54



H1-55

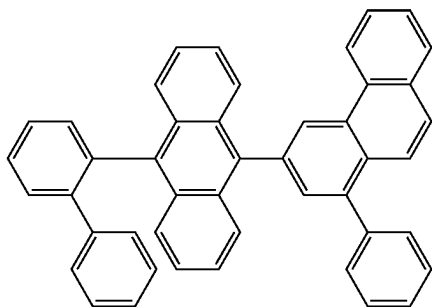


H1-56

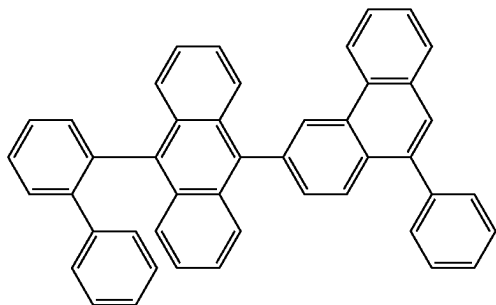


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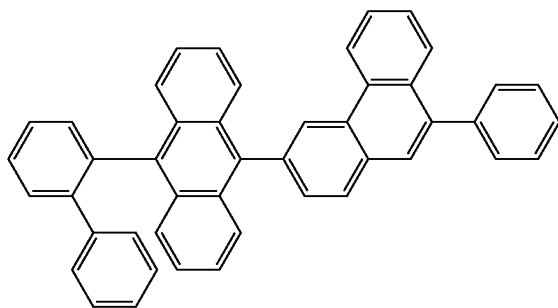
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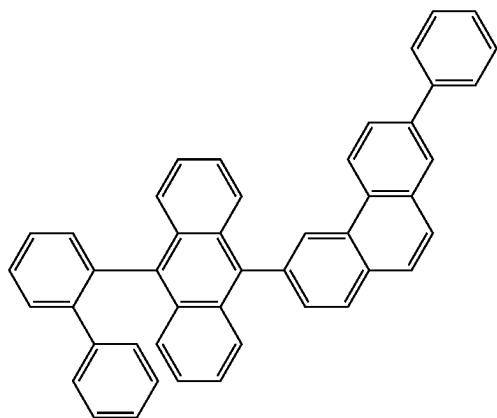
H1-58



H1-59

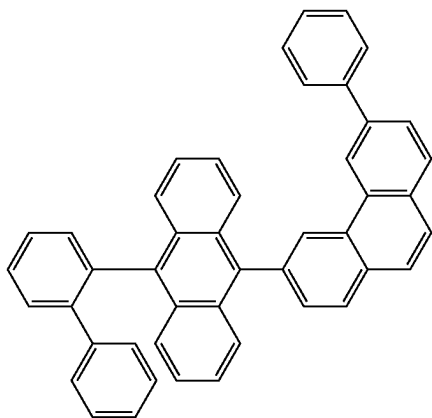


H1-60

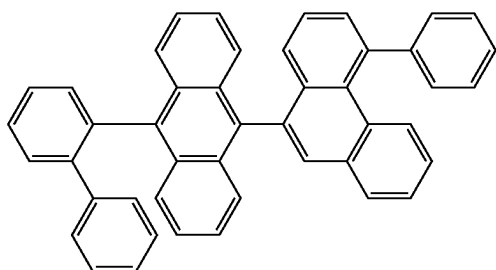


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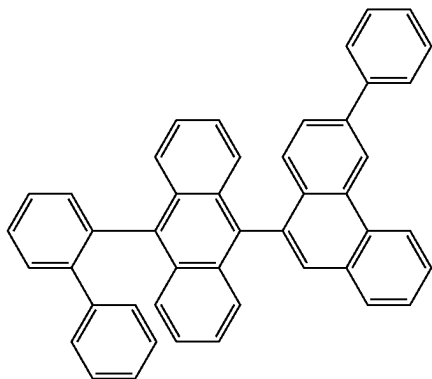
H1-61



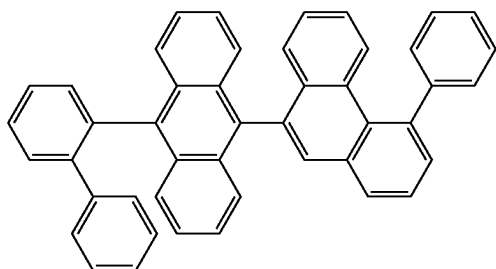
H1-62



H1-63

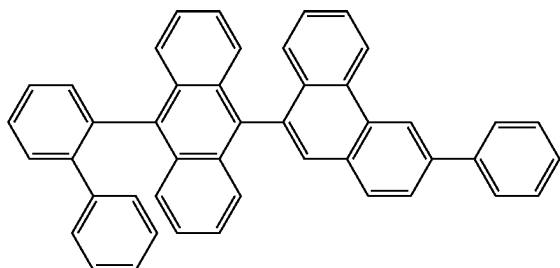


H1-64

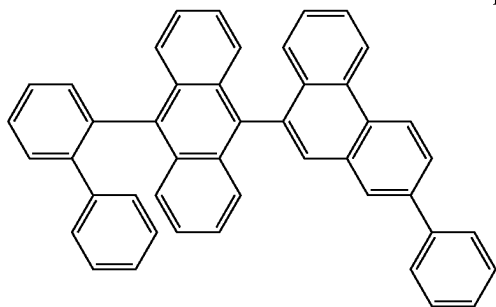


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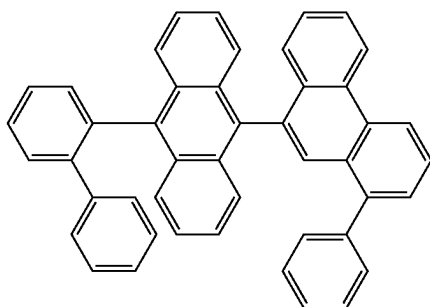
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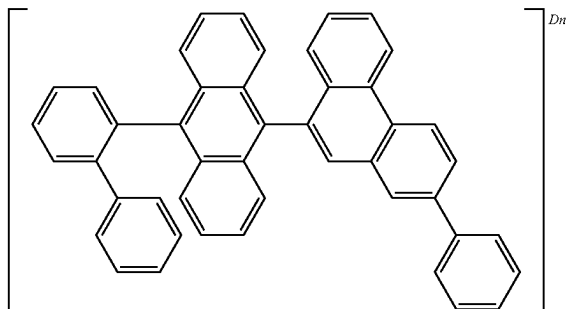
H1-66



H1-67

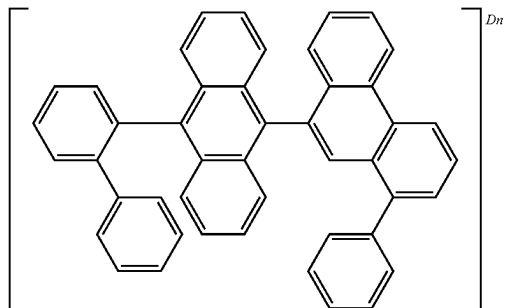


H1-68

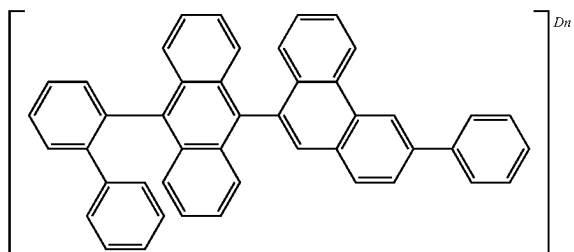


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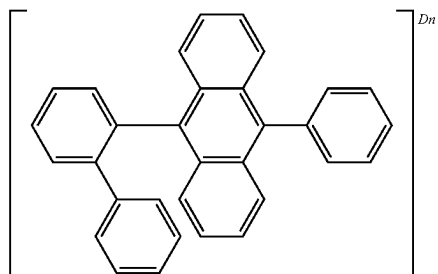
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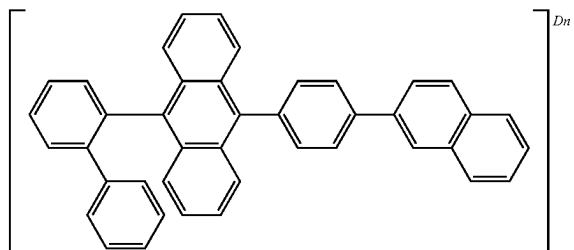
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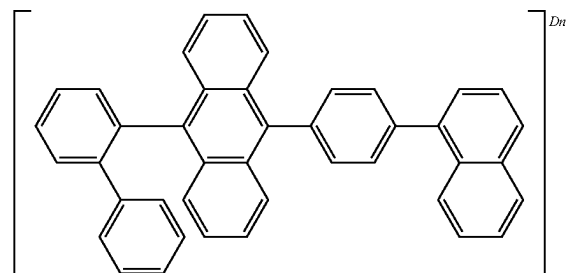
H1-71



H1-72

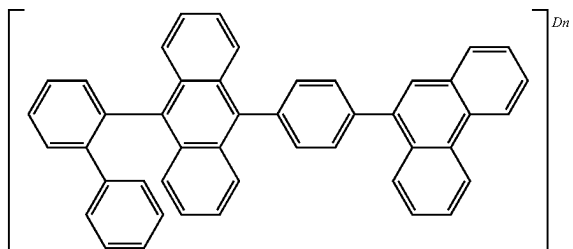


H1-73

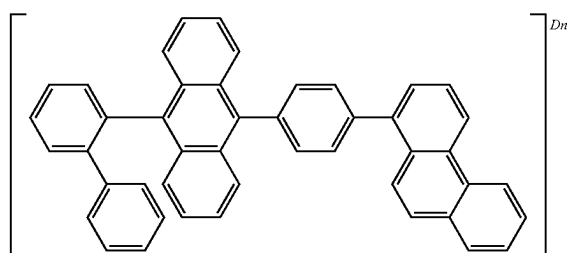


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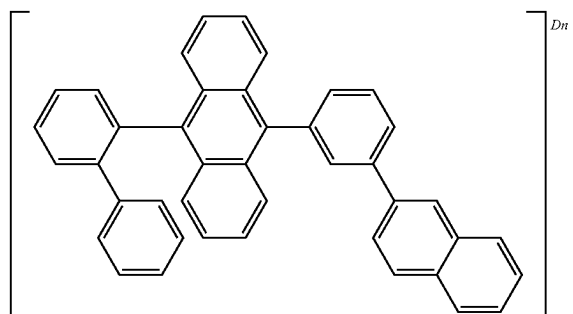
H1-74



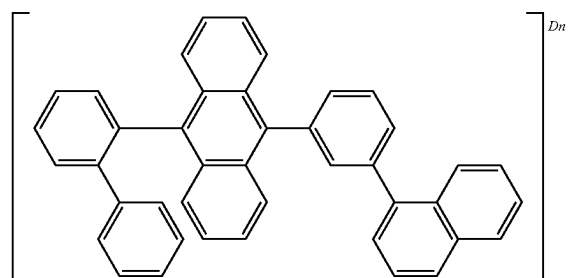
H1-75



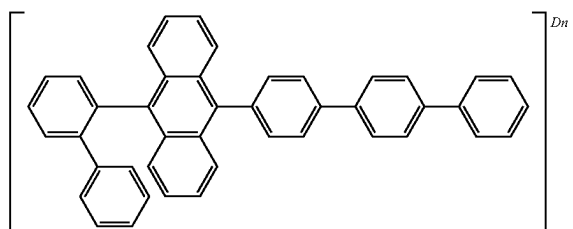
H1-76



H1-77

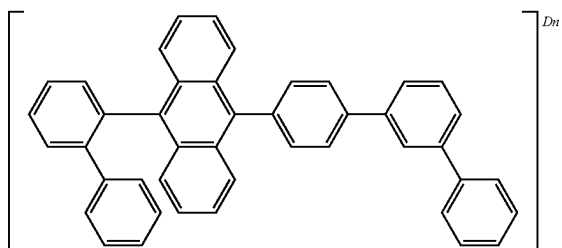


H1-78

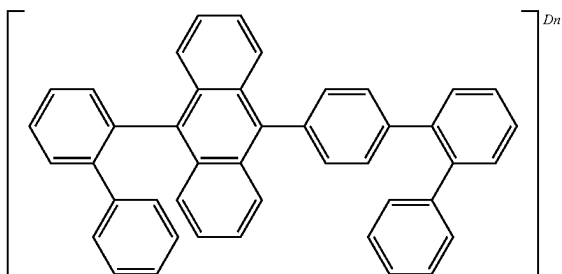


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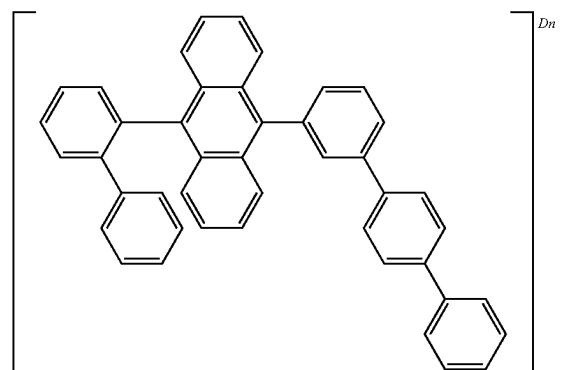
H1-79



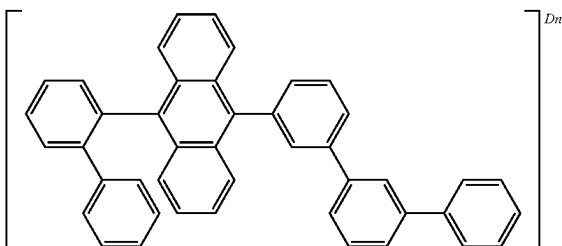
H1-80



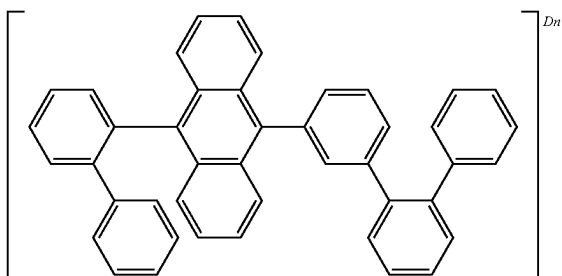
H1-81



H1-82

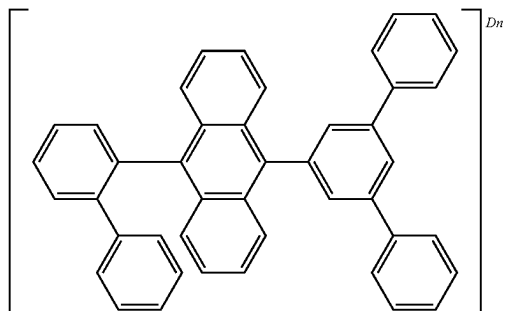


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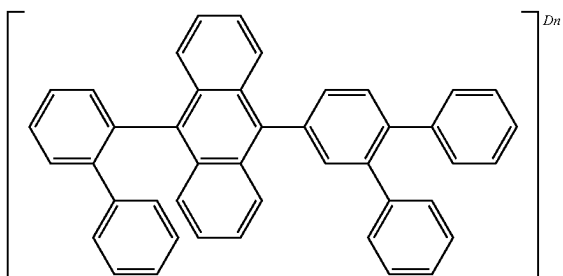


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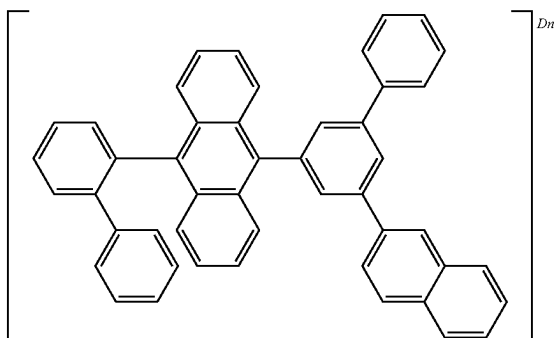
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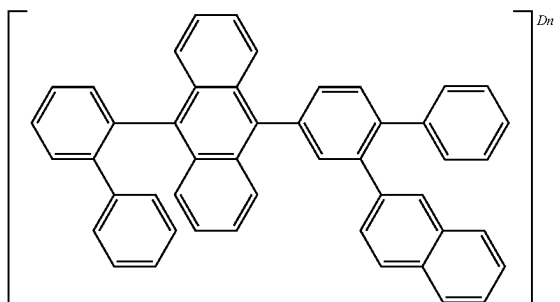
H1-85



H1-86

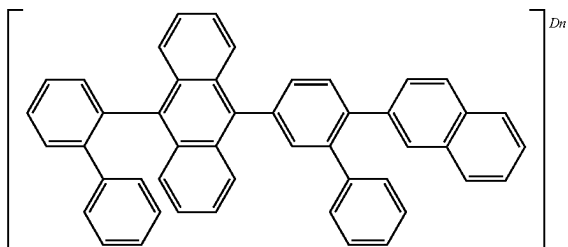


H1-87

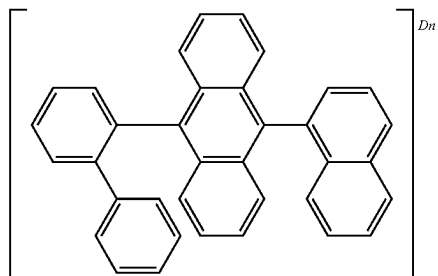


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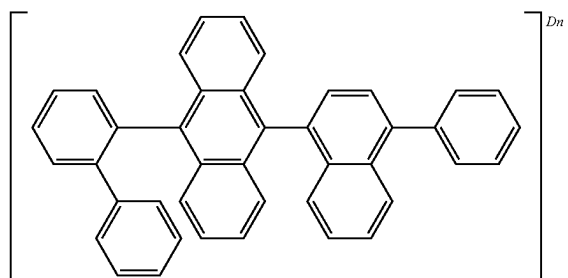
H1-88



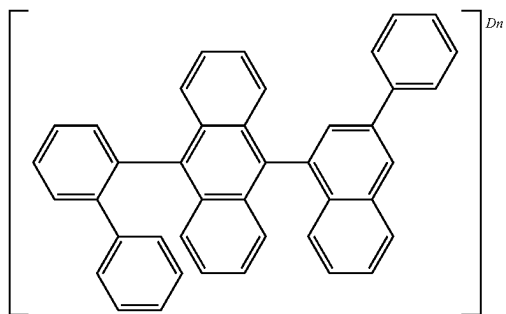
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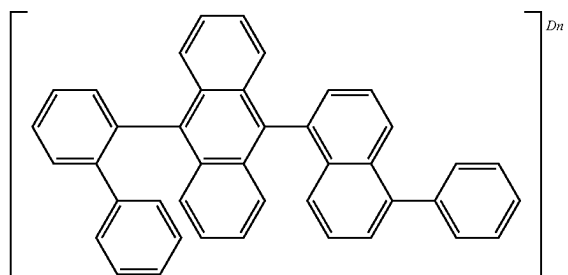
H1-90



H1-91

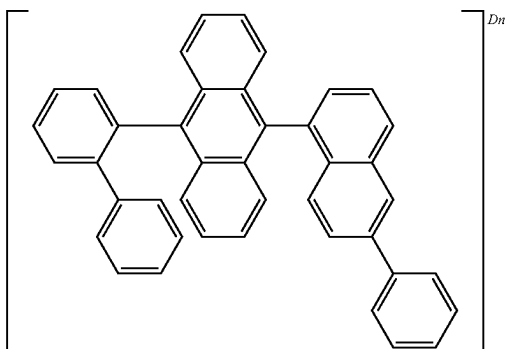


H1-92

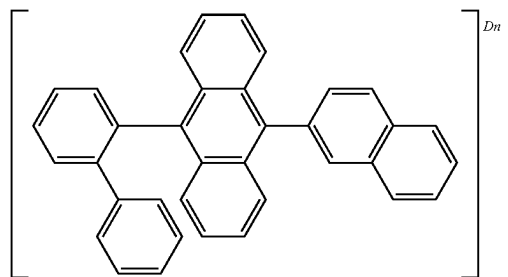


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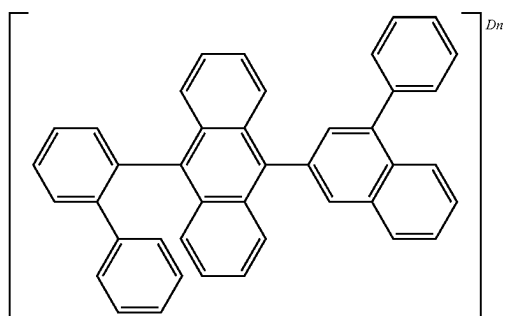
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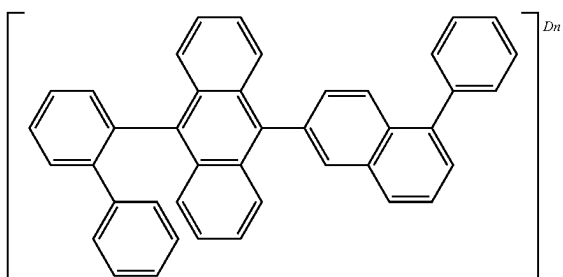
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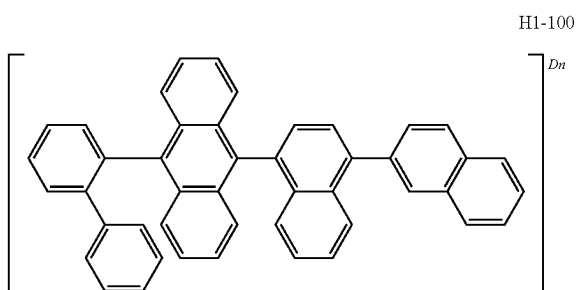
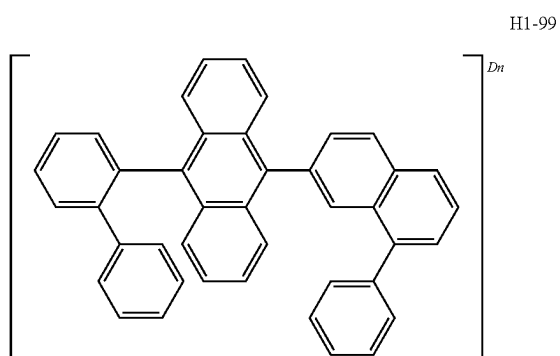
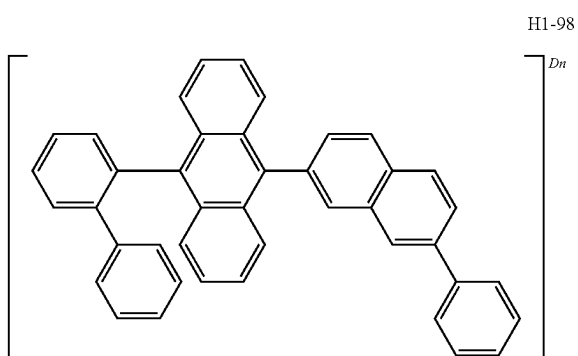
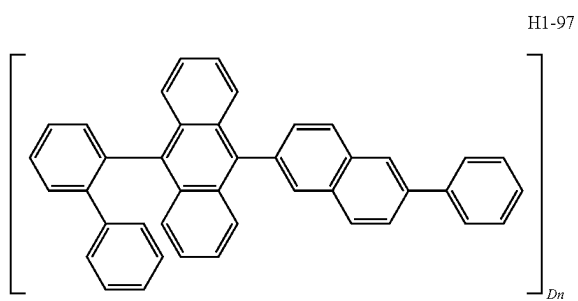
H1-95



H1-96

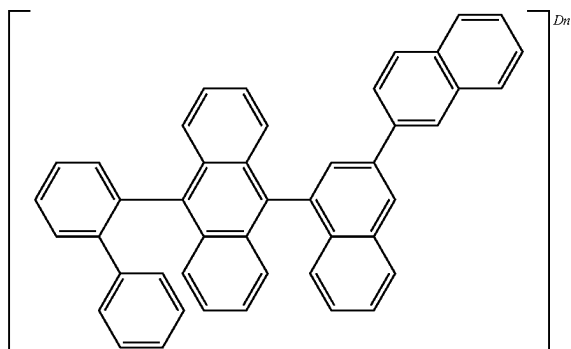


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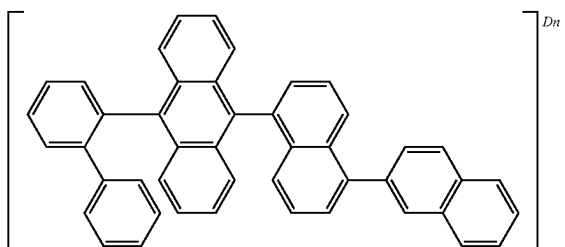


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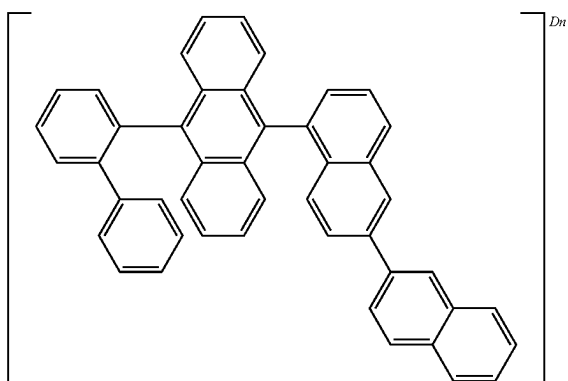
H1-101



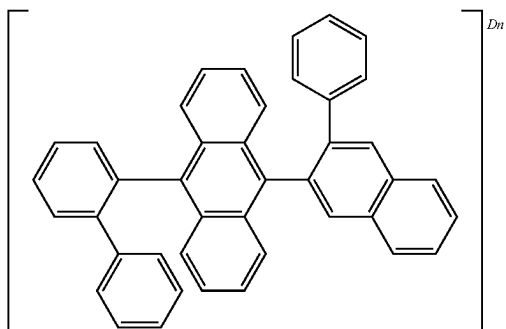
H1-102



H1-103

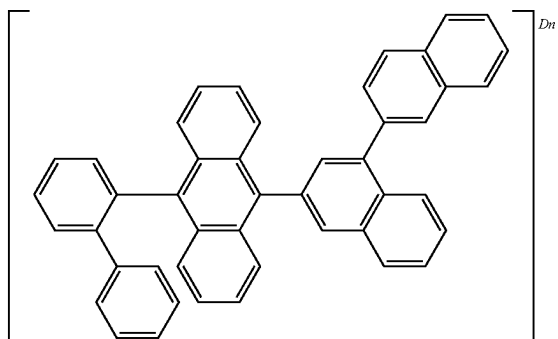


H1-104

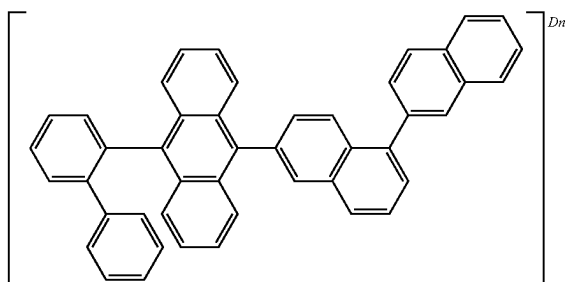


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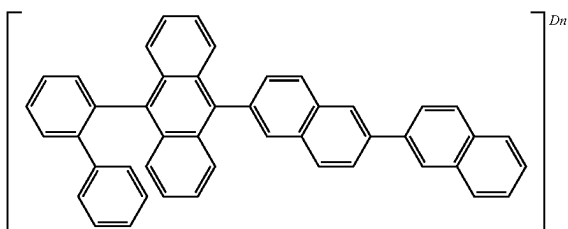
H1-105



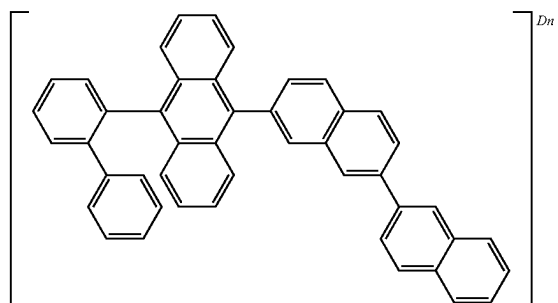
H1-106



H1-107

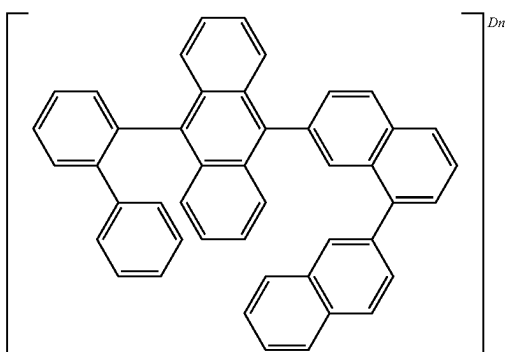


H1-108

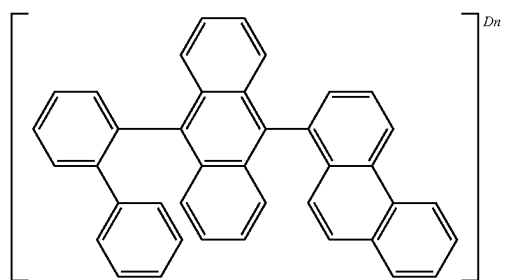


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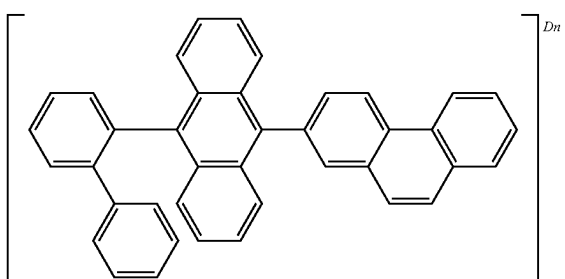
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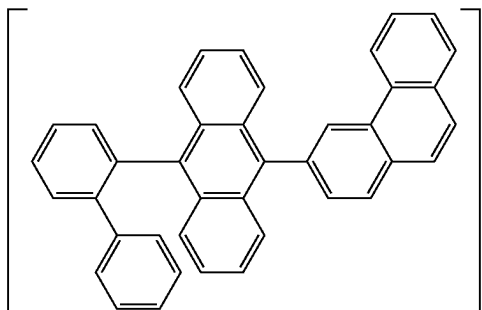
H1-110



H1-111

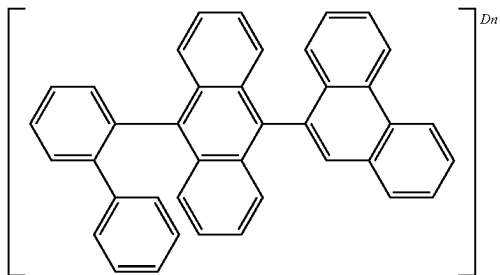


H1-112

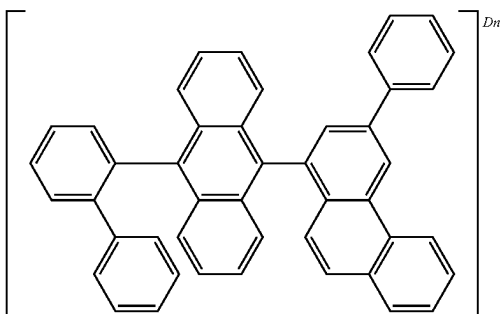


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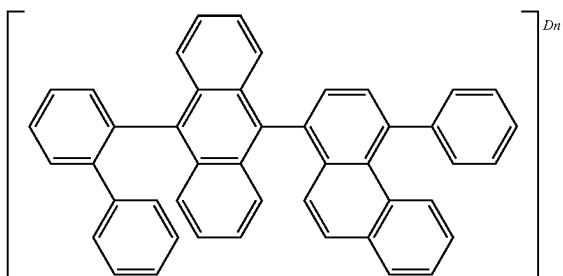
H1-113



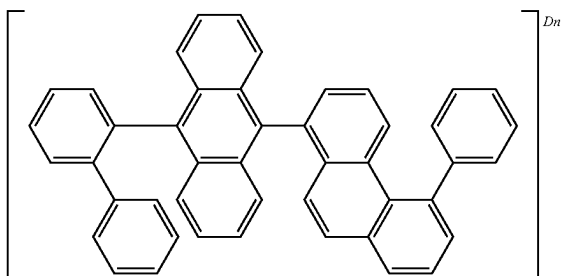
H1-114



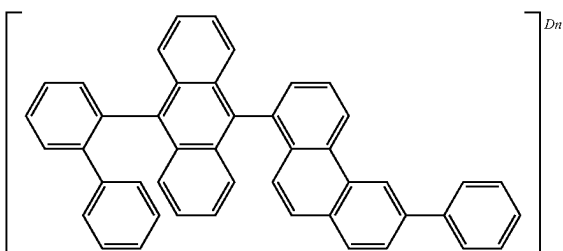
H1-115



H1-116

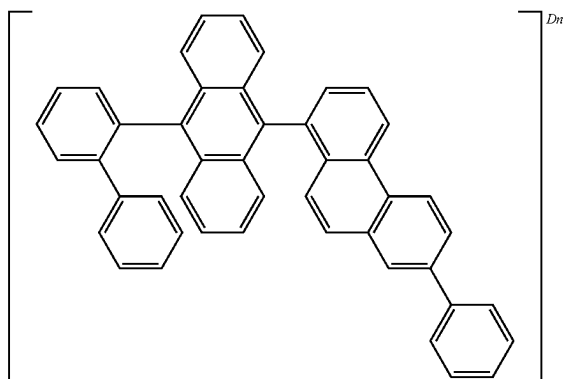


H1-117

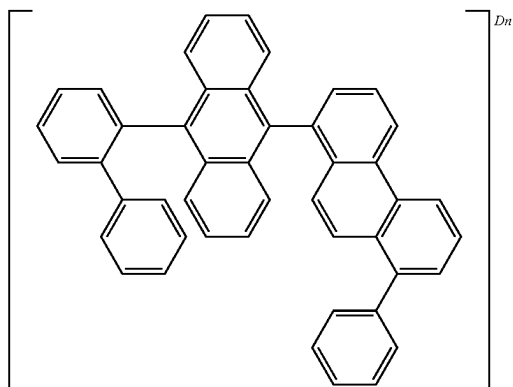


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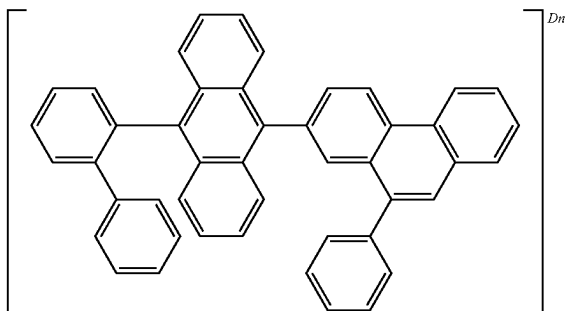
H1-118



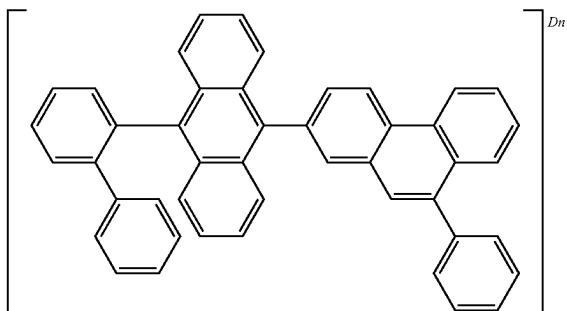
H1-119



H1-120

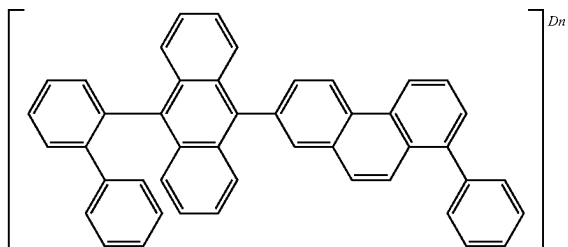


H1-121

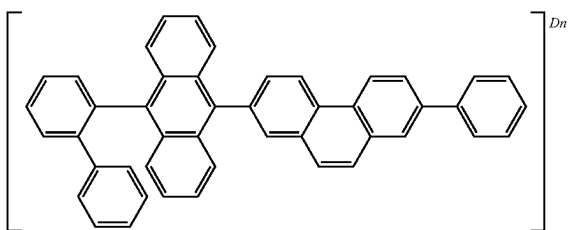


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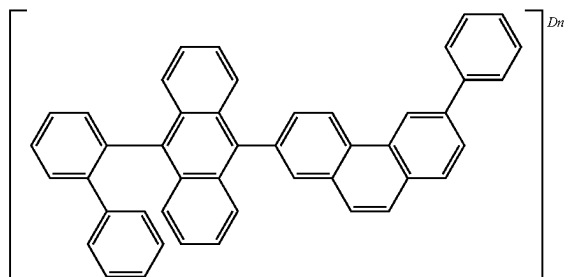
H1-122



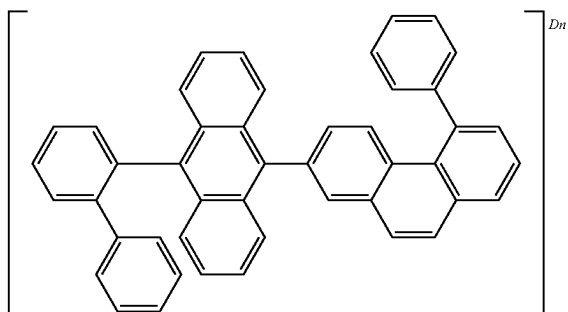
H1-123



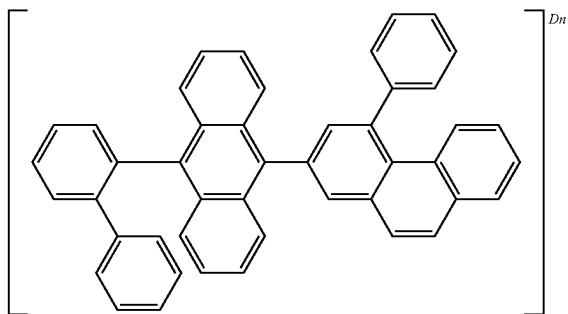
H1-124



H1-125

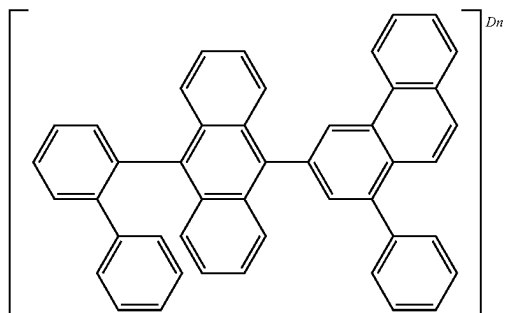


H1-126

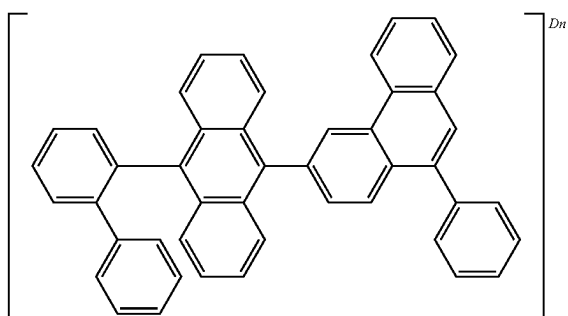


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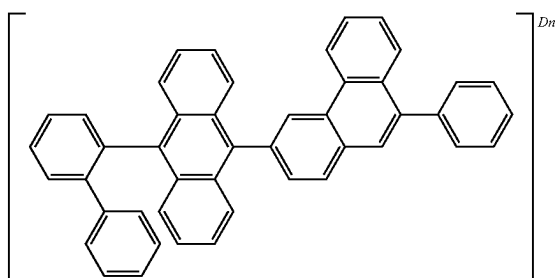
H1-127



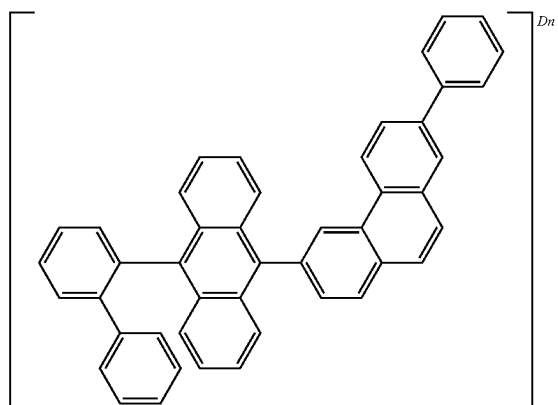
H1-128



H1-129

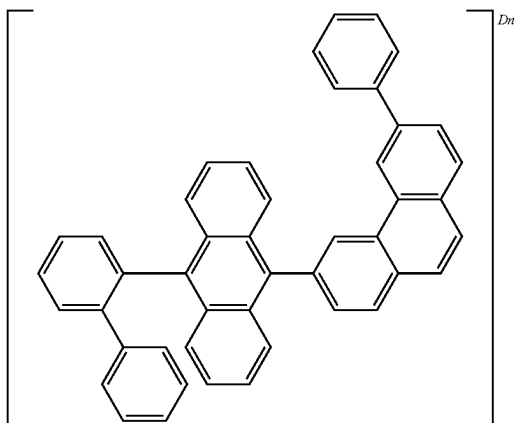


H1-130

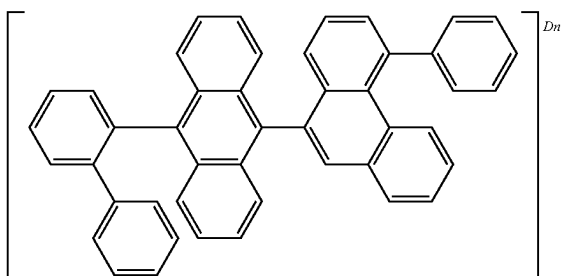


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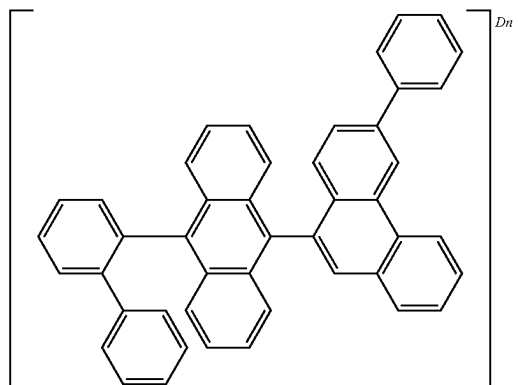
H1-131



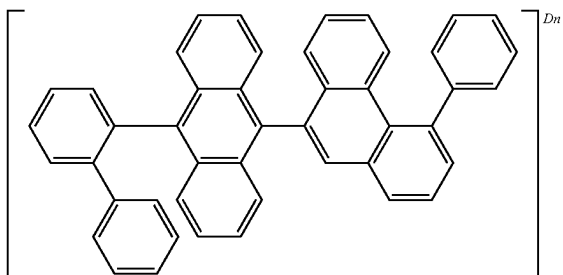
H1-132



H1-133

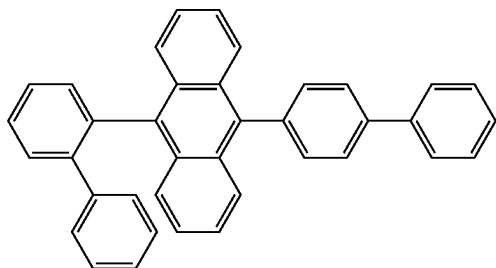


H1-134

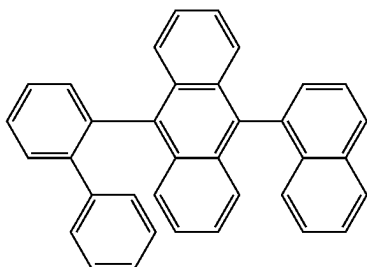


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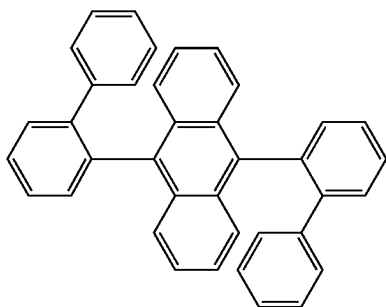
H1-135



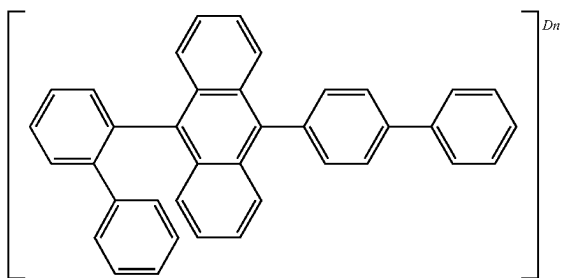
H1-136



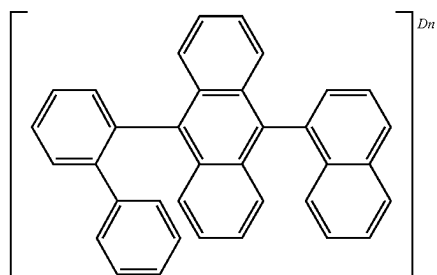
H1-137



H1-138



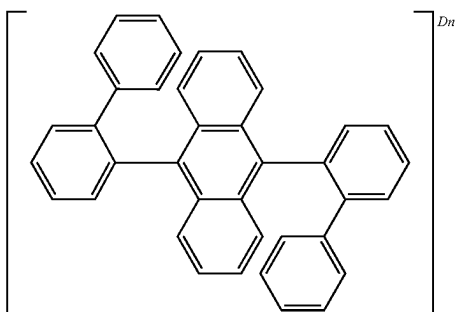
H1-139



and

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H1-140

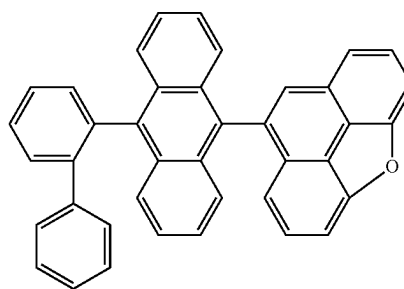


wherein D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to the maximum number of hydrogens in the compound.

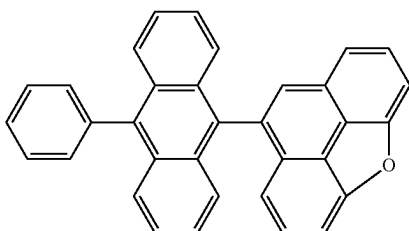
10. The plurality of host materials according to claim 1, wherein the compound represented by Formula 2 is at least one selected from the following compounds:

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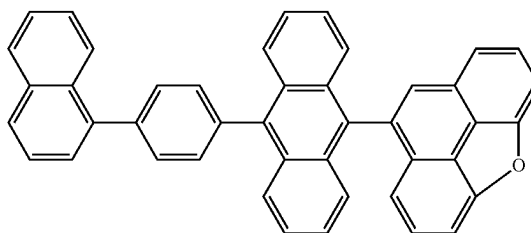
H2-4



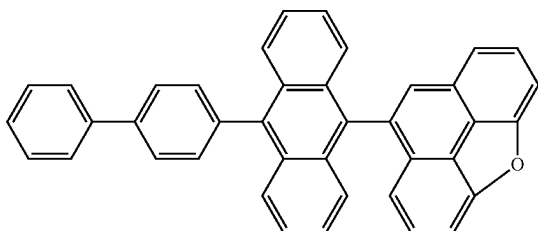
H2-1



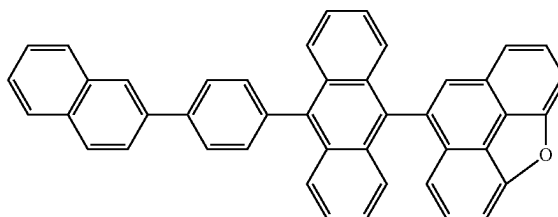
H2-5



H2-2

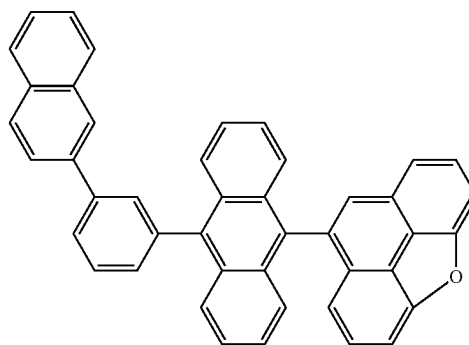
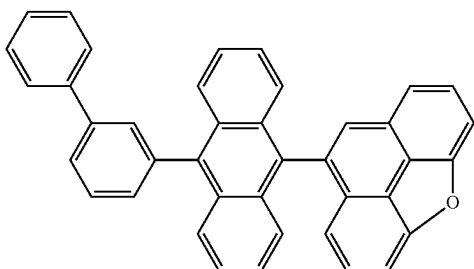


H2-6



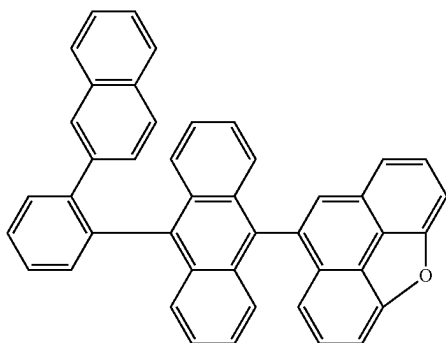
H2-7

H2-3

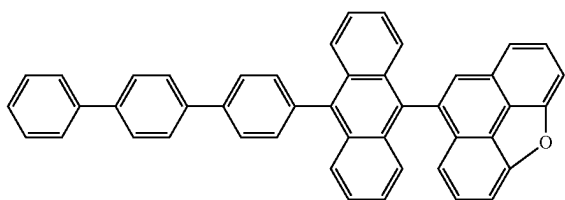


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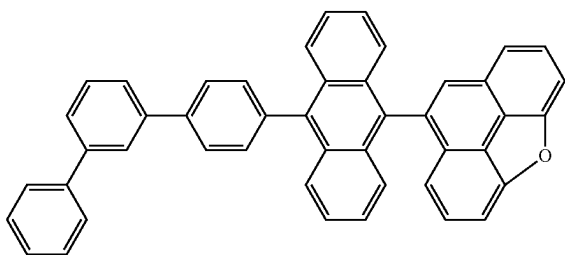
H2-8



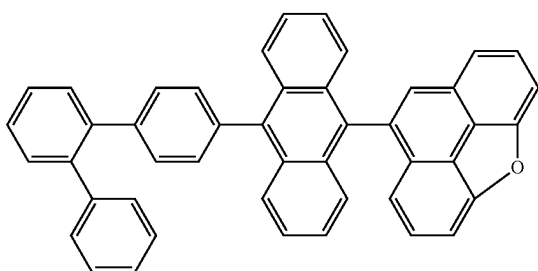
H2-9



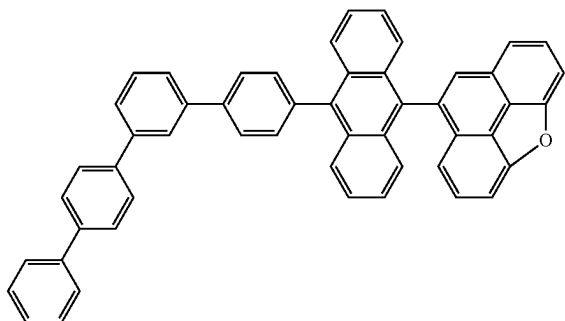
H2-10



H2-11

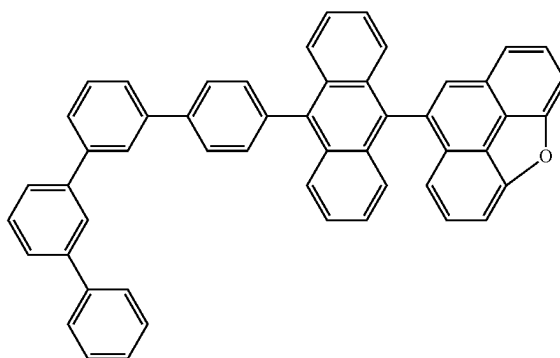


H2-12

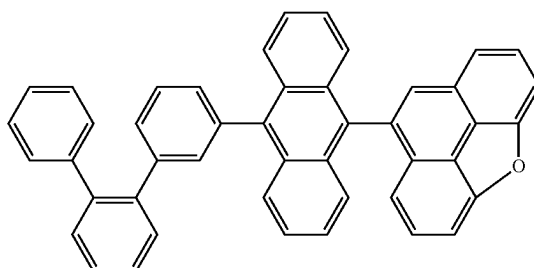


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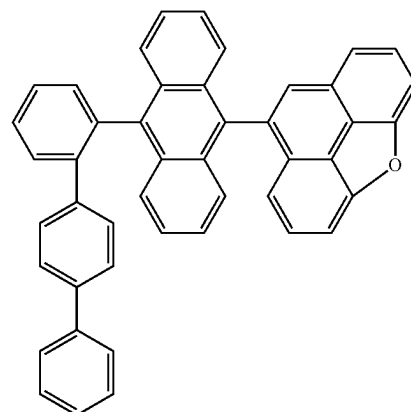
H2-13



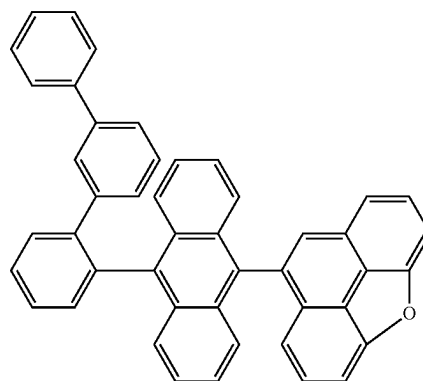
H2-14



H2-15

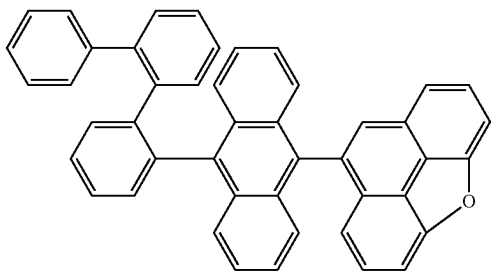


H2-16

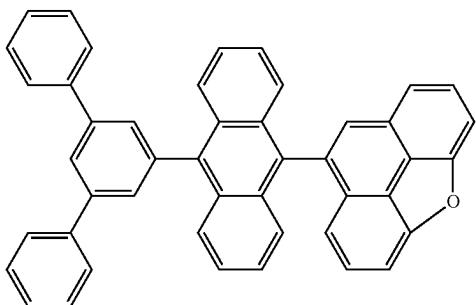


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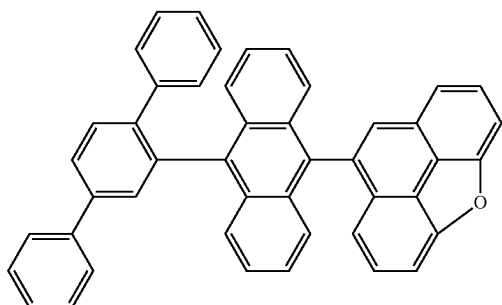
H2-17



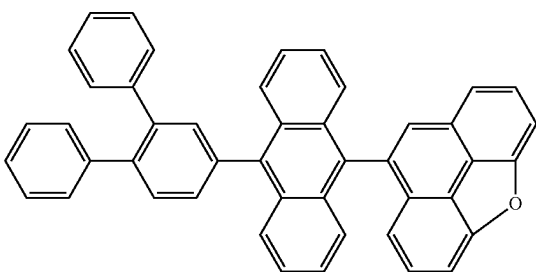
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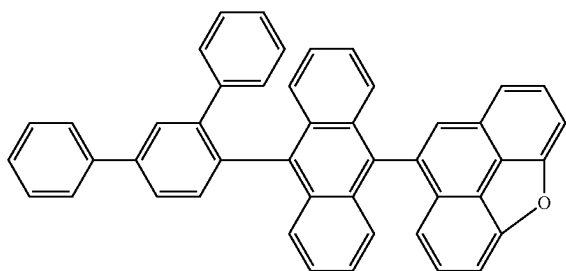
H2-19



H2-20

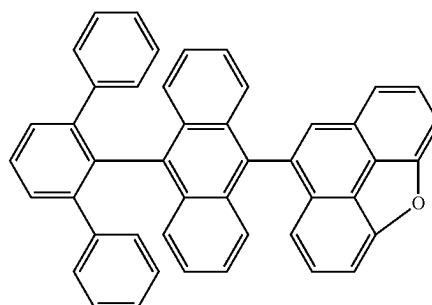


H2-21

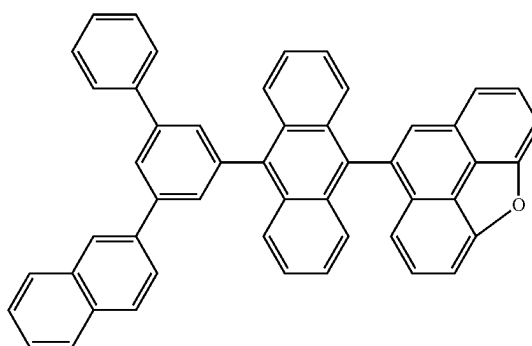


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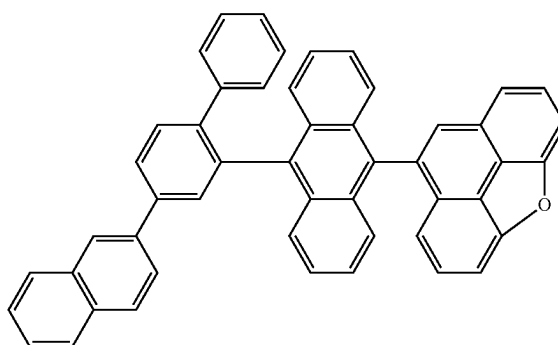
H2-22



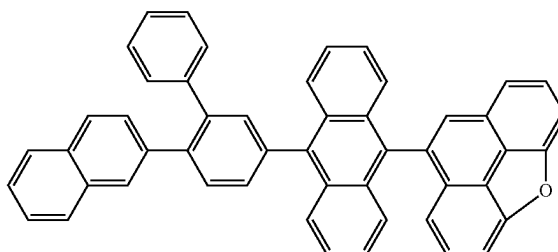
H2-23



H2-24

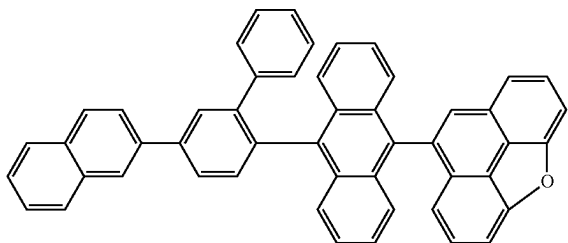


H2-25

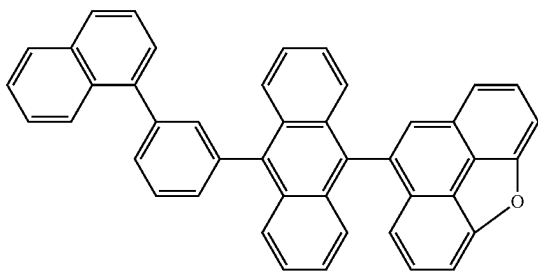


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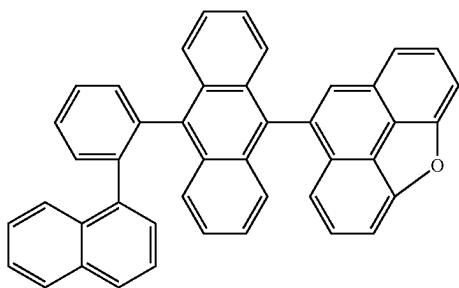
H2-26



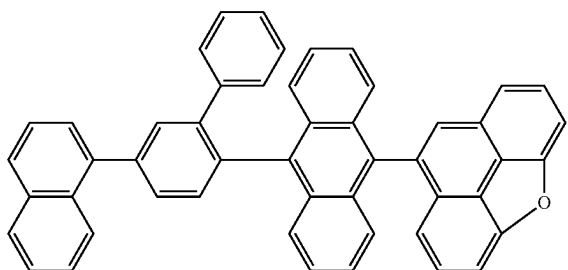
H2-27



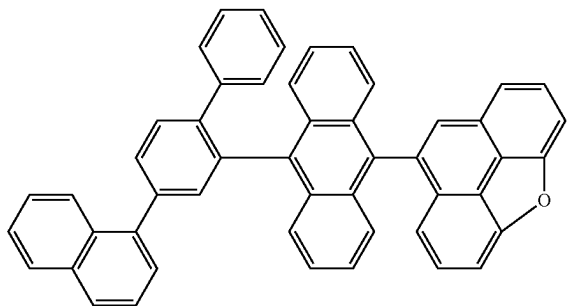
H2-28



H2-29

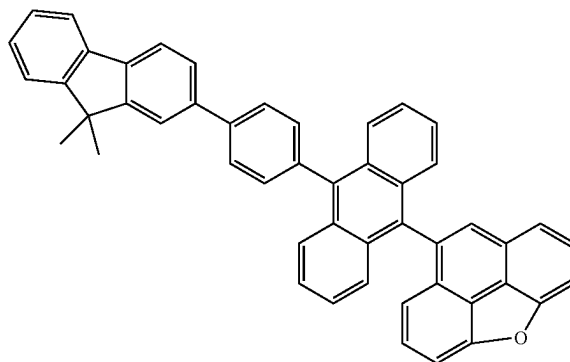


H2-30

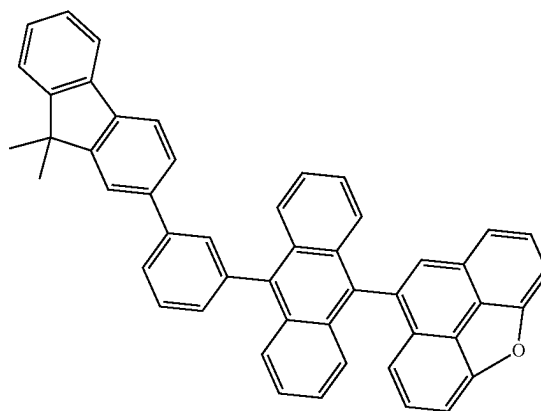


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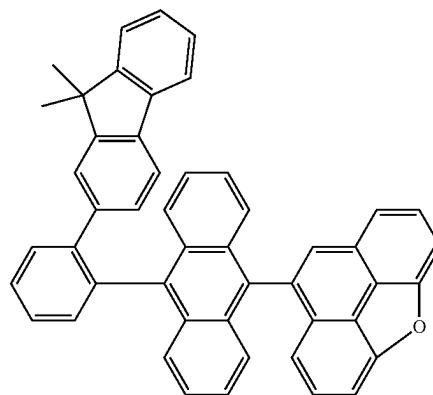
H2-31



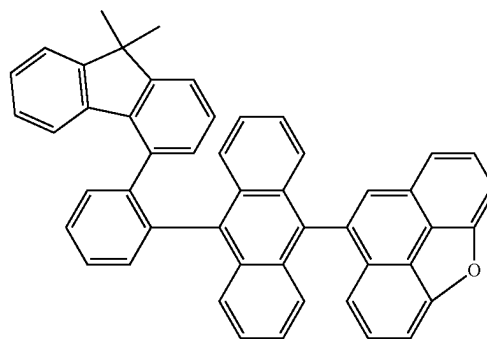
H2-32



H2-33

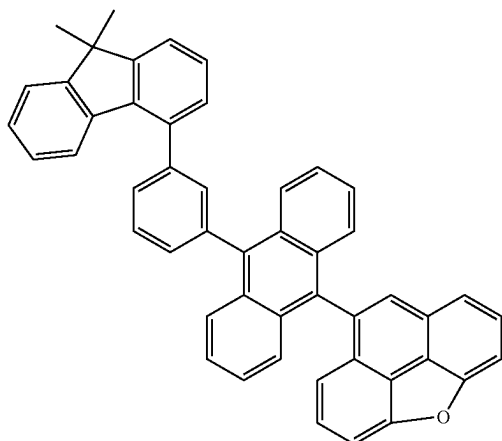


H2-34



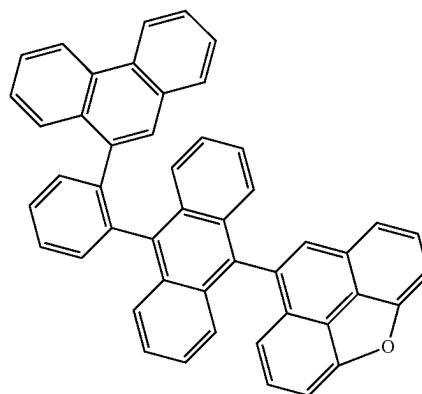
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H2-35



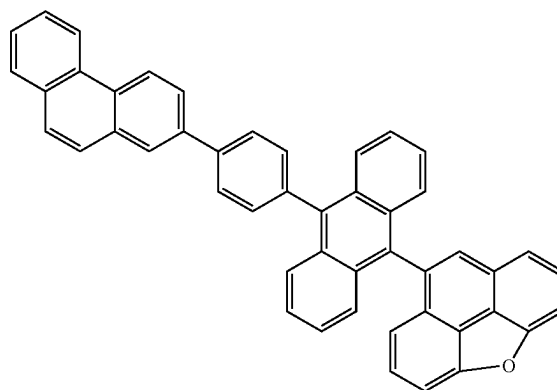
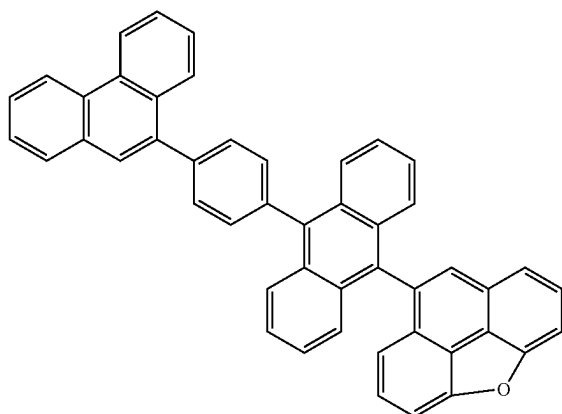
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H2-38

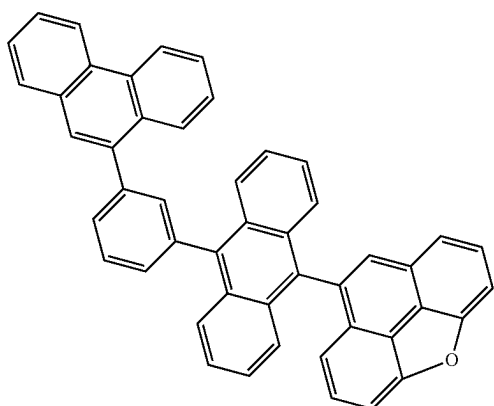


H2-39

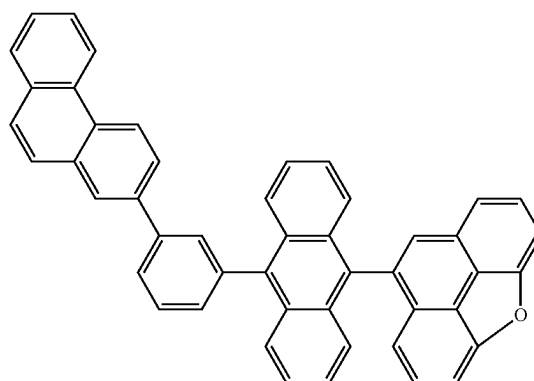
H2-36



H2-37

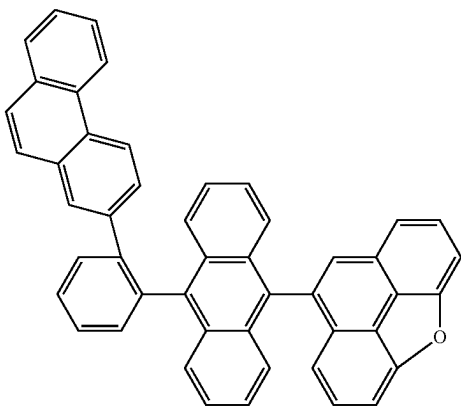


H2-40



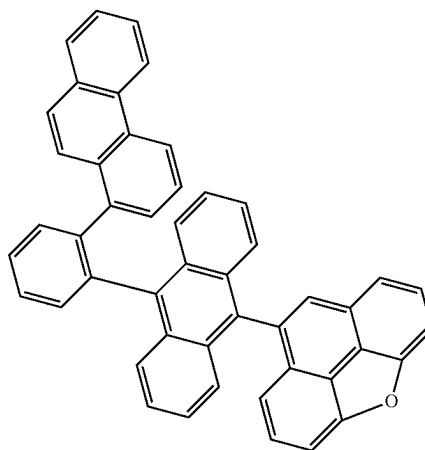
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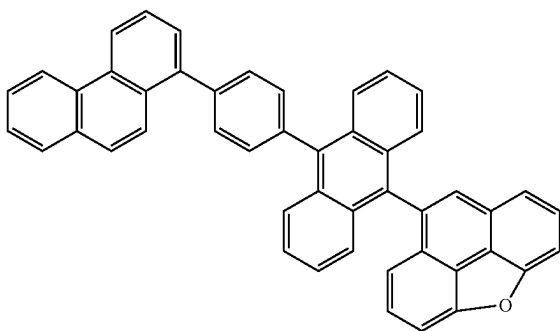


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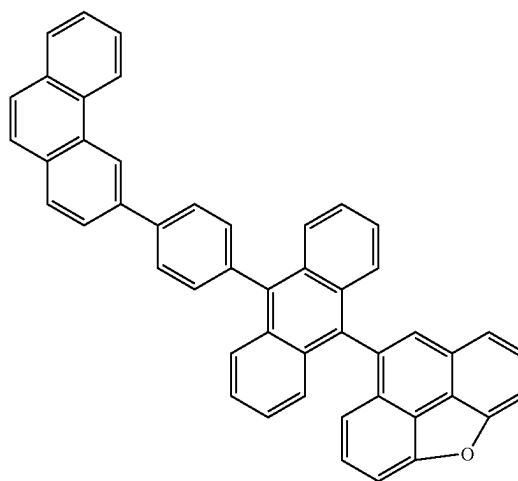
H2-44



H2-42

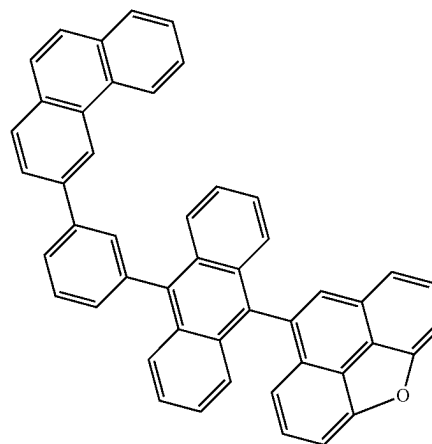
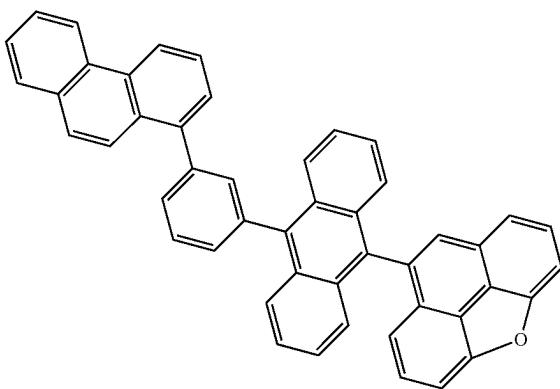


H2-45



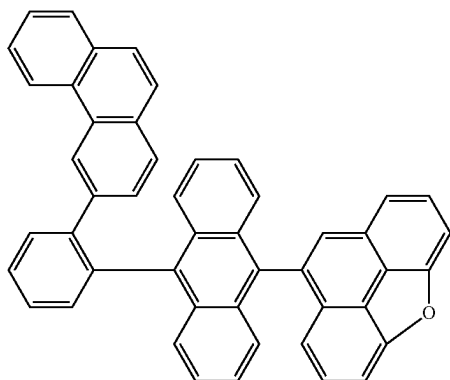
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H2-43



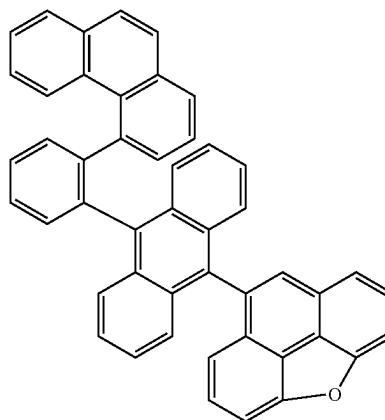
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H2-47

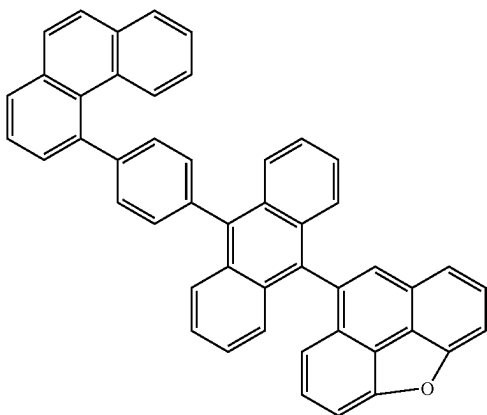


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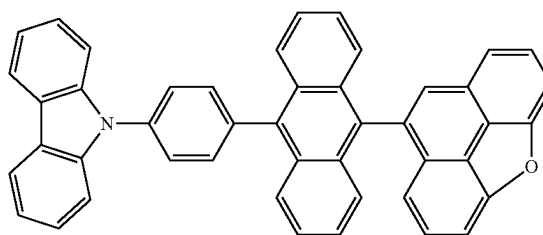
H2-50



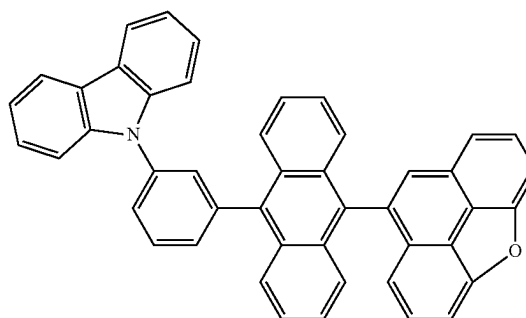
H2-48



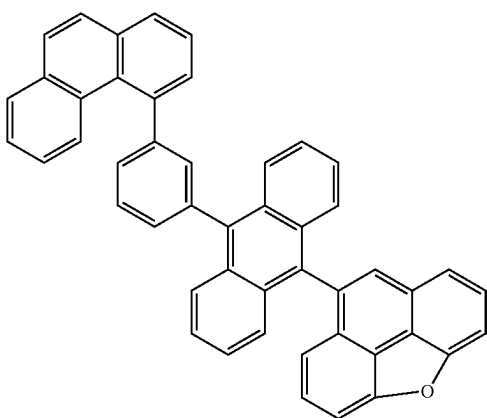
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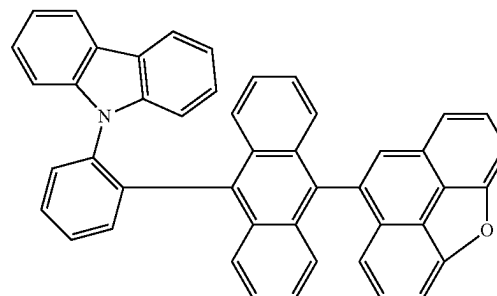
H2-52



H2-49

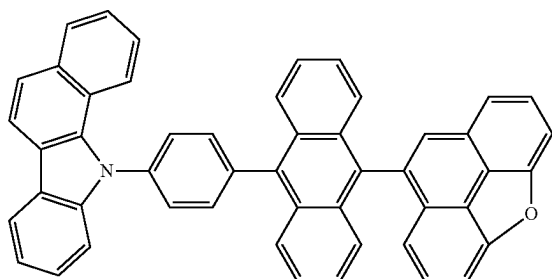


H2-53

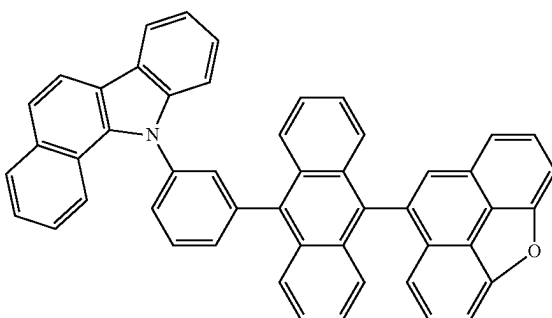


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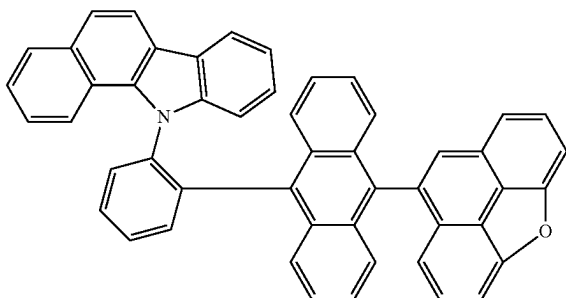
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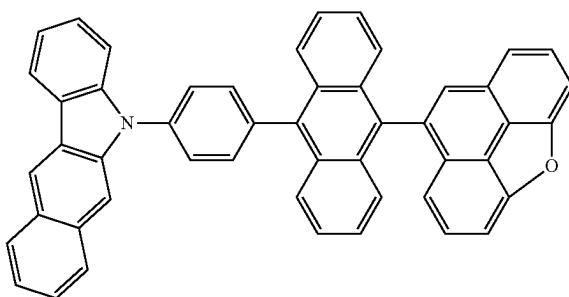
H2-55



H2-56

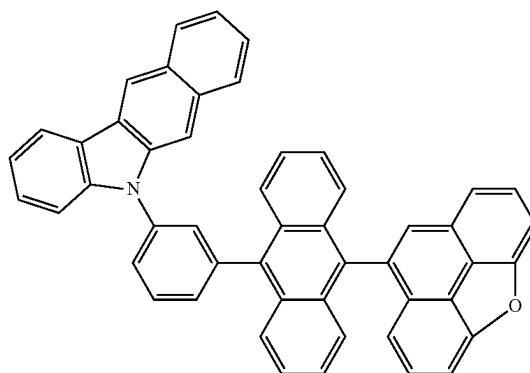


H2-57

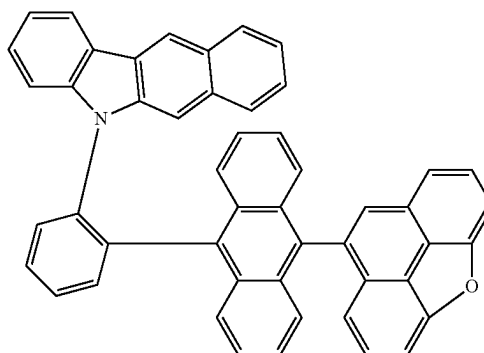


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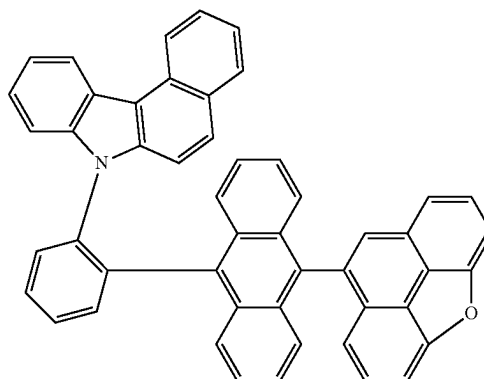
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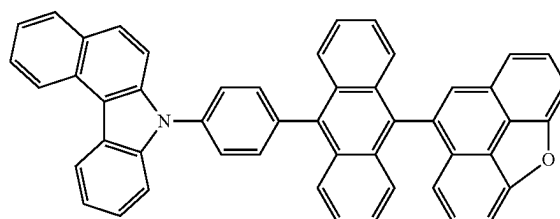
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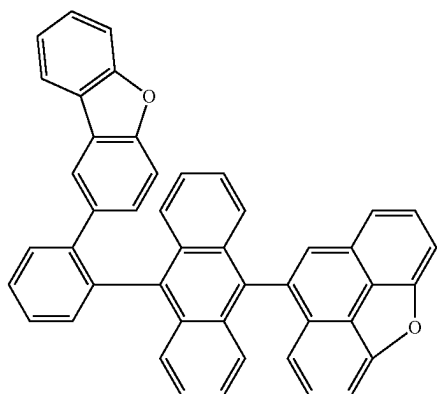
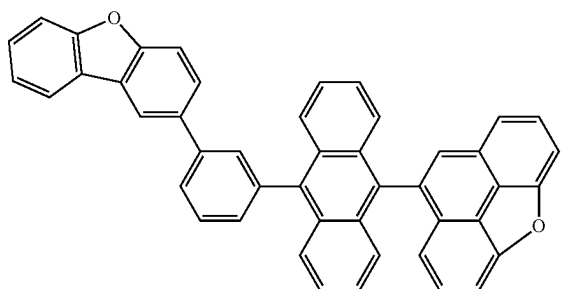
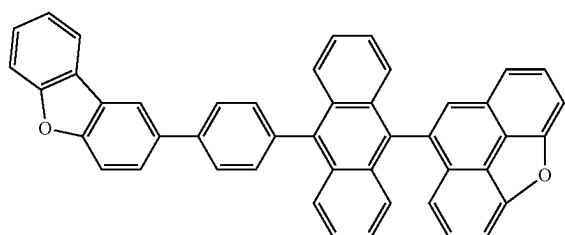
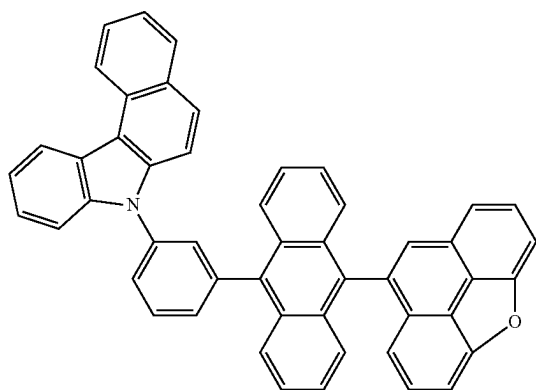
H2-60



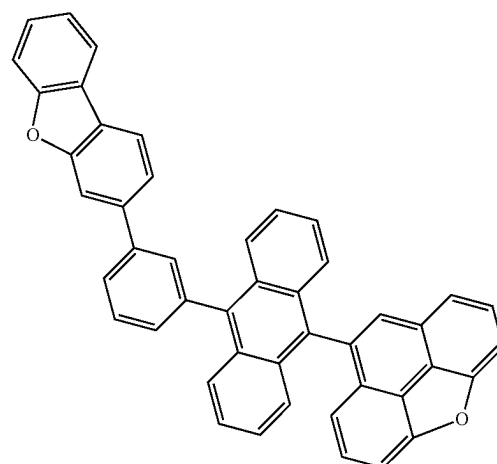
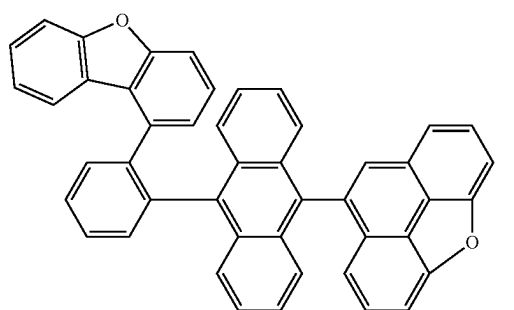
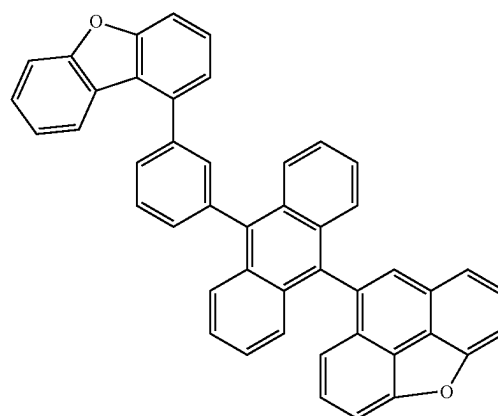
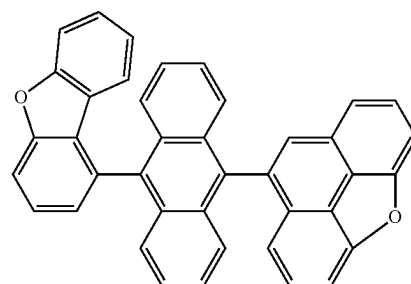
H2-61



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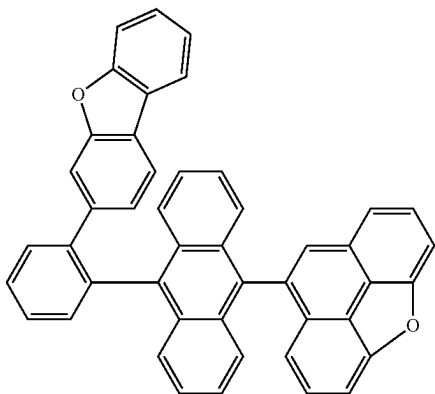


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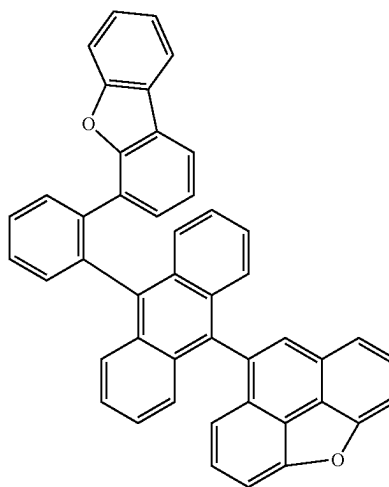
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H2-70



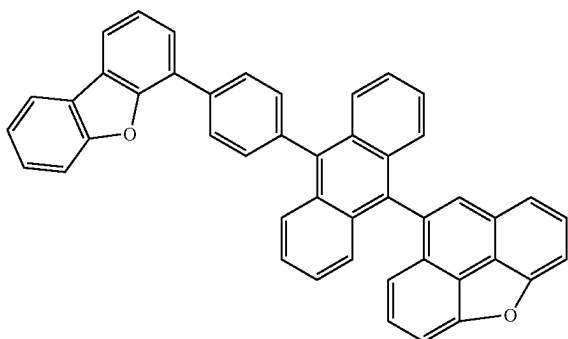
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H2-73

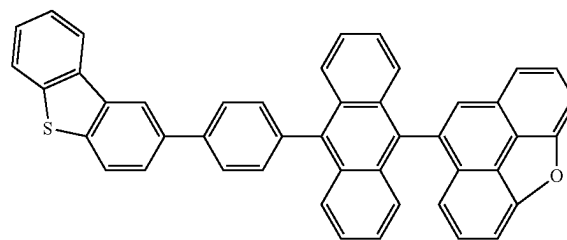
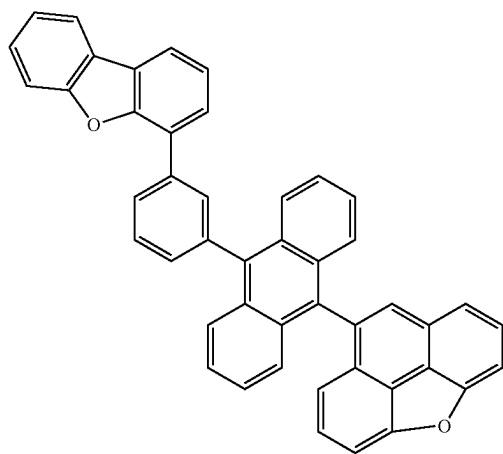


H2-74

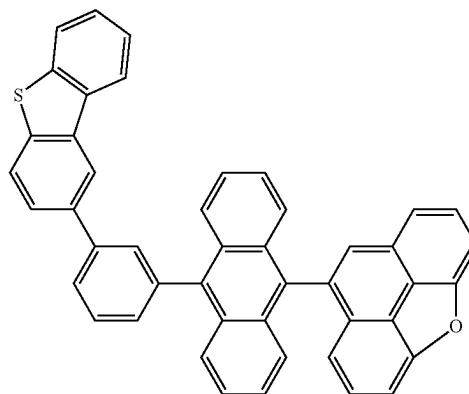
H2-71



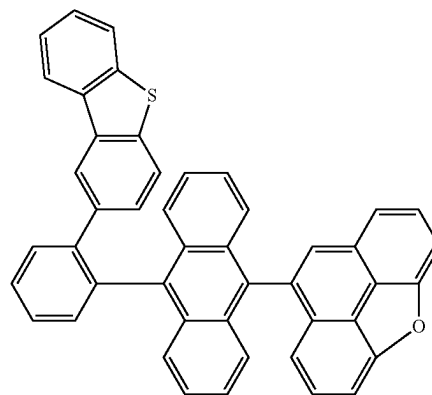
H2-72



H2-75

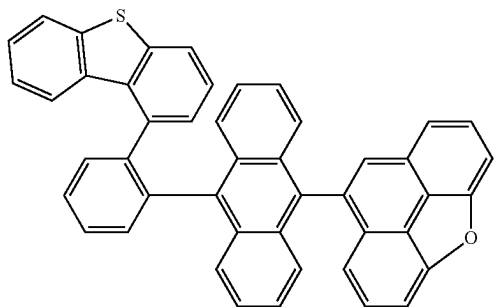


H2-76



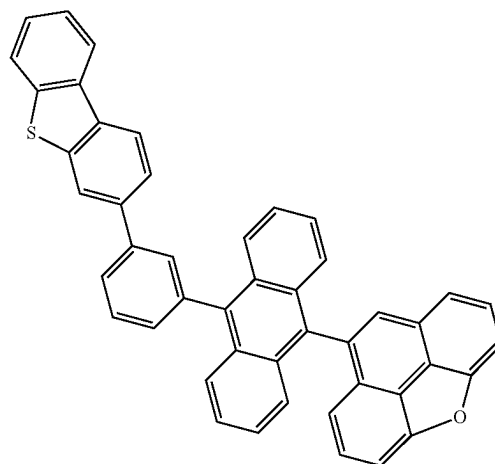
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H2-77

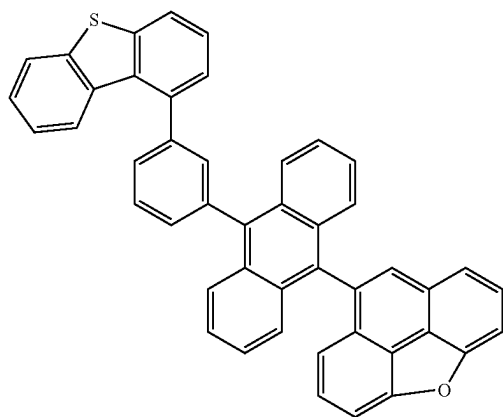


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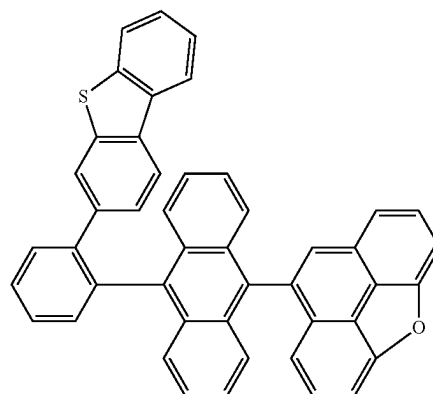
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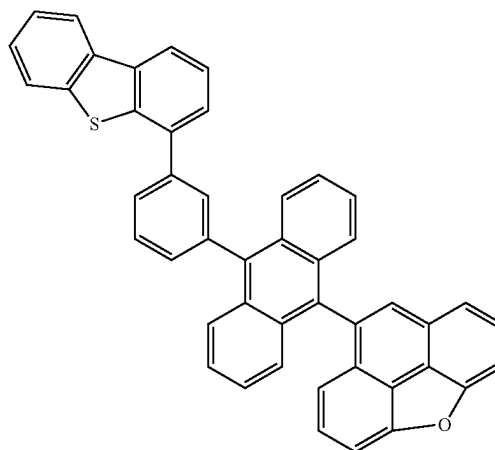
H2-78



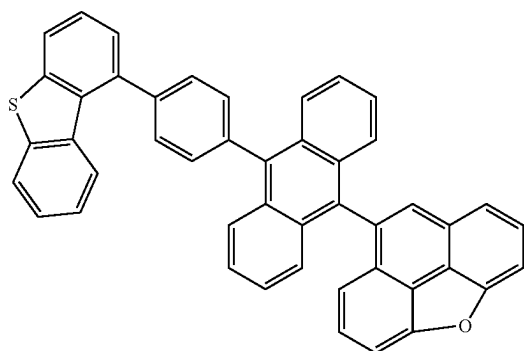
H2-81



H2-82

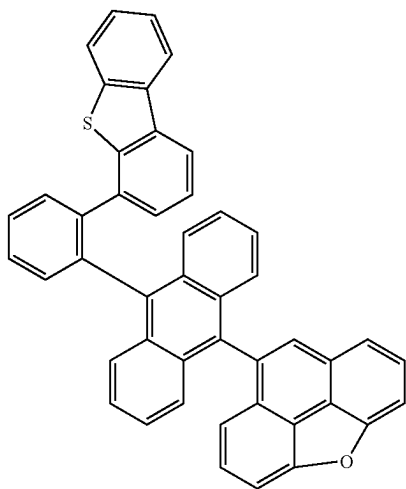


H2-79

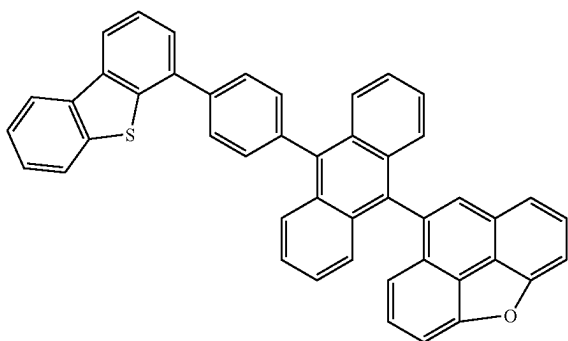


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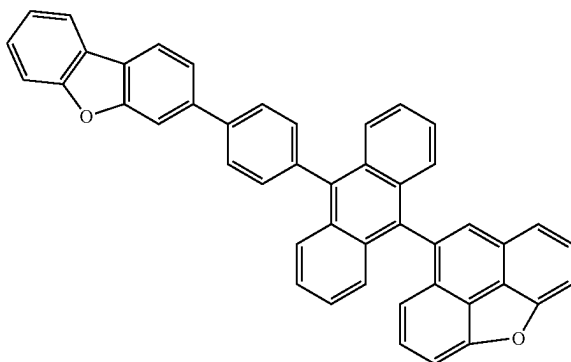
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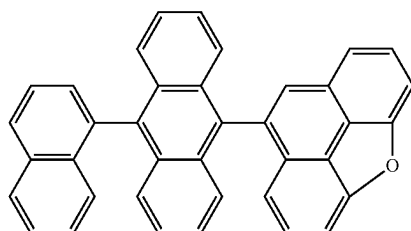
H2-84



H2-85

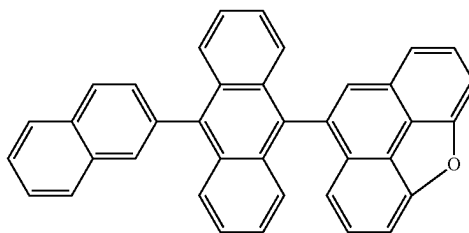


H2-86

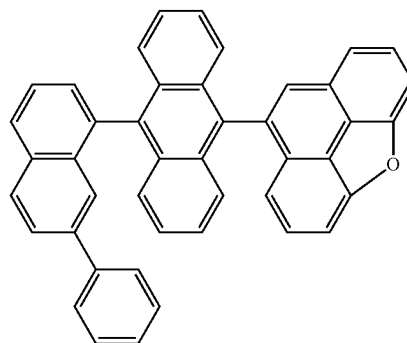


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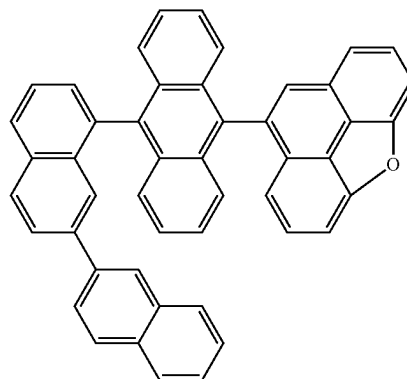
H2-87



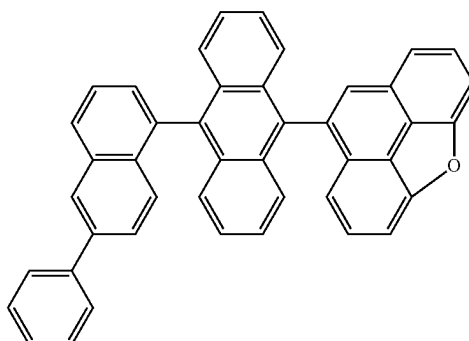
H2-88



H2-89

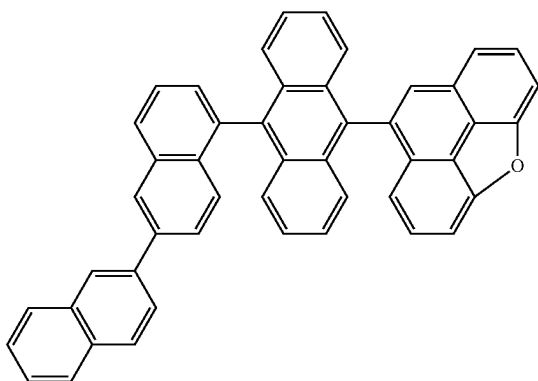


H2-90

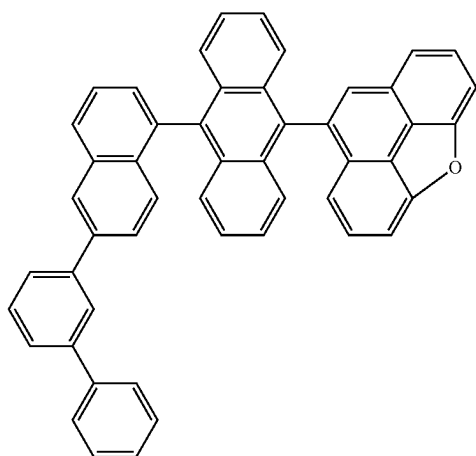


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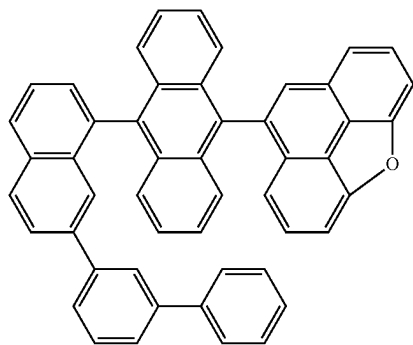
H2-91



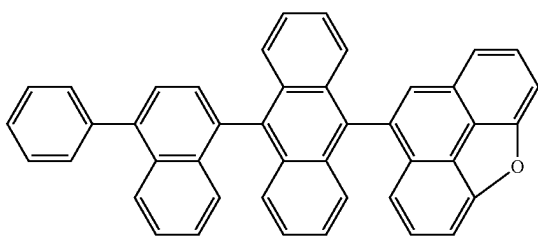
H2-92



H2-93

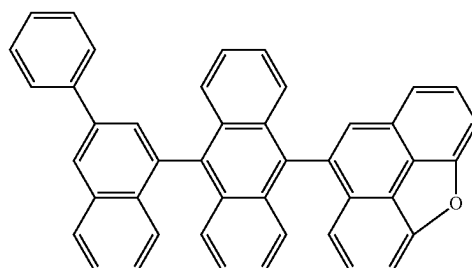


H2-94

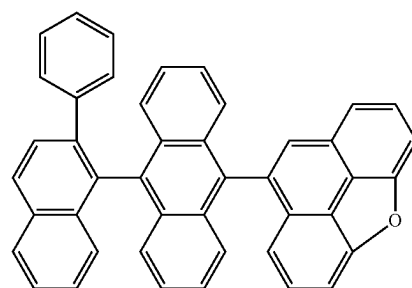


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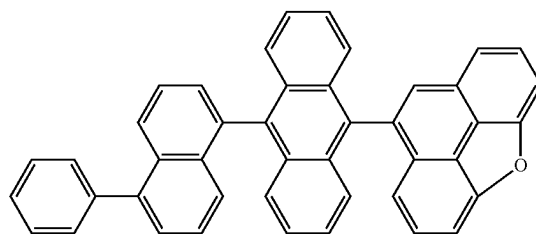
H2-95



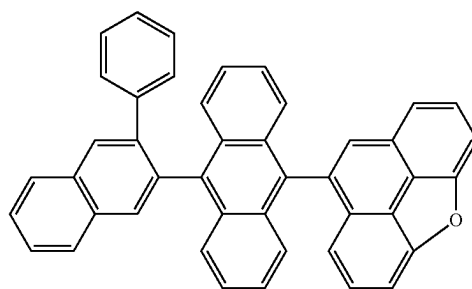
H2-96



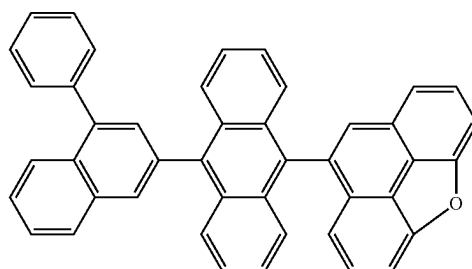
H2-97



H2-98

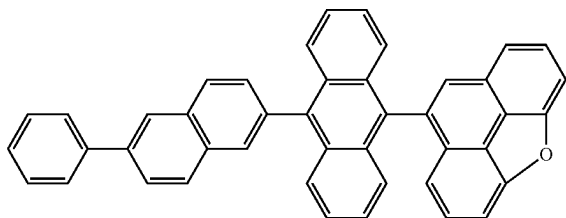


H2-99

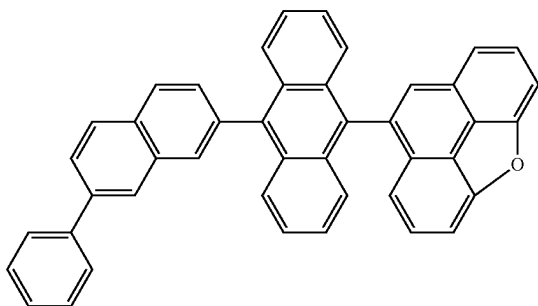


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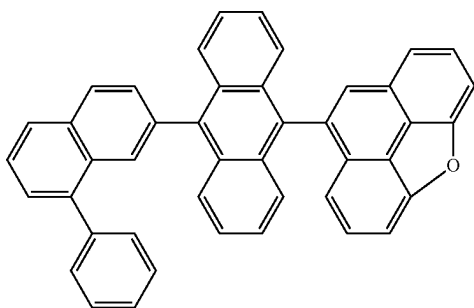
H2-100



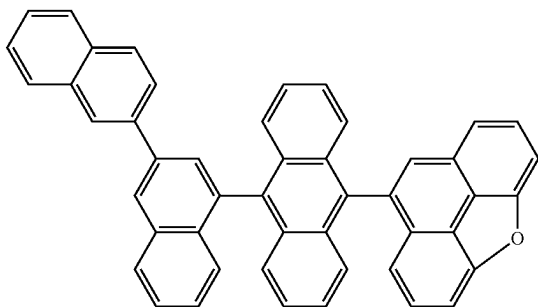
H2-101



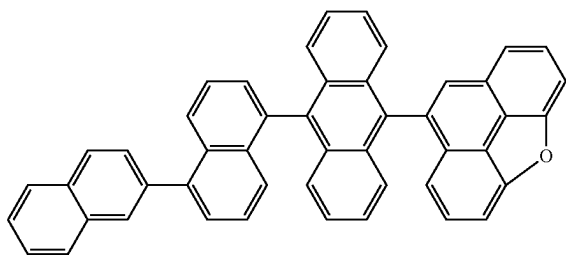
H2-102



H2-103

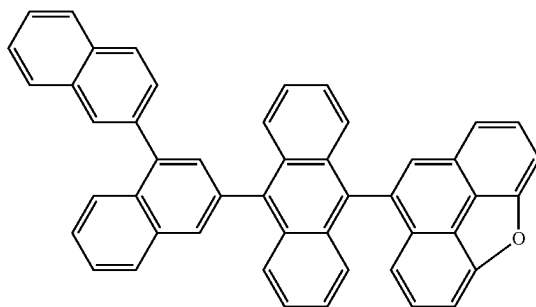


H2-104

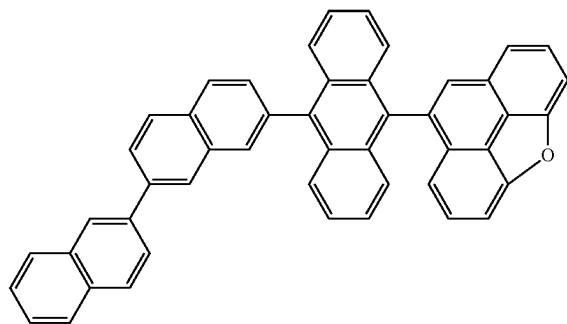


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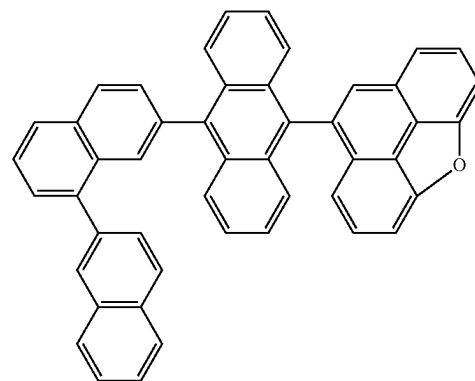
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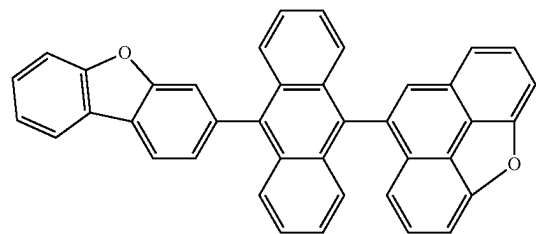
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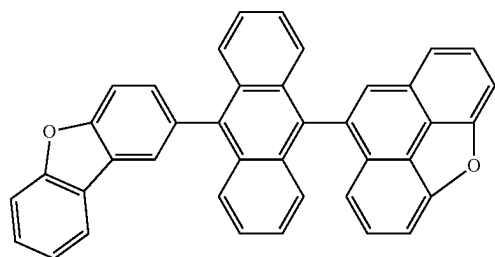
H2-107



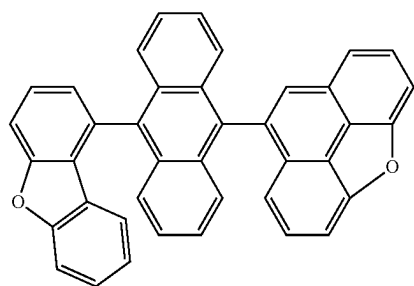
H2-108



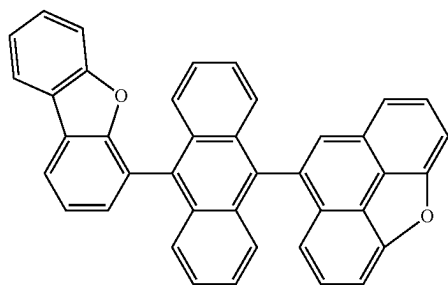
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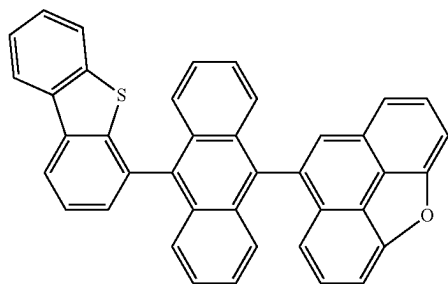
H2-109



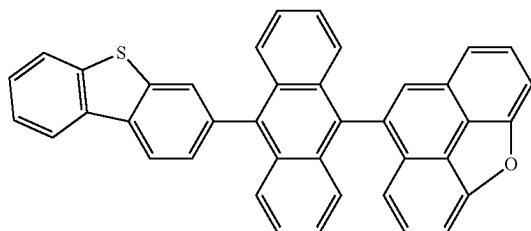
H2-110



H2-111

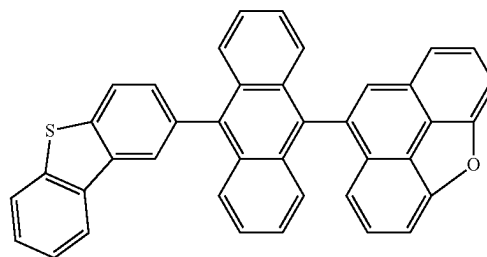


H2-112

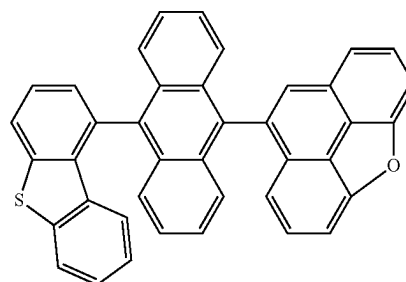


H2-113

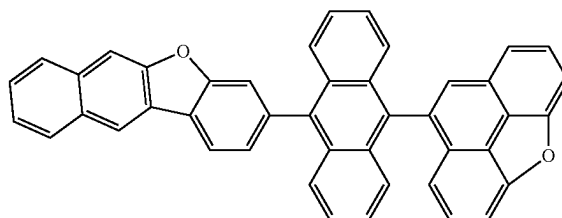
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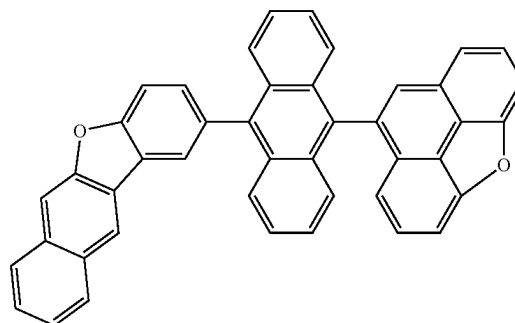
H2-114



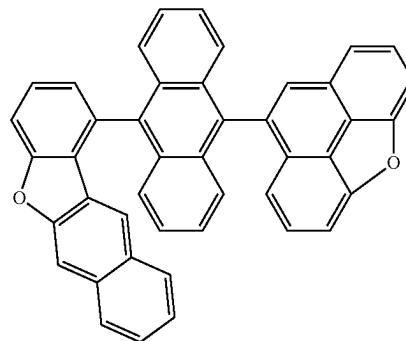
H2-115



H2-116



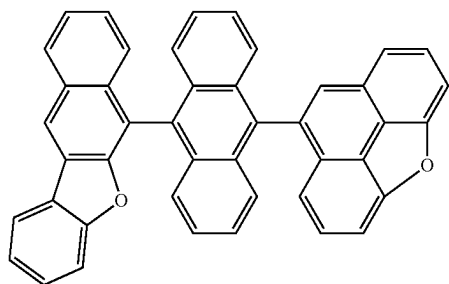
H2-117



H2-118

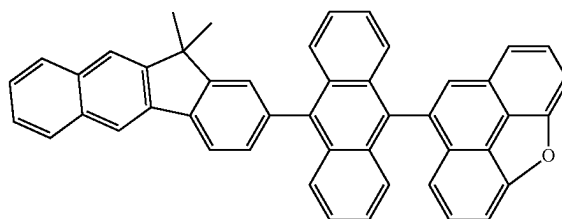
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H2-119



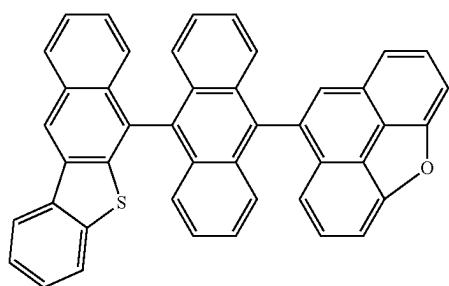
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H2-124

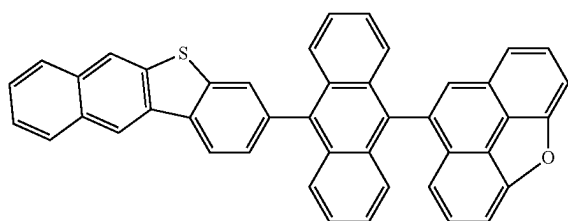


H2-125

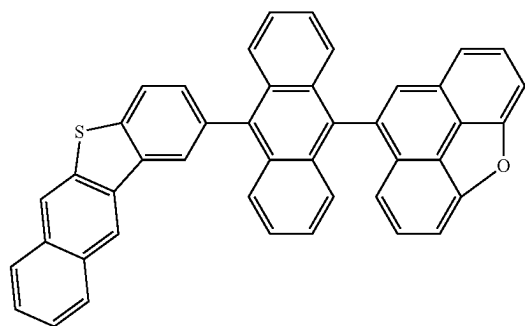
H2-120



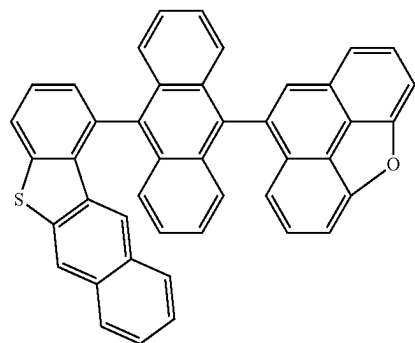
H2-121



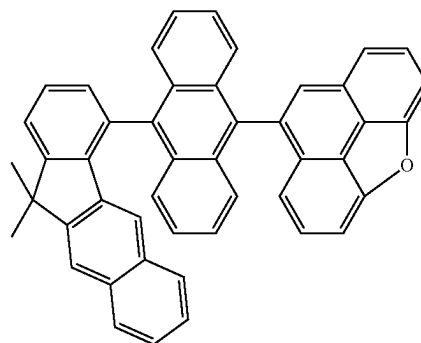
H2-122



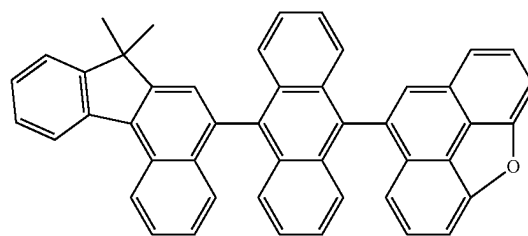
H2-123



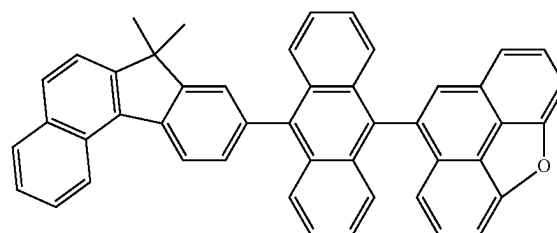
H2-126



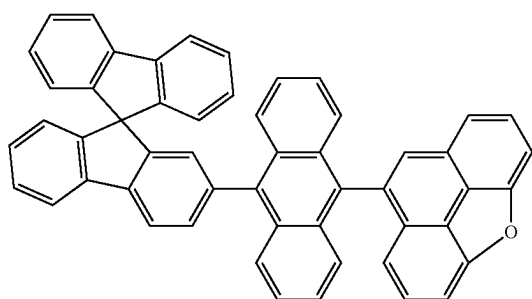
H2-127



H2-128

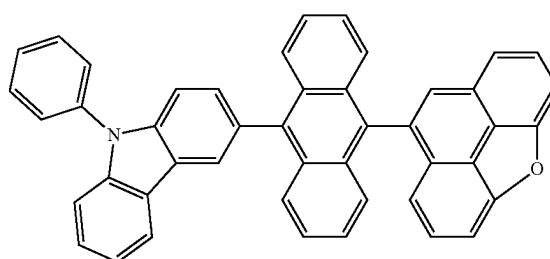


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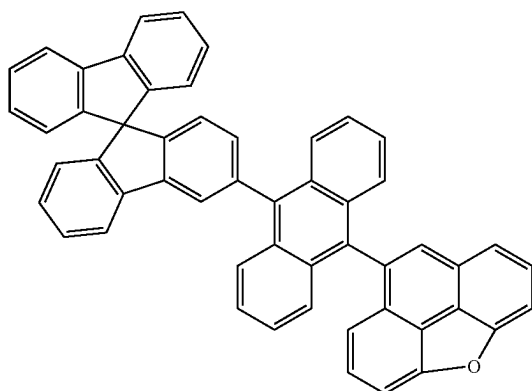


H2-129

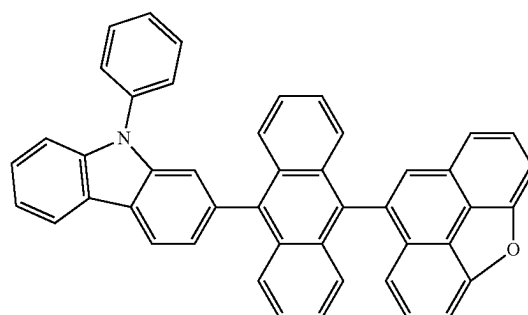
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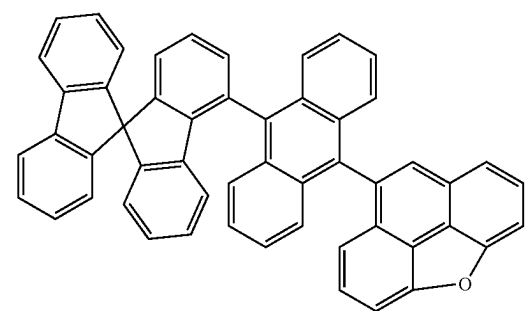
H2-133



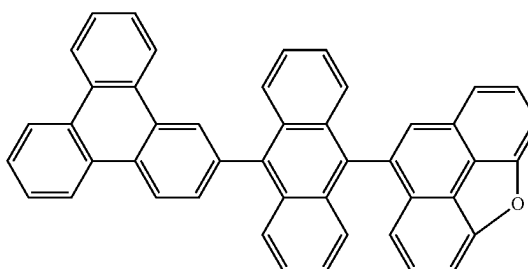
H2-130



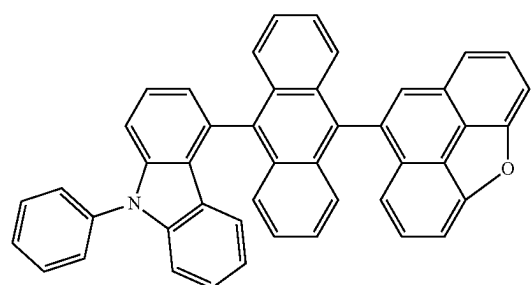
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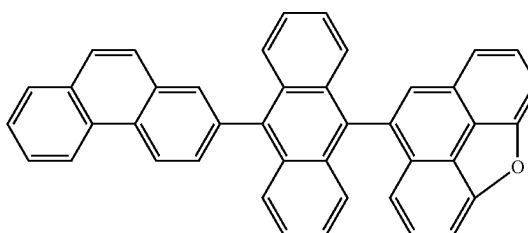
H2-131



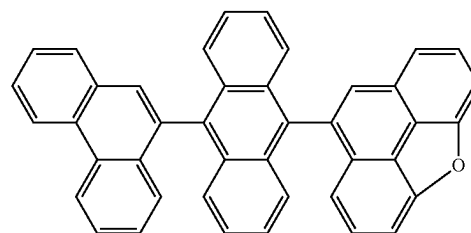
H2-135



H2-132

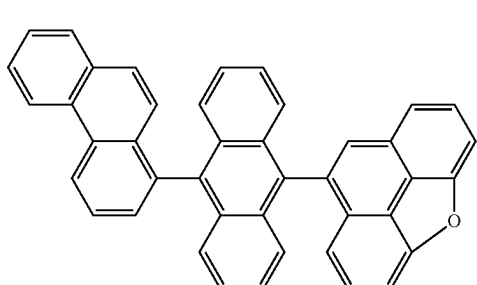


H2-136

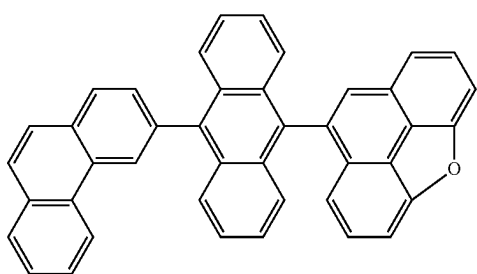


H2-137

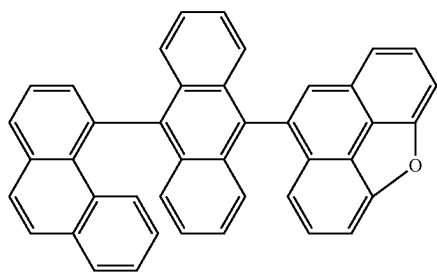
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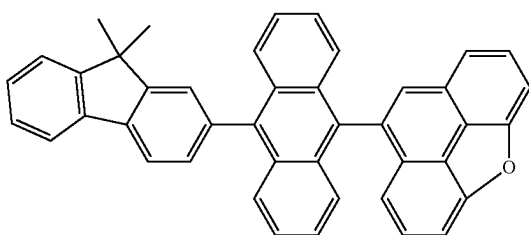
H2-138



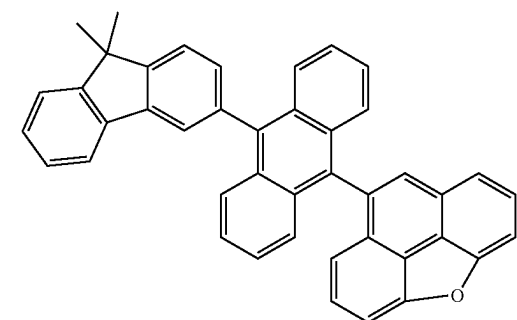
H2-139



H2-140

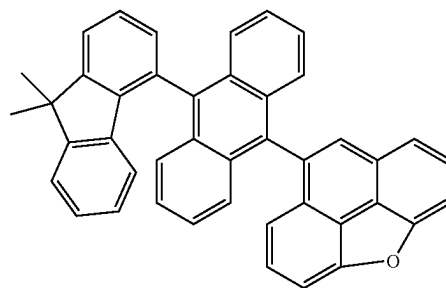


H2-141

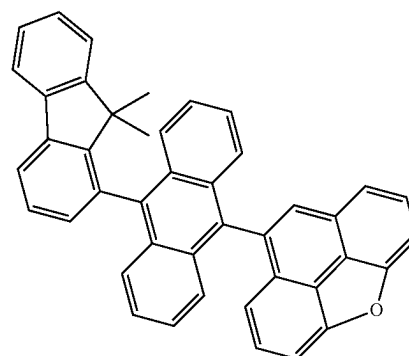


H2-142

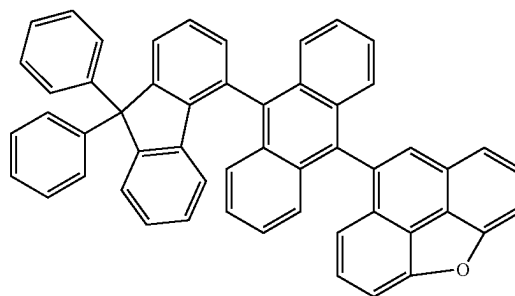
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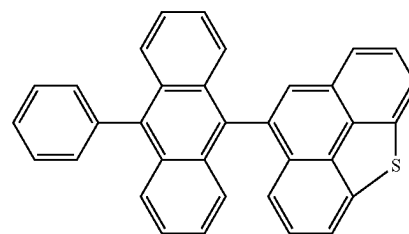
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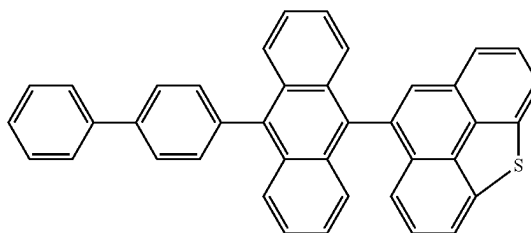
H2-144



H2-145

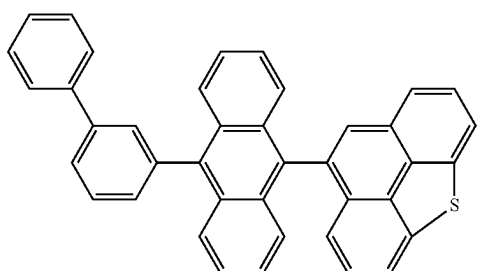


H2-146

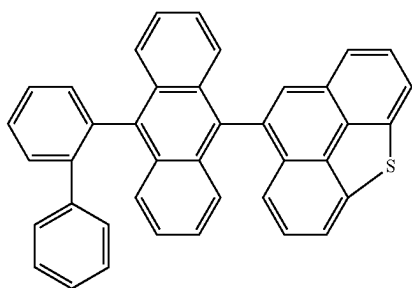


H2-147

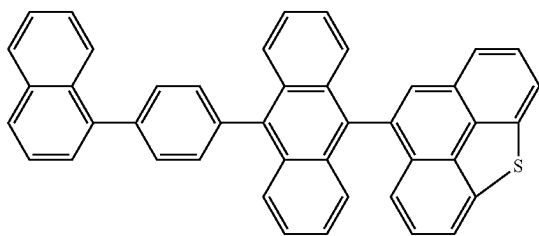
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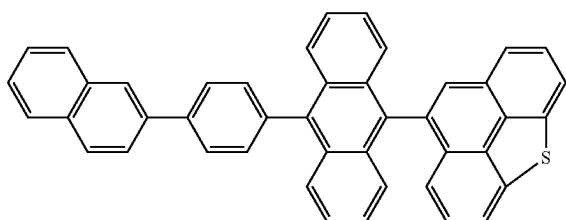
H2-148



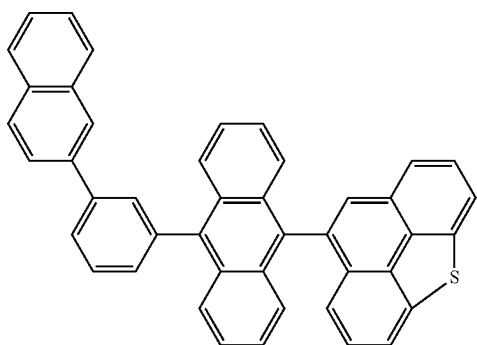
H2-149



H2-150

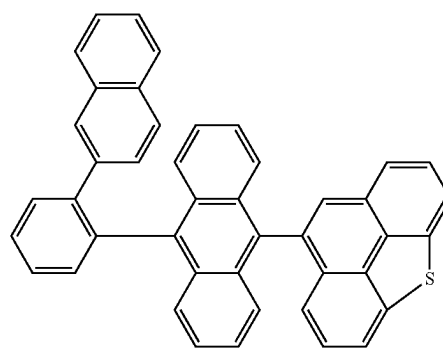


H2-151

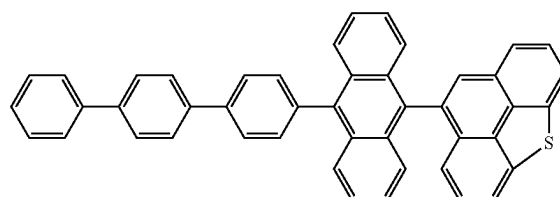


H2-152

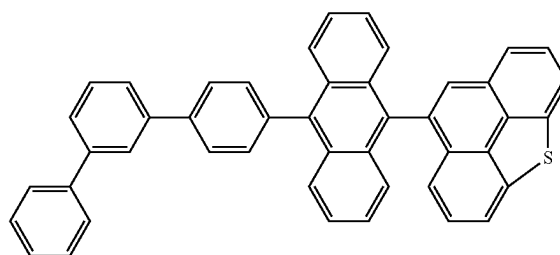
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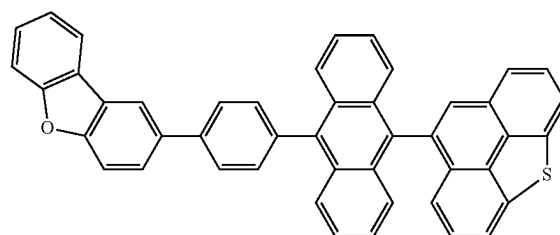
H2-153



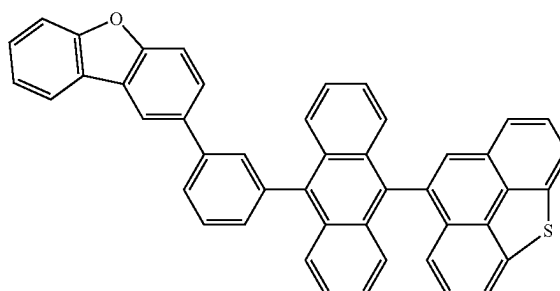
H2-154



H2-155



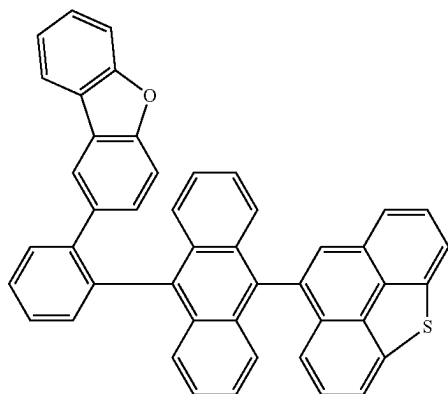
H2-156



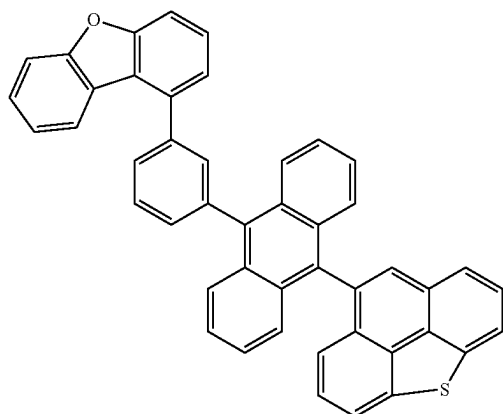
H2-157

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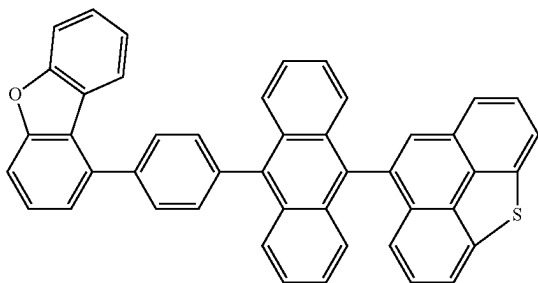
H2-158



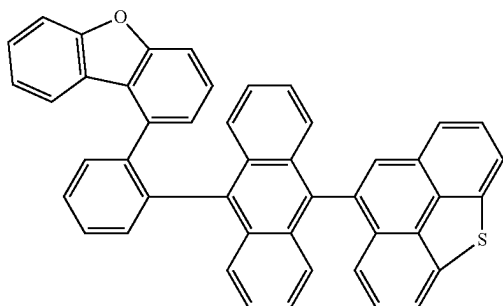
H2-159



H2-160

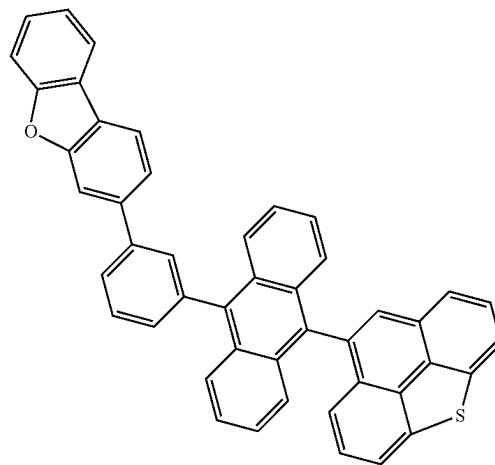


H2-161

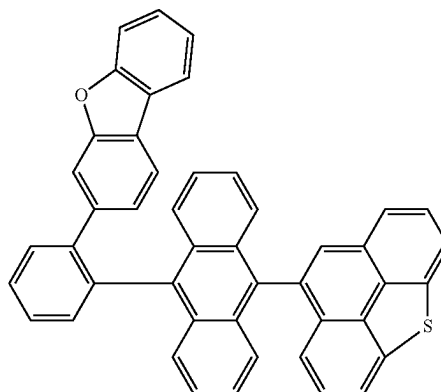


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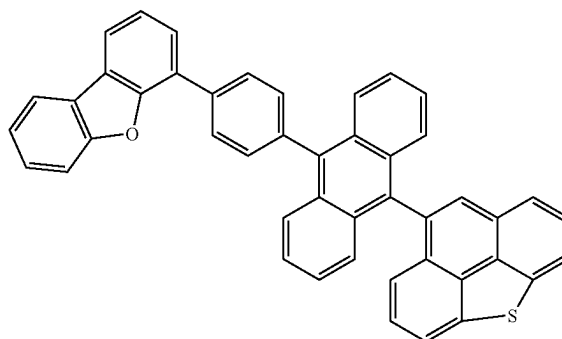
H2-162



H2-163

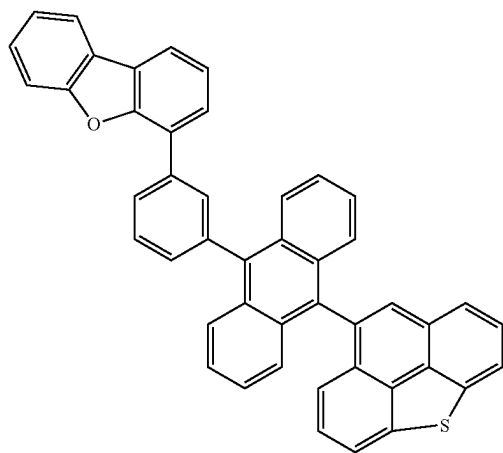


H2-164



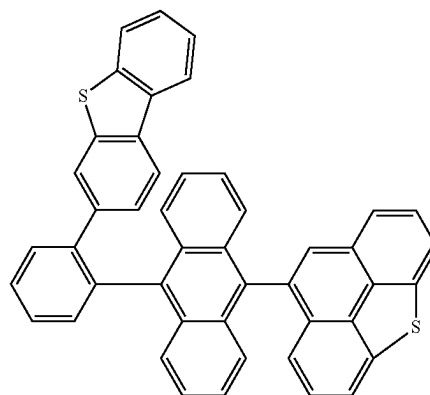
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H2-165

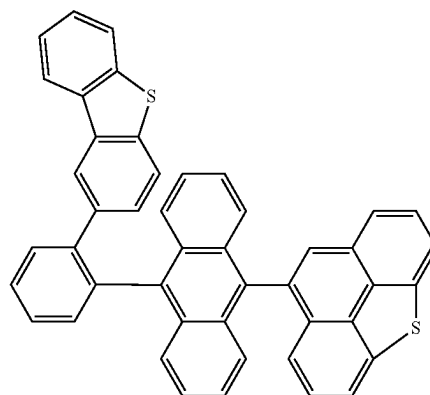


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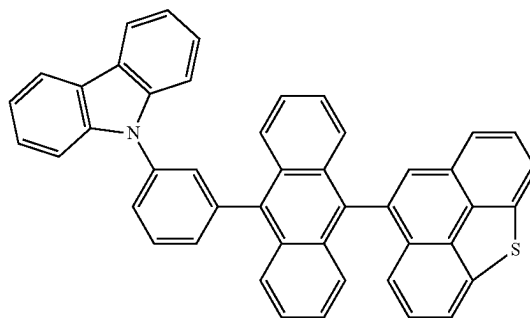
H2-168



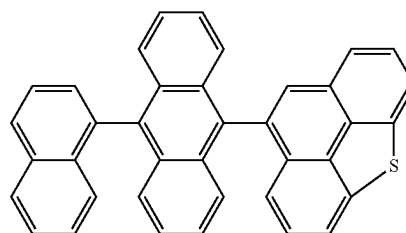
H2-169



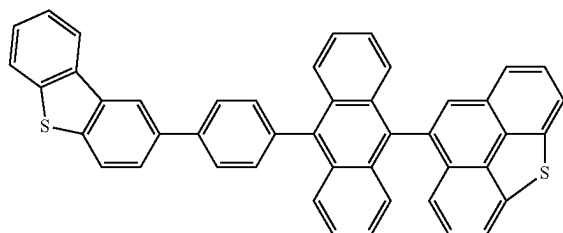
H2-170



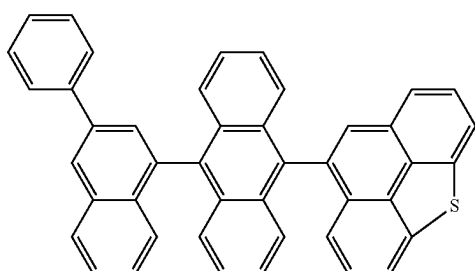
H2-171



H2-167

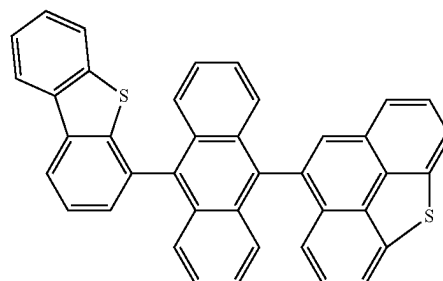


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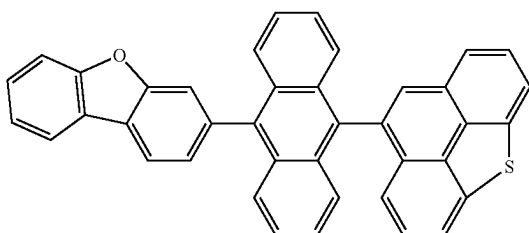


H2-172

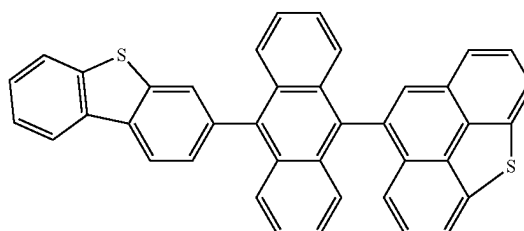
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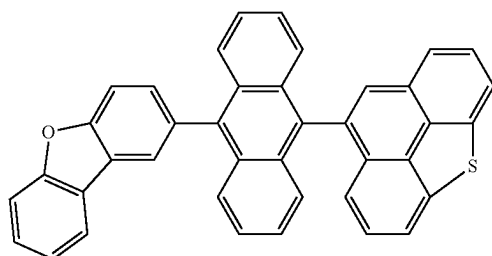
H2-177



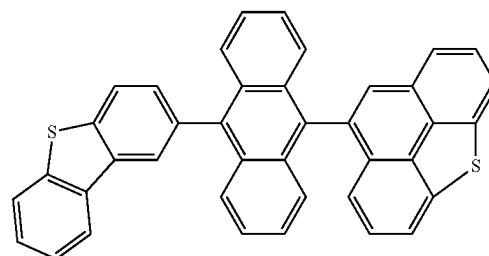
H2-173



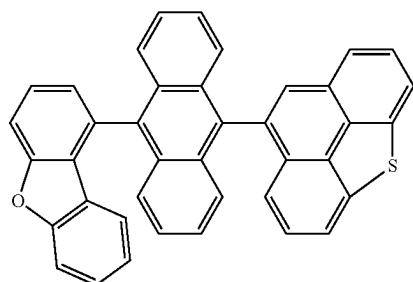
H2-178



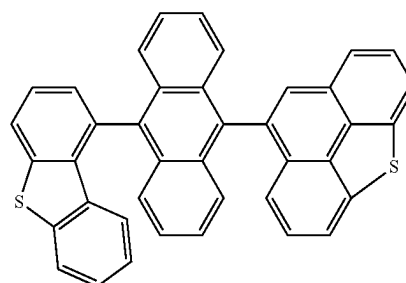
H2-174



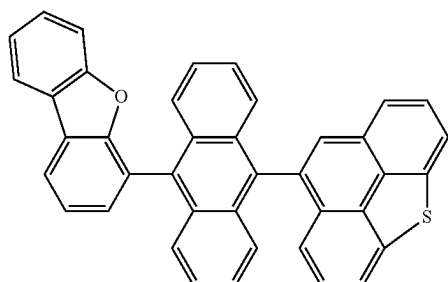
H2-179



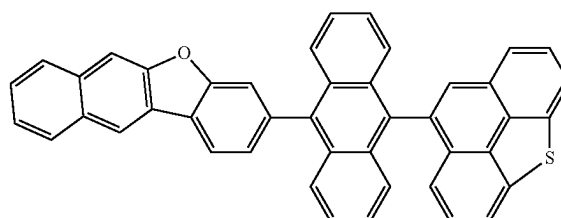
H2-175



H2-180

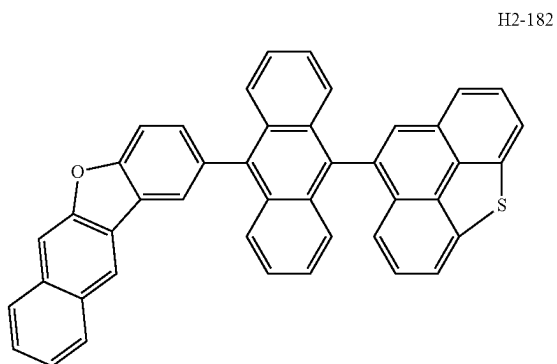


H2-176

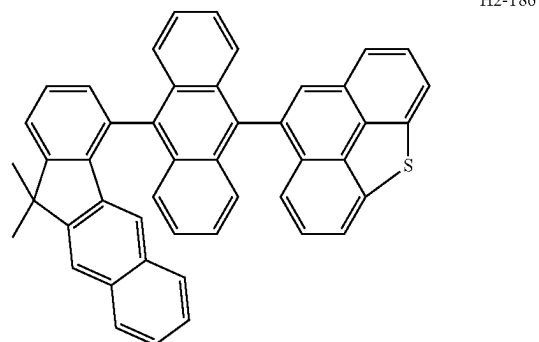


H2-181

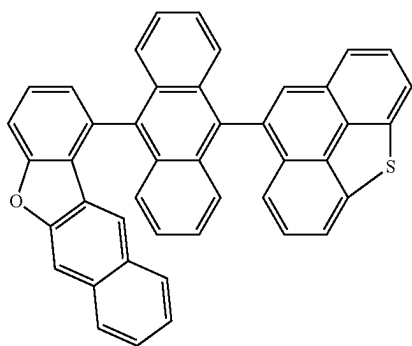
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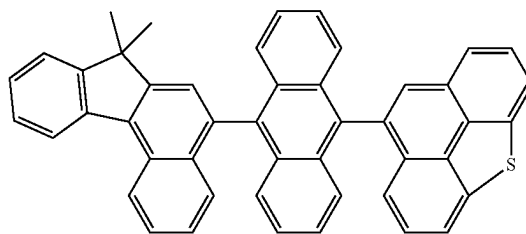
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H2-183

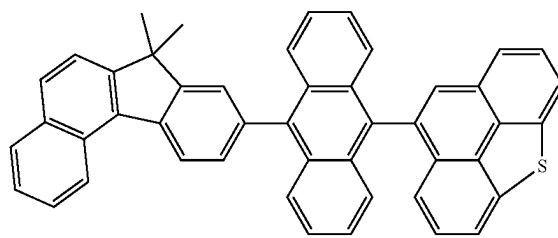
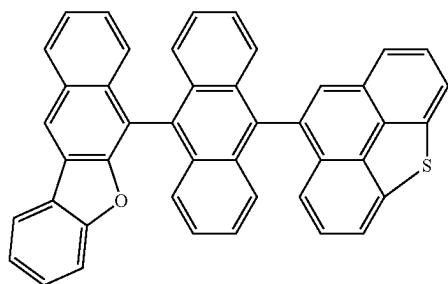


H2-187



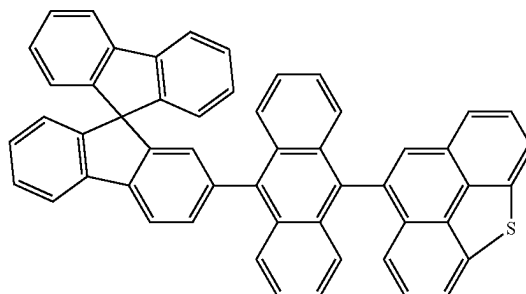
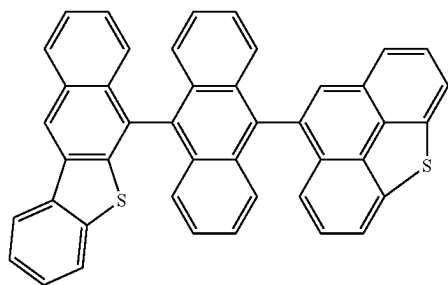
H2-188

H2-184



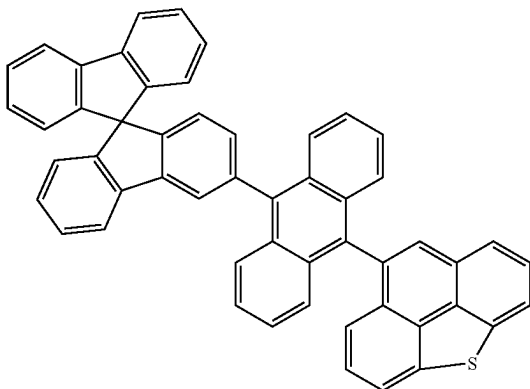
H2-189

H2-185

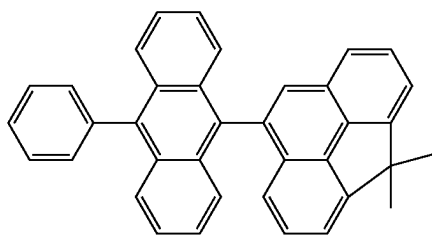


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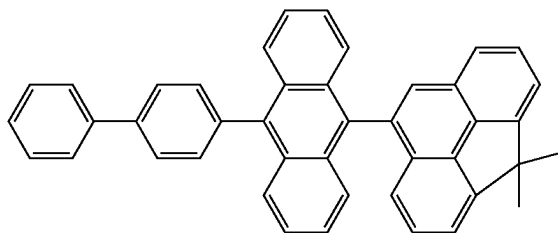
H2-190



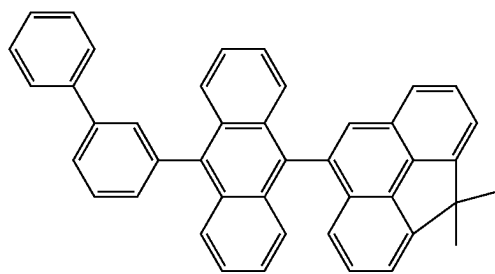
H2-191



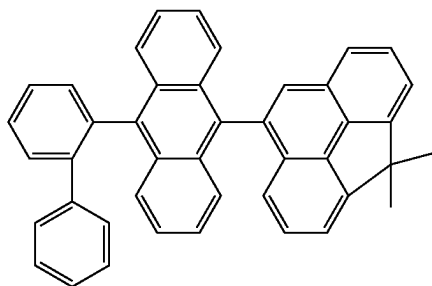
H2-192



H2-193

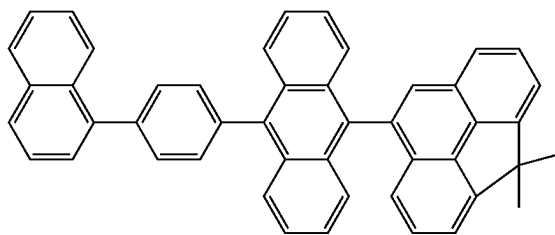


H2-194

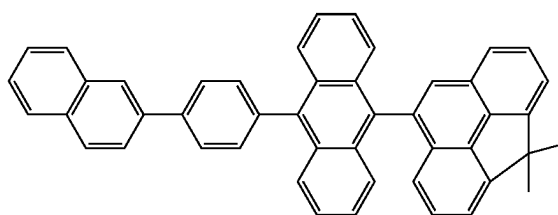


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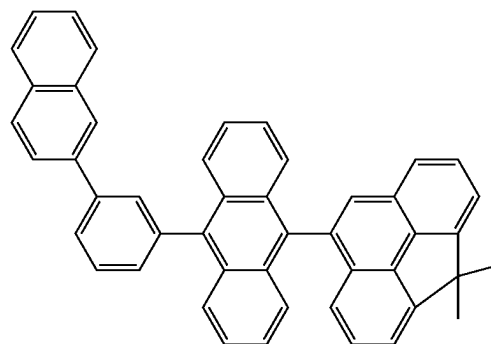
H2-195



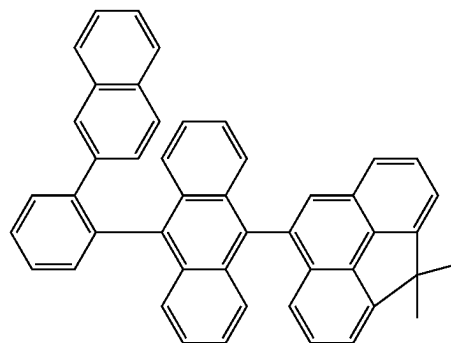
H2-196



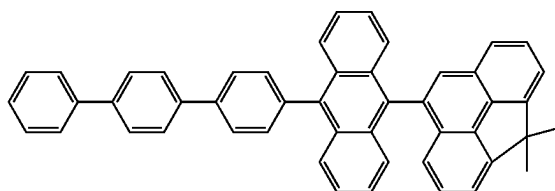
H2-197



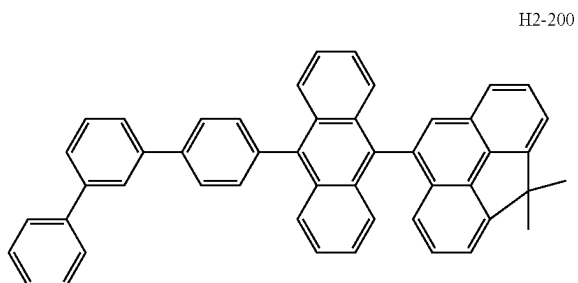
H2-198



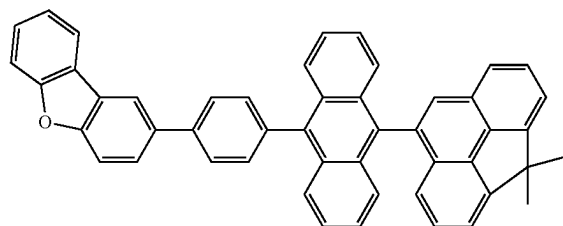
H2-199



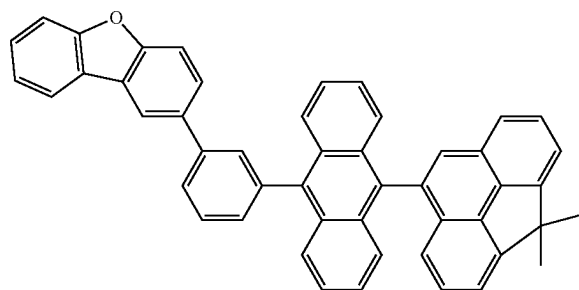
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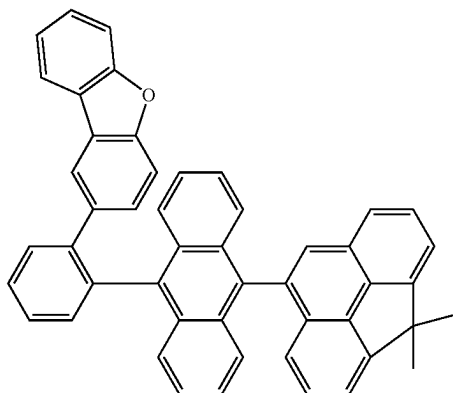
H2-201



H2-202

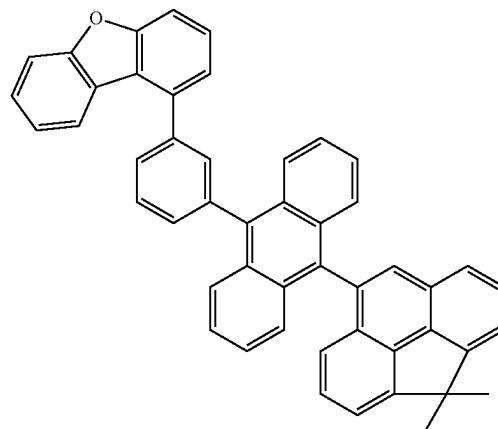


H2-203

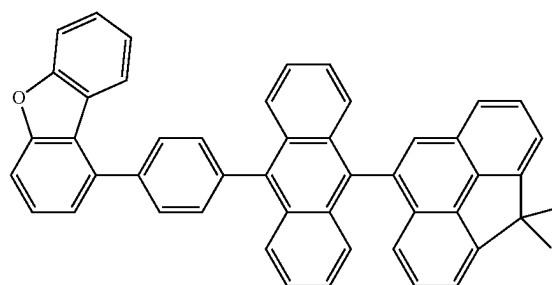


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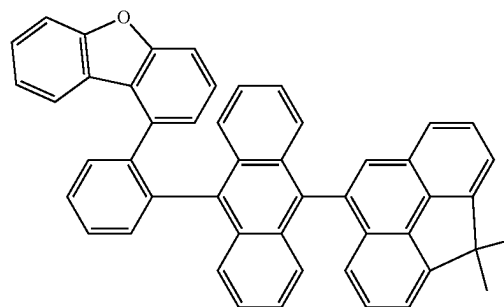
H2-204



H2-205

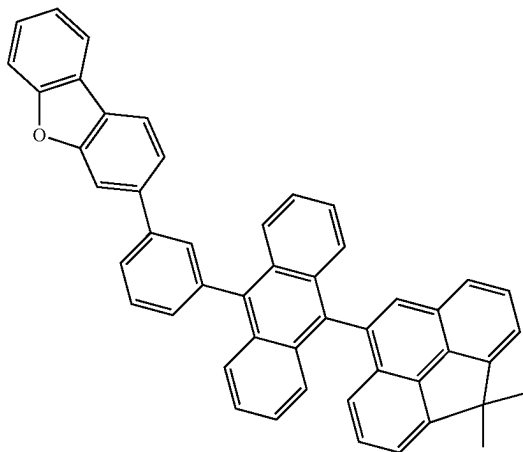


H2-206



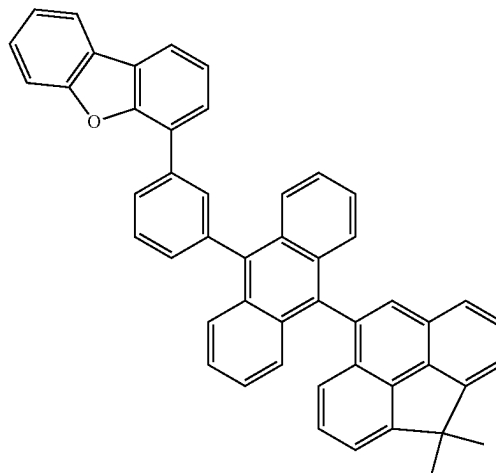
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H2-207

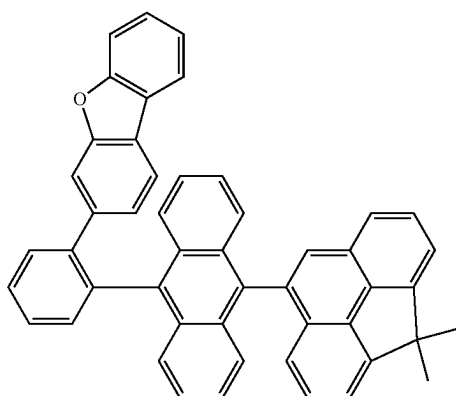


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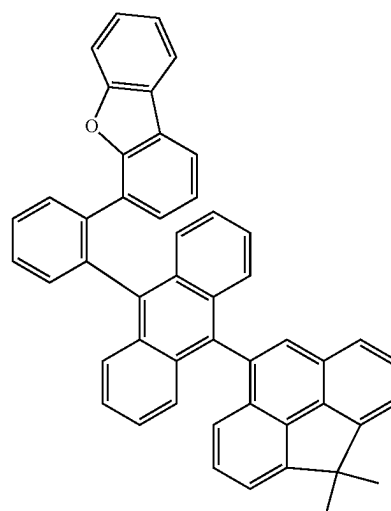
H2-210



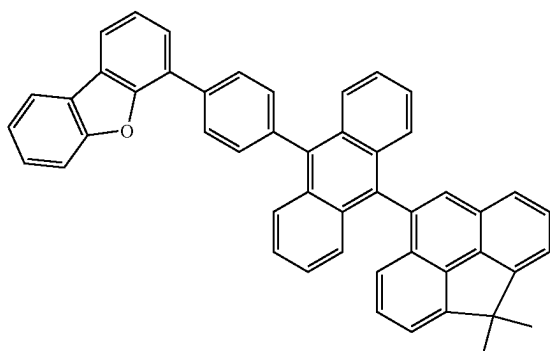
H2-208



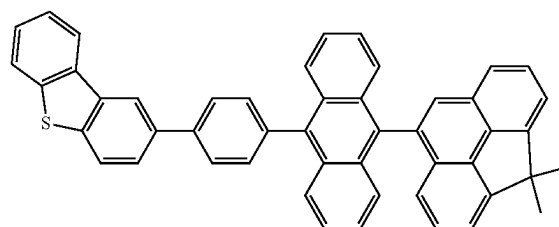
H2-211



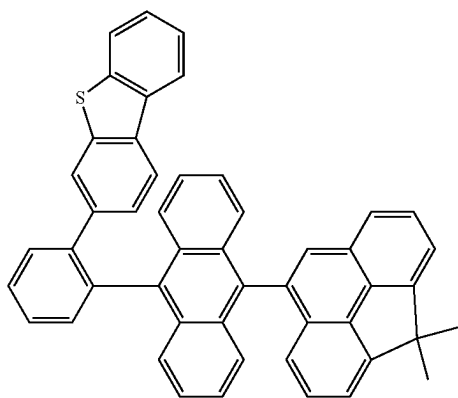
H2-209



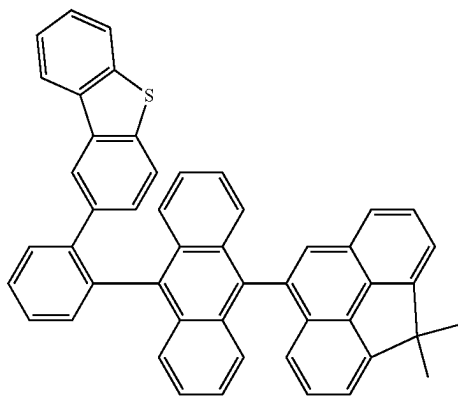
H2-212



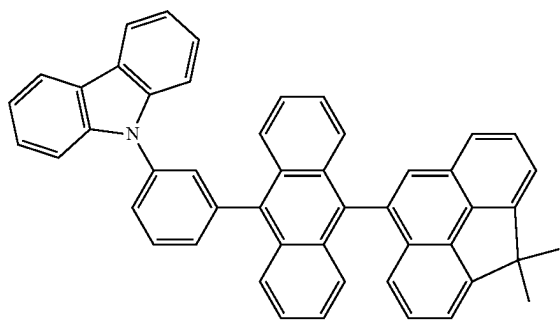
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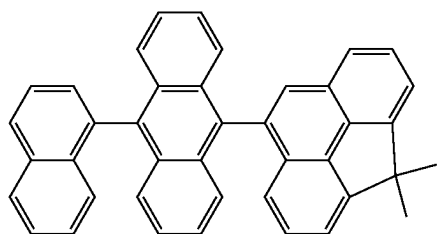
H2-213



H2-214

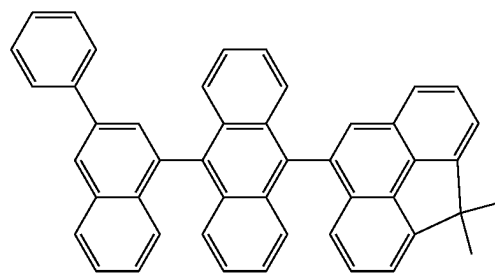


H2-215

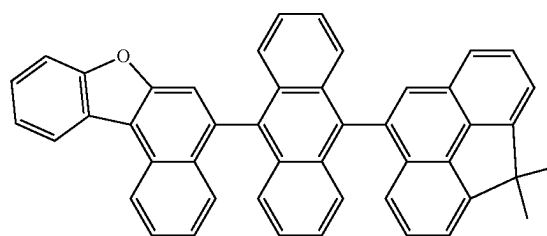


H2-216

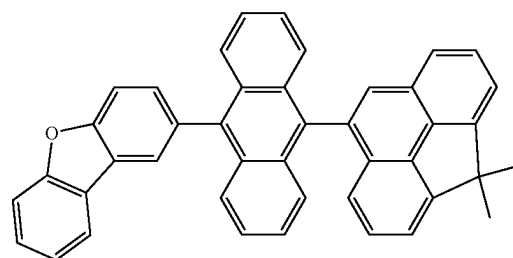
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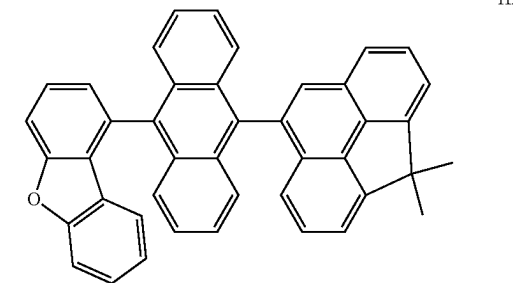
H2-217



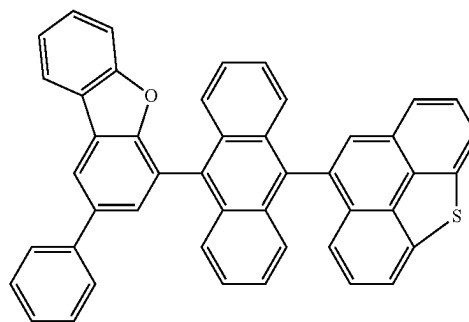
H2-218



H2-219



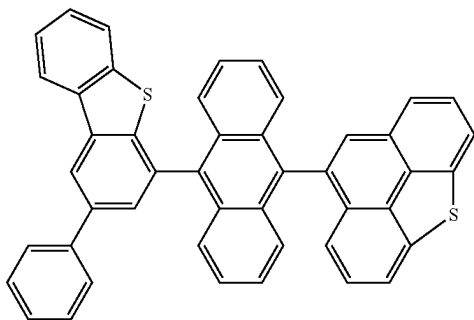
H2-220



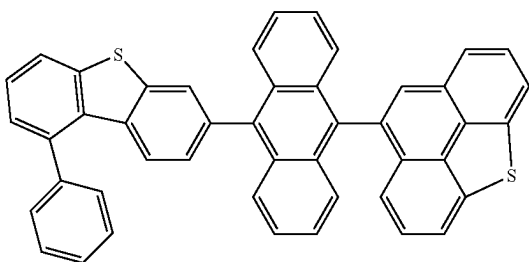
H2-221

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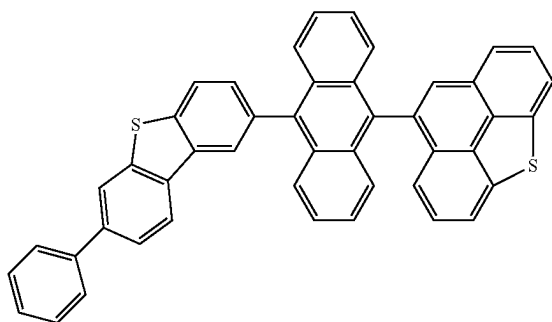
H2-222



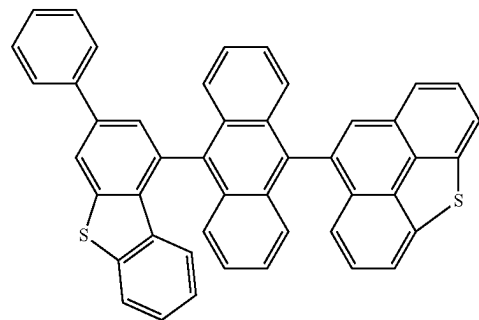
H2-223



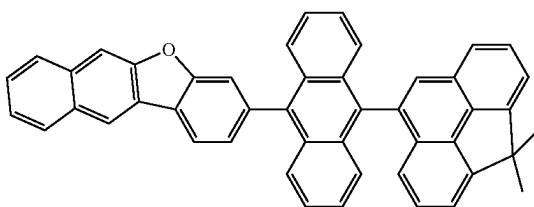
H2-224



H2-226

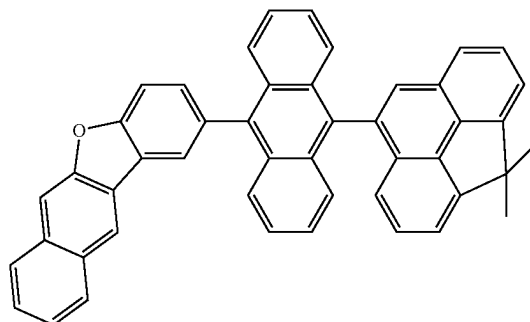


H2-227

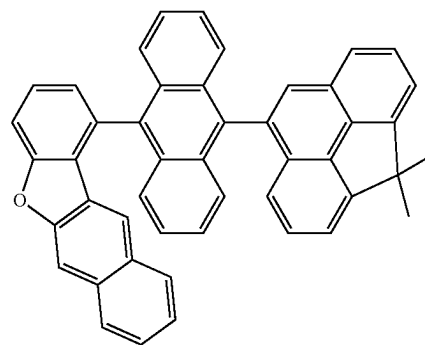


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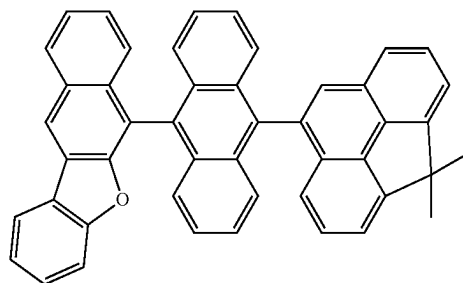
H2-227



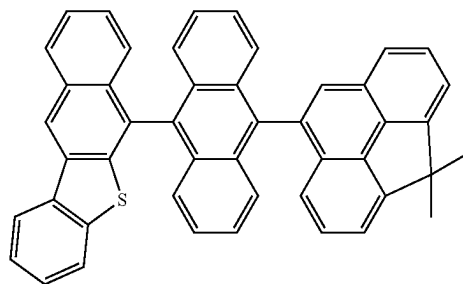
H2-228



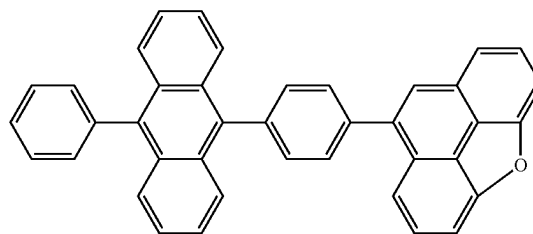
H2-229



H2-230

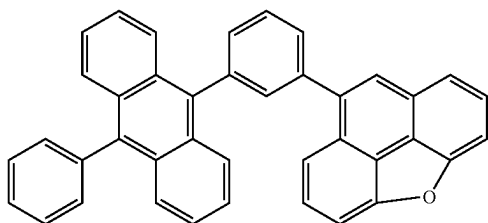


H2-231

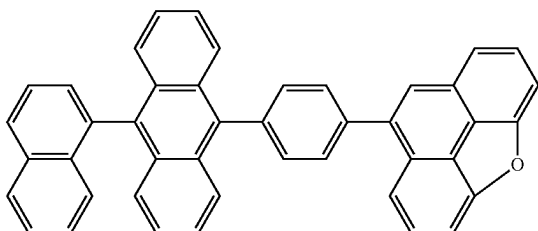


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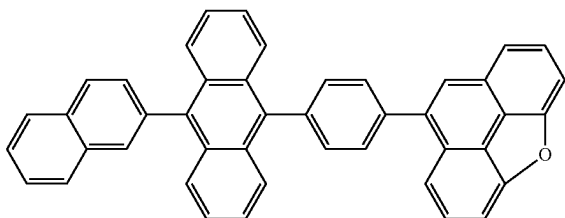
H2-232



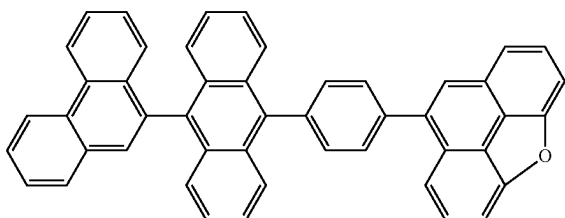
H2-233



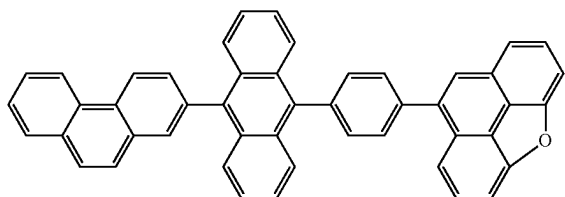
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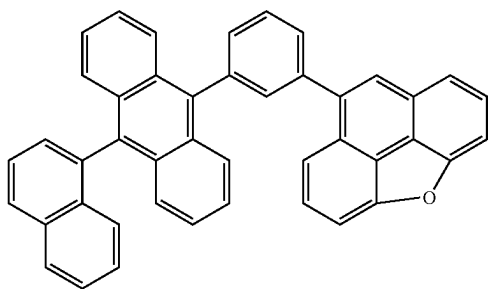
H2-235



H2-236

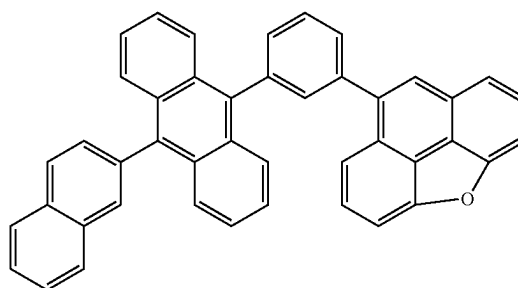


H2-237

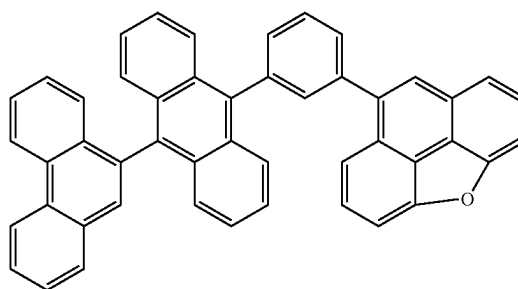


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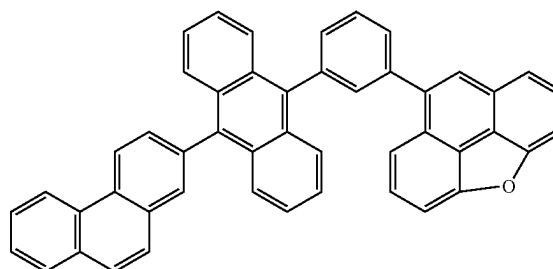
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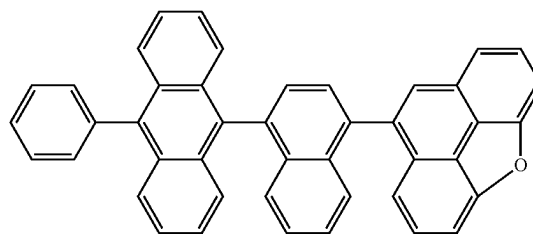
H2-239



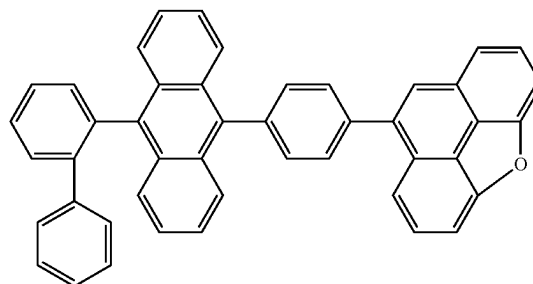
H2-240



H2-241

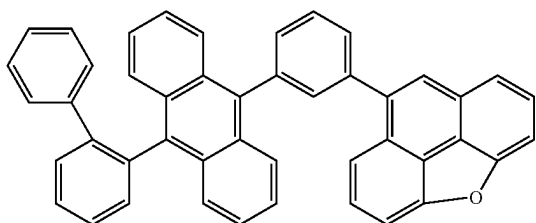


H2-242

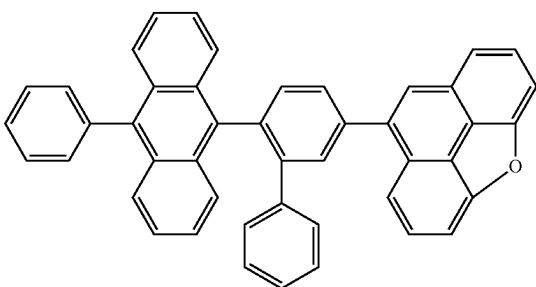


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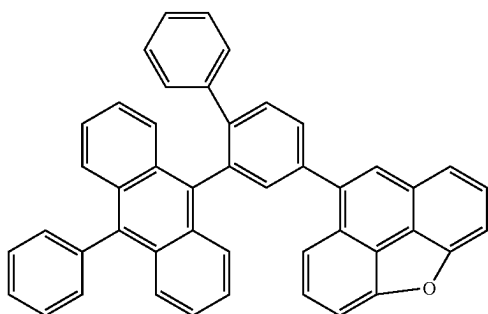
H2-243



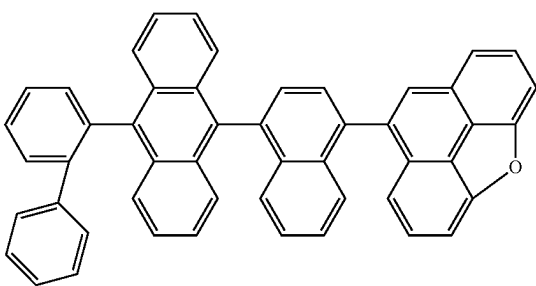
H2-244



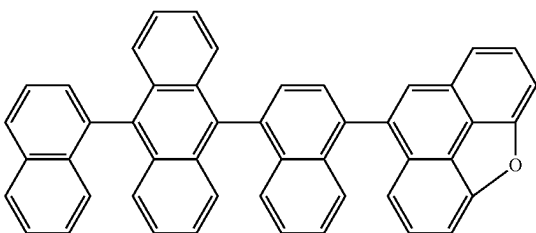
H2-245



H2-246

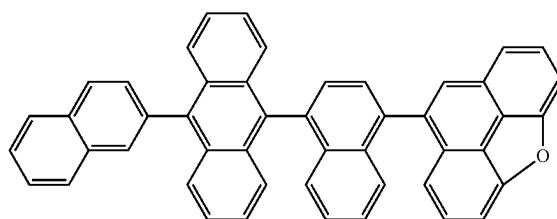


H2-247

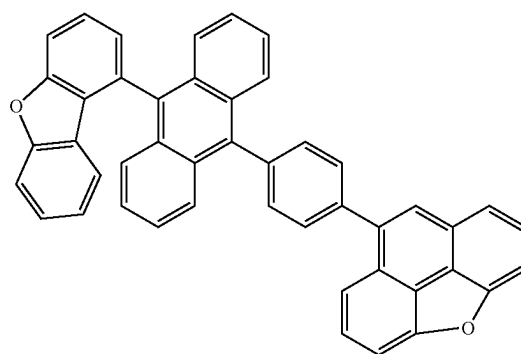


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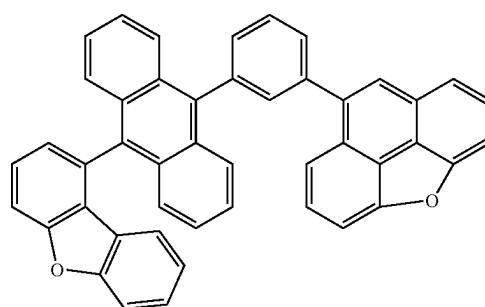
H2-248



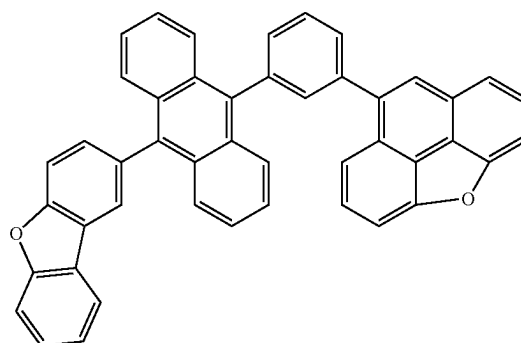
H2-249



H2-250

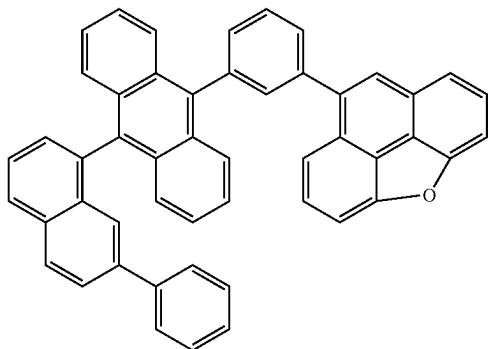


H2-251

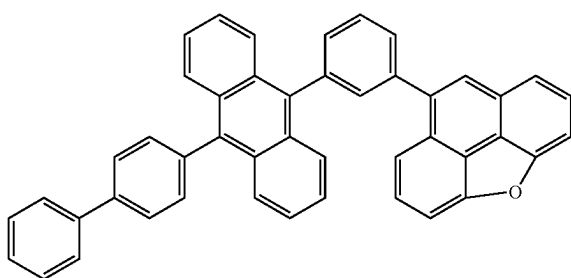


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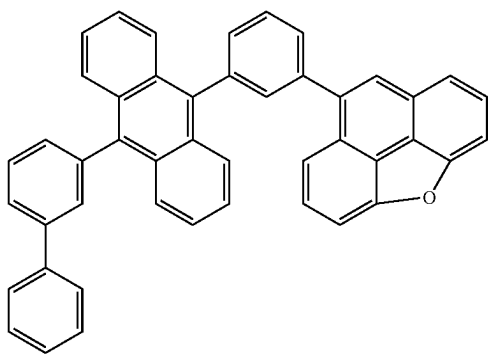
H2-252



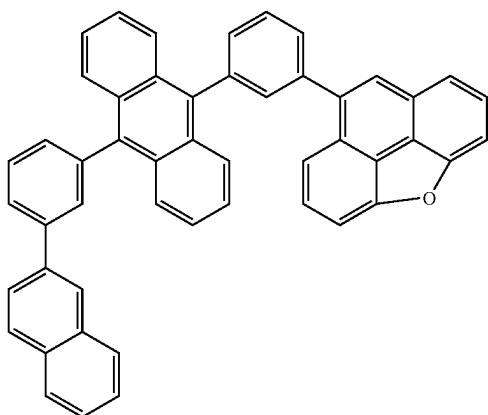
H2-253



H2-254

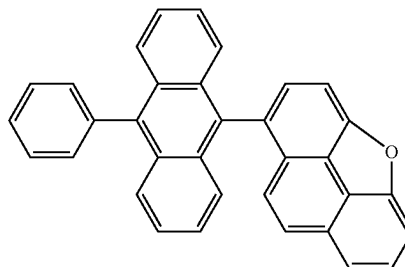


H2-255

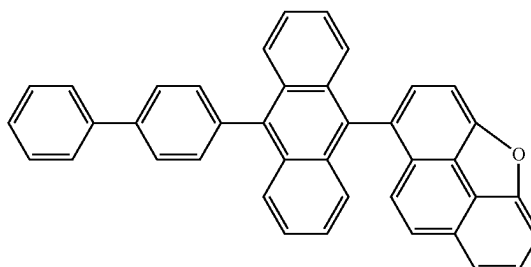


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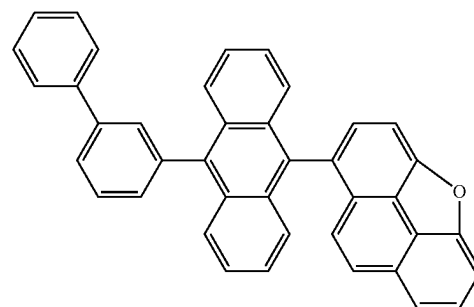
H2-256



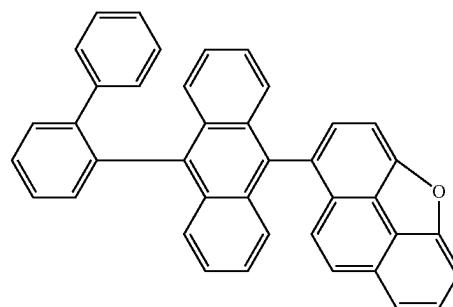
H2-257



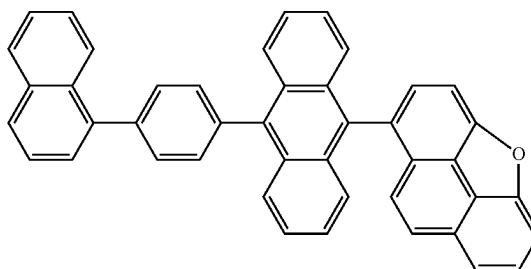
H2-258



H2-259

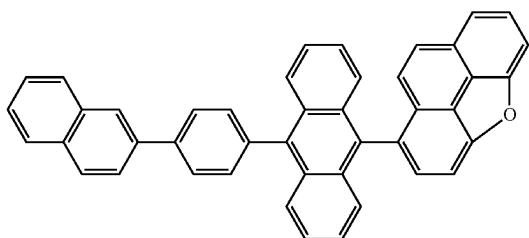


H2-260

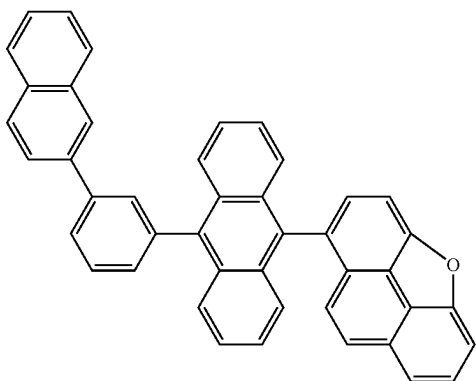


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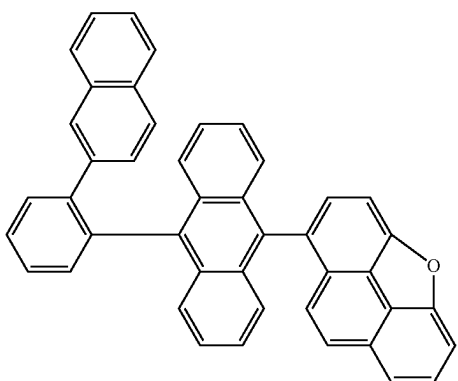
H2-261



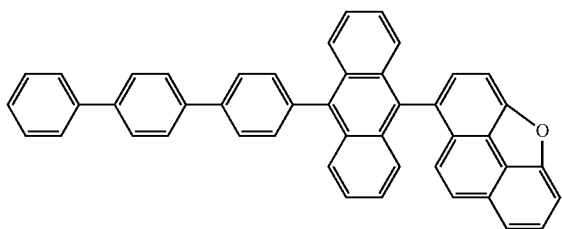
H2-262



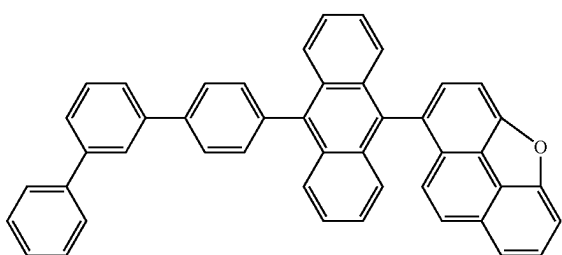
H2-263



H2-264

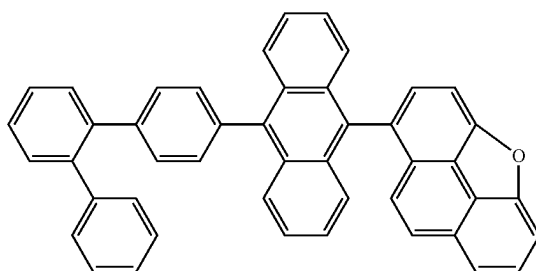


H2-265

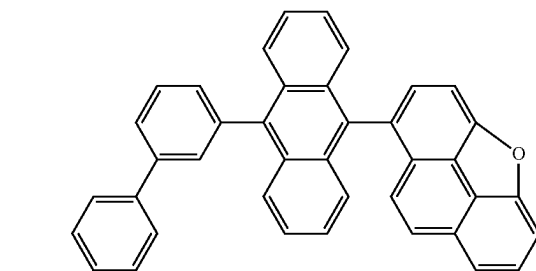


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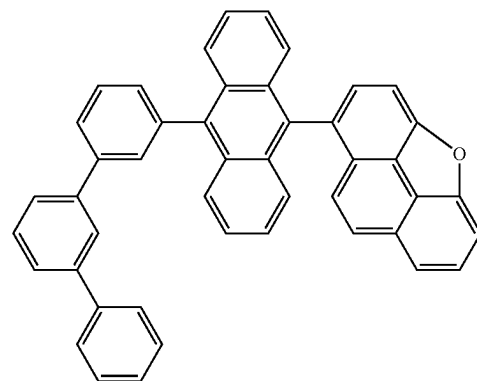
H2-266



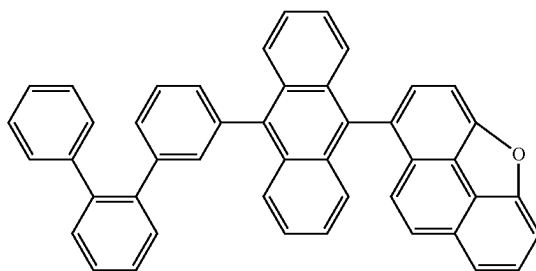
H2-267



H2-268

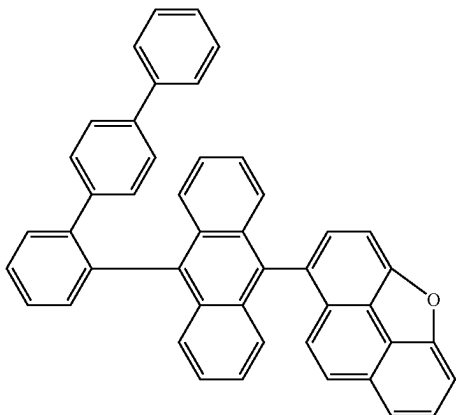


H2-269



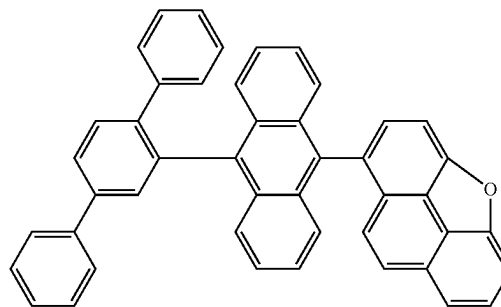
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H2-270

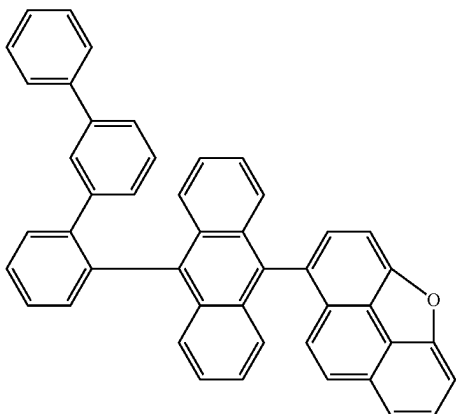


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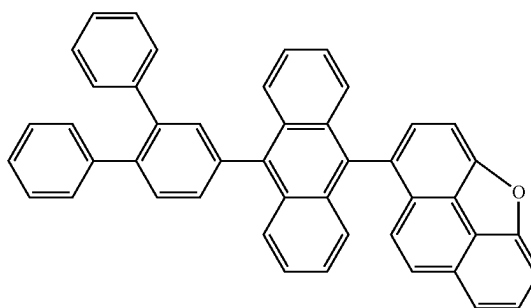
H2-274



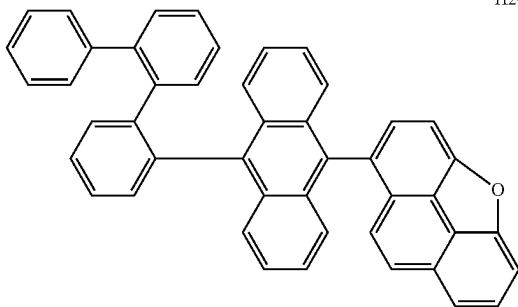
H2-271



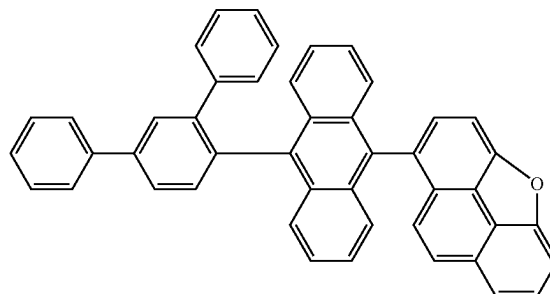
H2-275



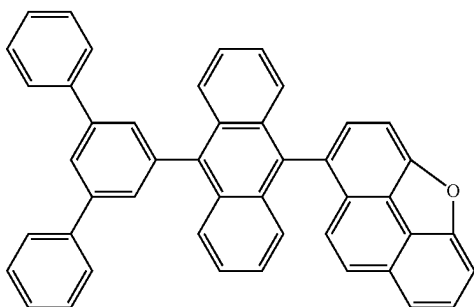
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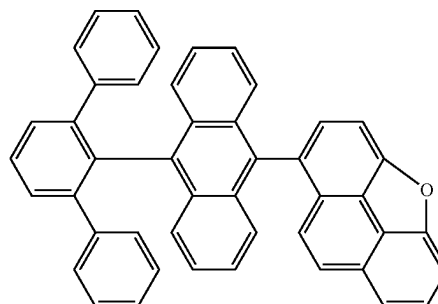
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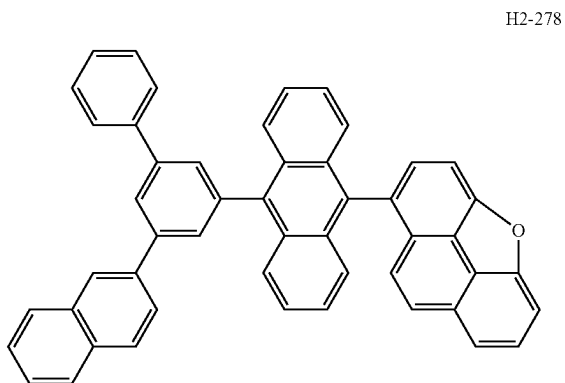
H2-273



H2-277

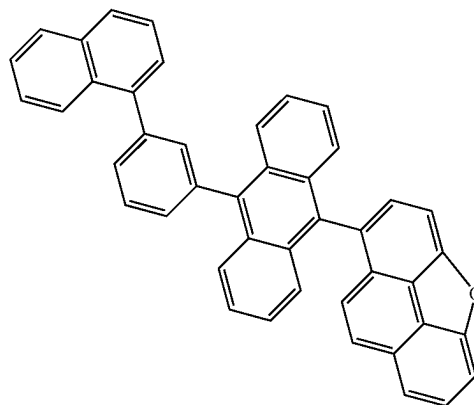


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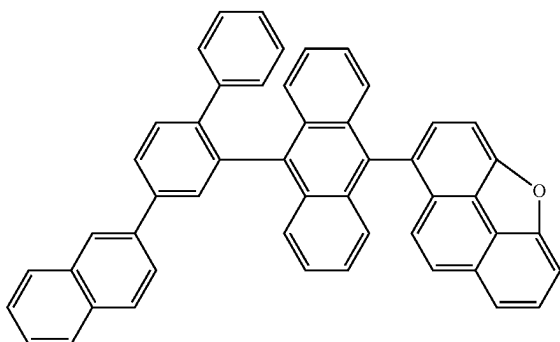


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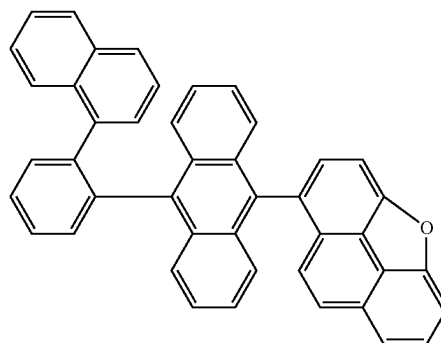
H2-282



H2-279

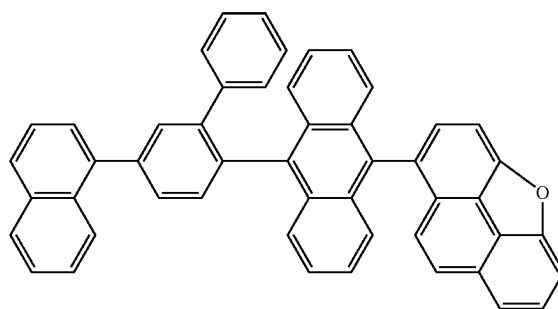
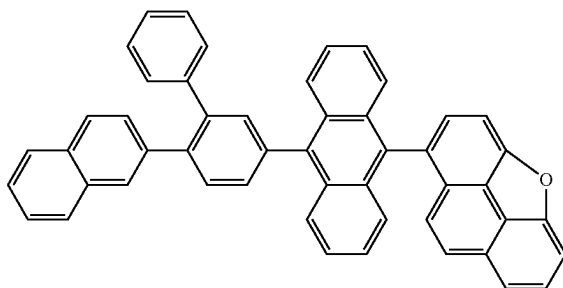


H2-283



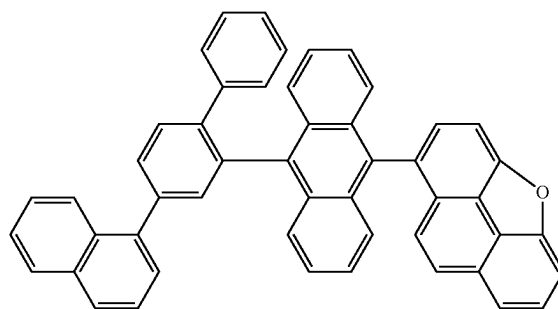
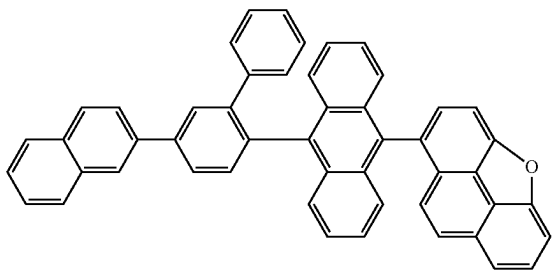
H2-284

H2-280



H2-285

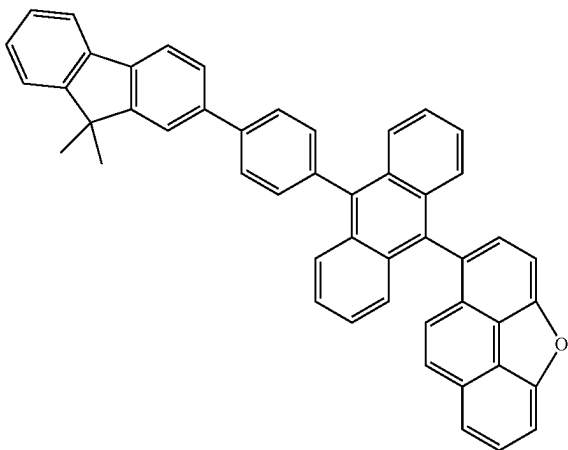
H2-281



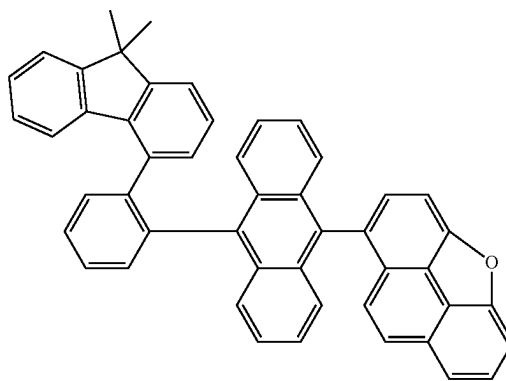
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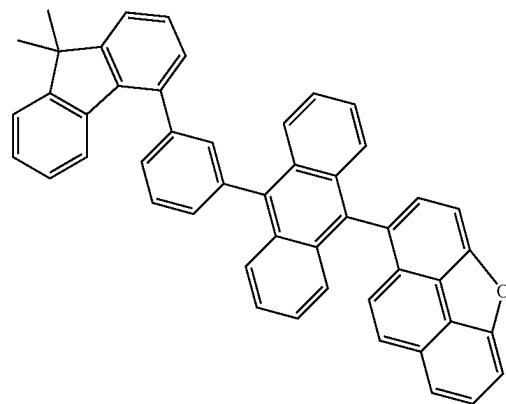
H2-286



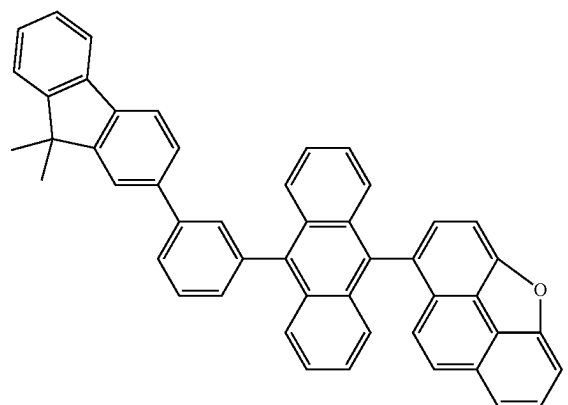
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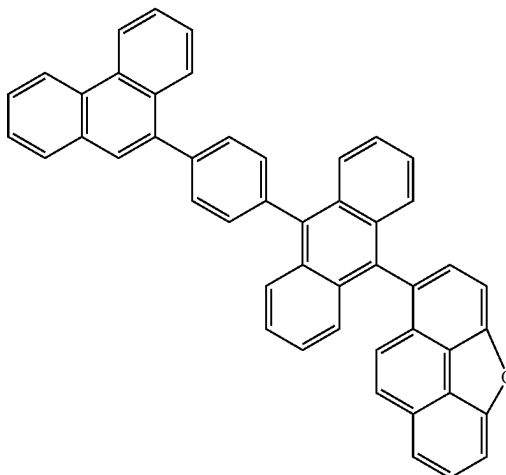
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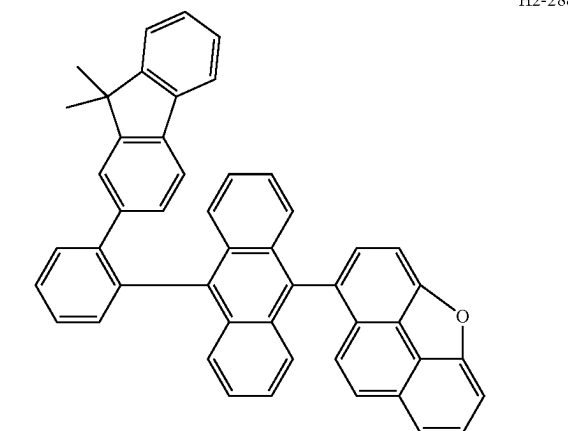
H2-287



H2-291

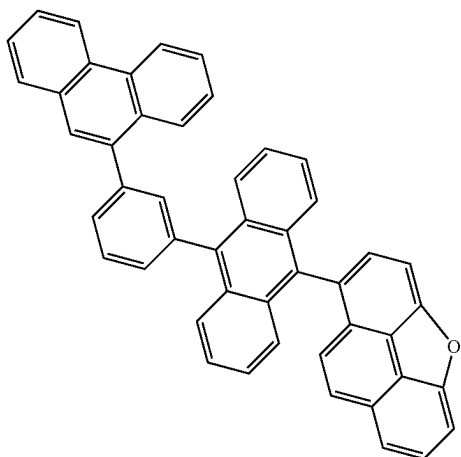


H2-288



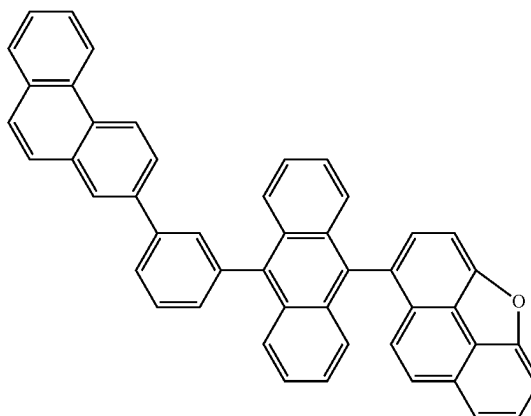
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H2-292

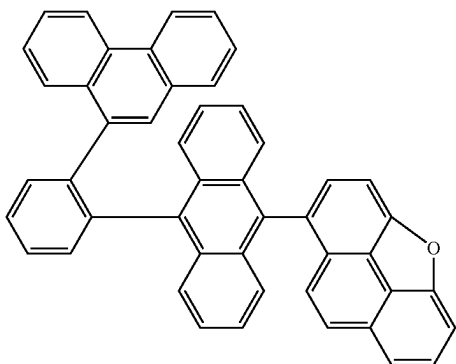


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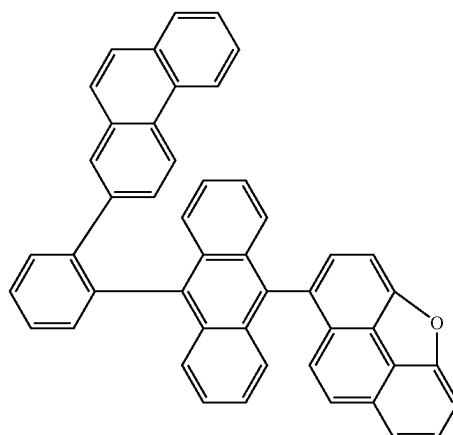
H2-295



H2-293

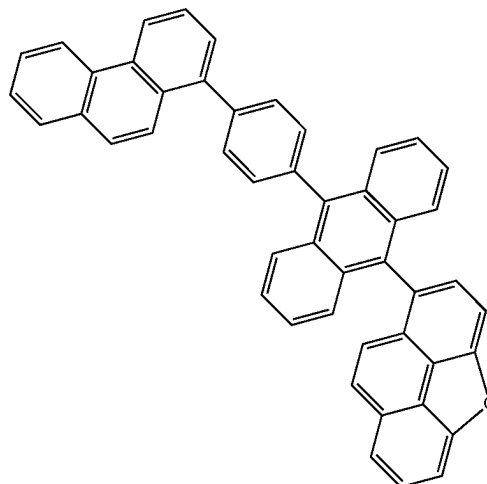
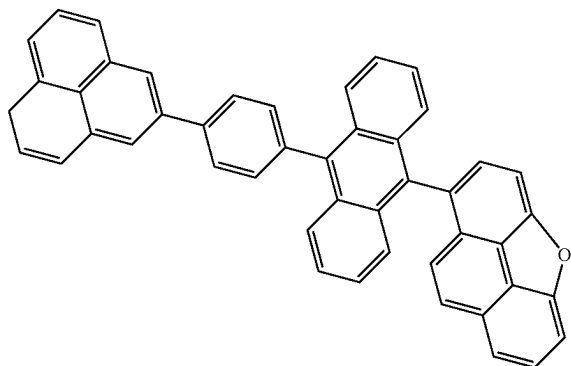


H2-296



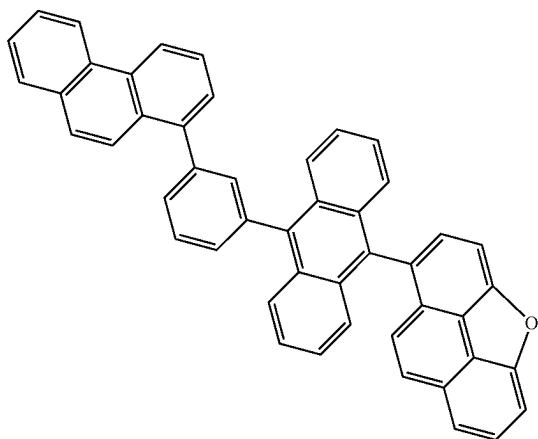
H2-297

H2-294



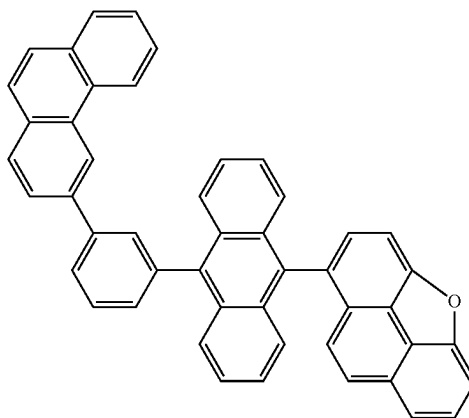
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H2-298

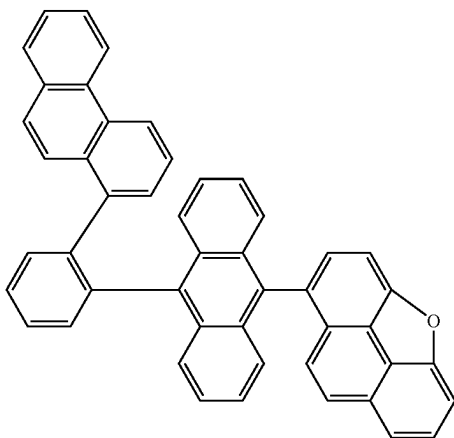


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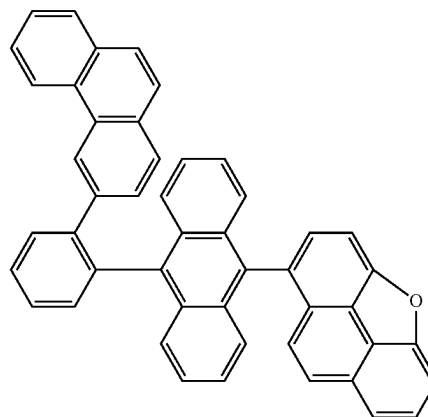
H2-301



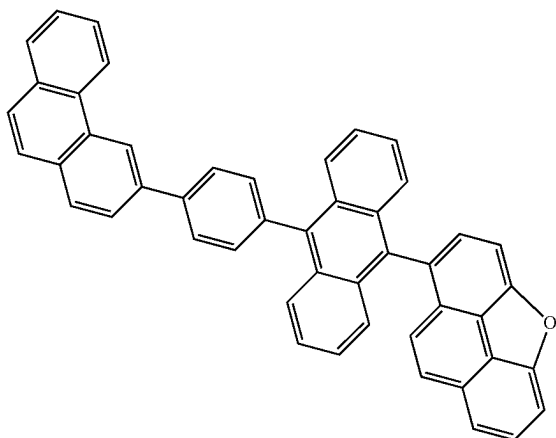
H2-299



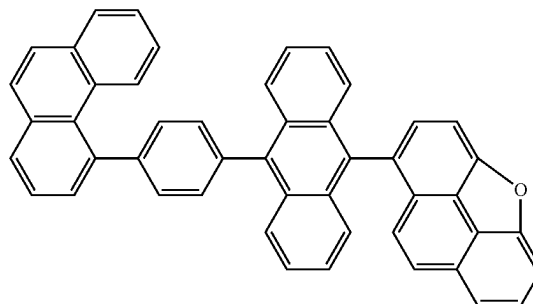
H2-302



H2-300

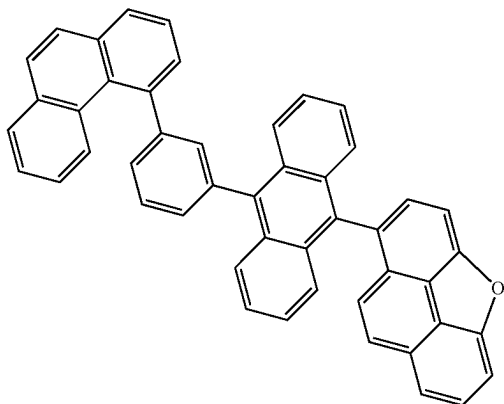


H2-303

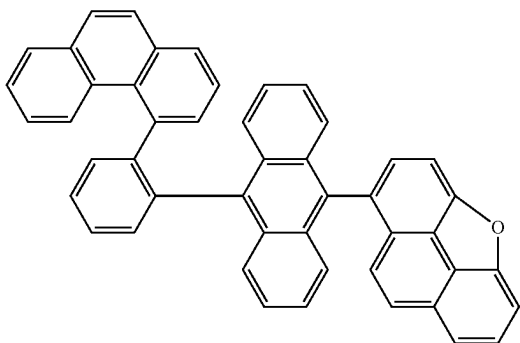


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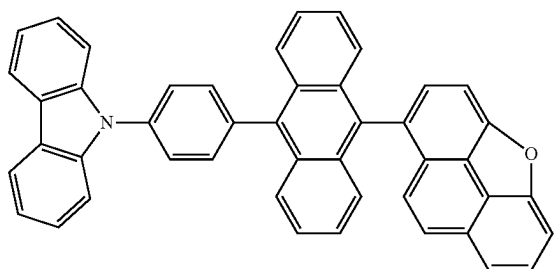
H2-304



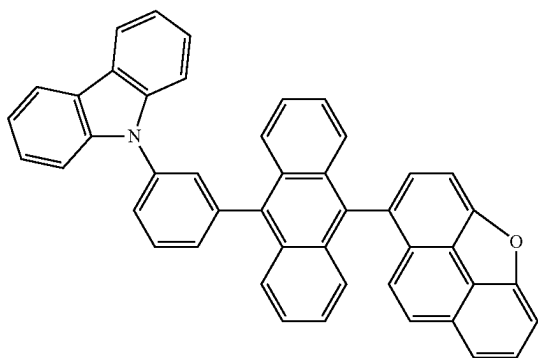
H2-305



H2-306

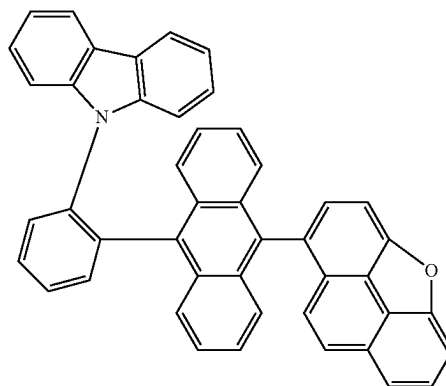


H2-307

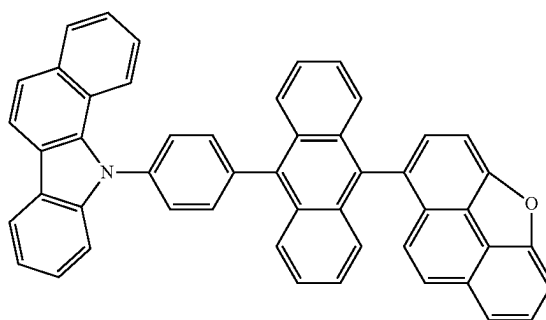


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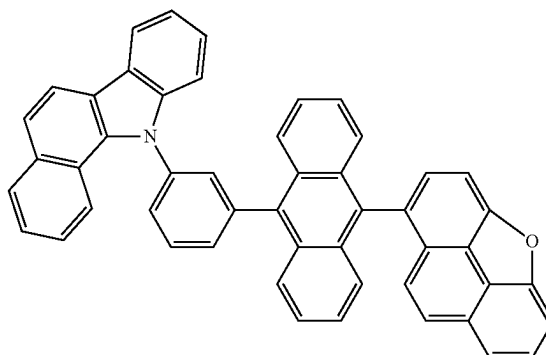
H2-308



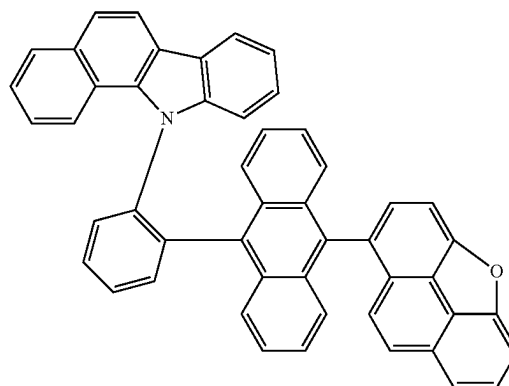
H2-309



H2-310

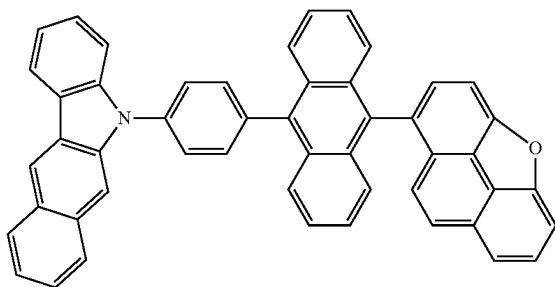


H2-311



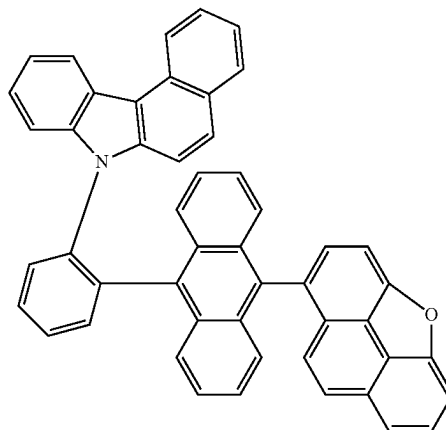
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H2-312



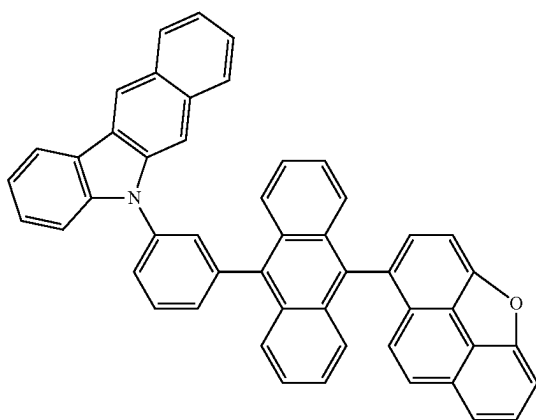
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H2-315

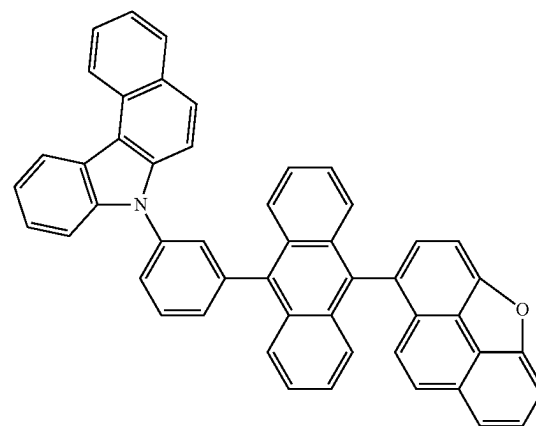


H2-316

H2-313

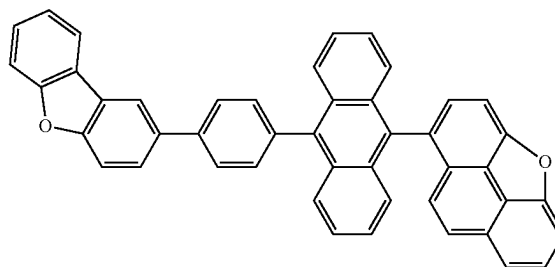
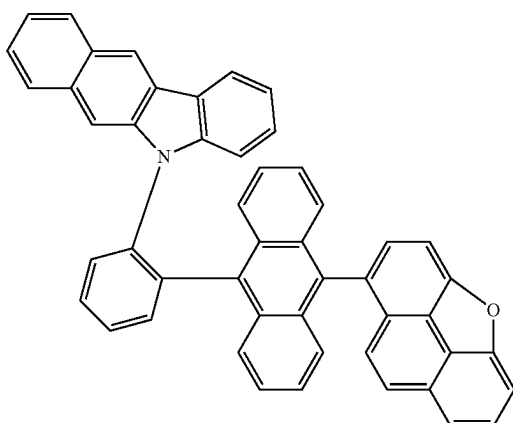


H2-317



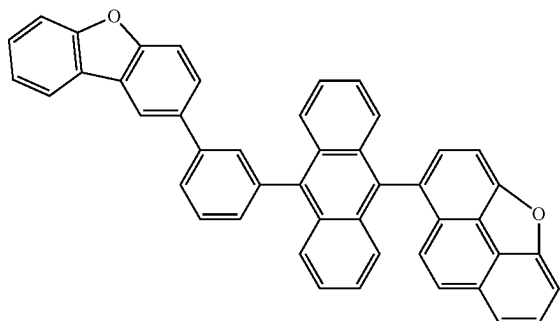
H2-318

H2-314

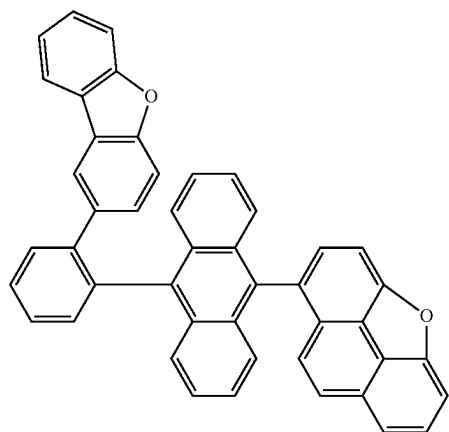


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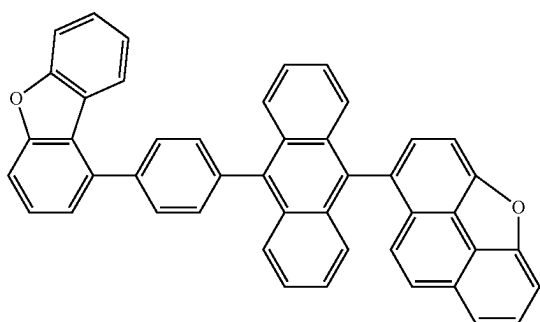
H2-319



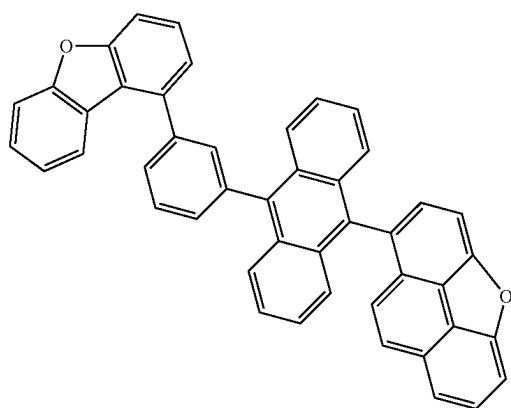
H2-320



H2-321

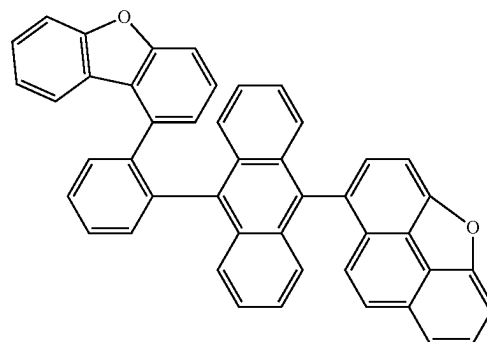


H2-322

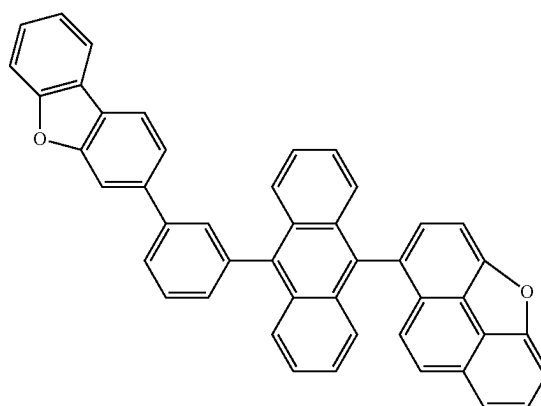


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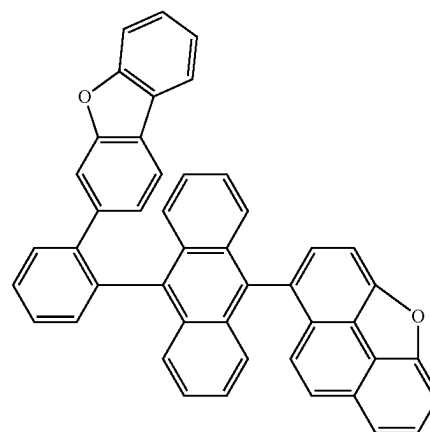
H2-323



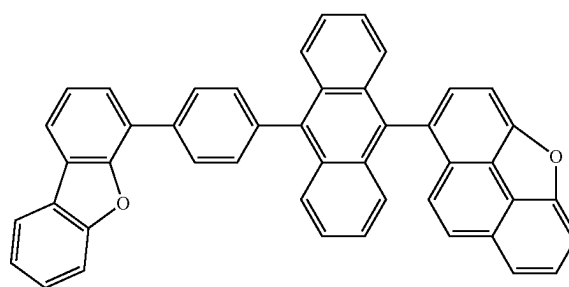
H2-324



H2-325

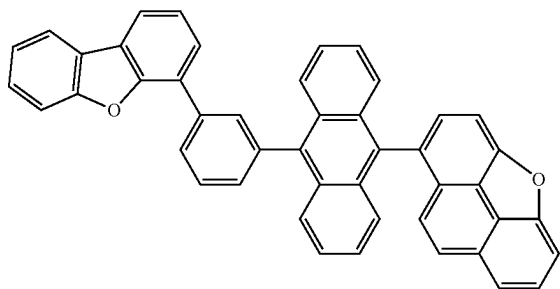


H2-326

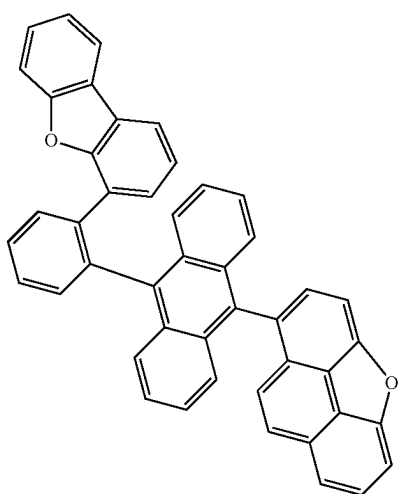


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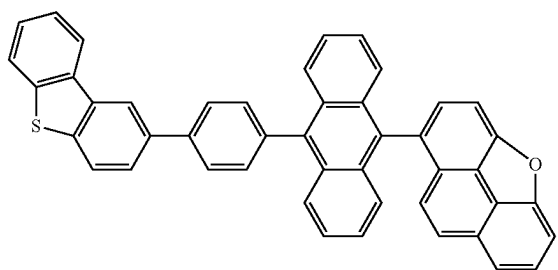
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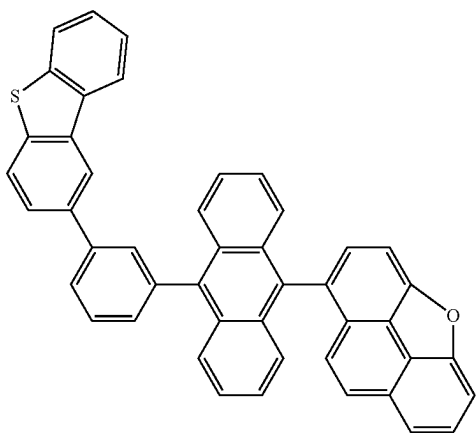
H2-328



H2-329

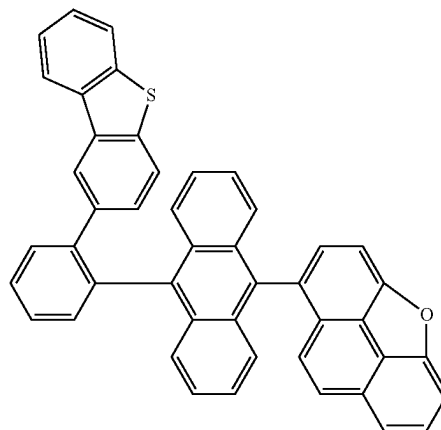


H2-330

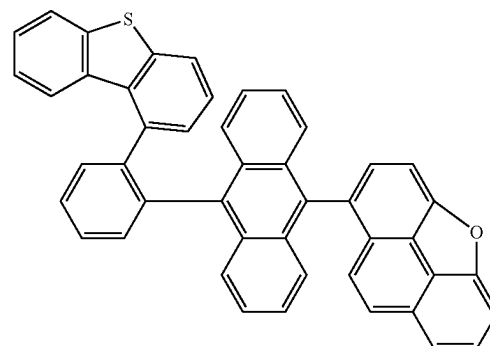


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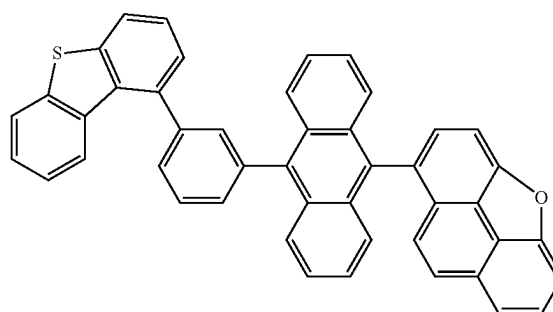
H2-331



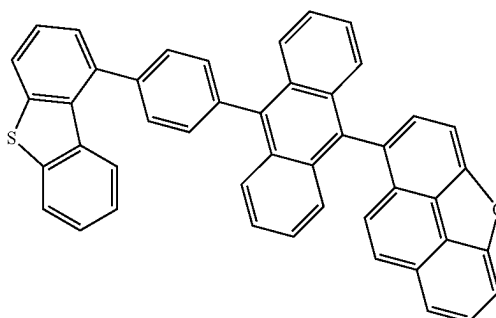
H2-332



H2-333

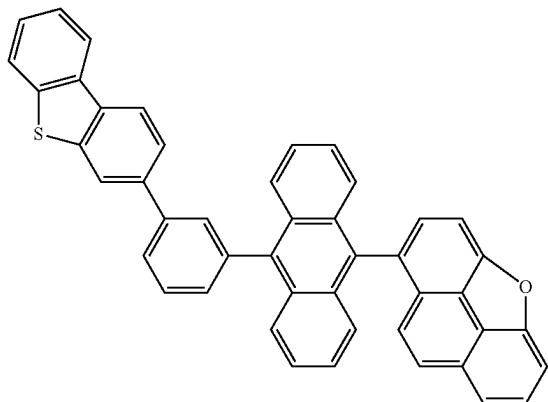


H2-334



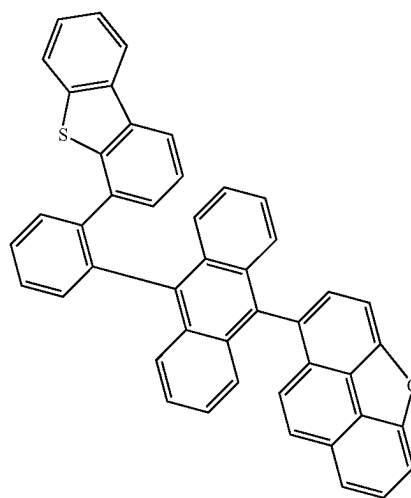
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H2-335

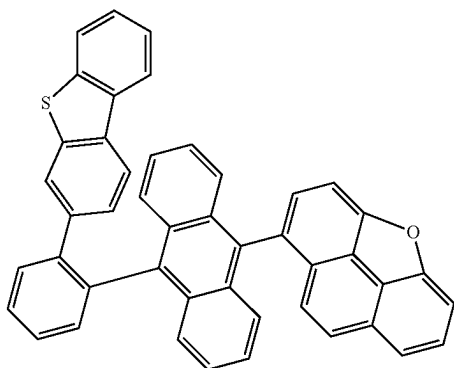


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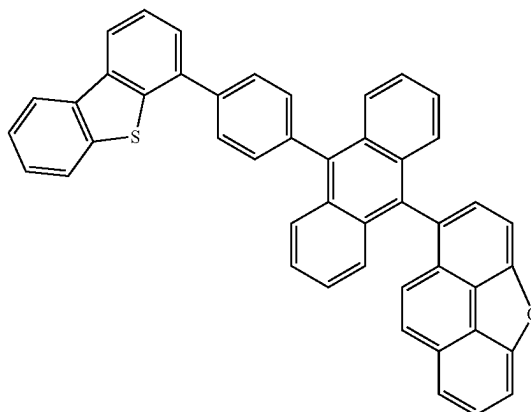
H2-338



H2-336

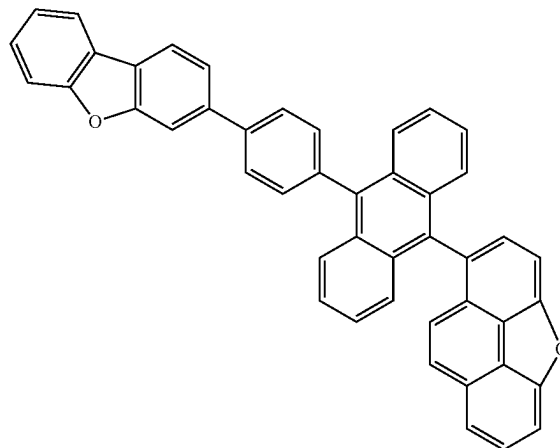
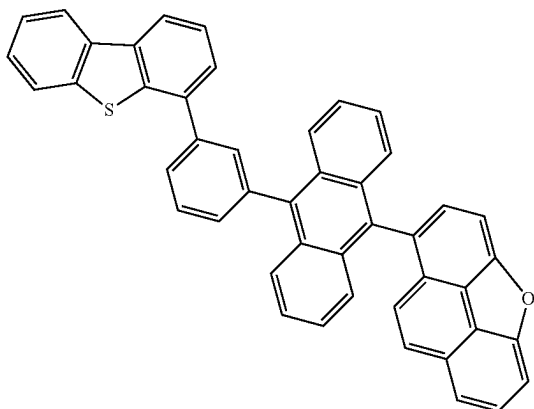


H2-339

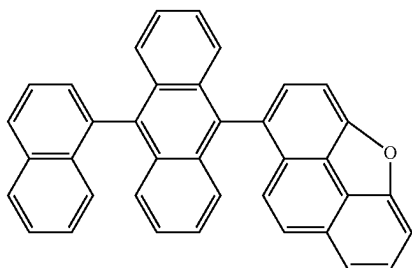


H2-340

H2-337

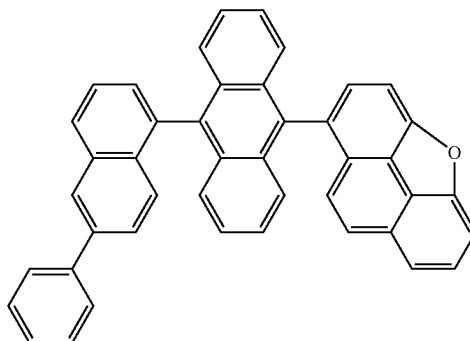


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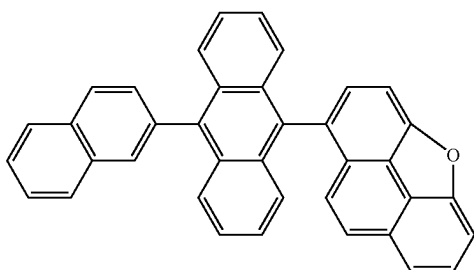


H2-341

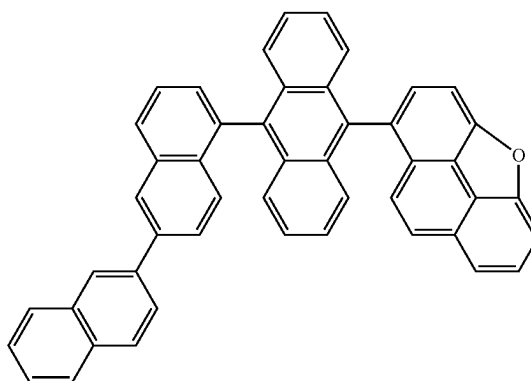
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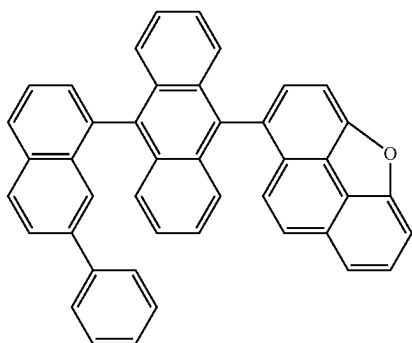
H2-345



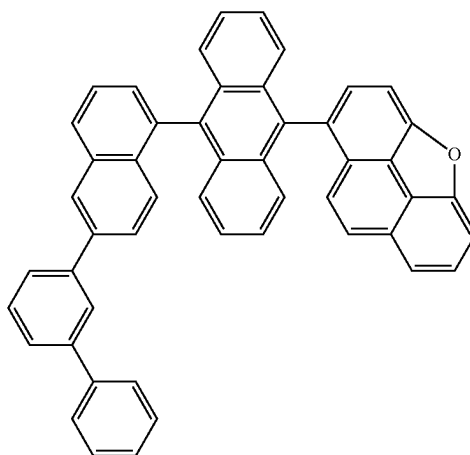
H2-342



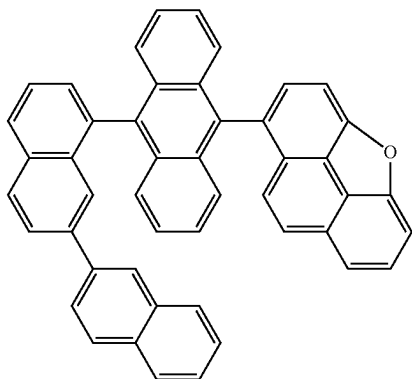
H2-346



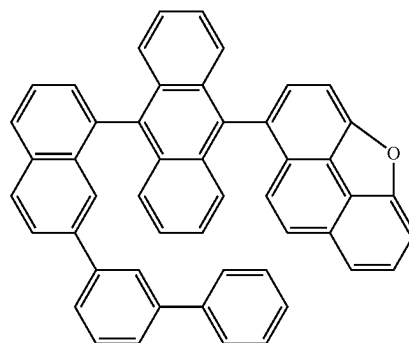
H2-343



H2-347



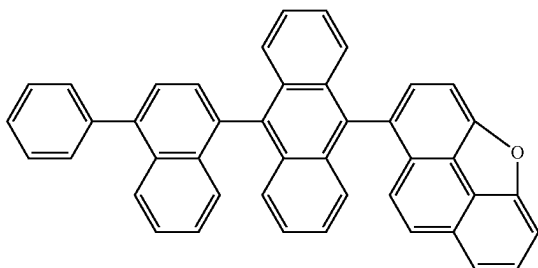
H2-344



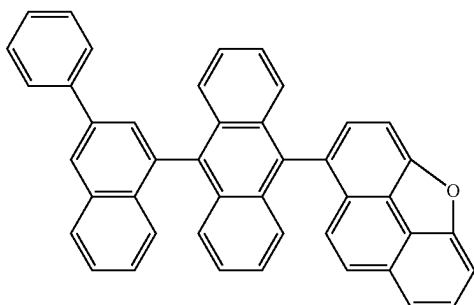
H2-348

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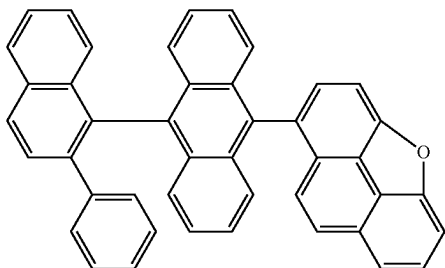
H2-349



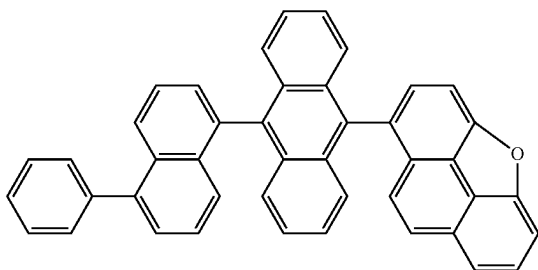
H2-350



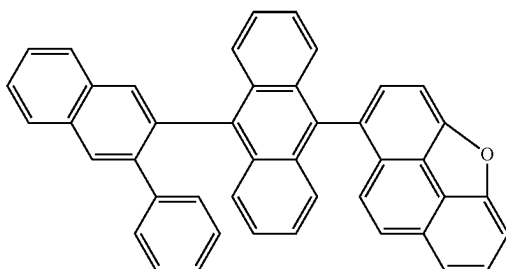
H2-351



H2-352

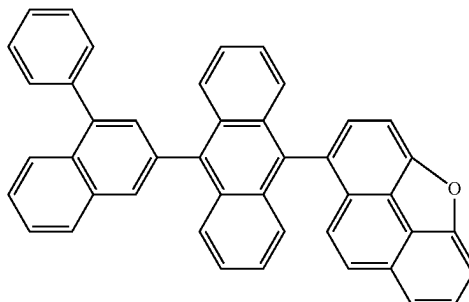


H2-353

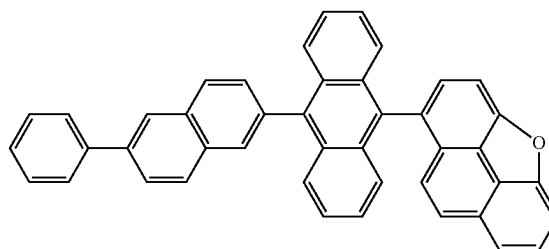


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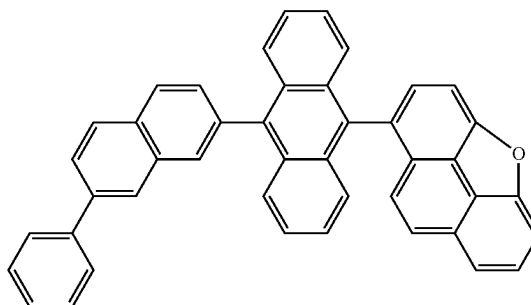
H2-354



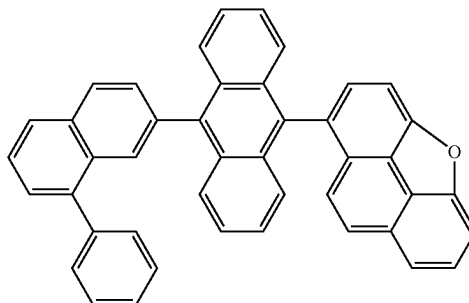
H2-355



H2-356

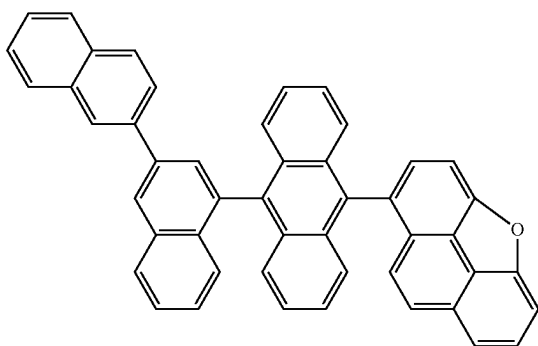


H2-357

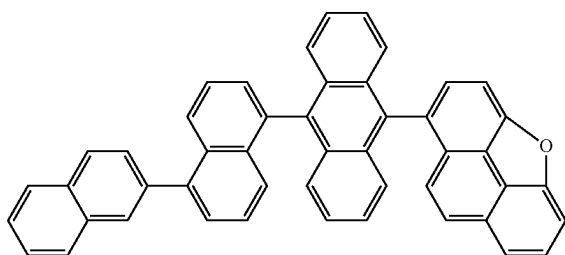


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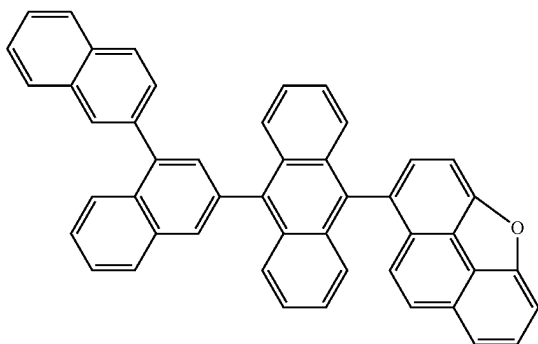
H2-358



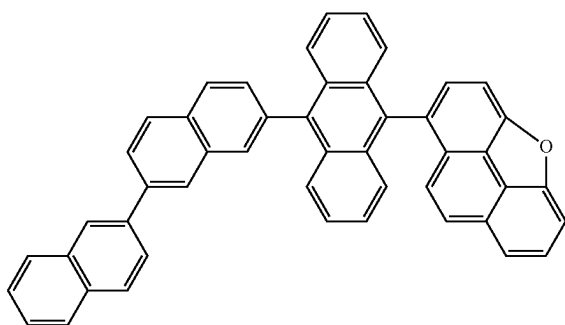
H2-359



H2-360

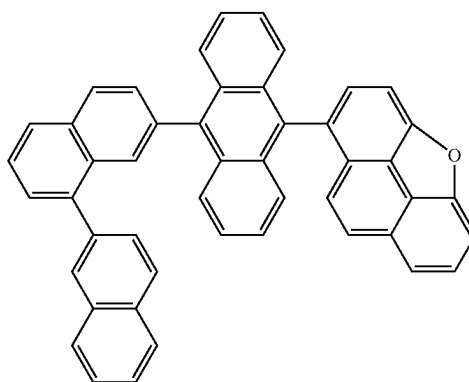


H2-361

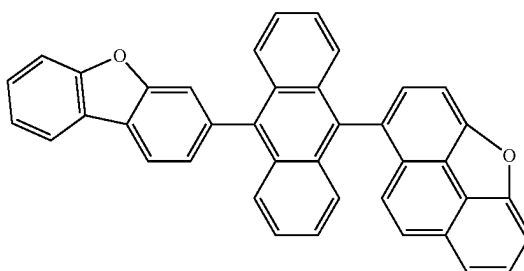


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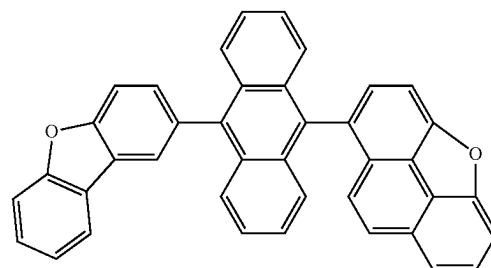
H2-362



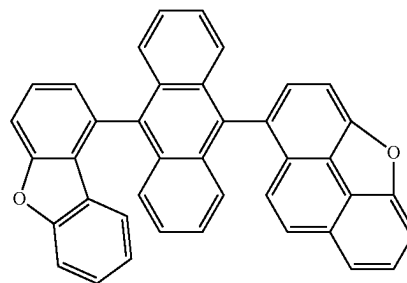
H2-363



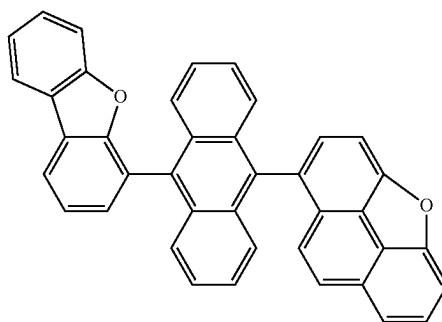
H2-364



H2-365

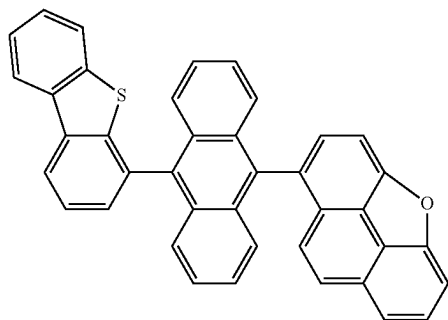


H2-366

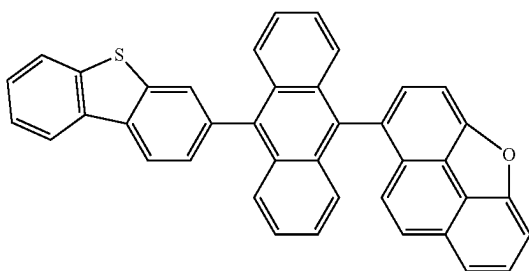


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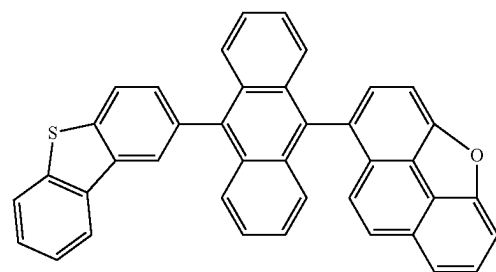
H2-367



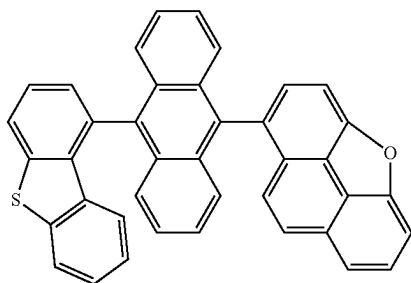
H2-368



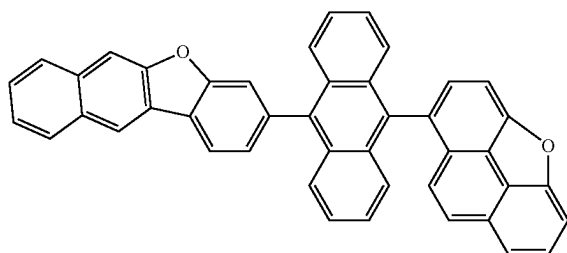
H2-369



H2-370

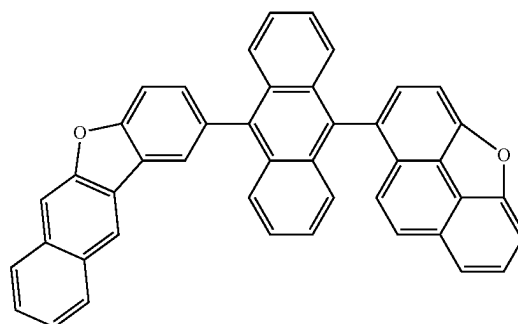


H2-371

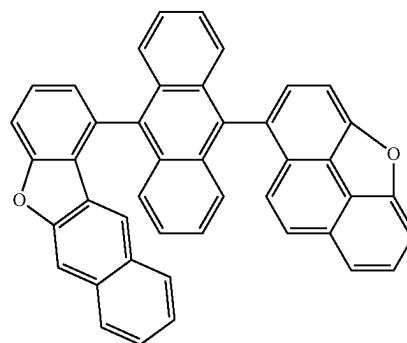


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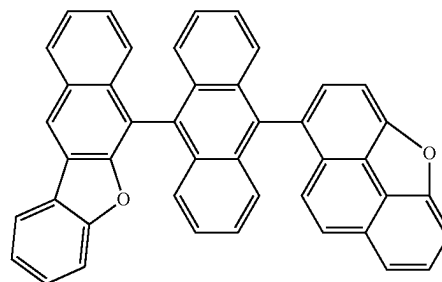
H2-372



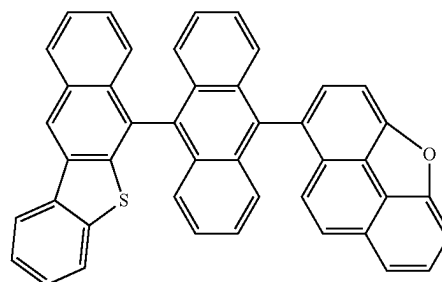
H2-373



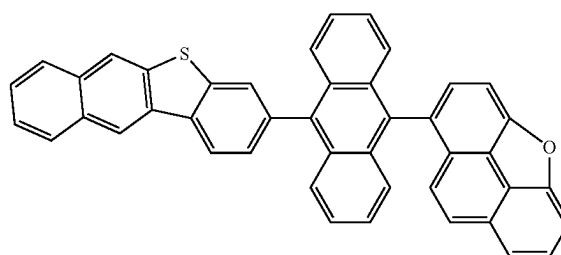
H2-374



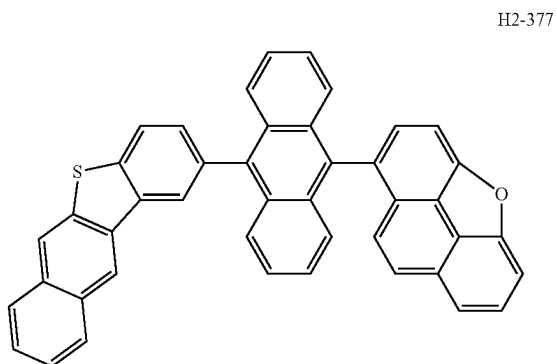
H2-375



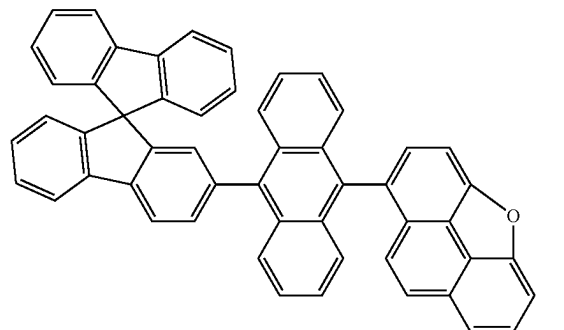
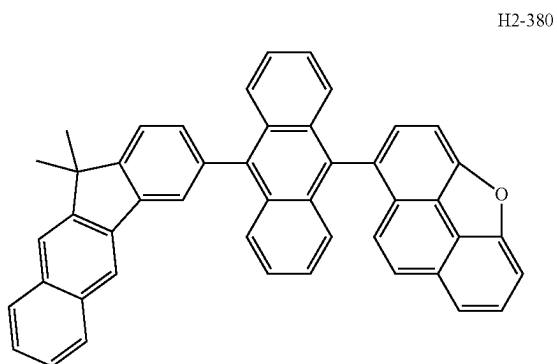
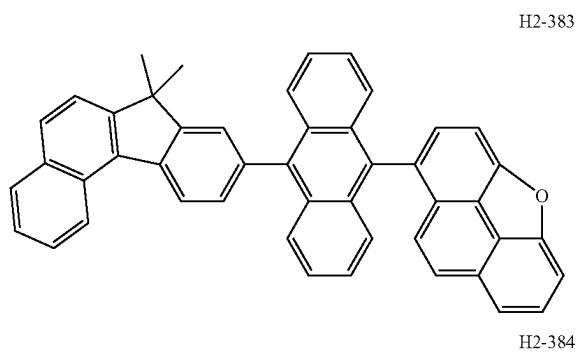
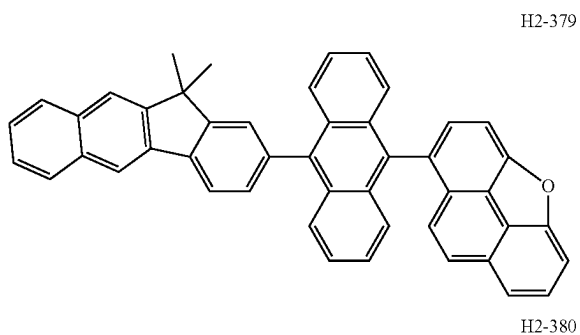
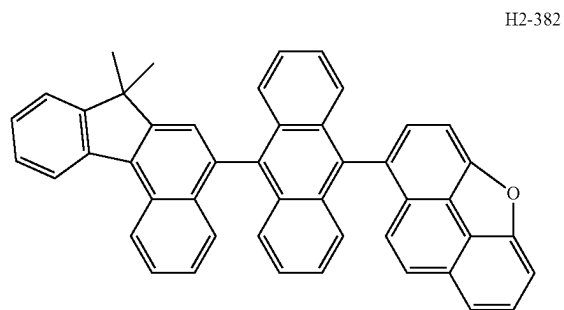
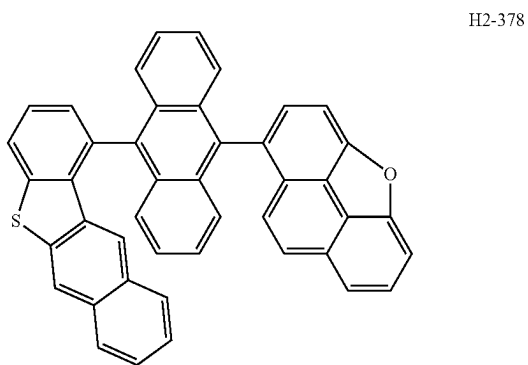
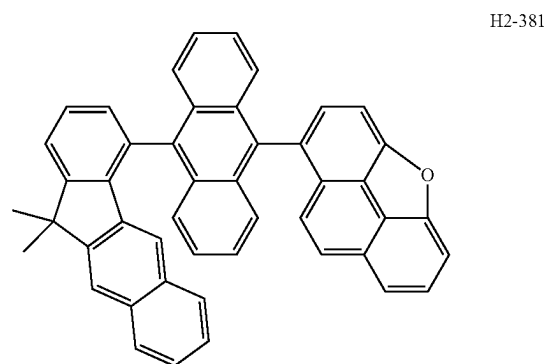
H2-376



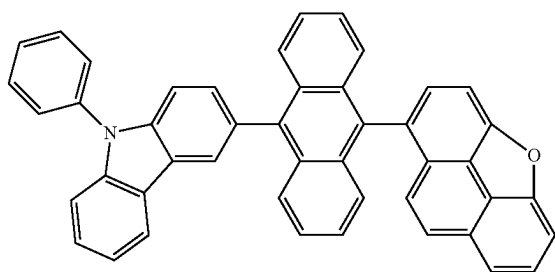
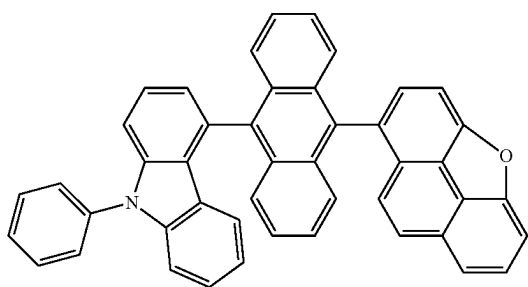
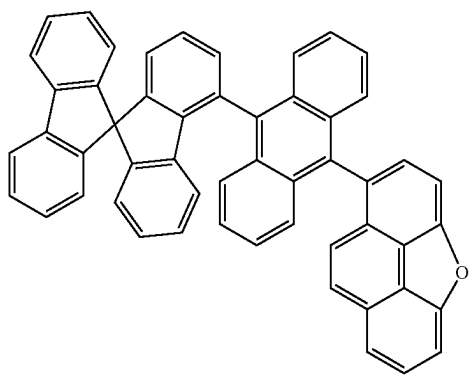
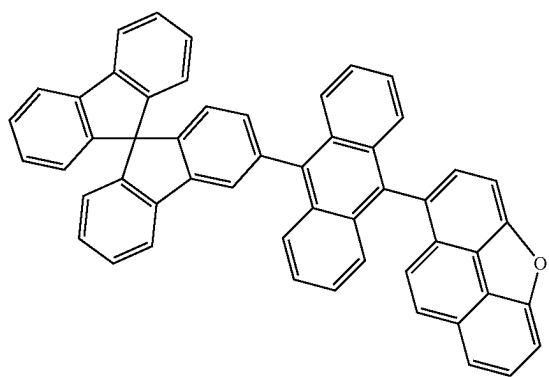
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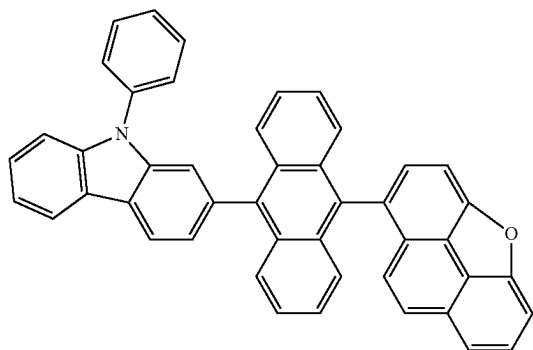


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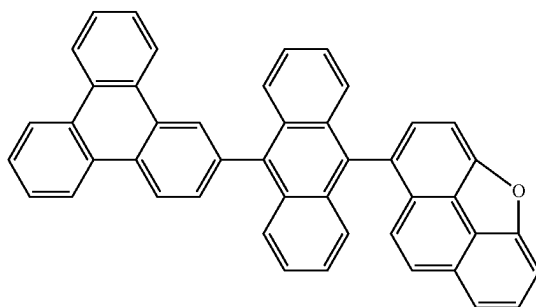


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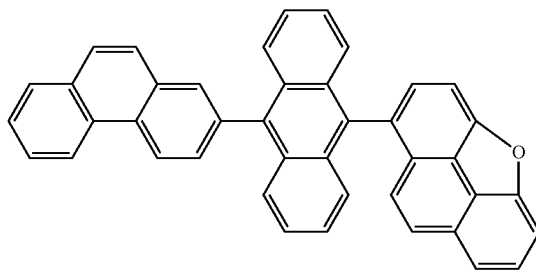
H2-389



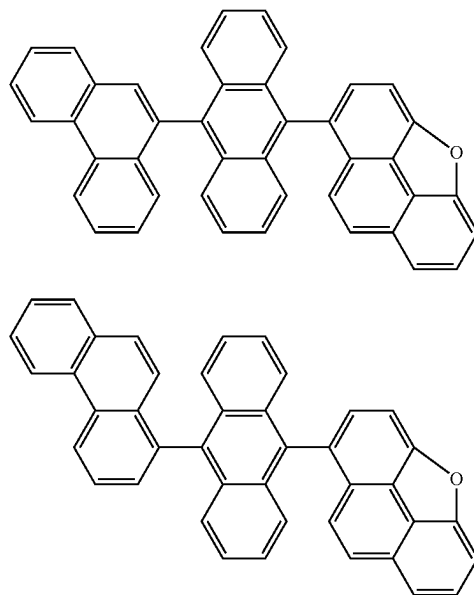
H2-390



H2-392

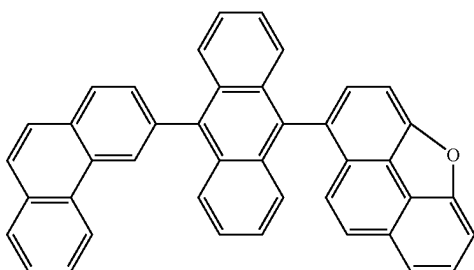


H2-393

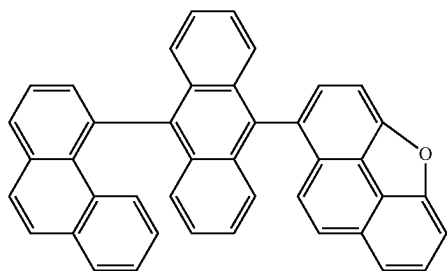


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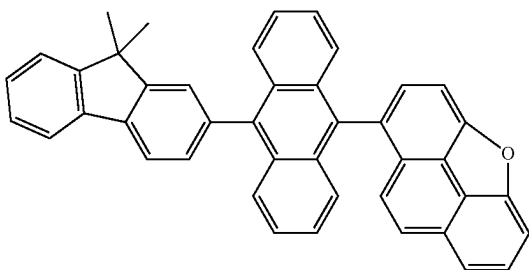
H2-394



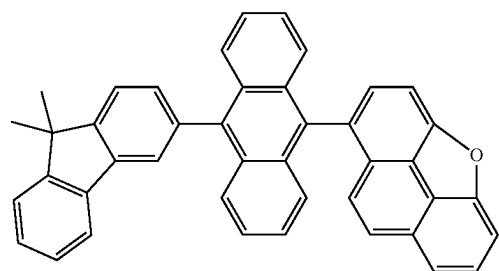
H2-395



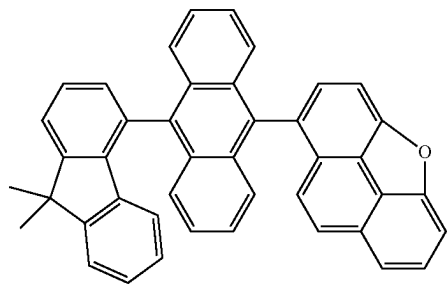
H2-396



H2-397

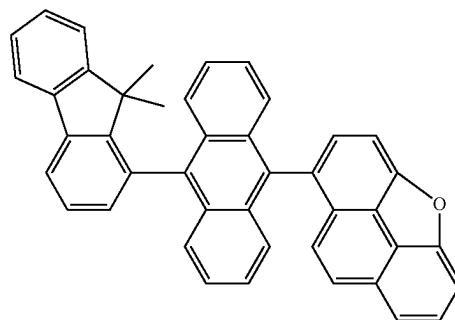


H2-398

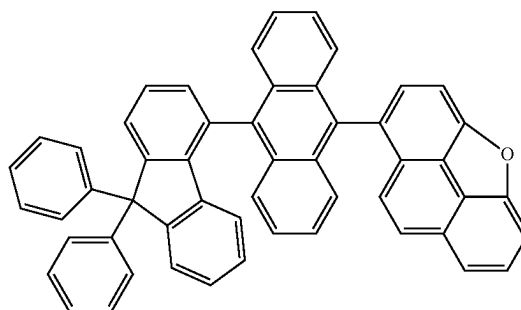


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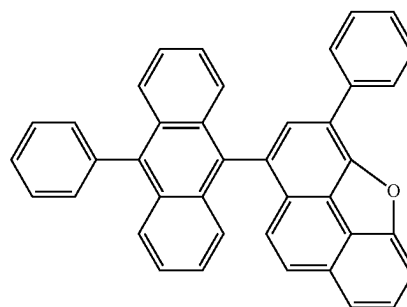
H2-399



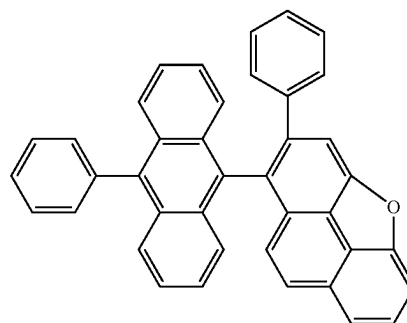
H2-400



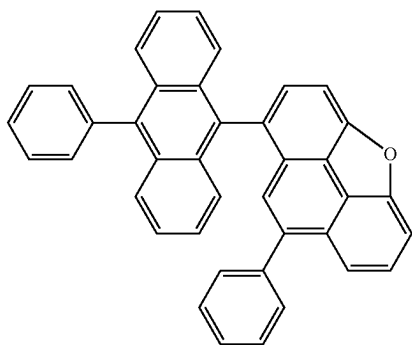
H2-401



H2-402

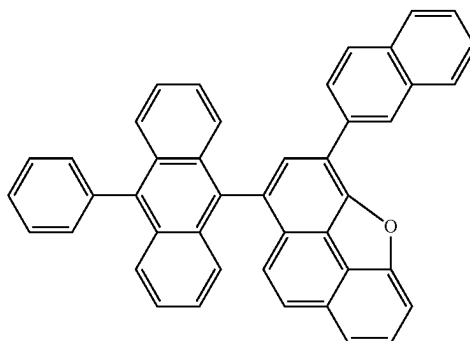


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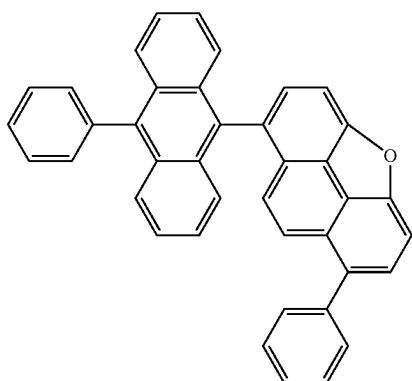


H2-403

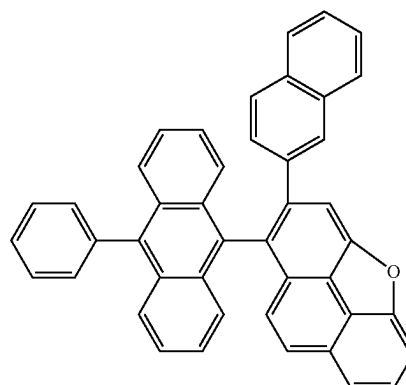
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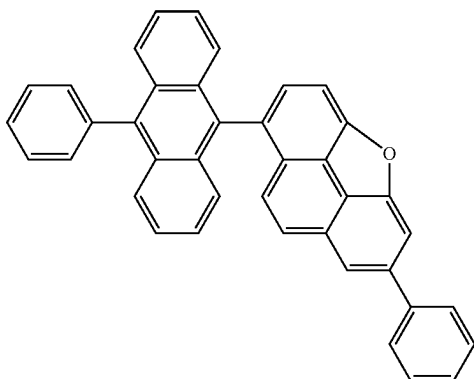
H2-407



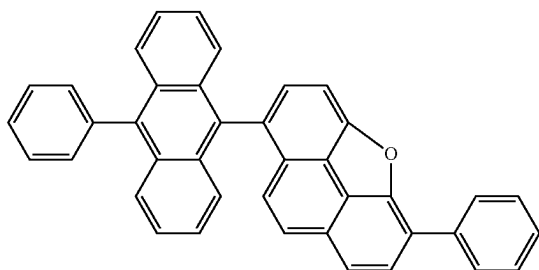
H2-404



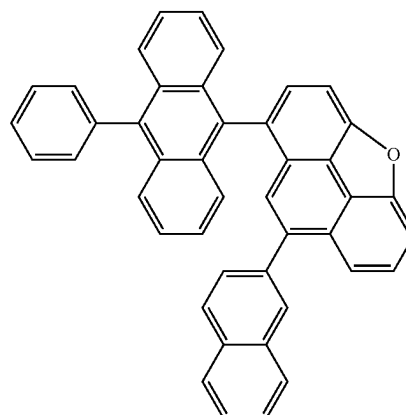
H2-408



H2-405



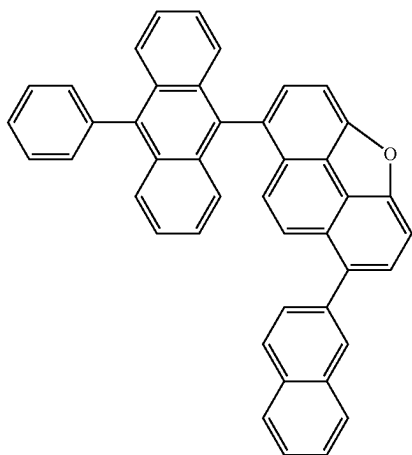
H2-406



H2-409

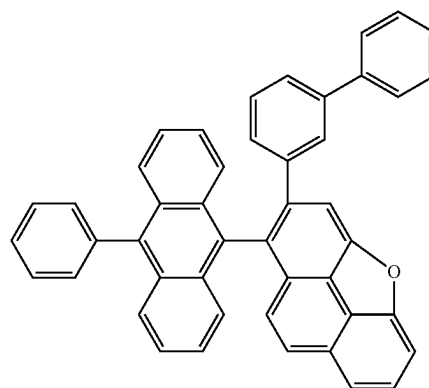
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H2-410

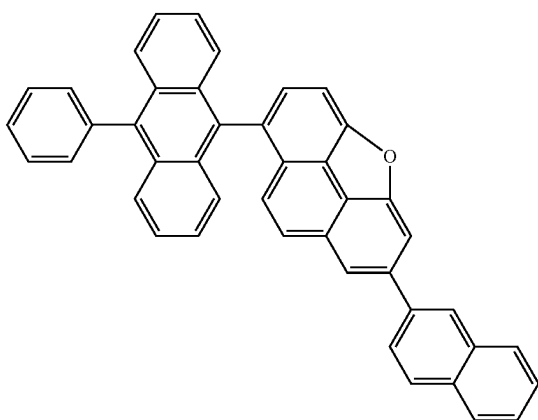


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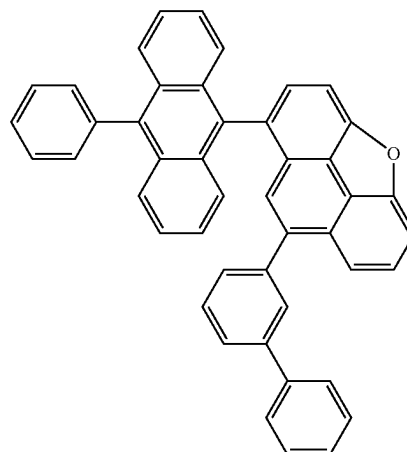
H2-414



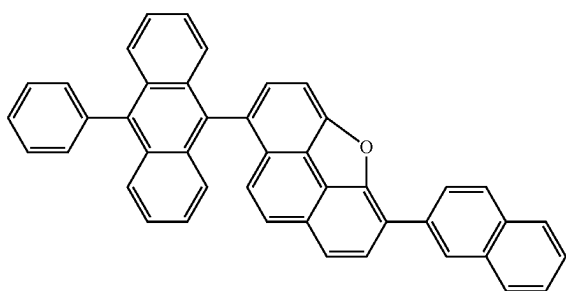
H2-411



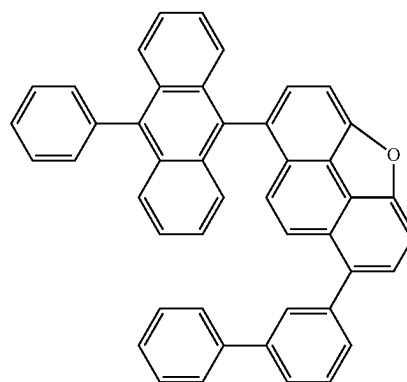
H2-415



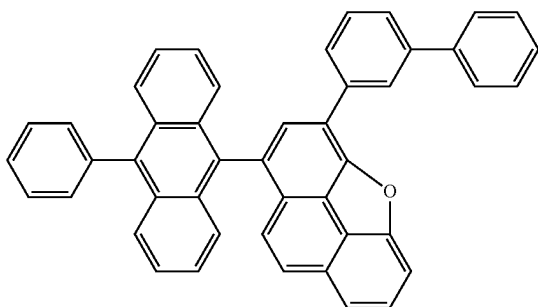
H2-412



H2-416

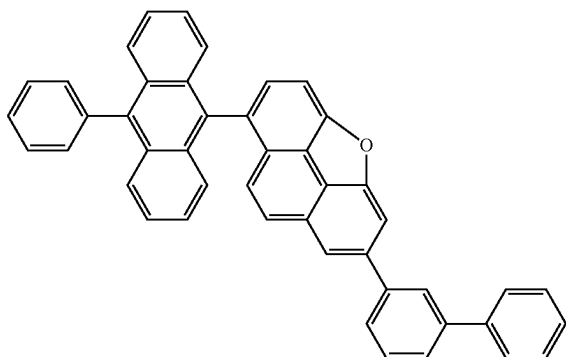


H2-413



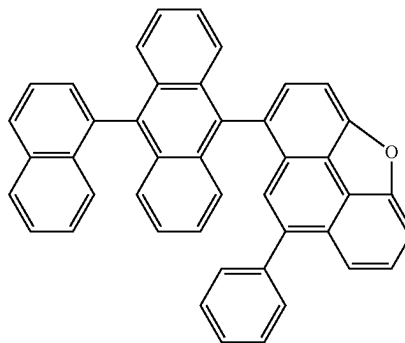
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H2-417

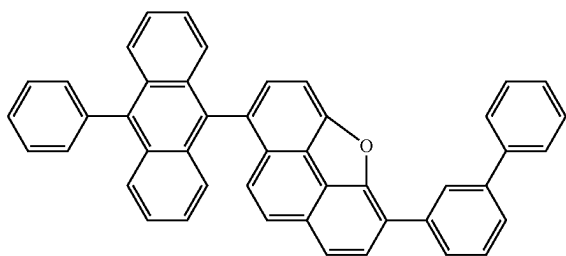


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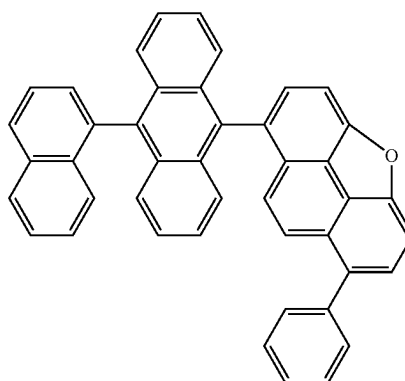
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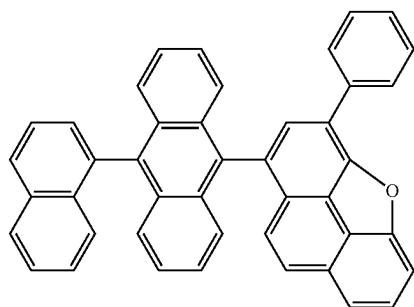
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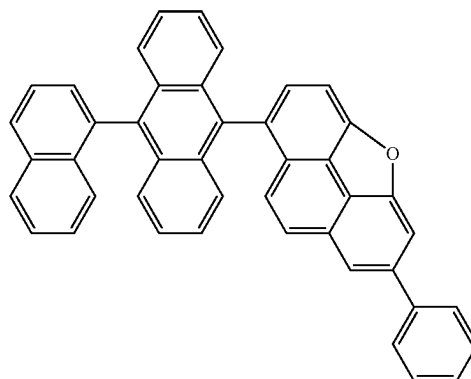
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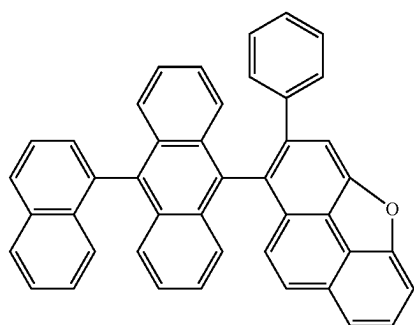
H2-419



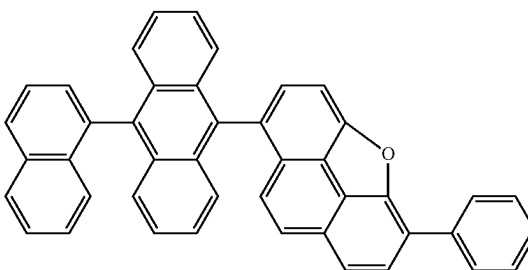
H2-423



H2-420

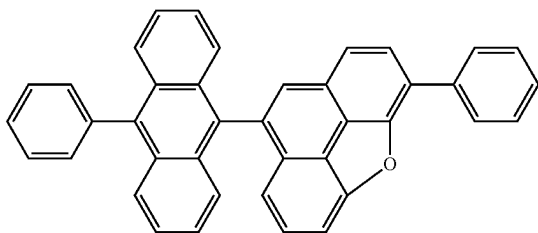


H2-424

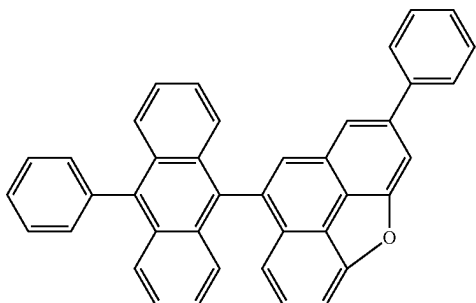


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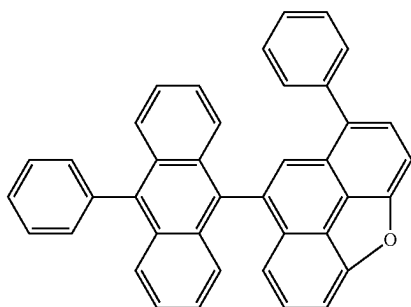
H2-425



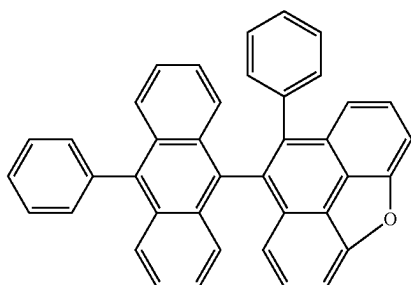
H2-426



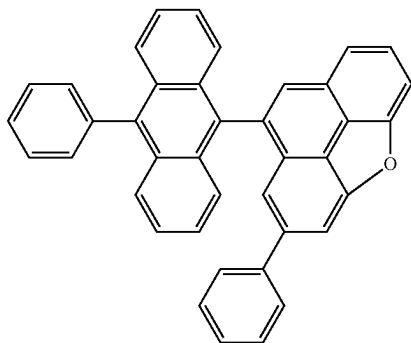
H2-427



H2-428

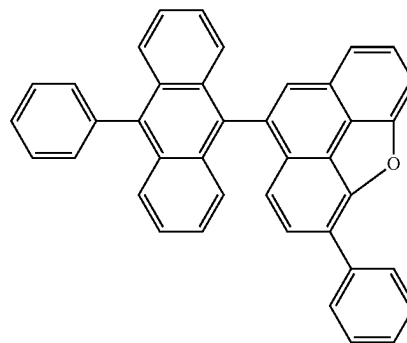


H2-429

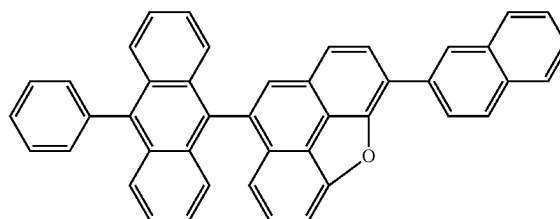


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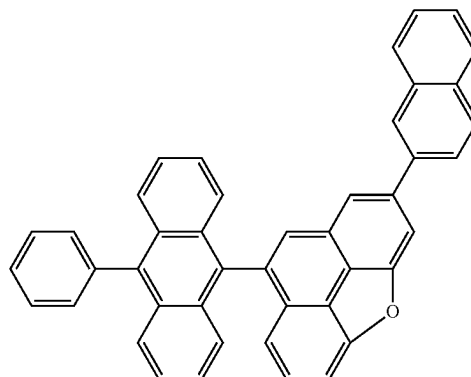
H2-430



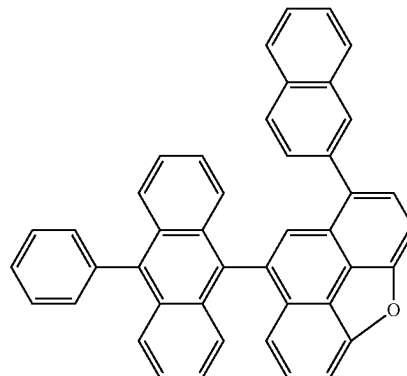
H2-431



H2-432

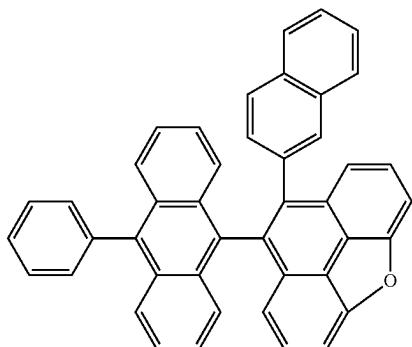


H2-433

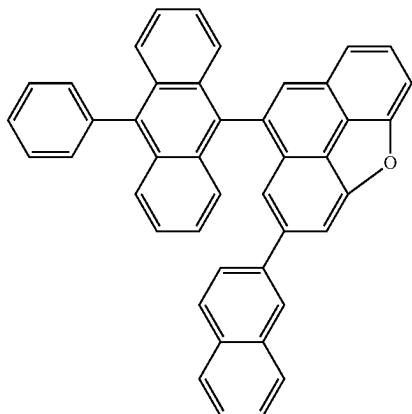


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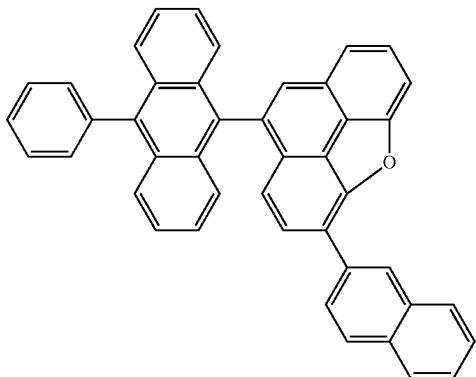
H2-434



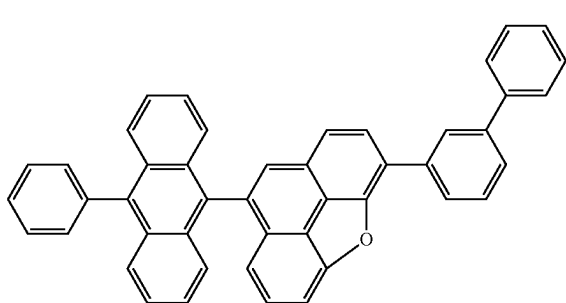
H2-435



H2-436

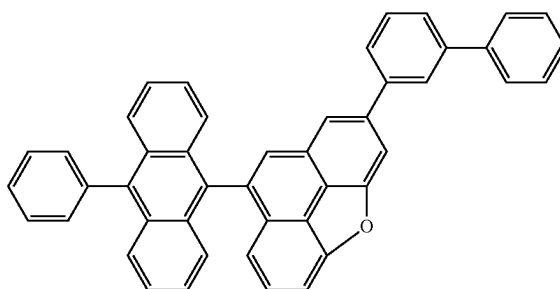


H2-437

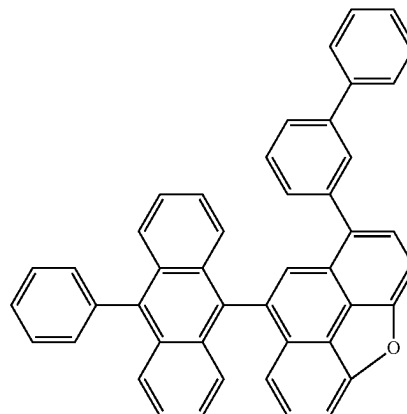


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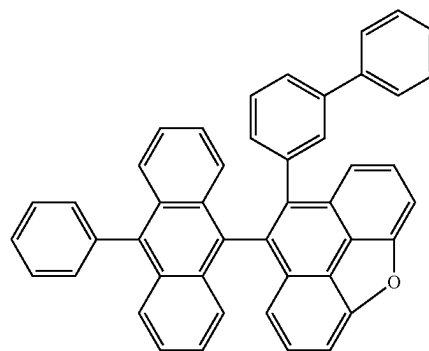
H2-438



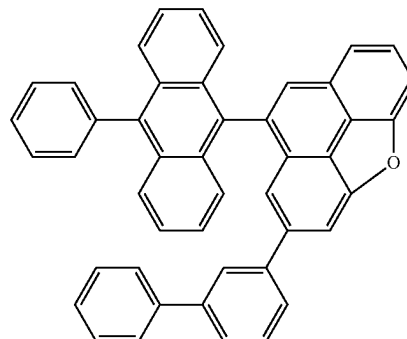
H2-439



H2-440

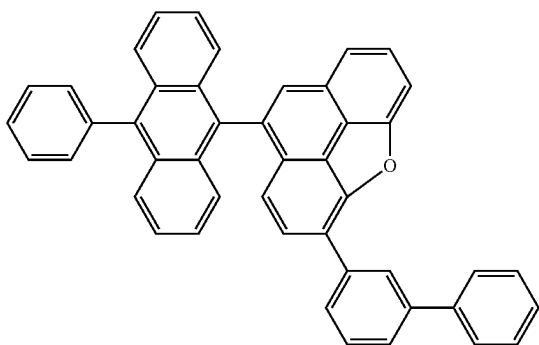


H2-441

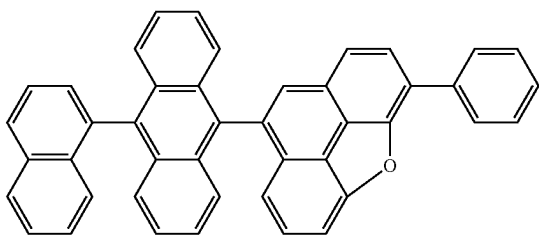


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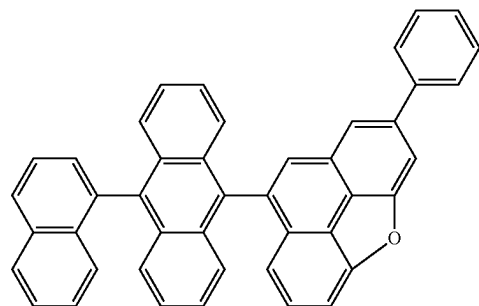
H2-442



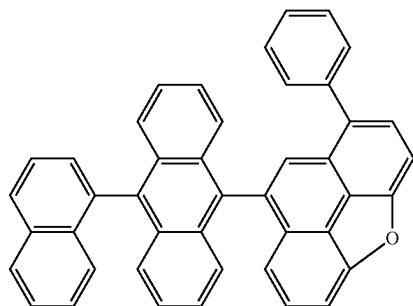
H2-443



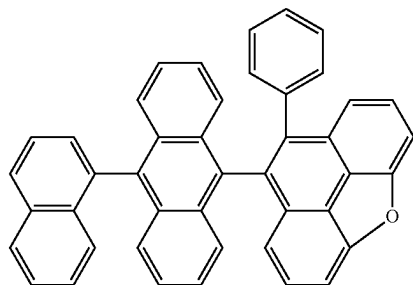
H2-444



H2-445

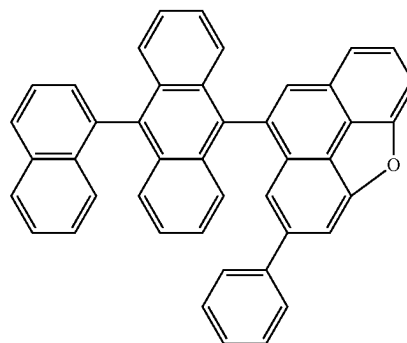


H2-446

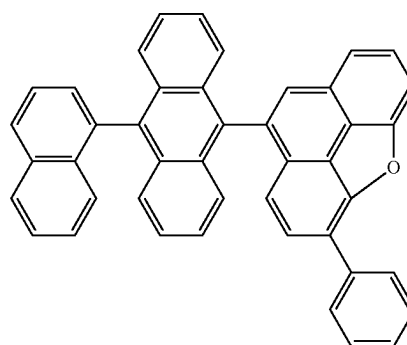


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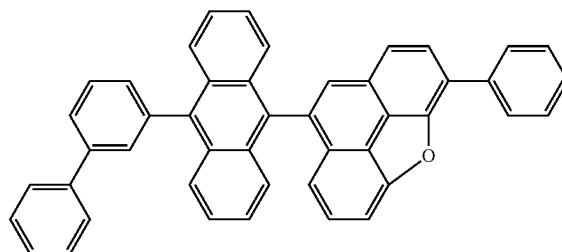
H2-447



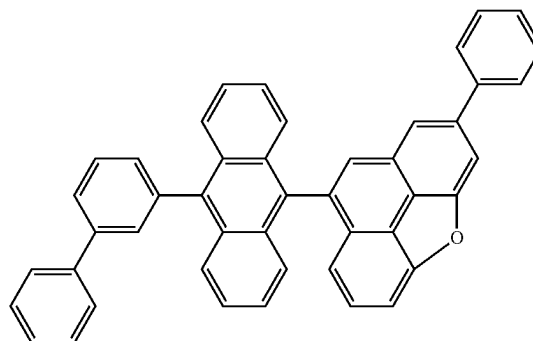
H2-448



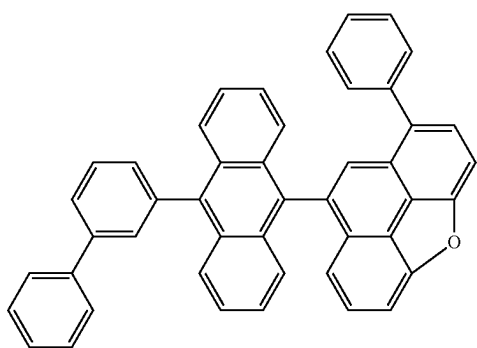
H2-449



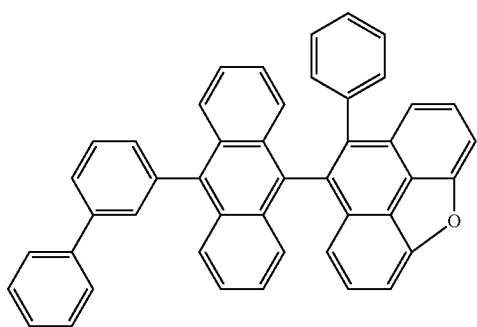
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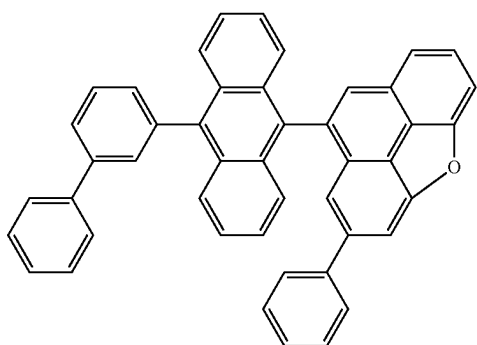
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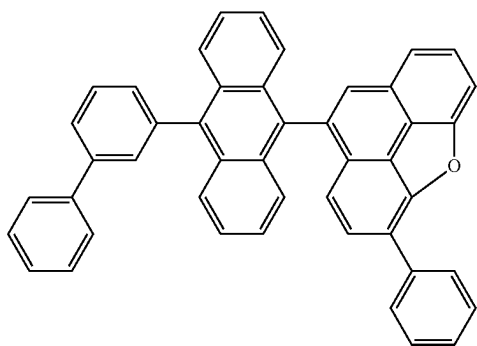
H2-451



H2-452

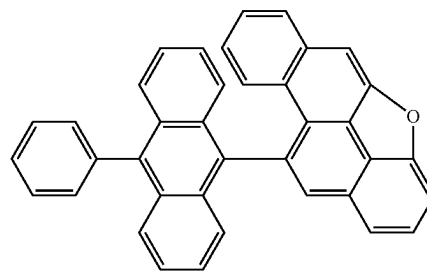


H2-453

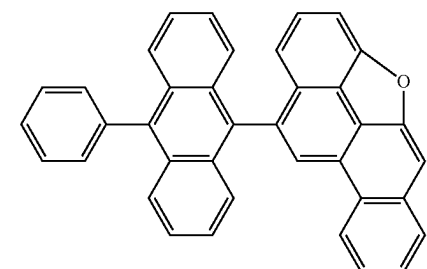


H2-454

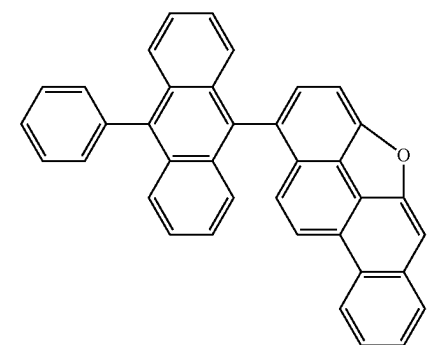
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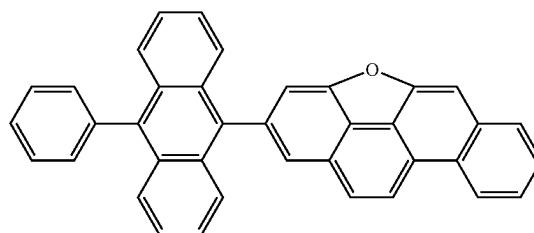
H2-455



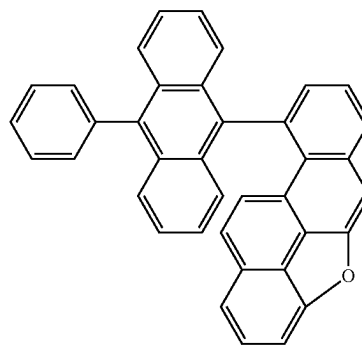
H2-456



H2-457



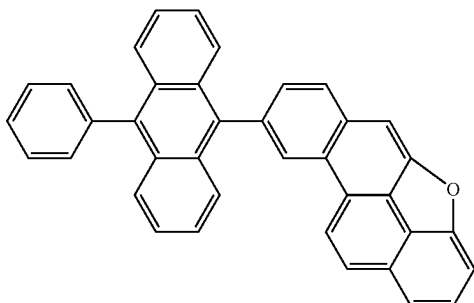
H2-458



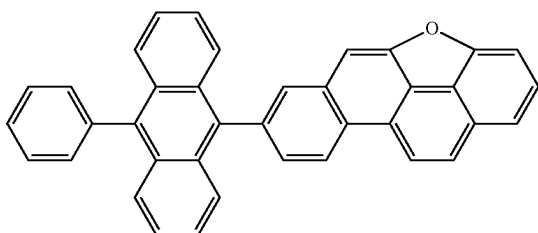
H2-459

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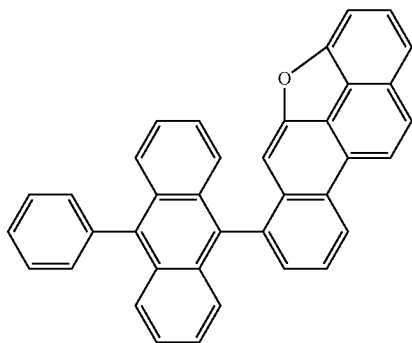
H2-460



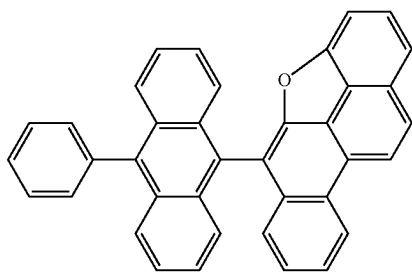
H2-461



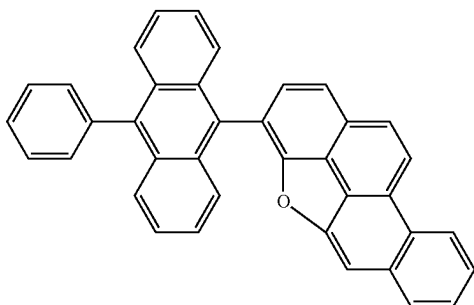
H2-462



H2-463

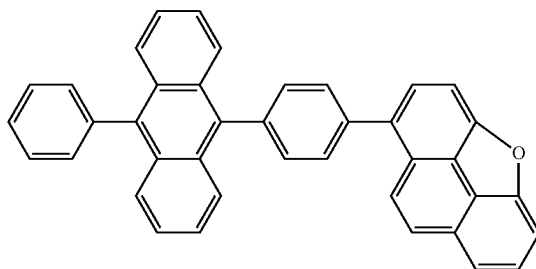


H2-464

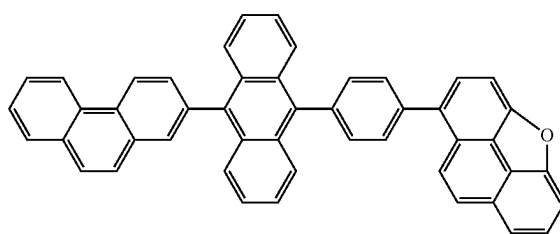


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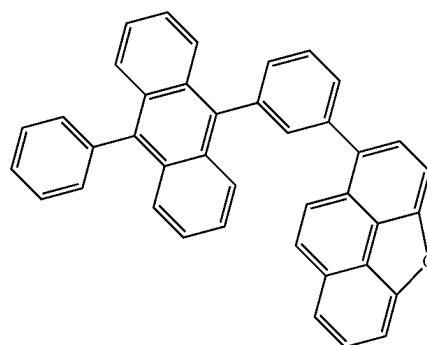
H2-465



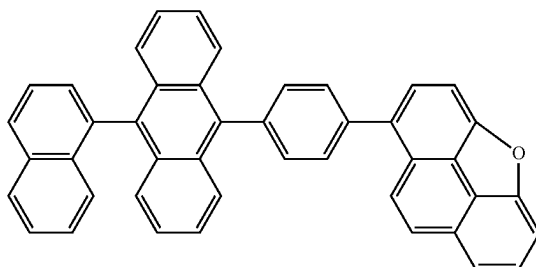
H2-466



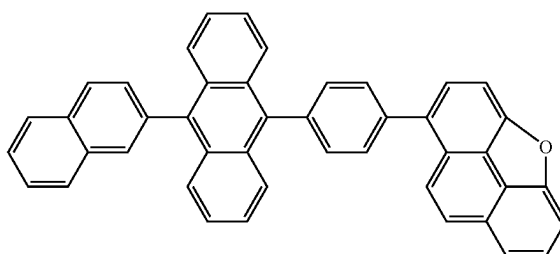
H2-467



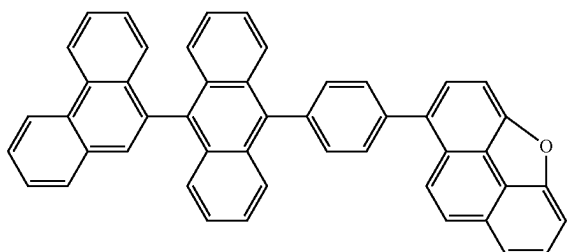
H2-468



H2-469

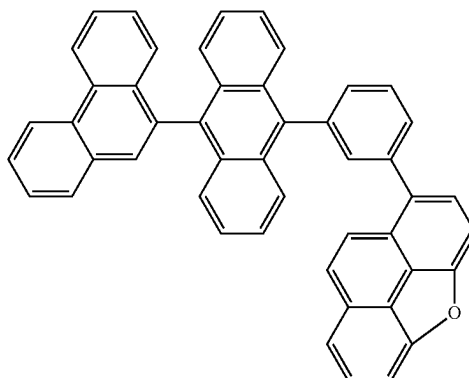


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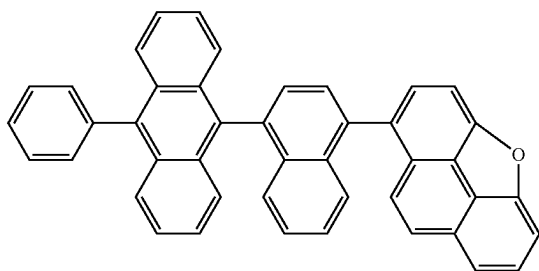


H2-470

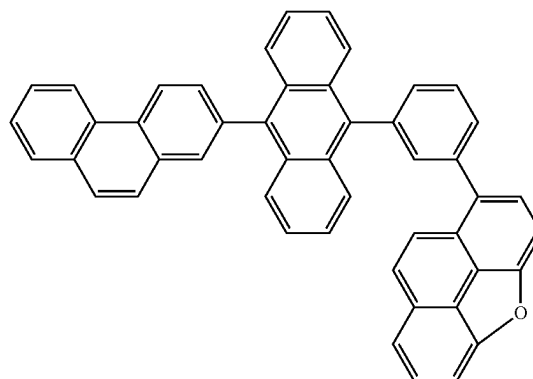
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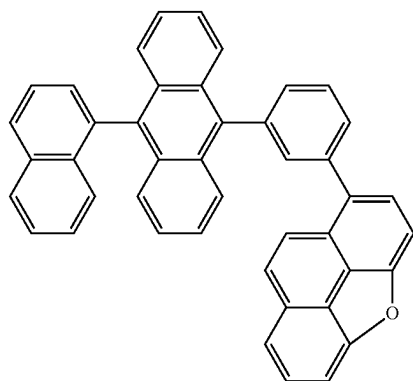
H2-474



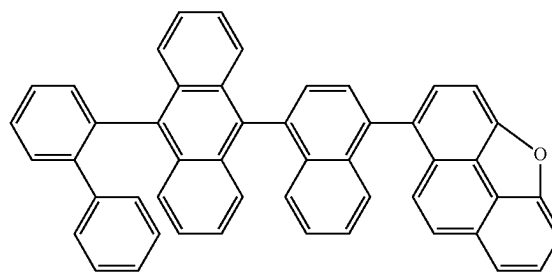
H2-471



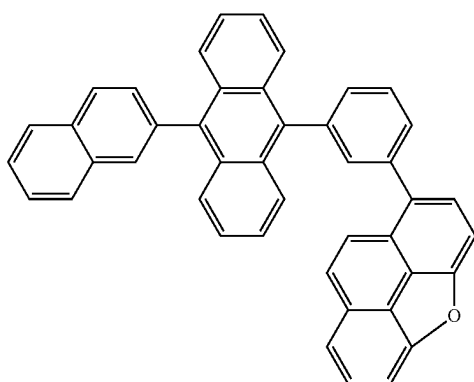
H2-475



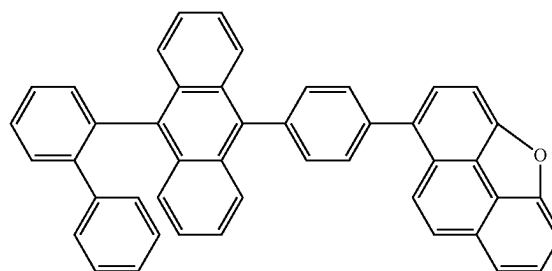
H2-472



H2-476

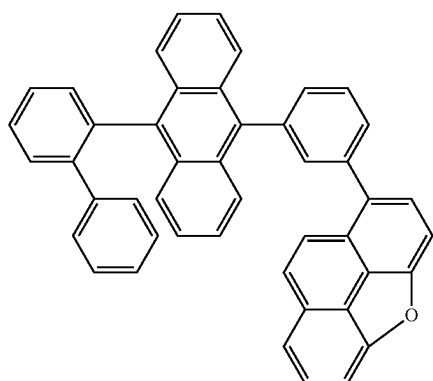


H2-473



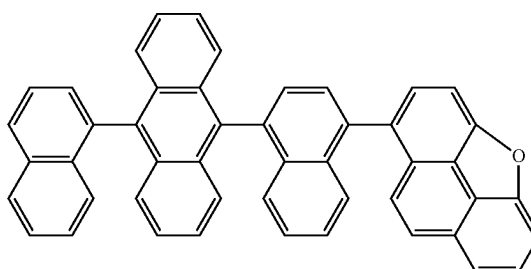
H2-477

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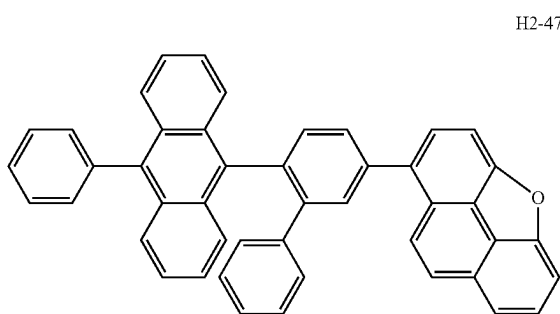


H2-478

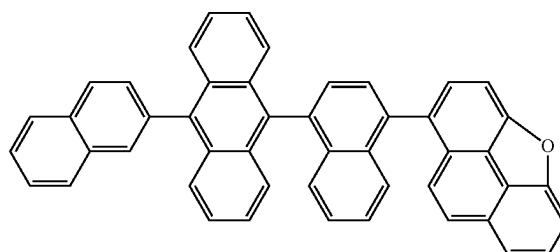
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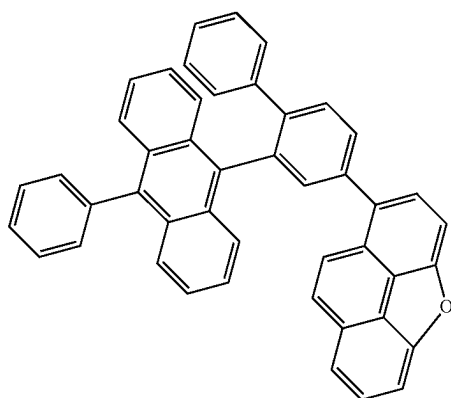
H2-482



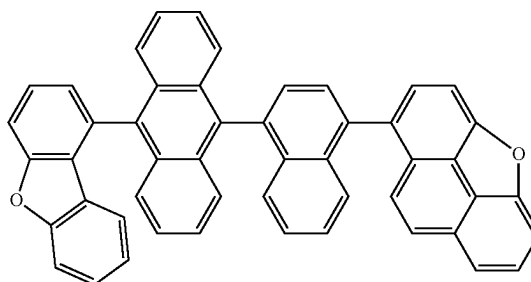
H2-479



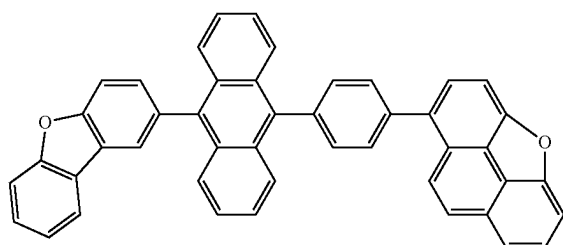
H2-483



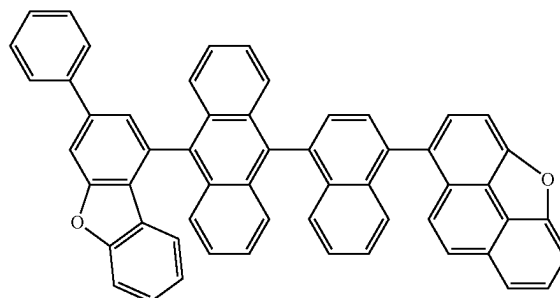
H2-480



H2-484



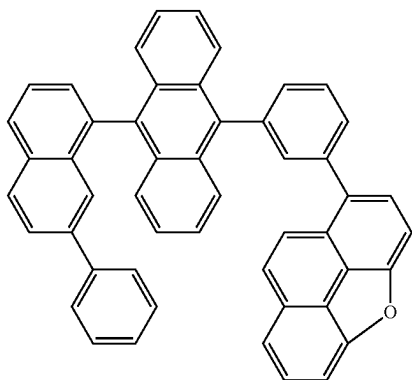
H2-481



H2-485

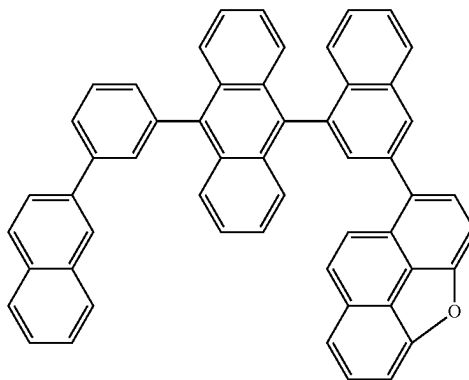
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H2-486

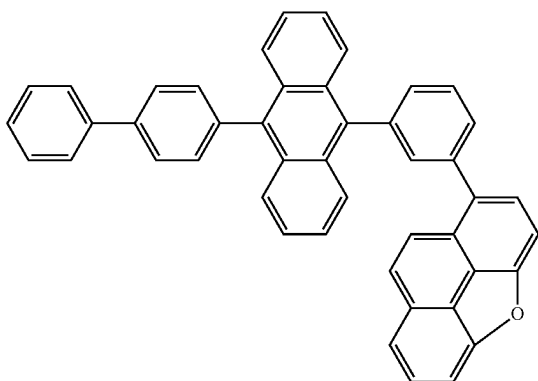


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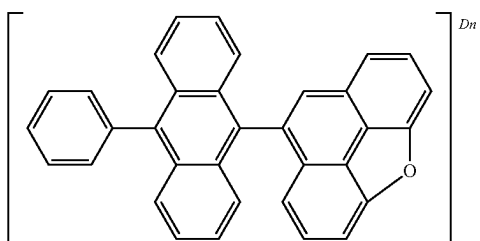
H2-490



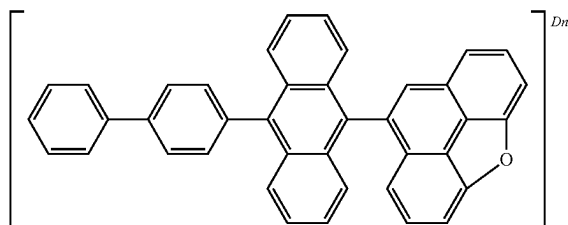
H2-487



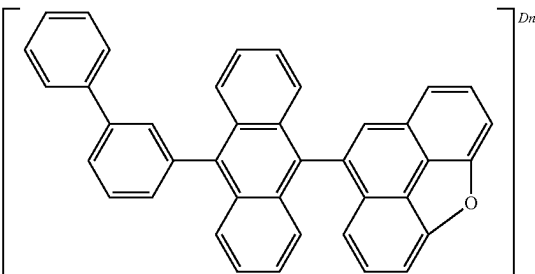
H2-488



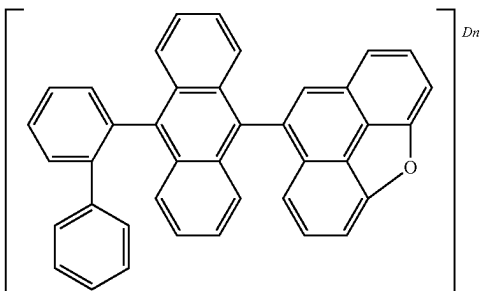
H2-491



H2-492

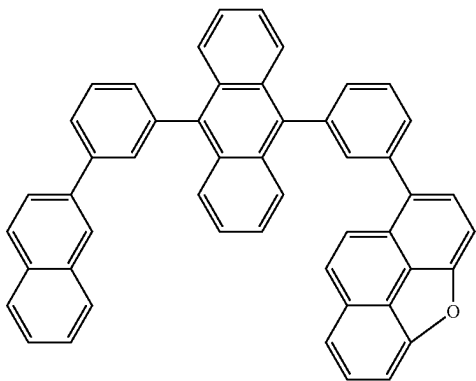


H2-493



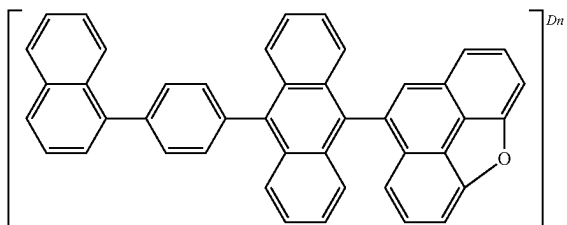
H2-494

H2-489

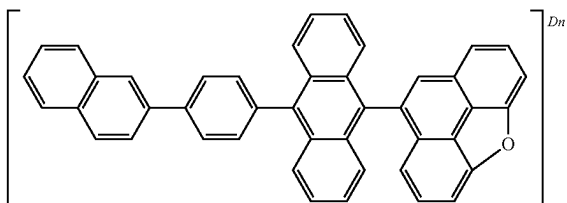


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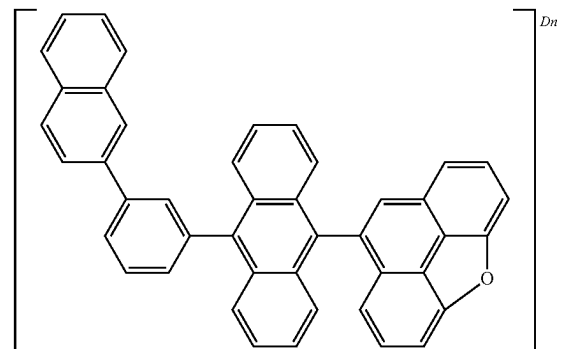
H2-495



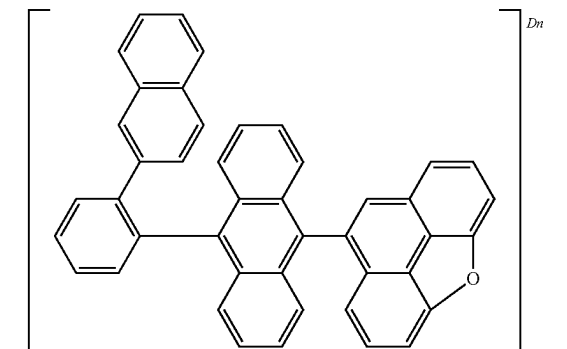
H2-496



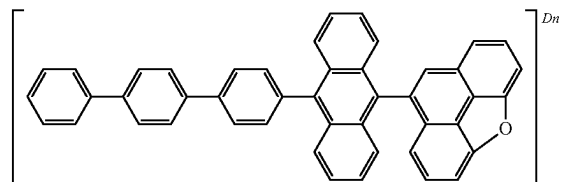
H2-497



H2-498

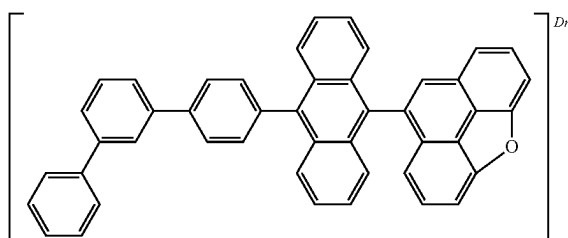


H2-499

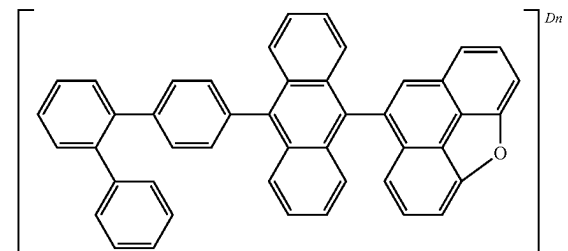


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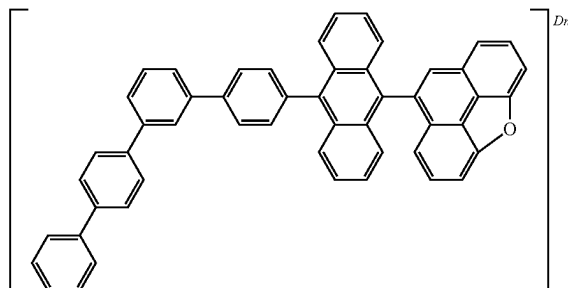
H2-500



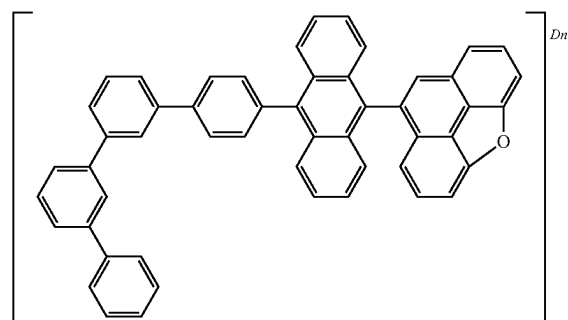
H2-501



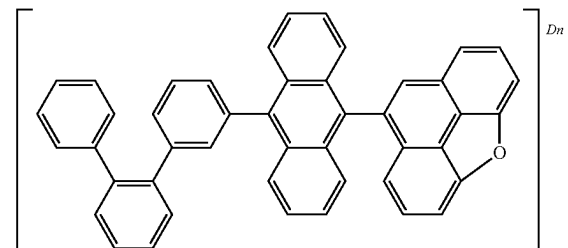
H2-502



H2-503

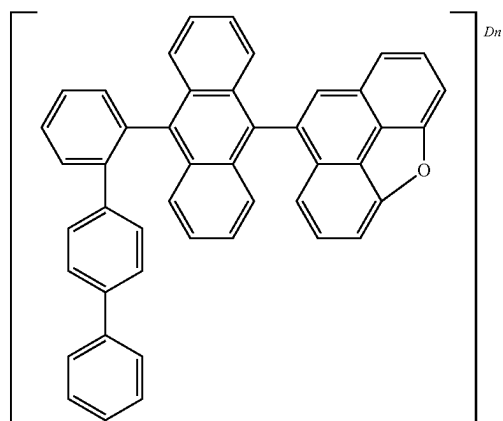


H2-504

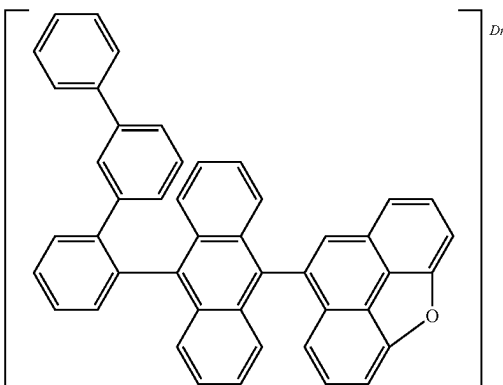


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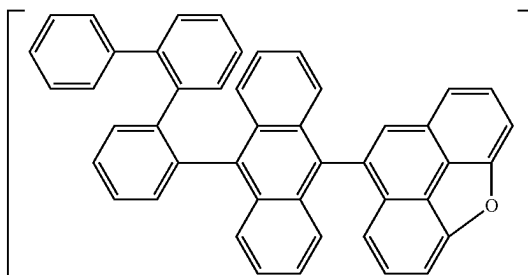
H2-505



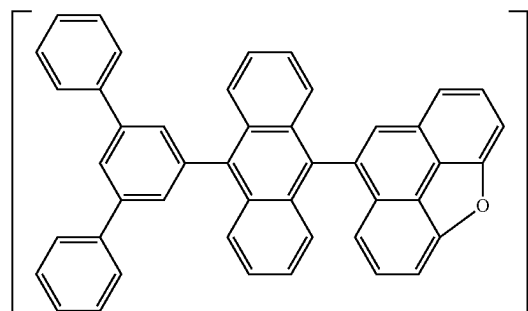
H2-506



H2-507

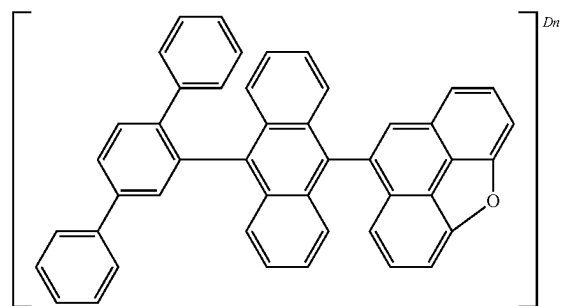


H2-508

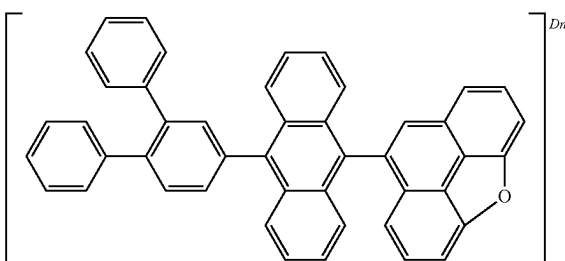


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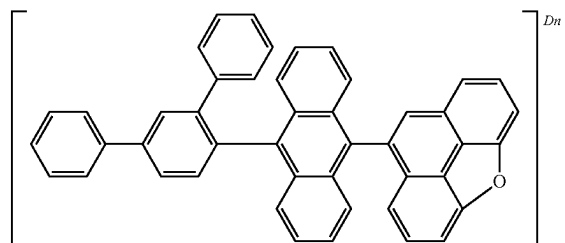
H2-509



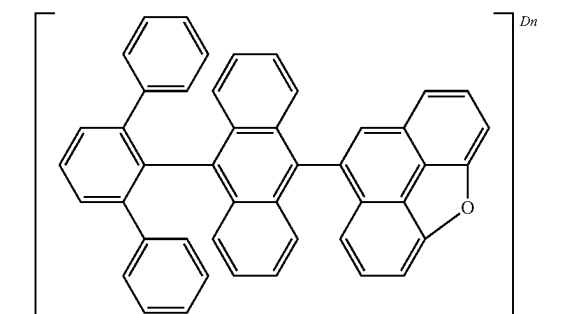
H2-510



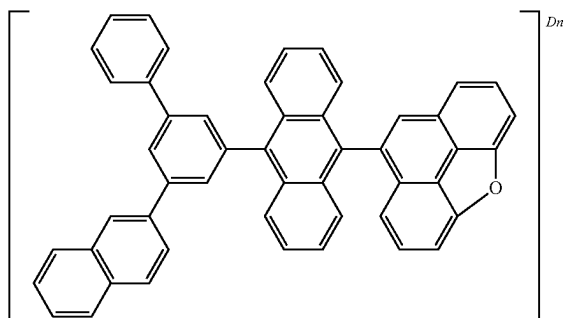
H2-511



H2-512

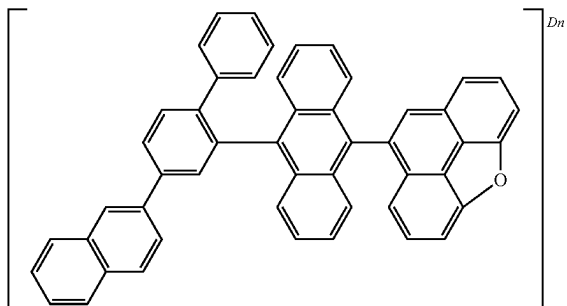


H2-513

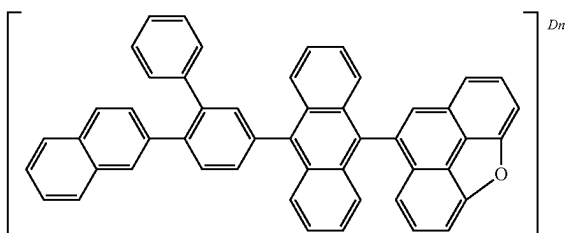


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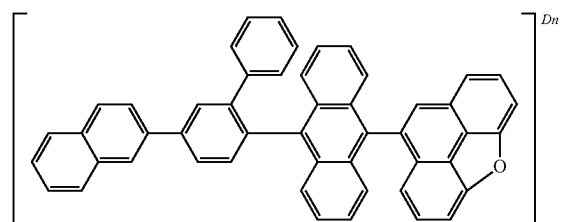
H2-514



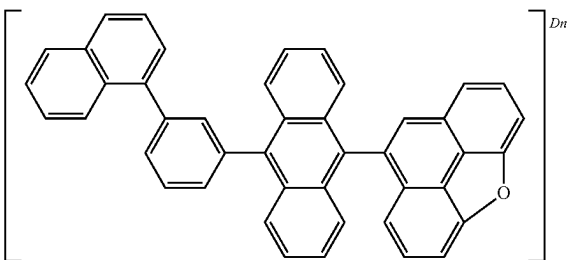
H2-515



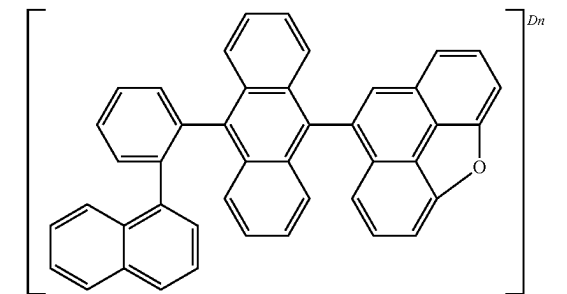
H2-516



H2-517

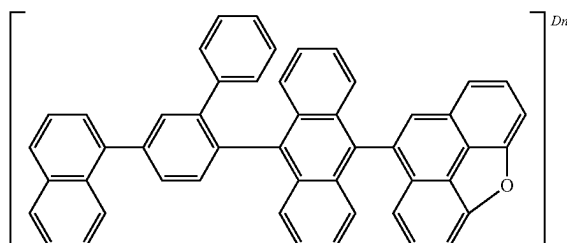


H2-518

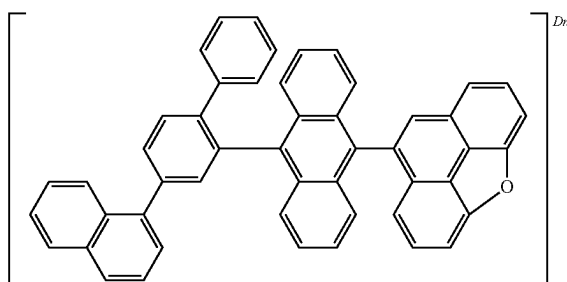


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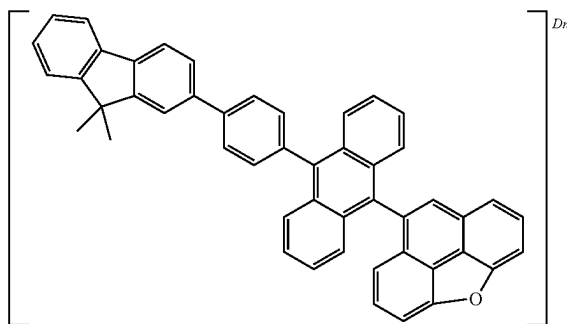
H2-519



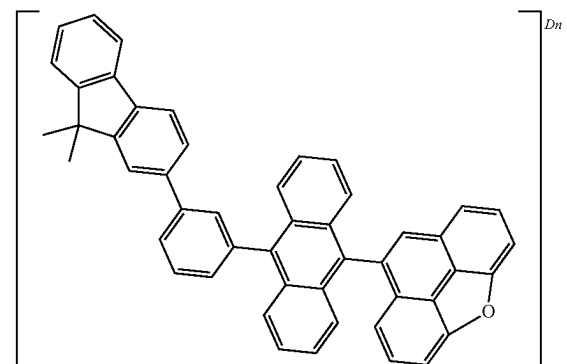
H2-520



H2-521

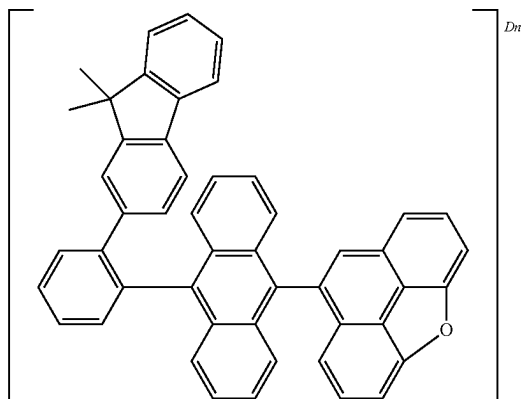


H2-522



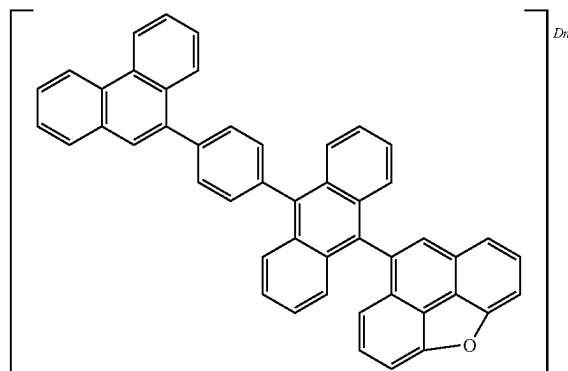
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H2-523

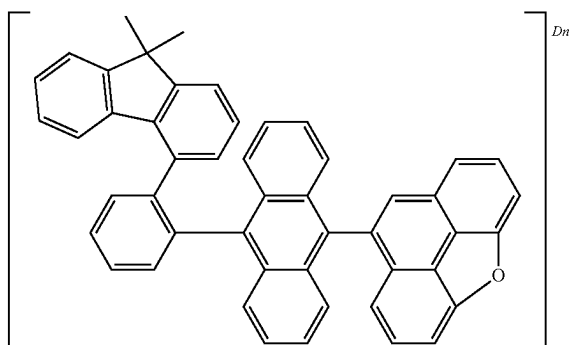


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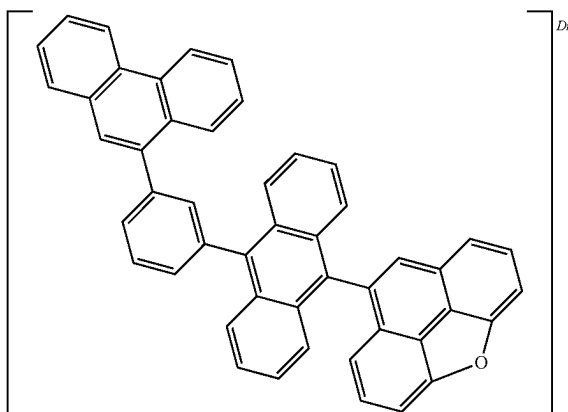
H2-526



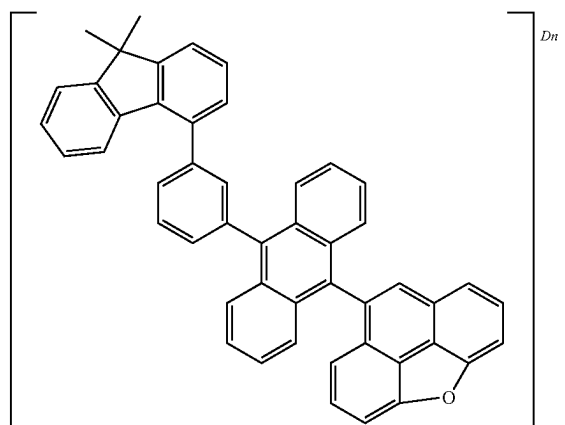
H2-524



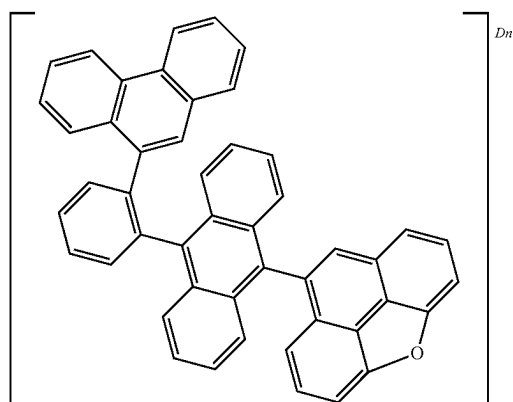
H2-527



H2-525

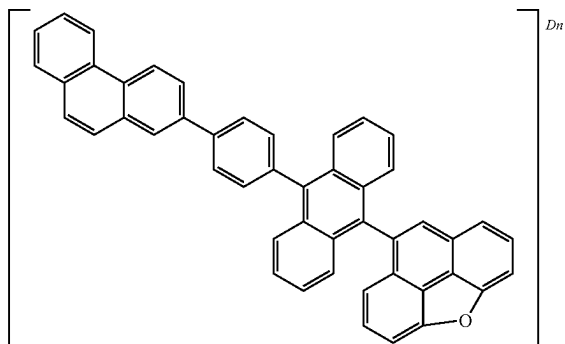


H2-528



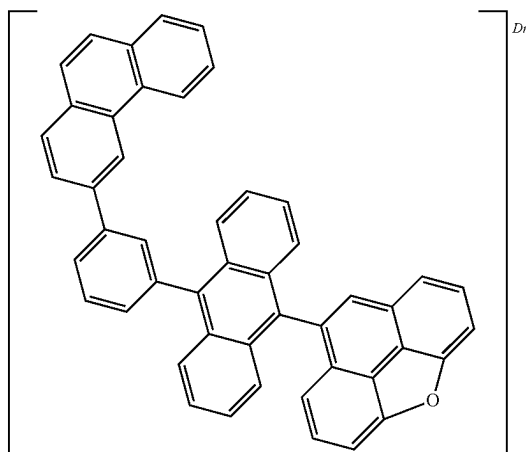
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H2-529

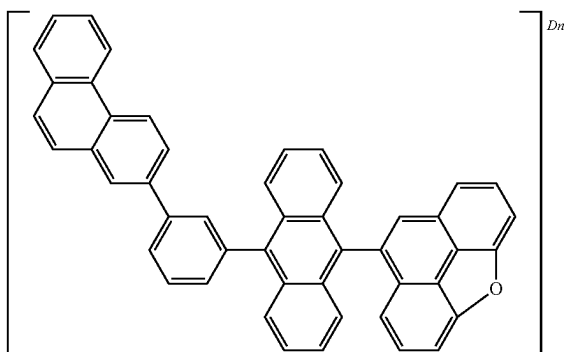


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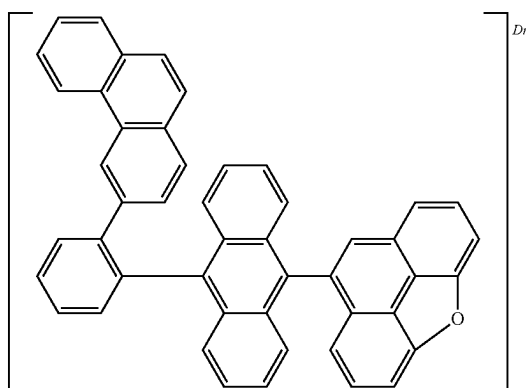
H2-536



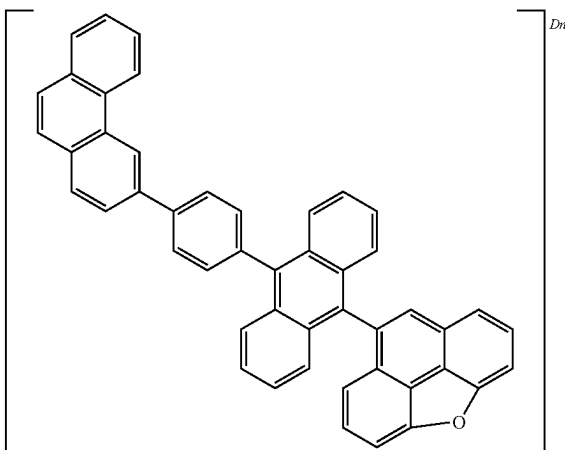
H2-530



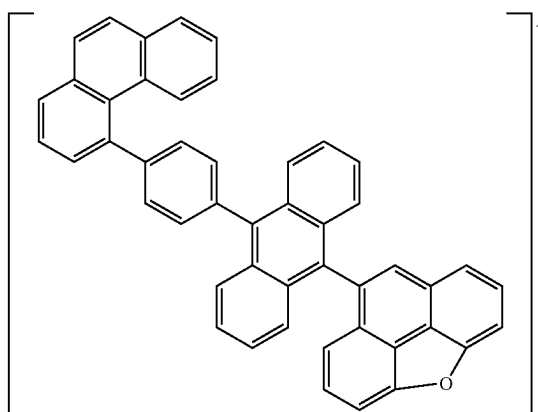
H2-537



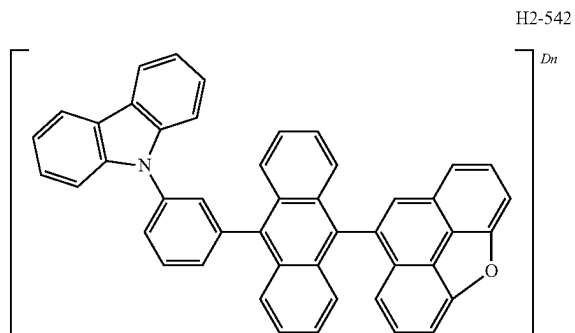
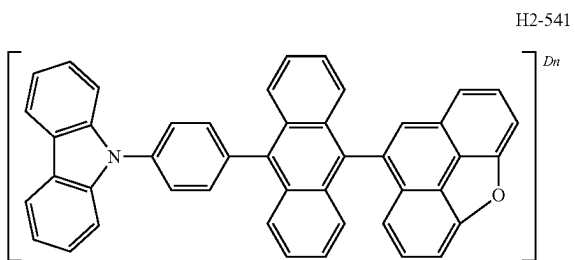
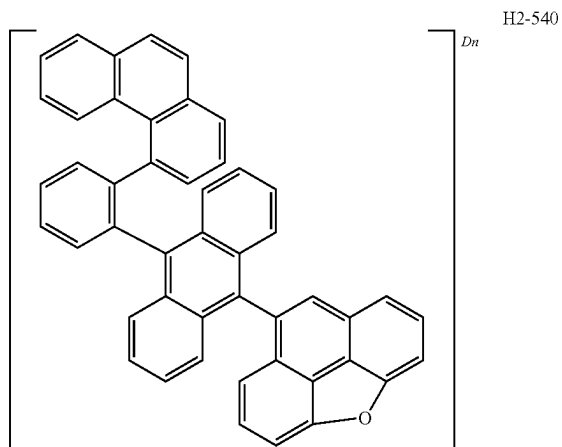
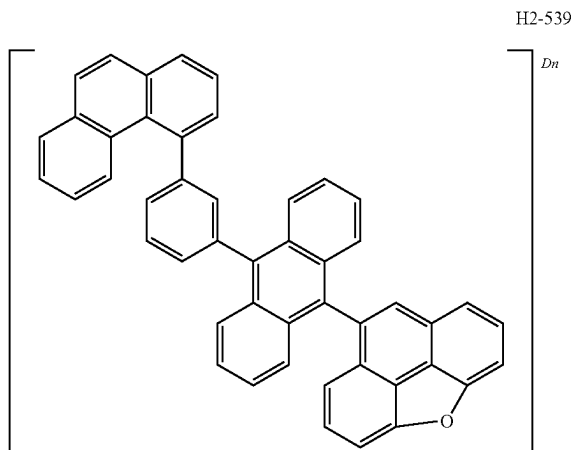
H2-535



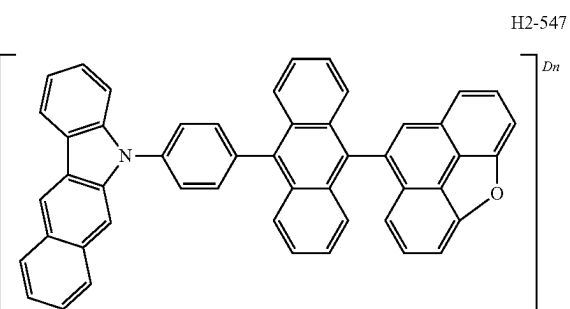
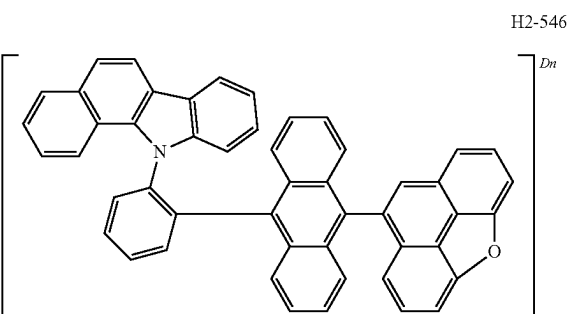
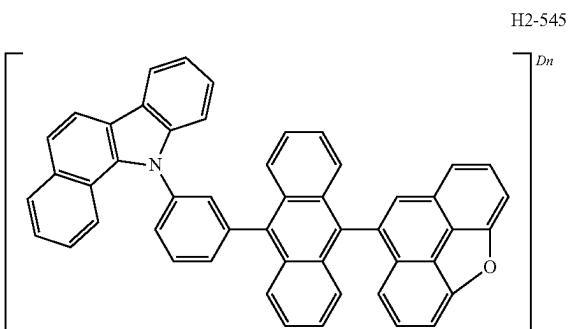
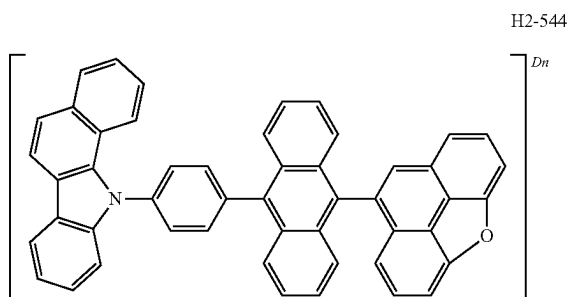
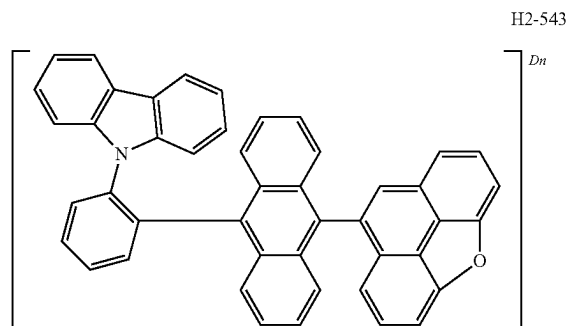
H2-538



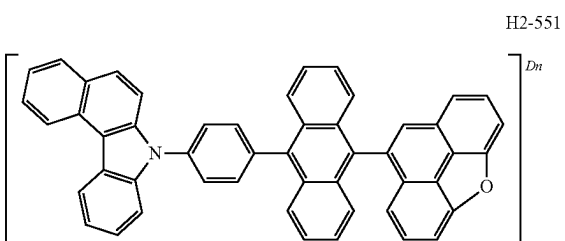
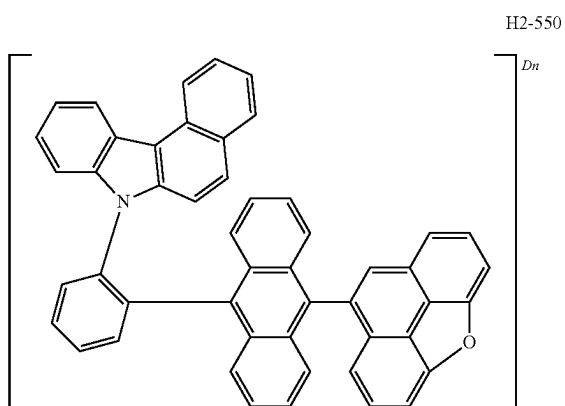
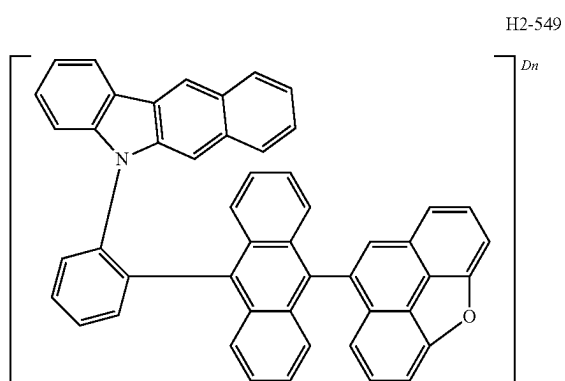
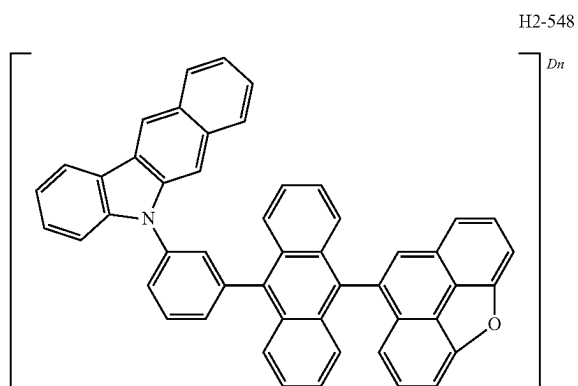
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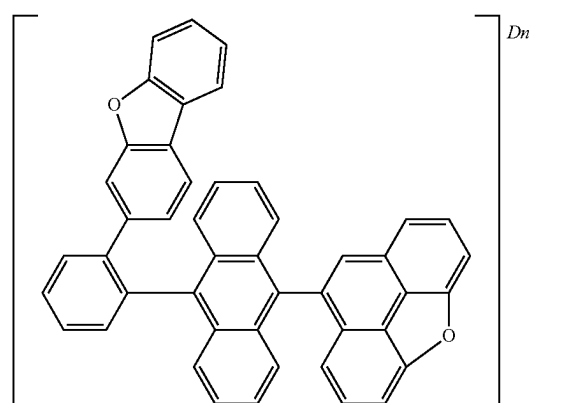
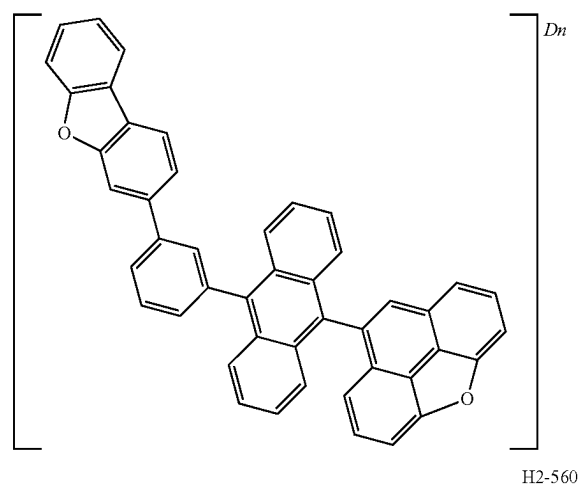
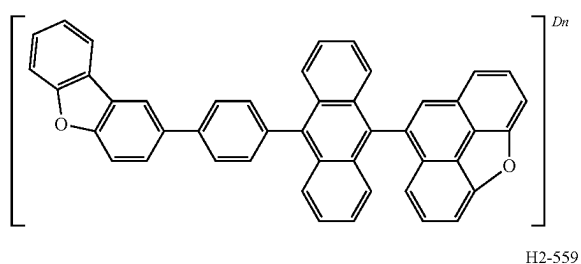
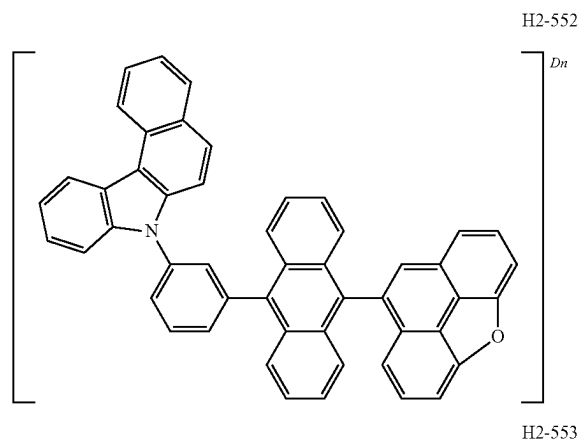
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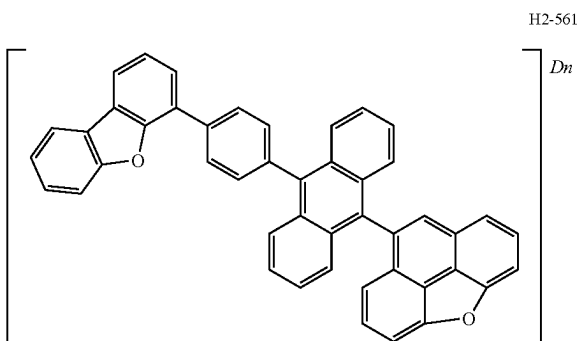
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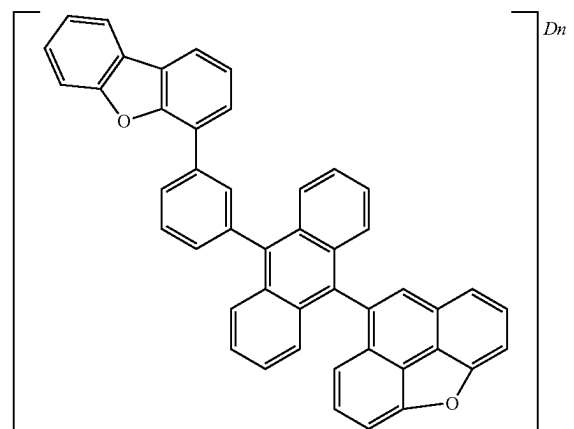
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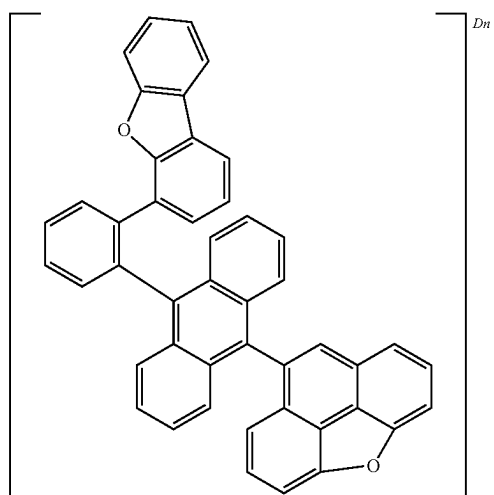
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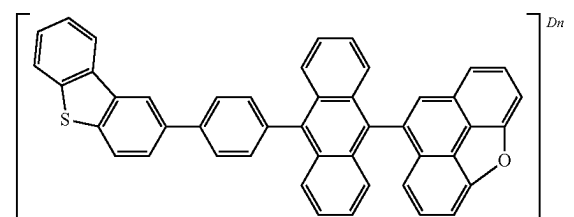
H2-562



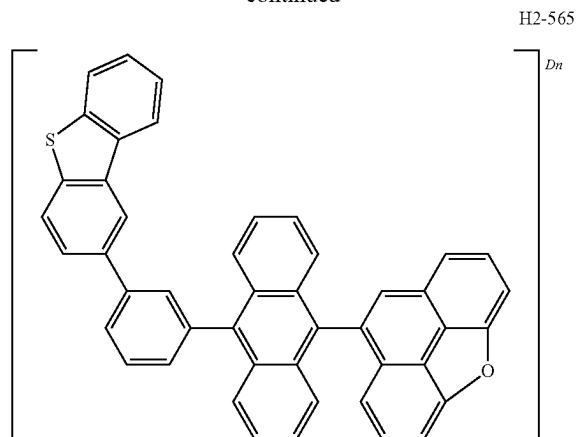
H2-563



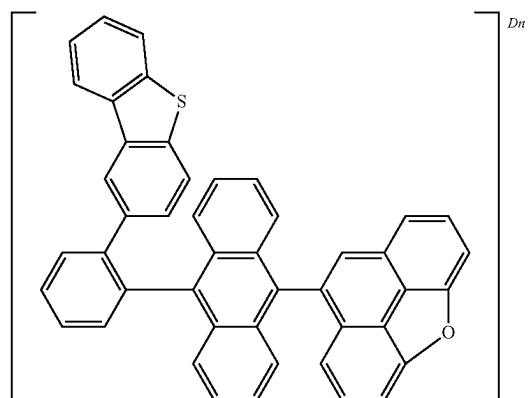
H2-564



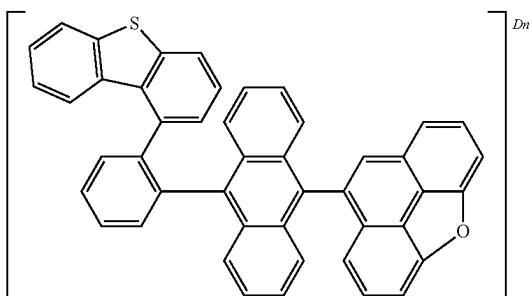
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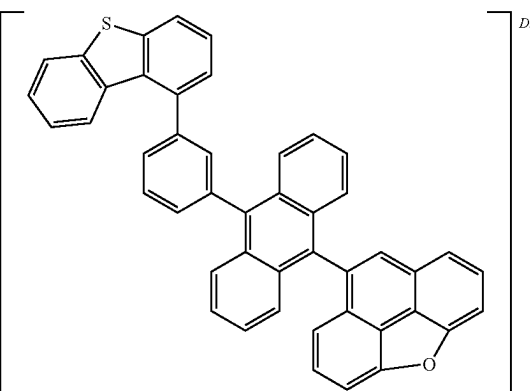
H2-566



H2-567

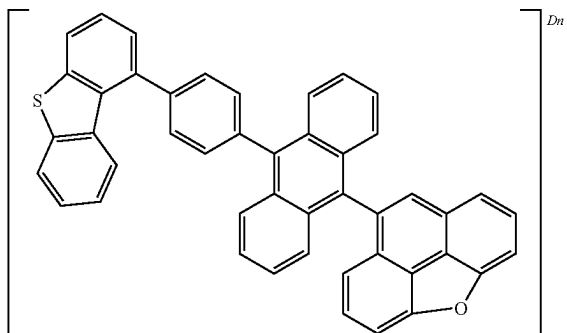


H2-568



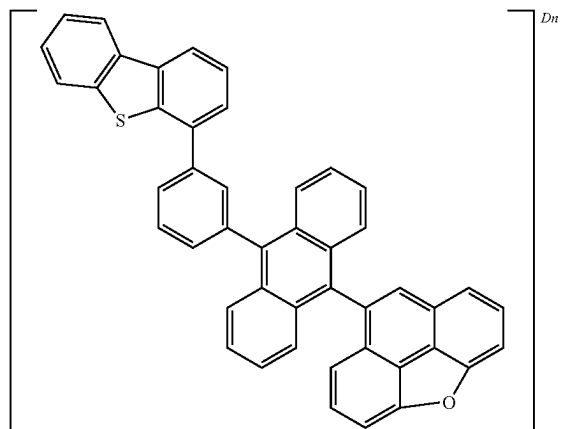
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H2-569

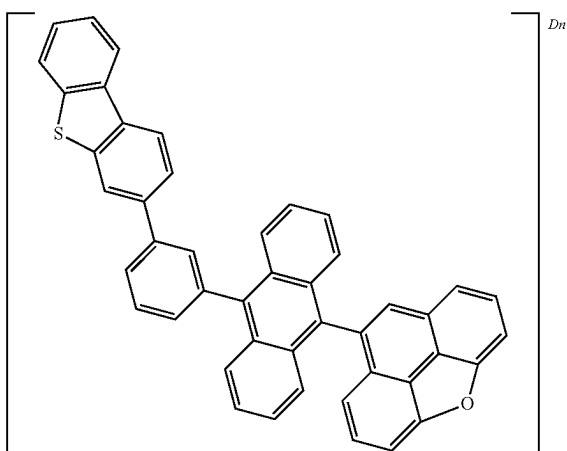


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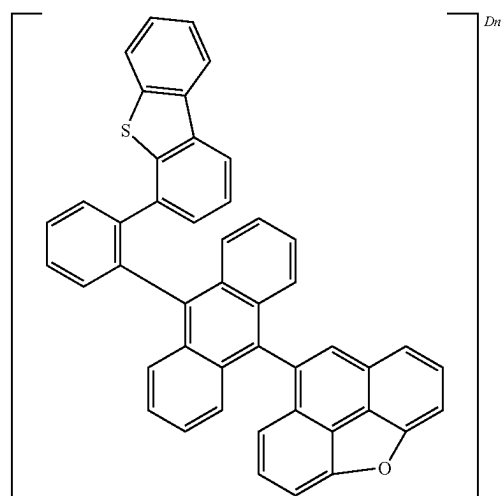
H2-572



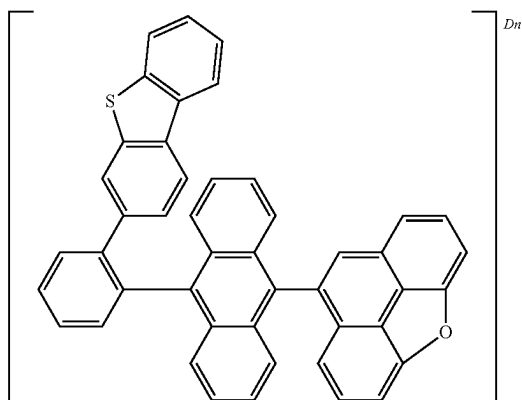
H2-570



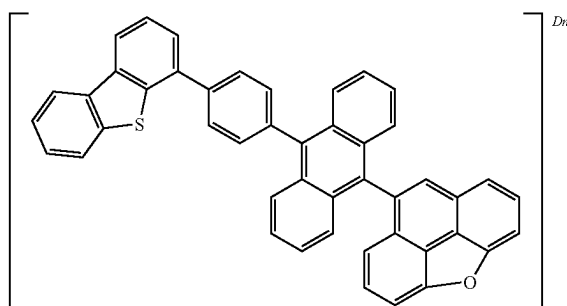
H2-573



H2-571

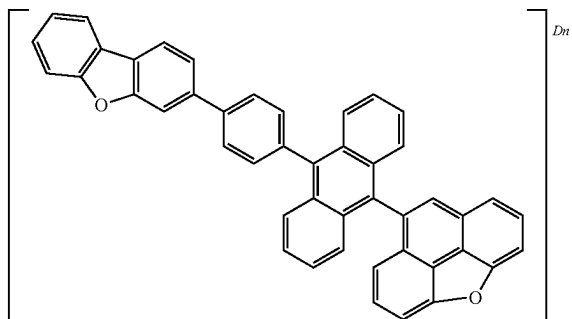


H2-574



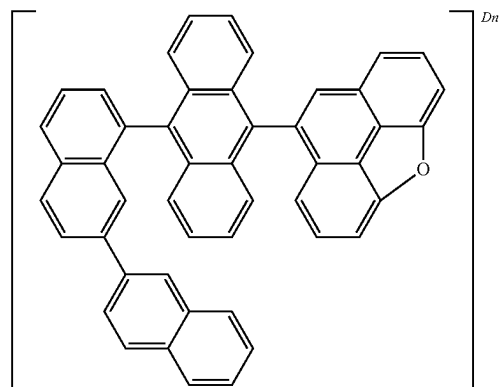
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H2-575

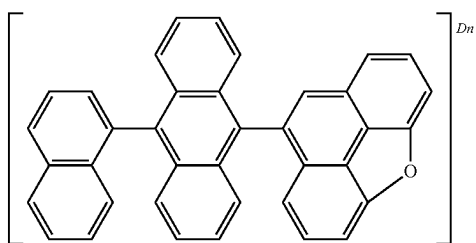


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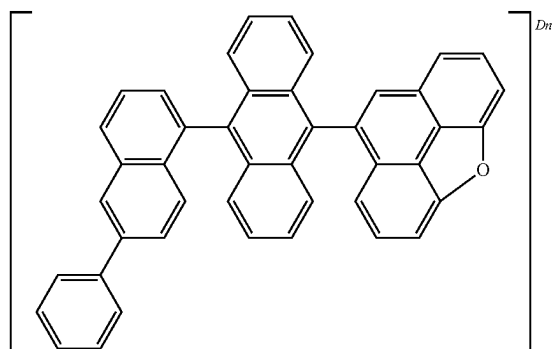
H2-579



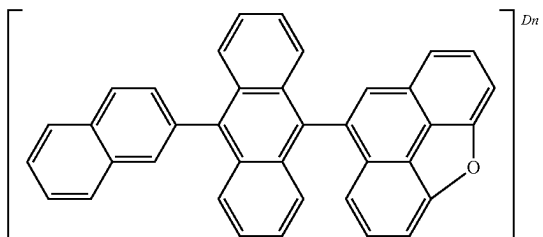
H2-576



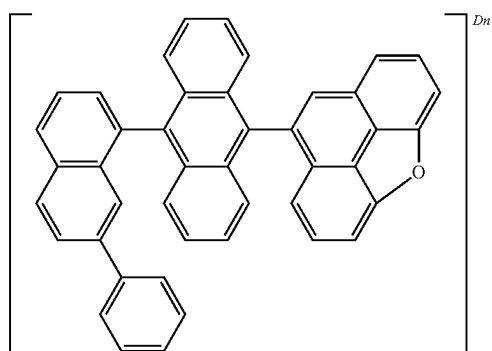
H2-580



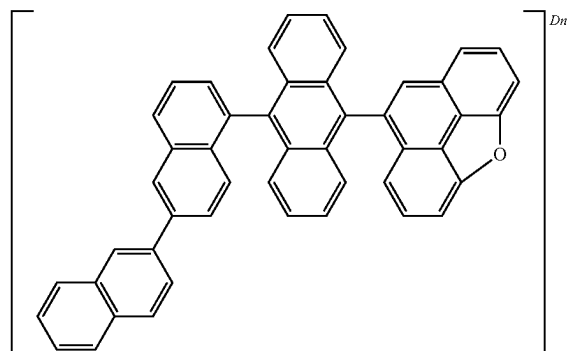
H2-577



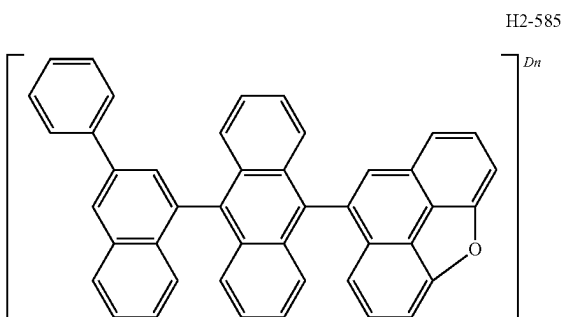
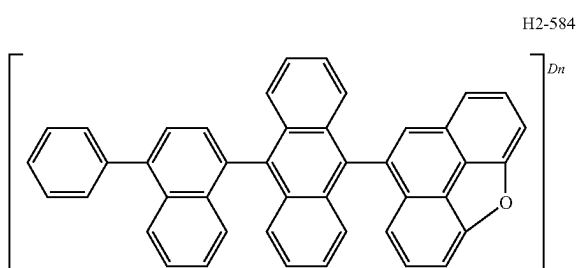
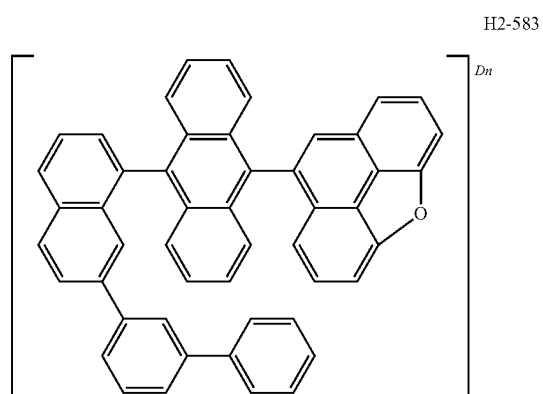
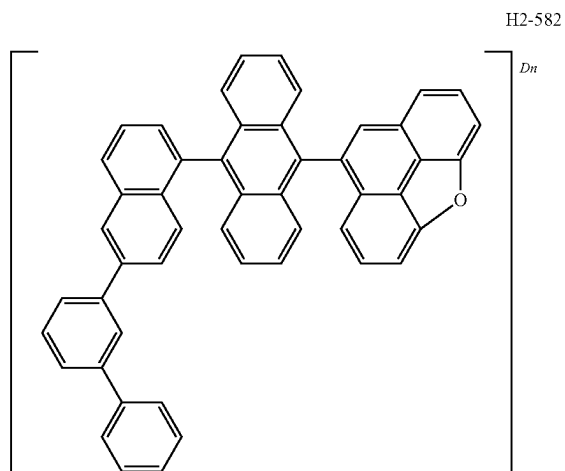
H2-578



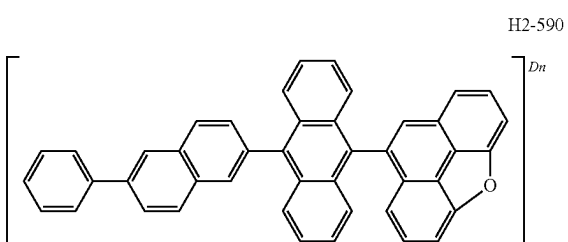
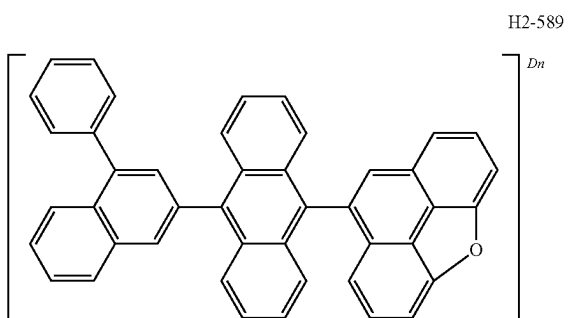
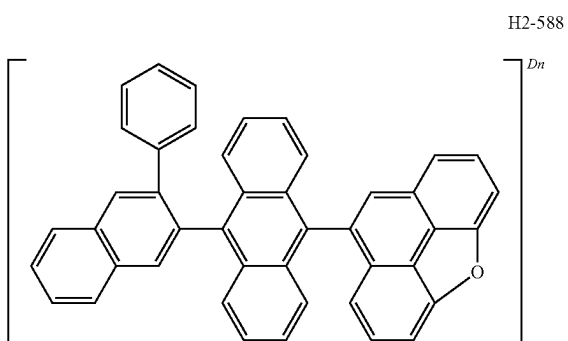
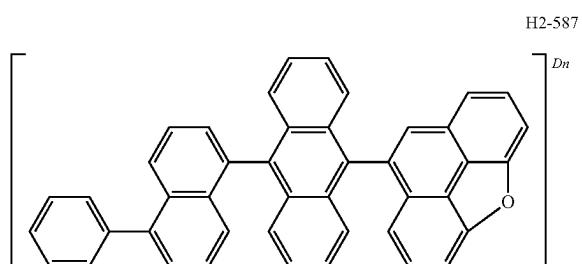
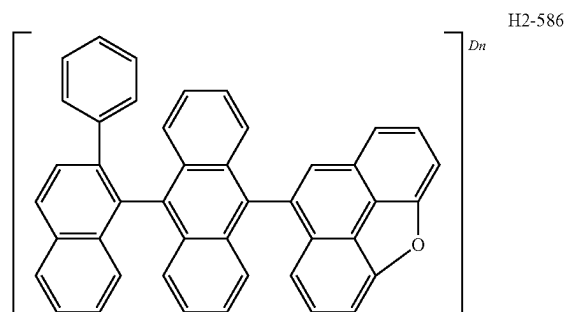
H2-581



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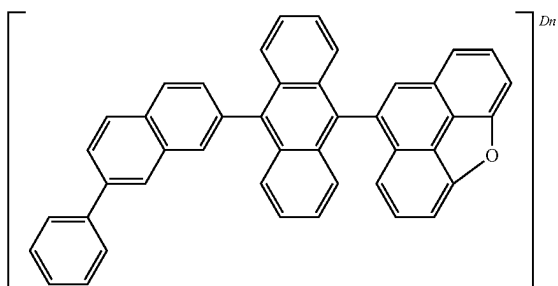


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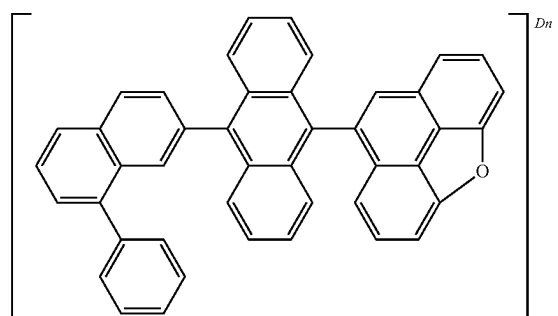


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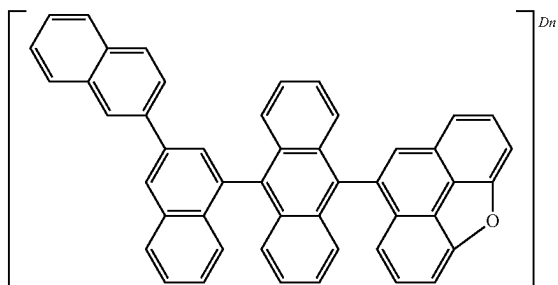
H2-591



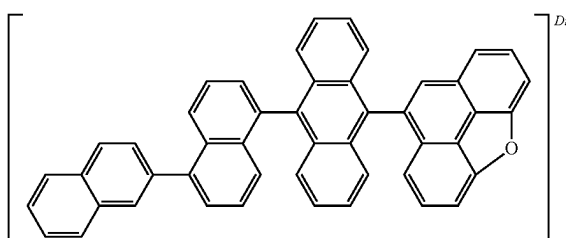
H2-592



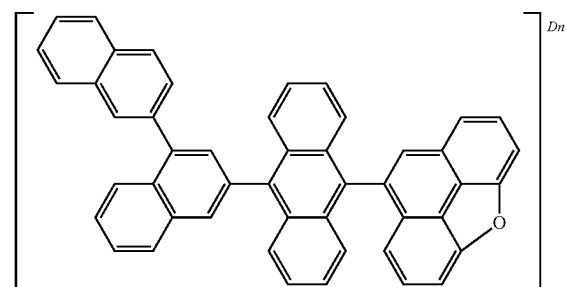
H2-593



H2-594

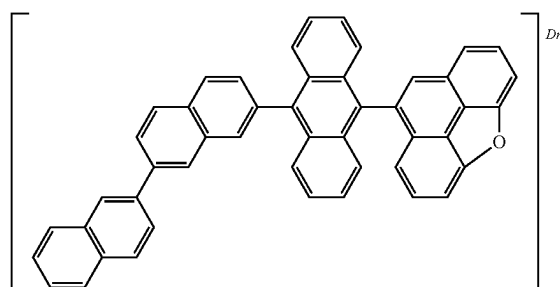


H2-595

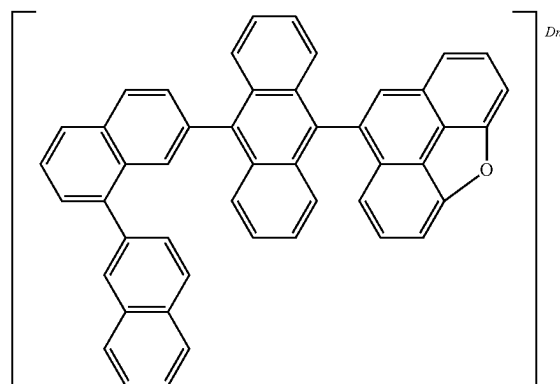


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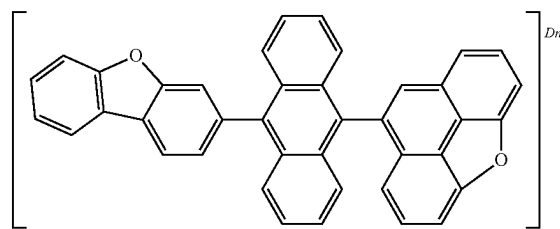
H2-596



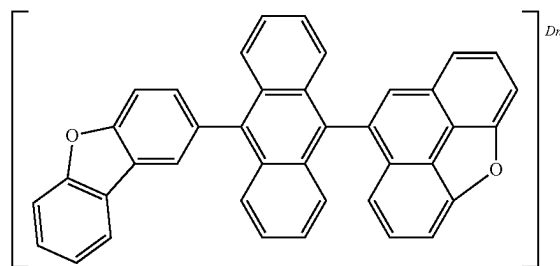
H2-597



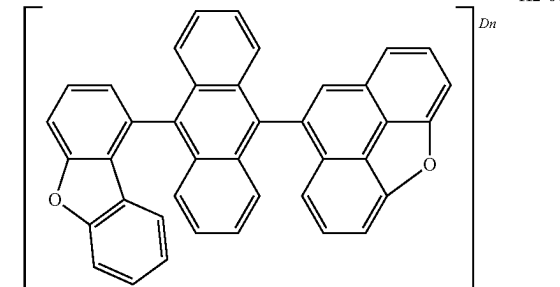
H2-598



H2-599

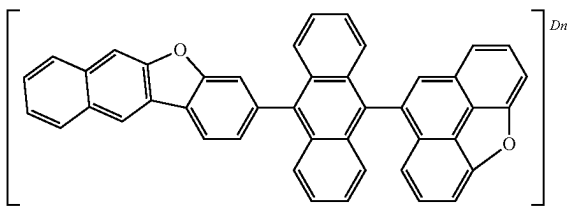


H2-600

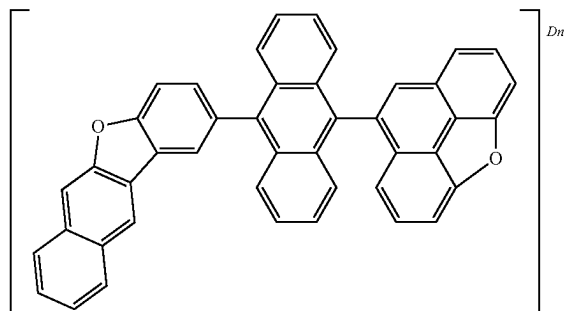


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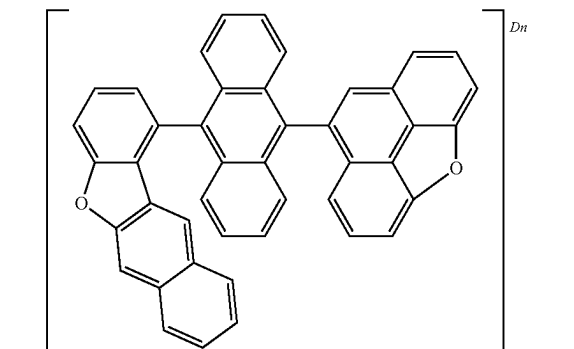
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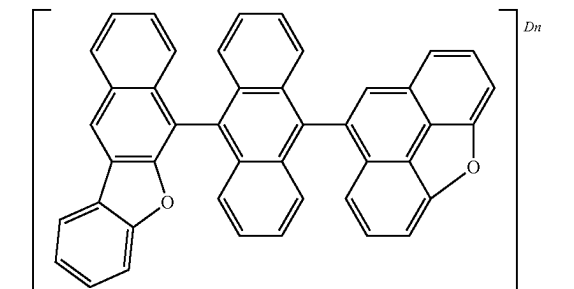
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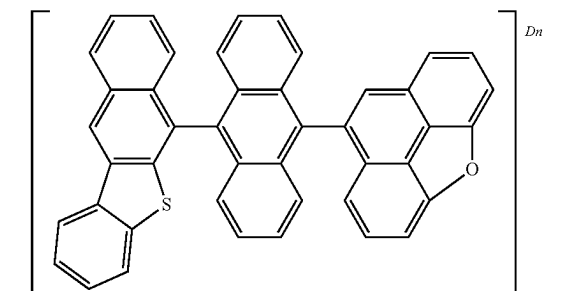
H2-603



H2-604

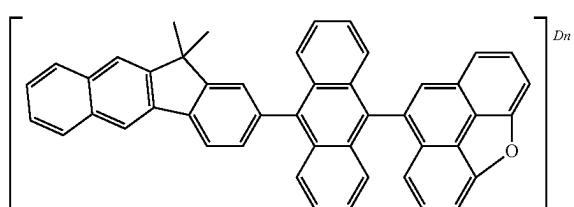


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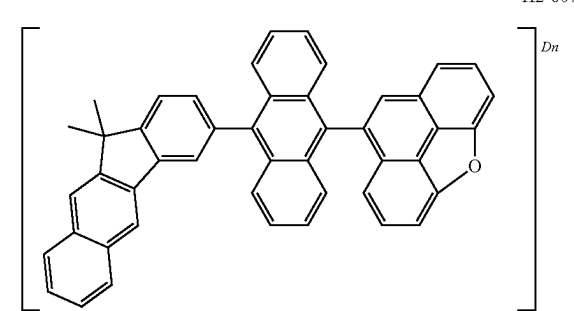


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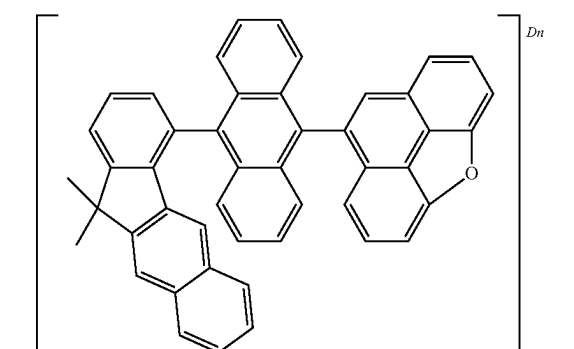
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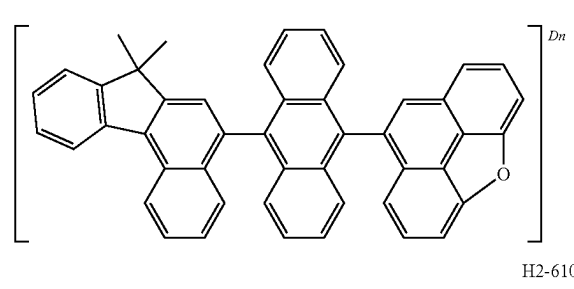
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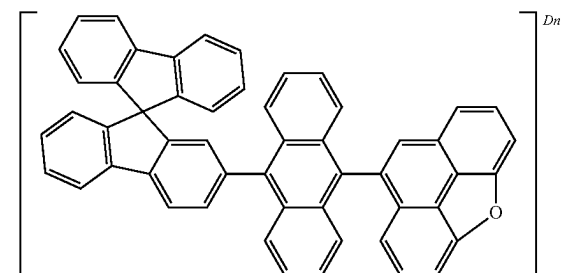
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H2-609

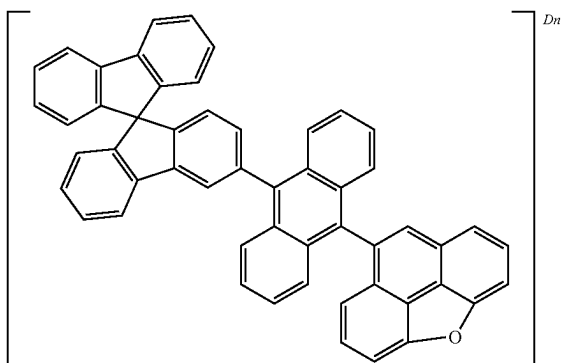


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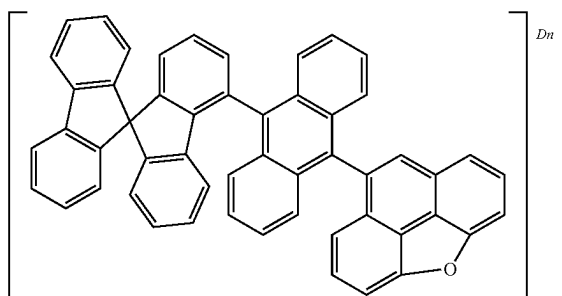


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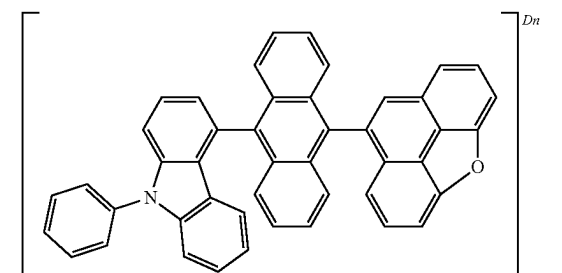
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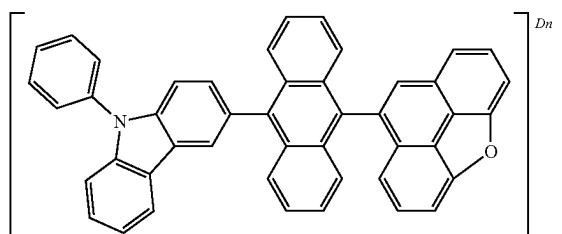
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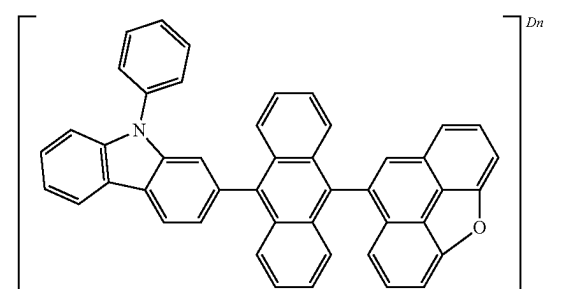
H2-613



H2-614

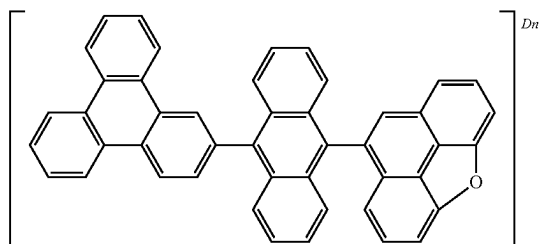


H2-615

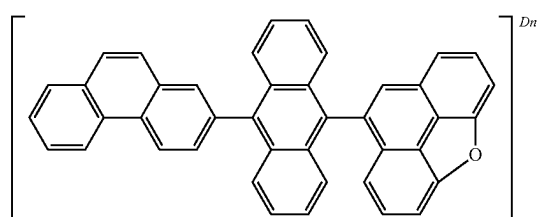


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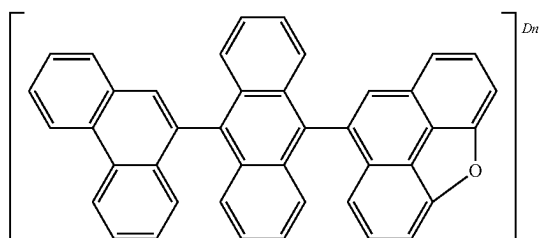
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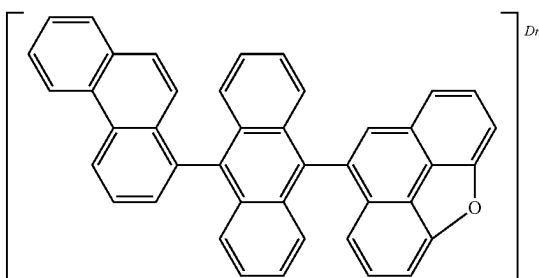
H2-617



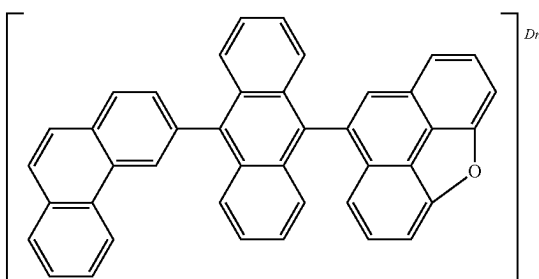
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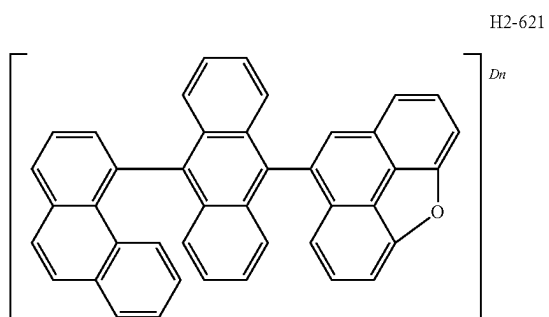
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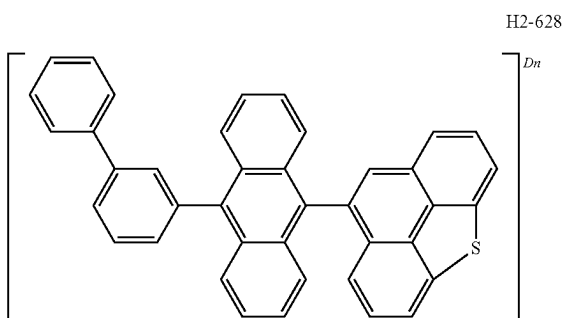
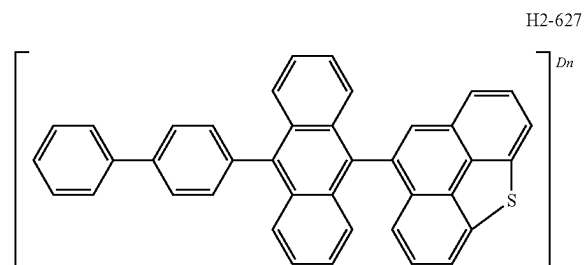
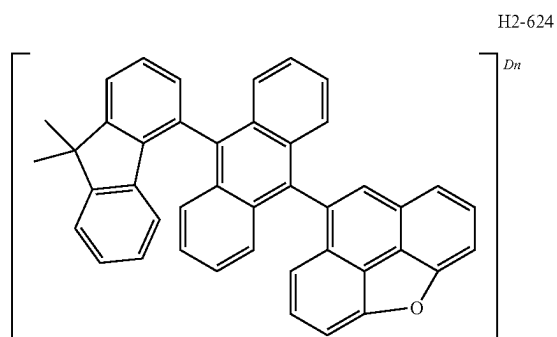
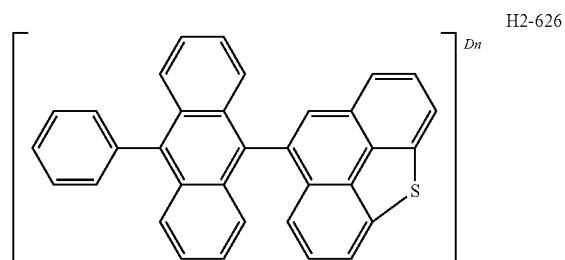
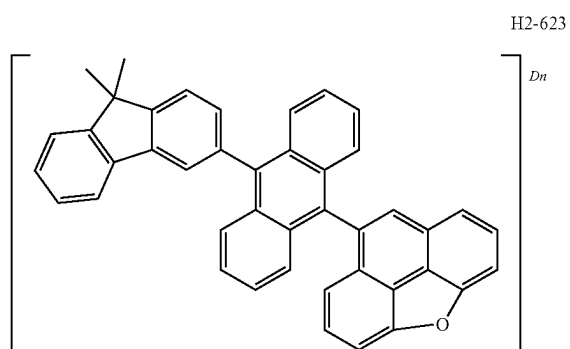
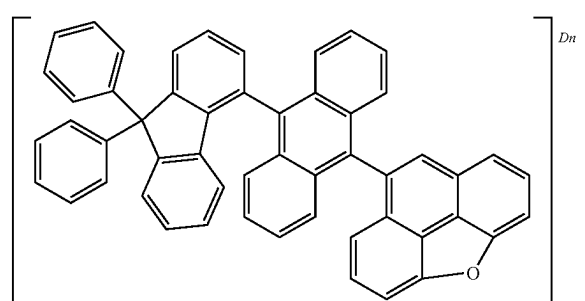
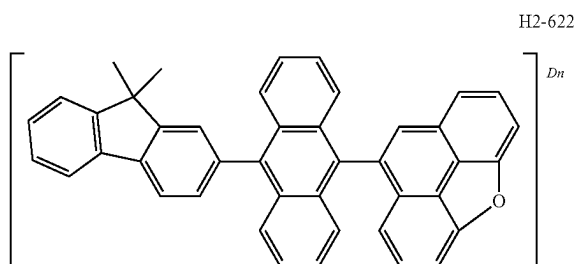
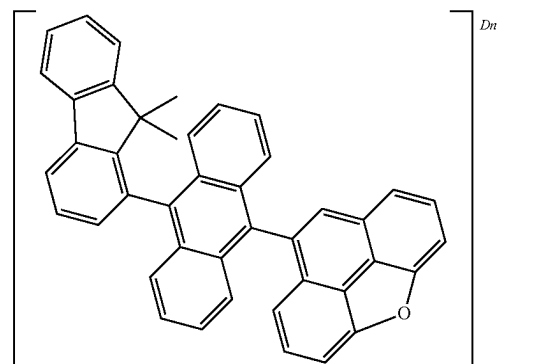
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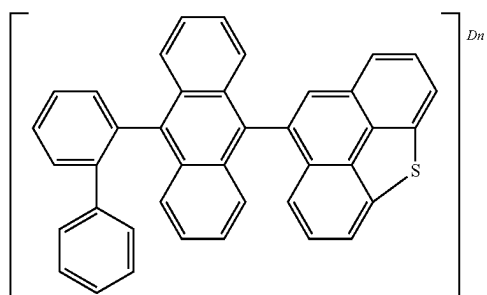
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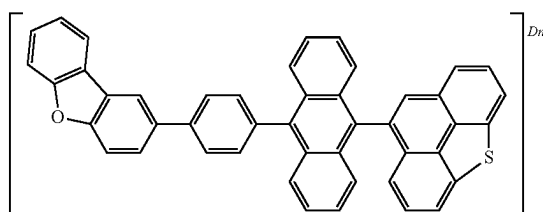


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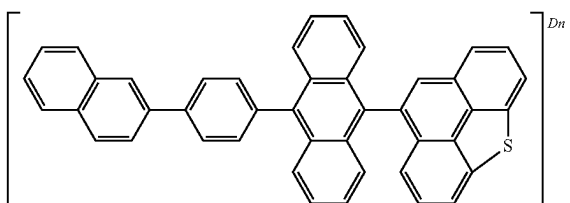


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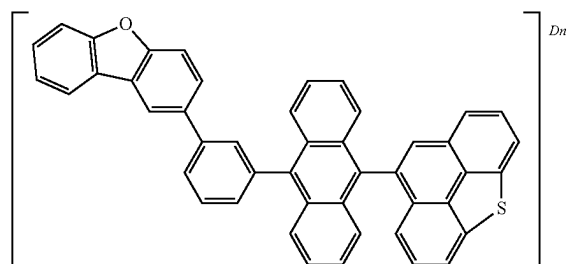
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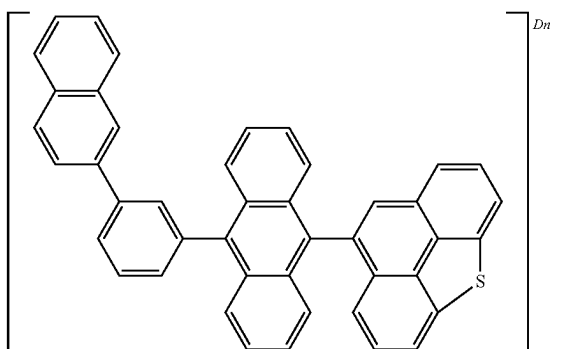
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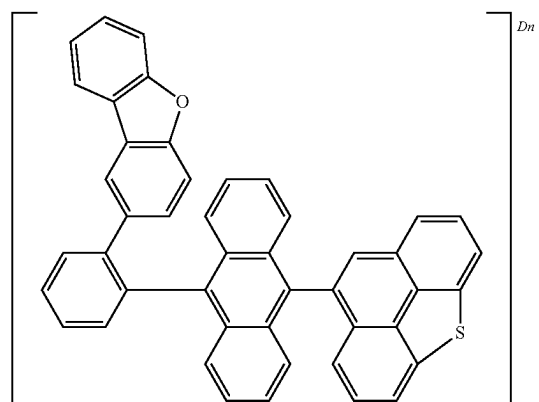
H2-630



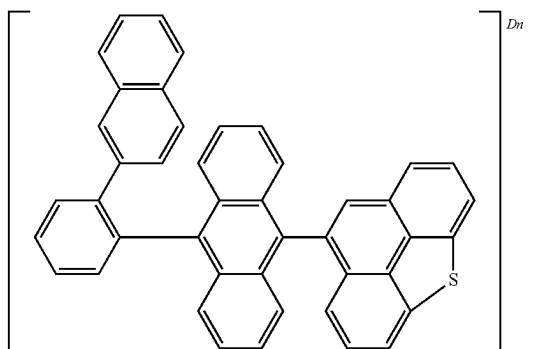
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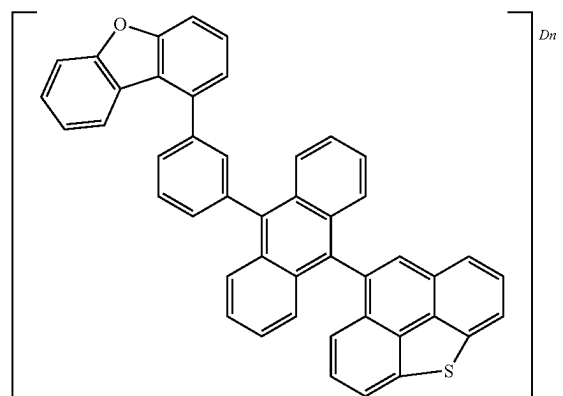
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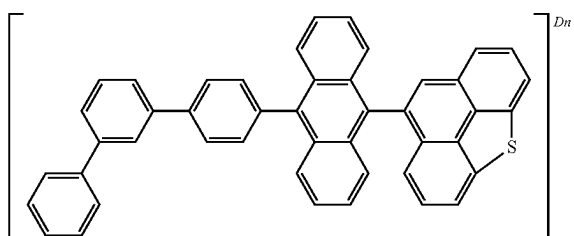
H2-636



H2-632



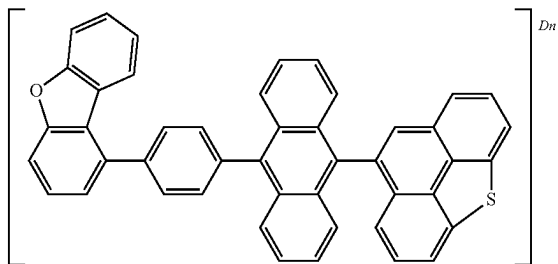
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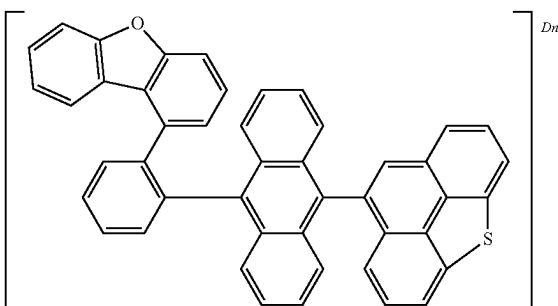
H2-633

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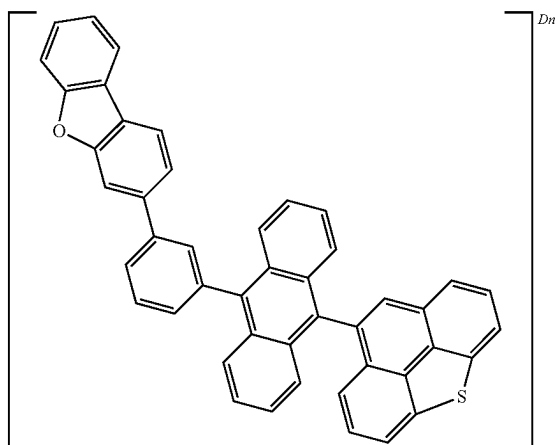
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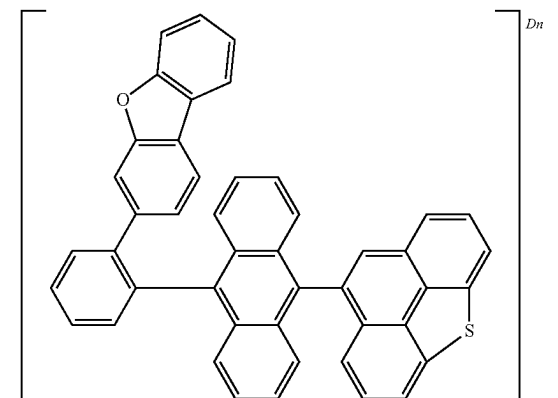
H2-639



H2-640

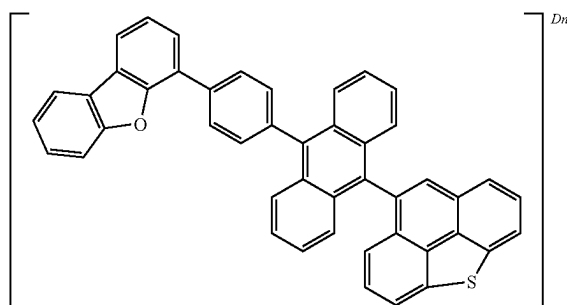


H2-641

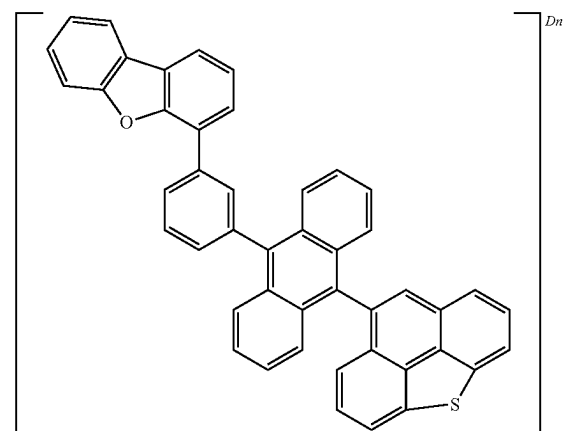


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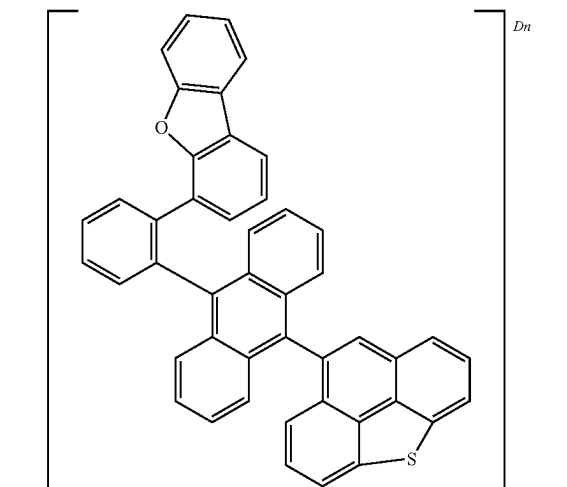
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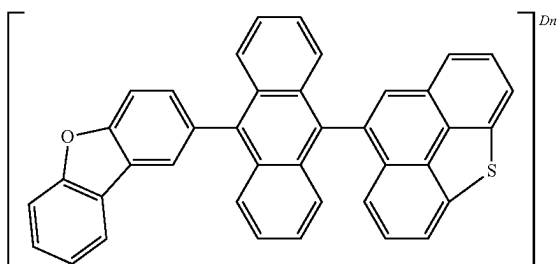
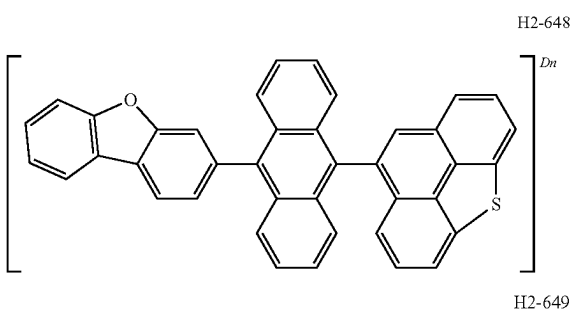
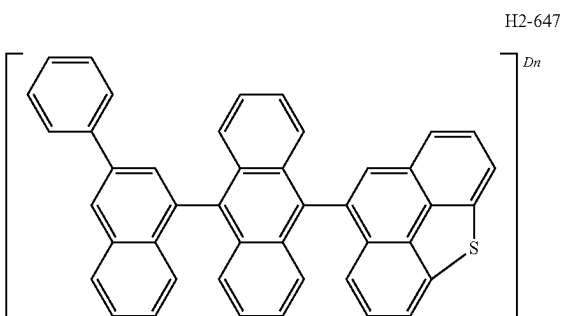
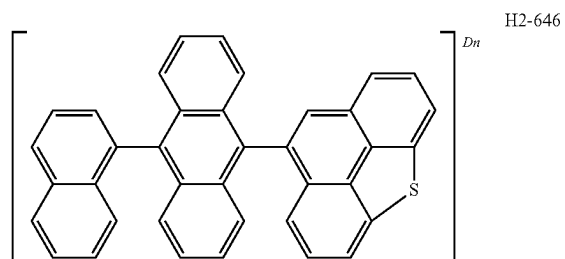
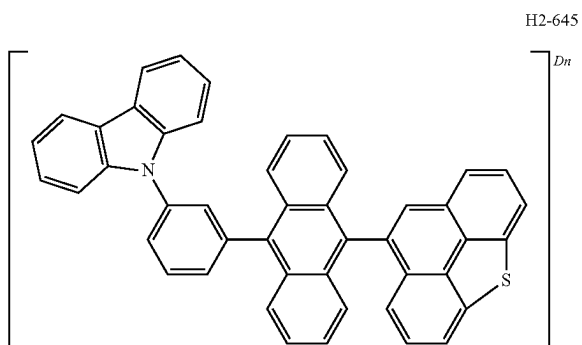
H2-643



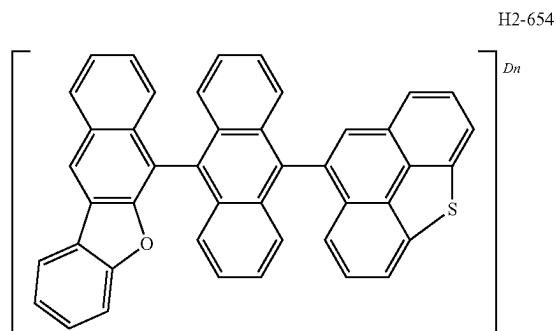
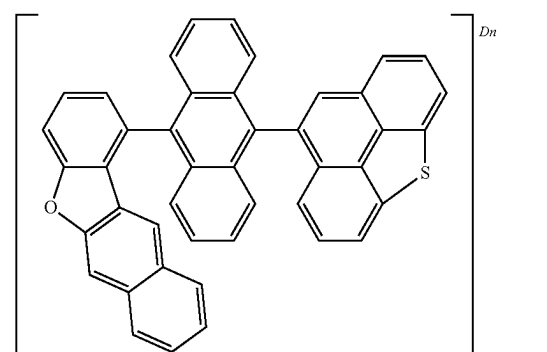
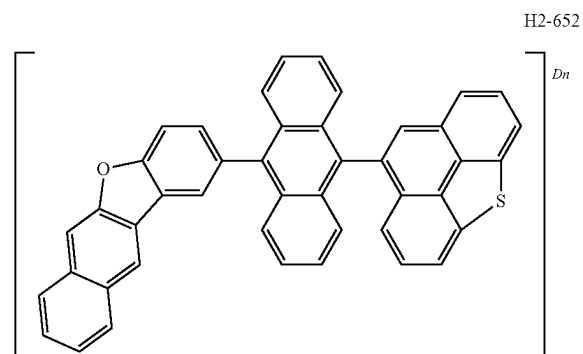
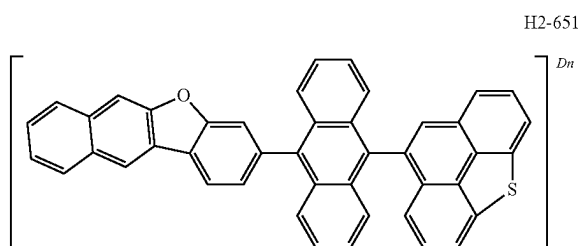
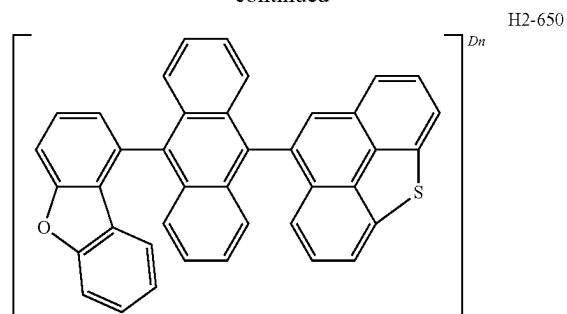
H2-644



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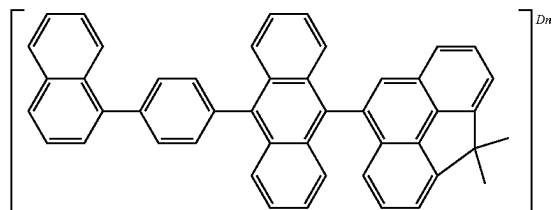


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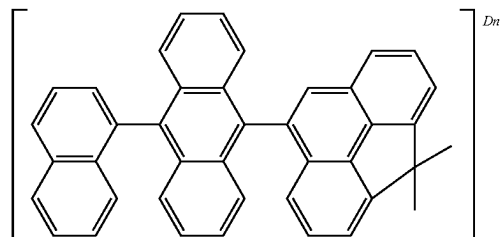


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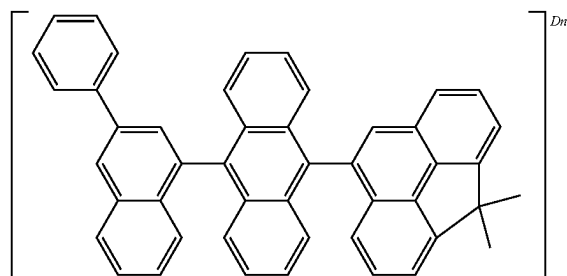
H2-660



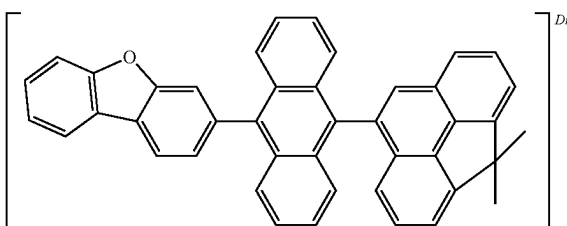
H2-661



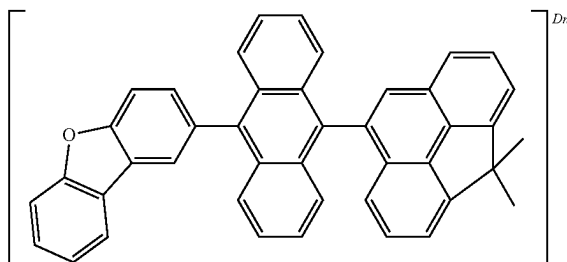
H2-662



H2-663

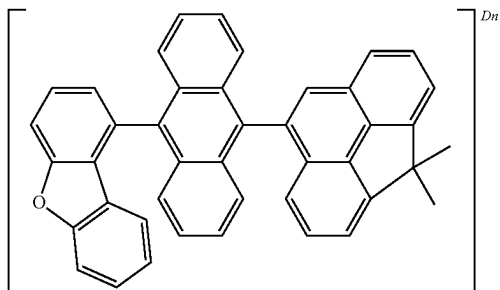


H2-664

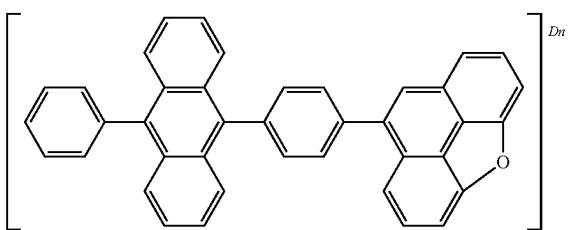


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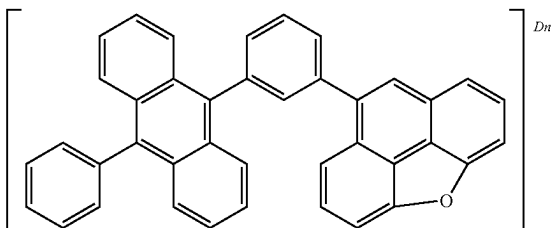
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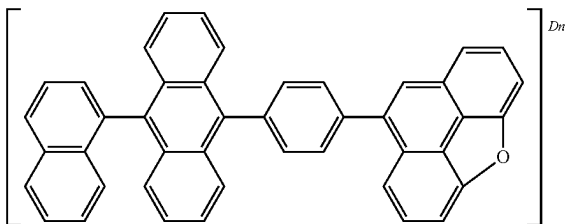
H2-666



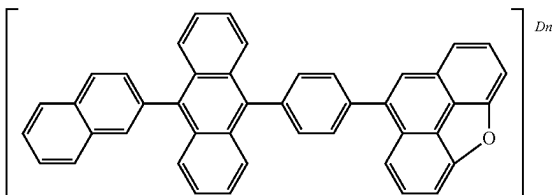
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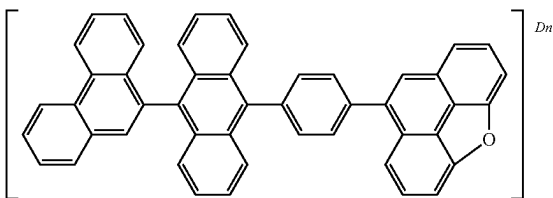
H2-668



H2-669

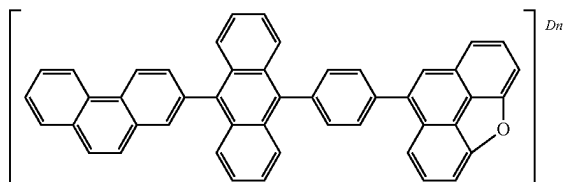


H2-670

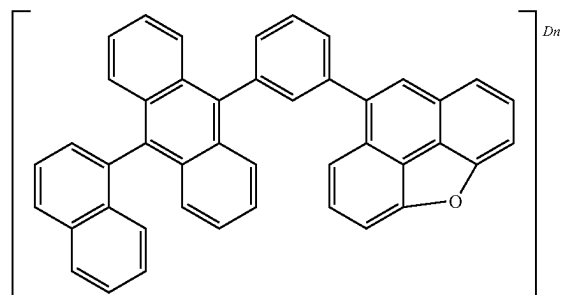


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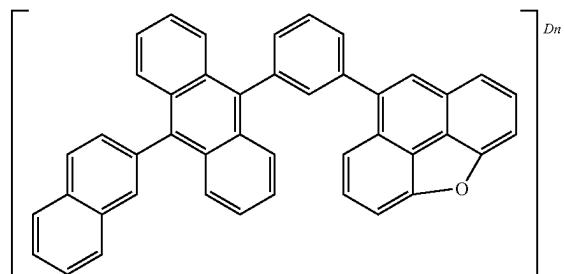
H2-671



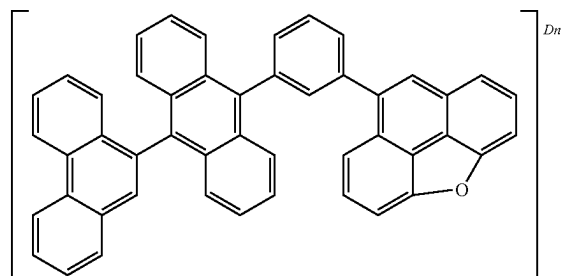
H2-672



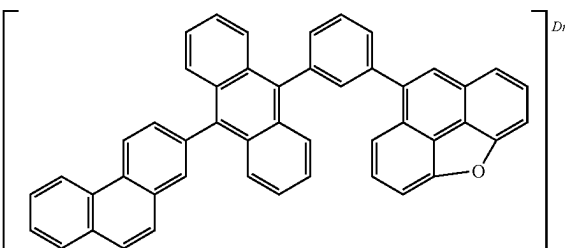
H2-673



H2-674

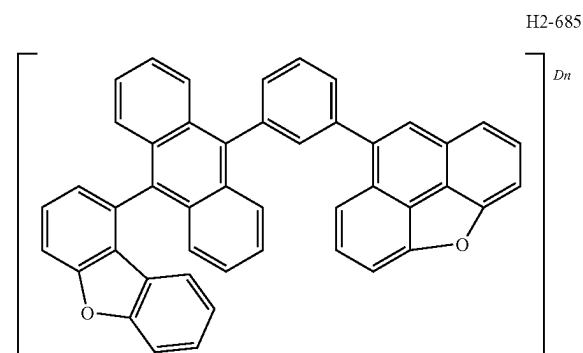
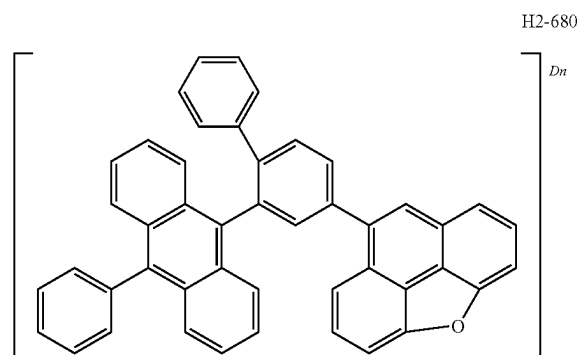
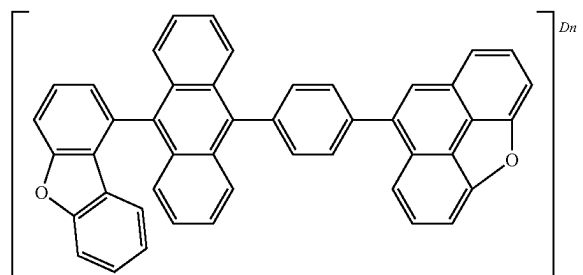
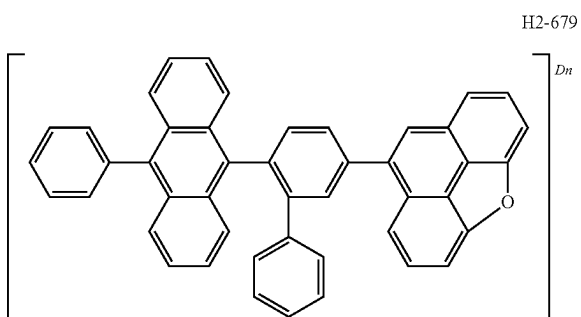
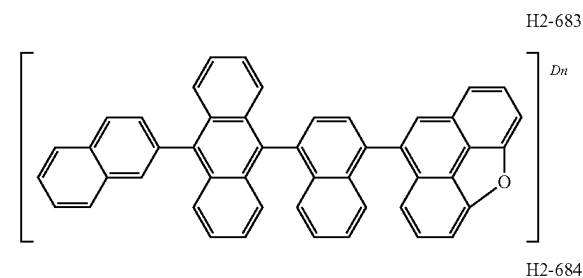
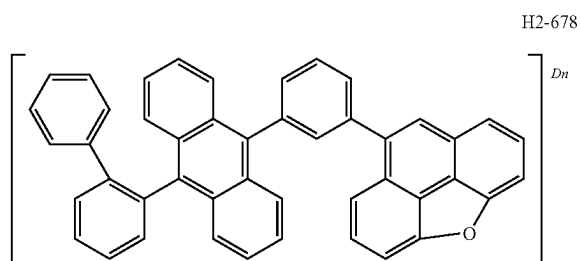
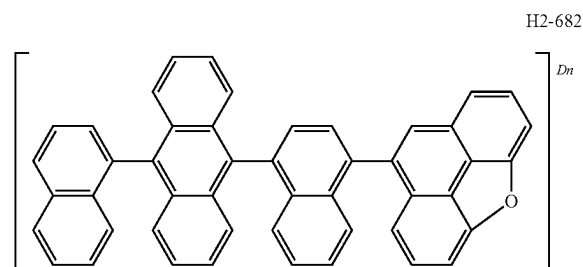
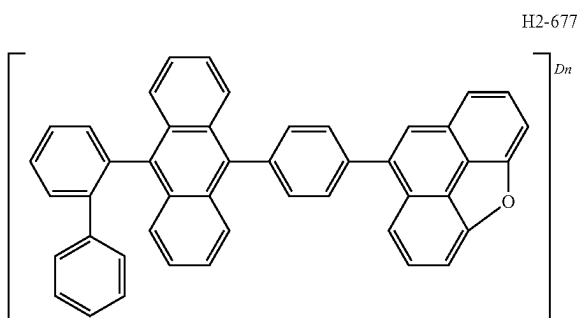
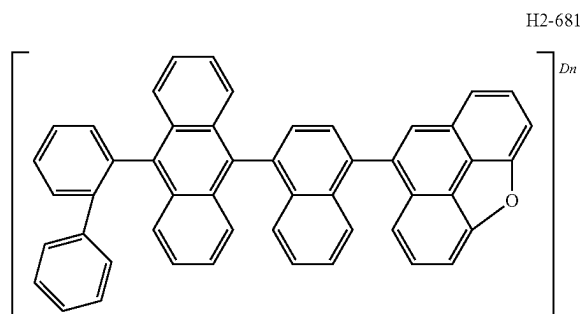
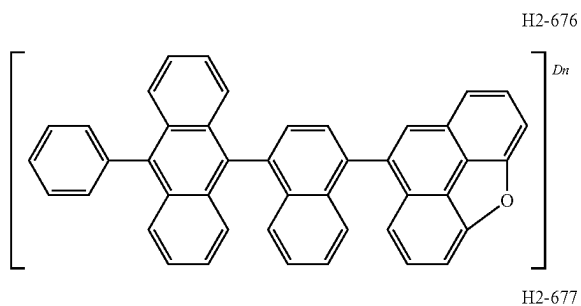


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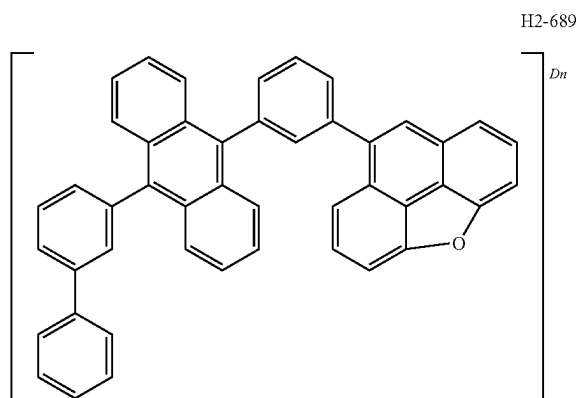
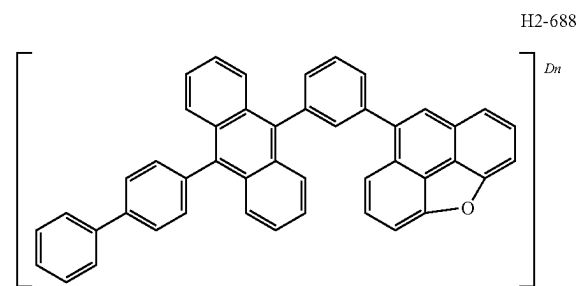
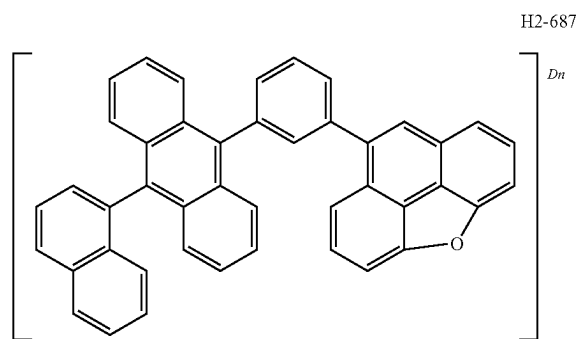
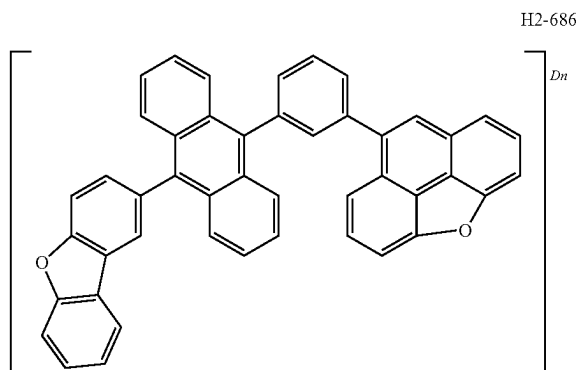


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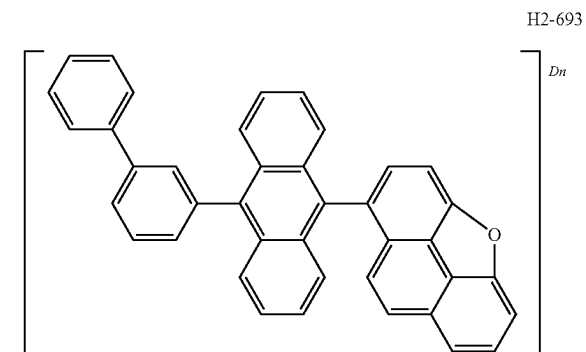
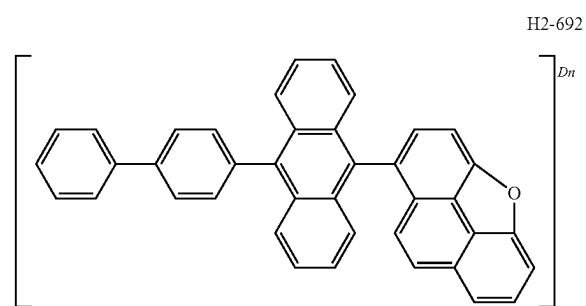
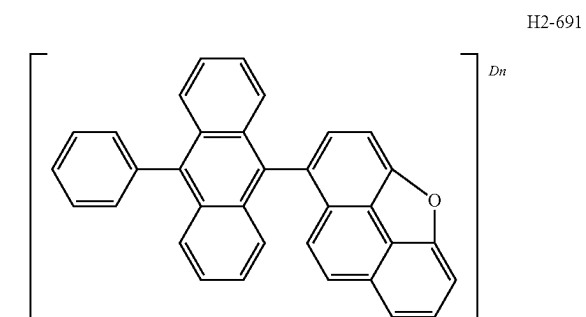
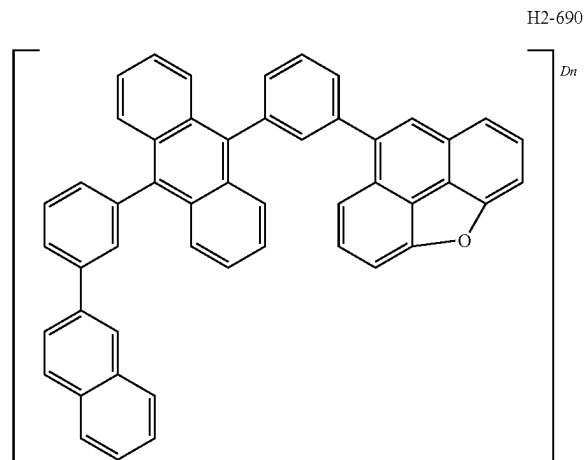
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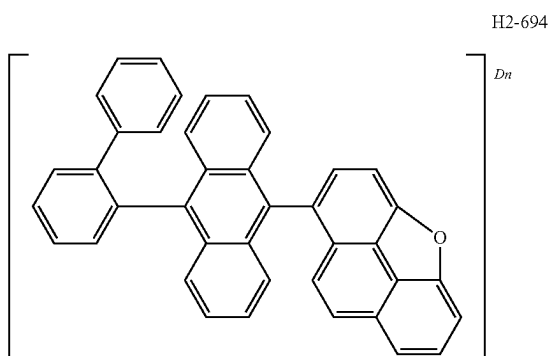
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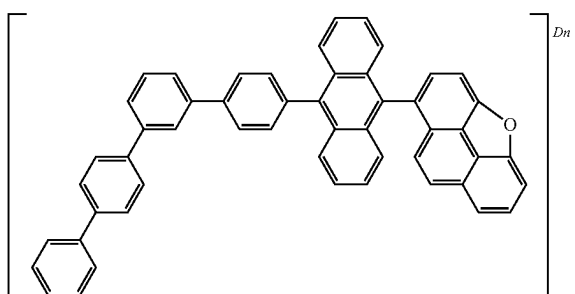
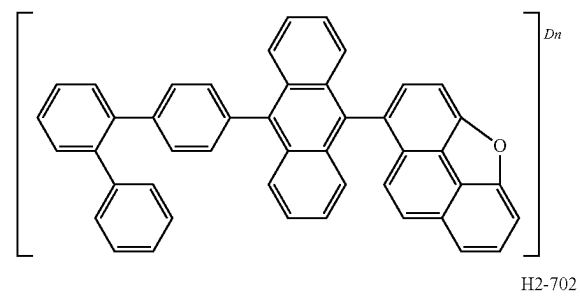
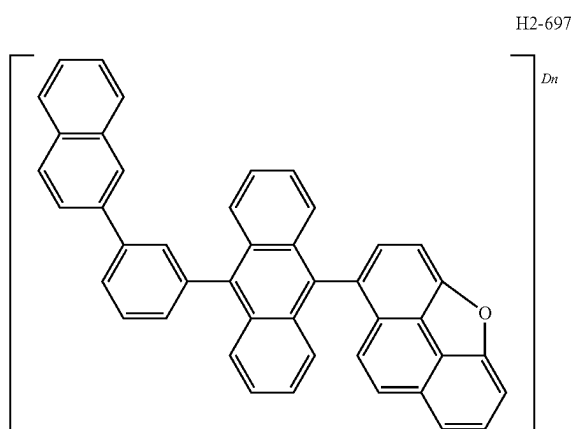
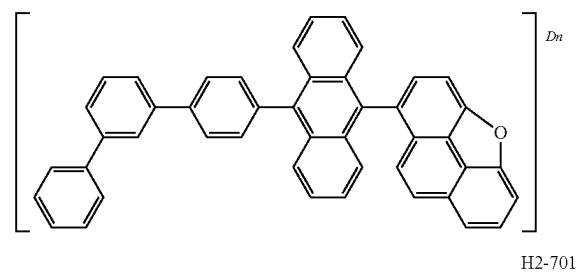
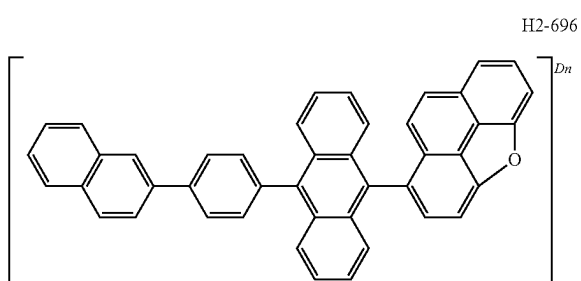
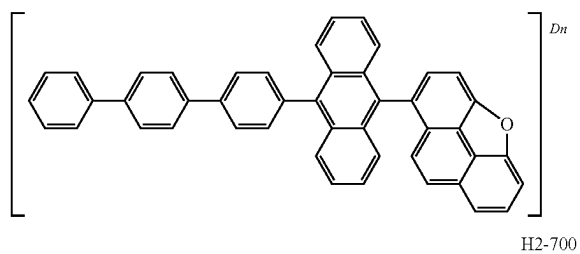
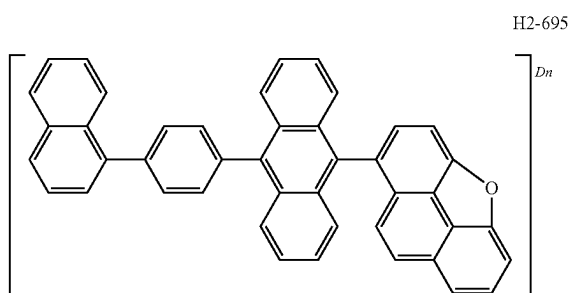
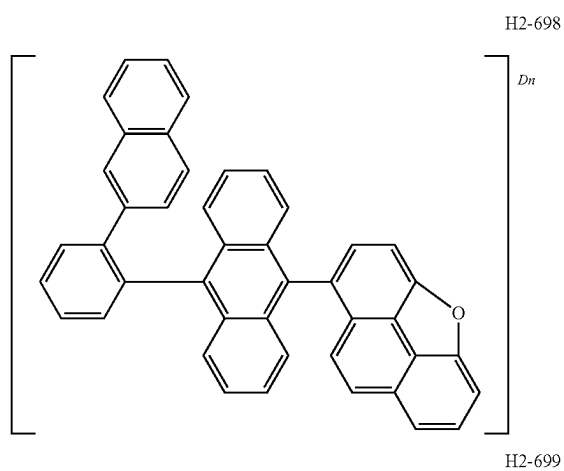
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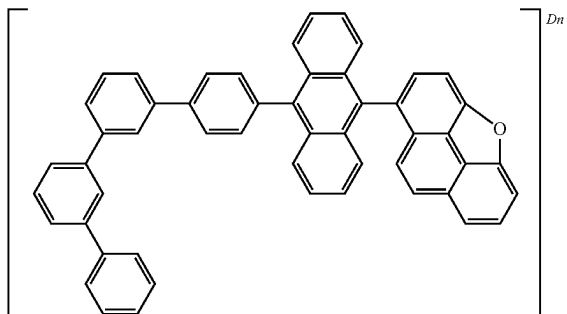


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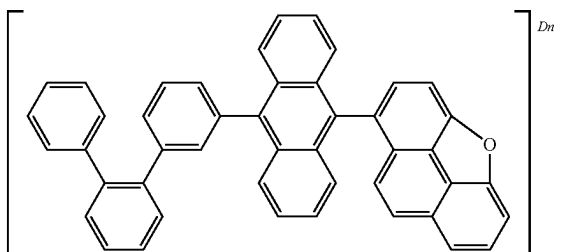


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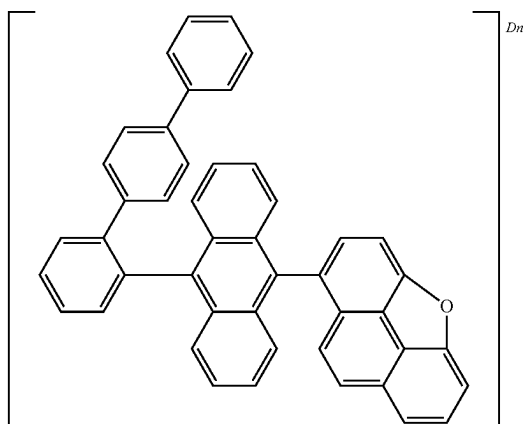
H2-703



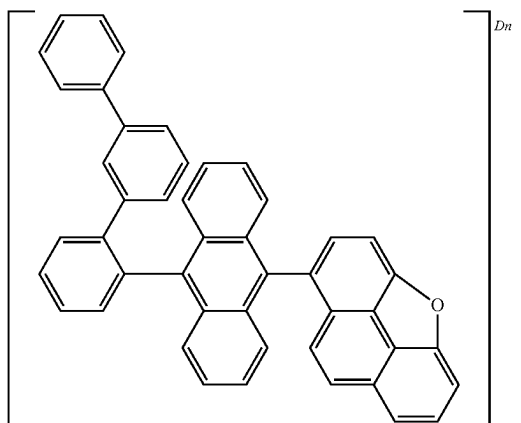
H2-704



H2-705

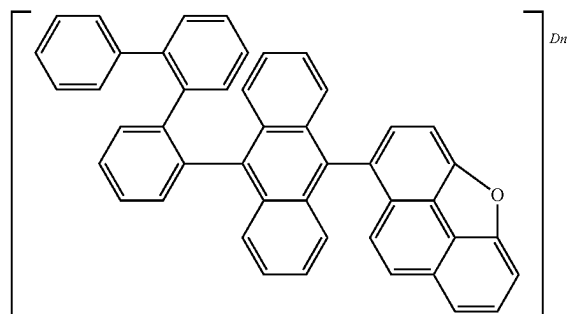


H2-706

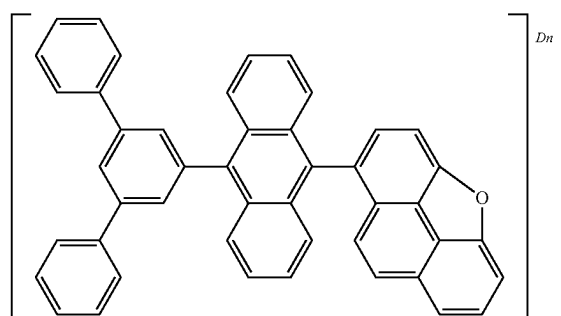


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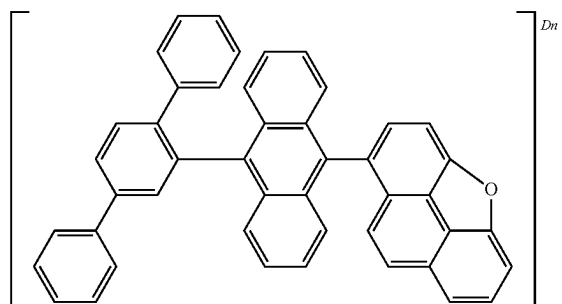
H2-707



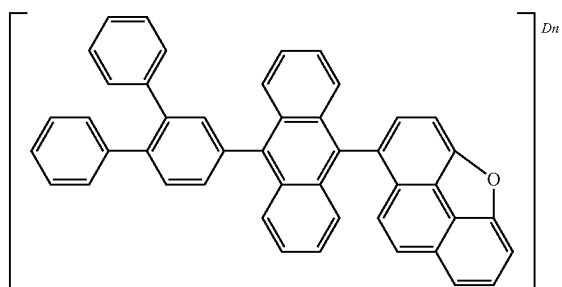
H2-708



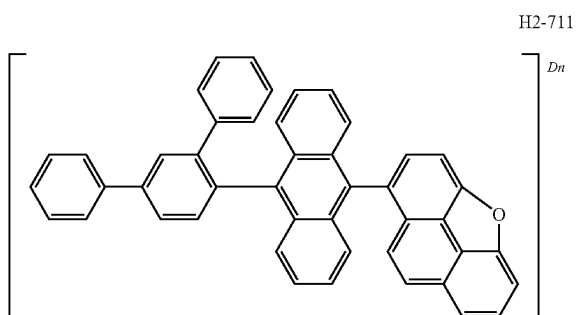
H2-709



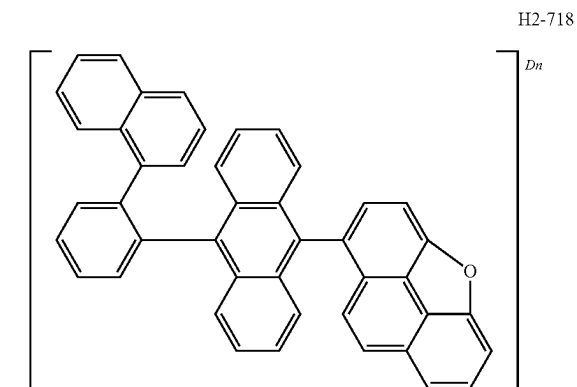
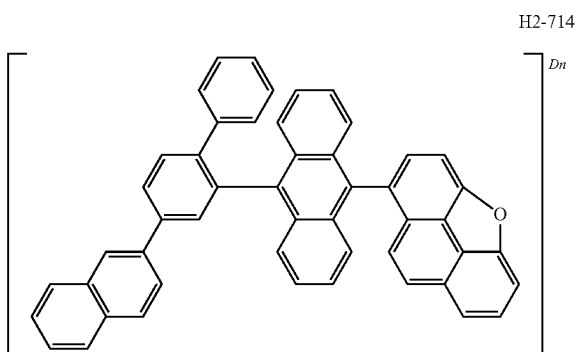
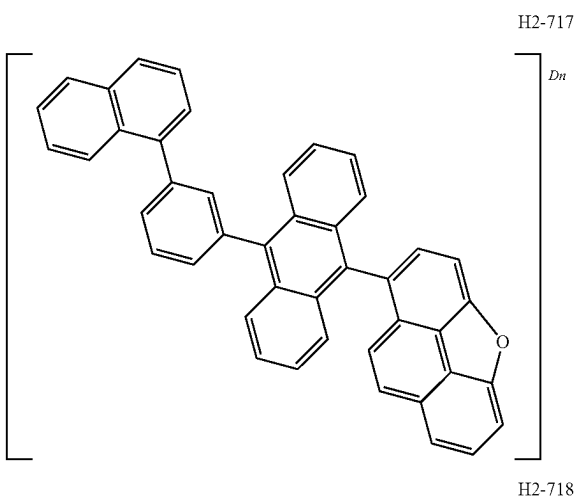
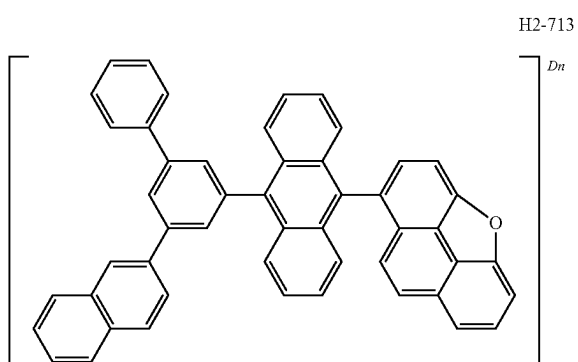
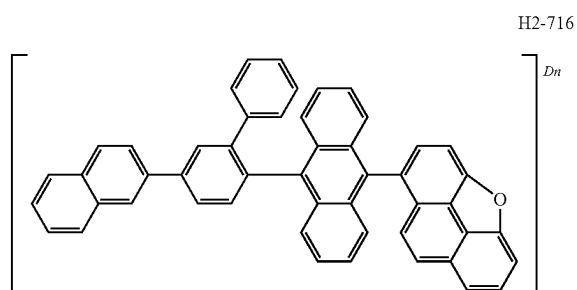
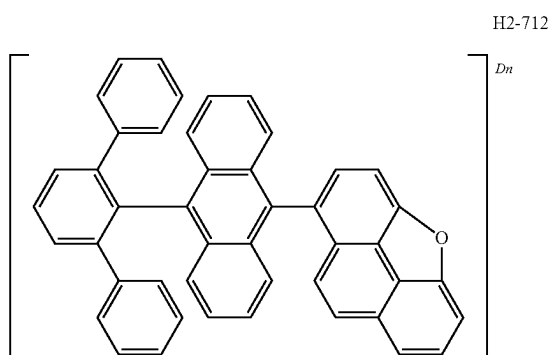
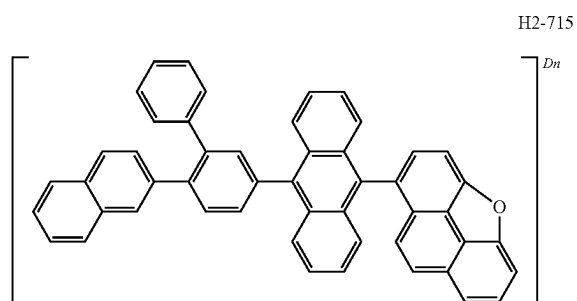
H2-710



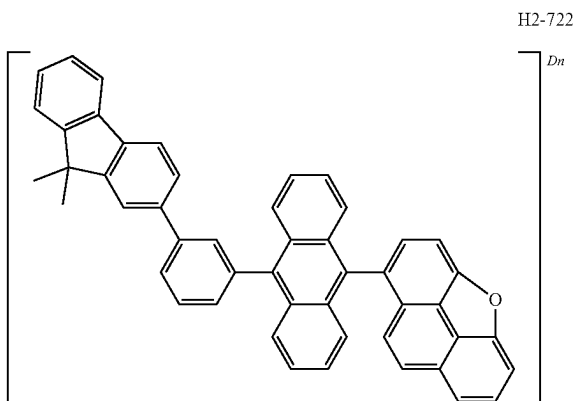
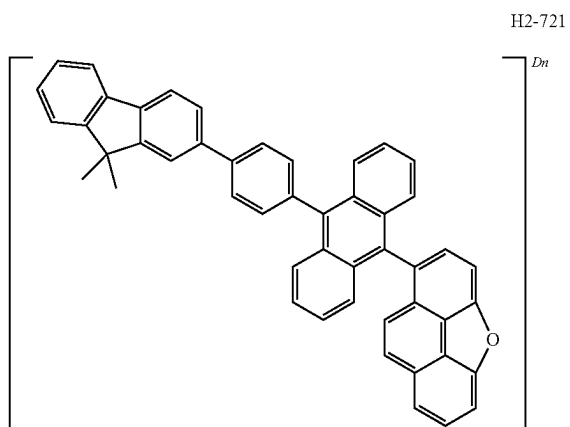
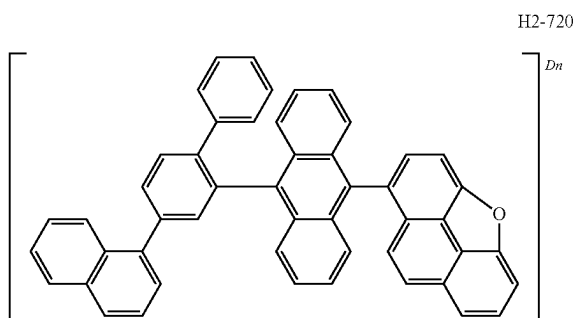
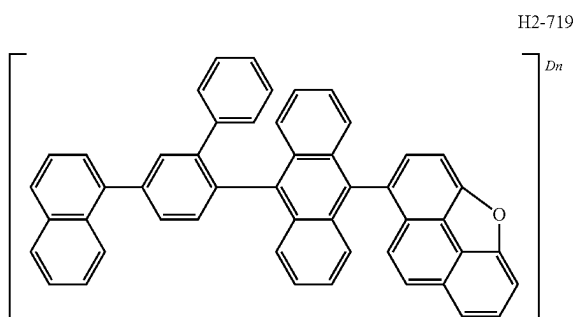
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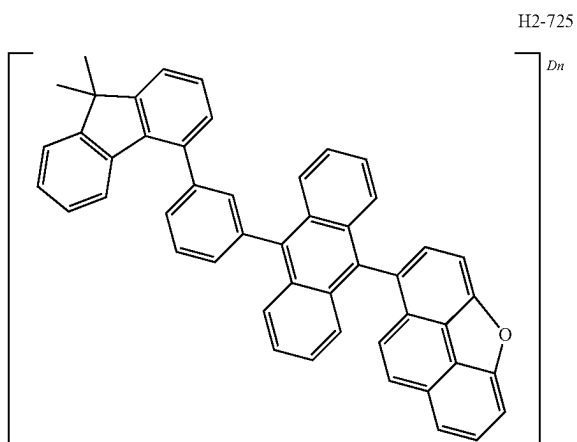
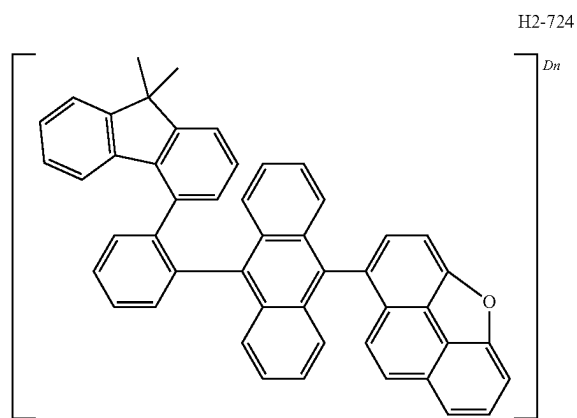
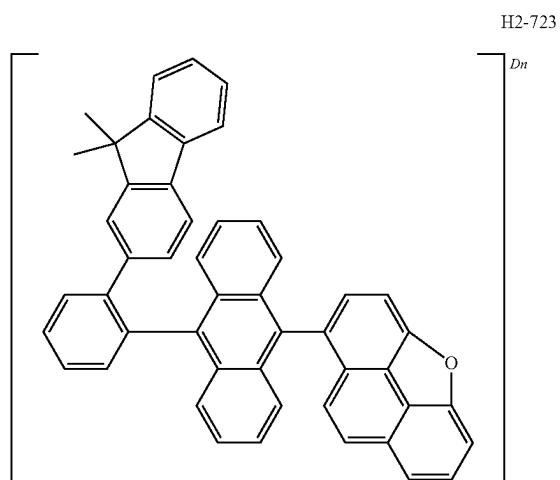
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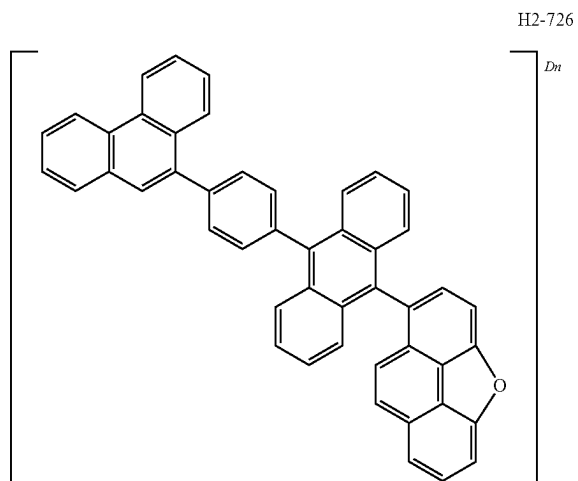
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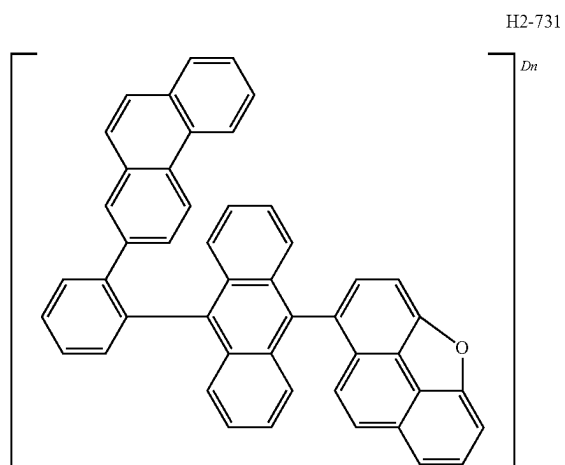
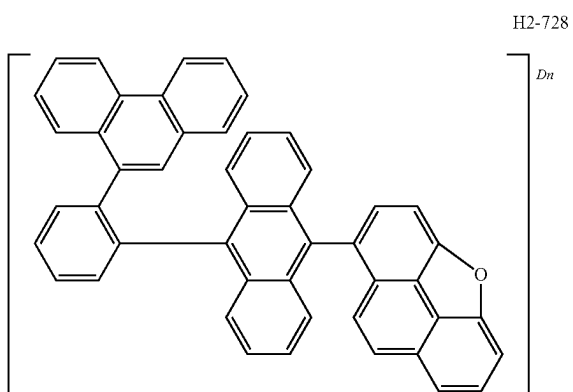
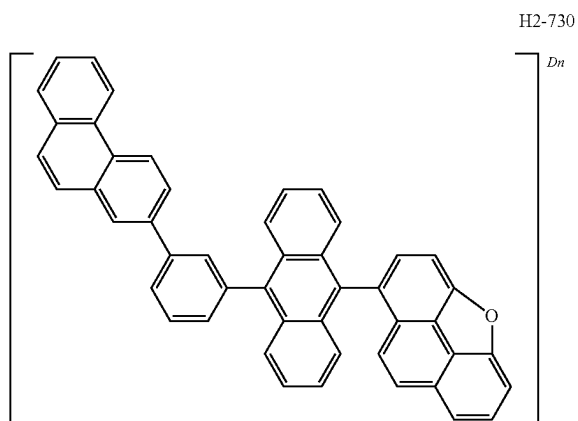
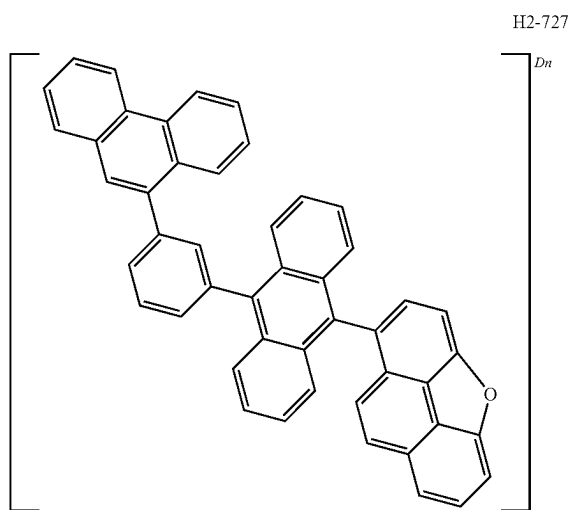
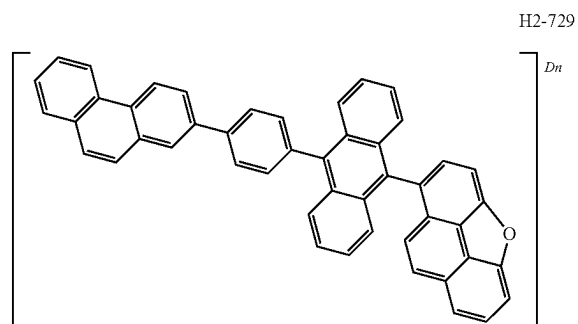
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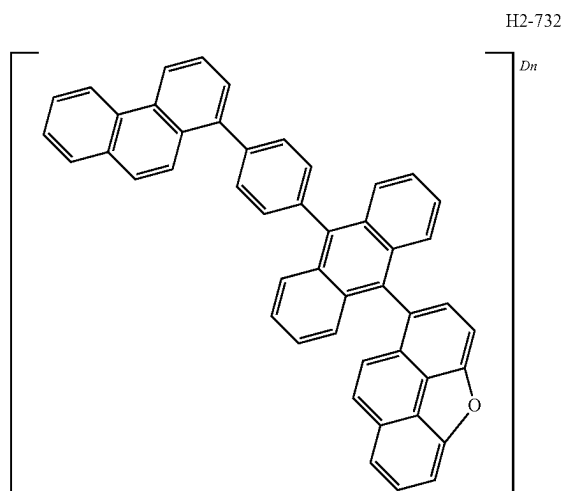
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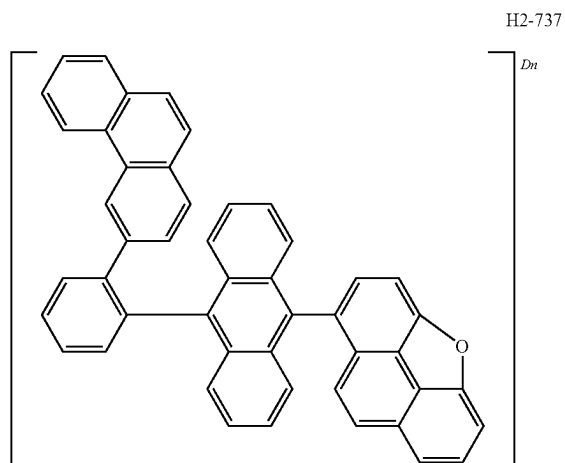
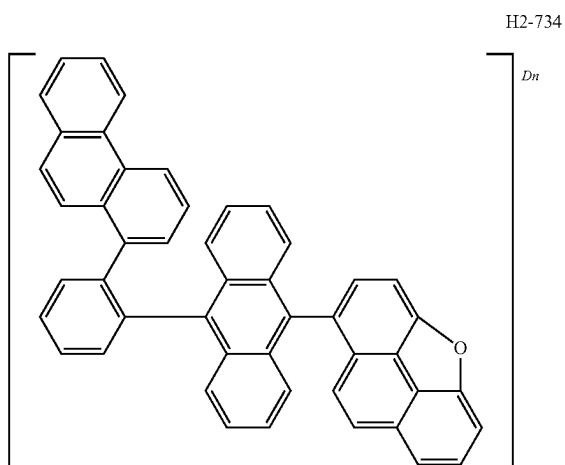
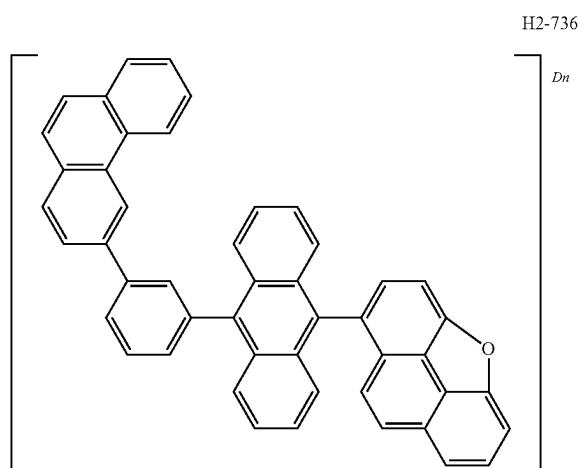
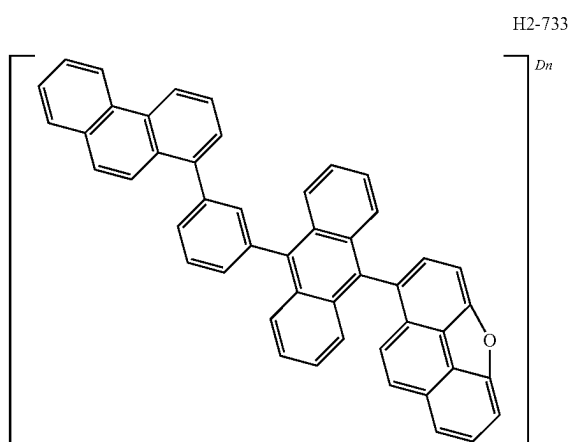
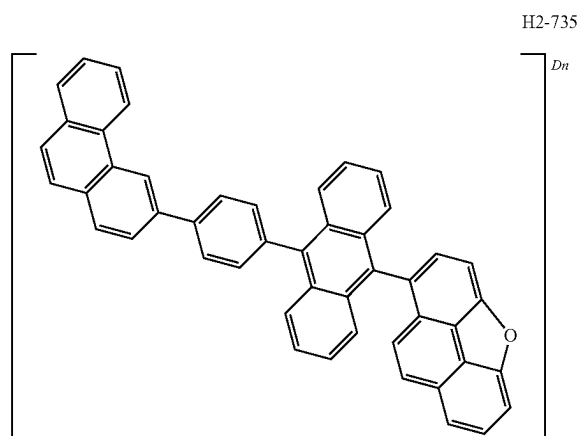
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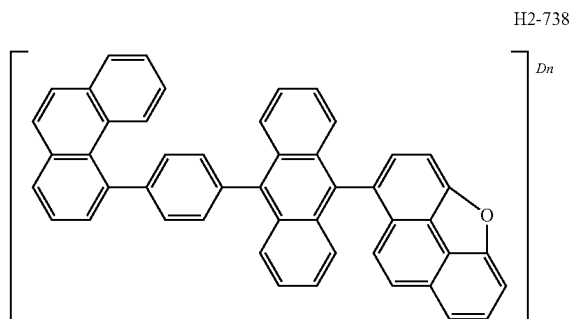
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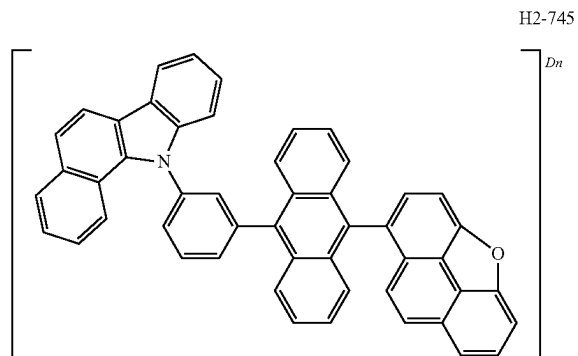
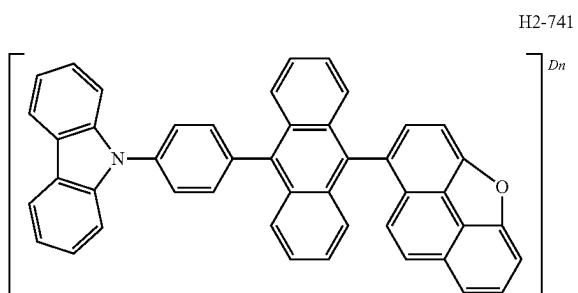
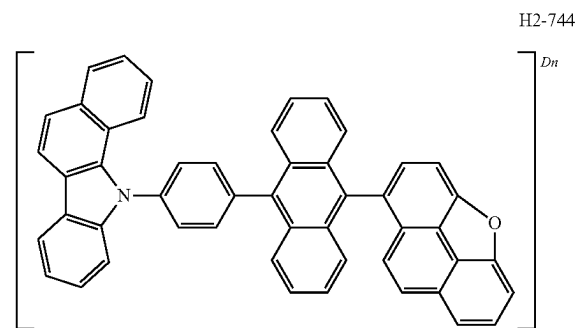
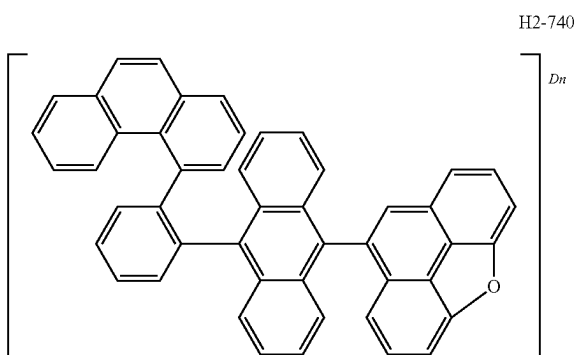
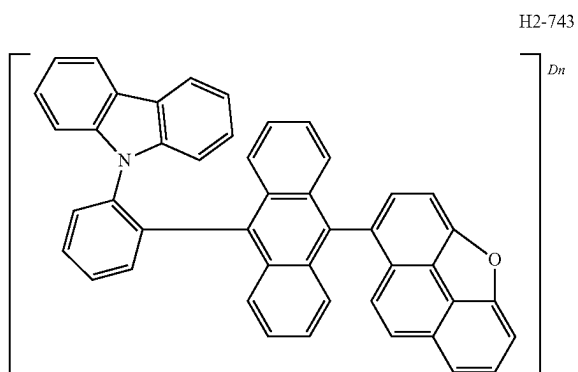
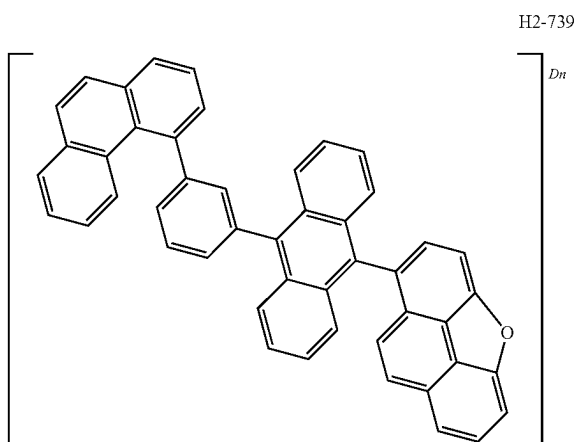
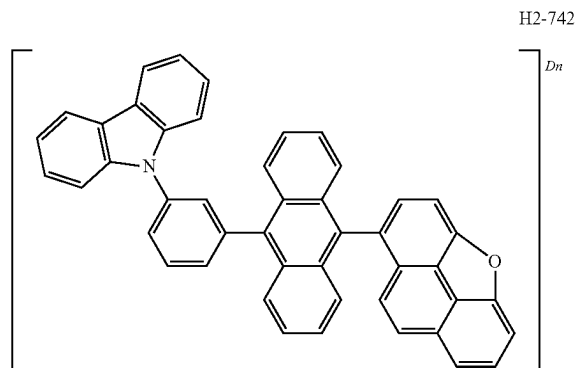
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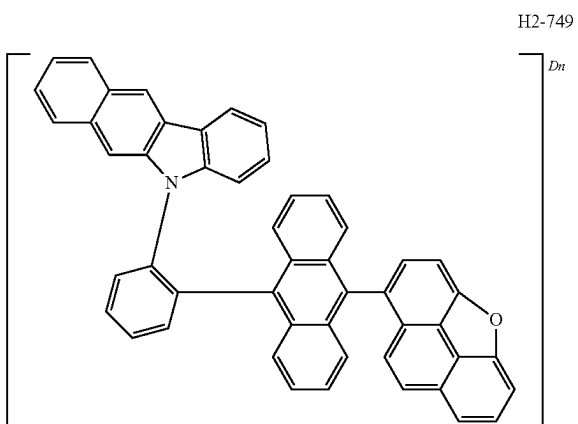
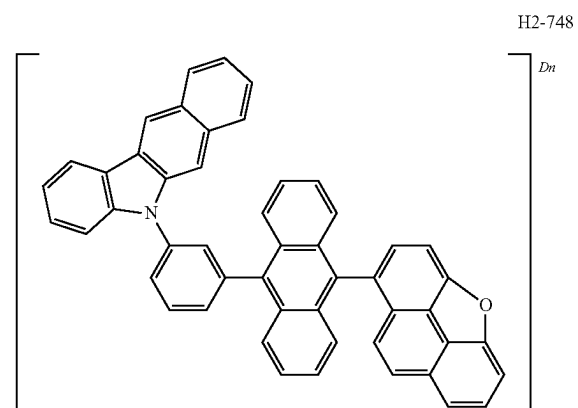
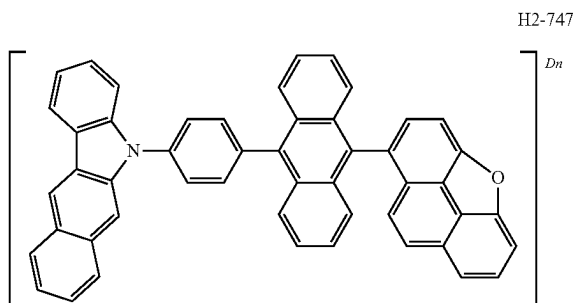
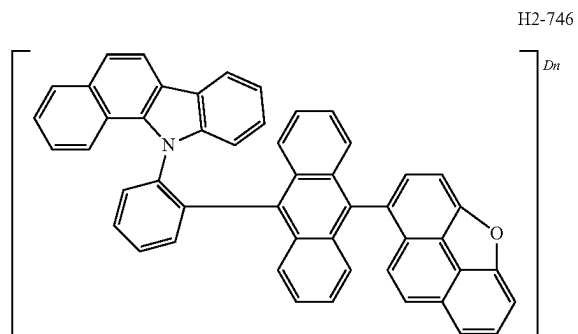
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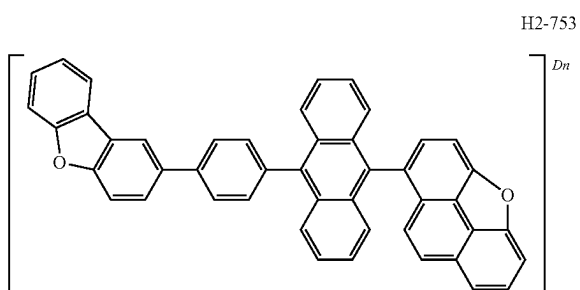
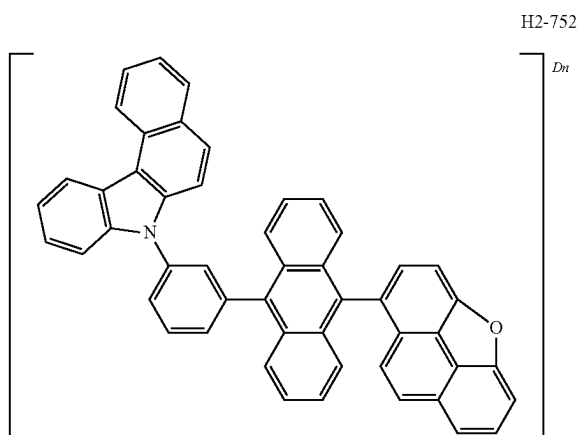
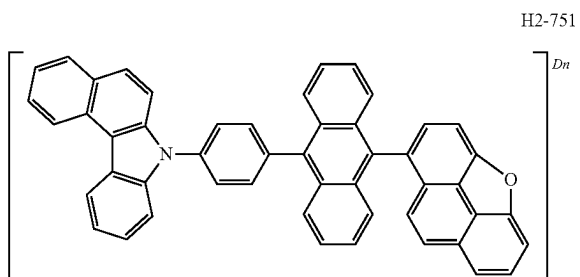
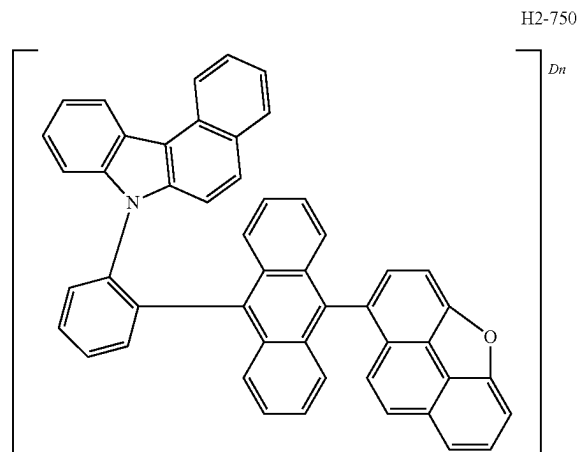
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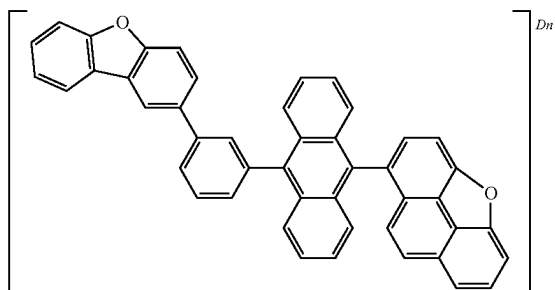


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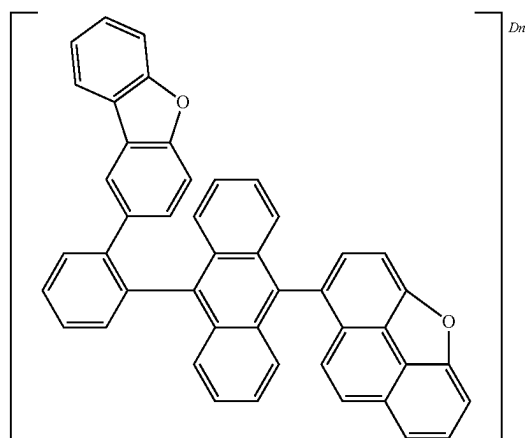


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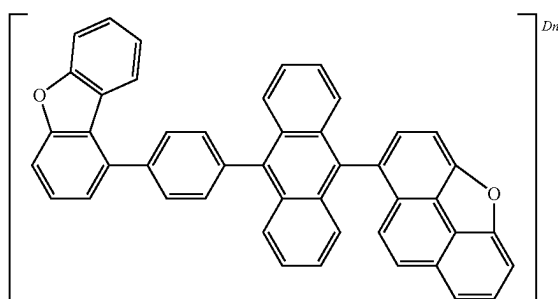
H2-754



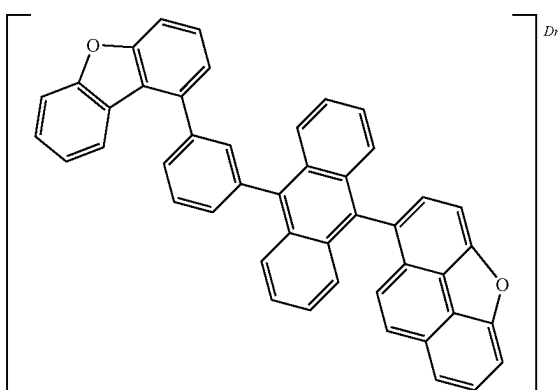
H2-755



H2-756

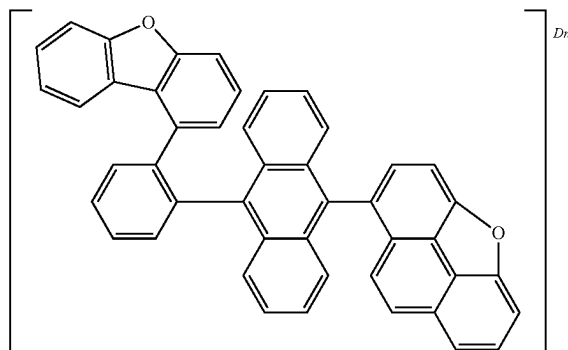


H2-757

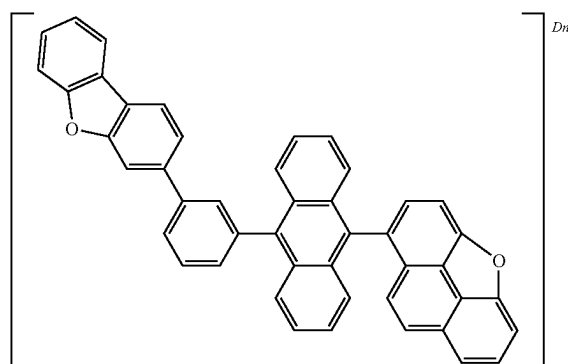


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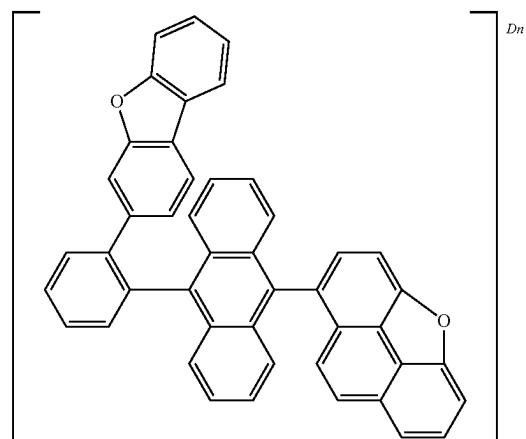
H2-758



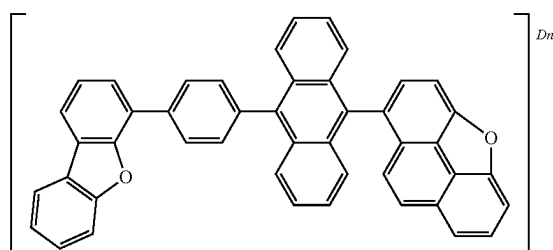
H2-759



H2-760

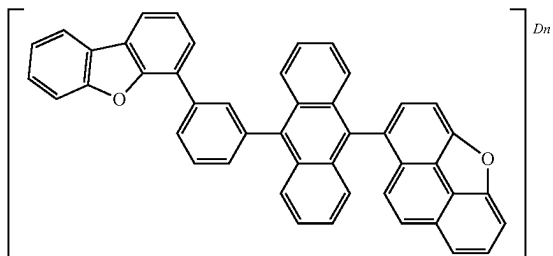


H2-761

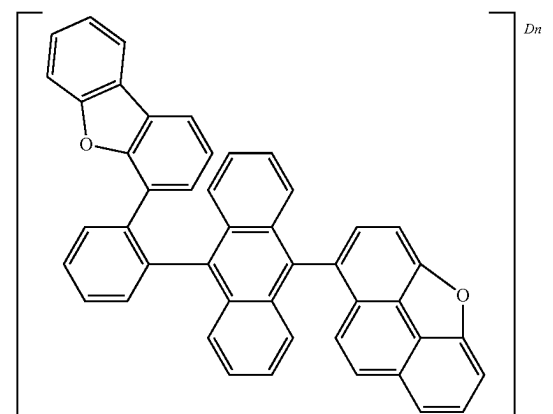


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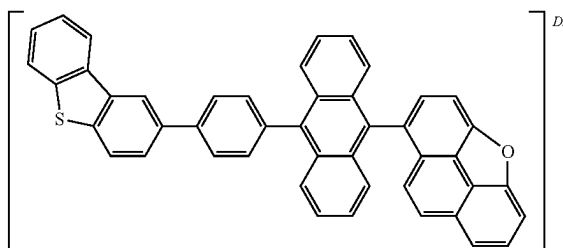
H2-762



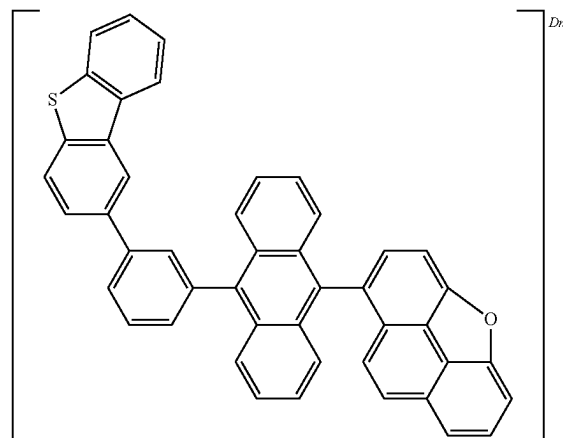
H2-763



H2-764

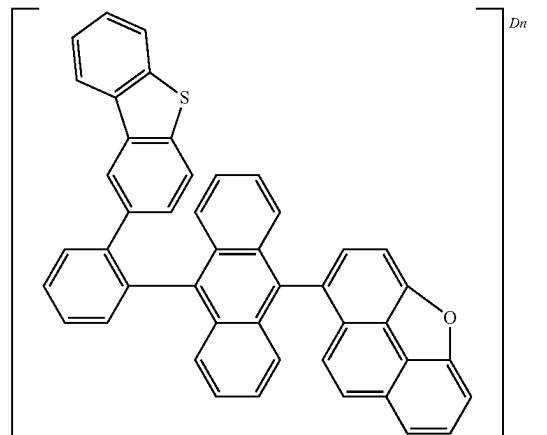


H2-765

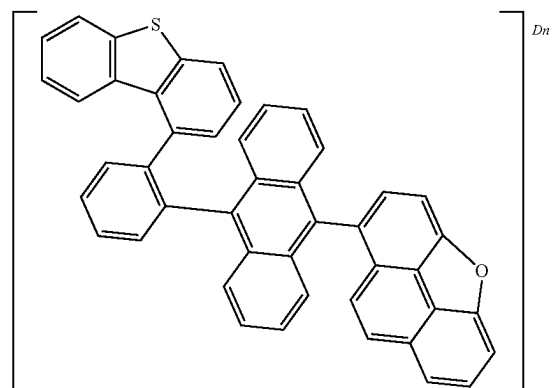


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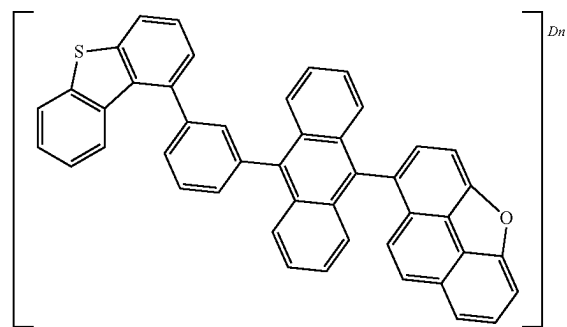
H2-766



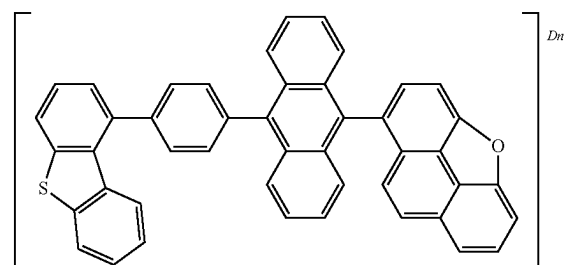
H2-767



H2-768

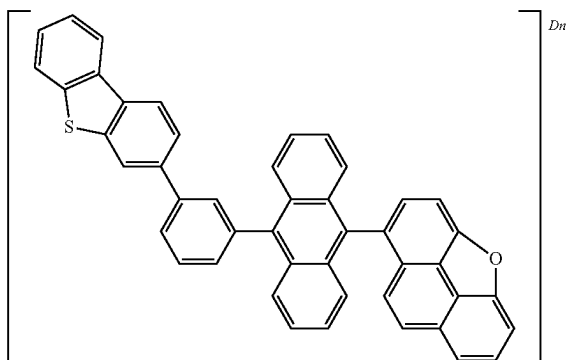


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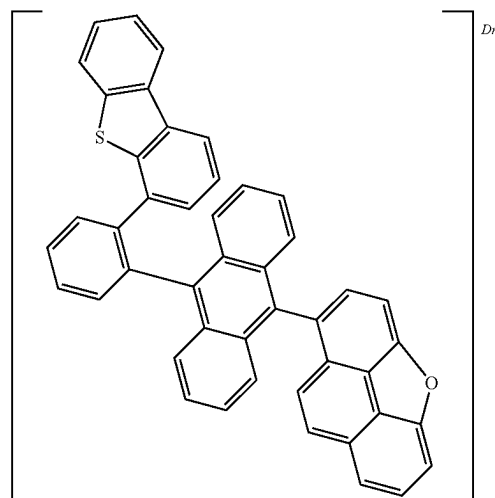
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H2-770

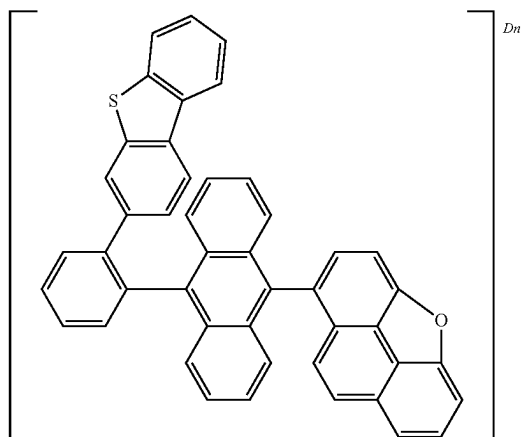


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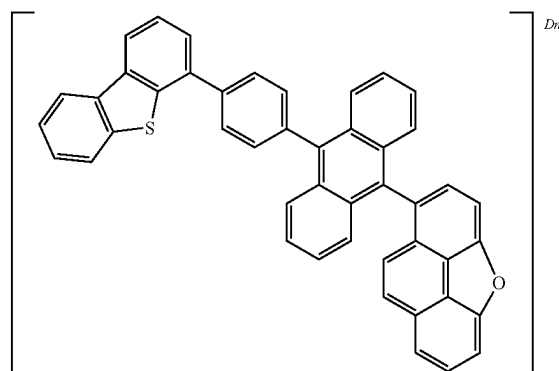
H2-773



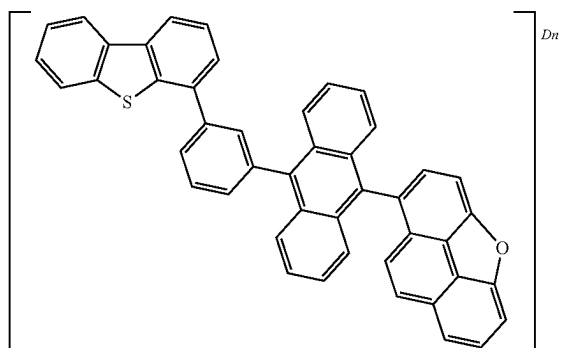
H2-771



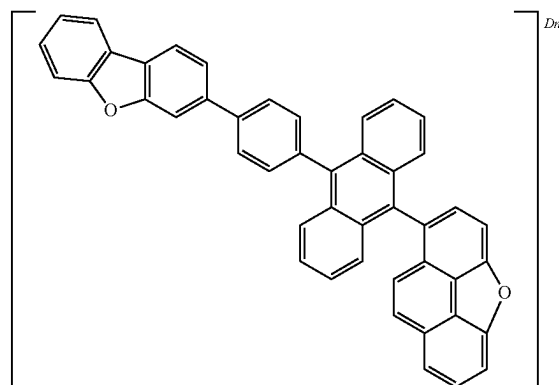
H2-774



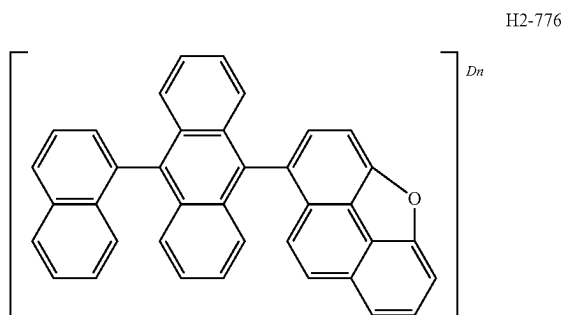
H2-772



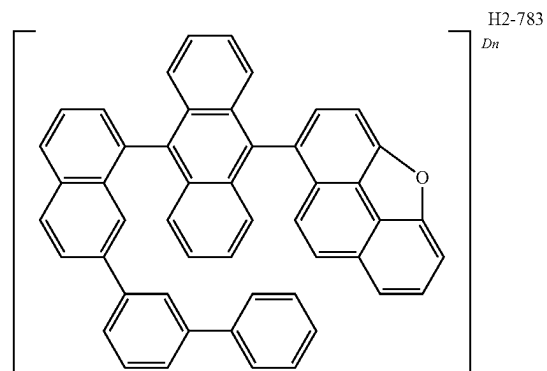
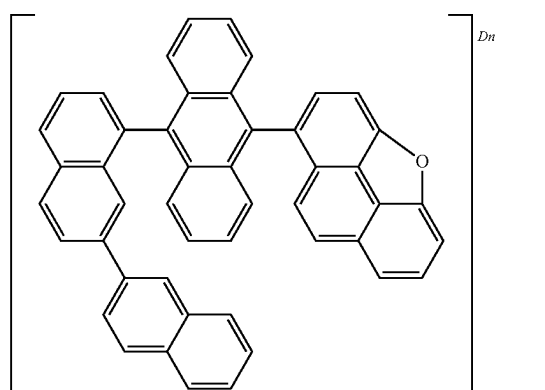
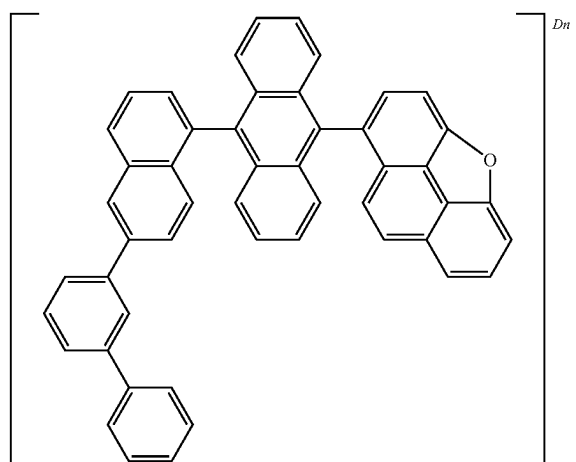
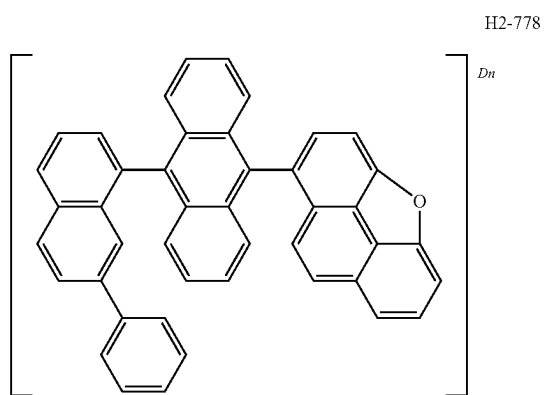
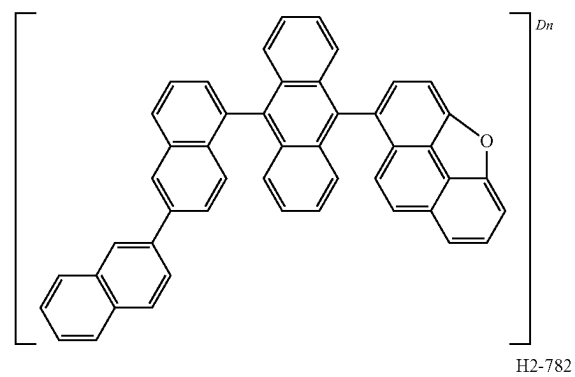
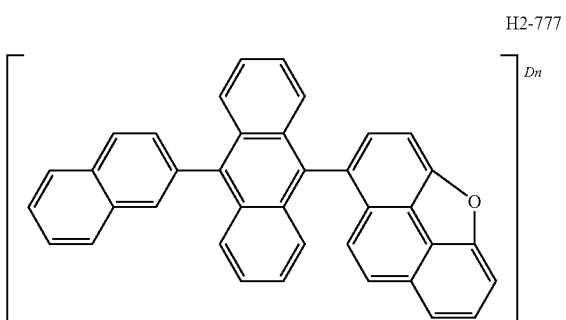
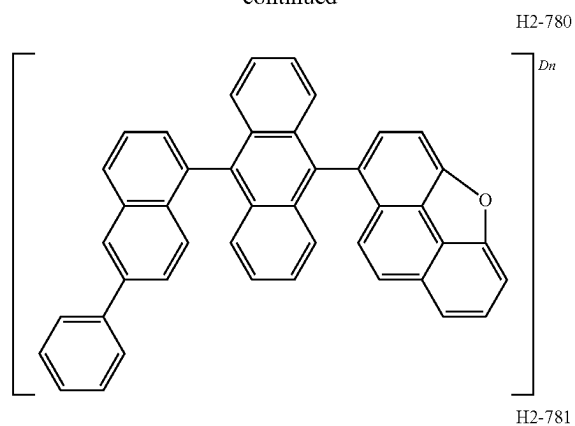
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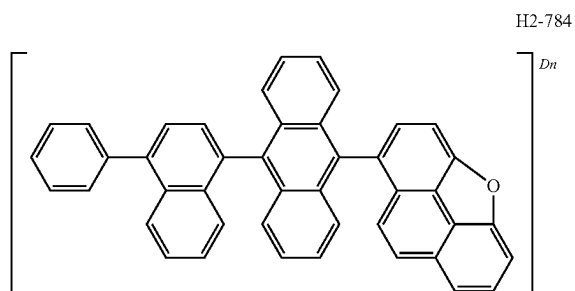
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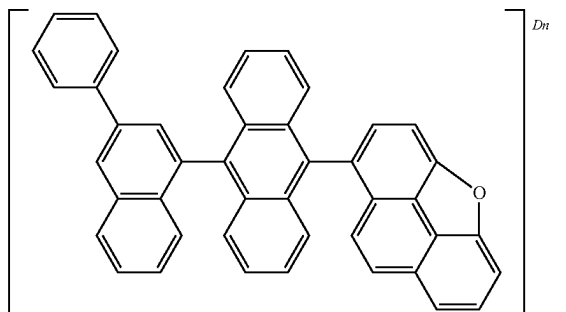
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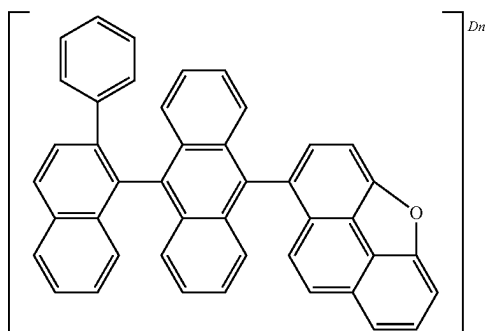
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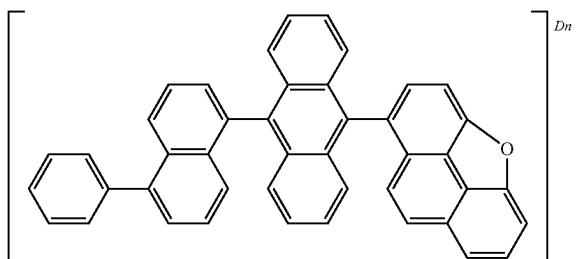
H2-785



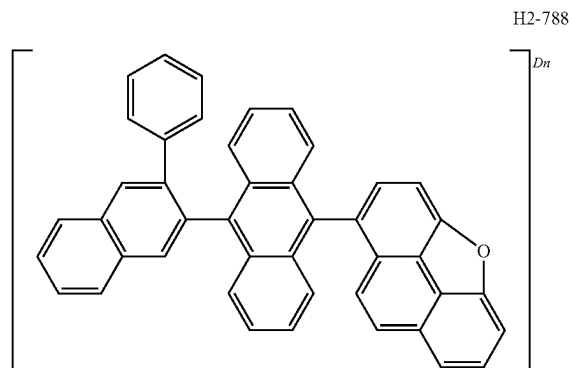
H2-786



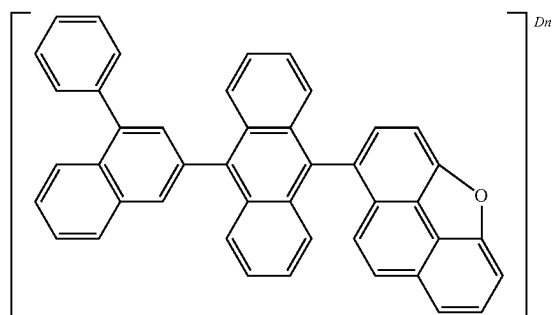
H2-787



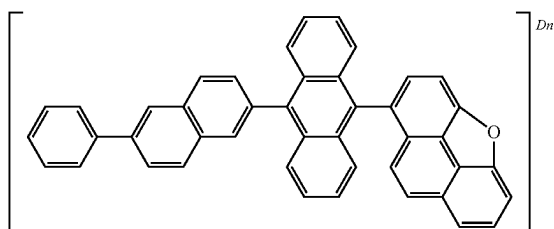
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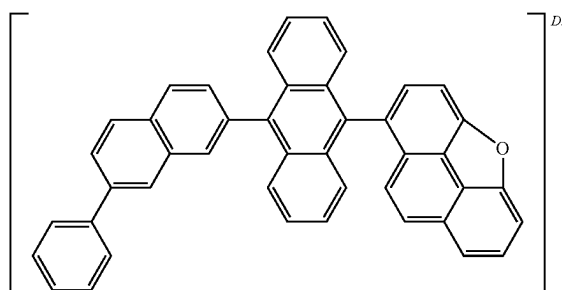
H2-789



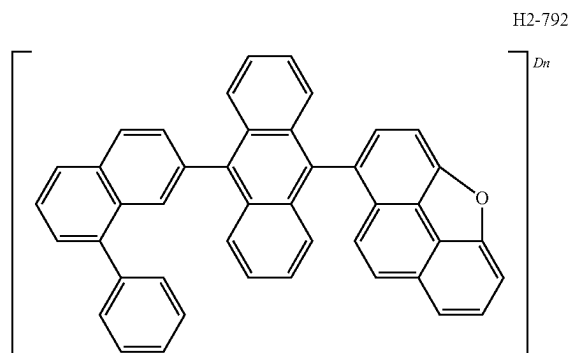
H2-790



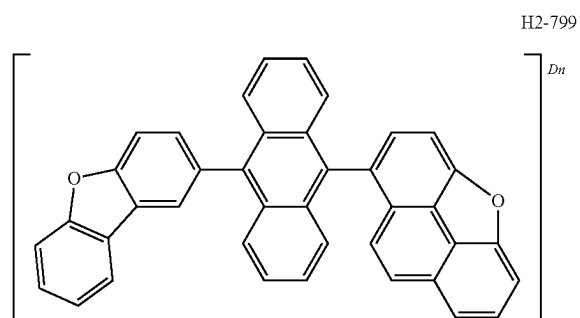
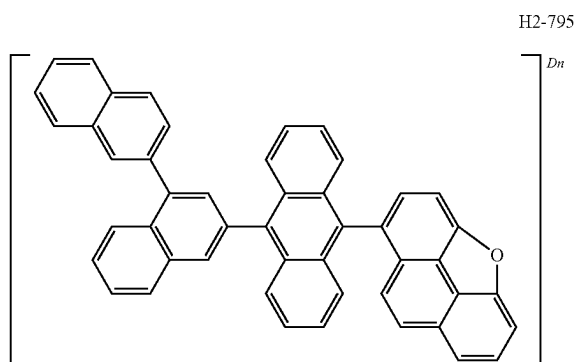
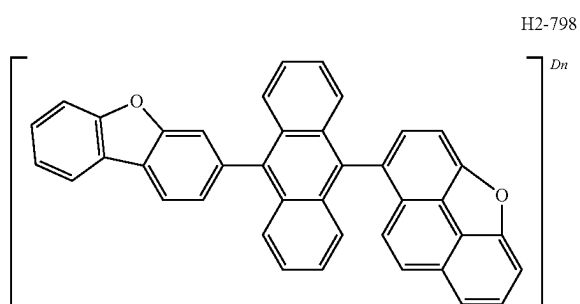
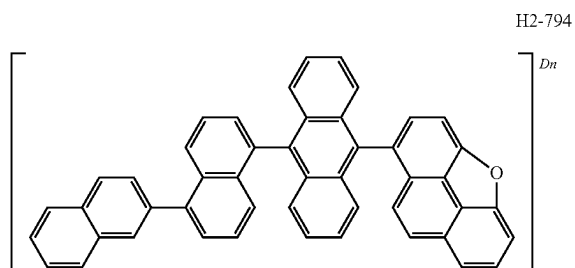
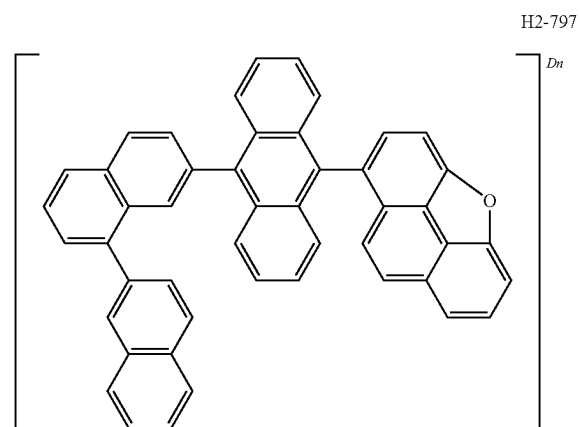
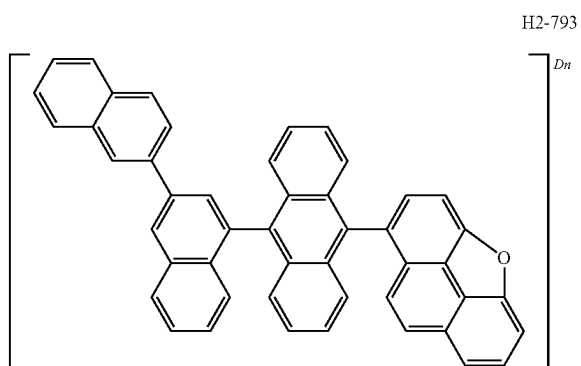
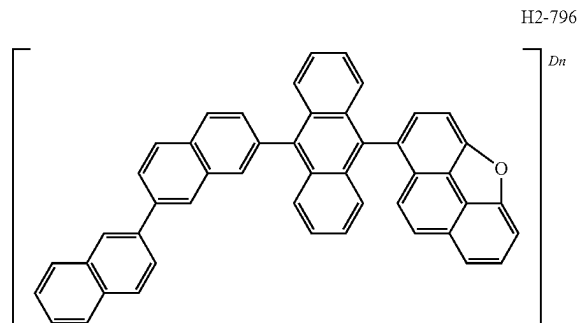
H2-791



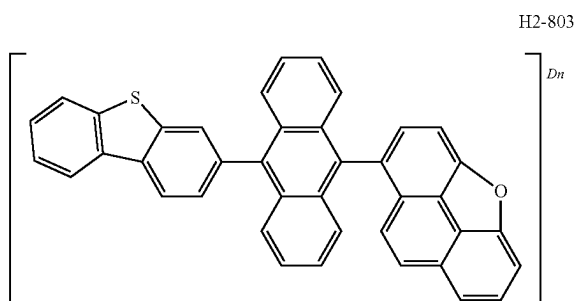
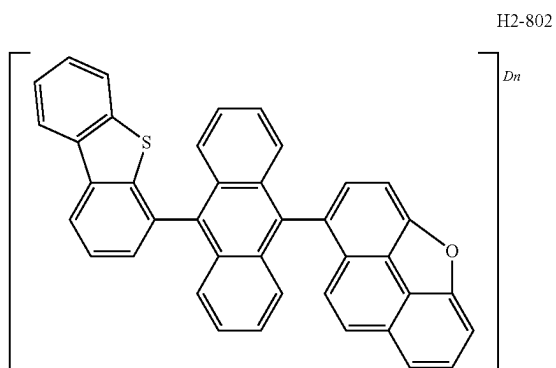
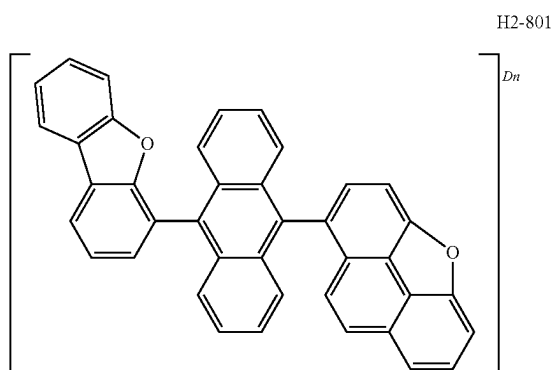
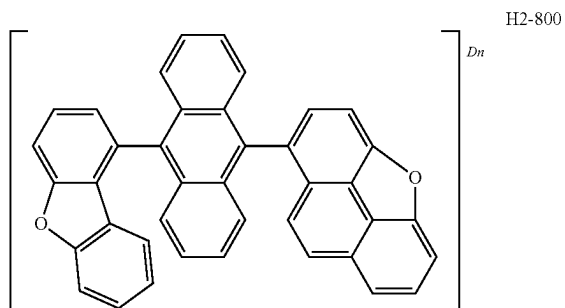
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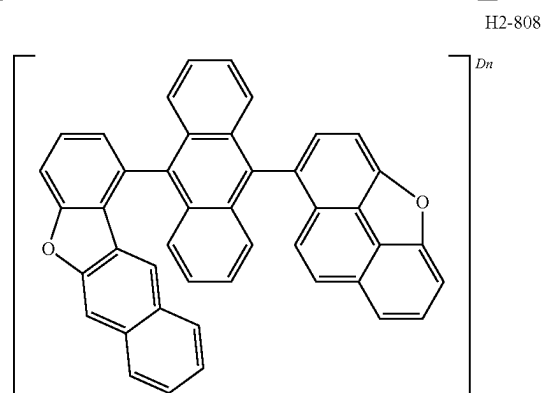
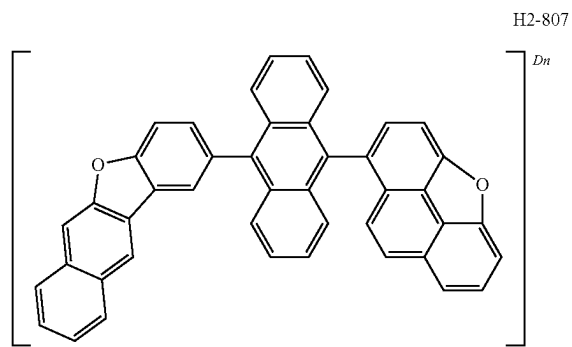
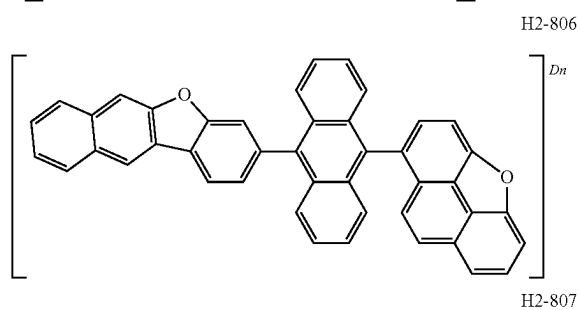
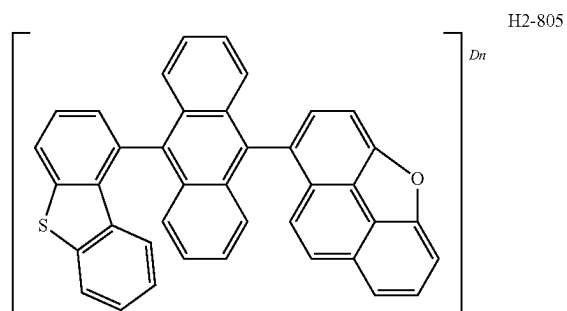
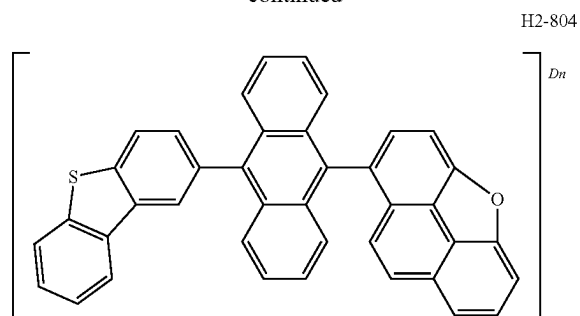
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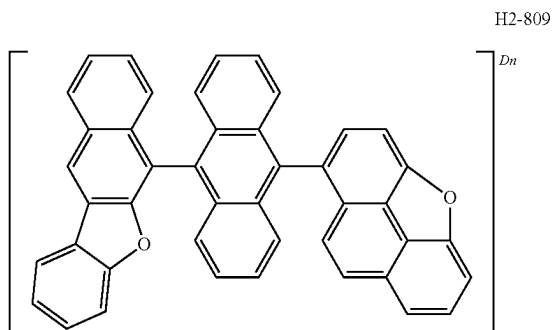
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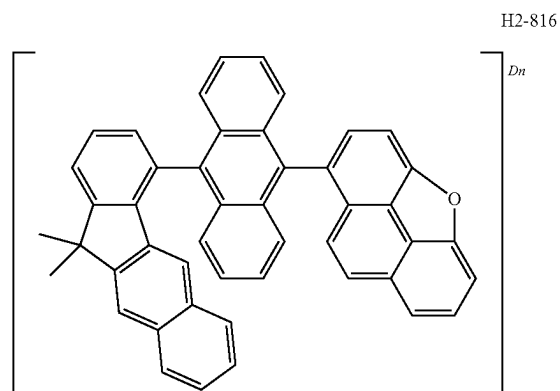
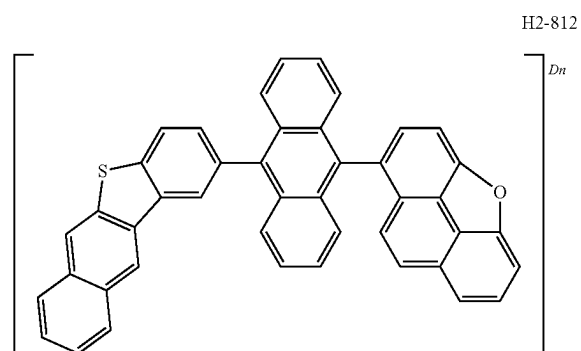
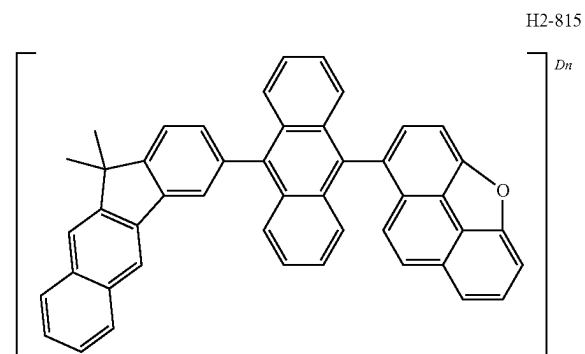
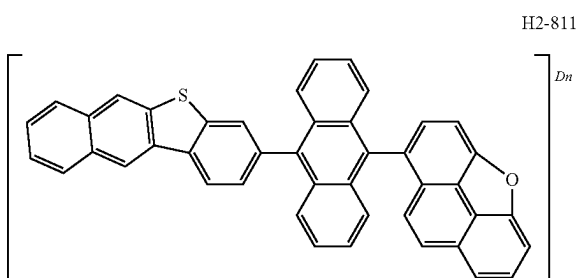
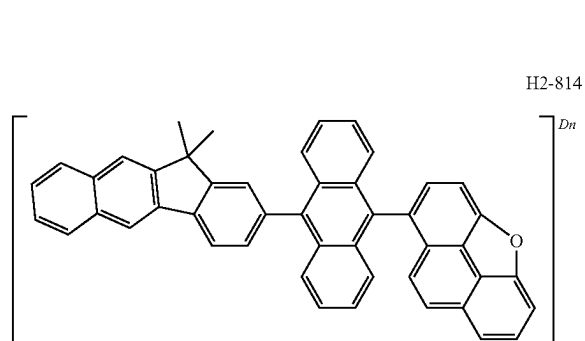
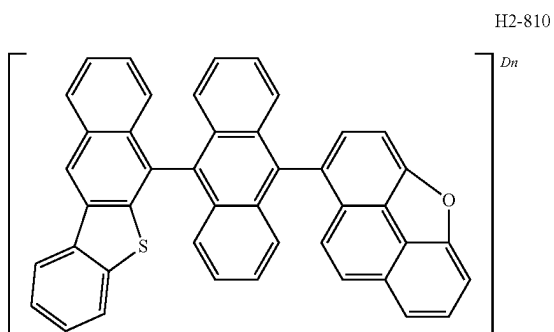
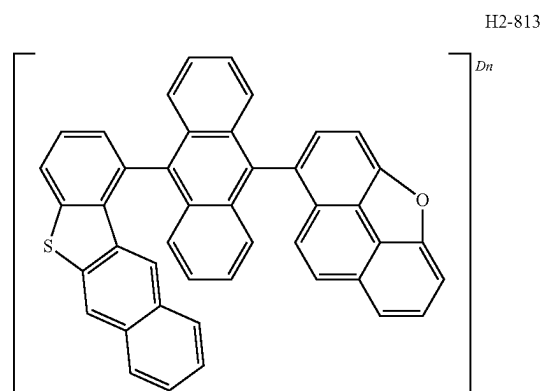
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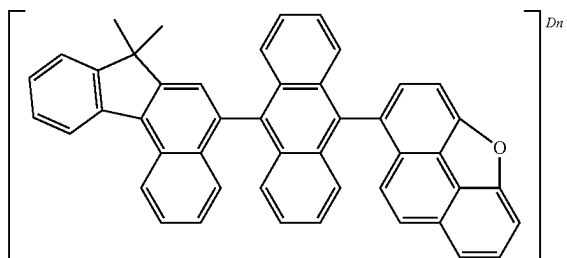


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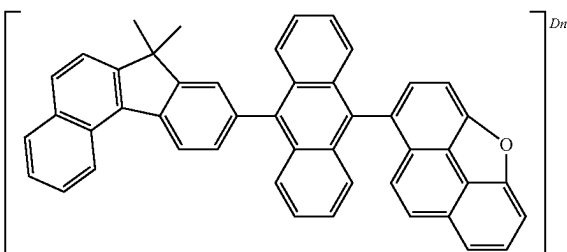


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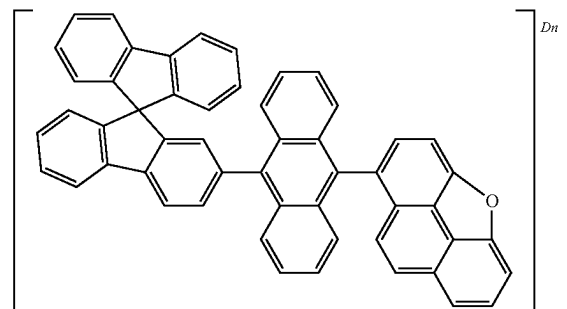
H2-817



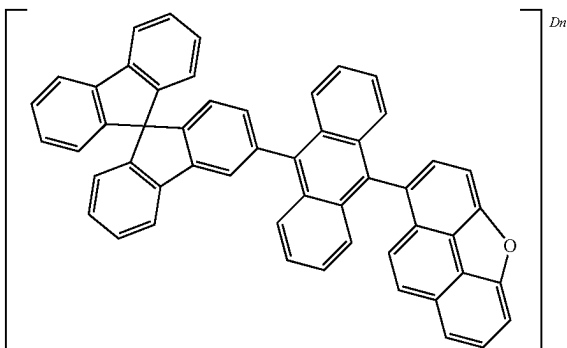
H2-818



H2-819

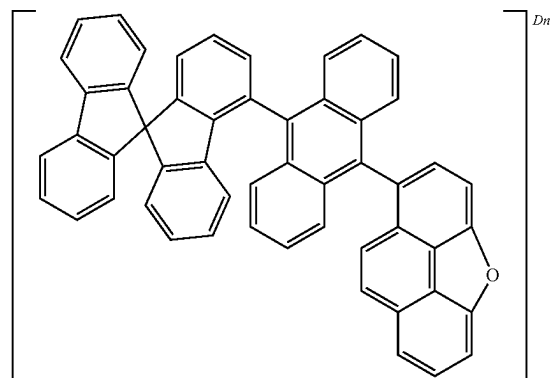


H2-820

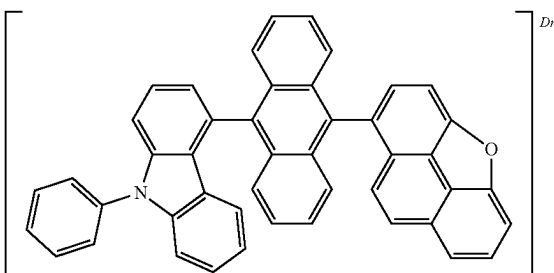


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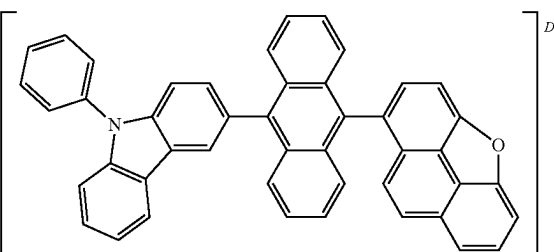
H2-821



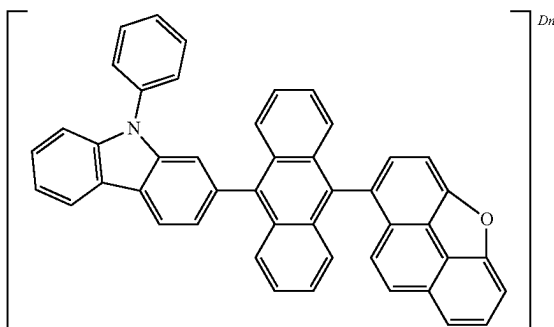
H2-822



H2-823

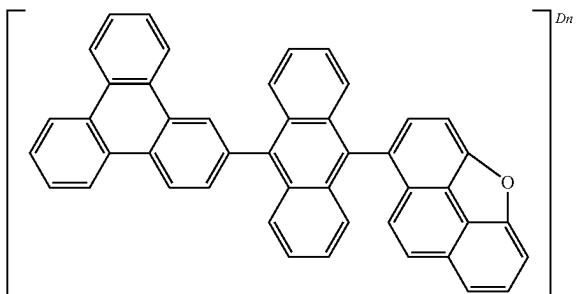


H2-824



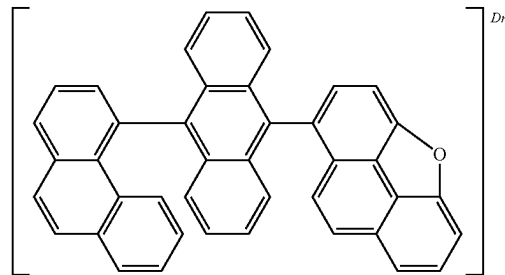
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H2-825

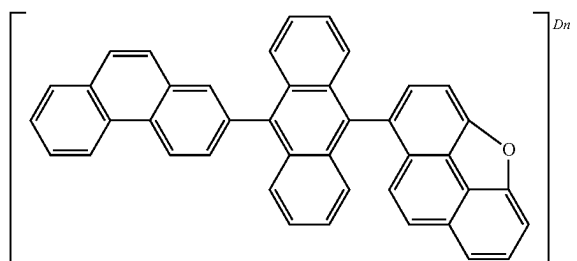


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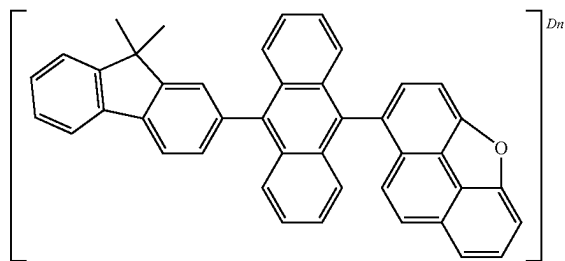
H2-830



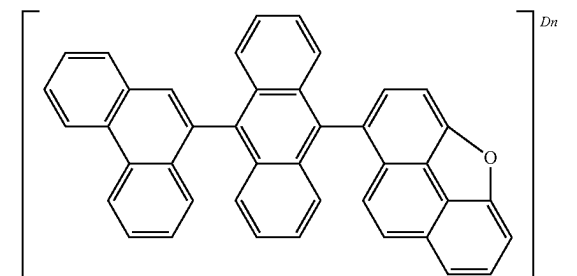
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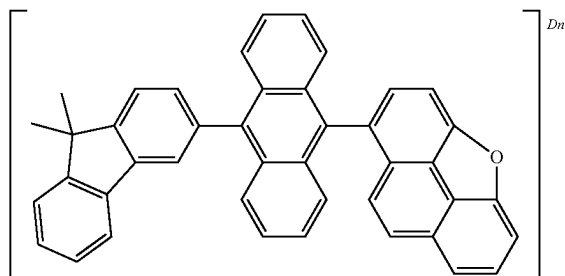
H2-831



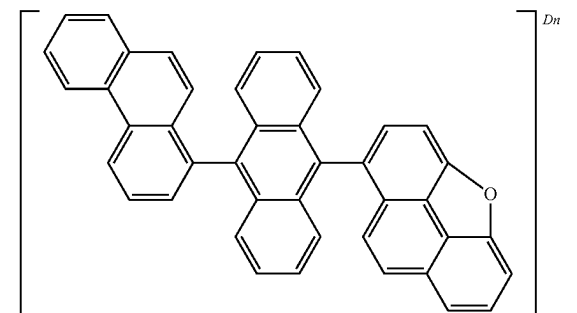
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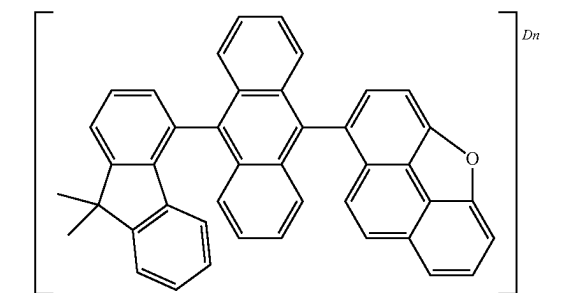
H2-832



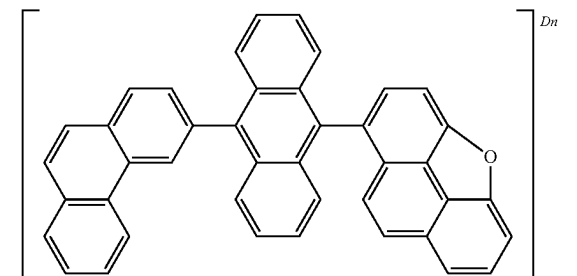
H2-828



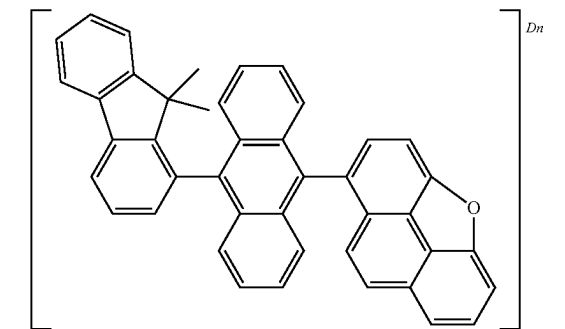
H2-833



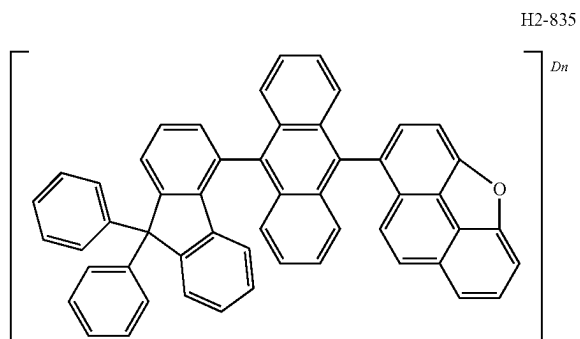
H2-829



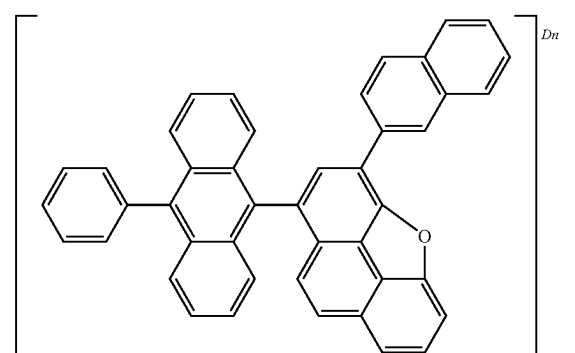
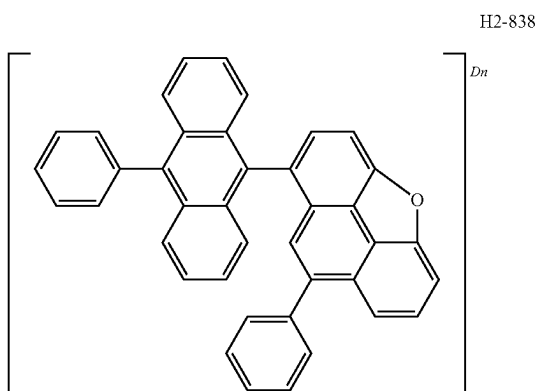
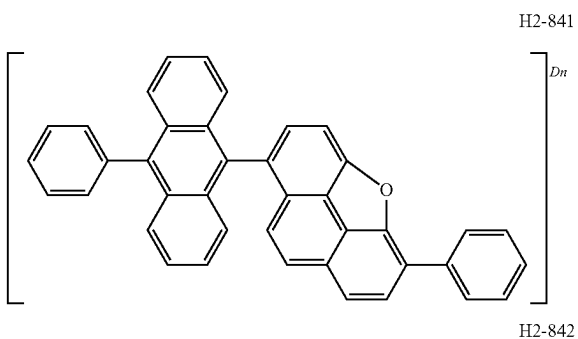
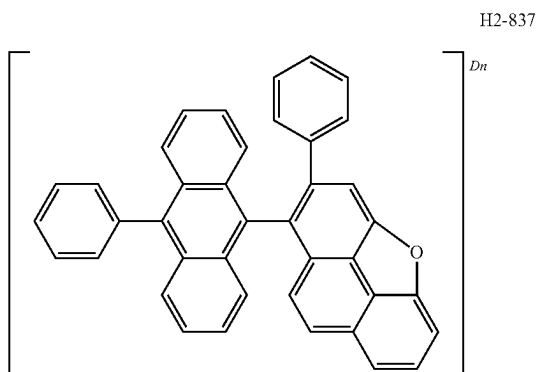
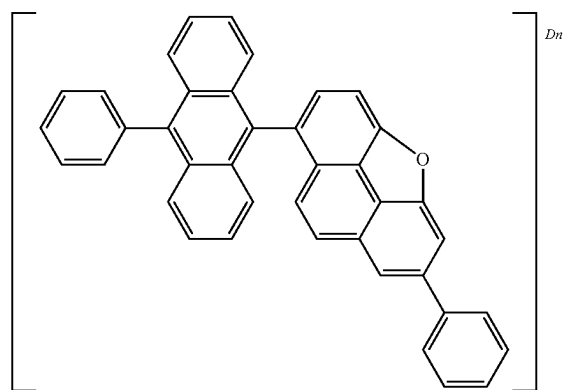
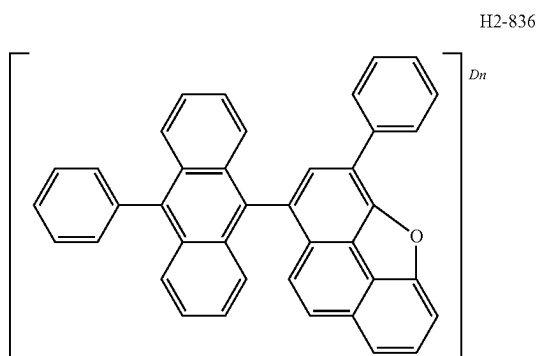
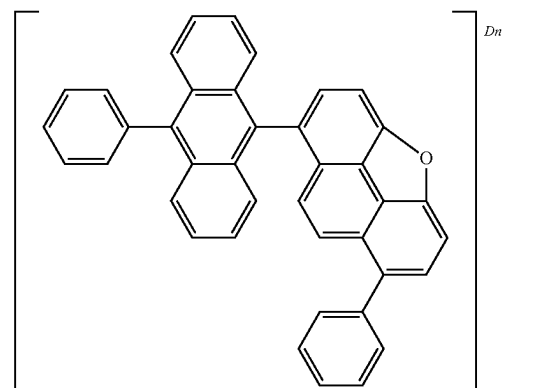
H2-834



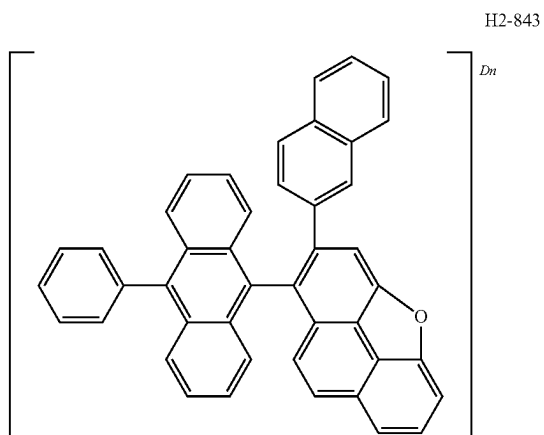
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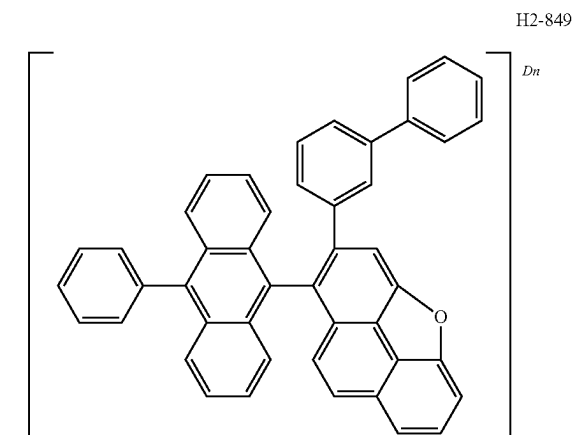
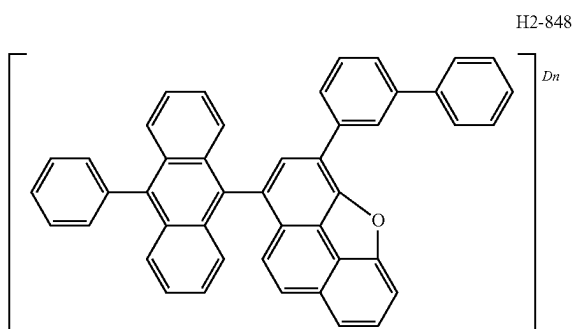
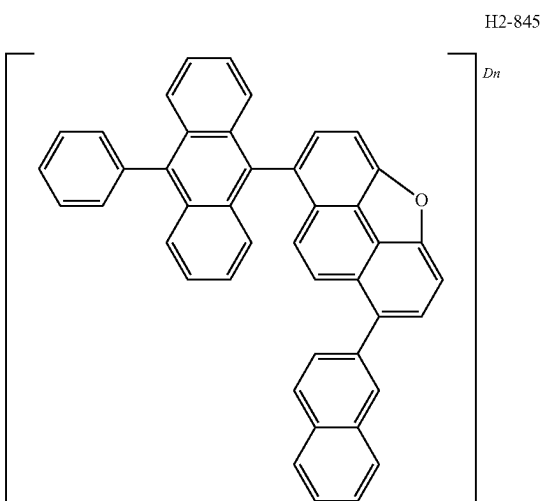
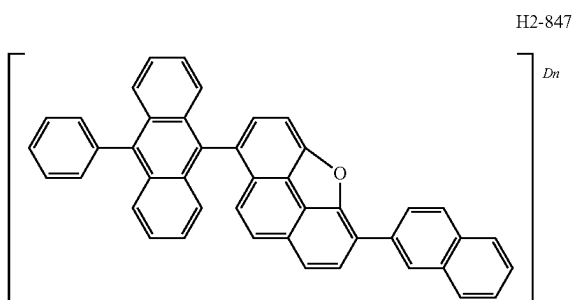
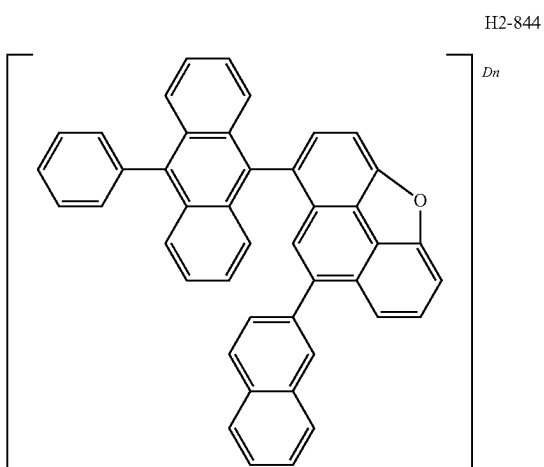
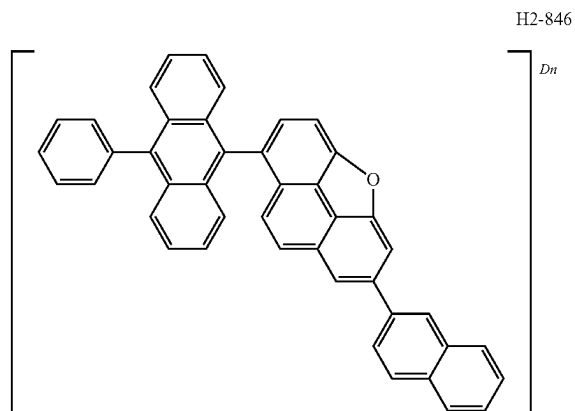
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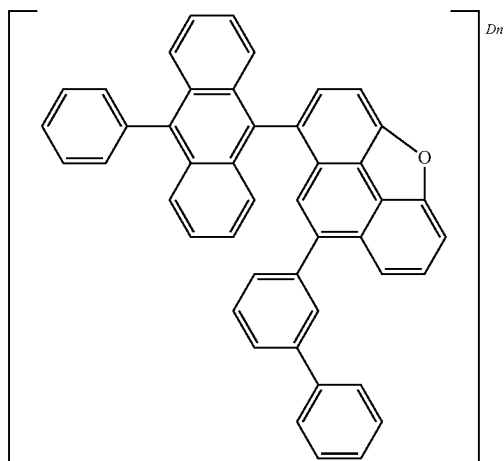


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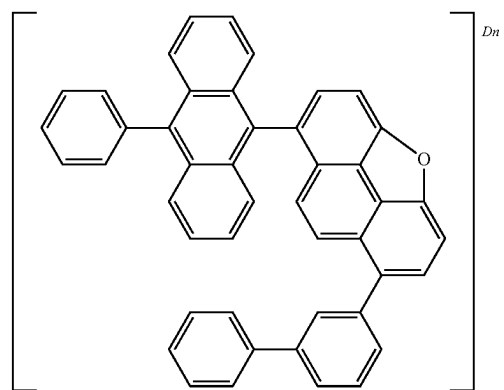


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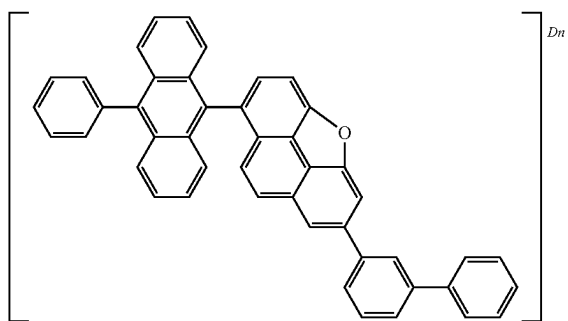
H2-850



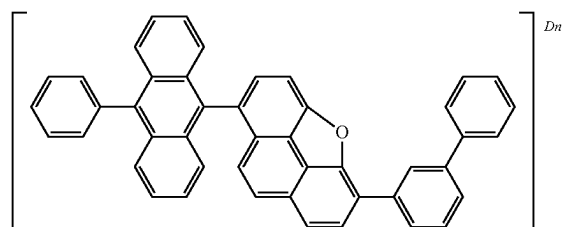
H2-851



H2-852

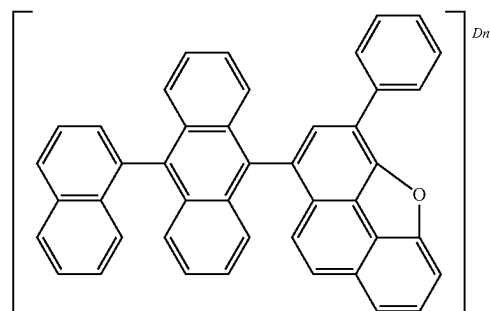


H2-853

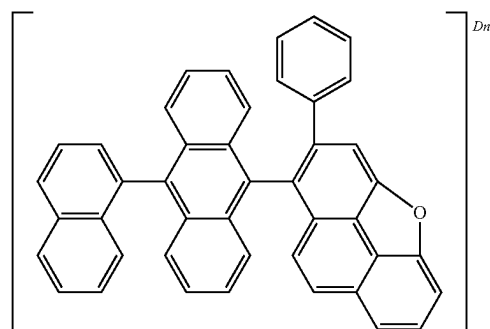


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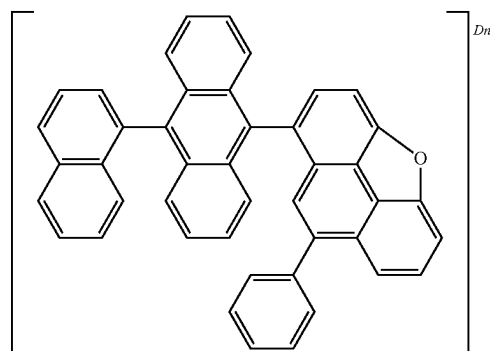
H2-854



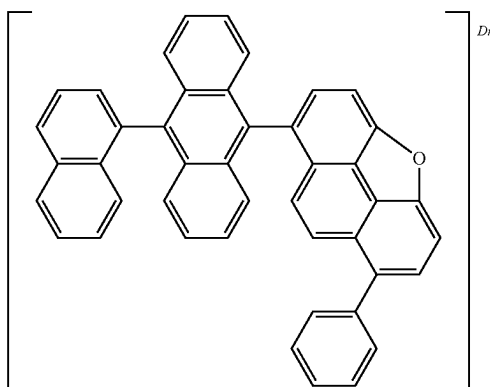
H2-855



H2-856

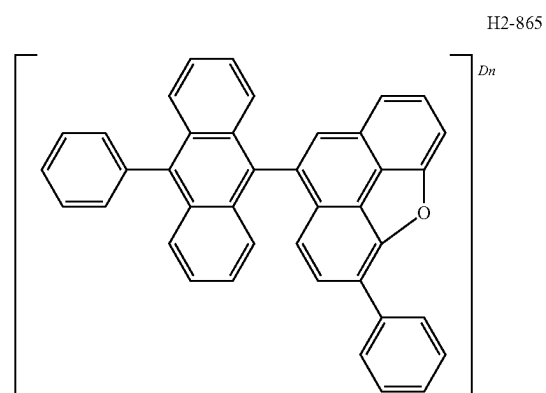
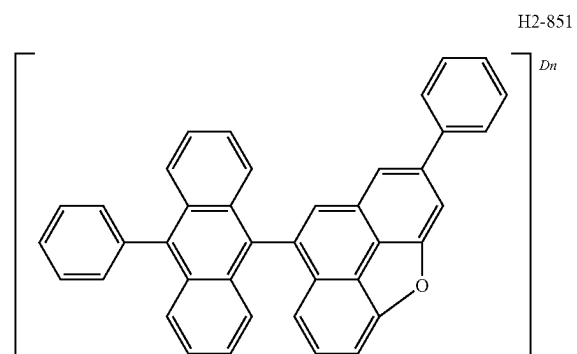
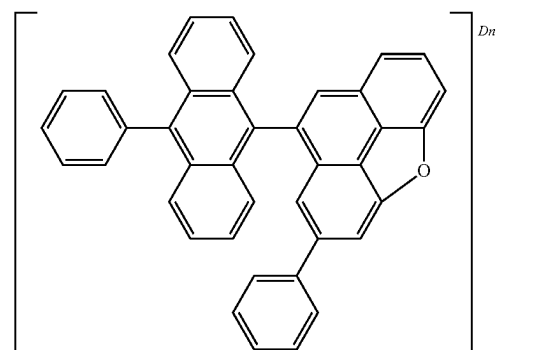
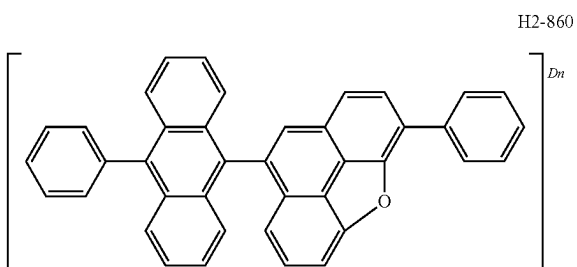
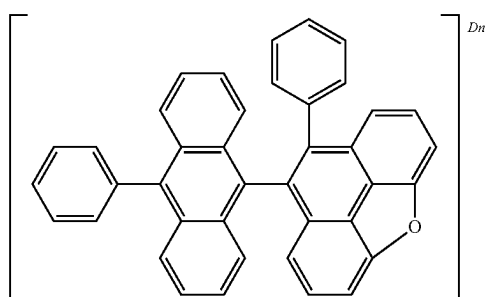
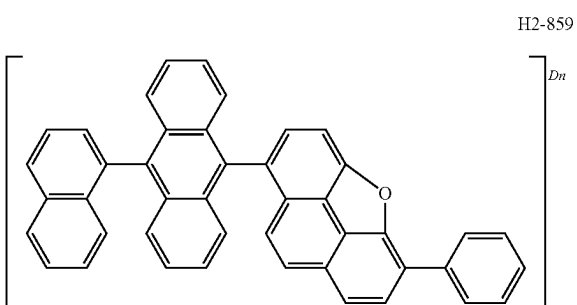
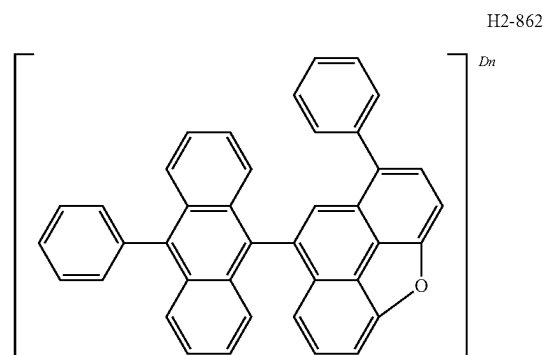
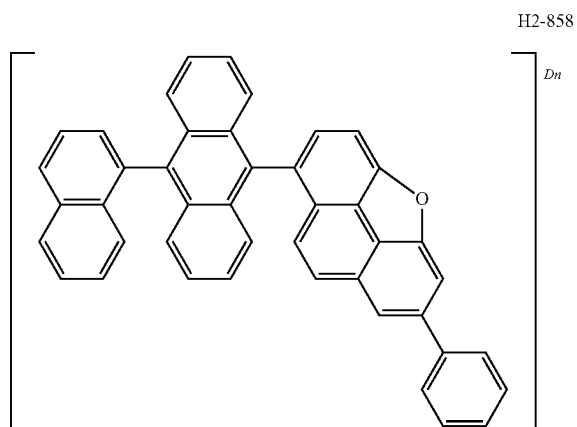


H2-857

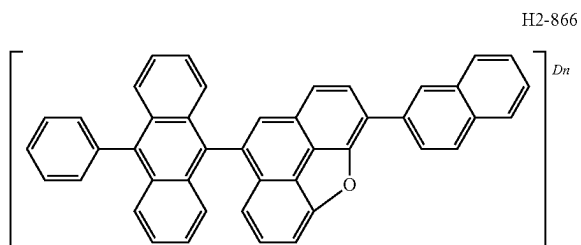


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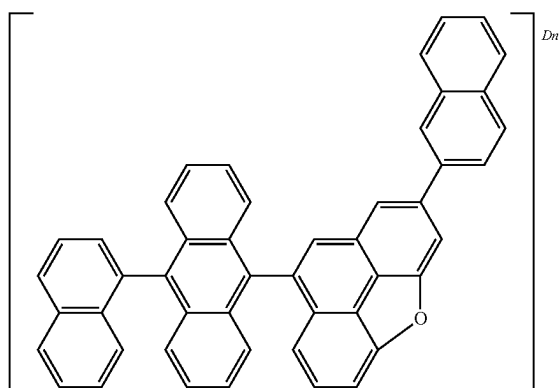
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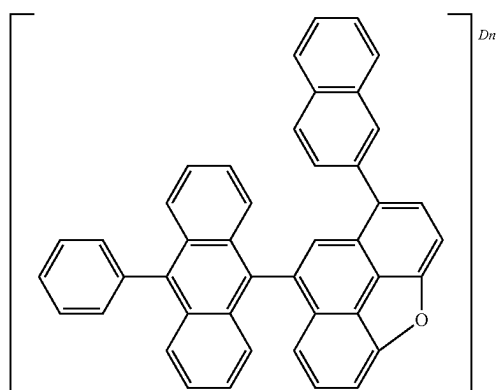
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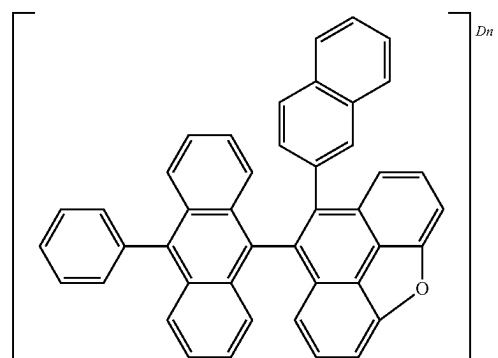
H2-867



H2-868

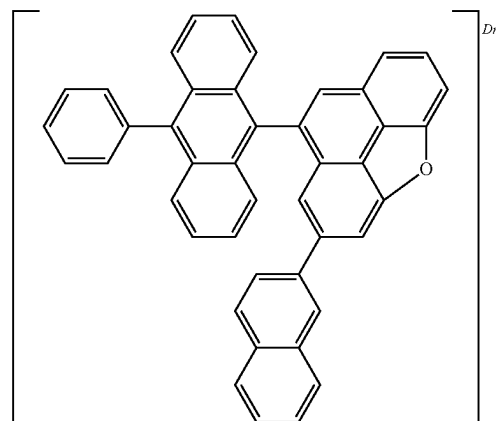


H2-869

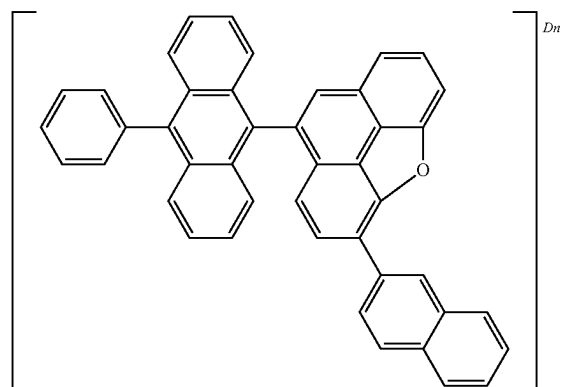


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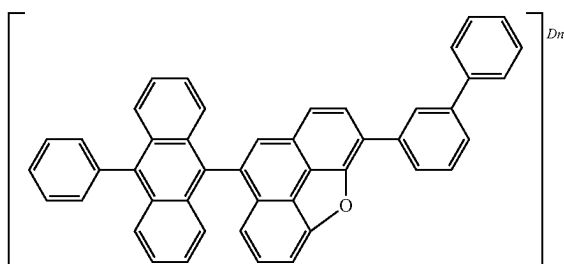
H2-870



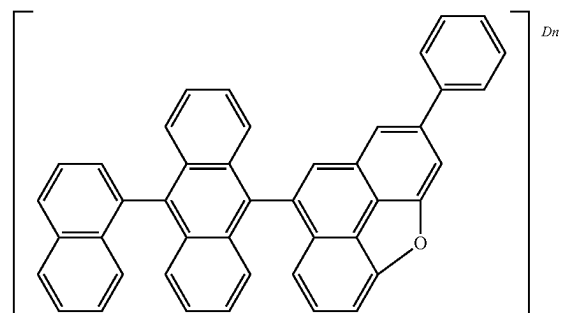
H2-871



H2-872

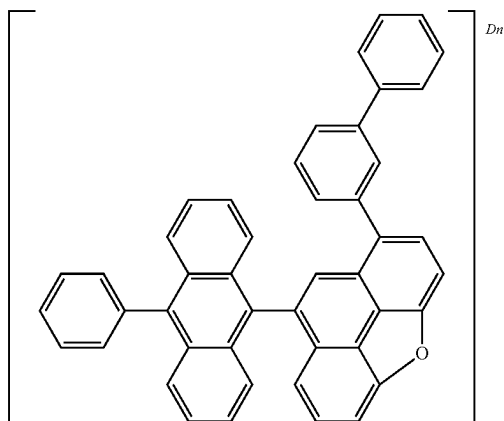


H2-873



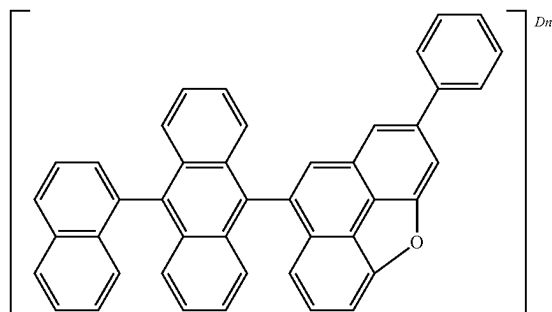
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H2-874

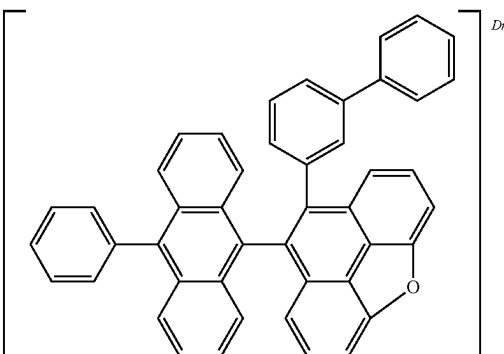


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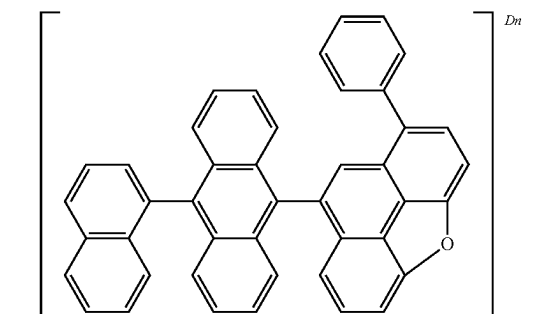
H2-878



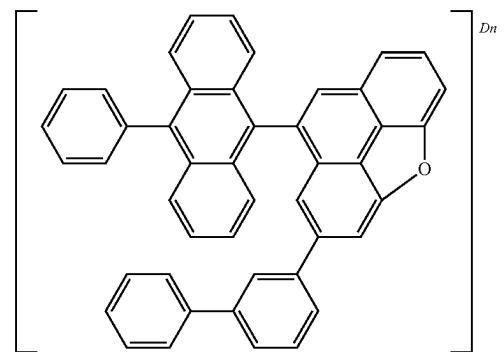
H2-875



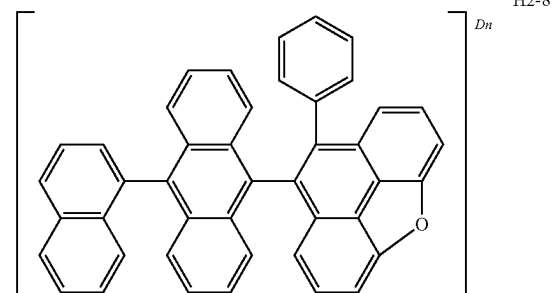
H2-880



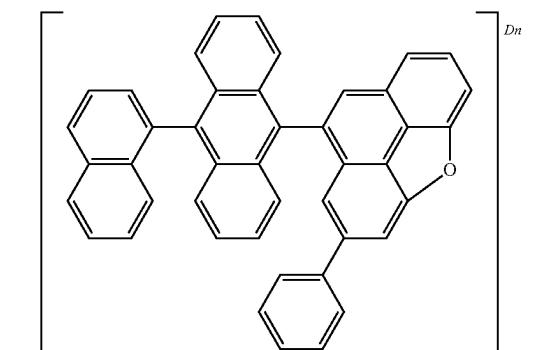
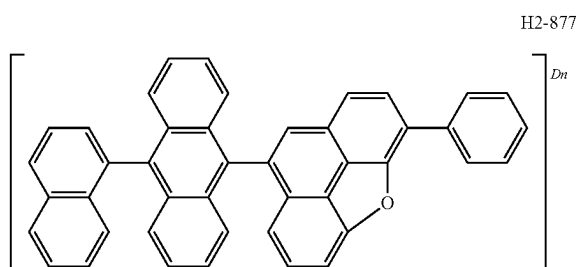
H2-876



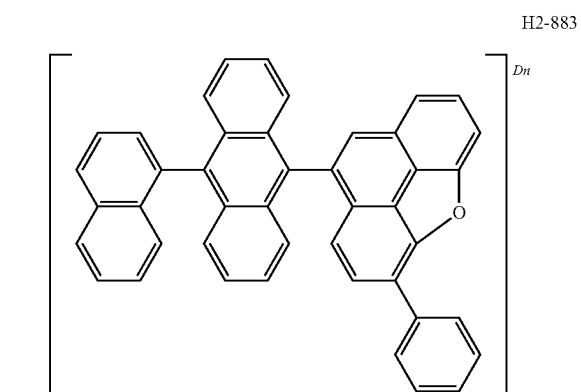
H2-881



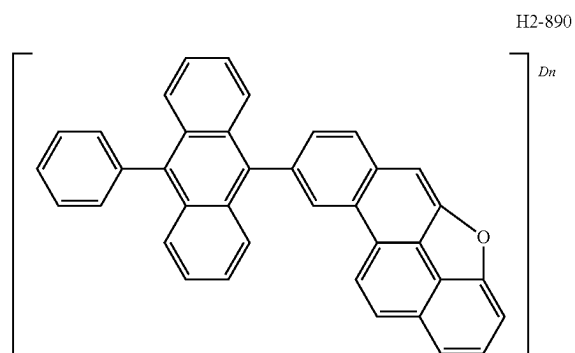
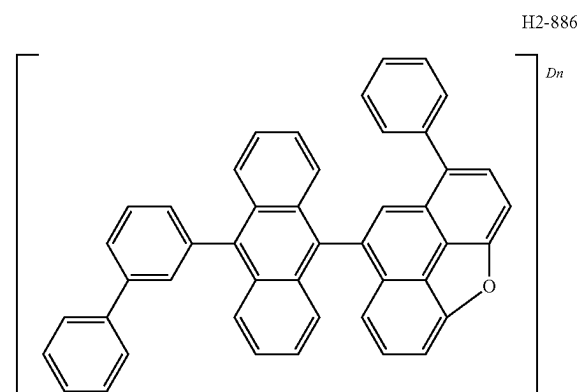
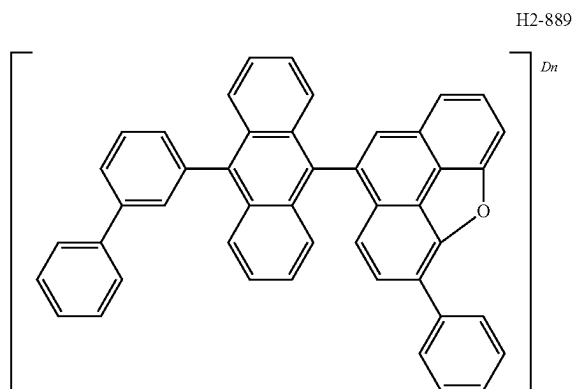
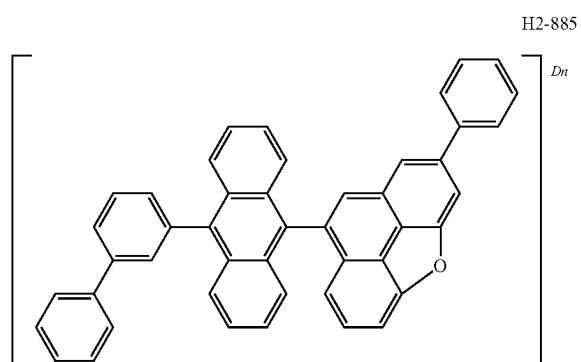
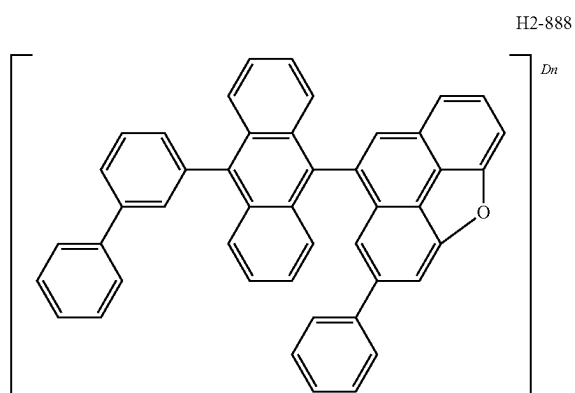
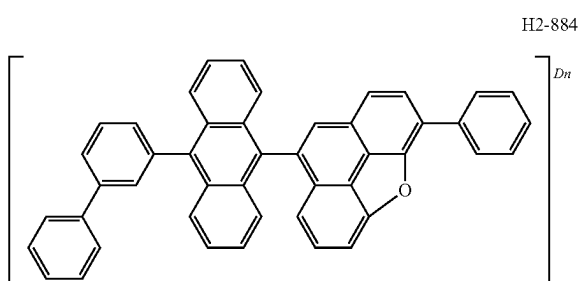
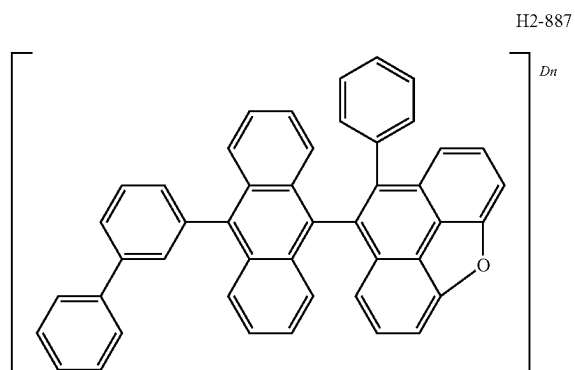
H2-882



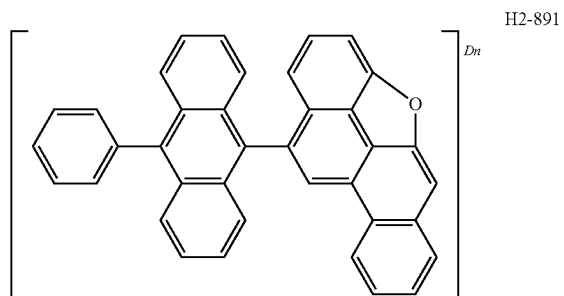
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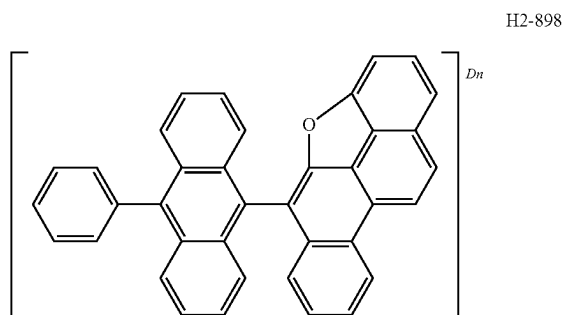
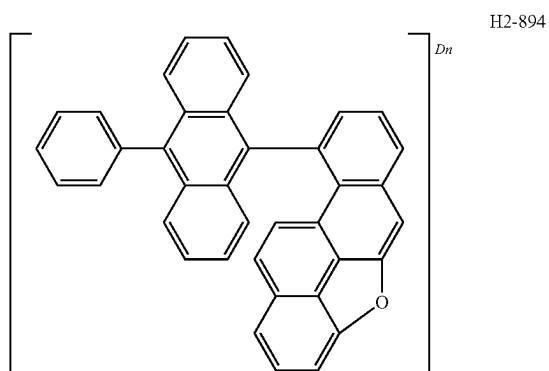
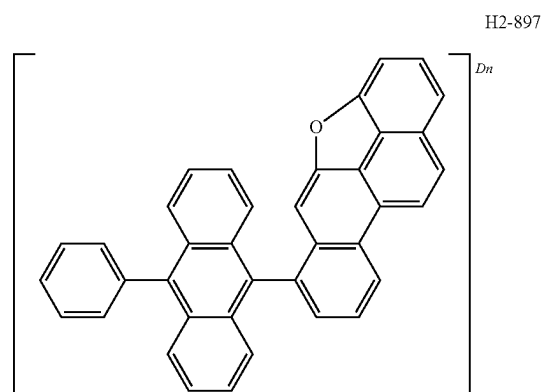
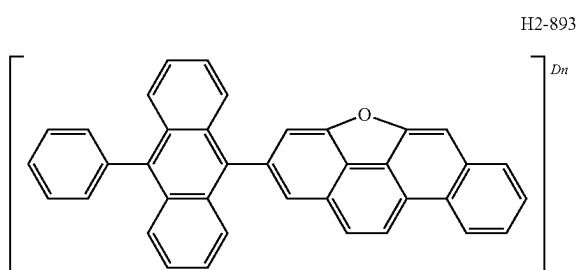
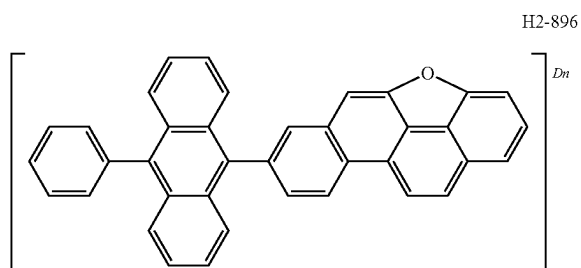
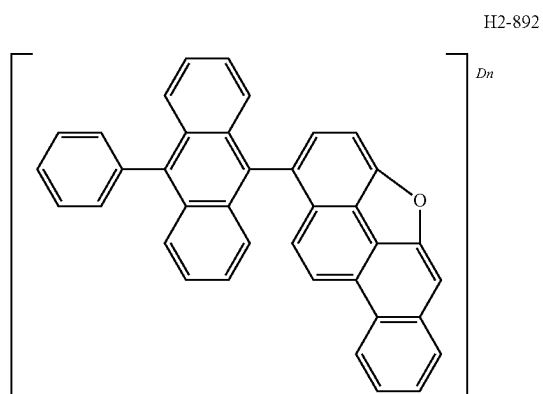
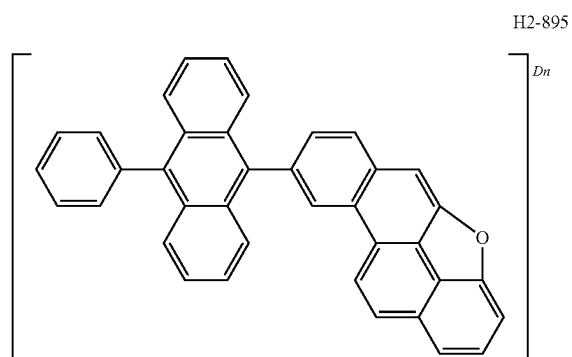
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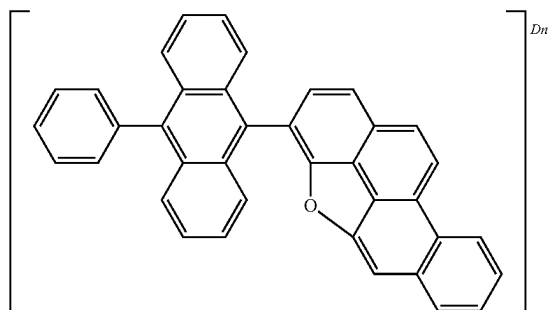


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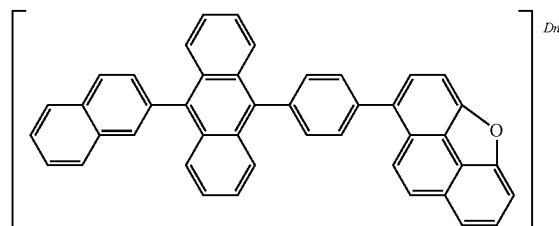
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H2-899

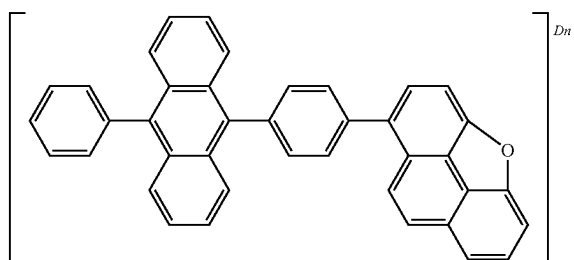


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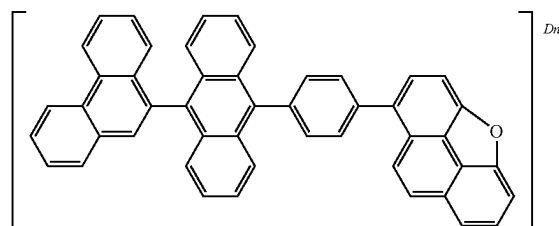
H2-904



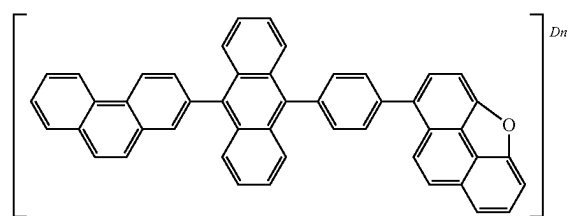
H2-900



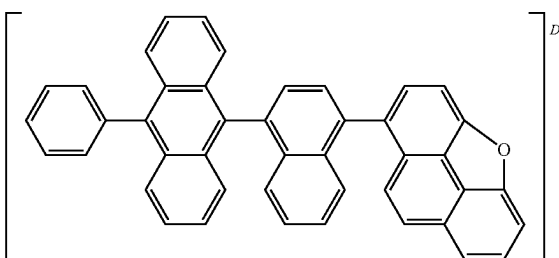
H2-905



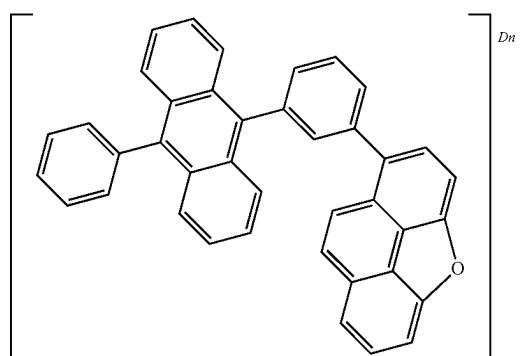
H2-901



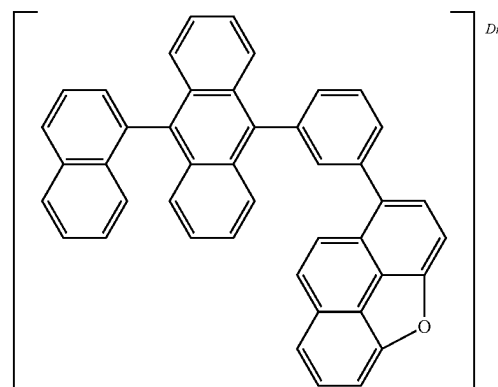
H2-906



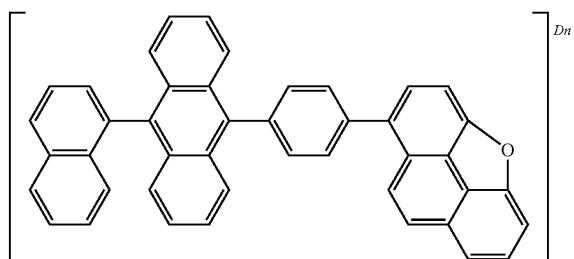
H2-902



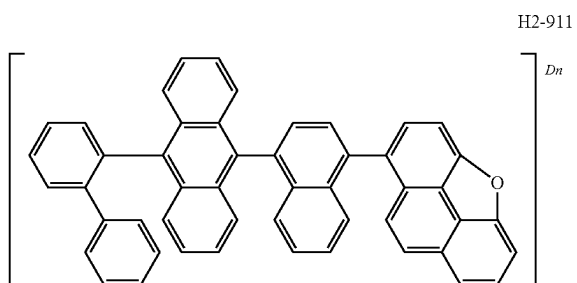
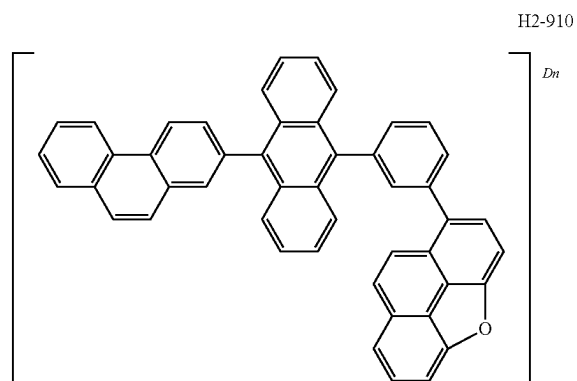
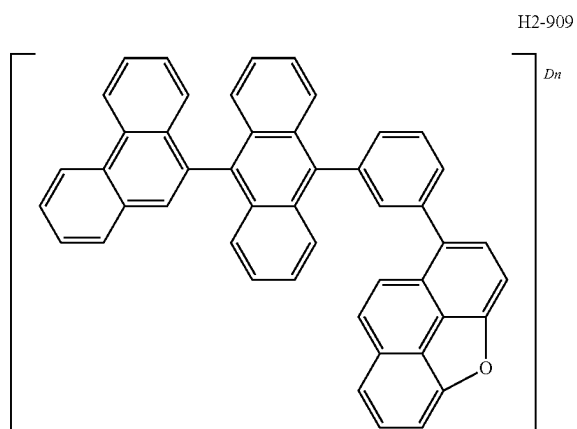
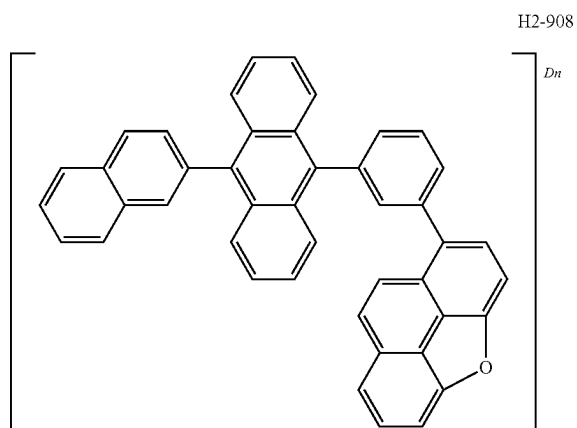
H2-907



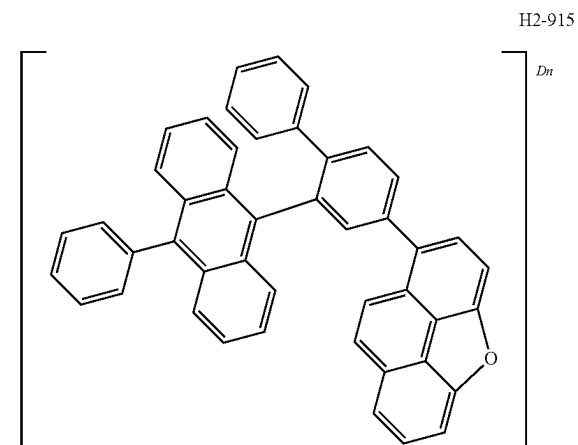
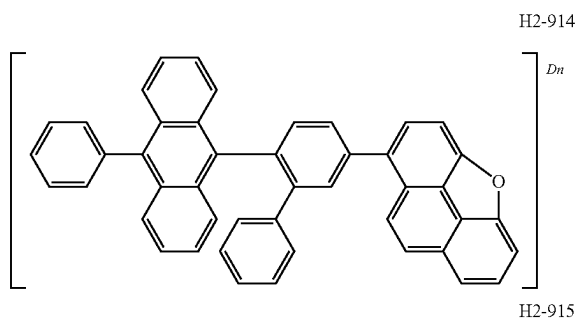
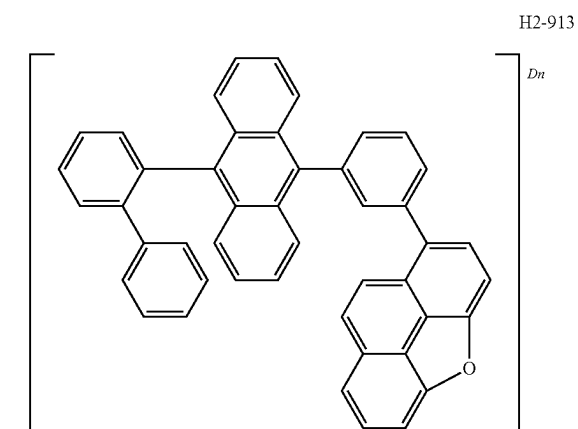
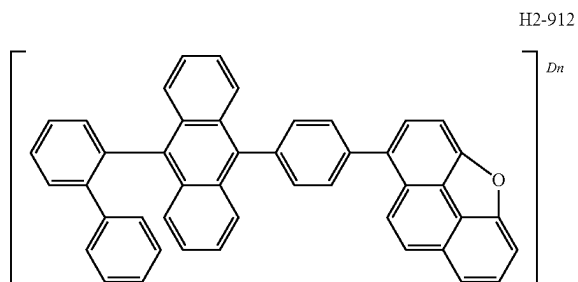
H2-903



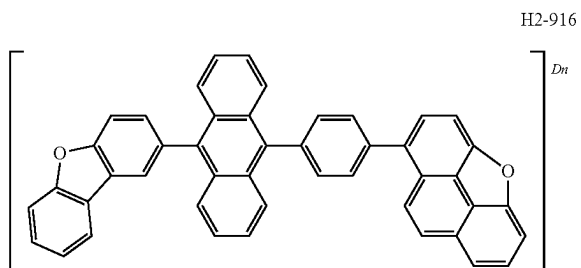
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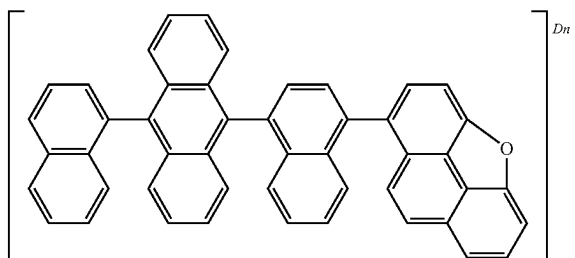
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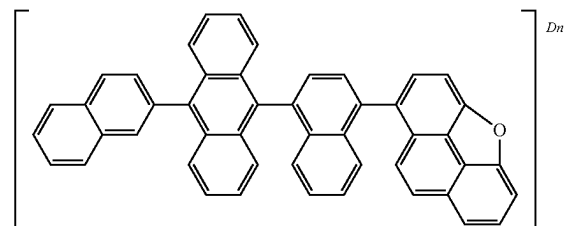
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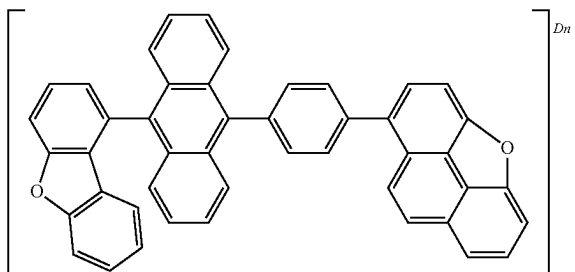
H2-917



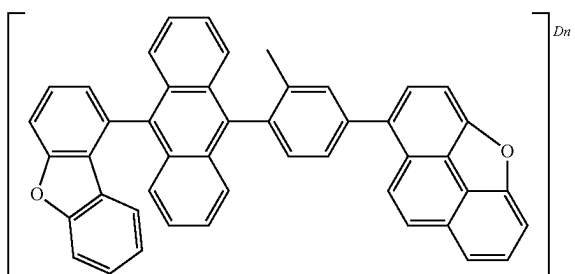
H2-918



H2-919

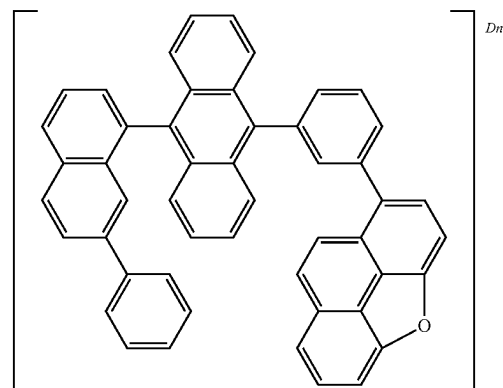


H2-920

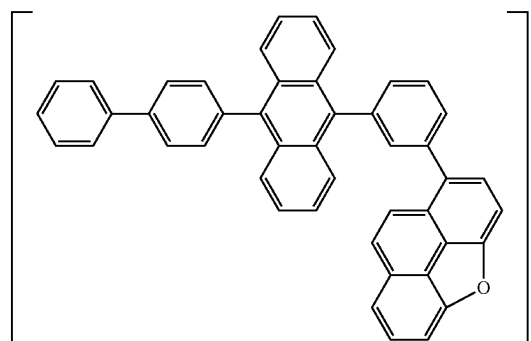


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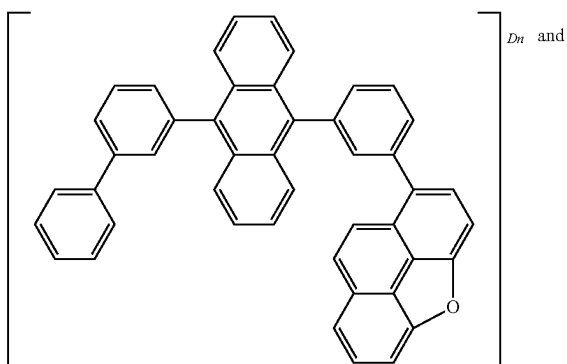
H2-921



H2-922

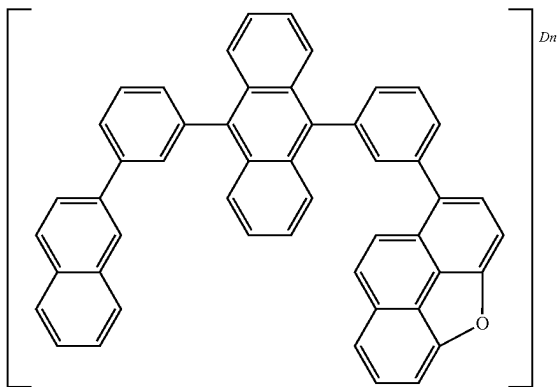


H2-923



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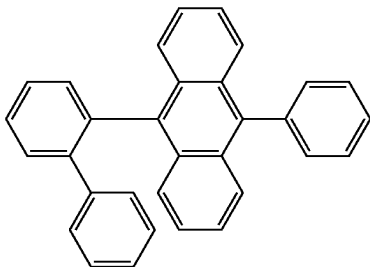
H2-924



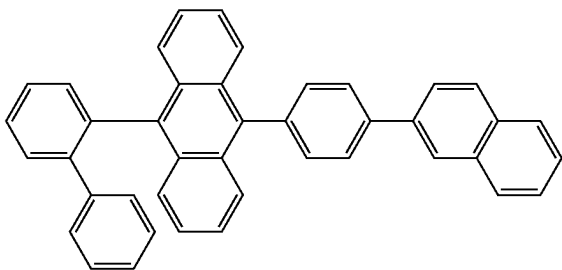
wherein D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to the maximum number of hydrogens in the compound.

11. The plurality of host materials according to claim 1, wherein the compound represented by Formula 2 is at least one selected from the following compounds:

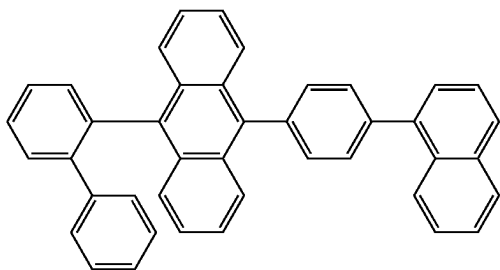
H1-1



H1-2

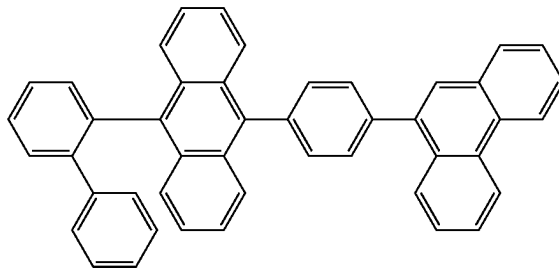


H1-3

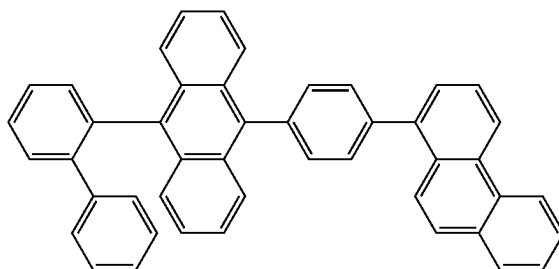


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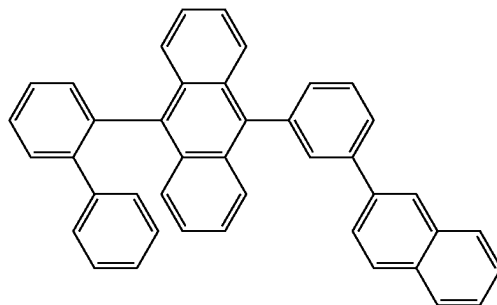
H1-4



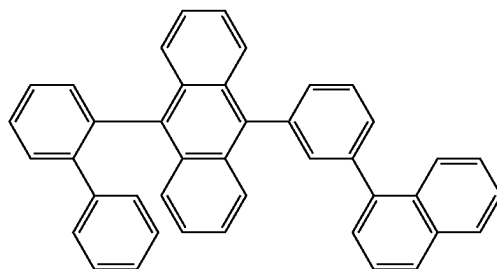
H1-5



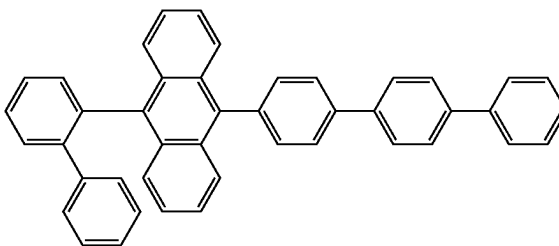
H1-6



H1-7

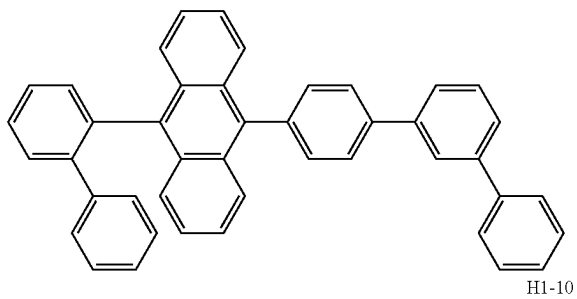


H1-8

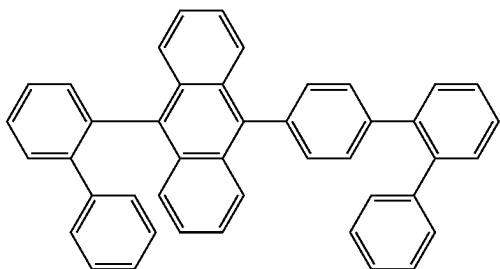


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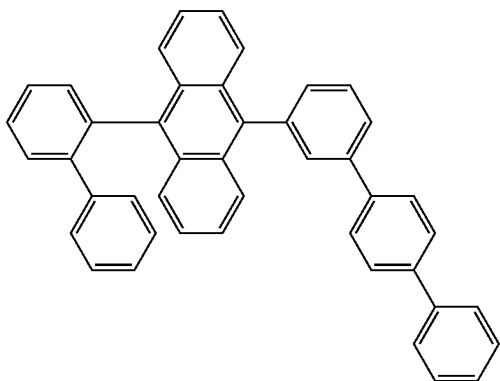
H1-9



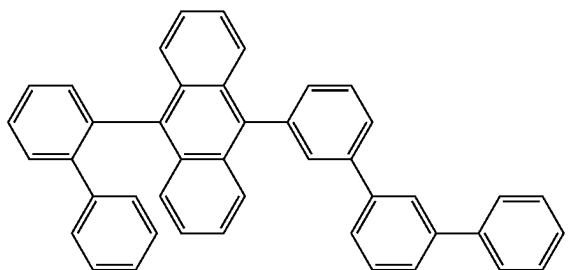
H1-10



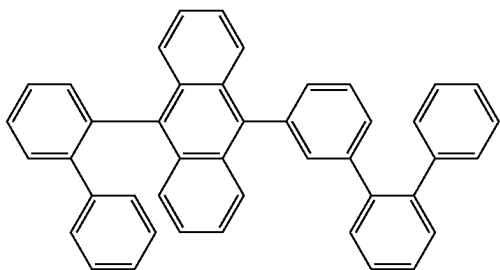
H1-11



H1-12

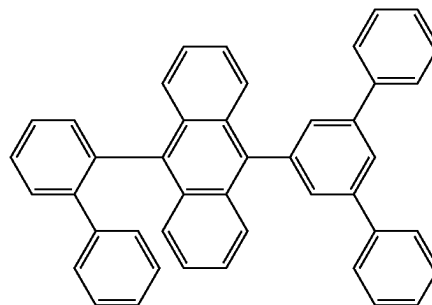


H1-13

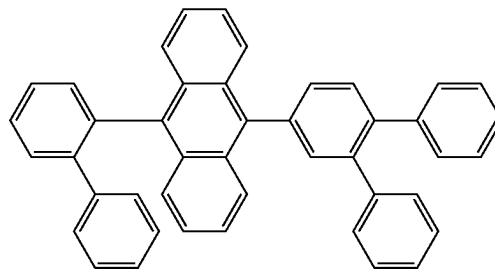


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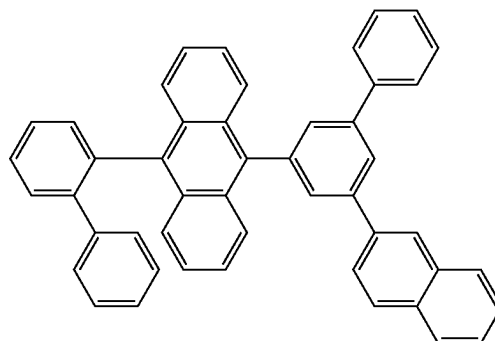
H1-14



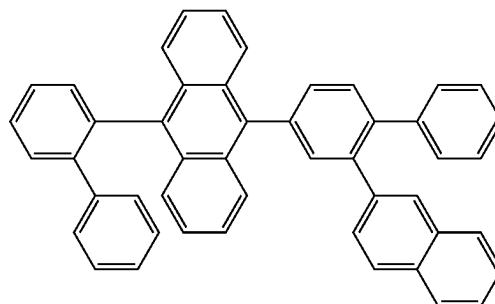
H1-15



H1-16

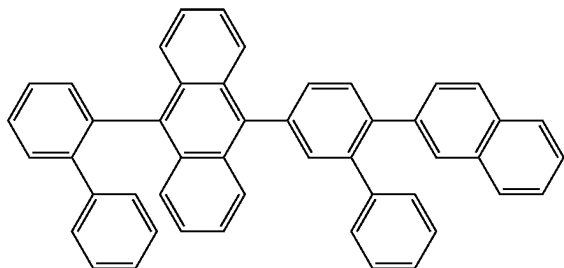


H1-17

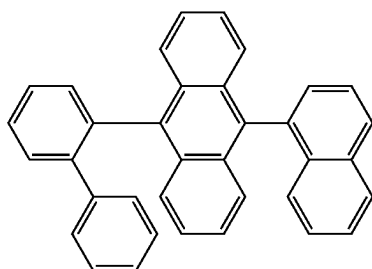


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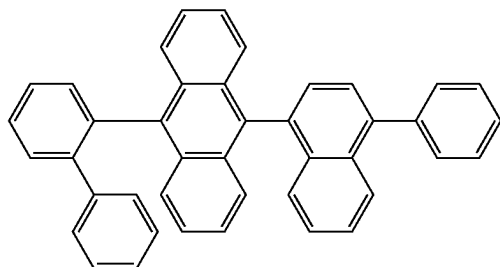
H1-18



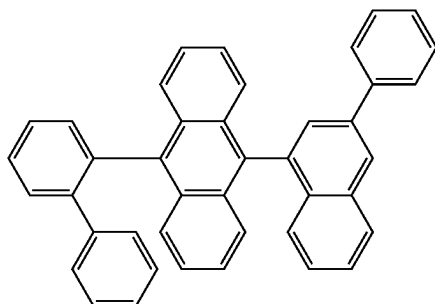
H1-19



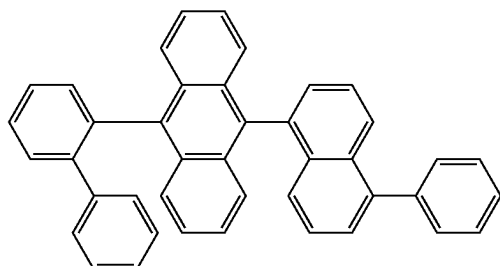
H1-20



H1-21

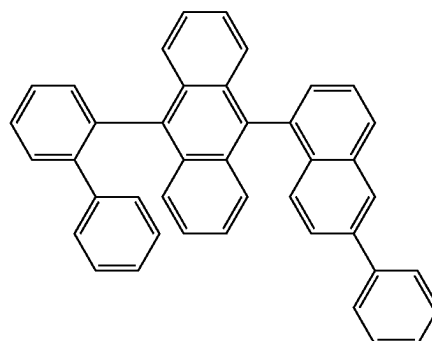


H1-22

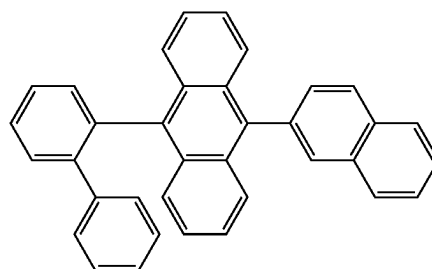


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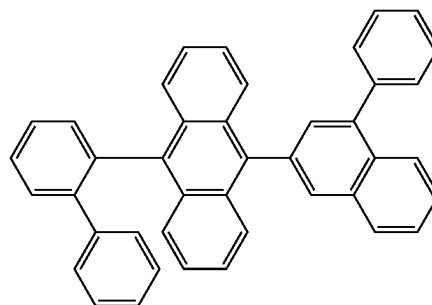
H1-23



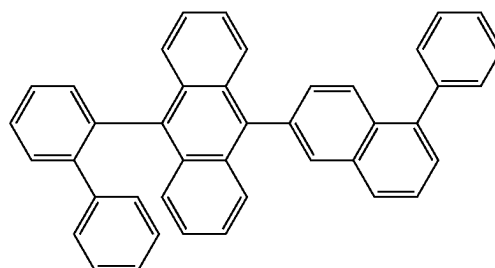
H1-24



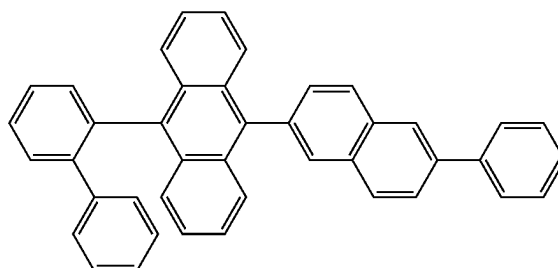
H1-25



H1-26

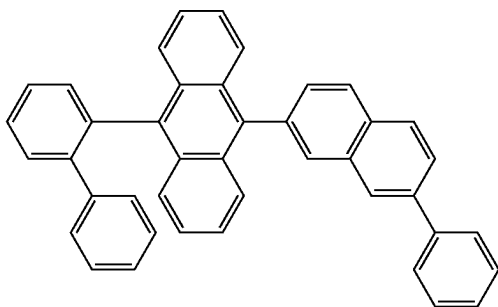


H1-27



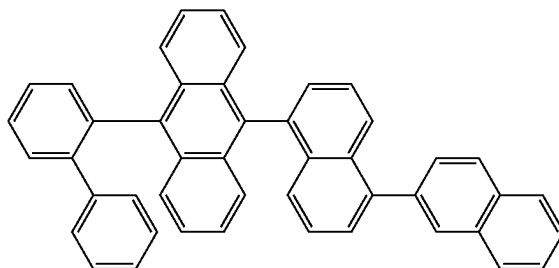
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H1-28



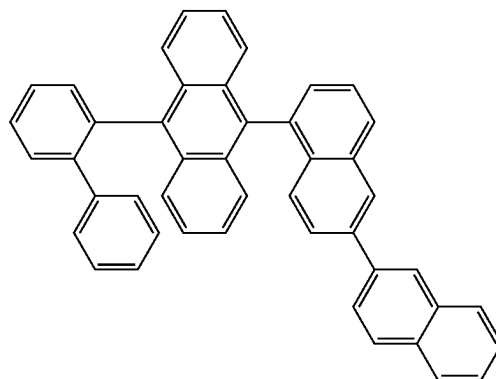
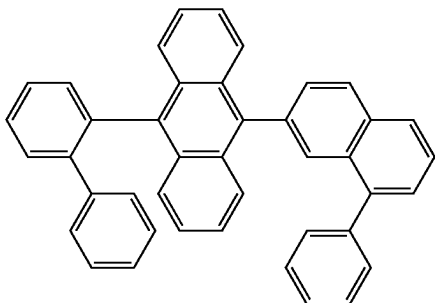
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H1-32



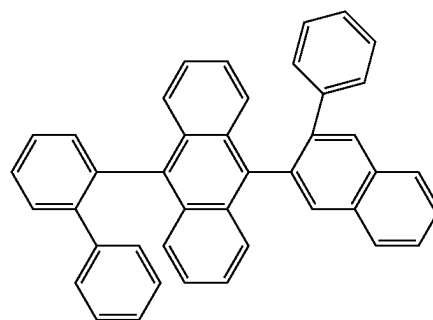
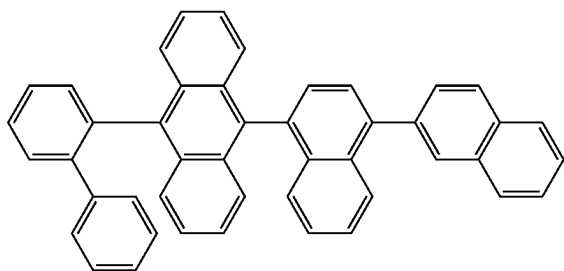
H1-33

H1-29



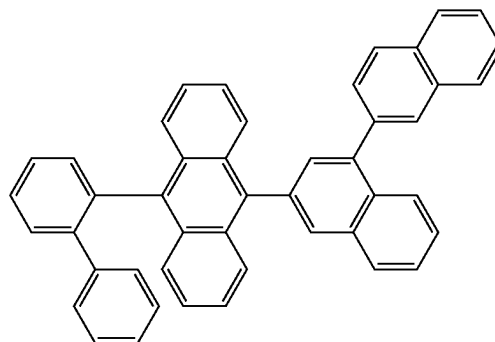
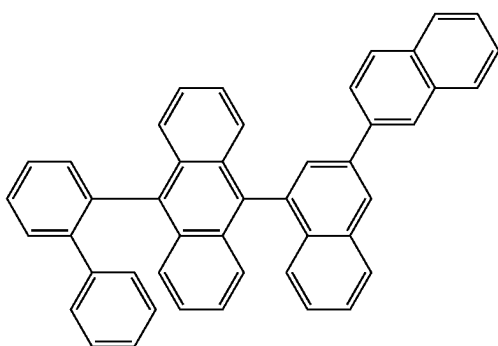
H1-30

H1-34

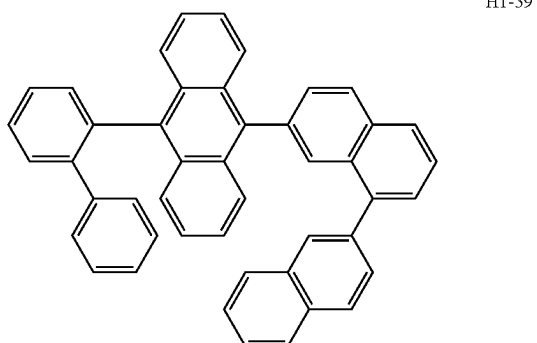
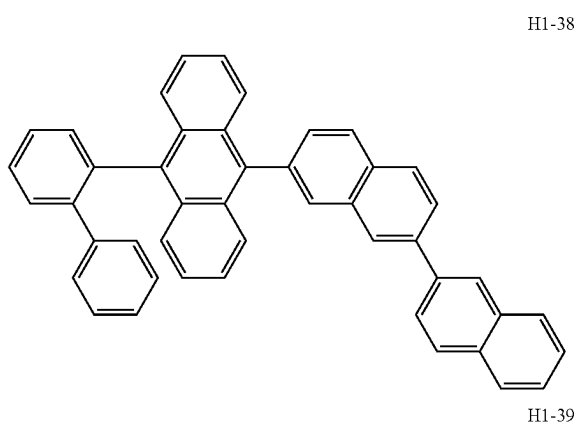
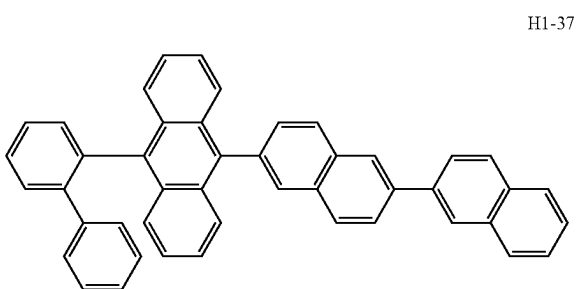
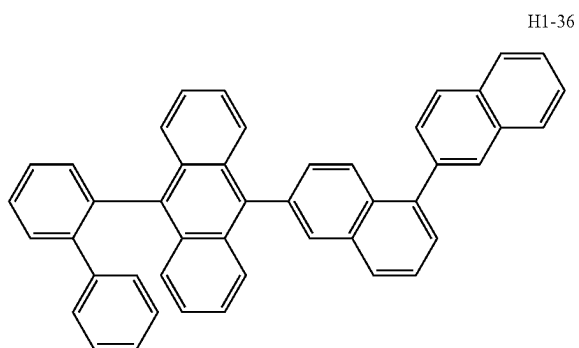


H1-31

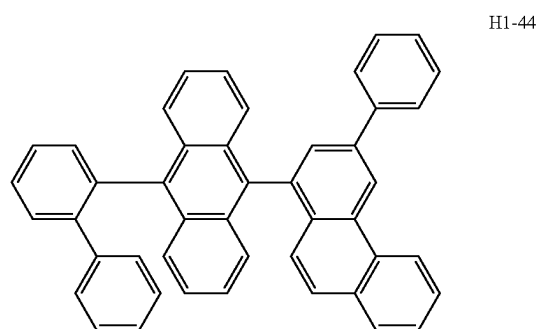
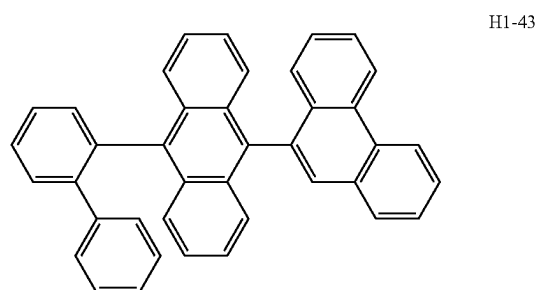
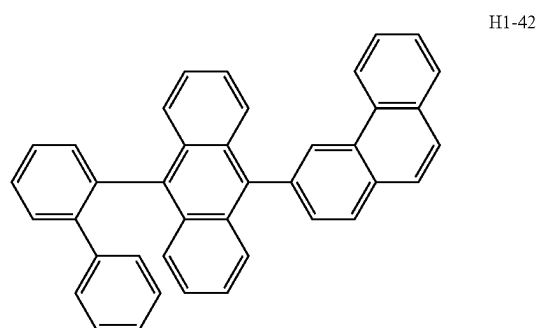
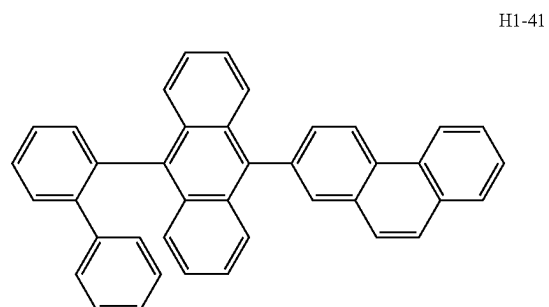
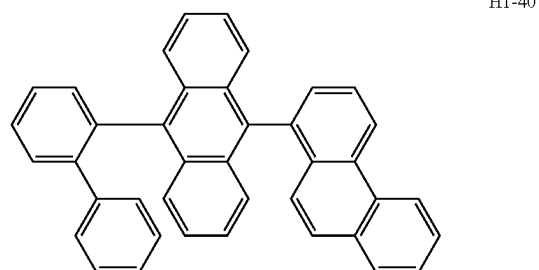
H1-35



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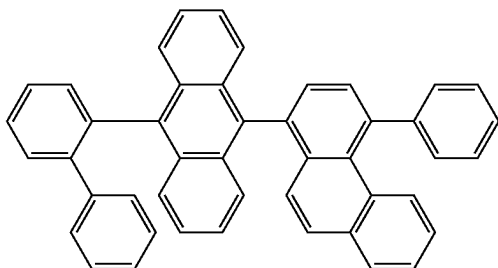


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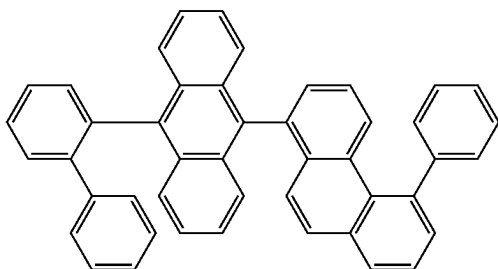


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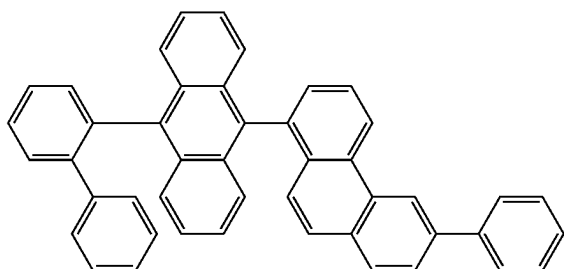
H1-45



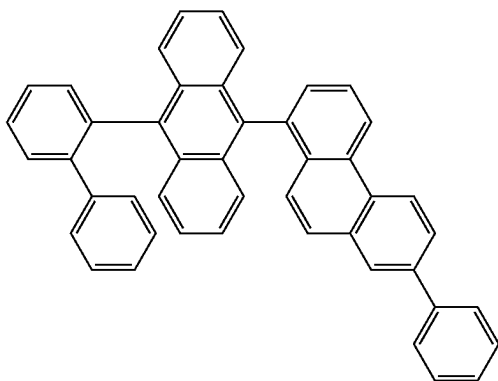
H1-46



H1-47

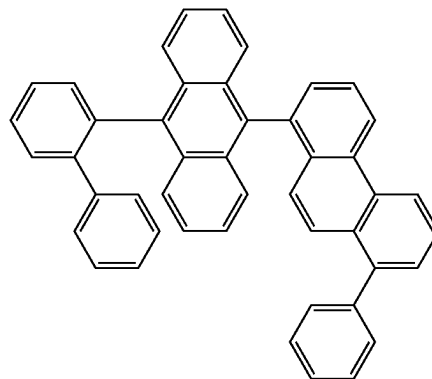


H1-48

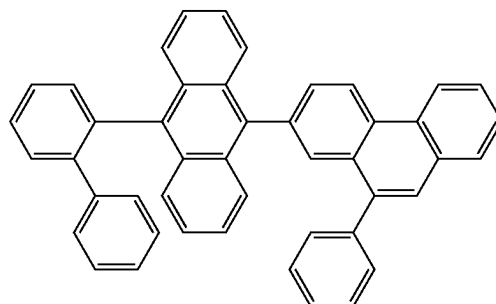


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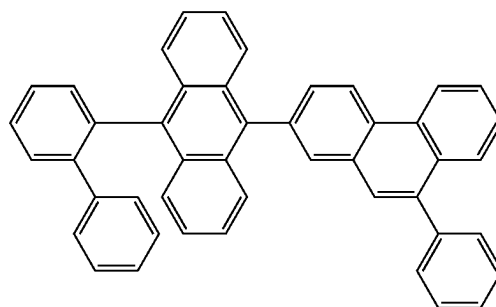
H1-49



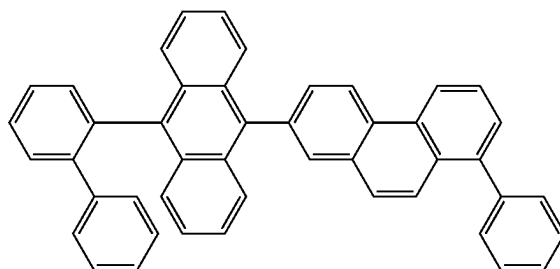
H1-50



H1-51

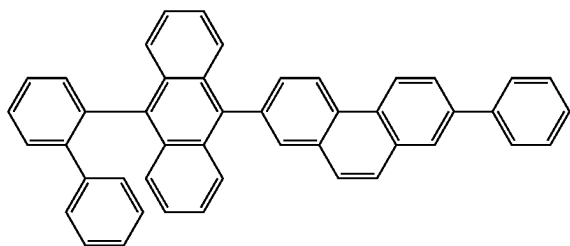


H1-52



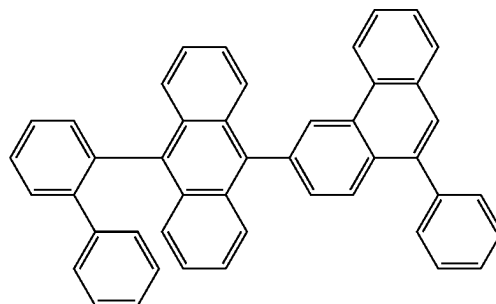
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H1-53

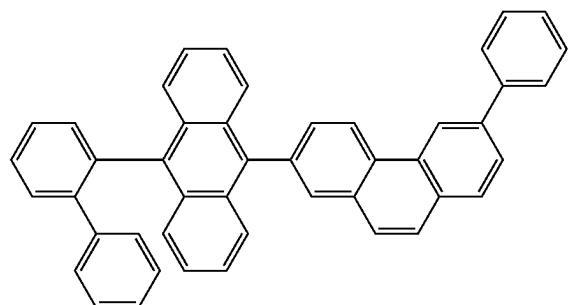


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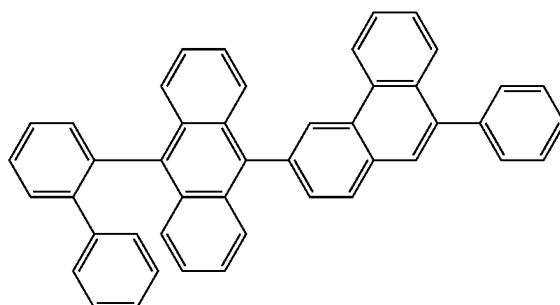
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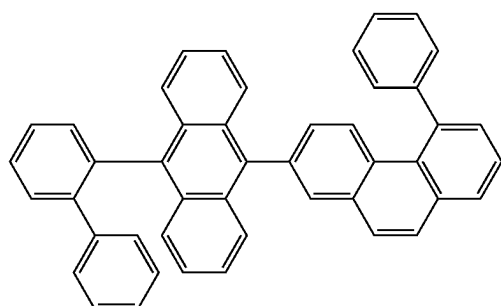
H1-54



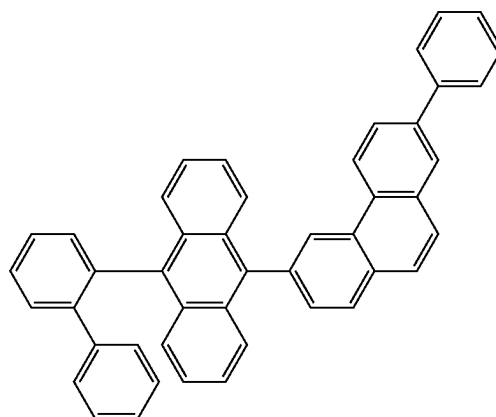
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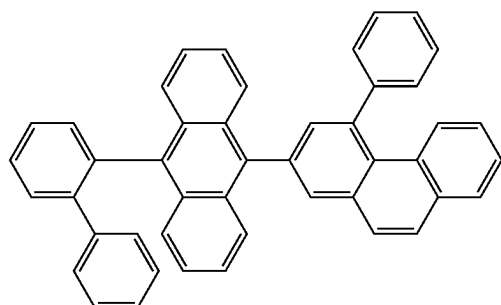
H1-55



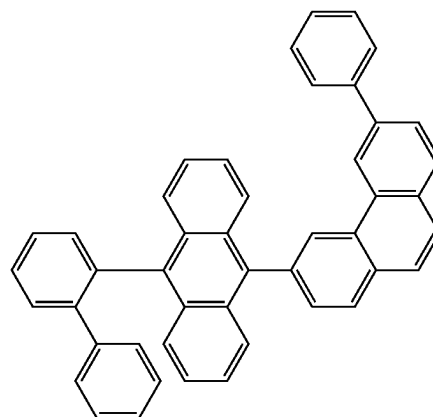
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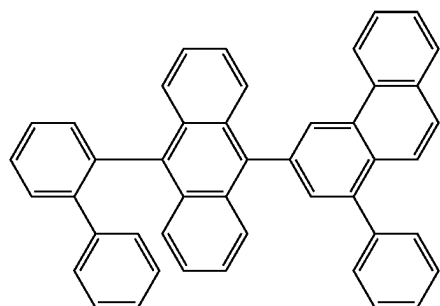
H1-56



H1-61

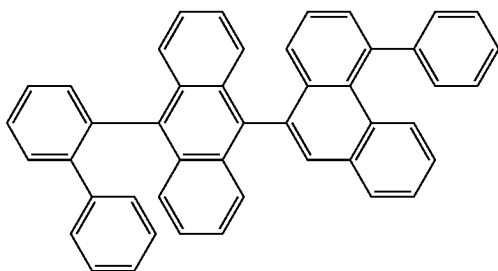


H1-57



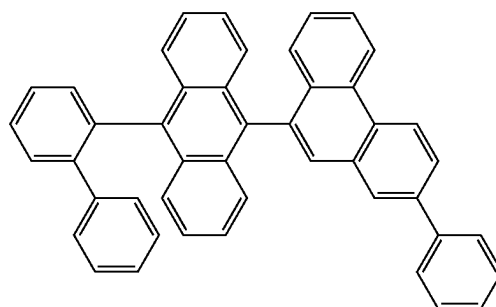
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H1-62

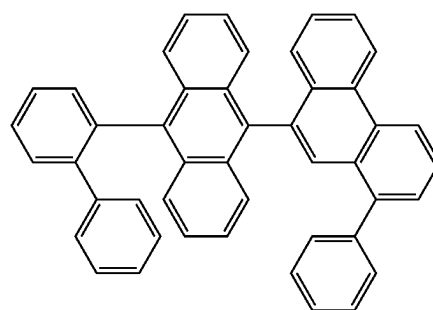


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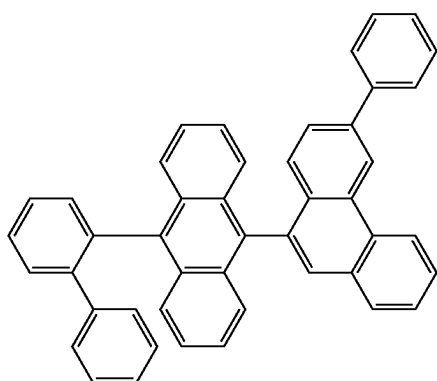
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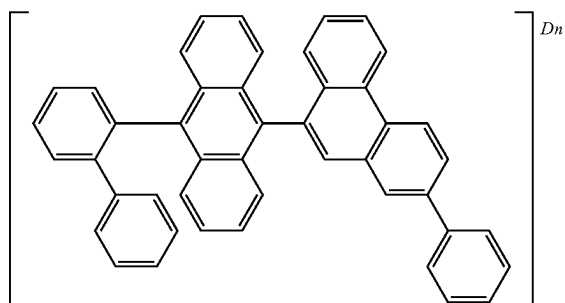
H1-67



H1-63

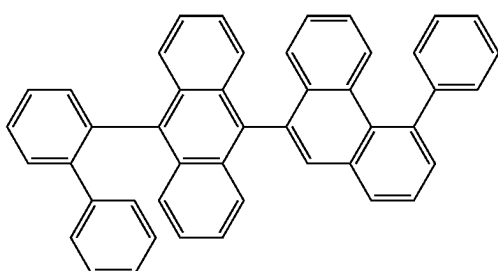


H1-68

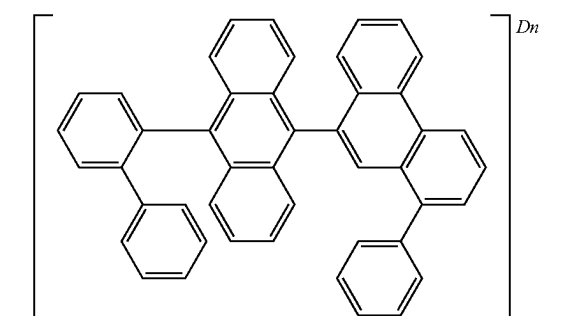
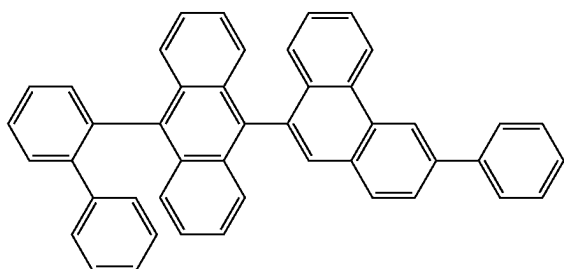


H1-69

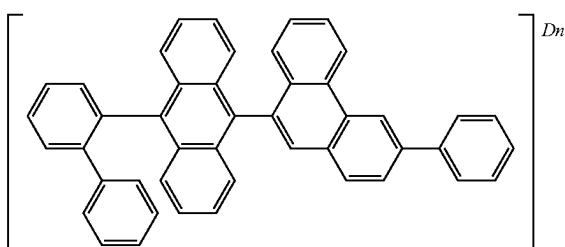
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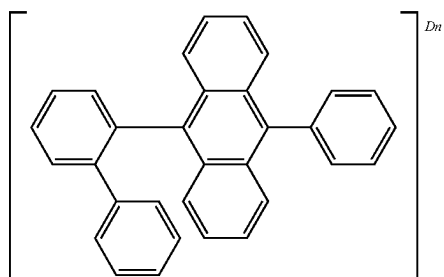
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H1-70

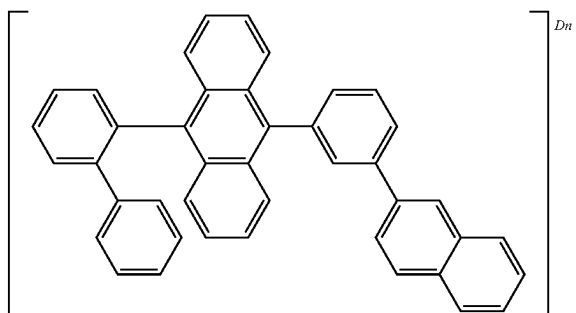


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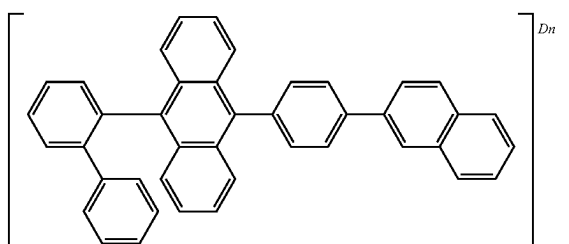


H1-71

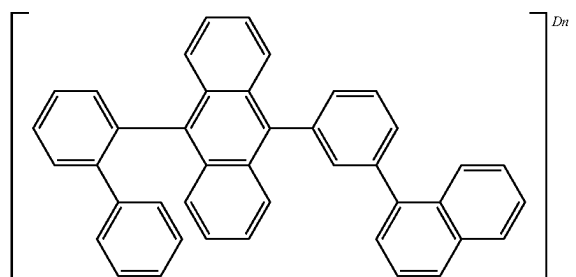
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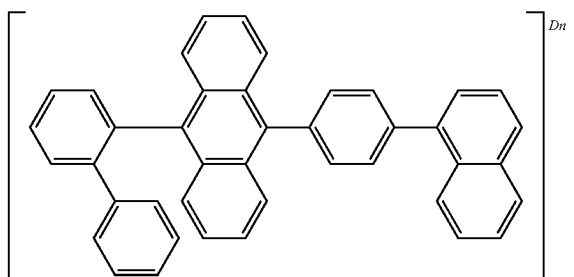
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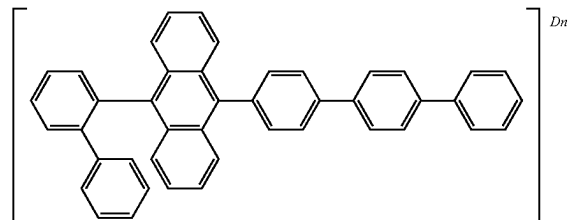
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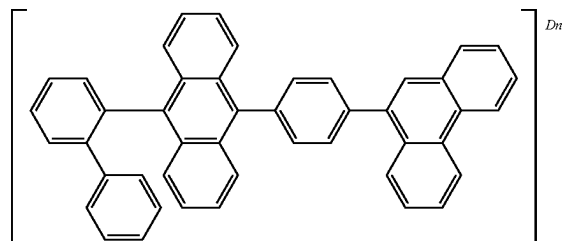
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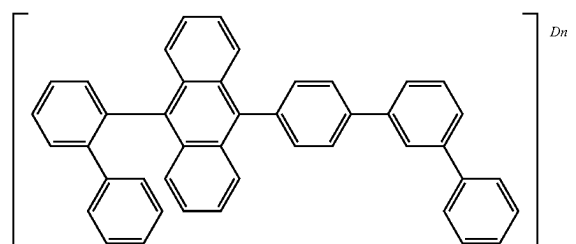
H1-73



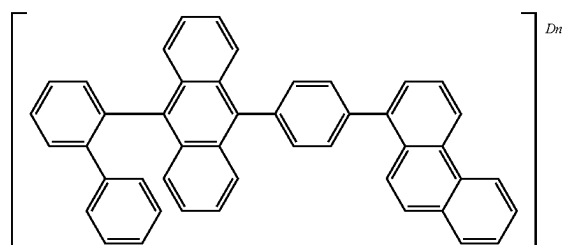
H1-78



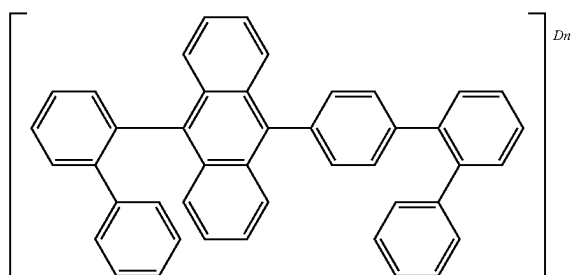
H1-74



H1-79

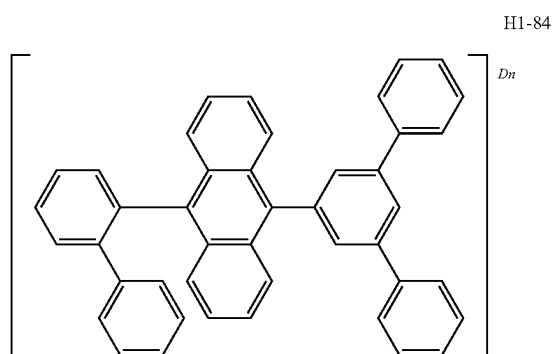
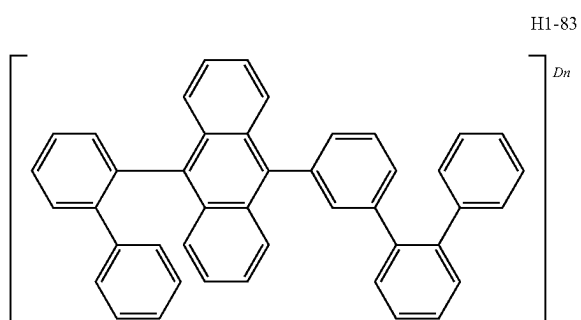
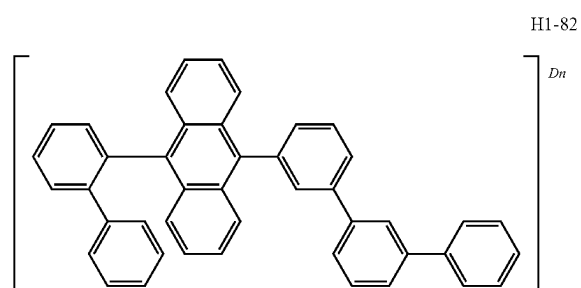
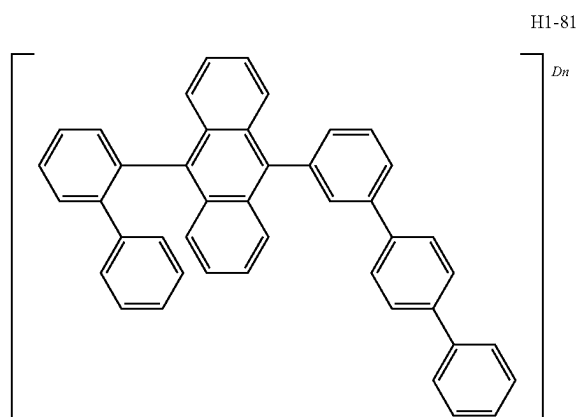


H1-75

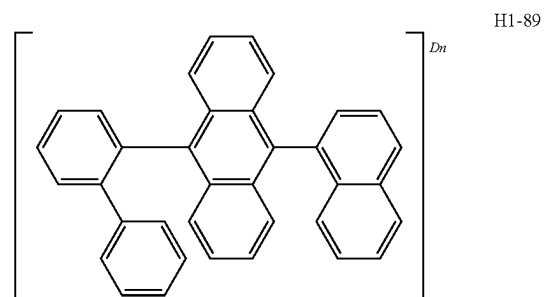
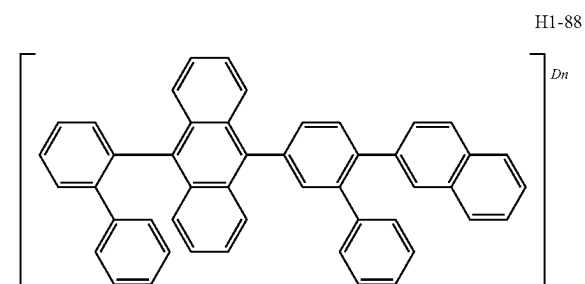
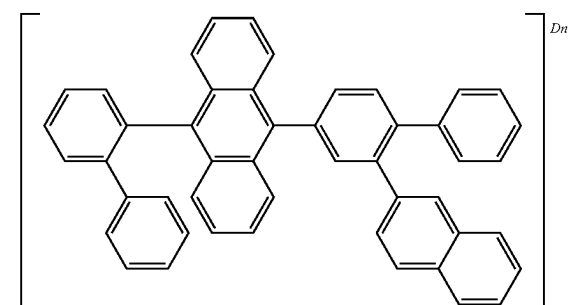
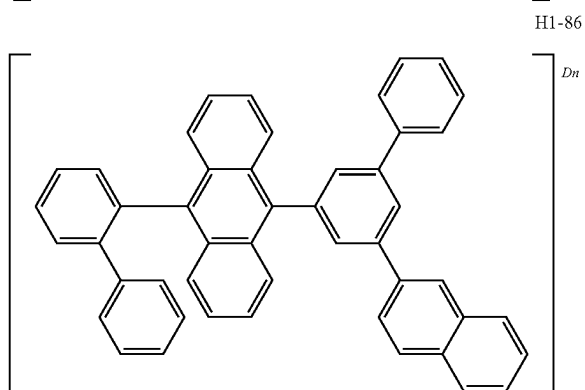
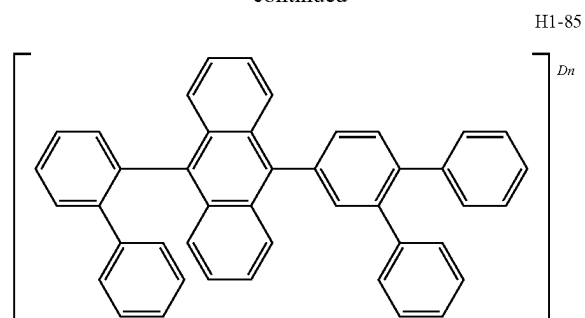


H1-80

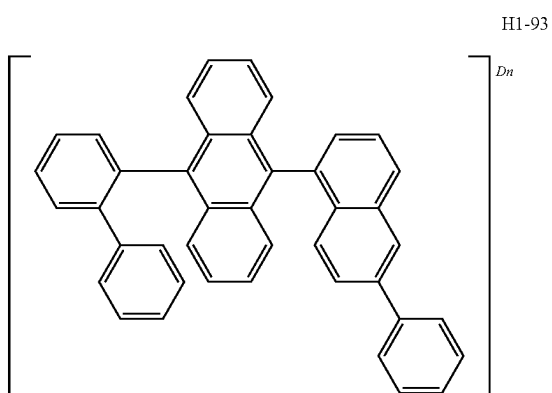
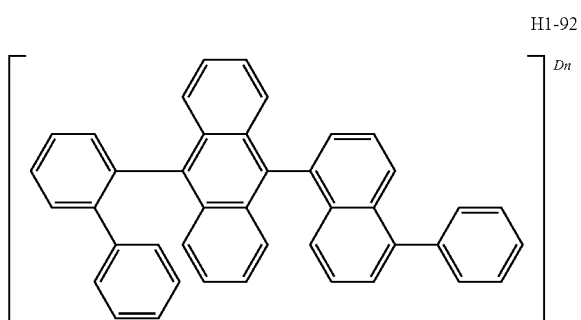
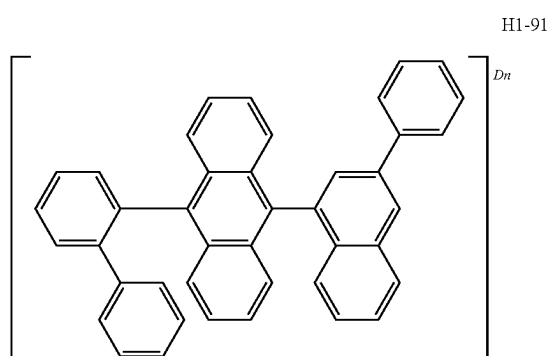
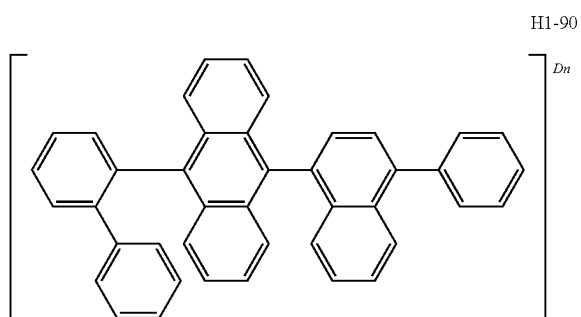
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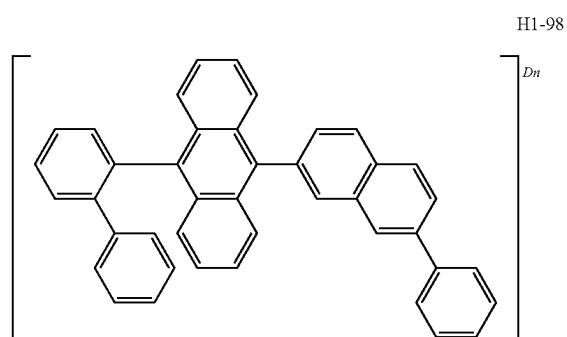
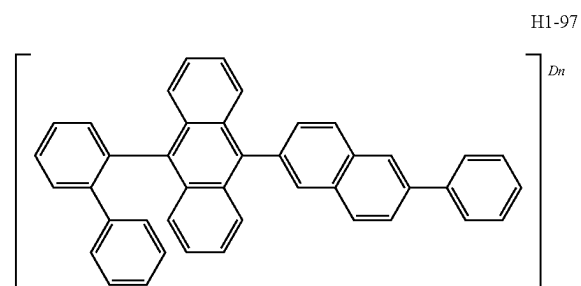
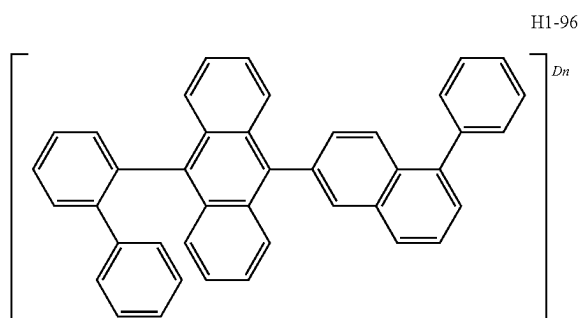
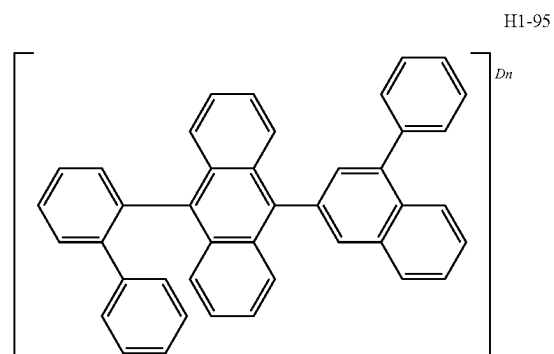
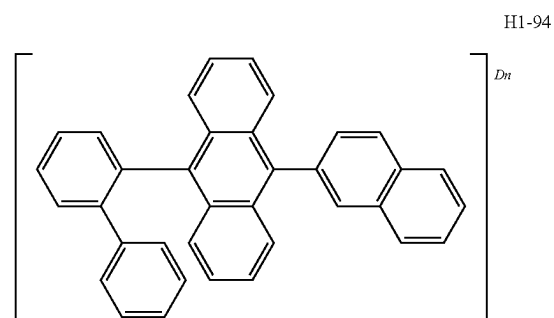
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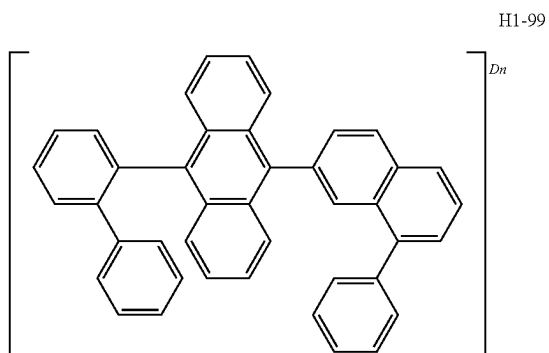
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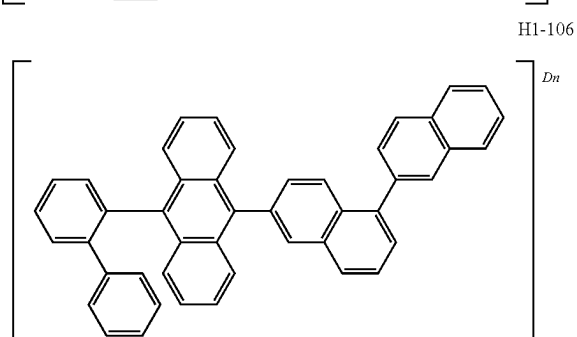
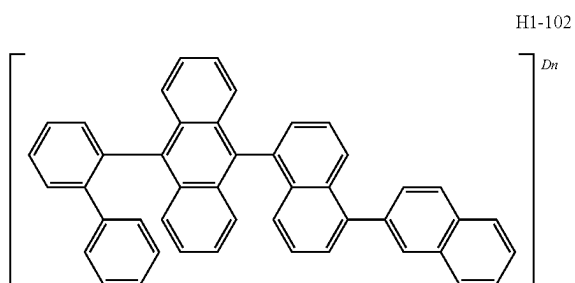
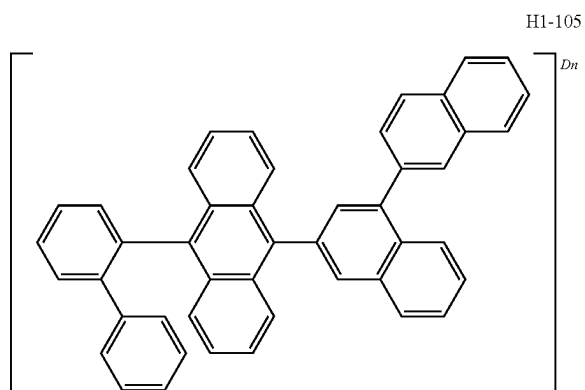
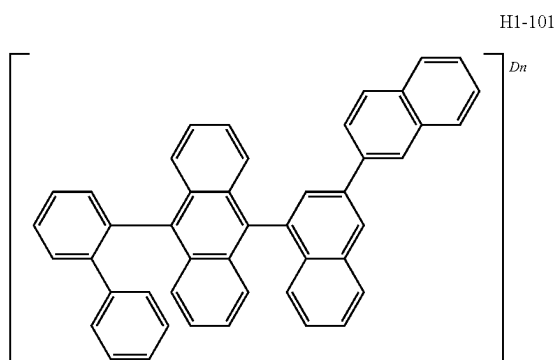
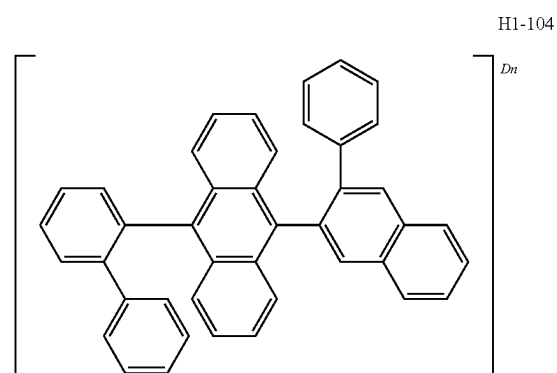
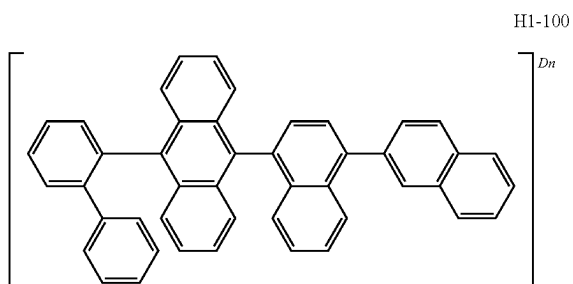
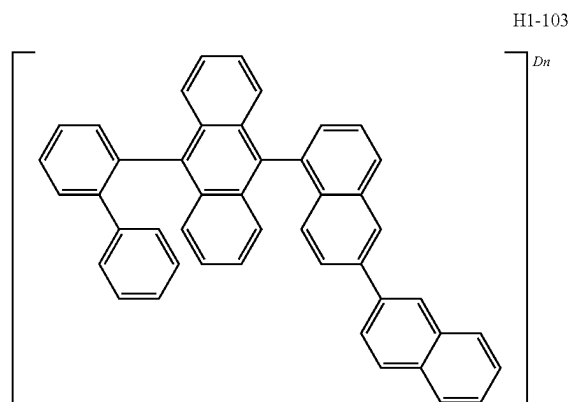
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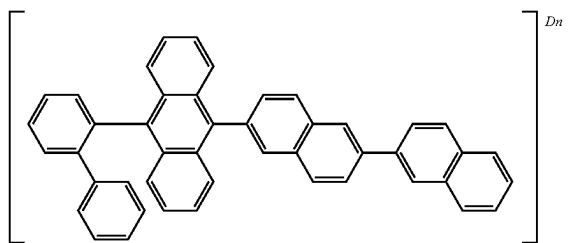


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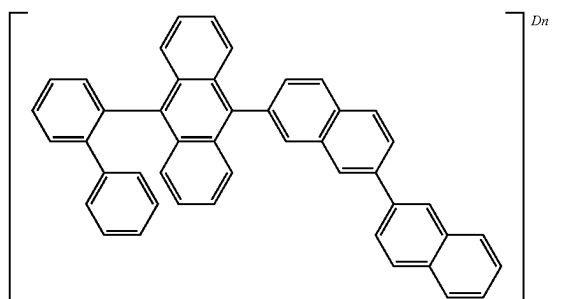


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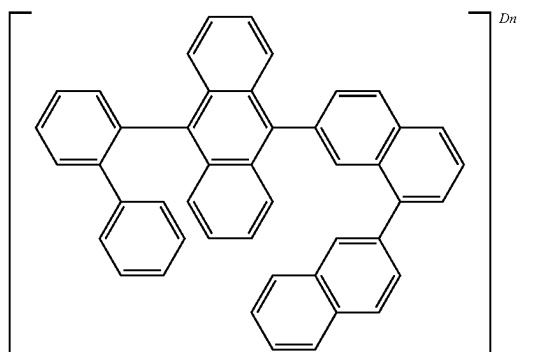
H1-107



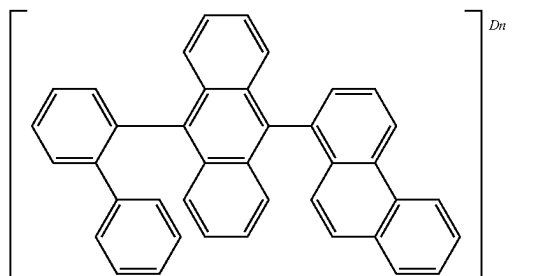
H1-108



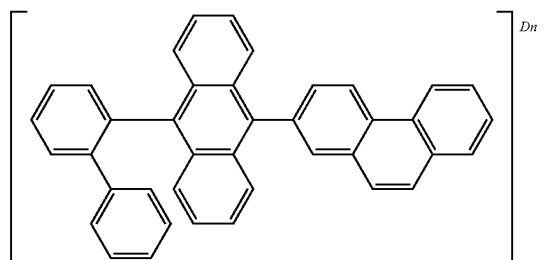
H1-109



H1-110

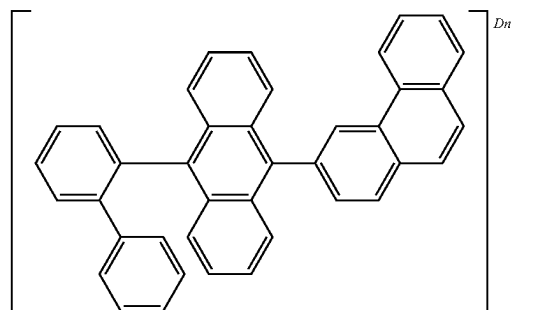


H1-111

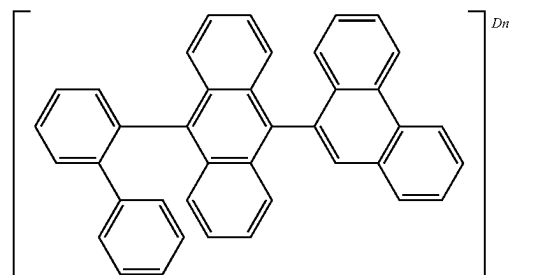


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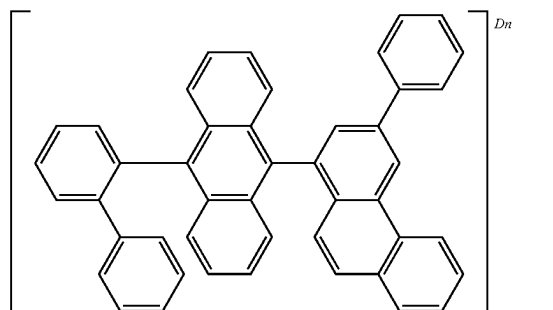
H1-112



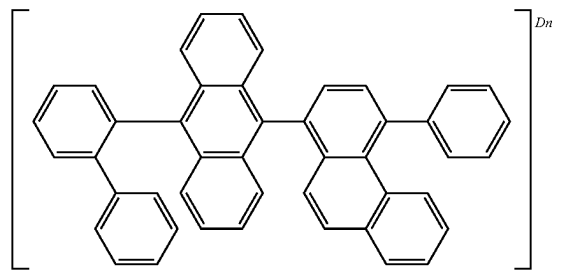
H1-113



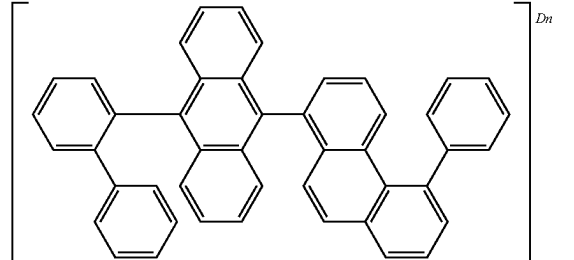
H1-114



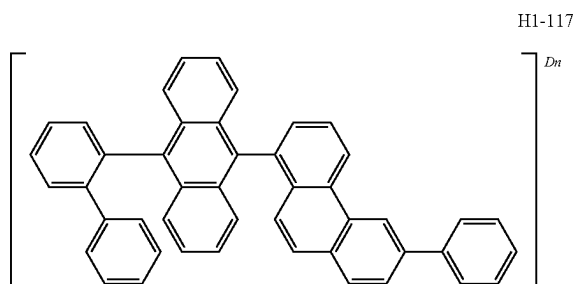
H1-115



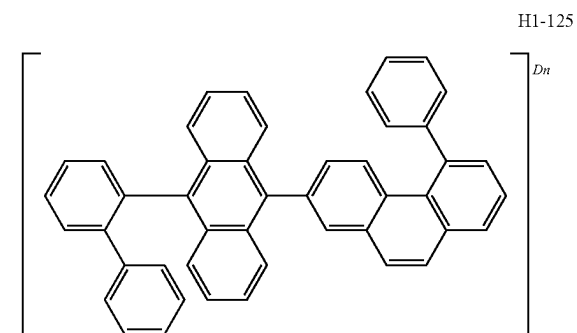
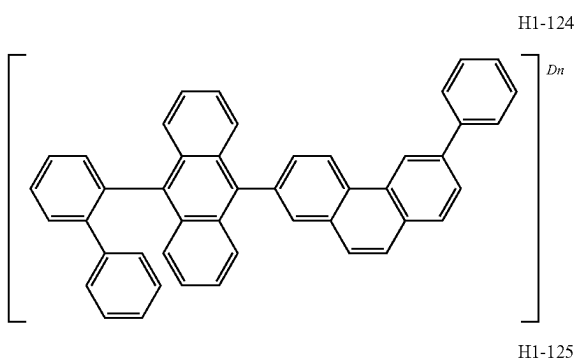
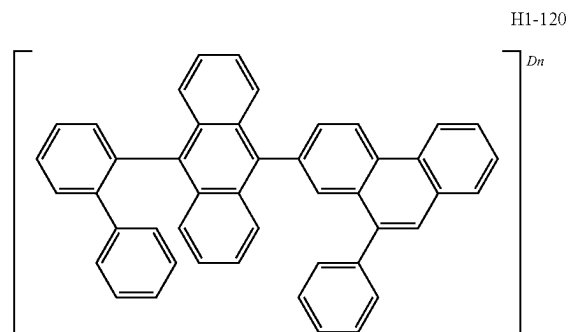
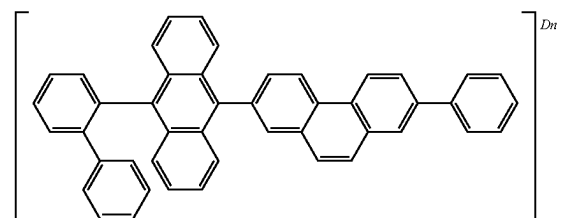
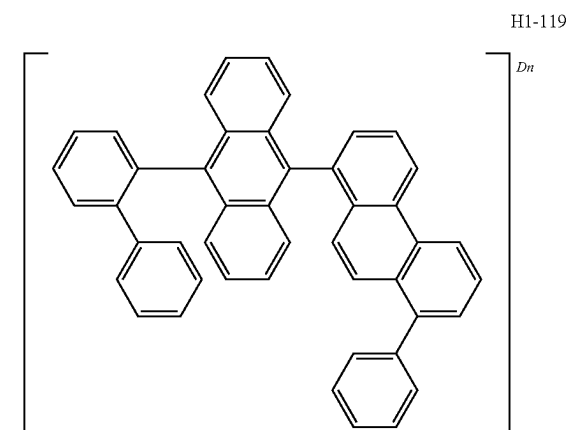
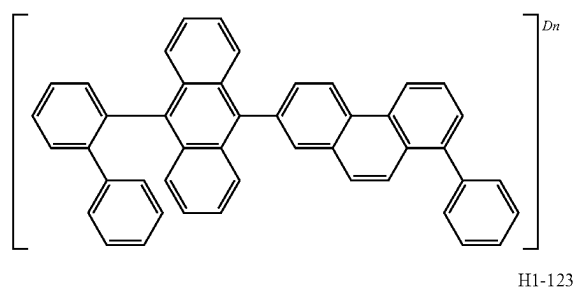
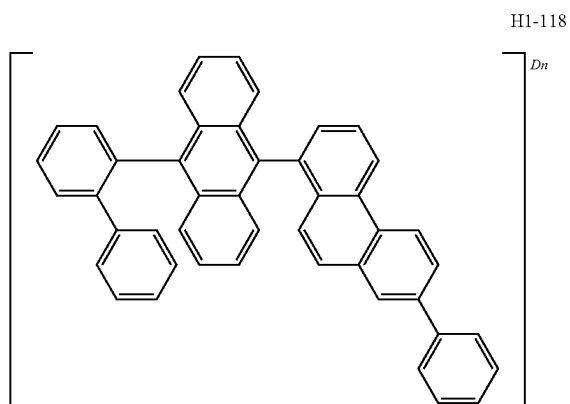
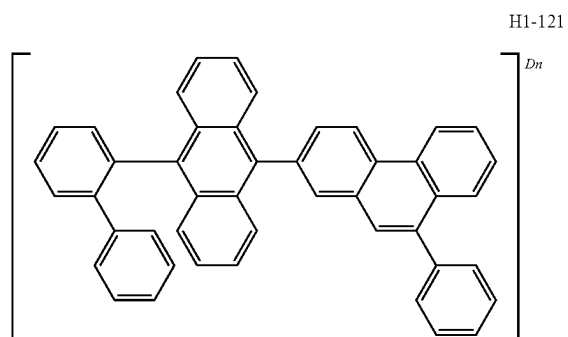
H1-116



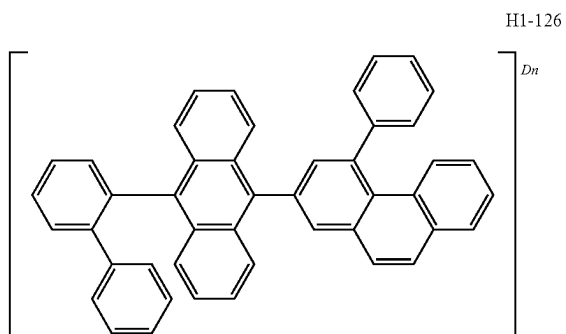
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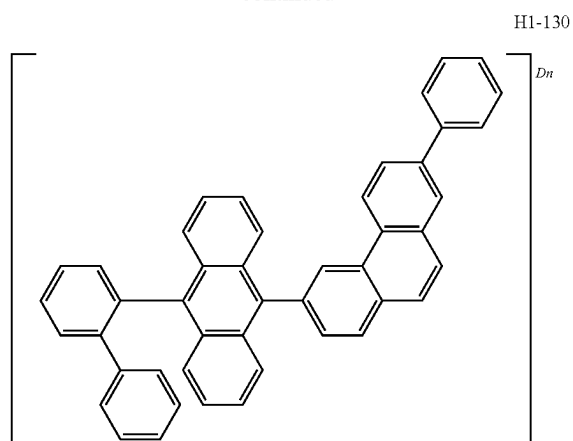
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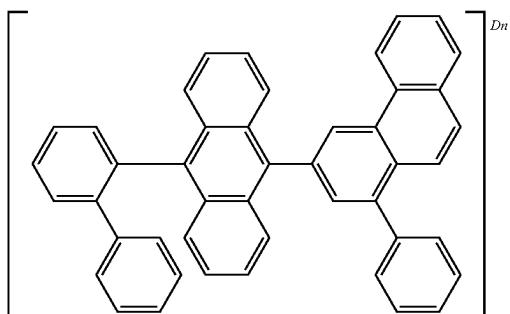
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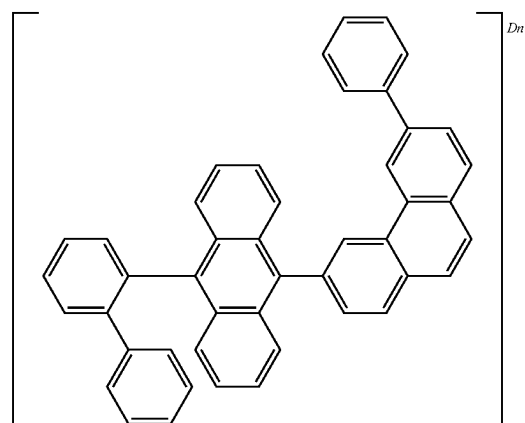
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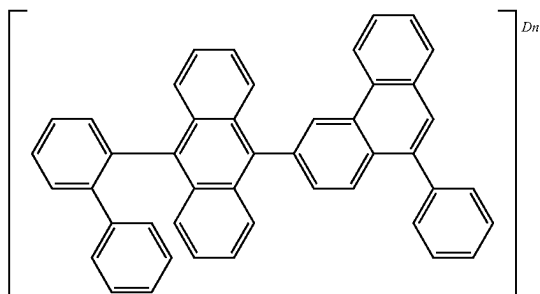
H1-127



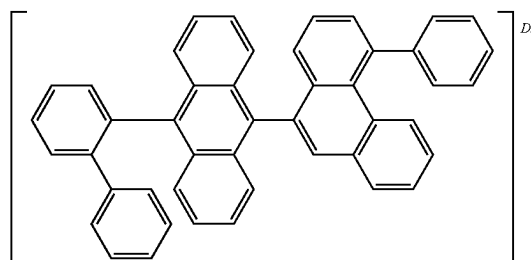
H1-131



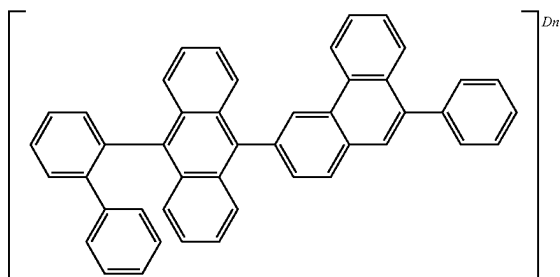
H1-128



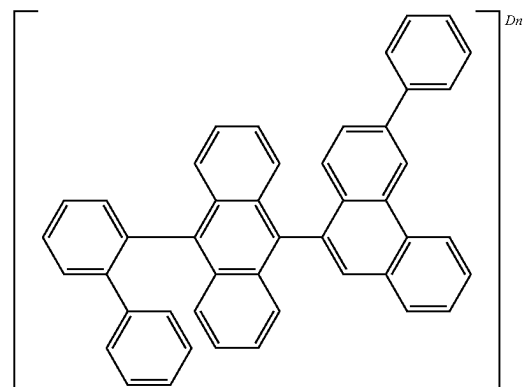
H1-132



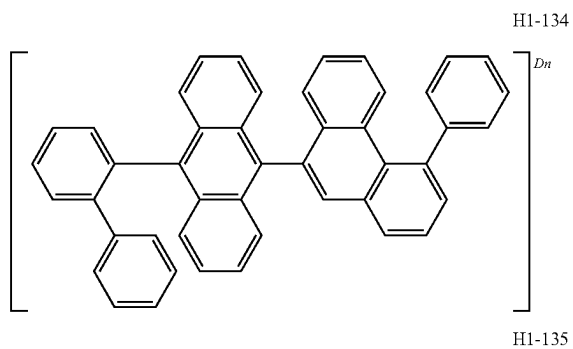
H1-129



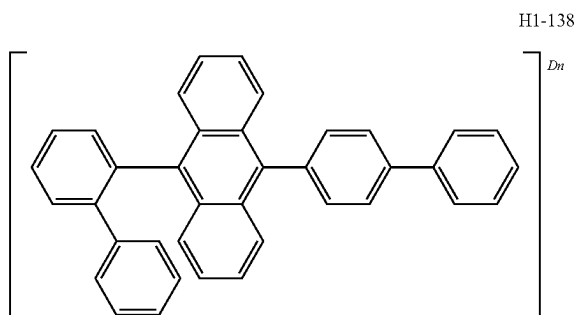
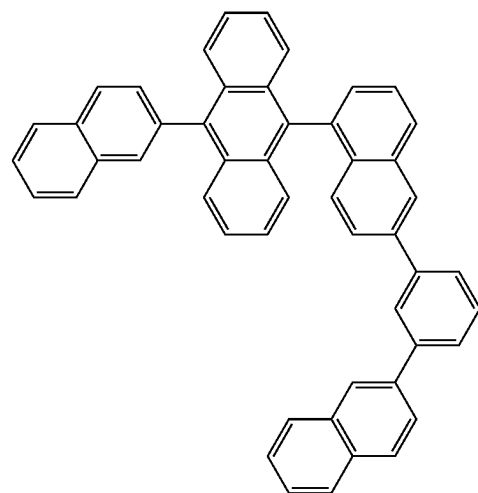
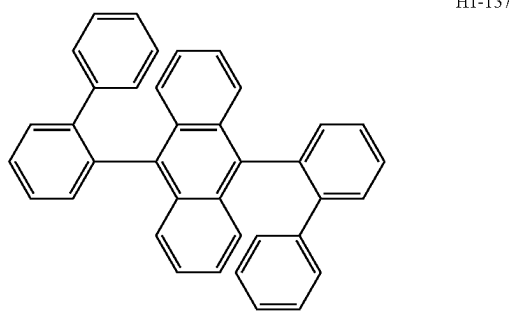
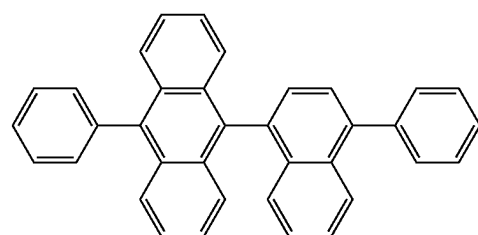
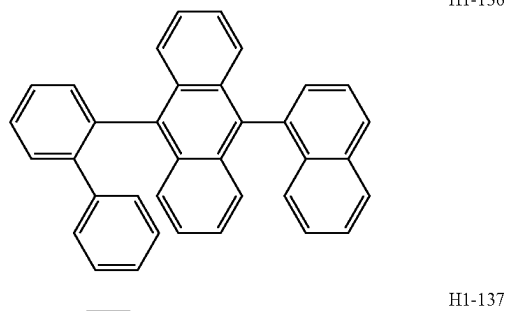
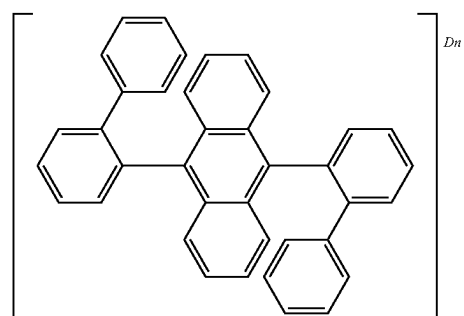
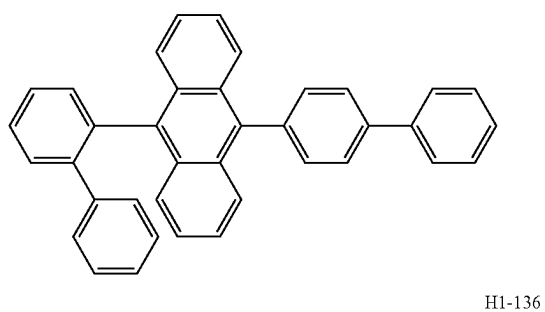
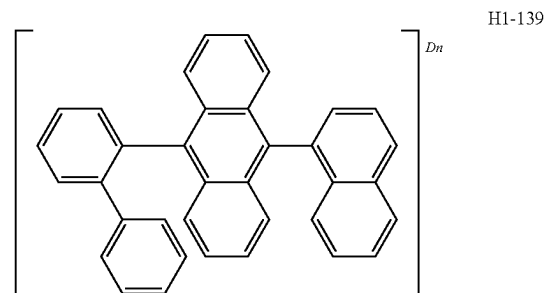
H1-133



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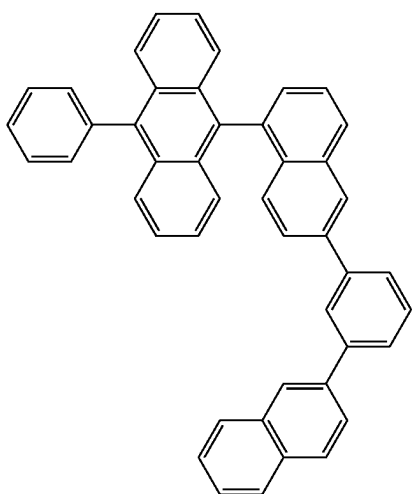


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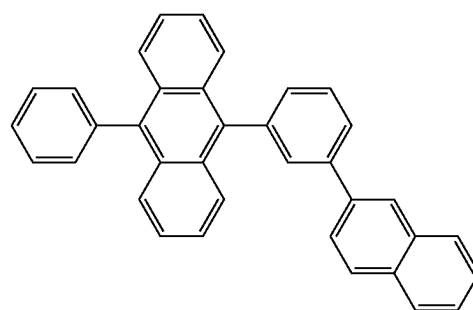


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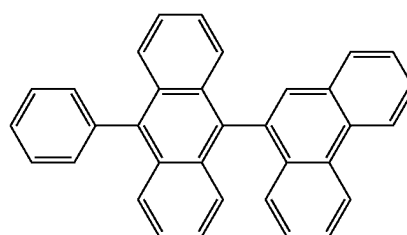
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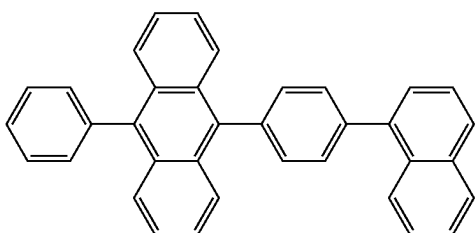
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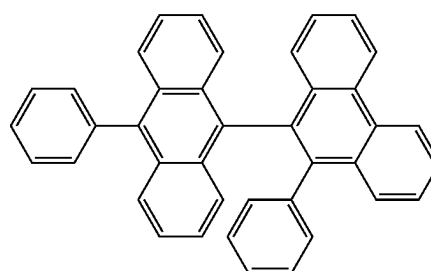
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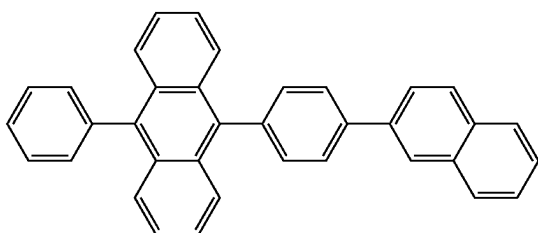
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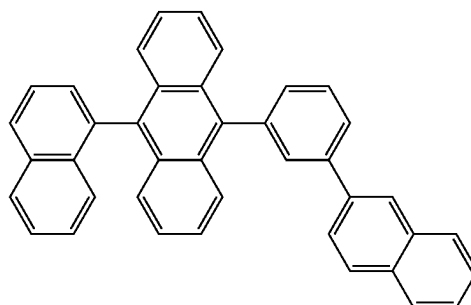
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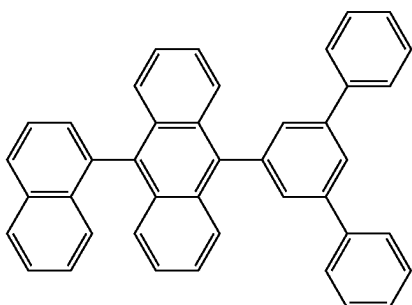
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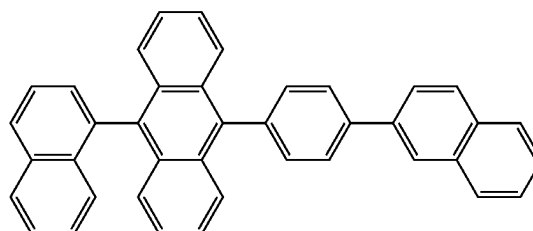
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H4-10



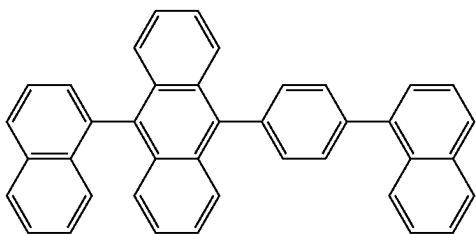
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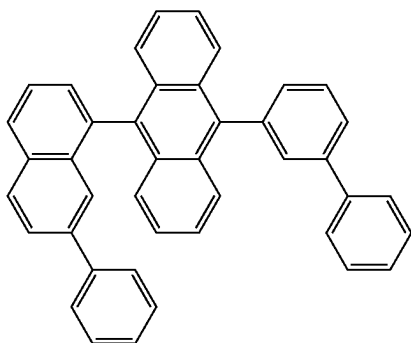
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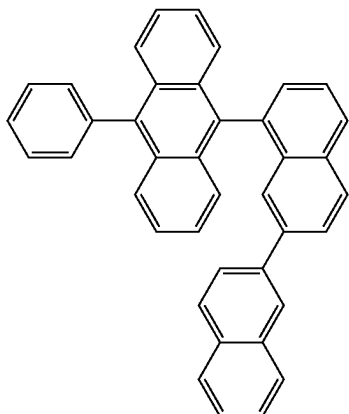
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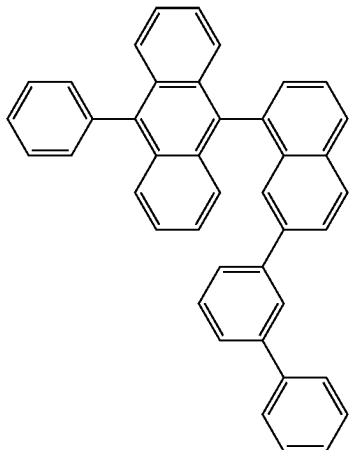
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H4-14

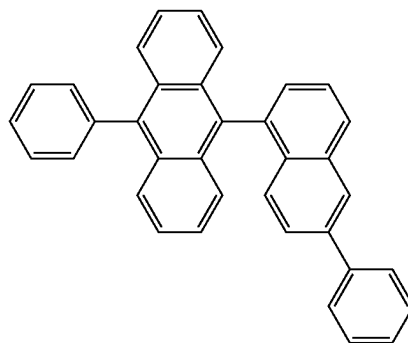


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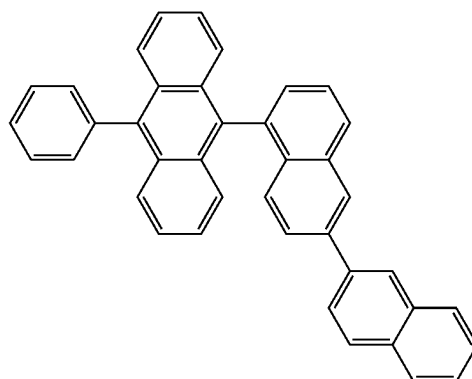


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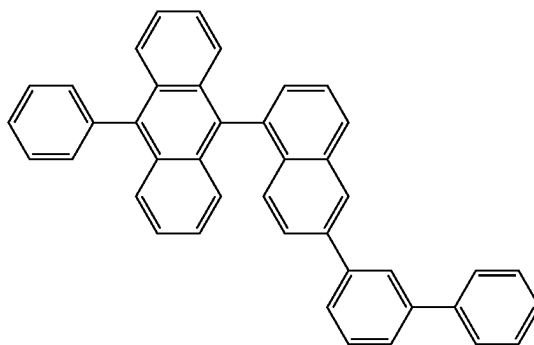
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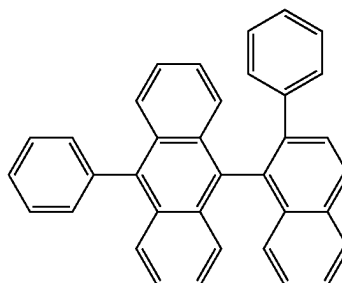
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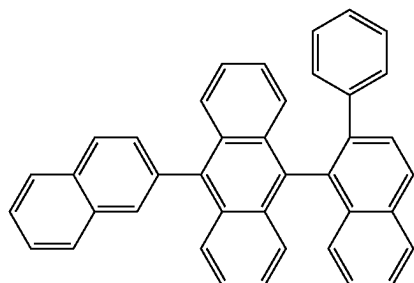
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H4-19

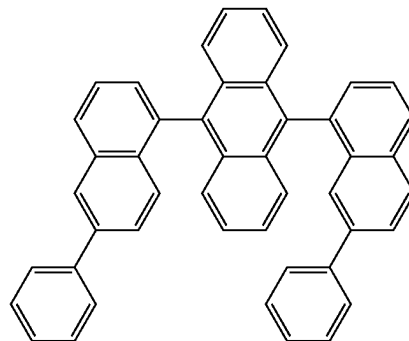


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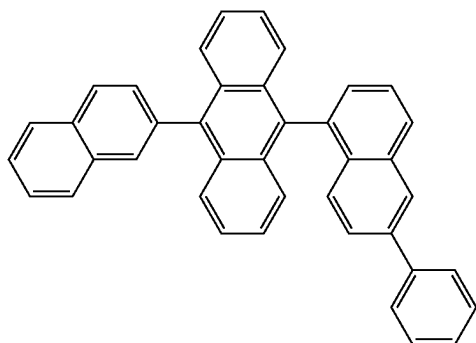


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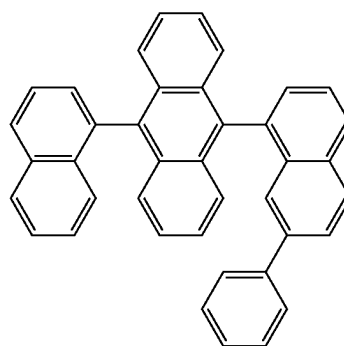
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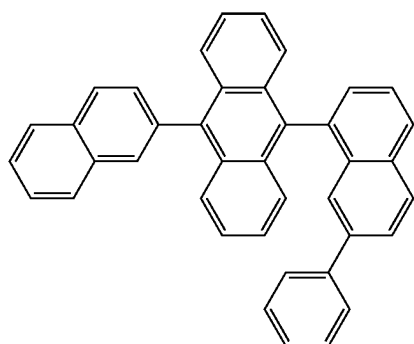
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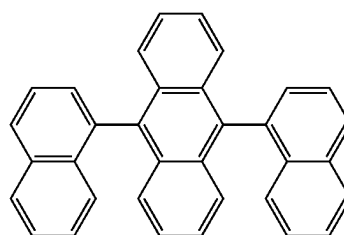
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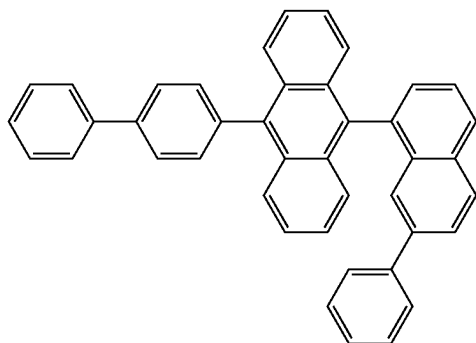
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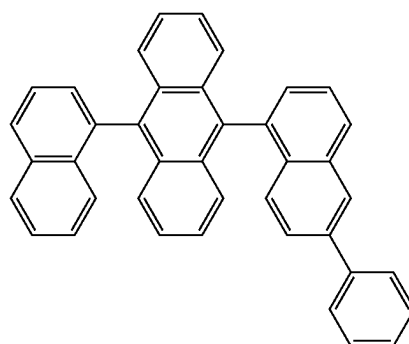
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H4-26

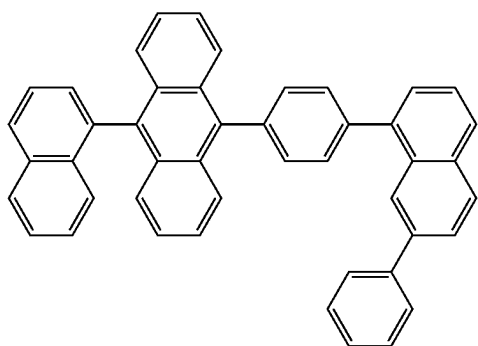


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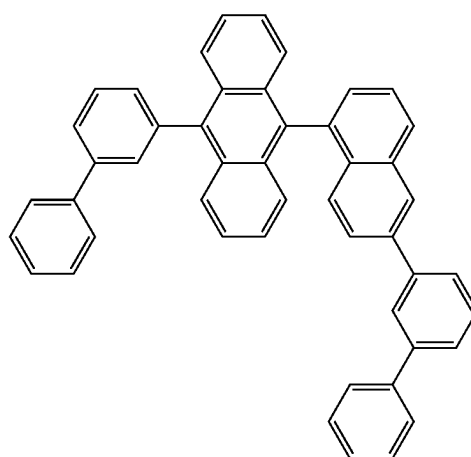
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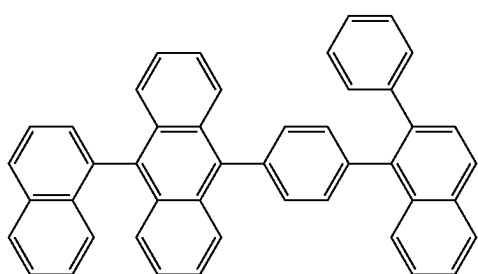


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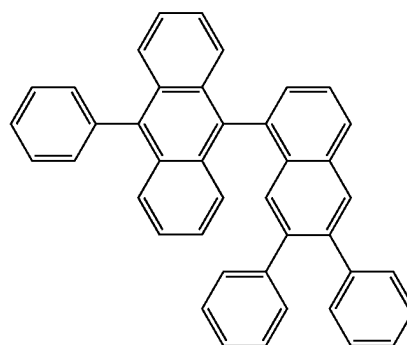
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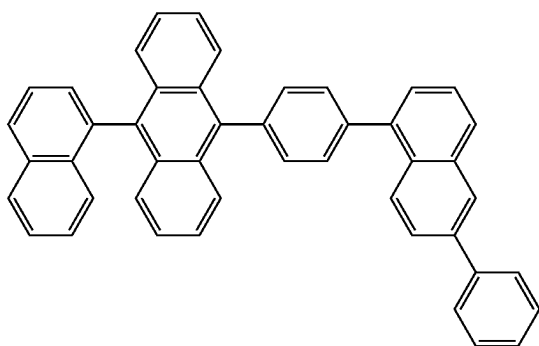
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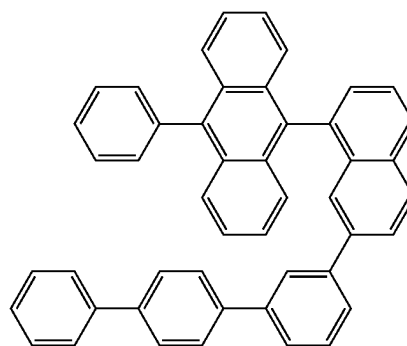
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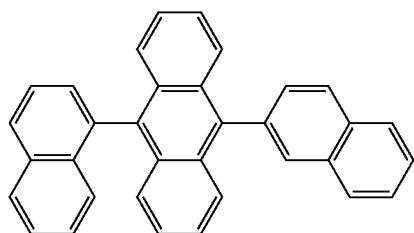
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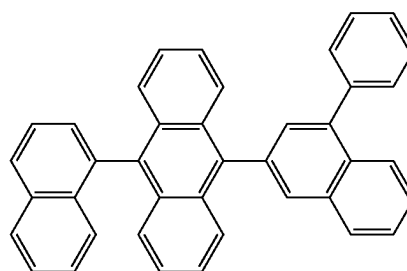
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H4-34

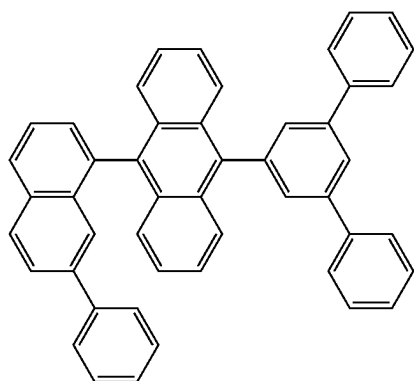


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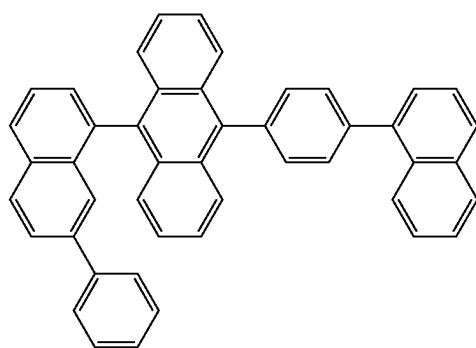
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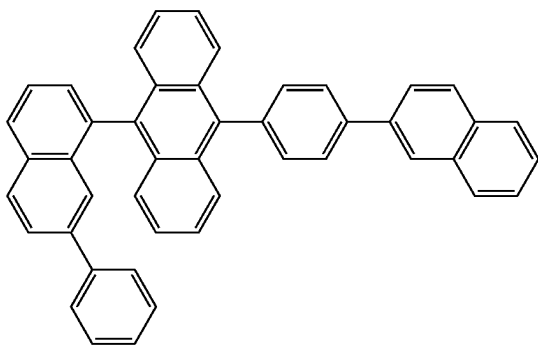
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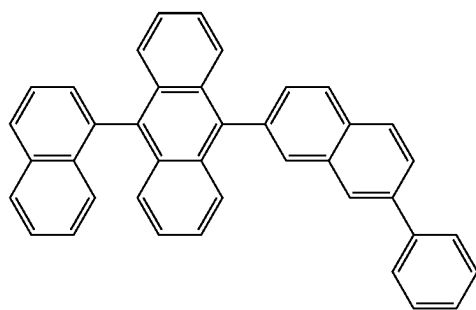


H4-40

H4-37

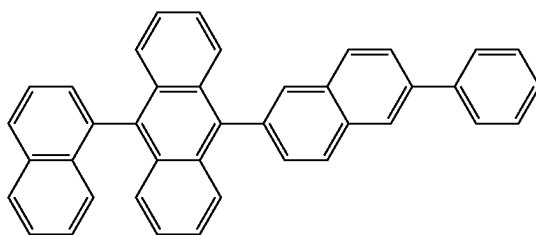
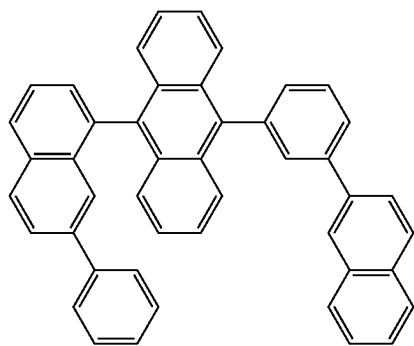


H4-41



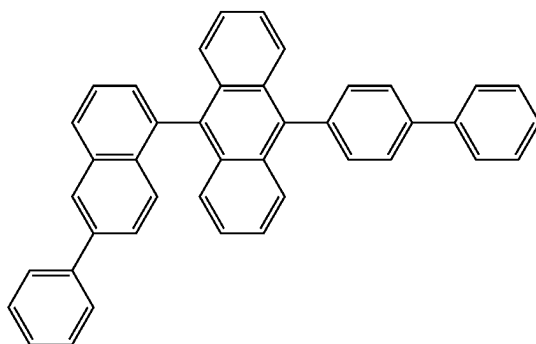
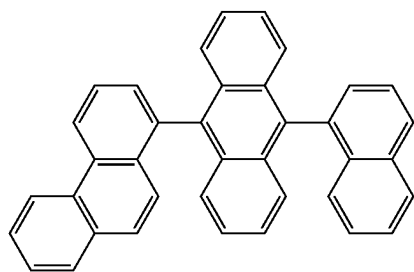
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H4-38

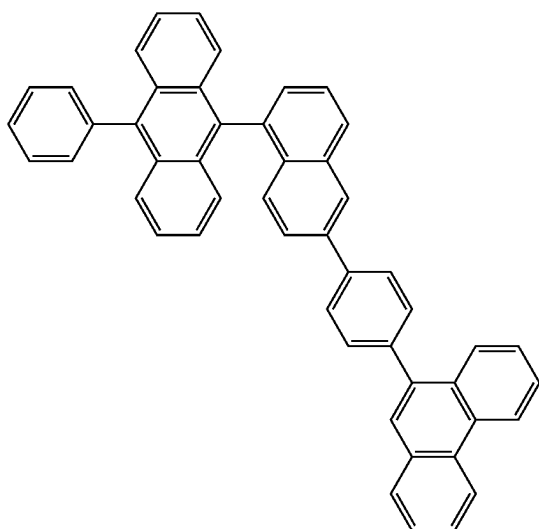


H4-43

H4-39

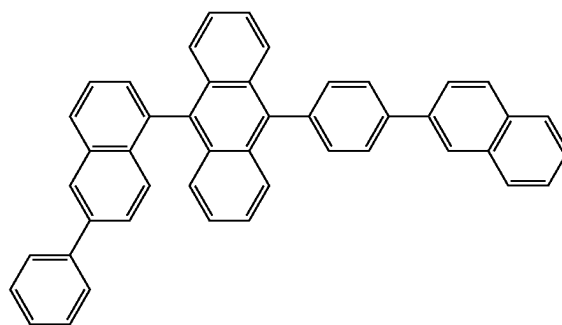


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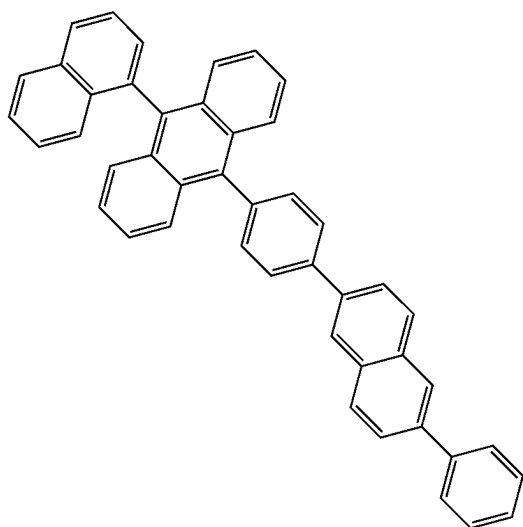


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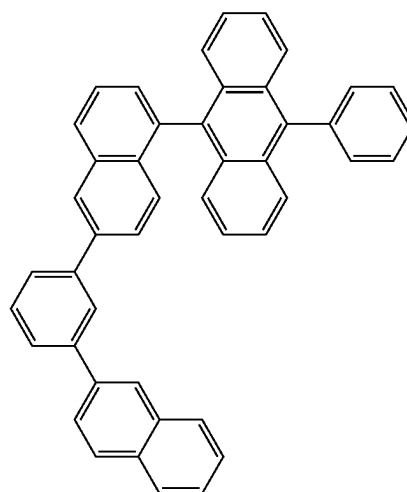
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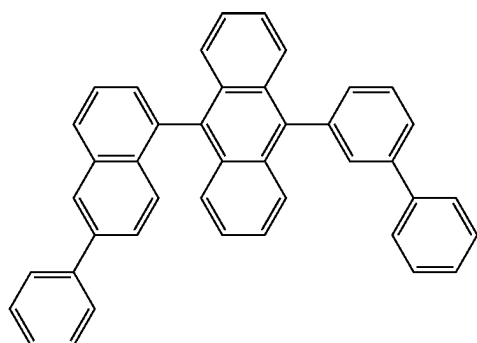
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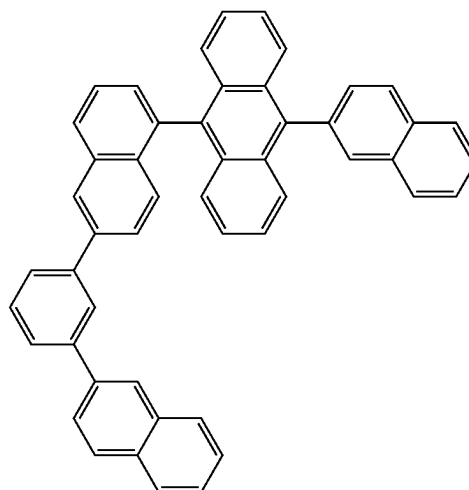
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H4-48

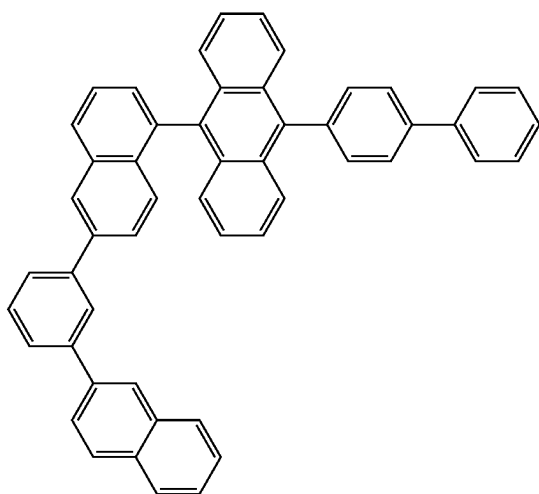


H4-46



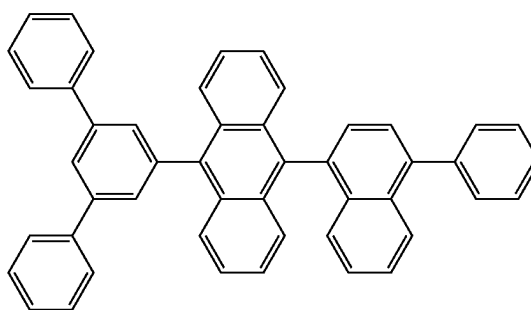
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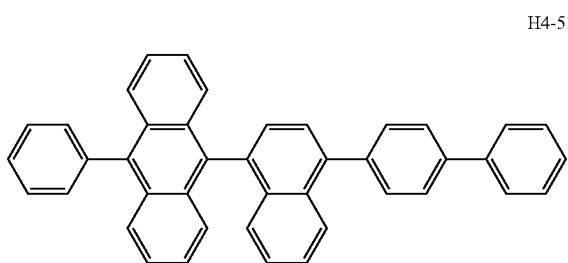


H4-50

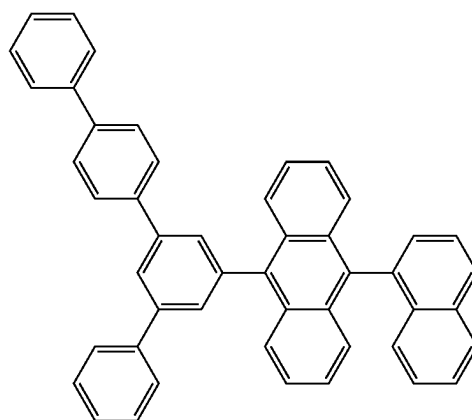
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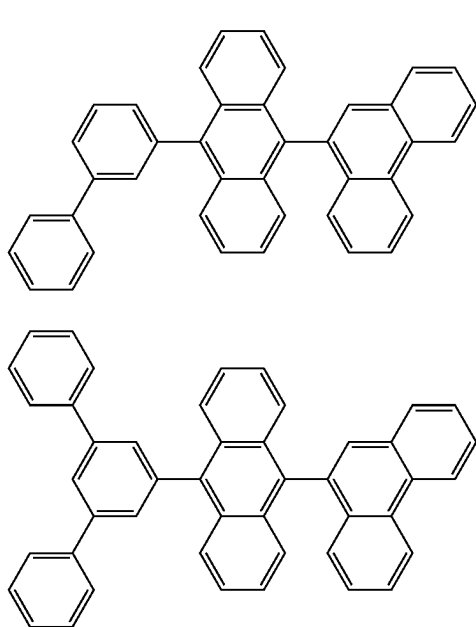
H4-54



H4-51

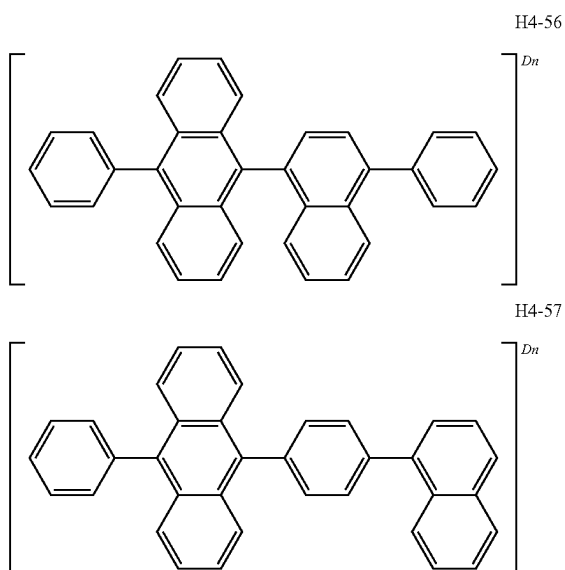


H4-55



H4-52

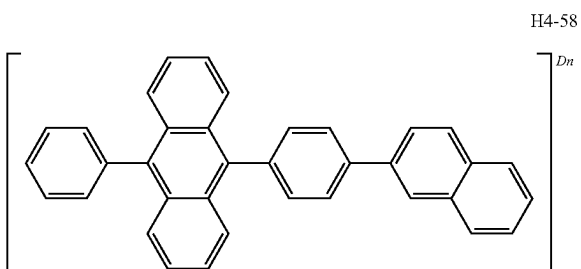
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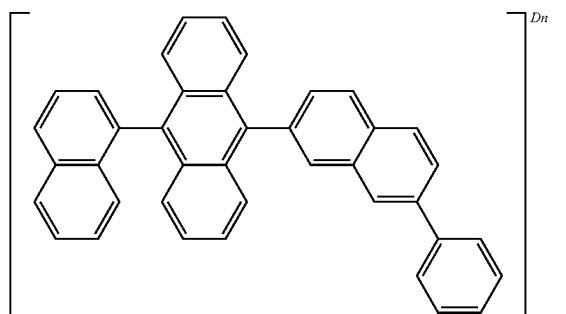
H4-56

H4-57

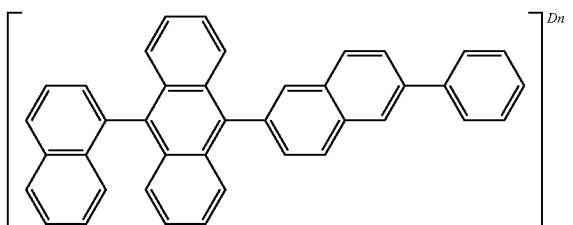
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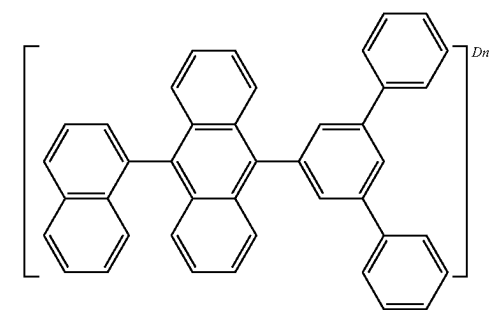
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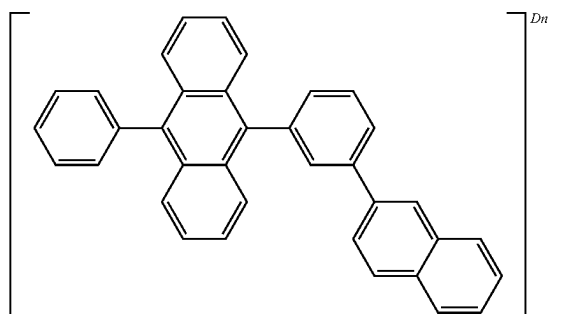
H4-61



H4-63

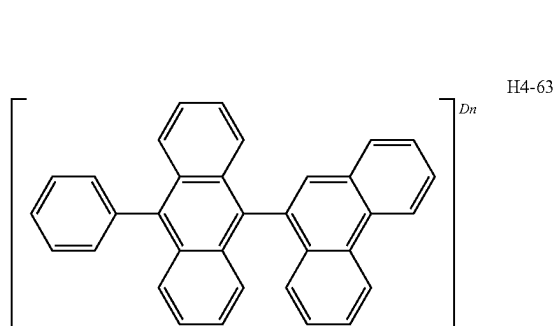


H4-65

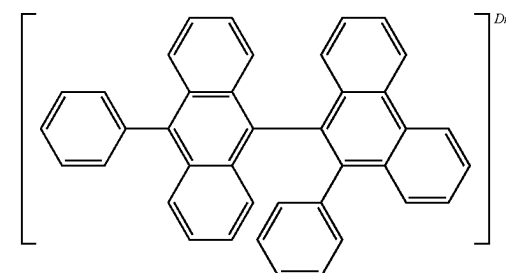


H4-67

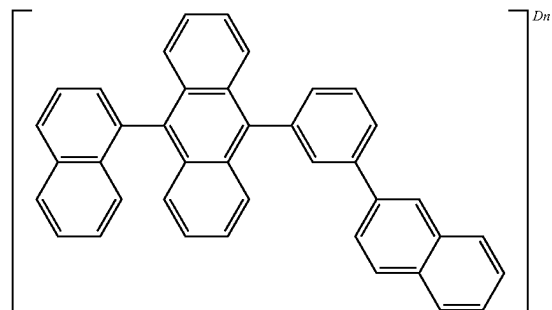
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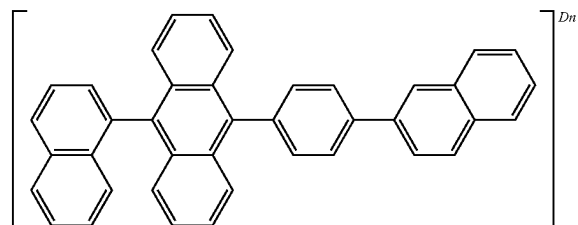
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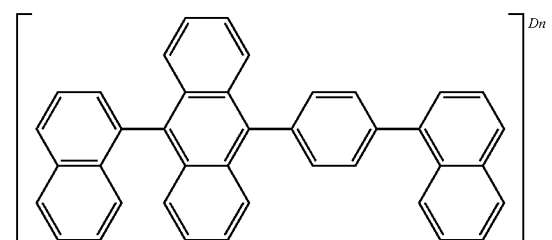
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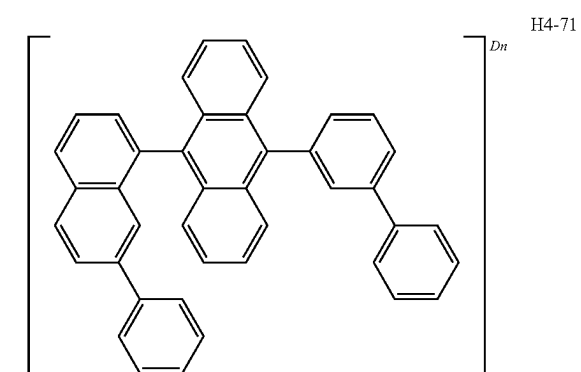
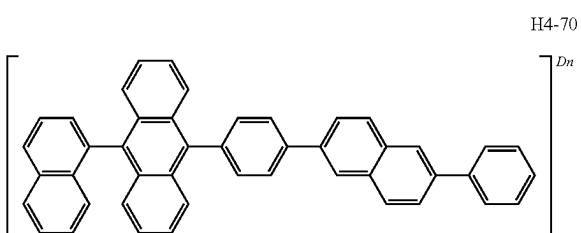
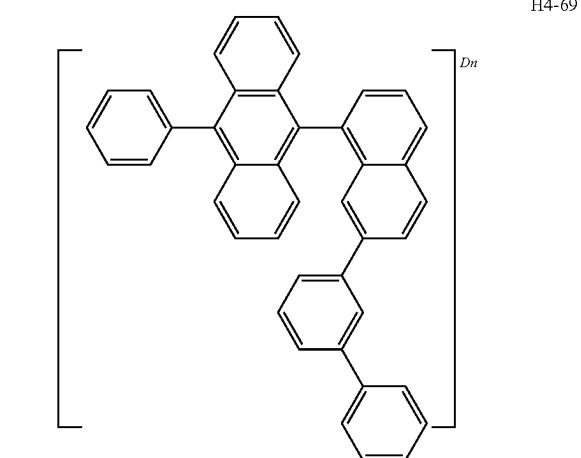
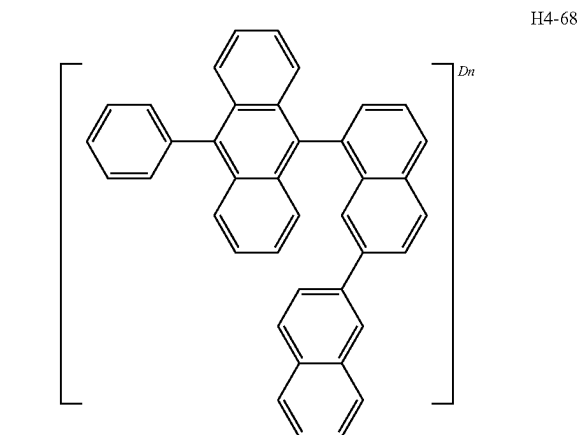
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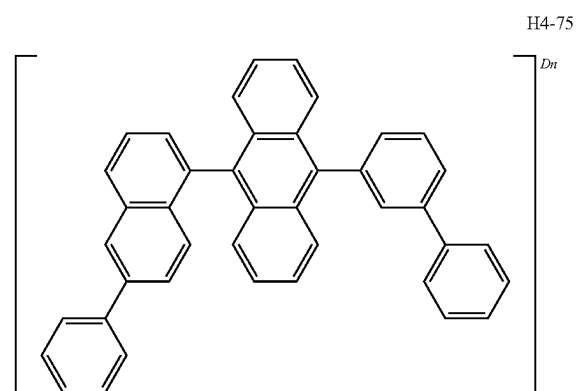
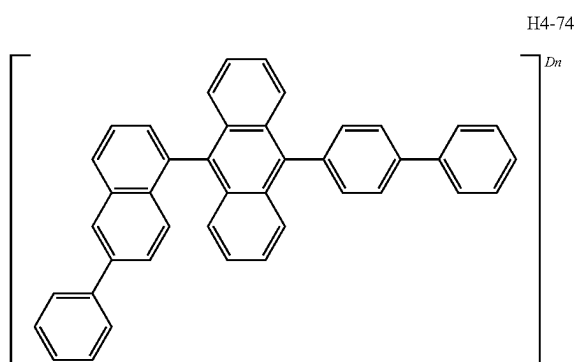
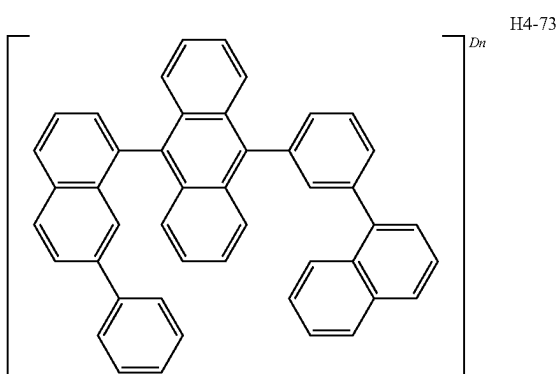
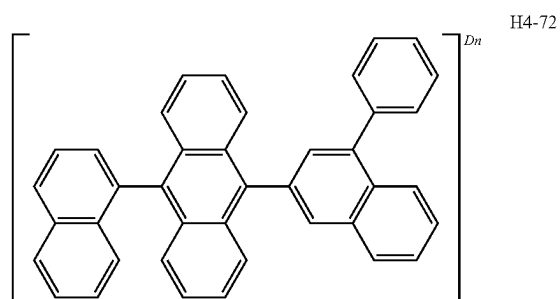
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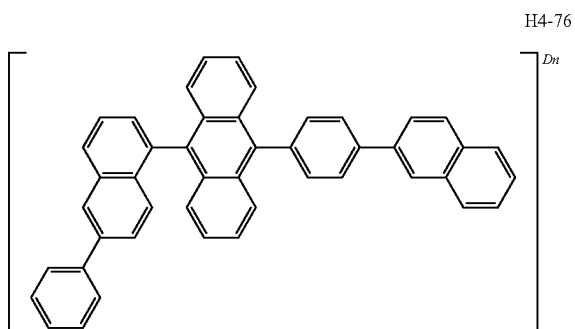
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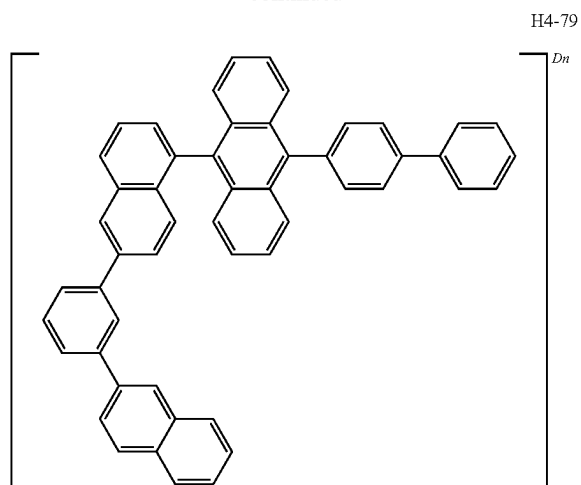
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H4-77

D_n

H4-80

D_n

H4-81

D_n

H4-78

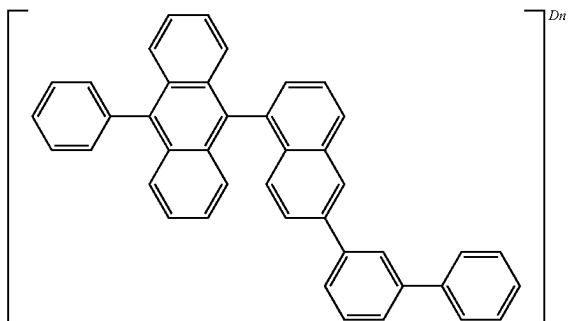
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H4-82

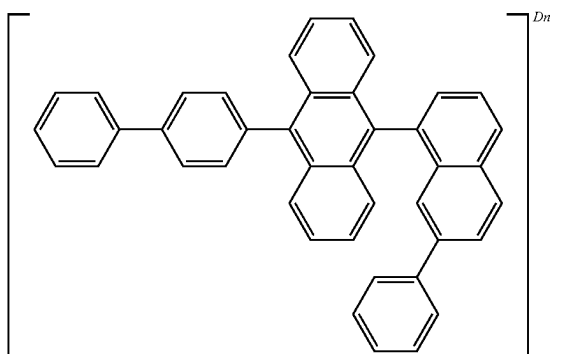
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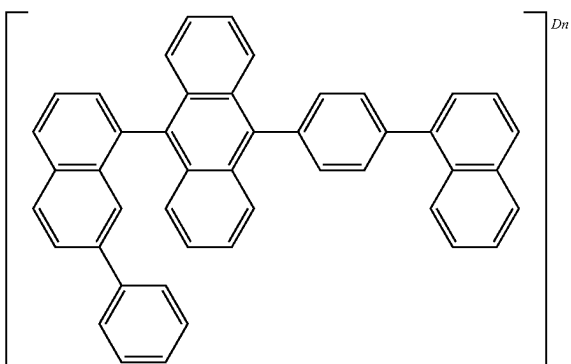
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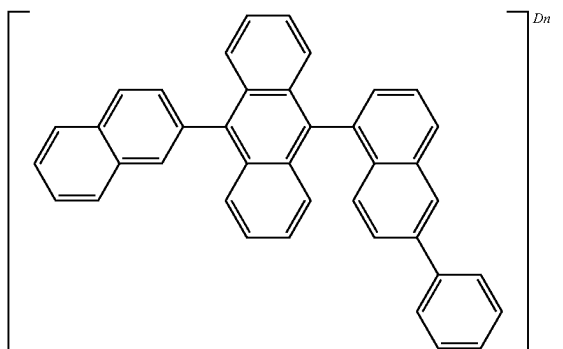
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H4-85

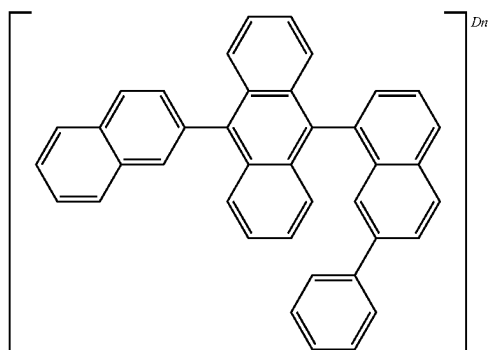


H4-86

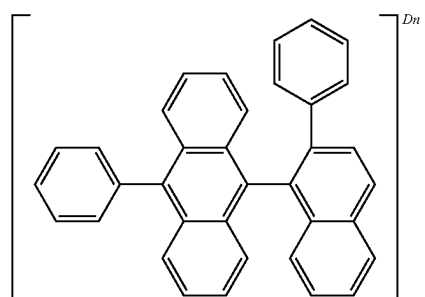


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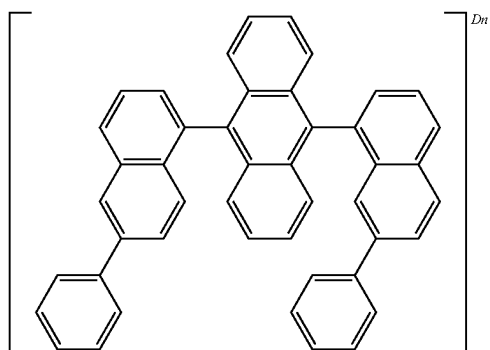
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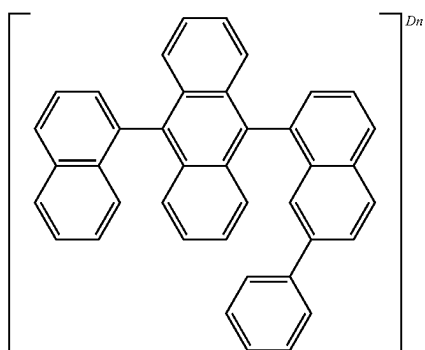
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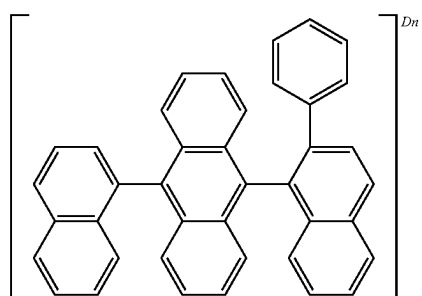
H4-89



H4-90

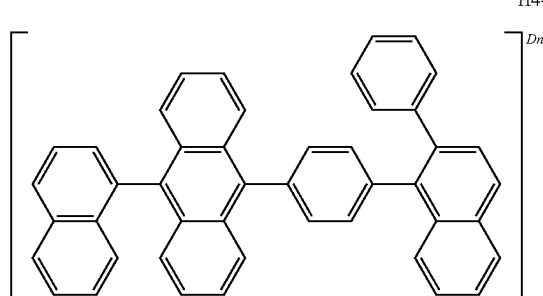


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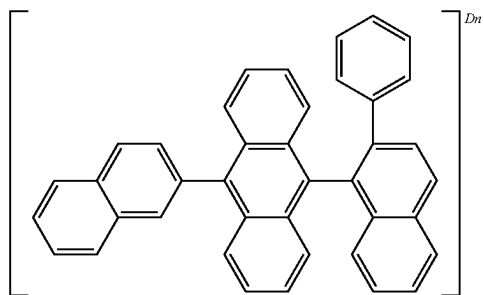


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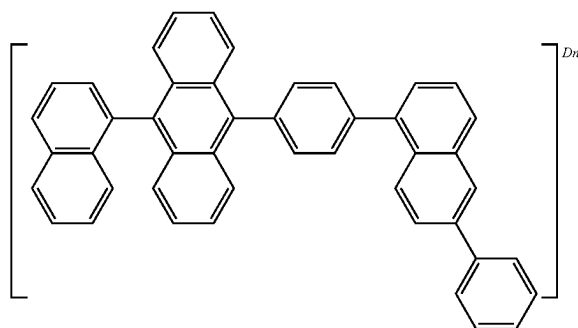
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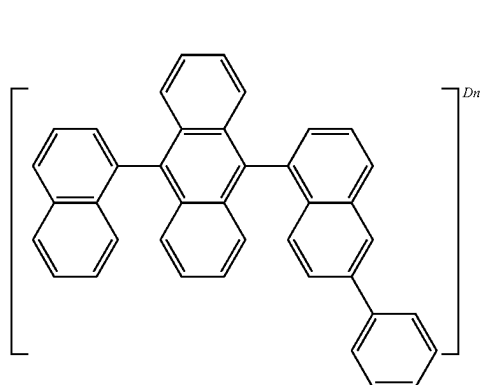
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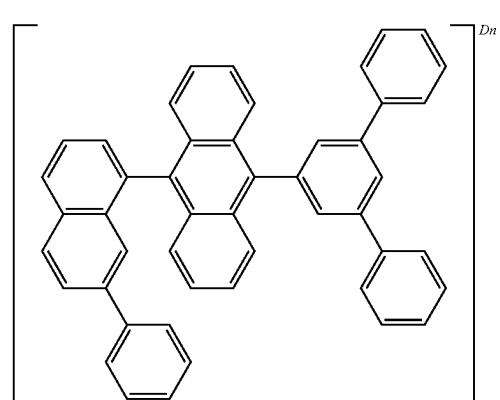
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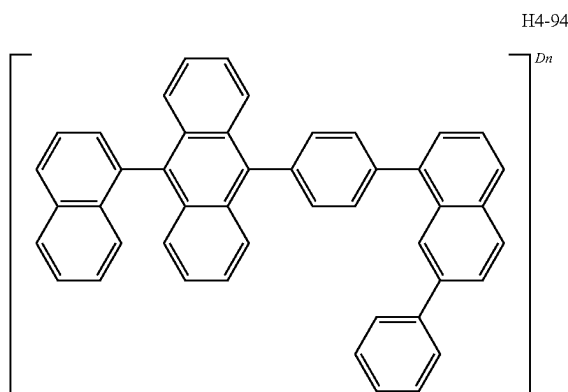
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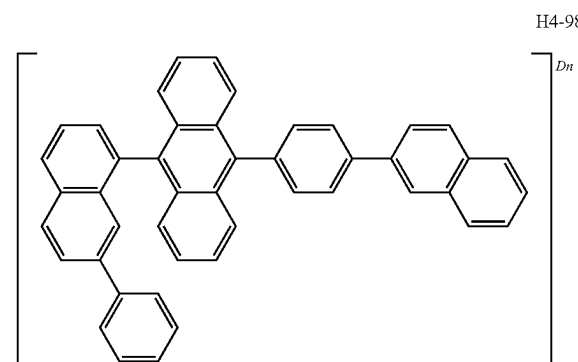
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H4-97

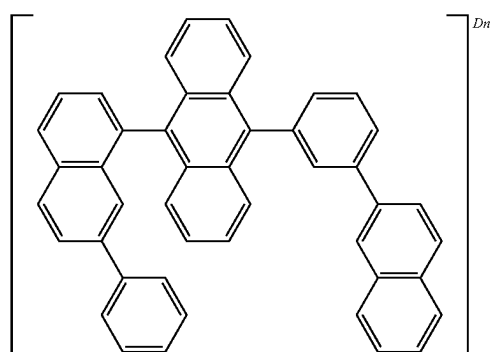


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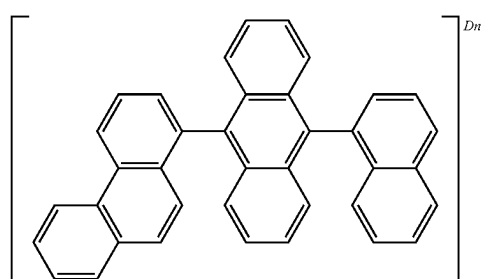


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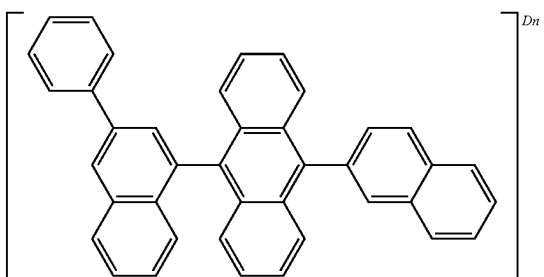
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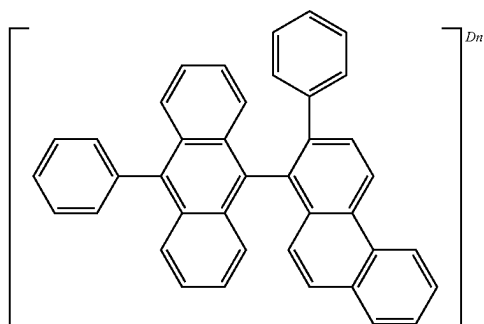
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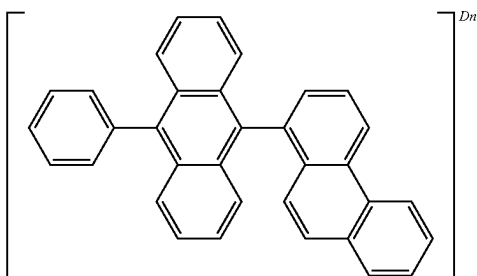
H4-100



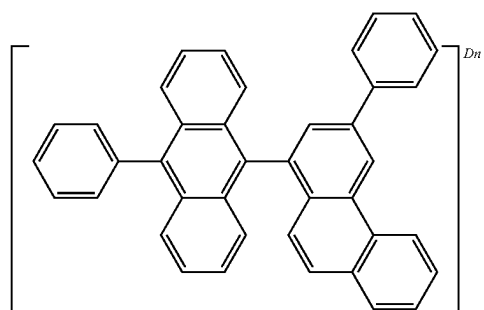
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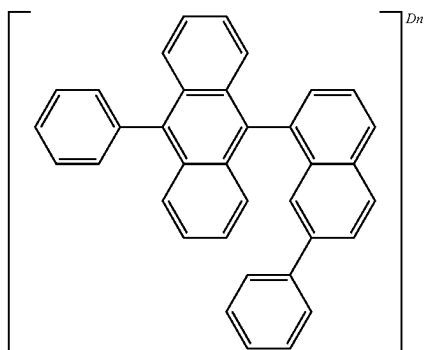
H4-101



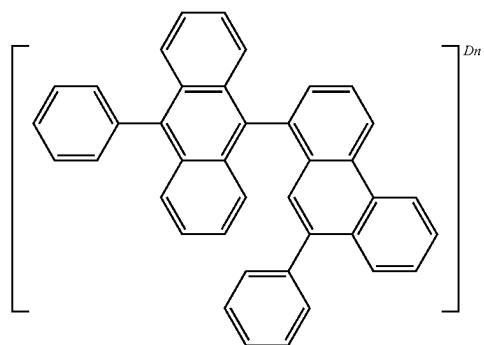
H4-106



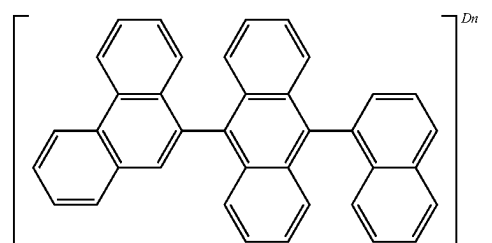
H4-102



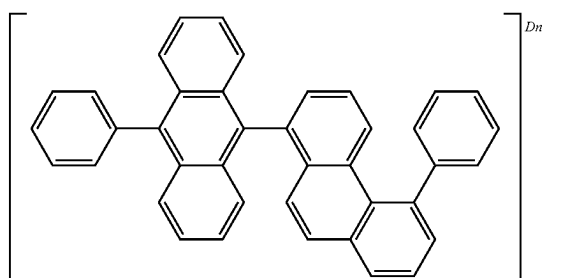
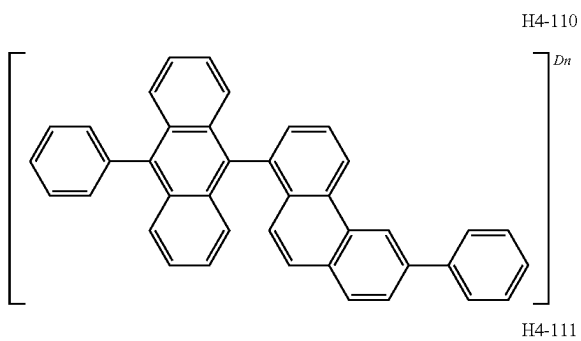
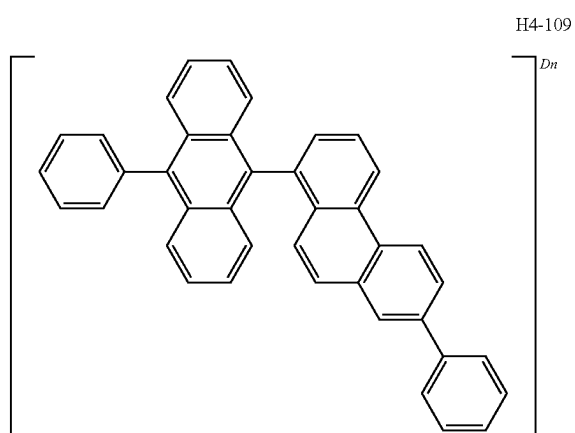
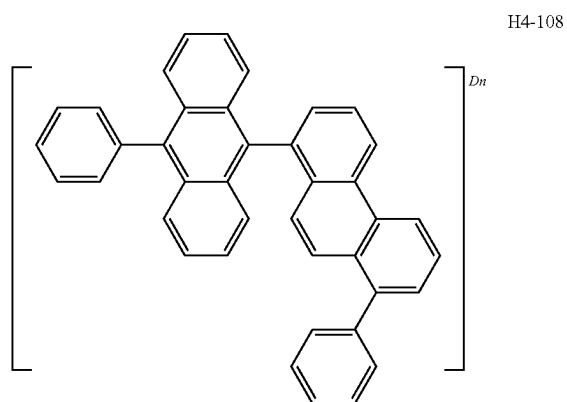
H4-107



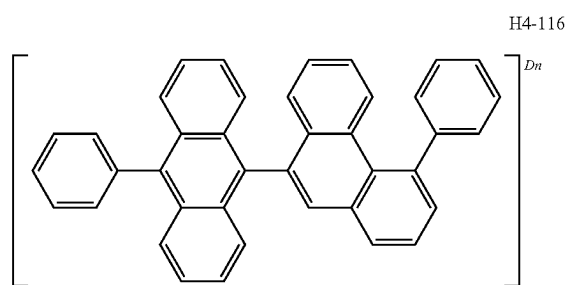
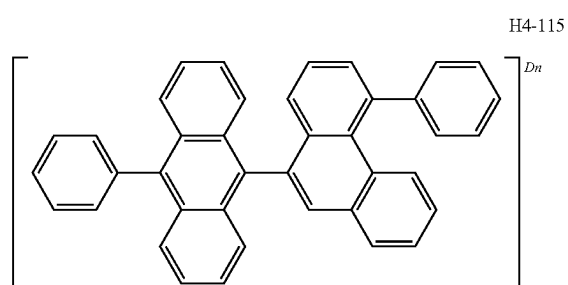
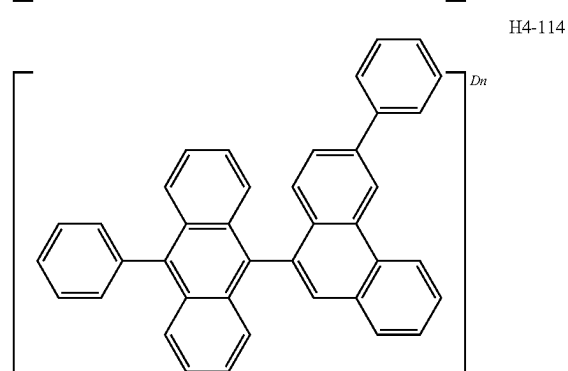
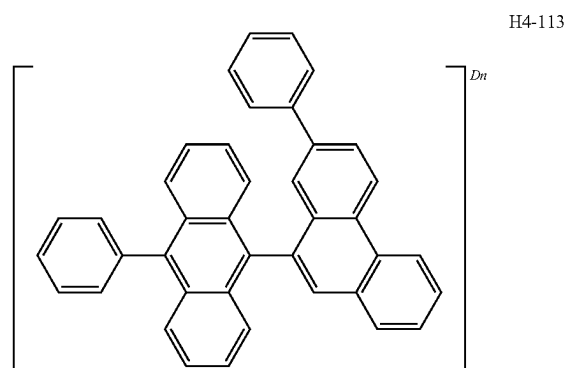
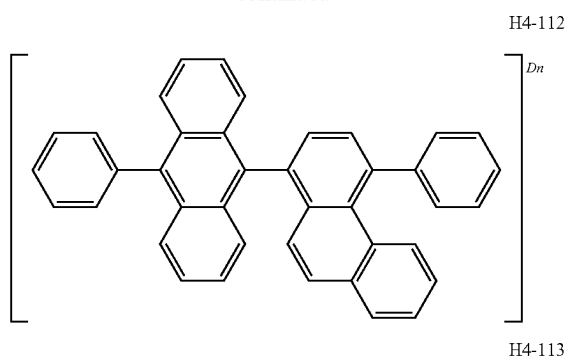
H4-103



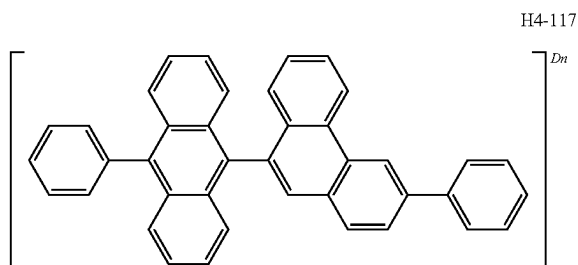
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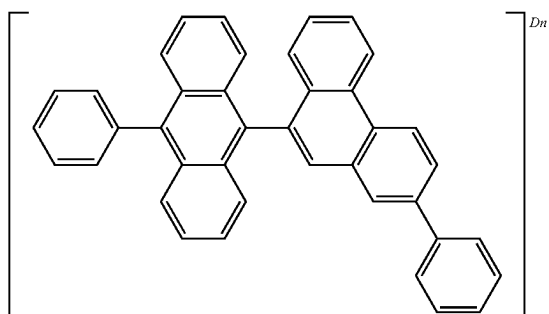
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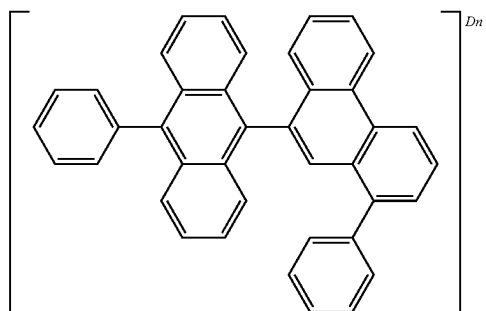
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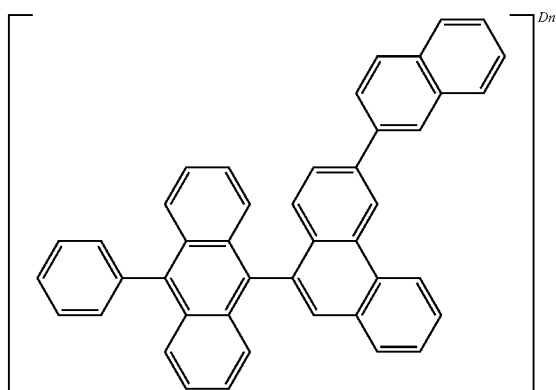
H4-118



H4-119

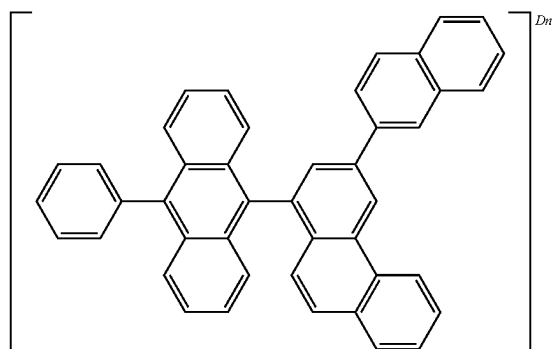


H4-120

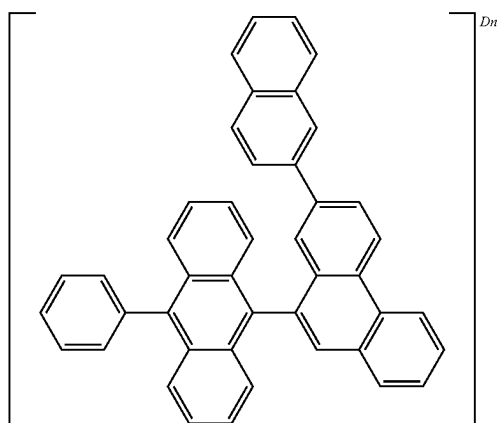


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H4-121

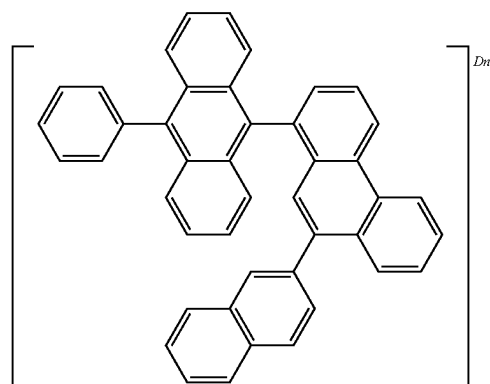


H4-122



and

H4-123

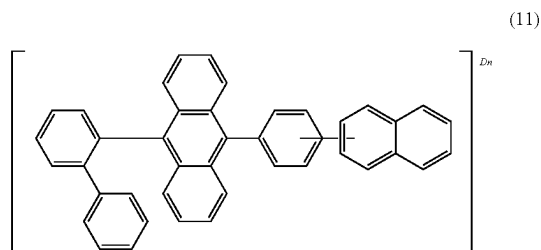


wherein D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to the maximum number of hydrogens in the compound.

12. An organic electroluminescent device comprising an anode; a cathode; and at least one light-emitting layer between the anode and the cathode, wherein at least one layer of the light-emitting layers comprises the plurality of host materials according to claim 1.

13. The organic electroluminescent device according to claim 12, comprising an anode; a cathode; and at least two light-emitting layers between the anode and the cathode, wherein at least one layer of the light-emitting layers comprises the plurality of host materials according to claim 1.

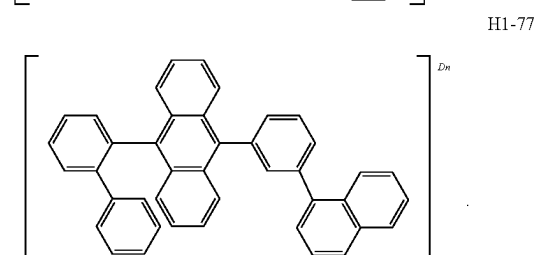
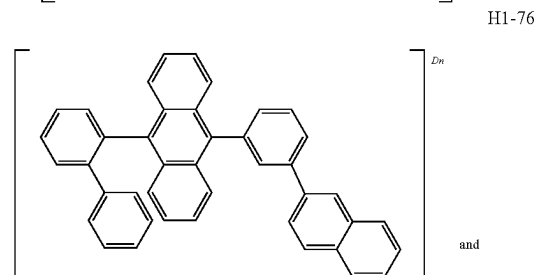
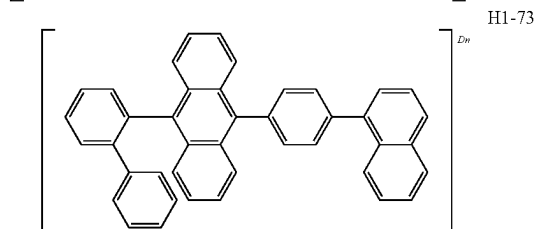
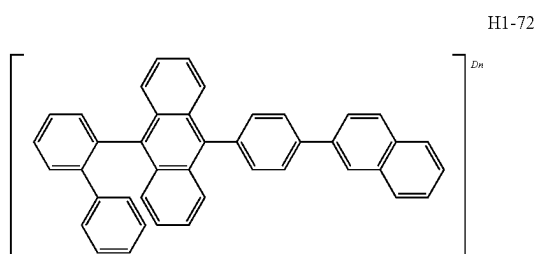
14. An organic electroluminescent compound represented by the following Formula 11:



wherein D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to 28.

15. The organic electroluminescent compound according to claim 14, wherein n is an integer from 8 to 27.

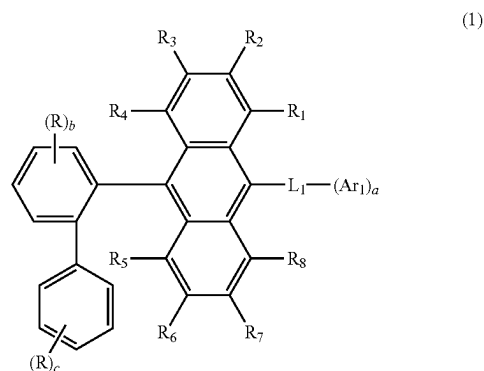
16. The organic electroluminescent compound according to claim 14, wherein the compound represented by Formula 11 is selected from the following compounds:



17. An organic electroluminescent device comprising an anode; a cathode; and at least one light-emitting layer between the anode and the cathode, wherein at least one layer of the light-emitting layers comprises the organic electroluminescent compound according to claim 14.

18. The organic electroluminescent device according to claim 17, comprising an anode; a cathode; and at least two light-emitting layers between the anode and the cathode, wherein at least one layer of the light-emitting layers comprises the organic electroluminescent compound according to claim 14.

19. An organic electroluminescent device comprising an anode; a cathode; and at least two light-emitting layers between the anode and the cathode, wherein at least one layer of the light-emitting layers comprises an organic electroluminescent compound represented by the following Formula 1 as a host material:



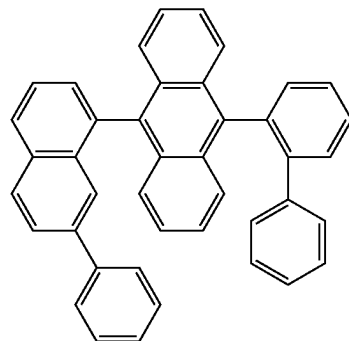
in Formula 1,

L_1 represents a single bond, a phenylene unsubstituted or substituted with deuterium, a naphthylene unsubstituted or substituted with deuterium, or a phenanthrylene unsubstituted or substituted with deuterium;

Ar_1 each independently represents a substituted or unsubstituted (C_6 - C_{30})aryl, or a substituted or unsubstituted (3- to 30-membered)heteroaryl;

R and R_1 to R_8 each independently represent hydrogen or deuterium; and

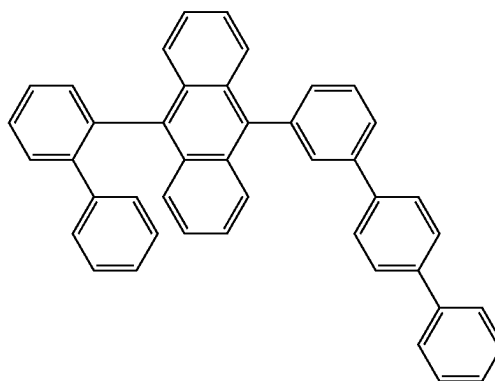
a represents an integer of 1 or 2, b represents an integer of 4, and c represents an integer of 5; and each Ar_1 may be the same as or different from each other; with a proviso that Formula 1 is not represented as



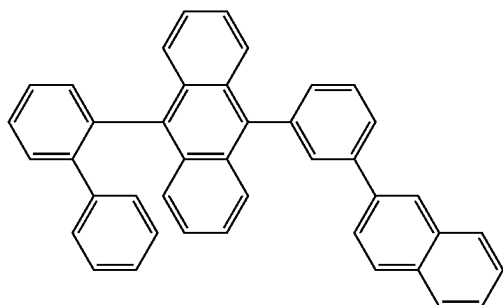
20. The organic electroluminescent device according to claim 19, wherein the compound represented by Formula 1 is selected from the following compounds:

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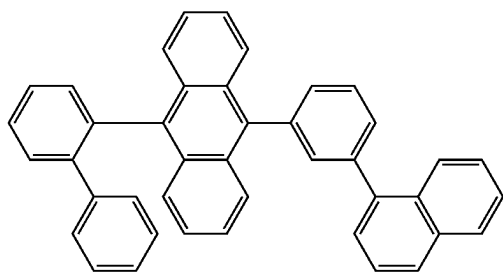
H1-11



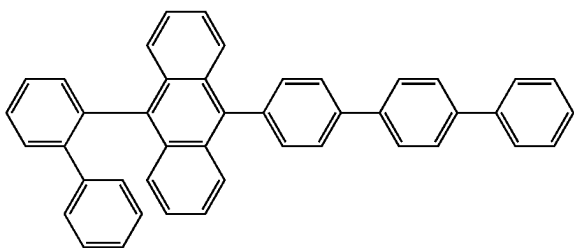
H1-6



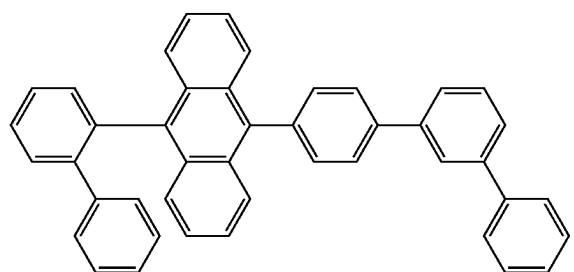
H1-7



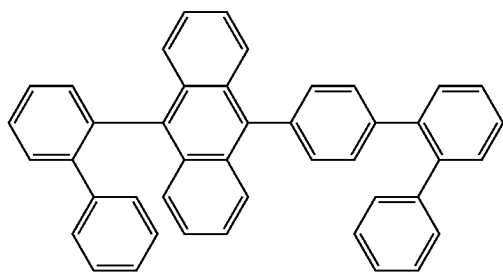
H1-8



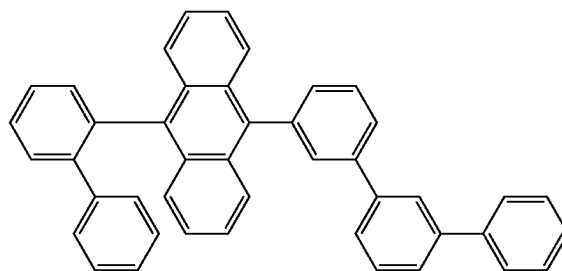
H1-9



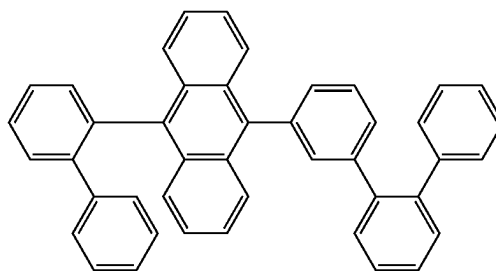
H1-10



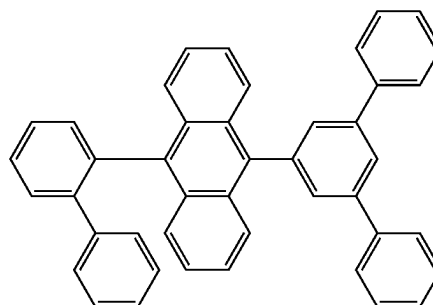
H1-12



H1-13

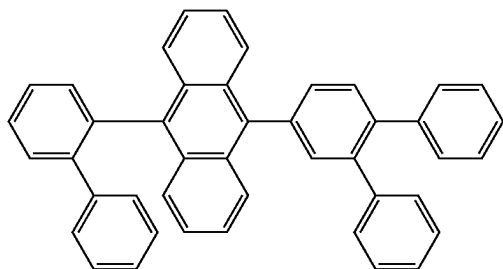


H1-14

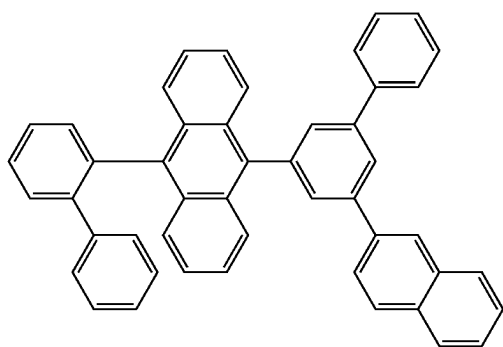


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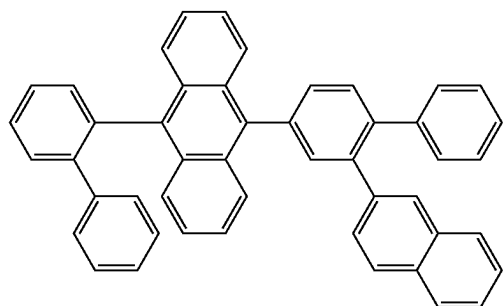
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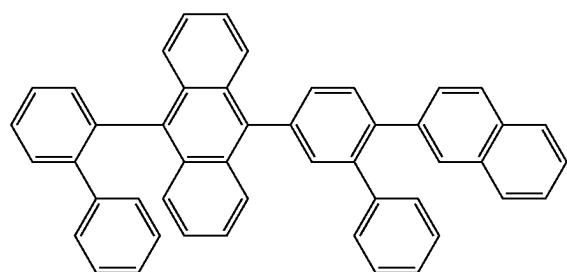
H1-16



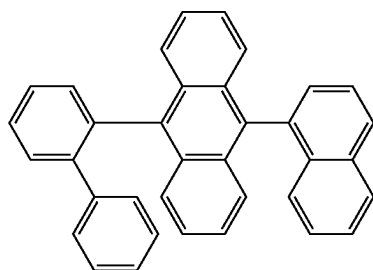
H1-17



H1-18

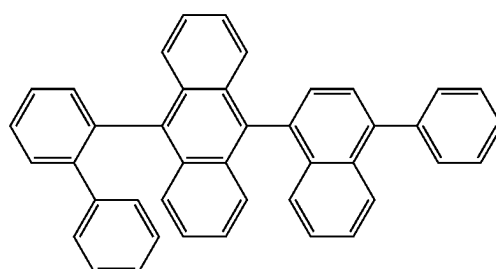


H1-19

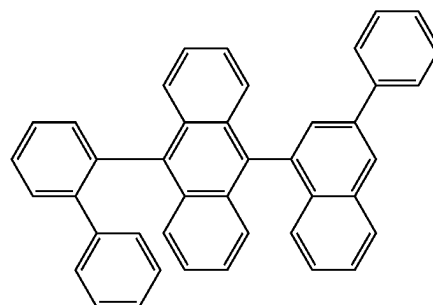


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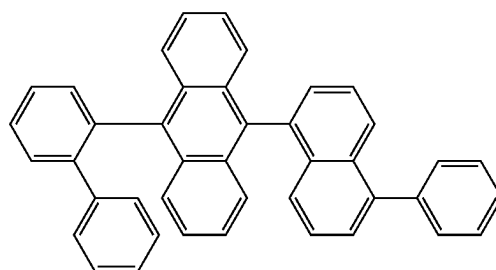
H1-20



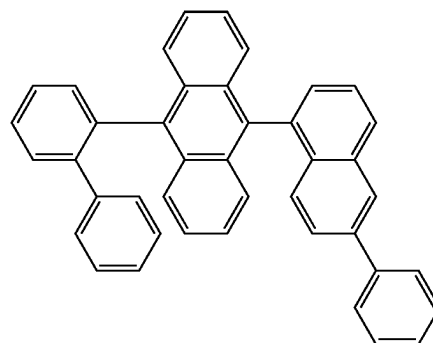
H1-21



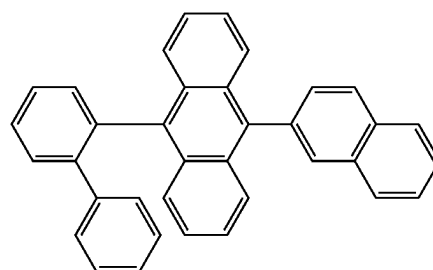
H1-22



H1-23

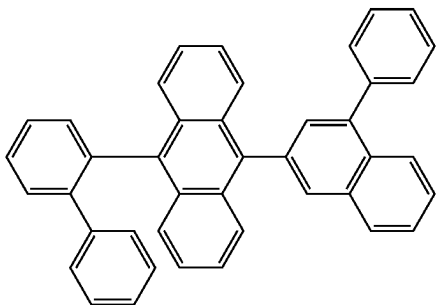


H1-24

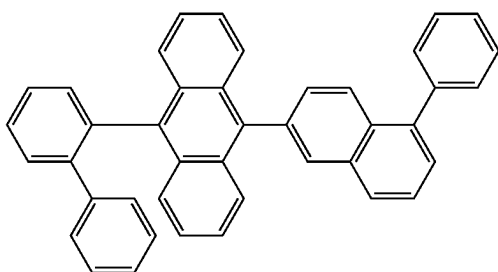


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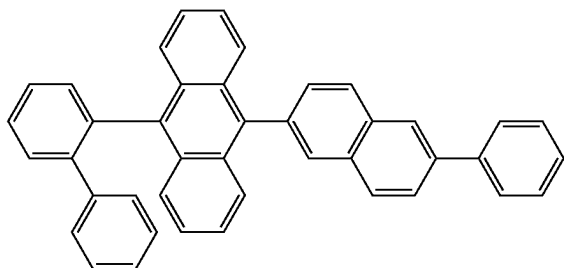
H1-25



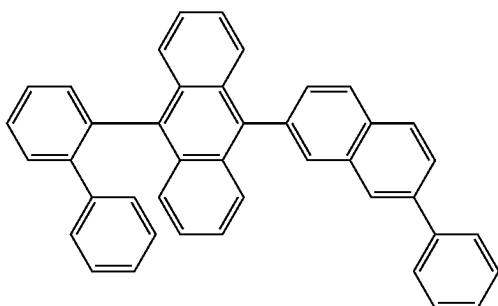
H1-26



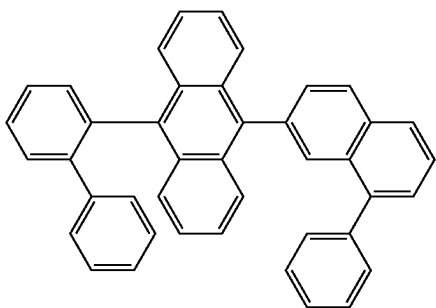
H1-27



H1-28

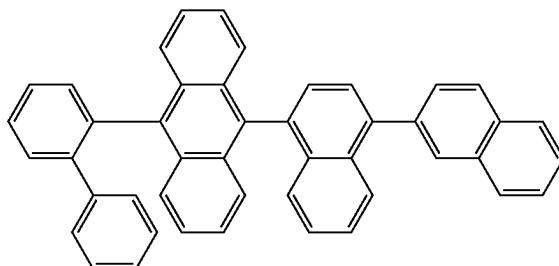


H1-29

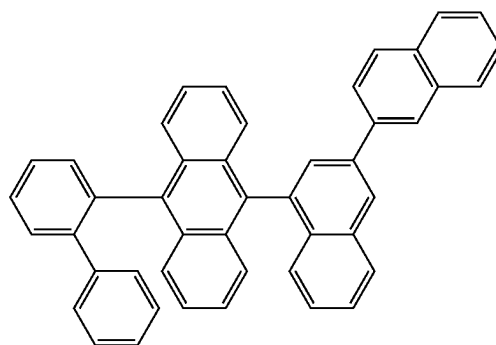


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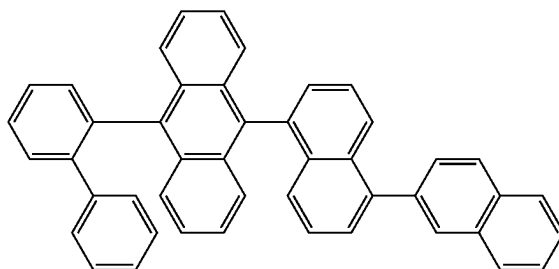
H1-30



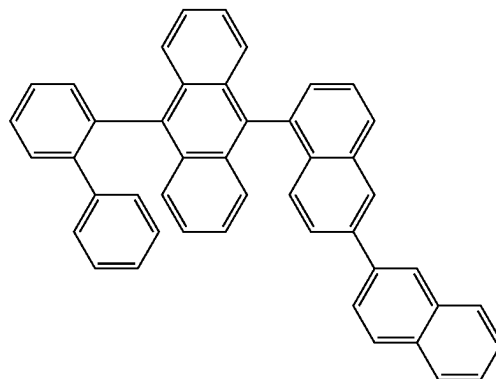
H1-31



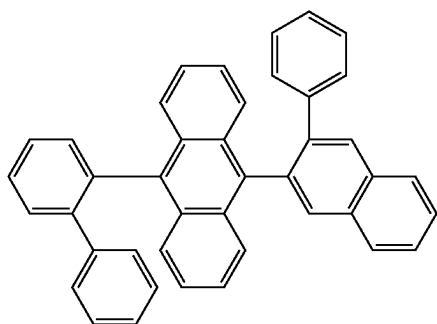
H1-32



H1-33

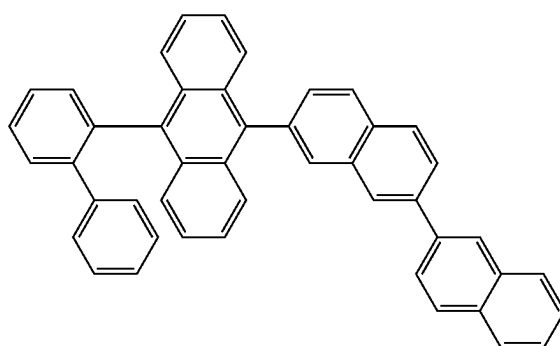


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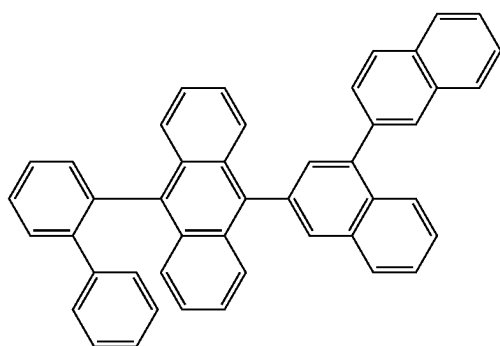


H1-34

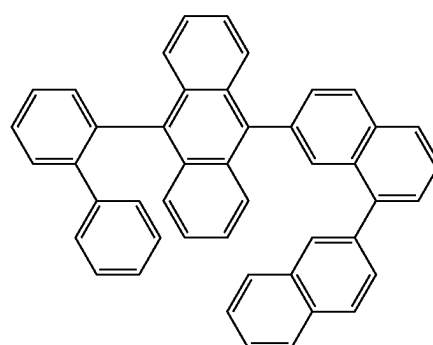
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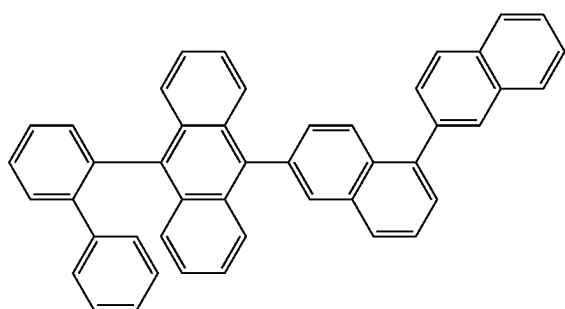
H1-38



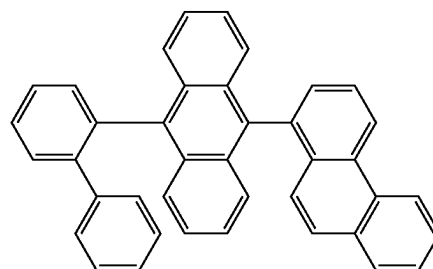
H1-35



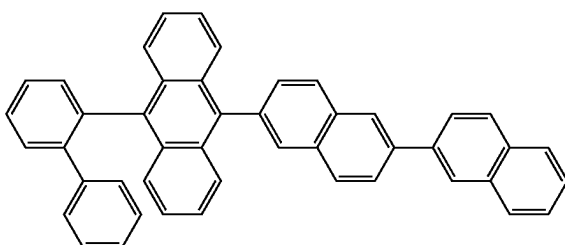
H1-39



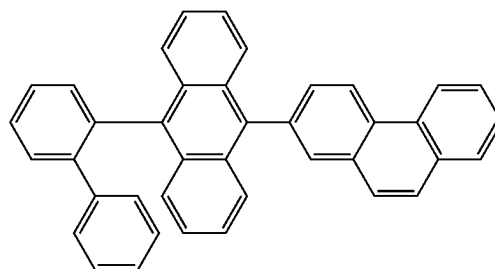
H1-36



H1-40



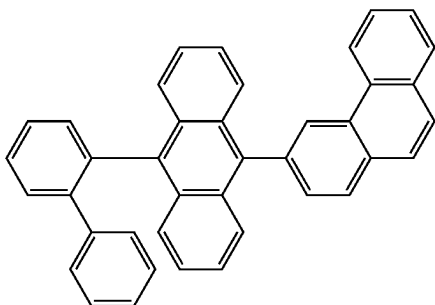
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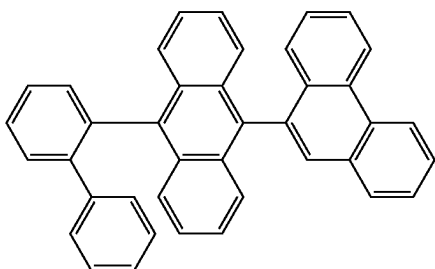
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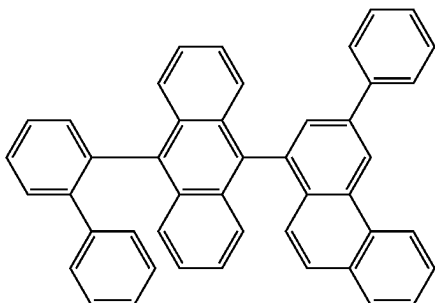
H1-42



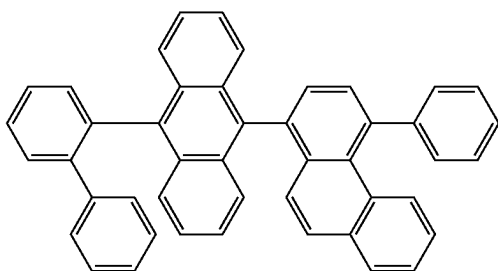
H1-43



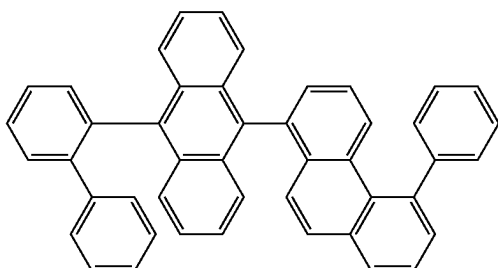
H1-44



H1-45

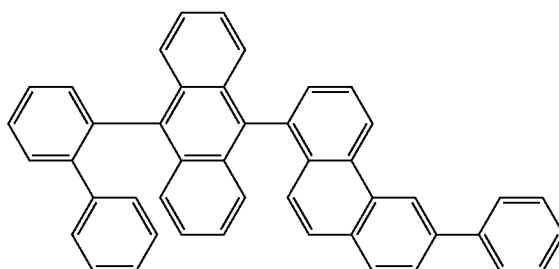


H1-46

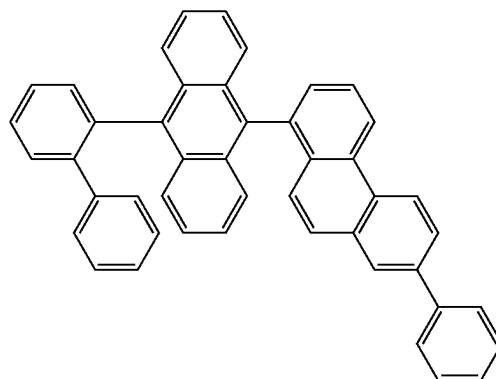


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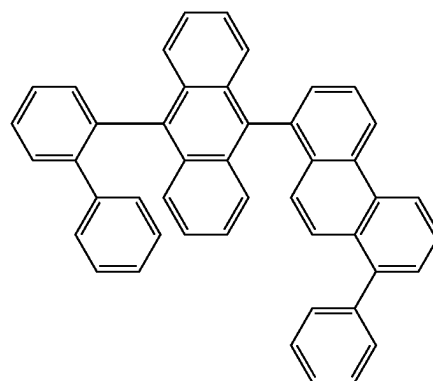
H1-47



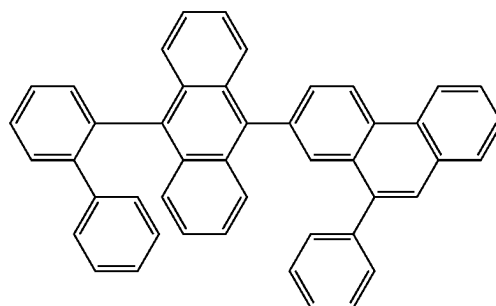
H1-48



H1-49

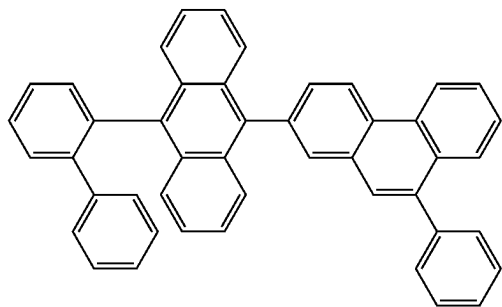


H1-50

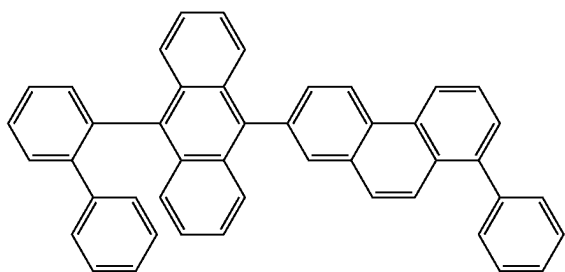


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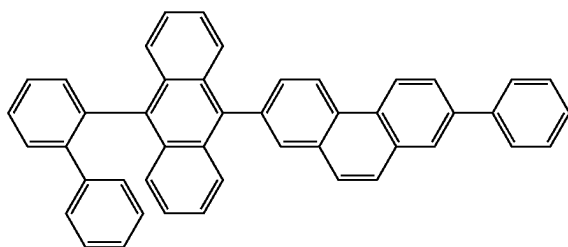
H1-51



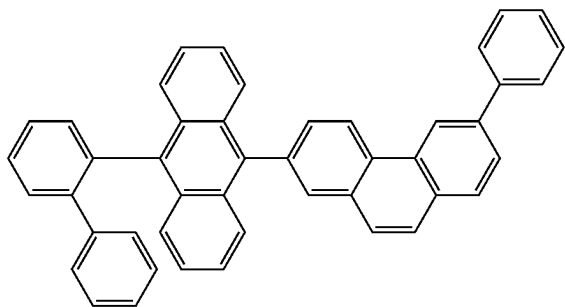
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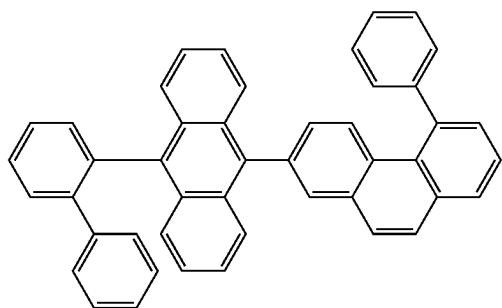
H1-53



H1-54

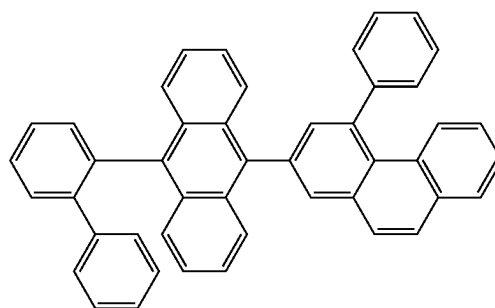


H1-55

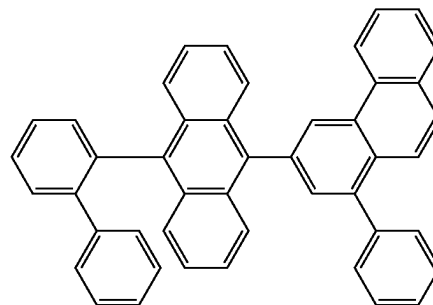


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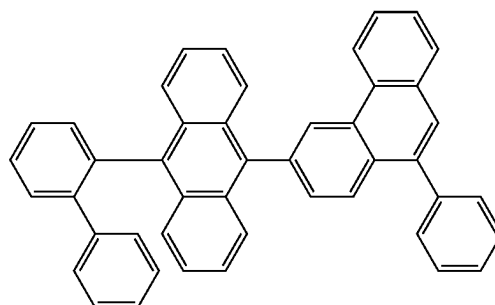
H1-56



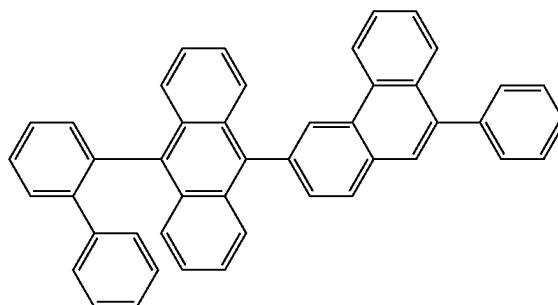
H1-57



H1-58

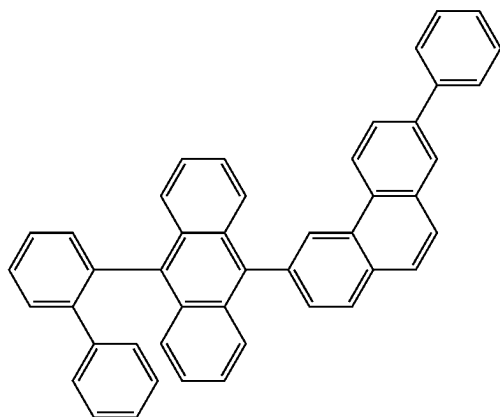


H1-59



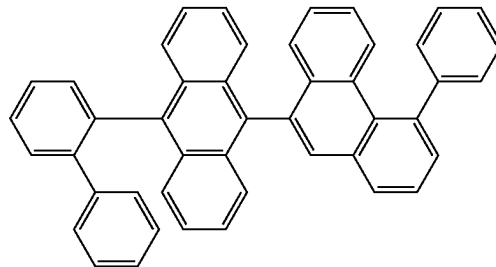
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H1-60

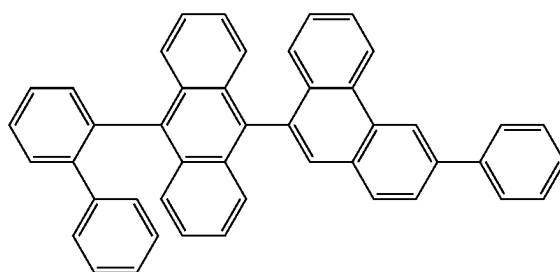


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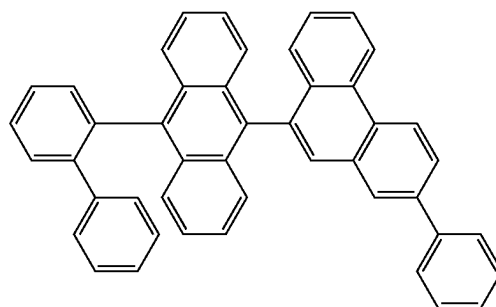
H1-64



H1-65

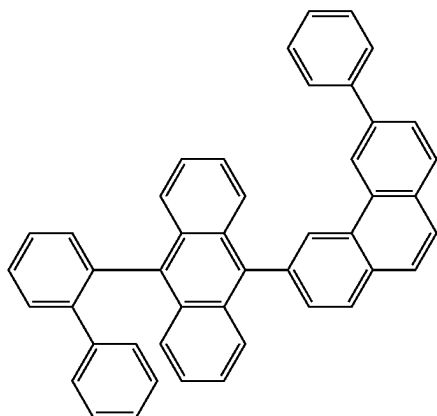


H1-66

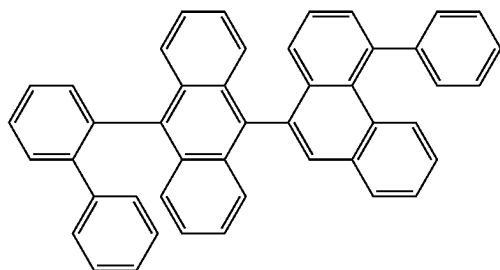


H1-67

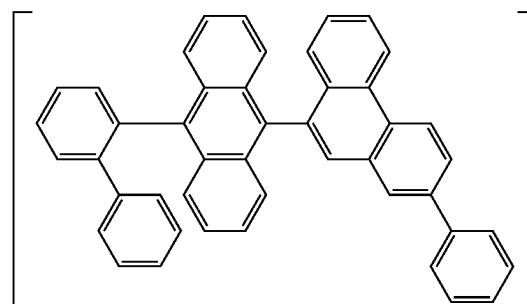
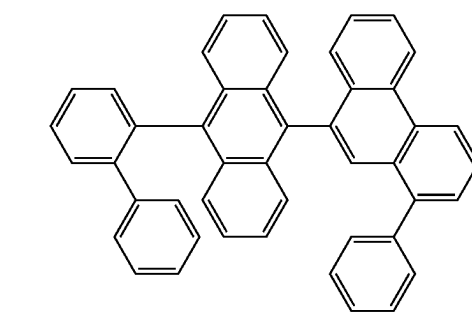
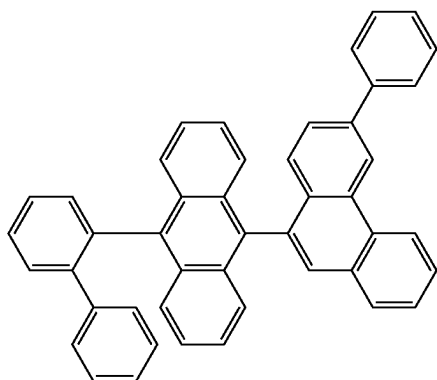
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H1-62

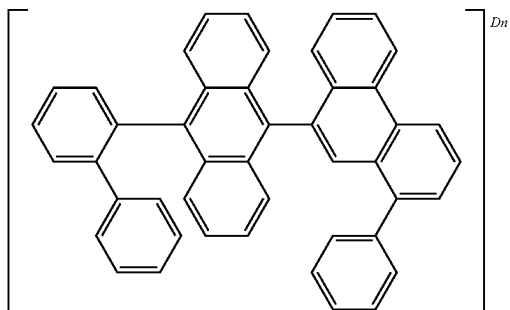


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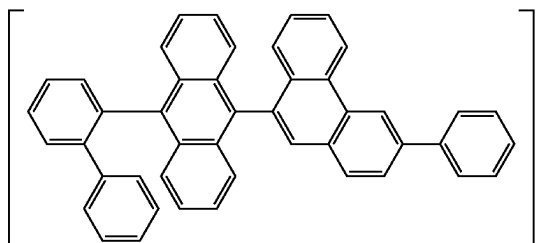


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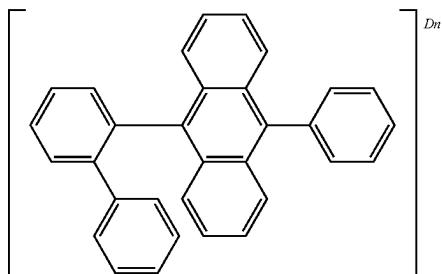
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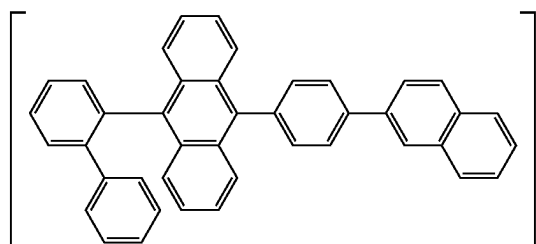
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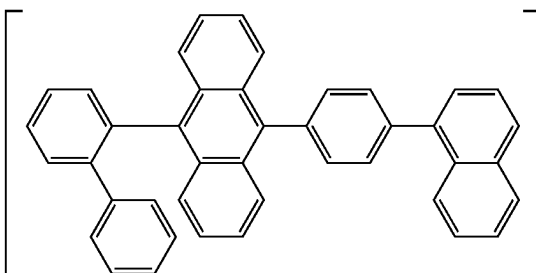
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H1-72

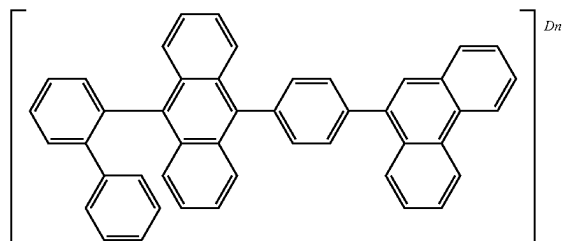


H1-73

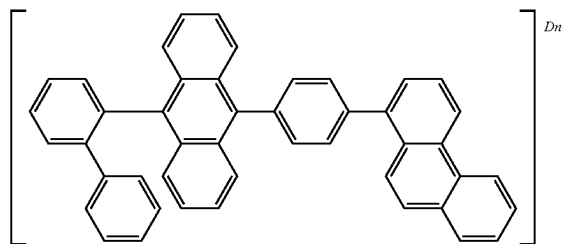


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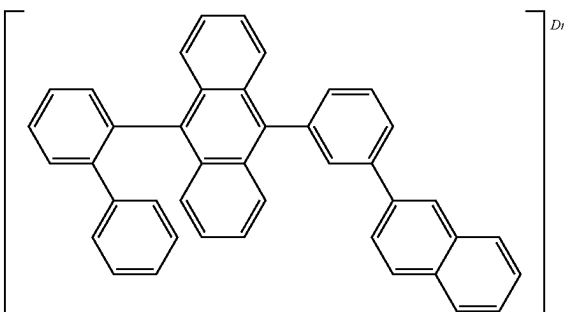
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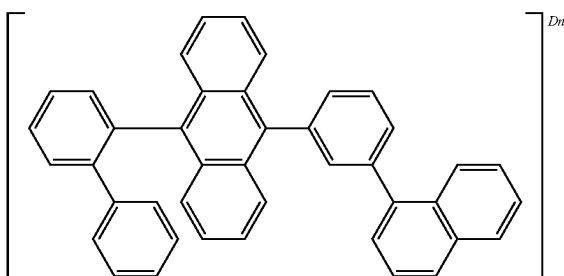
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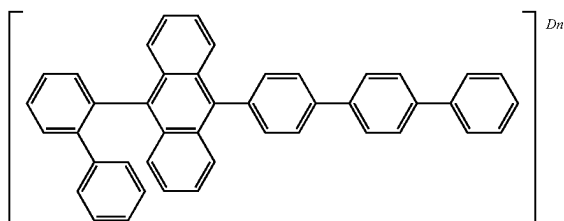
H1-76



H1-77

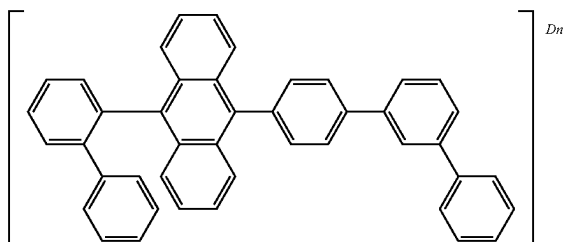


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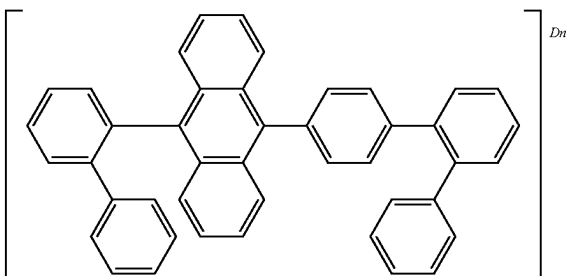


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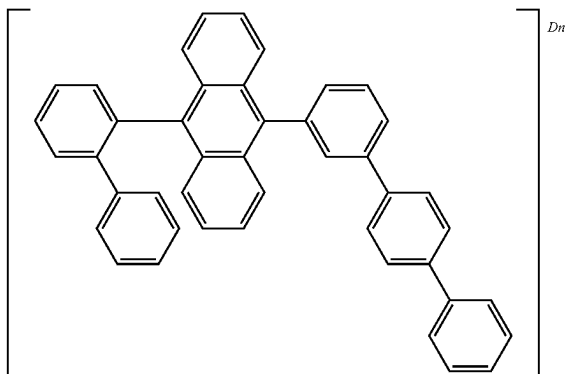
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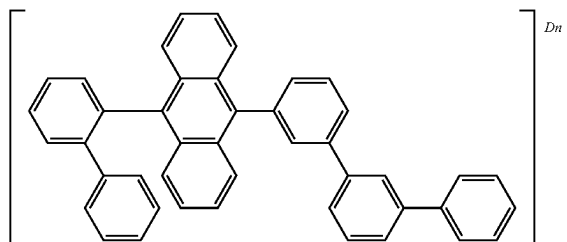
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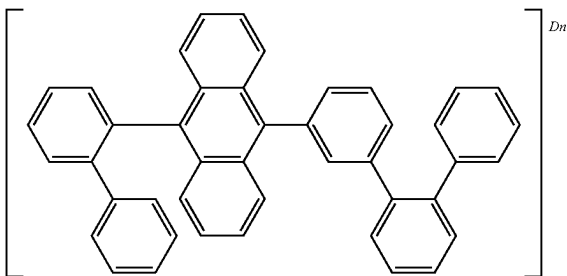
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H1-82

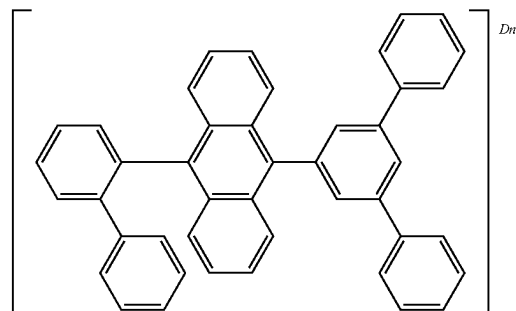


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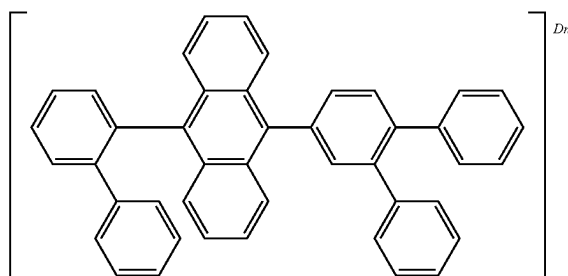


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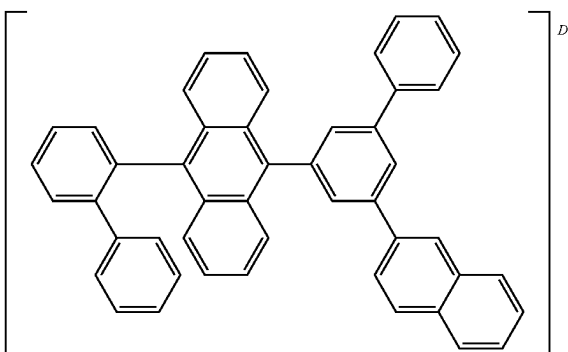
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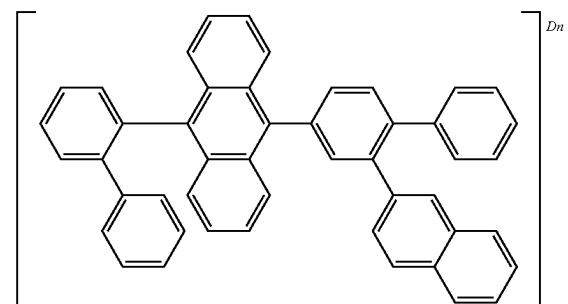
H1-85



H1-86

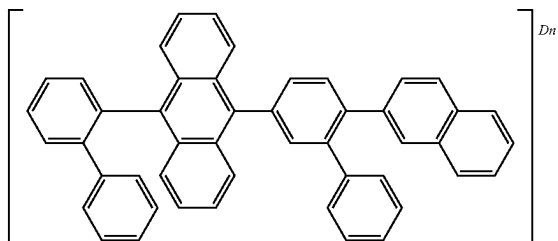


H1-87

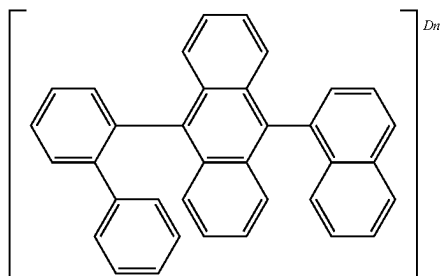


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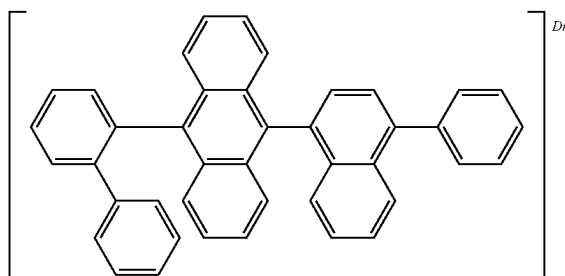
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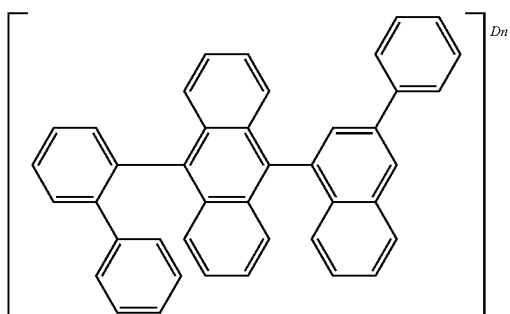
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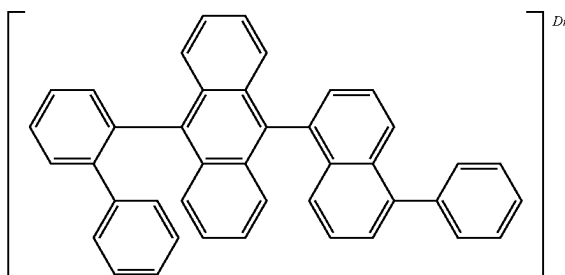
H1-90



H1-91

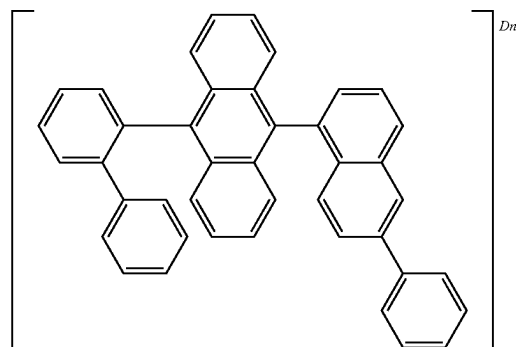


H1-92

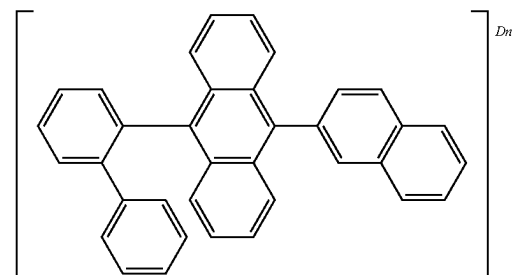


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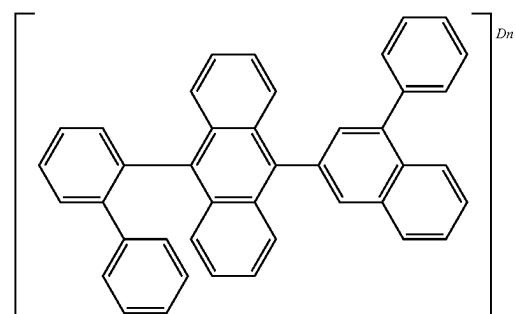
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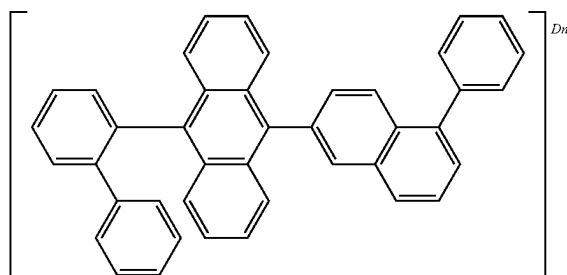
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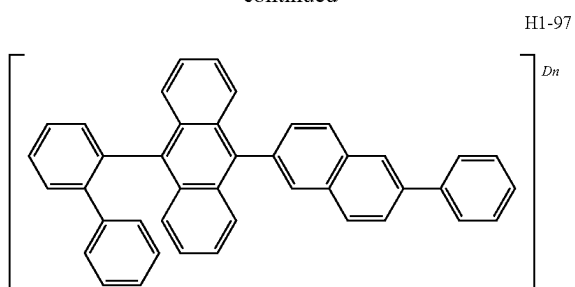
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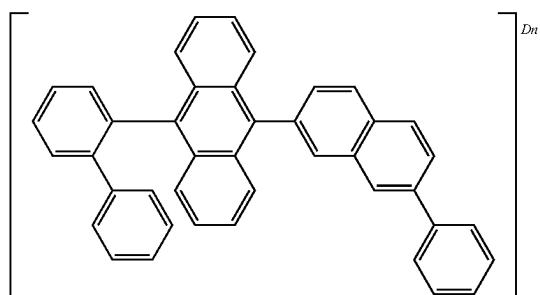
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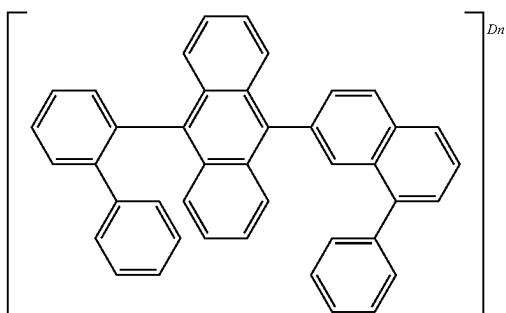
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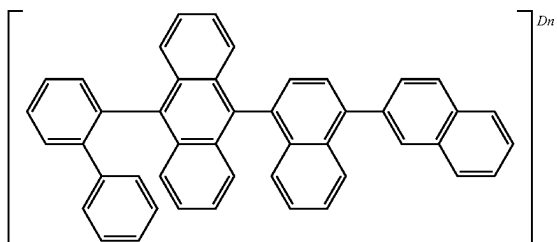
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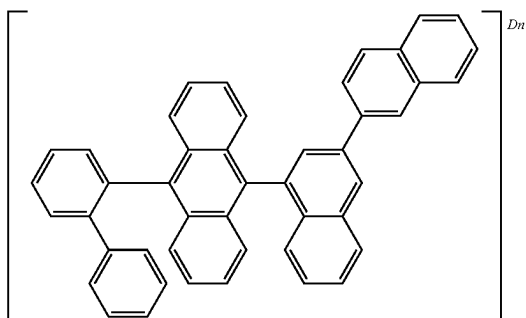
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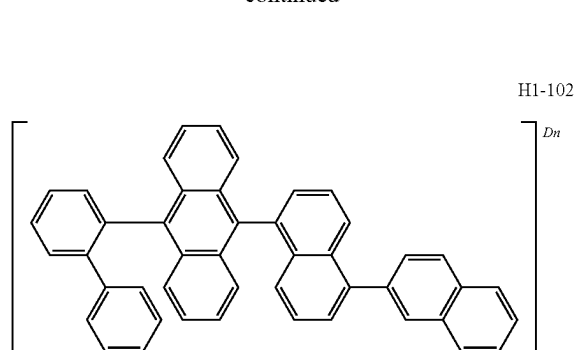
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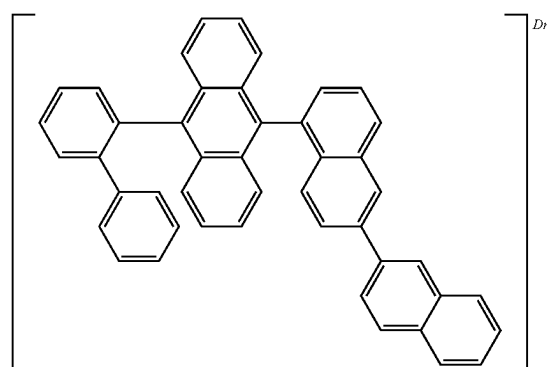
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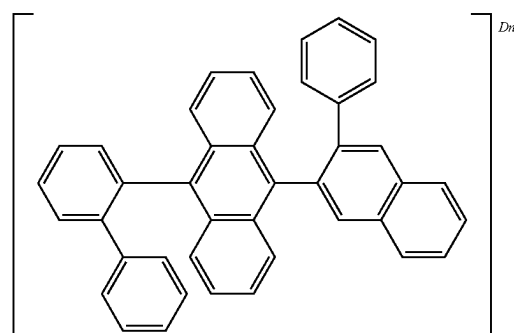
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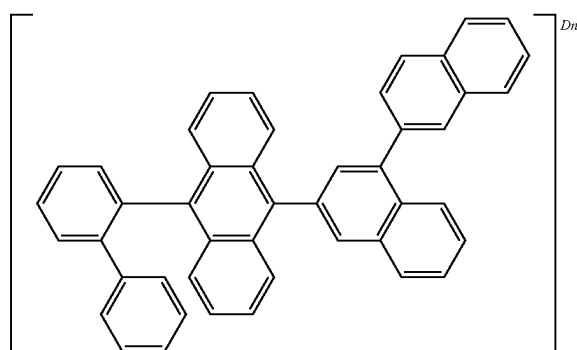
H1-103



H1-104

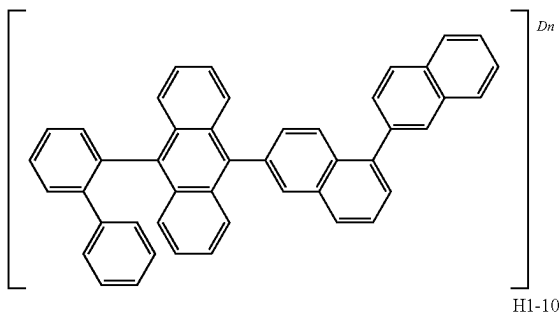


H1-105

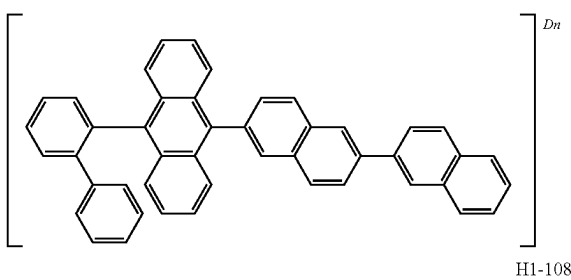


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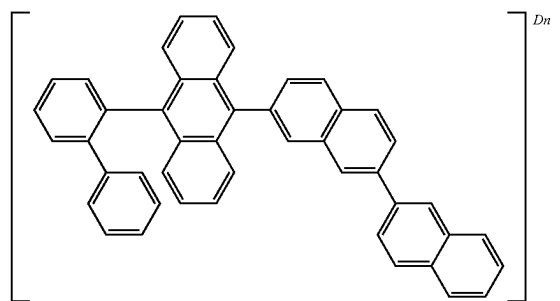
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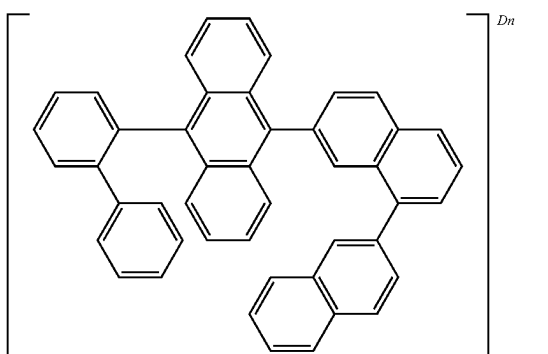
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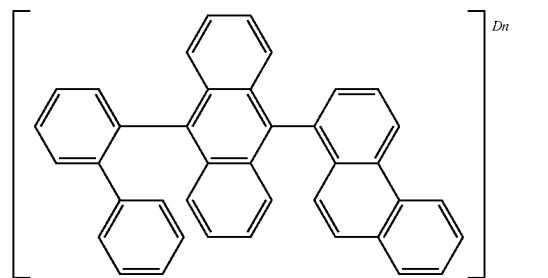
H1-108



H1-109

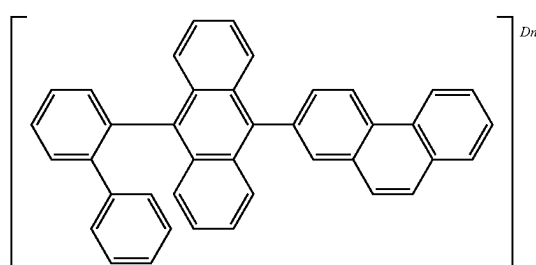


H1-110

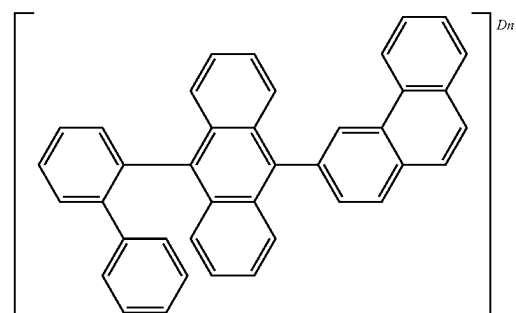


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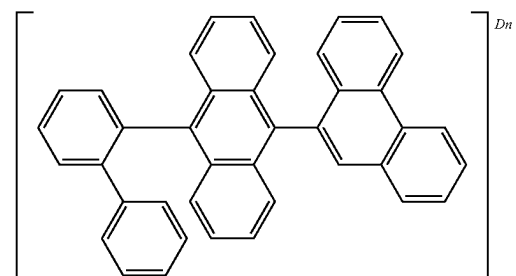
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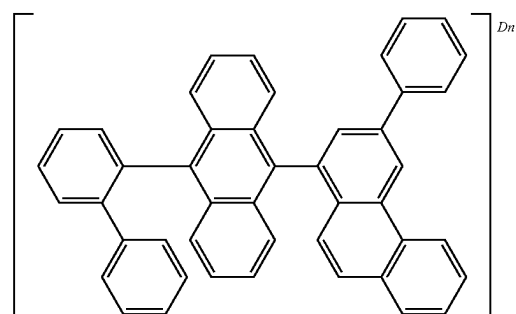
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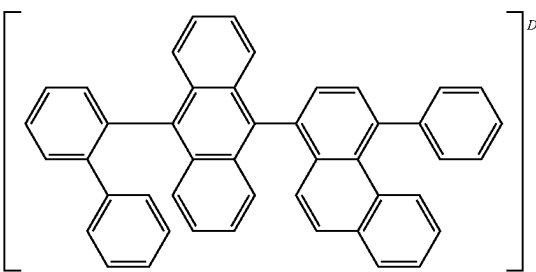
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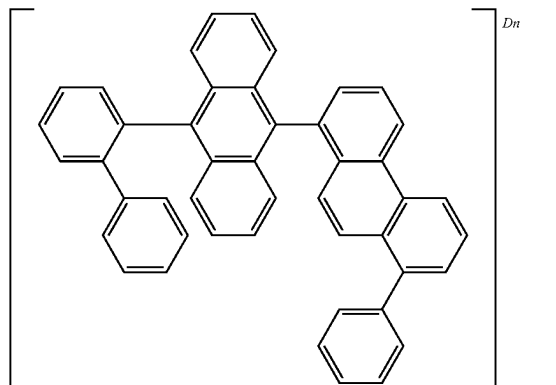
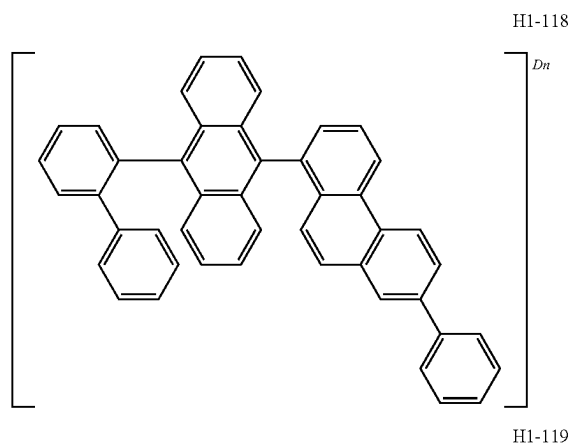
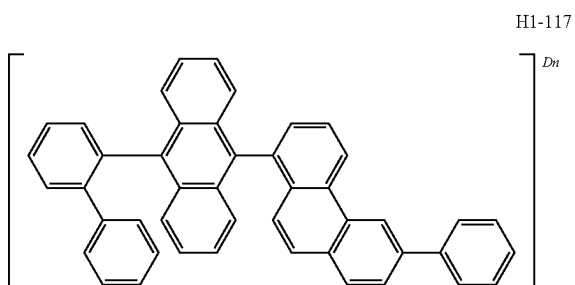
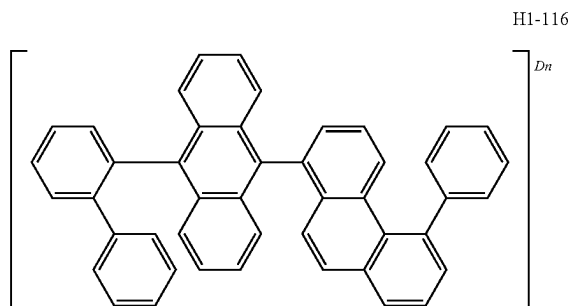
H1-114



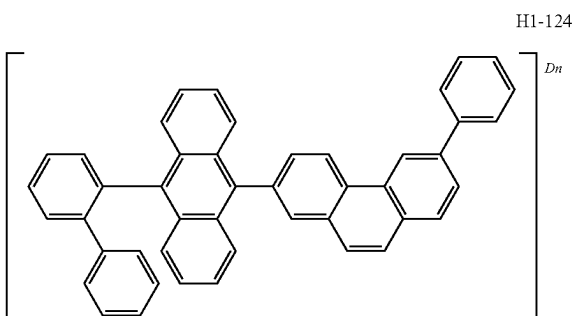
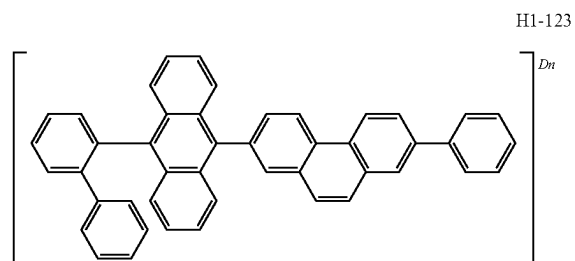
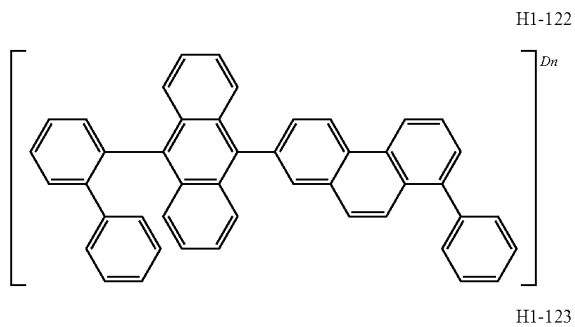
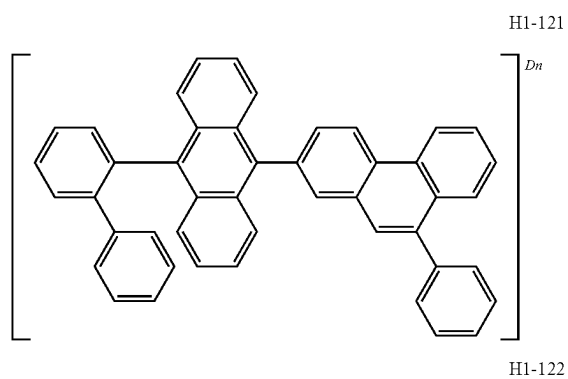
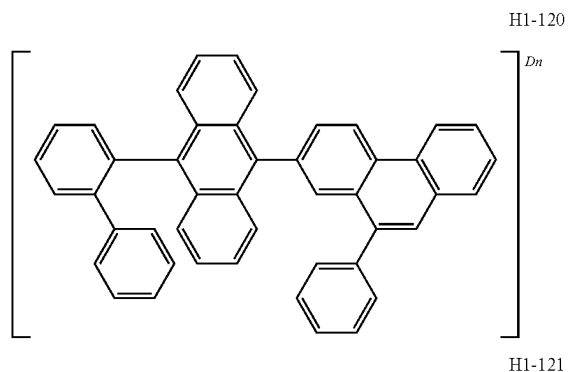
H1-115



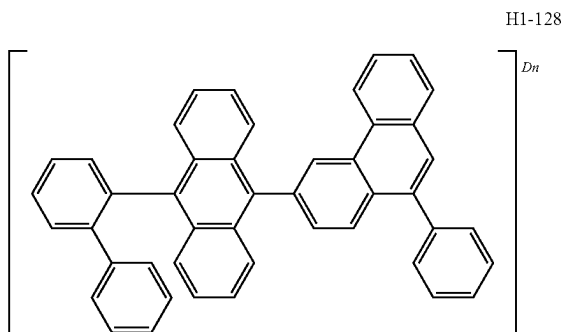
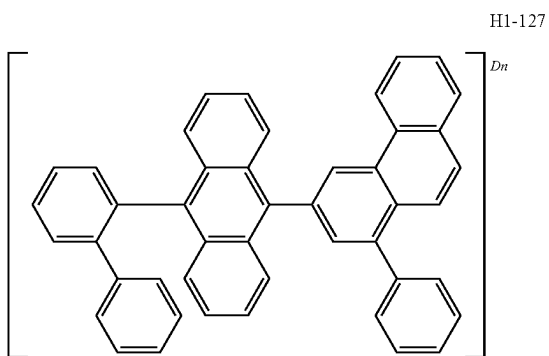
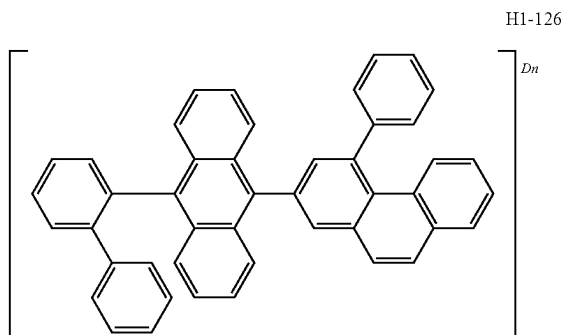
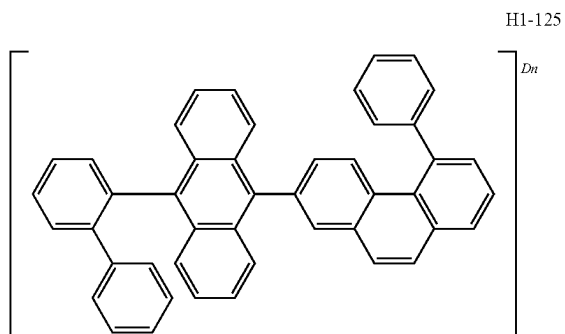
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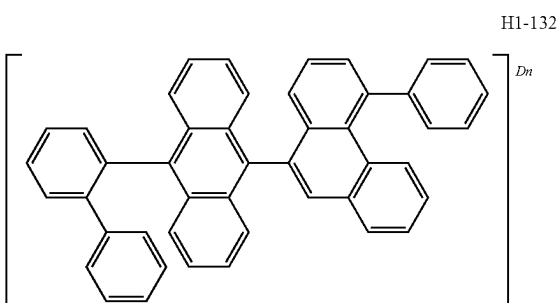
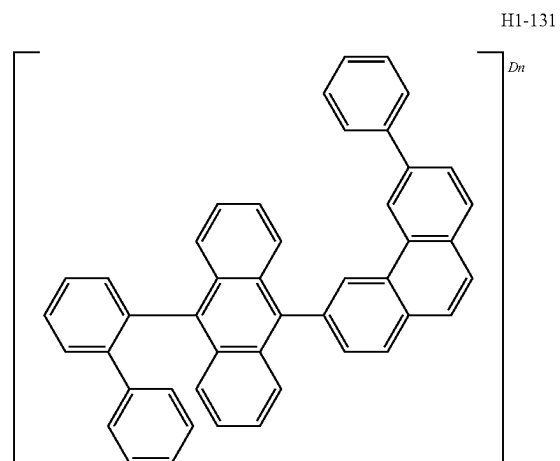
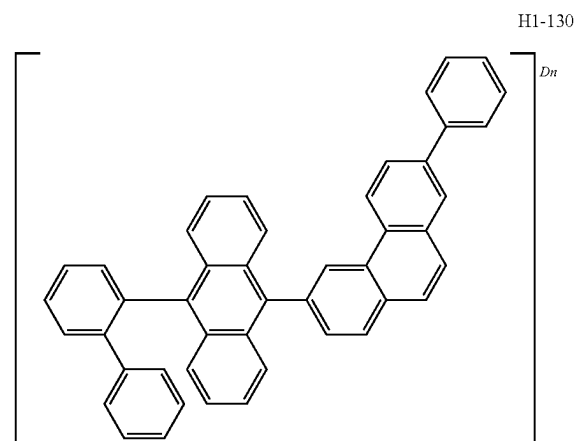
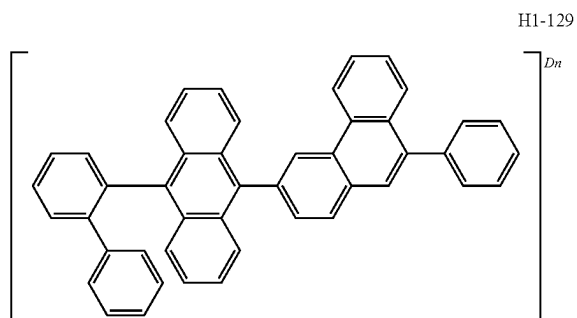
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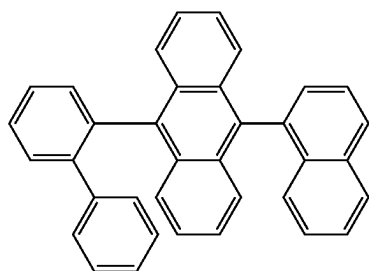
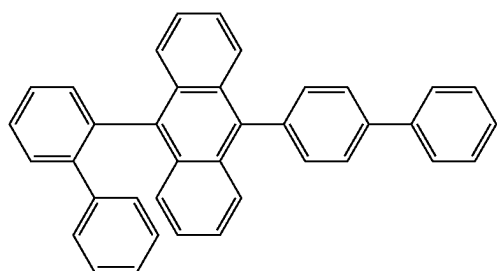
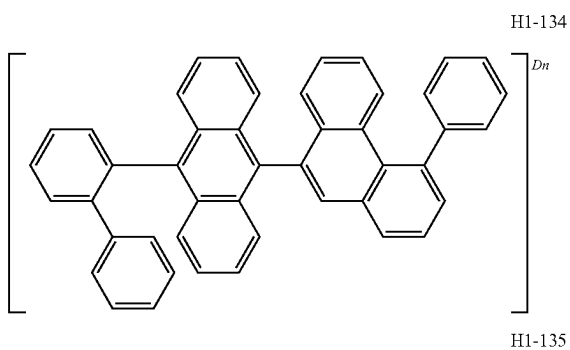
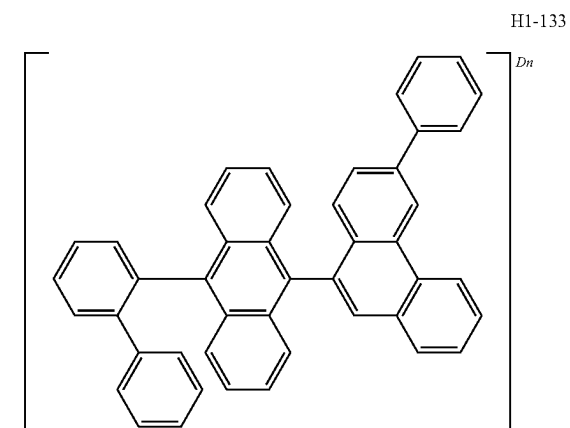
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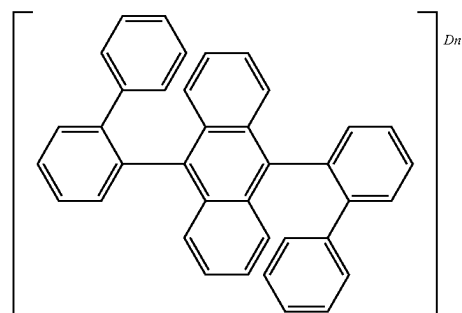
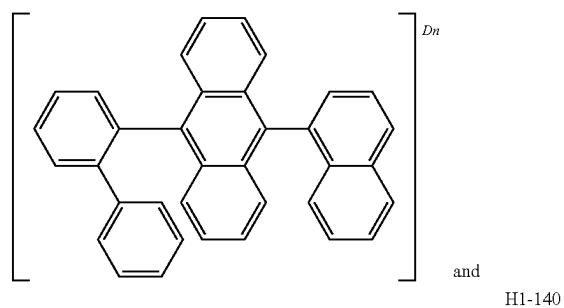
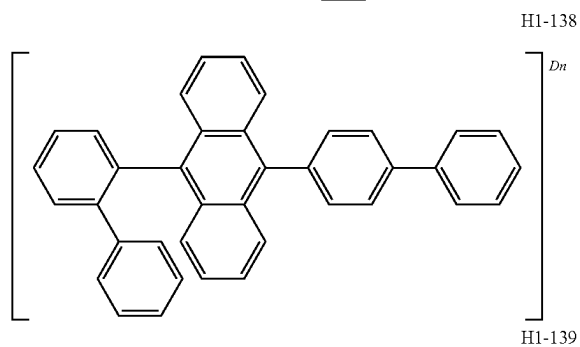
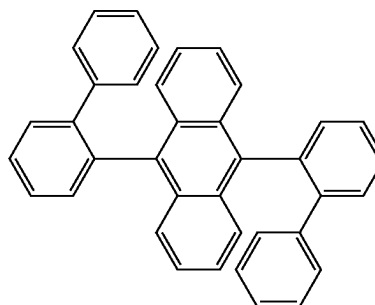
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wherein D_n signifies that n hydrogens are replaced with deuterium, and n is an integer from 1 to the maximum number of hydrogens in the compound.

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